BECOMING A DOCTOR OF

INTERNATIONAL ENERGY AND NATURAL RESOURCES LAW

DOCTORAL TEXT AND ADVANCED PROFESSIONAL REFERENCE



TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

BECOMING A DOCTOR OF INTERNATIONAL ENERGY AND NATURAL RESOURCES LAW

BECOMING A DOCTOR OF

International Energy and Natural Resources Law

A Doctoral Text and Advanced Professional Reference on Resource Sovereignty, Energy Transactions, Investment Protection, Environmental and Climate Governance, and International Dispute Resolution

TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

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DEDICATION

To the scholars, regulators, negotiators, advocates, and arbitrators whose rigorous work turns the world's energy and natural resources into lawful, accountable, and equitable development —

and to the communities whose lands, waters, and futures that work must always serve.

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Foreword

Few fields of law touch the lives of so many people as directly as the law governing energy and natural resources. The electricity that lights a classroom, the fuel that moves goods to market, the minerals inside the telephone in a student's pocket, the gas flared at a wellhead, and the revenues that fund — or fail to fund — schools and hospitals are all shaped by legal rules: by constitutions and statutes, by treaties and contracts, by the decisions of regulators, courts, and arbitral tribunals. Yet the field has long been studied in fragments. Petroleum law, mining law, electricity regulation, environmental law, investment law, and human rights law have often been taught in separate courses and written about in separate literatures, even though, in practice, every significant energy or resource project engages all of them at once.

This book brings these fragments together. It offers an integrated, advanced treatment of international energy and natural resources law, written for those who aspire to master the field at the highest level — doctoral candidates, advanced post-graduate students, academics, practitioners, regulators, judges, arbitrators, and policymakers. Its thirty chapters move from the jurisprudential and institutional foundations of the field through the law of petroleum, gas, and mining; electricity, renewable energy, and emerging technologies; international trade, investment, and contracts; environmental and climate law; human rights and resource justice; comparative law, geopolitics, and dispute resolution; and, finally, the methods of doctoral scholarship, the craft of law reform, and the future of the discipline.

Three features of the work deserve particular mention. First, it is rigorously grounded in primary authority. Throughout, the reader is directed to the treaties, statutes, cases, and awards that constitute the law, and is encouraged to verify every proposition against its source. Second, it is genuinely comparative and international while remaining attentive to Africa, and to Nigeria in particular. The Petroleum Industry Act 2021, the Electricity Act 2023, the Climate Change Act 2021, and the regional frameworks of ECOWAS, the West African Power Pool, and the African Continental Free Trade Area are examined alongside the law of the United States, the United Kingdom, the European Union, the Middle

East, Asia, and Latin America. Third, it is designed for active learning. Each chapter contains learning objectives, case studies, practical exercises — contract drafting, treaty interpretation, regulatory compliance reviews, simulated arbitrations, judicial decision analysis, and legislative drafting — and doctoral research questions. The hypothetical exercises are clearly identified as such, so that readers may practise the skills of the profession without confusing simulation with reality.

The book appears at a moment of profound transformation. The energy transition, the race for critical minerals, the digitalisation of energy systems, the recognition of climate change as a legal and financial risk, the growing insistence on the rights of communities, and the fragmentation of the international order are reshaping every part of the field. Those who master its law will be called upon to help societies navigate these changes wisely and fairly. It is our hope that this book will help prepare them for that responsibility.

We commend this work to its readers with confidence, and with the reminder that it is offered as a scholarly resource. It does not itself confer any academic qualification, and it is not legal advice. Its value will lie in the use its readers make of it — in the rigour of their research, the integrity of their practice, and the quality of the laws and institutions they help to build.

The President/Chancellor
Triune Biblical University Global Extension USA

Preface

Purpose of the Book

This book has been written to provide a comprehensive, advanced, and integrated text on international energy and natural resources law, suitable for doctoral study and for professionals working at the highest levels of the field. Its title — *Becoming a Doctor of International Energy and Natural Resources Law* — reflects its aspiration: to take the reader from a sound foundation in the principles of the field to the frontier of current scholarship, and to equip the reader with the analytical and methodological skills needed to contribute original knowledge to it. The title describes a journey of learning and research, not a credential; as stated throughout, a doctorate can be conferred only by an authorised institution following supervised research and examination.

Scope and Approach

The book addresses the full range of energy and natural resources — oil, natural gas, coal, minerals (including critical and transition minerals), electricity, nuclear energy, renewable energy, hydrogen, carbon capture and storage, water, and marine resources — and the full range of legal disciplines that govern them: public international law, international economic law, environmental and climate law, human rights law, constitutional and administrative law, contract and commercial law, tax and finance law, and the law of dispute resolution. It is organised in eight Parts and thirty chapters, each comprising eight numbered sections, followed by twenty appendices of practical and research materials.

The approach is at once doctrinal, comparative, and contextual. Doctrinal, because a command of the authoritative sources — treaties, legislation, cases, and awards — is the foundation of legal expertise. Comparative, because energy and resource law is practised across jurisdictions, and because much can be learned from the experience of different legal systems. Contextual, because energy and resource law cannot be understood without an appreciation of the economic, technical, political, environmental, and social realities in which it operates, and of the gap that often exists between law in the books and law in action.

The book gives sustained attention to Africa, and to Nigeria in particular, not as an afterthought but as an integral part of the analysis. Africa holds a large share of the world's critical minerals and renewable energy potential and significant hydrocarbon resources; it is home to most of the world's people without access to electricity; and it has been the setting for many of the field's most important legal developments and controversies, from the litigation over oil pollution in the Niger Delta to the reform of mining codes and the creation of continental free trade. Nigeria's recent reforms — the Petroleum Industry Act 2021, the Electricity Act 2023, the Climate Change Act 2021, and associated fiscal and institutional changes — are examined in depth.

Pedagogical Design

Each chapter follows a consistent structure designed to support advanced learning:

- **Learning objectives** identify the knowledge and skills the chapter seeks to develop.
- An **introduction** situates the chapter within the book and the field.
- **Eight numbered sections** provide substantive analysis, drawing on primary authorities and comparative perspectives.
- **Figures and tables** summarise complex frameworks and relationships.
- **Case studies** examine leading cases, awards, and events in depth.
- A **practical exercise** — contract drafting, treaty interpretation, regulatory compliance review, simulated arbitration, judicial decision analysis, legislative drafting, environmental liability assessment, or comparative research — develops professional skills. Exercises involving invented parties or facts are expressly labelled **hypothetical**.
- A **summary** recapitulates the chapter's key points.
- **Doctoral research questions** identify open questions suitable for advanced research.
- **Further reading** directs the reader to authoritative scholarship.

A Note on Currency and Verification

Energy and resource law changes rapidly. The law is stated as understood at the time of writing in 2026. Recent developments — pending legislation, ongoing litigation, arbitral proceedings, and evolving international negotiations — have been described with appropriate caution, and the reader should verify the current status of every authority before relying on it. The book adopts British spelling and, for legal citation, conventions broadly consistent with the Oxford University Standard for the Citation of Legal Authorities (OSCOLA).

A Note on Neutrality

Energy and resource law raises questions on which reasonable people disagree: the pace of the transition away from fossil fuels, the balance between investor protection and the right to regulate, the role of the state in resource industries, and the obligations of companies towards communities. The book seeks to present competing positions fairly and to ground its analysis in law and evidence, while not shrinking from evaluation where the authorities and evidence support it. Readers are encouraged to form their own judgements and to test them against the materials presented.

Acknowledgements

A work of this scope depends on the contributions of many people and institutions, and it is a pleasure to acknowledge them.

We are grateful, first, to the scholars of energy and natural resources law, international law, environmental law, and international economic law whose research forms the foundation of this book. The further reading lists at the end of each chapter and the bibliography record only part of this debt. The work of the international organisations, courts, tribunals, regulators, and research institutes that publish their decisions, reports, data, and analyses openly has also been indispensable; open access to legal materials is a public good on which the study of law depends.

We thank the faculty, research staff, and graduate students of Triune Biblical University Global Extension USA whose discussions, comments, and questions helped shape the structure and content of the book, and whose engagement with earlier versions of its materials improved its clarity and usefulness. We are grateful to the practitioners, regulators, and arbitrators who shared their insights into how energy and resource law operates in practice, and to the colleagues in Nigeria, elsewhere in Africa, and around the world who drew our attention to developments in their jurisdictions.

We acknowledge the editorial, design, and production work that made the book possible: the preparation of the figures and diagrams, the typesetting, the compilation of the tables and indexes, and the careful checking of references.

Finally, we acknowledge the communities living alongside energy and resource projects, whose experiences — of benefit and of harm — give the law examined in this book its human significance, and whose voices have increasingly and rightly shaped its development.

Responsibility for any errors or omissions remains with the author. Readers who identify errors or wish to suggest improvements for future editions are warmly invited to communicate them to the publisher.

About the Author

Triune Biblical University Global Extension USA is the author and publisher of this work. The institution offers programmes of study and research through a global extension model designed to serve learners across multiple countries and regions, including professionals who combine advanced study with careers in government, industry, practice, and civil society.

This book was prepared as part of the institution's commitment to advanced scholarship in law and governance. It was developed as a comprehensive resource for doctoral and advanced postgraduate study in international energy and natural resources law, and for the professional development of practitioners, regulators, policymakers, and researchers working in the field.

The work reflects the institution's emphasis on:

- **Rigour** — grounding analysis in primary legal authorities, careful reasoning, and transparent research methods;
- **Global and comparative perspective** — examining the law of multiple jurisdictions and international regimes, with sustained attention to Africa and the developing world;
- **Practical relevance** — linking doctrine to the skills of drafting, negotiation, regulation, advocacy, and dispute resolution;
- **Integrity** — upholding the highest standards of academic honesty, including the accurate representation of sources and the clear labelling of hypothetical material; and
- **Public responsibility** — recognising that energy and resource law affects the welfare of societies and communities, and that its practitioners and scholars bear corresponding responsibilities.

As stated on the copyright page, this book is an academic resource. It does not by itself confer any academic qualification, and it does not constitute legal advice.

Abbreviations

Table 0.1 Principal abbreviations used in this book.

Abbreviation	Meaning
AfCFTA	African Continental Free Trade Area
AfDB	African Development Bank
AI	Artificial intelligence
AIPN	Association of International Energy Negotiators (formerly Association of International Petroleum Negotiators)
AMV	Africa Mining Vision
AU	African Union
BBNJ	Agreement under UNCLOS on the Conservation and Sustainable Use of Marine Biological Diversity of Areas beyond National Jurisdiction
BIT	Bilateral investment treaty
CBAM	Carbon Border Adjustment Mechanism (EU)
CBD	Convention on Biological Diversity
CBDR-RC	Common but differentiated responsibilities and respective capabilities
CCS / CCUS	Carbon capture and storage / carbon capture, utilisation and storage
CDA	Community development agreement
COP	Conference of the Parties
CRMA	Critical Raw Materials Act (EU)
CSDDD	Corporate Sustainability Due Diligence Directive (EU)
CSRD	Corporate Sustainability Reporting Directive (EU)
DisCo / GenCo	Distribution company / generation company (Nigeria)
ECHR	European Convention on Human Rights
ECOWAS	Economic Community of West African States
ECT	Energy Charter Treaty
ECtHR	European Court of Human Rights
EEZ	Exclusive economic zone

Abbreviation	Meaning
EIA	Environmental impact assessment
EITI	Extractive Industries Transparency Initiative
ESG	Environmental, social and governance
ETS	Emissions trading system
FEED	Front-end engineering design
FET	Fair and equitable treatment
FPIC	Free, prior and informed consent
GATT	General Agreement on Tariffs and Trade
GHG	Greenhouse gas
HCDT	Host Communities Development Trust (Nigeria)
HGA	Host government agreement
IACtHR	Inter-American Court of Human Rights
IAEA	International Atomic Energy Agency
ICC	International Chamber of Commerce
ICJ	International Court of Justice
ICSID	International Centre for Settlement of Investment Disputes
IEA	International Energy Agency
IFC	International Finance Corporation
ILC	International Law Commission
ILO	International Labour Organization
IMO	International Maritime Organization
IOC	International oil company
IPCC	Intergovernmental Panel on Climate Change
IPP	Independent power producer
IRENA	International Renewable Energy Agency
ISA	International Seabed Authority
ISDS	Investor–state dispute settlement
ISSB	International Sustainability Standards Board
ITLOS	International Tribunal for the Law of the Sea
JOA	Joint operating agreement
JETP	Just Energy Transition Partnership
LCIA	London Court of International Arbitration
LNG	Liquefied natural gas
MFN	Most-favoured-nation
MYTO	Multi-Year Tariff Order (Nigeria)
NDC	Nationally determined contribution
NERC	Nigerian Electricity Regulatory Commission

Abbreviation	Meaning
NEITI	Nigeria Extractive Industries Transparency Initiative
NGFS	Network for Greening the Financial System
NMDPRA	Nigerian Midstream and Downstream Petroleum Regulatory Authority
NNPC Ltd	Nigerian National Petroleum Company Limited
NOC	National oil company
NUPRC	Nigerian Upstream Petroleum Regulatory Commission
OECD	Organisation for Economic Co-operation and Development
OPEC	Organization of the Petroleum Exporting Countries
PIA	Petroleum Industry Act 2021 (Nigeria)
PPA	Power purchase agreement
PSC	Production sharing contract
PSNR	Permanent sovereignty over natural resources
RIA	Regulatory impact assessment
SDG	Sustainable Development Goal
SEC	Securities and Exchange Commission (United States)
SPV	Special purpose vehicle
TCFD	Task Force on Climate-related Financial Disclosures
TCN	Transmission Company of Nigeria
UNCITRAL	United Nations Commission on International Trade Law
UNCLOS	United Nations Convention on the Law of the Sea 1982
UNDRIP	United Nations Declaration on the Rights of Indigenous Peoples
UNEP	United Nations Environment Programme
UNFCCC	United Nations Framework Convention on Climate Change
UNGPs	United Nations Guiding Principles on Business and Human Rights
VCLT	Vienna Convention on the Law of Treaties 1969
WAPP	West African Power Pool
WTO	World Trade Organization

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Introduction

The Field

International energy and natural resources law is the body of international, regional, transnational, and national legal rules, principles, institutions, and practices that govern the ownership, exploration, development, production, transformation, transportation, trade, consumption, and conservation of energy and natural resources, and the allocation of the benefits, risks, and burdens that these activities generate. It is not a single, codified body of law. It is, rather, a field defined by its subject matter, drawing on many legal disciplines: public international law (sovereignty, treaties, state responsibility, the law of the sea), international economic law (trade and investment), international environmental and climate law, international human rights law, constitutional and administrative law, contract and commercial law, property law, tax and public finance law, corporate law, and the law of dispute resolution.

The field matters because energy and natural resources are foundational to economic life, central to state revenues in many countries, decisive for the climate and the environment, and deeply entangled with questions of power, conflict, and justice. Its law determines who owns resources and on what terms they may be exploited; how rents are shared between states, investors, and communities; how markets for oil, gas, electricity, and minerals are structured and regulated; how environmental harm is prevented and remedied; how the rights of affected people are protected; how disputes are resolved; and how societies manage the transition from an energy system dominated by fossil fuels to one built on low-carbon sources.

Why This Book, and Why Now

Three developments make an integrated, advanced treatment of the field especially timely.

First, the **energy transition** is transforming the field's subject matter. The law of petroleum and coal, which long dominated the discipline, must now be studied alongside the law of renewable energy, storage, hydrogen, carbon

capture, and critical minerals — and in the shadow of climate obligations that, as the International Court of Justice's 2025 advisory opinion makes clear, bear directly on decisions about fossil fuel production. The questions of managed decline, stranded assets, decommissioning, and just transition have moved from the margins to the centre.

Second, the field's **centre of gravity is shifting**. Most of the growth in energy demand is now in emerging and developing economies; Africa holds much of the world's renewable potential and transition minerals and most of the world's people without electricity; and producer states in Africa, Latin America, and Asia are asserting greater control over their resources and greater value from them. The perspectives and legal experience of these regions demand attention commensurate with their importance.

Third, the field's **legal architecture is in flux**. Investment treaty protection is being reformed, and in some cases abandoned, as illustrated by the withdrawal of the European Union and many of its member states from the Energy Charter Treaty. Trade law is under strain from green industrial policy, carbon border measures, and geopolitical rivalry. Human rights and corporate accountability are being translated into binding due diligence obligations. Climate litigation is expanding. Dispute resolution is being reshaped. At the national level, countries including Nigeria have undertaken fundamental reforms of their petroleum and electricity laws.

Structure of the Book

The book is organised in eight Parts.

Part I — Jurisprudential and Institutional Foundations (Chapters 1–4) establishes the conceptual framework. Chapter 1 examines the nature, scope, and evolution of the field, from the early concessions to the energy transition. Chapter 2 analyses its sources and principles — treaties, custom, general principles, soft law, and the principles of permanent sovereignty, prevention, precaution, sustainable development, and common but differentiated responsibilities. Chapter 3 addresses resource sovereignty, state ownership, and constitutional governance, including models of ownership, the role of national resource companies, and constitutional resource provisions. Chapter 4 examines the international institutions and regulatory systems that govern energy, from OPEC and the IEA to the IAEA, IRENA, and regional bodies.

Part II — Petroleum, Gas, and Extractive Industries Law (Chapters 5–8) treats the law of the extractive industries. Chapter 5 examines upstream petroleum law, licensing, and the principal contractual models. Chapter 6 addresses natural gas law, LNG transactions, and international gas markets. Chapter 7

treats mining law and mineral governance. Chapter 8 examines critical minerals, strategic resources, and supply chain security.

Part III — Electricity, Renewable Energy, and Emerging Technologies (Chapters 9–12) addresses the law of the power sector and new technologies: electricity law and market regulation (Chapter 9), renewable energy law (Chapter 10), nuclear energy law (Chapter 11), and hydrogen, storage, carbon capture, offshore renewables, smart grids, and regulatory sandboxes (Chapter 12).

Part IV — International Trade, Investment, and Energy Contracts (Chapters 13–16) examines the international economic law of energy: trade law (Chapter 13), investment law, including the Energy Charter Treaty and its modernisation (Chapter 14), energy contracts and risk allocation (Chapter 15), and project finance, taxation, and fiscal governance (Chapter 16).

Part V — Environmental Law, Climate Governance, and Resource Sustainability (Chapters 17–20) addresses environmental protection: international environmental law (Chapter 17), climate change law and the energy transition (Chapter 18), water resources law (Chapter 19), and marine resources, offshore energy, and the law of the sea, including deep seabed mining (Chapter 20).

Part VI — Human Rights, Corporate Responsibility, and Resource Justice (Chapters 21–23) examines the human dimension: human rights, indigenous peoples, and extractive industry accountability (Chapter 21); corporate governance, compliance, and anti-corruption (Chapter 22); and energy justice, resource equity, and sustainable development (Chapter 23).

Part VII — Comparative Law, Geopolitics, and International Dispute Resolution (Chapters 24–27) provides comparative perspective (Chapter 24), examines energy geopolitics and security (Chapter 25), and analyses international arbitration (Chapter 26) and litigation, judicial review, and regulatory enforcement (Chapter 27).

Part VIII — Doctoral Scholarship, Legal Reform, and Professional Leadership (Chapters 28–30) turns to the reader's own contribution: advanced research methodology (Chapter 28), law reform, regulatory design, and policy leadership (Chapter 29), and the future of the field (Chapter 30).

The **appendices** provide practical and research materials: selected treaties; model and annotated agreements (petroleum, production sharing, gas sales, renewable power purchase); compliance checklists and frameworks (mining, environmental and social impact assessment, human rights due diligence, climate and transition compliance); risk allocation, finance, and fiscal templates; arbitration clauses and procedural documents; comparative regulatory frameworks; templates and criteria for doctoral proposals and dissertations; a research, citation,

and case analysis guide; selected judicial decisions; drafting and advisory exercises; a terminology glossary; and further research topics.

Distinguishing Overlapping Themes

Some themes recur across chapters, and the book has been designed so that each treatment adds a distinct perspective rather than repeating another. For example, resource sovereignty is examined as a constitutional and ownership question in Chapter 3 and as a matter of investment protection and expropriation in Chapter 14; contractual models are introduced as instruments of upstream regulation in Chapter 5 and analysed as instruments of commercial risk allocation in Chapter 15; investor–state arbitration is examined substantively in Chapter 14 and procedurally in Chapter 26; climate obligations are analysed as a regime in Chapter 18 and as the subject of litigation in Chapter 27; hydropower is treated as a matter of energy regulation in Chapter 10 and of water allocation and transboundary governance in Chapter 19; and environmental principles are introduced in Chapter 17 and applied to marine resources in Chapter 20. Cross-references guide the reader between related treatments.

Africa and Nigeria

Africa, and Nigeria in particular, receive sustained attention throughout. The book examines Nigeria's Petroleum Industry Act 2021 and its institutional, fiscal, and host community provisions; the Electricity Act 2023 and the decentralisation of electricity regulation to the states; the Climate Change Act 2021; the Nigerian Minerals and Mining Act 2007; the litigation over oil pollution in the Niger Delta in Nigerian, English, and Dutch courts and before the African Commission on Human and Peoples' Rights; and the role of NEITI. It examines regional frameworks, including ECOWAS, the West African Power Pool, the African Continental Free Trade Area, and the Africa Mining Vision, and the experience of other African states in mining, petroleum, and renewable energy governance.

A Note on Sources and Hypotheticals

The book is grounded in primary authority. Where it describes treaties, legislation, cases, and awards, it does so on the basis of the authoritative texts and reliable reports, and recent or uncertain developments are described with appropriate caution. The practical exercises frequently use invented states, companies, communities, and disputes to allow readers to practise professional skills. **All such material is expressly labelled hypothetical**; it does not describe real events, and any resemblance to actual persons, companies, or disputes is coincidental.

How to Use This Book

Intended Readers

The book is written for several audiences, each of whom may use it differently.

- **Doctoral candidates and advanced postgraduate students** may use it as a comprehensive foundation for their studies, as a guide to the literature and authorities, and as a source of research questions. Part VIII and Appendices N, O, P, and T are designed specifically to support the design, conduct, and writing of doctoral research.
- **Academics and teachers** may use individual chapters or Parts as the basis for advanced courses and seminars, and the practical exercises as the basis for assessments, simulations, and moot proceedings.
- **Practitioners** — lawyers, in-house counsel, and advisers — may use it as a reference work on the legal frameworks, contractual structures, and dispute resolution mechanisms of the field, and the appendices as checklists and starting points for drafting.
- **Regulators, policymakers, and legislators** may use it to understand comparative approaches to regulation and law reform, particularly through Chapters 4, 9, 24, and 29 and Appendices J and M.
- **Arbitrators and judges** may use it as a guide to the substantive law and procedural practice of energy disputes, particularly through Chapters 14, 15, 26, and 27 and Appendices I and Q.

Reading Paths

The chapters are designed to be read in sequence, building from foundations to frontiers, but each can also be read on its own, with cross-references indicating where related material is treated. Suggested reading paths include:

Table 0.2 Suggested reading paths through the book.

Focus	Suggested chapters
Foundations of the field	1, 2, 3, 4

Focus	Suggested chapters
Upstream oil, gas, and mining	3, 5, 6, 7, 8, 15, 16
Power sector and the energy transition	9, 10, 11, 12, 18, 23, 29
International economic law of energy	13, 14, 15, 16, 26
Environment, climate, and sustainability	2, 17, 18, 19, 20, 27
Human rights and accountability	21, 22, 23, 27
Africa and Nigeria	3, 5, 7, 9, 21, 23, 24, 29
Disputes and litigation	14, 15, 26, 27
Doctoral research	28, 29, 30 and Appendices N–P, T

Features of Each Chapter

Learning objectives at the start of each chapter identify what the reader should be able to do after studying it. Readers may find it helpful to return to them after completing the chapter as a check on their understanding.

Numbered sections (for example, 14.3) allow precise cross-referencing. Each chapter has eight sections.

Figures and tables summarise frameworks, processes, and comparisons. They are listed in the List of Figures and are numbered by chapter.

Case studies, set apart in shaded boxes, examine leading cases, awards, and events in depth. Readers are encouraged to read the primary sources — the judgments and awards themselves — alongside the case studies.

Comparative and practice boxes highlight the experience of particular jurisdictions and points of professional practice.

Practical exercises develop professional skills. They vary by subject: contract drafting, treaty interpretation, regulatory compliance reviews, simulated arbitration proceedings, judicial decision analysis, legislative drafting, environmental liability assessments, and comparative legal research. Most exercises use invented parties and facts, and **every such exercise is labelled hypothetical**. Exercises may be completed individually or in groups; many are suitable for role-play and moot formats.

Summaries recapitulate the chapter's key points and may be used for revision.

Doctoral research questions identify open questions suitable for advanced research. They are intended as starting points: a viable doctoral question must be refined through a review of the literature, consideration of method and sources, and discussion with supervisors (see Chapter 28 and Appendix N).

Further reading lists direct the reader to authoritative scholarship and primary sources. Where an edition may have been updated since writing, readers should consult the latest edition.

Reference Apparatus

The front matter contains tables of treaties and conventions, cases and arbitral awards, and statutes and regulatory instruments, so that readers can locate the treatment of particular authorities; to avoid duplication, these tables are not repeated at the end of the book. The back matter contains the bibliography and indexes of legal concepts, of institutions and organisations, and a subject index.

Citation and Verification

Citations broadly follow the Oxford University Standard for the Citation of Legal Authorities (OSCOLA). The law is stated as understood in 2026. Because energy and resource law changes rapidly, **readers must verify the current status of every authority before relying on it** — checking for amendments, repeals, appeals, annulments, and subsequent developments — using official sources and reliable databases. Appendix P offers guidance on legal research and verification.

Important Notices

This book does not confer a doctorate or any other academic qualification. It is a scholarly resource designed to support advanced study and research. Doctoral standing can be achieved only through enrolment in a duly authorised programme, supervised original research, and successful examination in accordance with the rules of the awarding institution.

This book is not legal advice. It analyses law and policy for purposes of education and research. Readers facing a specific legal question should consult a qualified practitioner in the relevant jurisdiction.

Hypothetical material is fictitious. Invented states, companies, communities, and disputes used in exercises do not describe real events, and any resemblance to actual persons, companies, or disputes is coincidental.



PART I

Jurisprudential and Institutional Foundations

The conceptual architecture of the discipline: its definition and history, its sources and principles, the doctrine of permanent sovereignty over natural resources, and the institutions through which global energy governance is organised.

1 CHAPTER

The Nature, Scope, and Evolution of International Energy and Natural Resources Law

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Define international energy and natural resources law as a field of legal study and distinguish it from adjacent fields, including public international law, international economic law, and international environmental law.
2. Trace the historical development of resource governance from the early concession era through decolonisation, the New International Economic Order, liberalisation, and the contemporary energy transition.
3. Explain how the internationalisation of energy markets has transformed the regulation of exploration, production, transportation, and trade.
4. Analyse the interaction between public law, private law, and transnational legal relations in energy and resource projects.
5. Evaluate the significance of legal pluralism and competing regulatory authorities in global resource governance.
6. Identify the principal foundational debates that structure doctoral research in the field.

Introduction

Energy and natural resources sit at the junction of almost every question that modern legal systems are asked to answer. Who owns what lies beneath the soil and the seabed? Under what conditions may a sovereign state grant rights to explore for and extract those resources, and on what terms may it later change its mind? How should the economic rent produced by a non-renewable asset be shared between the state, the investor, the host community, and future generations? What obligations attach to the transport of oil and gas across national frontiers, to the generation and trade of electricity, or to the emissions released

when fossil fuels are burned? How should disputes between powerful states and powerful corporations be resolved, and by whom? Each of these questions is, at bottom, a legal question, and the body of law that has developed to answer them is the subject of this book.

International energy and natural resources law is a young discipline in name but an old one in substance. The legal instruments through which states have granted, regulated, taxed, and expropriated mineral and petroleum rights have a history that stretches back more than a century, and the doctrines that frame those transactions — sovereignty, contract, property, state responsibility, and the protection of aliens — are older still. What is new is the recognition that these instruments and doctrines constitute a coherent field of study, with its own sources, principles, institutions, and methods, and with a distinctive set of intellectual problems that cannot be adequately addressed from within any single traditional legal category.

This opening chapter lays the foundation for the remainder of the book. It begins by defining the field and identifying its boundaries. It then traces the historical development of resource governance, explains the internationalisation of energy markets and regulation, and examines the interplay between public law, private law, and transnational legal relations. It considers the relationship between energy law and international economic law, the phenomenon of legal pluralism, and the contemporary challenges confronting global resource governance. The chapter closes by setting out the foundational debates that animate doctoral research and that recur throughout the chapters that follow.

The approach throughout is analytical rather than descriptive. A doctoral candidate in this field must be able to do more than recite the content of treaties and statutes; he or she must be able to explain why the law takes the form that it does, to identify the tensions and contradictions within it, to evaluate competing interpretations, and to propose reasoned reforms. The chapter therefore emphasises the conceptual architecture of the field — the ideas that hold it together and the fault lines along which it is contested.

1.1 Defining Energy and Natural Resources Law

The Problem of Definition

Any attempt to define a field of law must confront a familiar difficulty: legal categories are not natural kinds. They are constructed by scholars, practitioners, courts, and legislators for particular purposes, and their boundaries shift as those purposes change. “Energy law” and “natural resources law” are no exception. Neither phrase appears as a term of art in any multilateral treaty of general application, and neither corresponds to a single body of legislation in most domestic legal systems. Instead, each describes a cluster of rules, principles, institutions, and

practices that share a common subject matter — the ownership, development, transformation, trade, and use of energy sources and natural resources — and that are united by a common set of policy concerns.

A useful working definition, adopted for the purposes of this book, is the following:

> International energy and natural resources law is the body of rules, principles, institutions, and practices — derived from public international law, international economic law, comparative domestic law, and transnational commercial practice — that governs the allocation of rights over energy sources and natural resources, the conduct of activities involving their exploration, development, transformation, transport, trade, and use, and the distribution of the benefits and burdens arising from those activities, in situations that involve more than one legal order.

Several features of this definition deserve comment. First, it is *functional* rather than formal: it defines the field by reference to its subject matter and the problems it addresses, not by reference to a particular source of law. Second, it is *pluralist*: it expressly acknowledges that the relevant rules derive from multiple sources, including domestic legislation and private contracts as well as treaties and custom. Third, it is *distributive*: it identifies the allocation of benefits and burdens as a central concern, reflecting the fact that energy and resource law is fundamentally about who gets what, when, and how. Fourth, it is *transnational*: the field is concerned with situations that involve more than one legal order, whether because a resource straddles a boundary, because an investor is foreign, because a commodity is traded internationally, or because environmental effects cross frontiers.

Energy Law and Natural Resources Law: One Field or Two?

The conjunction "energy and natural resources" requires justification. Energy law, narrowly understood, is concerned with the production, conversion, transmission, distribution, and consumption of energy, whether derived from fossil fuels, nuclear fission, or renewable sources. Natural resources law, narrowly understood, is concerned with the ownership, management, and exploitation of resources found in nature, including minerals, petroleum, water, forests, fisheries, and biological diversity. The two fields overlap substantially — petroleum and natural gas are both natural resources and energy sources — but neither is wholly contained within the other. Renewable electricity generated from wind or solar radiation is an energy source but is not usually described as a natural resource in the extractive sense; copper and lithium are natural resources but are not themselves sources of energy, although they are indispensable to its transmission and storage.

There are strong reasons for treating the two fields together. Historically, the law governing petroleum developed out of the law governing hard-rock minerals, and many of the foundational instruments — concessions, licences, royalties, and the doctrine of permanent sovereignty — are common to both. Functionally, the energy transition is collapsing the distinction: the decarbonisation of energy systems depends on the extraction of critical minerals on an unprecedented scale, on access to land and water for renewable installations, and on the use of geological formations for the storage of carbon dioxide and hydrogen. Institutionally, the same ministries, regulators, and courts often have jurisdiction over both energy and minerals. Doctrinally, the same principles of international law — sovereignty over natural resources, the prohibition of transboundary harm, the protection of foreign investment, and the rights of indigenous peoples — govern both.

This book therefore treats energy law and natural resources law as a single integrated field, while recognising that particular chapters will emphasise one or the other. The integrated approach has the further advantage of exposing the connections between subjects that are too often studied in isolation: between petroleum fiscal regimes and renewable energy incentives, for example, or between mining law and climate law.

Relationship with Adjacent Fields

The field overlaps with several established branches of law, and a doctoral candidate must be able to locate it precisely in relation to each.

Public international law. The general rules of public international law — on sources, treaties, state responsibility, jurisdiction, immunity, and dispute settlement — supply the framework within which international energy and resource law operates. The doctrine of permanent sovereignty over natural resources, discussed in Chapter 2, is a principle of general international law. The law of the sea, international watercourse law, and international environmental law are specialised branches of public international law with direct application to resource activities.

International economic law. The law of international trade, investment, finance, and taxation governs the cross-border movement of energy commodities, capital, and revenues. The World Trade Organization (WTO) agreements apply to trade in energy goods and, to a more limited extent, services; bilateral and multilateral investment treaties protect foreign investors in the energy and mining sectors; and international tax rules shape the fiscal treatment of resource projects. The relationship between energy law and international economic law is examined in section 1.5 below.

International environmental and climate law. Resource activities are among the most significant sources of environmental harm, and international

environmental law imposes obligations of prevention, precaution, assessment, and cooperation that constrain the manner in which they may be carried out. Climate change law, in particular, is reshaping energy law by setting long-term objectives for the reduction of greenhouse gas emissions that are incompatible with the indefinite expansion of fossil fuel production.

Comparative and domestic law. Most of the rules that govern energy and resource activities on a day-to-day basis are domestic: constitutional provisions on resource ownership, petroleum and mining statutes, electricity regulation, environmental legislation, and tax codes. International energy law cannot be understood without close attention to these domestic regimes and to the patterns of convergence and divergence among them, which are explored in Chapter 24.

Transnational commercial law. Energy and resource projects are structured through complex networks of private contracts — joint operating agreements, sales and purchase agreements, engineering and construction contracts, financing agreements — that are frequently governed by English or New York law and subject to international commercial arbitration. The model contracts and standard terms produced by industry associations constitute a form of transnational private ordering that interacts with, and sometimes substitutes for, state regulation.

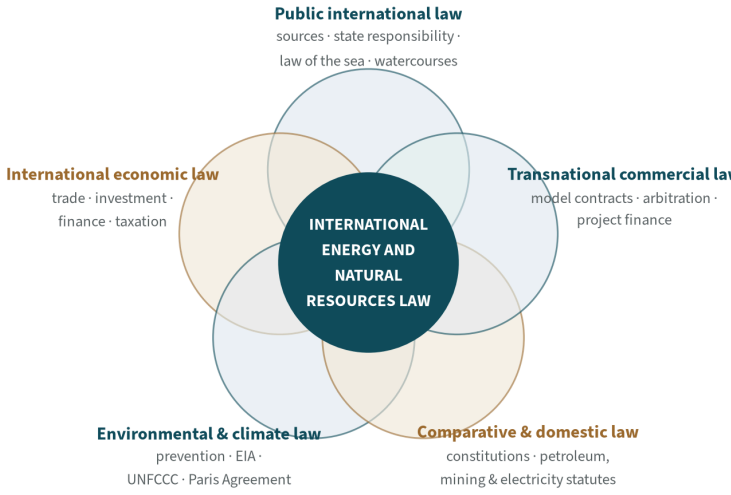


Figure 1.1 The constituent fields of international energy and natural resources law. The discipline is defined by its subject matter and draws on five overlapping bodies of law. *Source:* prepared for this edition.

The Scope of the Field

The scope of the field, as treated in this book, may be described along three axes.

The first axis is the *resource or energy source* concerned: petroleum (crude oil and natural gas), coal, hard-rock minerals including critical minerals, uranium and nuclear energy, renewable energy sources (solar, wind, hydropower, geothermal, biomass, marine), hydrogen and other energy carriers, electricity, water, and marine resources.

The second axis is the *stage of the value chain*: the allocation of rights to explore and develop resources (upstream), the transport and processing of those resources (midstream), and their sale, distribution, and consumption (downstream), together with the decommissioning of installations and the remediation of affected environments at the end of the project life cycle.

The third axis is the *legal function*: the allocation of property and access rights, the regulation of conduct, the protection of investment, the allocation of commercial risk by contract, the raising and distribution of revenue, the protection of the environment and human rights, and the resolution of disputes.

The chapters of this book are organised principally along the third axis, while particular chapters address specific resources (Chapters 5 to 12) and specific legal functions (Chapters 13 to 27).

Is There a "Lex Petrolea"?

One of the most contested questions in the definition of the field concerns the existence of a distinct transnational legal order governing the international petroleum industry — sometimes described as the *lex petrolea*. The idea has a long pedigree. In the arbitral award in *Kuwait v American Independent Oil Company (Aminoil)*, the tribunal referred to the existence of a body of practice in the oil industry that informed the interpretation of the concession, while stopping short of endorsing an autonomous legal system.[1] Proponents of the *lex petrolea* argue that the repeated use of standard contractual forms, the consistent practice of industry participants, the jurisprudence of arbitral tribunals, and the model contracts of bodies such as the Association of International Energy Negotiators (AIEN) have generated a body of transnational norms that operates partly independently of any national legal system.[2]

Critics respond that the concept is either trivially true — industry practice obviously influences the interpretation of industry contracts — or dangerously

[1] *Government of the State of Kuwait v American Independent Oil Company (Aminoil)*, Award of 24 March 1982, (1982) 21 ILM 976.

[2] See R. Doak Bishop, "International Arbitration of Petroleum Disputes: The Development of a *Lex Petrolea*" (1998) 23 *Yearbook of Commercial Arbitration* 1131; and Thomas C.C. Childs, "Update on *Lex Petrolea*: The Continuing Development of Customary Law Relating to International Oil and Gas Exploration and Production" (2011) 4 *Journal of World Energy Law and Business* 214.

overstated, in that it purports to detach contractual relations from the legal systems that ultimately give them force, and that it tends to privilege the interests of international oil companies over those of host states and affected communities. The debate is significant for doctoral researchers because it raises fundamental questions about the nature of law, the legitimacy of private ordering, and the relationship between state and non-state sources of normativity. It is revisited in Chapter 2 in the context of soft law and industry standards and in Chapter 26 in the context of arbitral practice.

1.2 Historical Development of Resource Governance

A historical perspective is indispensable to an understanding of contemporary energy and resource law. Many of the field's defining doctrines — the stabilisation clause, the doctrine of permanent sovereignty, the standard of compensation for expropriation, the treatment of production sharing contracts — were forged in specific historical controversies, and their meaning cannot be grasped without appreciating the circumstances in which they arose. This section traces five broad phases in the development of resource governance.

Phase One: The Classical Concession Era (c. 1900–1950)

The modern international petroleum industry was built on the classical concession. In 1901, William Knox D'Arcy obtained from the Shah of Persia a concession conferring the exclusive right to explore for, produce, and export petroleum over approximately three-quarters of the territory of Persia for a period of sixty years, in return for a modest initial payment and a share of annual net profits. The D'Arcy concession became the template for a generation of agreements across the Middle East, Latin America, and Asia.

The classical concession had several characteristic features. It covered vast areas, often a substantial proportion of the host state's territory. It endured for very long periods, frequently between sixty and ninety-nine years. It conferred upon the concessionaire not merely the right to extract petroleum but ownership of the petroleum once produced, together with broad managerial control over all aspects of operations. The host state's financial return was limited, typically to a fixed royalty per ton of production, and the state had little or no involvement in decisions concerning the rate of production, the pricing of crude oil, or the allocation of markets. The concessions were frequently negotiated between unequal parties, in circumstances in which the host state lacked the geological information, technical expertise, and bargaining power necessary to secure favourable terms.

The legal character of these instruments was ambiguous. From the perspective of the international oil companies and their home states, a concession was a contract that created vested rights protected by international law, and any unilateral modification or termination by the host state constituted a wrong for which

compensation was owed. From the perspective of host states, the concession was an act of sovereign authority — an administrative grant — that the state could revise in the public interest. This tension between contract and sovereignty is one of the enduring themes of the field.

During this period the industry was dominated by a small number of vertically integrated companies — later known as the "Seven Sisters" — which controlled exploration, production, refining, transportation, and marketing. Arrangements such as the Red Line Agreement of 1928, under which the partners in the Iraq Petroleum Company agreed not to compete with one another for concessions within a defined area of the former Ottoman Empire, and the Achnacarry or "As-Is" Agreement of the same year, which sought to stabilise market shares, illustrate the degree to which the international oil market was organised through private arrangements rather than public regulation.

Phase Two: Nationalisation and the Assertion of Sovereignty (1938–1960s)

The first major challenge to the concession system came from Latin America. In 1938, Mexico nationalised the holdings of foreign oil companies following a prolonged labour dispute and a confrontation with the companies over compliance with a decision of the Mexican Supreme Court. The Mexican nationalisation prompted a celebrated exchange of diplomatic notes in which the United States Secretary of State, Cordell Hull, asserted that international law required "prompt, adequate, and effective" compensation for the expropriation of foreign property — the formula that became known as the Hull standard. Mexico rejected the claim that international law imposed any such requirement, maintaining that compensation was a matter for domestic law. The dispute was eventually settled by agreement, but the doctrinal disagreement it exposed persisted for decades.

The second major challenge came in Iran. In 1951, the Iranian parliament enacted legislation nationalising the assets of the Anglo-Iranian Oil Company. The United Kingdom instituted proceedings before the International Court of Justice, but the Court held that it lacked jurisdiction because the relevant Iranian declaration accepting the Court's compulsory jurisdiction was limited to disputes concerning treaties, and the concession of 1933 was not a treaty between Iran and the United Kingdom.[3] The *Anglo-Iranian Oil Co* case is significant for its holding that a concession agreement between a state and a foreign company is not an international treaty, notwithstanding the involvement of the company's home state in its negotiation.

[3] *Anglo-Iranian Oil Co (United Kingdom v Iran)*, Preliminary Objection, Judgment, ICJ Reports 1952, p. 93.

The same period witnessed the gradual replacement of fixed royalty arrangements with profit-sharing formulas. Beginning with Venezuela in 1943 and spreading to Saudi Arabia in 1950, the so-called "fifty-fifty" principle — under which the host government received approximately half of the profits from petroleum operations through a combination of royalties and income taxes — became the industry norm. The fifty-fifty principle represented a significant shift in the distribution of resource rents, but it left the structure of the concession — and the managerial control of the companies — largely intact.

Phase Three: Decolonisation and the New International Economic Order (1960–1980)

The third phase was shaped by decolonisation and the emergence of a large number of newly independent states determined to assert control over their natural resources. Three developments were decisive.

First, in September 1960, Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela founded the Organization of the Petroleum Exporting Countries (OPEC). OPEC's initial objective was to resist reductions in posted prices unilaterally imposed by the companies, but it rapidly became a forum through which producing states coordinated their policies on pricing, taxation, and participation.

Second, the General Assembly of the United Nations adopted a series of resolutions articulating the principle of permanent sovereignty over natural resources. The most important of these, Resolution 1803 (XVII) of 14 December 1962, declared that "the right of peoples and nations to permanent sovereignty over their natural wealth and resources must be exercised in the interest of their national development and of the well-being of the people of the State concerned", while also providing that in cases of nationalisation "the owner shall be paid appropriate compensation, in accordance with the rules in force in the State taking such measures in the exercise of its sovereignty and in accordance with international law", and that foreign investment agreements "freely entered into by or between sovereign States shall be observed in good faith".[4] Resolution 1803 represented a carefully negotiated compromise between capital-exporting and capital-importing states, and it is widely regarded as reflecting customary international law in significant respects.

Third, in the early 1970s, producing states moved from profit-sharing to participation and outright nationalisation. Libya nationalised the interests of several foreign companies between 1971 and 1974; Algeria took a majority interest in French oil companies in 1971; Iraq nationalised the Iraq Petroleum Company in 1972; and the states of the Persian Gulf negotiated progressive participation

[4] UN General Assembly Resolution 1803 (XVII), "Permanent Sovereignty over Natural Resources", 14 December 1962, paras 1, 4, and 8.

agreements that culminated in full state ownership of their producing industries. The oil price increases of 1973–1974, associated with the Arab oil embargo and OPEC's assertion of control over pricing, transformed the economics of the industry and the balance of power between producing states and consuming states.

In 1974, the General Assembly adopted the Declaration on the Establishment of a New International Economic Order and the Charter of Economic Rights and Duties of States.[5] Article 2 of the Charter asserted the right of every state "to nationalize, expropriate or transfer ownership of foreign property, in which case appropriate compensation should be paid by the State adopting such measures, taking into account its relevant laws and regulations and all circumstances that the State considers pertinent", and provided that any controversy over compensation "shall be settled under the domestic law of the nationalizing State and by its tribunals", unless otherwise agreed. Unlike Resolution 1803, the Charter omitted any reference to international law as the governing standard for compensation and was adopted over the opposition or abstention of most capital-exporting states. Its status as a statement of customary international law has accordingly been doubted.

The legal consequences of the nationalisations of this period were addressed in a series of landmark arbitral awards, often referred to collectively as the "Libyan oil arbitrations" — *BP v Libya* (1973), *Texaco v Libya* (1977), and *LIAMCO v Libya* (1977) — together with the *Aminoil* award of 1982. These awards, which are examined in detail in Chapters 3 and 14, grappled with the legal status of concession contracts, the effect of stabilisation clauses, the applicable law, and the appropriate standard of compensation. Their reasoning remains foundational to the modern law of foreign investment.

[5] UN General Assembly Resolution 3201 (S-VI), "Declaration on the Establishment of a New International Economic Order", 1 May 1974; UN General Assembly Resolution 3281 (XXIX), "Charter of Economic Rights and Duties of States", 12 December 1974.



Figure 1.2 Selected milestones in the development of international energy and natural resources law, 1901–2025. *Source:* prepared for this edition.

Phase Four: Liberalisation, Investment Protection, and Market Integration (1980–2005)

The fourth phase was characterised by a reversal of the statist tendencies of the preceding period. The collapse of oil prices in the mid-1980s, the debt crisis in the developing world, the end of the Cold War, and the ascendancy of market-oriented economic policies led many states to reopen their resource sectors to foreign investment. Production sharing contracts — first used in Indonesia in the 1960s — became the dominant contractual form in many developing producer states, allowing governments to retain formal ownership of petroleum while engaging foreign companies as contractors entitled to a share of production. Mining codes across Africa and Latin America were rewritten, often with the support of international financial institutions, to attract foreign capital through security of tenure, fiscal stability, and liberal repatriation of profits.

At the same time, a dense network of bilateral investment treaties (BITs) emerged. From a few hundred in the 1980s, the number of BITs grew to more than two thousand by the early 2000s. These treaties conferred upon foreign investors substantive protections — fair and equitable treatment, full protection and security, protection against expropriation without compensation, national

and most-favoured-nation treatment — and, crucially, the right to bring claims directly against host states before international arbitral tribunals, typically under the auspices of the International Centre for Settlement of Investment Disputes (ICSID) established by the Washington Convention of 1965. The extension of investor–state arbitration transformed the enforcement of investor rights by removing the need for diplomatic protection by the investor's home state.

In the energy sector specifically, the Energy Charter Treaty (ECT), signed in Lisbon in December 1994 and in force since April 1998, established a multilateral framework for investment protection, trade, transit, and energy efficiency among the states of Europe and the former Soviet Union, together with a number of states in Asia. The ECT became the most frequently invoked investment treaty in the world, a fact that would later make it a central battleground in debates over the compatibility of investment protection with climate policy.

The liberalisation of domestic energy markets proceeded in parallel. In the electricity and gas sectors, the United Kingdom privatised its state-owned utilities in the late 1980s and 1990s; the United States introduced open access to transmission networks through Federal Energy Regulatory Commission (FERC) Order No. 888 of 1996; and the European Union adopted successive legislative packages, beginning with the first Electricity Directive of 1996, designed to create an internal energy market through the unbundling of vertically integrated utilities and the introduction of competition in generation and supply.

Phase Five: Resource Nationalism, Sustainability, and the Energy Transition (2005–present)

The fifth phase, which continues today, is marked by a complex and sometimes contradictory set of trends.

First, the commodity price boom of the 2000s prompted a revival of resource nationalism. Venezuela, Bolivia, Ecuador, and Russia, among others, increased state participation, renegotiated contracts, or expropriated foreign interests, generating a new wave of investment arbitrations, several of which resulted in awards of billions of United States dollars. Resource nationalism also took subtler forms, including increased royalties and windfall taxes, local content requirements, export restrictions on unprocessed minerals, and mandatory state participation in new projects.

Second, the international community increasingly emphasised transparency, good governance, and the equitable distribution of resource revenues. The Extractive Industries Transparency Initiative (EITI), launched in 2002 and now implemented in more than fifty countries, requires the disclosure of payments by companies to governments and of revenues received by governments from the extractive sector. Supply chain due diligence standards for conflict minerals, notably

the OECD *Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas*, sought to address the link between mineral extraction and armed conflict. The United Nations *Guiding Principles on Business and Human Rights*, endorsed by the Human Rights Council in 2011, articulated the responsibility of business enterprises, including extractive companies, to respect human rights.

Third, and most consequentially, climate change has moved from the margins to the centre of energy law. The Paris Agreement of 2015 set the objective of holding the increase in global average temperature to well below 2°C above pre-industrial levels and pursuing efforts to limit the increase to 1.5°C. At the twenty-eighth Conference of the Parties to the United Nations Framework Convention on Climate Change (COP28) in Dubai in 2023, the parties for the first time called in the outcome of the first global stocktake for "transitioning away from fossil fuels in energy systems, in a just, orderly and equitable manner". Climate litigation has proliferated in domestic courts, regional human rights bodies, and international tribunals, culminating in advisory opinions from the International Tribunal for the Law of the Sea (2024), the Inter-American Court of Human Rights (2025), and the International Court of Justice (2025) on the obligations of states in respect of climate change. These developments are examined in Chapter 18.

Fourth, the energy transition has created new legal frontiers: the regulation of renewable energy, hydrogen, energy storage, carbon capture and storage, and offshore wind; the governance of critical minerals essential to batteries, electric vehicles, and grid infrastructure; and the management of stranded assets and the decommissioning of fossil fuel infrastructure. Simultaneously, geopolitical events — notably the Russian invasion of Ukraine in February 2022 and the resulting disruption of European gas supply — have restored energy security to the forefront of policy and have demonstrated that the transition will be shaped as much by security considerations as by environmental ones.

KEY DOCTRINAL POINT

The history of resource governance is not a linear progression from one paradigm to another. Elements of each phase coexist in contemporary law. Classical concession-style licences remain in use in many jurisdictions; the doctrine of permanent sovereignty retains its normative force; investment treaties continue to protect investors even as they are reformed or terminated; and the imperatives of the energy transition are layered on top of these older structures rather than replacing them. A doctoral analysis must therefore be historically literate: it must recognise which layer of the law a given rule belongs to and how that rule has been reinterpreted over time.

1.3 Internationalisation of Energy Markets and Resource Regulation

The Global Character of Energy Markets

Energy markets are among the most internationalised of all economic sectors. Crude oil is a global commodity, traded on international exchanges and priced by reference to benchmark grades such as Brent and West Texas Intermediate. A substantial proportion of the world's oil production crosses at least one international border before it is consumed, often passing through maritime chokepoints such as the Strait of Hormuz and the Strait of Malacca. Natural gas markets, historically regional because of the constraints of pipeline transport, have become increasingly integrated through the growth of the liquefied natural gas (LNG) trade, which permits gas to be shipped across oceans and redirected in response to price signals. Coal is traded internationally in large volumes. Electricity, while still predominantly national, is increasingly traded across borders through interconnectors and regional power pools. The supply chains for renewable energy technologies — solar panels, wind turbines, batteries — are global, with manufacturing concentrated in a small number of states.

The internationalisation of markets has legal consequences. Domestic regulatory decisions in one state — on export restrictions, subsidies, environmental standards, or sanctions — can have profound effects on prices and supplies in others. Commercial disputes arising from cross-border transactions require rules on applicable law, jurisdiction, and the enforcement of judgments and awards. Infrastructure that crosses borders, such as pipelines and interconnectors, requires intergovernmental agreements to address transit, tariffs, and the allocation of regulatory authority.

Mechanisms of Internationalisation

The internationalisation of energy and resource regulation has proceeded through several mechanisms.

Treaty-making. States have concluded treaties addressing particular aspects of energy and resource activities: investment protection (BITs and the ECT), trade (the WTO agreements and regional trade agreements), transboundary pipelines (for example, the intergovernmental agreements underlying the Baku–Tbilisi–Ceyhan pipeline and the West African Gas Pipeline), nuclear safety and liability, the law of the sea, and environmental protection.

Institutional cooperation. International organisations such as the International Energy Agency (IEA), the International Atomic Energy Agency (IAEA), the International Renewable Energy Agency (IRENA), OPEC, and the Gas Exporting Countries Forum provide forums for the coordination of national policies, the exchange of information, and, in some cases, the adoption of binding

standards. Regional bodies — the Energy Community in South-East Europe, the West African Power Pool, the Southern African Power Pool — promote regional market integration.

Regulatory convergence. States often borrow regulatory models from one another, whether through voluntary emulation, the conditions attached to development finance, or the technical assistance of international organisations. The spread of independent regulatory agencies, of unbundling requirements in electricity, and of model petroleum agreements reflects this process of regulatory transplantation, which is examined in Chapter 24.

Private ordering. The standard contracts produced by industry associations — the AIEN Model Joint Operating Agreement, the International Federation of Consulting Engineers (FIDIC) suites of construction contracts, and the master agreements used in energy trading — create transnational norms that operate across jurisdictions.

Extraterritorial regulation. Some states regulate conduct outside their territory through legislation with extraterritorial effect. Examples include anti-corruption statutes such as the United States Foreign Corrupt Practices Act of 1977 and the United Kingdom Bribery Act 2010, sanctions regimes targeting the energy sectors of particular states, and supply chain due diligence obligations such as those contained in the European Union's Corporate Sustainability Due Diligence Directive and Batteries Regulation. These are discussed in Chapters 2 and 22.

The Limits of Internationalisation

Notwithstanding these developments, energy and resource regulation remains in important respects national. There is no comprehensive multilateral energy treaty of universal application; the ECT, the most ambitious attempt to create such an instrument, never attracted the participation of major producers and consumers such as the United States, China, Saudi Arabia, or Brazil, and its membership has contracted in recent years following withdrawals by the European Union and several of its member states. Petroleum, mining, and electricity remain subject primarily to domestic legislation, and the doctrine of permanent sovereignty ensures that the fundamental decisions about whether, when, and on what terms to develop natural resources are reserved to the state in which they are located. The internationalisation of energy law is thus partial, uneven, and contested — characteristics that make it a fertile subject for doctoral research.

1.4 Public Law, Private Law, and Transnational Legal Relations

The Public–Private Divide in Energy and Resource Law

One of the distinctive features of energy and resource law is the extent to which it straddles the conventional divide between public and private law. On the public side, the state acts as sovereign: it owns or controls subsurface resources, grants licences, sets fiscal terms, regulates conduct, protects the environment, and resolves disputes through its courts. On the private side, the state frequently acts as a commercial party: it enters into contracts with investors, participates in joint ventures through national oil or mining companies, sells resources on international markets, and borrows money to finance its share of project costs.

The dual character of the state — as sovereign regulator and as contracting party — generates persistent legal difficulties. When a state that has entered into a contract with a foreign investor subsequently changes its tax laws or environmental regulations in a manner that adversely affects the investor, is it acting as a sovereign exercising its regulatory powers, or as a contracting party in breach of its contractual obligations? The answer may determine whether the investor has a remedy, which tribunal has jurisdiction, and what law applies. The distinction between acts *jure imperii* and acts *jure gestionis*, familiar from the law of state immunity, and the distinction in investment law between a mere breach of contract and a breach of treaty, are both attempts to grapple with this problem.

The State Contract

The "state contract" — a contract between a state (or a state entity) and a foreign private party — is the central legal instrument of international resource development. Petroleum concessions, production sharing contracts, mining agreements, host government agreements for pipelines, and power purchase agreements with state-owned utilities are all species of state contract.

The legal status of state contracts has been debated for more than a century. Three principal positions may be distinguished.

The first position holds that state contracts are governed exclusively by the domestic law of the host state. On this view, the state retains its sovereign power to modify the contract by legislation, and the investor's protection lies only in such guarantees as domestic law provides. This position was associated with the Calvo doctrine in Latin America and with the claims of the New International Economic Order.

The second position holds that state contracts may be "internationalised" — that is, subjected to international law or to general principles of law — by the choice of the parties, by the inclusion of stabilisation and arbitration clauses, or by the nature of the contract as an economic development agreement. This

position found its fullest expression in the award of Sole Arbitrator René-Jean Dupuy in *Texaco v Libya*, which held that the concession contracts at issue had been internationalised and that Libya's nationalisation measures were a breach of its contractual obligations under international law.[6]

The third position, now widely accepted, holds that the contract is governed by the law chosen by the parties (often the host state's law, sometimes supplemented by international law or general principles), but that the investor may also enjoy independent protection under international law — most importantly through investment treaties — which may be invoked even where the contract itself is governed by domestic law. The distinction between contract claims and treaty claims, articulated by the *ad hoc* Committee in *Compañía de Aguas del Aconquija SA and Vivendi Universal v Argentina* (*Vivendi I*), is now a cornerstone of investment arbitration: "whether there has been a breach of the BIT and whether there has been a breach of contract are different questions".[7]

Transnational Legal Relations

Beyond the public–private divide lies a further category of relations that is neither purely public nor purely private, neither purely national nor purely international. Philip Jessup, writing in 1956, coined the term "transnational law" to include "all law which regulates actions or events that transcend national frontiers", embracing both public and private international law as well as "other rules which do not wholly fit into such standard categories".[8] Energy and resource activities exemplify transnational legal relations. A single offshore gas project may involve a host government agreement with the state, a production sharing contract with the national oil company, a joint operating agreement among the co-venturers, engineering contracts with international contractors, LNG sales agreements with buyers in several countries, financing agreements with commercial banks and export credit agencies, and an investment treaty between the host state and the home states of the investors. Each relationship is governed by a different combination of national and international rules, and disputes may be resolved in different forums.

The transnational character of these relations has three important implications. First, it means that no single legal order governs the project as a whole; the applicable law must be determined relationship by relationship. Second, it creates the risk of conflicting obligations, as when an environmental obligation imposed by domestic law conflicts with a stabilisation clause in a contract, or a

[6] *Texaco Overseas Petroleum Company and California Asiatic Oil Company v Government of the Libyan Arab Republic*, Award on the Merits of 19 January 1977, (1978) 17 ILM 1; (1979) 53 ILR 389.

[7] *Compañía de Aguas del Aconquija SA and Vivendi Universal v Argentine Republic*, ICSID Case No ARB/97/3, Decision on Annulment, 3 July 2002, para 96.

[8] Philip C. Jessup, *Transnational Law* (Yale University Press 1956), p. 2.

sanctions regime imposed by one state conflicts with a contractual obligation to supply. Third, it confers significant influence upon private actors — companies, banks, industry associations, arbitrators — in the creation and interpretation of norms, raising questions of legitimacy and accountability that recur throughout this book.

1.5 The Relationship Between Energy Law and International Economic Law

Energy as an Exception in Trade Law

The relationship between energy law and international economic law is close but historically uneasy. When the General Agreement on Tariffs and Trade (GATT) was negotiated in 1947, energy products were not excluded from its scope, but in practice trade in oil and gas was largely left outside the multilateral trading system. Many major producers were not parties to the GATT; the trade in crude oil was conducted largely within vertically integrated companies; and energy policy was regarded as so closely bound up with national security that states were unwilling to subject it to multilateral discipline. The GATT security exception in Article XXI, which permits measures "relating to fissionable materials or the materials from which they are derived" and measures "taken in time of war or other emergency in international relations", reinforced this sense of energy as a domain of sovereign prerogative.

The establishment of the WTO in 1995 and the accession of major energy producers — including Saudi Arabia in 2005 and the Russian Federation in 2012 — brought energy trade more firmly within the multilateral framework. WTO disputes have since addressed export restrictions on raw materials (*China — Raw Materials* and *China — Rare Earths*), local content requirements in renewable energy programmes (*Canada — Renewable Energy/Feed-in Tariff Program* and *India — Solar Cells*), and the European Union's energy market regulation (*EU — Energy Package*). These disputes are examined in Chapter 13.

Investment Law as the Primary Interface

If trade law has historically played a limited role in energy, investment law has played a central one. Energy and mining projects are capital-intensive, long-lived, and dependent on the stability of the host state's legal and fiscal regime; they are therefore particularly vulnerable to what economists call the "obsolescing bargain" — the tendency for the host state's bargaining power to increase once the investor's capital has been sunk. Investment treaties and investor–state arbitration were designed in large part to mitigate this risk. Statistics compiled by ICSID consistently show that the oil, gas, and mining sector, together with electric power

and other energy, account for a larger share of registered cases than any other economic sector.

The obsolescing bargain theory, associated with the work of Raymond Vernon, explains much of the dynamic of resource governance. At the time of the initial investment decision, the host state competes with other states for scarce capital and expertise and must offer attractive terms. Once the investment has been made, however, the investor's assets are immobile and the host state may seek to renegotiate the terms in its favour — through increased taxation, contract renegotiation, or expropriation. Investors anticipate this risk and demand protections: stabilisation clauses, international arbitration, investment treaty coverage, and political risk insurance. The law of international energy investment can be understood, in significant part, as an attempt to manage the obsolescing bargain.

Convergence and Tension

The relationship between energy law and international economic law is characterised by both convergence and tension. Convergence occurs where economic law supports energy policy objectives — for example, where investment protection encourages the capital flows necessary for the energy transition. Tension occurs where economic law constrains the regulatory autonomy that states need to pursue energy and climate objectives — for example, where investors challenge the phase-out of coal-fired power or the withdrawal of renewable energy subsidies, or where trade rules are invoked against local content requirements designed to build domestic manufacturing capacity. The phase-out of coal-fired power in the Netherlands gave rise to ECT claims by RWE and Uniper (the latter subsequently withdrawn), and the retroactive reduction of renewable energy incentives in Spain generated dozens of ECT arbitrations. The resulting debates about regulatory chill, the right to regulate, and the reform of investment treaties are examined in Chapter 14.

1.6 Legal Pluralism and Competing Regulatory Authorities

The Concept of Legal Pluralism

Legal pluralism describes situations in which two or more legal orders coexist in the same social field. Originally developed by legal anthropologists to describe the coexistence of state law and customary law in colonial and post-colonial societies, the concept has been extended to describe the coexistence of national, international, regional, and private normative orders in a globalised world. Energy and resource governance is a paradigmatic site of legal pluralism.

At least five layers of normative authority may be identified in a typical resource project:

1. **Customary and community norms.** In many resource-rich regions, land and resources are held under customary tenure systems that predate, and

coexist with, state law. In Nigeria, for example, customary land tenure was overlaid by the Land Use Act 1978, which vested land in each state in the Governor, while section 44(3) of the Constitution of the Federal Republic of Nigeria 1999 vests the entire property in and control of all minerals, mineral oils, and natural gas in the Government of the Federation. Communities in the Niger Delta nevertheless continue to assert claims to land and resources grounded in custom, and these claims shape the conduct of operations and the content of community development agreements.

2. **Subnational law.** In federal states, constitutional authority over resources may be divided between the federal government and the constituent units. In Canada, the provinces own and regulate most natural resources under section 92A of the Constitution Act 1867 (inserted in 1982); in Australia, the states own onshore minerals and petroleum; in the United States, mineral rights may be held privately, by the states, or by the federal government. In Nigeria, the Electricity Act 2023, enacted following a constitutional amendment, empowers states to regulate electricity markets within their territories.
3. **National law.** Constitutional provisions, petroleum and mining statutes, environmental legislation, tax law, and administrative law form the core of the regulatory framework.
4. **International and regional law.** Treaties, customary international law, and the law of regional organisations (such as the European Union, the Economic Community of West African States, and the African Union) impose obligations on states and, in some cases, confer rights on private parties.
5. **Transnational private regulation.** Industry standards, model contracts, voluntary codes of conduct, certification schemes, and the environmental and social performance standards of lenders — notably the International Finance Corporation (IFC) *Performance Standards on Environmental and Social Sustainability* and the Equator Principles adopted by commercial banks — impose requirements on companies that may exceed those of national law.

Sources of Conflict

The coexistence of multiple normative orders generates conflicts of several kinds. *Vertical* conflicts arise between different levels of authority — for example, between federal and state governments over the ownership of offshore resources, as in Nigeria's resource-control litigation, in which the Supreme Court held in *Attorney-General of the Federation v Attorney-General of Abia State* that for the purposes of revenue allocation the seaward boundary of a littoral state was the

low-water mark of its land surface.[9] *Horizontal* conflicts arise between different regimes at the same level — for example, between investment treaty obligations and environmental obligations. *Public–private* conflicts arise between state regulation and private standards or contracts. *Jurisdictional* conflicts arise between courts and tribunals asserting authority over the same dispute, as when an investor pursues parallel proceedings before domestic courts, commercial arbitration, and investment treaty arbitration.

Managing Pluralism

Legal systems have developed a range of techniques for managing these conflicts: hierarchical rules of priority (constitutional supremacy, the primacy of European Union law), principles of interpretation (systemic integration under Article 31(3)(c) of the Vienna Convention on the Law of Treaties, the presumption against conflict), doctrines of comity and deference (the margin of appreciation, the police powers doctrine in investment law), procedural mechanisms (fork-in-the-road and waiver clauses, *lis pendens*, *res judicata*), and institutional coordination (memoranda of understanding between regulators, joint commissions).

The International Law Commission's study of the fragmentation of international law, completed in 2006 under the chairmanship of Martti Koskenniemi, remains the most authoritative analysis of these techniques at the international level.[10] For energy and resource law, the key insight of the fragmentation debate is that specialised regimes — investment law, trade law, environmental law, human rights law — do not operate in isolation; they remain embedded in general international law, and their interpretation must take account of other relevant rules applicable between the parties.

> **COMPARATIVE PERSPECTIVE**
>
> **Federalism and resource authority in three jurisdictions.** In *Canada*, section 92A of the Constitution Act 1867 confirms provincial legislative authority over the exploration, development, conservation, and management of non-renewable natural resources, subject to federal paramountcy for interprovincial export. In *Nigeria*, section 44(3) of the 1999 Constitution vests ownership and control of minerals and petroleum in the Federal Government, and item 39 of the Exclusive Legislative List reserves "mines and minerals, including oil fields, oil mining, geological surveys and natural gas" to the National Assembly; the derivation principle in section 162(2) allocates not less than thirteen per cent of revenue accruing from natural resources

[9] *Attorney-General of the Federation v Attorney-General of Abia State & 35 Others* (2002) 6 NWLR (Pt 764) 542 (Supreme Court of Nigeria).

[10] International Law Commission, *Fragmentation of International Law: Difficulties Arising from the Diversification and Expansion of International Law*, Report of the Study Group finalised by Martti Koskenniemi, UN Doc A/CN.4/L.682 (13 April 2006).

to the states of origin. In the *United States*, subsurface minerals may be privately owned and severed from the surface estate, with federal and state governments owning substantial mineral estates in their own right and regulating activities on federal and state lands respectively. These differences shape the bargaining power of communities, the structure of fiscal regimes, and the incidence of resource conflicts.

1.7 Contemporary Challenges in Global Resource Governance

The field confronts a set of challenges that are without precedent in their scale and complexity. Six are of particular importance.

The Energy Transition and Stranded Assets

The first challenge is the transition from an energy system dominated by fossil fuels to one based predominantly on low-carbon sources. The legal implications are profound. Petroleum and coal licences granted for decades may become economically or legally unviable before their expiry, raising questions about compensation, decommissioning liability, and the protection of legitimate expectations. Investors in fossil fuel infrastructure may seek to rely on investment treaties to challenge phase-out measures. Producer states whose economies depend on hydrocarbon revenues face fiscal crises and must manage the social consequences of transition. The concept of "stranded assets" — assets that suffer from premature write-downs or devaluation as a result of the transition — has migrated from financial analysis into law, influencing disclosure obligations, directors' duties, and the interpretation of investment protections.

Critical Minerals and the New Extractivism

The second challenge arises from the paradox that decarbonisation requires more extraction, not less. The IEA has repeatedly projected that demand for minerals such as lithium, cobalt, nickel, copper, graphite, and rare earth elements will grow severalfold under scenarios consistent with the Paris Agreement. These minerals are geographically concentrated, and their processing is concentrated still further. States have responded with industrial policies, export restrictions, strategic stockpiles, and bilateral critical mineral agreements. Communities in mining regions confront the risk that the green transition will reproduce the environmental degradation and human rights abuses associated with earlier extractive booms. Chapter 8 addresses these issues in detail.

Energy Access and Energy Poverty

The third challenge is that a substantial proportion of humanity still lacks access to modern energy services. Hundreds of millions of people, the majority in sub-Saharan Africa, lack access to electricity, and far more lack access to clean cooking fuels. Sustainable Development Goal 7 commits states to ensure access

to affordable, reliable, sustainable, and modern energy for all by 2030. Achieving universal access while simultaneously decarbonising energy systems requires legal and regulatory frameworks that mobilise investment in decentralised and grid-based solutions, protect consumers, and allocate the costs of transition fairly. The concept of energy justice, explored in Chapter 23, provides a framework for analysing these distributional questions.

Geopolitics and Energy Security

The fourth challenge is the re-emergence of energy as an instrument of geopolitical competition. The disruption of gas supplies to Europe following the invasion of Ukraine, the use of sanctions against major energy exporters, attacks on energy infrastructure, and competition over critical mineral supply chains have demonstrated that energy security remains a central concern of state policy. The law governing sanctions, export controls, emergency powers, strategic reserves, and the protection of critical infrastructure — examined in Chapter 25 — has accordingly grown in importance.

Legitimacy of Investor–State Dispute Settlement

The fifth challenge concerns the legitimacy of the system of investor–state dispute settlement (ISDS). Critics argue that ISDS privileges foreign investors over domestic stakeholders, constrains the regulatory autonomy of states, produces inconsistent decisions, and lacks transparency and accountability. Reform efforts proceed in several forums: the United Nations Commission on International Trade Law (UNCITRAL) Working Group III, the European Union's proposal for a multilateral investment court, the modernisation of the ECT, and the renegotiation or termination of BITs by individual states. The outcome of these reforms will shape the legal environment for energy investment for decades.

Technological Change and Digitalisation

The sixth challenge arises from technological change. Hydrogen, advanced nuclear reactors, carbon capture, battery storage, offshore floating wind, and deep-sea mining all raise novel regulatory questions. Digitalisation — smart meters, distributed energy resources, automated trading, artificial intelligence in grid management — raises questions of data ownership, cybersecurity, and liability. Regulatory frameworks designed for centralised, fossil-fuel-based systems must be adapted to decentralised, digital, and decarbonised systems. Chapters 12 and 30 address these developments.

1.8 Doctoral Research Perspectives and Foundational Debates

The Nature of Doctoral Contribution

Doctoral research is distinguished from other forms of legal writing by the requirement of an original contribution to knowledge. In international energy

and natural resources law, such a contribution may take several forms: the identification and analysis of a new legal problem; a new interpretation of existing rules or cases; the application of a theoretical framework to a body of law; a comparative study revealing previously unrecognised patterns; an empirical investigation of how the law operates in practice; or a reasoned proposal for reform. Chapter 28 discusses research methodology in detail. Here, the purpose is to identify the foundational debates within which most doctoral research in the field is situated.

Debate One: Sovereignty versus Stability

The most enduring debate in the field concerns the tension between the sovereign right of states to regulate their natural resources in the public interest and the need of investors for legal and fiscal stability. The debate manifests itself in disputes over stabilisation clauses, the standard of compensation for expropriation, the scope of the fair and equitable treatment standard, and the legitimacy of contract renegotiation. Is the doctrine of permanent sovereignty compatible with binding long-term commitments to foreign investors? Can a state contract away its legislative power? How should the law balance the legitimate expectations of investors against the evolving needs of the host society?

Debate Two: Contract versus Regulation

A second debate concerns the relative roles of contract and regulation in governing resource activities. In some jurisdictions, the terms of resource development are set primarily by individually negotiated contracts; in others, primarily by general legislation and regulation applicable to all licensees. Each approach has advantages and drawbacks. Contracts offer flexibility and can be tailored to the circumstances of particular projects, but they are vulnerable to corruption, asymmetries of information, and lack of transparency. Regulation offers predictability, equality of treatment, and democratic accountability, but may be less responsive to project-specific risks. The trend in many jurisdictions, including Nigeria under the Petroleum Industry Act 2021, has been towards greater reliance on legislative frameworks, but contracts remain central.

Debate Three: Efficiency versus Equity

A third debate concerns the objectives of resource law. Economic approaches emphasise efficiency — the maximisation of resource rents, the minimisation of transaction costs, and the allocation of risk to the party best able to bear it. Equity-oriented approaches emphasise the fair distribution of benefits and burdens among states, investors, communities, and future generations, and the protection of the rights of affected persons. The "resource curse" literature, which examines why resource-rich states often experience slower growth, weaker institutions, and greater conflict than resource-poor states, has highlighted the importance of

institutional quality and has prompted reforms focused on transparency, revenue management, and accountability.

Debate Four: Development versus Sustainability

A fourth debate concerns the relationship between resource development and environmental sustainability. Developing states argue that they have a right to exploit their resources to finance development, as industrialised states did before them, and that the burden of climate mitigation should fall primarily on those states responsible for historical emissions. This argument is reflected in the principle of common but differentiated responsibilities and respective capabilities, articulated in the United Nations Framework Convention on Climate Change. At the same time, the physical limits of the carbon budget mean that a substantial proportion of known fossil fuel reserves cannot be burned if the temperature goals of the Paris Agreement are to be met. How should the right to development be reconciled with the imperative of decarbonisation? Who should bear the costs of leaving resources in the ground?

Debate Five: State-Centrism versus Transnational Governance

A fifth debate concerns the appropriate unit of analysis. Traditional approaches treat the state as the principal actor and source of law. Transnational approaches emphasise the role of corporations, international organisations, civil society, communities, and private standard-setters in creating and enforcing norms. The rise of corporate climate litigation, supply chain due diligence obligations, and private sustainability standards suggests that governance is increasingly polycentric. Doctoral research must decide how to account for this complexity without losing analytical clarity.

LEADING AUTHORITY

Texaco Overseas Petroleum Co v Libyan Arab Republic (1977)

Facts. Libya had granted deeds of concession to two American companies under its Petroleum Law of 1955. The concessions contained a stabilisation clause providing that the contractual rights expressly created by the concession should not be altered except by mutual consent, and an arbitration clause. In 1973 and 1974 Libya nationalised the companies' interests. The companies invoked arbitration; Libya refused to participate, and a sole arbitrator was appointed by the President of the International Court of Justice.

Holding. Sole Arbitrator Dupuy held that the concessions were internationalised contracts; that they were governed by principles of Libyan law common to international law and, in the absence of such common principles, by general principles of law; that the stabilisation clause was valid and binding; and that Libya had breached its obligations. Notably, he ordered *restitutio in integrum* — specific performance — as the primary remedy, although the parties subsequently settled.

Significance. The award is the high-water mark of the internationalisation theory. Its analysis of Resolution 1803 as reflecting customary law, and its treatment of the later Charter of Economic Rights and Duties of States as lacking that status, remain influential. Its remedy of restitution has rarely been followed; later tribunals, beginning with *LIAMCO* and *Aminoil*, preferred damages.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

The Kestrel Basin concession (hypothetical). The Republic of Aranda, a fictional coastal state, granted a thirty-year petroleum concession in 1998 to Northwind Energy plc, a company incorporated in a fictional state with which Aranda has a bilateral investment treaty. The concession contains (i) a choice of Arandan law "supplemented by such principles of international law as may be applicable"; (ii) a "freezing" stabilisation clause; and (iii) ICC arbitration seated in Geneva. In 2025, following discoveries of significant gas reserves and a change of government, Aranda enacts a Petroleum Revenue Act that raises the royalty rate from 8 to 15 per cent, imposes a windfall profits tax, and requires all existing concessionaires to convert to production sharing contracts within two years.

Prepare a memorandum of not more than 2,500 words that: (a) identifies each legal order potentially applicable to the dispute; (b) explains the competing characterisations of the measures as sovereign regulation or contractual breach; (c) assesses the legal effect of the stabilisation clause in the light of the doctrine of permanent sovereignty; and (d) recommends a dispute resolution strategy for Northwind, identifying the risks of parallel proceedings. *All parties, places, and instruments in this exercise are fictitious.*

CHAPTER SUMMARY

- International energy and natural resources law is a functional, pluralist, distributive, and transnational field concerned with the allocation of rights over energy sources and natural resources, the regulation of activities involving them, and the distribution of the resulting benefits and burdens.
- The field integrates energy law and natural resources law, a combination justified by shared history, doctrine, institutions, and — increasingly — the material interdependence created by the energy transition.
- Its history may be organised into five overlapping phases: the classical concession era, nationalisation, decolonisation and the New International Economic Order, liberalisation and investment protection, and the current era of resource nationalism, sustainability, and transition.
- Energy markets are highly internationalised, but energy regulation remains predominantly national; internationalisation proceeds through treaties, institutions, regulatory convergence, private ordering, and extraterritorial regulation.

- The dual character of the state as sovereign and contracting party is the source of many of the field's enduring problems, notably the status of state contracts and the distinction between contract and treaty claims.
- Legal pluralism is a defining feature of resource governance, requiring techniques of conflict management drawn from constitutional, international, and private law.
- Doctoral research is situated within five foundational debates: sovereignty versus stability, contract versus regulation, efficiency versus equity, development versus sustainability, and state-centrism versus transnational governance.

DOCTORAL RESEARCH QUESTIONS

1. To what extent does the concept of *lex petrolea* describe a genuine transnational legal order, and what are the implications of its existence or non-existence for the legitimacy of investor–state arbitration?
2. Has the energy transition rendered the integrated study of energy and natural resources law analytically indispensable, or does it obscure important differences between extractive and non-extractive energy sources?
3. How should the obsolescing bargain theory be revised to account for renewable energy projects, in which the host state's bargaining power may evolve differently from that in petroleum projects?
4. Can the techniques developed to manage the fragmentation of international law adequately address conflicts between investment protection and climate obligations, or is structural reform required?
5. What role should customary land tenure and community norms play in the allocation of resource rights in states, such as Nigeria, where the Constitution vests mineral ownership in the federal government?

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2 CHAPTER

Sources, Principles, and Institutions of International Energy Law

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Identify the formal sources of international energy and natural resources law and explain how each operates in the energy and resource context.
2. Analyse the content, legal status, and limits of the principle of permanent sovereignty over natural resources.
3. Apply the rules of state responsibility to conduct affecting energy investments, infrastructure, and the environment.
4. Explain the principles of jurisdiction, including the conditions under which states may regulate extraterritorially.
5. Evaluate the contribution of international organisations, soft law, industry standards, and corporate governance instruments to the field.
6. Assess the role of international courts and arbitral tribunals and analyse conflicts between domestic and international legal obligations.

Introduction

The preceding chapter described international energy and natural resources law as a pluralist field drawing upon multiple bodies of law. This chapter examines the materials from which that law is made — its sources — and the general principles and institutions through which it operates. It begins with the formal sources of international law identified in Article 38(1) of the Statute of the International Court of Justice and considers how each functions in the energy context. It then examines the principle that more than any other defines the field: permanent sovereignty over natural resources. The chapter proceeds to the law of state responsibility, the principles of jurisdiction, and the growing role of international organisations, soft law, and industry standards. It concludes with an analysis of

the institutions that adjudicate energy disputes and of the persistent problem of conflicts between domestic and international legal obligations.

A preliminary observation is in order. Students trained in a domestic legal system sometimes approach international law expecting a hierarchy of norms comparable to that of a national constitution, with clearly defined law-making institutions and a supreme court at the apex. International law does not work in that way. It is a decentralised system in which the subjects of the law — principally states — are also its makers, its interpreters, and, to a considerable extent, its enforcers. The sources of international law are therefore best understood not as a hierarchy but as a set of recognised processes through which states, and increasingly other actors, generate obligations. This decentralisation is particularly pronounced in the energy field, where there is no single governing treaty and where much of the normative work is done by contracts, standards, and the decisions of arbitral tribunals.

2.1 Treaties, Custom, and General Principles of International Law

Article 38 of the Statute of the International Court of Justice

Article 38(1) of the Statute of the International Court of Justice directs the Court, in deciding disputes in accordance with international law, to apply:

> (a) international conventions, whether general or particular, establishing rules expressly recognized by the contesting states;
> (b) international custom, as evidence of a general practice accepted as law;
> (c) the general principles of law recognized by civilized nations;
> (d) subject to the provisions of Article 59, judicial decisions and the teachings of the most highly qualified publicists of the various nations, as subsidiary means for the determination of rules of law.

Although Article 38 is formally a direction to the Court, it is universally regarded as an authoritative statement of the sources of international law. The archaic reference to "civilized nations" is today understood to refer to the community of nations generally. Article 38 does not establish a formal hierarchy among treaties, custom, and general principles, although in practice a treaty, as *lex specialis*, will usually prevail over a conflicting rule of custom as between the parties to it, subject always to the overriding force of peremptory norms (*jus cogens*).

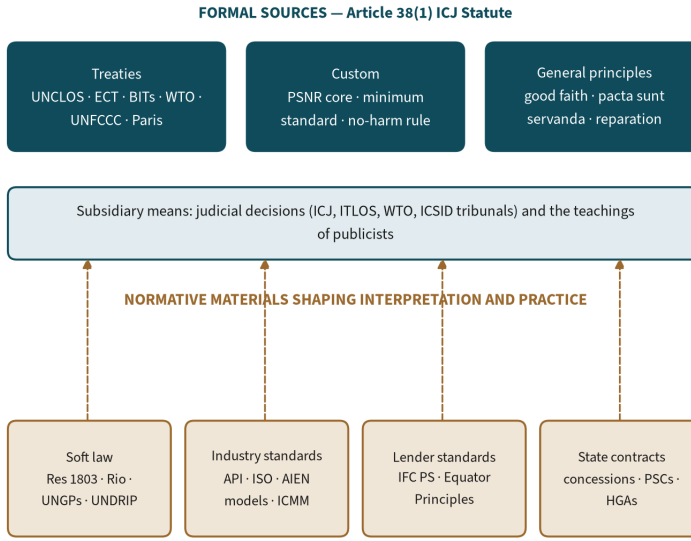


Figure 2.1 The sources of international energy and natural resources law. Formal sources under Article 38(1) of the ICJ Statute are supplemented by soft law, contracts, and transnational standards that shape practice and interpretation. *Source:* prepared for this edition.

Treaties

Treaties are the most important source of international energy law in terms of volume and specificity. The Vienna Convention on the Law of Treaties 1969 (VCLT) defines a treaty as "an international agreement concluded between States in written form and governed by international law, whether embodied in a single instrument or in two or more related instruments and whatever its particular designation". Treaties relevant to energy and natural resources may be classified as follows.

Multilateral treaties of general application that apply to energy and resource activities without being specifically directed at them. These include the United Nations Convention on the Law of the Sea 1982 (UNCLOS), the WTO agreements, the ICSID Convention 1965, the New York Convention on the Recognition and Enforcement of Foreign Arbitral Awards 1958, the United Nations Framework Convention on Climate Change 1992 (UNFCCC), the Paris Agreement 2015, and the Convention on Biological Diversity 1992.

Multilateral treaties specific to energy. The Energy Charter Treaty 1994 is the principal example of a multilateral treaty dedicated to energy. Others include the Statute of the International Atomic Energy Agency 1956, the Agreement on an International Energy Program 1974 (which established the IEA), the Statute of the International Renewable Energy Agency 2009, and the suite of nuclear

conventions on safety, security, liability, and waste management considered in Chapter 11.

Regional treaties, including the Treaty Establishing the Energy Community 2005, the ECOWAS Energy Protocol 2003 (modelled on the ECT), and the energy provisions of the Treaty on the Functioning of the European Union, notably Article 194.

Bilateral treaties, including more than two thousand bilateral investment treaties, tax treaties, maritime boundary agreements, joint development agreements, unitisation agreements for transboundary reservoirs, and intergovernmental agreements for cross-border pipelines and interconnectors.

Treaty Interpretation

Because so much of the field turns on the interpretation of treaties — investment treaties in particular — a doctoral candidate must have a thorough command of the rules of interpretation in Articles 31 to 33 of the VCLT, which reflect customary international law. Article 31(1) provides that a treaty "shall be interpreted in good faith in accordance with the ordinary meaning to be given to the terms of the treaty in their context and in the light of its object and purpose". Article 31(3) requires account to be taken, together with the context, of subsequent agreements between the parties regarding interpretation, subsequent practice establishing agreement regarding interpretation, and "any relevant rules of international law applicable in the relations between the parties". Article 32 permits recourse to supplementary means of interpretation, including the preparatory work of the treaty and the circumstances of its conclusion, to confirm the meaning resulting from Article 31 or to determine the meaning when that interpretation leaves it ambiguous or obscure or leads to a manifestly absurd or unreasonable result.

Three interpretive issues recur in energy disputes. The first is the weight to be given to the *object and purpose* of investment treaties: tribunals that emphasise the promotion and protection of investment, often stated in treaty preambles, have tended to adopt expansive interpretations of investor protections, while others insist that the object and purpose include a balance between investment protection and the host state's right to regulate. The second is the use of *subsequent agreements and practice*, including joint interpretive statements issued by treaty parties, such as the interpretive note issued by the NAFTA Free Trade Commission in 2001 limiting the minimum standard of treatment to the customary international law minimum standard. The third is *systemic integration* under Article 31(3)(c), which permits tribunals to take account of environmental, human rights, and climate obligations in interpreting investment treaties.

Customary International Law

Customary international law arises from a general practice of states accepted as law (*opinio juris*). Both elements must be established: practice that is sufficiently widespread, representative, and consistent, and the acceptance of that practice as legally required or permitted. The International Law Commission's *Draft Conclusions on Identification of Customary International Law*, adopted in 2018, provide authoritative guidance on the evidence of each element.

Customary international law performs several functions in the energy field. It supplies rules of general application where no treaty exists, such as the rules on the protection of foreign property and the standard of compensation for expropriation, the obligation to prevent significant transboundary harm, and the rules on state responsibility. It fills gaps in treaty regimes and informs their interpretation. And in some areas — notably the minimum standard of treatment of aliens — treaty provisions expressly incorporate customary standards.

The identification of custom in the energy field is often difficult. Investment treaties are numerous and broadly similar in content, prompting debate over whether they have generated or codified customary rules. In *Barcelona Traction*, the International Court of Justice observed that, in the field of foreign investment, the law had developed through bilateral and multilateral agreements rather than general rules,[1] and this remains a contested question. The dominant view is that the proliferation of BITs has not of itself transformed their substantive standards into custom, because states conclude such treaties to obtain reciprocal benefits rather than out of a sense of legal obligation, although the treaties may serve as evidence of the content of the customary minimum standard.

General Principles of Law

General principles of law recognised by the community of nations serve primarily to fill gaps in treaty and customary law. The International Law Commission adopted *Draft Conclusions on General Principles of Law* in 2023, distinguishing between principles derived from national legal systems and principles that may be formed within the international legal system. General principles of particular importance in the energy field include good faith, *pacta sunt servanda*, the prohibition of abuse of rights, the principle of unjust enrichment, estoppel, the principle that no one may profit from his own wrong, the obligation to make reparation for injury caused by a wrongful act, and the procedural principles of due process and equality of arms.

General principles played a central role in the arbitral jurisprudence of the concession era. In the *Abu Dhabi* arbitration of 1951, Lord Asquith of Bishopstone,

[1] *Barcelona Traction, Light and Power Company, Limited (Belgium v Spain)*, Second Phase, Judgment, ICJ Reports 1970, p. 3, paras 89–90.

acting as umpire, declined to apply the law of Abu Dhabi on the ground that no settled body of legal principles existed there that could be applied to the construction of modern commercial instruments, and instead applied principles "rooted in the good sense and common practice of the generality of civilised nations — a sort of 'modern law of nature'".[2] In *Sapphire International Petroleums v National Iranian Oil Company* (1963), the arbitrator similarly applied general principles of law.[3] Such reasoning reflected attitudes towards non-Western legal systems that are now rightly regarded as unacceptable, and modern tribunals generally apply the law chosen by the parties or the law of the host state, supplemented where appropriate by international law.

Subsidiary Means: Judicial Decisions and Teachings

Judicial decisions and the teachings of highly qualified publicists are subsidiary means for the determination of rules of law. There is no formal doctrine of binding precedent in international law; Article 59 of the ICJ Statute provides that decisions of the Court have no binding force except between the parties and in respect of the particular case. Nevertheless, the decisions of the ICJ, the International Tribunal for the Law of the Sea, the WTO Appellate Body, and investment tribunals carry significant persuasive authority, and tribunals routinely cite earlier awards. In investment arbitration, a *jurisprudence constante* has emerged on many issues, although significant divergences persist. The influence of scholarly writing is also considerable: the works of authors such as Nico Schrijver on permanent sovereignty, Christoph Schreuer on the ICSID Convention, and Philippe Sands on environmental law are frequently cited by tribunals.

2.2 Permanent Sovereignty over Natural Resources

Origins and Development

The principle of permanent sovereignty over natural resources (PSNR) is the cornerstone of international resource law. It emerged in the 1950s from the efforts of newly independent and developing states to secure control over their natural wealth, which in many cases remained subject to concessions granted by colonial administrations or on terms that the host states regarded as inequitable. The principle was first articulated in General Assembly Resolution 523 (VI) of 1952 on integrated economic development and commercial agreements, and Resolution 626 (VII) of the same year, which affirmed the right of peoples "freely to use and exploit their natural wealth and resources". A Commission on Permanent Sover-

[2] *Petroleum Development (Trucial Coast) Ltd v Sheikh of Abu Dhabi*, Award of September 1951, (1952) 1 ICLQ 247.

[3] *Sapphire International Petroleums Ltd v National Iranian Oil Company*, Award of 15 March 1963, (1967) 35 ILR 136.

eignty over Natural Resources was established in 1958, and its work culminated in Resolution 1803 (XVII) of 1962.

The principle was incorporated into common Article 1(2) of the International Covenant on Civil and Political Rights and the International Covenant on Economic, Social and Cultural Rights 1966, which provides:

> All peoples may, for their own ends, freely dispose of their natural wealth and resources without prejudice to any obligations arising out of international economic co-operation, based upon the principle of mutual benefit, and international law. In no case may a people be deprived of its own means of subsistence.

Article 21 of the African Charter on Human and Peoples' Rights 1981 is more emphatic still, providing that "all peoples shall freely dispose of their wealth and natural resources", that this right "shall be exercised in the exclusive interest of the people", and that "in no case shall a people be deprived of it". The provision also obliges states parties to "undertake to eliminate all forms of foreign economic exploitation particularly that practised by international monopolies so as to enable their peoples to fully benefit from the advantages derived from their national resources".

Content of the Principle

PSNR comprises a bundle of rights and, increasingly, duties. The principal rights include:

- the right to possess, use, and dispose of natural resources;
- the right to determine freely the conditions of exploration, development, and disposition of resources;
- the right to regulate foreign investment and to supervise the activities of foreign investors;
- the right to nationalise or expropriate foreign property, subject to the payment of appropriate compensation;
- the right to settle disputes in accordance with national law, subject to agreements to the contrary;
- the right to manage resources in pursuance of national environmental and developmental policy.

Schrijver's influential study demonstrated that the principle has evolved from a primarily rights-based claim asserted by developing states against foreign interests into a more balanced concept that also encompasses duties: the duty to exercise sovereignty in the interest of national development and the well-being of the people; the duty to respect the rights and interests of indigenous peoples; the duty to observe international agreements in good faith; the duty to treat foreign

investors fairly; and the duty to manage resources sustainably and to avoid transboundary environmental harm.[4]

Legal Status

The legal status of PSNR has been considered by the International Court of Justice. In *Armed Activities on the Territory of the Congo (Democratic Republic of the Congo v Uganda)*, the Court recognised that Resolution 1803 and the related instruments reflected customary international law in respect of the principle of permanent sovereignty, although it held that the principle was not applicable to the specific situation of looting, pillage, and exploitation of natural resources by members of the army of a state militarily intervening in another state, which fell instead to be addressed under the law of belligerent occupation and international humanitarian law.[5] In its 2022 judgment on reparations in the same case, the Court awarded the Democratic Republic of the Congo a global sum in respect of damage to natural resources.

It is generally accepted that the core of PSNR — the right of states to control and regulate their natural resources, and to expropriate foreign property for a public purpose on payment of compensation — forms part of customary international law. More contested are the claims, associated with the Charter of Economic Rights and Duties of States, that compensation is to be determined solely by domestic law and that disputes are to be resolved exclusively by domestic courts. The arbitral tribunal in *Texaco v Libya* held that Resolution 1803, adopted by a large majority including both developed and developing states, reflected customary law, whereas the relevant provisions of the Charter, adopted over the opposition of the principal capital-exporting states, did not.

PSNR and Contractual Commitments

The relationship between PSNR and contractual commitments is one of the field's central problems. Can a state, in the exercise of its sovereignty, bind itself not to exercise its sovereign powers in the future — for example, by agreeing to a stabilisation clause that freezes the applicable law? The arbitral jurisprudence has generally answered this question in the affirmative, on the ground that the power to enter into binding commitments is itself an attribute of sovereignty. As the sole arbitrator in *Texaco v Libya* put it, "the State, by entering into an international agreement with a private person, has not alienated its sovereignty but has made use of it".[6] On this view, PSNR does not authorise a state to disregard its contrac-

[4] Nico Schrijver, *Sovereignty over Natural Resources: Balancing Rights and Duties* (Cambridge University Press 1997), especially chs 9–10.

[5] *Armed Activities on the Territory of the Congo (Democratic Republic of the Congo v Uganda)*, Judgment, ICJ Reports 2005, p. 168, para 244; Reparations, Judgment, ICJ Reports 2022, p. 13.

[6] *Texaco Overseas Petroleum Company v Libyan Arab Republic*, Award of 19 January 1977, (1979) 53 ILR 389, para 68.

tual obligations; it merely affirms the state's ultimate authority over its resources, which must be exercised in accordance with international law, including the obligation to pay compensation for any taking.

The *Aminoil* tribunal adopted a more nuanced position. It accepted that stabilisation clauses were valid but held that, in the absence of an express prohibition on nationalisation of particular duration, the clause in the Aminoil concession did not prohibit nationalisation as such; rather, its effect was to be taken into account in determining the compensation payable. The tribunal also emphasised that the concession had undergone significant evolution over time, through renegotiation and the acceptance by the company of changes in the fiscal regime, such that the contract had been transformed and its original stabilisation clause could not be read in isolation from that evolution. The approach of *Aminoil* — treating stabilisation clauses as relevant to compensation rather than as an absolute bar to sovereign action — has been influential in later practice and is examined further in Chapters 14 and 15.

KEY DOCTRINAL POINT

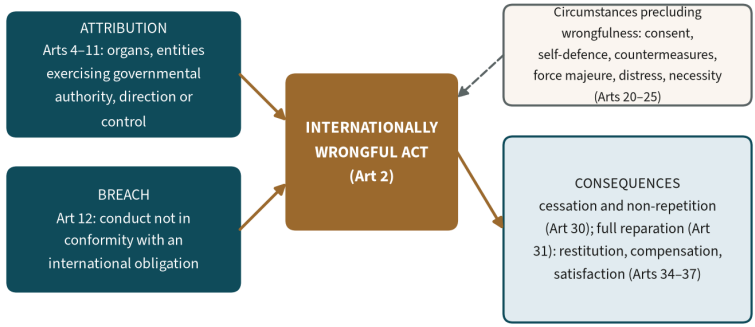
Permanent sovereignty over natural resources is not an unlimited licence to disregard commitments. In its modern form it is a principle of balanced rights and duties: the state controls its resources and may regulate and expropriate them in the public interest, but it must exercise that control in good faith, for the benefit of its people, in accordance with its international obligations, and subject to the payment of compensation where property is taken.

2.3 State Responsibility and International Legal Obligations

The Articles on State Responsibility

The law of state responsibility determines when a state has committed an internationally wrongful act and what consequences follow. Its authoritative statement is the International Law Commission's *Articles on Responsibility of States for Internationally Wrongful Acts* (ARSIWA), adopted in 2001 and commended to governments by the General Assembly. Although ARSIWA is not a treaty, many of its provisions are recognised as reflecting customary international law, and they are routinely applied by the ICJ and by investment tribunals.

Article 1 of ARSIWA states the basic principle: "Every internationally wrongful act of a State entails the international responsibility of that State." Under Article 2, an internationally wrongful act exists when conduct consisting of an action or omission (a) is attributable to the state under international law and (b) constitutes a breach of an international obligation of the state.



Chorzów Factory (1928): reparation must, as far as possible, wipe out all the consequences of the illegal act

Figure 2.2 Elements and consequences of state responsibility under the ILC Articles (2001). *Source:* prepared for this edition.

Attribution

Attribution is of particular importance in the energy sector, where states frequently act through state-owned enterprises, national oil companies, regulatory agencies, and subnational authorities. ARSIWA provides several grounds of attribution:

- **Organs of the state** (Article 4). The conduct of any state organ — legislative, executive, judicial, or other, whether central or territorial — is attributable to the state. A federal state is therefore responsible for the conduct of its constituent units, and a state is responsible for the decisions of its courts and regulators.
- **Persons or entities exercising elements of governmental authority** (Article 5). The conduct of an entity that is not a state organ but is empowered by law to exercise elements of governmental authority is attributable to the state when the entity acts in that capacity. A national oil company that is empowered to grant licences, collect royalties, or regulate operations exercises governmental authority in performing those functions.
- **Conduct directed or controlled by the state** (Article 8). The conduct of a person or group is attributable to the state if it acts on the instructions of, or under the direction or control of, the state in carrying out the conduct.
- **Conduct acknowledged and adopted by the state** (Article 11).

The attribution of the conduct of national oil companies and state-owned utilities has been extensively litigated. In *Noble Ventures v Romania*, the tribunal held that acts of a state privatisation agency were attributable to Romania because the

agency exercised governmental authority.[7] In *EDF (Services) Limited v Romania*, the tribunal examined whether the conduct of state-owned companies controlled by the state was attributable under Articles 5 and 8.[8] The general principle that emerges from the jurisprudence is that the separate legal personality of a state enterprise does not of itself shield the state from responsibility, but neither does state ownership of itself render the enterprise's commercial conduct attributable. The analysis turns on the function performed and the degree of state direction or control over the specific conduct at issue. Investment treaties sometimes contain specific provisions on the treatment of state enterprises, and the ECT, in Article 22, requires each contracting party to ensure that any state enterprise that it maintains or establishes conducts its activities in a manner consistent with the party's obligations under Part III of the treaty.

Breach of an International Obligation

A breach occurs when an act of a state is not in conformity with what is required of it by an international obligation, regardless of the origin or character of the obligation (Article 12). In the energy field, relevant obligations may arise under investment treaties, trade agreements, environmental treaties, human rights treaties, the law of the sea, and customary international law. A state may also incur responsibility for failing to exercise due diligence to prevent harm caused by private actors — for example, failing to prevent pollution from private oil operations from causing transboundary harm, or failing to provide full protection and security to a foreign investor's installations against attack.

The relationship between breach of contract and breach of international law is a recurring issue. A mere breach of a contract between a state and a foreign investor does not, of itself, constitute a breach of international law. Something more is required: an exercise of sovereign power (*puissance publique*) that goes beyond the conduct of an ordinary contracting party, a denial of justice in the resolution of the contractual dispute, or the operation of an "umbrella clause" in an investment treaty elevating contractual undertakings to treaty obligations. These issues are examined in Chapter 14.

Circumstances Precluding Wrongfulness

ARSIWA identifies circumstances that preclude the wrongfulness of conduct that would otherwise breach an international obligation: consent (Article 20), self-defence (Article 21), countermeasures (Article 22), *force majeure* (Article 23), distress (Article 24), and necessity (Article 25).

[7] *Noble Ventures, Inc v Romania*, ICSID Case No ARB/01/11, Award, 12 October 2005, paras 68–86.

[8] *EDF (Services) Limited v Romania*, ICSID Case No ARB/05/13, Award, 8 October 2009, paras 185–213.

The defence of **necessity** has been extensively litigated in the energy sector, particularly in the arbitrations arising from Argentina's economic crisis of 2001–2002, during which Argentina froze utility tariffs, converted dollar-denominated tariffs into pesos, and suspended tariff adjustment mechanisms that foreign investors in its gas and electricity sectors had been guaranteed. Article 25 permits a state to invoke necessity only if the act "is the only way for the State to safeguard an essential interest against a grave and imminent peril" and "does not seriously impair an essential interest of the State or States towards which the obligation exists, or of the international community as a whole"; necessity may not be invoked if the obligation excludes it or if the state has contributed to the situation of necessity.

The Argentine cases produced divergent results. In *CMS Gas Transmission Co v Argentina*, the tribunal rejected Argentina's necessity defence, finding that the measures were not the only way to respond to the crisis and that Argentina had contributed to the situation.[9] In *LG&E Energy Corp v Argentina*, by contrast, the tribunal accepted that a state of necessity existed for a defined period between December 2001 and April 2003, during which Argentina was exempt from liability.[10] The *ad hoc* annulment committee in *CMS* criticised the tribunal for conflating the customary necessity defence with the non-precluded measures clause in Article XI of the United States–Argentina BIT, while declining to annul the award on that ground.[11] The inconsistency of these outcomes on substantially similar facts became a prominent illustration of the legitimacy concerns surrounding investment arbitration.

Consequences: Cessation and Reparation

A state responsible for an internationally wrongful act is under an obligation to cease the act, if it is continuing, and to offer appropriate assurances and guarantees of non-repetition (Article 30). It is also under an obligation to make full reparation for the injury caused (Article 31). The classic statement of the reparation principle is that of the Permanent Court of International Justice in the *Factory at Chorzów* case:

> The essential principle contained in the actual notion of an illegal act ... is that reparation must, as far as possible, wipe out all the consequences of the illegal act

[9] *CMS Gas Transmission Company v Argentine Republic*, ICSID Case No ARB/01/8, Award, 12 May 2005, paras 315–331.

[10] *LG&E Energy Corp, LG&E Capital Corp and LG&E International Inc v Argentine Republic*, ICSID Case No ARB/02/1, Decision on Liability, 3 October 2006, paras 226–266.

[11] *CMS Gas Transmission Company v Argentine Republic*, ICSID Case No ARB/01/8, Decision of the ad hoc Committee on the Application for Annulment, 25 September 2007, paras 128–136.

> and re-establish the situation which would, in all probability, have existed if that act had not been committed.[12]

Reparation may take the form of restitution, compensation, or satisfaction, either singly or in combination (Article 34). Restitution — re-establishing the situation that existed before the wrongful act — is the primary form, but in practice it is rarely ordered in energy disputes, both because of the practical difficulty of compelling a sovereign to reverse a nationalisation and because of respect for sovereignty. Compensation, covering financially assessable damage including loss of profits insofar as established (Article 36), is the usual remedy. The valuation of compensation for energy assets — using discounted cash flow, comparable transactions, or asset-based methods — is one of the most technically demanding aspects of energy arbitration and is examined in Chapter 26.

2.4 Jurisdiction and Extraterritorial Regulation

Principles of Jurisdiction

Jurisdiction in international law refers to the authority of a state to prescribe rules (prescriptive or legislative jurisdiction), to apply those rules through its courts (adjudicative jurisdiction), and to enforce them (enforcement jurisdiction). The bases of prescriptive jurisdiction are well established:

- **Territoriality.** A state may regulate conduct occurring within its territory. The subjective territorial principle covers conduct initiated within the territory; the objective territorial principle covers conduct completed or producing its effects within it.
- **Nationality.** A state may regulate the conduct of its nationals, including corporations incorporated under its law, wherever they act.
- **Protective principle.** A state may regulate conduct abroad that threatens its security or essential governmental functions.
- **Universality.** For certain grave international crimes, any state may exercise jurisdiction.
- **Passive personality.** A state may, in limited circumstances, regulate conduct abroad that harms its nationals.

The so-called **effects doctrine**, developed in United States antitrust law, permits the regulation of conduct abroad that has substantial, direct, and foreseeable effects within the regulating state. Its application has been controversial, particularly when used to regulate the conduct of foreign companies abroad.

Enforcement jurisdiction is strictly territorial. In the *Lotus* case, the Permanent Court of International Justice held that a state "may not exercise its power in any

[12] *Factory at Chorzów (Germany v Poland)*, Merits, 1928 PCIJ Series A, No 17, p. 47.

form in the territory of another State" without a permissive rule to that effect, while affirming broad discretion as to prescriptive jurisdiction.[13]

Jurisdiction over Maritime Resources

Jurisdiction over offshore energy and resources is governed by UNCLOS, which establishes a graduated system of maritime zones. The coastal state has full sovereignty over its internal waters and territorial sea (up to 12 nautical miles from the baselines), subject to the right of innocent passage. In the exclusive economic zone (EEZ), extending to 200 nautical miles, the coastal state has sovereign rights for the purpose of exploring and exploiting, conserving, and managing natural resources, and with regard to other activities for the economic exploitation and exploration of the zone, "such as the production of energy from the water, currents and winds" (Article 56). On the continental shelf, the coastal state exercises sovereign rights for the purpose of exploring it and exploiting its natural resources (Article 77); these rights are exclusive and do not depend on occupation or express proclamation. Beyond national jurisdiction, the mineral resources of the seabed — "the Area" — are the common heritage of mankind and are administered by the International Seabed Authority. These rules are examined in Chapter 20.

Extraterritorial Regulation in the Energy Sector

Extraterritorial regulation has become increasingly important in the energy sector. Four categories may be distinguished.

Anti-corruption legislation. The United States Foreign Corrupt Practices Act 1977 prohibits the bribery of foreign officials by issuers of securities listed in the United States, domestic concerns, and persons acting within United States territory. The United Kingdom Bribery Act 2010 creates a corporate offence of failure to prevent bribery that applies to any organisation that carries on a business or part of a business in the United Kingdom, regardless of where the bribery occurs. Both statutes have been enforced extensively against energy and mining companies. The OECD Anti-Bribery Convention 1997 requires parties to criminalise the bribery of foreign public officials.

Sanctions. Economic sanctions targeting the energy sectors of particular states — including Iran, Russia, and Venezuela — are a prominent instrument of foreign policy. Some sanctions regimes have "secondary" effects, threatening penalties against foreign persons who engage in specified transactions with sanctioned parties even where there is no connection with the sanctioning state's territory. The European Union has adopted a "Blocking Statute" (Council Regulation (EC) No 2271/96) prohibiting EU persons from complying with certain extraterritorial sanctions. Sanctions are examined in Chapters 22 and 25.

[13] *SS Lotus (France v Turkey)*, 1927 PCIJ Series A, No 10, pp. 18–19.

Supply chain due diligence. A growing body of legislation imposes due diligence obligations on companies with respect to their global supply chains. Section 1502 of the United States Dodd–Frank Act 2010 required disclosure of the use of conflict minerals from the Democratic Republic of the Congo and adjoining countries. The European Union's Conflict Minerals Regulation (EU) 2017/821 imposes due diligence obligations on importers of tin, tantalum, tungsten, and gold. The EU Batteries Regulation (EU) 2023/1542 requires due diligence regarding raw materials used in batteries, and the Corporate Sustainability Due Diligence Directive (EU) 2024/1760 establishes broader human rights and environmental due diligence obligations for large companies, subject to amendments under the European Union's 2025 simplification initiatives.

Climate and environmental measures. The European Union's Carbon Border Adjustment Mechanism (CBAM), established by Regulation (EU) 2023/956, requires importers of specified carbon-intensive goods — including iron and steel, aluminium, cement, fertilisers, electricity, and hydrogen — to purchase certificates reflecting the embedded emissions of those goods, with the definitive regime applying from 2026. The EU Methane Regulation (EU) 2024/1787 imposes monitoring, reporting, and verification requirements on fossil fuel imports. Such measures regulate production processes occurring abroad by attaching conditions to market access, and their compatibility with international trade law is examined in Chapter 13.

PRACTICE NOTE

When advising a multinational energy company on extraterritorial exposure, counsel should map, for each jurisdiction in which the company is incorporated, listed, or carries on business, the anti-corruption, sanctions, export-control, and due diligence regimes that may apply to conduct anywhere in the group. Conflicts between regimes — for example, between a home-state sanction and a host-state contractual obligation to supply — should be identified at the contract negotiation stage and addressed through sanctions clauses, termination rights, and *force majeure* provisions.

2.5 International Organizations and Specialized Energy Institutions

The Fragmented Institutional Landscape

There is no single international organisation with comprehensive competence over energy. Instead, the institutional landscape is fragmented among organisations with specialised mandates, overlapping memberships, and varying legal powers. This fragmentation reflects the political sensitivity of energy, the divergent interests of producers and consumers, and the historical development of

energy governance through issue-specific responses to particular crises. Chapter 4 provides a detailed analysis of the principal energy institutions. This section introduces the main categories and their legal significance as sources and agents of international energy law.

Categories of Institutions

Consumer-oriented institutions. The International Energy Agency was established in 1974 within the framework of the OECD in response to the 1973–1974 oil crisis. Its founding treaty, the Agreement on an International Energy Program, obliges members to maintain emergency oil stocks equivalent to at least ninety days of net imports and to participate in coordinated emergency response measures.

Producer-oriented institutions. OPEC, established in 1960, and the Gas Exporting Countries Forum, established as an intergovernmental organisation in 2008, coordinate the policies of producing states. Since 2016, OPEC and a group of non-OPEC producers including the Russian Federation have coordinated production through the "OPEC+" arrangements under the Declaration of Cooperation.

Technology- and sector-specific institutions. The International Atomic Energy Agency (IAEA), established in 1957, promotes the peaceful uses of nuclear energy and administers safeguards to verify that nuclear material is not diverted to weapons. The International Renewable Energy Agency (IRENA), established in 2009 with headquarters in Abu Dhabi, promotes the adoption of renewable energy.

Treaty-based energy organisations. The Energy Charter Conference is the governing body of the ECT. The Energy Community, established by treaty in 2005, extends the European Union's internal energy market rules to contracting parties in South-East Europe and the Black Sea region, including Ukraine.

Dialogue forums. The International Energy Forum, headquartered in Riyadh, provides a forum for dialogue between producers and consumers, and administers the Joint Organisations Data Initiative (JODI) on oil and gas statistics.

General international organisations with energy competences. The United Nations and its agencies (including the United Nations Environment Programme, the UNFCCC secretariat, and UN-Energy), the World Bank Group (including the IFC and the Multilateral Investment Guarantee Agency), regional development banks, the WTO, and the OECD all exercise significant influence over energy policy.

Regional organisations. The European Union exercises extensive legislative competence over energy under Article 194 of the Treaty on the Functioning of

the European Union. In Africa, the African Union's Agenda 2063, the African Energy Commission (AFREC), the ECOWAS Regional Electricity Regulatory Authority, the West African Power Pool, and the Southern African Power Pool promote regional cooperation.

Legal Powers and Outputs

The legal powers of these institutions vary greatly. Most energy organisations lack the power to adopt decisions binding on their members, and their influence derives from information, expertise, coordination, and soft law. Exceptions include the IEA's emergency response mechanism, the IAEA's safeguards agreements (which are binding treaties between the Agency and individual states), the Energy Community's dispute settlement procedure, and, most significantly, the European Union, whose regulations and directives are binding on member states and, in the case of regulations, directly applicable.

The outputs of international organisations contribute to international energy law in several ways: they may be the forum for the negotiation of treaties; their resolutions and declarations may constitute evidence of customary international law or contribute to its formation; their standards and guidelines may be incorporated into domestic law, contracts, or investment conditions; and their data and analysis frame policy debates.

2.6 Soft Law, Industry Standards, and Corporate Governance

The Concept of Soft Law

"Soft law" refers to normative instruments that are not legally binding in themselves but that nonetheless influence the conduct of states and other actors and may acquire legal effect through incorporation, interpretation, or the formation of custom. Examples include General Assembly resolutions, declarations adopted at international conferences, codes of conduct, guidelines, recommendations of international organisations, and voluntary standards adopted by industry or multi-stakeholder initiatives. The concept is contested: some scholars reject it as a contradiction in terms, insisting on a binary distinction between law and non-law, while others regard it as an indispensable description of the graduated normativity characteristic of contemporary international governance.

Soft law is especially significant in energy and resource governance for several reasons. States are often reluctant to accept binding obligations in a field so closely tied to sovereignty and security. Many relevant actors — corporations, financial institutions, communities — are not subjects of international law in the traditional sense and cannot be parties to treaties. Technological and market conditions change rapidly, making flexible instruments preferable to treaties that are difficult to amend. And soft law can serve as a laboratory for norms that may later harden into binding law.

Principal Soft Law Instruments

The principal soft law instruments relevant to energy and natural resources include:

- **UN General Assembly Resolution 1803 (XVII) (1962)** on permanent sovereignty over natural resources, which, although a resolution, is widely regarded as reflecting customary international law.
- **The Rio Declaration on Environment and Development (1992)**, whose principles — including prevention (Principle 2), precaution (Principle 15), the polluter-pays principle (Principle 16), and environmental impact assessment (Principle 17) — have been incorporated into numerous treaties and domestic laws.
- **The UN Guiding Principles on Business and Human Rights (2011)**, establishing the "Protect, Respect and Remedy" framework.
- **The OECD Guidelines for Multinational Enterprises on Responsible Business Conduct**, revised most recently in 2023, which are backed by a system of National Contact Points that handle complaints ("specific instances") against enterprises.
- **The UN Declaration on the Rights of Indigenous Peoples (2007)**, which articulates the principle of free, prior, and informed consent.
- **The Voluntary Principles on Security and Human Rights (2000)**, a multi-stakeholder initiative addressing the use of public and private security forces by extractive companies.
- **The Natural Resource Charter**, setting out principles for the governance of natural resources for the benefit of citizens.
- **The Santiago Principles (2008)**, Generally Accepted Principles and Practices for sovereign wealth funds, many of which are funded by resource revenues.

Industry Standards and Transnational Private Regulation

Beyond intergovernmental soft law lies a dense body of industry standards and private regulation. Technical standards developed by the American Petroleum Institute (API), the International Organization for Standardization (ISO), and the International Electrotechnical Commission (IEC) govern the design, construction, and operation of energy installations and are frequently incorporated by reference into regulation and contracts. Model contracts developed by industry associations, including the AIEN Model Joint Operating Agreement and Model Unitization and Unit Operating Agreement, standardise commercial relations. The International Association of Oil & Gas Producers (IOGP) publishes guidance on health, safety, and environmental management. In mining, the International Council on Mining and Metals (ICMM) has adopted Mining Principles

that bind its member companies, and the Global Industry Standard on Tailings Management (2020) was developed following the catastrophic collapse of the Brumadinho tailings dam in Brazil in 2019.

Financial institutions exercise considerable regulatory influence through the conditions they attach to project finance. The IFC Performance Standards (2012) establish requirements on environmental and social risk management, labour, resource efficiency, community health and safety, land acquisition and involuntary resettlement, biodiversity, indigenous peoples, and cultural heritage. The Equator Principles, adopted by commercial banks, apply the IFC Performance Standards to project finance transactions above specified thresholds. Because many major energy and mining projects depend on international finance, these standards often impose more demanding requirements than domestic law.

Corporate Governance as a Source of Norms

Corporate governance instruments — codes of conduct, sustainability policies, human rights policies, and climate commitments — are increasingly treated as sources of legal obligation. In *Milieudefensie v Royal Dutch Shell*, the District Court of The Hague in 2021 relied on the UN Guiding Principles and other soft law instruments to give content to the unwritten standard of care under Dutch tort law and ordered Shell to reduce the net carbon dioxide emissions of the Shell group by 45 per cent by 2030 relative to 2019 levels.[14] On appeal, the Hague Court of Appeal in November 2024 set aside the specific reduction order, holding that there was insufficient consensus in climate science on a specific reduction percentage applicable to an individual company, while affirming that companies such as Shell have an obligation to contribute to the reduction of greenhouse gas emissions and that this obligation can be derived from the standard of care informed by human rights and soft law instruments.[15] The case illustrates how soft law can acquire legal force through domestic private law.

The Hardening of Soft Law

Soft law norms can harden into binding law through several pathways: incorporation into treaties (the Rio principles in the UNFCCC and the Convention on Biological Diversity); incorporation into domestic legislation (the UN Guiding Principles in mandatory human rights due diligence statutes); incorporation into contracts (the IFC Performance Standards in loan agreements and host government agreements); judicial reliance (as in the *Shell* litigation); and the formation of customary international law. The study of these pathways is a rich field for doctoral research.

[14] *Milieudefensie et al v Royal Dutch Shell plc*, District Court of The Hague, 26 May 2021, ECLI:NL:RBDHA:2021:5339.

[15] *Shell plc v Milieudefensie et al*, Court of Appeal of The Hague, Judgment of 12 November 2024.

2.7 International Judicial and Arbitral Institutions

The International Court of Justice

The International Court of Justice, the principal judicial organ of the United Nations, has decided numerous cases bearing on energy and natural resources, including maritime delimitation disputes involving offshore petroleum resources (*North Sea Continental Shelf*, 1969; *Land and Maritime Boundary between Cameroon and Nigeria*, 2002; *Maritime Delimitation in the Black Sea*, 2009; *Territorial and Maritime Dispute (Nicaragua v Colombia)*, 2012; *Maritime Delimitation in the Indian Ocean (Somalia v Kenya)*, 2021), disputes over shared watercourses and hydropower (*Gabčíkovo-Nagymaros Project*, 1997; *Pulp Mills on the River Uruguay*, 2010; *Dispute over the Status and Use of the Waters of the Silala*, 2022), and disputes concerning the protection of foreign investment (*Barcelona Traction*, 1970; *Elettronica Sicula SpA (ELSI)*, 1989). Its advisory opinion of 23 July 2025 on the *Obligations of States in respect of Climate Change* is of profound significance for energy law and is analysed in Chapter 18.

The Court's jurisdiction in contentious cases depends on the consent of states, given through special agreement, compromissory clauses in treaties, or optional clause declarations under Article 36(2) of its Statute. Because private parties have no standing, the Court plays a limited role in investor–state disputes, which are resolved principally through arbitration.

The International Tribunal for the Law of the Sea and Annex VII Tribunals

The International Tribunal for the Law of the Sea (ITLOS), established under Annex VI of UNCLOS and seated in Hamburg, has jurisdiction over disputes concerning the interpretation or application of the Convention. Its Special Chamber decided the maritime boundary dispute between Ghana and Côte d'Ivoire in 2017, a case driven by offshore petroleum discoveries, in which the Chamber also rejected Côte d'Ivoire's claim that Ghana had violated its sovereign rights by conducting oil activities in the disputed area.[16] Its Seabed Disputes Chamber delivered an important advisory opinion in 2011 on the responsibilities of states sponsoring activities in the Area, holding that sponsoring states have obligations of due diligence and direct obligations, including the obligation to apply a precautionary approach.[17] In 2024 the full Tribunal delivered an advisory

[16] *Dispute Concerning Delimitation of the Maritime Boundary between Ghana and Côte d'Ivoire in the Atlantic Ocean (Ghana/Côte d'Ivoire)*, ITLOS Case No 23, Judgment of the Special Chamber, 23 September 2017.

[17] *Responsibilities and Obligations of States Sponsoring Persons and Entities with respect to Activities in the Area*, Advisory Opinion, ITLOS Seabed Disputes Chamber, 1 February 2011, ITLOS Reports 2011, p. 10.

opinion holding that anthropogenic greenhouse gas emissions constitute pollution of the marine environment within the meaning of UNCLOS.

Arbitral tribunals constituted under Annex VII of UNCLOS have decided several disputes with energy dimensions, including the *South China Sea Arbitration (Philippines v China)* of 2016, which addressed the legal status of maritime features and the lawfulness of Chinese activities, including interference with Philippine petroleum exploration.

The WTO Dispute Settlement System

The WTO Dispute Settlement Body, comprising panels and the Appellate Body, has decided several disputes concerning energy goods and renewable energy policies. Since December 2019, the Appellate Body has been unable to hear appeals because the United States has blocked the appointment of new members. Many WTO members have joined the Multi-Party Interim Appeal Arbitration Arrangement (MPIA) under Article 25 of the Dispute Settlement Understanding as an interim substitute, but appeals "into the void" by members not party to the MPIA can prevent panel reports from being adopted. This crisis has weakened the enforcement of trade disciplines in the energy sector.

Investment Arbitration Institutions

Investor–state disputes in the energy sector are resolved principally through arbitration under the auspices of:

- **ICSID**, established by the Convention on the Settlement of Investment Disputes between States and Nationals of Other States 1965, with its own self-contained system of annulment rather than review by national courts, and enforcement provisions requiring contracting states to recognise awards as if they were final judgments of their own courts (Article 54);
- **the Permanent Court of Arbitration (PCA)** in The Hague, which administers many arbitrations conducted under the UNCITRAL Arbitration Rules, including the *Yukos* arbitrations and numerous ECT cases;
- **the Arbitration Institute of the Stockholm Chamber of Commerce (SCC)**, designated as one of the forums for investor–state disputes under Article 26 of the ECT;
- **the International Chamber of Commerce (ICC) International Court of Arbitration**, which administers many commercial and some investment disputes;
- **the London Court of International Arbitration (LCIA)** and regional centres, including the Singapore International Arbitration Centre, the Cairo Regional Centre for International Commercial Arbitration, and the Lagos Court of Arbitration.

The institutional framework, procedural rules, and practice of energy arbitration are examined in Chapter 26.

Regional Courts and Human Rights Bodies

Regional courts and human rights bodies have become increasingly important forums for energy-related disputes. The Court of Justice of the European Union has decided cases of enormous significance for intra-EU investment arbitration (*Achmea*, 2018; *Komstroy*, 2021; *PL Holdings*, 2021) and energy market regulation. The European Court of Human Rights, in *Verein KlimaSeniorinnen Schweiz v Switzerland* (2024), held that Article 8 of the European Convention on Human Rights encompasses a right to effective protection by state authorities from the serious adverse effects of climate change.[18] The African Commission on Human and Peoples' Rights, in *Social and Economic Rights Action Center (SERAC) and Center for Economic and Social Rights v Nigeria* (2001), found that Nigeria had violated the African Charter through the conduct of the state oil company and its failure to regulate the activities of oil companies in Ogoniland.[19] The ECOWAS Community Court of Justice, in *SERAP v Federal Republic of Nigeria* (2012), held Nigeria responsible for failing to protect the right to a satisfactory environment in the Niger Delta against the activities of oil companies.

2.8 Conflicts Between Domestic and International Legal Obligations

Monism, Dualism, and Their Limits

The relationship between international law and domestic law is traditionally described in terms of monism and dualism. Monist systems treat international law and domestic law as parts of a single legal order, such that treaties may have direct effect in domestic law upon ratification. Dualist systems treat the two as separate, such that treaties must be incorporated by legislation before they can be applied by domestic courts. Nigeria follows a dualist approach for treaties: section 12(1) of the Constitution of the Federal Republic of Nigeria 1999 provides that "no treaty between the Federation and any other country shall have the force of law except to the extent to which any such treaty has been enacted into law by the National Assembly". In *Abacha v Fawehinmi*, the Supreme Court of Nigeria held that the African Charter on Human and Peoples' Rights, having been domesticated by the African Charter on Human and Peoples' Rights (Ratification and Enforcement)

[18] *Verein KlimaSeniorinnen Schweiz and Others v Switzerland*, App No 53600/20, European Court of Human Rights (Grand Chamber), Judgment of 9 April 2024.

[19] *Social and Economic Rights Action Center (SERAC) and Center for Economic and Social Rights (CESR) v Nigeria*, Communication No 155/96, African Commission on Human and Peoples' Rights, Decision of 27 October 2001.

Act, was part of Nigerian law, although subordinate to the Constitution.[20] The United Kingdom is likewise dualist, whereas the Netherlands and many civil law states are closer to the monist model.

The International Law Perspective

From the perspective of international law, the classification of a state's domestic system is irrelevant to its international obligations. Article 27 of the VCLT provides that "a party may not invoke the provisions of its internal law as justification for its failure to perform a treaty". Article 3 of ARSIWA provides that the characterisation of an act of a state as internationally wrongful is governed by international law and is not affected by its characterisation as lawful by internal law. Accordingly, a state cannot escape liability under an investment treaty by pointing to the lawfulness of its measures under its own constitution, nor can it rely on a domestic court decision annulling a contract as a defence if the decision itself amounts to a denial of justice.

Typical Conflicts in the Energy Sector

Conflicts between domestic and international obligations arise in several recurring situations in the energy sector:

- **Legislative change and stabilisation.** A state enacts new fiscal or environmental legislation that is valid under its constitution but conflicts with stabilisation commitments in contracts or with investment treaty protections.
- **Judicial decisions.** Domestic courts annul or refuse to enforce contracts or arbitral awards, or issue anti-arbitration injunctions, in circumstances that international tribunals may regard as breaches of treaty obligations or of the New York Convention.
- **Constitutional limits on contracting.** Contracts concluded by state entities may be void under domestic law for failure to obtain required approvals, raising questions about whether the state can rely on its own law to avoid its obligations. Article 46 of the VCLT, by analogy, permits a state to invoke a violation of internal law regarding competence to conclude treaties only where the violation was manifest and concerned a rule of fundamental importance.
- **Climate and environmental obligations.** Domestic courts may order a state to adopt more stringent climate measures, as the Supreme Court of the Netherlands did in *State of the Netherlands v Urgenda Foundation* (2019),[21] potentially generating claims by investors whose investments are affected by the resulting measures.

[20] *Abacha v Fawehinmi* (2000) 6 NWLR (Pt 660) 228 (Supreme Court of Nigeria).

[21] *State of the Netherlands v Urgenda Foundation*, Supreme Court of the Netherlands, 20 December 2019, ECLI:NL:HR:2019:2007.

- **Human rights obligations and investment protection.** A state's obligations to protect the rights of indigenous peoples or the right to water may conflict with commitments to investors.

The Intra-EU Investment Arbitration Conflict

The most dramatic recent example of conflict between legal orders arose from intra-EU investment arbitration under the ECT. In *Slovak Republic v Achmea BV* (2018), the Court of Justice of the European Union held that arbitration clauses in intra-EU BITs were incompatible with EU law because they removed disputes potentially concerning the interpretation or application of EU law from the EU judicial system.[22] In *Republic of Moldova v Komstroy LLC* (2021), the Court extended this reasoning to the ECT, holding that Article 26 of the ECT must be interpreted as not applicable to disputes between a member state and an investor of another member state concerning an investment made by the latter in the first member state.[23]

Investment tribunals constituted under the ECT have overwhelmingly rejected the intra-EU objection, holding that they derive their jurisdiction from the ECT as a treaty of international law and that EU law cannot override the consent expressed in it. The result has been a direct conflict between legal orders: awards rendered against EU member states in favour of EU investors (many arising from Spain's withdrawal of renewable energy incentives) have been upheld by arbitral tribunals and enforced or recognised by courts in some non-EU jurisdictions such as the United States, Australia, and the United Kingdom, while courts within the European Union have refused enforcement and the European Commission has treated payment of such awards as potentially constituting unlawful state aid. The European Union, Euratom, and a number of member states subsequently withdrew from the ECT. This saga is a vivid case study in legal pluralism and is examined further in Chapters 14 and 26.

LEADING AUTHORITY

Factory at Chorzów (Germany v Poland), PCIJ, 1928

Facts. Following the transfer of Upper Silesia to Poland after the First World War, Poland took possession of a nitrate factory at Chorzów owned by German companies, in breach of the Geneva Convention of 1922 concerning Upper Silesia. Germany brought proceedings seeking reparation.

Holding. The Permanent Court held that it is a principle of international law that the breach of an engagement involves an obligation to make reparation in an adequate

[22] Case C-284/16, *Slowakische Republik v Achmea BV*, EU:C:2018:158 (CJEU, Grand Chamber, 6 March 2018).

[23] Case C-741/19, *République de Moldavie v Komstroy LLC*, EU:C:2021:655 (CJEU, Grand Chamber, 2 September 2021).

form, and that reparation must, as far as possible, wipe out all the consequences of the illegal act. It distinguished between lawful expropriation, which requires payment of the value of the undertaking at the time of dispossession plus interest, and unlawful expropriation, for which restitution in kind, or payment of a sum corresponding to the value that restitution would bear, is required — potentially including the value of the property at the date of the judgment rather than the date of the taking.

Significance for energy law. *Chorzów* remains the foundation of the law of reparation. Investment tribunals in energy cases — including *ADC v Hungary* (2006), *ConocoPhillips v Venezuela* (2019), and *Yukos Universal v Russia* (2014) — have relied on the distinction between lawful and unlawful expropriation to determine the valuation date and the measure of compensation, with major financial consequences.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Attributing the conduct of a national oil company (hypothetical). The fictional Federal Republic of Calvaria owns 100 per cent of Calvaria National Petroleum Company Ltd (CNPC), a company incorporated under the Calvarian Companies Act. Under the Calvarian Petroleum Act, CNPC holds all state participating interests in joint ventures and is empowered to approve work programmes and budgets on behalf of the state. In 2024, CNPC refused to approve a joint venture work programme submitted by its foreign partner, Delta Ridge Resources Inc, citing "national energy security priorities" communicated in a letter from the Minister of Petroleum. Delta Ridge alleges that the refusal was designed to force it to sell its interest at an undervalue to a company connected to senior officials.

Advise Delta Ridge on (a) whether CNPC's conduct is attributable to Calvaria under ARSIWA Articles 4, 5, and 8; (b) whether the refusal could constitute a breach of an investment treaty obligation as opposed to a mere contractual breach; and (c) what evidence should be gathered to support attribution. *All entities and facts are fictitious.*

CHAPTER SUMMARY

- The formal sources of international energy law are treaties, custom, and general principles, with judicial decisions and scholarly writings as subsidiary means of determining the law; no formal hierarchy exists, but treaties operate as *lex specialis* and peremptory norms prevail over all.
- Treaty interpretation under VCLT Articles 31–33 is central to the field, particularly in investment disputes; systemic integration under Article 31(3)(c) allows environmental and human rights obligations to inform the interpretation of investment treaties.
- Permanent sovereignty over natural resources is a principle of customary international law encompassing both rights and duties; it does not authorise a state

to disregard its commitments, but affirms the state's ultimate authority subject to international law.

- The ILC Articles on State Responsibility govern attribution, breach, defences, and reparation; the attribution of national oil company conduct and the necessity defence have generated extensive energy jurisprudence.
- Jurisdiction is primarily territorial, but extraterritorial regulation through anti-corruption, sanctions, supply chain, and carbon border measures increasingly shapes energy activities.
- International organisations, soft law, industry standards, lender requirements, and corporate governance instruments constitute a significant layer of normative authority, which may harden into binding law.
- Conflicts between domestic and international obligations are resolved, from the international perspective, by the rule that internal law cannot justify non-performance; the intra-EU arbitration saga illustrates the limits of that rule when legal orders collide.

DOCTORAL RESEARCH QUESTIONS

1. Has the proliferation of investment treaties containing similar substantive standards contributed to the formation of customary international law on the treatment of foreign energy investments? What methodology should be used to answer this question after the ILC's 2018 Draft Conclusions?
2. To what extent does the modern, duty-oriented conception of permanent sovereignty over natural resources constrain the freedom of states to grant long-term fossil fuel licences in the light of climate obligations?
3. Should the conduct of national oil companies be subject to a special rule of attribution, given their hybrid commercial and governmental functions?
4. What are the conditions under which soft law instruments such as the UN Guiding Principles acquire binding force through domestic tort law, and is this process democratically legitimate?
5. Does the intra-EU arbitration conflict demonstrate the failure of the techniques for managing fragmentation identified by the International Law Commission?

FURTHER READING

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3 CHAPTER

Resource Sovereignty, State Ownership, and Constitutional Governance

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the legal foundations of resource sovereignty at the international and constitutional levels.
2. Compare the principal models of subsurface resource ownership — state ownership, the accession (landowner) system, and hybrid regimes — and their consequences for licensing and revenue.
3. Analyse the public trust doctrine and related concepts of state stewardship over natural resources.
4. Evaluate the allocation of resource authority in federal systems, with particular reference to Nigeria, Canada, Australia, and the United States.
5. Assess the rights of indigenous peoples and local communities in resource governance.
6. Analyse resource nationalism, expropriation, nationalisation, and the standards of compensation under international law.

Introduction

Chapter 2 examined permanent sovereignty over natural resources as a principle of international law. This chapter turns to the ways in which that sovereignty is organised and exercised within states. International law recognises that a state controls the natural resources within its territory and continental shelf, but it is silent on the question of how ownership and authority are to be distributed internally: whether resources belong to the state, to the owners of the surface, to subnational units, or to communities; which organs of government may grant rights to explore and exploit them; and how the revenues they generate are to be shared. These questions are answered by constitutional law, statute, and custom, and the answers vary greatly among states.

The internal organisation of resource sovereignty has profound practical consequences. It determines who an investor must negotiate with, what kind of title the investor obtains, who receives royalties, and which courts have jurisdiction over disputes. It shapes the political economy of resource-rich states, influencing the incidence of conflict, the quality of governance, and the distribution of wealth. And it determines the extent to which affected communities can participate in decisions about resource development and share in its benefits.

The chapter proceeds in eight sections. It begins with the legal foundations of resource sovereignty, then compares models of constitutional ownership of subsurface resources. It examines the public trust doctrine, federalism and subnational authority, and the position of indigenous peoples. It analyses resource nationalism and the law of expropriation and compensation, and concludes with a comparative survey of constitutional approaches.

3.1 Legal Foundations of Resource Sovereignty

Territorial Sovereignty and Resource Control

Resource sovereignty is, at its root, an incident of territorial sovereignty. A state's sovereignty over its land territory extends to the subsoil beneath it, without any fixed depth limit, and to the airspace above it. Sovereignty over the territorial sea likewise extends to the seabed and subsoil (UNCLOS, Article 2). Beyond the territorial sea, the coastal state enjoys not sovereignty but "sovereign rights" over the natural resources of the EEZ and the continental shelf — functional rights that are exclusive but limited to the purposes specified in the Convention. The distinction between sovereignty and sovereign rights is important: in the EEZ and on the continental shelf, other states enjoy freedoms of navigation, overflight, and the laying of submarine cables and pipelines, subject to the coastal state's rights.

In the classical conception, the state's territorial sovereignty entitled it to deal with its resources as it saw fit, subject only to obligations it had freely assumed and to the general rules protecting the property of aliens. The doctrine of permanent sovereignty over natural resources, discussed in Chapter 2, reinforced this conception and gave it a new political meaning in the context of decolonisation, emphasising that sovereignty over resources belongs to the people of the state and must be exercised for their development and well-being.

The People as Bearers of Sovereignty

A significant development in the modern law of resource sovereignty is the recognition that sovereignty over natural resources is vested not merely in the state as an abstract entity but in the people. Common Article 1(2) of the International Covenants refers to the right of "all peoples" to freely dispose of their natural wealth and resources. Article 21 of the African Charter on Human and Peoples' Rights similarly vests the right in "peoples" and requires that it be exercised "in the

exclusive interest of the people". Many national constitutions declare that natural resources belong to the people, with the state acting as trustee or custodian. Article 257(6) of the Constitution of Ghana 1992, for example, vests every mineral in its natural state in the President "on behalf of, and in trust for, the people of Ghana". Article 62(1)(f) of the Constitution of Kenya 2010 classifies all minerals and mineral oils as public land, and Article 62(3) provides that such public land vests in and is held by the national government in trust for the people of Kenya.

The recognition of the people as bearers of resource sovereignty has several legal implications. It supports arguments that resource revenues must be managed transparently and for the public benefit. It provides a normative basis for the participation of citizens and communities in resource decisions. And it has been invoked to challenge contracts concluded by unrepresentative or corrupt governments, raising difficult questions about the continuity of state obligations following changes of government — an issue examined in the context of odious debt and illegitimate contracts.

The Limits of Sovereignty

Resource sovereignty is subject to limits imposed by international law. These include:

- obligations assumed by treaty, including investment treaties, trade agreements, and environmental agreements;
- the customary obligation to prevent significant transboundary harm, recognised in *Trail Smelter* and affirmed by the International Court of Justice in its *Legality of the Threat or Use of Nuclear Weapons* advisory opinion and in *Pulp Mills*;
- the obligation to cooperate in the management of shared natural resources, such as transboundary watercourses and straddling petroleum deposits;
- the international minimum standard of treatment of aliens and the rules governing expropriation;
- human rights obligations, including the rights of indigenous peoples;
- obligations under the law of armed conflict regarding the exploitation of resources in occupied territory.

These limits do not negate sovereignty; they define the conditions under which it must be exercised.

3.2 Constitutional Ownership of Subsurface Resources

Three Models of Ownership

National legal systems allocate ownership of subsurface resources according to three broad models.

The regalian or domanial system (state ownership). Under this model, which has its origins in Roman law, medieval European royal prerogatives over precious metals, and the Spanish colonial *derecho minero*, subsurface minerals and petroleum belong to the state (historically, the Crown or the sovereign) irrespective of the ownership of the surface. The state grants rights to explore and exploit through licences, concessions, or contracts. The regalian system prevails in the great majority of states, including virtually all of Africa, Latin America, the Middle East, and Asia, and in Europe. In the United Kingdom, petroleum in its natural condition in strata was vested in the Crown by the Petroleum (Production) Act 1934, a vesting now continued by section 2 of the Petroleum Act 1998; coal was nationalised in 1946 and is now vested in the Coal Authority; and gold and silver have long been "royal mines" belonging to the Crown.

The accession or landowner system. Under this model, derived from the common law maxim *cuius est solum, eius est usque ad caelum et ad inferos* ("whoever owns the soil owns it up to the heavens and down to the depths"), the owner of the surface also owns the minerals beneath it, unless they have been severed by conveyance or reserved by the state. The accession system prevails in the United States (for privately owned land) and was historically important in parts of Canada. It is the principal reason why private mineral ownership, mineral leasing between private parties, and royalty interests owned by individuals are characteristic features of American oil and gas law.

The *res nullius* and claim systems. Historically, some systems treated minerals as unowned until discovered and appropriated, giving rise to "free mining" or claim-staking regimes under which the first person to discover and stake a claim acquired rights. The United States General Mining Law of 1872 retains elements of this approach for hard-rock minerals on federal public lands, permitting citizens to locate claims on valuable mineral deposits. Free entry systems have been modified in many Canadian provinces to require consultation with indigenous peoples before claims are registered.

	Regalian / state ownership	Accession / landowner	Claim / free entry
Owner of minerals	State, nation or people (e.g. Nigeria s 44(3), UK, Mexico art 27)	Surface owner unless severed (US private land)	Unowned until located and claimed (US Mining Law 1872)
Grant instrument	Licence, lease, concession, PSC, service contract	Private lease negotiated with mineral owner	Location and registration of claim; patent
Rent capture	Royalty, tax, profit oil, state participation	Bonus and royalty to owner; state taxes	Limited royalty; taxation
Regulator's role	Owner and regulator: opens acreage, sets terms	Prevent waste, protect correlative rights	Record claims, environmental control
Typical issues	Contract stability, transparency, community benefit	Rule of capture, pooling, unitisation	Conflicts with indigenous rights, environment

Figure 3.1 Models of subsurface resource ownership and their typical legal consequences. *Source:* prepared for this edition.

The Rule of Capture and Its Consequences

The accession system, applied to fugacious resources such as oil and gas that migrate through porous rock, gave rise in the United States to the "rule of capture": a landowner owns all the oil and gas produced from wells drilled on his or her land, even if some of that oil and gas migrated from beneath neighbouring land. In *Westmoreland & Cambria Natural Gas Co v De Witt* (1889), the Supreme Court of Pennsylvania analogised oil and gas to wild animals (*ferae naturae*), which belong to the person who captures them. The rule of capture encouraged a race to drill, leading to wasteful over-production, the premature loss of reservoir pressure, and the dissipation of resources.

Terence Daintith's study of the rule of capture demonstrates how this legal doctrine shaped the organisation of the world oil industry, prompting the development of conservation regulation (well spacing, production prorationing, and compulsory pooling and unitisation) in the United States, and influencing the decision of many other states to adopt state ownership and licensing systems that would avoid the inefficiencies of fragmented ownership.[1] The doctrine of correlative rights, developed to mitigate the rule of capture, recognises that each owner

[1] Terence Daintith, *Finders Keepers? How the Law of Capture Shaped the World Oil Industry* (RFF Press 2010).

of land overlying a common reservoir has a right to a fair opportunity to produce its just and equitable share of the oil and gas, and a duty not to injure the reservoir or waste its resources.

Constitutional Vesting Clauses

In states following the regalian model, ownership is typically vested in the state by a constitutional provision. Some illustrative examples follow.

Nigeria. Section 44(3) of the Constitution of the Federal Republic of Nigeria 1999 provides:

> Notwithstanding the foregoing provisions of this section, the entire property in and control of all minerals, mineral oils and natural gas in under or upon any land in Nigeria or in, under or upon the territorial waters and the Exclusive Economic Zone of Nigeria shall vest in the Government of the Federation and shall be managed in such manner as may be prescribed by the National Assembly.

This vesting is reiterated in section 1 of the Petroleum Industry Act 2021 and section 1 of the Nigerian Minerals and Mining Act 2007. The Land Use Act 1978 separately vests all land comprised in the territory of each state in the Governor of that state, to be held in trust and administered for the use and common benefit of all Nigerians, with holders of land enjoying rights of occupancy rather than ownership. The combination of federal ownership of minerals and state control of land, overlaid on customary tenure, generates complex legal relationships that are examined further in section 3.4.

Mexico. Article 27 of the Mexican Constitution of 1917 vests in the Nation direct ownership of all minerals and hydrocarbons, declares that this ownership is inalienable and imprescriptible, and — following the constitutional reforms of 2013 and their partial reversal in 2024–2025 — regulates the participation of private parties in the hydrocarbon and electricity sectors.

Bolivia. Article 349 of the Constitution of Bolivia 2009 declares that natural resources are the property and direct domain, indivisible and imprescriptible, of the Bolivian people, and that their administration corresponds to the state in the collective interest.

Norway. Norway, which has no comparable constitutional clause, relies on statute: section 1-1 of the Petroleum Act 1996 provides that the Norwegian state has the proprietary right to subsea petroleum deposits and the exclusive right to resource management.

Consequences of the Ownership Model

The choice of ownership model has significant consequences. Under state ownership, the state is both owner and regulator; it captures the resource rent

through royalties, taxes, and production shares; and it decides which areas will be opened for exploration and on what terms. The relationship between the state and the investor is typically governed by a licence, concession, or contract, and the investor's rights are those conferred by that instrument. Under the accession system, the landowner captures a substantial portion of the rent through lease bonuses and royalties; the state's revenues derive principally from taxation; and the regulator's role is primarily to prevent waste, protect correlative rights, and safeguard the environment. Political economy research suggests that the distribution of resource rents to private landowners in the United States has helped sustain public support for oil and gas development in producing regions, whereas the concentration of rents in the state under the regalian model may contribute to the disconnection between resource extraction and local benefits that characterises many producer states.

COMPARATIVE PERSPECTIVE

Ownership and community benefit. In the United States, a farmer in North Dakota who owns the mineral estate beneath his land may receive royalties of 12.5 to 20 per cent or more of production value from a shale well, a direct private share of the resource rent. In Nigeria, a farmer in the Niger Delta whose land overlies a producing field owns neither the petroleum nor, under the Land Use Act, the land itself in the full sense; he is entitled to compensation for surface rights and crops affected by operations, and his community may benefit from the host community development trust established under Chapter 3 of the Petroleum Industry Act 2021. The contrast illustrates how the ownership model shapes the distribution of benefits and, consequently, local attitudes towards resource development.

3.3 Public Trust Doctrine and State Stewardship

Origins of the Public Trust Doctrine

The public trust doctrine holds that certain natural resources are held by the state in trust for the public and may not be alienated or managed in a manner inconsistent with the public interest. The doctrine traces its origins to Roman law, which recognised that certain things — the air, running water, the sea, and the shores of the sea — were common to all (*res communes*) under the Institutes of Justinian. It was received into English common law in relation to tidal waters and the foreshore, which the Crown held subject to the public rights of navigation and fishing, and it was developed most fully in the United States.

The leading American authority is *Illinois Central Railroad Co v Illinois* (1892), in which the United States Supreme Court held that the State of Illinois could not irrevocably convey to a private railroad company title to the submerged lands of Chicago's harbour on Lake Michigan, because those lands were held

in trust for the people for purposes of navigation, commerce, and fishing.[2] The Court reasoned that a state could no more abdicate its trust over property in which the whole people are interested than it could abdicate its police powers. The doctrine was revitalised by Joseph Sax's influential 1970 article, which proposed it as a tool for environmental protection, and was extended by the Supreme Court of California in *National Audubon Society v Superior Court* (1983) to require the state to consider the public trust in allocating water rights affecting Mono Lake.[3]

The Doctrine Beyond the United States

The public trust doctrine has been received and adapted in several other jurisdictions. The Supreme Court of India, in *M.C. Mehta v Kamal Nath* (1997), held that the public trust doctrine was part of Indian law and that the state, as trustee of natural resources, was under a legal duty to protect them; it quashed a lease of riparian forest land granted to a motel company.[4] Courts in Pakistan, the Philippines, Kenya, and South Africa have invoked the doctrine or analogous concepts. Section 3 of South Africa's Mineral and Petroleum Resources Development Act 28 of 2002 provides that "mineral and petroleum resources are the common heritage of all the people of South Africa and the State is the custodian thereof for the benefit of all South Africans", a formulation that replaced private mineral ownership with state custodianship. The Constitutional Court of South Africa, in *Agri South Africa v Minister for Minerals and Energy* (2013), held that this transition did not amount to an expropriation of private mineral rights requiring compensation, because the state had not acquired the rights for itself but had become their custodian.[5]

Stewardship and Intergenerational Equity

The public trust doctrine is closely related to the concept of intergenerational equity, which holds that each generation holds the natural and cultural resources of the planet in trust for future generations. Edith Brown Weiss's formulation of the principle of intergenerational equity — comprising conservation of options, conservation of quality, and conservation of access — has been influential in international environmental law.[6] In the context of non-renewable resources, intergenerational equity raises questions about the rate of extraction, the investment of resource revenues, and the obligation to leave future generations with comparable

[2] *Illinois Central Railroad Co v Illinois*, 146 US 387 (1892).

[3] *National Audubon Society v Superior Court of Alpine County*, 33 Cal 3d 419 (1983); see Joseph L. Sax, "The Public Trust Doctrine in Natural Resource Law: Effective Judicial Intervention" (1970) 68 *Michigan Law Review* 471.

[4] *M.C. Mehta v Kamal Nath* (1997) 1 SCC 388 (Supreme Court of India).

[5] *Agri South Africa v Minister for Minerals and Energy* [2013] ZACC 9; 2013 (4) SA 1 (CC).

[6] Edith Brown Weiss, *In Fairness to Future Generations: International Law, Common Patrimony, and Intergenerational Equity* (Transnational/United Nations University 1989).

opportunities. Sovereign wealth funds, such as Norway's Government Pension Fund Global and Nigeria's Sovereign Investment Authority, established by the Nigeria Sovereign Investment Authority (Establishment, etc.) Act 2011 with a Future Generations Fund among its constituent funds, represent institutional responses to this concern.

Climate litigants have invoked the public trust doctrine to argue that governments hold the atmosphere in trust and are obliged to protect it. In *Juliana v United States*, young plaintiffs relied in part on the public trust doctrine to claim that the federal government's promotion of fossil fuels violated their constitutional rights; the Ninth Circuit Court of Appeals held in 2020 that the plaintiffs lacked standing because the relief sought was beyond the power of an Article III court to provide, and the litigation was ultimately terminated after the Supreme Court declined review in 2025.[7] Courts in other jurisdictions have been more receptive to stewardship-based reasoning, as discussed in Chapter 18.

Legal Consequences of Stewardship

Where resources are held in trust, several legal consequences may follow. The state may be prohibited from alienating trust resources entirely or from granting rights that substantially impair the public interest. Grants of rights may be subject to implied conditions requiring their exercise consistently with the trust. Courts may review state decisions affecting trust resources with heightened scrutiny. And citizens may have standing to enforce the trust. These consequences can create tension with the security of tenure sought by investors, particularly where a court later concludes that a grant was inconsistent with the trust. Investors and their advisers should therefore assess, as part of legal due diligence, whether the host jurisdiction recognises a public trust doctrine or constitutional stewardship obligations that could affect the validity or scope of resource rights.

3.4 Federalism and Subnational Resource Authority

The Federal Dilemma

In federal states, the allocation of authority over natural resources between the federal government and the constituent units is among the most politically charged constitutional questions. Resources are often geographically concentrated, so that their ownership and revenues may disproportionately benefit particular regions. Federal ownership may be perceived by producing regions as the appropriation of their wealth by the centre; subnational ownership may produce stark inequalities between resource-rich and resource-poor regions and undermine national cohesion. The resolution of this dilemma varies widely.

[7] *Juliana v United States*, 947 F 3d 1159 (9th Cir 2020).

Nigeria: Federal Ownership, Derivation, and Resource Control

Nigeria provides a paradigmatic example of the federal dilemma. Before independence and during the First Republic, the regions enjoyed significant fiscal autonomy and the principle of derivation — the allocation of revenue to the region from which it was derived — played a major role in revenue sharing. Following the civil war and the rise of petroleum as the dominant source of public revenue, the federal government progressively centralised control. The Petroleum Act 1969 vested ownership of petroleum in the state (the Federal Military Government), and successive constitutions confirmed federal ownership. The derivation percentage fell sharply in the 1970s and 1980s.

The 1999 Constitution restored a minimum derivation component: section 162(2) provides that "the principle of derivation shall be constantly reflected in any approved formula as being not less than thirteen per cent of the revenue accruing to the Federation Account directly from any natural resources". The question of whether offshore revenues should be counted as derived from littoral states gave rise to the resource control litigation. In *Attorney-General of the Federation v Attorney-General of Abia State & 35 Others* (2002), the Supreme Court held that the seaward boundary of a littoral state for the purposes of the derivation principle was the low-water mark of its land surface (or the seaward limit of inland waters), so that revenues from offshore production on the continental shelf and in the EEZ accrued to the Federation rather than to the states. The decision was politically controversial and was effectively reversed in part by the Allocation of Revenue (Abolition of Dichotomy in the Application of the Principle of Derivation) Act 2004, which provided that the 200-metre water depth isobath would be the limit for the purposes of derivation, thereby treating shallow-water offshore production as derived from the adjoining littoral states.

The demand for "resource control" — greater control by producing states and communities over the resources found in their territories and the revenues derived from them — remains a central theme of Nigerian politics. It has been associated with the Ogoni struggle of the 1990s, the Kaiama Declaration of 1998, militancy in the Niger Delta in the 2000s, and the amnesty programme of 2009. The Petroleum Industry Act 2021 responded in part by establishing host community development trusts funded by an annual contribution of three per cent of the actual annual operating expenditure of settlors (holders of interests in petroleum prospecting licences and petroleum mining leases) in the preceding financial year (section 240(2)), although critics argue that this falls far short of meaningful resource control.

Electricity Federalism in Nigeria

In contrast to petroleum, electricity in Nigeria has undergone significant decentralisation. The Constitution (Fifth Alteration) (No. 33) Act 2023 amended the Exclusive and Concurrent Legislative Lists to permit states to legislate on electricity generation, transmission, and distribution within their territories. The Electricity Act 2023, which repealed the Electric Power Sector Reform Act 2005, provides a framework under which states may establish their own electricity markets and regulators, with the Nigerian Electricity Regulatory Commission transferring regulatory oversight of intrastate markets to state regulators once established. A number of states, including Enugu, Ekiti, Ondo, Imo, Oyo, Edo, Kogi, and Lagos, have enacted electricity laws and established regulatory commissions, and regulatory oversight has been transferred to several of them. This process of electricity federalism is examined in Chapter 9.

Canada: Provincial Ownership

In Canada, the provinces own the natural resources within their boundaries by virtue of section 109 of the Constitution Act 1867 (extended to the prairie provinces by the Natural Resources Transfer Agreements of 1930). Section 92A, added by the Constitution Act 1982 following prolonged federal–provincial conflict over energy pricing and taxation (notably the National Energy Program of 1980), confirms exclusive provincial legislative authority over exploration, development, conservation, and management of non-renewable natural resources, forestry resources, and electrical energy, and permits provinces to legislate on the export of such resources to other parts of Canada, subject to federal paramountcy. The federal government retains authority over offshore resources (confirmed by the Supreme Court of Canada in *Reference re Newfoundland Continental Shelf* (1984)), interprovincial and international pipelines, nuclear energy, and matters of national concern. Offshore resources off Newfoundland and Labrador and Nova Scotia are managed jointly through accords implemented by federal and provincial legislation.

The Supreme Court of Canada in *References re Greenhouse Gas Pollution Pricing Act* (2021) upheld federal carbon pricing legislation under the "peace, order and good government" power as a matter of national concern,[8] while in *Reference re Impact Assessment Act* (2023) it held that significant parts of the federal impact assessment regime were unconstitutional because they reached beyond federal jurisdiction into matters of provincial competence.[9]

[8] *References re Greenhouse Gas Pollution Pricing Act*, 2021 SCC 11.

[9] *Reference re Impact Assessment Act*, 2023 SCC 23.

Australia: State Ownership Onshore and Offshore Arrangements

In Australia, the states own onshore minerals and petroleum and regulate their development. Offshore, following the High Court's decision in *New South Wales v Commonwealth* (the *Seas and Submerged Lands Case*) (1975), which upheld Commonwealth sovereignty over the territorial sea and continental shelf, the Offshore Constitutional Settlement of 1979 allocated jurisdiction over the first three nautical miles of the territorial sea to the states, with the Commonwealth retaining authority beyond. Petroleum activities in Commonwealth waters are governed by the Offshore Petroleum and Greenhouse Gas Storage Act 2006, administered by the National Offshore Petroleum Titles Administrator and regulated for safety and environmental matters by the National Offshore Petroleum Safety and Environmental Management Authority (NOPSEMA).

The United States: Layered Ownership

In the United States, mineral ownership is layered among private parties, states, the federal government, and Native American tribes. The federal government owns the mineral estate under approximately 700 million acres of land and administers leasing under the Mineral Leasing Act 1920 and, for the outer continental shelf beyond state waters (generally three nautical miles), under the Outer Continental Shelf Lands Act 1953. The Submerged Lands Act 1953 confirmed state title to submerged lands within three nautical miles of the coast (three marine leagues for Texas and the Gulf coast of Florida). Regulation of oil and gas production on private and state lands is primarily a matter of state law, administered by bodies such as the Railroad Commission of Texas.

3.5 Indigenous Peoples and Resource Governance

The Encounter between Resource Development and Indigenous Rights

A large proportion of the world's remaining mineral, petroleum, and renewable energy resources lies on or beneath lands traditionally owned, occupied, or used by indigenous peoples. The encounter between resource development and indigenous rights has been one of the most significant sources of legal conflict in the field, and the recognition of indigenous rights has fundamentally altered the legal landscape for resource projects in many jurisdictions. Chapter 21 provides a detailed treatment of the international legal framework; this section focuses on the constitutional and property law dimensions.

Recognition of Indigenous Title and Rights

The recognition of indigenous land rights has proceeded through constitutional provisions, legislation, judicial decisions, and treaty arrangements. In Australia, the High Court in *Mabo v Queensland (No 2)* (1992) rejected the doctrine of *terra*

nullius and recognised native title at common law,[10] a recognition subsequently regulated by the Native Title Act 1993, which confers upon native title holders a "right to negotiate" with respect to certain future acts, including the grant of mining tenements. In *Wik Peoples v Queensland* (1996), the High Court held that pastoral leases did not necessarily extinguish native title.

In Canada, section 35 of the Constitution Act 1982 recognises and affirms existing aboriginal and treaty rights. The Supreme Court of Canada recognised aboriginal title in *Delgamuukw v British Columbia* (1997) and, for the first time, made a declaration of aboriginal title over a specific territory in *Tsilhqot'in Nation v British Columbia* (2014), holding that aboriginal title confers the right to decide how the land will be used and to enjoy its economic benefits, and that governments seeking to use titled land without consent must justify the infringement.[11] The Court has also developed a duty to consult and, where appropriate, accommodate aboriginal peoples when the Crown contemplates conduct that might adversely affect asserted or established rights (*Haida Nation v British Columbia (Minister of Forests)*, 2004). These duties have major implications for pipeline and mining projects; in *Clyde River (Hamlet) v Petroleum Geo-Services Inc* (2017), the Supreme Court quashed an authorisation for offshore seismic testing because the consultation with the Inuit had been inadequate.

In Latin America, the Inter-American Court of Human Rights has developed a substantial jurisprudence on indigenous and tribal peoples' property rights under Article 21 of the American Convention on Human Rights. In *Saramaka People v Suriname* (2007), the Court held that the state must ensure the effective participation of the Saramaka people in development or investment plans within their territory, guarantee that they receive a reasonable benefit, and ensure that no concession is issued without a prior environmental and social impact assessment; for large-scale projects with a major impact, the state has a duty to obtain their free, prior, and informed consent.[12] In *Kichwa Indigenous People of Sarayaku v Ecuador* (2012), the Court found Ecuador responsible for permitting a petroleum company to carry out seismic exploration on Sarayaku territory without prior consultation.

In Africa, the African Commission on Human and Peoples' Rights in *Centre for Minority Rights Development (Kenya) and Minority Rights Group International on behalf of Endorois Welfare Council v Kenya* (2010) found that Kenya had violated the rights of the Endorois community by evicting them from their ancestral land around Lake Bogoria without adequate consultation or compensation,

[10] *Mabo v Queensland (No 2)* (1992) 175 CLR 1 (High Court of Australia).

[11] *Tsilhqot'in Nation v British Columbia*, 2014 SCC 44, [2014] 2 SCR 257.

[12] *Saramaka People v Suriname*, Inter-American Court of Human Rights, Judgment of 28 November 2007, Series C No 172, paras 129–137.

and the African Court on Human and Peoples' Rights in *African Commission on Human and Peoples' Rights v Republic of Kenya* (the *Ogiek* case, 2017) found violations of the Ogiek people's rights to their ancestral land in the Mau Forest.

Indigenous Ownership of Subsurface Resources

A distinct question is whether indigenous peoples own the subsurface resources beneath their lands. In most regalian systems, the answer is no: the state's ownership of minerals extends beneath indigenous lands, and indigenous rights are limited to the surface and to rights of consultation, compensation, and benefit-sharing. There are exceptions. In the United States, tribes own the mineral estate beneath many reservation lands, and mineral development is governed by the Indian Mineral Leasing Act 1938 and the Indian Mineral Development Act 1982. In Australia's Northern Territory, the Aboriginal Land Rights (Northern Territory) Act 1976 confers a veto over exploration on Aboriginal land, subject to a national interest override. In Canada, the content of aboriginal title, as described in *Delgamuukw*, includes subsurface mineral rights.

The question has particular resonance in Africa, where the distinction between indigenous peoples and other local communities is often contested and where customary communities assert claims to resources beneath their lands. Under Nigerian law, such claims are foreclosed by section 44(3) of the Constitution, but communities may assert rights to compensation, environmental protection, and participation in benefit-sharing arrangements.

3.6 Resource Nationalism and Regulatory Reform

Defining Resource Nationalism

"Resource nationalism" describes the tendency of governments to assert greater control over natural resources and a larger share of the economic benefits they generate. The term is often used pejoratively by investors and commentators to describe measures perceived as opportunistic or confiscatory, but it can also describe legitimate efforts by states to correct inequitable bargains, to capture a fair share of windfall profits, and to promote national development. A more neutral analytical definition treats resource nationalism as a spectrum of state measures designed to increase state control over, or state revenue from, the resource sector, ranging from fiscal adjustments at one end to outright expropriation at the other.

Resource nationalism is cyclical. It tends to intensify during periods of high commodity prices, when the gap between the returns earned by investors and those anticipated at the time of investment widens and governments come under political pressure to capture windfall gains. It tends to recede during periods of low prices, when states compete to attract investment and are more willing to offer favourable terms. This cyclicality reflects the obsolescing bargain described in Chapter 1.

Forms of Resource Nationalism

Resource nationalism takes a variety of forms, which may be classified by increasing intensity:

1. **Fiscal measures**: increases in royalty rates, income tax rates, or special resource taxes; the introduction of windfall profit taxes or resource rent taxes; restrictions on cost recovery or deductions; and changes to the treatment of transfer pricing.
2. **Local content and beneficiation requirements**: obligations to employ nationals, procure goods and services locally, process minerals domestically before export, or transfer technology. Nigeria's Nigerian Oil and Gas Industry Content Development Act 2010 is a prominent example; Indonesia's ban on the export of unprocessed nickel ore, in force since January 2020, illustrates the beneficiation model.
3. **Export restrictions and domestic supply obligations**: requirements to supply a portion of production to the domestic market, often at below-market prices (domestic market obligations), or restrictions on exports. Nigeria's Petroleum Industry Act 2021 establishes a domestic crude oil supply obligation and a domestic gas delivery obligation.
4. **Increased state participation**: mandatory carried or paid interests for the state or national resource company in new or existing projects; requirements that a minimum percentage of shares be held by nationals. Tanzania's Natural Wealth and Resources (Permanent Sovereignty) Act 2017 and Natural Wealth and Resources Contracts (Review and Re-Negotiation of Unconscionable Terms) Act 2017 combined participation requirements with a statutory power to renegotiate "unconscionable" terms, and the Mining (Amendment) Act 2017 provided for a non-dilutable free carried interest of not less than sixteen per cent for the government in mining companies.
5. **Contract renegotiation and conversion**: unilateral or coerced renegotiation of contracts, or the mandatory conversion of existing contracts to new forms (for example, from concessions to production sharing contracts or service contracts).
6. **Expropriation and nationalisation**: the outright taking of investor assets, with or without compensation.

Drivers and Justifications

Several factors drive resource nationalism: high commodity prices and perceived windfall profits; the perception that existing contracts were negotiated on unfair terms, under duress, or as a result of corruption; fiscal pressures on governments; the political salience of resource sovereignty; the desire to build national industrial capacity; and, increasingly, concerns about energy and resource security. Governments frequently justify their measures by reference to permanent sovereignty,

the public interest, and the need to correct asymmetries of information and bargaining power at the time of contracting.

Regulatory Reform: The Nigerian Petroleum Industry Act 2021

Not all reform of resource governance is properly characterised as resource nationalism. Many states have undertaken comprehensive reforms designed to modernise their legal frameworks, improve governance, and attract investment. Nigeria's Petroleum Industry Act 2021 (PIA), the product of nearly two decades of legislative effort, is a leading example. The PIA:

- restructured the institutional framework by establishing two regulators — the Nigerian Upstream Petroleum Regulatory Commission (NUPRC) and the Nigerian Midstream and Downstream Petroleum Regulatory Authority (NMDPRA) — and providing for the incorporation of the Nigerian National Petroleum Corporation as a limited liability company, Nigerian National Petroleum Company Limited (NNPC Ltd), operating on commercial principles;
- replaced the oil prospecting licence and oil mining lease with petroleum prospecting licences and petroleum mining leases, with provisions on relinquishment, work commitments, and the conversion of existing licences;
- introduced a new fiscal regime comprising a hydrocarbon tax on upstream crude oil profits, companies income tax, and production-based royalties, with incentives for gas and frontier exploration;
- established host community development trusts;
- created frameworks for environmental remediation and decommissioning funds; and
- established the Frontier Exploration Fund and the Midstream and Downstream Gas Infrastructure Fund.

The PIA illustrates the trend towards replacing discretionary, contract-based governance with a comprehensive statutory framework. Its implementation and the conversion of existing licences to the new regime are examined in Chapters 5 and 24.

3.7 Expropriation, Nationalization, and Compensation

Terminology

"Expropriation" refers to the taking of property by the state. "Nationalisation" usually refers to large-scale takings of entire industries or sectors as part of a programme of economic or social reform. "Confiscation" refers to a taking without compensation. "Indirect expropriation" or "creeping expropriation" refers to measures that do not formally transfer title but substantially deprive the owner of the use, enjoyment, or economic value of the property. The distinction

between indirect expropriation and non-compensable regulation is one of the most difficult questions in international investment law and is examined in detail in Chapter 14. This section focuses on the general international law framework, with particular reference to its development through energy disputes.

Conditions of Lawful Expropriation

Customary international law, as reflected in Resolution 1803 and in the practice of investment treaties, permits a state to expropriate foreign property provided that the expropriation is:

1. for a public purpose;
2. non-discriminatory;
3. carried out in accordance with due process of law; and
4. accompanied by the payment of compensation.

These conditions are reflected, with variations, in virtually every investment treaty. Article 13 of the ECT, for example, prohibits expropriation of investments of investors of another contracting party except where the expropriation is "(a) for a purpose which is in the public interest; (b) not discriminatory; (c) carried out under due process of law; and (d) accompanied by the payment of prompt, adequate and effective compensation".

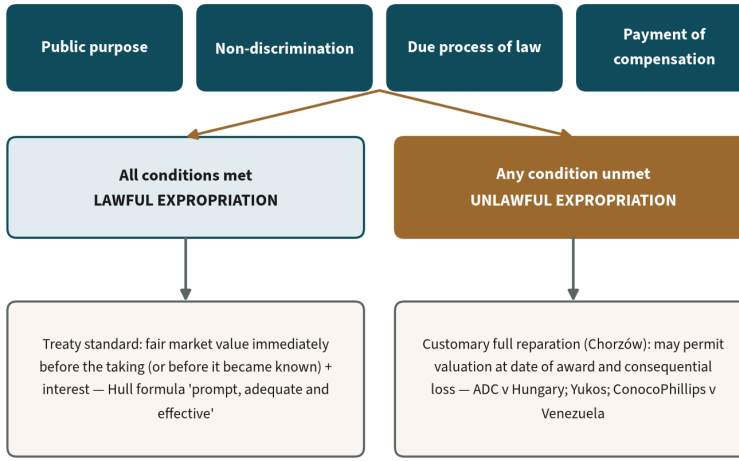


Figure 3.2 The four conditions of lawful expropriation and their consequences for compensation. *Source:* prepared for this edition.

The Standard of Compensation

The standard of compensation has been the most contested aspect of the law of expropriation. Three formulations compete.

The Hull standard — "prompt, adequate and effective" compensation — requires compensation equivalent to the full market value of the expropriated property, paid without delay and in convertible currency. It was championed by capital-exporting states and is reflected in most modern investment treaties, often with the further specification that compensation shall be equivalent to the fair market value of the investment immediately before the expropriation or before it became publicly known, whichever is earlier.

The "appropriate compensation" standard of Resolution 1803 and the Charter of Economic Rights and Duties of States permits account to be taken of a range of circumstances, potentially justifying compensation below full market value. The *Aminoil* tribunal applied this standard, holding that the "legitimate expectations" of the parties, as reflected in the evolution of their contractual relationship, informed the determination of appropriate compensation, and that compensation should take account of a reasonable rate of return rather than speculative profits.

Partial or no compensation has been asserted in the context of large-scale nationalisations forming part of fundamental social and economic reform, on the ground that full compensation would make such reform impossible. This position has not commanded general acceptance as a rule of international law, although lump-sum settlements negotiated between states have frequently provided for compensation at substantially less than full value.

In treaty-based arbitration today, the debate has largely been resolved in favour of the full market value standard for lawful expropriation, because the treaties themselves specify it. The more significant debates now concern valuation methodology, the valuation date, and the distinction between lawful and unlawful expropriation.

Lawful and Unlawful Expropriation

Following *Chorzów Factory*, tribunals distinguish between lawful expropriation, for which compensation is due under the treaty standard (typically fair market value at the date of taking), and unlawful expropriation, for which full reparation is due under customary international law. The distinction matters because full reparation may require valuation at the date of the award rather than the date of taking, allowing the claimant to benefit from subsequent increases in value — for example, increases in oil prices.

In *ADC Affiliate Ltd v Hungary* (2006), the tribunal held that because Hungary's expropriation was unlawful, the claimants were entitled to the value of their investment at the date of the award, which was considerably higher

than at the date of the taking.[13] In *Yukos Universal Ltd v Russian Federation* (2014), the tribunal found that Russia's measures against Yukos amounted to an unlawful expropriation and awarded damages of approximately USD 50 billion to the former majority shareholders, using a valuation date chosen to give the claimants the benefit of the higher value.[14] In *ConocoPhillips v Venezuela* (2019), the tribunal, having previously held that Venezuela's taking of ConocoPhillips' interests in heavy oil projects was unlawful because Venezuela had failed to negotiate compensation in good faith, awarded approximately USD 8.5 billion plus interest.[15] By contrast, in *Venezuela Holdings BV (Mobil) v Venezuela* (2014), the tribunal found that the claimants had not established that the expropriation was unlawful merely because the compensation offered was inadequate, and accordingly applied the treaty standard of compensation for lawful expropriation, valuing the investment as of the date of the taking.

The Yukos awards were set aside by the District Court of The Hague in 2016 on jurisdictional grounds relating to the provisional application of the ECT, reinstated by the Hague Court of Appeal in 2020, and subjected to further proceedings, with the Supreme Court of the Netherlands in 2021 remitting the question of alleged fraud in the arbitration to the Amsterdam Court of Appeal, which rejected Russia's fraud arguments in 2024. The saga illustrates the complexity and duration of enforcement proceedings against a sovereign that resists payment.

Compensation for Natural Resource Projects: Valuation Issues

The valuation of expropriated natural resource projects presents distinctive difficulties. The value of a petroleum or mining project depends on the volume of reserves, future commodity prices, production costs, fiscal terms, and the risk profile of the host state, each of which is uncertain. Tribunals have generally accepted the discounted cash flow (DCF) method for producing assets with an established record of profitability, while being sceptical of DCF valuations for projects at an early stage of development. In *Tethyan Copper Company Pty Limited v Pakistan* (2019), the tribunal awarded approximately USD 4 billion in damages and interest for the denial of a mining lease for the Reko Diq copper-gold deposit, valuing the project by DCF despite the fact that it had not yet entered production — a decision that attracted criticism and was followed by a negotiated

[13] *ADC Affiliate Limited and ADC & ADMC Management Limited v Republic of Hungary*, ICSID Case No ARB/03/16, Award, 2 October 2006, paras 493–499.

[14] *Yukos Universal Limited (Isle of Man) v Russian Federation*, PCA Case No AA 227, Final Award, 18 July 2014.

[15] *ConocoPhillips Petrozuata BV, ConocoPhillips Hamaca BV and ConocoPhillips Gulf of Paria BV v Bolivarian Republic of Venezuela*, ICSID Case No ARB/07/30, Award, 8 March 2019.

settlement in 2022 involving the reconstitution of the project with new participation arrangements.[16]

The Iran–United States Claims Tribunal and Lump-Sum Settlements

A substantial body of jurisprudence on the expropriation of energy assets was generated by the Iran–United States Claims Tribunal, established by the Algiers Accords of January 1981 following the Iranian revolution and the hostage crisis. Many of the claims arose from the nationalisation of the Iranian oil industry and the termination of petroleum agreements after 1979. In *Amoco International Finance Corporation v Iran* (1987), Chamber Three distinguished between lawful and unlawful expropriation and held that, where the taking was lawful, the compensation owed was the value of the undertaking as a going concern at the date of the taking, excluding the speculative future profits that would have been recoverable as damages for an unlawful taking.[17] In *Phillips Petroleum Company Iran v Iran* (1989), the Tribunal held that the claimant was entitled to compensation for the expropriation of its contractual rights under a joint structure agreement with the National Iranian Oil Company, and that the full value of those rights was to be determined using a discounted cash flow analysis informed by contemporaneous price expectations.[18] In *Starrett Housing Corporation v Iran* (1983), the Tribunal recognised that measures taken by a state can interfere with property rights to such an extent that the rights are rendered so useless that they must be deemed to have been expropriated, even though the state does not purport to have expropriated them and legal title remains with the original owner — a foundational statement of the concept of indirect expropriation.

Outside institutional adjudication, many nationalisation disputes have been resolved by lump-sum agreements between the expropriating state and the home state of the investors, under which the expropriating state pays a global sum that the home state distributes among its nationals through a domestic claims commission. The United States Foreign Claims Settlement Commission and the United Kingdom Foreign Compensation Commission administered such distributions in respect of nationalisations in Eastern Europe, Cuba, and elsewhere. The compensation provided under lump-sum agreements has frequently been considerably less than the full value of the property taken. Whether such settlements constitute state practice supporting a standard of less-than-full compensation, or merely represent negotiated compromises without *opinio juris*, remains debated;

[16] *Tethyan Copper Company Pty Limited v Islamic Republic of Pakistan*, ICSID Case No ARB/12/1, Award, 12 July 2019.

[17] *Amoco International Finance Corporation v Islamic Republic of Iran*, Partial Award No 310-56-3, 14 July 1987, 15 Iran-US CTR 189.

[18] *Phillips Petroleum Company Iran v Islamic Republic of Iran*, Award No 425-39-2, 29 June 1989, 21 Iran-US CTR 79.

the prevailing view is that they reflect pragmatic settlement rather than a belief that international law requires no more.

Due Process and Judicial Review of Expropriation

The due process condition requires that an expropriation be carried out in accordance with the procedures established by domestic law and with the fundamental international standards of procedural fairness, including reasonable advance notice, a fair hearing, and the opportunity to challenge the measure before an independent and impartial body. In *ADC v Hungary*, the tribunal held that due process "demands an actual and substantive legal procedure for a foreign investor to raise its claims against the depriving actions already taken or about to be taken against it", including a reasonable chance within a reasonable time to claim its legitimate rights and have its claims heard.[19] Domestic constitutional law frequently imposes additional requirements. Section 44(1) of the Nigerian Constitution, for instance, prohibits the compulsory acquisition of any interest in moveable or immoveable property except in the manner and for the purposes prescribed by a law that, among other things, requires the prompt payment of compensation and gives the person claiming compensation a right of access to a court or tribunal for the determination of his interest and the amount of compensation. Section 25 of the Nigerian Investment Promotion Commission Act (Cap N117, Laws of the Federation of Nigeria 2004) further provides that no enterprise shall be nationalised or expropriated by any government of the Federation, and that no person who owns the enterprise shall be compelled by law to surrender his interest to any other person, save where the acquisition is in the national interest or for a public purpose and under a law that provides for payment of fair and adequate compensation and a right of access to the courts.

3.8 Comparative Constitutional Approaches

A Typology of Constitutional Approaches

Constitutions address natural resources in strikingly different ways. A comparative survey reveals at least five approaches:

1. **Silence.** Some constitutions, including the Constitution of the United States, do not address natural resource ownership at all, leaving the matter to property law and legislation.
2. **Vesting clauses.** Many constitutions vest ownership of minerals and petroleum in the state, the nation, or the people (Nigeria, Mexico, Ghana).

[19] *ADC Affiliate Limited v Republic of Hungary*, ICSID Case No ARB/03/16, Award, 2 October 2006, para 435.

3. **Federal allocation clauses.** Federal constitutions allocate authority between levels of government (Canada's sections 92A and 109; Nigeria's Exclusive Legislative List and derivation clause).
4. **Stewardship and sustainability clauses.** Some constitutions impose duties of sustainable management. Article 112 of the Constitution of Norway (formerly Article 110b) provides that natural resources shall be managed on the basis of comprehensive long-term considerations that will safeguard this right for future generations as well. The Supreme Court of Norway in *Greenpeace Nordic and Nature and Youth v Ministry of Petroleum and Energy* (2020) held that Article 112 is a substantive right but that the threshold for judicial intervention in parliamentary decisions is high, and it declined to invalidate Arctic petroleum licences; the subsequent *Greenpeace Nordic v Norway* proceedings before the Oslo District Court in 2024 held certain field development approvals invalid for failure to assess combustion emissions.
5. **Rights of nature.** The Constitution of Ecuador 2008 (Articles 71–74) recognises the rights of nature (*Pacha Mama*) to exist, persist, maintain, and regenerate its vital cycles. The Constitutional Court of Ecuador in the *Los Cedros* case (2021) held that mining concessions in the Los Cedros protected forest violated the rights of nature.

Revenue-Sharing Clauses

Constitutions in resource-rich federations frequently prescribe revenue-sharing arrangements. Nigeria's thirteen per cent derivation principle has been noted. The Constitution of Iraq 2005, Article 112, provides that the federal government, with the producing governorates and regional governments, shall undertake the management of oil and gas extracted from present fields, with revenues distributed in a fair manner in proportion to the population distribution — a provision whose ambiguity has fuelled protracted disputes between Baghdad and the Kurdistan Regional Government, including the 2022 ruling of Iraq's Federal Supreme Court declaring the Kurdistan Region's oil and gas law unconstitutional, and the 2023 ICC arbitral award in Iraq's favour concerning exports through the Iraq–Türkiye pipeline. The Constitution of Kenya 2010 requires legislation providing for the sharing of benefits from natural resources between the national government, counties, and local communities, implemented in part by section 58 of the Petroleum Act 2019, which allocates 20 per cent of the national government's share of profit petroleum to the county government and 5 per cent to the local community.

Constitutional Governance of Resource Revenues

A further dimension of constitutional governance concerns the management of the revenues generated by resource extraction. Because petroleum and mineral revenues are volatile, finite, and often very large relative to the rest of the economy, they create risks of macroeconomic instability, pro-cyclical spending, and the diversion of public funds. Constitutional and quasi-constitutional responses fall into four categories.

Fiscal rules limit the use of resource revenues in the annual budget. Norway's fiscal rule, adopted in 2001, provides that the structural non-oil deficit should over time correspond to the expected real return on the Government Pension Fund Global, now estimated at three per cent. Chile's structural balance rule adjusts fiscal targets for the long-term price of copper. Nigeria's Fiscal Responsibility Act 2007 introduced a budget framework based on a reference oil price, with revenues above that price saved in the Excess Crude Account, an arrangement whose legal basis was repeatedly contested between the federal and state governments.

Sovereign wealth funds set aside resource revenues for stabilisation, saving, or development purposes. Their governance is guided by the Santiago Principles. The Nigeria Sovereign Investment Authority Act 2011 established three funds — the Stabilisation Fund, the Future Generations Fund, and the Nigeria Infrastructure Fund — each with defined investment mandates. Ghana's Petroleum Revenue Management Act 2011 (Act 815) established the Ghana Stabilisation Fund and the Ghana Heritage Fund and created a Public Interest and Accountability Committee to monitor compliance, an innovative example of statutory civil society oversight.

Earmarking and distribution rules allocate revenues to particular purposes or populations. Some states have experimented with direct distribution of resource revenues to citizens, as in the Alaska Permanent Fund Dividend, paid annually to qualifying residents since 1982.

Transparency and audit obligations require the publication of resource revenues and contracts and their independent audit. The Nigeria Extractive Industries Transparency Initiative (NEITI) Act 2007 made Nigeria the first EITI implementing country to give the initiative a statutory basis, requiring periodic audits of the extractive sector and the publication of the results. Section 3 of the Act sets out NEITI's objectives, including ensuring due process and transparency in payments made by extractive industry companies to the federal government and its agencies.

These arrangements form part of the constitutional governance of resources in a broad sense, and their effectiveness depends on enforcement mechanisms — legislative oversight, independent audit institutions, judicial review, and civil

society monitoring. Doctoral research on the resource curse has increasingly focused on the design and performance of such arrangements.

Lessons from Comparison

Comparative analysis suggests several lessons. First, the formal ownership model matters less than the quality of the institutions that administer it; state ownership can be associated with excellent governance (Norway) or with chronic mismanagement. Second, revenue-sharing arrangements that leave producing regions and communities with a visible and meaningful share of benefits tend to reduce conflict, but must be designed to avoid creating new inequalities or local governance failures. Third, constitutional entrenchment of sustainability and stewardship duties can provide a basis for judicial review of resource decisions, although courts have generally been cautious about intervening in what they regard as policy matters. Fourth, the recognition of indigenous and community rights through constitutional, judicial, and treaty mechanisms has become a defining feature of resource governance, with profound implications for project development.

LEADING AUTHORITY

Government of the State of Kuwait v American Independent Oil Company (Aminoil), 1982

Facts. Aminoil held a sixty-year concession granted in 1948 by the Ruler of Kuwait over the Kuwaiti share of the Neutral Zone. The concession contained stabilisation clauses providing that the Ruler would not annul the agreement by general or special legislation or administrative measures, and that no alteration should be made except by mutual agreement. Over the years, the fiscal terms were revised several times. In 1977, Kuwait terminated the concession by decree and took over Aminoil's assets.

Holding. The tribunal (Professor Paul Reuter, President; Professor Hamed Sultan; Sir Gerald Fitzmaurice) held that the nationalisation was lawful. The stabilisation clauses did not prohibit nationalisation because such a far-reaching limitation on sovereignty would require express stipulation and would have to cover a relatively limited period; moreover, the contract had evolved through renegotiation. However, the stabilisation clauses gave rise to legitimate expectations relevant to the assessment of compensation, which was to be "appropriate" and was calculated by reference to the value of the undertaking and a reasonable rate of return. Kuwait was ordered to pay approximately USD 179 million, including interest.

Significance. Aminoil is the leading authority for the proposition that stabilisation clauses operate primarily to affect compensation rather than to invalidate sovereign acts. Its emphasis on the evolving nature of long-term contracts and on legitimate expectations anticipated modern debates over fair and equitable treatment.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Constitutional design for a new petroleum producer (hypothetical). The fictional Republic of Tamara, a federal state with five provinces, has recently discovered substantial onshore and offshore petroleum reserves. Its constitution is silent on resource ownership. A constitutional review commission has asked you to prepare a 3,000-word advisory paper addressing: (a) the choice between federal and provincial ownership of onshore and offshore resources; (b) the design of a revenue-sharing formula, including any derivation component and its treatment of offshore production; (c) the incorporation of stewardship and intergenerational equity duties; (d) the recognition of the rights of customary landholding communities; and (e) mechanisms to reduce the risk of resource-related conflict. Draw on comparative experience from at least four jurisdictions. *Tamara and its circumstances are fictitious.*

CHAPTER SUMMARY

- Resource sovereignty is an incident of territorial sovereignty, extended by UNCLOS to sovereign rights over the resources of the EEZ and continental shelf; modern instruments increasingly locate that sovereignty in the people, with the state as trustee.
- Subsurface ownership follows three broad models — regalian (state), accession (landowner), and claim-based systems — with major consequences for licensing, revenue distribution, and community attitudes.
- The public trust doctrine and related stewardship concepts impose fiduciary-type duties on states, enforceable in some jurisdictions by citizens and capable of affecting the security of investor rights.
- Federal systems allocate resource authority in widely differing ways; Nigeria's combination of federal ownership, a thirteen per cent derivation floor, the onshore/offshore dichotomy litigation, and recent electricity decentralisation illustrates the political sensitivity of the federal dilemma.
- Indigenous rights to land, consultation, consent, and benefit-sharing have been recognised through constitutional provisions, judicial decisions, and regional human rights jurisprudence, transforming the legal environment for resource projects.
- Resource nationalism is cyclical and takes forms ranging from fiscal adjustments to expropriation; it must be distinguished from good-faith regulatory reform such as Nigeria's Petroleum Industry Act 2021.
- Lawful expropriation requires public purpose, non-discrimination, due process, and compensation; the distinction between lawful and unlawful expropriation determines the measure and valuation date of compensation, with very large financial consequences in energy cases.

DOCTORAL RESEARCH QUESTIONS

1. Does the constitutional recognition of the people as bearers of resource sovereignty create enforceable obligations regarding the management of resource revenues, and how might such obligations be judicially enforced?
2. To what extent has the public trust doctrine been, or could it be, received into Nigerian law as a constraint on the grant of petroleum and mining rights?
3. Evaluate whether host community development trusts under the Petroleum Industry Act 2021 represent a meaningful response to demands for resource control, using comparative evidence from benefit-sharing regimes elsewhere.
4. Is the distinction between lawful and unlawful expropriation, with its consequences for valuation date, theoretically justified, or does it produce windfalls for claimants inconsistent with the compensatory purpose of reparation?
5. How should the valuation of undeveloped resource projects in investment arbitration account for the risk that climate policy will render reserves unburnable?

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4 CHAPTER

International Energy Institutions, Regulatory Systems, and Governance

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Describe the architecture of global energy governance and explain why it is fragmented and polycentric.
2. Analyse the legal foundations, mandates, powers, and limitations of the International Energy Agency, OPEC, the International Renewable Energy Agency, and the International Atomic Energy Agency.
3. Evaluate the role of regional energy regulatory institutions, including those of the European Union, the Energy Community, and African regional bodies.
4. Explain the rationale for independent regulatory agencies and the mechanisms by which their accountability is secured.
5. Assess the prospects for regulatory cooperation and institutional reform in global energy governance.

Introduction

Institutions matter. The rules of international energy law described in the previous chapters do not apply themselves; they are negotiated, interpreted, monitored, and enforced by institutions — international organisations, regional bodies, national regulators, courts, and tribunals. The design of these institutions shapes the content and effectiveness of the law. An institution with a narrow mandate, limited membership, and no binding powers will produce different outcomes from one with broad competence, universal membership, and the authority to adopt binding decisions.

Global energy governance is distinctive for the absence of a central institution. Unlike trade, which is governed by the WTO, or international civil aviation, which is governed by the International Civil Aviation Organization, energy has no single global organisation with comprehensive competence. Instead, governance

is distributed among a constellation of institutions with specialised mandates, overlapping memberships, and divergent interests. This chapter maps that constellation, examines the principal institutions in detail, and considers the role of national regulators and the prospects for reform.

4.1 Global Energy Governance Architecture

The Concept of Global Energy Governance

"Global energy governance" refers to the collective efforts of states and non-state actors to manage energy-related problems that transcend national borders through rules, institutions, and processes. The problems that global energy governance addresses include the security of supply for importing states and security of demand for exporting states, the stability of energy prices, the prevention of nuclear proliferation, the mitigation of climate change, the promotion of energy access in developing countries, the governance of transboundary infrastructure, and the protection of foreign investment.

Scholars have characterised the architecture of global energy governance as a "regime complex" — a set of partially overlapping and non-hierarchical institutions governing a particular issue area. The concept, developed by Kal Raustiala, David Victor, Robert Keohane, and others, captures the absence of a single overarching regime and the presence of multiple institutions whose rules and activities interact, sometimes reinforcing and sometimes conflicting with one another.

Why Is Energy Governance Fragmented?

Several factors explain the fragmentation of global energy governance.

Divergent interests. Producers and consumers have fundamentally opposed interests regarding prices: producers seek high and stable prices, consumers seek low and stable prices. These opposing interests have produced separate institutions — OPEC for producers, the IEA for consumers — rather than a single inclusive organisation.

Sovereignty sensitivities. Energy resources are closely tied to national sovereignty and security. States have been reluctant to cede authority over energy policy to international institutions, preferring arrangements that preserve national discretion.

Path dependence. Energy institutions were created in response to specific crises — the nuclear arms race (IAEA), the oil shocks of the 1970s (IEA), the collapse of the Soviet Union (Energy Charter), the climate crisis (UNFCCC, IRENA) — and each bears the imprint of the circumstances of its creation.

Technological diversity. Different energy sources raise different governance challenges: nuclear energy raises proliferation and safety concerns; oil raises supply security and price stability concerns; renewables raise technology diffusion and investment concerns. Specialised institutions have emerged for each.

Shifting centres of gravity. The growth of energy demand in emerging economies, particularly China and India, has shifted the centre of gravity of global energy markets away from the OECD states that dominated the institutions created in the twentieth century. Existing institutions have struggled to accommodate these new actors.

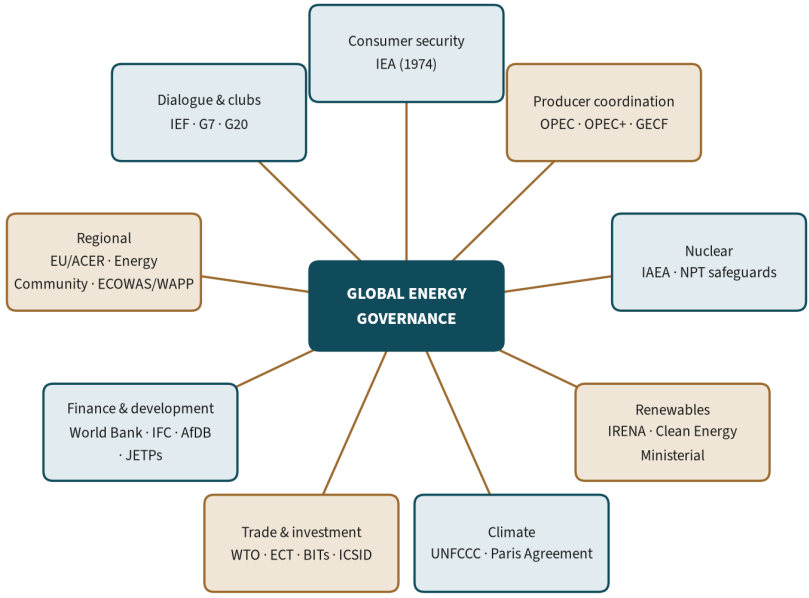


Figure 4.1 The polycentric architecture of global energy governance. *Source:* prepared for this edition.

Mapping the Architecture

The architecture of global energy governance may be mapped along two dimensions: the *functional* dimension, which identifies the governance function performed (security, markets, investment, technology, environment, development, nuclear non-proliferation), and the *membership* dimension, which distinguishes between universal, plurilateral, and regional institutions. Table 4.1 summarises the principal institutions.

Table 4.1 Principal international energy institutions. Membership figures are approximate and change over time; readers should consult the institutions' official sources for current data.

Institution	Founded	Legal basis	Member-ship	Core function	Binding pow-ers
International Energy Agency (IEA)	1974	Agreement on an International En-ergy Program	32 mem-ber coun-tries (OECD states) plus as-sociation countries	Oil supply secu-rity; energy data and analysis; clean energy tran-sitions	Emergency stockholding and response obligations
OPEC	1960	OPEC Statute	12 mem-ber coun-tries (2025)	Coordination of petroleum poli-cies; market sta-bilisation	Production decisions po-litically bind-ing; no exter-nal enforce-ment
International Renewable Energy Agency (IRENA)	2009	IRENA Statute	Approx. 170 members	Promotion of re-newable energy	None (advi-sory)
International Atomic En-ergy Agency (IAEA)	1957	IAEA Statute	Approx. 180 member states	Peaceful uses of nuclear energy; safeguards; safety standards	Safeguards agreements; standards binding when incorporated
Energy Char-ter Confer-ence	1994	Energy Charter Treaty	Contract-ing par-ties (con-tracted after EU with-drawal)	Investment pro-tection; transit; trade	Treaty obliga-tions; in-vestor–state arbitration
International Energy Forum (IEF)	1991	IEF Charter (char-ter 2011)	Approx. 70 coun-tries	Producer–con-sumer dialogue; JODI data	None
Gas Exporting Countries Fo-rum (GECF)	2008	GECF Statute	Gas-ex-porting states	Coordination of gas producer in-terests	None
UNFCCC / Paris Agree-ment	1992 / 2015	Treaties	Near-uni-versal	Climate mitiga-tion and adapta-tion	Procedural obligations; NDC cycle

Informal Governance: G7, G20, and Clubs

In addition to formal international organisations, informal groupings play a significant role in energy governance. The Group of Seven (G7) and the Group of Twenty (G20) have repeatedly addressed energy security, fossil fuel subsidies, and clean energy transitions. At the G20 summit in Pittsburgh in 2009, leaders committed to "rationalize and phase out over the medium term inefficient fossil

fuel subsidies that encourage wasteful consumption". The G7's Climate Club, launched in 2022–2023, seeks to accelerate the decarbonisation of industry. The Clean Energy Ministerial and Mission Innovation bring together energy ministers to promote clean energy technologies. Just Energy Transition Partnerships (JETPs), concluded between groups of donor states and South Africa (2021), Indonesia (2022), Vietnam (2022), and Senegal (2023), combine public and private finance to support the transition away from coal. Such informal arrangements lack legal personality and binding powers but can mobilise political commitment and finance.

4.2 The International Energy Agency and Its Institutional Role

Origins and Legal Foundation

The International Energy Agency was created in November 1974 in response to the oil embargo imposed by Arab members of OPEC in October 1973 following the Yom Kippur War and the associated quadrupling of oil prices. The embargo exposed the vulnerability of industrialised oil-importing states and the absence of any mechanism for coordinating their response. At the initiative of the United States, sixteen OECD states signed the Agreement on an International Energy Program (IEP Agreement) on 18 November 1974, establishing the IEA as an autonomous agency within the framework of the OECD.

The IEP Agreement is a treaty. Membership of the IEA is open only to members of the OECD that are willing and able to meet the requirements of the emergency response mechanism, including the holding of oil stocks. This requirement has been a significant barrier to the membership of major emerging economies such as China and India, which are not OECD members. To address this limitation, the IEA launched an "association" initiative in 2015, under which China, India, Indonesia, Brazil, South Africa, and other states have become association countries participating in IEA work without assuming the obligations of full membership.

Emergency Response Mechanism

The core legal obligation imposed by the IEP Agreement is the maintenance of emergency oil reserves. Article 2 requires each participating country to maintain emergency reserves sufficient to sustain consumption for at least sixty days with no net oil imports, a requirement subsequently increased to ninety days of net imports. The reserves may be held by governments (as in the United States Strategic Petroleum Reserve), by dedicated stockholding agencies, or by industry under obligation.

The IEP Agreement also established an emergency sharing system, triggered when the group or an individual participating country sustains or can reasonably be expected to sustain a reduction in the daily rate of its oil supplies of at least seven

per cent of the average daily rate of its final consumption during a base period. Upon activation, participating countries are required to implement demand restraint measures and to share available oil. In practice, the formal emergency sharing system has never been activated. Instead, the IEA has relied on "coordinated emergency response measures" (CERM), a more flexible mechanism under which members make stocks available in coordinated releases. Coordinated stock releases were undertaken in 1991 (Gulf War), 2005 (Hurricanes Katrina and Rita), 2011 (Libyan civil war), and twice in 2022 (following the Russian invasion of Ukraine), the latter being the largest in the Agency's history.

Evolution of the IEA's Mandate

Although established principally to address oil supply security, the IEA's mandate has broadened considerably. Its *World Energy Outlook*, published annually, is among the most influential sources of energy data and scenarios. In 2021, the IEA published *Net Zero by 2050: A Roadmap for the Global Energy Sector*, which concluded that achieving net zero emissions by 2050 would require no new oil and gas fields approved for development beyond those already committed as of 2021, and no new coal mines or mine extensions. The report was controversial among producers and some IEA members, illustrating the tensions between the Agency's historical mission of supply security and its newer role in promoting clean energy transitions. In 2022, IEA ministers mandated the Agency to help countries achieve net zero emissions, and the Agency has developed significant work programmes on critical minerals security, including the Critical Minerals Security Programme.

The IEA and Critical Minerals Security

The IEA's recent expansion into critical minerals illustrates how institutions created for one security problem adapt to another. The Agency's 2021 special report, *The Role of Critical Minerals in Clean Energy Transitions*, and its subsequent annual *Global Critical Minerals Outlook* documented the high geographical concentration of the extraction and, especially, the processing of minerals such as lithium, cobalt, nickel, graphite, and rare earth elements, and warned that supply disruptions could slow the energy transition and raise its cost. In 2024, IEA members launched the Critical Minerals Security Programme, which includes work on supply diversification, market transparency, and the feasibility of stockholding arrangements for critical minerals analogous in some respects to emergency oil stocks.

The analogy with oil is, however, imperfect. Oil is a fungible commodity traded on deep and liquid markets, whereas many critical minerals are traded in thin, opaque markets with product specifications tailored to individual buyers; stockholding is therefore more complex and potentially more expensive. More-

over, the security risks associated with critical minerals arise less from sudden supply interruptions than from structural concentration and the use of export controls as instruments of industrial and geopolitical strategy. Whether the IEA's legal toolkit, built around the IEP Agreement's oil-specific obligations, can be extended to minerals without treaty amendment is a significant institutional question. Chapter 8 examines the substantive law of critical minerals.

Legal Assessment

The IEA is a hybrid institution. Its emergency response obligations are legally binding treaty commitments, but its broader influence derives from its analytical authority, convening power, and technical expertise rather than from legal compulsion. Its limited membership constrains its legitimacy as a global institution, and its identification with consuming states limits its acceptance by producers. Nevertheless, it remains the most important institution for consumer-country cooperation and an indispensable source of the data on which energy policy and legal analysis depend.

4.3 OPEC and Petroleum Market Coordination

Origins and Legal Foundation

The Organization of the Petroleum Exporting Countries was founded at the Baghdad Conference of 10–14 September 1960 by Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela. Its Statute, adopted in 1961 and subsequently amended, defines its principal aims as "the co-ordination and unification of the petroleum policies of Member Countries and the determination of the best means for safeguarding their interests, individually and collectively" and the devising of "ways and means of ensuring the stabilization of prices in international oil markets with a view to eliminating harmful and unnecessary fluctuations" (Article 2). OPEC is headquartered in Vienna. Its membership has varied over time; Nigeria joined in 1971. Qatar left in 2019, Ecuador in 2020, and Angola in 2024.

The supreme authority of OPEC is the Conference, composed of delegations of the member countries, which meets at least twice a year and takes decisions by unanimity of full members (Article 11). The Board of Governors directs the management of the Organization, and the Secretariat, headed by the Secretary General, carries out executive functions.

Production Quotas and Market Management

OPEC's most visible function is the setting of production ceilings for its members. The system of production quotas was formally introduced in 1982 in response to falling prices and has been the principal instrument of OPEC market management since then. Quota decisions are adopted by the Conference and are politically binding on members, but the Statute provides no mechanism

for enforcing compliance, and over-production by members has been a recurring feature of OPEC history.

Since December 2016, OPEC members have coordinated production with ten non-OPEC producers, including the Russian Federation, Kazakhstan, Mexico, and Oman, under the Declaration of Cooperation, in an arrangement known as "OPEC+". In July 2019, the participants adopted the Charter of Cooperation, providing a framework for continued cooperation. OPEC+ has implemented significant production cuts, including an unprecedented cut of 9.7 million barrels per day agreed in April 2020 in response to the collapse in demand caused by the COVID-19 pandemic.

Quota Allocation and Baseline Disputes

The allocation of production quotas among members is among the most contentious aspects of OPEC's work, because the "baseline" from which each member's reductions are calculated determines its share of the collective market. Disputes over baselines have repeatedly strained the organisation. In 2023, OPEC+ participants agreed that the maximum sustainable production capacities of several members would be assessed by independent sources for the purpose of setting baselines for 2025, reflecting disagreement over whether some African producers, which had been persistently unable to produce up to their quotas, should have their baselines reduced. Angola announced its withdrawal from OPEC in December 2023, citing disagreement with its assigned production level.

For Nigeria, the quota system interacts with domestic production constraints. For extended periods in the early 2020s, Nigeria produced well below its OPEC allocation as a result of crude oil theft, pipeline vandalism, underinvestment, and the divestment of onshore assets by international oil companies. These constraints meant that the binding limit on Nigerian production was operational rather than legal, and they focused domestic policy attention on the security of pipeline infrastructure and the regulatory framework for onshore asset transfers under the Petroleum Industry Act 2021. The episode demonstrates that the legal significance of an international institution's decisions depends on the material circumstances of its members.

The Legal Character of OPEC Decisions

The legal character of OPEC Conference resolutions has never been authoritatively determined. They are adopted by the organ of an international organisation established by treaty, and in that sense they are acts of international law. Yet the Statute does not expressly confer on the Conference the power to adopt decisions binding on members regarding production, and members have consistently treated quota decisions as political commitments rather than legal obligations enforceable by sanctions. The OPEC+ Declaration of Cooperation and Charter

of Cooperation are, by their terms and the practice of the participants, non-binding instruments. The absence of legal enforcement is consistent with the general preference of states for flexibility in matters touching resource sovereignty, and it explains why OPEC's effectiveness has historically depended on the willingness of one or more members — usually Saudi Arabia — to act as "swing producer" by adjusting its own output to stabilise the market.

Legal Issues Surrounding OPEC

OPEC's activities raise several legal questions. The first concerns competition law. OPEC's coordination of production would, if carried out by private companies, constitute a cartel prohibited by the competition laws of most states. Private plaintiffs in the United States have sought to sue OPEC and its members under the Sherman Act. In *International Association of Machinists and Aerospace Workers v OPEC* (1981), the Ninth Circuit Court of Appeals affirmed the dismissal of such a suit on the basis of the act of state doctrine.[1] Legislation known as the "No Oil Producing and Exporting Cartels Act" (NOPEC), which would remove sovereign immunity and the act of state defence for such claims, has been introduced repeatedly in the United States Congress but has not been enacted.

The second concerns trade law. Production quotas are, in effect, restrictions on the production and therefore the export of oil, raising questions of compatibility with Article XI of the GATT, which prohibits quantitative restrictions on exports. Most OPEC members are members of the WTO, but no WTO challenge to OPEC production quotas has been brought. Arguments that quotas could be justified under Article XX(g) as measures relating to the conservation of exhaustible natural resources, or that production restrictions differ legally from export restrictions, have been advanced but not tested.

The third concerns the relationship between OPEC decisions and contractual obligations. Production quotas may require the restriction of production by joint ventures involving foreign companies, raising questions about whether quota compliance constitutes *force majeure* or a change in law under production sharing contracts and joint operating agreements. Many Nigerian production sharing contracts and joint venture arrangements contain provisions allocating the burden of OPEC-mandated production restrictions.

OPEC and the Energy Transition

OPEC faces a fundamental strategic challenge in the energy transition. If global oil demand peaks and declines, the value of members' reserves will fall, and competition for market share may intensify. OPEC has consistently argued that oil will remain a significant part of the global energy mix for decades and has warned

[1] *International Association of Machinists and Aerospace Workers v Organization of Petroleum Exporting Countries*, 649 F 2d 1354 (9th Cir 1981).

against underinvestment in oil supply. Within the UNFCCC process, OPEC members — particularly Saudi Arabia — have often resisted language calling for a phase-out of fossil fuels, and the formulation adopted at COP28, calling for "transitioning away from fossil fuels in energy systems", reflected a compromise in which producer-state resistance played a central role.

4.4 The International Renewable Energy Agency

Establishment and Legal Foundation

The International Renewable Energy Agency was established by the Statute of IRENA, opened for signature in Bonn on 26 January 2009 and entering into force on 8 July 2010. Its headquarters are in Masdar City, Abu Dhabi, in the United Arab Emirates — a symbolically significant choice, locating the principal international institution for renewable energy in a major oil-exporting state. Membership is open to all states members of the United Nations and to regional intergovernmental economic integration organisations; the European Union is a member. With approximately 170 members, IRENA has one of the broadest memberships of any energy institution.

Article II of the Statute defines IRENA's objective as promoting "the widespread and increased adoption and the sustainable use of all forms of renewable energy", taking into account national and domestic priorities and benefits derived from a combined approach of renewable energy and energy efficiency measures, and the contribution of renewable energy to environmental preservation, climate protection, economic growth and social cohesion, access to and security of energy supply, regional development, and intergenerational responsibility. Article III defines renewable energy to include bioenergy, geothermal energy, hydropower, ocean energy (including tidal, wave, and ocean thermal energy), solar energy, and wind energy.

Functions and Activities

IRENA's functions, set out in Article IV, are advisory and facilitative. They include analysing and systematising renewable energy practices, policy instruments, investment mechanisms, and best practices; providing policy advice and assistance to members on request; promoting knowledge and technology transfer; providing capacity building and education; and stimulating research and networking. IRENA publishes influential reports, including the *World Energy Transitions Outlook*, annual *Renewable Capacity Statistics*, and *Renewable Power Generation Costs*, the last of which has documented the dramatic decline in the costs of solar photovoltaic and wind power over the past decade.

IRENA has no power to adopt binding decisions. Its influence derives from its analytical work, its convening role, and its support for member states in designing renewable energy policies. In the context of COP28 in 2023, IRENA played a

central role in tracking progress towards the pledge, endorsed in the global stocktake decision, to triple global renewable energy capacity and double the global average annual rate of energy efficiency improvements by 2030.

Assessment

IRENA represents a distinctive model of international energy governance: a universal-membership institution with a promotional rather than regulatory mandate. Its strengths lie in its legitimacy, its expertise, and its capacity to support developing states. Its limitations lie in the absence of binding powers and of mechanisms for addressing the legal barriers to renewable energy deployment — such as trade disputes over local content requirements and investment disputes over the withdrawal of incentives — which are governed by other regimes.

4.5 The International Atomic Energy Agency

Establishment and Mandate

The International Atomic Energy Agency was established by its Statute, approved on 23 October 1956 and entering into force on 29 July 1957. Its creation followed President Eisenhower's "Atoms for Peace" address to the United Nations General Assembly in December 1953, which proposed an international agency to promote the peaceful uses of atomic energy while preventing the diversion of nuclear materials to military purposes. The IAEA is headquartered in Vienna and is an autonomous organisation within the United Nations system, reporting annually to the General Assembly and, where appropriate, to the Security Council.

Article II of the Statute defines the Agency's dual objective: to "seek to accelerate and enlarge the contribution of atomic energy to peace, health and prosperity throughout the world" and to "ensure, so far as it is able, that assistance provided by it or at its request or under its supervision or control is not used in such a way as to further any military purpose". This dual mandate — promotion and control — has shaped the Agency's work and generated tensions between its developmental and verification roles.

Safeguards

The IAEA's most significant legal function is the application of safeguards to verify that nuclear material is not diverted from peaceful uses to nuclear weapons or other nuclear explosive devices. Under Article III of the Treaty on the Non-Proliferation of Nuclear Weapons 1968 (NPT), each non-nuclear-weapon state party undertakes to accept safeguards, as set forth in an agreement to be negotiated and concluded with the IAEA, for the exclusive purpose of verification of the fulfilment of its obligations under the Treaty. These comprehensive safeguards agreements, based on a model document known as INFCIRC/153 (Corrected),

apply to all source or special fissionable material in all peaceful nuclear activities within the territory of the state.

Following the discovery in 1991 of Iraq's clandestine nuclear weapons programme, which comprehensive safeguards had failed to detect, the IAEA Board of Governors approved in 1997 the Model Additional Protocol (INFCIRC/540), which significantly expands the information that states must provide and the access that inspectors enjoy, including complementary access to sites. The Additional Protocol is voluntary but has been concluded by a large majority of states with comprehensive safeguards agreements.

Where the Agency finds that a state is in non-compliance with its safeguards agreement, the Board of Governors may report the non-compliance to all members and to the Security Council and General Assembly (Statute, Article XII.C). The Board found Iran in non-compliance in 2005, leading to referral to the Security Council and the adoption of sanctions resolutions, and later to the Joint Comprehensive Plan of Action of 2015, under which the IAEA was entrusted with extensive monitoring and verification responsibilities. The Board again found Iran in breach of its safeguards obligations in June 2025.

Safety Standards

Article III.A.6 of the Statute authorises the Agency "to establish or adopt ... standards of safety for protection of health and minimization of danger to life and property". The IAEA Safety Standards Series — comprising Safety Fundamentals, Safety Requirements, and Safety Guides — is not binding in itself, but the standards are binding on the Agency in relation to its own operations and on states in relation to operations assisted by the Agency, and they are widely incorporated into national regulation. The Convention on Nuclear Safety 1994 and the Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management 1997, both adopted under IAEA auspices, establish binding obligations reinforced by peer review mechanisms. These instruments are examined in Chapter 11.

Nuclear Newcomers and the Milestones Approach

A growing number of states without existing nuclear power programmes — including several in Africa — have expressed interest in developing nuclear energy. The IAEA supports such "newcomer" states through its Milestones Approach, which identifies nineteen infrastructure issues (including the national position, nuclear safety, management, funding and financing, the legal framework, safeguards, the regulatory framework, radiation protection, the electrical grid, human resource development, stakeholder involvement, site and supporting facilities, environmental protection, emergency planning, nuclear security, the nuclear fuel cycle, radioactive waste management, industrial involvement, and procurement)

that must be addressed across three phases of programme development, each concluding with a milestone. The Agency conducts Integrated Nuclear Infrastructure Review (INIR) missions to assess progress.

The legal framework is an explicit component of the Milestones Approach. A newcomer state is expected to adhere to the principal international nuclear instruments and to enact comprehensive national nuclear legislation establishing an independent regulatory body, licensing requirements, nuclear liability rules, and provisions on safeguards, security, and waste management. In Nigeria, the Nuclear Safety and Radiation Protection Act 1995 established the Nigerian Nuclear Regulatory Authority as the regulator for nuclear safety and radiological protection, while the Nigeria Atomic Energy Commission, established under the Nigeria Atomic Energy Commission Act 1976, is responsible for promoting the development of atomic energy. Ghana, Kenya, and Uganda are among the other African states that have engaged extensively with the IAEA on nuclear power infrastructure. The interaction between IAEA guidance and national legislation is a good illustration of how the soft law outputs of an international organisation shape domestic legal frameworks.

Assessment

The IAEA is the most legally powerful of the energy-specific international organisations, by virtue of its safeguards function and its role in the non-proliferation regime. Its experience demonstrates both the possibilities and the limits of international verification: safeguards have been strengthened in response to failures, but the Agency depends on the cooperation of states and on the Security Council for enforcement. Its promotional role has expanded with renewed interest in nuclear energy as a low-carbon source, including small modular reactors.

4.6 Regional Energy Regulatory Institutions

The European Union

The European Union possesses the most developed regional energy governance system in the world. Article 194 of the Treaty on the Functioning of the European Union, introduced by the Treaty of Lisbon in 2009, provides that Union policy on energy shall aim, in a spirit of solidarity between member states, to ensure the functioning of the energy market, ensure security of energy supply, promote energy efficiency and energy saving and the development of new and renewable forms of energy, and promote the interconnection of energy networks. Article 194(2) preserves "a Member State's right to determine the conditions for exploiting its energy resources, its choice between different energy sources and the general structure of its energy supply", a significant limitation on Union competence that reflects the sovereignty sensitivities discussed above.

EU energy law has developed through successive legislative packages. The Third Energy Package of 2009 required the unbundling of transmission system operators from generation and supply activities, strengthened the independence of national regulatory authorities, and established the Agency for the Cooperation of Energy Regulators (ACER) and the European Networks of Transmission System Operators for Electricity and Gas (ENTSO-E and ENTSOG). The Clean Energy for All Europeans Package of 2019 updated the framework for electricity markets, renewable energy, energy efficiency, and governance, including the Regulation on the Governance of the Energy Union and Climate Action (EU) 2018/1999, which requires member states to adopt integrated national energy and climate plans. The "Fit for 55" package, adopted between 2021 and 2023, aligned EU energy law with the target of reducing net greenhouse gas emissions by at least 55 per cent by 2030 relative to 1990 levels, established by the European Climate Law, Regulation (EU) 2021/1119, which also enshrines the objective of climate neutrality by 2050. The REPowerEU plan of 2022 sought to end dependence on Russian fossil fuels, and in 2024 the Union adopted reforms of electricity market design and a hydrogen and decarbonised gas market package.

The Court of Justice of the European Union has played a major role in shaping EU energy law, including through cases on unbundling, the independence of regulators, state aid for renewable energy (*PreussenElektra*, 2001), and the principle of energy solidarity. In *Germany v Poland* (2021), the Court held that the principle of energy solidarity in Article 194(1) is a legally binding principle that requires the EU institutions and member states to take into account the interests of other member states in energy security, annulling a Commission decision on the OPAL gas pipeline that had failed to examine the effects on Poland's security of supply.[2]

The Energy Community

The Energy Community, established by the Treaty Establishing the Energy Community signed in Athens on 25 October 2005, extends the EU's internal energy market rules to contracting parties in South-East Europe and the Black Sea region, currently comprising Albania, Bosnia and Herzegovina, Georgia, Kosovo (designation without prejudice to positions on status), Moldova, Montenegro, North Macedonia, Serbia, and Ukraine. Contracting parties undertake to implement the relevant EU *acquis* in energy, environment, competition, and renewables. The Energy Community has a Secretariat in Vienna and a dispute settlement procedure under which the Ministerial Council may declare that a contracting party has failed to comply with its obligations. The Energy Community has been

[2] Case C-848/19 P, *Federal Republic of Germany v Republic of Poland*, EU:C:2021:598 (CJEU, 15 July 2021).

a significant vehicle for the integration of Ukraine's energy system with that of the European Union, which culminated in the synchronisation of the Ukrainian and Moldovan electricity grids with the continental European grid in March 2022.

African Regional Institutions

Africa has developed a growing network of regional energy institutions, reflecting both the continent's vast energy resources and the imperative of expanding energy access.

The African Union. The AU's Agenda 2063 identifies energy as a priority, and the African Energy Commission (AFREC), established by a Convention adopted in 2001 and in force since 2008, is the AU's specialised agency for energy. The Programme for Infrastructure Development in Africa (PIDA) identifies priority regional energy projects. The African Single Electricity Market (AfSEM), launched in 2021, aims to create the world's largest electricity market by integrating Africa's regional power pools. The Agreement Establishing the African Continental Free Trade Area (AfCFTA), signed in 2018 and in force since 2019, has implications for trade in energy goods and services.

ECOWAS. In West Africa, the Economic Community of West African States has established the ECOWAS Energy Protocol (2003), modelled on the ECT, which provides for investment protection, trade, and transit in the energy sector; the ECOWAS Regional Electricity Regulatory Authority (ERERA), established in 2008 and based in Accra, which regulates cross-border electricity exchanges; the West African Power Pool (WAPP), a specialised institution of ECOWAS established in 1999 and given its current status in 2006, which coordinates the development of regional transmission infrastructure and the regional electricity market; and the ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), based in Praia. The West African Gas Pipeline, which transports Nigerian gas to Benin, Togo, and Ghana, is governed by a treaty concluded among the four states in 2003 and by an International Project Agreement with the pipeline company. The proposed Nigeria–Morocco Gas Pipeline and the Trans-Saharan Gas Pipeline would extend this infrastructure substantially.

Southern and Eastern Africa. The Southern African Power Pool (SAPP), established in 1995, is the most mature of the African power pools and operates a competitive regional electricity market. The Regional Electricity Regulators Association of Southern Africa (RERA) supports regulatory cooperation. The Eastern Africa Power Pool (EAPP) and the Central African Power Pool (CAPP) pursue similar objectives in their regions.

Development Finance Institutions as Energy Governors

Multilateral development banks and other development finance institutions are among the most influential actors in energy governance in developing countries,

although they are rarely described as energy institutions. Their influence operates through several channels: the conditions attached to policy-based lending, which have historically encouraged power sector unbundling, privatisation, and the establishment of independent regulators; the environmental and social standards attached to project finance, notably the World Bank's Environmental and Social Framework (2018) and the IFC Performance Standards; the provision of guarantees and political risk insurance, including through the Multilateral Investment Guarantee Agency; and the provision of technical assistance for the drafting of legislation and model contracts.

Development banks have also become central to the financing of energy access. Mission 300, launched by the World Bank Group and the African Development Bank in 2024, aims to connect 300 million people in Africa to electricity by 2030 through a combination of grid extension, mini-grids, and off-grid solutions. At the Mission 300 Africa Energy Summit held in Dar es Salaam in January 2025, participating governments presented national energy compacts setting out policy and regulatory reforms intended to attract investment. The African Development Bank's Desert to Power initiative seeks to develop solar generation across the Sahel. Such initiatives blend finance with policy conditionality, and their legal dimension — the reforms to tariff regulation, licensing, and procurement that they require — is a significant but understudied aspect of energy law in Africa.

The energy lending policies of development banks have also become a site of contestation over fossil fuel finance. The World Bank Group announced in 2017 that it would cease financing upstream oil and gas after 2019, save in exceptional circumstances, and the European Investment Bank's 2019 energy lending policy ended financing for most unabated fossil fuel projects. Developing-country governments, including Nigeria, have argued that such policies unfairly constrain the use of natural gas as a transition fuel in countries with low per capita emissions and large energy access deficits, an argument that connects institutional policy to the principle of common but differentiated responsibilities discussed in Chapter 18.

Other Regional Arrangements

Elsewhere, regional cooperation in energy takes a variety of forms. The Association of Southeast Asian Nations (ASEAN) has pursued the ASEAN Power Grid and the Trans-ASEAN Gas Pipeline projects through memoranda of understanding. In Latin America, the Central American Electrical Interconnection System (SIEPAC) is underpinned by the Framework Treaty for the Electricity Market of Central America (1996), which established a regional regulator, the Comisión Regional de Interconexión Eléctrica (CRIE). The Gulf Cooperation Council Interconnection Authority operates a regional transmission grid linking the six GCC states.

COMPARATIVE PERSPECTIVE

Regional integration models. The European Union represents a *supranational* model, in which binding legislation, an independent regional regulator (ACER), and a supreme court (the CJEU) drive deep market integration. The Energy Community represents an *acquis-export* model, extending EU rules to neighbouring states through treaty commitments. ECOWAS and SADC represent *intergovernmental* models, in which regional regulators and power pools coordinate national systems with more limited binding authority, and in which the pace of integration depends heavily on the construction of physical infrastructure and on the financial health of national utilities. The choice of model reflects not only political preferences regarding sovereignty but also the maturity of national markets and institutions.

4.7 Independent Regulators and Administrative Accountability

The Rise of the Independent Regulator

At the national level, the most significant institutional development in energy governance over the past four decades has been the spread of independent regulatory agencies. Before the liberalisation era, the electricity and gas sectors in most countries were dominated by vertically integrated, state-owned monopolies that were, in effect, self-regulating, subject to ministerial oversight. The petroleum sector was typically regulated by a ministry, sometimes acting through a national oil company that combined commercial and regulatory functions.

The liberalisation and privatisation of energy markets, beginning in the United Kingdom, Chile, and New Zealand in the 1980s and spreading globally in the 1990s, required new regulatory institutions. Where private companies own and operate natural monopoly networks, or where competition is introduced into generation and supply, an independent regulator is needed to set tariffs, ensure non-discriminatory network access, license market participants, monitor market conduct, and protect consumers. The model of the independent regulatory agency, long established in the United States in the form of state public utility commissions and the Federal Power Commission (later the Federal Energy Regulatory Commission), was transplanted across the world, frequently with the encouragement of the World Bank and other development institutions.

In Nigeria, the Electric Power Sector Reform Act 2005 established the Nigerian Electricity Regulatory Commission (NERC), now continued under the Electricity Act 2023. The Petroleum Industry Act 2021 established the NUPRC and the NMDPRA as regulators separate from the commercial NNPC Ltd, replacing the Department of Petroleum Resources, the Petroleum Products Pricing Regulatory Agency, and the Petroleum Equalisation Fund (Management) Board.

The Rationale for Independence

Several justifications have been advanced for the independence of energy regulators.

Credible commitment. Energy infrastructure is capital-intensive and long-lived, and investors require assurance that tariffs will allow them to recover their investment and earn a reasonable return. Political control of tariffs creates a risk that governments will hold down prices for electoral reasons, expropriating investors' sunk capital. Delegating tariff-setting to an independent regulator with a statutory mandate and secure tenure is a means by which governments can credibly commit not to behave opportunistically.

Expertise. Energy regulation is technically complex, requiring expertise in economics, engineering, accounting, and law. Specialised agencies can develop and retain such expertise more effectively than ministries.

Separation of roles. Where the state owns energy companies, independence separates the state's role as owner from its role as regulator, reducing conflicts of interest and promoting a level playing field between state-owned and private participants. The separation of the regulatory and commercial functions of national oil companies is a central theme of petroleum sector reform.

Insulation from capture. Independence is also intended to protect regulators from capture by the industries they regulate, although independence from government does not of itself guarantee independence from industry.

Elements of Independence

The independence of a regulator depends on several institutional features:

- **Legal foundation**: establishment by statute with a clearly defined mandate, rather than by executive decision;
- **Appointment and tenure**: appointment through a transparent process, fixed terms of office, and protection against removal except for defined cause;
- **Financial autonomy**: funding through levies on regulated entities or secure budgetary allocations, rather than discretionary appropriations;
- **Decision-making autonomy**: authority to take decisions without ministerial direction on individual matters, subject to general policy guidance;
- **Organisational autonomy**: control over staffing and internal organisation.

EU law imposes detailed independence requirements on national regulatory authorities. Under Article 57 of Directive (EU) 2019/944 on common rules for the internal market for electricity, member states must guarantee the independence of regulatory authorities and ensure that they exercise their powers impartially and transparently; regulatory staff and management must act independently from any market interest and must not seek or take direct instructions from any government or other public or private entity when carrying out regulatory tasks.

In *Commission v Germany* (2021), the Court of Justice held that Germany had failed to fulfil its obligations by conferring on the federal government powers to determine tariff methodologies that EU law reserved exclusively to the national regulatory authority.[3]

The Governance of National Oil Companies

National oil companies (NOCs) control the majority of the world's proven oil and gas reserves and are therefore among the most significant institutions in global energy governance, even though they are corporate entities rather than regulators. Their governance raises distinctive legal questions concerning the separation of commercial, policy, and regulatory functions; the relationship between the NOC and the state treasury; transparency and accountability; and the application of corporate law to entities whose shareholder is the state.

The "Norwegian model" is frequently cited as a benchmark. Under this model, policy functions are exercised by the Ministry of Energy, regulatory and resource management functions by a specialised agency (the Norwegian Offshore Directorate, as the Norwegian Petroleum Directorate was renamed in 2024), and commercial functions by a company operating on market terms (Equinor, formerly Statoil, partially listed since 2001), with the state's direct financial interests in petroleum licences managed separately by a wholly state-owned company, Petoro. Comparative research has questioned whether the Norwegian separation of functions is a necessary condition of good NOC performance, observing that it presupposes strong institutional capacity, and that some NOCs perform well under integrated models while some separated models perform poorly.

Nigeria's Petroleum Industry Act 2021 moved decisively towards a separation of functions. Section 53 provides for the incorporation of NNPC Limited under the Companies and Allied Matters Act, with ownership of its shares vested in the Federal Government and held by the Ministry of Finance Incorporated and the Ministry of Petroleum Resources on behalf of the Federation. NNPC Ltd is to operate as a commercial entity without recourse to government funding and to pay dividends to its shareholders, and the Act contemplates that its shares may in future be offered to the public. Regulatory functions previously exercised by NNPC divisions and the Department of Petroleum Resources were transferred to the NUPRC and the NMDPRA. Whether this formal separation will produce the commercial discipline and transparency its drafters intended depends on implementation, including the robustness of corporate governance at NNPC Ltd, the publication of audited accounts, and the treatment of legacy obligations such as fuel subsidy arrears.

[3] Case C-718/18, *European Commission v Federal Republic of Germany*, EU:C:2021:662 (CJEU, 2 September 2021).

Administrative Accountability

Independence must be balanced by accountability. An independent regulator exercises significant public power — it sets prices paid by millions of consumers, grants and revokes licences, and imposes penalties — and in a democratic system such power must be subject to mechanisms of accountability. These mechanisms include:

- **Legislative oversight**: reporting obligations to the legislature, appearances before parliamentary committees, and legislative approval of budgets;
- **Judicial review**: the right of affected persons to challenge regulatory decisions before courts or specialised tribunals on grounds of illegality, irrationality, and procedural impropriety, examined further in Chapter 27;
- **Procedural requirements**: obligations to consult on proposed decisions, to give reasons, to publish decisions, and to act transparently;
- **Appeals**: specialised appeal mechanisms, such as the Competition and Markets Authority's role in hearing appeals against certain decisions of Ofgem in the United Kingdom;
- **Performance review**: periodic review of regulatory performance by audit institutions or independent bodies.

The relationship between independence and accountability is a subject of ongoing scholarly debate. Some scholars emphasise the democratic deficit of independent agencies and argue for stronger political control; others emphasise the importance of independence for investment and argue that accountability is best secured through procedural transparency and judicial review. In the United States, the Supreme Court's decision in *Loper Bright Enterprises v Raimondo* (2024), overruling the *Chevron* doctrine of judicial deference to reasonable agency interpretations of ambiguous statutes, and its articulation of the "major questions doctrine" in *West Virginia v Environmental Protection Agency* (2022), have significantly altered the balance between regulatory agencies and courts, with major implications for energy and environmental regulation.[4]

Regulatory Independence in Practice: The Developing-Country Experience

The transplantation of the independent regulator model to developing countries has produced mixed results. Formal independence enshrined in legislation has often been undermined in practice by political interference in appointments, ministerial pressure on tariff decisions, inadequate funding, and limited technical capacity. Studies of electricity regulation in Africa have found that regulators frequently lack the authority to set cost-reflective tariffs in the face of political

[4] *Loper Bright Enterprises v Raimondo*, 603 US 369 (2024); *West Virginia v Environmental Protection Agency*, 597 US 697 (2022).

resistance, leading to utility financial deficits, underinvestment, and poor service quality. The Nigerian experience illustrates these challenges: NERC's efforts to implement cost-reflective tariffs under the Multi-Year Tariff Order framework have repeatedly been constrained by political considerations, and the accumulation of tariff shortfalls has required substantial government subsidies. The introduction of a "Band A" tariff structure in 2024, under which customers receiving at least twenty hours of supply daily pay higher, more cost-reflective tariffs, represents an attempt to align tariffs with service quality.

These experiences suggest that formal independence is a necessary but not sufficient condition for effective regulation. It must be supported by political commitment, adequate resources, technical capacity, and a broader governance environment that respects the rule of law. Scholars such as Brian Levy and Pablo Spiller have emphasised that regulatory design must be adapted to the institutional endowment of each country, and that in countries with weak checks and balances, contractual commitments and international guarantees may be needed to supplement regulatory independence.[5]

4.8 Regulatory Cooperation and Institutional Reform

Forms of Regulatory Cooperation

Regulatory cooperation in the energy sector takes several forms. At the bilateral level, regulators conclude memoranda of understanding to exchange information, coordinate decisions on cross-border infrastructure, and share expertise. At the regional level, regulators participate in associations such as the Council of European Energy Regulators (CEER), the Energy Regulators Regional Association (ERRA) for Central and Eastern Europe and beyond, the African Forum for Utility Regulators (AFUR), and RERA. At the global level, the International Confederation of Energy Regulators (ICER) provides a forum for cooperation among regional associations.

Regulatory cooperation is particularly important for cross-border infrastructure. The development of an interconnector or cross-border pipeline requires decisions by regulators in each jurisdiction on matters such as cost allocation, access rules, and tariffs. Within the European Union, Regulation (EU) 2022/869 on guidelines for trans-European energy infrastructure provides for the identification of projects of common interest and a framework for cross-border cost allocation decisions, with ACER empowered to decide where national regulators cannot agree.

[5] Brian Levy and Pablo T. Spiller, "The Institutional Foundations of Regulatory Commitment: A Comparative Analysis of Telecommunications Regulation" (1994) 10 *Journal of Law, Economics, and Organization* 201.

Proposals for Institutional Reform

The fragmentation of global energy governance has prompted numerous proposals for reform. These include:

Expanding the IEA. Proposals to open IEA membership to non-OECD states, particularly China and India, would require amendment of the IEP Agreement and the OECD framework. The association initiative represents an incremental step in this direction.

A World Energy Organization. Some scholars have proposed the creation of a comprehensive world energy organisation, analogous to the WTO, with universal membership and competence over energy trade, investment, and security. Such proposals face formidable political obstacles, given the divergence of producer and consumer interests and the reluctance of states to cede sovereignty over energy.

Strengthening the energy dimension of the climate regime. The UNFCCC and Paris Agreement are not energy institutions, but the decisions of the Conference of the Parties increasingly address energy directly, as in the COP28 global stocktake decision. Some scholars argue that the climate regime should become the primary forum for global energy governance; others caution that it lacks the expertise and instruments to address energy security and market issues.

Reforming the Energy Charter Treaty. The modernisation of the ECT, negotiated between 2020 and 2022 and adopted by the Energy Charter Conference in December 2024, sought to align the treaty with the Paris Agreement by permitting parties to exclude protection for fossil fuel investments, clarifying the right to regulate, and refining investment protection standards. The withdrawals of the European Union, Euratom, and several European states, and the continuing participation of other contracting parties, have left the treaty's future uncertain. These developments are examined in Chapter 14.

Networked governance. An alternative to institutional consolidation is to improve coordination among existing institutions through networked governance: joint initiatives, memoranda of understanding, shared data platforms, and the alignment of mandates. The joint work of the IEA, IRENA, the World Bank, the World Health Organization, and UN statistical bodies on tracking progress towards Sustainable Development Goal 7 illustrates this approach.

Data, Transparency, and Institutional Legitimacy

A final dimension of institutional reform concerns data and transparency. Energy governance depends on reliable data on production, consumption, stocks, prices, reserves, emissions, and investment. Much of that data is held by governments and companies that may have strategic or commercial reasons to withhold it, and the quality of data varies widely among states. The Joint Organisations Data Initiative, administered by the International Energy Forum with the participation

of APEC, Eurostat, GECF, IEA, OLADE, OPEC, and the United Nations Statistics Division, seeks to improve the transparency of oil and gas statistics. The EITI Standard requires implementing countries to disclose information along the extractive industry value chain, including contracts, licences, beneficial ownership, production, revenues, and the allocation of revenues. Satellite monitoring of methane emissions, gas flaring, and infrastructure is increasingly supplementing official data, as in the United Nations Environment Programme's International Methane Emissions Observatory.

Data has legal significance. Regulatory decisions, arbitral damages assessments, and climate litigation all depend on evidence, and the authority of institutions such as the IEA rests heavily on the credibility of their data and scenarios. Contests over data — over the reliability of emissions inventories, the assumptions of energy scenarios, or the reserves on which valuations depend — are therefore contests over institutional legitimacy. For doctoral researchers, the political economy of energy data is a promising and relatively unexplored field.

Institutional Reform and the Energy Transition

The energy transition poses fundamental challenges for energy institutions. Institutions designed for a world of centralised fossil fuel systems must adapt to a world of decentralised, digital, and decarbonised systems. Consumer-oriented institutions concerned with oil supply security must address new security challenges arising from critical mineral supply chains, cyber threats, and the variability of renewable generation. Producer-oriented institutions must address the prospect of declining demand. Regulators must design markets that accommodate variable renewable generation, storage, demand response, and distributed resources, and must balance the costs of transition against affordability for consumers. The capacity of institutions to adapt to these challenges will significantly influence the pace and fairness of the transition.

LEADING AUTHORITY

Federal Republic of Germany v Republic of Poland (OPAL), CJEU, 2021

Facts. In 2016, the European Commission approved a modification of the exemption from third-party access and tariff regulation previously granted to the OPAL pipeline, the German onshore extension of the Nord Stream pipeline, allowing Gazprom to use a greater share of OPAL's capacity. Poland challenged the decision before the General Court, arguing that it threatened Poland's security of gas supply by diverting transit flows from pipelines crossing Polish territory.

Holding. The General Court annulled the Commission's decision in 2019 for breach of the principle of energy solidarity. On appeal by Germany, the Court of Justice confirmed that energy solidarity under Article 194(1) TFEU is a general principle of EU law, which entails rights and obligations both for the EU and for the member

states and requires that the interests of all stakeholders liable to be affected be taken into account. The Commission had failed to examine the effects of its decision on the energy security of Poland and other member states.
Significance. The judgment transformed solidarity from a political aspiration into a justiciable legal principle that constrains EU energy decisions. It illustrates the role of courts in regional energy governance and the legal relevance of energy security considerations.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Designing an independent regulator (hypothetical). The fictional State of Morenia, a constituent state of a federal republic, has acquired legislative competence over its intrastate electricity market following a constitutional amendment. The Morenia State Assembly has asked you to prepare draft provisions for a Morenia Electricity Regulatory Commission Law addressing: (a) the Commission's statutory objectives, ranked in order of priority; (b) appointment, tenure, and removal of commissioners; (c) funding arrangements; (d) the relationship between the Commission and the State Commissioner for Energy, including the extent of permissible policy directions; (e) consultation, reason-giving, and publication obligations; (f) appeal and judicial review mechanisms; and (g) cooperation with the federal regulator during the transition of regulatory oversight. Accompany the draft provisions with a commentary of not more than 2,000 words explaining your choices by reference to comparative practice. *Morenia is fictitious.*

CHAPTER SUMMARY

- Global energy governance is a fragmented, polycentric regime complex, reflecting divergent producer and consumer interests, sovereignty sensitivities, path dependence, technological diversity, and shifting centres of economic gravity.
- The IEA, created in 1974, imposes binding emergency stockholding and response obligations on its OECD members and exercises broad analytical influence, but its limited membership constrains its global legitimacy.
- OPEC and the OPEC+ arrangements coordinate producer policies through politically binding but unenforceable production decisions, raising unresolved questions under competition and trade law.
- IRENA has near-universal membership and a promotional mandate without binding powers; the IAEA combines promotion with legally significant safeguards and safety functions.
- Regional institutions range from the supranational EU model to intergovernmental African power pools; the CJEU's recognition of energy solidarity as a binding principle illustrates the legal depth of EU integration.
- Independent regulators are justified by credible commitment, expertise, separation of roles, and insulation from capture, but must be accountable through

legislative oversight, judicial review, and procedural transparency; formal independence is often undermined in practice in developing countries.
- Reform proposals include expanding the IEA, creating a world energy organisation, strengthening the climate regime's energy role, modernising the ECT, and improving networked governance.

DOCTORAL RESEARCH QUESTIONS

1. Does the concept of a "regime complex" adequately explain the fragmentation of global energy governance, or does it underestimate the role of power asymmetries among states?
2. Assess the compatibility of OPEC+ production restrictions with Articles XI and XX(g) of the GATT. Why has no WTO member brought a challenge?
3. To what extent has the formal independence of energy regulators in sub-Saharan Africa translated into decisional independence in tariff-setting? Design a methodology to measure the gap.
4. How should the emergence of state-level electricity regulators in Nigeria be coordinated with federal regulation to avoid regulatory fragmentation in the national grid?
5. Could the energy solidarity principle articulated in *Germany v Poland* serve as a model for regional energy governance in ECOWAS?

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PART II

Petroleum, Gas, and Extractive Industries Law

The law governing the upstream petroleum industry, natural gas and LNG transactions, hard-rock mining, and the critical minerals on which the energy transition depends.

5 CHAPTER

International Petroleum Law and Upstream Resource Development

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the legal classification of petroleum resources and the stages of the upstream value chain.
2. Analyse the nature of exploration and production rights and the legal character of petroleum titles.
3. Compare licensing and concession systems with production sharing contracts and service contracts, and evaluate their respective allocation of risk and reward.
4. Analyse the structure and key provisions of joint operating agreements.
5. Evaluate the roles of national oil companies and state participation in upstream petroleum.
6. Explain the legal framework for unitisation and the development of transboundary reservoirs.
7. Analyse the legal regime governing decommissioning and abandonment liabilities.

Introduction

The upstream petroleum industry — the exploration for and production of crude oil and natural gas — is the historical core of international energy law. It was in the upstream sector that the foundational instruments of the field were developed: the concession, the licence, the production sharing contract, the joint operating agreement, and the unitisation agreement. It was disputes over upstream rights that generated the classic arbitral awards on stabilisation, internationalisation, and expropriation. And it is in the upstream sector that the tension between the sovereign interests of host states and the commercial interests of international oil companies is most acute.

The upstream sector has distinctive economic characteristics that shape its legal regime. Exploration is a high-risk activity: most exploration wells fail to discover commercial quantities of petroleum, and the costs of a single deep-water well can exceed one hundred million United States dollars. Development requires enormous capital investment, often billions of dollars, before any revenue is earned. Production continues for decades, during which prices, costs, technologies, and political conditions change unpredictably. And the resource is non-renewable and owned, in most jurisdictions, by the state. The legal instruments of the upstream sector are designed to allocate the risks and rewards of this activity between the state, as resource owner, and the investor, as provider of capital and technology.

This chapter examines the legal framework of upstream petroleum. It begins with the legal classification of petroleum resources, proceeds to the nature of exploration and production rights, and analyses the principal contractual and licensing systems. It then examines joint operating agreements, national oil companies, unitisation, and decommissioning. Throughout, Nigerian law — particularly the Petroleum Industry Act 2021 (PIA) — is used as a principal reference point, alongside the laws of the United Kingdom, Norway, and other leading jurisdictions.

5.1 Legal Classification of Petroleum Resources

What Is "Petroleum"?

The definition of "petroleum" in legislation and contracts determines the scope of rights granted and obligations imposed. Definitions vary, but most encompass naturally occurring hydrocarbons in liquid, gaseous, or solid form. Section 318 of the PIA defines "petroleum" as "crude oil, natural gas and condensates", and separately defines "crude oil" as "mineral oil, asphalt, ozokerite and all kinds of hydrocarbons and bitumen, whether in solid or in liquid form, in their natural state or obtained from oil shales, tar sands or any other mineral or rock, whether gaseous or not" and "natural gas" as "gas obtained from boreholes and wells and consisting primarily of hydrocarbons". Section 1(1) of the United Kingdom's Petroleum Act 1998 defines petroleum to include "any mineral oil or relative hydrocarbon and natural gas existing in its natural condition in strata", but expressly excludes coal and bituminous shales and other stratified deposits from which oil can be extracted by destructive distillation.

The precise definition can have significant legal consequences. Whether coal-bed methane, shale gas, oil shale, or helium falls within the definition of petroleum determines which licensing regime applies and who has the right to exploit the resource. The distinction between crude oil and natural gas matters because many

legal regimes treat them differently for fiscal purposes, ownership of associated and non-associated gas, and domestic supply obligations.

Associated and Non-Associated Gas

Natural gas may occur either in association with crude oil (associated gas), dissolved in the oil or as a gas cap above it, or in reservoirs that contain predominantly gas (non-associated gas). The distinction is legally important. Historically, many petroleum agreements granted rights to crude oil with only limited provision for associated gas, which was often flared because no market existed for it. Nigeria has long been among the world's largest gas flaring countries. The Associated Gas Re-Injection Act 1979 prohibited gas flaring without ministerial permission after 1 January 1984, but permission was routinely granted subject to payment of nominal fees. Section 104 of the PIA prohibits the flaring or venting of natural gas except in specified circumstances such as emergencies, or under an exemption granted by the NUPRC for facility start-up or for strategic operational reasons, and requires licensees and lessees to submit a natural gas flare elimination and monetisation plan. The Act further provides that payments of flare penalties shall not be cost-recoverable or tax-deductible. The Flare Gas (Prevention of Waste and Pollution) Regulations 2018 established the Nigerian Gas Flare Commercialisation Programme, under which flare gas is auctioned to third parties for commercial use.

Conventional and Unconventional Resources

A further classification distinguishes conventional resources — petroleum that can be produced from reservoirs by natural pressure, pumping, or conventional secondary recovery — from unconventional resources, such as shale oil and gas, tight gas, coal-bed methane, oil sands, and gas hydrates, which require specialised techniques such as horizontal drilling and hydraulic fracturing. The shale revolution in the United States, which transformed that country into the world's largest producer of both oil and natural gas, was made possible not only by technology but also by the American legal framework of private mineral ownership, which permitted rapid leasing and development without the need for state licensing. Elsewhere, unconventional resources have been subject to moratoria or bans on hydraulic fracturing, including in France (Law No 2011-835 of 13 July 2011), and, for a period, in the United Kingdom, where a moratorium imposed in 2019 was briefly lifted in 2022 and then reimposed.

Reserves Classification

The classification of petroleum reserves has legal and financial significance, since reserves underpin company valuations, borrowing capacity, and, in disputes, the assessment of damages. The Petroleum Resources Management System (PRMS), developed by the Society of Petroleum Engineers and other professional organ-

isations, classifies petroleum quantities according to their commerciality and the degree of uncertainty in recoverable volumes: reserves (proved, probable, and possible, commonly abbreviated as 1P, 2P, and 3P), contingent resources, and prospective resources. Securities regulators impose disclosure requirements based on reserves definitions; in the United States, the Securities and Exchange Commission's rules on oil and gas reserves disclosure apply to listed companies. The distinction between proved reserves and more speculative categories is frequently central to the valuation of expropriated petroleum assets in arbitration.

5.2 Exploration and Production Rights

The Nature of Petroleum Titles

In jurisdictions where petroleum is owned by the state, the right of an investor to explore for and produce petroleum derives from a grant by the state. The legal nature of that grant varies. It may be characterised as:

- an **administrative licence**, a permission granted by the state in the exercise of public power, which creates rights and obligations defined by statute and regulation and which may be varied or revoked in accordance with administrative law;
- a **contract**, a bilateral agreement between the state (or its agency) and the investor, creating reciprocal obligations enforceable in accordance with contract law;
- a **property right**, a real right in the petroleum or in the land, such as a lease or *profit à prendre*, enforceable against third parties; or
- a **hybrid** combining elements of each.

The characterisation has practical consequences. If the title is an administrative licence, the state may be able to vary its terms by legislation or regulation, subject to the protection afforded by administrative law and any applicable investment treaty. If it is a contract, unilateral variation will constitute a breach. If it is a property right, it may be mortgaged, assigned, and protected against third-party interference, and its taking may constitute expropriation.

In the United Kingdom, a petroleum licence granted under the Petroleum Act 1998 is a licence in the sense of a statutory permission, but it incorporates "model clauses" prescribed by regulation which take effect as terms of the licence and have a contractual character. The licence confers exclusive rights to "search and bore for, and get" petroleum within the licensed area. In *Shell UK Ltd v Enterprise Oil plc* and related litigation, English courts have treated licences as conferring valuable proprietary interests capable of assignment, subject to the consent of the regulator.

In Nigeria, petroleum titles under the PIA comprise the petroleum exploration licence (non-exclusive, for reconnaissance), the petroleum prospecting licence

(exclusive, for exploration and appraisal), and the petroleum mining lease (exclusive, for production). Under the PIA, a petroleum prospecting licence confers the exclusive right to drill exploration and appraisal wells and to carry out test production, while a petroleum mining lease confers the exclusive right to win, work, carry away, and dispose of petroleum. These titles are statutory instruments granted by the Minister on the recommendation of the NUPRC following a competitive bid round, and their terms are supplemented by the regulations made under the Act and by any associated contracts with NNPC Ltd.

Allocation of Rights: Competitive Bidding and Discretionary Award

States allocate petroleum rights through competitive bidding, discretionary award (negotiation), or first-come, first-served application systems. Competitive bidding, in which rights are awarded on the basis of bids comprising signature bonuses, work programme commitments, royalty or profit-share terms, or a combination of these, is now the dominant method in most producing states and is generally regarded as more transparent and efficient. The PIA requires that petroleum prospecting licences and petroleum mining leases be granted through a "fully transparent and competitive bid process" in accordance with regulations and guidelines issued by the NUPRC, subject to limited exceptions. Discretionary award was historically widespread in Nigeria and elsewhere and was associated with allegations of corruption, most notoriously in the award of Oil Prospecting Licence (OPL) 245, which generated criminal proceedings in Italy and Nigeria and civil litigation in England.

The design of bidding systems involves trade-offs. Signature bonuses generate immediate revenue for the state but may deter exploration in high-risk areas; work programme bidding encourages exploration activity but may lead to over-commitment; profit-share bidding defers state revenue but aligns state returns with project outcomes. Norway's licensing rounds, conducted through the Awards in Predefined Areas (APA) system for mature areas and numbered licensing rounds for frontier areas, award licences on the basis of discretionary criteria (technical competence, financial capacity, and the quality of work programmes) rather than monetary bids, reflecting a policy choice to capture rent through taxation rather than through bonuses.

Duration, Relinquishment, and Work Obligations

Petroleum titles are granted for defined periods. An exploration phase, typically lasting between three and ten years and often divided into sub-phases, is followed, upon discovery of a commercial field and approval of a development plan, by a production phase lasting twenty to thirty years or more. Under the PIA, a petroleum prospecting licence for onshore and shallow water acreage is granted for an initial exploration period of three years, which may be extended by a

further three years, with longer periods available for deep offshore and frontier acreage; a petroleum mining lease is granted for a term not exceeding twenty years, renewable in accordance with the Act.

Titles typically impose minimum work obligations — commitments to acquire seismic data and drill a specified number of wells within defined periods — backed by financial guarantees. Failure to perform the work programme may result in forfeiture of the guarantee and revocation of the title. Relinquishment provisions require the titleholder to surrender portions of the licensed area at specified intervals, ensuring that acreage that is not actively explored is returned to the state for re-allocation. The PIA provides for the relinquishment of fifty per cent of the area of a petroleum mining lease ten years after the grant, subject to specified conditions, a provision designed to address the problem of large, under-explored acreage held under legacy oil mining leases.

Assignment and Change of Control

Petroleum titles are valuable assets that are frequently transferred through farm-outs, sales, and corporate acquisitions. Because the identity, competence, and financial capacity of the titleholder are matters of public interest, most regimes require the consent of the state or regulator to any assignment of an interest in a title and, increasingly, to any change in control of the titleholder. The PIA prohibits the assignment, novation, or transfer of an interest in a licence or lease, or any change of control of the holder, without the prior consent of the Minister on the recommendation of the NUPRC. The divestment of onshore assets by international oil companies to Nigerian independents between 2021 and 2025 has generated significant regulatory scrutiny under these provisions, with the NUPRC requiring assurances concerning the purchasers' technical and financial capacity, their assumption of decommissioning and environmental liabilities, and the protection of host communities.

5.3 Petroleum Licensing and Concession Systems

Typology of Petroleum Arrangements

Petroleum arrangements between host states and investors are conventionally classified into two families: *concessionary* (or licensing) systems and *contractual* systems. Contractual systems are further divided into production sharing contracts and service contracts. The classification turns principally on the ownership of produced petroleum and the form of the state's financial return.

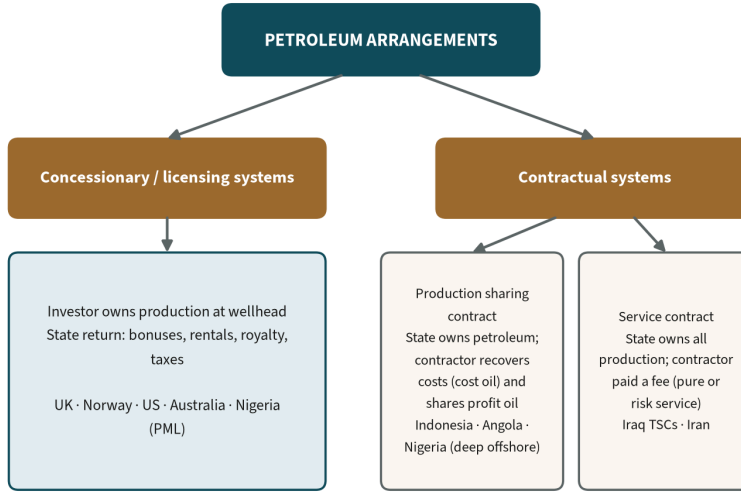


Figure 5.1 Principal families of petroleum arrangements, classified by ownership of production and the form of the state's return. *Source:* prepared for this edition.

Under a **concessionary or licensing system**, the investor acquires the exclusive right to explore for and produce petroleum within a defined area and owns the petroleum produced at the wellhead (or at another defined point), subject to the payment of royalties and taxes to the state. The state's return derives from fiscal instruments — bonuses, rentals, royalties, and taxes — rather than from a share of physical production. Modern licensing systems are used in the United Kingdom, Norway, the United States (for federal and state leases), Australia, Brazil (for its concession regime), and many other jurisdictions. Nigeria's historic oil mining lease system and the PIA's petroleum mining lease are licensing instruments, although in Nigeria they have been combined with joint ventures and production sharing contracts involving the national oil company.

Under a **production sharing contract (PSC)**, the state (usually acting through its national oil company) retains ownership of the petroleum, and the investor acts as a contractor that bears the exploration and development risk and costs. In return, the contractor is entitled to recover its costs from a portion of production ("cost oil") and to receive a share of the remaining production ("profit oil"), with the balance going to the state. Title to the contractor's share passes at a defined delivery point.

Under a **service contract**, the state retains ownership of all production, and the contractor is paid a fee for its services, either in cash or in kind. In a *pure* service contract, the fee is fixed and the state bears the risk; in a *risk* service contract, the contractor bears the exploration risk and is remunerated only if production

is achieved. Service contracts are used where constitutional or political considerations preclude any transfer of ownership of petroleum to foreign companies, notably in Iran (whose buyback contracts and the subsequent Iran Petroleum Contract reflect constitutional prohibitions on foreign ownership), Iraq (whose technical service contracts for major fields awarded from 2009 paid a fixed remuneration fee per barrel), and Mexico (prior to the 2013 reforms).

The Modern Licence

The modern petroleum licence differs fundamentally from the classical concession described in Chapter 1. Where the classical concession covered vast areas for very long periods with minimal state control, the modern licence covers a defined block for a limited exploration period, imposes work obligations and relinquishment requirements, subjects the licensee to comprehensive regulatory control over operations, and secures a substantial share of the resource rent for the state through a progressive fiscal regime. The modern licence is embedded in a statutory framework that regulates health and safety, environmental protection, resource management, decommissioning, and local content.

In the United Kingdom, for example, the North Sea Transition Authority (NSTA, formerly the Oil and Gas Authority) regulates licensing and resource management under the Petroleum Act 1998, the Energy Act 2016, and the model clauses; the Health and Safety Executive regulates safety under the Offshore Installations (Safety Case) Regulations 2015; the Offshore Petroleum Regulator for Environment and Decommissioning (OPRED), within the Department for Energy Security and Net Zero, regulates environmental matters and decommissioning; and HM Revenue and Customs administers a fiscal regime comprising ring-fence corporation tax, a supplementary charge, and the Energy Profits Levy introduced in 2022. Since 2021, the NSTA has been required to apply its "Strategy", which includes a central obligation on licensees and others to take the steps necessary to secure that the maximum value of economically recoverable petroleum is recovered from the strata beneath relevant UK waters, while also assisting the Secretary of State in meeting the net zero target.

Advantages and Disadvantages of Licensing

Licensing systems have several advantages. They are relatively simple to administer, since the state's return derives from fiscal instruments applied through the tax system. They provide investors with clear title to production, facilitating financing. They are compatible with general legislation applicable to all licensees, promoting transparency and equality of treatment. And they keep the state at arm's length from commercial operations, reducing the need for state technical capacity.

Their principal disadvantages, from the host state's perspective, are that the state has limited direct control over operations and limited access to information; that fiscal regimes based on royalties and taxes may fail to capture the full resource rent, especially when prices rise unexpectedly; and that the transfer of ownership of petroleum to foreign companies may be politically sensitive.

Local Content Obligations in Upstream Petroleum

Modern upstream regimes almost invariably impose local content obligations — requirements that licensees employ nationals, procure goods and services from domestic suppliers, use domestic facilities, and transfer technology and skills. Local content policies seek to convert petroleum extraction from an enclave activity into a driver of broader industrial development. They raise legal questions under trade law (where they discriminate against imported goods), investment law (where they impose performance requirements prohibited by some treaties), and domestic administrative law (where their enforcement is discretionary).

Nigeria's Nigerian Oil and Gas Industry Content Development Act 2010 (the NOGICD Act) is among the most comprehensive local content statutes in the world. It provides that Nigerian independent operators shall be given first consideration in the award of oil blocks, oil field licences, oil lifting licences, and all projects for which contracts are to be awarded in the Nigerian oil and gas industry, subject to the fulfilment of conditions specified by the Minister; that exclusive consideration shall be given to Nigerian indigenous service companies that demonstrate ownership of equipment, Nigerian personnel, and capacity to execute the work in bids for land and swamp operating areas; and that operators shall submit Nigerian content plans with every application for a licence, permit, or interest, demonstrating compliance with the minimum levels of Nigerian content set out in the Schedule to the Act for specified goods and services. The Act established the Nigerian Content Development and Monitoring Board to implement its provisions, and the Nigerian Content Development Fund, financed by a levy of one per cent of the value of every contract awarded in the upstream sector. Presidential directives issued in 2024 extended local content expectations to aspects of procurement and financing in the oil and gas industry.

Comparable statutes exist in Ghana (the Petroleum (Local Content and Local Participation) Regulations 2013, L.I. 2204), Angola, Brazil, Kazakhstan, and many other producer states. Empirical research suggests that local content policies are most effective when targets are realistic, linked to measurable capacity-building, and backed by investment in education, infrastructure, and access to finance; overly ambitious or poorly enforced targets may increase costs without delivering lasting capability. The design of local content provisions in licences and contracts — including the definitions of "local" goods, services, and companies,

the measurement methodology, the consequences of non-compliance, and the interaction with stabilisation clauses — is a significant area of legal practice.

5.4 Production Sharing Contracts

Origins

The production sharing contract was pioneered in Indonesia in the mid-1960s by the state oil company Permina (later Pertamina) under the leadership of Ibnu Sutowo, in response to political pressure to end the concession system and assert national control over petroleum. The Indonesian model, which provided for the state to retain ownership and management control while the foreign contractor bore the costs and risks and recovered them from production, was adopted by many other states in Asia, Africa, and Latin America. By the 1990s, the PSC had become the predominant form of petroleum arrangement in the developing world.

In Nigeria, PSCs were introduced in 1973 with Ashland Oil and became the primary vehicle for deep offshore development following the 1993 licensing round. The Deep Offshore and Inland Basin Production Sharing Contracts Act 1999 (as amended in 2019) established the fiscal framework for deep offshore PSCs, including royalty rates that decline with water depth. The 2019 amendment, which increased royalty rates and introduced a price-based royalty, was enacted after the federal government had lost significant revenue because the 1999 Act's provision for review of the royalty rates once oil prices exceeded USD 20 per barrel had never been implemented. The PIA now governs new upstream arrangements, while preserving the terms of existing PSCs subject to the conversion provisions discussed below.

Structure and Key Provisions

A typical PSC contains the following key provisions:

Parties and contract area. The PSC is concluded between the state or national oil company and the contractor (often a consortium of companies). It defines the contract area, usually by reference to geographical coordinates.

Term and phases. The PSC specifies an exploration period, divided into phases with associated work commitments, and a production period that commences upon a declaration of commerciality and approval of a development plan.

Work programme and budget. The contractor must submit annual work programmes and budgets for approval by a management committee on which the national oil company is represented.

Management committee. The management committee, composed of representatives of the national oil company and the contractor, supervises operations. The allocation of voting rights between the parties determines the degree of state control.

Cost recovery. The contractor is entitled to recover its exploration, development, and operating costs from a defined percentage of production — the "cost oil" or "cost recovery limit" — commonly between 50 and 80 per cent of production in any period. Costs that exceed the limit may usually be carried forward to future periods. Accounting procedures annexed to the PSC define which costs are recoverable and how they are to be allocated.

Profit oil split. Production remaining after royalty and cost oil — "profit oil" — is divided between the state and the contractor in accordance with an agreed formula. The split may be fixed or, more commonly, may vary according to a sliding scale based on cumulative production, daily production rates, the contractor's rate of return, or an "R-factor" (the ratio of the contractor's cumulative revenues to its cumulative costs).

Royalty and taxation. Many PSCs also require payment of a royalty and subject the contractor's share of profit oil to income tax. In some PSCs, the national oil company pays the contractor's income tax out of the state's share of profit oil (a "pay-on-behalf" arrangement), to enable the contractor to claim foreign tax credits in its home jurisdiction.

Title and lifting. Title to the contractor's share of production passes at the delivery point, and lifting arrangements govern the allocation of cargoes.

Ownership of assets. Title to equipment and facilities typically passes to the state either upon acquisition or upon full cost recovery, although the contractor retains the right to use them for operations.

Domestic supply obligations, local content, training, and stabilisation clauses are also common.

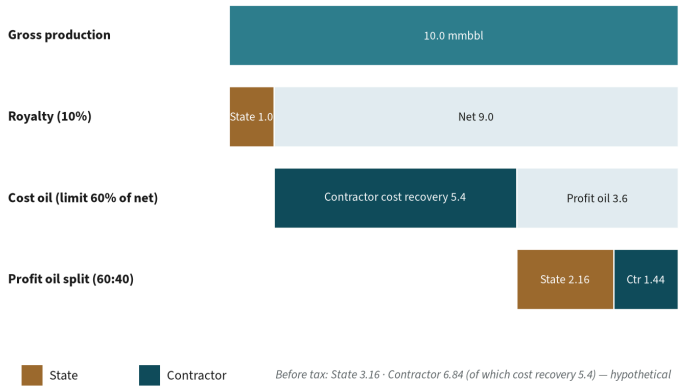


Figure 5.2 Illustrative production sharing waterfall. The percentages are hypothetical and are used solely to illustrate the mechanics of allocation. *Source:* prepared for this edition.

A Worked Illustration

The mechanics of a PSC may be illustrated with a simplified, hypothetical example. Suppose that in a given year a field produces 10 million barrels of oil, the PSC provides for a royalty of 10 per cent, a cost recovery limit of 60 per cent of production after royalty, and a profit oil split of 60:40 in favour of the state, and that the contractor has recoverable costs of 7 million barrels' worth for the year.

1. Royalty: 10 per cent of 10 million = 1 million barrels to the state, leaving 9 million barrels.
2. Cost oil: the cost recovery limit is 60 per cent of 9 million = 5.4 million barrels. The contractor recovers 5.4 million barrels; the unrecovered balance of 1.6 million barrels' worth is carried forward.
3. Profit oil: 9 million minus 5.4 million = 3.6 million barrels, of which the state receives 60 per cent (2.16 million) and the contractor 40 per cent (1.44 million).
4. Tax: if the contractor's profit oil is subject to income tax at, say, 30 per cent, a further portion of the contractor's share is payable to the state.

In this illustration, before tax the state receives 3.16 million barrels (31.6 per cent of production) and the contractor 6.84 million barrels, of which 5.4 million represent cost recovery. The "government take" — the share of the economic rent (revenues minus costs) accruing to the state — is a more meaningful measure than the share of gross production, and is the standard metric for comparing fiscal regimes. Chapter 16 examines fiscal design in more detail.

Advantages, Disadvantages, and Disputes

PSCs offer host states several advantages: formal ownership of petroleum is retained, satisfying political and constitutional requirements; the national oil company participates in management and gains access to information and expertise; and the state's share of production rises with profitability under progressive profit-sharing formulas. For investors, PSCs offer the right to recover costs from production and a share of production that can be booked as reserves under some accounting rules.

The disadvantages include administrative complexity, especially in auditing cost recovery; the incentive for contractors to "gold-plate" costs, since higher recoverable costs increase their share of production; and the potential for disputes over cost allocation, the interpretation of profit oil formulas, and the lifting of entitlements. The Nigerian experience includes significant disputes concerning the deep offshore PSCs. In the dispute concerning the Oil Mining Lease 133 (Erha) PSC, an arbitral tribunal in 2011 upheld claims by the contractors, Esso Exploration and Production Nigeria Ltd and Shell Nigeria Exploration and Production Co Ltd, against NNPC concerning the lifting of tax oil and royalty

oil. NNPC applied to the Federal High Court in Abuja, which set the award aside on the basis that the dispute concerned tax matters that were not arbitrable under Nigerian law, generating further appeals in Nigeria and parallel proceedings in the United States to enforce the award. The litigation illustrates the jurisdictional complexities that arise when fiscal provisions are embedded in contracts and the tension between the finality of arbitral awards and national courts' views on arbitrability.

The PIA and the Conversion of Existing Arrangements

The PIA introduced a framework for the voluntary conversion of existing oil prospecting licences and oil mining leases, including those underlying PSCs and joint ventures, to the new petroleum prospecting licences and petroleum mining leases. Conversion is incentivised by more favourable fiscal terms under the new regime, while holders that do not convert continue under their existing terms until expiry but cannot benefit from the new fiscal incentives and face restrictions on renewal. The Act also required the conversion of NNPC's joint venture interests into incorporated joint ventures within a specified period after its commencement. These transitional provisions illustrate a characteristic technique of petroleum law reform: the use of incentives rather than compulsion to migrate investors from legacy arrangements, thereby reducing the risk of investment treaty claims.

Stabilisation and Economic Equilibrium in PSCs

Because PSCs typically run for decades, contractors seek protection against changes in the legal and fiscal regime that would reduce the economic value of their entitlements. PSCs therefore frequently contain stabilisation provisions, which may take one of three principal forms. *Freezing clauses* provide that the law applicable to the contract is the law in force at the date of signature, and that subsequent legislation will not apply to the contractor to the extent that it is inconsistent with the contract. *Economic equilibrium clauses* permit the application of new laws but require the parties to restore the original economic balance of the contract, by adjusting fiscal terms or providing compensation, where a change in law materially affects the contractor's economic position. *Hybrid clauses* combine elements of both. Many modern PSCs prefer economic equilibrium clauses, because they preserve the state's legislative freedom while protecting the investor's economic expectations, and because freezing clauses have been criticised for insulating investors from environmental, labour, and human rights regulation.

The interaction between stabilisation and human rights has attracted particular attention. A study prepared for the United Nations Special Representative on Business and Human Rights in 2008 found that stabilisation clauses in many investment contracts in developing countries were drafted broadly enough to

exempt investors from new social and environmental legislation. The *Principles for Responsible Contracts*, endorsed by the Human Rights Council in 2011 as an addendum to the UN Guiding Principles, recommend that stabilisation clauses be drafted so as not to interfere with the state's good-faith efforts to implement laws, regulations, or policies in a non-discriminatory manner in order to meet its human rights obligations. Stabilisation clauses are examined further in Chapters 14 and 15.

Marginal Fields

Many mature producing provinces contain discovered but undeveloped fields that are too small to justify development by the original licensee. To promote their development by smaller companies, several jurisdictions have created special "marginal field" regimes. In Nigeria, the Petroleum (Amendment) Act 1996 introduced the marginal field programme, empowering the President to farm out marginal fields within an oil mining lease where they had been left unattended for not less than ten years from the date of first discovery. The programme facilitated the emergence of a cohort of Nigerian independent producers. The 2020–2021 marginal field bid round, conducted under the Marginal Field Operations (Fiscal Regime) Regulations and the Guidelines for Farm-out and Operation of Marginal Fields, awarded fifty-seven fields to Nigerian companies. The PIA continues to provide for the declaration and farm-out of fields that are not being developed, subject to the NUPRC's approval. Marginal field regimes illustrate how licensing law can be used as an instrument of industrial policy, but they also generate disputes with the original leaseholders concerning the identification of fields, access to infrastructure, and the terms of farm-out agreements.

Service Contracts in Practice

Service contracts, although less common than licences and PSCs, deserve attention because of their use in some of the world's largest oil provinces. The Iraqi technical service contracts awarded in 2009 and 2010 for super-giant fields such as Rumaila, West Qurna, and Majnoon paid contractors a fixed remuneration fee per barrel of incremental production above an agreed baseline, plus recovery of costs, with fees subject to an R-factor adjustment. Because contractors earned the same fee per barrel regardless of oil prices, the contracts transferred almost the entire price upside to the state, but they also left contractors exposed to production disruptions, export capacity constraints, and delays in cost recovery payments. Several contracts were subsequently renegotiated to lower production targets. The experience illustrates that the allocation of risk and reward in petroleum arrangements is determined not by the label attached to the contract but by the detailed economic and legal terms.

5.5 Joint Operating Agreements

Purpose of the Joint Operating Agreement

Petroleum titles are commonly held by several companies jointly, in order to spread risk, share capital costs, combine technical capabilities, and satisfy state participation requirements. The relationship among the co-venturers is governed by a joint operating agreement (JOA), a contract that establishes the rules for conducting joint operations, sharing costs and production, making decisions, and addressing default and withdrawal. The JOA does not usually create a partnership or a separate legal entity; most JOAs expressly provide that the parties' rights and obligations are several and not joint or collective, and that nothing in the agreement shall be construed as creating a partnership. The co-venturers hold undivided interests in the title as tenants in common (or the civil law equivalent), each entitled to take and separately dispose of its share of production.

The most widely used international model is the AIEN Model International Operating Agreement, first published in 1990 and most recently revised in 2012 (with subsequent updates). National models also exist, notably the UK Oil and Gas Authority (now NSTA) Master Deed and model JOAs developed under the industry's Standard Agreements initiative. In Nigeria, the historic NNPC joint ventures with international oil companies were governed by JOAs based in part on the AIEN model and in part on bespoke terms.

Key Provisions

Scope and participating interests. The JOA defines the joint property, the contract area, and each party's participating interest, which determines its share of costs, production, and voting power.

The operator. One party is designated as operator, responsible for conducting joint operations on behalf of all parties. The operator acts as an agent of the non-operators in some respects but is typically not entitled to profit from its role, recovering its costs through the joint account (including overhead charges) on a "no gain, no loss" basis. The JOA defines the operator's duties — commonly to conduct operations in a diligent, safe, and efficient manner in accordance with good oilfield practice — and limits its liability. A key feature of most international JOAs is that the operator is liable to the non-operators only for losses resulting from its "gross negligence or wilful misconduct", and in some forms only where such conduct is that of senior supervisory personnel. The interpretation of these terms has been considered in English cases, and their meaning depends heavily on the precise contractual wording.

Operating committee. Decisions on work programmes, budgets, and major operational matters are taken by an operating committee in which each party is represented. The JOA sets out voting thresholds, often requiring a majority

of participating interests together with a minimum number of parties ("pass marks"), and identifies matters requiring unanimity.

Work programmes, budgets, and authorisations for expenditure. The operator prepares annual work programmes and budgets for approval, and individual items of expenditure above specified thresholds require authorisations for expenditure (AFEs).

Sole risk and non-consent operations. Where a party wishes to undertake an operation that the operating committee has not approved, the JOA may permit it to proceed at its sole risk and cost. A party that elects not to participate in an approved operation ("non-consent") may be excluded from the benefits of that operation unless it pays a substantial premium — often several hundred per cent of the costs it avoided — to re-enter. These provisions balance the interests of parties wishing to proceed against those of parties wishing to limit risk.

Default. If a party fails to pay its share of costs ("cash calls"), the JOA provides remedies for the non-defaulting parties, which may include suspension of the defaulting party's rights to vote and to receive production, and ultimately forfeiture of its participating interest. Forfeiture provisions have been challenged as penalties. In *Cavendish Square Holding BV v Makdessi* and *ParkingEye Ltd v Beavis*, decided together in 2015, the United Kingdom Supreme Court reformulated the rule against penalties, holding that the test is whether the provision is a secondary obligation that imposes a detriment on the contract-breaker out of all proportion to any legitimate interest of the innocent party in the enforcement of the primary obligation.[1] In the oil and gas context, the legitimate interest of non-defaulting parties in ensuring the continuation of joint operations has generally been regarded as supporting the enforceability of forfeiture clauses, although each clause must be assessed on its terms.

Transfer and pre-emption. The JOA restricts transfers of participating interests, commonly requiring the consent of the other parties (not to be unreasonably withheld) and granting them pre-emption rights. Pre-emption provisions have been a frequent source of disputes, particularly regarding whether they are triggered by indirect transfers through changes of control of a party.

Withdrawal, decommissioning, and confidentiality. The JOA governs the right of a party to withdraw, the allocation of decommissioning costs (often addressed in a separate decommissioning security agreement), confidentiality, and dispute resolution.

[1] *Cavendish Square Holding BV v Makdessi; ParkingEye Ltd v Beavis* [2015] UKSC 67, [2016] AC 1172.

Joint Ventures with National Oil Companies

Where the state participates in upstream operations through its national oil company, the JOA governs the relationship between the national oil company and its private partners. In Nigeria, the joint venture arrangements between NNPC and international oil companies such as Shell, ExxonMobil, Chevron, TotalEnergies, and Eni historically required NNPC to fund its share of costs (typically 55 to 60 per cent) through cash calls. Chronic failure by NNPC to meet its cash call obligations, owing to budgetary constraints, led to underinvestment and the accumulation of arrears, prompting the development of alternative funding arrangements, including modified carry agreements, under which the private partners fund NNPC's share and recover it from production. The PIA's provision for the conversion of NNPC joint ventures into incorporated joint ventures, which could raise finance independently, was intended in part to address the cash call problem.

5.6 National Oil Companies and State Participation

Forms of State Participation

States participate in upstream petroleum in several ways:

1. **Full state operation**, in which the national oil company holds and operates all petroleum rights, as in Saudi Arabia (Saudi Aramco), Kuwait (Kuwait Petroleum Corporation), and Mexico (Pemex, historically).
2. **Majority or minority participation in joint ventures**, in which the national oil company holds an equity interest alongside private partners, either funding its share (paid participation) or being carried by the private partners through exploration (carried interest), as in Nigeria's joint ventures, Angola, and Ghana, where the Ghana National Petroleum Corporation holds a carried interest of not less than 15 per cent in petroleum agreements under the Petroleum (Exploration and Production) Act 2016 (Act 919).
3. **Contractual participation**, in which the national oil company is the contracting party under a PSC or service contract and participates in management through the management committee.
4. **Indirect participation through state-owned commercial companies or holding companies** that hold the state's direct financial interests, as in Norway's Petoro.

The Dual Role Problem

National oil companies frequently combine commercial and governmental functions: they may hold equity interests in joint ventures, act as the state's concessionaire or contracting party under PSCs, perform regulatory functions such as approving work programmes or monitoring compliance, market the state's share of production, and perform social or political functions such as fuel supply and

subsidy administration. This combination of roles creates conflicts of interest and impairs both commercial performance and regulatory effectiveness. A national oil company that is both a partner in a joint venture and the regulator of its partners cannot regulate impartially; one that is required to supply fuel at subsidised prices cannot operate on commercial principles.

The separation of the commercial and regulatory roles of national oil companies has been a central objective of petroleum sector reform, as discussed in Chapter 4 in the context of the Norwegian model and the PIA. The PIA's creation of NNPC Ltd as a commercial company without regulatory functions, and its vesting of regulatory functions in the NUPRC and NMDPRA, represents Nigeria's response to the dual role problem.

National Oil Companies Abroad

Many national oil companies have become international investors in their own right, operating outside their home states. Companies such as Saudi Aramco, Petronas, the China National Petroleum Corporation, the China National Offshore Oil Corporation, Equinor, and Petrobras hold significant upstream interests abroad. When national oil companies invest abroad, questions arise about their status as investors under investment treaties, their entitlement to sovereign immunity, and their susceptibility to the host state's competition and foreign investment screening laws. The ICSID Convention permits claims by "nationals of another Contracting State", and tribunals have held that state-owned companies may qualify as nationals unless they are acting as agents of the government or discharging an essentially governmental function — the test proposed by Aron Broches and applied in *Ceskoslovenska Obchodni Banka AS v Slovak Republic* (1999) and subsequent cases. In *Beijing Urban Construction Group Co Ltd v Republic of Yemen* (2017), the tribunal applied the Broches test to hold that a Chinese state-owned enterprise was a national for the purposes of the Convention.

5.7 Unitization and Cross-Border Reservoir Development

The Problem of Shared Reservoirs

Petroleum reservoirs do not respect licence boundaries or international borders. Where a reservoir straddles the boundary between two licence areas, or between two states, independent development by each owner would lead to the inefficiencies associated with the rule of capture: competitive drilling, wasteful duplication of infrastructure, loss of reservoir pressure, and reduced ultimate recovery. The solution is unitisation: the development of the reservoir as a single unit, with costs and production shared among the owners according to agreed "tract participation" percentages reflecting the proportion of the reservoir located within each area.

Domestic Unitisation

Within a single state, unitisation of a reservoir straddling licence boundaries may be achieved voluntarily or compelled by the regulator. Most petroleum regimes empower the regulator to require unitisation where it is necessary to secure the most effective recovery of petroleum. In the United Kingdom, the model clauses empower the Secretary of State (now acting through the NSTA) to require licensees to cooperate in the development of a field extending beyond the licensed area. In Nigeria, the PIA empowers the NUPRC to direct unitisation where a petroleum reservoir extends across licence boundaries and the holders cannot agree.

A unitisation and unit operating agreement (UUOA) establishes the unit, determines tract participations, appoints a unit operator, and provides for redeterminations of tract participations as geological knowledge improves. Redeterminations are a frequent source of disputes, because a change in tract participation can transfer very large values between the parties. UUOAs therefore contain detailed procedures for redetermination, often involving independent experts whose decisions are final and binding except for manifest error or fraud.

Transboundary Unitisation

Where a reservoir straddles an international boundary, unitisation requires cooperation between the states concerned. Customary international law does not clearly establish an obligation to unitise, but state practice, reflected in numerous bilateral agreements, demonstrates a strong preference for cooperation. Many maritime boundary agreements include a "mineral deposits clause", first appearing in the 1965 Agreement between the United Kingdom and Norway relating to the Delimitation of the Continental Shelf, which obliges the parties, where a single geological petroleum structure or field extends across the dividing line, to seek to reach agreement as to the manner in which the structure or field shall be most effectively exploited and the manner in which the proceeds shall be apportioned.

The Framework Agreement concerning Cross-Boundary Petroleum Co-operation between Norway and the United Kingdom of 2005 provides a comprehensive regime for the development of cross-boundary fields, including procedures for approval of unitisation agreements by both governments, apportionment of reserves, and the use of infrastructure on one side of the boundary to develop fields on the other. Similar arrangements exist in the Gulf of Mexico under the Agreement between the United States and Mexico concerning Transboundary Hydrocarbon Reservoirs in the Gulf of Mexico (2012).

Joint Development Zones

Where states cannot agree on a maritime boundary, they may establish a joint development zone (JDZ) in the disputed area, allowing petroleum activities to proceed without prejudice to their respective claims. The Nigeria–São Tomé and Príncipe Joint Development Zone, established by a treaty signed in Abuja in 2001, is administered by a Joint Ministerial Council and a Joint Development Authority, with revenues shared 60:40 in favour of Nigeria. Other examples include the Malaysia–Thailand Joint Development Area (1979) and the arrangements in the Timor Sea between Australia and Timor-Leste, which were superseded by the maritime boundary treaty of 2018 concluded following the first-ever compulsory conciliation under Annex V of UNCLOS. Joint development is examined further in Chapter 20.

5.8 Decommissioning and Abandonment Liabilities

The Decommissioning Challenge

At the end of their productive lives, petroleum installations must be decommissioned: wells must be plugged and abandoned, platforms and pipelines removed or made safe, and sites remediated. Decommissioning is technically complex and extremely expensive. Estimates of the total cost of decommissioning offshore infrastructure on the United Kingdom Continental Shelf alone run to tens of billions of pounds over the coming decades, and globally the liability runs to hundreds of billions of dollars. The energy transition is likely to accelerate decommissioning as fields become uneconomic earlier than anticipated.

Decommissioning raises four principal legal questions: what standard of decommissioning is required; who is liable to carry it out and pay for it; how the state can ensure that funds will be available when decommissioning becomes due; and how liability is affected by transfers of interests during the life of a field.

International Standards

UNCLOS Article 60(3) provides that "any installations or structures which are abandoned or disused shall be removed to ensure safety of navigation, taking into account any generally accepted international standards established in this regard by the competent international organization", and that "such removal shall also have due regard to fishing, the protection of the marine environment and the rights and duties of other States". The competent organisation is the International Maritime Organization, whose *Guidelines and Standards for the Removal of Offshore Installations and Structures on the Continental Shelf and in the Exclusive Economic Zone* (1989) require the removal of installations standing in less than 75 metres of water and weighing less than 4,000 tonnes in air (excluding the deck and superstructure), and of all installations placed after 1 January 1998 in waters of

less than 100 metres depth, while permitting partial removal of larger installations subject to specified conditions.

In the North-East Atlantic, OSPAR Decision 98/3 on the Disposal of Disused Offshore Installations, adopted in 1998 following the controversy over Shell's proposal to dispose of the Brent Spar storage buoy in deep water, prohibits the dumping and leaving wholly or partly in place of disused offshore installations, subject to derogations for the footings of large steel jackets weighing more than 10,000 tonnes and for concrete gravity-based installations, where the competent authority is satisfied that there are significant reasons why an alternative disposal option is preferable to reuse, recycling, or final disposal on land. The Brent Spar episode illustrates the role of public opinion and civil society in shaping decommissioning law.

The London Convention 1972 and its 1996 Protocol on the prevention of marine pollution by dumping also regulate the disposal of platforms at sea.

Domestic Regimes

In the United Kingdom, Part IV of the Petroleum Act 1998 establishes a detailed regime under which the Secretary of State may require persons responsible for offshore installations and pipelines to submit decommissioning programmes for approval. Section 29 notices may be served on a wide class of persons, including current and former licensees, owners of installations, and companies associated with them. The Energy Act 2008 extended the power to require security for decommissioning liabilities. Crucially, liability under Part IV can extend to former licensees: a company that has sold its interest may be called upon to carry out decommissioning if the purchaser defaults. This "residual liability" regime gives the state a powerful protection against insolvency of current operators but complicates transactions, since sellers must manage the risk of being called back.

In Nigeria, the PIA requires the holder of a petroleum mining lease to submit a decommissioning and abandonment plan and to establish a decommissioning and abandonment fund, to be maintained with a financial institution and managed in accordance with regulations issued by the NUPRC. Contributions to the fund are to be made over the life of the field, ensuring that adequate resources will be available. The PIA also provides that the obligations relating to decommissioning and abandonment are not extinguished by the transfer of an interest without the express agreement of the regulator, and the NUPRC's regulations on decommissioning and abandonment set out detailed requirements. The adequacy of decommissioning provisioning has been a central issue in the review of onshore divestments by international oil companies, given the age of much of the onshore infrastructure in the Niger Delta and the legacy of environmental contamination.

Liability Allocation in Contracts

Contracts allocate decommissioning liabilities among co-venturers and between buyers and sellers. JOAs typically provide that decommissioning costs are shared in proportion to participating interests at the time decommissioning is carried out, but require parties to provide security in advance — often through decommissioning security agreements (DSAs) — to protect against the default of any party. In asset sales, the seller typically seeks an indemnity from the buyer for decommissioning liabilities, supported by security such as letters of credit or parent company guarantees, because the seller may remain exposed to residual statutory liability. Disputes over the adequacy of security and the calculation of decommissioning cost estimates are frequent.

Decommissioning and the Energy Transition

The energy transition creates new opportunities and challenges for decommissioning. Infrastructure that would otherwise be removed may be repurposed for carbon dioxide transport and storage, hydrogen production or transport, or offshore wind support. Such repurposing requires legal frameworks that address the transfer of liability for legacy infrastructure, the long-term liability for stored carbon dioxide, and the interaction between petroleum, carbon storage, and renewable energy licensing regimes. The United Kingdom's Energy Act 2023 and the European Union's CCS Directive 2009/31/EC address some of these questions, examined in Chapter 12. At the same time, the risk that fields will be abandoned prematurely, by operators lacking the resources to decommission them properly, raises concerns about stranded liabilities that may ultimately fall on the state and taxpayers.

Onshore Abandonment and Legacy Pollution

Onshore, decommissioning intersects with the remediation of legacy pollution. In the Niger Delta, decades of oil spills, gas flaring, and inadequate decommissioning of wells and pipelines have left extensive environmental damage. The United Nations Environment Programme's *Environmental Assessment of Ogoniland* (2011) found widespread contamination of land, groundwater, and surface water and recommended a clean-up programme that it estimated would take up to thirty years. The Hydrocarbon Pollution Remediation Project (HYPREP) was established by the Federal Government in 2016 to implement the recommendations. The PIA's environmental remediation fund provisions, discussed in Chapter 17, are designed to ensure that operators provide for the remediation of environmental damage arising from their operations. The legal liability of parent companies for pollution caused by their subsidiaries in Nigeria has been addressed by English and Dutch courts, notably in *Okpabi v Royal Dutch Shell plc* (2021) and *Four Nigerian Farmers v Shell* (2021), considered in Chapter 21.

LEADING AUTHORITY

Ghana/Côte d'Ivoire Maritime Delimitation (ITLOS Special Chamber, 2017) — petroleum activities in disputed areas

Facts. Ghana had for decades licensed petroleum activities up to a line it regarded as the customary equidistance boundary with Côte d'Ivoire, and significant discoveries had been made, including the TEN fields. Côte d'Ivoire contested the boundary and claimed that Ghana's unilateral petroleum activities in the disputed area violated Côte d'Ivoire's sovereign rights and Article 83(3) of UNCLOS, which requires states, pending agreement on delimitation, to "make every effort to enter into provisional arrangements of a practical nature" and "not to jeopardize or hamper the reaching of the final agreement". In 2015 the Special Chamber had prescribed provisional measures requiring Ghana not to undertake new drilling in the disputed area.

Holding. The Special Chamber delimited the boundary using the equidistance/relevant circumstances methodology, producing a line close to that claimed by Ghana. It rejected Ghana's argument that a tacit agreement on the boundary had arisen from the parties' oil practice, but also held that Ghana had not violated Côte d'Ivoire's sovereign rights, because the activities had been carried out in an area that was ultimately attributed to Ghana, and that Ghana had not violated Article 83(3).

Significance. The decision is important for upstream petroleum practice in disputed maritime areas. It confirms that oil practice alone rarely establishes a tacit maritime boundary, and it illustrates the risks — including provisional measures halting new drilling — that investors face when operating in contested waters.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Allocating risk under a production sharing contract (hypothetical). The fictional Republic of Kandara is preparing a model PSC for its first deep-water licensing round. You are counsel to the Kandaran Ministry of Petroleum. Prepare (a) an annotated term sheet of the key fiscal provisions (royalty, cost recovery limit, profit oil split based on an R-factor, taxation, and state participation), explaining the policy rationale for each choice; (b) a short model calculation applying your terms to a hypothetical field producing 50,000 barrels per day at an assumed price of USD 70 per barrel, with stated assumptions; and (c) a note identifying the three provisions most likely to generate disputes, with drafting recommendations to reduce that risk. *Kandara, its institutions, and all figures are hypothetical and for teaching purposes only.*

CHAPTER SUMMARY

- Petroleum is defined by statute and contract; the distinctions between crude oil and gas, associated and non-associated gas, and conventional and unconven-

tional resources have significant legal and fiscal consequences, as illustrated by Nigeria's gas flaring provisions.

- Petroleum titles may be characterised as administrative licences, contracts, property rights, or hybrids; the characterisation affects the state's power to vary terms and the investor's protection.
- Competitive and transparent allocation of rights is increasingly required by law, as under the PIA, in reaction to the abuses associated with discretionary award.
- Petroleum arrangements fall into concessionary/licensing systems, production sharing contracts, and service contracts, distinguished by ownership of production and the form of the state's return; the economic substance matters more than the label.
- Joint operating agreements govern relations among co-venturers, including the operator's role and liability, operating committee decisions, sole risk, default, and transfers; NNPC cash call problems illustrate the difficulties of state participation.
- National oil companies' dual commercial and regulatory roles create conflicts of interest that reforms such as the PIA seek to resolve.
- Shared reservoirs require unitisation; transboundary fields are governed by mineral deposits clauses, framework agreements, and joint development zones.
- Decommissioning is governed by UNCLOS, IMO guidelines, regional instruments such as OSPAR Decision 98/3, and domestic regimes that increasingly require advance funding and impose residual liability.

DOCTORAL RESEARCH QUESTIONS

1. Is the conventional classification of petroleum arrangements into licences, PSCs, and service contracts analytically useful, given that fiscal terms can make any form economically equivalent to another?
2. Evaluate the PIA's incentive-based conversion regime as a technique for migrating legacy arrangements without triggering investment treaty claims.
3. Has state practice in maritime boundary agreements given rise to a customary obligation to cooperate in, or to unitise, the development of transboundary petroleum deposits?
4. How should residual decommissioning liability be designed to protect the state without deterring the transfer of mature assets to new operators better placed to extend field life?
5. Should international law impose obligations on states to ensure that fields are not abandoned without adequate remediation, and what would be the source of such obligations?

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6 CHAPTER

Natural Gas Law, LNG Transactions, and International Gas Markets

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the physical, economic, and commercial characteristics of natural gas that distinguish its legal regime from that of crude oil.
2. Analyse the regulation of gas exploration and production, including gas-specific licensing, domestic supply obligations, and flaring rules.
3. Evaluate the legal models for gas transportation and pipeline access, including negotiated and regulated third-party access and unbundling.
4. Compare the principal structures used to develop liquefied natural gas (LNG) projects and identify the legal risks associated with each.
5. Analyse the key provisions of long-term gas sales and purchase agreements, including take-or-pay, delivery, quality, and destination provisions.
6. Explain the law and practice of price review and the mechanisms by which gas contracts accommodate changing market conditions.
7. Evaluate the international legal framework governing cross-border gas pipelines and transit.
8. Assess the causes and resolution of gas supply disputes and the legal limits on regulatory intervention in gas markets.

Introduction

Natural gas occupies a distinctive and increasingly contested position in international energy law. For much of the twentieth century it was treated as an unwanted by-product of oil production, flared or reinjected because there was no economical way to bring it to market. Over the past five decades it has become a global commodity in its own right, supplying almost a quarter of world primary energy, fuelling a large share of electricity generation, providing heat for homes and industry, and serving as a feedstock for fertilisers and petrochemicals. Its lower carbon intensity relative to coal and oil has led many governments to describe

it as a "bridge" or "transition" fuel, while its methane emissions and long-lived infrastructure have led critics to warn that new gas investment risks locking in emissions incompatible with the goals of the Paris Agreement. The legal regime of natural gas must therefore accommodate both its role in energy security and its contested position in the energy transition.

The legal regime of natural gas differs from that of oil in ways that flow directly from the physical properties of the commodity. Oil is a dense liquid that can be stored easily and transported cheaply by tanker, rail, or pipeline to almost any market; this fungibility gave rise, from the 1980s onwards, to a global oil market with transparent spot and futures prices. Gas, by contrast, has a very low energy density at atmospheric pressure. It must be transported either through high-pressure pipelines, which are fixed, capital-intensive, and tie producer and consumer together for decades, or as liquefied natural gas, which requires cooling the gas to approximately minus 162 degrees Celsius, shipping it in specialised vessels, and regasifying it at the destination. Both modes of transport require very large, specific, and long-lived investments. As a consequence, gas markets historically developed as regional rather than global markets, organised around long-term contracts that allocated the risks of these investments between sellers and buyers, and subject to extensive regulation of the natural monopoly elements of the supply chain.

Three transformations over the past four decades have reshaped the legal regime of gas. The first is liberalisation: the introduction of competition in gas supply in North America from the 1980s, in the United Kingdom from the 1990s, and in the European Union from the late 1990s onwards, requiring the unbundling of transmission from supply and the creation of rights of third-party access to pipelines. The second is the growth of the LNG trade, which has expanded from a handful of point-to-point projects linking specific producers with Japanese and European buyers into a global market involving dozens of exporting and importing countries, with an increasing share of spot and short-term trade and greater price convergence among regions. The third is the politicisation of gas supply, illustrated most dramatically by the Russian Federation's curtailment of gas deliveries to Europe in 2021 and 2022, which precipitated an energy crisis, extraordinary regulatory interventions, and a fundamental restructuring of European gas supply.

This chapter examines the legal regime of natural gas from wellhead to burner tip. It begins with the legal and commercial characteristics of gas (section 6.1) and the regulation of gas exploration and production (section 6.2). It then examines gas transportation and pipeline access (section 6.3), the structures used to develop LNG projects (section 6.4), long-term gas sales and purchase agreements (section 6.5), and the mechanisms of price review, take-or-pay, and contractual

flexibility (section 6.6). It concludes with the legal framework for cross-border gas infrastructure and transit (section 6.7) and the law of gas supply disputes and regulatory intervention (section 6.8). The treatment of submarine pipelines under the law of the sea is reserved for Chapter 20, and the geopolitics of pipeline dependence for Chapter 25; this chapter focuses on the commercial and regulatory law of gas.

6.1 Legal and Commercial Characteristics of Natural Gas

Physical Properties and Their Legal Consequences

Natural gas is a mixture of hydrocarbon gases composed predominantly of methane, together with varying proportions of heavier hydrocarbons (ethane, propane, butane, and pentanes), and non-hydrocarbon components such as carbon dioxide, nitrogen, hydrogen sulphide, and water vapour. Before gas can be delivered to a transmission system, it must usually be processed to remove impurities and to extract natural gas liquids, which are valuable products in their own right. The composition of gas varies from field to field, and its energy content — its calorific or heating value — varies accordingly. Gas is therefore sold not by volume alone but by energy content, commonly measured in British thermal units (Btu), therms, gigajoules, or kilowatt hours, and gas contracts and network codes specify detailed quality specifications, including ranges for calorific value, the Wobbe index (a measure of the interchangeability of gases in burners), and maximum levels of contaminants.

These physical characteristics have several legal consequences. First, the specification of quality and the allocation of responsibility for off-specification gas are central issues in gas contracts and network codes. A seller that delivers gas outside the contractual specification may be in breach, and the buyer may be entitled to reject it, while a transmission system operator may refuse to accept gas that would compromise the safety or integrity of the network. Second, the measurement of gas — volume, pressure, temperature, and composition — requires agreed metering standards and procedures for resolving measurement disputes. Third, because gas from different sources is commingled in pipelines and storage, title to gas is usually a matter of contractual allocation rather than physical identification; the gas a buyer receives is not physically the same gas that its seller delivered. Gas law therefore relies heavily on concepts such as nominations, allocations, and balancing, which translate physical flows into contractual entitlements.

Natural Monopoly and Network Characteristics

High-pressure transmission pipelines and low-pressure distribution networks exhibit strong economies of scale and are commonly regarded as natural monopolies: it is usually more efficient for a single network to serve a given area than for competing networks to be duplicated. Natural monopoly provides the classic

justification for economic regulation: in the absence of competition, the owner of a network could charge monopoly prices or refuse access to competitors in upstream production or downstream supply. Gas law therefore distinguishes between the potentially competitive segments of the supply chain — production, wholesale trading, and retail supply — and the monopoly segments — transmission, distribution, and, in some cases, storage and LNG terminals — which are subject to regulation of tariffs, access, and investment.

Asset Specificity and the Long-Term Contract

The economics of gas infrastructure are characterised by high asset specificity. A pipeline linking a particular field to a particular market, or an LNG liquefaction plant built to process gas from a particular reservoir, has little value in any alternative use. Once the investment is made, its owner is exposed to opportunistic behaviour by its counterparties — the "hold-up" problem analysed in transaction cost economics by Oliver Williamson and others. A producer that has built a pipeline to a single buyer may find the buyer demanding lower prices once the pipeline is complete; a buyer that has built power stations relying on a single supplier may find the supplier demanding higher prices.

The long-term gas contract emerged as the principal legal device for managing this risk. By committing the buyer to purchase, or pay for, specified quantities of gas over periods of fifteen to twenty-five years, and the seller to deliver them, the long-term contract provides the revenue certainty necessary to finance the infrastructure, while price indexation and price review mechanisms allow the price to adjust to changing market conditions without reopening the fundamental bargain. The long-term contract is thus a form of private governance structure for a relationship in which neither party can easily exit. Its central provisions — take-or-pay, destination restrictions, price indexation, and price review — are examined in sections 6.5 and 6.6.

Regional Markets and Pricing Systems

Partly because of the costs of transport, gas markets developed historically as regional markets with distinct pricing systems. Four broad systems may be identified.

Gas-on-gas competition prevails in North America, the United Kingdom, and much of continental Europe, where liberalised markets with liquid trading hubs determine prices through the interaction of supply and demand. The principal hubs are Henry Hub in Louisiana, which serves as the delivery point for the New York Mercantile Exchange natural gas futures contract; the National Balancing Point (NBP) in Great Britain, a virtual trading point within the National Gas transmission system; and the Title Transfer Facility (TTF) in the Netherlands, which has become the benchmark for continental European gas prices.

Oil indexation was the traditional pricing method in continental Europe and remains significant in Asia. Under oil-indexed contracts, the gas price is linked by formula to the prices of competing fuels, typically crude oil or oil products. In Asian LNG contracts, the price has commonly been expressed as a percentage ("slope") of the Japan Customs-Cleared Crude price (the "JCC" or "Japanese Crude Cocktail") or, increasingly, Brent crude, sometimes with an "S-curve" that moderates the effect of very high or very low oil prices. The rationale for oil indexation, which originated in the pricing of Groningen gas in the Netherlands in the 1960s, was the "market value" or "replacement value" principle: gas should be priced at a level that allowed it to compete with the alternative fuel the buyer would otherwise use.

Regulated pricing prevails in many producing countries and some importing countries, where governments set gas prices administratively, often below the cost of supply, to support domestic industry and power generation or to protect consumers. Regulated domestic gas prices have been a significant obstacle to gas development in many countries, including Nigeria, where the historically low regulated domestic price discouraged investment in gas supply to the domestic market.

Spot and short-term LNG pricing has grown with the expansion of the LNG trade, and Asian spot LNG is now commonly priced by reference to the Japan Korea Marker (JKM), an assessed price published by a price reporting agency, while increasing quantities of LNG from the United States are sold on the basis of Henry Hub prices plus a fixed liquefaction fee.

The International Gas Union's periodic wholesale gas price surveys have documented a long-term shift from oil indexation towards gas-on-gas competition, particularly in Europe, although the extreme price volatility of hub prices during the 2021–2023 crisis led some buyers to reconsider the value of oil-linked contracts as a hedge.

Gas in the Energy Transition

The role of gas in the energy transition is legally significant because it shapes the policy frameworks within which gas investment takes place. The European Union's Taxonomy Regulation, through a Complementary Climate Delegated Act adopted in 2022, classifies certain gas-fired electricity generation and heat activities as transitional activities that may qualify as environmentally sustainable, subject to stringent emissions thresholds and conditions, including a commitment to switch to renewable or low-carbon gases by 2035. The inclusion of gas (and nuclear) in the taxonomy was controversial and was challenged before the General Court of the European Union by Austria and by environmental organisations; in September 2025 the General Court dismissed Austria's action. The

EU's Regulation on the reduction of methane emissions in the energy sector, adopted in 2024, imposes measurement, reporting, and verification obligations on oil, gas, and coal operators, restricts routine venting and flaring, and will progressively apply equivalent monitoring requirements to imported fossil fuels — a measure with significant implications for exporting states, including Nigeria. At the international level, the outcome of the first global stocktake under the Paris Agreement, adopted at COP28 in Dubai in 2023, called on parties to contribute to "transitioning away from fossil fuels in energy systems, in a just, orderly and equitable manner", while recognising that "transitional fuels can play a role in facilitating the energy transition while ensuring energy security".

For gas-producing developing countries, the transition raises profound questions of equity. African governments, including Nigeria, have argued that gas is essential to expanding energy access, industrialisation, and clean cooking, and that it would be unjust for countries that bear little historical responsibility for climate change to be denied financing for gas development. Nigeria declared 2021–2030 the "Decade of Gas", reflecting a strategy of using its large gas reserves — among the largest in Africa — to drive domestic industrialisation and power generation. The tension between this position and the climate policies of international lenders, which have restricted financing for upstream gas, is examined further in Chapters 18 and 23.

6.2 Gas Exploration and Production Regulation

The Historical Neglect of Gas in Petroleum Legislation

Petroleum legislation was designed historically around oil. Early concessions and petroleum statutes often made little distinction between oil and gas, or treated gas as an incidental by-product. As a result, petroleum regimes frequently failed to provide adequate incentives for gas development. Gas discovered in the course of oil exploration might be reinjected, flared, or left undeveloped because the fiscal terms designed for oil made gas development uneconomic, because there was no domestic market or infrastructure, or because the rights to gas were unclear. The development of gas-specific legal provisions — governing rights to non-associated gas, the fiscal treatment of gas, the obligations of producers to supply gas to the domestic market, and the restriction of flaring — has been a significant feature of petroleum law reform over the past three decades.

Rights to Gas Under Petroleum Titles

Petroleum titles usually confer rights to both oil and gas within the licence area, but they may differ in their treatment of the two. Some regimes provide for separate gas licences or for distinct terms applicable to non-associated gas discoveries, recognising that gas development requires longer periods for market development and a different fiscal structure. A common device is the extended "gas holding" or

"gas retention" period, which allows a licensee that has discovered non-associated gas but cannot yet develop it commercially to retain the discovery area for a defined period while it seeks a market, without being required to relinquish it. Without such provisions, licensees may be forced to relinquish gas discoveries before markets have developed.

Under Nigeria's Petroleum Industry Act 2021, the general framework for petroleum prospecting licences and petroleum mining leases applies to both oil and gas, but the Act contains specific provisions recognising the distinct characteristics of gas. Holders of petroleum prospecting licences that discover non-associated gas may be granted a period to establish its commerciality before conversion to a lease, and the Act's fiscal provisions treat gas production more favourably than oil, notably by subjecting gas to companies income tax rather than the hydrocarbon tax applicable to crude oil, and by providing reduced royalty rates for gas, with lower rates for gas delivered to the domestic market. These provisions reflect the policy of the Act to promote the development of gas for domestic use.

Domestic Gas Supply Obligations

Many gas-producing states impose domestic supply obligations requiring producers to make a proportion of their gas available for the domestic market, often at regulated prices. Such obligations reflect the policy of ensuring that national resources contribute to domestic energy supply and industrial development, rather than being exported exclusively. They may, however, deter investment if the regulated price is below the cost of supply or the export price.

In Nigeria, the Nigerian Gas Master Plan of 2008 and the National Domestic Gas Supply and Pricing Regulations 2008 introduced a domestic gas delivery obligation requiring producers to supply specified volumes of gas to the domestic market. The PIA continues this approach, empowering the Nigerian Midstream and Downstream Petroleum Regulatory Authority (NMDPRA) to determine the domestic gas delivery obligation and to set a domestic base price for gas supplied to the strategic sectors (notably power generation) until a willing-buyer, willing-seller market develops. In Western Australia, the Domestic Gas Reservation Policy, introduced in 2006, requires LNG export projects to make gas equivalent to fifteen per cent of their LNG production available to the domestic market, and the federal Australian Domestic Gas Security Mechanism, introduced in 2017, empowers the government to restrict LNG exports where there is a domestic shortfall. In 2022 and 2023, Australia introduced emergency price caps on wholesale gas and a mandatory code of conduct for the east coast gas market in response to the price surge triggered by the European energy crisis. Similar obligations exist in Indonesia, Egypt, Israel, and other producing countries.

Domestic supply obligations raise legal questions in international trade law, since they may function as export restrictions, and in investment law, since their introduction or tightening after an investment is made may affect the investor's legitimate expectations. They are discussed further in Chapter 13 in the context of export restrictions.

Gas Flaring Regulation

Gas flaring — the burning of associated gas at the wellsite — wastes a valuable resource, emits carbon dioxide and black carbon, and exposes nearby communities to air pollution, noise, heat, and light. The World Bank's Global Gas Flaring Tracker reports that over 140 billion cubic metres of gas are flared globally each year, with Russia, Iraq, Iran, the United States, Venezuela, Algeria, Libya, and Nigeria among the largest flaring countries. Venting — the direct release of unburned gas, predominantly methane — is even more damaging to the climate, since methane is a far more potent greenhouse gas than carbon dioxide over short time horizons.

Legal approaches to flaring include outright prohibitions with exceptions for safety and emergencies, flare fees or penalties, requirements for gas utilisation plans as a condition of development approval, and mechanisms to allocate flare gas to third parties that can commercialise it. Nigeria's experience with flaring regulation, examined in Chapter 5, illustrates both the difficulties of enforcement and the evolution of policy from prohibition to commercialisation through the Flare Gas (Prevention of Waste and Pollution) Regulations 2018 and the Nigerian Gas Flare Commercialisation Programme. The *Gbemre* case, also discussed in Chapter 5 and Chapter 21, established the human rights dimension of flaring in Nigerian law.

Internationally, the World Bank's Zero Routine Flaring by 2030 Initiative, launched in 2015, has been endorsed by governments, oil companies, and development institutions, which commit not to engage in routine flaring in new oil field developments and to end legacy routine flaring as soon as possible and no later than 2030. The Global Methane Pledge, launched at COP26 in 2021, commits participating countries to collective action to reduce global anthropogenic methane emissions by at least thirty per cent from 2020 levels by 2030. The Oil and Gas Methane Partnership 2.0 (OGMP 2.0), administered by the United Nations Environment Programme, has established a reporting framework for company methane emissions that has been incorporated by reference into the EU Methane Regulation. These instruments are largely voluntary, but they are increasingly being translated into binding domestic and regional law.

Gas Processing and Natural Gas Liquids

Gas processing — the separation of natural gas liquids, removal of impurities, and conditioning of gas to pipeline specification — occupies a regulatory borderland between upstream and midstream activity. In some regimes, processing facilities associated with a specific field are treated as part of upstream operations under the petroleum licence; in others, they require separate licences. The PIA treats gas processing as a midstream activity requiring a licence from the NMDPRA, distinct from the upstream licence regulated by the Nigerian Upstream Petroleum Regulatory Commission (NUPRC). The classification affects the applicable fiscal regime, the identity of the regulator, and the rights of third parties to access the facilities. Access to upstream infrastructure, including processing facilities, by third parties is examined in section 6.3.

Fiscal Treatment of Gas

Because gas development is typically less profitable than oil development, owing to higher transport costs, lower prices per unit of energy, and the need for market development, many petroleum fiscal regimes provide more favourable terms for gas. Common incentives include lower royalty rates, exemption from or reduced rates of resource rent taxes, accelerated depreciation for gas infrastructure, and tax holidays for gas utilisation projects. Nigeria's Associated Gas Framework Agreement of 1992 and the incentives for gas utilisation in the Companies Income Tax Act, now carried forward in modified form in the PIA, are examples. The design of gas fiscal incentives must balance the need to make gas development commercially viable against the risk of conferring windfall gains on projects that would have proceeded without them, and against the need to avoid distortions between oil and gas production that might encourage artificial classification of production. Fiscal regimes generally are examined in Chapter 16.

6.3 Gas Transportation and Pipeline Access

The Gas Supply Chain and Its Regulation

The physical supply chain for pipeline gas comprises upstream gathering pipelines, which collect gas from wells and convey it to processing facilities; high-pressure transmission pipelines, which carry processed gas over long distances; storage facilities, including depleted reservoirs, salt caverns, and aquifers, which balance seasonal and short-term fluctuations in demand; and low-pressure distribution networks, which deliver gas to end-users. LNG terminals, considered in section 6.4, perform an analogous function at the interface between maritime and pipeline transport. Each element raises distinct regulatory issues, but the central questions are the same: who may own and operate the infrastructure; on what terms must the owner grant access to third parties; how are tariffs determined; and how is new investment encouraged.

Models of Industry Structure

Historically, gas industries in most countries were organised as vertically integrated monopolies, whether state-owned (as in France, Italy, and much of the developing world) or privately owned but regulated (as in the United States). A single company, or a small number of regional companies, purchased gas from producers, transported it, and sold it to distributors or end-users. Liberalisation sought to introduce competition in the potentially competitive segments of the chain by separating them from the monopoly networks and requiring network owners to provide access to competing suppliers on non-discriminatory terms.

The United States led this process. Under the Natural Gas Act of 1938, interstate pipelines were regulated by the Federal Power Commission (now the Federal Energy Regulatory Commission, FERC) as merchants that bought and resold gas. Following the partial decontrol of wellhead prices under the Natural Gas Policy Act of 1978, FERC Order No 436 (1985) encouraged pipelines to offer open-access transportation services, and Order No 636 (1992) required interstate pipelines to unbundle their sales and transportation services, transforming them into pure transporters and creating a competitive wholesale gas market. The United Kingdom followed with the privatisation of British Gas in 1986, the Gas Act 1995, and the separation of transportation from supply, producing one of the most liquid gas markets in the world. The European Union adopted a sequence of liberalisation directives, culminating in the Third Energy Package of 2009, comprising Directive 2009/73/EC concerning common rules for the internal market in natural gas and Regulation (EC) No 715/2009 on conditions for access to the natural gas transmission networks. In 2024, the EU adopted the Hydrogen and Decarbonised Gas Market Package — Directive (EU) 2024/1788 and Regulation (EU) 2024/1789 — which recasts the gas market rules and extends them to the emerging hydrogen sector.

Third-Party Access

Third-party access (TPA) is the right of a person other than the owner of a pipeline or other facility to use its capacity to transport gas. TPA may be negotiated or regulated.

Under **negotiated TPA**, the owner and the access seeker negotiate the terms of access commercially, subject to a general obligation to negotiate in good faith and not to discriminate, and to the oversight of a regulator or competition authority in the event of a dispute. Negotiated TPA is common for upstream pipelines and processing facilities, where the circumstances of each access request are highly specific. In the United Kingdom, for example, the Energy Act 2011 and the Infrastructure Code of Practice govern access to upstream oil and gas infra-

structure on the UK Continental Shelf, and the North Sea Transition Authority may determine the terms of access in default of agreement.

Under **regulated TPA**, the terms of access, including tariffs, are published in advance and approved or set by an independent regulator. Regulated TPA is the norm for transmission and distribution networks in the European Union, where Directive 2009/73/EC (now the recast Directive) requires member states to ensure a system of third-party access to transmission and distribution systems and LNG facilities based on published tariffs, applicable to all eligible customers and applied objectively and without discrimination. Access to storage may be either negotiated or regulated, at the option of the member state.

Access may be refused on limited grounds. The EU directive permits a natural gas undertaking to refuse access on the basis of lack of capacity, or where access would prevent it from carrying out public service obligations, or on the basis of serious economic and financial difficulties with take-or-pay contracts. Refusals must be duly substantiated.

Unbundling

Third-party access is unlikely to be effective if the network owner is also a supplier of gas in competition with the access seekers, because it will have both the incentive and the means to discriminate against them — for example, by underinvesting in capacity, manipulating information, or delaying connections. Liberalisation has therefore been accompanied by requirements for the *unbundling* of network activities from supply and production activities. Unbundling may take several forms, of increasing stringency:

1. **Accounting unbundling**, requiring separate accounts for network and supply activities, to prevent cross-subsidisation.
2. **Functional or legal unbundling**, requiring network activities to be carried out by a separate legal entity with independent management, although it may remain within a vertically integrated group.
3. **Independent system operator (ISO)**, under which the integrated group retains ownership of the network but its operation is entrusted to an independent operator.
4. **Independent transmission operator (ITO)**, under which the network remains within the integrated group but is subject to detailed rules ensuring its independence in management and decision-making.
5. **Ownership unbundling**, requiring the separation of ownership of transmission networks from production and supply interests.

The Third Energy Package required member states to choose among ownership unbundling, the ISO model, and the ITO model for transmission system operators, and the choice was the subject of intense political controversy, with France

and Germany defending the ITO model to protect their integrated incumbents. The "third country clause" in Article 11 of the Directive required certification of transmission system operators controlled by persons from third countries, permitting refusal of certification where the grant would put at risk the security of energy supply of the member state or the Union — a provision directed in large part at Gazprom.

Tariff Regulation

The regulation of transportation tariffs seeks to allow network owners to recover their efficiently incurred costs, including a reasonable return on capital, while preventing monopoly pricing. The principal methodologies are *cost-of-service* or *rate-of-return* regulation, which sets tariffs to recover a defined revenue requirement based on operating costs, depreciation, and a return on the regulated asset base; and *incentive regulation*, such as price caps or revenue caps of the "RPI–X" type pioneered in the United Kingdom, under which tariffs are set for multi-year periods and the network owner retains the benefits of efficiency improvements. Tariff structures may be *distance-related* (point-to-point), *postage stamp* (uniform irrespective of distance), or *entry–exit*, under which users book capacity separately at entry and exit points and gas can flow freely within the system between them. The EU Network Code on Harmonised Transmission Tariff Structures for Gas (Regulation (EU) 2017/460) requires entry–exit systems and sets out detailed rules for reference price methodologies.

Pipeline Regulation in Nigeria

Nigeria's gas transportation infrastructure has historically been owned and operated by the Nigerian Gas Company, a subsidiary of NNPC, now succeeded within NNPC Ltd's corporate structure. The PIA establishes a framework for midstream gas regulation by the NMDPRA, including licensing of gas transportation pipelines, gas distribution systems, gas processing facilities, and gas storage; the establishment of a Gas Transportation Network Code governing access to the transportation network; open access to pipelines and facilities on a non-discriminatory basis, subject to available capacity; and the regulation of tariffs. The Act requires the separation of gas transportation network operation from gas wholesale and supply activities, reflecting the European unbundling model. Major infrastructure projects, including the Ajaokuta–Kaduna–Kano (AKK) pipeline and the Obiafu–Obrikom–Oben (OB3) pipeline, are intended to connect gas resources in the south with demand centres in the north, but have faced delays associated with financing, security, and construction challenges.

6.4 Liquefied Natural Gas Project Structures

The LNG Value Chain

The LNG value chain comprises upstream production of feed gas; liquefaction, in which the gas is treated and cooled to approximately minus 162 degrees Celsius, reducing its volume by a factor of around six hundred; shipping in specialised LNG carriers; and regasification at import terminals, either onshore or on floating storage and regasification units (FSRUs). Each stage requires very large capital investments: a large liquefaction plant may cost tens of billions of United States dollars, a modern LNG carrier more than two hundred million dollars, and a regasification terminal several hundred million dollars or more. The capital intensity of the chain, together with its specificity and long life, means that LNG projects have historically required long-term contracts covering most of their output before financing can be secured.

Physical chain (top) and principal legal instruments (bottom)

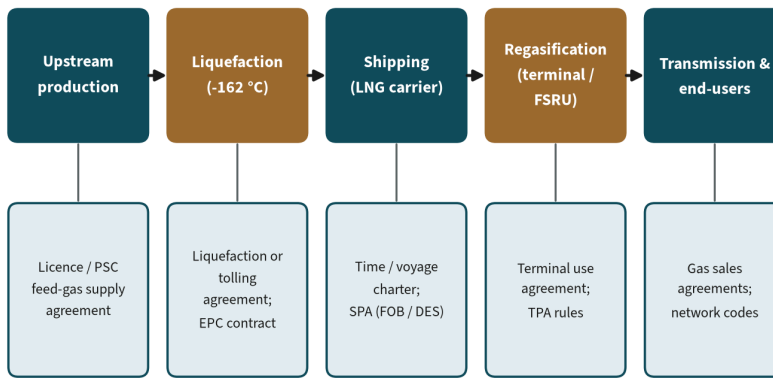


Figure 6.1 The LNG value chain and the principal contracts that link its stages. *Source:* prepared for this edition.

Principal Project Structures

LNG export projects have been organised according to several structures, distinguished by the relationship between ownership of the upstream gas, the liquefaction facilities, and the LNG product.

The integrated structure. Under the integrated structure, the same parties (in the same proportions) own the upstream gas reserves and the liquefaction plant, and sell LNG directly to buyers. The upstream and liquefaction activities are treated as a single project for fiscal and commercial purposes. The structure aligns the interests of upstream and midstream investors and avoids the need for

a transfer price between them, but it requires all investors to participate in both segments. Many projects in Qatar, Australia, and elsewhere have used integrated structures.

The merchant structure. Under the merchant structure, the liquefaction company is owned separately from the upstream producers (or by them in different proportions). It purchases feed gas from the producers under gas supply agreements, liquefies it, and sells LNG to buyers on its own account, taking the margin between its purchase and sales prices. Nigeria LNG Limited (NLNG), owned by NNPC (49 per cent), Shell (25.6 per cent), TotalEnergies (15 per cent), and Eni (10.4 per cent), operates on a broadly merchant basis, purchasing feed gas from upstream joint ventures and selling LNG under long-term contracts to buyers in Europe and elsewhere. The merchant structure requires a negotiated feed gas price, which allocates value between upstream and midstream, and it may raise fiscal issues where the liquefaction company is subject to a different tax regime from the upstream producers.

The tolling structure. Under the tolling structure, the liquefaction company does not take title to the gas. It provides liquefaction services to the gas owners (or to LNG offtakers) in return for a tolling fee, which is typically fixed and payable regardless of whether capacity is used, providing a utility-like return to the plant owners. The gas owners retain title through the liquefaction process and sell the LNG themselves. The tolling model was used for later trains of Atlantic LNG in Trinidad and Tobago and for several of the export projects developed in the United States from the mid-2010s, such as Freeport LNG, whose capacity is contracted to offtakers under liquefaction tolling agreements. Other United States projects, such as Cheniere's Sabine Pass, sell LNG under sale and purchase agreements priced at a percentage of Henry Hub plus a fixed fee, which replicate the economics of tolling. Under the United States model, buyers enter into long-term "liquefaction services" or sale and purchase agreements under which they pay a fixed liquefaction fee for capacity and purchase feed gas at Henry Hub prices plus a percentage, giving them flexibility to lift cargoes or cancel them (paying the fixed fee) and to resell LNG anywhere in the world.

Hybrid structures. Many projects combine elements of these structures, and the choice depends on the fiscal regime, the identity and objectives of the investors, the availability of financing, and the requirements of the host state.

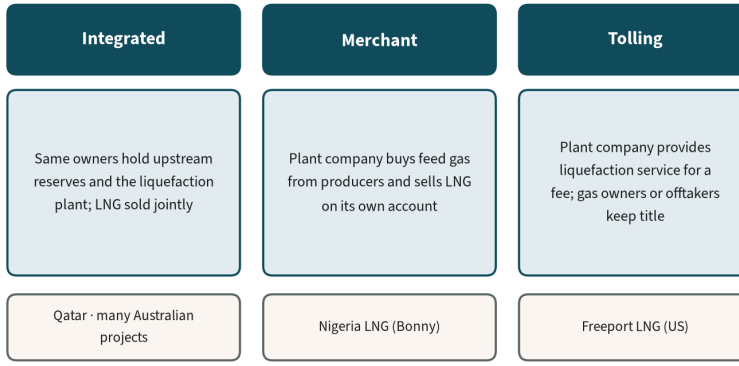


Figure 6.2 Integrated, merchant, and tolling structures for LNG export projects. *Source:* prepared for this edition.

The Host State Framework

LNG projects typically require a dedicated legal framework, often in the form of a project-specific statute or host government agreement, providing fiscal terms, guarantees against expropriation, stability of the legal regime, rights to export LNG, and rights to retain export proceeds offshore. The Nigeria LNG (Fiscal Incentives, Guarantees and Assurances) Act 1990 is an example: it conferred on NLNG a package of fiscal incentives, guarantees, and assurances, including exemption from certain taxes for an initial period, a guarantee against nationalisation, the right to maintain offshore accounts, and a commitment that the Act could not be amended without the consent of the company's shareholders. The NLNG Act illustrates both the attractions and the controversies of project-specific legislation. When the Nigerian Maritime Administration and Safety Agency (NIMASA) sought to impose levies on NLNG's shipping activities, and when the Federal Government sought to subject NLNG to contributions to the Niger Delta Development Commission, the company invoked the protections of the Act, generating litigation concerning the extent to which a statute can bind future legislatures. The question whether a legislature may validly restrict its own future legislative competence by statute raises fundamental constitutional issues; in common law systems influenced by the doctrine of parliamentary sovereignty, such entrenchment is generally ineffective as a matter of domestic law, although it may still give rise to liability under contract or investment treaties.

Shipping

LNG shipping arrangements depend on the delivery terms of the sales contract. Under a *free on board* (FOB) contract, the buyer takes delivery at the loading port and is responsible for shipping; under a *delivered ex-ship* (DES) or *cost, insurance and freight* (CIF) contract, the seller is responsible for shipping and delivers the LNG at the discharge port. LNG carriers are chartered under long-term time charters, voyage charters, or, increasingly, short-term charters for spot cargoes. The legal issues include the compatibility of vessels with terminals (ship–shore compatibility), the allocation of boil-off gas (LNG that evaporates during the voyage), demurrage, and the application of maritime conventions.

Regasification and FSRUs

The legal structure of regasification terminals varies. Some terminals are owned by the buyers that use them; others are owned by independent operators that offer capacity to multiple users under terminal use agreements, subject to regulated or negotiated third-party access. In the European Union, LNG terminals are subject to regulated third-party access, but major new infrastructure may obtain an exemption under Article 36 of Directive 2009/73/EC (now carried forward in the recast) where the investment enhances competition and security of supply and would not take place without the exemption. Many European terminals, including the Adriatic LNG terminal in Italy and the Gate terminal in Rotterdam, have obtained exemptions.

Floating storage and regasification units have dramatically reduced the time and cost required to establish LNG import capacity, enabling countries such as Egypt, Pakistan, Bangladesh, and, following the 2022 crisis, Germany to begin importing LNG within months rather than years. Germany's LNG Acceleration Act (LNG-Beschleunigungsgesetz) of May 2022 expedited permitting for FSRUs and onshore terminals by, among other things, temporarily exempting projects from certain environmental impact assessment requirements, illustrating the use of emergency legislation to accelerate energy infrastructure.

LNG Export Authorisation

Some states require specific authorisation for the export of LNG. In the United States, section 3 of the Natural Gas Act requires authorisation by the Department of Energy for exports of natural gas. Exports to countries with which the United States has a free trade agreement requiring national treatment for trade in natural gas are deemed to be consistent with the public interest and must be granted without modification or delay; exports to non-FTA countries must be authorised unless the Department finds that they would not be consistent with the public interest. In January 2024, the Department announced a temporary pause on pending decisions on non-FTA exports while it updated the economic and envi-

ronmental analyses underlying its public interest determinations; the pause was lifted in January 2025 by the incoming administration. The episode illustrates the regulatory risk facing LNG projects, whose commercial viability depends on export authorisation, and the growing incorporation of climate considerations into export licensing.

6.5 Long-Term Gas Sales and Purchase Agreements

Function and Forms

The gas sales and purchase agreement (GSPA, or SPA in the LNG context) is the central commercial instrument of the gas industry. It allocates between seller and buyer the volume risk (the risk that the buyer will not need, or the seller will not be able to supply, the contracted quantities), the price risk (the risk that market prices will move against one party), and the operational risks of delivery. Long-term GSPAs of fifteen to twenty-five years' duration underpin the financing of most major pipeline and LNG export projects. Medium-term contracts (two to ten years), short-term contracts (up to two years), and spot transactions have grown in importance with the development of liquid hubs and the LNG spot market, and are commonly concluded under master agreements that set out general terms, with individual transactions confirmed by short confirmation notices.

Several standard forms are used in the industry. For trading on European hubs, the European Federation of Energy Traders (EFET) General Agreement Concerning the Delivery and Acceptance of Natural Gas is widely used; in North America, the North American Energy Standards Board (NAESB) Base Contract for Sale and Purchase of Natural Gas performs a similar function. For LNG, the Association of International Energy Negotiators (AIEN) has published a Model Master LNG Sale and Purchase Agreement, and the International Group of Liquefied Natural Gas Importers (GIIGNL) has published a Master LNG Sale and Purchase Agreement for spot and short-term trades. Long-term SPAs, by contrast, are typically bespoke, although they follow well-established structures. Appendix D contains a model international gas sales agreement with annotations.

Contract Quantity Provisions

The quantity provisions of a long-term GSPA typically define:

- the **Annual Contract Quantity (ACQ)**, the quantity of gas the seller undertakes to make available and the buyer undertakes to take or pay for in each contract year;
- the **Daily Contract Quantity (DCQ)**, usually the ACQ divided by the number of days in the year, and the **Maximum Daily Quantity**, often 110 or 120 per cent of the DCQ, which defines the buyer's daily flexibility;

- **downward quantity tolerance** or **downward flexibility**, permitting the buyer to reduce its annual nominations by a specified percentage without incurring take-or-pay liability;
- adjustments to the ACQ for gas not taken because of the seller's failure to deliver, force majeure, or the buyer's rejection of off-specification gas; and
- the **build-up period** at the start of the contract and the **plateau period** during which the ACQ remains constant.

LNG SPAs specify quantities in terms of the Annual Contract Quantity expressed in tonnes or million British thermal units, and contain an **Annual Delivery Programme** procedure under which the parties agree before each contract year on the schedule of cargoes, subject to adjustments for operational reasons. The seller may have rights to reduce quantities for scheduled maintenance of the liquefaction facilities, and the buyer may have rights to divert or cancel cargoes, depending on the terms of the contract.

Delivery Point and Transfer of Title and Risk

The GSPA defines the delivery point at which title and risk pass from seller to buyer. In pipeline contracts, the delivery point may be at the outlet of the seller's processing facilities, at a border crossing, at a trading hub, or at the buyer's facilities. In LNG contracts, the delivery point depends on whether the contract is FOB (title and risk pass at the loading port, usually as the LNG passes the flange connecting the loading arms to the vessel's manifold) or DES (title and risk pass at the discharge port). Under a CIF contract, risk passes at the loading port but the seller arranges and pays for shipping and insurance. The choice affects which party controls the vessel and, consequently, the ability to divert cargoes.

Quality and Measurement

GSPAs specify the quality of gas to be delivered and the consequences of off-specification deliveries. The buyer may be entitled to reject off-specification gas, or to accept it subject to a price reduction or compensation for costs incurred in treating it. The allocation of risk for off-specification gas is particularly important where the gas is to be used in sensitive equipment, such as gas turbines, or injected into networks with strict specifications. Measurement provisions define the methods, standards, and equipment to be used for measuring quantity and quality, the procedures for calibration and testing, and the resolution of measurement disputes, often by reference to an independent surveyor in the LNG context.

Destination Clauses and Diversion Rights

Traditional long-term gas contracts frequently contained **destination clauses** (or territorial restriction clauses) prohibiting the buyer from reselling gas outside a defined territory, and **profit-sharing mechanisms** requiring the buyer to share with the seller any profit from resale to a different destination. Sellers defended

these provisions as necessary to prevent arbitrage between markets in which they sold at different prices, and to protect their pricing strategies. Buyers and competition authorities criticised them as restrictions on competition and on the development of integrated markets.

In the European Union, destination clauses in contracts with EU buyers have been treated as restrictions of competition contrary to Article 101 of the Treaty on the Functioning of the European Union (TFEU). From the early 2000s, the European Commission pursued investigations concerning the contracts of suppliers including Gazprom, Sonatrach, and Nigeria LNG, resulting in the removal of destination clauses from their contracts with European buyers. In its 2018 decision accepting commitments from Gazprom in Case AT.39816 (*Upstream Gas Supplies in Central and Eastern Europe*), the Commission addressed concerns that Gazprom had pursued an overall strategy of partitioning Central and Eastern European gas markets, including through territorial restrictions, and Gazprom committed to remove contractual barriers to the cross-border resale of gas and to facilitate gas flows to isolated markets. In 2018 the Commission also opened an investigation into whether Qatar Petroleum's long-term LNG supply agreements with European importers contained territorial restrictions.

In Asia, the Japan Fair Trade Commission published a report in 2017 concluding that destination restrictions in FOB LNG contracts were likely to violate Japan's Antimonopoly Act, and that profit-sharing mechanisms and certain restrictions in DES contracts were also problematic. The report has encouraged Japanese buyers to negotiate greater flexibility in new contracts, contributing, together with the growth of United States LNG supplied on destination-free FOB terms, to the increasing liquidity and flexibility of the global LNG market.

Force Majeure and Seller's Failure to Deliver

GSPAs contain detailed force majeure provisions excusing performance where a party is prevented from performing by events beyond its reasonable control. In the gas context, force majeure clauses often enumerate specific events, such as failures of facilities, governmental action, and natural disasters, and may treat certain events — such as failures of the seller's upstream supply or of the buyer's downstream market — as either within or outside the clause depending on the parties' negotiated allocation of risk. Force majeure in energy contracts is examined in Chapter 15; here it is sufficient to note that the treatment of upstream and transportation failures has been a recurring source of disputes in gas contracts, and that the 2022 crisis generated significant litigation concerning Gazprom's declaration of force majeure in relation to deliveries to European buyers.

GSPAs also provide remedies for the seller's failure to deliver contracted quantities otherwise than as a result of force majeure. These commonly include

a price discount on shortfall quantities, sometimes expressed as a percentage of the contract price, and may be the buyer's exclusive remedy for shortfalls up to a specified level. The adequacy of such remedies was tested during the 2022 crisis, when replacement gas cost many times the contract price; buyers argued that the contractual shortfall remedies did not apply to deliberate non-performance, and sought full damages in arbitration.

6.6 Price Review, Take-or-Pay, and Contractual Flexibility

Take-or-Pay

The **take-or-pay** clause is the defining feature of the long-term gas contract. It obliges the buyer to pay for a minimum quantity of gas in each contract year (the "take-or-pay quantity", often 80 to 90 per cent of the ACQ, after allowing for downward flexibility) whether or not it actually takes delivery of that quantity. If the buyer takes less than the minimum, it must pay for the shortfall, but it typically acquires a right to take the paid-for but untaken gas ("make-up gas") in subsequent years, usually after it has taken the ACQ for that year, and sometimes at the price applicable at the time of payment. Conversely, if the buyer takes more than the ACQ in a year, it may accumulate "carry-forward" credits that reduce its minimum take obligation in future years.

The economic function of take-or-pay is to allocate volume risk to the buyer, providing the seller with predictable revenues sufficient to service the debt raised to finance the production and transportation facilities. In return, the buyer obtains security of supply over the long term. Lenders to gas projects typically require a substantial proportion of the project's output to be committed under take-or-pay contracts with creditworthy buyers before they will provide financing.

The enforceability of take-or-pay clauses has been challenged on the ground that they are penalties. Under English law, the rule against penalties applies only to clauses that operate upon a breach of contract; a clause that provides for payment as a primary obligation in the event of a contractual contingency, rather than a breach, is not subject to the rule. In *M&J Polymers Ltd v Imerys Minerals Ltd*, the High Court considered a take-or-pay clause in a supply contract and held that, although the rule against penalties was potentially applicable on the particular construction of that clause, the clause was commercially justifiable and not a penalty.[1] Following the restatement of the penalty rule in *Cavendish Square Holding BV v Makdessi* (discussed in Chapter 5), the predominant view is that a properly drafted take-or-pay clause, which creates a primary obligation to pay for gas made available, does not engage the penalty rule at all, and that even if it did, it would usually be justified by the seller's legitimate interest in recovering its

[1] *M&J Polymers Ltd v Imerys Minerals Ltd* [2008] EWHC 344 (Comm), [2008] 1 Lloyd's Rep 541.

investment. In the United States, courts have generally upheld take-or-pay clauses, treating them as alternative performance obligations, although extensive litigation followed the collapse of gas prices in the 1980s, when pipeline companies that had contracted to take gas at high prices sought to escape their take-or-pay obligations.

Price Indexation

The price provisions of a long-term GSPA typically combine a base price with an indexation formula that adjusts the price periodically to reflect changes in the value of gas. In oil-indexed contracts, the formula links the gas price to the prices of crude oil, gas oil, or fuel oil, often averaged over a preceding period (for example, six or nine months) and applied with a time lag, smoothing the effect of short-term volatility. In hub-indexed contracts, the price is linked to a hub price, such as TTF or NBP, perhaps with a fixed discount or premium. Hybrid formulas combine oil and hub indexation.

Price Review

Because long-term contracts run for decades, and because the indexation formula may cease to reflect the market value of gas as markets evolve, most long-term GSPAs contain a **price review** (or price reopener) clause permitting either party to request a revision of the price formula at specified intervals (commonly every three or five years) and, in some cases, at any time upon the occurrence of exceptional changes in market circumstances ("wildcard" reviews). Price review clauses typically provide that if, since the last price review, there has been a significant change in the economic circumstances in the buyer's market beyond the control of the parties, and the contract price no longer reflects the value of gas in that market, the parties shall negotiate in good faith a revision of the price formula. If they fail to agree within a specified period, either party may refer the matter to arbitration or to an expert.

Price review clauses raise distinctive legal issues. The *trigger* question is whether the conditions for a revision are satisfied — whether there has been a sufficiently significant change in circumstances of the specified kind. The *revision* question is what the revised price should be — what price reflects the value of gas in the buyer's market, often measured by reference to the prices paid by comparable buyers under comparable contracts or by reference to the "netback" value of gas from the price of competing fuels. The *scope* question is what the tribunal is empowered to change — whether it may alter only the base price, or also the indexation, the weighting of indices, or the introduction of hub indexation.

The wave of price review arbitrations that followed the decoupling of European hub prices from oil-indexed contract prices after 2009 — when the development of hub trading and the arrival of LNG intended for the United States market led to hub prices falling well below oil-indexed prices — trans-

formed the European gas market. European buyers, committed to purchasing large quantities at oil-indexed prices while their customers could buy cheaper gas at the hubs, invoked price review clauses against their suppliers, including Gazprom, Statoil (now Equinor), and Sonatrach. Many disputes were settled through renegotiation, often introducing a significant element of hub indexation; others were decided by arbitral tribunals, which in a number of reported cases accepted that the emergence of hub prices was a relevant change in circumstances and introduced hub indexation into the contract formula. Because most price review awards are confidential, the jurisprudence is known primarily through practitioners' commentaries and the limited number of awards that have entered the public domain through court proceedings.

One such case is *Atlantic LNG Co of Trinidad and Tobago v Gas Natural Aprovisionamientos SDG SA*, in which a price review award in favour of Gas Natural concerning LNG supplied from Trinidad was challenged before the United States District Court for the Southern District of New York. The court confirmed the award in 2008, rejecting the argument that the tribunal had exceeded its powers in determining the revised price.[2] The case illustrates the deference shown by courts in leading arbitral seats to tribunals' interpretation of price review clauses.

Hardship and Price Review Compared

Price review clauses perform a function similar to hardship clauses in general commercial contracts and to doctrines such as *imprévision* in French administrative law, *Wegfall der Geschäftsgrundlage* (now codified in § 313 of the German Civil Code), and Article 6.2.2 of the UNIDROIT Principles of International Commercial Contracts. They differ, however, in that they are designed to operate routinely and periodically, to keep the price aligned with the market, rather than to address exceptional events that fundamentally alter the equilibrium of the contract. A price review tribunal is not concerned with whether performance has become excessively onerous for one party, but with whether the contract price reflects the market value of gas. Chapter 15 examines hardship clauses in energy contracts generally.

Contractual Flexibility in a Changing Market

The trend in gas contracting over the past two decades has been towards greater flexibility: shorter contract durations, smaller volumes, greater downward quantity tolerance, removal of destination restrictions, hub indexation, and rights to divert and cancel cargoes. In the LNG market, portfolio players — international oil companies and trading houses that aggregate supply from multiple projects and sell to multiple buyers — have intermediated between sellers seeking long-term commitments and buyers seeking flexibility.

[2] *Atlantic LNG Co of Trinidad and Tobago v Gas Natural Aprovisionamientos SDG SA*, (SDNY 2008).

The 2021–2023 crisis complicated this trend. The extreme volatility of hub prices and the scarcity of uncommitted LNG led some buyers, particularly in Asia and in Europe, to sign long-term contracts again, including twenty-year contracts with Qatari and United States suppliers, while European buyers sought to balance supply security against the risk that gas demand would fall over the life of the contract as a result of decarbonisation policies. The resulting contracts frequently include provisions addressing the energy transition, such as rights to reduce volumes in response to changes in climate policy, commitments to reduce methane emissions across the supply chain, and provisions for the future supply of low-carbon gases. The capacity of contract law to accommodate the energy transition is examined further in Chapter 15.

6.7 Cross-Border Gas Infrastructure and Transit

The Distinctive Legal Problem of Cross-Border Pipelines

A cross-border gas pipeline links a producing state, a consuming state, and frequently one or more transit states through a single, fixed, and very expensive physical asset. Once built, the pipeline cannot be relocated, and each state through which it passes acquires a degree of leverage over the others: the producing state can withhold supply, the consuming state can refuse to purchase or delay payment, and the transit state can interrupt flows, demand higher tariffs, or divert gas for its own use. Unlike LNG, which can be redirected to alternative markets, pipeline gas offers no easy exit. The legal framework for cross-border pipelines must therefore address not only the ordinary questions of project finance and regulation but also the political risks that arise from the interdependence of sovereign states.

The legal architecture of cross-border pipelines typically operates at three levels: an **intergovernmental agreement** (IGA) among the states concerned, establishing the principles governing the project, the rights and obligations of the states, and the framework for dispute resolution; **host government agreements** (HGAs) between each state and the project company, setting out the detailed legal and fiscal regime applicable in each territory; and the commercial contracts among the project participants, shippers, and buyers. This structure was developed and refined in the Baku–Tbilisi–Ceyhan (BTC) oil pipeline project, completed in 2006, and has been applied to gas pipelines including the South Caucasus Pipeline, the Trans-Anatolian Natural Gas Pipeline (TANAP), and the Trans Adriatic Pipeline (TAP), which together form the Southern Gas Corridor bringing gas from Azerbaijan to Europe.

International Law of Transit

General international law does not confer a comprehensive right of transit for energy across the territory of another state. Several treaty regimes, however, address transit.

Article V of the General Agreement on Tariffs and Trade 1994 provides that there shall be "freedom of transit through the territory of each contracting party, via the routes most convenient for international transit, for traffic in transit to or from the territory of other contracting parties", and that such traffic shall not be subject to unnecessary delays or restrictions and shall be exempt from customs duties and transit duties, other than charges commensurate with the administrative expenses entailed by transit or with the cost of services rendered. Whether Article V applies to fixed infrastructure such as pipelines, as distinct from goods moving by conventional means of transport, has been debated. The WTO panel in *Russia — Measures Concerning Traffic in Transit* (2019), which concerned Russian restrictions on road and rail transit from Ukraine, is significant chiefly for its interpretation of the national security exception in Article XXI, but it confirmed the justiciability of Article XXI defences, a matter of considerable importance for energy trade (discussed in Chapters 13 and 25).

Article 7 of the Energy Charter Treaty provides that each contracting party shall take the necessary measures to facilitate the transit of energy materials and products consistent with the principle of freedom of transit and without distinction as to the origin, destination, or ownership of such energy materials and products or discrimination as to pricing on the basis of such distinctions, and without imposing any unreasonable delays, restrictions, or charges. Article 7(6) provides that a contracting party through whose area energy materials and products transit shall not, in the event of a dispute over any matter arising from that transit, interrupt or reduce the existing flow prior to the conclusion of the conciliation procedure set out in Article 7(7), except where specifically provided in a contract or other agreement or permitted in accordance with the conclusion of the conciliator. Article 7 does not, however, oblige a contracting party to permit the construction or modification of energy transport facilities or to allow new or additional transit through existing facilities that it demonstrates would endanger the security or efficiency of its energy systems.

Negotiations on a more detailed **Energy Charter Transit Protocol** began in 2000 but were suspended after Russia, which had signed but never ratified the ECT and which terminated its provisional application in 2009, objected to key provisions, including the treatment of existing long-term transit contracts and the "right of first refusal" for holders of expiring transit capacity. The failure of the Transit Protocol, and Russia's withdrawal from provisional application of the ECT, left European gas transit governed largely by bilateral arrangements

and by EU law. The modernisation of the ECT adopted in December 2024, and the withdrawal of the EU and many of its member states from the Treaty, are discussed in Chapters 4 and 14.

Article 125 of the United Nations Convention on the Law of the Sea confers on land-locked states a right of access to and from the sea and freedom of transit through the territory of transit states by all means of transport, the terms and modalities of which are to be agreed through bilateral, subregional, or regional agreements. Submarine pipelines on the continental shelf and in the exclusive economic zone are governed by Articles 58 and 79 of UNCLOS, examined in Chapter 20.

Transit Tariffs and Transit Disputes

Transit states are compensated for the use of their territory through transit fees or tariffs, which may be set by reference to the costs of providing the service, the value of the gas transported, or the bargaining power of the parties. Transit tariffs have been a recurrent source of disputes. The gas disputes between Russia and Ukraine in January 2006 and January 2009 — the latter resulting in a complete cessation of gas flows through Ukraine to south-eastern Europe for almost two weeks in mid-winter — arose from linked disputes over the price of gas supplied to Ukraine, the tariff for transit of Russian gas to Europe, and Ukraine's accumulated debts. They exposed the vulnerability of European consumers to disputes between producer and transit states and prompted the European Union to adopt measures to strengthen security of supply, including the requirement of reverse-flow capacity on interconnectors, and to support the diversification of supply routes. The 2009 crisis was resolved by a set of ten-year contracts between Gazprom and Naftogaz, the disputes under which are examined in the casebox at the end of this chapter. Gas transit through Ukraine to the EU ended on 1 January 2025, when the transit agreement concluded in 2019 expired and was not renewed.

Regional Gas Pipeline Frameworks in Africa

Africa offers several examples of regional and transnational gas pipeline frameworks. The **West African Gas Pipeline** (WAGP), which transports Nigerian gas to Benin, Togo, and Ghana, is governed by the Treaty on the West African Gas Pipeline Project signed by the four states in 2003 and by an International Project Agreement between the states and the West African Gas Pipeline Company. The Treaty established the West African Gas Pipeline Authority as a regional regulator, with powers over tariffs and access, and provided for a harmonised legal and fiscal regime for the pipeline across the four states. The WAGP has suffered from supply interruptions, attributable in part to supply shortfalls in Nigeria and to damage to the pipeline, and from payment arrears by downstream buyers, illustrating the

challenges of regional gas integration where domestic supply and payment discipline are weak.

Other initiatives include the proposed **Nigeria–Morocco Gas Pipeline** (also referred to as the African Atlantic Gas Pipeline), intended to transport Nigerian gas along the West African coast to Morocco and potentially to Europe, for which intergovernmental memoranda have been signed with numerous ECOWAS states and Mauritania; and the **Trans-Saharan Gas Pipeline**, intended to link Nigeria with Algeria via Niger, for which the three governments signed a memorandum of understanding in 2022. These projects face formidable challenges of financing, security, and coordination, but they illustrate the aspiration to use gas infrastructure as an instrument of regional integration. The ECOWAS Energy Protocol of 2003, modelled on the Energy Charter Treaty, contains provisions on investment protection and transit, and the African Continental Free Trade Area (AfCFTA) may in time provide a broader framework for cross-border energy trade (examined in Chapter 13).

Cross-Border LNG: The Greater Tortue Ahmeyim Project

Cross-border cooperation is not confined to pipelines. The Greater Tortue Ahmeyim (GTA) gas field straddles the maritime boundary between Mauritania and Senegal. In 2018, the two governments concluded an Inter-Governmental Cooperation Agreement providing for the joint development of the field as a single unit, with resources and revenues shared equally in the initial phase, subject to future redeterminations, and for the liquefaction of the gas on a floating LNG vessel located at a hub near the maritime boundary. The project, operated by BP with Kosmos Energy and the national oil companies of both states, illustrates the application of the unitisation principles examined in Chapter 5 to a transboundary LNG development, and the use of a bilateral treaty framework to provide the legal stability required for financing in states with no prior history of petroleum production. First LNG cargoes from the project were exported in 2025.

Elsewhere in Africa, the Mozambique LNG project in the Rovuma Basin illustrates the vulnerability of LNG projects to security risks: TotalEnergies declared force majeure in April 2021 following insurgent attacks on the town of Palma near the project site, suspending construction for an extended period, while the Coral South floating LNG project, operated by Eni, began production in 2022. The force majeure declaration, and the subsequent review of the human rights situation around the project commissioned by the operator, raise issues examined in Chapters 15 and 21.

Host Government Agreements and Human Rights

Host government agreements for cross-border pipelines have attracted criticism for conferring extensive stabilisation protections on project companies, poten-

tially restricting the ability of host states to adopt new environmental, labour, and human rights legislation. In response to such criticism concerning the BTC pipeline, the project participants executed a Human Rights Undertaking in 2003 clarifying that the stabilisation provisions would not be invoked to challenge host state regulation in the fields of health, safety, environment, and human rights that was consistent with international standards. The episode contributed to the development of the *Principles for Responsible Contracts* discussed in Chapter 5 and has influenced the drafting of later agreements.

6.8 Gas Supply Disputes and Regulatory Intervention

Typology of Gas Disputes

Gas disputes may be classified into several categories: price review disputes between sellers and buyers under long-term contracts (examined in section 6.6); disputes over volumes, including take-or-pay claims and claims for failure to deliver; disputes over quality and measurement; transit disputes; disputes between investors and host states arising from regulatory measures, such as changes in fiscal terms, domestic supply obligations, or export restrictions; and disputes arising from emergency interventions in gas markets. Many of these disputes are resolved through international commercial arbitration under the rules of the ICC, the LCIA, or the Arbitration Institute of the Stockholm Chamber of Commerce (SCC), or through investment treaty arbitration.

Supply Failure and Diversion: The Egyptian Disputes

The supply failures in Egypt following the political upheavals of 2011 generated a series of significant awards. Egypt had committed to supply gas for export, both by pipeline to Israel and Jordan through the Arab Gas Pipeline and the East Mediterranean Gas (EMG) pipeline, and to LNG export plants at Damietta and Idku. As domestic demand outstripped production and the pipeline to Israel was repeatedly attacked in the Sinai, Egypt diverted gas to the domestic market and terminated the EMG supply contract in 2012.

The resulting claims included an ICC arbitration brought by the Israel Electric Corporation against the Egyptian General Petroleum Corporation (EGPC) and the Egyptian Natural Gas Holding Company (EGAS), in which the tribunal in 2015 awarded the Israel Electric Corporation approximately USD 1.76 billion; and the ICSID case *Ampal-American Israel Corp v Egypt*, in which the tribunal found that Egypt had breached the fair and equitable treatment standard by failing to protect the pipeline and by the manner in which the supply contract was terminated. In *Unión Fenosa Gas SA v Arab Republic of Egypt* (2018), an ICSID tribunal held that Egypt had breached the fair and equitable treatment standard of the Spain–Egypt bilateral investment treaty by failing to supply gas to the Damietta LNG plant, in which the claimant held an interest, and awarded damages of

approximately USD 2 billion. The tribunal rejected Egypt's defence of necessity, holding that Egypt had contributed to the situation of necessity through its domestic gas pricing policies, which had encouraged domestic consumption. These disputes were eventually resolved through settlements, including arrangements that facilitated the resumption of LNG exports from Damietta and the reversal of the EMG pipeline to carry Israeli gas to Egypt.

The Egyptian disputes illustrate the tension between export commitments and domestic supply obligations, the limits of the necessity defence in investment law (examined in Chapter 14), and the way in which regional gas flows can reverse dramatically as production patterns change.

The European Gas Crisis of 2021–2023

The most significant gas supply crisis of the modern era began in 2021, when Gazprom reduced deliveries to European buyers to the minimum contractual levels and ceased selling additional volumes on the spot market, while European storage levels fell. Following the Russian invasion of Ukraine in February 2022, Russia required European buyers to pay for gas in roubles through a mechanism established by Presidential Decree No 172 of 31 March 2022, and suspended supplies to buyers that refused to comply, including buyers in Poland, Bulgaria, Finland, Denmark, and the Netherlands. Gazprom progressively reduced flows through the Nord Stream pipeline, citing technical problems with turbines, before halting them entirely in August 2022. In September 2022, explosions damaged both Nord Stream pipelines. European hub prices reached unprecedented levels, peaking in August 2022 at more than ten times their historical average.

The crisis generated extensive commercial arbitration between European buyers and Gazprom. In 2024, the German importer Uniper announced that an arbitral tribunal had awarded it damages of more than EUR 13 billion for Gazprom's failure to deliver contracted gas, and that it had been granted the right to terminate its long-term contracts. Other European buyers obtained similar awards. Gazprom responded by obtaining anti-arbitration and anti-suit injunctions from Russian courts under Articles 248.1 and 248.2 of the Russian Arbitrazh Procedure Code, introduced in 2020, which assert exclusive jurisdiction of Russian courts over disputes involving sanctioned Russian persons. The enforcement of awards against Gazprom's assets outside Russia raises complex questions examined in Chapter 26.

Regulatory Intervention: The European Emergency Measures

The European Union responded to the crisis with an unprecedented series of emergency measures, many adopted under Article 122 TFEU, which permits the Council to adopt measures appropriate to the economic situation "in a spirit of solidarity between Member States", in particular if severe difficulties arise in the

supply of certain products, notably in the area of energy. The principal measures included:

- **Regulation (EU) 2022/1032** on gas storage, amending the Security of Gas Supply Regulation (EU) 2017/1938, which required member states to fill their underground storage facilities to 80 per cent of capacity by 1 November 2022 and 90 per cent by 1 November in subsequent years;
- **Council Regulation (EU) 2022/1369** on coordinated demand-reduction measures, which established a voluntary target to reduce gas consumption by 15 per cent compared with the average of the previous five years, with the possibility of a mandatory target in a "Union alert";
- **Council Regulation (EU) 2022/2576** on enhancing solidarity, which established the AggregateEU mechanism for joint purchasing of gas and default solidarity rules;
- **Council Regulation (EU) 2022/2578** establishing a market correction mechanism, which capped prices for TTF front-month derivatives in defined circumstances; and
- the **REPowerEU** plan of May 2022, which set the objective of ending EU dependence on Russian fossil fuels, and subsequent legislative proposals in 2025 to phase out remaining imports of Russian gas.

At national level, member states nationalised or recapitalised gas importers facing insolvency as a result of the need to purchase replacement gas at spot prices to meet their supply obligations — notably Germany's stabilisation of Uniper, in which the federal government acquired a stake of approximately 99 per cent — placed Russian-owned subsidiaries under trusteeship, as Germany did with Gazprom Germania (later renamed SEFE), and introduced price caps, windfall taxes, and subsidies for consumers.

The Principle of Energy Solidarity

The crisis gave new prominence to the principle of energy solidarity in Article 194(1) TFEU, which provides that EU energy policy shall aim "in a spirit of solidarity between Member States" to ensure the functioning of the energy market, ensure security of energy supply, promote energy efficiency and renewable energy, and promote the interconnection of networks. In *Germany v Poland* (Case C-848/19 P), decided in July 2021, the Court of Justice confirmed the judgment of the General Court in the OPAL case (examined as a casebox in Chapter 4), holding that the principle of energy solidarity is a legally binding principle that entails rights and obligations for both the EU and its member states, and that EU institutions must assess, in adopting measures in the energy field, whether there is a risk to the energy interests of member states. The principle has been invoked in support of the emergency measures adopted in 2022 and 2023.

Limits on Regulatory Intervention

Regulatory interventions in gas markets, however justified by emergency, raise legal questions concerning their compatibility with property rights, contract rights, investment treaties, and the rule of law. Windfall taxes and price caps have been challenged before national courts and the Court of Justice of the European Union, and some interventions have given rise to investment treaty claims. The legal limits of emergency intervention, including the doctrine of necessity in international law and the treatment of emergency measures under investment treaties, are examined in Chapters 14 and 25. The general lesson of the crisis for gas law is that the commercial framework of long-term contracts and liberalised markets operates within a political and security environment that can override it, and that the law must provide mechanisms both for decisive intervention in emergencies and for the protection of legitimate expectations and the restoration of normal market functioning.

Gas Supply Security Regulation Beyond Europe

The European experience is the most fully developed, but security of gas supply regulation exists in many jurisdictions. In Japan and Korea, which depend almost entirely on imported LNG, governments have encouraged buyers to maintain diversified portfolios of long-term contracts and have established strategic buffer LNG arrangements; Japan introduced a "strategic buffer LNG" mechanism in 2023 under which the state-backed Japan Organization for Metals and Energy Security (JOGMEC) may procure LNG for use in emergencies. China has imposed gas storage obligations on suppliers and local governments following winter shortages in 2017 and 2018, when an aggressive coal-to-gas switching policy outpaced supply. In the United States, the Federal Energy Regulatory Commission and the North American Electric Reliability Corporation have examined the interdependence of gas and electricity systems following Winter Storm Uri in Texas in February 2021, during which freezing of gas production and processing facilities contributed to the failure of the Texas electricity system and the loss of life. These experiences demonstrate that gas security is a function not only of supply sources but also of the physical resilience of infrastructure and the coordination of gas and power markets.

In Nigeria, the principal security of supply challenges are domestic: inadequate gas supply to power stations, attributable to pipeline vandalism, insufficient processing and transportation capacity, payment arrears in the electricity value chain, and the historic preference of producers for export markets that pay in foreign currency. The PIA's domestic gas delivery obligation, the Gas Transportation Network Code, and the initiatives of the Decade of Gas are intended to address these challenges, but their success depends on resolving the liquidity crisis in the

electricity sector examined in Chapter 9, on the security of pipeline infrastructure, and on the gradual movement of domestic gas prices towards cost-reflective levels.

LEADING AUTHORITY

The Gazprom–Naftogaz Arbitrations (SCC, 2017–2018)

Background. Following the January 2009 gas crisis, Gazprom and Naftogaz of Ukraine concluded two ten-year contracts: a supply contract, under which Gazprom sold gas to Naftogaz for consumption in Ukraine, and a transit contract, under which Naftogaz transported Russian gas across Ukraine to European buyers. The supply contract contained a take-or-pay obligation for large annual quantities at an oil-indexed price that, following the fall in oil prices and the deterioration of relations after 2014, Naftogaz regarded as far above the market value of gas. In 2014 the parties commenced arbitration under the Rules of the Arbitration Institute of the Stockholm Chamber of Commerce, with claims by Gazprom for unpaid gas and take-or-pay amounts exceeding USD 30 billion, and claims by Naftogaz for a revision of the supply contract price, the removal of the take-or-pay obligation and the prohibition on re-export, and compensation for Gazprom's failure to deliver contracted transit volumes.

Awards. In the supply arbitration, the tribunal's final award of December 2017 substantially reduced Gazprom's take-or-pay claim, revised the contract price to reflect the market in continental Europe, and invalidated the prohibition on re-export, while ordering Naftogaz to pay for gas delivered but unpaid. In the transit arbitration, the tribunal's final award of February 2018 held that Gazprom had failed to deliver the agreed transit volumes and awarded Naftogaz damages of approximately USD 4.6 billion. After netting off the amounts awarded in the two proceedings, Gazprom owed Naftogaz approximately USD 2.56 billion. Gazprom's challenges before the Swedish courts were unsuccessful, and Naftogaz commenced enforcement proceedings against Gazprom's assets in several European jurisdictions. In December 2019, alongside the conclusion of a new five-year transit arrangement, Gazprom paid Naftogaz approximately USD 2.9 billion in settlement of the awards, and the parties withdrew pending claims.

Significance. The arbitrations are among the largest commercial arbitrations in history. They demonstrate the willingness of arbitral tribunals to revise price and volume provisions in long-term gas contracts where market circumstances have fundamentally changed and the contract so permits, and to sever export restrictions incompatible with competition law. They also illustrate the interaction between commercial arbitration and geopolitics: the enforcement of the awards and their eventual settlement were intertwined with intergovernmental negotiations over the future of gas transit through Ukraine, mediated by the European Commission.

PRACTICE NOTE

Drafting checklist: key protections for a buyer in a long-term LNG SPA

1. Downward quantity tolerance sufficient to absorb foreseeable demand fluctuations, including those resulting from decarbonisation policies.
2. Destination flexibility (FOB delivery or DES with diversion rights) and the absence of profit-sharing on resale, subject to applicable competition law.
3. A price review clause with a clear trigger, a defined reference market, express authority for the tribunal to change the indexation, and a time limit for determination.
4. Shortfall remedies that are not exclusive in cases of wilful or deliberate non-delivery, and a cap on the seller's ability to invoke upstream supply failures as force majeure.
5. Quality specifications compatible with the buyer's terminal and downstream network, with clear allocation of the costs of treating off-specification LNG.
6. Methane intensity reporting and reduction commitments across the supply chain, aligned with applicable import regulations such as the EU Methane Regulation.
7. Termination rights for prolonged force majeure and for sanctions that render performance unlawful, and a neutral seat of arbitration in a New York Convention state.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Simulated price review arbitration (hypothetical). *All parties, facts, and figures in this exercise are fictitious and created solely for teaching purposes.* Coastal Gas Exporters Ltd, a fictional liquefaction company in the fictional Republic of Ardenia, supplies 2 million tonnes per annum of LNG to Northern Utilities AG, a fictional utility in the fictional state of Valdora, under a twenty-year DES SPA concluded in 2015. The price is 14 per cent of the Brent crude oil price (three-month average) plus USD 0.50 per MMBtu. Clause 12 permits either party, every four years, to request a price review if "there has been a significant change in circumstances in the Buyer's Market, beyond the control of the Parties, as a result of which the Contract Price no longer reflects the value of LNG in the Buyer's Market", and provides that in default of agreement the tribunal "shall determine a revised Contract Price that reflects the value of LNG in the Buyer's Market". In 2025, Northern Utilities requests a price review, arguing that the Valdoran market is now priced predominantly by reference to TTF and that hub prices have been consistently below the contract price. Coastal Gas Exporters argues that the shift to hub pricing predated the 2021 review and that the 2021–2023 crisis showed that oil-indexed prices provide valuable security.

Divide into teams representing the claimant, the respondent, and the tribunal. (a) Prepare written submissions of not more than 2,000 words on the trigger question and the revision question, identifying the evidence each party would need (for example, comparable contract data and expert evidence on market value). (b) Conduct a thirty-minute oral hearing. (c) As the tribunal, draft the dispositive section of a

reasoned award, addressing whether the tribunal has authority to introduce hub indexation into the price formula.

CHAPTER SUMMARY

- The physical properties of natural gas — low energy density, the need for fixed transport infrastructure, and variable composition — explain its distinctive legal regime: long-term contracts, regulation of network monopolies, and detailed provisions on quality, measurement, and balancing.
- Gas markets have developed as regional markets with distinct pricing systems — gas-on-gas competition at hubs such as Henry Hub, NBP, and TTF; oil indexation; regulated pricing; and spot LNG — although LNG trade has promoted convergence.
- Gas-specific upstream regulation addresses rights to non-associated gas, domestic supply obligations, fiscal incentives, and the restriction of flaring and venting, increasingly through binding methane regulation.
- Liberalisation has required third-party access to pipelines and LNG terminals, unbundling of network and supply activities, and regulation of tariffs, models that Nigeria's PIA has adopted for the midstream gas sector.
- LNG projects are structured as integrated, merchant, tolling, or hybrid projects, within host-state frameworks that frequently include project-specific statutes or agreements.
- Long-term gas sales agreements allocate volume risk through take-or-pay clauses and price risk through indexation and price review clauses; competition law has curtailed destination restrictions.
- Cross-border pipelines are governed by intergovernmental and host government agreements and by transit regimes in GATT Article V and ECT Article 7; African initiatives such as the WAGP illustrate the promise and difficulties of regional integration.
- Gas supply disputes — from the Egyptian supply failures to the European crisis of 2021–2023 — reveal the interaction between commercial contracts, investment protection, emergency regulation, and geopolitics.

DOCTORAL RESEARCH QUESTIONS

1. Has the take-or-pay clause outlived its economic rationale in liberalised gas markets, or does the experience of the 2021–2023 crisis demonstrate its continuing value for security of supply?
2. To what extent do price review tribunals act as market regulators, and what are the implications of the confidentiality of their awards for the development of a coherent body of gas law?

3. Evaluate the adequacy of GATT Article V and ECT Article 7 as frameworks for energy transit in the light of the Russia–Ukraine transit experience and the failure of the Energy Charter Transit Protocol.
4. How should the law reconcile domestic gas supply obligations in producing developing countries with export commitments and investment protection, in the light of the Egyptian experience?
5. Does the principle of energy solidarity in Article 194 TFEU, as interpreted in *Germany v Poland*, offer a model for regional energy integration in West Africa?
6. Can the extraterritorial reach of import-side methane regulation, such as the EU Methane Regulation, be reconciled with the principle of common but differentiated responsibilities and with WTO law?

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7 CHAPTER

Mining Law and the Governance of Mineral Resources

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the legal foundations of mineral rights and the principal systems for allocating mining titles.
2. Analyse the structure of exploration permits, mining licences, and the cadastral systems that administer them.
3. Evaluate the role of mining concessions and investment agreements and the controversies surrounding their stability provisions.
4. Identify the environmental and social obligations imposed on mining operations by domestic and international law.
5. Assess legal mechanisms for community development and benefit sharing, including community development agreements.
6. Analyse the regulation of mine closure, rehabilitation, and financial assurance.
7. Evaluate approaches to the regulation and formalisation of artisanal and small-scale mining.
8. Critically assess international standards for responsible mining and their interaction with domestic law.

Introduction

Mining is one of humanity's oldest economic activities and one of its most legally complex. Minerals extracted from the earth supply the metals of modern infrastructure and technology, the fertilisers of agriculture, the materials of construction, and, increasingly, the components of the energy transition: copper for electrical networks, lithium, cobalt, nickel, and graphite for batteries, rare earth elements for permanent magnets in wind turbines and electric motors, and silver and silicon for solar panels. At the same time, mining transforms landscapes, consumes vast quantities of water and energy, generates enormous volumes of waste rock and tailings, displaces communities, and has been associated in many

countries with conflict, corruption, and human rights abuses. The law of mining must reconcile the interests of states seeking revenue and development, investors seeking secure and profitable titles, communities seeking protection and a fair share of benefits, and the broader public interest in environmental protection.

Mining law shares many features with petroleum law, examined in Chapter 5: both regulate the extraction of non-renewable subsurface resources, both are founded on the state's ownership or control of those resources (examined in Chapter 3), and both rely on systems of titles, fiscal regimes, and contracts to allocate rights and revenues between states and investors. But mining differs from petroleum in important respects. Mineral deposits are fixed in place and do not migrate across boundaries in the manner of petroleum reservoirs, so the rule of capture and unitisation are rarely relevant. The diversity of minerals — from bulk commodities such as iron ore and coal, to base metals such as copper and zinc, to precious metals and gemstones, to industrial minerals and construction materials — and the diversity of operators — from multinational corporations operating giant open-pit mines to millions of individual artisanal miners using hand tools — produce a far more heterogeneous regulatory landscape. Mining also has a much larger physical footprint than most petroleum operations and a longer environmental legacy, since tailings facilities and waste dumps must be managed in perpetuity after mining ends.

The modern history of mining law in developing countries has been characterised by successive waves of reform. In the 1960s and 1970s, many newly independent states nationalised their mining industries, as Zambia did with its copper mines in 1969–1970 and Zaire (now the Democratic Republic of the Congo) did with the Union Minière du Haut-Katanga in 1967. The poor performance of many state mining enterprises, combined with the debt crises of the 1980s, led in the 1990s to a wave of liberalising reforms, often supported by the World Bank, which privatised state enterprises and adopted investor-friendly mining codes offering secure titles, low royalties, and generous fiscal incentives. The commodity price boom of the 2000s, driven by Chinese demand, led many states to conclude that these codes had transferred too great a share of mineral rents to investors, prompting a third wave of reforms — revised mining codes in the Democratic Republic of the Congo (2018), Tanzania (2017), Zambia, Guinea, and elsewhere — that increased royalties and state participation, curtailed stability guarantees, and strengthened local content and community development requirements. The Africa Mining Vision, adopted by African Union heads of state in 2009, articulates the aspiration for "transparent, equitable and optimal exploitation of mineral resources to underpin broad-based sustainable growth and socio-economic development".

This chapter examines the core institutions of mining law. It begins with mineral rights and mining titles (section 7.1), exploration permits and mining licences (section 7.2), and mining concessions and investment agreements (section 7.3). It then examines environmental and social obligations (section 7.4), community development and benefit sharing (section 7.5), and mine closure and rehabilitation (section 7.6). It concludes with the regulation of artisanal and small-scale mining (section 7.7) and international standards for responsible mining (section 7.8). Chapter 8 addresses the distinctive legal issues raised by critical minerals, and Chapter 20 considers deep-seabed mining under the law of the sea.

7.1 Mineral Rights and Mining Titles

Ownership of Minerals

The starting point of any mining regime is the ownership of minerals in their natural state. As explained in Chapter 3, legal systems allocate subsurface mineral ownership according to two broad models: the *accession* model, under which minerals belong to the owner of the surface land (the position at common law, retained for most minerals in the United States in respect of private land); and the *domanial* or *regalian* model, under which minerals are owned or controlled by the state irrespective of surface ownership (the position in most civil law jurisdictions and in almost all developing countries). Even where the accession principle applies, certain minerals are often reserved to the Crown or state: in the United Kingdom, gold and silver in their natural state belong to the Crown under the *Case of Mines* (1568), and coal was nationalised under the Coal Industry Nationalisation Act 1946 and is now vested in the Coal Authority.

In Nigeria, section 44(3) of the Constitution of the Federal Republic of Nigeria 1999 vests "the entire property in and control of all minerals, mineral oils and natural gas in, under or upon any land in Nigeria or in, under or upon the territorial waters and the Exclusive Economic Zone of Nigeria" in the Government of the Federation, and "mines and minerals, including oil fields, oil mining, geological surveys and natural gas" appear in the Exclusive Legislative List. Section 1 of the Nigerian Minerals and Mining Act 2007 reiterates that the property in and control of all mineral resources in, under, or upon any land in Nigeria is vested in the Government of the Federation for and on behalf of the people of Nigeria. The exclusive federal competence over minerals has been controversial in a federation where states and communities bear the environmental and social costs of mining, and it has complicated the regulation of artisanal mining, which is often more effectively supervised at local level. Proposals to devolve control of solid minerals to the states have featured in constitutional reform debates.

In South Africa, the Mineral and Petroleum Resources Development Act 2002 (MPRDA) made a fundamental change. Before the Act, mineral rights in

South Africa could be privately owned and severed from surface ownership, and were concentrated in the hands of a small number of predominantly white-owned mining houses. Section 3(1) of the MPRDA declares that mineral and petroleum resources are "the common heritage of all the people of South Africa" and that the state is "the custodian thereof for the benefit of all South Africans". The Act abolished private ownership of unexploited mineral rights and required holders of "old order rights" to convert them into new order rights granted by the state within transitional periods. In *Agri South Africa v Minister for Minerals and Energy* (2013), the Constitutional Court held by a majority that the MPRDA did not effect an expropriation of the old order rights requiring compensation, because the state had not acquired the rights for itself but had become their custodian.[1] The decision is significant for the constitutional theory of resource ownership and for the distinction between deprivation and expropriation, examined in Chapter 3.

The Nature of Mining Titles

Where minerals belong to the state, a person may lawfully explore for or extract them only by virtue of a title granted by the state. As with petroleum titles, the legal nature of mining titles varies among legal systems. In civil law systems influenced by French law, mining titles have traditionally been characterised as concessions — administrative acts conferring real rights in immovable property that may be mortgaged, transferred with consent, and protected against arbitrary withdrawal. In common law systems, mining leases and licences are statutory rights, the nature of which depends on the construction of the statute, but which are commonly treated as property for the purposes of constitutional protection and as capable of being mortgaged and transferred. In many African mining codes, mining titles are expressly declared to be real rights or immovable property, distinct from the ownership of the surface land, in order to facilitate their use as security for project finance.

Mining titles usually confer the exclusive right to carry out the specified activity within the title area for the specified minerals, together with ancillary rights to use land, water, and timber, to construct infrastructure, and to dispose of the minerals produced. They are subject to obligations: to carry out work programmes, to pay fees, royalties, and taxes, to comply with environmental, health, safety, and labour requirements, to report to the regulator, and to rehabilitate the land at the end of operations.

Systems for Allocating Mining Titles

Mining regimes employ three principal methods for allocating titles: first-come, first-served; competitive tender; and administrative discretion.

[1] *Agri South Africa v Minister for Minerals and Energy* [2013] ZACC 9, 2013 (4) SA 1 (CC).

Under the **first-come, first-served** system, the first qualified applicant to apply for an available area is entitled to the title, provided that it meets the statutory requirements. The system has its origins in the "free mining" tradition of the American West, under which prospectors could stake claims on public land by physically marking the boundaries and recording the claim; the United States General Mining Law of 1872 still permits citizens to locate claims to "valuable mineral deposits" on open federal lands. In modern regimes, the first-come, first-served system is typically administered through an electronic mining cadastre that records applications in order of receipt and grants titles to the first applicant that satisfies objective criteria. Its advantages are simplicity, transparency, and the reduction of administrative discretion, which limits opportunities for corruption. It is well suited to exploration licences over areas where geological information is limited and the value of the area is uncertain.

Under the **competitive tender** system, titles over areas of known prospectivity — for example, areas where the state has carried out geological surveys or where previous exploration has identified a deposit — are allocated through a competitive bidding process, with bids evaluated on the basis of work programmes, financial terms, technical capacity, or a combination of these. Competitive tender allows the state to capture a greater share of the value of known deposits, but requires greater administrative capacity and robust procedures to prevent manipulation.

Under **administrative discretion**, the minister or another authority grants titles on the basis of a discretionary assessment of the merits of applications. Discretionary systems have been associated with corruption and the allocation of rights to politically connected persons, and the trend in mining law reform has been to reduce discretion and to rely on objective criteria administered by an independent cadastre.

Many regimes combine these methods: first-come, first-served for exploration licences in open areas, competitive tender for areas of known prospectivity, and some residual discretion for strategic deposits.

The Mining Cadastre

The mining cadastre — the official register of mining titles and applications — is the central administrative institution of a modern mining regime. A well-functioning cadastre records the location, holder, type, status, and obligations of every title; prevents overlapping grants; processes applications according to transparent rules; monitors compliance with title obligations, such as payment of annual fees and fulfilment of work programmes; and makes information publicly available. Electronic cadastres, often implemented with support from the World Bank and other development partners, have been introduced in many African

states, including Nigeria, Ghana, Tanzania, Zambia, the Democratic Republic of the Congo, and Mozambique, many offering online portals that enable investors and the public to view title information.

In Nigeria, the Minerals and Mining Act 2007 established the Mining Cadastre Office (MCO) as the agency responsible for the administration of mineral titles and the maintenance of the cadastral register. The Act provides for applications to be processed on a first-come, first-served basis, and the MCO operates an electronic mining cadastre system (eMC+), which enables online applications and public access to the cadastral map. The Act provides for the revocation of titles for failure to pay annual service fees or to comply with work obligations, and the MCO has periodically revoked large numbers of dormant titles held for speculative purposes.

Surface Rights and Land Access

Because mineral ownership is usually separated from surface ownership, mining law must regulate the relationship between title holders and surface owners or occupiers. Mining codes typically confer on title holders a right of access to the land, subject to obligations to obtain the consent of, or give notice to, surface owners and occupiers; to pay compensation for disturbance of surface rights, damage to crops, and loss of use of land; and in some cases to acquire the land through compulsory acquisition by the state. The adequacy of compensation, the treatment of customary land rights, and the relationship between mining law and land law are recurrent sources of conflict, particularly in Africa, where much land is held under customary tenure without formal registration.

In Nigeria, the Land Use Act 1978 vests all land in each state in the governor, to be held in trust for the use and common benefit of all Nigerians, and the Minerals and Mining Act 2007 provides that land required for mining purposes shall have priority over other uses, subject to the payment of compensation to the owners or occupiers for disturbance of surface rights, damage to crops, and other loss. In South Africa, the Constitutional Court held in *Maledu v Itereleng Bakgatla Mineral Resources (Pty) Ltd* (2018) that the holder of a mining right could not evict a community holding informal land rights protected by the Interim Protection of Informal Land Rights Act 1996 without complying with that Act, which requires the consent of the community or the expropriation of the rights with compensation.[2] In *Baleni v Minister of Mineral Resources* (2018), concerning the proposed Xolobeni titanium mine on the Wild Coast, the High Court held that the Minister could not grant a mining right over land occupied by the Umgungundlovu community under informal rights without obtaining the community's

[2] *Maledu v Itereleng Bakgatla Mineral Resources (Pty) Ltd* [2018] ZACC 41, 2019 (2) SA 1 (CC).

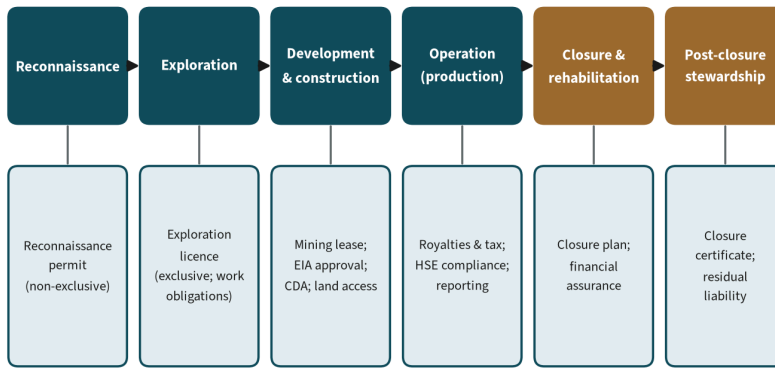
full and informed consent.[3] These decisions are significant for the emergence of community consent requirements in domestic mining law, examined further in Chapter 21.

7.2 Exploration Permits and Mining Licences

The Sequence of Mining Titles

Most modern mining codes establish a sequence of titles corresponding to the stages of the mining life cycle: reconnaissance, exploration, development and production, and closure. Each stage involves different activities, risks, and levels of investment, and the titles granted for each stage differ in their duration, area, rights, and obligations. The holder of an exploration title that discovers a commercially viable deposit typically has a right — subject to conditions — to obtain a mining title over the deposit. This "security of tenure", which links the exploration investment to the right to exploit any resulting discovery, is regarded by investors as the most important feature of a mining regime: no rational company will invest millions of dollars in exploration if the state may refuse a mining licence and award the deposit to someone else.

Stages of the mining life cycle (top) and the corresponding legal instruments (bottom)



Security of tenure links exploration to the right to mine; closure obligations are planned and secured from the outset

Figure 7.1 The mining life cycle and the corresponding sequence of legal titles and approvals. *Source:* prepared for this edition.

Reconnaissance Permits

Reconnaissance is the preliminary search for mineral occurrences by geological, geophysical, and geochemical surveys, remote sensing, and surface sampling,

[3] *Baleni v Minister of Mineral Resources* [2018] ZAGPPHC 829, 2019 (2) SA 453 (GP).

without significant ground disturbance. Reconnaissance permits are typically non-exclusive, cover large areas, and are granted for short periods (one or two years). Under the Nigerian Minerals and Mining Act 2007, a reconnaissance permit confers a non-exclusive right to search for minerals by surface geological and geophysical surveys, and does not entitle the holder to engage in drilling, excavation, or other subsurface activities.

Exploration Licences

Exploration involves detailed investigation of a defined area to locate and evaluate mineral deposits, including drilling, trenching, bulk sampling, and feasibility studies. Exploration licences are exclusive, cover defined areas (often subject to a maximum size), and are granted for initial periods of three to five years, renewable for further periods subject to the fulfilment of work obligations and the relinquishment of a proportion of the area on renewal. Work obligations may be expressed as minimum expenditure commitments or as minimum work programmes. Relinquishment requirements prevent the holding of large areas for speculative purposes and return unexplored land to the state for allocation to others.

Under the Nigerian Act, an exploration licence confers the exclusive right to conduct exploration for the minerals specified in the licence within the licence area. It is granted for an initial period of three years, renewable twice for periods of two years each, subject to the holder having complied with its obligations, and the licence area is subject to maximum size limits. The holder is required to commence exploration within a specified period, to expend minimum amounts on exploration, to submit reports, and to pay annual service fees. Upon discovery of a commercially viable deposit, the holder may apply for a mining lease or a small-scale mining lease.

Exploration licensing raises recurrent legal issues: the transferability of licences (which is usually subject to ministerial consent), the treatment of licences held by "juniors" — small exploration companies, often listed on the Toronto, Australian, or London AIM stock exchanges, that finance exploration through equity markets and seek to sell discoveries to larger companies; the extent of the holder's right to a mining licence; and the confidentiality of exploration data, which must usually be submitted to the state geological survey and may be released after a period.

Mining Licences and Leases

A mining licence or lease confers the exclusive right to develop and operate a mine within a defined area and to sell the minerals produced. Mining licences are granted for long periods — commonly twenty to thirty years, or the estimated life of the mine — renewable for further periods. The application for a mining licence

must usually be supported by a feasibility study demonstrating the technical and economic viability of the deposit, a mining plan, an environmental impact assessment and environmental management plan, a community development plan, and evidence of financial and technical capacity.

Under the Nigerian Act, a mining lease is granted for a period of up to twenty-five years, renewable every twenty-four years, over an area not exceeding a specified maximum. The holder must commence development within a specified period of grant, comply with an approved Environmental Impact Assessment and an Environmental Protection and Rehabilitation Programme, enter into a community development agreement with the host community (discussed in section 7.5), pay royalties and other fees, and comply with the Act's provisions on health, safety, and employment. The Act also provides for small-scale mining leases, quarry leases (for the extraction of construction materials such as granite, marble, and laterite), and water use permits.

The Right to Proceed from Exploration to Mining

The extent to which an exploration licence holder is entitled to a mining licence is critical. Some regimes confer an automatic right to a mining licence upon satisfaction of objective criteria; others confer a right to apply, leaving the state a discretion to refuse on grounds such as environmental considerations, the public interest, or inadequate financial terms. A discretion to refuse undermines security of tenure but preserves the state's ability to protect the public interest.

The tension between security of tenure and regulatory discretion has generated major investment disputes. In *Crystallex International Corp v Bolivarian Republic of Venezuela* (2016), an ICSID tribunal held that Venezuela had breached the fair and equitable treatment standard and expropriated the claimant's investment in the Las Cristinas gold deposit by refusing an environmental permit required for exploitation and subsequently rescinding the mining contract, and awarded damages of approximately USD 1.2 billion plus interest.[4] In *Tethyan Copper Co Pty Ltd v Islamic Republic of Pakistan* (2019), concerning the Reko Diq copper-gold deposit in Balochistan, an ICSID tribunal held that the refusal of a mining lease to the claimant, after it had carried out extensive exploration and a feasibility study, breached the fair and equitable treatment standard of the Australia–Pakistan bilateral investment treaty and constituted an expropriation, and awarded approximately USD 5.9 billion in damages and interest — one of the largest awards in ICSID history.[5] The dispute was resolved in 2022 through a settlement

[4] *Crystallex International Corp v Bolivarian Republic of Venezuela*, ICSID Case No ARB(AF)/11/2, Award (4 April 2016).

[5] *Tethyan Copper Co Pty Ltd v Islamic Republic of Pakistan*, ICSID Case No ARB/12/1, Award (12 July 2019).

under which Barrick Gold, one of the claimant's shareholders, agreed to develop the project in partnership with the federal and provincial governments.

These awards are frequently cited by investors as demonstrating the importance of security of tenure, and by critics of investment arbitration as demonstrating the risk that awards of speculative future profits impose enormous liabilities on developing countries. The methodology of damages in such cases, including the use of discounted cash flow valuation for projects that have never produced, is examined in Chapter 26.

Transfer and Change of Control

Mining titles are valuable assets that are frequently bought and sold, both directly and through transactions involving the shares of title-holding companies. Most mining codes require ministerial consent to the transfer of titles, and increasingly to changes of control of title holders, in order to ensure that transferees meet the qualification requirements and to enable the state to capture a share of the gains through capital gains taxation. Indirect transfers — the sale of shares in an offshore holding company that indirectly owns a mining title — have been a particular concern, since they may allow substantial gains to escape taxation in the host state. Several states, including Uganda, Tanzania, Mozambique, and India, have introduced legislation to tax indirect transfers, and the issue is examined in Chapter 16. The Platform for Collaboration on Tax (comprising the IMF, OECD, United Nations, and World Bank) published a toolkit on the taxation of offshore indirect transfers in 2020.

7.3 Mining Concessions and Investment Agreements

The Role of Mining Agreements

Many large mining projects, particularly in developing countries, are governed not only by the general mining code but also by a project-specific agreement between the state and the investor — variously called a mining agreement, mineral development agreement, mining convention, or investment agreement. Such agreements may supplement the code by providing details specific to the project, or derogate from it by granting special fiscal terms, stability guarantees, or other incentives. In some states, mining agreements must be ratified by the legislature, giving them the force of law, as in Liberia, Sierra Leone, and Ghana (where the Constitution requires parliamentary ratification of agreements relating to the grant of rights to minerals).

Mining agreements typically address: the project area and title; development obligations and timetable; fiscal terms, including royalty rates, income tax rates, depreciation, and import duty exemptions; state participation; foreign exchange and the right to maintain offshore accounts; employment and local procurement; infrastructure, including access to rail, ports, and power; community develop-

ment; environmental obligations; stabilisation; and dispute resolution, usually by international arbitration.

Stabilisation in Mining Agreements

Stabilisation clauses in mining agreements are designed to protect investors from adverse changes in the legal and fiscal regime over the long life of the project. Mining codes themselves may offer stability guarantees: the 2002 Mining Code of the Democratic Republic of the Congo, for example, guaranteed holders of mining rights that the fiscal and customs regime in force at the time of grant would remain unchanged for ten years. Ghana's Minerals and Mining Act 2006 (Act 703) permits the Minister to enter into stability agreements guaranteeing that the holder will not be adversely affected by changes in law for a period not exceeding fifteen years, and development agreements for investments exceeding a specified threshold.

The effectiveness of stabilisation in mining has been sorely tested by the "resource nationalism" of the past two decades. When commodity prices rose dramatically in the 2000s, many states concluded that stabilised fiscal terms negotiated in a period of low prices were inequitable, and sought to renegotiate or override them. The DRC's 2018 Mining Code, which increased the royalty rate for cobalt and other "strategic substances" to ten per cent, introduced a super-profits tax, and reduced the stability period, applied the new provisions to existing title holders notwithstanding the stability guarantee in the 2002 Code, provoking protests from major mining companies. Tanzania's Natural Wealth and Resources (Permanent Sovereignty) Act 2017 and the Natural Wealth and Resources Contracts (Review and Re-negotiation of Unconscionable Terms) Act 2017 required the review of agreements containing "unconscionable" terms, prohibited the submission of disputes relating to natural resources to international arbitration, and asserted the permanent sovereignty of the people of Tanzania over natural resources. In the same year, the Tanzanian government imposed an export ban on gold and copper concentrates and issued tax assessments of approximately USD 190 billion against Acacia Mining, a subsidiary of Barrick Gold. The dispute was resolved through a framework agreement in 2019 under which the government acquired a sixteen per cent free-carried interest in Barrick's Tanzanian mines and a fifty per cent share of economic benefits, and Barrick paid USD 300 million in settlement of the tax claims.

The Renegotiation of Mining Agreements

The renegotiation of mining agreements raises fundamental questions about the balance between *pacta sunt servanda* and the state's sovereign right to regulate in the public interest. Investors argue that stability is essential to attract long-term investment and that states should be held to their bargains; states argue that

agreements concluded under conditions of information asymmetry, weak negotiating capacity, or corruption may lack legitimacy, and that circumstances may change so fundamentally that the original bargain becomes inequitable. Several states have conducted systematic reviews of mining agreements: Liberia reviewed concessions concluded during its civil war; Guinea conducted a review under its 2011 Mining Code, leading to the revocation of the rights of BSG Resources over the Simandou iron ore deposit on the ground that they had been obtained through corruption; and the DRC reviewed sixty-one mining contracts in 2007–2008.

The Guinea case illustrates the relevance of corruption to the validity of mining titles. BSGR commenced ICSID arbitration against Guinea following the revocation of its rights, and Vale, which had paid BSGR USD 500 million for a fifty-one per cent interest in the project, commenced LCIA arbitration against BSGR. In 2019 an LCIA tribunal found that BSGR had obtained its rights through bribery and fraudulently misrepresented the position to Vale, and ordered BSGR to pay Vale approximately USD 2 billion. The English courts subsequently enforced the award. The ICSID proceedings were settled in 2019 through an agreement under which BSGR relinquished its claims to Simandou. The Simandou project, divided into blocks held by Rio Tinto and a consortium including Chinese interests, began exporting iron ore in late 2025, illustrating both the immense value of major mineral deposits and the decades-long delays that legal and political disputes can cause.

Model Mining Development Agreement

In response to concerns about the imbalance of negotiating capacity in mining agreements, the International Bar Association's Mining Law Committee published a Model Mine Development Agreement (MMDA) in 2011, designed as a template that reflects a balance of the interests of states, investors, and communities. The MMDA includes provisions on community development, environmental and social impact assessment, transparency, and dispute resolution, and is accompanied by commentary explaining the policy choices involved. Other resources include the Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development (IGF) Mining Policy Framework of 2013, the African Minerals Development Centre's Country Mining Vision guidebook, and the negotiation support initiatives of the African Legal Support Facility, established by the African Development Bank to provide legal advice to African governments in complex commercial negotiations, including mining agreements.

Mining Investment Disputes and Social Licence

A significant body of investment arbitration concerns the interaction between mining investments and community opposition. In *Bear Creek Mining Corp v*

Republic of Peru (2017), the claimant had obtained rights to the Santa Ana silver project near the border with Bolivia, but the government revoked the decree authorising the project following violent protests by local Aymara communities. The tribunal held that the revocation constituted an indirect expropriation under the Canada–Peru free trade agreement, rejecting Peru's argument that the claimant's failure to obtain a "social licence to operate" should defeat or reduce its claim, although the tribunal limited damages to the amounts actually invested.[6] A partially dissenting arbitrator, Professor Philippe Sands, would have reduced damages on the ground that the investor had contributed to the social unrest by failing to engage adequately with the affected communities. In *Copper Mesa Mining Corp v Republic of Ecuador* (2016), the tribunal reduced the damages awarded to the investor by thirty per cent on account of contributory fault, because the investor's contractors had used armed security personnel against protesters. In *Eco Oro Minerals Corp v Republic of Colombia* (2021), the tribunal held that Colombia's prohibition of mining in the Santurbán páramo, a high-altitude wetland ecosystem, breached the minimum standard of treatment under the Canada–Colombia FTA, but declined to find that the general environmental exception in the treaty precluded liability, and later awarded only limited damages. In *Gabriel Resources Ltd v Romania* (2024), concerning the Roșia Montană gold project, the tribunal dismissed the investor's claims, finding that the investor had not secured the social licence necessary for the project and that the state's conduct did not breach the treaty.

These cases illustrate the growing significance of community acceptance in mining investment law and the extent to which arbitral tribunals are prepared to consider the investor's own conduct in assessing liability and damages. They are examined further in Chapters 14 and 21.

7.4 Environmental and Social Obligations

The Environmental Footprint of Mining

Mining has profound environmental impacts across its life cycle. Exploration may involve the clearing of vegetation, the construction of access roads, and drilling. Development and operation involve the removal of overburden, the excavation of open pits or underground workings, the disposal of waste rock, the processing of ore using chemicals such as cyanide (in gold processing) and sulphuric acid (in copper heap leaching), the consumption of large volumes of water and energy, and the storage of tailings — the finely ground residue remaining after the extraction of valuable minerals — in tailings storage facilities that may be among the largest engineered structures on earth. Acid mine drainage, which occurs when sulphide

[6] *Bear Creek Mining Corp v Republic of Peru*, ICSID Case No ARB/14/21, Award (30 November 2017).

minerals exposed by mining react with air and water to produce sulphuric acid that mobilises heavy metals, can contaminate watercourses for centuries. The catastrophic failure of tailings dams — at Mount Polley in British Columbia in 2014, at Fundão (Mariana) in Brazil in 2015, and at Córrego do Feijão (Brumadinho) in Brazil in 2019, the last of which killed 270 people — has demonstrated the scale of the risks.

Environmental Impact Assessment and Permitting

Virtually all modern mining codes and environmental statutes require an environmental impact assessment (EIA), or an environmental and social impact assessment (ESIA), before a mining licence is granted or operations commence. The EIA process, examined generally in Chapter 17, requires the proponent to identify and evaluate the likely environmental and social impacts of the project, to consider alternatives, to propose measures to avoid, minimise, mitigate, and compensate for adverse impacts, and to consult the public and affected communities. The outcome is typically an environmental management plan or programme, the implementation of which becomes a condition of the licence.

In Nigeria, the Environmental Impact Assessment Act 1992 (now the Environmental Impact Assessment Act, Cap E12, Laws of the Federation of Nigeria 2004) requires an EIA for mining projects, and the Minerals and Mining Act 2007 requires the holder of a mining lease, small-scale mining lease, or quarry lease to submit an Environmental Impact Assessment statement and an Environmental Protection and Rehabilitation Programme (EPRP) to the Mines Environmental Compliance Department of the Ministry of Solid Minerals Development. The Federal Ministry of Environment administers the EIA Act, and the National Environmental Standards and Regulations Enforcement Agency (NESREA) has issued regulations on mining and processing of coal, ores, and industrial minerals.

Water, Biodiversity, and Protected Areas

Mining's demands on water are a growing source of conflict, particularly in arid regions such as the Atacama Desert in Chile, where copper and lithium mining compete with agriculture and indigenous communities for scarce water. Mining laws increasingly require water use permits, impose limits on extraction, and require the treatment of discharges. In Chile, the 2022 reform of the Water Code (Law No 21.435) recognised access to water for human consumption and sanitation as a human right and limited the duration of new water rights, with significant implications for mining.

Many jurisdictions prohibit or restrict mining in protected areas, such as national parks, wetlands, and areas of high biodiversity value. Colombia's prohibition of mining in páramo ecosystems, upheld by its Constitutional Court in a 2016 decision and challenged in the *Eco Oro* arbitration discussed in section

7.3, is an example. International instruments such as the Ramsar Convention on Wetlands (1971), the World Heritage Convention (1972), and the Convention on Biological Diversity (1992) inform these restrictions, and the Kunming-Montreal Global Biodiversity Framework, adopted in December 2022, sets the target of effectively conserving thirty per cent of the world's terrestrial and marine areas by 2030 (the "30 by 30" target), which may constrain the availability of land for mining. The International Council on Mining and Metals (ICMM) committed its members in 2003 not to explore or mine in World Heritage properties, a commitment that has been influential in the industry.

Tailings Governance

The tailings dam disasters of the past decade have transformed the governance of tailings facilities. Following the Brumadinho disaster, Brazil enacted Law No 14.066 of 2020, amending its National Dam Safety Policy, which prohibits the construction of "upstream" tailings dams — a design, in which successive embankments are built on top of previously deposited tailings, that was used at both Fundão and Brumadinho — and requires the decommissioning of existing upstream dams. Chile and Peru had prohibited upstream dams in seismically active areas some years earlier.

At the international level, the **Global Industry Standard on Tailings Management** (GISTM), published in August 2020 following an independent review co-convened by the ICMM, the United Nations Environment Programme, and the Principles for Responsible Investment, establishes requirements for the safe management of tailings facilities, with the goal of zero harm to people and the environment. It requires operators to take a precautionary approach, to design and operate facilities to minimise risk, to establish accountability at board level, to engage meaningfully with affected communities, to prepare for emergencies, and to disclose information about their facilities. ICMM members committed to implementing the standard at their highest-consequence facilities by August 2023 and at all other facilities by August 2025. Although the GISTM is a voluntary industry standard, it is increasingly referenced by regulators, investors, and insurers, and its requirements may inform the standard of care in negligence litigation.

Litigation Arising from Mining Disasters

The legal consequences of the Fundão disaster illustrate the multiple layers of liability arising from mining catastrophes. The collapse of the dam, operated by Samarco Mineração SA, a joint venture between Vale and BHP, released about forty million cubic metres of tailings into the Doce River basin, killing nineteen people and causing extensive environmental and economic damage along hundreds of kilometres of river to the Atlantic coast. In Brazil, the companies entered into a settlement in 2016 establishing the Renova Foundation to carry out reme-

diation and compensation programmes, and in October 2024 the companies and Brazilian public authorities signed a revised settlement valued at approximately BRL 170 billion. In England, more than six hundred thousand claimants, including municipalities, businesses, and individuals, brought proceedings against BHP's English and Australian parent companies. The Court of Appeal in *Municipio de Mariana v BHP Group plc* (2022) reversed the High Court's decision striking out the claims as an abuse of process, allowing them to proceed,[7] and a liability trial took place in 2024 and 2025. The case is one of the largest group actions ever brought in the English courts and is examined further, alongside parent company liability generally, in Chapters 21 and 27.

Social Obligations

Mining codes and licences impose a range of social obligations on title holders, including obligations to employ and train nationals, to procure goods and services locally, to respect labour and occupational health and safety standards, to protect cultural heritage, and to contribute to community development. The destruction by Rio Tinto in May 2020 of the 46,000-year-old rock shelters at Juukan Gorge in the Pilbara region of Western Australia, sacred to the Puutu Kunti Kurrama and Pinikura peoples, although lawful under a ministerial consent granted under the Aboriginal Heritage Act 1972 (WA), led to the resignation of Rio Tinto's chief executive, a parliamentary inquiry, and the enactment of the Aboriginal Cultural Heritage Act 2021 (WA). That Act was itself repealed in 2023 after objections from landholders that its requirements were unworkable, and the 1972 Act was restored with amendments. The episode illustrates the gap between legal compliance and social acceptability, and the speed with which reputational and investor pressures can transform corporate practice.

Occupational health and safety in mining is governed by domestic legislation and by international standards, notably the ILO Safety and Health in Mines Convention 1995 (No 176), which requires ratifying states to adopt and enforce laws ensuring safe working conditions in mines, to establish inspection systems, and to confer rights on workers, including the right to remove themselves from danger. Labour rights in extractive industries are examined in Chapter 21.

7.5 Community Development and Benefit Sharing

The Rationale for Benefit Sharing

Communities located near mining operations bear a disproportionate share of the costs of mining — loss of land and livelihoods, environmental degradation, social disruption, and health risks — while the principal benefits, in the form of royalties and taxes, accrue to the central government. This mismatch fuels

[7] *Municipio de Mariana v BHP Group (UK) Ltd* [2022] EWCA Civ 951.

conflict between companies, communities, and governments, and has led to the development of legal mechanisms for sharing the benefits of mining with affected communities. Benefit sharing may take the form of fiscal transfers from central to local government, direct payments or equity participation for communities, local employment and procurement commitments, and investments in community infrastructure and services. These mechanisms aim not only to compensate communities for the costs they bear but also to secure their support for mining operations — the "social licence to operate".

Community Development Agreements

The community development agreement (CDA) — also called an impact and benefit agreement (IBA) in Canada, an indigenous land use agreement in Australia, or a social and labour plan in South Africa — is a contract between a mining company and a host community that sets out the company's obligations to the community and the community's support for the project. CDAs may be voluntary or mandated by law.

Section 116 of Nigeria's Minerals and Mining Act 2007 requires the holder of a mining lease, small-scale mining lease, or quarry lease to conclude a community development agreement with the host community before commencing development. The CDA must contain undertakings with respect to the social and economic contribution that the project will make to the sustainability of the community, and may address educational scholarships, apprenticeships, technical training, and employment opportunities for indigenes of the community; financial or other forms of contributory support for infrastructural development and maintenance, such as education, health, or other community services, roads, water, and power; assistance with the creation, development, and support of small-scale and micro-enterprises; agricultural product marketing; and methods and procedures of environmental and socio-economic management and local governance enhancement. The agreement must be reviewed every five years, and the Mines Inspectorate Department of the Ministry monitors compliance. The Nigerian Minerals and Mining Regulations 2011 elaborate the procedures for negotiating CDAs.

The CDA model has been adopted in many other African mining codes, including those of Sierra Leone (Mines and Minerals Act 2009), Guinea (Mining Code 2011, as amended in 2013), Kenya (Mining Act 2016), and Mozambique. The Kenyan Mining Act 2016 also provides that a share of royalties shall be distributed to county governments and communities: seventy per cent to the national government, twenty per cent to the county government, and ten per cent to the community where the mining operations occur.

Challenges in Negotiating and Implementing CDAs

The effectiveness of CDAs depends on several factors that are frequently absent in practice. The first is the identification of the community: mining impacts may extend across multiple villages, ethnic groups, and administrative units, and the determination of which communities are parties to the agreement, and who represents them, is contentious. Traditional leaders may lack democratic legitimacy or may capture benefits for themselves or their families, and women, youth, and marginalised groups may be excluded. The second is capacity: communities typically lack the technical, legal, and financial expertise to negotiate effectively with mining companies, and the law rarely provides for independent advice to be funded. The third is enforceability: CDAs are often vaguely worded, lacking clear obligations, timeframes, and dispute resolution mechanisms, and communities may lack the means to enforce them through the courts. The fourth is sustainability: community development projects may be poorly designed, may create dependency, or may cease when the mine closes.

Good practice guidance, such as the World Bank's *Mining Community Development Agreements: Source Book* (2012), emphasises the importance of inclusive representation, independent advice for communities, clear and measurable commitments, transparent governance of community funds, monitoring and evaluation, and accessible grievance mechanisms.

Fiscal Decentralisation and Revenue Sharing

In addition to direct agreements, benefit sharing occurs through the allocation of mineral revenues to subnational governments. Nigeria's federal revenue allocation formula provides for thirteen per cent of revenues derived from natural resources to be allocated to the states from which they derive (the "derivation principle" in the proviso to section 162(2) of the Constitution), although in practice the derivation principle has been applied principally to oil revenues. In Ghana, the Minerals Development Fund Act 2016 (Act 912) provides for a share of mineral royalties to be allocated to mining communities and local authorities. In Peru, the *canon minero* allocates fifty per cent of the income tax paid by mining companies to regional and local governments in the producing areas. The design of revenue sharing must balance the claims of producing regions against those of the nation as a whole, and must address the capacity of subnational governments to manage revenues effectively. Resource revenue distribution is examined further in Chapter 23.

Equity Participation by Communities

Some regimes provide for community equity participation in mining projects. South Africa's Mining Charter, issued under the MPRDA, has required black economic empowerment ownership of mining companies, including a share for

host communities and employees. The 2018 Mining Charter required new mining rights to have a minimum of thirty per cent black economic empowerment shareholding, including a carried interest of five per cent for host communities. In *Minerals Council of South Africa v Minister of Mineral Resources and Energy* (2021), the High Court held that the Mining Charter was a policy instrument rather than binding legislation and set aside several of its provisions, including those purporting to require the continuation of empowerment ownership after historical transactions had been completed.[8] The case illustrates the limits of policy instruments as a basis for legally binding obligations.

7.6 Mine Closure and Rehabilitation

The Closure Challenge

Every mine eventually closes, whether because the ore body is exhausted, because commodity prices fall below the cost of production, or because of accidents, disputes, or regulatory action. Closure is not the end of a mine's legal life. Open pits, underground workings, waste rock dumps, and tailings facilities remain, and may require monitoring, maintenance, and treatment of water for decades or even in perpetuity. Historically, many mines were simply abandoned at the end of their lives, leaving a legacy of contaminated land and water, unsafe structures, and stranded communities. Thousands of abandoned mines in the United States, Canada, Australia, South Africa, and Europe, and many more in developing countries, impose ongoing costs on governments and communities. In Nigeria, the legacy of colonial and post-colonial tin and columbite mining on the Jos Plateau has left extensive areas of mine ponds and spoil heaps that have affected agriculture and public health. The modern law of mine closure seeks to ensure that closure is planned from the outset, that the costs are borne by the operator rather than the public, and that land is rehabilitated to a safe, stable, and productive condition.

Closure Planning

Modern mining regimes require mining operators to prepare closure plans as part of their applications for mining licences, and to update them periodically during the life of the mine. A closure plan typically identifies the post-mining land use objectives, agreed where possible with affected communities and authorities; describes the measures for decommissioning infrastructure, stabilising landforms, revegetating disturbed land, managing water, and securing tailings facilities; estimates the cost of closure; and establishes criteria by which successful completion of closure will be assessed. Progressive rehabilitation — the rehabilitation of

[8] *Minerals Council of South Africa v Minister of Mineral Resources and Energy* (High Court, Gauteng Division, Pretoria, 21 September 2021).

disturbed areas during the life of the mine, rather than at the end — is increasingly required, since it reduces the eventual closure liability and demonstrates the operator's commitment.

Closure plans must also address the social dimensions of closure: the loss of employment, the loss of revenue for local governments and suppliers, and the loss of services, such as health clinics, schools, and water supply, that the mine may have provided to the community. Social closure planning seeks to reduce the dependence of communities on the mine during its operating life and to support economic diversification. The ICMM's *Integrated Mine Closure: Good Practice Guide* (2nd edn, 2019) addresses both environmental and social dimensions.

Financial Assurance

The central legal problem of mine closure is ensuring that funds are available to carry out closure when it is required. Operators may become insolvent, particularly at the end of a mine's life when revenues decline; parent companies may dispose of subsidiaries holding closure liabilities; and assets may be sold to under-capitalised purchasers. Mining laws therefore require operators to provide financial assurance — security for the estimated cost of closure — which the state may call upon if the operator fails to carry out its obligations. Forms of financial assurance include cash deposits, trust funds, bank guarantees and letters of credit, insurance bonds, and, in some jurisdictions, corporate guarantees or self-bonding (reliance on the financial strength of the operator itself). Self-bonding has been criticised as inadequate, following the bankruptcies of several major United States coal companies in 2015 and 2016, which left states facing large unfunded reclamation liabilities.

In Nigeria, the Minerals and Mining Act 2007 requires the holder of a mineral title to contribute to an Environmental Protection and Rehabilitation Fund, established to guarantee the environmental obligations of holders, in amounts determined by reference to the approved EPRP. South Africa's National Environmental Management Act 1998, as amended, and the Financial Provisioning Regulations made under it require holders of mining rights to make financial provision for the rehabilitation and remediation of environmental impacts, including latent and residual impacts, and to assess and adjust the provision annually. Western Australia's Mining Rehabilitation Fund Act 2012 established a pooled fund to which most tenement holders contribute an annual levy, used to rehabilitate abandoned mine sites, replacing the previous system of unconditional performance bonds for most tenements.

Relinquishment and Completion Criteria

A critical question in mine closure law is when the operator's obligations end. Most regimes require the operator to demonstrate that closure objectives have

been met and to obtain a certificate of closure or relinquishment from the regulator, after which the financial assurance is released and responsibility for the site passes to the state. Regulators are often reluctant to accept relinquishment, because of the risk of latent impacts emerging later, while operators seek certainty. Some regimes provide for residual liability funds or for long-term stewardship arrangements to manage sites that require perpetual care, such as those with acid mine drainage requiring permanent water treatment. The development of clear, measurable, and agreed completion criteria is essential to the effective operation of the closure regime.

Closure and the Energy Transition

The energy transition creates both new closure challenges and new opportunities. The phase-out of coal mining in many countries — in Germany, the Coal Phase-out Act (Kohleausstiegsgesetz) of 2020 provides for the end of coal-fired power generation by 2038 at the latest, and in the United Kingdom the last coal-fired power station closed in 2024 — requires the closure and rehabilitation of large mining areas and support for affected communities, issues addressed by just transition policies examined in Chapters 18 and 23. Former mine sites may be repurposed for renewable energy generation, pumped hydro storage, or other uses, and mine waste may contain recoverable critical minerals. The reprocessing of tailings to recover minerals and the regulation of "mining waste as a resource" are emerging areas of law, examined in Chapter 8.

7.7 Artisanal and Small-Scale Mining Regulation

The Scale and Significance of ASM

Artisanal and small-scale mining (ASM) — mining by individuals, groups, families, or cooperatives with minimal mechanisation, often in the informal sector — is a major economic activity in many developing countries. The World Bank has estimated that around forty-five million people are directly engaged in ASM worldwide, and many more depend on it indirectly. ASM produces a significant share of the world's gold, tin, tantalum, tungsten, coloured gemstones, and diamonds, and a substantial share of the world's cobalt is produced by artisanal miners in the Democratic Republic of the Congo. For many rural households, ASM is a vital source of income and a buffer against poverty and agricultural failure.

At the same time, ASM is associated with serious problems: dangerous working conditions and frequent fatalities; child labour; environmental degradation, including deforestation, river siltation, and mercury pollution; conflict and the financing of armed groups; tax evasion and smuggling; and conflict with large-scale mining companies over overlapping rights. The use of mercury to amalgamate gold in artisanal and small-scale gold mining (ASGM) is the largest source of an-

thropogenic mercury emissions globally. In Nigeria, the lead poisoning epidemic in Zamfara State in 2010, caused by the artisanal processing of lead-rich gold ore in villages, killed several hundred children and was one of the worst lead poisoning outbreaks in modern history, prompting a major remediation effort.

Legal Approaches to ASM

Historically, many mining laws ignored ASM or treated it as illegal, focusing exclusively on large-scale industrial mining. Criminalisation has generally failed: it drives ASM underground, exposes miners to extortion and violence, deprives the state of revenue, and makes it impossible to address environmental and safety problems. The prevailing approach in contemporary mining law is formalisation — bringing ASM within the legal framework through simplified licensing, technical and financial support, and integration into formal supply chains.

Formalisation strategies typically include: the creation of a distinct category of ASM licences or permits, with simplified application procedures, lower fees, and reduced technical requirements; the designation of areas reserved for ASM; the encouragement of cooperatives or associations to facilitate licensing, training, and access to finance; the provision of geological information, extension services, and equipment; the establishment of official buying centres to purchase ASM production, particularly of gold, at fair prices; and the integration of environmental, health, and safety requirements adapted to the scale of operations.

Nigeria's Minerals and Mining Act 2007 provides for the grant of small-scale mining leases to individuals, cooperatives, and companies, and requires artisanal miners to form cooperatives in order to obtain titles. The Act established the Artisanal and Small-Scale Mining Department within the Ministry to provide extension services, training, and support. The Presidential Artisanal Gold Development Initiative (PAGDI), launched in 2019, sought to formalise artisanal gold mining by establishing a framework under which the Central Bank of Nigeria would purchase gold from artisanal miners through licensed buying centres for inclusion in the country's external reserves. In 2024 the federal government established a Mining Marshals unit within the Nigeria Security and Civil Defence Corps to combat illegal mining, reflecting concern over the links between illegal mining, insecurity, and the financing of armed groups in the north-west.

In Ghana, illegal artisanal mining, known as *galamsey*, has become a major political issue because of its devastating impact on rivers and forests. The Minerals and Mining (Amendment) Act 2019 (Act 995) increased penalties for illegal mining, and the government has periodically imposed bans on small-scale mining and deployed military task forces, with limited long-term success. The experience demonstrates the difficulty of reconciling the livelihood needs of artisanal miners with environmental protection where governance is weak.

The Minamata Convention

The **Minamata Convention on Mercury**, adopted in 2013 and in force since 2017, addresses mercury use in ASGM. Article 7 requires each party with ASGM within its territory that is more than insignificant to take steps to reduce, and where feasible eliminate, the use of mercury and mercury compounds in, and the emissions and releases to the environment of mercury from, such mining and processing. Such parties must develop and implement a national action plan in accordance with Annex C, which must include national objectives and reduction targets, actions to eliminate the worst practices (whole ore amalgamation, open burning of amalgam, burning of amalgam in residential areas, and cyanide leaching in sediment, ore, or tailings to which mercury has been added without first removing the mercury), steps to facilitate formalisation, and strategies for promoting the reduction of mercury use. Nigeria is a party to the Convention and has developed a national action plan for ASGM.

ASM and Large-Scale Mining

Conflicts between ASM and large-scale mining (LSM) are common, particularly where large-scale companies obtain titles over areas historically worked by artisanal miners. Such conflicts have led to violence, including forced evictions and abuses by security forces, examined in Chapter 21. Approaches to managing ASM–LSM coexistence include designating portions of large-scale concessions for ASM, establishing partnerships in which large-scale companies purchase ASM production or provide technical support, and resolving disputes through dialogue mechanisms. The Voluntary Principles on Security and Human Rights, examined in section 7.8, address the conduct of security forces in such situations.

7.8 International Standards for Responsible Mining

The Proliferation of Standards

Mining is governed not only by domestic law and international treaties but by a dense and growing body of international standards developed by international financial institutions, industry associations, multi-stakeholder initiatives, and civil society. These standards address environmental and social performance, human rights, transparency, supply chain due diligence, and governance. Although most are not legally binding in themselves, they acquire legal significance in several ways: through incorporation into loan agreements and investment conditions; through incorporation into domestic law and licence conditions; through reference in contracts between companies and their customers; through their use by courts and tribunals as evidence of the standard of care or of good industry practice; and through the reputational and market pressures that lead companies to commit to them publicly. This section surveys the principal standards; supply chain due diligence standards specific to minerals, notably the OECD Due

Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas, are examined in Chapter 8.

International Finance Corporation Performance Standards and the Equator Principles

The **IFC Performance Standards on Environmental and Social Sustainability**, adopted by the International Finance Corporation (the private sector arm of the World Bank Group) and most recently revised in 2012, are the most influential environmental and social standards for project finance in emerging markets. They comprise eight standards: PS1 (Assessment and Management of Environmental and Social Risks and Impacts); PS2 (Labor and Working Conditions); PS3 (Resource Efficiency and Pollution Prevention); PS4 (Community Health, Safety, and Security); PS5 (Land Acquisition and Involuntary Resettlement); PS6 (Biodiversity Conservation and Sustainable Management of Living Natural Resources); PS7 (Indigenous Peoples); and PS8 (Cultural Heritage). The Performance Standards are supported by Guidance Notes and by the World Bank Group's Environmental, Health, and Safety (EHS) Guidelines, including industry-specific guidelines for mining. Compliance with the Performance Standards is a condition of IFC financing, and the IFC's Compliance Advisor Ombudsman (CAO) receives complaints from communities affected by IFC-financed projects.

The **Equator Principles**, first adopted in 2003 by a group of commercial banks and now in their fourth iteration (EP4, effective from October 2020), commit participating financial institutions — over one hundred banks in more than thirty countries — to apply the IFC Performance Standards (and, in designated countries with robust governance, host country law) to project finance transactions above a specified threshold. EP4 strengthened requirements in relation to human rights, climate change risk assessment, and free, prior, and informed consent of indigenous peoples. Through the Equator Principles, the IFC Performance Standards have become a de facto global benchmark for the environmental and social performance of major mining projects, regardless of whether the IFC itself is involved.

The decision of the United States Supreme Court in *Jam v International Finance Corp* (2019), holding that international organisations such as the IFC enjoy only the same restrictive immunity from suit as foreign states under the Foreign Sovereign Immunities Act, opened the possibility of litigation against the IFC in United States courts by communities harmed by projects it financed, although the subsequent proceedings in that case, concerning a coal-fired power plant in India, were dismissed on other grounds.[9]

[9] *Jam v International Finance Corp*, 586 US 199 (2019).

Industry Standards

The **International Council on Mining and Metals** (ICMM), whose members include most of the world's largest mining companies, has adopted Mining Principles comprising ten principles and associated performance expectations covering ethical business, decision-making, human rights, risk management, health and safety, environmental performance, conservation of biodiversity, responsible production, social performance, and stakeholder engagement. Members must implement the principles, report on their performance, and obtain independent assurance. ICMM has also adopted position statements on indigenous peoples and mining (committing members to work to obtain the consent of indigenous peoples for new projects), on climate change, on water stewardship, and on tailings governance, and co-convened the development of the Global Industry Standard on Tailings Management discussed in section 7.4.

National industry initiatives include the Mining Association of Canada's **Towards Sustainable Mining** (TSM) programme, launched in 2004 and adopted by mining associations in several other countries, which requires member companies to assess and report their performance against protocols on tailings management, indigenous and community relationships, biodiversity, climate change, safety and health, crisis management, water stewardship, and preventing child and forced labour, with external verification.

Multi-Stakeholder Standards

The **Initiative for Responsible Mining Assurance** (IRMA), established in 2006 by a coalition of non-governmental organisations, affected communities, labour unions, mining companies, and purchasers of minerals, has developed the IRMA Standard for Responsible Mining, which provides for independent third-party audits of mine sites against comprehensive requirements covering business integrity, planning for positive legacies, social responsibility, and environmental responsibility. IRMA is distinctive in its equal governance by six stakeholder houses and in its requirement for site-level rather than corporate-level assessment. Purchasers of minerals, including automotive and electronics manufacturers, increasingly require their suppliers to undergo IRMA audits.

The **Extractive Industries Transparency Initiative** (EITI), launched in 2003, requires implementing countries to publish information on the extractive sector value chain, including the award of contracts and licences, beneficial ownership, production, revenue collection, revenue allocation, and social and economic spending, and to reconcile company payments with government receipts. Nigeria implements the EITI through the Nigeria Extractive Industries Transparency Initiative (NEITI) Act 2007, which makes Nigeria one of the few countries to have given EITI a statutory basis, and NEITI publishes regular audits

of the oil, gas, and solid minerals sectors. The EITI is examined further in Chapter 16 in the context of revenue transparency.

The **Kimberley Process Certification Scheme**, established in 2003 following United Nations General Assembly resolutions on "conflict diamonds" that financed civil wars in Angola, Sierra Leone, Liberia, and the Democratic Republic of the Congo, requires participating states to ensure that shipments of rough diamonds are accompanied by government-validated certificates attesting that they are conflict-free, and to trade rough diamonds only with other participants. The Kimberley Process has been criticised for its narrow definition of conflict diamonds, which is limited to diamonds used by rebel movements to finance wars against legitimate governments and does not extend to human rights abuses by governments or security forces, and for its consensus-based decision-making, which has blocked proposals for reform.

The **Voluntary Principles on Security and Human Rights**, adopted in 2000 by the governments of the United States and the United Kingdom, extractive companies, and non-governmental organisations, provide guidance to extractive companies on maintaining the safety and security of their operations within a framework that ensures respect for human rights. They address risk assessment, relations with public security forces (police and military), and relations with private security providers, and they have been incorporated into some host government agreements, security arrangements, and financing conditions.

Regional Frameworks

At the regional level, the **Africa Mining Vision** (2009) and its Action Plan provide a policy framework for African mining that emphasises linkages between mining and the broader economy, beneficiation and value addition, artisanal mining, human and institutional capacity, and the management of mineral revenues. The African Minerals Development Centre, established to support implementation, has developed guidance on Country Mining Visions. In West Africa, the **ECOWAS Directive on the Harmonisation of Guiding Principles and Policies in the Mining Sector** (C/DIR.3/05/09), adopted in 2009, requires member states to harmonise their mining laws in accordance with common principles concerning human rights, transparency, environmental protection, community development, and the rights of local communities, including requirements for free, prior, and informed consent of local communities before exploration and prior to each subsequent phase of mining operations. The ECOWAS Directive is notable as one of the first binding regional instruments to require community consent in mining, although its implementation in member states has been uneven. The **ECOWAS Mining and Minerals Development Policy** of 2011 elaborates these principles.

Evaluating the Standards Landscape

The proliferation of standards raises questions of legitimacy, coherence, and effectiveness. Standards developed by industry associations may be criticised as self-serving or lacking independent verification; multi-stakeholder standards may be more credible but are costly to implement; and the coexistence of multiple overlapping standards imposes compliance burdens and may confuse stakeholders. Standards are, moreover, most effective in relation to large companies with reputational exposure and access to international capital markets, and least effective in relation to state-owned enterprises, companies from jurisdictions with weaker accountability mechanisms, and the informal sector. Their legal status is ambiguous: they are not law in the traditional sense, yet they shape behaviour, inform legal standards, and may become binding through contract and regulation. The relationship between such transnational private regulation and formal law, analysed in Chapter 2, is one of the most important areas for doctoral research in natural resources law.

LEADING AUTHORITY

Reko Diq: From the Supreme Court of Pakistan to ICSID (2013–2022)

Facts. In 1993, the Balochistan Development Authority concluded the Chagai Hills Exploration Joint Venture Agreement (CHEJVA) with BHP Minerals for exploration of the Reko Diq area. BHP's interest was later acquired by Tethyan Copper Company, owned jointly by Barrick Gold and Antofagasta. Tethyan discovered a world-class copper-gold deposit and, after completing a feasibility study, applied in 2011 for a mining lease, which the Balochistan authorities refused.

Domestic proceedings. In January 2013, the Supreme Court of Pakistan, in proceedings brought by public interest petitioners, declared the CHEJVA void *ab initio* on the ground that it had been concluded in violation of the applicable mining rules and in excess of the authority of the officials who signed it.

International proceedings. Tethyan commenced ICSID arbitration under the Australia–Pakistan bilateral investment treaty. In its 2017 decision on jurisdiction and liability, the tribunal held that the refusal of the mining lease was motivated by the provincial government's intention to develop the project itself and breached the fair and equitable treatment standard and the prohibition of unlawful expropriation; it rejected Pakistan's allegations that the investment had been procured through corruption, finding them unproven. In July 2019, the tribunal awarded damages of approximately USD 4.1 billion plus interest, bringing the total to approximately USD 5.9 billion. Pakistan's application for annulment was pending when, in 2022, the parties reached a settlement under which Barrick would develop the project with the federal and Balochistan governments and Pakistani state-owned enterprises, and the award would be discharged. The Supreme Court of Pakistan, in an advisory opinion in December 2022, held that the new arrangements were not contrary to law or to its 2013 judgment.

Significance. The case illustrates the interaction between domestic and international legal orders in mining disputes: a domestic judgment invalidating a mining agreement did not prevent an international tribunal from finding the state liable for breaching treaty protections. It raises questions about the valuation of undeveloped mining projects, the role of corruption allegations in investment arbitration, and the use of settlements to convert awards into renegotiated partnerships.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Regulatory compliance review of a mining lease application (hypothetical). *The company, community, and facts in this exercise are fictitious.* Ogbo Minerals Ltd, a fictional Nigerian company with a foreign parent, holds an exploration licence over 40 square kilometres in a fictional local government area in Nigeria's north-central zone and has identified a commercially viable lithium-bearing pegmatite deposit. Part of the deposit is worked by an estimated 1,500 artisanal miners organised loosely into three informal groups, and a seasonal stream used by two villages for drinking water crosses the area. The company has prepared a feasibility study and wishes to apply for a mining lease.

Acting as counsel to the company, prepare a compliance memorandum (not more than 3,000 words) that (a) identifies the approvals, documents, and agreements required under Nigerian law before development may commence, with reference to the Minerals and Mining Act 2007, the Nigerian Minerals and Mining Regulations 2011, and the Environmental Impact Assessment Act; (b) advises on the negotiation of a community development agreement, including identification of the host community and representation of women and youth; (c) proposes a strategy for engagement with the artisanal miners consistent with Nigerian law and the IFC Performance Standards; and (d) identifies the international standards that prospective purchasers of the company's lithium concentrate are likely to require it to meet.

CHAPTER SUMMARY

- Mineral ownership is vested in the state in most jurisdictions; in Nigeria, section 44(3) of the Constitution and section 1 of the Minerals and Mining Act 2007 vest minerals in the Federal Government, while South Africa's MPRDA treats minerals as the common heritage of the people under state custodianship.
- Mining titles follow the mining life cycle — reconnaissance, exploration, mining — and are allocated by first-come, first-served, competitive tender, or discretion, ideally through a transparent electronic cadastre.
- Security of tenure, linking exploration to the right to mine, is central to investment; its denial has generated major awards, including *Crystallex* and *Tethyan Copper*.

- Mining agreements and stability clauses have been challenged by waves of resource nationalism, as in the DRC's 2018 Code and Tanzania's 2017 legislation; corruption can vitiate titles, as in the Simandou dispute.
- Environmental obligations include EIA, water and biodiversity protection, and tailings governance, transformed by the Brumadinho and Mariana disasters and the GISTM; social obligations include cultural heritage protection, highlighted by Juukan Gorge.
- Community development agreements, mandated by section 116 of the Nigerian Act, and revenue sharing mechanisms seek to share mining benefits with affected communities, but face challenges of representation, capacity, and enforceability.
- Mine closure law requires closure planning, financial assurance, and completion criteria to prevent public liability for abandoned mines.
- Artisanal and small-scale mining is best addressed through formalisation; the Minamata Convention requires action plans to reduce mercury use in ASGM.
- International standards — IFC Performance Standards, Equator Principles, ICMM, IRMA, EITI, Kimberley Process, Voluntary Principles, and the ECOWAS Mining Directive — increasingly shape mining practice and law.

DOCTORAL RESEARCH QUESTIONS

1. Is security of tenure compatible with the state's continuing duty to regulate mining in the public interest? How should mining codes structure the transition from exploration to mining to balance these interests?
2. Evaluate the effectiveness of mandatory community development agreements under section 116 of Nigeria's Minerals and Mining Act 2007, using empirical evidence from at least two mining regions.
3. To what extent should investment tribunals take account of an investor's failure to obtain a "social licence to operate" in determining liability and damages? Analyse *Bear Creek*, *Copper Mesa*, and *Gabriel Resources*.
4. Has the Global Industry Standard on Tailings Management altered the legal standard of care owed by mining companies and their parent companies to communities downstream of tailings facilities?
5. Assess the ECOWAS Mining Directive's requirement of community consent as a model for regional harmonisation, and explain the gap between its terms and member state implementation.
6. Should the formalisation of artisanal mining be pursued through mining law, labour law, environmental law, or a dedicated legal framework?

FURTHER READING

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8 CHAPTER

Critical Minerals, Strategic Resources, and Supply Chain Security

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain how states and international organisations define and classify "critical" and "strategic" minerals, and the legal consequences of such classification.
2. Analyse the distinctive legal and policy issues associated with lithium, cobalt, nickel, copper, and rare earth elements.
3. Evaluate the legal regimes governing state ownership of strategic minerals and export controls on minerals and processing technologies, including their compatibility with international trade law.
4. Assess the use of industrial policy to secure mineral supply chains, including the EU Critical Raw Materials Act and United States legislation.
5. Analyse mandatory and voluntary frameworks for mineral supply chain due diligence.
6. Identify the human rights and labour risks associated with critical mineral production and the legal mechanisms for addressing them.
7. Evaluate the regulation of mineral processing, recycling, and the circular economy.
8. Critically assess the emerging architecture of critical mineral agreements and the geopolitical competition for mineral resources.

Introduction

The energy transition is, among other things, a transition from a fuel-intensive energy system to a material-intensive one. A conventional car requires little more than steel, aluminium, and copper; an electric car requires, in addition, around two hundred kilograms of minerals for its battery and motor, including lithium, nickel, cobalt, manganese, graphite, and rare earth elements. An onshore wind plant requires around nine times more mineral resources than a gas-fired power plant of the same capacity. Solar photovoltaic panels require silicon, silver, copper,

and, in some technologies, tellurium, indium, and gallium. Electricity networks, which must expand dramatically to accommodate renewable generation and electrification, require vast quantities of copper and aluminium. The International Energy Agency has projected that, in scenarios consistent with the goals of the Paris Agreement, demand for minerals for clean energy technologies could increase several-fold by 2040, with demand for lithium potentially growing by a factor of more than ten.

The minerals on which the transition depends are characterised by a combination of features that generate distinctive legal and political challenges. Their production is geographically concentrated: the Democratic Republic of the Congo produces around seventy per cent of the world's cobalt; Australia, Chile, and China dominate lithium production; Indonesia has become the world's largest producer of nickel; and China accounts for around sixty to seventy per cent of rare earth mining. Processing is even more concentrated: China refines the majority of the world's lithium, cobalt, graphite, and rare earths, and controls most of the world's capacity to manufacture battery cells and permanent magnets. Supply is relatively inelastic, because new mines typically take ten to fifteen years or more from discovery to production. Many deposits are located in countries with weak governance, high levels of poverty, and histories of conflict, and in or near territories of indigenous peoples. Production is associated with serious environmental impacts and, in some contexts, with child labour, forced labour, and other human rights abuses.

These features have transformed critical minerals from a niche concern of defence planners and materials scientists into a central preoccupation of energy, trade, industrial, and foreign policy. Governments in importing states have adopted critical mineral strategies, legislation, and international partnerships aimed at diversifying supply and reducing dependence on China. Governments in producing states, particularly in Africa and Latin America, have sought to capture a greater share of the value of their mineral resources by requiring domestic processing, restricting exports of unprocessed ores, increasing state participation, and renegotiating agreements. China, for its part, has used export controls on minerals and processing technologies as instruments of economic statecraft. The legal regime of critical minerals sits at the intersection of mining law (examined in Chapter 7), trade law (Chapter 13), investment law (Chapter 14), human rights law (Chapter 21), and the law of geopolitics and national security (Chapter 25), and it is evolving with extraordinary speed.

This chapter examines the legal classification of critical and strategic minerals (section 8.1); the distinctive issues associated with lithium, cobalt, nickel, copper, and rare earth elements (section 8.2); strategic mineral ownership and export controls (section 8.3); resource security and industrial policy (section 8.4); supply

chain due diligence (section 8.5); human rights and labour risks in mineral production (section 8.6); processing, recycling, and circular economy regulation (section 8.7); and geopolitical competition and critical mineral agreements (section 8.8). The institutional dimension of critical minerals governance, including the role of the International Energy Agency, was introduced in Chapter 4.

8.1 Legal Classification of Critical and Strategic Minerals

The Concept of Criticality

There is no universally accepted definition of a "critical mineral". The concept of criticality, as developed in the policy literature, generally combines two dimensions: the *economic importance* of a mineral — the extent to which it is essential to economic activity, technologies, or national security, and the difficulty of substituting it — and its *supply risk* — the likelihood of disruption to its supply, arising from geological scarcity, geographic concentration of production and processing, political instability or poor governance in producing countries, export restrictions, low rates of recycling, and similar factors. A mineral is "critical" if it scores highly on both dimensions. The related concept of a "strategic" mineral, which has its origins in military planning during and after the Second World War, emphasises importance for national defence or for strategic sectors of the economy, such as the energy transition, digital technologies, and aerospace.

Criticality assessments are contingent and dynamic. They depend on the perspective of the assessing state (a mineral critical to an importing state may be of little concern to a producing state), on technological developments (new battery chemistries may reduce or increase demand for particular minerals), and on market and geopolitical conditions. Most jurisdictions therefore review their lists periodically.

Classification in Major Jurisdictions

The European Union. The European Commission first published a list of critical raw materials in 2011 and has updated it every three years since. The Critical Raw Materials Act — Regulation (EU) 2024/1252, which entered into force in May 2024 — placed the classification on a legislative footing. Annex II lists thirty-four critical raw materials, assessed on the basis of economic importance and supply risk according to a methodology set out in the Regulation, and Annex I lists seventeen strategic raw materials, selected from among the critical raw materials on the basis of their importance for technologies supporting the green and digital transitions and defence and aerospace applications, the projected growth in demand, and the difficulty of increasing production. The strategic list includes lithium, nickel (battery grade), cobalt, natural graphite, manganese (battery grade), copper, rare earth elements for magnets, gallium, germanium, silicon metal, tungsten, and platinum group metals. The classification has legal

consequences: strategic raw materials are the subject of the Act's benchmarks for domestic capacity and diversification and are eligible for strategic project status, examined in section 8.4.

The United States. In the United States, the Energy Act of 2020 defines a "critical mineral" as a non-fuel mineral or mineral material essential to the economic or national security of the United States, the supply chain of which is vulnerable to disruption, and that serves an essential function in the manufacturing of a product, the absence of which would have significant consequences for the economic or national security of the United States. The Act requires the Secretary of the Interior, acting through the United States Geological Survey (USGS), to publish and periodically review a list of critical minerals. The 2022 list identified fifty critical minerals, and the USGS's subsequent review in 2025 expanded the list, adding minerals including copper. Separately, the Department of Energy maintains a list of "critical materials" for energy, and the Department of Defense manages the National Defense Stockpile of strategic and critical materials under the Strategic and Critical Materials Stock Piling Act. The classification has legal consequences in, among other things, the eligibility requirements for the clean vehicle tax credit under section 30D of the Internal Revenue Code (as amended by the Inflation Reduction Act of 2022), streamlined permitting, and federal financing.

Other jurisdictions. Australia, Canada, Japan, India, the United Kingdom, South Korea, and China have also adopted lists of critical or strategic minerals. China's National Mineral Resources Plan (2016–2020) identified twenty-four strategic minerals, including oil, gas, coal, rare earths, lithium, cobalt, and graphite. Producing states have increasingly adopted their own classifications, often for different purposes: whereas importing states classify minerals as critical to identify supply risks, producing states classify them as strategic to justify enhanced state control, higher royalties, and local processing requirements. The Democratic Republic of the Congo's Mining Code of 2018 permits the government to declare minerals "strategic substances" subject to a ten per cent royalty, and in 2018 the government so classified cobalt, germanium, and columbite-tantalite (coltan).

The African Perspective

African institutions have increasingly articulated their own understanding of critical minerals, emphasising their potential to drive industrialisation rather than merely their role in supplying importing economies. The African Union's Africa Mining Vision (2009), discussed in Chapter 7, already emphasised beneficiation and linkages, and the African Union has since developed an African Green Minerals Strategy aimed at leveraging the continent's mineral endowment for green industrialisation, regional value chains, and economic transformation. The Africa

Finance Corporation, the African Development Bank, and the United Nations Economic Commission for Africa have supported initiatives to develop regional battery and electric vehicle value chains, such as the cooperation agreement between the Democratic Republic of the Congo and Zambia, signed in 2022, to develop a battery precursor value chain linking Congolese cobalt and Zambian copper and manganese.

In Nigeria, the Ministry of Solid Minerals Development has identified a number of minerals, including lithium, as priority minerals for development, and the federal government has adopted a policy of promoting local processing of lithium and other minerals rather than the export of raw ore. The first lithium processing plants in Nigeria, owned by Chinese investors and located in Kaduna and Nasarawa States, began operations in 2024.

Legal Consequences of Classification

Classification as critical or strategic may have a range of legal consequences, depending on the jurisdiction:

- **Permitting**: projects involving critical minerals may benefit from expedited permitting procedures, as under the EU Critical Raw Materials Act's strategic project regime and various United States initiatives.
- **Financing**: projects may be eligible for public financing, loan guarantees, or grants, as under Title III of the United States Defense Production Act and the programmes of export credit agencies.
- **Trade**: classification may support export restrictions in producing states or tariff and procurement preferences in importing states.
- **Investment screening**: foreign investment in critical mineral assets may be subject to enhanced scrutiny on national security grounds, as under the Canadian government's November 2022 policy on investments by state-owned enterprises in critical minerals, which led to orders requiring three Chinese companies to divest their interests in Canadian lithium companies.
- **Fiscal**: producing states may impose higher royalties on strategic minerals, as in the DRC.
- **Stockpiling**: governments may maintain strategic stockpiles of critical minerals, analogous to strategic petroleum reserves (examined in Chapter 25).

Because the label "critical" or "strategic" may trigger significant legal consequences, the methodology and transparency of classification are themselves legally significant, and the classification decisions of administrative agencies may be subject to judicial review.

8.2 Lithium, Cobalt, Nickel, Copper, and Rare Earth Elements

Each of the principal energy transition minerals has a distinctive geology, market structure, geography of production, and set of legal issues. This section examines five of the most significant.

Lithium

Geology and production. Lithium is produced from two principal types of deposit: hard-rock deposits, principally spodumene-bearing pegmatites mined in Australia (the world's largest producer), Zimbabwe, Nigeria, Brazil, and China; and brine deposits in the salt flats (*salares*) of the "lithium triangle" of Argentina, Bolivia, and Chile, from which lithium-rich brine is pumped into evaporation ponds and concentrated over many months. Direct lithium extraction technologies, which extract lithium from brine without evaporation ponds, are being developed and may reduce water consumption and environmental impacts. Lithium is processed into lithium carbonate or lithium hydroxide for use in battery cathodes, and the majority of the world's lithium chemical processing capacity is located in China.



Figure 8.1 Lithium brine evaporation ponds in a high-altitude salt flat. Illustrative image.

Legal regimes. Lithium has been subject to distinctive legal regimes in several Latin American states, reflecting its classification as a strategic resource. In Chile, lithium was declared a strategic mineral and "non-concessible" (not subject to ordinary mining concessions) by Decree-Law No 2.886 of 1979, on account of its potential use in nuclear fusion; lithium can be exploited only by the state, by state enterprises, or under special administrative contracts (*contratos especiales de operación de litio*, CEOLs) or leases granted by the state. The leases held by the private companies SQM and Albemarle over the Salar de Atacama, granted through the state development agency CORFO, have been the subject of repeated

renegotiation and litigation. In April 2023, the Chilean government announced a National Lithium Strategy providing for state control of strategic salt flats through public-private partnerships in which the state holds a majority, and in 2024 the state copper company Codelco concluded an agreement with SQM to form a joint venture, majority-owned by Codelco from 2031, for the exploitation of the Salar de Atacama.

In Bolivia, which holds some of the world's largest lithium resources in the Salar de Uyuni, the Constitution of 2009 and the state's model of resource sovereignty have reserved lithium exploitation to the state company Yacimientos de Litio Bolivianos (YLB), which has sought partnerships with foreign companies for direct lithium extraction technologies. In Mexico, a 2022 reform of the Mining Law declared lithium to be of public utility, reserved its exploration, exploitation, and use to the state, and created a state lithium company (LitioMx). In Argentina, by contrast, a federal country in which mineral resources belong to the provinces under Article 124 of the National Constitution, lithium is subject to ordinary mining concessions, and a combination of provincial ownership, a liberal investment regime, and the incentive regime for large investments (RIGI) introduced in 2024 has attracted substantial foreign investment.

Africa. Africa's lithium resources, principally in Zimbabwe, Namibia, the Democratic Republic of the Congo, Mali, Ghana, and Nigeria, have attracted significant investment, much of it from Chinese companies. In December 2022 Zimbabwe banned the export of unprocessed lithium ore, permitting exports only of concentrates, and announced its intention to ban the export of concentrates from 2027 to require domestic refining. Namibia similarly banned the export of unprocessed lithium ore in 2023. Ghana's first lithium mining lease, granted in 2023 to Barari DV Ghana (a subsidiary of the Australian company Atlantic Lithium) for the Ewoyaa project, provided for a royalty of ten per cent, a thirteen per cent free-carried interest for the state, and a further six per cent equity stake for the Minerals Income Investment Fund, and its ratification by Parliament was delayed amid debates over whether the terms secured sufficient value for Ghana. The DRC's Manono deposit, among the largest hard-rock lithium deposits in the world, has been the subject of a dispute between the Australian company AVZ Minerals and the Congolese state and state mining company Cominière, including ICC and ICSID proceedings.

Cobalt

Production and market. Cobalt is produced predominantly as a by-product of copper mining (and, to a lesser extent, nickel mining), which means that its supply responds primarily to copper and nickel prices rather than to cobalt demand. The Democratic Republic of the Congo, principally in the Copperbelt provinces of Lualaba and Haut-Katanga, accounts for around seventy per cent of global

production, and a substantial share of that production comes from mines owned wholly or partly by Chinese companies, notably CMOC Group's Tenke Fungurume and Kisanfu mines. A significant share of Congolese cobalt is produced by artisanal miners, often working in hazardous conditions, including in some cases children.

Legal issues. The legal issues associated with cobalt include the fiscal terms of the DRC's 2018 Mining Code (including its classification of cobalt as a strategic substance); the renegotiation of the "Sicomines" infrastructure-for-minerals agreement concluded in 2008 between the DRC and a consortium of Chinese companies, under which the Chinese partners were to finance infrastructure in exchange for mining rights, renegotiated in 2024 to increase the DRC's share of benefits; the formalisation of artisanal cobalt mining, including through the creation in 2019 of the state-owned Entreprise Générale du Cobalt (EGC), which holds a monopoly on the purchase, processing, and export of artisanal cobalt; and the due diligence obligations of companies sourcing cobalt, examined in section 8.5. In February 2025, in response to a collapse in cobalt prices caused by oversupply, the DRC government suspended cobalt exports for four months, subsequently extending the suspension and then replacing it with an export quota system — an example of a producing state using export restrictions as an instrument of market management, analogous to production quotas in the oil market.

Nickel

Production and market. Nickel is used in stainless steel and, increasingly, in high-nickel battery cathodes. Indonesia has become the dominant producer, accounting for around half of global mined supply following a dramatic expansion of production since 2020, predominantly from laterite deposits processed through rotary kiln-electric furnace and high-pressure acid leaching (HPAL) technologies, in industrial parks developed largely with Chinese investment. Other significant producers include the Philippines, Russia, New Caledonia, Canada, and Australia. The expansion of Indonesian supply has caused prices to fall sharply, forcing the closure of mines in Australia and New Caledonia.

Legal issues. Indonesia's nickel policy is the most prominent example of a resource-rich state using export restrictions to drive downstream industrialisation. Indonesia's Mining Law No 4 of 2009 required domestic processing and refining of minerals; after a partial relaxation, the government banned exports of nickel ore with effect from January 2020. The ban attracted substantial investment in nickel smelting and refining in Indonesia, increasing the value of Indonesian nickel exports dramatically, but it was challenged by the European Union before the WTO in *Indonesia — Measures Relating to Raw Materials* (DS592). In November 2022, the panel found that the export ban was inconsistent with Article XI:1 of

the GATT 1994 and could not be justified under Article XI:2(a), as a measure temporarily applied to prevent or relieve a critical shortage, or under Article XX(d), as a measure necessary to secure compliance with laws not inconsistent with the GATT. Indonesia appealed the report to the non-functioning Appellate Body, preventing its adoption — an "appeal into the void" — and has maintained the ban. The dispute is examined further in section 8.3 and in Chapter 13.

Indonesia's nickel boom has also raised serious environmental and social concerns, including deforestation, coastal and marine pollution from mine runoff and tailings, the reliance of smelters on captive coal-fired power plants, labour conditions and fatal accidents at smelters, and impacts on indigenous communities. These concerns have led some purchasers to scrutinise the environmental and social performance of Indonesian nickel suppliers, and the question whether Indonesian nickel processed with Chinese investment qualifies for incentives under the United States Inflation Reduction Act has been politically contested.

Copper

Production and market. Copper is the archetypal transition metal, essential for electricity generation, transmission, and distribution, electric vehicles, and renewable energy installations. Chile and Peru are the largest producers, followed by the Democratic Republic of the Congo, China, the United States, Australia, Russia, Zambia, and Mexico. Copper demand is projected to grow strongly, and many analysts anticipate a structural deficit by the end of the 2020s, as ore grades in existing mines decline and the development of new mines faces permitting delays, community opposition, water constraints, and rising costs.

Legal issues. The legal issues associated with copper reflect the challenges of developing large new mines. The closure of the Cobre Panamá mine illustrates the risks. In 2023, the government of Panama and First Quantum Minerals concluded a revised concession contract for the mine, one of the largest new copper mines in the world, following a 2017 Supreme Court ruling that the original concession law was unconstitutional. The new contract, approved by the National Assembly as Law 406, provoked mass protests over environmental concerns and perceived inadequate benefits for Panama. In November 2023, the Supreme Court of Panama unanimously declared Law 406 unconstitutional, and the government ordered the closure of the mine. First Quantum commenced arbitration proceedings under the contract and under the Canada–Panama free trade agreement, although it subsequently suspended the proceedings to facilitate negotiations with the government. The case illustrates the vulnerability of mining projects to constitutional challenges and social opposition, and the interaction between domestic constitutional law and international investment protection.

In Chile, the Mining Royalty Law (Law No 21.591) of 2023 introduced a new royalty on large copper producers, combining an *ad valorem* component with a component based on operating margins, with an effective tax burden capped at a specified percentage of operating margin. In Zambia, frequent changes in mining fiscal terms have been a source of tension with investors, and in 2022 the government introduced changes to make mineral royalties deductible for income tax purposes in order to attract investment.

Rare Earth Elements

Production and market. The rare earth elements comprise the fifteen lanthanides, together with scandium and yttrium. They are not, in fact, geologically rare, but economically viable concentrations are uncommon, and their separation from one another is chemically complex, environmentally hazardous (generating radioactive and acidic waste), and technically demanding. Neodymium, praseodymium, dysprosium, and terbium are used in high-performance permanent magnets for wind turbines, electric vehicle motors, electronics, and defence systems. China dominates every stage of the rare earth supply chain: it accounts for around sixty to seventy per cent of mining, around ninety per cent of separation and refining, and a similar proportion of magnet manufacturing. Other producers include the United States (Mountain Pass, California), Australia (Mount Weld, with processing by Lynas in Malaysia and Australia), Myanmar (where unregulated mining in Kachin State has caused severe environmental damage), and, increasingly, Brazil, Vietnam, and several African countries.

Legal issues. Rare earths have been at the centre of trade disputes and export control controversies. In *China — Measures Related to the Exportation of Rare Earths, Tungsten, and Molybdenum* (2014), the WTO Appellate Body upheld findings that China's export duties and export quotas on rare earths were inconsistent with its WTO obligations and not justified on conservation grounds, examined in section 8.3. China's subsequent export controls on rare earth processing technologies (2023) and on rare earth elements and magnets (2025) have prompted urgent efforts by importing states to develop alternative supply chains, including through public financing, offtake agreements, and price floors. In 2025, the United States Department of Defense agreed to acquire an equity stake in MP Materials, the owner of the Mountain Pass mine, and to guarantee a minimum price for its neodymium-praseodymium output — an unprecedented intervention illustrating the shift towards state capitalism in critical mineral supply chains.

8.3 Strategic Mineral Ownership and Export Controls

State Ownership and Control of Strategic Minerals

As examined in Chapter 3, almost all states assert ownership or control over subsurface minerals. In relation to minerals classified as strategic, however, many

states go further, reserving exploitation to the state or to state enterprises, requiring majority state participation, or subjecting the minerals to special regimes. The Chilean, Bolivian, and Mexican regimes for lithium, examined in section 8.2, are examples, as are the DRC's regime for strategic substances and the reservation of radioactive minerals (uranium and thorium) to the state in many jurisdictions. Such reservations are an exercise of permanent sovereignty over natural resources and are, in principle, consistent with international law, provided that they are applied prospectively or, where they affect existing rights, in compliance with the requirements for lawful expropriation, including compensation (examined in Chapters 3 and 14).

State ownership of strategic minerals is often justified on grounds of national security, the capture of resource rents, and the promotion of downstream industrialisation. Its effectiveness depends on the capacity and governance of state enterprises, which have frequently struggled to attract the capital and technology necessary to develop complex projects. Bolivia's experience with lithium, where state control has not yet produced significant output despite immense resources, illustrates the difficulties; Chile's model of public-private partnerships under state control represents an attempt to combine state ownership with private capital and expertise.

Export Restrictions by Producing States

Producing states have increasingly used export restrictions — bans, quotas, export taxes, and licensing requirements — to promote domestic processing, conserve resources, manage prices, or raise revenue. The Organisation for Economic Co-operation and Development (OECD), which maintains an inventory of export restrictions on industrial raw materials, has documented a more than five-fold increase in such restrictions between 2009 and 2020, with the largest increases in critical minerals. Examples include Indonesia's nickel ore export ban, Zimbabwe's and Namibia's restrictions on unprocessed lithium, the DRC's cobalt export suspension and quotas, and the restrictions on the export of unprocessed bauxite in Guinea and other states.

International Trade Law Disciplines

Export restrictions are subject to the disciplines of international trade law, examined generally in Chapter 13. The central provision is **Article XI:1 of the GATT 1994**, which prohibits "prohibitions or restrictions other than duties, taxes or other charges, whether made effective through quotas, import or export licences or other measures" on the exportation of any product destined for the territory of another WTO member. Export *duties*, by contrast, are not prohibited by the GATT, although some members, notably China, accepted commitments to eliminate export duties in their accession protocols.

Export restrictions may be justified under several exceptions:

- **Article XI:2(a)** permits export prohibitions or restrictions "temporarily applied to prevent or relieve critical shortages of foodstuffs or other products essential to the exporting contracting party".
- **Article XX(g)** permits measures "relating to the conservation of exhaustible natural resources if such measures are made effective in conjunction with restrictions on domestic production or consumption", subject to the requirements of the chapeau of Article XX that measures not be applied in a manner that would constitute arbitrary or unjustifiable discrimination or a disguised restriction on international trade.
- **Article XX(i)** permits measures "involving restrictions on exports of domestic materials necessary to ensure essential quantities of such materials to a domestic processing industry during periods when the domestic price of such materials is held below the world price as part of a governmental stabilization plan", subject to conditions.
- **Article XXI(b)** permits measures that a member "considers necessary for the protection of its essential security interests", including those "taken in time of war or other emergency in international relations".

The WTO dispute settlement bodies have interpreted these exceptions narrowly in the context of mineral export restrictions. In *China — Measures Related to the Exportation of Various Raw Materials* (2012), concerning Chinese export duties and quotas on bauxite, coke, fluorspar, magnesium, manganese, silicon carbide, silicon metal, yellow phosphorus, and zinc, the Appellate Body held that China could not invoke Article XX to justify export duties inconsistent with paragraph 11.3 of its Accession Protocol, because that paragraph contained no reference to the GATT exceptions. It also held that a measure is "temporarily applied" under Article XI:2(a) only if it is applied for a limited time to bridge a passing need, and that a "critical shortage" refers to deficiencies in quantity that are crucial or that reach a vitally important or decisive stage. In *China — Rare Earths* (2014), the Appellate Body upheld the panel's findings that China's export quotas on rare earths, tungsten, and molybdenum were not justified under Article XX(g), because China had not demonstrated that the quotas were made effective in conjunction with restrictions on domestic production or consumption in an even-handed manner; on the contrary, the quotas appeared designed to secure supply for domestic users at the expense of foreign users. China removed the export duties and quotas in 2015.

In *Indonesia — Raw Materials* (DS592), discussed in section 8.2, the panel held that Indonesia's nickel ore export ban was a prohibition contrary to Article XI:1, that it was not "temporarily applied" to relieve a "critical shortage" within

Article XI:2(a), since it was intended to continue indefinitely, and that it was not justified under Article XX(d). The panel's report has not been adopted owing to Indonesia's appeal to the non-functioning Appellate Body.

These decisions establish that WTO law significantly constrains the use of export restrictions to promote downstream processing. Critics argue that the rules are asymmetric: they constrain the industrial policy tools available to resource-rich developing countries, which historically were used by today's industrialised countries during their own development, while importing states are relatively free to use subsidies and other measures to build processing capacity. The legitimacy of the WTO disciplines on export restrictions, and the scope for reform, are significant themes for doctoral research, explored in Chapter 13.

Export Controls on Minerals and Technologies by China

China's use of export controls on critical minerals and processing technologies has transformed the landscape. The Export Control Law of the People's Republic of China (2020) established a comprehensive framework for controlling exports of dual-use items, military products, nuclear materials, and other items related to national security and interests, and authorised countermeasures against countries that abuse export controls to endanger China's national security and interests. Under this framework and related regulations, including the Regulations on the Export Control of Dual-Use Items (2024), China has imposed licensing requirements on exports of:

- **gallium and germanium** and their compounds, from August 2023;
- **graphite**, including certain high-purity natural and synthetic graphite products, from December 2023;
- **rare earth extraction, separation, and magnet manufacturing technologies**, added to the Catalogue of Technologies Prohibited or Restricted from Export in December 2023;
- **antimony** and superhard materials, from September 2024;
- **seven medium and heavy rare earth elements** (samarium, gadolinium, terbium, dysprosium, lutetium, scandium, and yttrium) and related magnets, from April 2025.

In December 2024, China prohibited exports of gallium, germanium, antimony, and superhard materials to the United States in response to United States export controls on semiconductors. In October 2025, China announced a substantial expansion of its rare earth export controls, including provisions asserting jurisdiction over foreign-made products containing Chinese-origin rare earths or produced with Chinese technology; following a meeting between the presidents of China and the United States at the end of October 2025, China announced that it would suspend the implementation of the expanded measures for one year.

These measures, and their partial suspension as part of a broader trade negotiation, illustrate the use of critical minerals as instruments of economic statecraft and the vulnerability of importing states to concentrated supply chains. China has justified the measures as necessary for the protection of national security and the fulfilment of non-proliferation obligations; importing states have characterised them as economic coercion. Their consistency with WTO law would turn largely on the interpretation of Article XXI, examined in Chapters 13 and 25.

Export Controls by Importing States and Investment Screening

Importing states also use export controls and investment screening in relation to critical minerals. The United States, the European Union, Japan, and other states control exports of technologies related to semiconductor manufacturing, advanced batteries, and other strategic sectors, and the United States' foreign direct product rules extend these controls extraterritorially. Investment screening regimes — such as the Committee on Foreign Investment in the United States (CFIUS) under the Foreign Investment Risk Review Modernization Act of 2018, the EU Foreign Direct Investment Screening Regulation (EU) 2019/452 (the revision of which was under negotiation in 2025), the United Kingdom's National Security and Investment Act 2021, and the Australian Foreign Acquisitions and Takeovers Act 1975 — have been used to block or condition acquisitions of critical mineral assets by foreign investors, particularly Chinese state-linked companies. Australia has blocked several Chinese investments in rare earth and lithium companies on national interest grounds. The use of investment screening in the mineral sector raises questions about its consistency with investment treaties and with the principle of non-discrimination, and about the risk that screening becomes an instrument of geopolitical rivalry rather than a narrowly targeted security measure.

8.4 Resource Security and Industrial Policy

From Market Reliance to Industrial Policy

For several decades, the dominant approach of industrialised states to mineral supply was reliance on global markets: minerals were commodities to be purchased wherever they were cheapest, and the state's role was limited to maintaining open trade and, in some cases, strategic stockpiles for defence purposes. The concentration of supply chains in China, China's demonstrated willingness to use export restrictions, the supply chain disruptions of the COVID-19 pandemic, and the Russian invasion of Ukraine have led to a fundamental shift. Industrialised states now pursue active industrial policies aimed at building domestic and allied capacity across the mineral value chain — mining, processing, component manufacturing, and recycling — using legislation, subsidies, public financing, permitting reforms, and international partnerships.

The EU Critical Raw Materials Act

The EU Critical Raw Materials Act (Regulation (EU) 2024/1252) is the most comprehensive legislative framework for critical mineral security. Its principal elements are:

Benchmarks. Article 5 establishes benchmarks for strengthening the different stages of the strategic raw materials value chain, providing that by 2030 the Union's capacity should be sufficient for extraction to produce at least ten per cent, processing to produce at least forty per cent, and recycling to produce at least twenty-five per cent of the Union's annual consumption of strategic raw materials, and that the Union should diversify its imports so that no more than sixty-five per cent of its annual consumption of each strategic raw material at any relevant stage of processing comes from a single third country. The benchmarks are not legally binding targets for member states, but they guide the implementation of the Act.

Strategic projects. The Act establishes a procedure for the recognition of strategic projects — projects that contribute meaningfully to the security of the Union's supply of strategic raw materials, are technically feasible, are implemented sustainably, and, for projects in third countries, are mutually beneficial for the Union and the third country concerned. Strategic projects benefit from priority status in permitting, with maximum permit-granting periods (twenty-seven months for extraction projects and fifteen months for processing and recycling projects within the Union), a presumption that they are in the public interest for the purposes of certain environmental assessments, and support in accessing finance. In March 2025, the Commission selected the first forty-seven strategic projects within the Union, and in June 2025 it selected thirteen strategic projects in third countries, including projects in Africa and Latin America.

Monitoring and risk preparedness. The Act requires the Commission and member states to monitor supply chains, to conduct stress tests, and to coordinate strategic stocks, and requires large companies manufacturing strategic technologies to carry out risk assessments of their supply chains.

Circularity and sustainability. The Act requires member states to adopt national programmes to increase the collection and recycling of critical raw materials, and requires certain products to disclose information on the recyclability and recycled content of permanent magnets.

International partnerships. The Act provides a framework for strategic partnerships with third countries, building on the memoranda of understanding concluded by the EU with Canada, Ukraine, Kazakhstan, Namibia, Argentina, Chile, Zambia, the Democratic Republic of the Congo, Greenland, Rwanda, Norway, Uzbekistan, Australia, and others.

United States Industrial Policy

The United States has adopted a range of measures to build domestic critical mineral supply chains. The **Inflation Reduction Act of 2022** introduced, among other incentives, a clean vehicle tax credit under section 30D of the Internal Revenue Code, part of which was conditional on a rising percentage of the value of critical minerals in the vehicle's battery being extracted or processed in the United States or in a country with which the United States has a free trade agreement, or recycled in North America, and excluded vehicles whose batteries contained critical minerals extracted, processed, or recycled by a "foreign entity of concern". The FTA requirement led the United States to conclude "critical minerals agreements" with Japan (March 2023) and to negotiate similar agreements with the European Union and the United Kingdom, raising questions about whether such limited agreements qualify as "free trade agreements". The clean vehicle tax credit was terminated for vehicles acquired after 30 September 2025 by the One Big Beautiful Bill Act of July 2025, illustrating the vulnerability of industrial policy to political change.

The **Defense Production Act** of 1950, Title III of which authorises the President to provide financial assistance to expand domestic productive capacity for materials essential to national defence, has been invoked to support critical mineral projects, and in 2022 President Biden authorised its use for battery minerals. In March 2025, President Trump issued an Executive Order on Immediate Measures to Increase American Mineral Production, directing agencies to expedite permitting for priority mineral projects and to expand the use of Defense Production Act authorities and federal financing. The Department of Energy's Loan Programs Office, the Development Finance Corporation, and the Export-Import Bank of the United States have provided financing for domestic and overseas critical mineral projects, and in 2025 the federal government took equity stakes in several critical mineral companies, including MP Materials and the lithium developer Lithium Americas.

Other Jurisdictions

Japan has pursued mineral security for decades through the Japan Organization for Metals and Energy Security (JOGMEC), which provides equity financing, loan guarantees, and exploration support for overseas mineral projects and maintains national stockpiles. Following the Chinese restrictions on rare earth exports to Japan in 2010, during a territorial dispute, JOGMEC and Japanese companies invested in the Australian rare earth producer Lynas, helping to establish one of the few significant sources of rare earths outside China. Australia's Critical Minerals Strategy 2023–2030 and its Critical Minerals Facility, administered by Export Finance Australia, provide financing for critical mineral projects, and the Future Made in Australia Act 2024 introduced production tax incentives for

critical mineral processing. In October 2025, Australia and the United States concluded a framework agreement on critical minerals providing for joint investment in projects. Canada's Critical Minerals Strategy (2022) and India's National Critical Mineral Mission (2025) pursue similar objectives.

Industrial Policy in Producing States

Producing states pursue their own industrial policies, aimed at capturing a greater share of the value chain. These include requirements for domestic processing (export bans on unprocessed ores, as in Indonesia, Zimbabwe, and Namibia), local content and beneficiation requirements, state participation, and incentives for investment in processing and manufacturing. The success of Indonesia's nickel strategy in attracting downstream investment has inspired other producing states, but its replicability is uncertain: Indonesia combined its export ban with a large domestic market, abundant coal-fired power, and massive Chinese investment, and its strategy has generated significant environmental and social costs. Producing states must also navigate the constraints of WTO law and investment treaties, and the risk that export restrictions will simply divert investment to other sources.

The Tension Between Security and Sustainability

Industrial policies aimed at accelerating mineral supply create tensions with environmental protection, indigenous rights, and community consent. Streamlined permitting, presumptions of public interest, and political pressure to approve projects quickly may compromise the rigour of environmental assessment and the quality of consultation. The EU Critical Raw Materials Act seeks to address this tension by requiring that strategic projects be implemented sustainably, including in compliance with environmental and social standards, but civil society organisations have criticised the selection of strategic projects for insufficient transparency and inadequate consideration of community opposition. The reconciliation of mineral security with sustainability and human rights is one of the defining legal challenges of the energy transition.

8.5 Supply Chain Due Diligence

The Concept of Supply Chain Due Diligence

Mineral supply chains are long, complex, and opaque. A cobalt atom in a smartphone battery may have been mined by an artisanal miner in the Congolese Copperbelt, sold to a trader at a local depot, processed at a refinery in the DRC or China, converted into cathode material in China, South Korea, or Japan, incorporated into a battery cell, and assembled into a device sold by a brand headquartered in the United States or Europe. At each stage, minerals from different sources are mixed, and the connection between the final product and the condi-

tions of extraction is obscured. Supply chain due diligence is the process by which companies identify, assess, prevent, mitigate, and account for the adverse impacts associated with their supply chains, including impacts caused by their suppliers and their suppliers' suppliers.

The concept derives from the UN Guiding Principles on Business and Human Rights (2011), examined in Chapter 21, which provide that business enterprises should carry out human rights due diligence to identify, prevent, mitigate, and account for how they address their adverse human rights impacts, including impacts directly linked to their operations, products, or services by their business relationships. In the mineral sector, due diligence was first developed in response to the financing of armed conflict in the eastern Democratic Republic of the Congo by the trade in tin, tantalum, tungsten, and gold (the "3TG" or "conflict minerals").

The OECD Due Diligence Guidance

The **OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas**, first adopted in 2010 and approved as an OECD Council Recommendation in 2011 (now in its third edition, 2016), is the leading international standard for mineral supply chain due diligence. It establishes a five-step framework:

1. **Establish strong company management systems**, including a supply chain policy consistent with the model policy in Annex II of the Guidance, internal structures to support due diligence, a system of controls and transparency over the supply chain (including chain of custody or traceability systems), and a grievance mechanism.
2. **Identify and assess risks in the supply chain**, including by mapping the supply chain and assessing the conditions of extraction, trade, handling, and export.
3. **Design and implement a strategy to respond to identified risks**, which may involve continuing trade while pursuing measurable risk mitigation, temporarily suspending trade, or disengaging from a supplier.
4. **Carry out independent third-party audits** of due diligence practices at identified points in the supply chain, notably smelters and refiners.
5. **Report annually** on supply chain due diligence.

The Guidance distinguishes between "upstream" companies (from the mine to smelters and refiners) and "downstream" companies (from smelters and refiners to retailers), recognising that smelters and refiners constitute a "choke point" in the supply chain through which minerals from many sources pass and at which audits can be most effectively focused. Annex II identifies the most serious risks — including torture and cruel, inhuman, or degrading treatment, forced or com-

pulsory labour, the worst forms of child labour, gross human rights violations, war crimes, direct or indirect support to non-state armed groups, and bribery and money laundering — and the expected responses to them. Although originally developed for 3TG, the Guidance applies to all minerals, and supplements have addressed specific issues. The OECD has also developed guidance for responsible business conduct in other sectors, including the OECD Due Diligence Guidance for Responsible Business Conduct (2018).

Mandatory Due Diligence: Dodd-Frank Section 1502

The first mandatory mineral due diligence regime was **section 1502 of the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010** in the United States, which required companies listed on United States exchanges to determine whether their products contained 3TG originating in the DRC or adjoining countries and, if so, to exercise due diligence on the source and chain of custody of those minerals and to file a report with the Securities and Exchange Commission (SEC). The SEC's implementing rule, adopted in 2012, required companies to describe products that had not been found to be "DRC conflict free". In *National Association of Manufacturers v SEC* (2014, reaffirmed 2015), the United States Court of Appeals for the District of Columbia Circuit held that the requirement to describe products as not "DRC conflict free" violated the First Amendment by compelling speech, while upholding other aspects of the rule.[1] Section 1502 was praised for raising awareness and driving industry due diligence, but was also criticised for causing a *de facto* embargo on minerals from the eastern DRC, which harmed the livelihoods of artisanal miners without proportionately reducing conflict financing.

The EU Conflict Minerals Regulation

The **EU Conflict Minerals Regulation** — Regulation (EU) 2017/821, which became fully applicable on 1 January 2021 — requires Union importers of 3TG ores, concentrates, and metals above specified volume thresholds to comply with supply chain due diligence obligations aligned with the OECD Guidance, including management systems, risk management, third-party audits, and disclosure. Unlike Dodd-Frank, the Regulation applies to minerals from any conflict-affected and high-risk area, not only the DRC region, and focuses on upstream importers rather than downstream manufacturers. Member state authorities are responsible for enforcement, and the Commission has published an indicative list of conflict-affected and high-risk areas.

[1] *National Association of Manufacturers v Securities and Exchange Commission*, 748 F.3d 359 (DC Cir 2014), reaffirmed on rehearing, 800 F.3d 518 (DC Cir 2015).

The EU Batteries Regulation

The **EU Batteries Regulation** — Regulation (EU) 2023/1542 — extends mandatory due diligence to the principal battery minerals. It requires economic operators placing batteries on the Union market, subject to a turnover threshold, to adopt and implement battery due diligence policies in respect of cobalt, natural graphite, lithium, and nickel, and chemical compounds based on them, covering risks related to human rights, labour rights, the environment, and community life, in line with internationally recognised due diligence instruments including the UN Guiding Principles and the OECD Guidance. Operators must have their due diligence policies verified by a notified body and must make information publicly available. The due diligence obligations were originally due to apply from August 2025, but their application was postponed to 2027 by an amendment adopted in 2025 as part of the Commission's simplification agenda. The Regulation also introduces a "battery passport" — an electronic record containing information on each industrial and electric vehicle battery, including its composition, carbon footprint, and recycled content — and requirements on carbon footprint declarations, recycled content, and recycling efficiency, examined in section 8.7.

General Mandatory Human Rights Due Diligence

Beyond mineral-specific regimes, general mandatory human rights and environmental due diligence laws apply to companies in mineral supply chains. France's Duty of Vigilance Law (*Loi relative au devoir de vigilance*) of 2017 requires large French companies to establish, publish, and implement a vigilance plan identifying and preventing serious violations of human rights, health and safety, and the environment resulting from their activities and those of their subsidiaries, subcontractors, and suppliers. Germany's Supply Chain Due Diligence Act (*Lieferkettensorgfaltspflichtengesetz*), in force since 2023, imposes similar obligations on large companies operating in Germany. The EU **Corporate Sustainability Due Diligence Directive** — Directive (EU) 2024/1760 — requires large companies to identify, prevent, mitigate, and remediate adverse human rights and environmental impacts in their own operations, those of their subsidiaries, and those of their business partners in their "chain of activities", and to adopt climate transition plans. The Directive's transposition and application dates were postponed, and its scope and requirements were reduced, through the "Omnibus" simplification package negotiated in 2025. These general regimes are examined in Chapters 21 and 22.

Import Bans Based on Forced Labour

A further category of supply chain regulation consists of import bans on goods produced with forced labour. In the United States, section 307 of the Tariff Act of 1930 prohibits the importation of goods mined, produced, or manufactured

wholly or in part by forced labour, and the **Uyghur Forced Labor Prevention Act** of 2021 establishes a rebuttable presumption that goods mined, produced, or manufactured wholly or in part in the Xinjiang Uyghur Autonomous Region of China, or by entities on a designated list, are made with forced labour and are prohibited from importation. The Act has significant implications for supply chains of polysilicon (used in solar panels), aluminium, and battery materials. The EU **Forced Labour Regulation** — Regulation (EU) 2024/3015 — prohibits products made with forced labour from being placed on the Union market or exported from it, with application from December 2027.

Industry and Multi-Stakeholder Initiatives

Industry initiatives play a central role in implementing due diligence. The **Responsible Minerals Initiative** (RMI) operates the Responsible Minerals Assurance Process, which audits smelters and refiners against standards aligned with the OECD Guidance, and publishes lists of conformant facilities; it has developed standards for cobalt, mica, and other minerals in addition to 3TG. The **London Bullion Market Association** Responsible Gold Guidance requires refiners accredited to the London Good Delivery list to undergo annual audits. The **International Conference on the Great Lakes Region** (ICGLR), an intergovernmental organisation of twelve African states, operates a Regional Certification Mechanism for 3TG under its 2006 Protocol against the Illegal Exploitation of Natural Resources. Initiatives addressing artisanal cobalt include the Fair Cobalt Alliance and the cobalt standard of the DRC's Entreprise Générale du Cobalt.

Evaluating Due Diligence

Mineral supply chain due diligence has become a central feature of the legal regime of critical minerals, but its effectiveness is debated. Proponents argue that it has improved transparency, reduced the financing of armed groups in some areas, and established expectations of corporate responsibility. Critics argue that due diligence has too often become a box-ticking exercise focused on audits and certifications rather than on actual improvements in conditions at mine sites; that it may lead companies to disengage from high-risk areas, harming the communities that depend on mining rather than improving conditions; that traceability systems are vulnerable to fraud, as illustrated by allegations concerning the laundering of minerals from conflict areas through neighbouring countries; and that the costs fall disproportionately on artisanal and small-scale producers. The design of due diligence regimes that promote responsible engagement rather than disengagement is a key challenge for regulators.

8.6 Human Rights and Labour Risks in Mineral Production

The Spectrum of Risks

The production of critical minerals is associated with a wide spectrum of human rights and labour risks. These include the worst forms of child labour, particularly in artisanal mining; forced labour, including in state-sponsored labour transfer programmes; unsafe working conditions, resulting in deaths and injuries from collapses, accidents, and exposure to toxic substances; violations of the rights of indigenous peoples, including the right to free, prior, and informed consent; forced evictions and inadequate resettlement; violence by public and private security forces; environmental harms affecting the rights to health, water, and an adequate standard of living; and the financing of armed conflict. These risks are examined generally in Chapter 21; this section highlights their particular manifestations in critical mineral supply chains.

Child Labour and Artisanal Cobalt

In 2016, Amnesty International and African Resources Watch published *"This is what we die for": Human rights abuses in the Democratic Republic of the Congo power the global trade in cobalt*, documenting the use of child labour and hazardous working conditions in artisanal cobalt mining in the DRC and tracing cobalt from artisanal sources through processing companies to battery manufacturers and consumer brands. The report contributed to the development of industry due diligence for cobalt and to the inclusion of cobalt in the EU Batteries Regulation. In *Doe v Apple Inc*, a group of former child miners and the families of deceased child miners from the DRC sued Apple, Alphabet, Dell, Microsoft, and Tesla in the United States, alleging that the companies had knowingly benefited from forced child labour in their cobalt supply chains in violation of the Trafficking Victims Protection Reauthorization Act. The District Court dismissed the claims in 2021, and in March 2024 the Court of Appeals for the District of Columbia Circuit affirmed, holding that purchasing cobalt in the global market did not amount to participation in a "venture" with the suppliers that used forced labour within the meaning of the statute.[2] The case illustrates the difficulty of establishing the legal liability of downstream companies for abuses in their supply chains under existing law, and the significance of mandatory due diligence legislation in filling that gap.

Forced Labour and State-Sponsored Labour Programmes

Reports by researchers, journalists, and civil society organisations have documented the involvement of state-sponsored labour transfer programmes in the Xinjiang Uyghur Autonomous Region in the production of polysilicon, alu-

[2] *Doe v Apple Inc*, 96 F.4th 403 (DC Cir 2024).

minium, and other materials used in energy transition technologies. The United Nations Office of the High Commissioner for Human Rights, in its 2022 assessment of human rights concerns in Xinjiang, concluded that serious human rights violations had been committed and that the extent of arbitrary and discriminatory detention may constitute international crimes. Such programmes raise the risk that solar panels, batteries, and other clean energy products incorporate materials produced with forced labour. The Uyghur Forced Labor Prevention Act and the EU Forced Labour Regulation, examined in section 8.5, are the principal legal responses.

Indigenous Peoples and Lithium

A large share of known lithium, nickel, copper, and other transition mineral resources is located on or near the lands of indigenous peoples. A study published in *Nature Sustainability* in 2022 estimated that more than half of energy transition mineral projects are located on or near indigenous peoples' lands. In the lithium triangle, indigenous Atacameño (Lickanantay), Colla, and Kolla communities have raised concerns about the impact of brine extraction on scarce water resources, wetlands, and traditional livelihoods, and have challenged lithium projects before national courts and the Inter-American human rights system. In Argentina, communities in the Salinas Grandes and Laguna de Guayatayoc basin have invoked their right to free, prior, and informed consent under ILO Convention No 169 and the Argentine Constitution. In the United States, the Thacker Pass lithium project in Nevada, located near a site of a historic massacre of Paiute people, was challenged by indigenous groups, ranchers, and environmental organisations; the Court of Appeals for the Ninth Circuit in 2023 rejected the challenges to the project's federal approvals.[3] These disputes illustrate the tension between the urgency of the energy transition and the rights of indigenous peoples, which some commentators have described as a risk of "green colonialism" or "green extractivism".

Labour Conditions in Processing and Manufacturing

Labour risks are not confined to mining. Nickel smelters in Indonesia's industrial parks have experienced fatal accidents, including an explosion at a smelter in the Indonesia Morowali Industrial Park in December 2023 that killed around twenty workers, and reports have documented long working hours, inadequate safety measures, and restrictions on freedom of association. Battery manufacturing and recycling involve exposure to hazardous substances. The ILO's core conventions on freedom of association, forced labour, child labour, discrimination, and occupational safety and health, and the ILO Declaration on Fundamental Principles

[3] *Bartell Ranch LLC v McCullough* (9th Cir, July 2023) (memorandum disposition), affirming the District Court of Nevada.

and Rights at Work (as amended in 2022 to include a safe and healthy working environment), provide the international framework.

The UN Secretary-General's Panel

In recognition of the risks associated with the expansion of critical mineral production, the United Nations Secretary-General established in 2024 the **Panel on Critical Energy Transition Minerals**, comprising governments, intergovernmental organisations, industry, and civil society. The Panel's report, published in September 2024, set out seven guiding principles: that human rights must be at the core of all mineral value chains; that the integrity of the planet, its environment, and biodiversity must be safeguarded; that justice and equity must underpin mineral value chains; that development should be fostered through benefit sharing, value addition, and economic diversification; that investment, finance, and trade should be responsible and fair; that transparency, accountability, and anti-corruption measures are necessary to ensure good governance; and that multilateral and international cooperation should underpin global action. The Panel recommended, among other things, the establishment of a high-level expert group to advance the principles, a global traceability, transparency, and accountability framework, and a global mining legacies fund to address abandoned mines. Although not binding, the principles represent the first United Nations-endorsed framework specifically addressed to energy transition minerals.

8.7 Mineral Processing, Recycling, and Circular Economy Regulation

The Processing Bottleneck

The concentration of mineral processing — the conversion of ores and concentrates into refined metals and chemicals suitable for industrial use — is the most acute vulnerability in critical mineral supply chains. Even where mining is geographically diverse, processing may be concentrated in a single country: lithium mined in Australia is shipped as spodumene concentrate to China for conversion to lithium hydroxide; cobalt mined in the DRC is refined predominantly in China; and almost all rare earth separation and magnet manufacturing takes place in China. Processing is capital-intensive, technically demanding, energy-intensive, and environmentally hazardous, and the economics of new processing capacity outside China are challenged by the scale, experience, and low costs of Chinese producers, as well as by price volatility that Chinese oversupply can intensify.

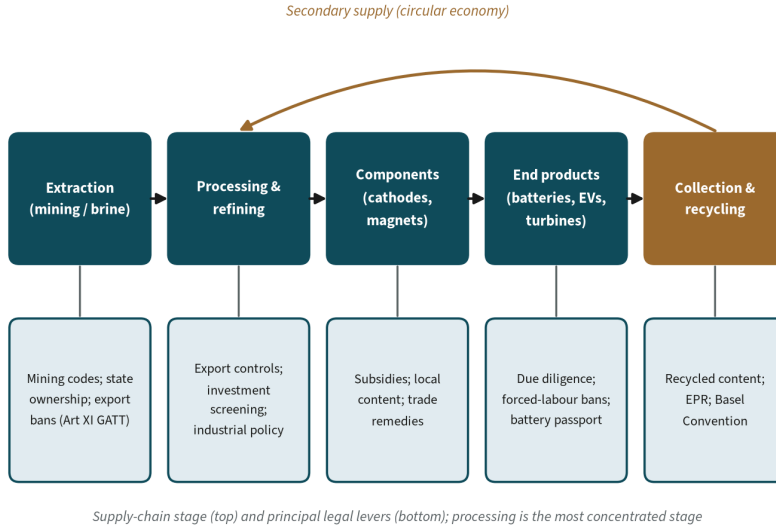


Figure 8.2 The critical mineral supply chain and the principal legal levers applied at each stage. *Source:* prepared for this edition.

The law relating to processing has several dimensions. Environmental regulation of processing facilities — governing emissions to air and water, waste management (including radioactive waste from rare earth processing), and energy use — is critical, and the strength of environmental regulation affects both the location of processing and its environmental footprint. The controversy over the Lynas rare earth processing plant in Kuantan, Malaysia, which faced protests and litigation over the management of low-level radioactive waste and was required by the Malaysian government to relocate its cracking and leaching operations to Australia, illustrates the issues. Trade and industrial policy measures — export restrictions on unprocessed ores in producing states and subsidies for processing in importing states — shape the location of processing, as examined in sections 8.3 and 8.4. Investment law protects processing investments, and investment screening may restrict foreign ownership of processing facilities.

The Circular Economy

The circular economy — an economic model in which materials are kept in use for as long as possible through reuse, repair, remanufacturing, and recycling, and waste is minimised — offers a partial solution to the supply challenges of critical minerals. Recycling of end-of-life batteries, electronics, magnets, and other products can provide a secondary source of critical minerals that is located in consuming countries, reduces the environmental impacts of primary mining, and reduces dependence on concentrated supply chains. However, recycling

cannot meet growing demand in the near term, because the stock of end-of-life products is small relative to projected demand, particularly for batteries, most of which are still in use. The IEA has projected that recycling could reduce primary supply requirements for lithium, cobalt, and nickel by a meaningful share by 2040 in scenarios with ambitious recycling policies, but primary mining will remain essential for decades.

The EU Batteries Regulation: Circularity Requirements

The EU Batteries Regulation (Regulation (EU) 2023/1542), introduced in section 8.5, is the most comprehensive regulatory framework for battery circularity. Its principal circularity provisions include:

- **Recycled content**: from 2031, industrial batteries with a capacity above 2 kWh, electric vehicle batteries, and certain other batteries must contain minimum shares of recovered cobalt (16 per cent), lead (85 per cent), lithium (6 per cent), and nickel (6 per cent), rising in 2036 to 26 per cent for cobalt, 12 per cent for lithium, and 15 per cent for nickel;
- **Recycling efficiency and material recovery**: recycling processes must achieve minimum recycling efficiencies (for lithium-based batteries, 65 per cent by the end of 2025 and 70 per cent by the end of 2030), and minimum material recovery rates (for lithium, 50 per cent by the end of 2027 and 80 per cent by the end of 2031; for cobalt, copper, lead, and nickel, 90 per cent by the end of 2027 and 95 per cent by the end of 2031);
- **Extended producer responsibility**: producers must organise and finance the collection and treatment of waste batteries, with rising collection targets for portable batteries;
- **Carbon footprint**: electric vehicle batteries and certain other batteries must be accompanied by a carbon footprint declaration and will become subject to maximum carbon footprint thresholds;
- **Battery passport**: from February 2027, electric vehicle batteries, industrial batteries above 2 kWh, and light means of transport batteries must have a digital battery passport.

The Regulation also addresses removability and replaceability of portable batteries and labelling requirements. Its extraterritorial effect is significant: any battery placed on the EU market, wherever manufactured, must comply, and the recycled content and carbon footprint requirements create incentives for battery manufacturers worldwide to adopt EU standards — an example of the "Brussels effect" in energy transition regulation.

Waste and the Transboundary Movement of E-Waste

The recycling of critical minerals depends on the collection and transport of waste products, which are subject to the law on hazardous waste. The **Basel Conven-**

tion on the Control of Transboundary Movements of Hazardous Wastes and their Disposal (1989) requires the prior informed consent of importing and transit states for transboundary movements of hazardous wastes, and prohibits exports to states that have prohibited imports. The Basel Ban Amendment, which entered into force in 2019, prohibits exports of hazardous waste from OECD and EU member states and Liechtenstein to other states for any purpose. Amendments to the Convention's Annexes adopted in 2022, which entered into force on 1 January 2025, extend the prior informed consent procedure to all electronic and electrical waste, whether or not hazardous. The **Bamako Convention** (1991) of the African Union goes further, prohibiting the import of hazardous wastes into Africa from non-parties. These regimes are designed to prevent the dumping of hazardous waste in developing countries, a practice exemplified by the 1988 Koko incident in Nigeria, in which several thousand tonnes of toxic waste from Italy were dumped in the port town of Koko in present-day Delta State, prompting Nigeria's Harmful Waste (Special Criminal Provisions, etc.) Act 1988. The legal challenge is to distinguish legitimate trade in recyclable materials, which can support the circular economy, from the export of waste for disposal or substandard recycling in countries lacking adequate facilities, which harms workers and communities, as in the informal e-waste processing sites of Agbogbloshie in Accra, Ghana, and parts of Lagos.

Mine Waste as a Resource

Mine wastes and tailings may contain recoverable quantities of critical minerals that were not economically extractable when the waste was produced. The reprocessing of tailings to recover cobalt, copper, rare earths, and other minerals is attracting investment, and the EU Critical Raw Materials Act requires member states to compile databases of closed extractive waste facilities with potential for recovery of critical raw materials. Reprocessing raises legal questions concerning ownership of tailings (whether they belong to the state as unworked minerals, to the operator as property, or to the owner of the land), the allocation of liability for historical contamination, and the applicability of mining, waste, and environmental law to reprocessing operations.

8.8 Geopolitical Competition and Critical Mineral Agreements

The Geopolitics of Critical Minerals

Critical minerals have become an arena of intense geopolitical competition. The principal axis of competition is between China, which has built dominance across critical mineral supply chains through decades of state-supported investment in domestic and overseas mining and processing, and the United States and its allies, which seek to reduce their dependence on Chinese supply. Producing states in Africa, Latin America, and Asia find themselves courted by competing powers,

which offers opportunities to negotiate better terms but also risks entanglement in great power rivalry. The geopolitics of critical minerals is examined further in Chapter 25 in the context of energy security; this section focuses on the legal instruments through which states cooperate and compete.

Types of Critical Mineral Agreements

A diverse and rapidly evolving array of international instruments addresses critical minerals:

Plurilateral partnerships among consumer states. The **Minerals Security Partnership** (MSP), launched in June 2022 by the United States and a group of partners including Australia, Canada, the European Union, Japan, South Korea, and the United Kingdom, aims to catalyse public and private investment in responsible critical mineral supply chains, including through a Minerals Security Partnership Finance Network. The **G7 Critical Minerals Action Plan**, adopted at the Kananaskis summit in June 2025, commits G7 members to cooperation on standards-based markets, investment, recycling, and innovation. The IEA's **Critical Minerals Security Programme**, launched in 2024, supports its members in assessing supply risks and coordinating responses.

Bilateral critical minerals agreements. The Agreement between the Government of the United States of America and the Government of Japan on Strengthening Critical Minerals Supply Chains (March 2023) commits the parties not to impose export duties on critical minerals traded between them, to promote labour rights and environmental protection in critical mineral supply chains, and to cooperate on investment screening and recycling. Its principal practical function was to enable Japan to be treated as an FTA partner for the purposes of the IRA clean vehicle credit.

Strategic partnerships between consumers and producers. The European Union has concluded memoranda of understanding on strategic partnerships on raw materials with numerous producing states, typically covering integration of value chains, cooperation on research and innovation, application of environmental, social, and governance standards, infrastructure development, and capacity building. The United States has supported the **Lobito Corridor**, a rail and port infrastructure project linking the copper and cobalt mining regions of the DRC and Zambia to the Angolan port of Lobito on the Atlantic coast, in partnership with the European Union, the African Development Bank, and the Africa Finance Corporation, as an alternative to routes to the Indian Ocean.

Minerals-for-security and minerals-for-reconstruction arrangements. In April 2025, the United States and Ukraine concluded an agreement establishing a United States–Ukraine Reconstruction Investment Fund, under which Ukraine contributes revenues from new licences for natural resources, including

critical minerals, oil, and gas, to a jointly governed fund that invests in Ukrainian projects. In June 2025, the DRC and Rwanda signed a peace agreement facilitated by the United States, accompanied by a regional economic integration framework addressing mineral supply chains in the Great Lakes region. These arrangements, which link access to minerals with security guarantees or peace processes, raise questions about the permanent sovereignty of states over their natural resources, the legitimacy of agreements concluded in conditions of conflict and dependence, and the risk of reproducing historical patterns of extraction.

Resource-for-infrastructure agreements. Chinese state-backed companies have concluded numerous agreements under which infrastructure is financed and constructed in exchange for mining rights or mineral offtake, notably the Sicomines agreement in the DRC (examined in section 8.2). Such agreements have been criticised for opacity, undervaluation of mineral rights, and inadequate benefits for host states, and several have been renegotiated.

The Legal Character of Critical Mineral Agreements

Many critical mineral instruments are memoranda of understanding or political declarations rather than legally binding treaties. Their non-binding character provides flexibility and enables rapid conclusion, but it limits their enforceability and their capacity to provide the legal certainty required for long-term investment. Where they are binding, as in the case of the United States–Japan agreement, they are often narrow in scope. The emerging architecture of critical mineral agreements is thus fragmented and largely soft, contrasting with the comprehensive multilateral frameworks that govern trade generally under the WTO. Some commentators have proposed a multilateral critical minerals agreement, or a critical minerals plurilateral within the WTO, to establish common rules on export restrictions, subsidies, standards, and investment; others have proposed producer-consumer dialogues analogous to the International Energy Forum in the oil sector.

Producer Coordination

Producing states have explored coordination to strengthen their bargaining position. Proposals have been made for a lithium producers' organisation among Argentina, Bolivia, and Chile, analogous to OPEC, and Indonesia has floated the idea of a nickel producers' organisation. African states have pursued coordination through the African Union and regional economic communities, including the development of regional value chains. Such coordination faces the obstacles that undermined earlier producer associations for copper, bauxite, and other minerals in the 1970s — divergent interests, the availability of substitutes, and the entry of new producers — but it reflects the determination of producing states to avoid a

repetition of the historical pattern in which resource wealth flowed outward while host countries remained dependent on raw material exports.

Assessment

The legal regime of critical minerals is in the midst of rapid and contested formation. It is shaped by the imperatives of the energy transition, the pursuit of national security, the aspirations of producing states for development and industrialisation, and the claims of communities, workers, and indigenous peoples for respect for their rights. The instruments that constitute the regime — national legislation, trade rules, export controls, industrial policy, due diligence obligations, and bilateral and plurilateral agreements — are fragmented and sometimes contradictory. A central question for international energy and natural resources law is whether this regime can be developed in a manner that secures mineral supply for the energy transition while respecting the permanent sovereignty of producing states, ensuring that they and their populations share equitably in the benefits, and protecting human rights and the environment.

LEADING AUTHORITY

Indonesia — Measures Relating to Raw Materials (WTO, DS592, Panel Report 2022)

Measures. Indonesia banned the export of nickel ore with effect from 1 January 2020, and required domestic processing of nickel ore before export. The European Union, whose stainless steel industry relied on imported nickel ore, challenged the measures.

Findings. The panel found that the export ban was a prohibition on exportation within the meaning of Article XI:1 of the GATT 1994. It rejected Indonesia's defence under Article XI:2(a), holding that the ban was not "temporarily applied", since Indonesia intended it to remain in place indefinitely, and that Indonesia had not demonstrated a "critical shortage" of nickel ore, given its very large reserves. It also rejected Indonesia's defence under Article XX(d), holding that the ban was not "necessary" to secure compliance with Indonesia's mining laws, since a requirement in domestic law could not itself justify a measure inconsistent with the GATT simply because the measure implemented it. Having rejected the defences, the panel did not need to address the domestic processing requirement in detail.

Aftermath. Indonesia appealed in December 2022 to the Appellate Body, which has been unable to hear appeals since December 2019 because of the blocking of appointments by the United States; the report therefore has not been adopted and is not binding. Indonesia has maintained the ban. The European Union's Enforcement Regulation (Regulation (EU) No 654/2014), as amended in 2021 by Regulation (EU) 2021/167, permits it to adopt countermeasures where a dispute is blocked by an appeal into the void, although it has not done so in this dispute.

Significance. The case is the leading recent authority on the application of GATT disciplines to export restrictions designed to promote downstream processing. It

confirms the narrow scope of Article XI:2(a) and Article XX(d), and illustrates the practical limits of WTO enforcement in the absence of a functioning Appellate Body. It has been widely discussed in debates about the policy space available to resource-rich developing countries.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Treaty interpretation and negotiation simulation: a critical minerals partnership (hypothetical). *The states, companies, and facts in this exercise are fictitious.* The fictional Republic of Kavuma, a West African state with large lithium and graphite deposits, is negotiating a "Strategic Partnership on Sustainable Raw Materials" with the fictional Northern Economic Union, a consumer bloc of industrialised states. Kavuma's negotiating objectives are to secure investment in domestic lithium refining, to retain the right to impose an export ban on unprocessed spodumene from 2028, and to secure financing for a rail corridor. The Union's objectives are to secure long-term supply of lithium concentrate, to ensure that Kavuma does not impose export duties or restrictions on partners, and to require compliance with environmental and human rights standards consistent with its due diligence legislation.

Working in two negotiating teams, (a) prepare a negotiating brief identifying each side's legal constraints, including under WTO law (assuming both are WTO members), any applicable investment treaties, and regional obligations such as the ECOWAS Mining Directive and the AfCFTA; (b) negotiate and draft the key provisions of a partnership instrument (no more than 1,500 words), deciding whether it should be legally binding; and (c) write an individual reflective memorandum (1,000 words) assessing whether the agreed text respects Kavuma's permanent sovereignty over natural resources.

CHAPTER SUMMARY

- "Critical" and "strategic" minerals are classified by reference to economic importance and supply risk; classifications such as those in the EU Critical Raw Materials Act and the United States Energy Act of 2020 carry significant legal consequences.
- Lithium, cobalt, nickel, copper, and rare earth elements each present distinctive legal issues, from Chile's state-controlled lithium regime to the DRC's cobalt export controls, Indonesia's nickel ore ban, Panama's Cobre Panamá litigation, and China's rare earth dominance.
- Export restrictions on minerals are constrained by GATT Article XI and narrowly interpreted exceptions, as in *China — Raw Materials*, *China — Rare Earths*, and *Indonesia — Raw Materials*; China's export controls on minerals and technologies have become instruments of economic statecraft.

- Importing states pursue industrial policies combining benchmarks, accelerated permitting, public finance, and partnerships, notably under the EU Critical Raw Materials Act and United States legislation.
- Supply chain due diligence, grounded in the OECD Guidance, has become mandatory under Dodd-Frank section 1502, the EU Conflict Minerals Regulation, the EU Batteries Regulation, and general due diligence laws, supplemented by forced labour import bans.
- Critical mineral production is associated with child labour, forced labour, indigenous rights violations, and unsafe conditions; liability of downstream companies remains difficult to establish, as *Doe v Apple* shows.
- Circular economy regulation, especially the EU Batteries Regulation, and the law on transboundary waste shape the recycling of critical minerals.
- Critical mineral agreements form a fragmented, largely soft-law architecture of partnerships, bilateral agreements, and minerals-for-security arrangements, raising questions about permanent sovereignty and equity.

DOCTORAL RESEARCH QUESTIONS

1. Should WTO law be reformed to provide greater policy space for resource-rich developing countries to use export restrictions to promote downstream processing? Evaluate the arguments in the light of *Indonesia — Raw Materials*.
2. Are China's export controls on critical minerals and technologies justifiable under GATT Article XXI, and what are the implications of the answer for the security exception generally?
3. Does mandatory supply chain due diligence improve conditions at mine sites, or does it primarily shift compliance costs and encourage disengagement from high-risk areas? Design an empirical study to test the question.
4. Analyse the legal character and legitimacy of minerals-for-security agreements, such as the United States–Ukraine Reconstruction Investment Fund, by reference to the principle of permanent sovereignty over natural resources.
5. To what extent does the EU Batteries Regulation exercise extraterritorial regulatory power, and is that power consistent with international law and with the principle of common but differentiated responsibilities?
6. Is there a case for a multilateral critical minerals agreement, and what institutional form should it take?

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PART III

Electricity, Renewable Energy, and Emerging Technologies

The law of electricity systems and power markets, renewable energy development, nuclear energy and its international regime, and the emerging technologies — hydrogen, storage, carbon capture, offshore renewables, and digital grids — that are reshaping energy law.

9 CHAPTER

International Electricity Law and Power Market Regulation

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the technical and economic characteristics of electricity that shape the legal structure of electricity industries.
2. Analyse the regulation of generation, transmission, and distribution, including licensing, tariff regulation, and the role of system operators.
3. Evaluate the models and legal instruments of electricity market liberalisation, with particular reference to the European Union, the United States, the United Kingdom, and Nigeria.
4. Assess the law of grid access and connection, including third-party access rights and the allocation of network capacity.
5. Analyse the structure and key provisions of power purchase agreements, including those underpinning independent power projects in developing countries.
6. Evaluate the legal frameworks for cross-border electricity trade, including the West African Power Pool and the EU internal electricity market.
7. Identify the legal mechanisms for consumer protection and energy affordability in electricity markets.
8. Analyse the regulation of market abuse and competition in electricity markets and the enforcement powers of regulators.

Introduction

Electricity is the energy carrier of modern life and, increasingly, of the energy transition. It powers homes, hospitals, schools, factories, and digital infrastructure; its reliable supply is a precondition of economic development and of the realisation of numerous human rights. The decarbonisation of energy systems depends overwhelmingly on electricity: on the replacement of fossil-fuelled generation with renewable and other low-carbon sources, and on the electrification of transport,

heating, and industrial processes that currently rely on direct combustion of fossil fuels. The International Energy Agency has described the current period as the beginning of an "age of electricity", in which global electricity demand is growing faster than overall energy demand, driven by industrial growth, air conditioning, electric vehicles, and data centres. At the same time, around seven hundred million people, the large majority of them in sub-Saharan Africa, still lack access to electricity, and hundreds of millions more have access only to unreliable supply.

The law of electricity has historically been predominantly domestic. Electricity systems were built as national or regional monopolies, regulated by domestic statutes and agencies, and international law played a limited role. That position has changed. Cross-border electricity trade is expanding through regional power pools and interconnected markets; foreign investment in generation and networks is protected by investment treaties and has given rise to numerous arbitrations; international development institutions have shaped the reform of electricity sectors across the developing world; the European Union has built a supranational electricity market governed by detailed legislation; and international climate law is driving the transformation of generation. International electricity law, as treated in this chapter, therefore comprises both the comparative law of domestic electricity regulation — the models and techniques that states have adopted and that circulate among them — and the international and regional legal frameworks that govern cross-border trade, investment, and cooperation.

This chapter begins with the legal structure of electricity industries (section 9.1) and the regulation of generation, transmission, and distribution (section 9.2). It then examines electricity market liberalisation (section 9.3), grid access and third-party rights (section 9.4), and power purchase agreements (section 9.5). It concludes with cross-border electricity trading (section 9.6), consumer protection and energy affordability (section 9.7), and market abuse, competition, and regulatory enforcement (section 9.8). Throughout, particular attention is paid to Nigeria's Electricity Act 2023, which fundamentally restructured the legal framework of the Nigerian electricity supply industry, and to the regional framework of the Economic Community of West African States (ECOWAS). Renewable energy law is examined in Chapter 10, and the technologies of the future grid — storage, smart grids, and digital infrastructure — in Chapter 12.

9.1 Legal Structure of Electricity Industries

The Physical Characteristics of Electricity

The legal structure of electricity industries is shaped by the physical characteristics of electricity, which distinguish it from every other commodity. Four characteristics are particularly important.

First, **electricity cannot be economically stored in large quantities**, at least with technologies widely deployed until recently. Supply and demand must therefore be balanced continuously, second by second, across the entire interconnected system. If generation falls short of demand, the frequency of the system falls, and if the imbalance is not corrected quickly, protective devices disconnect generators and loads, potentially causing a cascading blackout. The need for continuous balancing requires a central coordinator — the system operator — with authority to dispatch generation, procure reserves, and take emergency action. The development of battery storage at scale, examined in Chapter 12, is beginning to change this characteristic, but the need for real-time balancing remains fundamental.

Second, **electricity flows according to the laws of physics, not the terms of contracts**. Electricity injected into a meshed network flows along all available paths in inverse proportion to their impedance (Kirchhoff's laws), so that a contract for the sale of electricity from a generator in one location to a consumer in another does not determine the physical path that the electricity takes. This "loop flow" phenomenon means that transactions between two parties affect the network capacity available to others, including in neighbouring countries, and complicates the allocation of transmission rights and costs.

Third, **demand for electricity is highly variable and, in the short term, relatively insensitive to price**. Demand varies by time of day, day of the week, and season, and most consumers historically had no means of observing or responding to real-time prices. Generation capacity must therefore be sufficient to meet peak demand plus a reserve margin, and much capacity is used only for a few hours a year.

Fourth, **the networks through which electricity is transmitted and distributed are natural monopolies**, as in the gas sector (examined in Chapter 6). It would be inefficient to duplicate transmission lines or distribution networks, and network owners therefore enjoy market power that must be regulated.

The Electricity Value Chain

The electricity supply industry comprises four principal functions:

1. **Generation** — the production of electricity in power stations using fossil fuels, nuclear energy, hydropower, wind, solar, geothermal, biomass, or other sources.
2. **Transmission** — the transport of electricity at high voltage over long distances from generators to load centres, and the operation of the interconnected system.
3. **Distribution** — the transport of electricity at lower voltages from the transmission system to end-users.

4. **Supply** (or retail) — the sale of electricity to end-users, including metering, billing, and customer service.

To these should be added **system operation** — the real-time balancing of supply and demand and the dispatch of generation, which may be performed by the transmission owner or by an independent entity — and **market operation** — the administration of wholesale markets, settlement, and related functions.

Models of Industry Structure

The legal structure of electricity industries can be classified according to the degree of vertical integration and competition. A widely used typology, derived from the work of Sally Hunt and Graham Shuttleworth, distinguishes four models.

Model 1: Vertically integrated monopoly. A single utility — state-owned or privately owned but regulated — owns and operates generation, transmission, distribution, and supply within its territory, with no competition at any level. This was the dominant model worldwide until the 1980s and remains common in many developing countries and in parts of the United States.

Model 2: Single buyer. The integrated utility, or a separate entity, acts as the sole purchaser of electricity from generators, including independent power producers (IPPs) that sell under long-term power purchase agreements, and resells it to distributors or consumers. Competition exists only for the right to build new generation, typically through competitive procurement of PPAs. The single buyer model was widely adopted in Asia and Africa in the 1990s as a means of attracting private investment in generation without full liberalisation.

Model 3: Wholesale competition. Generators compete to sell electricity to distribution companies and large consumers through a wholesale market, which may be organised as a mandatory pool, a power exchange, or bilateral contracts, with non-discriminatory access to the transmission network. Distribution companies retain monopoly supply to small consumers.

Model 4: Retail competition. All consumers, including households, may choose their supplier, and distribution networks are open to competing retailers. This is the model required by EU law and adopted in the United Kingdom, Australia, parts of the United States, and other jurisdictions.

	Competition in generation	Wholesale buyer choice	Retail choice	Typical examples
1. Vertically integrated monopoly	No	No	No	Many developing states; SE United States
2. Single buyer	Yes — for new PPAs	No	No	1990s Asian & African IPP models
3. Wholesale competition	Yes	Yes	No	Transitional systems; Nigeria (in law)
4. Retail competition	Yes	Yes	Yes	EU; Great Britain; Texas; Australia

Figure 9.1 Models of electricity industry structure, from vertically integrated monopoly to retail competition. *Source:* prepared for this edition.

These models are ideal types; real electricity industries frequently combine elements of several. Many developing countries, including Nigeria, have adopted a transitional model in which a liberalised framework exists in law but the market operates in practice through regulated contracts and administered prices, pending the development of the conditions for effective competition.

Ownership

Electricity industries may be state-owned, privately owned, or mixed. State ownership was justified historically by the natural monopoly characteristics of networks, the strategic importance of electricity, the need for very large investments, and the goal of universal access. Privatisation, beginning in Chile in the 1980s and in England and Wales with the Electricity Act 1989, was justified by the expected efficiency gains of private ownership, the need to mobilise private capital, and the fiscal pressures facing governments. The results of privatisation have been mixed, and in recent years some jurisdictions have taken steps towards renationalisation or increased state ownership, as in France, where the state acquired full ownership of Électricité de France (EDF) in 2023, and in Germany, where municipalities have repurchased distribution networks.

The Nigerian Electricity Supply Industry: Structural Evolution

The Nigerian electricity supply industry illustrates the evolution from vertically integrated monopoly to an unbundled, privatised, and increasingly decentralised structure. From 1972, electricity was supplied by the National Electric Power Authority (NEPA), a vertically integrated state monopoly that became a byword

for unreliable supply. The National Electric Power Policy of 2001 and the Electric Power Sector Reform Act 2005 (EPSRA) provided for the unbundling of NEPA into the Power Holding Company of Nigeria (PHCN), and subsequently into eighteen successor companies — six generation companies (GenCos), eleven distribution companies (DisCos), and the Transmission Company of Nigeria (TCN) — and established the Nigerian Electricity Regulatory Commission (NERC) as an independent regulator. In November 2013, the federal government completed the privatisation of the GenCos and DisCos, while retaining ownership of TCN, whose management was contracted for a period to a private operator. The Nigerian Bulk Electricity Trading Plc (NBET) was established as a transitional bulk trader, purchasing power from GenCos and IPPs under PPAs and reselling it to DisCos under vesting contracts.

The privatised industry has faced severe challenges: insufficient gas supply to thermal power stations; transmission constraints; high technical and commercial losses; tariffs below cost-recovery levels, resulting in large subsidy obligations and a liquidity crisis in which DisCos failed to remit full payment for energy received; widespread estimated billing and inadequate metering; and the inability of many DisCos to attract investment. Several DisCos were taken over by their creditors, including the Asset Management Corporation of Nigeria (AMCON), following defaults on acquisition financing.

The Electricity Act 2023 and Constitutional Decentralisation

Two legal developments in 2023 transformed the structure of the Nigerian electricity industry. First, the Constitution of the Federal Republic of Nigeria (Fifth Alteration) Acts of 2023 amended the Exclusive and Concurrent Legislative Lists to empower state Houses of Assembly to make laws with respect to the generation, transmission, and distribution of electricity in areas covered by the national grid within the state, extending the existing state competence over areas not covered by the national grid. Secondly, the Electricity Act 2023, signed into law in June 2023, repealed EPSRA and established a new framework that recognises the competence of states to regulate electricity markets within their territories. Under the Act, a state that establishes its own electricity market and regulatory body assumes regulatory oversight of the market within the state, and NERC must transfer regulatory functions over that market to the state regulator. By 2025, a number of states — beginning with Enugu, followed by Ekiti, Ondo, Imo, Oyo, Edo, Kogi, Ogun, Niger, Plateau, and others — had enacted electricity laws and established state regulators, and NERC had transferred regulatory oversight to them.

The Electricity Act 2023 also provides for the establishment of the Nigerian Independent System Operator (NISO), separating system operation from the transmission business of TCN; for the promotion of renewable energy and

a requirement that generation companies include renewable sources in their generation mix; for the development of a competitive wholesale market; for the protection of consumers; and for integrated resource planning. The constitutional and statutory decentralisation of electricity regulation in Nigeria is one of the most significant recent developments in African electricity law, and raises novel questions concerning the relationship between federal and state regulation, the regulation of interstate transmission and trading, and the coherence of the national market. These questions are examined in section 9.2 and in Chapter 24.

9.2 Generation, Transmission, and Distribution Regulation

Licensing

Most electricity regimes require persons carrying out generation, transmission, distribution, supply, or trading activities to hold a licence (or, in some jurisdictions, an authorisation, permit, or concession) issued by the regulator or the ministry. Licensing serves several functions: it enables the state to verify that licensees have the technical, financial, and managerial capacity to perform their activities safely and reliably; it provides the legal instrument through which conditions — technical standards, tariff rules, consumer protection obligations, reporting requirements, and compliance with codes — are imposed; and it gives the regulator enforcement powers, including the ultimate sanction of revocation.

Licence conditions are typically of two kinds: standard conditions applicable to all licensees of a given class, and special conditions applicable to particular licensees. In Great Britain, for example, the Electricity Act 1989 provides for licences for generation, transmission, distribution, supply, interconnection, and the operation of the electricity system, each subject to standard conditions modifiable by the Gas and Electricity Markets Authority (acting through Ofgem) under statutory procedures. Licence conditions require licensees to comply with industry codes, such as the Grid Code, the Distribution Code, the Balancing and Settlement Code, and the Connection and Use of System Code, which set out detailed technical and commercial rules and are themselves subject to regulated modification procedures.

Under Nigeria's Electricity Act 2023, no person may construct, own, or operate an undertaking for generation (above a specified threshold capacity), transmission, system operation, distribution, supply, or trading of electricity without a licence issued by NERC or, within a state that has established its own market, by the state regulator, subject to the Act's provisions on interstate and national functions. Small generation facilities below the threshold capacity are exempt from licensing but may be required to obtain a permit. The Act empowers the regulator to impose licence conditions, to amend licences, and to suspend or cancel licences for breach.

Generation Regulation

In liberalised systems, generation is in principle a competitive activity, and regulation focuses on safety, environmental performance, and compliance with technical standards and market rules rather than on prices. Entry into generation is subject to licensing, planning, and environmental permitting, and may be subject to resource adequacy policies (examined in section 9.3). The environmental regulation of generation — emissions limits under instruments such as the EU Industrial Emissions Directive (Directive 2010/75/EU), carbon pricing under emissions trading systems, and restrictions on coal-fired generation — has become a principal driver of the generation mix.

In single-buyer systems, generation investment is determined by the procurement of PPAs, often through competitive tenders conducted by the single buyer or a procurement agency, and the regulation of generation is largely contractual. In vertically integrated systems, generation investment is determined by the utility's integrated resource planning, subject to regulatory approval.

Transmission Regulation

Transmission is a natural monopoly and is regulated in all systems. Transmission regulation addresses five principal matters.

Ownership and unbundling. As in the gas sector, liberalised electricity regimes require the separation of transmission from generation and supply, to prevent discrimination against competing generators. EU law, under Directive (EU) 2019/944, requires member states to apply ownership unbundling, the independent system operator (ISO) model, or the independent transmission operator (ITO) model to transmission system operators (TSOs), and the third country certification provisions apply to electricity as to gas.

Tariffs. Transmission tariffs must allow the transmission owner to recover efficient costs and earn a reasonable return, while sending efficient signals for the location of generation and load. Tariff methodologies are subject to regulatory approval, and the allocation of costs between generators and consumers and between locations is often contentious.

Investment and planning. The expansion of transmission is essential for the integration of renewable generation, which is often located far from load centres, and for the electrification of the economy. Transmission planning involves the identification of needs, the selection of projects, the allocation of costs, and the obtaining of permits. In the United States, FERC Order No 1000 (2011) required transmission providers to participate in regional transmission planning processes and to adopt cost allocation methods for regional projects, and Order No 1920 (2024) required long-term, forward-looking regional transmission planning over a twenty-year horizon that takes account of factors including changes in the

resource mix and policy requirements. In the European Union, the Ten-Year Network Development Plan prepared by the European Network of Transmission System Operators for Electricity (ENTSO-E) and the Trans-European Networks for Energy (TEN-E) Regulation (EU) 2022/869 identify projects of common interest that benefit from accelerated permitting and eligibility for EU financing.

System operation. The system operator is responsible for real-time balancing, dispatch, procurement of ancillary services (such as frequency response and reserves), management of congestion, and emergency actions. System operation may be performed by the TSO, by an ISO, or, in the United States, by regional transmission organisations (RTOs) and independent system operators that operate across the networks of multiple transmission owners. The independence of the system operator from market participants is essential to non-discriminatory dispatch. The establishment of the Nigerian Independent System Operator under the Electricity Act 2023 reflects this principle.

Reliability standards. System operators and transmission owners must comply with reliability standards governing the security of the system. In North America, the North American Electric Reliability Corporation (NERC), certified by FERC as the Electric Reliability Organization under the Energy Policy Act of 2005, develops and enforces mandatory reliability standards, which became legally binding following the Northeast blackout of August 2003. In the European Union, network codes adopted under Regulation (EU) 2019/943, such as the System Operation Guideline (Regulation (EU) 2017/1485), establish harmonised operational security requirements.

Distribution Regulation

Distribution networks are natural monopolies regulated by tariff controls, quality of service standards, and connection obligations. The role of distribution networks is changing fundamentally with the growth of distributed energy resources — rooftop solar, small-scale storage, electric vehicles, heat pumps, and demand response — which transform distribution networks from passive conduits delivering electricity in one direction into active systems managing bidirectional flows. Distribution system operators (DSOs) are increasingly required to procure flexibility services, to facilitate the connection of distributed resources, and to coordinate with TSOs. EU law, under Directive (EU) 2019/944, requires DSOs above a specified size to be legally and functionally unbundled from supply activities and empowers them to procure flexibility services through market-based procedures.

Tariff Regulation

Tariff regulation is the core of economic regulation of electricity networks and, in non-competitive systems, of end-user prices. The two principal approaches,

introduced in Chapter 6 in the context of gas, are cost-of-service (rate-of-return) regulation and incentive regulation. In Great Britain, Ofgem's RIIO framework ("Revenue = Incentives + Innovation + Outputs") sets revenue allowances for network companies over five-year price control periods, with incentives linked to outputs such as reliability, connections, customer satisfaction, and environmental performance. In the United States, state public utility commissions regulate retail rates and distribution, while FERC regulates wholesale rates and interstate transmission under the Federal Power Act of 1935, which requires that rates be "just and reasonable" and not unduly discriminatory.

In Nigeria, NERC has regulated end-user tariffs through the Multi-Year Tariff Order (MYTO) methodology, which sets tariffs for each DisCo on the basis of projected costs, including generation, transmission, and distribution costs, losses, and a return on investment. Because full cost-reflective tariffs would have been politically unacceptable, the federal government historically subsidised the gap between cost-reflective and allowed tariffs, accumulating large subsidy obligations. In April 2024, NERC introduced a significant increase in tariffs for customers in "Band A" — those receiving a minimum of twenty hours of supply per day — while maintaining lower tariffs for customers receiving less reliable supply. The service-based tariff structure links tariffs to the quality of supply, and DisCos that fail to deliver the specified hours of supply to Band A feeders must reclassify them, reducing the applicable tariff. The tariff reform illustrates the use of regulatory design to link price increases to service improvements and to address the political economy of tariff reform.

Federal and State Regulation

In federal systems, the allocation of regulatory competence between federal and state authorities is a fundamental feature of electricity law. In the United States, the Federal Power Act divides jurisdiction: FERC regulates interstate transmission and wholesale sales of electricity in interstate commerce, while states regulate retail sales, distribution, and generation facilities. The boundary has been the subject of frequent litigation. In *FERC v Electric Power Supply Association* (2016), the Supreme Court upheld FERC's jurisdiction to regulate compensation for demand response in wholesale markets, holding that demand response directly affects wholesale rates.[1] In *Hughes v Talen Energy Marketing LLC* (2016), decided the same year, the Court held that a Maryland programme guaranteeing a new generator a rate different from the wholesale market clearing price was pre-empted, because it set an interstate wholesale rate in contravention of FERC's exclusive authority.[2]

[1] *FERC v Electric Power Supply Association*, 577 US 260 (2016).
[2] *Hughes v Talen Energy Marketing LLC*, 578 US 150 (2016).

Nigeria's constitutional and statutory reforms of 2023 introduced a comparable division. Under the Electricity Act 2023, NERC retains regulatory oversight of interstate and international electricity matters, including the transmission network operated by TCN, system operation by NISO, and trading across state boundaries, while state regulators oversee markets within their states, including distribution and retail supply. The transition raises questions about the treatment of existing DisCo licences, many of which cover multiple states; the determination of tariffs where wholesale costs are regulated federally and retail tariffs by states; and the coherence of technical codes and standards. Several DisCos have established subsidiaries to hold licences in individual states, and the relationship between federal and state regulators will be shaped by the courts and by intergovernmental cooperation.

9.3 Electricity Market Liberalization

The Rationale for Liberalisation

Electricity market liberalisation — the introduction of competition in generation and supply, combined with the regulation of networks and the unbundling of competitive and monopoly activities — was driven by a combination of economic theory, technological change, and political ideology. Economists argued that the natural monopoly characteristics of electricity were confined to networks, and that competition in generation and supply would promote efficiency, innovation, and lower prices. The development of combined-cycle gas turbines, which could be built at smaller scale and lower cost than traditional coal and nuclear plants, reduced economies of scale in generation and lowered barriers to entry. The broader privatisation and deregulation movement of the 1980s and 1990s, and the influence of international financial institutions, spread the model across the world.

Pioneers: Chile, England and Wales, and Norway

Chile was the first country to liberalise its electricity industry, under the General Electricity Services Law of 1982, which introduced a wholesale market based on marginal cost dispatch and the separation of generation, transmission, and distribution. England and Wales followed with the Electricity Act 1989, which privatised the Central Electricity Generating Board and the area boards and established a mandatory wholesale pool, later replaced in 2001 by the New Electricity Trading Arrangements (NETA), based on bilateral contracts and a balancing mechanism, and extended to Great Britain as a whole in 2005 as the British Electricity Trading and Transmission Arrangements (BETTA). Norway's Energy Act of 1990 introduced competition while retaining substantial public ownership, and the Norwegian market became the nucleus of the Nordic power exchange, Nord Pool, which has expanded to cover the Nordic and Baltic countries.

The European Union Internal Electricity Market

The European Union has built a supranational internal electricity market through four successive legislative packages: the First Electricity Directive (96/92/EC), the Second Directive (2003/54/EC), the Third Energy Package (2009), and the Clean Energy for All Europeans package (2019), comprising Directive (EU) 2019/944 on common rules for the internal market for electricity and Regulation (EU) 2019/943 on the internal market for electricity, together with Regulation (EU) 2019/942 establishing the European Union Agency for the Cooperation of Energy Regulators (ACER). The legislation requires retail competition for all consumers, unbundling of networks, regulated third-party access, independent national regulatory authorities, and the integration of national markets through common rules for cross-border trading, examined in section 9.6.

In response to the energy price crisis of 2021–2023, the European Union adopted a reform of electricity market design in 2024, through Regulation (EU) 2024/1747 and Directive (EU) 2024/1711. The reform retained the marginal pricing model of the wholesale market but sought to shield consumers from short-term price volatility by promoting long-term contracts: it requires that direct price support schemes for new investments in renewable and nuclear generation take the form of two-way contracts for difference (under which the generator receives a top-up payment when the market price falls below a strike price and repays the difference when it rises above it), facilitates power purchase agreements, strengthens consumer rights to fixed-price contracts, and empowers the Council to declare a regional or Union-wide electricity price crisis permitting member states to set regulated prices below cost for households and small businesses.

The United States

The United States has a patchwork of market structures. FERC Order No 888 (1996) required public utilities owning interstate transmission to provide open-access transmission service on a non-discriminatory basis under an open access transmission tariff, and encouraged the formation of independent system operators. Order No 2000 (1999) encouraged the formation of regional transmission organisations. Today, RTOs and ISOs — including PJM Interconnection, the Midcontinent ISO, the California ISO, the Electric Reliability Council of Texas (ERCOT), the Southwest Power Pool, ISO New England, and the New York ISO — operate organised wholesale markets covering around two-thirds of United States electricity demand, while the South-east and much of the West retain vertically integrated utilities. Retail competition was introduced in some states, including Texas, Pennsylvania, Illinois, and much of the North-east, but abandoned or never adopted in others, particularly following the California electricity crisis of 2000–2001, in which flawed market design, market manipulation

by traders including Enron, and supply shortages caused soaring prices, rolling blackouts, and the bankruptcy of Pacific Gas and Electric.

Wholesale Market Design

Wholesale electricity markets may be organised in various ways. In a **mandatory pool**, all generation is sold into a central pool, which dispatches generation in order of bid price and pays all dispatched generators the marginal (clearing) price. In **bilateral markets**, generators and buyers contract directly, with a balancing mechanism operated by the system operator to resolve imbalances. In **power exchanges**, standardised products are traded anonymously on an exchange. Most markets combine these elements, with forward bilateral contracting, day-ahead and intraday auctions on power exchanges, and real-time balancing by the system operator.

A central feature of most wholesale markets is **marginal pricing**, under which all generators dispatched in a given period receive the price bid by the most expensive generator needed to meet demand. Marginal pricing provides efficient short-term dispatch signals, but it means that when gas-fired generators set the marginal price, all generators, including those with low costs such as renewable and nuclear plants, receive the high gas-based price. During the 2021–2023 crisis, this produced very high electricity prices and windfall profits for inframarginal generators, prompting the EU to adopt Council Regulation (EU) 2022/1854, which imposed a temporary cap on market revenues of inframarginal generators and a temporary solidarity contribution on fossil fuel companies, and prompting the electricity market design reform described above.

Markets also differ in their treatment of location. **Nodal pricing** (locational marginal pricing), used in most United States RTOs, sets different prices at each node of the network, reflecting transmission constraints and losses. **Zonal pricing**, used in Europe, sets a single price across large bidding zones, often corresponding to national borders, with congestion between zones managed through cross-border capacity allocation. The configuration of bidding zones is politically sensitive; Great Britain considered and, in 2025, rejected the introduction of zonal or nodal pricing in its Review of Electricity Market Arrangements.

Resource Adequacy and Capacity Mechanisms

A persistent concern in liberalised markets is whether energy-only markets — in which generators are paid only for the electricity they produce — provide sufficient incentives for investment in capacity needed to ensure security of supply at times of peak demand. Many jurisdictions have introduced **capacity mechanisms** that pay generators (and, increasingly, storage and demand response providers) for being available. Capacity mechanisms include capacity markets, strategic reserves, and capacity payments. In the European Union, capacity mech-

anisms are subject to the State aid rules and to conditions in Regulation (EU) 2019/943, including emission limits that exclude new coal-fired plants. In *Tempus Energy Ltd v European Commission* (2018), the General Court annulled the Commission's decision approving the British capacity market under the State aid rules without opening a formal investigation, on the ground that the Commission should have had doubts about the treatment of demand-side response; the British capacity market was suspended for around a year until the Commission re-approved it following a formal investigation, and in 2021 the Court of Justice set aside the General Court's judgment on appeal.

Liberalisation in Developing Countries

Electricity sector reform in developing countries, promoted from the 1990s by the World Bank and other development institutions, typically followed a standard sequence: corporatisation of the state utility, establishment of an independent regulator, introduction of IPPs, unbundling, privatisation of distribution and generation, and the eventual creation of a competitive wholesale market. Experience has shown that the standard model often fails to deliver its expected benefits in contexts characterised by tariffs below cost-recovery levels, weak payment discipline, high losses, limited regulatory capacity, and political interference. Scholars and practitioners have increasingly argued for "hybrid" models adapted to local conditions, which may retain significant state roles while introducing private participation and competition selectively. The Nigerian experience since privatisation in 2013 illustrates both the difficulty of implementing a liberalised model in the absence of financial viability and the potential of decentralisation to create new dynamics of reform.

9.4 Grid Access and Third-Party Rights

The Importance of Access

Access to the grid is the lifeblood of competition and of the energy transition. A generator that cannot connect to the network, or that cannot obtain the transmission capacity necessary to deliver its output to buyers, cannot compete; a consumer that cannot use the network to buy from a competing supplier cannot exercise choice. The law of grid access determines who may connect to and use the network, on what terms, at what cost, and with what priority when capacity is scarce. It has become one of the most important and contested areas of electricity law, as the queue of renewable and storage projects seeking connection has grown far faster than the capacity of networks to accommodate them.

Third-Party Access to Networks

Liberalised electricity regimes, like gas regimes, require network owners to grant third-party access to their networks on non-discriminatory terms. EU law, under

Article 6 of Directive (EU) 2019/944, requires member states to ensure a system of third-party access to transmission and distribution systems based on published tariffs, applicable to all customers and applied objectively and without discrimination between system users. Access may be refused only where the operator lacks the necessary capacity, and refusals must be duly substantiated. The Court of Justice has emphasised that the right of access is a fundamental element of the internal market. In *VEMW* (Case C-17/03, 2005), the Court held that the grant of priority access to cross-border transmission capacity to a former monopoly under pre-liberalisation contracts was discriminatory and contrary to the Electricity Directive, notwithstanding the age of the contracts.

In the United States, FERC Order No 888 required public utilities to provide open access transmission service under an open access transmission tariff, distinguishing between network integration service and point-to-point service, and requiring utilities to take transmission service for their own wholesale sales under the same tariff ("functional unbundling").

Connection Rights and Procedures

Connection to the network is distinct from use of the network. Connection procedures determine how a generator, storage facility, or large consumer obtains a physical connection to the grid, including the studies required to assess the impact of the connection on the network, the reinforcements needed, the allocation of connection costs, and the timeline. In many jurisdictions, connection queues have become a major bottleneck for the energy transition. In the United States, the capacity of generation and storage projects seeking interconnection grew to more than two thousand gigawatts by the mid-2020s — far exceeding existing installed capacity — and the median time from interconnection request to commercial operation lengthened to several years. FERC Order No 2023 (2023) reformed interconnection procedures by replacing the "first-come, first-served" serial study process with a "first-ready, first-served" cluster study process, imposing stricter financial commitments and site control requirements on applicants, and imposing penalties on transmission providers that fail to complete studies on time.

Great Britain faced a similar challenge, with a connection queue exceeding seven hundred gigawatts by 2024, many of the projects in it speculative or unlikely to proceed. Ofgem approved reforms in 2025 replacing the first-come, first-served approach with a process that prioritises projects that are ready to proceed and needed to meet the government's Clean Power 2030 objectives. The EU's Action Plan for Grids (2023) and the electricity market design reform of 2024 require transparency of available grid capacity and encourage flexible connection agreements, under which projects may connect more quickly in exchange for accepting curtailment in certain conditions.

Allocation of Connection Costs

The allocation of the costs of connecting new generation — between the connecting party and network users generally — affects the location and economics of generation investment. Under a **deep connection charging** approach, the connecting party pays the full cost of the connection assets and any wider network reinforcement required; under a **shallow** approach, it pays only for the assets connecting it to the nearest point of the network, with wider reinforcement costs socialised through network tariffs; and under **shallowish** approaches, it pays for some reinforcement. Deep charging sends strong locational signals but may deter investment, particularly by first movers in areas that require reinforcement; shallow charging encourages connection but may lead to inefficient location decisions.

Priority Dispatch and Curtailment

Renewable energy has historically benefited from **priority access** to the grid and **priority dispatch**, under which system operators must dispatch renewable generation ahead of conventional generation, subject to the security of the system. The EU Renewable Energy Directive 2009/28/EC required member states to provide priority or guaranteed access and priority dispatch for renewable electricity. The 2019 Clean Energy package narrowed priority dispatch to small installations and demonstration projects, reflecting the view that renewable generation had matured and should be integrated into markets, while requiring compensation for curtailment of renewable generation in specified circumstances under Article 13 of Regulation (EU) 2019/943. As the share of variable renewable generation grows, curtailment — the reduction of output from renewable generators because of network constraints or oversupply — has increased, and the allocation of the costs and risks of curtailment has become an important issue in grid access law and in PPAs.

Grid Access in Nigeria and Developing Countries

In Nigeria, the transmission network's limited capacity — wheeling capacity has frequently been below the available generation capacity — has constrained the evacuation of power from generation plants, resulting in "stranded" generation and payments to GenCos and IPPs for capacity that cannot be used. The Grid Code for the Nigeria Electricity Transmission System, issued by NERC, sets out the connection and operational requirements for users of the transmission system, and NERC's regulations on embedded generation, mini-grids, and eligible customers define access rights for distributed generators and large consumers. The **Eligible Customer Regulations**, first issued by NERC in 2017 and since revised, permit large consumers meeting specified thresholds to purchase electricity directly from generators or traders rather than from their local DisCo,

subject to payment of network charges — a limited form of retail competition. The **Mini-Grid Regulations** (2016, revised 2023) establish a framework for the development of isolated and interconnected mini-grids, including provisions for the treatment of mini-grids when the main grid arrives, and have supported significant investment in rural electrification, examined in Chapter 23.

Access for Distributed Energy Resources

Distributed energy resources (DERs) raise new questions of access. Consumers who install rooftop solar may wish to export surplus electricity to the grid ("prosumers"), and aggregators may wish to combine many small resources to participate in wholesale markets. EU law confers rights on "active customers" to generate, store, sell, and participate in flexibility and energy efficiency schemes, and on "citizen energy communities" and "renewable energy communities" to engage in generation, supply, aggregation, and storage. FERC Order No 2222 (2020) requires RTOs and ISOs to allow aggregations of distributed energy resources to participate in wholesale markets. The compensation of prosumers for exported electricity — through net metering, feed-in tariffs, or market-based arrangements — and the allocation of network costs between prosumers and other consumers have been contentious, as illustrated by debates over the revision of net metering in California in 2022 and 2023, which significantly reduced the compensation paid to new rooftop solar customers for exported electricity.

9.5 Power Purchase Agreements

Function and Types

A power purchase agreement (PPA) is a contract for the sale of electricity (and, often, associated capacity, ancillary services, and environmental attributes) by a generator to a buyer over a defined period. PPAs serve different functions in different market structures.

In **single-buyer systems** and developing countries, the PPA between an independent power producer and a state-owned utility or bulk purchaser is the foundation of private investment in generation. The long-term PPA — typically twenty to twenty-five years — provides the revenue certainty that lenders require to finance the project, and allocates the principal risks between the IPP and the offtaker, often with support from the government (examined in Chapter 16).

In **liberalised markets**, PPAs are used by generators and buyers to hedge price risk, and by corporate buyers to procure renewable electricity. **Corporate PPAs** — contracts between generators, usually of renewable energy, and corporate consumers such as technology companies, manufacturers, and retailers — have grown rapidly, driven by corporate sustainability commitments. Corporate PPAs may be **physical** (involving the delivery of electricity through the grid, often via a licensed supplier in a "sleeved" arrangement) or **virtual** or **financial** (a contract

for difference under which the parties settle the difference between an agreed strike price and the market price, without physical delivery, while the generator sells its output into the market and the buyer purchases its electricity from its supplier).

Appendix E contains a sample renewable energy PPA with annotations. The following discussion focuses on the long-term IPP PPA in developing countries, which raises the most significant legal issues.

Key Provisions of an IPP PPA

Capacity and energy payments. The tariff structure of an IPP PPA for a dispatchable plant typically comprises a **capacity payment** (or availability payment), payable for making the plant's capacity available to the offtaker regardless of whether it is dispatched, designed to cover the IPP's fixed costs, debt service, and return on equity; and an **energy payment**, payable for electricity actually delivered, designed to cover variable costs, principally fuel. The capacity payment is the economic equivalent of take-or-pay in gas contracts, allocating demand risk to the offtaker. For variable renewable generation, PPAs typically provide for payment per unit of energy delivered, with "deemed energy" payments where the plant is available but cannot deliver because of grid constraints or curtailment by the offtaker.

Dispatch. The PPA provides for the offtaker (or system operator) to dispatch the plant within specified technical parameters, and allocates the risk of non-dispatch.

Fuel supply. For thermal plants, the PPA allocates fuel supply risk. The IPP may procure fuel itself under a fuel supply agreement, with fuel costs passed through in the energy payment, or the offtaker may supply fuel under a tolling arrangement. In Nigeria, the unreliability of gas supply to power plants has been a major source of risk, and PPAs have addressed whether the IPP is entitled to capacity payments when it cannot generate because of gas shortfalls.

Currency and indexation. Because IPPs in developing countries typically finance projects in foreign currency, PPAs commonly denominate tariffs in, or index them to, the United States dollar, transferring currency risk to the offtaker. Large devaluations, such as those experienced by Nigeria in 2016 and 2023–2024, Egypt in 2016, and Pakistan, can render dollar-indexed tariffs unaffordable for the offtaker and consumers, creating pressure for renegotiation.

Change in law. PPAs typically protect the IPP against changes in law that increase its costs or reduce its revenues, through tariff adjustments or compensation.

Force majeure. PPAs distinguish between **natural force majeure** (events outside the control of either party, such as natural disasters), and **political force majeure** (events within the control of the government, such as changes in law,

expropriation, and political violence), and allocate the consequences differently: political force majeure events commonly entitle the IPP to continued capacity payments and, if prolonged, to terminate and receive a termination payment.

Termination and termination payments. The PPA specifies events of default by each party and the termination payments payable on termination. Upon termination for offtaker default or prolonged political force majeure, the termination payment typically covers outstanding debt and a return on equity; upon termination for IPP default, it may be limited to outstanding debt or less. Termination payments are often backed by government guarantees or put-and-call option agreements under which the government is obliged to purchase the plant.

Credit support. Because state-owned offtakers in many developing countries are not creditworthy, PPAs are supported by credit enhancement mechanisms: letters of credit for short-term payment obligations, escrow accounts, government guarantees or support agreements, and partial risk guarantees from multilateral development banks such as the World Bank's International Development Association (IDA) and the Multilateral Investment Guarantee Agency (MIGA).

Dispute resolution. PPAs typically provide for expert determination of technical disputes and international arbitration of other disputes.

The Azura-Edo IPP

The Azura-Edo IPP, a 461 MW open-cycle gas-fired power plant in Edo State that achieved commercial operation in 2018, was the first project-financed IPP in Nigeria to reach financial close under the post-privatisation framework. Its PPA with NBET was supported by a put-call option agreement with the federal government and by a partial risk guarantee from the World Bank backing NBET's payment obligations under a letter of credit, together with political risk insurance from MIGA. The project's financing structure became a template for subsequent projects, but its experience also illustrated the risks: as the liquidity crisis in the Nigerian electricity market deepened, the World Bank's guarantee was reportedly called upon to cover arrears in NBET's payments. The project illustrates the extent to which long-term PPAs in developing countries depend on sovereign and multilateral credit support to compensate for the weakness of the offtaker.

PPA Disputes and Renegotiation

Long-term PPAs have generated significant disputes, particularly where economic crises, currency devaluations, or changes in government have made tariffs appear excessive. In the aftermath of the Asian financial crisis of 1997–1998, the Indonesian state electricity company PLN sought to renegotiate or cancel numerous IPP PPAs concluded under the Suharto government. In *Himpurna California Energy Ltd v PT (Persero) Perusahaan Listrik Negara* (1999), an UNCITRAL tribunal

awarded the claimant more than USD 390 million for PLN's breach of an energy sales contract for a geothermal project, while reducing the damages claimed to take account of the economic crisis; when the award was not paid, the claimants recovered under political risk insurance from the Overseas Private Investment Corporation. In India, the Dabhol power project, developed by a consortium led by Enron, became the subject of disputes with the Maharashtra State Electricity Board in 2001 after the board refused to pay tariffs it regarded as excessive, leading to arbitrations, litigation in India and elsewhere, and eventually a settlement under which the plant was transferred to Indian state ownership.

More recently, Pakistan's government undertook in 2024 a programme of renegotiating or terminating PPAs with IPPs, citing the burden of capacity payments on consumers and the public finances, and several IPPs agreed to early termination or conversion of their contracts to "take-and-pay" terms under which payment is made only for energy delivered. Ghana renegotiated several IPP PPAs from 2020 onwards to reduce excess capacity charges. These episodes illustrate the tension between the sanctity of contracts necessary to attract investment and the political and economic pressures that arise when long-term PPAs impose burdens perceived as unsustainable. They also raise questions about the role of investment treaty protection and the extent to which renegotiation under pressure is consistent with the fair and equitable treatment standard, examined in Chapter 14.

Corporate PPAs and the Energy Transition

Corporate PPAs have become a major driver of renewable energy investment in Europe and North America and are growing in emerging markets. Their legal issues include the allocation of "shape" and "volume" risk (the mismatch between the generator's variable output and the buyer's demand profile); the treatment of curtailment and negative prices; the transfer of guarantees of origin or renewable energy certificates (examined in Chapter 10); change in law; credit support; and accounting treatment, including whether a virtual PPA must be accounted for as a derivative. The EU electricity market design reform of 2024 requires member states to remove unjustified barriers to PPAs and to ensure the availability of instruments, such as guarantee schemes, to reduce the financial risks of PPAs for buyers. In Nigeria, the eligible customer framework and the bilateral trading opportunities created by the Electricity Act 2023 have facilitated the emergence of corporate PPAs between generators and industrial consumers, including captive and embedded renewable projects.

9.6 Cross-Border Electricity Trading

Benefits and Challenges

Interconnecting national electricity systems offers substantial benefits. Interconnection enables countries to share generation reserves, reducing the capacity each

must maintain to ensure security of supply; to exploit differences in generation costs, resource endowments, and demand patterns, so that low-cost hydropower in one country can displace expensive thermal generation in another; to integrate variable renewable generation over larger areas, smoothing fluctuations in wind and solar output; and to provide mutual assistance in emergencies. For small developing countries, regional integration can make possible the development of large, low-cost generation projects that would be too large for any single national market.

Cross-border electricity trade nonetheless faces significant obstacles: the physical characteristics of electricity, which require close coordination of system operation across borders; differences in market structures, regulatory frameworks, and technical standards; concerns about dependence on foreign supply and the loss of national control; the creditworthiness of buyers; and the need to agree on the allocation of interconnector costs and capacity. The legal frameworks for cross-border trade range from bilateral agreements between utilities, through regional power pools with common rules, to fully integrated supranational markets.

The European Union Internal Electricity Market

The European Union has developed the world's most advanced framework for cross-border electricity trade. Regulation (EU) 2019/943 and the network codes and guidelines adopted under it establish common rules for cross-border trading, capacity allocation, and congestion management. The **Capacity Allocation and Congestion Management Guideline** (Regulation (EU) 2015/1222) established **single day-ahead coupling** (SDAC) and **single intraday coupling** (SIDC), under which cross-border transmission capacity is allocated implicitly together with energy through a common price coupling algorithm (EUPHEMIA), maximising the use of interconnectors and producing price convergence across bidding zones when capacity is not congested. Regulation (EU) 2019/943 requires TSOs to make available for cross-border trade at least seventy per cent of the transmission capacity of interconnectors and internal critical network elements, subject to transitional arrangements. ACER oversees the implementation of these rules, decides on certain cross-border matters where national regulators cannot agree, and monitors the market. The EU has also set an interconnection target of fifteen per cent of installed generation capacity for each member state by 2030.

Following the United Kingdom's withdrawal from the European Union, Great Britain left the EU's internal electricity market and market coupling arrangements. The Trade and Cooperation Agreement of 2020 contains an energy title that provides for cooperation on electricity trading over interconnectors and commits the parties to develop new arrangements for the efficient use of interconnectors, although the more efficient implicit allocation of capacity

has not yet been restored. The episode illustrates both the depth of integration achieved within the EU and the costs of exit.

The West African Power Pool

The **West African Power Pool** (WAPP) was established by Decision A/DEC.5/12/99 of the ECOWAS Authority of Heads of State and Government in 1999, and its Articles of Agreement were adopted in 2006, establishing it as a specialised institution of ECOWAS. The WAPP's mission is to integrate the national power systems of the fourteen continental ECOWAS member states into a unified regional electricity market, with the objective of providing reliable and affordable electricity. Its members are the national utilities, transmission companies, and generation companies of member states. The WAPP has developed a master plan of priority regional generation and transmission projects, including the **CLSG interconnection** linking Côte d'Ivoire, Liberia, Sierra Leone, and Guinea, the **OMVG interconnection** linking the Gambia River Basin Development Organisation states (Gambia, Guinea, Guinea-Bissau, and Senegal), and the **North Core** project linking Nigeria, Niger, Benin, Togo, and Burkina Faso.

The **ECOWAS Regional Electricity Regulatory Authority** (ERERA), established in 2008 by a Supplementary Act of the ECOWAS Authority, is the regional regulator, responsible for regulating cross-border electricity exchanges among member states, approving regional market rules, setting tariffs for regional transmission, and resolving disputes. The **ECOWAS Regional Electricity Market** was launched in phases, with the first phase commencing in 2018 on the basis of bilateral contracts and regional market rules approved by ERERA, and later phases envisaging a day-ahead market. The legal framework is supported by the ECOWAS Energy Protocol of 2003, examined in Chapter 6, and by the ECOWAS directive on the organisation of the regional electricity market.

Nigeria has historically been a significant exporter of electricity to neighbouring states, supplying Benin, Togo, and Niger under bilateral contracts. In August 2023, following the military coup in Niger and the imposition of sanctions by ECOWAS, Nigeria suspended electricity supply to Niger, illustrating the vulnerability of cross-border electricity trade to political crises and the use of energy interdependence as an instrument of regional pressure. Supply was restored in 2024 after ECOWAS lifted most sanctions. Nigeria's own domestic supply shortages have periodically generated political pressure to reduce exports, and the payment performance of foreign buyers has been a recurrent concern.

Other Regional Power Pools

The **Southern African Power Pool** (SAPP), established in 1995 under an intergovernmental memorandum of understanding among member states of the Southern African Development Community (SADC), was the first regional

power pool in Africa. It operates competitive day-ahead, intraday, and forward markets alongside bilateral contracts, and the **Regional Electricity Regulators Association of Southern Africa** (RERA) facilitates regulatory harmonisation. The **Eastern Africa Power Pool** (EAPP), established in 2005, and the **Central African Power Pool** (CAPP), established in 2003, are at earlier stages of development, while the **Comité Maghrébin de l'Électricité** coordinates North African interconnection.

At the continental level, the **African Single Electricity Market** (AfSEM), launched by the African Union in 2021 with the support of the European Union, aims to integrate the regional power pools into a single continental market by 2040, through a continental transmission master plan, harmonised regulatory frameworks, and continental market rules. The **Continental Power System Master Plan**, adopted in 2023, identifies priority interconnections. The AfSEM is linked to the African Continental Free Trade Area, which may provide a legal framework for trade in electricity services.

Elsewhere, the **ASEAN Power Grid**, established under ASEAN agreements and memoranda of understanding, has developed through bilateral interconnections and the Lao PDR–Thailand–Malaysia–Singapore Power Integration Project, the first multilateral electricity trade in ASEAN. The **Gulf Cooperation Council Interconnection Authority** (GCCIA) operates an interconnection linking the six GCC states, used principally for mutual support in emergencies. In South Asia, India's cross-border electricity trade with Bhutan, Nepal, and Bangladesh is governed by bilateral agreements and Indian regulations on cross-border trade.

International Law and Cross-Border Electricity

The international law governing cross-border electricity trade is relatively undeveloped. Electricity is generally treated as a good for the purposes of GATT 1994, although the classification of electricity transmission and distribution services under the General Agreement on Trade in Services is uncertain. The Energy Charter Treaty's transit provisions, examined in Chapter 6, apply to electricity as well as to gas. Regional frameworks such as the ECOWAS Energy Protocol and the EU legislation provide the most developed rules. The legal issues raised by long-distance interconnections — such as the proposed Xlinks project to transmit solar and wind power from Morocco to Great Britain through submarine cables, and the proposed interconnections between North Africa and Europe — include the regulatory treatment of interconnectors, the allocation of costs and capacity, security of supply, and the law of submarine cables, examined in Chapter 20.

9.7 Consumer Protection and Energy Affordability

Electricity as an Essential Service

Electricity is an essential service, without which households cannot light their homes, cook, heat or cool their dwellings, store food and medicines, communicate, study, or work. Access to affordable and reliable electricity is closely connected to the realisation of human rights, including the rights to an adequate standard of living, to housing, to health, and to education, as examined in Chapter 23. Sustainable Development Goal 7 calls for universal access to affordable, reliable, sustainable, and modern energy by 2030. Electricity consumers are, moreover, in a structurally weak position relative to suppliers: they depend on a continuous supply that they cannot easily replace, they face complex contracts and tariffs, and in many markets they have little or no choice of supplier. Electricity law therefore contains extensive provisions for consumer protection, which have become more important as liberalisation has exposed consumers to market risks and as the energy price crisis of 2021–2023 created severe affordability pressures.

Consumer Rights in EU Law

Directive (EU) 2019/944 establishes a comprehensive set of consumer rights, including: the right to a contract with specified information; the right to switch supplier within three weeks (and, from 2026, within twenty-four hours) free of charge; the right to clear billing information, at least once a year based on actual consumption; the right to a dynamic electricity price contract for consumers with smart meters; the right to request a smart meter; access to out-of-court dispute settlement mechanisms; and protection for vulnerable customers and against disconnection. Member states must define the concept of vulnerable customers, which may refer to energy poverty, and ensure that the rights and obligations linked to vulnerable customers are applied, including measures to prohibit disconnection in critical times. The 2024 market design reform strengthened these rights, including by giving consumers the right to fixed-price, fixed-term contracts and requiring suppliers to have appropriate hedging strategies to limit the risk that supplier failures harm consumers, and by establishing a framework for suppliers of last resort.

Price Regulation for Households

Many jurisdictions regulate retail electricity prices for households, either through comprehensive tariff regulation (in non-liberalised systems) or through price caps in liberalised markets. In Great Britain, the Domestic Gas and Electricity (Tariff Cap) Act 2018 required Ofgem to impose a cap on standard variable and default tariffs, which has been adjusted periodically by reference to wholesale and other costs. During the 2022 crisis, the cap rose dramatically, and the government

introduced the **Energy Price Guarantee**, which limited the unit rates paid by households below the level of the cap, with the government compensating suppliers for the difference. In the European Union, the Court of Justice held in *Federutility* (Case C-265/08, 2010) that public intervention in the price of supply of natural gas to final consumers is compatible with the internal market provided it pursues a general economic interest, is proportionate, and complies with conditions of transparency, non-discrimination, and limited duration; the reasoning has been applied to electricity, and the 2019 Directive codified the conditions under which member states may regulate prices for households and microenterprises, including energy poor and vulnerable households.

Consumer Protection in Nigeria

In Nigeria, consumer protection in the electricity sector has centred on metering, billing, quality of supply, and complaint handling. A large proportion of electricity consumers have historically been unmetered and subjected to **estimated billing**, which consumers regard as arbitrary and excessive. NERC has adopted a series of measures to address the metering gap, including the Meter Asset Provider Regulations (2018, revised), which permit independent meter asset providers to finance and install meters; the National Mass Metering Programme, launched in 2020; and orders capping estimated bills for unmetered customers by reference to the consumption of metered customers on the same feeder. The NERC Customer Service Standards of Performance for Distribution Companies, and the Customer Complaints Handling Standards and Procedures, establish timelines for addressing complaints, connecting new customers, and restoring supply, with compensation for failures. Consumers may escalate complaints to NERC Forum Offices, which operate as a quasi-judicial dispute resolution mechanism. The Electricity Act 2023 contains provisions on consumer protection and empowers the regulator to establish rules on customer rights and obligations, and the Federal Competition and Consumer Protection Act 2018 applies to the electricity sector.

The service-based tariff structure introduced by NERC, in which tariffs vary according to the hours of supply received by each feeder band, links consumer payments to service quality, and NERC has imposed sanctions on DisCos that failed to deliver the minimum hours of supply to Band A feeders. The effectiveness of these measures depends on the regulator's capacity to monitor service delivery and enforce compliance.

Disconnection and the Right to Electricity

The disconnection of electricity supply for non-payment raises acute questions of consumer protection and human rights. Many jurisdictions restrict disconnection: prohibiting it during winter months or extreme heat, for households with vulnerable members (such as those dependent on medical equipment), or

without adequate notice and opportunity to pay. In South Africa, the courts have developed a body of jurisprudence on the procedural fairness of electricity disconnections by municipalities. In *Joseph v City of Johannesburg* (2009), the Constitutional Court held that tenants who were not parties to the supply contract between their landlord and the municipality were nonetheless entitled to procedural fairness, including adequate notice, before their supply was disconnected, because the municipality's obligation to provide basic services gave rise to a public law relationship with the users of those services.[3] The Court's reasoning rests on the constitutional duty of local government to ensure the provision of services to communities in a sustainable manner.

The prolonged "load shedding" — scheduled rotational power cuts — imposed by South Africa's state utility Eskom because of generation shortfalls also gave rise to litigation. In *United Democratic Movement v Eskom Holdings SOC Ltd* (2023), the High Court in Pretoria held that the state's failure to address the impact of load shedding on hospitals, clinics, schools, and police stations infringed constitutional rights, and ordered the Minister of Public Enterprises to take steps to ensure sufficient supply to such facilities. The case illustrates the potential of constitutional litigation to translate the human rights dimensions of electricity supply into enforceable obligations, a theme examined in Chapters 23 and 27.

Energy Poverty and Social Tariffs

Energy poverty — the inability of households to secure adequate energy services at affordable cost — affects households in both developed and developing countries. Legal and policy responses include social or "lifeline" tariffs, under which low-income or low-consumption households pay reduced prices for an initial block of consumption; direct income support; energy efficiency programmes targeting low-income households; and protections against disconnection. Nigeria's tariff structure has included a lifeline tariff for residential customers with very low consumption. The EU's Social Climate Fund, established by Regulation (EU) 2023/955 to address the social impacts of the extension of carbon pricing to buildings and road transport, will support vulnerable households from 2026. The design of social tariffs must balance the goal of protecting vulnerable consumers against the risks of subsidising consumption by better-off households and undermining the financial viability of utilities. Energy poverty and universal access are examined further in Chapter 23.

[3] *Joseph v City of Johannesburg* [2009] ZACC 30, 2010 (4) SA 55 (CC).

9.8 Market Abuse, Competition, and Regulatory Enforcement

Vulnerability of Electricity Markets to Abuse

Electricity markets are unusually vulnerable to the exercise of market power and to manipulation. Because electricity cannot be stored economically at scale and demand is inelastic in the short term, a generator that withholds capacity at times of tight supply can raise prices dramatically; because the marginal generator sets the price for all, even a small generator may have significant market power at peak times; and because markets are governed by complex rules, traders may exploit flaws in market design to profit at the expense of other participants and consumers. The California electricity crisis of 2000–2001, examined in the casebox below, demonstrated the scale of harm that market abuse can inflict, and led to the development of specialised regimes for the prevention and sanctioning of market manipulation in energy markets.

Market Manipulation and Insider Trading: REMIT and United States Law

In the European Union, the **Regulation on Wholesale Energy Market Integrity and Transparency** (REMIT) — Regulation (EU) No 1227/2011, amended by Regulation (EU) 2024/1106 — prohibits insider trading and market manipulation in wholesale energy markets, requires market participants to publish inside information in a timely manner, and requires the reporting of transactions and orders to ACER. Market manipulation includes transactions or orders that give false or misleading signals as to supply, demand, or price, that secure the price at an artificial level, or that employ fictitious devices or any other form of deception. ACER monitors trading activity, and national regulatory authorities investigate and enforce the prohibitions, imposing penalties under national law. The 2024 amendments strengthened ACER's investigatory powers in cross-border cases, extended REMIT to algorithmic trading, and harmonised penalties. Enforcement decisions have included substantial fines for wash trades, layering, and the withholding of capacity, such as the fine imposed by the French Energy Regulatory Commission's dispute settlement and sanctions committee (CoRDiS) on VITOL in 2018 for manipulation of gas markets, and decisions by Ofgem against generators for charging excessive prices in the balancing mechanism.

In the United States, the **Energy Policy Act of 2005** amended the Federal Power Act and the Natural Gas Act to prohibit the use of any manipulative or deceptive device or contrivance in connection with the purchase or sale of electricity or natural gas subject to FERC's jurisdiction, and gave FERC authority to impose civil penalties of up to USD 1 million per day per violation (adjusted for inflation). FERC's Anti-Manipulation Rule, modelled on Rule 10b-5 under

the Securities Exchange Act of 1934, has been the basis of numerous enforcement actions. In *FERC v Barclays Bank plc*, FERC assessed a civil penalty of USD 435 million against Barclays and four of its traders for manipulating electricity prices in western United States markets between 2006 and 2008 by trading physical products to benefit financial swap positions; the matter was settled in 2017 for USD 105 million. The Commodity Futures Trading Commission has parallel jurisdiction over manipulation in energy futures and swaps.

Competition Law in Electricity Markets

General competition law applies to electricity markets alongside sector-specific regulation. In the European Union, Article 102 TFEU, which prohibits the abuse of a dominant position, has been applied to incumbent electricity companies in a series of cases. In 2008, the European Commission accepted commitments from E.ON to divest generation capacity and its extra-high-voltage transmission network in Germany, following an investigation into suspected strategic withholding of generation capacity to raise prices and discrimination in the procurement of balancing services. In 2010, the Commission accepted commitments from EDF to release capacity tied up in long-term supply contracts with large industrial customers in France, which allegedly foreclosed the market to competing suppliers. In *DEI v Commission*, concerning the Greek incumbent's exclusive access to lignite deposits, the Commission found that Greece had infringed Article 106 in conjunction with Article 102 by granting the state-owned Public Power Corporation privileged access to lignite, and the Court of Justice in 2014 upheld the Commission's approach, holding that it was not necessary to prove actual abuse where state measures created a situation of inequality of opportunity between economic operators.[4] The **State aid rules** of Articles 107 and 108 TFEU are also of central importance, governing the compatibility of renewable support schemes, capacity mechanisms, and emergency interventions, as examined in Chapter 10.

In the United Kingdom, the Competition and Markets Authority's **Energy Market Investigation** (2014–2016) found that a combination of features of the retail market, including weak customer engagement, gave suppliers a position of unilateral market power over their inactive customers, resulting in detriment of around £1.4 billion per year, and recommended remedies including a temporary price cap for customers on prepayment meters. The subsequent legislative price cap for default tariffs, described in section 9.7, went further than the CMA's recommendations.

In Nigeria, the Federal Competition and Consumer Protection Act 2018 established the Federal Competition and Consumer Protection Commission

[4] *European Commission v DEI*, Case C-553/12 P, EU:C:2014:2083.

(FCCPC) with powers over anti-competitive conduct and consumer protection across the economy, while the Electricity Act 2023 confers on NERC responsibility for promoting competition and preventing abuse of market power in the electricity industry. The coordination of sector regulators and competition authorities, examined in Chapter 4, is a recurrent challenge.

Regulatory Enforcement Powers

Effective regulation depends on effective enforcement. Electricity regulators typically possess powers to require information, to investigate suspected breaches of licence conditions, codes, and legislation, to issue compliance orders, to impose financial penalties, to require the payment of compensation to consumers, and to modify, suspend, or revoke licences. In Great Britain, Ofgem may impose penalties of up to ten per cent of a licensee's turnover for breaches of licence conditions or relevant requirements, and has frequently agreed settlements under which licensees make payments to consumer redress funds. In Nigeria, the Electricity Act 2023 empowers NERC to impose fines and sanctions on licensees for breaches of the Act, regulations, codes, and licence conditions, and NERC has periodically imposed fines on DisCos for violations such as failure to meet metering obligations, overbilling, and failure to deliver required hours of supply.

The exercise of enforcement powers is subject to administrative law principles of legality, procedural fairness, rationality, and proportionality, and enforcement decisions are subject to appeal or judicial review. The design of enforcement — the balance between deterrence and compliance-oriented approaches, the use of settlements and consumer redress, and the transparency of enforcement decisions — has been the subject of extensive regulatory scholarship, including the theory of "responsive regulation" developed by Ian Ayres and John Braithwaite, which advocates an "enforcement pyramid" in which regulators escalate from persuasion to increasingly severe sanctions in response to non-compliance. Regulatory enforcement and judicial review are examined further in Chapter 27.

LEADING AUTHORITY

The California Electricity Crisis (2000–2001) and the Law of Market Manipulation

Background. California's restructuring legislation of 1996 (Assembly Bill 1890) required the state's investor-owned utilities to divest much of their generation and to buy electricity in a spot market operated by the California Power Exchange, while capping retail prices for a transitional period and discouraging long-term contracts. In 2000, a combination of rising demand, drought reducing hydroelectric imports, rising gas prices, and capacity withdrawals caused wholesale prices to rise to many times their historical levels, while retail rates remained frozen.

Manipulation. Investigations by FERC and others subsequently revealed that traders, notably at Enron, had exploited flaws in market rules through strategies given names such as "Death Star", "Get Shorty", and "Ricochet", which involved scheduling fictitious congestion to collect congestion relief payments, selling ancillary services they did not have, and moving electricity out of and back into California to evade price caps. Generators were also found to have withheld capacity to raise prices.

Consequences. Pacific Gas and Electric filed for bankruptcy in April 2001, Southern California Edison came close to insolvency, and the state government purchased power on behalf of the utilities at very high prices under long-term contracts. Rolling blackouts affected millions of customers. FERC ordered refunds, and extensive litigation followed. Enron collapsed in December 2001 amid accounting fraud, and several of its executives and traders were convicted of criminal offences.

Legal significance. The crisis led directly to the anti-manipulation provisions of the Energy Policy Act of 2005, to reforms of market design in California and elsewhere, including the introduction of locational marginal pricing and market power mitigation measures, and to a lasting scepticism about retail price caps combined with spot market exposure. Its lessons informed the design of REMIT in the European Union. It remains the paradigmatic illustration of the importance of market design and enforcement in electricity markets.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Legislative drafting: a state electricity market law (hypothetical). *The state and facts are fictitious, although the exercise is set within the actual framework of Nigeria's Constitution and Electricity Act 2023.* The fictional Nigerian state of Oroke, with a population of five million, receives electricity from a single DisCo whose licence covers Oroke and two neighbouring states. The state government wishes to establish its own electricity market under the Electricity Act 2023, with the objectives of increasing metering, attracting investment in embedded generation and mini-grids, protecting vulnerable consumers, and ensuring a smooth transition of regulatory oversight from NERC.

Prepare (a) a policy memorandum (1,500 words) identifying the constitutional and statutory basis for the state's competence, the matters that remain within federal competence, and the principal transitional issues; (b) the draft text of the key provisions of an "Oroke State Electricity Market Law" (approximately 2,000 words), including provisions establishing a state regulatory commission, licensing, tariff methodology, consumer protection, and the transition of existing licences; and (c) a short note (500 words) assessing the risks of regulatory fragmentation and proposing mechanisms for coordination with NERC and other states.

CHAPTER SUMMARY

- The physical characteristics of electricity — the need for continuous balancing, flows governed by physics rather than contract, inelastic demand, and natural monopoly networks — shape its legal structure.
- Electricity industries range from vertically integrated monopolies through single-buyer and wholesale competition models to full retail competition; Nigeria's industry has moved from the NEPA monopoly through privatisation to constitutional and statutory decentralisation under the Electricity Act 2023.
- Regulation of generation, transmission, and distribution involves licensing, unbundling, tariff regulation, planning, system operation, and reliability standards, with significant federal–state allocation issues in the United States and now Nigeria.
- Liberalisation has produced diverse market designs; the EU's 2024 reform retains marginal pricing while promoting long-term contracts and two-way contracts for difference.
- Grid access law governs third-party access, connection procedures, cost allocation, priority dispatch, and access for distributed resources; connection queues have become a major bottleneck for the energy transition.
- Power purchase agreements allocate risk through capacity payments, currency indexation, change in law, force majeure, and termination provisions, supported by sovereign and multilateral credit enhancement; their renegotiation under economic pressure is a recurrent source of disputes.
- Cross-border trade is governed by the EU internal market, the WAPP and ERERA in West Africa, other regional power pools, and the emerging African Single Electricity Market.
- Consumer protection addresses billing, metering, switching, price regulation, disconnection, and energy poverty, with constitutional litigation emerging as a means of enforcement.
- Market abuse is addressed through REMIT, United States anti-manipulation law, competition law, and regulatory enforcement powers.

DOCTORAL RESEARCH QUESTIONS

1. Is the standard model of electricity liberalisation appropriate for developing countries, or should reform pursue "hybrid" models adapted to local conditions? Evaluate with reference to Nigeria's experience since 2013.
2. Analyse the constitutional and legal implications of the decentralisation of electricity regulation in Nigeria under the 2023 constitutional amendments and the Electricity Act 2023, drawing comparisons with the federal–state division in the United States.
3. Does marginal pricing remain an appropriate foundation for wholesale electricity markets in systems dominated by zero-marginal-cost renewable generation?

4. To what extent can regional electricity regulators such as ERERA exercise effective authority over national utilities, and what legal reforms would strengthen regional market integration in West Africa?
5. How should the law balance the sanctity of long-term PPAs against the need to renegotiate contracts that have become unaffordable as a result of macroeconomic shocks?
6. Is there a justiciable right to electricity? Evaluate South African and other jurisprudence and its potential application in Nigeria.

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10 CHAPTER

Renewable Energy Law and Sustainable Power Development

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the international, regional, and domestic legal foundations of renewable energy development, including targets, definitions, and institutional frameworks.
2. Analyse the licensing and land rights issues associated with utility-scale and distributed solar energy, with particular reference to customary land tenure.
3. Evaluate the permitting and environmental compliance requirements for onshore and offshore wind energy, including the protection of wildlife and the rights of indigenous peoples.
4. Assess the legal framework for hydropower development and the allocation of water resources at the project level.
5. Identify the distinctive legal issues raised by geothermal and biomass energy, including resource classification and sustainability criteria.
6. Compare the principal renewable energy support mechanisms and evaluate their legal design and compatibility with State aid and trade law.
7. Analyse the law governing renewable energy certificates and attribute markets, including guarantees of origin and corporate claims.
8. Critically assess the jurisprudence of renewable energy investment disputes, including the Spanish and Italian arbitrations and their enforcement.

Introduction

Renewable energy has moved from the margins to the centre of energy law. In the space of two decades, wind and solar power have been transformed from expensive niche technologies dependent on generous subsidies into the cheapest sources of new electricity generation in most of the world. Renewable sources now account for the large majority of new generating capacity installed globally each year, and at the 2023 United Nations Climate Change Conference (COP28) in Dubai, the

parties to the Paris Agreement called for a tripling of global renewable energy capacity and a doubling of the global average annual rate of energy efficiency improvements by 2030. Renewable energy is also central to the achievement of universal energy access: decentralised solar systems and mini-grids offer the most cost-effective means of electrifying many of the rural communities in sub-Saharan Africa that the grid has not reached.

The law has been both a driver and a constraint of this transformation. Legal instruments — renewable energy targets, feed-in tariffs, renewable portfolio standards, auctions, tax credits, priority grid access, and permitting regimes — created the markets that drove down costs through deployment and learning. At the same time, legal obstacles — lengthy and uncertain permitting, conflicts over land and resources, inadequate grid access, regulatory instability, and opposition from affected communities — have slowed deployment. And the retroactive reduction of renewable support schemes by several European governments in the aftermath of the financial crisis generated one of the largest waves of investment treaty arbitration in history, testing the limits of investor protection against regulatory change.

The legal issues raised by renewable energy differ from those of fossil fuels in important ways. Renewable resources — sunlight, wind, flowing water, geothermal heat, and biomass — are not "owned" in the same way as subsurface minerals, and their legal status has required new conceptual frameworks. Renewable projects have low operating costs but high upfront capital costs, so their economics depend critically on the cost of capital and therefore on regulatory stability. Many renewable resources are variable, requiring new approaches to grid integration and market design. And the land-intensive character of solar and wind energy, the impacts of hydropower on rivers, and the sustainability concerns surrounding bioenergy generate conflicts with other land uses, biodiversity, and the rights of local and indigenous communities.

This chapter examines the legal foundations of renewable energy development (section 10.1), the licensing and land rights issues of solar energy (section 10.2), wind energy permitting and environmental compliance (section 10.3), hydropower law and water resource allocation (section 10.4), geothermal and biomass energy regulation (section 10.5), renewable energy incentives and support mechanisms (section 10.6), renewable energy certificates and attribute markets (section 10.7), and renewable energy investment disputes (section 10.8). The integration of renewable energy into electricity markets is addressed in Chapter 9, the emerging technologies of storage, hydrogen, and offshore renewables in Chapter 12, and the transboundary dimensions of hydropower in Chapter 19.

10.1 Legal Foundations of Renewable Energy Development

Defining Renewable Energy

Legal definitions of renewable energy matter because they determine eligibility for support schemes, targets, priority access, and certificates. Most definitions enumerate the sources considered renewable. Article 2(1) of the EU Renewable Energy Directive (Directive (EU) 2018/2001, as amended by Directive (EU) 2023/2413) defines "energy from renewable sources" as energy from renewable non-fossil sources, namely wind, solar (solar thermal and solar photovoltaic) and geothermal energy, osmotic energy, ambient energy, tide, wave and other ocean energy, hydropower, biomass, landfill gas, sewage treatment plant gas, and biogas. The Statute of the International Renewable Energy Agency (IRENA), adopted in 2009, defines renewable energy in Article III to include all forms of energy produced from renewable sources in a sustainable manner, including bioenergy, geothermal energy, hydropower, ocean energy (including tidal, wave, and ocean thermal energy), solar energy, and wind energy.

Definitions raise contested questions at the margins. Is large hydropower renewable, given its environmental and social impacts? Is biomass renewable if its harvesting causes deforestation or releases more carbon than it saves over relevant time horizons? Is waste incineration renewable? Should nuclear energy, which is low-carbon but not renewable, be included in "clean energy" targets? Different legal systems answer these questions differently, and the trend has been to supplement definitions with sustainability criteria, as in the EU's sustainability and greenhouse gas saving criteria for biomass, examined in section 10.5, and to adopt broader "clean" or "low-carbon" categories alongside "renewable" categories.

International Foundations

There is no comprehensive international treaty on renewable energy. The international legal foundations of renewable energy are found in several instruments.

Climate change law. The UNFCCC, the Kyoto Protocol, and the Paris Agreement, examined in Chapter 18, provide the overarching framework. Although the Paris Agreement does not specifically mention renewable energy in its operative provisions, the nationally determined contributions of most parties include renewable energy targets or measures, and the COP28 decision on the first global stocktake called on parties to contribute to tripling renewable energy capacity globally by 2030. Article 4(1)(c) of the UNFCCC commits all parties to promote and cooperate in the development, application, and diffusion of technologies that control, reduce, or prevent anthropogenic emissions of greenhouse gases in all relevant sectors, including energy.

The Sustainable Development Goals. SDG 7, adopted by the United Nations General Assembly in 2015, calls for access to affordable, reliable, sustainable,

and modern energy for all by 2030, with target 7.2 calling for a substantial increase in the share of renewable energy in the global energy mix.

IRENA. The Statute of the International Renewable Energy Agency, which entered into force in 2010, established IRENA as an intergovernmental organisation to promote the widespread and increased adoption and sustainable use of all forms of renewable energy. IRENA's institutional role is examined in Chapter 4.

The International Solar Alliance. The Framework Agreement on the Establishment of the International Solar Alliance, launched by India and France at COP21 in 2015 and in force since 2017, establishes a treaty-based organisation of countries committed to promoting solar energy, particularly in countries located between the Tropics of Cancer and Capricorn, through the aggregation of demand, the mobilisation of finance, and capacity building. Nigeria is a party.

The Energy Charter Treaty. Article 19 of the ECT requires contracting parties to strive to minimise harmful environmental impacts of energy activities, including by "promoting and using renewable energy sources", and the ECT's investment protection provisions have been the basis of most renewable energy investment arbitrations, examined in section 10.8.

Regional and Domestic Targets

Renewable energy targets are the most widespread legal instrument of renewable energy policy, adopted by almost every country. Targets may be expressed as shares of final energy consumption, electricity generation, or installed capacity, and may be binding or indicative.

The European Union has adopted binding targets for renewable energy since the Renewable Energy Directive of 2009 (Directive 2009/28/EC), which set a binding target of twenty per cent of gross final energy consumption by 2020, broken down into binding national targets. The recast Directive of 2018 set a binding Union-level target of at least thirty-two per cent by 2030, without binding national targets, relying on national energy and climate plans and a governance mechanism. Directive (EU) 2023/2413 ("RED III") raised the binding Union target to at least 42.5 per cent by 2030, with an indicative aspiration of forty-five per cent, and introduced sub-targets for transport, buildings, industry, and district heating and cooling, including a binding target for the use of renewable fuels of non-biological origin (principally renewable hydrogen) in industry.

In Africa, the ECOWAS Renewable Energy Policy, adopted in 2013 and implemented through the ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE) in Praia, Cape Verde, set targets for increasing the share of renewable energy in the regional electricity mix, including large hydropower, and member states adopted National Renewable Energy Action Plans. The African Union's Agenda 2063 and the Africa Renewable Energy Initiative (AREI),

launched at COP21, aimed to install substantial new renewable capacity across the continent.

The Nigerian Framework

Nigeria's legal and policy framework for renewable energy has developed incrementally. The **National Renewable Energy and Energy Efficiency Policy** (NREEEP), approved in 2015, set out the government's objectives for renewable energy, and the **Nigeria Renewable Energy Master Plan** set targets for the contribution of renewables to electricity supply. NERC's **Feed-in Tariff Regulations for Renewable Energy Sourced Electricity** (2015) provided for feed-in tariffs for small hydropower, wind, solar, and biomass, and required DisCos to procure a proportion of their electricity from renewable sources, although implementation was limited. In 2016, NBET signed PPAs with fourteen solar IPPs for a combined capacity of around 1,125 MW, but most of these projects failed to reach financial close, owing largely to disagreements over tariffs, the absence of adequate payment security, and the liquidity crisis in the electricity market.

The **Rural Electrification Agency** (REA), established under EPSRA and continued under the Electricity Act 2023, administers the Rural Electrification Fund and implements programmes, including the World Bank-supported Nigeria Electrification Project and the Distributed Access through Renewable Energy Scale-up (DARES) programme, that have supported the deployment of solar mini-grids and stand-alone solar systems. The **Climate Change Act 2021** set a target of net zero emissions between 2050 and 2070 and established the National Council on Climate Change, and the **Energy Transition Plan** of 2022 set out a pathway to net zero by 2060, with a substantial expansion of solar capacity. The **Electricity Act 2023** contains specific provisions on renewable energy, including obligations on the regulator to promote renewable energy, provision for renewable purchase obligations, incentives for renewable energy, and the treatment of renewable energy in integrated resource planning. These developments are examined further in Chapter 24.

The Legal Status of Renewable Resources

A distinctive conceptual question is whether, and by whom, renewable resources can be owned. Wind and sunlight are generally regarded as *res communes* — things common to all, incapable of ownership in their natural state. The legal question therefore concerns not ownership of the resource itself, but rights to access and use it: rights to use land on which solar panels and wind turbines are located, protection against interference with access to the resource (for example, by structures that shade solar panels or obstruct wind flows), and the allocation of offshore areas for renewable development. Some jurisdictions have legislated to protect access to solar or wind resources: several states of the United States have enacted

"solar access" laws permitting the creation of solar easements or protecting solar installations from shading, and the law of nuisance and easements may be invoked to protect access.

Offshore, the coastal state's sovereign rights in the exclusive economic zone under Article 56(1)(a) of UNCLOS extend expressly to "the production of energy from the water, currents and winds", providing the international legal basis for offshore wind and ocean energy development, examined in Chapters 12 and 20. Water used for hydropower and geothermal resources are, by contrast, subject to well-developed legal regimes of public ownership and allocation, examined in sections 10.4 and 10.5.

10.2 Solar Energy Licensing and Land Rights

The Solar Revolution

Solar photovoltaic (PV) energy has experienced the most dramatic cost reductions of any energy technology: the cost of solar modules has fallen by around ninety per cent over the past decade, and utility-scale solar is now among the cheapest sources of new electricity in most regions. Solar energy is deployed at every scale: utility-scale solar farms of hundreds or thousands of megawatts; commercial and industrial installations on rooftops and land; residential rooftop systems; solar mini-grids serving rural communities; and stand-alone solar home systems and appliances. Each scale raises different legal issues.

Licensing and Permitting of Utility-Scale Solar

Utility-scale solar projects typically require a generation licence (where the capacity exceeds the licensing threshold), land rights, planning or land use permission, environmental assessment, grid connection, and, for projects selling to a utility, a PPA. The environmental impacts of solar farms are generally modest compared with other generation technologies, but they include land use change, habitat loss, visual impacts, glare, and the use of water for cleaning panels in arid areas. Large solar projects typically require an EIA, and in some jurisdictions they benefit from streamlined permitting. Under the EU's RED III, member states must designate "renewables acceleration areas" in which renewable projects benefit from simplified and accelerated permitting, including exemption from project-level environmental assessment where a strategic environmental assessment has been carried out, and must ensure that permit-granting procedures for projects in such areas do not exceed twelve months for onshore projects. RED III also provides that the planning, construction, and operation of renewable energy plants are presumed to be in the overriding public interest for the purposes of certain environmental assessments under EU nature conservation law, until climate neutrality is achieved.

In Nigeria, utility-scale solar projects above the licensing threshold require a generation licence from NERC (or, where applicable, a state regulator), an EIA approved by the Federal Ministry of Environment, land rights secured under the Land Use Act, and a grid connection agreement with TCN or a DisCo. Projects intended to sell to the grid require a PPA with a buyer — historically NBET, and increasingly DisCos, eligible customers, or traders.

Land Rights for Solar Projects

Utility-scale solar projects are land-intensive, requiring around one to two hectares per megawatt, depending on technology and location. Developers must secure long-term rights to the land for the life of the project (typically twenty-five to thirty-five years), sufficient to satisfy lenders. The form of these rights depends on the land tenure system: freehold purchase, long-term leasehold, rights of occupancy, easements, or concessions over public land.

In jurisdictions with customary land tenure, such as most of sub-Saharan Africa, securing land rights for solar projects raises distinctive challenges. Land may be held communally by families, clans, or communities under customary law, often without formal registration; rights may be overlapping and contested; and the identification of the persons entitled to consent to a transfer may be uncertain. In Nigeria, the **Land Use Act 1978** vests all land in each state in the governor, who may grant statutory rights of occupancy over urban land, while local governments may grant customary rights of occupancy over non-urban land. Holders of customary rights may transfer their interests only with the consent of the governor (for statutory rights) or the local government (for customary rights), and the requirement of governor's consent has been a notorious source of delay and cost. Developers of solar projects in Nigeria must therefore negotiate with customary landholders and communities, obtain consent to the transfer or grant of rights of occupancy, and register their interests, while managing the risk of competing claims.

The **Voluntary Guidelines on the Responsible Governance of Tenure of Land, Fisheries and Forests** (VGGT), endorsed by the Committee on World Food Security in 2012, and **IFC Performance Standard 5** on land acquisition and involuntary resettlement provide international standards for responsible land acquisition, emphasising the recognition of legitimate tenure rights (including customary rights), consultation, the avoidance of involuntary resettlement where possible, and fair compensation. Large solar projects in Africa, such as the Noor Ouarzazate complex in Morocco, which was built on land acquired from a local collective of the Ait Oukrour Toundout community under Moroccan law on collective lands, have attracted scrutiny of the adequacy of consultation and compensation. These issues are examined further in Chapter 21.

Agrivoltaics and Competing Land Uses

The land requirements of solar energy create potential conflicts with agriculture, particularly in densely populated areas or regions with high-quality agricultural land. Agrivoltaics — the co-location of solar panels and agricultural production, for example by mounting panels on elevated structures over crops or grazing land — offers a means of reducing conflict, and some jurisdictions have adopted specific legal frameworks for agrivoltaic projects. France's Law No 2023-175 on the acceleration of renewable energy production defined agrivoltaic installations and established conditions under which they may be located on agricultural land, requiring that they provide a direct service to agricultural production, such as improving agronomic potential or protection against hazards, and do not significantly reduce agricultural output.

Distributed Solar and Mini-Grids

Distributed solar — rooftop systems and small ground-mounted installations — raises different legal issues: the right of consumers to install generation and connect it to the grid; net metering and compensation for exports; building regulations and fire safety standards; and the treatment of distributed generation in network tariffs. In Nigeria, NERC's Regulations on Embedded Generation (2012) and its regulatory instruments on net metering govern the connection of distributed generation to distribution networks and the treatment of exports.

Solar mini-grids are particularly important in Nigeria and other countries with low rates of electrification. The NERC **Mini-Grid Regulations** (2016, revised 2023) distinguish between isolated mini-grids (not connected to the distribution network) and interconnected mini-grids (connected to an existing distribution network, often to serve underserved customers), set thresholds for permits and registration, allow mini-grid operators to charge cost-reflective tariffs approved by the regulator, and address the consequences of the arrival of the main grid, including options for the mini-grid to convert to a small power distributor or to be compensated for its assets. These provisions are critical to investor confidence, because the risk of grid arrival rendering the mini-grid uneconomic is one of the principal risks of mini-grid investment.

Stand-alone solar systems and solar home systems, often sold on "pay-as-you-go" terms using mobile money, have brought electricity to millions of households in sub-Saharan Africa. Their legal issues include consumer protection (product quality standards, warranties, transparent pricing, and responsible debt collection), the regulation of mobile payments, import duties and taxes (several countries have exempted solar products from import duties and value added tax), and the management of electronic waste from end-of-life batteries and panels.

The quality standards developed by the World Bank Group's Lighting Global programme (now VeraSol) have been widely adopted as a reference.

10.3 Wind Energy Permitting and Environmental Compliance

Onshore and Offshore Wind

Wind energy is the second largest source of renewable electricity after hydropower and a cornerstone of decarbonisation strategies in Europe, China, the United States, and increasingly elsewhere. Onshore wind is a mature and low-cost technology; offshore wind, which benefits from stronger and more consistent winds and fewer conflicts with land uses, has grown rapidly in northern Europe and China and is expanding to the United States, Asia, and other regions, although rising costs, supply chain constraints, and interest rates led to the cancellation or delay of several projects in 2023 and 2024. Wind energy raises distinctive permitting issues relating to its visual and landscape impacts, noise, impacts on birds and bats, interference with radar and aviation, and, offshore, interactions with fisheries, shipping, marine ecosystems, and military activities.

Permitting of Onshore Wind

Onshore wind projects typically require planning permission or development consent, environmental assessment, a generation licence, grid connection, and, depending on the jurisdiction, consents relating to aviation, radar, and wildlife. Permitting has been widely identified as the principal bottleneck for onshore wind deployment. In Europe, permitting procedures for onshore wind projects historically took many years, and a substantial pipeline of projects was blocked in permitting or litigation. The EU's Council Regulation (EU) 2022/2577, adopted as an emergency measure in December 2022, laid down temporary rules to accelerate the deployment of renewable energy, including a presumption that renewable energy projects are in the overriding public interest; RED III made many of these measures permanent, requiring maximum permitting periods and the designation of renewables acceleration areas.

Germany's experience illustrates the significance of legal reform. Onshore wind deployment in Germany slowed dramatically after 2017, owing to the introduction of auctions, restrictive distance rules (notably Bavaria's "10H rule", which required a distance of ten times the height of the turbine from residential buildings), and litigation. The 2023 amendments to the Renewable Energy Sources Act (EEG 2023) declared that the construction and operation of renewable energy installations are in the overriding public interest and serve public security (§ 2 EEG), and the Wind-on-Land Act (Wind-an-Land-Gesetz) of 2022 required the federal states to designate two per cent of the national land area for onshore wind by 2032, overriding restrictive distance rules where states fail

to meet their area targets. Deployment and permitting accelerated substantially thereafter.

Wildlife Protection

The protection of birds and bats is a recurrent issue in wind energy permitting. In the European Union, the **Birds Directive** (Directive 2009/147/EC) and the **Habitats Directive** (Council Directive 92/43/EEC) require the protection of wild birds and of species and habitats of Community interest. Article 6(3) of the Habitats Directive requires an "appropriate assessment" of any plan or project likely to have a significant effect on a Natura 2000 site, and permits approval only if the project will not adversely affect the integrity of the site, unless, under Article 6(4), there are imperative reasons of overriding public interest, no alternative solutions, and compensatory measures are taken. Article 12 of the Habitats Directive and Article 5 of the Birds Directive prohibit the deliberate killing or disturbance of protected species. The Court of Justice has interpreted these prohibitions strictly; in *Föreningen Skydda Skogen* (Joined Cases C-473/19 and C-474/19, 2021), it held that the prohibitions apply irrespective of the conservation status of the species concerned and are not limited to activities aimed at killing or disturbing. The presumption of overriding public interest introduced by RED III is intended to facilitate the derogations necessary for renewable energy projects.

In the United States, the Migratory Bird Treaty Act of 1918, the Bald and Golden Eagle Protection Act of 1940, and the Endangered Species Act of 1973 protect birds and other wildlife, and wind energy operators may obtain incidental take permits for eagles. In 2022, a wind energy company pleaded guilty to violations of the Migratory Bird Treaty Act in connection with the deaths of eagles at its facilities, illustrating the criminal liability risks of wildlife impacts.

Wind Energy and Indigenous Rights: The Fosen Case

The most significant judicial decision on the relationship between wind energy and indigenous rights is the judgment of the Supreme Court of Norway in the *Fosen* case (2021). The Norwegian government had granted licences and expropriation permits for the Storheia and Roan wind farms on the Fosen peninsula, among the largest onshore wind developments in Europe, located in an area used by Sámi reindeer herders for winter grazing. The reindeer herders argued that the wind farms would interfere with their ability to practise their culture, in violation of Article 27 of the International Covenant on Civil and Political Rights, which protects the right of persons belonging to minorities, in community with other members of their group, to enjoy their own culture. In October 2021, the Supreme Court held unanimously that the licences and expropriation decisions were invalid because the wind farms would have a significant negative effect on the

herders' ability to practise reindeer husbandry, in violation of Article 27, and that the mitigation measures proposed were insufficient. The Court rejected the argument that the interference could be justified by the public interest in renewable energy, noting that the green transition could be achieved through other means that did not infringe Sámi rights.

The judgment did not specify the consequences for the turbines, which had been constructed and were operating. Following protests by Sámi activists and environmentalists in 2023, which drew attention to the continued operation of the turbines more than five hundred days after the judgment, the Norwegian government apologised and negotiated settlements with the affected reindeer herding groups, concluded in 2023 and 2024, providing for compensation, additional winter grazing land, and other measures allowing the wind farms to continue operating. The *Fosen* case is a landmark in the emerging law of "green" or "just" transition, demonstrating that renewable energy projects are subject to the same human rights constraints as fossil fuel projects, and that courts may invalidate projects that disproportionately burden indigenous peoples. It is examined further in Chapter 21.

Comparable conflicts have arisen elsewhere. In Mexico, wind farm development in the Isthmus of Tehuantepec in Oaxaca, one of the windiest regions in the world, has been contested by Zapotec and Ikoots indigenous communities, and the Supreme Court of Justice of the Nation in 2018 addressed the adequacy of the consultation conducted for the Eólica del Sur project, upholding the consultation in that case while affirming the right to prior consultation. In Kenya, the Lake Turkana Wind Power project, the largest wind farm in Africa when commissioned in 2019, faced litigation by local communities challenging the acquisition of the land on which it was built; in 2021, the Environment and Land Court at Meru found that the process by which the land had been set aside was irregular and ordered the regularisation of the land acquisition in accordance with the Community Land Act, without ordering the removal of the project.

Offshore Wind Leasing and Permitting

Offshore wind development requires a legal framework for the allocation of rights to the seabed and water column, environmental assessment, and coordination with other marine users. Coastal states allocate offshore wind rights through several models. In the **leasing model** used in the United Kingdom, the Crown Estate (for England, Wales, and Northern Ireland) and Crown Estate Scotland grant seabed leases through competitive leasing rounds, and developers must separately obtain development consent and a route to market (typically a contract for difference). In the **centralised tender model** used in Denmark, the Netherlands, and Germany for certain sites, the state pre-investigates and pre-consents sites and tenders them, together with grid connection, to developers, in some cases through

"negative bidding" or criteria-based tenders. In the United States, the Bureau of Ocean Energy Management (BOEM) leases areas of the outer continental shelf under the Outer Continental Shelf Lands Act, as amended by the Energy Policy Act of 2005, through competitive lease sales, and developers must obtain approval of construction and operations plans following environmental review under the National Environmental Policy Act. In 2025, the federal government suspended new offshore wind leasing and halted work on certain projects under construction, giving rise to litigation and illustrating the regulatory risks of offshore wind in the United States.

Offshore wind permitting addresses impacts on marine mammals (particularly from pile-driving noise), seabirds, fish and benthic habitats, fisheries, navigation, aviation, defence, and cultural heritage, and requires coordination through marine spatial planning, which allocates marine space among competing uses. The EU's Maritime Spatial Planning Directive (Directive 2014/89/EU) requires member states to establish maritime spatial plans. Offshore wind and the law of the sea are examined in Chapters 12 and 20.

Decommissioning and Repowering

Wind turbines have operating lives of twenty to thirty years, and the first generation of wind farms in Europe and the United States is reaching the end of its life. Decommissioning raises issues of financial security (analogous to those in mining and petroleum, examined in Chapters 5 and 7), the recycling of turbine blades (which are made of composite materials that are difficult to recycle, and which several European countries have banned from landfill), and site restoration. **Repowering** — the replacement of old turbines with fewer, larger, and more efficient turbines on the same site — can substantially increase output, and RED III requires member states to facilitate repowering through simplified permitting.

10.4 Hydropower Law and Water Resource Allocation

The Role and Impacts of Hydropower

Hydropower is the largest source of renewable electricity globally and the backbone of electricity systems in many countries, including Norway, Brazil, Canada, Paraguay, Ethiopia, the Democratic Republic of the Congo, and Zambia. Reservoir hydropower provides dispatchable generation and storage that can balance variable wind and solar output, and pumped storage hydropower is the largest form of grid-scale electricity storage. In Nigeria, the Kainji, Jebba, and Shiroro hydropower plants on the Niger and Kaduna rivers, and the more recent Zungeru plant, contribute a significant share of grid generation.

At the same time, large dams have serious environmental and social impacts: displacement of communities (the World Commission on Dams estimated in 2000 that large dams had displaced between forty and eighty million people

worldwide), loss of livelihoods dependent on rivers, disruption of fisheries and sediment flows, inundation of forests and cultural heritage, greenhouse gas emissions from reservoirs in tropical areas, and risks of catastrophic dam failure. The report of the **World Commission on Dams**, *Dams and Development: A New Framework for Decision-Making* (2000), proposed a rights-and-risks approach to dam decision-making, emphasising gaining public acceptance, comprehensive options assessment, addressing existing dams, sustaining rivers and livelihoods, recognising entitlements and sharing benefits, ensuring compliance, and sharing rivers for peace, development, and security. Although its recommendations are not binding, they have influenced the policies of development institutions and the industry's **Hydropower Sustainability Standard**, launched in 2021 by the Hydropower Sustainability Council, which provides for independent certification of hydropower projects against environmental, social, and governance criteria.

This section examines the domestic legal framework for hydropower projects — water rights, licensing, concessions, and dam safety. The international law of transboundary watercourses, and the basin-level management of hydropower on shared rivers, are examined in Chapter 19.

Water Rights and Allocation for Hydropower

Hydropower depends on the right to use water. Legal systems allocate water rights according to several models: **riparian rights**, under which owners of land bordering a watercourse are entitled to reasonable use of the water (the traditional common law position); **prior appropriation**, under which rights are allocated on a "first in time, first in right" basis to those who put water to beneficial use (the position in the western United States); and **public ownership and administrative allocation**, under which water is owned or controlled by the state and allocated through permits or licences (the position in most modern water codes, including those of most developing countries). In most jurisdictions, hydropower developers require a water use permit or licence specifying the quantity of water that may be abstracted or diverted, the conditions of use (including minimum environmental flows), and the duration of the right.

In Nigeria, the **Water Resources Act 1993** (Cap W2, Laws of the Federation of Nigeria 2004) vests the right to the use and control of all surface water and groundwater, and of all water in any watercourse affecting more than one state, in the Government of the Federation, and provides for the licensing of water use, including for hydropower. The **Minerals and Mining Act 2007** separately provides for water use permits for mining. The River Basin Development Authorities, established by statute, are responsible for the development of water resources within their basins, and the **Nigeria Hydrological Services Agency** monitors water resources. A National Water Resources Bill, intended to modernise the framework, has been the subject of controversy, in part because of concerns from

states about the concentration of control over water resources in the federal government.

Environmental Flows

The concept of **environmental flows** — the quantity, timing, and quality of water flows required to sustain freshwater ecosystems and the livelihoods that depend on them — has become a central feature of modern water and hydropower law. Water licences for hydropower increasingly specify minimum flows to be released downstream, and require operators to manage releases to mimic natural flow patterns. The EU Water Framework Directive (Directive 2000/60/EC) requires member states to achieve "good ecological status" of water bodies, which has required the revision of hydropower licences to provide adequate flows and fish passage. In *Commission v Austria* (Case C-346/14, 2016), the Court of Justice upheld Austria's authorisation of a hydropower plant on the Schwarze Sulm river, notwithstanding the deterioration of the river's status, because the national authorities had properly applied the derogation for projects of overriding public interest under Article 4(7) of the Directive. In *Bund für Umwelt und Naturschutz Deutschland* (Case C-461/13, 2015, the "Weser" case), the Court held that member states must refuse authorisation for projects that may cause deterioration of the status of a body of surface water, unless a derogation applies, interpreting "deterioration" to include deterioration of a single quality element.

Hydropower Concessions and Licences

Hydropower plants are typically developed under concessions or licences granted by the state for long periods — thirty to seventy-five years or more — reflecting the long life of dams. Concessions specify the rights and obligations of the concessionaire, including water use, operation, maintenance, dam safety, environmental obligations, payment of royalties or fees, and the reversion of assets to the state at the end of the concession. In Norway, the Industrial Licensing Act of 1917 introduced the principle of "reversion" (*hjemfall*), under which hydropower plants owned by private companies revert to the state without compensation at the end of the licence period; the principle was upheld by the EFTA Court in *EFTA Surveillance Authority v Norway* (Case E-2/06, 2007) in so far as it applied to public as well as private owners, after Norway amended its legislation to remove discrimination against private and foreign owners.

In France, the renewal of expiring hydropower concessions, many of them held by EDF, has been the subject of a long-running dispute with the European Commission, which considers that EU public procurement and concession law requires competitive tendering, while successive French governments have resisted opening the concessions to competition. In the United States, non-federal hydropower plants are licensed by FERC under the Federal Power Act for

terms of thirty to fifty years, and relicensing proceedings involve consideration of environmental conditions imposed by federal and state agencies.

In Nigeria, the Kainji and Jebba hydropower plants were concessioned in 2013 to Mainstream Energy Solutions Ltd, and the Shiroro plant to North South Power Company Ltd, under thirty-year concession agreements as part of the privatisation of the power sector. The 700 MW Zungeru hydropower plant, constructed by a Chinese consortium with Chinese financing, was concessioned in 2023 to Mainstream Energy Solutions for a period of thirty years. The proposed **Mambilla hydropower project**, a planned 3,050 MW project in Taraba State, has been delayed for decades and became the subject of an ICC arbitration brought by Sunrise Power and Transmission Company Ltd against the Federal Government of Nigeria, alleging breach of a contract to develop the project — an illustration of the legal risks associated with large hydropower projects.

Dam Safety

The failure of dams can cause catastrophic loss of life and property, as the failures of the Derna dams in Libya in September 2023, which killed thousands of people, demonstrated. Dam safety legislation requires owners and operators of dams to design, construct, operate, maintain, and inspect dams according to specified standards, to prepare emergency action plans, and to coordinate releases with downstream authorities. In Nigeria, the release of water from the Lagdo Dam in neighbouring Cameroon has been associated with severe flooding in Nigeria, notably in 2012 and 2022, raising questions about transboundary coordination examined in Chapter 19. The International Commission on Large Dams (ICOLD) publishes technical bulletins on dam safety that are widely used as standards.

10.5 Geothermal and Biomass Energy Regulation

Geothermal Energy

Geothermal energy — heat derived from the earth's interior — can be used to generate electricity, where high-temperature resources are available, typically in tectonically active regions, and for direct heating and cooling. Geothermal power provides firm, dispatchable, low-carbon electricity, and is a major source of supply in Kenya, Iceland, El Salvador, New Zealand, the Philippines, and Indonesia. Kenya, which has developed the Olkaria geothermal fields in the East African Rift, generates a substantial share of its electricity from geothermal sources and is among the world's leading geothermal producers. Emerging technologies, including enhanced geothermal systems and closed-loop systems, may extend geothermal development beyond traditional high-temperature regions.

Legal classification. The legal classification of geothermal resources is a threshold question with significant consequences. Geothermal resources may

be classified as minerals (and governed by mining law), as water (and governed by water law), or as a sui generis resource (governed by specific geothermal legislation). Classification determines who owns the resource, which authority regulates it, and which legal regime applies. In the United States, the Geothermal Steam Act of 1970 governs geothermal leasing on federal lands, and state laws variously classify geothermal resources as minerals, water, or a distinct resource. In Indonesia, geothermal resources were historically regulated as a mining activity, which meant that geothermal exploration was prohibited in conservation forests under the forestry laws governing mining; Law No 21 of 2014 on Geothermal reclassified geothermal energy as a non-mining activity, enabling development in conservation forest areas subject to environmental permits, and transferred licensing authority for geothermal power from regional governments to the central government. Kenya's Energy Act 2019, which replaced the Geothermal Resources Act 1982, vests geothermal resources in the national government and provides for licensing by the Cabinet Secretary.

Licensing and risk. Geothermal development involves high upfront exploration risk: drilling exploratory wells is expensive, and the characteristics of the resource cannot be confirmed until wells are drilled. Many governments therefore undertake or finance early-stage exploration through state geothermal companies (such as Kenya's Geothermal Development Company) or risk mitigation facilities (such as the Geothermal Risk Mitigation Facility for Eastern Africa, supported by the African Union Commission and development partners), and then offer proven steam fields to private developers. Geothermal licences typically follow a sequence of exploration and exploitation permits similar to that in mining and petroleum law.

Environmental issues. Geothermal development can cause induced seismicity (particularly with enhanced geothermal systems), land subsidence, emissions of hydrogen sulphide and other gases, and impacts on surface thermal features. The suspension of an enhanced geothermal project in Pohang, South Korea, after it was found by a government commission to have triggered a magnitude 5.4 earthquake in 2017, and the abandonment of a geothermal project in Basel, Switzerland, after induced earthquakes in 2006, illustrate the risks. Geothermal projects in Kenya's Rift Valley have also raised issues of land acquisition and resettlement affecting Maasai communities, which were the subject of complaints to the World Bank's Inspection Panel concerning the Olkaria IV project.

Biomass and Bioenergy

Bioenergy — energy derived from biomass, including wood, agricultural residues, energy crops, and organic waste — is the largest source of renewable energy globally when traditional uses of biomass for cooking and heating are included. Modern bioenergy includes the combustion of biomass for electricity and heat,

the production of liquid biofuels (ethanol and biodiesel) for transport, and the production of biogas and biomethane from anaerobic digestion. Traditional biomass use — the burning of wood, charcoal, crop residues, and dung for cooking in inefficient stoves — remains the primary source of cooking energy for around two billion people, predominantly in sub-Saharan Africa and South Asia, causing household air pollution that the World Health Organization estimates contributes to millions of premature deaths annually, and contributing to deforestation. The transition to clean cooking is examined in Chapter 23.

Sustainability concerns. The renewable and low-carbon credentials of bioenergy depend critically on how the biomass is produced. The combustion of wood releases carbon dioxide at the point of combustion, and the carbon is reabsorbed only as replacement trees grow, which may take decades; if biomass is sourced from primary forests or causes deforestation, its net climate impact may be negative over relevant time horizons. Biofuels produced from food crops may compete with food production and drive land use change, including the conversion of forests and grasslands to agriculture elsewhere ("indirect land use change", or ILUC), which may negate their emissions savings. These concerns have led to the development of sustainability criteria in law.

The EU sustainability criteria. The EU Renewable Energy Directive establishes sustainability and greenhouse gas emissions saving criteria that biofuels, bioliquids, and biomass fuels must satisfy in order to count towards renewable energy targets, to be eligible for financial support, and to count towards renewable energy obligations. Under Article 29 of Directive (EU) 2018/2001, as amended by RED III, the criteria prohibit the sourcing of agricultural biomass from land with high biodiversity value (primary forests, highly biodiverse grasslands, and protected areas) or high carbon stock (wetlands, continuously forested areas, and peatland); impose requirements on forest biomass to ensure legal harvesting, regeneration, protection of biodiversity and soil, and maintenance of carbon stocks; and require minimum greenhouse gas emissions savings compared with fossil fuels. The ILUC Directive (Directive (EU) 2015/1513) capped the contribution of biofuels produced from food and feed crops towards transport targets, and the Commission's Delegated Regulation (EU) 2019/807 identified palm oil as a high-ILUC-risk feedstock, to be phased out from counting towards targets by 2030. The phase-out of palm oil biofuels was challenged by Malaysia and Indonesia before the WTO in *EU — Palm Oil (Malaysia)* (DS600) and *EU — Palm Oil (Indonesia)* (DS593); in March 2024, the panel in the Malaysian case accepted the legitimacy of the EU's objective of addressing ILUC, while finding deficiencies in certain aspects of the design and application of the measures, and a panel report in the Indonesian case followed in early 2025.

Biofuel mandates. Many countries mandate the blending of biofuels with petrol and diesel. The United States Renewable Fuel Standard, established under the Energy Policy Act of 2005 and expanded by the Energy Independence and Security Act of 2007, requires the blending of specified volumes of renewable fuels, administered through a system of tradable Renewable Identification Numbers. Brazil's long-standing ethanol programme and blending mandates have made it a world leader in sugarcane ethanol. Nigeria's **Official Gazette of the Nigerian Bio-fuel Policy and Incentives** (2007) set targets for blending ethanol and biodiesel with petroleum products, but implementation has been limited.

Waste-to-energy and biogas. The conversion of municipal solid waste, agricultural residues, and sewage into energy through incineration, gasification, or anaerobic digestion is regulated by waste law, air quality law, and energy law. EU law establishes a waste hierarchy, under Directive 2008/98/EC, that prioritises prevention, reuse, and recycling over energy recovery, and waste-to-energy plants are subject to emission limits under the Industrial Emissions Directive.

10.6 Renewable Energy Incentives and Support Mechanisms

The Rationale for Support

Public support for renewable energy has been justified on several grounds: the failure of markets to price the environmental costs of fossil fuels, particularly greenhouse gas emissions, which places renewable energy at an artificial disadvantage; the learning-by-doing benefits of deployment, which drive down costs for all future users but cannot be fully captured by early adopters; energy security and diversification; industrial policy and job creation; and the need to accelerate the transition to meet climate targets. As the costs of wind and solar have fallen, the rationale for direct subsidies has weakened in many markets, and the focus of support has shifted from subsidising generation costs to reducing investment risk and the cost of capital, which is the principal determinant of the cost of renewable electricity.

	What the state fixes	Who bears price risk	Typical instruments / examples
Feed-in tariff	Price per MWh	Consumers / state	German EEG 2000; Nigeria FiT Regs 2015
Feed-in premium	Premium over market	Shared	EU schemes after 2014 State aid guidelines
Quota + certificates	Quantity (obligation)	Generator	US RPS; UK Renewables Obligation
Auction + PPA / CfD	Quantity; price by bid	Shared (two-way CfD)	South Africa REIPPPP; UK CfD
Fiscal incentives	Tax credit or relief	Taxpayers	US ITC/PTC; 45Y/48E; duty waivers

Figure 10.1 Principal renewable energy support mechanisms and the allocation of price and volume risk. *Source:* prepared for this edition.

Feed-in Tariffs

A **feed-in tariff** (FiT) guarantees eligible renewable generators a fixed price per unit of electricity fed into the grid for a long period (typically fifteen to twenty years), often combined with an obligation on network operators or suppliers to purchase the electricity. The German Electricity Feed-in Act (*Stromeinspeisungsgesetz*) of 1990 and Renewable Energy Sources Act (*Erneuerbare-Energien-Gesetz*, EEG) of 2000 established the model, which was adopted by numerous countries and is credited with driving the early growth of wind and solar energy. FiTs provide high investment certainty, but they require the regulator to set the tariff administratively, with the risk of overcompensation if costs fall faster than anticipated — as occurred with solar PV in Spain, Italy, the Czech Republic, and Germany in the late 2000s, leading to "solar booms" and escalating costs for consumers, and ultimately to retroactive tariff reductions that generated the investment disputes examined in section 10.8.

A **feed-in premium** (FiP) pays generators a premium on top of the market price, either fixed or sliding (varying with the market price to provide a target level of revenue). Feed-in premiums expose generators to market signals while providing revenue support, and EU State aid guidelines from 2014 required member states to move from feed-in tariffs to market-based premiums for all but small installations.

Quota Obligations and Tradable Certificates

Under a **quota obligation** — known in the United States as a **renewable portfolio standard** (RPS) and in the United Kingdom, until 2017, as the **Renewables Obligation** — electricity suppliers or other obligated parties must source a specified and generally rising proportion of their electricity from renewable sources, demonstrating compliance by surrendering tradable green certificates issued to renewable generators. The price of certificates is determined by the market, and generators receive revenue from the sale of both electricity and certificates. Quota systems leave the quantity of renewable energy to be set by the regulator and the price to be determined by the market, inverting the logic of feed-in tariffs. They have been criticised for exposing generators to volatile certificate prices, increasing the cost of capital. More than half of the states of the United States, together with the District of Columbia, have adopted RPS or clean energy standards, many with targets of one hundred per cent clean electricity by 2040 or 2050.

Auctions and Contracts for Difference

Competitive auctions (or tenders) have become the dominant mechanism for allocating support to utility-scale renewable energy worldwide. Under an auction, the government or a designated agency invites bids for the right to build a specified quantity of renewable capacity or generation, and awards support — usually a long-term PPA or contract for difference — to the bidders offering the lowest price, sometimes combined with other criteria. Auctions combine the investment certainty of long-term contracts with competitive price discovery, and have driven dramatic reductions in the prices of wind and solar electricity in countries including India, Brazil, Mexico, the United Arab Emirates, Saudi Arabia, and Morocco.

South Africa's **Renewable Energy Independent Power Producer Procurement Programme** (REIPPPP), launched in 2011, is widely regarded as one of the most successful renewable procurement programmes in a developing country. Through successive bid windows, it procured several gigawatts of wind, solar, and other renewable capacity at falling prices, combining price competition with economic development criteria (local content, job creation, and community ownership) and a robust contractual framework of twenty-year PPAs with Eskom backed by government support agreements. Its success was attributed to transparent procurement, clear rules, credible institutions (the IPP Office, a unit established jointly by the Department of Energy, the National Treasury, and the Development Bank of Southern Africa), and bankable documentation, although its momentum was interrupted between 2015 and 2018 when Eskom delayed signing PPAs.

In the United Kingdom, the **Contracts for Difference** (CfD) scheme, established under the Energy Act 2013, awards fifteen-year (now twenty-year for some

technologies) private law contracts between generators and the Low Carbon Contracts Company, a government-owned counterparty, through competitive allocation rounds. Under a two-way CfD, the generator receives the difference between a "strike price" and a market reference price when the market price is below the strike price, and pays the difference when the market price is above it. During the 2022 crisis, generators with CfDs paid substantial sums back to the counterparty, reducing consumer bills. As noted in Chapter 9, the EU's 2024 electricity market design reform requires direct price support for new renewable and nuclear investment to take the form of two-way CfDs, subject to exceptions.

Fiscal Incentives

Tax incentives are a major instrument of renewable energy support, particularly in the United States, where the federal **investment tax credit** (ITC) and **production tax credit** (PTC) were the principal instruments of support for decades. The Inflation Reduction Act of 2022 extended and restructured these credits, replacing them from 2025 with technology-neutral clean electricity investment and production credits (sections 48E and 45Y of the Internal Revenue Code), with bonus credits for meeting domestic content, energy community, and prevailing wage and apprenticeship requirements, and introduced "transferability", permitting credits to be sold to unrelated taxpayers. The One Big Beautiful Bill Act of July 2025 significantly curtailed these incentives for wind and solar energy, terminating the credits for wind and solar facilities placed in service after 2027 unless construction began within a short period after enactment, and imposing restrictions related to "prohibited foreign entities" — an illustration of the vulnerability of fiscal incentives to political change. Developing countries commonly use import duty and VAT exemptions for renewable energy equipment, tax holidays, and accelerated depreciation; Nigeria has granted pioneer status incentives and import duty waivers for certain renewable energy equipment.

State Aid Control in the European Union

In the European Union, renewable energy support schemes funded through state resources are subject to the State aid rules of Articles 107 and 108 TFEU. The Court of Justice's jurisprudence on whether support schemes constitute State aid has been complex. In *PreussenElektra* (Case C-379/98, 2001), the Court held that the German obligation on electricity suppliers to purchase renewable electricity at minimum prices did not involve the transfer of state resources and therefore did not constitute State aid. In *Vent De Colère* (Case C-262/12, 2013), it held that the French scheme, under which the additional costs of purchase obligations were financed by a levy on consumers collected and administered under state control, involved state resources. In *Germany v Commission* (Case C-405/16 P, 2019), the Court set aside the General Court's judgment and annulled the Commission's

decision classifying the EEG 2012 as State aid, holding that the sums generated by the EEG surcharge were not under public control and therefore did not constitute state resources. Where schemes do constitute State aid, they must be notified and approved by the Commission under the **Guidelines on State aid for climate, environmental protection and energy** (CEEAG), adopted in 2022, which require, among other things, competitive allocation of aid and a demonstrated incentive effect.

The interaction of national support schemes with the free movement of goods was considered in *Ålands Vindkraft AB v Energimyndigheten* (Case C-573/12, 2014), in which the Court held that a Swedish green certificate scheme that awarded certificates only to producers located in Sweden constituted a measure having equivalent effect to a quantitative restriction on imports, but was justified by the objective of promoting renewable energy, given that the Renewable Energy Directive expressly permitted member states to limit support to domestic production. The WTO compatibility of renewable support schemes with local content requirements, considered in *Canada — Renewable Energy/Feed-in Tariff Program* (2013) and *India — Solar Cells* (2016), is examined in Chapter 13.

10.7 Renewable Energy Certificates and Attribute Markets

The Concept of Environmental Attributes

Once electricity enters the grid, electrons from renewable and fossil sources are physically indistinguishable, and it is impossible to trace the electricity consumed by a particular customer to a particular generator. Renewable energy certificates address this problem by separating the "environmental attributes" of renewable generation — its renewable origin and associated characteristics — from the physical electricity, and embodying them in a tradable instrument. A certificate typically represents one megawatt-hour of electricity generated from a renewable source and records information about the source, location, technology, date of generation, and the installation. Certificates enable consumers, suppliers, and companies to substantiate claims that their electricity is renewable, to comply with disclosure or quota obligations, and to account for the emissions associated with their electricity consumption.

Certificates serve different legal functions in different systems, and it is important to distinguish between them. **Compliance certificates** are used to demonstrate compliance with quota obligations (such as renewable portfolio standards and the former UK Renewables Obligation, examined in section 10.6). **Disclosure certificates** — notably guarantees of origin in the European Union — are used to substantiate the information that suppliers must give consumers about the sources of their electricity. **Voluntary certificates** are purchased by

companies and individuals to support voluntary claims of renewable energy use, for example under corporate sustainability commitments.

Guarantees of Origin in the European Union

Article 19 of the Renewable Energy Directive requires member states to ensure that the origin of electricity from renewable sources can be guaranteed, for the purpose of proving to final customers the share or quantity of renewable energy in a supplier's energy mix, by issuing **guarantees of origin** (GOs) in response to a request from a producer. Each guarantee of origin has a standard size of one megawatt-hour, may be used only once, and must be used within twelve months of the production of the corresponding unit of energy; GOs that have not been cancelled expire no later than eighteen months after production. Member states must recognise GOs issued by other member states, and may refuse recognition only where they have well-founded doubts about their accuracy, reliability, or veracity. RED III extended the GO system to renewable gases, including hydrogen. Under the Electricity Directive, suppliers must disclose the share of each energy source in their overall fuel mix, and must use GOs to demonstrate the share of renewable energy. The technical standard for GOs is the European Energy Certificate System (EECS), administered by the Association of Issuing Bodies, which links national registries through a hub enabling cross-border transfers.

Article 19(2) of the Directive, as amended, also provides that member states may decide not to issue GOs to producers that receive financial support from a support scheme, or to issue them and transfer their value to the market, to avoid double compensation — an issue that has generated debate about whether certificates from subsidised plants should be sold to consumers who might believe that their purchases contribute to new renewable investment.

Certificate Systems Elsewhere

In the United States, **renewable energy certificates** (RECs) are used both for compliance with state renewable portfolio standards and for voluntary claims, tracked through regional tracking systems. Australia's large-scale generation certificates, created under the Renewable Energy (Electricity) Act 2000, are used for compliance with the Renewable Energy Target. In countries without national certificate systems, including most of Africa, Asia, and Latin America, the **I-REC Standard**, administered by a non-profit foundation, provides an international framework for issuing and tracking certificates through local issuers, and has been used in a growing number of African countries. The development of certificate systems in developing countries enables domestic and foreign companies to procure renewable electricity credibly, and may support corporate renewable procurement and investment, but requires robust registries, independent verification, and safeguards against double counting.

Corporate Renewable Procurement and Claims

Corporate demand has become an important driver of renewable energy investment. Companies with commitments to use renewable electricity — many of them members of the **RE100** initiative, which brings together companies committed to sourcing one hundred per cent renewable electricity — procure renewable electricity through several means: on-site generation; **corporate PPAs**, under which the company contracts directly with a renewable generator for its output and associated certificates; green tariffs offered by utilities; and the purchase of "unbundled" certificates separately from electricity. Corporate PPAs may be **physical** (involving the delivery of electricity to the buyer through the grid) or **virtual** or **financial** (a contract for difference settled financially against the market price, with the transfer of certificates). RED II required member states to remove unjustified barriers to corporate renewable PPAs, and the EU's 2024 electricity market reform requires member states to facilitate PPAs, including through state-backed guarantee schemes addressing the credit risk of buyers.

The accounting rules for corporate claims are set by the **Greenhouse Gas Protocol**, a private standard-setting initiative of the World Resources Institute and the World Business Council for Sustainable Development. Its Scope 2 Guidance (2015) requires companies to report emissions from purchased electricity using two methods: a **location-based method**, reflecting the average emissions intensity of the grid in the locations where consumption occurs, and a **market-based method**, reflecting the emissions associated with contractual instruments such as certificates and PPAs, subject to quality criteria. The market-based method has been criticised for allowing companies to claim zero emissions from electricity by purchasing cheap unbundled certificates from existing plants, which may have no effect on actual emissions or on investment in new renewable capacity ("additionality"), and from distant locations or different times of day from the company's consumption. In response, some companies have adopted "24/7 carbon-free energy" commitments, matching their consumption on an hourly basis with carbon-free generation in the same grid region, and the Greenhouse Gas Protocol has consulted on revisions to the Scope 2 Guidance, including proposals for hourly matching and requirements that certificates be sourced from the same market as consumption.

The integrity of corporate claims has become a legal as well as a reputational issue. In the European Union, Directive (EU) 2024/825 on empowering consumers for the green transition, which applies from September 2026, prohibits generic environmental claims that cannot be substantiated and claims, based on the offsetting of greenhouse gas emissions, that a product has a neutral, reduced, or positive impact on the environment. Consumer protection authorities and courts in several jurisdictions have taken action against misleading "green" claims,

and the use of renewable energy certificates to support claims may be scrutinised under consumer protection and unfair competition law. The relationship between renewable certificates and carbon credits, which are distinct instruments representing emission reductions or removals and are governed by the carbon market rules examined in Chapter 18, is a recurrent source of confusion: a renewable energy certificate attests to the renewable origin of a unit of electricity, while a carbon credit represents a tonne of carbon dioxide equivalent reduced or removed relative to a baseline, and the same project should not generate both instruments for the same attribute without safeguards against double counting.

10.8 Renewable Energy Investment Disputes

The Context: Retroactive Changes to Support Schemes

The renewable energy investment disputes arose from a common pattern. In the mid-2000s, several European states — notably Spain, Italy, and the Czech Republic — introduced generous feed-in tariffs to attract investment in renewable energy, particularly solar PV. The tariffs, combined with rapidly falling costs of solar panels, generated investment booms far exceeding government projections, and the cost of support, borne by consumers or the state, escalated rapidly. Following the global financial crisis of 2008, and facing large deficits in their electricity systems, these governments enacted measures reducing the support available to existing installations, including caps on the hours of production eligible for support, levies on revenues, and, in Spain's case, the replacement of the feed-in tariff regime by a new regime guaranteeing a "reasonable return" on investment calculated over the regulatory life of installations. Investors, many of them foreign, brought claims under the Energy Charter Treaty and bilateral investment treaties, arguing that the measures breached the obligation to accord fair and equitable treatment (FET), including by frustrating their legitimate expectations of regulatory stability, and constituted indirect expropriation.

The Spanish Arbitrations

Spain's original regime, set out principally in Royal Decree 661/2007, offered renewable installations registered by specified deadlines a regulated tariff for their entire operational life, with tariffs for solar PV substantially above market prices. In 2010, Spain introduced initial measures limiting the hours of production eligible for the tariff and imposing a grid access charge. Between 2012 and 2014, it enacted more radical measures: a seven per cent tax on the value of electricity production (Law 15/2012); the elimination of the inflation adjustment; and, by Royal Decree-law 9/2013, Law 24/2013, Royal Decree 413/2014, and Ministerial Order IET/1045/2014, the abolition of the RD 661/2007 regime and its replacement by a "specific remuneration" regime designed to provide a "reasonable return" for a "standard installation", initially set at around 7.4 per cent pre-tax,

over the regulatory life of the installation, taking into account the remuneration already received.

Spain faced around fifty arbitration claims, making it one of the most frequently sued states in the world. The outcomes varied, but investors prevailed in the majority of cases decided on the merits in respect of the 2013–2014 measures.

Claims dismissed. In *Charanne BV and Construction Investments SARL v Spain* (SCC, 2016), the first award, the tribunal dismissed claims concerning the 2010 measures, holding that the investors could not have had a legitimate expectation that the regulatory framework would remain unchanged in the absence of a specific commitment, that RD 661/2007 was a general regulation, and that the 2010 measures were not unreasonable or disproportionate. In *Isolux Infrastructure Netherlands BV v Spain* (SCC, 2016), the tribunal dismissed claims concerning the 2013–2014 measures, holding that the investor, which had invested in 2012 after the first changes and when further reform was foreseeable, could not have had a legitimate expectation that the regime would remain unchanged. In *Stadtwerke München GmbH and others v Spain* (ICSID, 2019), the tribunal similarly found that a diligent investor would have known that the regime could be modified, and that the new regime was not unreasonable.

Claims upheld. In *Eiser Infrastructure Ltd and Energía Solar Luxembourg SÀRL v Spain* (ICSID, 2017), examined in the case box below, the tribunal found that Spain had breached the FET standard by fundamentally and abruptly altering the regulatory regime, and awarded around EUR 128 million. Subsequent tribunals, including in *Novenergia II v Spain* (SCC, 2018), *Masdar Solar & Wind Cooperatief UA v Spain* (ICSID, 2018), *Antin Infrastructure Services Luxembourg SARL v Spain* (ICSID, 2018), *RREEF Infrastructure v Spain* (ICSID, 2019), *NextEra Energy Global Holdings BV v Spain* (ICSID, 2019), and *Cube Infrastructure Fund SICAV v Spain* (ICSID, 2019), found breaches of the FET standard, with awards ranging from tens to hundreds of millions of euros. The reasoning varied. Some tribunals, such as *Masdar*, found that specific assurances — including letters from the Spanish authorities confirming that the installations qualified for the RD 661/2007 tariff for their operational life — created legitimate expectations. Others, such as *RREEF*, found no legitimate expectation that the specific tariff would be preserved, but held that the investors were entitled to expect a reasonable return, and that Spain had breached the FET standard to the extent that the new regime failed to provide it, or applied retroactively by "clawing back" past remuneration.

The tax carve-out. Tribunals generally held that the seven per cent tax on the value of electricity production was a "taxation measure" within the meaning of Article 21 of the ECT, which excludes taxation measures from the scope of the treaty's protections (subject to exceptions for expropriation and certain non-dis-

crimination obligations), and therefore declined jurisdiction over claims relating to it, notwithstanding the investors' argument that the tax was in substance a disguised reduction of the tariff.

::: casebox {title="Eiser Infrastructure Ltd and Energía Solar Luxembourg SÀRL v Kingdom of Spain (ICSID, 2017)"} **Facts.** The claimants, a United Kingdom investment fund and its Luxembourg affiliate, invested in three concentrated solar power plants in Spain in 2007, relying on the regulated tariff regime of Royal Decree 661/2007. Following the 2013–2014 reforms, the plants' revenues fell dramatically, and the claimants alleged that the plants could no longer service their project finance debt.

Decision. The tribunal held that Article 10(1) of the ECT, which requires contracting parties to accord fair and equitable treatment, did not give investors a right to regulatory stability in the sense of a regulatory regime frozen in time; states retain the right to modify their regulatory regimes to meet changing circumstances in the public interest. However, the FET obligation protected investors against the "total and unreasonable" change of the regime on which they had relied in making long-term investments. Spain's new regime had been applied to existing installations, on the basis of a "standard installation" model that disregarded the claimants' actual costs and the capital-intensive character of their investment, and had eliminated the essential characteristics of the regime on which they had relied, depriving them of essentially all of the value of their investment. The tribunal awarded around EUR 128 million, using a discounted cash flow valuation.

Annulment. In 2020, an ICSID *ad hoc* committee annulled the award in its entirety because one of the arbitrators, appointed by the claimants, had failed to disclose long-standing professional connections with the firm of damages experts that had given evidence for the claimants, which the committee found could have had a material effect on the award. The claimants subsequently resubmitted their claims.

Significance. *Eiser* was the first award against Spain and set the template for many subsequent awards. Its distinction between permissible regulatory change and the "radical" or "fundamental" alteration of a regime on which investors had relied influenced the jurisprudence on regulatory stability under the ECT. Its annulment is a leading illustration of the importance of arbitrator disclosure, a theme examined in Chapter 26. :::

The Italian and Czech Arbitrations

Italy introduced a series of *Conto Energia* decrees offering feed-in tariffs for solar PV, and in 2014, by the so-called *Spalma-incentivi* decree, required existing installations to choose between reduced tariffs spread over a longer period and

other options reducing their incentives. Investors brought claims under the ECT. In *Blusun SA and others v Italy* (ICSID, 2016), the tribunal dismissed the claims, holding that the investors had not demonstrated that Italy had made specific commitments that the incentives would not be altered and that, in any event, the failure of the project was not caused by the measures. Other tribunals reached different results: in *Greentech Energy Systems A/S and others v Italy* (SCC, 2018) and *CEF Energia BV v Italy* (SCC, 2019), tribunals found that Italy had breached the FET standard by reducing incentives that had been specifically guaranteed to the investors' installations by agreements with the state agency responsible for administering the incentives, while in *Eskosol SpA in liquidazione v Italy* (ICSID, 2020), the tribunal dismissed the claim.

The Czech Republic introduced a levy on revenues from solar installations commissioned in 2009 and 2010 (the "solar levy") and withdrew certain tax exemptions. Claims under the ECT and BITs were largely unsuccessful: in *Antaris Solar GmbH and Dr Michael Göde v Czech Republic* (PCA, 2018) and in *JSW Solar (zwei) GmbH and others v Czech Republic* (PCA, 2019), the tribunals dismissed the claims, finding that the investors could not legitimately expect the incentives to remain unchanged, that the Czech Republic had acted reasonably in response to the solar boom, and that the investors, many of whom invested late in the boom, had been on notice of possible changes.

Beyond Europe, *Mesa Power Group LLC v Canada* (UNCITRAL, NAFTA, 2016) concerned the allocation of feed-in tariff contracts under Ontario's Green Energy Act, which included domestic content requirements found inconsistent with WTO law in *Canada — Renewable Energy*. The tribunal dismissed the claims, holding, among other things, that the domestic content requirements fell within the procurement exception of NAFTA Article 1108.

Doctrinal Themes

The renewable energy arbitrations have produced a substantial body of jurisprudence on several doctrinal questions.

Legitimate expectations and general legislation. A central question is whether general legislation — as distinct from specific commitments made to the investor — can create legitimate expectations protected by the FET standard. The prevailing view, reflected in *Charanne*, *Blusun*, and many subsequent awards, is that general legislation does not, by itself, create a legitimate expectation that it will not be changed; states retain the right to regulate. However, tribunals have differed as to whether legislation that specifically targets a defined class of investors and promises them particular treatment for a defined period (such as RD 661/2007's promise of tariffs for the operational life of registered installations)

can create legitimate expectations, and as to the significance of registration and other individualised acts.

Due diligence and foreseeability. Tribunals have consistently held that investors must conduct reasonable due diligence into the legal framework and its stability, including judicial decisions (for example, Spanish Supreme Court judgments upholding earlier regulatory changes), and that expectations of stability are not legitimate if regulatory changes were reasonably foreseeable at the time of investment. The timing of investment has therefore been critical: investors who invested late, after warnings of reform, have generally fared worse.

Proportionality and the reasonable return. Many tribunals have assessed whether the measures were reasonable and proportionate, considering the public interest in addressing the tariff deficit and the impact on investors. Spain's argument that its new regime satisfied investors' legitimate expectations by guaranteeing a reasonable return was accepted by some tribunals and rejected by others, depending on whether the reasonable return was in fact achieved and whether the regime clawed back past remuneration.

Valuation of damages. The arbitrations generated extensive debate over the valuation of damages, including the use of discounted cash flow methods for operating assets, the appropriate discount rate, the useful life of installations, and the treatment of regulatory risk. These questions are examined in Chapter 26.

The Intra-EU Objection and Enforcement

Most of the claimants against Spain and Italy were investors from other EU member states, and the disputes became entangled in the controversy over intra-EU investment arbitration. In *Slovak Republic v Achmea BV* (Case C-284/16, 2018), the Court of Justice held that investor–state arbitration clauses in intra-EU BITs are incompatible with EU law, because they remove disputes that may concern the interpretation or application of EU law from the EU judicial system. In *Republic of Moldova v Komstroy LLC* (Case C-741/19, 2021), the Court held that Article 26 of the ECT must be interpreted as not applicable to disputes between a member state and an investor of another member state concerning an investment in the first member state. In 2017, the Commission adopted a decision concluding that Spain's renewable energy support scheme constituted State aid, and in 2025 it adopted a decision concluding that the payment of the award in *Antin* would constitute unlawful State aid. Arbitral tribunals have, almost without exception, rejected the intra-EU objection, holding that their jurisdiction derives from the ECT as a matter of international law, and that EU law does not override the clear terms of the treaty.

The result has been a conflict between the obligations of EU member states under EU law and their obligations under the ECT and the ICSID Convention,

which has played out in enforcement proceedings outside the European Union. In *Kingdom of Spain v Infrastructure Services Luxembourg Sàrl* [2023] HCA 11, the High Court of Australia held that Spain's agreement to Article 54 of the ICSID Convention, which requires contracting states to recognise ICSID awards as binding and enforce the pecuniary obligations imposed by them as if they were final judgments of their own courts, amounted to a waiver of immunity from the jurisdiction of Australian courts to recognise and enforce the award, although not of immunity from execution against particular assets. Courts in the United Kingdom and the United States have also rejected Spain's immunity objections at various stages of enforcement proceedings; in *NextEra Energy Global Holdings BV v Kingdom of Spain* (2024), the United States Court of Appeals for the District of Columbia Circuit held that federal courts had jurisdiction to enforce intra-EU ECT awards against Spain under the arbitration exception to the Foreign Sovereign Immunities Act, and reversed injunctions that had been granted to prevent award creditors from pursuing proceedings in Europe. Further proceedings have continued, and the conflict between EU law and the international obligations of member states remains unresolved.

In response to the arbitrations, Spain adopted Royal Decree-law 17/2019, which offered installations whose owners renounced or discontinued arbitration claims a fixed reasonable return of around 7.4 per cent until 2031. The European Union and many of its member states have since withdrawn from the ECT, a development examined in Chapter 14.

Lessons for Regulatory Design

The renewable energy arbitrations offer important lessons for the design of renewable energy support. First, support schemes should be designed with mechanisms that allow costs to be controlled — such as volume caps, automatic degression of tariffs, and competitive allocation — so that governments are not forced into retroactive changes. Second, governments should distinguish clearly between general policy statements and specific commitments, and should avoid individualised assurances that they may not be able to honour. Third, changes to support schemes should, as far as possible, apply prospectively to new installations, with transitional arrangements for existing installations. Fourth, investors should conduct thorough due diligence into the stability of the legal framework, and price regulatory risk accordingly. These lessons are relevant to developing countries, including Nigeria, that seek to attract private investment in renewable energy through long-term contracts and support schemes.

::: exercise {title="Practical Exercise: The Savania Solar Support Reform and the Lake Ilera Mini-Grid (HYPOTHETICAL)"} *This exercise is entirely hypothetical. The Republic of Savania, the Lake Ilera community, and all companies*

named are fictional, and any resemblance to real states, communities, or companies is coincidental.

Part A — Retroactive reform. In 2017, the Republic of Savania, a party to the Energy Charter Treaty, enacted the Solar Development Law, which guaranteed solar installations registered before 31 December 2019 a fixed tariff of USD 0.22 per kWh "for twenty years from commissioning". The Ministry of Energy issued registration certificates to each installation stating that it was "entitled to the regulated tariff for twenty years". The law contained no mechanism for controlling the volume of installations. Over 3,000 MW was registered against a forecast of 800 MW. By 2023, the cost of support had created a deficit equal to two per cent of GDP. In 2024, Savania enacted the Electricity Sustainability Law, which (i) replaced the fixed tariff for all installations, including existing ones, with a regime providing a "reasonable return" of 6.5 per cent pre-tax over the regulatory life of a "standard installation", calculated by taking into account revenues received since commissioning; and (ii) imposed a five per cent levy on the gross revenues of all generators.

Helios Infrastructure Partners GmbH (HYPOTHETICAL), incorporated in a non-EU ECT contracting party, invested in 2018 in a 150 MW solar plant financed with seventy per cent debt. Its due diligence report noted "political discussion of tariff costs" but concluded that the risk of retroactive change was low given the registration certificate.

1. Advise Helios on the strength of a claim for breach of Article 10(1) of the ECT, considering the case law on legitimate expectations, specific commitments, due diligence, and proportionality.
2. Advise Savania on its defences, including the argument that the reasonable-return regime satisfies any legitimate expectation, and the significance of the "clawback" of past revenues.
3. Can Helios challenge the five per cent levy under the ECT? Consider Article 21.
4. Propose an alternative reform that Savania could have adopted to control costs while minimising its exposure to claims.

Part B — Mini-grid and grid arrival. In 2021, Lumen Mini-Grids Ltd (HYPOTHETICAL) obtained a permit from the Savania Electricity Regulatory Commission to build and operate an isolated 500 kW solar-hybrid mini-grid serving 2,000 households in the Lake Ilera community, under regulations allowing a cost-reflective tariff. The permit stated that if the main grid reached the community, Lumen could elect either to become a small distributor purchasing bulk power from the distribution company, or to sell its distribution assets to the distribution company "at a fair valuation determined by the Commission". In 2025, the distribution company extended its network to Lake Ilera, began

offering connections at a tariff subsidised to forty per cent below Lumen's tariff, and refused to negotiate either option.

5. Identify Lumen's legal remedies under the regulatory framework, including the role of the Commission.
6. What principles should govern the "fair valuation" of Lumen's assets?
7. How should mini-grid regulations be designed to protect both investors and consumers when the main grid arrives? Draft a model clause of no more than 250 words. :::

CHAPTER SUMMARY

- Renewable energy has become the dominant source of new generating capacity worldwide. The law has driven its growth through targets, support mechanisms, and grid access, and has constrained it through permitting delays, land conflicts, and regulatory instability.
- There is no comprehensive international treaty on renewable energy. Its international foundations lie in climate law, SDG 7, the IRENA Statute, the International Solar Alliance, and the ECT. The EU's Renewable Energy Directive, as amended in 2023, sets a binding Union target of at least 42.5 per cent by 2030.
- Wind and sunlight are generally regarded as things common to all. The law therefore concerns rights to use land and to access the resource, together with the allocation of offshore areas under the coastal state's EEZ rights.
- Utility-scale solar raises land rights issues, particularly where land is held under customary tenure. Mini-grid regulations, such as Nigeria's, must address cost-reflective tariffs and the consequences of the main grid's arrival.
- Wind permitting addresses landscape, wildlife, and indigenous rights. The *Fosen* judgment of the Supreme Court of Norway demonstrates that renewable projects are subject to human rights constraints.
- Hydropower law governs water rights, environmental flows, long-term concessions, reversion, and dam safety. Geothermal law turns on the classification of the resource, and bioenergy is subject to sustainability criteria addressing land use change.
- Support mechanisms include feed-in tariffs and premiums, quota obligations with certificates, auctions with long-term contracts or CfDs, and fiscal incentives. Auctions have become dominant, and EU support schemes are subject to State aid control.
- Certificates separate environmental attributes from physical electricity. They serve compliance, disclosure, and voluntary functions, and the integrity of corporate claims is increasingly subject to legal scrutiny.
- The Spanish, Italian, and Czech arbitrations established that general legislation does not, by itself, freeze the regulatory framework. However, fundamental and retroactive changes that defeat specific commitments or a reasonable

return may breach the FET standard. Their enforcement has exposed a conflict between EU law and the ECT and ICSID Convention.

DOCTORAL RESEARCH QUESTIONS

1. Does the *Fosen* judgment imply a general principle that the public interest in decarbonisation cannot justify interference with indigenous peoples' cultural rights? How should courts balance climate objectives against the rights of affected communities in the energy transition?
2. To what extent did the design of European feed-in tariff schemes, rather than the subsequent reforms, cause the renewable energy investment disputes? What does this imply for the allocation of responsibility between states and investors?
3. Evaluate the coherence of arbitral jurisprudence on legitimate expectations arising from general legislation in the Spanish, Italian, and Czech renewable energy cases. Is a consistent doctrine discernible?
4. Should corporate claims to use renewable electricity be regulated by law, and if so, should the law require additionality, hourly matching, or geographic deliverability?
5. Critically assess the adequacy of the Nigerian legal framework for mini-grids and distributed solar in achieving universal electricity access. What reforms would better align investor protection, consumer protection, and the expansion of the main grid?
6. Is the classification of geothermal resources as minerals, water, or a sui generis resource merely technical, or does it reflect deeper choices about ownership and the distribution of resource benefits?

FURTHER READING

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11 CHAPTER

Nuclear Energy Law, Safety, and International Responsibility

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the legal basis of the right to the peaceful uses of nuclear energy and its relationship with non-proliferation obligations.
2. Describe the architecture of international nuclear law, including the IAEA Statute, the principal conventions, and the role of IAEA safety standards as soft law.
3. Analyse the legal requirements of nuclear safety, with particular attention to the independence of the regulatory body and the lessons of Chernobyl and Fukushima.
4. Explain the system of IAEA safeguards, including comprehensive safeguards agreements and the Additional Protocol, and evaluate its effectiveness in contested cases.
5. Assess the international and domestic law of nuclear security and physical protection.
6. Analyse the principles of the international nuclear liability regimes and evaluate their adequacy in the light of major accidents.
7. Evaluate the legal frameworks for radioactive waste management and decommissioning, including their financing.
8. Identify the emerging legal issues raised by small modular and advanced reactors, and the legal preparations required of nuclear newcomer states, including those in Africa.

Introduction

Nuclear energy occupies a unique position in energy law. It is a low-carbon source of firm electricity that provides around a tenth of the world's electricity and a much larger share of its low-carbon electricity. It is also the only energy technology whose materials and know-how can be diverted to the production of weapons

of mass destruction, and whose accidents can cause radioactive contamination across national borders and over generations. These characteristics have produced a body of law that is more internationalised, more technically detailed, and more closely bound up with questions of international peace and security than any other branch of energy law.

The legal framework of nuclear energy is commonly described by reference to "three Ss" — safety, security, and safeguards — to which a fourth pillar, liability, is usually added, together with the management of radioactive waste and decommissioning. **Safety** concerns the protection of workers, the public, and the environment from the harmful effects of ionising radiation, including the prevention and mitigation of accidents. **Security** concerns the prevention and detection of, and response to, theft, sabotage, unauthorised access, and other malicious acts involving nuclear material and facilities. **Safeguards** concern the verification that nuclear material is not diverted from peaceful uses to nuclear weapons. **Liability** concerns compensation for damage caused by nuclear incidents. Each pillar has its own international instruments, institutions, and domestic legal requirements.

After decades of stagnation following the accidents at Three Mile Island (1979) and Chernobyl (1986), and renewed doubt after the Fukushima Daiichi accident (2011), nuclear energy has experienced a revival of interest driven by climate commitments, energy security concerns following the Russian invasion of Ukraine, and the electricity demand of data centres. At COP28 in 2023, a group of states launched a declaration aiming to triple global nuclear energy capacity by 2050; the European Union included nuclear energy, subject to conditions, in its taxonomy of sustainable activities; the World Bank announced in 2025 that it would resume financing nuclear energy projects; and technology companies have concluded long-term contracts to purchase nuclear electricity, including from the restart of a closed reactor. At the same time, several countries, including Germany, have phased out nuclear power, and the costs and delays of large nuclear construction projects in Europe and the United States have been severe. Several African states, including Egypt, Ghana, Kenya, Nigeria, and Uganda, have pursued nuclear power programmes, and Egypt is constructing the continent's second nuclear power plant, after South Africa's Koeberg.

This chapter examines the peaceful uses of nuclear energy (section 11.1), the international nuclear legal framework (section 11.2), nuclear safety and regulatory independence (section 11.3), safeguards and non-proliferation (section 11.4), nuclear security (section 11.5), nuclear liability (section 11.6), radioactive waste and decommissioning (section 11.7), and small modular reactors and emerging legal issues (section 11.8).

	Objective	Principal instruments	Institutions / review
Safety	Protect people & environment from radiation; prevent accidents	CNS 1994; Joint Convention 1997; IAEA Safety Standards	National regulator; CNS peer review; IRRS
Security	Prevent theft, sabotage & malicious acts	CPPNM & 2005 Amendment; ICSANT; UNSCR 1540	Competent authority; IPPAS
Safeguards	Verify non-diversion to weapons	NPT Art III; INFCIRC/153; Additional Protocol	IAEA Board; UN Security Council
Liability	Compensate victims of nuclear damage	Paris/Brussels; Vienna; Joint Protocol; CSC	Courts of installation state
Waste & decommissioning	Avoid undue burdens on future generations	Joint Convention; Euratom Dir. 2011/70	Waste agencies; segregated funds

Figure 11.1 The pillars of nuclear energy law: objectives, principal instruments, and institutions. *Source:* prepared for this edition.

11.1 Peaceful Uses of Nuclear Energy

From Atoms for Peace to the Right to Peaceful Uses

The international law of nuclear energy originated in the effort, after the bombings of Hiroshima and Nagasaki in 1945, to reconcile the promise of civil nuclear technology with the danger of nuclear weapons proliferation. The first resolution of the United Nations General Assembly, adopted in January 1946, established the United Nations Atomic Energy Commission to make proposals for the control of atomic energy "to the extent necessary to ensure its use only for peaceful purposes". The proposals for international ownership and control of nuclear activities presented by the United States (the Baruch Plan) failed amid the onset of the Cold War. In his "Atoms for Peace" address to the General Assembly in December 1953, President Eisenhower proposed an international agency to which states would contribute fissile material for peaceful uses, leading to the adoption of the Statute of the International Atomic Energy Agency in 1956 and its entry into force in 1957.

The bargain at the heart of the international nuclear order was codified in the **Treaty on the Non-Proliferation of Nuclear Weapons** (NPT), opened for signature in 1968 and in force since 1970. The NPT distinguishes between

"nuclear-weapon States" — defined in Article IX(3) as states that manufactured and exploded a nuclear weapon or other nuclear explosive device before 1 January 1967 (the United States, the Soviet Union (now Russia), the United Kingdom, France, and China) — and "non-nuclear-weapon States". Under Articles I and II, nuclear-weapon states undertake not to transfer nuclear weapons or assist non-nuclear-weapon states to acquire them, and non-nuclear-weapon states undertake not to receive, manufacture, or otherwise acquire nuclear weapons. Under Article III, non-nuclear-weapon states undertake to accept IAEA safeguards on all source and special fissionable material in all their peaceful nuclear activities. Under Article VI, all parties undertake to pursue negotiations in good faith on effective measures relating to cessation of the nuclear arms race and to nuclear disarmament.

Article IV of the NPT

Article IV(1) of the NPT provides that nothing in the Treaty shall be interpreted as affecting "the inalienable right of all the Parties to the Treaty to develop research, production and use of nuclear energy for peaceful purposes without discrimination and in conformity with articles I and II of this Treaty". Article IV(2) commits parties to facilitate, and grants them the right to participate in, the fullest possible exchange of equipment, materials, and scientific and technological information for the peaceful uses of nuclear energy, and requires parties in a position to do so to cooperate in contributing to the further development of applications of nuclear energy for peaceful purposes, "especially in the territories of non-nuclear-weapon States Party to the Treaty, with due consideration for the needs of the developing areas of the world".

The interpretation of Article IV has been contested. Many developing states regard the "inalienable right" as encompassing the entire nuclear fuel cycle, including uranium enrichment and spent fuel reprocessing, which are "sensitive" technologies because they can produce weapons-usable material. Supplier states have sought to restrict the spread of enrichment and reprocessing through export controls, coordinated through the **Nuclear Suppliers Group** (NSG), an informal arrangement established in 1974–1975 after India's first nuclear test, whose guidelines restrict the transfer of sensitive technologies and, since 1992, require full-scope safeguards as a condition of the supply of nuclear materials and equipment to non-nuclear-weapon states. The controversy over Iran's enrichment programme, examined in section 11.4, was in significant part a dispute over the scope of Article IV. The qualification that the right must be exercised "in conformity with articles I and II", and, by implication, with the safeguards obligations of Article III, means that the right is not unconditional.

Peaceful Applications Beyond Electricity

Although this chapter focuses on nuclear power, the peaceful uses of nuclear technology extend far beyond electricity generation. Nuclear and radiation technologies are used in medicine (diagnostic imaging and cancer radiotherapy), agriculture (the sterile insect technique and plant mutation breeding), industry (non-destructive testing and sterilisation), water resource management (isotope hydrology), and scientific research. The IAEA's technical cooperation programme supports these applications in developing states, and they constitute a substantial part of the nuclear activities regulated in most countries, including Nigeria. The legal frameworks for the safety and security of radioactive sources used in these applications, including the IAEA's Code of Conduct on the Safety and Security of Radioactive Sources (2003), are an important component of nuclear law.

The Decision to Pursue Nuclear Power

A state's decision to introduce nuclear power is a long-term commitment requiring the development of a comprehensive legal and institutional infrastructure. The IAEA's **Milestones Approach**, set out in its publication *Milestones in the Development of a National Infrastructure for Nuclear Power*, identifies nineteen infrastructure issues — including national position, nuclear safety, management, funding and financing, legal framework, safeguards, regulatory framework, radiation protection, electrical grid, human resource development, stakeholder involvement, site and supporting facilities, environmental protection, emergency planning, nuclear security, nuclear fuel cycle, radioactive waste management, industrial involvement, and procurement — to be addressed in three phases, from the consideration of the nuclear option through the preparation for construction to the activities to implement the first plant. The IAEA conducts Integrated Nuclear Infrastructure Review (INIR) missions to assess newcomer states' progress, and several African states, including Nigeria, Ghana, Kenya, Egypt, Uganda, and Morocco, have hosted such missions.

The decision to pursue nuclear power raises legal questions beyond technical regulation: democratic legitimacy and public participation (the Aarhus Convention's Compliance Committee and the parties' Meeting have addressed the application of public participation requirements to decisions on nuclear power, including decisions to extend the lifetime of reactors); transboundary consultation under the Espoo Convention on Environmental Impact Assessment in a Transboundary Context (1991), whose Implementation Committee and parties have considered the application of the Convention to the lifetime extension of nuclear power plants; and the compatibility of nuclear power with constitutional rights and environmental principles. The Court of Justice's judgment in *Inter-Environnement Wallonie and Bond Beter Leefmilieu Vlaanderen* (Case C-411/17, 2019) held that the extension of the operating life of Belgian nuclear

reactors required an environmental impact assessment under the EIA Directive and an appropriate assessment under the Habitats Directive, illustrating the application of general environmental law to nuclear decisions.

11.2 International Nuclear Legal Frameworks

The International Atomic Energy Agency

The IAEA, an autonomous international organisation within the United Nations family with its headquarters in Vienna, is the central institution of international nuclear law. Its objectives, set out in Article II of its Statute, are to "seek to accelerate and enlarge the contribution of atomic energy to peace, health and prosperity throughout the world" and to "ensure, so far as it is able, that assistance provided by it or at its request or under its supervision or control is not used in such a way as to further any military purpose". The dual mandate of promotion and control has been a source of tension, and critics have questioned whether an organisation charged with promoting nuclear energy can effectively regulate it.

The IAEA's statutory functions under Article III include encouraging research and development, facilitating the exchange of information, establishing and administering safeguards, and establishing or adopting "standards of safety for protection of health and minimization of danger to life and property", and providing for the application of these standards. The IAEA's governing organs are the General Conference of all member states and the Board of Governors of thirty-five member states, which has the power under Article XII(C) of the Statute to report a member state's non-compliance with its safeguards obligations to the Security Council and the General Assembly. The institutional structure of the IAEA is examined further in Chapter 4.

The Treaty Framework

International nuclear law consists of a network of treaties, most adopted in response to specific events, together with a large body of non-binding standards and guidance. The principal treaties may be grouped by pillar.

Safety. The *Convention on Nuclear Safety* (CNS, 1994; in force 1996) applies to land-based civil nuclear power plants and requires parties to establish a legislative and regulatory framework, an independent regulatory body, and safety requirements for siting, design, construction, and operation. The *Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management* (1997; in force 2001) applies analogous obligations to spent fuel and radioactive waste. Both are "incentive" conventions, relying on national reporting and peer review at review meetings of the parties rather than on enforcement.

Emergency preparedness. The *Convention on Early Notification of a Nuclear Accident* (1986) and the *Convention on Assistance in the Case of a Nuclear Accident*

or Radiological Emergency (1986), both adopted within months of the Chernobyl accident, require states to notify potentially affected states and the IAEA of nuclear accidents with transboundary effects and establish a framework for international assistance.

Security. The *Convention on the Physical Protection of Nuclear Material* (CPPNM, 1980; in force 1987), as amended in 2005 (the amendment in force since 2016), and the *International Convention for the Suppression of Acts of Nuclear Terrorism* (ICSANT, 2005; in force 2007), examined in section 11.5.

Safeguards and non-proliferation. The NPT, safeguards agreements concluded between states and the IAEA, regional treaties establishing nuclear-weapon-free zones (including the African Nuclear-Weapon-Free Zone Treaty, or Treaty of Pelindaba, signed in 1996 and in force since 2009), and the Treaty on the Prohibition of Nuclear Weapons (2017; in force 2021), examined in section 11.4.

Liability. The *Paris Convention on Third Party Liability in the Field of Nuclear Energy* (1960), the *Brussels Supplementary Convention* (1963), the *Vienna Convention on Civil Liability for Nuclear Damage* (1963), the *Joint Protocol* (1988), the protocols amending the Vienna (1997) and Paris and Brussels (2004) Conventions, and the *Convention on Supplementary Compensation for Nuclear Damage* (1997; in force 2015), examined in section 11.6.

IAEA Safety Standards as Soft Law

The most detailed and practically important rules of international nuclear safety law are found not in treaties but in the **IAEA Safety Standards Series**, adopted under Article III(A)(6) of the IAEA Statute. The standards are arranged in a hierarchy: the *Fundamental Safety Principles* (Safety Fundamentals, SF-1, 2006), which set out the fundamental safety objective — "to protect people and the environment from harmful effects of ionizing radiation" — and ten safety principles; *Safety Requirements*, which establish the requirements that must be met to ensure safety, expressed as "shall" statements; and *Safety Guides*, which provide recommendations on how to comply with the requirements. The ten principles of SF-1 address responsibility for safety (which rests with the licensee), the role of government (including an independent regulatory body), leadership and management for safety, justification of facilities and activities, optimisation of protection, limitation of risks to individuals, protection of present and future generations, prevention of accidents, emergency preparedness and response, and protective actions to reduce existing or unregulated radiation risks.

The Safety Standards are not binding on states as such. However, they are binding on the IAEA in relation to its own operations, and on states in relation to operations assisted by the IAEA under project and supply agreements. They acquire legal effect through incorporation into national law, which is widespread,

and through reference in treaties: the CNS and the Joint Convention refer to internationally endorsed safety principles, and the review process under those conventions uses the standards as benchmarks. In the European Union, the Euratom Nuclear Safety Directive (Council Directive 2009/71/Euratom, as amended by Council Directive 2014/87/Euratom) gives binding effect to principles derived from the standards. The IAEA standards are therefore a leading example of the transformation of soft law into hard obligations through domestic and regional incorporation, a theme examined in Chapter 2.

Euratom

The *Treaty establishing the European Atomic Energy Community* (Euratom Treaty), signed in Rome in 1957 alongside the Treaty establishing the European Economic Community, established a distinct regional legal order for nuclear energy among the member states of the European Union. The Euratom Treaty provides for the promotion of research, the establishment of uniform safety standards for the protection of workers and the public (Articles 30–39), the supply of nuclear materials through the Euratom Supply Agency, which has a right of option on ores, source materials, and special fissile materials produced in member states and an exclusive right to conclude supply contracts (Article 52), a regional safeguards system administered by the Commission (Chapter VII), and the ownership by the Community of special fissile materials (Article 86). On the basis of Articles 31 and 32 of the Treaty, the Council has adopted the Basic Safety Standards Directive (Council Directive 2013/59/Euratom), the Nuclear Safety Directive, and the Radioactive Waste and Spent Fuel Directive (Council Directive 2011/70/Euratom). In *Commission v Council* (Case C-29/99, 2002), the Court of Justice held that the Community had competence in the field of nuclear installation safety under the Euratom Treaty, opening the way for the adoption of the Nuclear Safety Directive.

The Euratom Treaty also requires member states to notify the Commission of investment projects in the nuclear field (Article 41) and to provide general data on plans for the disposal of radioactive waste (Article 37). In *Austria v Commission* (Case C-594/18 P, 2020), the Court of Justice dismissed Austria's appeal against the General Court's judgment upholding the Commission's approval, under the State aid rules, of United Kingdom support for the Hinkley Point C nuclear power plant, holding that the promotion of nuclear energy was a legitimate objective of common interest under the Euratom Treaty and that State aid for nuclear power need not pursue an objective of common interest beyond that. In September 2025, the General Court dismissed Austria's action challenging the inclusion of nuclear energy and gas in the EU taxonomy of sustainable activities by Commission Delegated Regulation (EU) 2022/1214.

The Nigerian Framework

Nigeria's nuclear legal framework is based principally on the **Nigeria Atomic Energy Commission Act 1976**, which established the Nigeria Atomic Energy Commission (NAEC) as the body responsible for the promotion and development of atomic energy, including nuclear power, and the **Nuclear Safety and Radiation Protection Act 1995**, which established the **Nigerian Nuclear Regulatory Authority** (NNRA) as the regulatory body responsible for nuclear safety, radiation protection, and the regulation of nuclear and radiation activities. The NNRA has adopted regulations on matters including radiation safety, the security of radioactive sources, and the licensing of nuclear installations. Nigeria operates a research reactor, the Nigeria Research Reactor-1 (NIRR-1), a miniature neutron source reactor at the Centre for Energy Research and Training of Ahmadu Bello University, Zaria, which was converted from highly enriched to low enriched uranium fuel in 2018 with the assistance of the IAEA, China, and the United States — an achievement in nuclear security that removed weapons-usable material from Nigeria. Nigeria has concluded intergovernmental agreements with the Russian Federation on cooperation in the construction of nuclear power plants, and has hosted IAEA missions to review its nuclear infrastructure. Nigeria is a party to the NPT, the Treaty of Pelindaba, and several of the IAEA conventions. The separation of the promotional functions of NAEC from the regulatory functions of the NNRA reflects the requirement of regulatory independence examined in section 11.3.

11.3 Nuclear Safety and Regulatory Independence

The Responsibility for Safety

The first principle of nuclear safety law is that the **prime responsibility for safety rests with the person or organisation responsible for facilities and activities that give rise to radiation risks** — that is, the licensee or operator. This principle, set out in Principle 1 of the IAEA Safety Fundamentals and in Article 9 of the Convention on Nuclear Safety, means that the regulator does not assume responsibility for safety by licensing and inspecting a facility; the operator remains responsible at all times, and must establish a management system that gives safety overriding priority. The principle is reinforced by the concept of **safety culture**, defined by the International Nuclear Safety Advisory Group after Chernobyl as the assembly of characteristics and attitudes in organisations and individuals which establishes that, as an overriding priority, nuclear plant safety issues receive the attention warranted by their significance.

The Requirements of the Convention on Nuclear Safety

The CNS imposes obligations on contracting parties concerning the legislative and regulatory framework (Article 7), the regulatory body (Article 8), responsi-

bility of the licence holder (Article 9), priority to safety (Article 10), financial and human resources (Article 11), human factors (Article 12), quality assurance (Article 13), assessment and verification of safety (Article 14), radiation protection (Article 15), emergency preparedness (Article 16), siting (Article 17), design and construction (Article 18), and operation (Article 19). The legislative and regulatory framework must provide for national safety requirements, a system of licensing, a system of regulatory inspection and assessment, and the enforcement of applicable regulations and licence terms, including suspension, modification, or revocation.

Article 18 requires that the design and construction of a nuclear installation provide for several reliable levels and methods of protection — the concept of **defence in depth** — against the release of radioactive materials, with a view to preventing the occurrence of accidents and mitigating their radiological consequences. Defence in depth is the foundational concept of nuclear safety engineering: it requires multiple independent and redundant layers of protection, so that the failure of one layer is compensated by another, ranging from conservative design and high-quality operation, through control of abnormal operation and engineered safety features, to the control of severe accidents and off-site emergency response.

The CNS is implemented through a peer review process. Each contracting party submits a national report on the measures taken to implement its obligations, which is reviewed by the other parties at review meetings held every three years. Following the Fukushima accident, the parties adopted the **Vienna Declaration on Nuclear Safety** (2015), which set out principles for the implementation of the CNS's objective of preventing accidents and mitigating radiological consequences: new nuclear power plants are to be designed, sited, and constructed with the objective of preventing accidents and, should an accident occur, mitigating possible releases of radionuclides causing long-term off-site contamination and avoiding early radioactive releases or releases large enough to require long-term protective measures and actions; comprehensive and systematic safety assessments are to be carried out periodically for existing installations, and reasonably practicable safety improvements implemented in a timely manner; and national requirements are to take into account the relevant IAEA safety standards. The Declaration was the compromise outcome of a Swiss proposal to amend the CNS, which did not attract sufficient support.

Regulatory Independence

The independence of the regulatory body is the cornerstone of nuclear safety governance. Article 8(2) of the CNS requires each contracting party to take the appropriate steps to ensure "an effective separation between the functions of the regulatory body and those of any other body or organization concerned with

the promotion or utilization of nuclear energy". Principle 2 of the IAEA Safety Fundamentals requires an "effectively independent" regulatory body, and the IAEA's General Safety Requirements Part 1, *Governmental, Legal and Regulatory Framework for Safety*, elaborates the elements of independence: the regulatory body must be functionally separate from entities having responsibilities or interests that could unduly influence its decision-making; it must have adequate legal authority, competence, and financial and human resources; and it must be able to make regulatory judgments and decisions free from undue influence, including political influence and pressure from licensees.

In the European Union, Article 5 of the Euratom Nuclear Safety Directive, as amended in 2014, requires member states to ensure that the competent regulatory authority is functionally separate from any other body or organisation concerned with the promotion or utilisation of nuclear energy, and that it does not seek or take instructions from any such body or organisation when carrying out its regulatory tasks; that it makes regulatory decisions founded on robust and transparent safety-related requirements; that it is given dedicated and appropriate budget allocations; and that it employs an appropriate number of staff with the necessary qualifications.

Independence is a matter of substance, not merely form. Regulatory independence may be compromised by structural arrangements (such as the location of the regulator within a ministry responsible for promoting nuclear energy), by financial dependence, by the movement of personnel between regulator and industry, and by a culture of deference. The **IAEA Integrated Regulatory Review Service** (IRRS) provides peer reviews of national regulatory infrastructure against IAEA standards, and IRRS missions frequently comment on the independence of regulators.

::: casebox {title="The Fukushima Daiichi Accident (2011) and Regulatory Capture"} **Facts.** On 11 March 2011, the Great East Japan Earthquake and the tsunami that followed it struck the Fukushima Daiichi Nuclear Power Station, operated by the Tokyo Electric Power Company (TEPCO). The tsunami, which exceeded the height of the plant's sea wall, flooded the plant and disabled the emergency diesel generators and electrical systems. The resulting loss of cooling led to core meltdowns in three reactors, hydrogen explosions, and the release of large quantities of radioactive material. More than one hundred thousand people were evacuated from the surrounding area. The accident was rated at Level 7, the highest level, on the International Nuclear and Radiological Event Scale — the only accident other than Chernobyl to receive that rating.

Findings. The Fukushima Nuclear Accident Independent Investigation Commission established by the National Diet of Japan concluded in its 2012 report that the accident was "a profoundly manmade disaster — that could

and should have been foreseen and prevented". It found that TEPCO and the regulators had been aware of the risk of a tsunami exceeding the design basis but had failed to take action, and that the relationship between the regulators and TEPCO amounted to "regulatory capture": the regulators had lacked independence from the industry and from the ministry that promoted nuclear power, and had deferred to the operator's expertise and interests. The Nuclear and Industrial Safety Agency, the principal regulator, had been located within the Ministry of Economy, Trade and Industry, which was responsible for promoting nuclear energy.

Legal reforms. Japan abolished the Nuclear and Industrial Safety Agency and in 2012 established the **Nuclear Regulation Authority** (NRA) as an independent commission attached to the Ministry of the Environment, adopted new regulatory requirements incorporating severe accident measures and stricter tsunami and seismic standards, and required all reactors to undergo review against the new standards before restarting. Internationally, the accident led to the IAEA Action Plan on Nuclear Safety (2011), "stress tests" of nuclear power plants in the European Union, the 2014 amendment of the Euratom Nuclear Safety Directive, and the Vienna Declaration on Nuclear Safety.

Litigation. The accident generated extensive litigation in Japan. In June 2022, the Supreme Court of Japan held in cases brought by evacuees that the State was not liable for damages, because the tsunami was far larger than anticipated and the accident could not have been prevented even if the State had exercised its regulatory powers to require TEPCO to take countermeasures. In July 2022, the Tokyo District Court, in a shareholders' derivative action, ordered four former TEPCO executives to pay the company around 13 trillion yen in damages for failing to take measures against the foreseeable risk of a tsunami, a judgment subject to appeal. Compensation for victims under the Act on Compensation for Nuclear Damage is examined in section 11.6.

Significance. Fukushima demonstrated that regulatory independence is a substantive safety requirement and not a formality, and that the assumptions underlying the design basis of nuclear installations must be continuously reassessed in the light of new knowledge about external hazards. :::

Licensing and Safety Assessment

National nuclear safety law establishes a system of licensing that governs each stage in the life of a nuclear installation: site evaluation, design, construction, commissioning, operation, and decommissioning. Licensing decisions are based on the regulator's review of the applicant's safety case — a comprehensive set of documents and analyses demonstrating that the installation will meet safety requirements, including a deterministic safety analysis of design basis accidents

and a probabilistic safety assessment estimating the frequency of core damage and large releases. Many legal systems also require **periodic safety reviews**, typically every ten years, in which the operator reassesses the safety of the installation against current standards and identifies reasonably practicable improvements.

Courts generally defer to the technical judgments of nuclear regulators, while scrutinising the legality of their procedures. In *Vermont Yankee Nuclear Power Corp v Natural Resources Defense Council* 435 US 519 (1978), the United States Supreme Court held that courts may not impose procedural requirements on agencies, including the Nuclear Regulatory Commission, beyond those required by the Administrative Procedure Act and other statutes. In *Baltimore Gas & Electric Co v Natural Resources Defense Council* 462 US 87 (1983), it upheld the Commission's assumption, for the purposes of licensing decisions, that the permanent storage of nuclear waste would have no significant environmental impact, emphasising that a reviewing court must be at its most deferential when an agency makes predictions within its area of special expertise at the frontiers of science. In India, the Supreme Court in *G Sundarrajan v Union of India* (2013) declined to halt the commissioning of the Kudankulam nuclear power plant, holding that the safety of the plant was a matter for the expert regulator, but issued directions on safety, the disposal of spent fuel, and the welfare of the local population. In Japan, district courts have on several occasions issued injunctions suspending the operation of reactors on safety grounds, which have generally been overturned on appeal, illustrating both the potential and the limits of judicial supervision of nuclear safety.

The law of nuclear safety also intersects with the law of nuisance and transboundary harm. In *Land Oberösterreich v ČEZ as* (Case C-115/08, 2009), the Court of Justice held that Austria could not refuse to take into account, in proceedings brought by the Austrian province of Upper Austria seeking an injunction against nuisance from the Temelín nuclear power plant in the Czech Republic, the official authorisation granted to the plant in another member state, because to do so would discriminate on grounds of nationality contrary to the Euratom Treaty.

Nuclear Safety in Armed Conflict: Zaporizhzhia

The Russian occupation of the Zaporizhzhia Nuclear Power Plant in Ukraine — the largest nuclear power plant in Europe — in March 2022 confronted nuclear safety law with an unprecedented situation: a nuclear power plant operated under military occupation in an active war zone, subject to shelling, repeated losses of off-site power, and the detention and intimidation of its staff. The IAEA Director General articulated "seven indispensable pillars" for ensuring nuclear safety and security during armed conflict, derived from IAEA safety standards and security guidance, including physical integrity, the functioning of safety and

security systems, the ability of staff to fulfil their duties without undue pressure, secure off-site power supply, uninterrupted supply chains, effective radiation monitoring and emergency arrangements, and reliable communication with the regulator. In May 2023, he set out to the United Nations Security Council five concrete principles for the protection of the plant, including that there should be no attack from or against the plant and that it should not be used as a base for heavy weapons. The IAEA established a continuous presence of its experts at the plant from September 2022. Under international humanitarian law, Article 56 of Additional Protocol I to the Geneva Conventions (1977) provides special protection for works and installations containing dangerous forces, including nuclear electrical generating stations, which are not to be made the object of attack, even where they are military objectives, if such attack may cause the release of dangerous forces and consequent severe losses among the civilian population, subject to limited exceptions. The protection of energy infrastructure in armed conflict is examined further in Chapter 25.

11.4 Safeguards and Non-Proliferation Obligations

The Concept of Safeguards

IAEA safeguards are a system of technical measures through which the IAEA verifies that states are honouring their international legal obligations to use nuclear material and technology only for peaceful purposes. Safeguards comprise nuclear material accountancy (the maintenance of records and reports of nuclear material quantities and movements), containment and surveillance (seals and cameras), and inspections and other verification activities at facilities and other locations. The IAEA's safeguards conclusions, reported annually in the Safeguards Implementation Report, state whether the IAEA has found any indication of the diversion of declared nuclear material and, for states with additional protocols, whether it has found any indication of undeclared nuclear material or activities.

Types of Safeguards Agreements

Safeguards are implemented under agreements concluded between the IAEA and states. There are three principal types.

Comprehensive safeguards agreements (CSAs), concluded by non-nuclear-weapon states party to the NPT in accordance with Article III, are based on the IAEA document *The Structure and Content of Agreements between the Agency and States Required in Connection with the Treaty on the Non-Proliferation of Nuclear Weapons* (INFCIRC/153 (Corrected), 1972). Under a CSA, the state undertakes to accept safeguards on all source or special fissionable material in all peaceful nuclear activities within its territory, under its jurisdiction, or carried out under its control anywhere, for the exclusive purpose of verifying that such material is not diverted to nuclear weapons or other nuclear explosive devices.

The state must establish a State System of Accounting for and Control of nuclear material (SSAC), declare its nuclear material and facilities, and accept inspections. Many states with minimal or no nuclear material have concluded **small quantities protocols** suspending most of the detailed provisions of their CSAs; in 2005, the IAEA Board modified the standard text of the protocol to reinstate certain requirements, including the provision of design information on new facilities and the possibility of inspections.

Item-specific safeguards agreements, based on INFCIRC/66/Rev.2, apply to specified nuclear material, facilities, or equipment, and are concluded with states not party to the NPT — India, Pakistan, and Israel.

Voluntary offer agreements have been concluded by the five nuclear-weapon states, under which they offer some or all of their civil nuclear facilities for safeguards.

The Additional Protocol

The discovery after the 1991 Gulf War of an extensive clandestine nuclear weapons programme in Iraq — a party to the NPT with a CSA in force — revealed that safeguards focused on the verification of declared material were inadequate to detect undeclared activities. The IAEA responded with a programme of strengthened safeguards, culminating in the **Model Additional Protocol** (INFCIRC/540 (Corrected)), approved by the Board of Governors in 1997. Under an additional protocol, a state must provide an expanded declaration of its nuclear-related activities, including nuclear fuel cycle-related research and development, uranium mines and concentration plants, the manufacture of specified equipment, and the export of specified equipment and material; and must grant the IAEA "complementary access" to any location on the site of a facility, to other locations listed in the declaration, and to other locations specified by the IAEA to resolve questions or inconsistencies, on short notice. The additional protocol enables the IAEA to provide credible assurance of the absence of undeclared nuclear material and activities. Although a large majority of states with CSAs have brought additional protocols into force, the additional protocol is not legally required by the NPT, and some states have declined to conclude one. The NSG and some supplier states have debated whether the additional protocol should be a condition of supply.

Non-Compliance and the Security Council

Where the IAEA Board finds that a state is in non-compliance with its safeguards agreement, it may, under Article XII(C) of the Statute, report the non-compliance to the Security Council and the General Assembly. The Security Council may then take action under Chapter VII of the United Nations Charter, as it did in respect of Iraq, the Democratic People's Republic of Korea (DPRK), and Iran.

The DPRK announced its withdrawal from the NPT in 2003 under Article X, which permits withdrawal on three months' notice if extraordinary events related to the subject matter of the Treaty have jeopardised the supreme interests of the withdrawing state, and subsequently conducted nuclear weapons tests; the legal validity of the withdrawal remains contested, and the Security Council has imposed sanctions under Chapter VII.

Iran and the Joint Comprehensive Plan of Action

The case of Iran is the most extensively litigated and negotiated safeguards controversy. In 2003, the IAEA reported that Iran had failed to declare certain nuclear material and activities, including enrichment-related activities, in breach of its CSA. In 2005, the Board found Iran in non-compliance and in 2006 reported the matter to the Security Council, which adopted a series of resolutions under Chapter VII demanding that Iran suspend enrichment and imposing sanctions. In 2015, Iran concluded the **Joint Comprehensive Plan of Action** (JCPOA) with China, France, Germany, Russia, the United Kingdom, the United States, and the European Union, under which Iran accepted limits on its enrichment activities and stockpile, provisional application of its additional protocol, and enhanced monitoring, in exchange for the lifting of nuclear-related sanctions. The Security Council endorsed the JCPOA in Resolution 2231 (2015), which terminated the earlier resolutions while providing a "snapback" mechanism by which a JCPOA participant could trigger the reimposition of the previous sanctions.

The United States withdrew from the JCPOA in 2018 and reimposed its own sanctions, and Iran subsequently exceeded the JCPOA's limits, enriched uranium to sixty per cent, and reduced its cooperation with the IAEA. In June 2025, the IAEA Board found that Iran was in non-compliance with its safeguards obligations, and Israel and the United States carried out military strikes on Iranian nuclear facilities, raising grave questions under the law on the use of force; Iran subsequently suspended cooperation with the IAEA. In August 2025, France, Germany, and the United Kingdom invoked the snapback mechanism, and the previous United Nations sanctions were reimposed at the end of September 2025, although Russia and China contested the legality of the snapback. The Iranian case illustrates the dependence of the safeguards system on political support and the limits of verification without cooperation, and it is a case study in the interaction between technical verification, treaty interpretation, and the use of force.

Nuclear-Weapon-Free Zones and the Prohibition Treaty

Regional nuclear-weapon-free zones complement the NPT. The **Treaty of Pelindaba**, which established the African Nuclear-Weapon-Free Zone, prohibits the research, development, manufacture, stockpiling, acquisition, possession, and testing of nuclear explosive devices in the zone and the dumping of radioactive

waste; requires parties to conclude CSAs with the IAEA; requires the maintenance of the highest standards of physical protection; and prohibits armed attack on nuclear installations. It established the **African Commission on Nuclear Energy** (AFCONE), headquartered in South Africa, to ensure compliance and promote the peaceful uses of nuclear energy in Africa. Other zones cover Latin America and the Caribbean (Treaty of Tlatelolco, 1967), the South Pacific (Treaty of Rarotonga, 1985), South-East Asia (Treaty of Bangkok, 1995), and Central Asia (Treaty of Semipalatinsk, 2006).

The **Treaty on the Prohibition of Nuclear Weapons** (TPNW), adopted in 2017 and in force since January 2021, prohibits states parties from developing, testing, producing, acquiring, possessing, stockpiling, using, or threatening to use nuclear weapons. None of the nuclear-armed states is a party, and the relationship between the TPNW and the NPT is contested. Article 3 of the TPNW requires states parties to maintain, at a minimum, their existing safeguards obligations, and Article 4 provides for safeguards for states that eliminate their nuclear weapons. The legality of the threat or use of nuclear weapons under general international law was addressed by the International Court of Justice in its Advisory Opinion on the *Legality of the Threat or Use of Nuclear Weapons* (1996), in which it held, among other things, that there exists an obligation to pursue in good faith and bring to a conclusion negotiations leading to nuclear disarmament in all its aspects under strict and effective international control.

11.5 Nuclear Security and Physical Protection

The Concept of Nuclear Security

Nuclear security is defined in IAEA guidance as the prevention and detection of, and response to, criminal or intentional unauthorised acts involving or directed at nuclear material, other radioactive material, associated facilities, or associated activities. Its concerns include the theft of nuclear material for the purpose of constructing a nuclear explosive device; the theft of radioactive sources for use in a radiological dispersal device ("dirty bomb"); the sabotage of nuclear facilities or transports with a view to causing radiological release; and cyberattacks on the digital systems of nuclear facilities. Unlike safety, which addresses accidental events, security addresses deliberate acts by adversaries, and the two are complementary: measures designed for safety and for security must be coordinated so that each supports rather than undermines the other — for example, safety requires easy egress in an emergency, while security requires controlled access.

Nuclear security rose to the top of the international agenda after the terrorist attacks of 11 September 2001, which raised concerns about the potential for terrorists to acquire nuclear or radiological material or to attack nuclear facilities. The **Nuclear Security Summits** convened between 2010 and 2016 produced na-

tional commitments to minimise the use of highly enriched uranium, strengthen physical protection, and ratify the relevant conventions, and contributed to the entry into force of the 2005 Amendment to the CPPNM.

The Convention on the Physical Protection of Nuclear Material

The **Convention on the Physical Protection of Nuclear Material** (CPPNM), adopted in 1980 and in force since 1987, originally applied only to nuclear material used for peaceful purposes in international transport. It required parties to ensure that nuclear material in international transport within their territory, or on board ships or aircraft under their jurisdiction, was protected at specified levels; prohibited the export or import of nuclear material unless assurances of such protection had been received; and required parties to criminalise specified offences, including theft, embezzlement, and the threat to use nuclear material to cause death or serious injury or substantial property damage, and to establish jurisdiction over and either prosecute or extradite alleged offenders.

The **Amendment to the CPPNM**, adopted in 2005 and in force since 8 May 2016, substantially expanded the scope of the Convention. The amended Convention, sometimes called the Convention on the Physical Protection of Nuclear Material and Nuclear Facilities, extends to nuclear material in domestic use, storage, and transport, and to the protection of nuclear facilities against sabotage. It requires each party to establish, implement, and maintain an appropriate physical protection regime, consisting of a legislative and regulatory framework, a competent authority, and the responsibility of licence holders, and sets out twelve "fundamental principles" of physical protection, including the responsibility of the state, the responsibility of licence holders, legislative and regulatory framework, competent authority, security culture, threat evaluation (the "design basis threat"), a graded approach, defence in depth, quality assurance, contingency plans, and confidentiality. It adds new offences, including sabotage of a nuclear facility and the smuggling of nuclear material, and requires parties to cooperate in locating and recovering stolen or smuggled material.

The International Convention for the Suppression of Acts of Nuclear Terrorism

The **International Convention for the Suppression of Acts of Nuclear Terrorism** (ICSANT), adopted by the United Nations General Assembly in 2005 and in force since 2007, is one of the sectoral counter-terrorism conventions negotiated under United Nations auspices. It requires parties to criminalise the possession of radioactive material or the making or possession of a device with the intent to cause death or serious bodily injury, substantial damage to property or the environment, or to compel a natural or legal person, an international organisation, or a state to do or refrain from doing an act; the use of radioactive material or

a device, or the use of or damage to a nuclear facility in a manner which releases or risks the release of radioactive material, with such intent; and threats and demands relating to such acts. It applies the "prosecute or extradite" principle and provides for international cooperation. ICSANT applies to radioactive material in general, not only nuclear material, and to acts by any person, but excludes the activities of armed forces during an armed conflict, which are governed by international humanitarian law.

United Nations Security Council Resolution 1540

Security Council Resolution 1540 (2004), adopted under Chapter VII of the United Nations Charter, decided that all states shall refrain from providing any form of support to non-state actors that attempt to develop, acquire, manufacture, possess, transport, transfer, or use nuclear, chemical, or biological weapons and their means of delivery, and shall adopt and enforce appropriate effective laws prohibiting such activities, and appropriate effective measures to account for and secure such items, develop effective border controls, and establish export and trans-shipment controls. Resolution 1540 is notable as an instance of the Security Council imposing general legislative obligations on all states, rather than responding to a specific situation, and it has been described as "legislative" in character. Its implementation is monitored by a committee of the Security Council.

The Security of Radioactive Sources

Radioactive sources used in medicine, industry, and research present significant security risks because they are widely distributed, often poorly controlled, and potentially usable in radiological dispersal devices. Incidents involving lost or abandoned sources — such as the incident at Goiânia, Brazil, in 1987, in which an abandoned radiotherapy source was opened by scrap metal scavengers, causing several deaths and widespread contamination — illustrate the dangers. The IAEA's **Code of Conduct on the Safety and Security of Radioactive Sources** (2003), together with its supplementary Guidance on the Import and Export of Radioactive Sources and Guidance on the Management of Disused Radioactive Sources, provides non-binding guidance that a large number of states have committed politically to implement, including through national registers of sources, licensing, and the safe and secure management of disused sources.

In Nigeria, the NNRA regulates the security of radioactive sources, which are widely used in the oil and gas industry for well logging and in industrial radiography, as well as in medicine. The theft of radioactive sources from oil and gas operations has occurred in Nigeria and elsewhere, illustrating the intersection of nuclear security with the regulation of the extractive industries examined in Chapter 5.

Nuclear Security Law in Practice

The IAEA's **Nuclear Security Series** provides guidance on the implementation of nuclear security, including the *Nuclear Security Fundamentals* (2013) and *Nuclear Security Recommendations* on physical protection, on radioactive material and associated facilities, and on material out of regulatory control. Unlike the Safety Standards, nuclear security guidance is formally treated as guidance rather than standards, reflecting the sensitivity of security matters as an aspect of national sovereignty. The IAEA's **International Physical Protection Advisory Service** (IPPAS) offers peer reviews of national physical protection regimes. The confidentiality of security information means that nuclear security law is less transparent than safety law, which complicates public scrutiny and accountability. Cybersecurity has become a growing dimension of nuclear security, as nuclear facilities increasingly rely on digital instrumentation and control systems, and is examined in the context of energy infrastructure more generally in Chapter 12.

11.6 Nuclear Liability and Compensation

The Rationale for a Special Liability Regime

The potential for nuclear accidents to cause catastrophic and transboundary damage, the difficulty of proving causation for radiation-induced illnesses that may manifest decades later, and the reluctance of suppliers and insurers to participate in the nuclear industry without protection from unlimited liability led, from the late 1950s, to the development of special liability regimes for nuclear damage. These regimes were designed to serve two objectives: to ensure adequate compensation for victims of nuclear incidents, and to enable the development of the nuclear industry by limiting and channelling liability so that it could be insured.

The Principles of Nuclear Liability Law

The international nuclear liability conventions are based on a set of common principles.

Exclusive liability of the operator (channelling). Liability for nuclear damage is channelled exclusively to the operator of the nuclear installation in which the nuclear incident occurs, or from which the nuclear material involved originated. No other person — including suppliers of equipment, designers, and constructors — is liable, and the operator's right of recourse against other persons is limited to cases where recourse is expressly provided for by contract or where the damage results from an act or omission done with intent to cause damage. Channelling simplifies claims for victims, who need only identify the operator, and protects suppliers, avoiding the need for every participant in the nuclear industry to insure against the same risk.

Strict liability. The operator's liability is strict (absolute), in the sense that it does not depend on proof of fault or negligence. Exonerations are limited — principally to damage caused by acts of armed conflict, hostilities, civil war, or insurrection, and, under the older conventions, grave natural disasters of an exceptional character (an exoneration removed by the revised conventions).

Limitation of liability in amount. The operator's liability may be limited in amount, in exchange for strict and exclusive liability, with the possibility of state or collective funding beyond the operator's liability.

Limitation in time. Claims must be brought within specified time limits, which the revised conventions have extended to thirty years from the incident for claims in respect of loss of life and personal injury, recognising the latency of radiation-induced illness.

Compulsory financial security. The operator must maintain insurance or other financial security covering its liability.

Exclusive jurisdiction. Jurisdiction over claims lies exclusively with the courts of the state in whose territory the nuclear incident occurred, ensuring a single forum for the equitable distribution of compensation among all victims.

Non-discrimination. Compensation must be distributed without discrimination based on nationality, domicile, or residence.

The Paris and Vienna Systems

The **Paris Convention on Third Party Liability in the Field of Nuclear Energy** (1960), adopted under the auspices of the OECD Nuclear Energy Agency, applies principally among Western European states. It was supplemented by the **Brussels Supplementary Convention** (1963), which provides additional tiers of compensation from public funds of the installation state and from collective contributions of the parties. The **Protocols of 2004** amending the Paris and Brussels Conventions, which entered into force on 1 January 2022, raised the minimum liability of the operator to EUR 700 million, increased total compensation available under the Brussels tiers to EUR 1.5 billion, broadened the definition of nuclear damage to include economic loss, the costs of measures of reinstatement of impaired environment, loss of income deriving from an economic interest in the use or enjoyment of the environment, and the costs of preventive measures, and extended the time limit for personal injury claims to thirty years.

The **Vienna Convention on Civil Liability for Nuclear Damage** (1963), adopted under the auspices of the IAEA, was intended to be of worldwide application and applies principally among states in Central and Eastern Europe, Latin America, and elsewhere. The **1997 Protocol** amending the Vienna Convention, in force since 2003, raised the minimum liability amount to not less than 300 million Special Drawing Rights, broadened the definition of nuclear damage in

similar terms to the 2004 Paris Protocol, and extended the geographical scope of the Convention to damage suffered wherever, subject to exceptions.

The **Joint Protocol relating to the Application of the Vienna Convention and the Paris Convention** (1988), in force since 1992 and adopted in response to the Chernobyl accident, links the two systems, so that victims in a state party to one convention may claim against an operator in a state party to the other.

The Convention on Supplementary Compensation

The **Convention on Supplementary Compensation for Nuclear Damage** (CSC), adopted in 1997 and in force since 2015, was intended to establish a global liability regime capable of attracting states that are not party to either the Paris or the Vienna Convention, including the United States. It may be joined by parties to the Paris or Vienna Conventions, and by states whose national law complies with the provisions of its Annex, a "grandfather clause" designed to accommodate the economic channelling of the United States Price-Anderson Act. It requires installation states to ensure the availability of at least 300 million SDRs and provides for an international fund of contributions from all parties, calculated principally by reference to installed nuclear capacity. The CSC's parties include the United States, Japan, India, Canada, and the United Arab Emirates, as well as several other states. Japan's accession in 2015 was motivated in part by the desire to facilitate the participation of international companies in the decommissioning of Fukushima Daiichi by providing them with channelling protection.

National Liability Regimes

The United States pioneered nuclear liability legislation with the **Price-Anderson Act** of 1957, which requires operators of nuclear power plants to maintain the maximum available private insurance and to participate in a retrospective premium scheme under which each operator contributes to the compensation of damage caused by an incident at any plant, creating a pool of more than USD 15 billion. Price-Anderson employs "economic channelling" — liability is not legally channelled to the operator, but the insurance covers any person liable — and provides for the consolidation of claims in federal courts. In *Duke Power Co v Carolina Environmental Study Group* 438 US 59 (1978), the Supreme Court upheld the constitutionality of the Act's limitation of liability, holding that it was rationally related to the legitimate objective of encouraging private participation in nuclear energy and that it provided a reasonably just substitute for the common law remedies it replaced.

Japan's **Act on Compensation for Nuclear Damage** (1961) imposes unlimited, strict, and exclusive liability on operators, with government support. Following Fukushima, TEPCO's compensation liabilities far exceeded its capac-

ity, and the **Nuclear Damage Compensation Facilitation Corporation** (later the Nuclear Damage Compensation and Decommissioning Facilitation Corporation) was established in 2011 to provide financial assistance to TEPCO, funded by government bonds and contributions from the nuclear industry. Compensation was paid through direct claims to TEPCO, through the government's Dispute Reconciliation Committee for Nuclear Damage Compensation, which issued guidelines on the categories of compensable damage and conducted alternative dispute resolution, and through litigation. The total compensation paid by TEPCO exceeded ten trillion yen.

India's **Civil Liability for Nuclear Damage Act 2010** departed from the principle of exclusive channelling by granting the operator, in section 17(b), a right of recourse against the supplier where the nuclear incident resulted from the supply of equipment or material with patent or latent defects or substandard services. This provision, a response to the Bhopal gas disaster of 1984 and the inadequacy of the compensation it produced, deterred foreign suppliers from participating in the Indian nuclear market. India ratified the CSC in 2016, and the government issued explanations of the provision's scope, but the compatibility of section 17(b) with the CSC and its implications for suppliers remained controversial.

Adequacy of the Liability Regimes

The international nuclear liability regime has been criticised on several grounds. The liability limits, even as revised, are modest in comparison with the costs of major accidents: the damage caused by Chernobyl and Fukushima has been estimated in the hundreds of billions of US dollars. Participation in the conventions is fragmented, and many states with nuclear power programmes, including China and some newcomer states, are party to none of them; the absence of a treaty relationship means that victims in one state may be unable to claim against an operator in another. Channelling has been criticised for removing the incentive for suppliers to ensure the safety of their products. And the conventions' exclusive jurisdiction provisions may leave victims in neighbouring states dependent on the courts of the installation state. For newcomer states, including those in Africa, adherence to a liability convention — commonly the revised Vienna Convention or the CSC — is a practical prerequisite for attracting foreign suppliers and financing, and the IAEA's International Expert Group on Nuclear Liability (INLEX) has recommended that states establish treaty relations among themselves through adherence to the revised conventions and the CSC.

11.7 Radioactive Waste and Decommissioning

The Challenge of Radioactive Waste

Radioactive waste is generated at every stage of the nuclear fuel cycle and by the use of radioactive materials in medicine, industry, and research. It is classified according to its radioactivity and the duration of its hazard: very low level and low level waste, which may be disposed of in near-surface facilities; intermediate level waste, which requires greater containment; and high level waste, principally spent nuclear fuel (where it is treated as waste rather than reprocessed) and the residues of reprocessing, which generates heat and remains hazardous for many thousands of years. The management of high level waste is the most intractable problem of nuclear energy, both technically and politically, and the long timescales involved raise distinctive questions of intergenerational equity.

The International Framework

The **Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management** (1997) is the only legally binding international instrument addressing the safety of spent fuel and radioactive waste management on a global scale. It applies to spent fuel and radioactive waste resulting from civilian applications, and requires contracting parties to ensure that at all stages of management individuals, society, and the environment are adequately protected against radiological hazards, including by taking into account interdependencies among the different steps of management, providing for effective protection of individuals, society, and the environment, taking into account biological, chemical, and other hazards, striving to avoid actions that impose reasonably predictable impacts on future generations greater than those permitted for the current generation, and aiming to avoid imposing undue burdens on future generations (Articles 4 and 11). The preamble affirms that radioactive waste should, as far as is compatible with the safety of the management of such material, be disposed of in the state in which it was generated, while recognising that in certain circumstances safe and efficient management might be fostered through agreements among parties to use facilities in one of them.

Article 27 of the Joint Convention regulates transboundary movements of spent fuel and radioactive waste, requiring prior notification to and consent of the state of destination, and prohibiting shipments to destinations south of latitude 60 degrees South for storage or disposal. The Bamako Convention on the Ban of the Import into Africa and the Control of Transboundary Movement and Management of Hazardous Wastes within Africa (1991) prohibits the import of hazardous wastes, including radioactive wastes, into Africa from non-parties, and the Treaty of Pelindaba prohibits the dumping of radioactive wastes in the African Nuclear-Weapon-Free Zone. The dumping of radioactive waste at sea is

prohibited by the 1996 Protocol to the London Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter (1972). The Joint Convention reflects the principle of national responsibility for radioactive waste, which is also expressed in Article 4 of the Euratom Radioactive Waste and Spent Fuel Directive (Council Directive 2011/70/Euratom), under which each member state has ultimate responsibility for management of the spent fuel and radioactive waste generated in it, and must establish a national policy and national programme. Article 4(4) of the Directive provides that radioactive waste shall be disposed of in the member state in which it was generated, unless an agreement has been concluded for the use of a disposal facility in another member state or a third country, subject to strict conditions.

Deep Geological Disposal

There is a broad scientific consensus that deep geological disposal — the emplacement of high level waste and spent fuel in engineered repositories several hundred metres underground in stable geological formations, relying on a combination of engineered and natural barriers to isolate the waste for the required period — is the most appropriate long-term solution for high level waste. Progress in implementing geological disposal has nevertheless been slow, owing principally to public opposition and the difficulty of securing the consent of host communities.

Finland is the first country to construct a deep geological repository for spent fuel. The Onkalo repository at Olkiluoto, developed by Posiva, received a construction licence in 2015 following a decision-in-principle ratified by Parliament in 2001, and commenced trial operations in 2024. The Finnish process has been regarded as a model because it combined a staged decision-making process, the right of the host municipality (Eurajoki) to veto the decision-in-principle, an independent and trusted regulator (the Radiation and Nuclear Safety Authority, STUK), and the long-term commitment of the waste producers. **Sweden** approved the construction of a repository at Forsmark in 2022, following a process in which the host municipality consented. **France** has pursued the Cigéo project at Bure under legislation adopted in 1991, 2006, and 2016, which introduced the concept of reversibility — the ability to retrieve waste for a period after emplacement — into the legal framework. The emphasis in these processes on voluntarism, consent-based siting, and staged decision-making reflects lessons learned from earlier failures.

The United States illustrates the consequences of a failed siting process. The **Nuclear Waste Policy Act** of 1982 established a federal responsibility for the disposal of spent fuel and required utilities to pay a fee into a Nuclear Waste Fund in exchange for a federal commitment to begin accepting spent fuel by 31 January 1998. The 1987 amendments to the Act designated Yucca Mountain in Nevada as the sole site to be characterised, over Nevada's opposition. The

Department of Energy applied for a construction licence in 2008, but the Obama administration sought to withdraw the application in 2010 and ceased funding. In *In re Aiken County* 725 F.3d 255 (DC Cir 2013), the Court of Appeals for the District of Columbia Circuit issued a writ of mandamus requiring the Nuclear Regulatory Commission to continue the licensing process with the funds available, holding that the Commission could not decline to comply with the statutory mandate on policy grounds. The project nevertheless remained stalled. The federal government's failure to accept spent fuel by 1998 has resulted in extensive litigation by utilities for breach of contract, in which the government has been held liable for billions of dollars in damages for the costs of on-site storage, and the fee was suspended in 2014 following a ruling of the D.C. Circuit. Efforts to license private interim storage facilities were addressed by the Supreme Court in *Nuclear Regulatory Commission v Texas* (2025), which held that the State of Texas and other challengers, which had not been parties to the licensing proceeding, were not entitled to seek judicial review of a licence granted for an interim storage facility.

Decommissioning

Decommissioning comprises the administrative and technical actions taken to allow the removal of some or all regulatory controls from a nuclear facility at the end of its life, including decontamination, dismantling, and the management of resulting waste. Strategies include **immediate dismantling**, in which decommissioning commences shortly after permanent shutdown, and **deferred dismantling** (safe enclosure), in which the facility is maintained in a safe condition for several decades to allow radioactive decay before dismantling. IAEA safety standards and the Euratom Directives increasingly favour immediate dismantling, on grounds including the availability of staff familiar with the facility and the avoidance of transferring burdens to future generations.

Financing. The legal regulation of decommissioning financing is critical, because decommissioning costs are incurred decades after the revenue from operation has been earned, and there is a risk that operators will become insolvent or that funds will be inadequate. The principal models are segregated external funds, in which operators make contributions during operation to an independently managed fund; internal provisions on the operator's balance sheet; and state funding. Article 9 of the Euratom Radioactive Waste Directive requires member states to ensure that the national framework requires that adequate financial resources be available for the implementation of national programmes, with due regard to the responsibility of waste generators. The United Kingdom's arrangements for new nuclear power stations, under the Energy Act 2008, require operators to have an approved Funded Decommissioning Programme before construction may begin.

The German Nuclear Phase-Out

Germany's phase-out of nuclear energy illustrates the legal consequences of the political decision to end nuclear power. In 2002, the government and the nuclear operators reached a consensus, implemented by legislation, under which each plant was allocated a residual quantity of electricity it could produce. In 2010, the government extended the operating lives of the plants, but following the Fukushima accident in 2011, the Thirteenth Amendment to the Atomic Energy Act revoked the extension and required all plants to close by the end of 2022. The operators challenged the amendment before the Federal Constitutional Court, and the Swedish state-owned company Vattenfall also brought a claim under the Energy Charter Treaty (*Vattenfall AB and others v Germany*, ICSID Case No ARB/12/12).

In its judgment of 6 December 2016, the Federal Constitutional Court held that the accelerated phase-out was largely compatible with the Basic Law: it did not constitute an expropriation, and it was, in principle, a permissible determination of the content and limits of property under Article 14 of the Basic Law, justified by the legislature's reassessment of the risks of nuclear energy after Fukushima. However, the Court held that the amendment was unconstitutional in so far as it failed to provide compensation for the residual electricity quantities allocated in 2002 that certain operators could not use, and for investments made in reliance on the 2010 extension. Germany enacted compensation legislation, and in 2021 reached a settlement with the operators, under which they received approximately EUR 2.4 billion in compensation, including around EUR 1.4 billion for Vattenfall, which discontinued its ECT claim. The last three German nuclear power plants closed in April 2023, after a brief extension due to the energy crisis.

Separately, the **Act on the Reorganisation of Responsibility in Nuclear Waste Management** (2017) transferred responsibility for the interim and final storage of radioactive waste from the operators to the state, in exchange for the payment by the operators of approximately EUR 24 billion into a public fund (KENFO), while the operators remained responsible for decommissioning and dismantling their plants and for packaging the waste. This arrangement illustrates a model for the allocation of long-term liability between private operators and the state.

11.8 Small Modular Reactors and Emerging Legal Issues

The Promise of Small Modular and Advanced Reactors

Small modular reactors (SMRs) are nuclear reactors with a power capacity of up to around 300 MW(e) per unit, designed to be manufactured in factories and transported to sites for installation, in contrast to large reactors constructed

on-site. Proponents claim that SMRs offer lower upfront capital costs, shorter construction times, enhanced safety through passive safety features and smaller radioactive inventories, siting flexibility (including at the sites of retiring coal plants and in remote locations), and suitability for smaller electricity grids, including those of many developing countries. **Advanced reactors** — including high-temperature gas-cooled reactors, molten salt reactors, sodium-cooled fast reactors, and microreactors of a few megawatts — employ non-light-water technologies with potential advantages in safety, efficiency, waste management, and industrial heat applications. Several SMR designs have progressed towards deployment: the BWRX-300 at Darlington in Ontario, Canada, received a licence to construct from the Canadian Nuclear Safety Commission in 2025; the United Kingdom selected a Rolls-Royce SMR design for deployment in 2025; and the floating nuclear power plant *Akademik Lomonosov*, comprising two small reactors mounted on a barge, has operated in the Russian Arctic since 2019–2020. Technology companies have concluded agreements to purchase electricity from SMRs and other advanced reactors to power data centres.

Licensing Challenges

Existing licensing frameworks were developed for large light-water reactors and may not be well suited to SMRs and advanced reactors, whose designs, safety features, and fuel cycles differ substantially. Regulators have responded with technology-inclusive and risk-informed approaches. In the United States, the **Nuclear Energy Innovation and Modernization Act** of 2019 required the Nuclear Regulatory Commission to develop a technology-inclusive regulatory framework for advanced reactors, and the **ADVANCE Act** of 2024 directed the Commission to modernise its mission and licensing processes, reduce fees for advanced reactor applicants, and expedite reviews. The NRC approved the design of NuScale's SMR in 2020 and an uprated version in 2025, the first SMR designs certified or approved in the United States. In the United Kingdom, the Generic Design Assessment process permits the assessment of reactor designs independently of specific sites.

Key licensing issues include the scope and size of **emergency planning zones** (which proponents argue can be reduced for SMRs, given their smaller inventories and passive safety, enabling siting closer to population centres and industrial users); **staffing requirements** and remote or autonomous operation; the treatment of multi-module plants; **security requirements** proportionate to the risk; and the qualification of **novel fuels**, such as high-assay low-enriched uranium (HALEU), which raises safeguards and security considerations.

International Harmonisation

The economic case for SMRs depends on serial factory production of standardised designs, which in turn depends on the ability to deploy the same design in multiple countries without extensive redesign to meet different national requirements. Licensing has traditionally been an exclusively national prerogative, and differences in national requirements are a significant barrier to SMR deployment. The IAEA launched the **Nuclear Harmonization and Standardization Initiative** (NHSI) in 2022 to promote regulatory collaboration and industrial standardisation for SMRs, and regulators have entered into cooperation arrangements, including agreements between the United States and Canadian regulators on joint reviews of advanced reactor designs. Proposals for mutual recognition of design assessments, or for an international design certification, raise questions of sovereignty and regulatory accountability: under the CNS and IAEA standards, responsibility for safety cannot be delegated by the host state, which must retain an independent and competent regulatory body capable of making its own judgments.

Transportable and Floating Reactors

Transportable reactors — including floating nuclear power plants and microreactors designed to be moved between sites — raise novel legal questions under international nuclear law, which was developed for stationary installations. These include the determination of the "installation state" for the purposes of safety, safeguards, and liability; the application of the liability conventions to reactors in transit or on the high seas or in the territorial sea of a foreign state; the application of the law of the sea to floating reactors (including innocent passage and the rights of coastal states); the allocation of safeguards responsibilities where a reactor is fuelled in one state and operated in another; and the security of reactors in transit. The IAEA has studied these questions, but the international legal framework remains incomplete.

Safeguards by Design and Proliferation Considerations

SMRs and advanced reactors raise safeguards considerations relating to the use of higher-enriched fuels, novel fuel forms (such as pebbles or molten salt fuels, which complicate item counting), long-life cores that are not refuelled for many years, and deployment in large numbers in many locations, including in states with limited nuclear infrastructure. The IAEA promotes "safeguards by design", the integration of safeguards considerations into the design of nuclear facilities from the earliest stages, and the participation of the IAEA in the design process.

Newcomer States and Nuclear Governance in Africa

SMRs are frequently marketed to newcomer states, including in Africa, as a means of introducing nuclear power at a scale compatible with smaller grids

and financing capacities. However, the IAEA Milestones Approach applies equally to SMRs: a newcomer state must establish a comprehensive legal framework covering safety, security, safeguards, and liability; an independent and competent regulator; human resources; and arrangements for waste management, decommissioning, and financing, regardless of the size of the reactor. The vendor-financed "build-own-operate" model, under which a foreign state-owned vendor finances, builds, owns, and operates a nuclear plant in the host state, as in Rosatom's Akkuyu project in Turkey, may reduce the burden on the host state but raises questions of long-term dependence, regulatory independence from a powerful foreign operator, and the allocation of liability for waste and decommissioning. Egypt's El Dabaa project, under construction with Rosatom since 2022 with four VVER-1200 reactors and financed largely by a Russian state loan, will be the largest nuclear project in Africa. South Africa's Koeberg plant, the only operating nuclear power plant in Africa, received a twenty-year extension of the operating licence of its first unit in 2024. The development of nuclear governance in Africa, including the role of AFCONE and regional cooperation in regulation, is a significant subject for doctoral research.

Financing, Taxonomy, and Market Design

The cost and financing of nuclear power are as important as its regulation. Large nuclear projects have high capital costs, long construction times, and a history of cost overruns, which make them difficult to finance in liberalised markets without state support. Governments have adopted a range of support mechanisms: CfDs, as used for Hinkley Point C in the United Kingdom; the **regulated asset base** (RAB) model, introduced in the United Kingdom by the Nuclear Energy (Financing) Act 2022 and applied to the Sizewell C project, under which investors receive a regulated return during construction funded by consumers; direct state ownership, as in France's full nationalisation of EDF in 2023; and state loans and guarantees. The inclusion of nuclear energy in the EU taxonomy, the removal of the World Bank's longstanding restriction on financing nuclear projects in 2025, and the pledges made at COP28 to triple nuclear capacity have improved the prospects of access to sustainable finance.

::: comparative {title="Comparative Perspective: Regulatory Models for Nuclear Newcomers"} Newcomer states have adopted different institutional models for nuclear regulation, each with distinctive strengths and risks.

- **The United Arab Emirates** enacted a comprehensive Federal Law by Decree No 6 of 2009 on the Peaceful Uses of Nuclear Energy, establishing the Federal Authority for Nuclear Regulation as an independent regulator before contracting for its first plant. It renounced domestic enrichment and reprocessing in a cooperation agreement with the United States, brought

an additional protocol into force, and acceded to the CSC and the Vienna Convention. The Barakah plant, built by a Korean consortium, entered commercial operation between 2021 and 2024. The UAE is frequently cited as a model of sequencing, establishing law and institutions before construction.

- **Turkey**, under the Akkuyu intergovernmental agreement with Russia (2010), adopted a build-own-operate model in which a Russian-owned project company owns and operates the plant. This model reduces the host state's financial burden but places heavy demands on the regulator's capacity to oversee a powerful foreign operator.
- **Bangladesh**, constructing the Rooppur plant with Russian financing and technology, enacted the Bangladesh Atomic Energy Regulatory Act 2012 to separate regulation from promotion. It has relied heavily on IAEA support and the vendor's regulatory documentation.
- **Ghana** created an independent Nuclear Regulatory Authority by the Nuclear Regulatory Authority Act 2015 (Act 895), separating regulation from the Ghana Atomic Energy Commission. It has established Nuclear Power Ghana as an owner-operator entity and has considered both large reactors and SMRs.
- **Nigeria** separated promotion (NAEC) from regulation (the NNRA) by statute in 1995, well before any decision to construct a power reactor. Commentators have identified the modernisation of the 1995 Act, the strengthening of the NNRA's financial independence, and adherence to a liability convention as priorities before any power reactor project proceeds.

The common lesson is that the legal framework must precede, rather than follow, the commitment to construct. The regulator's independence and competence also depend as much on funding, staffing, and culture as on statutory form. :::

::: exercise {title="Practical Exercise: The Republic of Kesena SMR Programme (HYPOTHETICAL)"} *This exercise is entirely hypothetical. The Republic of Kesena, the Kesena Atomic Energy Agency, NordReactor Corporation, and all other entities named are fictional.*

The Republic of Kesena is a West African state with an electricity grid of 4,500 MW peak demand. It is a party to the NPT and the Treaty of Pelindaba, and has a comprehensive safeguards agreement with a small quantities protocol but no additional protocol. It is not a party to any nuclear liability convention. Its Atomic Energy Act 1988 establishes the Kesena Atomic Energy Agency (KAEA). The KAEA is responsible for both "the promotion of nuclear energy" and "the licensing and inspection of nuclear facilities", and its Director General reports to the Minister of Energy, who chairs the national nuclear power programme.

The government has signed a memorandum of understanding with NordReactor Corporation (HYPOTHETICAL), a state-owned vendor from a non-African state. Under the memorandum, NordReactor will build, own, and operate two 150 MW(e) SMR units at Port Abeni, 40 km from the capital, and will sell electricity to the state utility under a twenty-five-year PPA. The draft intergovernmental agreement includes the following provisions:

- (a) "The Republic shall recognise the design certification issued by the vendor state's regulator as satisfying all Kesenan licensing requirements relating to reactor design."
- (b) "The Republic shall indemnify NordReactor and its suppliers against all claims for nuclear damage in excess of USD 50 million."
- (c) "Spent fuel shall be stored at the site for the life of the plant. Responsibility for its final disposal shall be agreed by the Parties in due course."
- (d) "NordReactor may, with the consent of the Minister, relocate a third floating SMR unit to Kesenan territorial waters to supply the Port Abeni industrial zone."

Answer the following:

1. Conduct a gap analysis of Kesena's legal framework against the Convention on Nuclear Safety, the IAEA Safety Fundamentals, and the IAEA Milestones Approach. What legislative and institutional reforms should be made before a construction licence is issued?
2. Is clause (a) compatible with the principle that the host state retains responsibility for safety and must maintain an independent and competent regulator? Redraft it to permit reliance on the vendor state's assessment without abdicating Kesena's regulatory responsibility.
3. Advise Kesena on whether to accede to the revised Vienna Convention, the CSC, or both. Evaluate clause (b) in the light of the principles of channelling and the position of potential victims in neighbouring Pelindaba states.
4. Evaluate clause (c) against the Joint Convention, the Treaty of Pelindaba, and the Bamako Convention. Draft an alternative clause addressing spent fuel management, decommissioning, and their financing.
5. What safeguards, security, and law of the sea issues are raised by clause (d)? Should Kesena bring an additional protocol into force and rescind its small quantities protocol before proceeding? :::

CHAPTER SUMMARY

- The international nuclear order rests on the bargain of the NPT. Non-nuclear-weapon states renounce nuclear weapons and accept safeguards, in exchange for the inalienable right to the peaceful uses of nuclear energy under Article IV, which must be exercised in conformity with non-proliferation obligations.

- International nuclear law is organised around safety, security, safeguards, and liability, together with waste management and decommissioning. Each pillar has its own conventions, and the IAEA, with its dual mandate of promotion and control, is the central institution.
- The detailed rules of nuclear safety are found in the IAEA Safety Standards. These are not binding as such but acquire legal effect through incorporation into national law, treaty review processes, and Euratom directives.
- The prime responsibility for safety rests with the operator. The Convention on Nuclear Safety requires an effective separation between the regulator and bodies promoting nuclear energy. Fukushima demonstrated that regulatory independence and the continuous reassessment of external hazards are substantive safety requirements.
- Safeguards agreements, strengthened by the Additional Protocol, verify the non-diversion of nuclear material. The cases of Iraq, the DPRK, and Iran show both the importance and the political limits of verification.
- The amended CPPNM, ICSANT, and Security Council Resolution 1540 form the core of nuclear security law. The security of radioactive sources is a significant practical concern, including in the oil and gas industry.
- The nuclear liability conventions channel strict, limited, and insured liability to the operator in a single forum. Their limits, fragmented participation, and the Indian departure from channelling are central issues, and newcomers need liability treaty relations to attract suppliers.
- Deep geological disposal is the accepted solution for high level waste, and Finland leads its implementation. Consent-based siting and secure financing of waste management and decommissioning are critical, as the United States and German experiences demonstrate.
- SMRs and advanced reactors raise issues of licensing modernisation, international harmonisation, emergency planning, transportable reactors, and safeguards by design. They do not relieve newcomer states of the obligation to establish a comprehensive legal and institutional infrastructure.

DOCTORAL RESEARCH QUESTIONS

1. Is the IAEA's dual mandate to promote and to regulate nuclear energy compatible with effective international oversight of nuclear safety? Should international nuclear safety standards become legally binding?
2. Evaluate the legal significance of the IAEA Board's findings of non-compliance and the invocation of the JCPOA snapback mechanism. What do the Iranian and North Korean cases reveal about the enforcement of non-proliferation obligations?

3. Does the principle of legal channelling in the nuclear liability conventions remain justified, or should suppliers bear liability for defects in the equipment and services they provide? Assess India's approach in this light.
4. How should international law respond to the operation of nuclear power plants in zones of armed conflict? Is there a need for a new instrument protecting nuclear installations in armed conflict?
5. Critically assess the readiness of the legal and institutional frameworks of one or more African states to host nuclear power, with reference to the IAEA Milestones Approach and the vendor-financed build-own-operate model.
6. Can international nuclear law accommodate transportable and floating reactors without amendment of the principal conventions? Identify the key gaps and propose solutions.

FURTHER READING

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- National Diet of Japan, Fukushima Nuclear Accident Independent Investigation Commission, *The Official Report* (2012).
- OECD Nuclear Energy Agency, *International Nuclear Law: History, Evolution and Outlook* (OECD 2010).
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12 CHAPTER

Hydrogen, Energy Storage, and Emerging Energy Technologies

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Analyse the regulatory classification of hydrogen and the legal frameworks governing its production, transport, and use.
2. Distinguish the principal hydrogen production pathways and evaluate the legal criteria — including additionality, temporal correlation, and carbon intensity thresholds — by which "renewable" and "low-carbon" hydrogen are defined.
3. Assess the law of hydrogen certification and the emerging framework for international trade in hydrogen and its derivatives.
4. Explain the legal treatment of electricity storage, including its definition, ownership restrictions, market participation, and network charging.
5. Evaluate the legal frameworks for carbon capture, utilisation, and storage, including the London Protocol, storage permitting, pore space rights, and long-term liability.
6. Analyse the international and domestic law applicable to offshore renewable energy technologies, including floating wind, wave and tidal energy, and offshore grids.
7. Assess the legal issues raised by smart grids and the digitalisation of energy systems, including data protection and cybersecurity.
8. Evaluate the use of regulatory sandboxes and other instruments for accommodating technological innovation in energy regulation.

Introduction

The decarbonisation of the energy system requires not only the replacement of fossil-fuelled electricity generation with renewable and other low-carbon sources, examined in Chapters 10 and 11, but also the transformation of the infrastructure, markets, and technologies through which energy is converted, stored, transported, and used. Electricity from variable wind and solar resources must be

stored or balanced; sectors that are difficult to electrify — steel, chemicals, fertilisers, shipping, and aviation — require low-carbon fuels and feedstocks; residual emissions from industrial processes must be captured and stored; offshore resources must be harnessed and connected; and an increasingly decentralised and digitalised energy system must be managed securely. Hydrogen, electricity storage, carbon capture and storage, offshore renewable technologies, and smart grids are the principal technological responses to these challenges.

These emerging technologies pose a common legal problem: the law was designed for an energy system with different technologies, actors, and boundaries. Electricity law presupposed a clear separation between generation, transmission, distribution, and consumption, which storage, which can act as each in turn, does not respect. Gas law was designed for natural gas, and its application to hydrogen raises questions of classification, network regulation, and safety. Mining and petroleum law regulates the extraction of substances from the subsurface, and must be adapted to regulate the injection of carbon dioxide into it. The law of the sea was developed for navigation, fishing, and seabed resources, and must accommodate large-scale offshore energy infrastructure. And energy regulation was designed for a system of large, centrally controlled assets, not a system of millions of connected devices exchanging data. The legal response has been a combination of the adaptation of existing frameworks, the creation of new regimes, and the use of experimental instruments such as regulatory sandboxes.

This chapter examines hydrogen production and regulatory classification (section 12.1), hydrogen production pathways and their legal definition (section 12.2), hydrogen certification and cross-border trade (section 12.3), battery storage and grid integration (section 12.4), carbon capture, utilisation, and storage (section 12.5), offshore renewable energy technologies (section 12.6), smart grids, digital infrastructure, and cybersecurity (section 12.7), and regulatory sandboxes and technological innovation (section 12.8).



Figure 12.1 The hydrogen value chain and the principal legal questions arising at each stage. *Source:* prepared for this edition.

12.1 Hydrogen Production and Regulatory Classification

The Role of Hydrogen in the Energy Transition

Hydrogen is the most abundant element in the universe, but it occurs on Earth almost entirely in compounds, principally water and hydrocarbons, from which it must be separated using energy. It is therefore an energy carrier rather than a primary energy source. Hydrogen has been produced and used for more than a century as an industrial feedstock, principally in oil refining and in the production of ammonia for fertilisers and methanol, and global production amounts to around one hundred million tonnes per year, almost all of it produced from natural gas and coal without carbon capture, resulting in substantial greenhouse gas emissions.

The interest in hydrogen as an instrument of decarbonisation rests on its potential, when produced with low emissions, to replace fossil fuels in applications that are difficult to electrify directly: as a feedstock and reducing agent in steel

production (replacing coking coal), as a feedstock in ammonia and methanol production, as a fuel for shipping (directly or through derivatives such as ammonia and methanol), aviation (through synthetic fuels), and heavy road transport, and as a medium for the long-duration storage of energy. Its role in other applications, such as residential heating and light passenger vehicles, is contested, because direct electrification is generally more efficient. Many governments have adopted national hydrogen strategies, and several resource-rich states, including Australia, Chile, Morocco, Namibia, Egypt, Oman, and Saudi Arabia, aspire to become exporters of hydrogen and its derivatives.

The legal framework for hydrogen is at an early stage of development. Hydrogen raises legal questions at each stage of its value chain: production (permitting, electricity and water supply, and the definition of "renewable" or "low-carbon" hydrogen); conversion (into ammonia, methanol, or other carriers); transport and storage (by pipeline, ship, or truck, and in underground salt caverns or depleted reservoirs); trade (certification and the treatment of hydrogen under trade law); and end use (safety standards, demand-side mandates, and support schemes).

The Problem of Classification

The threshold legal question is how hydrogen is to be classified. Existing legal categories — "gas", "natural gas", "fuel", "chemical", "dangerous substance", and "energy product" — have been defined with other substances in mind, and the classification of hydrogen determines which regulatory regimes apply to it.

Hydrogen as a gas. In some jurisdictions, hydrogen falls within the definition of "gas" for the purposes of gas legislation, bringing hydrogen networks within existing frameworks of licensing, network regulation, and safety. The United Kingdom's Gas Act 1986 defines "gas" broadly to include any substance in a gaseous state consisting wholly or mainly of methane, ethane, propane, butane, hydrogen, or carbon monoxide, among other things, so that the conveyance of hydrogen is, in principle, within the licensing regime. In other jurisdictions, gas legislation is limited to "natural gas" or "methane", and hydrogen is excluded.

Hydrogen as a chemical and dangerous substance. Hydrogen is a flammable gas with a wide flammability range and low ignition energy, and is subject to the legislation governing dangerous substances, industrial safety, and the prevention of major accidents. In the European Union, the **Seveso III Directive** (Directive 2012/18/EU) on the control of major-accident hazards involving dangerous substances applies to establishments where hydrogen is present above specified thresholds, and the **ATEX Directives** (Directives 2014/34/EU and 1999/92/EC) govern equipment and workplaces in potentially explosive atmospheres. Hydrogen is also classified under chemicals legislation, including the

EU's REACH Regulation, and under the international rules on the transport of dangerous goods.

Hydrogen in the EU gas market framework. The European Union adopted the **Hydrogen and Decarbonised Gas Market Package** in 2024, comprising Directive (EU) 2024/1788 on common rules for the internal markets for renewable gas, natural gas and hydrogen, and Regulation (EU) 2024/1789 on the internal markets for renewable gas, natural gas and hydrogen. The package establishes, for the first time, a dedicated regulatory framework for hydrogen networks, distinct from that for natural gas. It provides for the vertical unbundling of hydrogen network operators from hydrogen production and supply, using models analogous to those for gas and electricity examined in Chapters 6 and 9; for regulated third-party access to hydrogen networks from 2033, with negotiated access permitted until then; for the certification of hydrogen network operators; for the establishment of a European Network of Network Operators for Hydrogen (ENNOH); for integrated network planning; and for rules on the blending of hydrogen into natural gas networks, under which transmission operators must accept flows with up to two per cent hydrogen by volume at interconnection points. The package also defines "low-carbon hydrogen" — hydrogen whose energy content is derived from non-renewable sources and which meets a greenhouse gas emissions reduction threshold of seventy per cent compared with fossil fuel — and provides for its certification.

Hydrogen pipelines in the United States. The regulation of hydrogen pipelines in the United States illustrates the difficulties of fitting hydrogen into existing categories. Hydrogen pipelines are subject to safety regulation by the Pipeline and Hazardous Materials Safety Administration, but the federal authority responsible for their economic regulation is unclear: the Federal Energy Regulatory Commission's jurisdiction under the Natural Gas Act extends to "natural gas", which may not include pure hydrogen, while the Surface Transportation Board's residual jurisdiction over pipelines transporting commodities other than water, gas, or oil may extend to hydrogen. Blends of hydrogen with natural gas transported in natural gas pipelines are generally considered to remain within FERC's jurisdiction.

Production Permitting

The production of hydrogen by electrolysis requires an electrolyser — industrial equipment that uses electricity to split water into hydrogen and oxygen — together with a supply of electricity and water. Electrolysers are typically permitted as industrial installations, subject to land use, environmental, and major-accident hazard requirements. Large electrolysers above specified thresholds may be subject to the EU's Industrial Emissions Directive and environmental impact assessment, and RED III requires member states to ensure that permit-granting procedures

for renewable hydrogen production and for associated infrastructure are streamlined. Production from natural gas with carbon capture requires, in addition, the permitting of capture facilities and carbon dioxide transport and storage, examined in section 12.5.

Water. Electrolysis requires around nine litres of purified water per kilogram of hydrogen, in addition to cooling water. Although the water requirements of hydrogen production are modest relative to agriculture, large-scale production in water-scarce regions — including many of the regions with the best solar resources, such as North Africa, the Middle East, Namibia, and Australia — may require desalination, raising issues of water rights, the disposal of brine, and the allocation of scarce water between export industries and local needs. Water allocation law, examined in Chapter 10, therefore forms part of the legal framework for hydrogen.

Land and benefit-sharing. Export-scale hydrogen projects in developing countries involve the allocation of large areas of land for renewable generation, electrolysers, and port facilities, raising issues of land rights, environmental protection, and the sharing of benefits with host states and communities. Namibia's flagship green hydrogen project in the Tsau //Khaeb National Park near Lüderitz, for which the government concluded a feasibility and implementation agreement with a private developer and acquired an option to take an equity stake, has attracted debate about the location of an industrial project in a national park, the terms of the state's participation, and the distribution of benefits. These questions echo those examined in the context of extractive industries in Chapters 5 and 7, and resource nationalism and local content in the hydrogen sector are emerging themes of energy law in Africa.

Nigeria. Nigeria does not yet have a dedicated legal framework for hydrogen. Hydrogen production from natural gas is undertaken incidentally in refining and in the fertiliser industry, and the PIA's provisions on gas utilisation and midstream licensing may apply to some hydrogen-related activities. Nigeria's abundant natural gas, solar resources, and existing fertiliser and ammonia industries, together with the Energy Transition Plan's identification of hydrogen as a potential decarbonisation pathway, suggest that the development of a hydrogen strategy and legal framework is a subject of growing interest.

12.2 Green, Blue, and Other Hydrogen Production Pathways

The Colour Spectrum

Hydrogen production pathways are commonly described using a spectrum of colours. **Grey hydrogen** is produced from natural gas by steam methane reforming without carbon capture, and accounts for most current production; **brown** or **black hydrogen** is produced from coal by gasification. **Blue hydrogen**

is produced from natural gas with the capture and storage of a large share of the resulting carbon dioxide. **Green hydrogen** is produced by electrolysis using renewable electricity. **Pink** or **purple hydrogen** is produced by electrolysis using nuclear electricity. **Turquoise hydrogen** is produced by methane pyrolysis, which splits methane into hydrogen and solid carbon. **White** or **natural hydrogen** is hydrogen occurring naturally in geological formations, whose potential has attracted recent exploration interest.

The colour terminology is a useful shorthand, but it is legally imprecise and potentially misleading. The climate impact of hydrogen depends not on its colour but on its life-cycle greenhouse gas emissions, which vary within each colour: green hydrogen produced by electrolysers drawing electricity from a grid dominated by fossil fuels may have higher emissions than grey hydrogen, and blue hydrogen's emissions depend on the capture rate and the upstream methane emissions from natural gas production. Law and policy have therefore moved towards **carbon intensity** approaches, which define hydrogen by reference to maximum life-cycle emissions per unit of hydrogen, sometimes combined with criteria relating to the source of energy.

The EU Definition of Renewable Hydrogen

The European Union has adopted the most detailed legal definition of renewable hydrogen. Under RED II and RED III, hydrogen and other fuels produced from renewable electricity are classified as **renewable fuels of non-biological origin** (RFNBOs), which count towards renewable energy targets, including RED III's binding target that at least forty-two per cent of the hydrogen used in industry should be RFNBOs by 2030 and sixty per cent by 2035, and its sub-target for RFNBOs in transport. To qualify as RFNBOs, fuels must satisfy two Commission Delegated Regulations adopted in 2023.

Delegated Regulation (EU) 2023/1184 establishes rules for determining when electricity used to produce RFNBOs may be counted as fully renewable. Electricity taken directly from a renewable installation connected to the electrolyser may be counted as renewable if the installation came into operation no earlier than thirty-six months before the electrolyser. Electricity taken from the grid may be counted as fully renewable if the bidding zone in which the electrolyser is located has a share of renewable electricity above ninety per cent, if the emissions intensity of the electricity is below a specified threshold, or if the producer has concluded renewable PPAs that satisfy three criteria:

- **Additionality**: the renewable installations from which the electricity is sourced must have come into operation no earlier than thirty-six months before the electrolyser, and must not have received operating or investment aid, so that the hydrogen production is matched by new, unsubsidised

renewable capacity rather than diverting existing renewable electricity from other uses. Electrolysers in operation before 1 January 2028 are exempted from the additionality requirement until 1 January 2036.

- **Temporal correlation**: the renewable electricity must be produced in the same calendar month as the hydrogen until 31 December 2029, and in the same one-hour period from 1 January 2030.
- **Geographical correlation**: the renewable installation must be located in the same bidding zone as the electrolyser, in an interconnected bidding zone with higher electricity prices, or in an offshore bidding zone interconnected with the electrolyser's bidding zone.

Delegated Regulation (EU) 2023/1185 establishes the methodology for assessing the greenhouse gas emissions savings of RFNBOs and recycled carbon fuels, and sets a minimum emissions savings threshold of seventy per cent compared with a fossil fuel comparator of 94 grams of carbon dioxide equivalent per megajoule, equivalent to approximately 3.4 kilograms of carbon dioxide equivalent per kilogram of hydrogen.

The additionality, temporal, and geographical correlation requirements are intended to ensure that the production of renewable hydrogen does not increase emissions by diverting renewable electricity from the grid and increasing fossil generation elsewhere. Industry stakeholders have argued that they are excessively strict and increase the cost of renewable hydrogen, while environmental organisations have defended them as necessary to ensure environmental integrity. The rules apply to RFNBOs imported into the European Union as well as those produced within it, with significant implications for exporting states, as examined in section 12.3. In 2025, the Commission adopted a delegated act setting out the methodology for determining the greenhouse gas emissions of low-carbon fuels, including low-carbon hydrogen produced from natural gas with carbon capture or from nuclear electricity, under the Hydrogen and Decarbonised Gas Market Package.

The United States Clean Hydrogen Production Credit

In the United States, the Inflation Reduction Act of 2022 introduced a **clean hydrogen production tax credit** under section 45V of the Internal Revenue Code, with four tiers of credit value according to the life-cycle greenhouse gas emissions of the hydrogen, up to a maximum emissions rate of four kilograms of carbon dioxide equivalent per kilogram of hydrogen, determined by a "well-to-gate" analysis. Final regulations issued by the Treasury Department in January 2025 adopted three requirements for hydrogen produced using grid electricity to be credited with the emissions of the clean electricity procured through energy attribute certificates — incrementality (new clean generation), temporal matching

(annual matching until 2029, and hourly matching thereafter), and deliverability (generation in the same region) — closely analogous to the EU's criteria, while providing flexibilities for existing nuclear plants at risk of retirement and for states with robust clean electricity standards. The One Big Beautiful Bill Act of July 2025 curtailed the credit by limiting it to facilities beginning construction before 1 January 2028, an illustration of the political volatility of hydrogen support.

Other National Approaches

The **United Kingdom's Low Carbon Hydrogen Standard** sets a maximum threshold of twenty grams of carbon dioxide equivalent per megajoule of hydrogen (on a lower heating value basis) for hydrogen to be eligible for government support, applicable to all production pathways, including electrolysis and natural gas reforming with carbon capture. The United Kingdom supports low-carbon hydrogen production through the **Hydrogen Production Business Model**, a contract for difference-style mechanism under which producers receive the difference between a strike price and a reference price, and the first hydrogen allocation round resulted in contracts with eleven electrolytic hydrogen projects in 2023. **Japan's Hydrogen Society Promotion Act** of 2024 provides for fifteen-year price-gap support for low-carbon hydrogen and ammonia supply chains, including imports, meeting carbon intensity requirements. Other states, including India, China, Australia, and Chile, have adopted standards defining green or low-carbon hydrogen, some based on carbon intensity alone and others combining intensity with source-based criteria.

Natural Hydrogen and Legal Classification

The discovery of naturally occurring hydrogen accumulations — notably at Bourakébougou in Mali, where a well drilled for water in the 1980s was found to produce almost pure hydrogen, which has since been used to generate electricity for a village — has generated interest in exploration for natural hydrogen in several countries. Natural hydrogen raises the classification question in acute form: is it a mineral, a petroleum substance, a gas, or a sui generis resource, and which licensing regime applies to its exploration and production? South Australia amended its Petroleum and Geothermal Energy Act in 2021 to include hydrogen as a regulated substance, enabling the grant of exploration licences for natural hydrogen, and France amended its Mining Code to include natural hydrogen among the substances subject to mining titles. In most jurisdictions, including Nigeria, the classification of natural hydrogen is unaddressed, and the definitions of "petroleum" and "minerals" in the PIA and the Nigerian Minerals and Mining Act may not clearly extend to it.

12.3 Hydrogen Certification and Cross-Border Trade

The Function of Certification

Because the climate benefit of hydrogen depends on how it was produced, and because molecules of hydrogen produced by different pathways are physically identical, regulatory and commercial claims about low-carbon hydrogen depend on reliable certification of its origin and emissions. Certification serves several legal functions: demonstrating eligibility for support schemes and tax credits; demonstrating compliance with renewable energy targets and obligations; enabling consumers to make claims under emissions accounting frameworks; and enabling the differentiation of hydrogen in international trade. Certification systems may follow a **book-and-claim** model, analogous to guarantees of origin, under which certificates are traded separately from the physical product, or a **mass balance** model, under which certificates and the physical product move together along the supply chain, with the quantity of certified hydrogen withdrawn not exceeding the quantity injected. EU law requires the mass balance approach for the purposes of demonstrating compliance with the RFNBO and biofuel sustainability criteria under Article 30 of the Renewable Energy Directive, while guarantees of origin, which follow a book-and-claim model, are used for disclosure to consumers.

EU Certification Through Voluntary Schemes

Under Article 30 of the Renewable Energy Directive, economic operators may demonstrate compliance with the RFNBO criteria, as with the biofuel sustainability criteria, through national schemes or voluntary international schemes recognised by the Commission. The Commission has recognised a number of voluntary schemes for the certification of RFNBOs, which audit producers' compliance with the additionality, temporal and geographical correlation, and emissions savings requirements, including for producers located outside the European Union. Implementing Regulation (EU) 2022/996 sets out rules for verifying sustainability, greenhouse gas emissions saving, and low indirect land use change risk criteria, including the requirements for audits and the use of the Union Database, which tracks the transactions of certified fuels to prevent fraud and double counting.

International Standards

The proliferation of national definitions and certification schemes threatens to fragment the international market for hydrogen, increasing the costs of compliance for producers seeking to export to multiple markets. Efforts to develop internationally harmonised methodologies have been led by the **International Partnership for Hydrogen and Fuel Cells in the Economy** (IPHE), whose working paper on a methodology for determining the greenhouse gas emissions associated with the production of hydrogen informed the development of **ISO**

19870:2023, the first international standard on the methodology for determining greenhouse gas emissions associated with the production, conditioning, and transport of hydrogen to the consumption gate. The ISO standard provides a common accounting methodology, but does not set thresholds or determine which hydrogen is "clean", which remain matters for national and regional law. Mutual recognition of certification schemes between importing and exporting states has been proposed as a means of reducing fragmentation, and the European Union has explored cooperation with partner countries on certification.

Trade Arrangements and Support for Imports

Several importing states have developed instruments to support imports of renewable hydrogen and derivatives. Germany's **H2Global** mechanism, managed by an intermediary company (Hintco), uses a double-auction model: the intermediary concludes long-term purchase agreements with producers outside the European Union, selected through competitive tenders on the basis of the lowest price, and sells the products in the European market through short-term auctions to the highest bidder, with the difference between the purchase and sale prices funded by the German government. The first H2Global tender, concluded in 2024, resulted in a contract for the supply of renewable ammonia produced in Egypt. The European Union's **European Hydrogen Bank**, funded by the Innovation Fund, conducts auctions for fixed premium support for renewable hydrogen production in the European Economic Area, with the first auction in 2024 awarding around EUR 720 million to projects selected on the basis of the lowest premium requested, and is developing an international component to support imports.

Hydrogen and International Trade Law

International trade in hydrogen and its derivatives is subject to the WTO agreements examined in Chapter 13. Several questions arise. First, the **classification** of hydrogen and its derivatives in the Harmonized System — hydrogen is classified under heading 28.04 and ammonia under heading 28.14 — does not distinguish between low-carbon and high-carbon production, so that tariff differentiation based on production methods would require new subheadings or other instruments. Second, the **non-discrimination** obligations of the GATT raise the question whether low-carbon and high-carbon hydrogen are "like products" — a question that turns on the debate over whether non-product-related processes and production methods can be a basis for regulatory distinction, examined in Chapter 13. Third, the **TBT Agreement** applies to technical regulations and standards defining low-carbon hydrogen and the conformity assessment procedures by which compliance is certified, requiring them not to be more trade-restrictive than necessary to fulfil a legitimate objective and encouraging the use of international standards, such as ISO 19870. Fourth, **subsidies** for hydrogen

production, such as the US 45V credit and EU support schemes, may be subject to the disciplines of the Agreement on Subsidies and Countervailing Measures, particularly where they are conditioned on domestic content. Fifth, the EU **Carbon Border Adjustment Mechanism** (Regulation (EU) 2023/956) covers hydrogen and ammonia among its initial sectors, so that imports of carbon-intensive hydrogen and ammonia into the European Union will bear a carbon price from the definitive period beginning in 2026, as examined in Chapter 13.

Transport of Hydrogen and Derivatives

International trade in hydrogen requires its transport over long distances, by pipeline or by ship. Pipeline transport of hydrogen within Europe is being developed through the conversion of natural gas pipelines and the construction of new hydrogen pipelines, including the planned "European Hydrogen Backbone" and cross-border corridors such as the H2Med project linking the Iberian Peninsula with France and Germany, and the proposed SoutH2 Corridor linking North Africa and Italy with Austria and Germany. Such cross-border pipelines raise issues of intergovernmental agreements, transit, and regulatory coordination similar to those examined for natural gas pipelines in Chapter 6.

Shipping of hydrogen requires its conversion into a form with higher energy density: liquefied hydrogen (at minus 253 degrees Celsius), ammonia, methanol, or liquid organic hydrogen carriers. The *Suiso Frontier*, the world's first liquefied hydrogen carrier, completed a pilot voyage from Australia to Japan in 2022. The International Maritime Organization adopted interim recommendations for the carriage of liquefied hydrogen in bulk, pending amendment of the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk (IGC Code). Ammonia, which is already traded internationally in large quantities, is likely to be the principal carrier in the near term, but its toxicity raises safety and environmental issues in its handling, transport, and use as a fuel, which the IMO has begun to address through interim guidelines on the safety of ships using ammonia as fuel.

12.4 Battery Storage and Grid Integration

The Role of Electricity Storage

Electricity storage — the conversion of electrical energy into a form that can be stored and reconverted into electricity when needed — is increasingly central to the integration of variable renewable energy. Storage can shift energy from periods of surplus to periods of scarcity, provide fast-responding frequency regulation and other ancillary services, defer or replace investment in network reinforcement, provide backup supply, and enable consumers to increase their self-consumption of rooftop solar. **Pumped storage hydropower** remains the largest form of grid-scale storage globally, but **battery energy storage systems**, principally using

lithium-ion technology, have grown dramatically as their costs have fallen, and are now deployed at grid scale in many markets. Other technologies include compressed air, flywheels, thermal storage, and, for long durations, hydrogen.

The rapid growth of battery storage has been illustrated by projects such as the Hornsdale Power Reserve in South Australia, commissioned in 2017 following a series of blackouts, which demonstrated the capability of large batteries to provide fast frequency response and reduced the costs of frequency control services in the Australian National Electricity Market. In sub-Saharan Africa, batteries are integral to solar mini-grids and stand-alone solar systems, examined in Chapter 10, and to the management of grids subject to frequent outages, including in Nigeria, where behind-the-meter batteries increasingly substitute for diesel generators.

The Legal Characterisation of Storage

The principal legal difficulty presented by storage is that it does not fit within the categories of traditional electricity law. Storage consumes electricity when charging and produces it when discharging; it may be connected at transmission or distribution level or behind the customer's meter; and it may provide services to the market, to network operators, and to consumers. Early regulatory frameworks treated storage as generation (because it produces electricity), as consumption (because it draws electricity), or as both, with the result that storage could be subject to double charges — paying network charges and levies both when charging and when discharging — and to uncertainty about whether network operators could own it.

The European Union addressed these issues in the Clean Energy for All Europeans package. Article 2(59) of Directive (EU) 2019/944 defines "energy storage" in the electricity system as "deferring the final use of electricity to a moment later than when it was generated, or the conversion of electrical energy into a form of energy which can be stored, the storing of such energy, and the subsequent reconversion of such energy into electrical energy or use as another energy carrier". The definition recognises storage as a distinct activity, rather than as generation or consumption. Article 15(5) of the Directive requires member states to ensure that active customers who own energy storage facilities are not subject to double charges, including network charges, for stored electricity remaining within their premises or when providing flexibility services to system operators.

In Great Britain, storage was treated as a form of generation for licensing purposes, and the Energy Act 2023 amended the Electricity Act 1989 to define electricity storage expressly as a distinct subset of generation, providing legal clarity. In the United States, storage has been treated as a distinct resource for the purposes of wholesale market participation, as examined below. In Nigeria, the legal treatment of storage under the Electricity Act 2023 and NERC's regulations

is still developing, and a comprehensive framework for the licensing, market participation, and network charging of storage has yet to be established.

Ownership of Storage by Network Operators

Whether network operators — whose activities are regulated monopolies — may own and operate storage facilities is a significant question. Storage owned by network operators could defer network investment and support system security, but it could also distort competition in markets in which storage competes with generation and demand response to provide energy and flexibility services. The EU's unbundling principles, examined in Chapter 9, apply: Articles 36 and 54 of Directive (EU) 2019/944 provide that distribution and transmission system operators shall not own, develop, manage, or operate energy storage facilities, except where the facilities are "fully integrated network components" (such as capacitors that are used solely to maintain network stability) and the regulator has approved them, or where, following an open, transparent, and non-discriminatory tendering procedure, no other party has expressed interest in providing the storage services at reasonable cost, the facilities are necessary for the system operator's obligations, and the regulator has approved the derogation. The regulator must periodically reassess whether third parties are interested in owning the facilities, and if so, the system operator must phase out its storage activities.

Market Participation

The ability of storage to participate in wholesale electricity markets on equal terms with other resources depends on market rules designed for generators and loads. In the United States, the Federal Energy Regulatory Commission's **Order No 841** (2018) required regional transmission organisations and independent system operators to revise their market rules to establish participation models for electric storage resources, enabling them to provide all capacity, energy, and ancillary services that they are technically capable of providing, to be dispatched and to set market prices as both buyers and sellers, to account for their physical and operational characteristics, and to buy energy for later resale at the wholesale market price. In *National Association of Regulatory Utility Commissioners v FERC* 964 F.3d 1177 (DC Cir 2020), the Court of Appeals upheld Order No 841, rejecting the argument that FERC had exceeded its jurisdiction by preventing states from prohibiting storage resources connected at the distribution level from participating in wholesale markets, holding that the Order regulated practices directly affecting wholesale rates within FERC's jurisdiction under the Federal Power Act while preserving states' authority over the safety and reliability of the distribution system. **Order No 2222** (2020), examined in Chapter 9, extended wholesale market participation to aggregations of distributed energy resources, including behind-the-meter batteries.

Several jurisdictions have adopted **procurement mandates** for storage. California's Assembly Bill 2514 (2010) directed the California Public Utilities Commission to determine whether to adopt storage procurement targets, and the Commission subsequently required the state's investor-owned utilities to procure 1,325 MW of storage by 2020, which was followed by substantially larger procurement orders to maintain reliability as solar generation increased. Capacity markets, examined in Chapter 9, have also become an important source of revenue for storage, subject to "de-rating" factors that reflect the limited duration of battery storage.

Safety, Siting, and End of Life

Large battery installations raise safety issues, principally the risk of thermal runaway leading to fires that are difficult to extinguish and may release toxic gases. A major fire at a large battery storage facility at Moss Landing, California, in January 2025 led to local evacuations and renewed scrutiny of battery safety standards and siting. Safety is addressed through fire codes and standards, including the United States National Fire Protection Association's NFPA 855 standard for the installation of stationary energy storage systems and certification standards such as UL 9540, and through planning and permitting requirements. The end-of-life management of batteries — including their collection, reuse in "second life" applications, and recycling — is governed in the European Union by the Batteries Regulation (Regulation (EU) 2023/1542), examined in Chapter 8, which also imposes carbon footprint, recycled content, due diligence, and battery passport requirements on industrial batteries, including those used for grid storage.

12.5 Carbon Capture, Utilisation, and Storage

The Role of CCUS

Carbon capture, utilisation, and storage (CCUS) comprises technologies that capture carbon dioxide from large point sources — such as power plants, cement and steel works, hydrogen production, and natural gas processing — or directly from the atmosphere, and either use it in products or store it permanently in deep geological formations. The Intergovernmental Panel on Climate Change has identified CCUS as a necessary component of many pathways limiting global warming to 1.5°C or 2°C, particularly for the abatement of emissions from industrial processes, such as cement production, for which there are few alternatives, and for the removal of carbon dioxide from the atmosphere through bioenergy with carbon capture and storage (BECCS) and direct air capture with carbon storage (DACCS). CCUS is nevertheless controversial: critics argue that it has a history of underperformance and cost overruns, that it may prolong the use of fossil fuels, and that much captured carbon dioxide has been used for enhanced oil recovery, which increases oil production.

Commercial-scale geological storage of carbon dioxide has been undertaken since 1996 at the **Sleipner** field in the Norwegian North Sea, where carbon dioxide separated from natural gas has been injected into a saline aquifer beneath the seabed, partly in response to Norway's carbon tax on offshore emissions. The **Northern Lights** project, part of Norway's Longship project, began injecting carbon dioxide transported by ship from industrial emitters in 2025, providing the first commercial cross-border carbon dioxide transport and storage service.

The International Framework: The London Protocol

The injection of carbon dioxide into sub-seabed geological formations raised the question whether it constituted "dumping" prohibited by the international law of marine pollution. The 1996 **Protocol to the Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter** (the London Protocol) adopts a "reverse list" approach, prohibiting the dumping of all wastes or other matter except those listed in Annex 1. In 2006, the parties amended Annex 1 to include "carbon dioxide streams from carbon dioxide capture processes for sequestration", permitting the storage of carbon dioxide in sub-seabed geological formations, provided that the streams consist overwhelmingly of carbon dioxide, no wastes are added for the purpose of disposal, and the storage is subject to a permit and to the specific guidelines adopted by the parties. The **OSPAR Convention** for the protection of the marine environment of the North-East Atlantic was amended in 2007 to permit the storage of carbon dioxide in sub-soil geological formations.

Article 6 of the London Protocol prohibits the export of wastes or other matter to other countries for dumping or incineration at sea, which would have prevented the cross-border transport of carbon dioxide for sub-seabed storage. In 2009, the parties adopted an amendment to Article 6 permitting the export of carbon dioxide streams for storage, provided that an agreement or arrangement is entered into by the countries concerned, allocating permitting responsibilities. Because the 2009 amendment had not received sufficient acceptances to enter into force, the parties adopted a resolution in 2019 permitting its provisional application by parties that so declare, enabling cross-border projects such as Northern Lights to proceed on the basis of bilateral arrangements, such as those concluded between Norway and several European states.

Domestic Storage Regimes

The European Union. Directive 2009/31/EC on the geological storage of carbon dioxide (the CCS Directive) establishes a comprehensive legal framework for storage in the territory of member states, including their exclusive economic zones and continental shelves. Member states retain the right to determine the areas from which storage sites may be selected, including the right not to allow

any storage in parts or the whole of their territory. Exploration requires an exploration permit, and storage requires a storage permit, which may be issued only if the site has been characterised and assessed and found to present no significant risk of leakage or of environmental or health damage. Operators must monitor the storage site and take corrective measures in the event of leakage or significant irregularities, and must surrender emission allowances under the EU ETS for any leaked emissions. Following closure, responsibility for the site is transferred to the competent authority when all available evidence indicates that the stored carbon dioxide will be completely and permanently contained, a minimum period — normally not less than twenty years — has elapsed, a financial contribution has been made, and the site has been sealed and the injection facilities removed. Operators must provide financial security for their obligations. Captured and stored emissions are not counted as emissions under the EU ETS, providing an economic incentive for CCS in sectors covered by the ETS. The **Net-Zero Industry Act** (Regulation (EU) 2024/1735) set an objective of achieving an annual carbon dioxide injection capacity of at least fifty million tonnes in the Union by 2030, and imposes on holders of oil and gas production licences an obligation to contribute to that objective in proportion to their share of production.

The United Kingdom. The Energy Act 2008 established a licensing regime for offshore carbon dioxide storage, administered by the North Sea Transition Authority, and the Energy Act 2023 established a framework for the economic regulation of carbon dioxide transport and storage networks by Ofgem, together with revenue support mechanisms for industrial carbon capture, power with CCS, and hydrogen production.

The United States. The geological storage of carbon dioxide is regulated by the Environmental Protection Agency under the Safe Drinking Water Act's Underground Injection Control programme, which established in 2010 a dedicated "Class VI" well category for geological sequestration, with requirements relating to site characterisation, well construction, monitoring, financial responsibility, and post-injection site care; states may obtain primacy to administer the programme. The **section 45Q** tax credit, substantially enhanced by the Inflation Reduction Act, provides a credit per tonne of carbon oxide captured and stored or used, and has been a principal driver of CCUS investment.

Pore Space and Long-Term Liability

Two legal questions are distinctive to onshore geological storage. The first concerns the **ownership of pore space** — the void spaces in subsurface rock formations in which carbon dioxide is stored. In common law jurisdictions where land ownership extends to the subsurface, the question arises whether pore space belongs to the surface owner or the mineral owner, where these have been severed. Several states of the United States, including North Dakota, Montana,

and Wyoming, have enacted legislation vesting pore space in the surface owner, and provided for the "unitisation" or "amalgamation" of pore space, permitting storage operators to proceed with the consent of a specified proportion of owners, with compensation for non-consenting owners. In jurisdictions where the subsurface is owned by the state, as in most civil law and many developing countries, the state may grant storage rights through licences. In Nigeria, where minerals and petroleum are vested in the Federal Government, the ownership of pore space and the legal basis for storage licensing remain unaddressed, although the PIA's framework for petroleum operations might be adapted to regulate storage in depleted reservoirs.

The second concerns **long-term liability** for stored carbon dioxide, which must remain contained for centuries or millennia, far beyond the lifetime of the companies that inject it. Most regimes provide for the transfer of liability to the state after a period of post-closure monitoring, as in the EU CCS Directive, subject to conditions and financial contributions, and some, such as Australia's Offshore Petroleum and Greenhouse Gas Storage Act 2006, provide for the Commonwealth to assume long-term liability after a closing certificate is issued. The design of liability transfer must balance the need to provide investors with certainty against the risk of socialising the costs of leakage.

CCUS in Climate Law and Carbon Markets

CCS was accepted as an eligible project activity under the Kyoto Protocol's Clean Development Mechanism by a decision of the parties in 2011, subject to modalities and procedures addressing site selection, monitoring, liability, and financial provision. Under the Paris Agreement, carbon dioxide removals, including through BECCS and DACCS, may generate credits under the Article 6.4 mechanism, subject to standards on removals adopted by its Supervisory Body, as examined in Chapter 18. The treatment of CCUS in climate law raises questions about the permanence of storage, the accounting for emissions from enhanced oil recovery, and the risk that reliance on future removals may delay emission reductions.

12.6 Offshore Renewable Energy Technologies

The Expansion of Offshore Energy

The ocean offers vast renewable energy resources: offshore wind, which is stronger and more consistent than onshore wind; wave energy; tidal energy, derived from tidal ranges and tidal currents; ocean thermal energy conversion (OTEC), which exploits temperature differences between surface and deep waters in tropical regions; and salinity gradient energy. Offshore wind is the only one of these technologies that has achieved commercial maturity at scale, with tens of gigawatts installed, predominantly in the North Sea, the Baltic Sea, and the coastal waters

of China. **Floating offshore wind**, in which turbines are mounted on floating platforms moored to the seabed, extends the potential of offshore wind to deep waters, which account for most of the global offshore wind resource, including off the coasts of Japan, Korea, the western United States, Norway, and parts of Africa. Wave and tidal energy remain at the demonstration and early commercial stage, with projects such as the MeyGen tidal stream array in Scotland.

This section examines the legal framework governing offshore renewable technologies — their installation, the allocation of marine space, offshore grids, and decommissioning. The leasing and permitting of offshore wind have been introduced in Chapter 10, and the law of the sea relating to maritime zones, submarine cables, and seabed resources more generally is examined in Chapter 20.

The Law of the Sea Framework

The United Nations Convention on the Law of the Sea (UNCLOS) provides the international legal framework for offshore renewable energy. In the **territorial sea**, extending up to twelve nautical miles from the baselines, the coastal state exercises sovereignty, subject to the right of innocent passage of foreign ships, and may authorise offshore renewable installations under its domestic law. In the **exclusive economic zone**, extending up to two hundred nautical miles, Article 56(1)(a) grants the coastal state sovereign rights for the purpose of exploring and exploiting, conserving, and managing the natural resources of the zone, "and with regard to other activities for the economic exploitation and exploration of the zone, such as the production of energy from the water, currents and winds". Article 60 grants the coastal state the exclusive right to construct and to authorise and regulate the construction, operation, and use of artificial islands, installations, and structures for these purposes, and exclusive jurisdiction over them, including with regard to customs, fiscal, health, safety, and immigration laws and regulations.

Article 60 imposes obligations on the coastal state in respect of installations in the EEZ. Due notice must be given of their construction, and permanent means for giving warning of their presence must be maintained. Installations which are abandoned or disused must be removed to ensure safety of navigation, taking into account generally accepted international standards (the IMO's 1989 *Guidelines and Standards for the Removal of Offshore Installations and Structures on the Continental Shelf and in the Exclusive Economic Zone*). The coastal state may establish reasonable safety zones around installations, which shall not exceed a distance of five hundred metres around them, except as authorised by generally accepted international standards or as recommended by the competent international organisation. Installations may not be established where interference may be caused to the use of recognised sea lanes essential to international navigation. In exercising its rights in the EEZ, the coastal state must have due regard to the

rights and duties of other states, including the freedoms of navigation, overflight, and the laying of submarine cables and pipelines (Article 56(2) and Article 58).

The growth of large offshore wind farms raises questions about the adequacy of the five-hundred-metre limit on safety zones, the effect of large arrays on navigation and on the rights of other states, the extension of offshore renewable development beyond the EEZ onto the extended continental shelf (where the coastal state's rights under Part VI of UNCLOS relate to natural resources of the seabed and subsoil and may not extend to wind energy in the superjacent waters), and the protection of marine biodiversity. The **Agreement under UNCLOS on the Conservation and Sustainable Use of Marine Biological Diversity of Areas beyond National Jurisdiction** (BBNJ Agreement, 2023), which provides for environmental impact assessments of activities with potential effects in areas beyond national jurisdiction, may apply to the transboundary impacts of large offshore energy installations.

Maritime Spatial Planning

The allocation of marine space among competing uses — offshore energy, shipping, fishing, aquaculture, conservation, defence, cables and pipelines, and tourism — is increasingly governed by **maritime spatial planning**. The EU Maritime Spatial Planning Directive (Directive 2014/89/EU) required member states to establish maritime spatial plans by 2021, taking into account land-sea interactions, environmental, economic, social, and safety aspects, and cooperation with neighbouring states. Maritime spatial plans designate areas suitable for offshore renewable energy, and RED III requires member states to designate renewables acceleration areas, including offshore. The integration of offshore wind with conservation objectives has generated debate about the co-location of wind farms with marine protected areas and the potential for offshore wind foundations to act as artificial reefs.

Offshore Grids and Hybrid Projects

The scale of planned offshore wind development, particularly in the North Sea, has led to a shift from the radial connection of individual wind farms to shore towards **offshore grids** — networks of offshore transmission infrastructure that connect multiple wind farms and interconnect national electricity systems. **Hybrid projects** combine offshore wind connection and cross-border interconnection, allowing offshore wind to be exported to more than one country. These projects raise complex legal questions: the classification of hybrid assets as transmission or interconnection; the allocation of cross-border capacity (under EU rules requiring that at least seventy per cent of interconnection capacity be made available for cross-border trade, examined in Chapter 9); the definition of **offshore bidding zones**, in which offshore wind farms connected to a hybrid

asset receive the price of the offshore zone, rather than that of the onshore zone, which may reduce their revenues; the allocation of costs and benefits between states; and the governance of cross-border infrastructure.

In the **Esbjerg Declaration** of May 2022, Denmark, Germany, Belgium, and the Netherlands committed to developing the North Sea as a "green power plant of Europe", with a combined target of at least 65 GW of offshore wind by 2030 and 150 GW by 2050. In the **Ostend Declaration** of April 2023, nine North Seas states set targets of 120 GW by 2030 and 300 GW by 2050. The **North Seas Energy Cooperation**, an intergovernmental cooperation framework, supports coordination of offshore grid planning. The EU's TEN-E Regulation (Regulation (EU) 2022/869) provides for priority offshore grid corridors and for cooperation among member states in the development of offshore network development plans, and the 2024 electricity market reform addresses cost sharing and revenue arrangements for offshore hybrid projects. Denmark's plans for artificial "energy islands" in the North Sea and on Bornholm in the Baltic, intended as hubs for offshore wind and interconnectors, illustrate both the ambition and the difficulties of offshore grid development: Denmark put the North Sea energy island on hold after cost estimates rose substantially.

Wave, Tidal, and Ocean Thermal Energy

Wave and tidal energy technologies require legal frameworks for seabed leasing, consenting, and support. The United Kingdom's Contracts for Difference scheme has included ring-fenced budgets for tidal stream energy in recent allocation rounds, supporting deployment. The proposed Swansea Bay tidal lagoon, which would have used a seawall to harness the tidal range, was rejected by the United Kingdom government in 2018 on the ground that it did not offer value for money, illustrating the significance of support mechanisms and cost for novel technologies. Tidal range and tidal stream projects raise environmental issues relating to marine mammals, fish, and estuarine ecosystems, and require appropriate assessment under nature conservation law. OTEC, which is of interest to tropical island states, raises issues relating to the discharge of cold deep water and the location of installations in deep waters.

Offshore Renewables in Africa

Offshore renewable energy development in Africa is at an early stage. South Africa has identified significant offshore wind potential, and the World Bank's Offshore Wind Development Program has supported assessments of offshore wind resources in several African countries, including Morocco, Mauritania, South Africa, and Nigeria, and the development of roadmaps. Nigeria does not have a legal framework specifically addressing offshore renewable energy; offshore activities in its territorial sea and EEZ are governed principally by petroleum legis-

lation, the **Exclusive Economic Zone Act 1978**, and the **Territorial Waters Act 1967**, as amended, and the allocation of offshore areas for renewable energy would require consideration of the interaction with petroleum licensing under the PIA, coastal state jurisdiction, and federal–state relations. The development of offshore renewable frameworks in African states offers an opportunity to learn from the experience of other jurisdictions in the design of leasing, consenting, local content, and decommissioning rules.

Decommissioning of Offshore Renewable Installations

The decommissioning of offshore renewable installations is governed by Article 60(3) of UNCLOS, the IMO Guidelines of 1989, and, in the North-East Atlantic, the OSPAR Convention, together with domestic law. Domestic regimes typically require developers to prepare decommissioning programmes and to provide financial security. In the United Kingdom, the Energy Act 2004 requires persons responsible for offshore renewable energy installations to submit decommissioning programmes for approval and empowers the Secretary of State to require financial security. The decommissioning of offshore wind farms raises questions similar to those of offshore petroleum installations, examined in Chapter 5, including the extent of removal required (particularly of foundations, scour protection, and cables), the potential for "rigs-to-reefs" approaches, and the recycling of turbine blades.

12.7 Smart Grids, Digital Infrastructure, and Cybersecurity

The Digitalisation of Energy Systems

The electricity system is being transformed by digitalisation: smart meters that measure consumption at short intervals and communicate data to suppliers and network operators; sensors and automated controls on transmission and distribution networks; distributed energy resources — rooftop solar, batteries, electric vehicles, and heat pumps — that can be controlled remotely and aggregated; digital platforms for peer-to-peer trading and flexibility markets; and the use of artificial intelligence to forecast demand and renewable output and to optimise system operation. A **smart grid** is an electricity network that uses digital technologies to monitor and manage the flow of electricity from all generation sources to meet the varying demands of end users efficiently, reliably, and sustainably. At the same time, the rapid growth of electricity demand from data centres, driven by cloud computing and artificial intelligence, has created new pressures on electricity systems and new questions of grid access, cost allocation, and planning, examined in Chapter 9.

Digitalisation raises legal issues in four principal areas: the regulation of smart metering; the protection of energy data and privacy; the governance of data access and interoperability; and cybersecurity.

Smart Metering

The deployment of smart meters has been mandated or encouraged in many jurisdictions. In the European Union, the Third Energy Package required member states to ensure the implementation of intelligent metering systems where a cost-benefit assessment was positive, with a target of rolling out smart meters to at least eighty per cent of consumers by 2020. Articles 19 to 21 of Directive (EU) 2019/944 set out requirements for smart metering systems, including functionality, the right of consumers to access their data, data protection, cybersecurity, and interoperability, and the right of every consumer to request a smart meter where the roll-out is not positively assessed. In Great Britain, energy suppliers were required by licence conditions to take all reasonable steps to roll out smart meters to their domestic and small business customers, and the Data Communications Company, a licensed body, provides a national communications infrastructure for smart meters.

In Nigeria, metering is a critical issue for the electricity sector, because a large proportion of customers have been unmetered and charged through "estimated billing", which has been a major source of consumer grievance and revenue leakage. NERC's **Meter Asset Provider Regulations** (2018, subsequently revised) enabled third-party meter asset providers to finance, supply, and install meters for distribution companies, and the **National Mass Metering Programme**, launched in 2020, sought to provide meters free of charge to unmetered customers. NERC has issued orders capping estimated bills for unmetered customers, and the transition to prepayment and smart meters raises issues of consumer protection, data, and the security of revenue examined in Chapter 9.

Energy Data and Privacy

Smart meters and connected devices generate granular data on energy consumption that can reveal intimate details of private life, including occupancy patterns, daily routines, and the use of particular appliances. Energy data is therefore personal data subject to data protection law where it relates to identified or identifiable individuals. In the European Union, the **General Data Protection Regulation** (Regulation (EU) 2016/679) applies to the processing of energy data, requiring a lawful basis for processing, purpose limitation, data minimisation, and security, and the Electricity Directive requires member states to ensure that the processing of consumers' metering data complies with data protection law. In the Netherlands, the initial proposal for a mandatory smart meter roll-out was withdrawn in 2009 after the Senate expressed concerns that compulsory installation would infringe the right to respect for private life under Article 8 of the European Convention on Human Rights, and the roll-out proceeded on a voluntary basis with privacy safeguards. In Nigeria, the **Nigeria Data Protection Act 2023**, which established the Nigeria Data Protection Commission, governs

the processing of personal data, including by electricity distribution companies and meter providers.

Data Access and Interoperability

The value of energy data for innovation — enabling new services such as demand response, energy efficiency advice, and the aggregation of flexibility — depends on the ability of consumers and authorised third parties to access data in standardised, interoperable formats. Article 23 of Directive (EU) 2019/944 requires member states to specify rules on access to data of the final customer by eligible parties, ensuring efficient and secure data access and exchange, and Article 24 provides for the adoption of interoperability requirements and non-discriminatory and transparent procedures for access to data. The EU's **Data Act** (Regulation (EU) 2023/2854) establishes general rules on access to data generated by connected products, including smart energy devices, giving users the right to access and share such data with third parties. The governance of energy data raises tensions between innovation, competition, privacy, and security.

Cybersecurity

The digitalisation of energy systems has increased their exposure to cyberattacks, which can disrupt supply, damage equipment, and endanger safety. In December 2015, a cyberattack on distribution companies in western Ukraine caused power outages affecting around two hundred and twenty-five thousand customers — the first publicly confirmed cyberattack causing a power outage — and a further attack in December 2016, using malware designed specifically to attack electricity grid control systems, caused an outage in part of Kyiv. In May 2021, a ransomware attack on the information technology systems of **Colonial Pipeline**, which operates the largest refined products pipeline system in the United States, led the company to shut down the pipeline for several days, causing fuel shortages on the east coast; the company paid a ransom, part of which was subsequently recovered by the Department of Justice. In response, the Transportation Security Administration issued security directives imposing mandatory cybersecurity requirements on pipeline operators, illustrating the shift from voluntary to mandatory cybersecurity regulation in the energy sector.

Regulatory frameworks. In the United States, the North American Electric Reliability Corporation, the electric reliability organisation certified by FERC under section 215 of the Federal Power Act (added by the Energy Policy Act of 2005), develops mandatory **Critical Infrastructure Protection** (CIP) reliability standards, approved and enforced by FERC, which require owners and operators of the bulk electric system to identify critical cyber assets and implement security controls, with substantial penalties for violations. In the European Union, the **NIS2 Directive** (Directive (EU) 2022/2555) imposes

cybersecurity risk management and incident reporting obligations on essential and important entities, including in the energy sector (electricity, district heating, oil, gas, and hydrogen); the **Critical Entities Resilience Directive** (Directive (EU) 2022/2557) addresses the physical and other resilience of critical entities; and the **Network Code on Cybersecurity** for cross-border electricity flows (Commission Delegated Regulation (EU) 2024/1366) establishes sector-specific rules for cybersecurity risk assessment and management in the electricity sector. In Nigeria, the **Cybercrimes (Prohibition, Prevention, etc.) Act 2015**, as amended in 2024, provides for the designation and protection of critical national information infrastructure and criminalises attacks on it.

International law. Cyberattacks on energy infrastructure by states or state-sponsored actors raise questions of international law, including the prohibition of the use of force, the principle of non-intervention, state responsibility, and, in armed conflict, international humanitarian law. The reports of the United Nations Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security, endorsed by the General Assembly, affirmed in 2013 that international law, and in particular the United Nations Charter, applies to cyberspace, and in 2015 set out voluntary norms of responsible state behaviour, including that a state should not conduct or knowingly support activity contrary to its obligations under international law that intentionally damages critical infrastructure or otherwise impairs the use and operation of critical infrastructure to provide services to the public. The *Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations* (2017), prepared by an international group of experts, offers a scholarly restatement of the application of international law to cyber operations, including the protection of critical infrastructure. These questions are examined further in Chapter 25.

12.8 Regulatory Sandboxes and Technological Innovation

The Regulatory Challenge of Innovation

Energy regulation faces a recurrent dilemma in responding to technological innovation. Rules designed for existing technologies and business models may inadvertently prohibit, disadvantage, or create uncertainty for innovations — such as peer-to-peer trading, community energy, vehicle-to-grid services, local flexibility markets, and novel tariff structures — that do not fit established categories. Yet regulators cannot simply suspend the rules that protect consumers, ensure system security, and prevent the abuse of market power, and they often lack the information needed to design new rules for technologies whose characteristics and risks are not yet understood. This is a variant of the "Collingridge dilemma":

at an early stage, when a technology can still be shaped, its impacts are difficult to predict; once its impacts are known, it may be entrenched and difficult to regulate.

Several instruments have been developed to address this dilemma, including **experimentation clauses**, which authorise regulators or ministers to permit deviations from statutory requirements for pilot projects; **innovation funding** schemes, such as the network innovation allowances and competitions administered by Ofgem under the RIIO price control framework; **principles-based regulation**, which sets outcomes rather than prescriptive rules; and **regulatory sandboxes**.

The Concept of a Regulatory Sandbox

A regulatory sandbox is a framework established by a regulator that allows innovators to test new products, services, or business models in a real-world environment, for a limited period and subject to safeguards, with temporary relief from specified regulatory requirements or with regulatory guidance on how existing rules apply. The concept originated in financial regulation, with the United Kingdom Financial Conduct Authority's sandbox launched in 2016, and was rapidly adopted in the energy sector and elsewhere. The OECD's **Recommendation of the Council for Agile Regulatory Governance to Harness Innovation** (2021) encourages the use of experimental approaches, including sandboxes, as part of agile regulatory governance.

Energy regulatory sandboxes have been established in many jurisdictions. **Ofgem's Energy Regulation Sandbox**, launched in 2017, offers innovators bespoke guidance, comfort, and, where Ofgem has the power to do so, derogations from licence conditions and industry codes for trials, and has supported trials of peer-to-peer trading, local energy markets, and innovative tariffs. **Singapore's** Energy Market Authority launched a regulatory sandbox in 2017. **Germany** adopted the SINTEG Regulation (2017), which permitted participants in the "Smart Energy Showcases" programme to be compensated for certain economic disadvantages arising from trials, and subsequently developed the concept of "real-world laboratories" (*Reallabore*) for the energy transition. In the European Union, the **Net-Zero Industry Act** (Regulation (EU) 2024/1735) requires member states to establish regulatory sandboxes for net-zero technologies, and the AI Act (Regulation (EU) 2024/1689) requires the establishment of AI regulatory sandboxes, which may encompass energy applications.

::: casebox {title="Regulatory Sandboxing in the Australian National Electricity Market (2022)"} **Background.** In 2019, the Australian Energy Market Commission (AEMC) recommended the establishment of a regulatory sandboxing framework for the National Electricity Market. The Commission concluded that the existing regulatory framework, which is set out in detailed national legis-

lation and rules, contained barriers to the trialling of innovative technologies and business models. It also found that the regulators lacked the powers needed to grant relief for trials.

The framework. Legislation amending the National Energy Laws, enacted in 2022, together with the National Electricity Amendment (Regulatory sandboxing) Rule 2022, established a framework with three components:

- an **innovation enquiry service**, provided by the Australian Energy Regulator (AER), which offers guidance to innovators on how the existing framework applies to their proposals;
- **trial waivers**, which the AER may grant to exempt a trial project from specified provisions of the National Electricity Rules for a limited period, subject to conditions;
- **trial rules**, which the AEMC may make to enable trials that require exemptions beyond the AER's waiver powers, such as exemptions from requirements of the national legislation.

Trial projects must be genuinely innovative and must have the potential to contribute to the achievement of the national energy objective. They must be subject to appropriate consumer protections and must be of limited duration and scale.

Significance. The Australian framework is notable for giving sandboxing an express statutory basis, for distinguishing guidance, waivers, and rule-making, and for linking trials to the possibility of permanent rule changes where they succeed. It addresses the principle of legality by ensuring that regulators have clear authority to grant relief from binding legal requirements. That authority was absent from some earlier sandboxes, which could offer only guidance or comfort. :::

Legal Issues Raised by Sandboxes

Regulatory sandboxes raise several legal questions.

Legal basis. Under the principle of legality, a regulator may waive or modify binding legal requirements only if it has statutory authority to do so. Where a sandbox involves relief from requirements contained in primary legislation, express legislative authorisation is required, as in Australia. Where the regulator lacks such authority, it may be able to offer only guidance, informal comfort, or the non-enforcement of rules — which may provide limited legal certainty for participants and may raise questions about the legitimacy of selective non-enforcement.

Equal treatment and competition. The selection of participants for a sandbox, and the grant of exemptions to some market participants but not others, may confer competitive advantages and raise questions of equal treatment, non-discrimination, and, in the European Union, State aid. Sandboxes therefore

require transparent and objective eligibility criteria, open application processes, and publication of decisions.

Consumer protection. Consumers participating in trials may be exposed to risks — of higher prices, interruption of supply, or loss of the protections they would otherwise enjoy. Sandbox frameworks typically require informed consent, maintenance of core protections (such as protections for vulnerable customers and access to dispute resolution), exit arrangements returning consumers to standard arrangements at the end of a trial, and limits on the number of participating consumers.

From experiment to rule. The value of sandboxes lies not only in enabling individual trials but also in informing permanent regulatory change. Frameworks that provide for the systematic evaluation and publication of trial results, and for the consideration of rule changes where trials succeed, are more likely to achieve systemic reform. Conversely, sandboxes that merely grant temporary exemptions without a pathway to rule change may create a cycle of trials without lasting effect.

Regulatory capacity and capture. Sandboxes demand significant regulatory resources and close engagement between regulators and innovators, which raises concerns about regulatory capture and the regulator's ability to maintain its independence and impartiality.

Innovation and Regulation in Developing Countries

Regulatory sandboxes and innovation-friendly frameworks are of particular relevance to developing countries, where new technologies and business models — such as pay-as-you-go solar, mobile money payment for electricity, mini-grids, productive use appliances, and digital metering — have expanded energy access faster than traditional utility models, often operating in gaps or grey areas of the regulatory framework. In Nigeria, the **Nigeria Startup Act 2022** provides a framework for supporting technology start-ups, including through regulatory sandboxes, and financial regulators have adopted sandbox frameworks for financial technology, relevant to the payment systems used in energy access business models. The NERC Mini-Grid Regulations, examined in Chapter 10, are an example of a regulatory framework designed to accommodate a new delivery model with tailored, lighter-touch requirements. Regulators in developing countries face the challenge of encouraging innovation while protecting consumers, ensuring quality, and integrating new models with the expansion of the main grid, and comparative study of the design of energy sandboxes is a promising area for doctoral research.

::: exercise {title="Practical Exercise: The Daruna Green Ammonia Export Project (HYPOTHETICAL)"} *This exercise is entirely hypothetical. The Republic*

of Daruna, the Port Asaba Special Economic Zone, Solveig Hydrogen AS, and all other entities named are fictional.

The Republic of Daruna is a coastal West African state with excellent solar and wind resources, limited freshwater, and a national grid in which 70 per cent of electricity is generated from natural gas. Solveig Hydrogen AS (HYPOTHETICAL), a European developer, proposes a USD 8 billion project in the Port Asaba Special Economic Zone. The project comprises:

- 3 GW of solar and wind generation on 40,000 hectares of land, part of which is held under customary tenure by the Ilaro and Kemi communities;
- a 1.5 GW electrolyser;
- a desalination plant;
- an ammonia synthesis plant and export terminal.

The ammonia will be sold to European buyers seeking to qualify it as an RFNBO, including under an H2Global-style tender. Daruna has no hydrogen legislation. Its Electricity Act requires all generators above 1 MW to hold a licence and to sell to the national grid company, and its Water Act vests all water in the state.

Solveig proposes the following arrangements:

- (a) the renewable plants will be connected both directly to the electrolyser and to the national grid, so that surplus electricity can be sold to the grid and grid electricity can be used when renewable output is low;
- (b) a twenty-five-year host government agreement will grant Solveig exemption from the Electricity Act, a ten-year tax holiday, stabilisation of the legal framework, and the right to export all of the ammonia;
- (c) the government will acquire a 10 per cent carried interest in the project company.

Community groups and civil society organisations have argued that the project should supply electricity and water to local communities, which have low rates of access, and that the government should obtain a larger share of the benefits.

1. Identify the legislative reforms required to enable the project, including the classification of hydrogen and ammonia, licensing of production, electricity sector rules, water rights, and land acquisition.
2. Advise Solveig on whether the ammonia can qualify as an RFNBO under Delegated Regulation (EU) 2023/1184, given arrangement (a), and on the certification arrangements required. How do the additionality and temporal correlation requirements affect the project's design?
3. Evaluate arrangement (b) from the perspective of Daruna's long-term interests, with reference to the material on stabilisation clauses in Chapter 14 and on resource governance in Chapter 7. Draft alternative provisions that would better balance investor protection and the state's regulatory autonomy.

4. Propose legal mechanisms by which the project could contribute to domestic energy access and water supply and share benefits with the Ilaro and Kemi communities. Consider the risk of "green extractivism" and the requirements of free, prior, and informed consent discussed in Chapter 21.
5. Would a regulatory sandbox be an appropriate instrument for enabling a pilot phase of the project? Explain your answer. :::

CHAPTER SUMMARY

- Hydrogen, storage, CCUS, offshore renewables, and smart grids are essential to decarbonisation. Each strains legal categories designed for an energy system with different technologies and boundaries.
- The classification of hydrogen determines the applicable regimes of gas, chemical, and safety law. The EU's 2024 Hydrogen and Decarbonised Gas Market Package creates a dedicated framework for hydrogen networks, with unbundling and regulated third-party access from 2033.
- The colour terminology for hydrogen is legally imprecise, and law has moved towards carbon intensity criteria. The EU's RFNBO rules require additionality, temporal correlation, and geographical correlation. The US 45V credit adopted analogous requirements before it was curtailed in 2025.
- Certification, through book-and-claim or mass balance systems, underpins regulatory and commercial claims about hydrogen. International harmonisation through ISO 19870 and mutual recognition is incomplete. International trade in hydrogen raises questions under the GATT, the TBT Agreement, the SCM Agreement, and CBAM.
- EU law defines electricity storage as a distinct activity, protects storage owners against double charging, and restricts network operator ownership of storage. FERC Order No 841, upheld in *NARUC v FERC*, opened wholesale markets to storage.
- CCUS is governed by the London Protocol, whose 2006 and 2009 amendments permit sub-seabed storage and its export, and by domestic regimes such as the EU CCS Directive and the US Class VI programme. Pore space ownership and long-term liability are distinctive issues.
- UNCLOS grants coastal states sovereign rights over energy production from winds and currents in the EEZ, subject to obligations regarding installations, safety zones, navigation, and removal. Offshore grids and hybrid projects raise complex cross-border questions.
- Smart grids raise issues of smart metering, data protection, data access, and cybersecurity. Mandatory cybersecurity regulation of energy infrastructure has expanded, as the NERC CIP standards, NIS2, and the response to Colonial Pipeline show.

- Regulatory sandboxes enable controlled experimentation but require a clear legal basis, equal treatment, consumer protection, and a pathway from successful trials to permanent rule change.

DOCTORAL RESEARCH QUESTIONS

1. Is a carbon intensity approach to the definition of low-carbon hydrogen preferable to a source-based approach? Evaluate the EU's RFNBO criteria in the light of environmental integrity, cost, and the development interests of exporting states.
2. To what extent do the EU's rules on renewable hydrogen and their application to imports constitute a form of extraterritorial regulation, and are they compatible with WTO law?
3. Who should own pore space, and who should bear long-term liability for stored carbon dioxide? Compare the approaches of two or more jurisdictions and propose a model for a developing country.
4. Do the rights of coastal states under UNCLOS adequately accommodate large-scale offshore energy development, including beyond the EEZ? Is there a need for new international rules on offshore energy installations?
5. Critically assess whether existing international law adequately protects energy infrastructure against cyberattacks. What role should the norms of responsible state behaviour play?
6. Evaluate the potential of regulatory sandboxes to accelerate energy access in sub-Saharan Africa. What legal design features are necessary to protect consumers and ensure legitimacy?

FURTHER READING

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- Roggenkamp, Martha M and others (eds), *Energy Law in Europe* (Oxford University Press).
- Schmitt, Michael N (ed), *Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations* (Cambridge University Press 2017).
- Organisation for Economic Co-operation and Development, *Recommendation of the Council for Agile Regulatory Governance to Harness Innovation* (OECD 2021).

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PART IV

International Trade, Investment, and Energy Contracts

The rules of international trade and investment that govern cross-border energy commerce and protect energy investments, and the contractual and financial architecture — from production sharing to project finance — through which energy projects are structured, funded, and governed.

13 CHAPTER

International Trade Law and Cross-Border Energy Commerce

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the distinctive characteristics of trade in energy commodities and the historical relationship between energy and the multilateral trading system.
2. Analyse the application of the core WTO disciplines — most-favoured-nation treatment, national treatment, the prohibition of quantitative restrictions, freedom of transit, and the general and security exceptions — to energy trade.
3. Evaluate the law of subsidies and trade remedies as applied to energy, including renewable energy support with local content requirements and fossil fuel subsidies.
4. Assess the legality of export restrictions on energy commodities, including export duties, licensing, and dual pricing.
5. Analyse the disciplines on technical regulations, standards, and market access for energy goods and services.
6. Evaluate the energy provisions of regional trade agreements, including the African Continental Free Trade Area and ECOWAS instruments.
7. Assess the compatibility of carbon border adjustment measures, including the EU CBAM, with WTO law and the climate regime.
8. Critically analyse the jurisprudence of trade disputes involving energy and natural resources, and the consequences of the crisis of the WTO Appellate Body.

Introduction

Energy is the most traded commodity group in the world. Crude oil and refined products, natural gas (by pipeline and as LNG), coal, and, increasingly, electricity, hydrogen derivatives, and the equipment and minerals required for renewable energy cross national borders in enormous volumes. The rules governing this trade — on tariffs, quantitative restrictions, subsidies, transit, standards, and the treatment of foreign goods and services — shape the economics of energy production

and consumption and the distribution of the gains from trade between producing and consuming states. As the energy transition proceeds, trade law has become a central arena in which states contest the legality of industrial policies supporting clean energy manufacturing, carbon border measures, and export restrictions on critical minerals.

For much of the history of the multilateral trading system, energy occupied an anomalous position. Although the General Agreement on Tariffs and Trade of 1947 (GATT) made no exception for energy goods, energy trade was, in practice, largely outside the system's effective disciplines. Many major oil-exporting states were not parties to the GATT; international oil markets were dominated first by the major international oil companies and later by the Organization of the Petroleum Exporting Countries (OPEC); energy was widely regarded as a strategic sector subject to national security considerations; and the principal concerns of trade negotiators were import tariffs and restrictions in manufacturing and agriculture rather than export restrictions on natural resources. The creation of the World Trade Organization in 1995, the accession of major energy producers — including Saudi Arabia in 2005 and Russia in 2012 — and the increasing salience of energy and climate policy have brought energy firmly within the scope of WTO law.

This chapter examines energy commodities and international trade rules (section 13.1), the WTO disciplines relevant to energy (section 13.2), tariffs, subsidies, and trade remedies (section 13.3), export restrictions and resource security (section 13.4), technical standards and market access (section 13.5), regional trade agreements and energy integration (section 13.6), carbon border measures (section 13.7), and trade disputes involving energy and natural resources (section 13.8). The protection of foreign investment in the energy sector, which is governed by a distinct body of international investment law, is examined in Chapter 14; the trade law of critical minerals has been introduced in Chapter 8; and the trade law dimensions of national security are examined further in Chapter 25.

13.1 Energy Commodities and International Trade Rules

The Characteristics of Energy Trade

Trade in energy differs from trade in most other goods in several respects that affect the operation of trade rules.

Geographical concentration and the "resource curse". Energy resources are unevenly distributed. A relatively small number of states hold most of the world's proven reserves of oil and gas, and many of these states are heavily dependent on energy exports for government revenue and foreign exchange. Consuming states are correspondingly dependent on imports. This asymmetry gives

energy trade a geopolitical dimension and creates incentives for both exporting and importing states to intervene in markets.

Export-side concerns. Whereas the traditional concern of trade law has been import barriers, the central concerns in energy trade frequently relate to exports — export restrictions, export taxes, and production quotas that restrict supply and raise world prices, and dual pricing policies under which energy is sold domestically at prices below export prices. As examined in section 13.4, the GATT's disciplines on export measures are weaker than those on import measures.

Network dependence and transit. Pipeline gas and electricity are traded through fixed networks that often cross the territories of third states. Trade therefore depends on access to networks and on freedom of transit, and the owners of networks have significant market power. Transit disputes, such as those between Russia and Ukraine examined in Chapter 6, illustrate the vulnerability of networked energy trade.

State ownership and state trading. National oil companies and state-owned utilities play a dominant role in the energy sector in many states, raising questions about the application of trade disciplines to state-owned enterprises and state trading enterprises.

Environmental externalities. The production and consumption of fossil fuels generate greenhouse gas emissions and other environmental harms that are not reflected in market prices. Measures to address these externalities — carbon taxes, emissions standards, renewable energy support, and carbon border adjustments — have significant trade effects and raise questions of compatibility with trade rules.

Energy as Goods and Services

A preliminary question is whether energy is a good, a service, or both, because different WTO agreements govern trade in goods (the GATT and associated agreements) and trade in services (the General Agreement on Trade in Services, or GATS). Crude oil, petroleum products, natural gas, coal, and uranium are clearly goods, classified in Chapter 27 (mineral fuels and oils) and other chapters of the Harmonized Commodity Description and Coding System (HS), the international nomenclature for the classification of goods administered by the World Customs Organization. **Electricity** is classified in the HS under heading 27.16, and is generally treated as a good for the purposes of the GATT, although its intangible character and the inseparability of its supply from network services have led some to question the classification. The Court of Justice of the European Union treated electricity as a good for the purposes of the free movement of goods in *Almelo* (Case C-393/92, 1994).

Many activities associated with energy are services: exploration and drilling services, pipeline transportation, the transmission and distribution of electricity, energy trading, engineering and construction, and consulting. These are governed by the GATS, under which members' obligations depend substantially on their specific commitments. There is no specific sectoral classification for "energy services" in the services classification list used in the Uruguay Round, and energy-related services are scattered across several categories — services incidental to mining, pipeline transportation of fuels, services incidental to energy distribution, and wholesale and retail trade in fuels — with the result that commitments in energy services have been limited and uneven. Proposals to improve the classification of energy services were made in the Doha Round negotiations, without result.

The Historical Position of Energy in the GATT

The GATT of 1947 applied in principle to all goods, including energy. However, energy was largely absent from the agenda of the successive rounds of GATT negotiations, for several reasons. The Havana Charter of 1948, which would have established an International Trade Organization, contained a chapter on intergovernmental commodity agreements that would have permitted arrangements regulating the production and trade of primary commodities, but the Charter never entered into force. Major oil-exporting states, many of which achieved independence after 1947, did not join the GATT for many years. The formation of OPEC in 1960, and its assertion of control over production and pricing in the 1970s, took place outside the GATT framework. The industrialised states that dominated the GATT were importers of energy and did not seek disciplines on export restrictions, preferring to address energy security through the International Energy Agency, established in 1974, examined in Chapter 4. A persistent view held that the GATT did not apply to natural resources in the ground, or to the production decisions of sovereign states over their natural resources, reflecting the principle of permanent sovereignty over natural resources examined in Chapter 2.

Energy in the WTO

The Uruguay Round agreements, which established the WTO in 1995, apply to energy goods and services on the same terms as to other goods and services, and there is no specific WTO agreement on energy. The accession of major energy exporters has brought energy trade within the WTO in practice, and accession protocols have included energy-specific commitments. The accession of **Saudi Arabia** in 2005 involved negotiations on the dual pricing of natural gas liquids used as feedstock for the petrochemical industry, with Saudi Arabia undertaking that producers and distributors of natural gas liquids would operate on the basis

of normal commercial considerations and recover costs and a reasonable profit. The accession of **Russia** in 2012 included commitments on export duties for a range of products, and commitments on the pricing of natural gas, under which producers and distributors of natural gas would operate on the basis of normal commercial considerations, based on recovery of costs and profit.

The WTO has not, however, adopted rules addressing some of the most significant features of energy trade, such as OPEC's production coordination, investment in energy infrastructure, competition in network industries, and the relationship between trade and energy security. The Energy Charter Treaty, examined in Chapter 14, was designed in part to fill these gaps, applying GATT rules to energy trade among its parties (including those not members of the WTO, under its trade provisions as amended in 1998) and establishing specific obligations on transit and investment.

13.2 World Trade Organization Disciplines

Most-Favoured-Nation Treatment

The cornerstone of the multilateral trading system is the obligation of **most-favoured-nation** (MFN) treatment in Article I:1 of the GATT, which requires that any advantage, favour, privilege, or immunity granted by a member to any product originating in or destined for any other country be accorded immediately and unconditionally to the like product originating in or destined for the territories of all other members. The MFN obligation applies to customs duties and charges on imports and exports, the method of levying them, rules and formalities in connection with importation and exportation, and internal taxes and regulations. In the energy sector, the MFN obligation prohibits members from discriminating between imported energy products according to their country of origin — for example, by imposing higher tariffs on oil from one member than from another — or between exports according to destination. Measures that differentiate between energy products on the basis of their production methods, such as the carbon intensity of production, may raise MFN issues if they result in less favourable treatment of products from particular members, as examined in section 13.7.

National Treatment

Article III of the GATT requires members to accord imported products treatment no less favourable than that accorded to like domestic products in respect of internal taxes and other internal charges (Article III:2) and laws, regulations, and requirements affecting their internal sale, offering for sale, purchase, transportation, distribution, or use (Article III:4). The national treatment obligation is central to the legality of energy and environmental measures, including fuel taxes,

energy efficiency standards, renewable energy support schemes, and local content requirements.

The application of Article III depends on the concept of **"like products"**. In *EC — Asbestos* (2001), the Appellate Body held that the determination of likeness under Article III:4 is fundamentally a determination about the nature and extent of a competitive relationship between products, considering the products' physical properties, end uses, consumers' tastes and habits, and tariff classification, and that health risks associated with a product may be relevant to likeness in so far as they affect the competitive relationship, for example through consumers' perceptions. The question whether products that are physically identical but produced by methods with different environmental impacts — for example, electricity generated from coal and from solar energy, or steel produced with high and low emissions — can be regarded as "unlike" is one of the most contested issues in trade and environment law, known as the issue of **non-product-related processes and production methods** (NPR-PPMs).

Even where products are like, a measure violates Article III:4 only if it accords "less favourable treatment" to imported products. The Appellate Body held in *Korea — Various Measures on Beef* (2001) that less favourable treatment requires a modification of the conditions of competition to the detriment of imported products. In *US — Clove Cigarettes* (2012), interpreting the parallel national treatment obligation of the TBT Agreement, it held that a detrimental impact on imports does not establish less favourable treatment if it stems exclusively from a legitimate regulatory distinction — a reasoning not extended to Article III:4 of the GATT, under which regulatory justifications are to be considered under Article XX.

Quantitative Restrictions and State Trading

Article XI:1 of the GATT prohibits prohibitions or restrictions, other than duties, taxes, or other charges, on the importation or exportation of products, whether made effective through quotas, import or export licences, or other measures. The prohibition applies equally to imports and exports, and is therefore of particular importance in the energy sector, where export restrictions are common. Its application to export restrictions on energy and minerals is examined in section 13.4.

Article XVII of the GATT requires members that establish or maintain **state trading enterprises** — enterprises granted exclusive or special privileges — to ensure that they act, in their purchases or sales involving imports or exports, in a manner consistent with the general principles of non-discriminatory treatment, and make such purchases or sales solely in accordance with commercial considerations. The application of Article XVII to national oil companies and state-owned

utilities is uncertain, and in *Canada — Wheat Exports and Grain Imports* (2004), the Appellate Body held that Article XVII:1 does not impose a broad obligation on state trading enterprises to act commercially, but rather requires them to act in a non-discriminatory manner, with the commercial considerations requirement serving to define the content of non-discrimination.

Freedom of Transit

Article V of the GATT establishes the principle of **freedom of transit**. Article V:2 provides that there shall be freedom of transit through the territory of each member, via the routes most convenient for international transit, for traffic in transit to or from the territory of other members, and that no distinction shall be made based on the flag of vessels, the place of origin, departure, entry, exit, or destination, or on any circumstances relating to the ownership of goods, of vessels, or of other means of transport. Article V:3 prohibits unnecessary delays or restrictions on traffic in transit and exempts it from customs duties and transit duties, except charges for transportation or commensurate with administrative expenses or the cost of services rendered, and Article V:4 requires charges on traffic in transit to be reasonable.

The application of Article V to fixed infrastructure such as pipelines and electricity networks has been debated. The text refers to "routes most convenient for international transit", and it has been argued that the provision was designed for transport by road, rail, and waterway and does not require members to grant access to pipelines or to construct new transit infrastructure. In *Colombia — Ports of Entry* (2009), the panel held that Article V:2 requires members to extend freedom of transit via the routes most convenient for international transit, but did not address fixed infrastructure. In *Russia — Traffic in Transit* (2019), examined below, the panel considered measures affecting road and rail transit. The Energy Charter Treaty's transit provision, Article 7, examined in Chapter 6, was designed to address the specific features of energy transit through fixed infrastructure, and the WTO Trade Facilitation Agreement (2013, in force 2017) contains further provisions on freedom of transit.

The General Exceptions: Article XX

Article XX of the GATT permits members to adopt measures that would otherwise violate GATT obligations, provided that the measures fall within one of the listed exceptions and satisfy the requirements of the introductory clause ("chapeau"). The exceptions most relevant to energy and environmental measures are those for measures:

- (b) necessary to protect human, animal, or plant life or health;
- (d) necessary to secure compliance with laws or regulations which are not inconsistent with the provisions of the GATT;

- (g) relating to the conservation of exhaustible natural resources, if such measures are made effective in conjunction with restrictions on domestic production or consumption;
- (i) involving restrictions on exports of domestic materials necessary to ensure essential quantities of such materials to a domestic processing industry during periods when the domestic price of such materials is held below the world price as part of a governmental stabilisation plan;
- (j) essential to the acquisition or distribution of products in general or local short supply.

The chapeau requires that measures not be applied in a manner which would constitute a means of arbitrary or unjustifiable discrimination between countries where the same conditions prevail, or a disguised restriction on international trade.

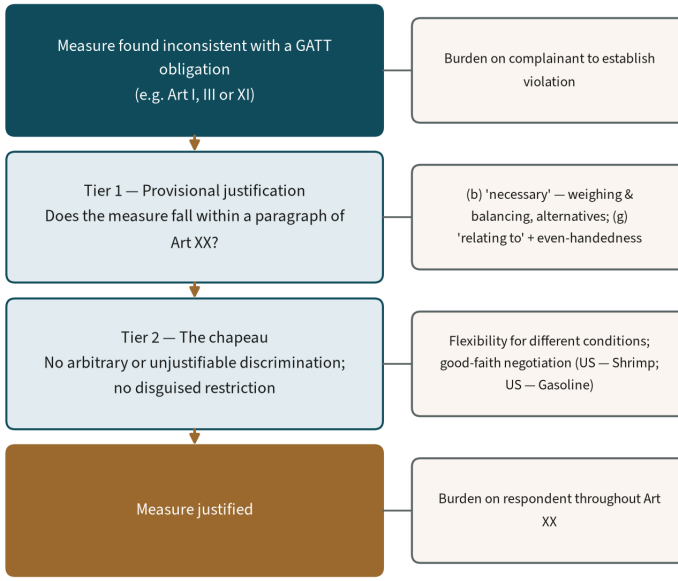


Figure 13.1 The two-tier test under Article XX of the GATT. *Source:* prepared for this edition.

The Appellate Body has established a **two-tier test** for the application of Article XX: first, the measure must be provisionally justified under one of the specific exceptions; second, it must satisfy the requirements of the chapeau. In *US — Gasoline* (1996), the first dispute decided by the Appellate Body, the United States had adopted regulations under the Clean Air Act establishing baselines for the cleanliness of reformulated and conventional gasoline. Domestic refiners could use individual baselines, while foreign refiners were generally required to use a

statutory baseline. The Appellate Body held that clean air was an exhaustible natural resource within the meaning of Article XX(g), and that the baseline rules "related to" its conservation because they bore a substantial relationship to that objective and were made effective in conjunction with restrictions on domestic gasoline. However, it found that the measures failed the chapeau because the United States had not pursued cooperative arrangements with the governments of foreign refiners and had not considered the costs imposed on foreign refiners, resulting in unjustifiable discrimination and a disguised restriction on international trade. The Appellate Body also emphasised that the GATT is "not to be read in clinical isolation from public international law", a statement of considerable significance for the interaction of trade law with other branches of international law, including environmental and energy law.

In *US — Shrimp* (1998), the Appellate Body held that living species, including sea turtles, could be exhaustible natural resources, interpreting the term in an evolutionary manner by reference to contemporary international environmental law, including UNCLOS, the Convention on Biological Diversity, and the Rio Declaration. It held that measures conditioning market access on whether exporting members had adopted policies comparable to those of the importing member could fall within Article XX, but found that the US measure failed the chapeau because of its rigid application and the failure to negotiate with all affected exporters. Following modifications, the revised measure was found to comply in the compliance proceedings (2001). In *Brazil — Retreaded Tyres* (2007), the Appellate Body held that a measure is "necessary" under Article XX(b) where it makes a material contribution to the achievement of its objective, considering the importance of the interests at stake and the trade-restrictiveness of the measure in comparison with reasonably available alternatives, and that measures addressing climate change and other environmental problems may contribute to their objectives over time.

The Security Exception: Article XXI

Article XXI(b) of the GATT provides that nothing in the Agreement shall be construed to prevent any member from taking any action "which it considers necessary for the protection of its essential security interests" relating to fissionable materials or the materials from which they are derived (sub-paragraph (i)), relating to the traffic in arms, ammunition, and implements of war and to such traffic in other goods and materials as is carried on directly or indirectly for the purpose of supplying a military establishment (sub-paragraph (ii)), or "taken in time of war or other emergency in international relations" (sub-paragraph (iii)). Article XXI is of particular relevance to energy, which is widely regarded as a matter of national security; sub-paragraph (i) expressly refers to fissionable materials, and

energy security measures may be justified as measures taken in an emergency in international relations.

For decades, it was debated whether Article XXI was "self-judging" — that is, whether the invoking member's determination was beyond review by WTO panels. In *Russia — Measures Concerning Traffic in Transit* (2019), Ukraine challenged Russian restrictions on the transit of goods from Ukraine across Russia to Kazakhstan and the Kyrgyz Republic. The panel held that Article XXI is not entirely self-judging: a panel has jurisdiction to determine whether the circumstances listed in the sub-paragraphs objectively exist, including whether there is an "emergency in international relations", and whether the member has articulated its essential security interests and acted in good faith, in the sense that the measures are not implausible as measures protective of those interests. It found that the situation between Russia and Ukraine since 2014 constituted an emergency in international relations and that Russia's measures were justified. Subsequent panels, including in *US — Steel and Aluminium Products* (2022) concerning the United States' tariffs under Section 232 of the Trade Expansion Act of 1962, applied the same approach and rejected the United States' invocation of Article XXI; the United States, which maintains that Article XXI is self-judging, appealed into the void. The security exception and energy is examined further in Chapter 25.

Other WTO Agreements

Several other WTO agreements are relevant to energy trade. The **Agreement on Subsidies and Countervailing Measures** (SCM Agreement) disciplines subsidies, including those for renewable energy and fossil fuels, examined in section 13.3. The **Agreement on Trade-Related Investment Measures** (TRIMs Agreement) prohibits investment measures inconsistent with Articles III and XI of the GATT, including local content requirements, which are common in the energy sector. The **Agreement on Technical Barriers to Trade** (TBT Agreement) governs technical regulations, standards, and conformity assessment procedures, examined in section 13.5. The **Agreement on Safeguards** and the **Anti-Dumping Agreement** govern trade remedies. The **GATS** governs trade in energy services. The plurilateral **Agreement on Government Procurement** (GPA), which applies only to its parties, governs procurement by covered entities, including in some cases energy utilities. The **Agreement on Trade-Related Aspects of Intellectual Property Rights** (TRIPS) governs intellectual property, including in clean energy technologies, the transfer of which has been the subject of debate in the climate negotiations.

Dispute Settlement and the Appellate Body Crisis

WTO disputes are resolved under the **Dispute Settlement Understanding** (DSU), which provides for consultations, adjudication by panels, appellate review by the Appellate Body, and the authorisation of countermeasures (the suspension of concessions) where a member fails to comply with adopted rulings. From December 2019, the Appellate Body ceased to function because the United States blocked the appointment of new members, citing concerns about the Appellate Body's alleged overreach, disregard of procedural deadlines, and treatment of its reports as precedent. As a result, a party that loses before a panel may "appeal into the void", preventing the adoption of the panel report and its enforcement. In response, a group of members established the **Multi-Party Interim Appeal Arbitration Arrangement** (MPIA) in 2020, under Article 25 of the DSU, which provides for appeal arbitration between participating members. The Appellate Body crisis has significantly weakened the enforcement of WTO rules, including in disputes concerning energy and natural resources, and several important energy-related panel reports — including in *EU — Energy Package* and *Indonesia — Raw Materials* — have been appealed into the void.

13.3 Tariffs, Subsidies, and Trade Remedies

Tariffs on Energy Goods

Tariffs on crude oil and natural gas are generally low or zero in most major importing states, because these states depend on imports and seek to minimise energy costs. Tariffs on refined petroleum products, coal, and energy-related equipment vary more widely. Tariffs on clean energy goods — solar panels, wind turbines, and their components — have been the subject of negotiations aimed at liberalisation. The **APEC List of Environmental Goods**, agreed in 2012, committed APEC economies to reduce applied tariffs on fifty-four environmental goods, including solar panels and wind turbines, to five per cent or less by 2015. Plurilateral negotiations on an **Environmental Goods Agreement** among a group of WTO members, launched in 2014, stalled in 2016. Tariffs on clean energy goods have, however, increased in some major markets in recent years as a result of trade remedies and industrial policy, including the substantial increases in US tariffs on Chinese solar cells, batteries, and electric vehicles announced in 2024.

The Law of Subsidies

The SCM Agreement defines a **subsidy** in Article 1 as a financial contribution by a government or public body (including direct transfers of funds, forgone revenue, the provision of goods or services other than general infrastructure, or the purchase of goods), or any form of income or price support, which confers a **benefit**. A subsidy is subject to the Agreement only if it is **specific** to an enterprise or industry or group of enterprises or industries (Article 2). The Agreement clas-

sifies subsidies into two categories: **prohibited subsidies** (Article 3) — subsidies contingent on export performance or on the use of domestic over imported goods — which are prohibited outright; and **actionable subsidies** (Article 5), which are challengeable if they cause adverse effects to the interests of other members, including injury to their domestic industry, nullification or impairment of benefits, or serious prejudice. A category of non-actionable subsidies, which included certain environmental subsidies, lapsed at the end of 1999.

Renewable energy subsidies. Renewable energy support schemes, including feed-in tariffs, tax credits, and grants, may constitute subsidies within the meaning of the SCM Agreement. The most significant questions concern the identification of a "benefit", the legality of local content requirements, and the treatment of support conditioned on domestic production.

In *Canada — Renewable Energy / Canada — Feed-in Tariff Program* (2013), Japan and the European Union challenged the feed-in tariff programme of the Canadian province of Ontario, which offered twenty-year contracts at guaranteed prices to generators of renewable electricity, subject to minimum requirements for domestic content in the equipment used. The Appellate Body upheld the panel's finding that the domestic content requirements violated Article III:4 of the GATT and Article 2.1 of the TRIMs Agreement, rejecting Canada's argument that the programme fell within the government procurement derogation in Article III:8(a) of the GATT, because the product procured by the government (electricity) was not in a competitive relationship with the product discriminated against (generation equipment). On the subsidy claim, the Appellate Body reversed the panel's finding that the complainants had failed to establish a benefit, holding that the benchmark for determining whether the government's purchase of electricity conferred a benefit must be a market that the government itself had created — a competitive wholesale market for electricity generated from wind and solar technologies, reflecting the government's legitimate choice of energy supply mix — rather than the overall wholesale electricity market, but was unable to complete the analysis for lack of factual findings. The Appellate Body's approach was regarded as recognising governments' freedom to create markets for renewable energy without automatically conferring a benefit, while preserving the discipline on local content requirements.

In *India — Certain Measures Relating to Solar Cells and Solar Modules* (2016), the United States challenged domestic content requirements in India's Jawaharlal Nehru National Solar Mission. The Appellate Body upheld the panel's findings that the requirements violated Article III:4 of the GATT and Article 2.1 of the TRIMs Agreement, that the government procurement derogation did not apply, and that the requirements were not justified under Article XX(j) (products in general or local short supply), because solar cells and modules were not in short

supply given the availability of imports, or under Article XX(d), because India had not identified a domestic or international legal obligation with which the measures secured compliance, India's climate obligations under the UNFCCC not being "laws or regulations" with direct effect in Indian law. In *United States — Certain Measures Relating to the Renewable Energy Sector* (2019), brought by India, the panel found that local content requirements and incentives in renewable energy programmes of several US states violated Article III:4 of the GATT; the United States appealed into the void.

These disputes established that local content requirements in renewable energy support schemes are inconsistent with WTO law, notwithstanding the climate objectives of the schemes. They have been criticised for impeding the use of industrial policy to build domestic political support for renewable energy, and the extensive local content and domestic manufacturing incentives adopted since 2022, including in the United States' Inflation Reduction Act — which conditioned the clean vehicle credit on the sourcing of battery components and critical minerals from the United States or its free trade agreement partners — have raised questions about the continuing effectiveness of these disciplines. In March 2024, China requested consultations with the United States concerning certain IRA tax credits (*United States — Certain Tax Credits under the Inflation Reduction Act*).

Fossil fuel subsidies. Governments provide extensive subsidies for the production and consumption of fossil fuels, estimated by the International Monetary Fund and the International Energy Agency at hundreds of billions of US dollars annually in explicit subsidies, and substantially more if unpriced externalities are included. Consumption subsidies, which keep domestic fuel prices below international market levels, are common in oil-producing and developing states, including, until 2023, Nigeria, where the petrol subsidy consumed a substantial share of government revenue before its removal in May 2023, examined in Chapter 24. The SCM Agreement has rarely been applied to fossil fuel subsidies. Consumption subsidies available to all consumers may not be "specific", and production subsidies have rarely been challenged, in part because many members provide them. A group of WTO members issued a ministerial statement on fossil fuel subsidy reform at the Eleventh Ministerial Conference in 2017, and an initiative on fossil fuel subsidy reform continues, but no binding disciplines have been agreed. The WTO Agreement on Fisheries Subsidies, adopted in 2022 and in force since September 2025, demonstrates that sector-specific subsidy disciplines driven by environmental objectives can be negotiated, and has been cited as a potential model. At COP26 in 2021, the parties to the Glasgow Climate Pact called for the phase-out of "inefficient fossil fuel subsidies", a commitment reaffirmed at COP28.

Trade Remedies in the Energy Sector

Trade remedies — anti-dumping duties, countervailing duties, and safeguard measures — have been used extensively in the clean energy sector. Under the Anti-Dumping Agreement, a member may impose anti-dumping duties on imports sold at less than normal value (dumping) which cause material injury to the domestic industry. Under Part V of the SCM Agreement, a member may impose countervailing duties to offset subsidies on imports causing injury. Under the Agreement on Safeguards, a member may temporarily restrict imports of a product whose increased imports cause or threaten serious injury to the domestic industry.

The solar industry has been the subject of numerous trade remedy actions. In 2013, the European Union imposed anti-dumping and countervailing duties on imports of solar panels and cells from China, combined with a price undertaking accepted from Chinese exporters; the measures expired in 2018. The United States has imposed anti-dumping and countervailing duties on solar cells and modules from China since 2012, extended to imports from other countries, including in South-East Asia, found to be used to circumvent the duties, and imposed a global safeguard measure on solar cells and modules in 2018 under Section 201 of the Trade Act of 1974. In October 2024, the European Union imposed definitive countervailing duties on imports of battery electric vehicles from China, following an investigation that found that the Chinese industry benefited from subsidies; China challenged the duties before the WTO.

Trade remedies have also been applied to biofuels. In *EU — Biodiesel (Argentina)* (2016), the Appellate Body held that the European Union had acted inconsistently with the Anti-Dumping Agreement by adjusting the costs of Argentine biodiesel producers to reflect international prices of soybeans rather than the lower domestic prices resulting from Argentina's export taxes on soybeans, because the producers' records reasonably reflected the costs associated with production, and the Anti-Dumping Agreement did not permit the investigating authority to disregard costs merely because they were lower than international prices owing to government intervention. The decision has implications for the treatment of distortions arising from export restrictions and dual pricing, examined in section 13.4.

13.4 Export Restrictions and Resource Security

The Asymmetry of Trade Disciplines on Exports

The multilateral trading system has historically focused on import barriers, and its disciplines on export measures are weaker and less developed. Article XI:1 of the GATT prohibits export prohibitions and restrictions, other than duties, taxes, or other charges, in the same terms as import restrictions. But **export duties** are not

prohibited by the GATT, which permits members to impose them subject only to the MFN obligation, and most members have not bound their export duties in their schedules of concessions. Export duties may be disciplined only through specific commitments, such as those undertaken by China, Russia, Viet Nam, and other acceding members in their accession protocols, or through regional trade agreements. This asymmetry reflects the historical bargaining dynamics of the GATT, in which the dominant importing states did not seek, and resource-exporting states resisted, disciplines on export measures.

The weakness of export disciplines has become increasingly significant as states have used export restrictions to promote domestic processing of raw materials (resource-based industrialisation), to stabilise domestic prices, to conserve resources, and to exert geopolitical leverage. The use of export restrictions on critical minerals — including by China, Indonesia, and others — has been examined in Chapter 8. This section examines the broader framework of export restrictions on energy commodities.

The Exceptions to Article XI

Article XI:2(a) permits "export prohibitions or restrictions temporarily applied to prevent or relieve critical shortages of foodstuffs or other products essential to the exporting contracting party". Article XX(g) permits measures relating to the conservation of exhaustible natural resources made effective in conjunction with restrictions on domestic production or consumption; Article XX(i) permits restrictions on exports of domestic materials necessary to ensure essential quantities to a domestic processing industry during periods when the domestic price is held below the world price as part of a governmental stabilisation plan, provided that such restrictions do not operate to increase the exports of or the protection afforded to the domestic industry; and Article XX(j) permits measures essential to the acquisition or distribution of products in general or local short supply.

The interpretation of these exceptions was developed in two disputes concerning Chinese export restrictions. In *China — Measures Related to the Exportation of Various Raw Materials* (2012), the Appellate Body upheld the panel's findings that China's export duties and quotas on bauxite, coke, fluorspar, magnesium, manganese, silicon carbide, silicon metal, yellow phosphorus, and zinc violated its accession commitments and Article XI:1, and held that China could not invoke Article XX to justify export duties that violated its accession protocol, which contained no express reference to Article XX. It interpreted Article XI:2(a) to permit restrictions only of limited duration to address situations of decisive importance, rather than long-term measures to conserve resources. In *China — Measures Related to the Exportation of Rare Earths, Tungsten, and Molybdenum* (2014), the Appellate Body again rejected China's Article XX(g) defence, holding that a measure must not only relate to conservation but must also be "made effective in

conjunction with" restrictions on domestic production or consumption — that is, the burdens of conservation must be shared even-handedly between foreign and domestic consumers — and that China's export quotas, which were not matched by comparable restrictions on domestic consumption, were designed to secure supplies for domestic industry rather than to conserve resources. These decisions, together with the panel report in *Indonesia — Raw Materials* (2022), examined in Chapter 8, establish that export restrictions designed to promote domestic processing are difficult to justify under the GATT's exceptions.

OPEC and Production Quotas

The most significant export-related measures in the energy sector are the production quotas agreed by the members of OPEC and, since 2016, the wider "OPEC+" group, which includes Russia and other non-OPEC producers under the Declaration of Cooperation. Production quotas restrict the supply of crude oil and thereby raise world prices, and their compatibility with WTO law has been debated.

The arguments that OPEC production quotas fall outside, or are justified under, the GATT include: first, that production quotas restrict production, not exports, and Article XI applies only to measures applied at the border; second, that the decisions of sovereign states on the exploitation of their natural resources fall within the principle of permanent sovereignty over natural resources; third, that production quotas may be justified under Article XX(g) as measures for the conservation of exhaustible natural resources, given that they restrict domestic production; and fourth, that OPEC decisions are not themselves measures of WTO members, but intergovernmental decisions implemented by members. The arguments to the contrary include that production quotas have the effect of restricting exports in a manner equivalent to an export quota, particularly for states that export most of their production, and that the term "other measures" in Article XI:1 is broad enough to cover them. No WTO member has challenged OPEC production quotas, reflecting the political sensitivity of the issue and the fact that many members, including major importers, also have interests in oil production. The application of competition law to OPEC has been attempted in the United States, but courts have dismissed antitrust claims against OPEC and its members on the basis of the act of state doctrine and sovereign immunity, as in *International Association of Machinists and Aerospace Workers v OPEC* 649 F.2d 1354 (9th Cir 1981). Legislative proposals to remove these obstacles (the "NOPEC" bill) have been introduced repeatedly in the United States Congress without enactment.

Dual Pricing

Dual pricing refers to the practice of selling energy domestically at prices lower than export prices, typically through export restrictions, export taxes, or price controls. Dual pricing benefits domestic consumers and energy-intensive industries, such as petrochemicals, fertilisers, and steel, which obtain feedstock at below world market prices. Importing states have argued that dual pricing constitutes a subsidy to downstream industries. Under the SCM Agreement, the provision of goods by a government at less than adequate remuneration may constitute a financial contribution and a benefit, and the question whether low domestic energy prices resulting from export restrictions constitute a subsidy has arisen in countervailing duty investigations. In *US — Softwood Lumber IV* (2004), the Appellate Body held that an investigating authority may use a benchmark other than private prices in the country of provision when those prices are distorted by the government's predominant role in the market, and in *US — Countervailing Measures on Certain Hot-Rolled Carbon Steel Flat Products from India* (2014), it addressed the provision of iron ore by a state-owned enterprise. However, the panel in *US — Measures Treating Export Restraints as Subsidies* (2001) held that an export restraint does not in itself constitute a "financial contribution" in the form of government-entrusted or directed provision of goods, and the Appellate Body in *US — Countervailing Duty Investigation on DRAMS* (2005) emphasised that entrustment or direction requires more than mere encouragement or policy pronouncements. As noted in section 13.3, in *EU — Biodiesel (Argentina)* (2016), the Appellate Body rejected the EU's adjustment of costs reflecting distortions caused by Argentine export taxes in anti-dumping investigations. The accession commitments of Saudi Arabia and Russia on the pricing of natural gas, noted in section 13.1, reflect the concerns of other members about dual pricing.

Export Licensing of LNG and Energy Security

Energy-importing states have also restricted exports of energy for reasons of energy security and domestic price stability. In the United States, the Natural Gas Act requires authorisation by the Department of Energy for exports of natural gas, including LNG. Exports to countries with which the United States has a free trade agreement requiring national treatment for trade in natural gas are deemed to be in the public interest and must be authorised without modification or delay, while exports to other countries must be authorised unless the Department finds that they are not consistent with the public interest. In January 2024, the Department paused the consideration of new authorisations of LNG exports to non-FTA countries while it updated its analysis of the economic and environmental effects of exports; the pause was challenged by several states and was lifted in January 2025. Some states have restricted exports of energy during crises: Australia introduced the Australian Domestic Gas Security Mechanism

in 2017, empowering the government to restrict LNG exports where there is a domestic shortfall, and China and India have periodically restricted exports of refined products and coal. Such measures may raise questions under Article XI of the GATT, which may be answered by Article XI:2(a) or Article XX(j) in cases of genuine critical shortage, or by Article XXI in emergencies in international relations.

Export Restrictions and Development

The debate over export restrictions has a significant development dimension. Many resource-rich developing states, including in Africa, regard export restrictions and export taxes on unprocessed raw materials as legitimate instruments for promoting domestic value addition, industrialisation, and economic diversification, and point to the historical use of such measures by now-developed states. The **Africa Mining Vision** adopted by the African Union in 2009 calls for the development of resource-based industrialisation through linkages, including the processing of minerals within Africa. Nigeria's policy of encouraging domestic refining of crude oil, including through the "domestic crude supply obligation" under the PIA, which requires producers to supply crude oil to domestic refineries on a willing-buyer, willing-seller basis, raises comparable questions about the relationship between resource-based industrialisation policies and trade disciplines. Whether WTO law should accommodate export restrictions for development purposes, through special and differential treatment or reinterpretation of the exceptions, is a significant question for doctoral research.

13.5 Technical Standards and Market Access

The TBT Agreement

The **Agreement on Technical Barriers to Trade** (TBT Agreement) governs technical regulations (mandatory requirements laying down product characteristics or their related processes and production methods), standards (voluntary requirements approved by a recognised body), and conformity assessment procedures (procedures to determine whether technical regulations or standards are fulfilled). Energy-related technical regulations and standards are pervasive: fuel quality specifications, energy efficiency standards for appliances and vehicles, emissions standards, safety standards for energy equipment, labelling requirements, and sustainability criteria for biofuels and hydrogen. Such measures pursue legitimate objectives but may create barriers to trade, especially where they differ between markets or are designed to favour domestic producers.

The TBT Agreement imposes several obligations on technical regulations. Article 2.1 requires members to accord to imported products treatment no less favourable than that accorded to like products of national origin and to like products originating in any other country. As noted in section 13.2, the Appellate

Body held in *US — Clove Cigarettes* (2012) that Article 2.1 does not prohibit detrimental impact on imports that stems exclusively from a legitimate regulatory distinction. In *US — COOL* (2012) and *US — Tuna II (Mexico)* (2012), it elaborated the test, examining whether the regulatory distinction was designed and applied in an even-handed manner. Article 2.2 requires that technical regulations not be prepared, adopted, or applied with a view to or with the effect of creating unnecessary obstacles to international trade, and not be more trade-restrictive than necessary to fulfil a legitimate objective, taking account of the risks non-fulfilment would create; legitimate objectives include national security requirements, the prevention of deceptive practices, and the protection of human health or safety, animal or plant life or health, or the environment. Article 2.4 requires members to use relevant international standards as a basis for their technical regulations, except when such standards would be ineffective or inappropriate for the fulfilment of the legitimate objectives pursued. Article 2.7 encourages members to accept as equivalent the technical regulations of other members if satisfied that they adequately fulfil the objectives of their own regulations, and Article 6 encourages the mutual recognition of conformity assessment.

The TBT Agreement's application to measures based on NPR-PPMs is contested, because the definition of a technical regulation refers to "product characteristics or their related processes and production methods", which has been interpreted by some as limited to processes and production methods that affect the characteristics of the final product. The question has significant implications for energy and climate measures based on the carbon intensity of production, such as low-carbon hydrogen standards, biofuel sustainability criteria, and low-carbon product standards for steel and cement. In *EC — Seal Products* (2014), the Appellate Body held that the EU's measure prohibiting the sale of seal products, subject to exceptions, was not a technical regulation because it did not lay down product characteristics, but did not resolve the broader question of NPR-PPMs.

Fuel Standards and Energy Efficiency Regulations

Fuel standards are among the most important energy-related technical regulations. Low carbon fuel standards, which require reductions in the average carbon intensity of transport fuels over their life cycle, have been adopted by California, Oregon, Washington, and British Columbia, and the EU's Fuel Quality Directive (Directive 98/70/EC, as amended) required fuel suppliers to reduce the life-cycle greenhouse gas intensity of fuels by six per cent by 2020. Such standards assign different carbon intensities to fuels according to their feedstock and production pathway, including higher values for fuels derived from oil sands and other high-intensity crude oils, and have raised trade concerns from producing states. Canada objected to proposals under the Fuel Quality Directive to assign a specific default value to fuels derived from oil sands, which was not adopted in the final

implementing measure. Within the United States, California's Low Carbon Fuel Standard was challenged under the dormant Commerce Clause, which prohibits state measures that discriminate against interstate commerce; in *Rocky Mountain Farmers Union v Corey* 730 F.3d 1070 (9th Cir 2013), the Court of Appeals held that the Standard's life-cycle analysis, which took account of emissions from transportation and electricity used in production, did not facially discriminate against out-of-state fuels.

Energy efficiency standards and labelling requirements for appliances, equipment, buildings, and vehicles are widespread and generally uncontroversial in trade terms, provided that they are applied without discrimination. The EU's Ecodesign for Sustainable Products Regulation (Regulation (EU) 2024/1781) extends ecodesign requirements beyond energy-related products to most physical goods placed on the EU market, including requirements relating to durability, repairability, recycled content, and carbon footprint, and requires the introduction of digital product passports. Such measures have extraterritorial effects on producers in exporting states, which must comply with them to access the EU market — a phenomenon sometimes described as the "Brussels effect".

Market Access for Energy Services

Market access for energy services under the GATS depends on members' specific commitments in their schedules, which specify the sectors in which a member undertakes market access and national treatment obligations and any limitations. Commitments in energy-related services have been limited, reflecting the sensitivity of the energy sector and the prevalence of state ownership and exclusive rights. Many members have made commitments in services incidental to mining and in some engineering and construction services, but fewer in pipeline transportation, electricity transmission and distribution, and energy trading. The GATS permits members to maintain monopolies and exclusive service suppliers, subject to the obligation in Article VIII to ensure that such suppliers do not act inconsistently with the member's MFN obligation and specific commitments, and do not abuse their monopoly position when competing in the supply of services outside the scope of their monopoly rights.

In *EU and its Member States — Certain Measures Relating to the Energy Sector* (DS476, panel report 2018), Russia challenged several elements of the EU's Third Energy Package, including the ownership unbundling requirements for transmission system operators, the "third-country certification" requirement applicable to operators controlled by persons from third countries (the so-called "Gazprom clause"), and the treatment of infrastructure exemptions and the OPAL pipeline. The panel found that some aspects of the measures, including certain aspects of the implementation of the infrastructure exemptions and the conditions imposed on the OPAL pipeline, were inconsistent with the GATT and GATS, but rejected

most of Russia's claims, including those against ownership unbundling, which it held did not violate the market access and national treatment obligations of the GATS in the circumstances, and, in the case of the third-country certification requirement, found a violation of the GATS national treatment obligation which was not justified under the public security exception in Article XIV. Both parties appealed, and the appeal has not been heard because of the Appellate Body crisis. The dispute illustrated the potential application of trade law to the structural regulation of energy markets.

Government Procurement

Public procurement in the energy sector — including the procurement of generation capacity, equipment, and construction services by state-owned utilities — is significant in value. Article III:8(a) of the GATT excludes from the national treatment obligation laws, regulations, or requirements governing the procurement by governmental agencies of products purchased for governmental purposes and not with a view to commercial resale. As the *Canada — Renewable Energy* and *India — Solar Cells* disputes established, this derogation is narrowly interpreted and does not cover local content requirements imposed on private generators selling electricity to state utilities. The plurilateral **Agreement on Government Procurement**, as revised in 2012, applies to procurement by covered entities of its parties above specified thresholds, and some parties have listed energy utilities as covered entities, although many have excluded them. Nigeria, like most African states, is not a party to the GPA.

13.6 Regional Trade Agreements and Energy Integration

Regional Trade Agreements in WTO Law

Regional trade agreements (RTAs) — free trade agreements and customs unions — are an exception to the MFN principle, permitted under Article XXIV of the GATT for trade in goods and Article V of the GATS for trade in services, provided that they meet specified conditions. Article XXIV:8 requires that duties and other restrictive regulations of commerce be eliminated with respect to "substantially all the trade" between the constituent territories, and Article XXIV:5 requires that the duties and other regulations of commerce applied to third parties not be higher or more restrictive than those applicable before the formation of the agreement. In *Turkey — Textiles* (1999), the Appellate Body held that Article XXIV may justify a measure inconsistent with other GATT provisions only if the measure is introduced upon the formation of an RTA that fully meets the requirements of Article XXIV, and the formation of the RTA would be prevented if the measure were not allowed. Hundreds of RTAs are in force, and many contain provisions on energy that go beyond WTO rules, addressing export duties, dual

pricing, transit, access to networks, state enterprises, investment, and regulatory cooperation.

Energy in North American Trade Agreements

The **North American Free Trade Agreement** (NAFTA, 1994) contained a dedicated chapter on energy and basic petrochemicals (Chapter 6), which incorporated GATT disciplines on import and export restrictions and, controversially, a "proportionality" clause (Article 605) under which a party that restricted exports of an energy good to another party for reasons of conservation, shortage, or price stabilisation was required to maintain the proportion of total supply exported to that party over the previous thirty-six months. Mexico was exempted from the proportionality clause, and reserved its constitutional monopoly over hydrocarbons and electricity. The proportionality clause was criticised in Canada as restricting its ability to control its energy resources.

The **United States–Mexico–Canada Agreement** (USMCA, in force 2020), which replaced NAFTA, contains no dedicated energy chapter and no proportionality clause. Mexico's commitments on energy reflect its 2013 constitutional reform, which opened the hydrocarbons and electricity sectors to private investment, but the USMCA includes provisions recognising Mexico's direct, inalienable, and imprescriptible ownership of hydrocarbons in the subsoil. The USMCA's investment chapter substantially narrowed the availability of investor–state arbitration, except for investors with "covered government contracts" in specified sectors, including oil and natural gas and power generation, who retain broader access to arbitration under Annex 14-E. In July 2022, the United States and Canada requested consultations with Mexico under the USMCA regarding Mexican energy policies alleged to favour the state-owned companies Petróleos Mexicanos (Pemex) and the Comisión Federal de Electricidad (CFE) over private and foreign energy companies, including measures under the 2021 amendments to Mexico's Electricity Industry Law. The episode illustrates the role of RTAs in disciplining energy nationalism.

European Union Agreements

The European Union's trade agreements increasingly contain dedicated chapters on energy and raw materials. The **EU–Ukraine Association Agreement** (2014), including its Deep and Comprehensive Free Trade Area, contains provisions on trade-related energy, including on transit, the prohibition of dual pricing, and regulatory approximation with EU energy law. The **EU–UK Trade and Cooperation Agreement** (2020) contains a heading on energy and raw materials, which addresses pricing, export monopolies, third-party access to networks, unbundling, and cooperation in electricity trading and offshore energy in the North Seas. The **EU–Chile Advanced Framework Agreement** (2023)

contains a chapter on energy and raw materials that prohibits dual pricing and export monopolies, addresses access to energy infrastructure, and promotes cooperation in renewable energy, hydrogen, and critical raw materials, including lithium and copper.

The **Treaty establishing the Energy Community** (2005, in force 2006) extends the EU's internal energy market acquis to non-EU states in South-East Europe and the Black Sea region, including the Western Balkans, Ukraine, Moldova, and Georgia. Contracting parties undertake to implement EU legislation on electricity, gas, renewable energy, energy efficiency, environment, and competition, subject to adaptation and monitoring by the Energy Community Secretariat, with a dispute settlement procedure that may result in a finding of breach by the Ministerial Council. The Energy Community is a distinctive model of regional energy integration through the export of a regulatory acquis.

The African Continental Free Trade Area

The **Agreement Establishing the African Continental Free Trade Area** (AfCFTA), signed in Kigali in March 2018 and in force since May 2019, with trading under its preferential arrangements commencing in January 2021, aims to create a single continental market for goods and services. It is implemented through protocols on trade in goods, trade in services, rules and procedures on the settlement of disputes, investment (adopted in 2023), competition policy, intellectual property rights, and digital trade, among others. Nigeria is a party.

The AfCFTA has potential significance for energy trade and investment in Africa. Intra-African trade in energy is limited by inadequate infrastructure, but the progressive elimination of tariffs on energy goods, including petroleum products and renewable energy equipment, and the liberalisation of energy-related services could support the development of regional value chains — for example, in the refining of crude oil, the production of fertilisers from natural gas, the manufacture of batteries using African critical minerals, and the assembly of solar equipment. The AfCFTA's rules of origin will determine whether products incorporating imported components qualify for preferential treatment. Energy services were not among the initial priority sectors for services negotiations, but may be addressed in subsequent rounds. The Protocol on Investment includes provisions on the right to regulate and on sustainable development, and the relationship between the AfCFTA and regional energy integration through power pools and the African Single Electricity Market (AfSEM), examined in Chapter 9, is an important area of development.

ECOWAS Instruments

In West Africa, the **Revised Treaty of the Economic Community of West African States** (1993) provides in Article 28 for cooperation in energy, includ-

ing the development of energy resources and the establishment of appropriate mechanisms for the collective resolution of the energy development problems of the Community. The **ECOWAS Energy Protocol** (A/P4/1/03), adopted in 2003 and modelled substantially on the Energy Charter Treaty, provides for the promotion and protection of investment in energy, freedom of energy transit, and access to energy infrastructure, and contains a dispute settlement provision that includes investor–state arbitration. The Protocol's practical application has been limited, but it represents an important regional legal framework for energy investment and trade. ECOWAS has also established a common external tariff (in force since 2015) and the ECOWAS Trade Liberalisation Scheme, which applies to trade in energy goods among member states.

The **Treaty on the West African Gas Pipeline Project** (2003), concluded among Benin, Ghana, Nigeria, and Togo, established the legal framework for the West African Gas Pipeline, which transports Nigerian gas to the three other states. The Treaty and the associated International Project Agreement established the West African Gas Pipeline Authority as the regulator of the pipeline, and provide for the harmonisation of the legal and fiscal frameworks applicable to it, transit, and dispute settlement. The pipeline has been affected by supply interruptions and payment disputes, which illustrate the challenges of regional energy infrastructure examined in Chapter 6. Proposals for the Nigeria–Morocco Gas Pipeline, which would traverse the territories of numerous West African states, would require an extensive intergovernmental legal framework.

Other Regional Arrangements

In Southern Africa, the **SADC Protocol on Energy** (1996) provides for cooperation in the development of energy and energy pooling, and established the framework for the Southern African Power Pool. In South-East Asia, the **ASEAN Plan of Action for Energy Cooperation** supports the ASEAN Power Grid and the Trans-ASEAN Gas Pipeline. In South America, regional integration of energy infrastructure has been pursued through bilateral treaties, such as those governing the Itaipu hydroelectric project between Brazil and Paraguay. These arrangements demonstrate a range of approaches to regional energy integration, from binding market integration (the Energy Community) to project-specific treaties (the West African Gas Pipeline) and cooperation frameworks (SADC and ASEAN).

13.7 Carbon Border Measures and Trade Compatibility

The Problem of Carbon Leakage

Where states adopt climate policies of differing stringency, carbon-intensive industries in states with stringent policies may face higher costs than competitors in states with less stringent policies, with two potential consequences: a loss of competitiveness for domestic industries, and **carbon leakage** — the relocation

of production, investment, or emissions to states with less stringent policies, undermining the environmental effectiveness of climate policy. Governments have addressed leakage risk principally through the free allocation of emission allowances to industries at risk, as under the EU Emissions Trading System, but free allocation weakens the incentive to reduce emissions and reduces revenues. **Carbon border adjustments** — measures that impose a carbon price on imports equivalent to that borne by domestic producers, and possibly rebate carbon costs on exports — offer an alternative approach.

The EU Carbon Border Adjustment Mechanism

The **Carbon Border Adjustment Mechanism** (CBAM), established by Regulation (EU) 2023/956, is the first comprehensive carbon border adjustment in force. It applies to imports into the European Union of goods in sectors at high risk of carbon leakage: cement, iron and steel, aluminium, fertilisers, electricity, and hydrogen, together with certain precursors and downstream products. During a **transitional period** from 1 October 2023 to 31 December 2025, importers were required to report the embedded emissions of imported goods, without financial obligations. From the beginning of the **definitive period** in 2026, importers must be authorised as CBAM declarants and must purchase and surrender CBAM certificates corresponding to the embedded emissions of the goods they import, with the price of certificates linked to the price of allowances under the EU ETS. Importers may deduct the carbon price effectively paid in the country of production, and the number of certificates to be surrendered is adjusted to reflect the free allocation still received by EU producers of the same goods. The free allocation for the CBAM sectors is to be phased out progressively between 2026 and 2034, as CBAM is phased in. Embedded emissions are to be calculated on the basis of actual emissions data, verified by accredited verifiers, or, where adequate data is unavailable, default values. Amendments adopted in 2025 as part of a simplification package introduced a de minimis threshold based on the mass of imported goods, exempting importers of small quantities, and deferred the sale of CBAM certificates.

Compatibility with WTO Law

The compatibility of CBAM with WTO law has been extensively debated. The analysis depends on how the measure is characterised.

Border tax adjustment or regulation. If CBAM is characterised as a border adjustment of an internal tax, it may be permitted under Article II:2(a) of the GATT, which permits members to impose on imports a charge equivalent to an internal tax imposed in respect of the like domestic product or an article from which the imported product has been manufactured, subject to the national treatment obligation of Article III:2. Whether the EU ETS, a cap-and-trade

system, constitutes an "internal tax" for this purpose is uncertain. If CBAM is characterised instead as an internal regulation, it would be assessed under Article III:4. Either way, the national treatment analysis turns on whether imported goods are treated less favourably than like domestic goods. The EU designed CBAM to mirror the ETS, so that importers bear a carbon cost equivalent to that borne by EU producers, net of free allocation. Critics argue that differences in the methodologies for calculating embedded emissions, the use of default values, and the administrative burdens imposed on importers may result in less favourable treatment.

Most-favoured-nation treatment. The deduction of carbon prices paid in the country of origin results in different treatment of imports according to the climate policies of exporting states, which may raise issues under Article I:1. The EU argues that the measure treats all imports according to the same objective criterion — embedded emissions net of carbon prices paid — and that differences in treatment reflect differences in products' carbon intensity and carbon costs rather than their origin.

Article XX. If CBAM is found inconsistent with Articles I, II, or III, the EU would seek to justify it under Article XX(g), as a measure relating to the conservation of exhaustible natural resources (the global atmosphere and climate system) made effective in conjunction with restrictions on domestic production or consumption (the EU ETS), or Article XX(b), as necessary to protect human, animal, or plant life or health. The provisional justification of a well-designed carbon border adjustment under Article XX(g) appears plausible in the light of *US — Gasoline* and *US — Shrimp*. The more difficult question is the chapeau, which requires that the measure not be applied in a manner constituting arbitrary or unjustifiable discrimination between countries where the same conditions prevail. Following *US — Shrimp*, a measure that conditions market access on exporting countries' policies must allow sufficient flexibility to take account of different conditions in different countries, and should be preceded by serious good faith efforts to negotiate. Questions arise as to whether CBAM adequately accounts for the different circumstances of developing countries, whether it recognises climate policies other than explicit carbon pricing, and whether the EU engaged in sufficient negotiation.

In 2025, Russia requested consultations with the European Union under the DSU concerning CBAM, initiating the first WTO dispute concerning the measure. Other states, including India, China, and South Africa, have expressed strong objections to CBAM, characterising it as a unilateral and discriminatory measure.

Compatibility with the Climate Regime

CBAM also raises questions under the international climate regime. Article 3(5) of the UNFCCC provides that measures taken to combat climate change, "including unilateral ones", should not constitute a means of arbitrary or unjustifiable discrimination or a disguised restriction on international trade — language that mirrors the chapeau of Article XX. The principle of common but differentiated responsibilities and respective capabilities, embodied in the UNFCCC and the Paris Agreement and examined in Chapter 18, has been invoked by developing states to argue that CBAM imposes the EU's climate policy on states that bear lesser responsibility for climate change and have fewer resources to decarbonise. The Paris Agreement's bottom-up architecture, under which each party determines its own contribution, is said to be in tension with a measure that effectively requires exporters to meet the carbon price of the importing state. The EU has responded that CBAM applies to embedded emissions irrespective of the policies of the exporting state, that it is necessary to maintain the integrity of its own climate policy, and that it has committed to support for decarbonisation in developing countries. CBAM revenues have not, however, been earmarked for climate finance for developing countries, a point of criticism.

Impact on Developing and African Countries

The economic impact of CBAM on developing countries depends on their exports of covered goods to the European Union and the carbon intensity of their production. Several African countries have significant exposure, including Mozambique (aluminium), South Africa (iron, steel, and aluminium), Egypt, Algeria, and Morocco (fertilisers and steel), and Zimbabwe (iron and steel). Studies have estimated that CBAM could reduce exports and GDP in affected African economies, while also creating incentives for investment in low-carbon production, including through Africa's renewable energy potential. Nigeria's exports of covered goods to the European Union are relatively limited, but CBAM may affect its fertiliser exports and its potential exports of hydrogen and ammonia, examined in Chapter 12. The application of CBAM to electricity imports affects states connected to the EU grid, including the Western Balkans, Ukraine, and potentially North African states planning to export renewable electricity to Europe by submarine cable.

Other Carbon Border Measures and Climate Clubs

The United Kingdom has legislated for its own CBAM, to take effect from January 2027, covering a similar range of sectors. Carbon border adjustments have been proposed in the United States, Canada, and elsewhere, but not adopted. The European Union and the United States negotiated from 2021 on a Global Arrangement on Sustainable Steel and Aluminium, intended to address both car-

bon intensity and non-market excess capacity, without agreement. The **Climate Club**, launched by the G7 in 2022, provides a forum for cooperation on industrial decarbonisation, including the development of common methodologies for measuring the carbon intensity of products, which might reduce the fragmentation of carbon border measures. The relationship between carbon border measures and the multilateral trading system has been discussed in the WTO Committee on Trade and Environment and in the Trade and Environmental Sustainability Structured Discussions, but no multilateral framework has been agreed.

13.8 Trade Disputes Involving Energy and Natural Resources

An Overview of the Jurisprudence

The body of WTO jurisprudence concerning energy and natural resources has grown substantially, although energy disputes remain a small proportion of the WTO's caseload. The principal disputes, several of which have been examined in earlier sections, may be grouped thematically.

Table 13.1 Principal WTO disputes concerning energy and natural resources.

Theme	Principal disputes	Key holdings
Environmental regulation of fuels	*US — Gasoline* (1996)	Clean air is an exhaustible natural resource; two-tier Article XX test; chapeau requires cooperation and even-handedness
Renewable energy support and local content	*Canada — Renewable Energy* (2013); *India — Solar Cells* (2016); *US — Renewable Energy* (panel, 2019)	Local content requirements violate Article III:4 and TRIMs; procurement derogation narrowly construed; benefit benchmark may be a government-created market
Export restrictions on raw materials	*China — Raw Materials* (2012); *China — Rare Earths* (2014); *Indonesia — Raw Materials* (panel, 2022)	Article XX unavailable for accession commitments on export duties without textual link; "even-handedness" required under XX(g); XI:2(a) limited to temporary measures
Biofuels and sustainability criteria	*EU — Biodiesel (Argentina)* (2016); *EU — Palm Oil (Malaysia)* (panel, 2024)	Costs distorted by export taxes cannot be disregarded in anti-dumping; ILUC-based criteria assessed under TBT Agreement
Energy market structure	*EU — Energy Package* (panel, 2018)	Ownership unbundling largely upheld; third-country certification inconsistent with GATS national treatment

Theme	Principal disputes	Key holdings
Transit and security	*Russia — Traffic in Transit* (2019); *US — Steel and Aluminium* (panel, 2022)	Article XXI not wholly self-judging; objective existence of emergency reviewable
Trade remedies in clean energy	Numerous anti-dumping and countervailing duty disputes concerning solar products, biodiesel, and electric vehicles	Methodological questions on subsidies, dumping margins, and market distortions

Cross-Cutting Themes

Several themes emerge from the jurisprudence.

Environmental objectives and trade disciplines. The jurisprudence since *US — Gasoline* has recognised that WTO members may pursue environmental objectives, including the conservation of natural resources and the protection of health, through trade-restrictive measures, provided that the measures are genuinely directed at those objectives and are applied in an even-handed and non-arbitrary manner. However, the energy disputes have also established that environmental objectives do not justify discrimination in favour of domestic industry, as the local content cases illustrate. The tension between the climate imperative and the WTO's non-discrimination disciplines is likely to intensify as states adopt industrial policies in support of clean energy manufacturing.

Export restrictions and resource sovereignty. The export restriction cases reflect a tension between the principle of permanent sovereignty over natural resources and the disciplines of the GATT. The Appellate Body acknowledged in *China — Rare Earths* that members retain sovereignty over their natural resources, but held that this sovereignty must be exercised consistently with their WTO obligations, which they have freely accepted. For resource-exporting developing states, the jurisprudence constrains the use of export restrictions as instruments of industrial policy.

The extraterritorial reach of importing-state regulation. Measures such as the EU's biofuel sustainability criteria, RFNBO rules, deforestation regulation, and CBAM regulate imports by reference to the conditions of production in exporting states. The jurisprudence, particularly *US — Shrimp*, permits such measures in principle, subject to the chapeau's requirements of flexibility and good faith negotiation, but their proliferation raises concerns among exporting states about unilateralism and the imposition of the regulatory preferences of large markets.

Security. The *Russia — Traffic in Transit* line of cases has opened the security exception to review, but the United States' rejection of that review, together with the increasing invocation of security in trade policy, including in energy and critical minerals, poses a significant challenge to the system.

The Consequences of the Appellate Body Crisis

The paralysis of the Appellate Body since December 2019 has had significant consequences for energy trade disputes. Several panel reports in energy-related disputes — including *EU — Energy Package*, *US — Renewable Energy*, *Indonesia — Raw Materials*, and *US — Steel and Aluminium* — have been appealed into the void, leaving the disputes unresolved and the panel findings without binding effect. The MPIA provides an alternative for its participants, which include the European Union, China, and many other members, but not the United States, India, Indonesia, or Russia. The weakening of WTO dispute settlement has encouraged states to resort to unilateral measures, to rely on bilateral and regional dispute settlement mechanisms, and to address energy trade issues through plurilateral and sectoral arrangements, such as critical minerals agreements. Reform of the dispute settlement system was the subject of negotiations aimed at a fully functioning system, but no agreement had been reached at the time of writing.

The Interaction of Trade and Investment Disputes

Energy measures frequently give rise to both trade and investment disputes. The Ontario feed-in tariff programme was challenged both before the WTO (*Canada — Renewable Energy*) and in investment arbitration under NAFTA (*Mesa Power v Canada*, examined in Chapter 10). The EU's Third Energy Package was challenged both before the WTO (*EU — Energy Package*) and, in the case of the Nord Stream 2 pipeline, under the Energy Charter Treaty (*Nord Stream 2 AG v European Union*). Trade and investment disputes differ in their parties (states only in the WTO, investors against states in investment arbitration), their remedies (prospective compliance in the WTO, compensation in investment arbitration), and their substantive standards, but their interaction raises questions of coherence, forum shopping, and the cumulative constraints on regulatory autonomy. Investment arbitration is examined in Chapters 14 and 26.

::: casebox {title="EU — Palm Oil (Malaysia) (WTO Panel, 2024)"} **Background.** As examined in Chapter 10, the EU's Renewable Energy Directive (RED II) limits the extent to which biofuels made from food and feed crops with high risk of indirect land use change (ILUC) may count towards renewable energy targets. These biofuels are to be phased out by 2030. Commission Delegated Regulation (EU) 2019/807 identified palm oil as the only high-ILUC-risk feedstock. The Delegated Regulation also established criteria for certifying "low ILUC-risk" biofuels, which would be exempt from the phase-out. France additionally excluded palm oil-based biofuels from a tax incentive. Malaysia, a major palm oil producer, challenged the measures under the TBT Agreement, the GATT, and the SCM Agreement.

Findings. The panel, reporting in March 2024, accepted that the EU measures pursued a legitimate objective of addressing the greenhouse gas emissions associated with ILUC, a form of environmental protection. It also accepted that the EU was in principle entitled to differentiate between feedstocks on the basis of ILUC risk. However, the panel found deficiencies in certain aspects of the design and implementation of the measures, including:

- the criteria and procedures for certifying low ILUC-risk biofuels;
- the EU's failure to conduct a timely review of the data underlying the high-ILUC-risk determination.

It found these aspects inconsistent with the EU's obligations. It also found aspects of the French measure inconsistent with the GATT. Neither party appealed, and the EU undertook to bring the measures into conformity.

Significance. The dispute illustrates how the TBT Agreement applies to sustainability criteria based on production methods and land use impacts. It shows that WTO panels are prepared to accept environmental differentiation based on life-cycle impacts, while scrutinising the evidence, design, and procedures used to implement it. The reasoning is of potential relevance to other measures that differentiate imports by their carbon intensity or environmental footprint, including RFNBO rules, deforestation regulation, and CBAM. A parallel dispute brought by Indonesia (DS593) resulted in a panel report in early 2025. :::

::: exercise {title="Practical Exercise: The Republic of Tawana's Energy Trade Measures (HYPOTHETICAL)"} *This exercise is entirely hypothetical. The Republic of Tawana, the Republic of Norland, and all companies named are fictional.*

The Republic of Tawana is a WTO member and a party to the AfCFTA. It is a significant producer of natural gas and an exporter of nitrogen fertilisers, 60 per cent of which are sold to the European Union. To promote industrialisation and clean energy, Tawana's government has adopted three measures.

- **Measure 1 — Solar auction with local content.** Tawana's state utility will procure 2,000 MW of solar capacity through auctions. Winning bidders receive twenty-year PPAs. Bidders must source at least 40 per cent of the value of solar modules and inverters from manufacturers located in Tawana, and those meeting a 60 per cent threshold receive a tariff premium of 5 per cent.
- **Measure 2 — Gas export duty and domestic supply obligation.** Tawana imposes an export duty of 25 per cent on LNG and requires producers to sell 30 per cent of their output to domestic fertiliser and power producers at a regulated price equal to half the export parity price. Tawana's accession protocol contains no commitments on export duties.
- **Measure 3 — Response to CBAM.** Tawana's fertiliser exporters face CBAM obligations from 2026. Tawana is considering introducing a domes-

tic carbon tax of USD 15 per tonne on fertiliser production so that the tax can be deducted under CBAM, and challenging CBAM at the WTO.

The Republic of Norland (HYPOTHETICAL), an exporter of solar modules and a major importer of fertilisers, is considering WTO proceedings against Measures 1 and 2. Norland is also considering imposing countervailing duties on fertiliser imports from Tawana.

1. Advise Norland on the strength of claims against Measure 1 under Article III:4 of the GATT, the TRIMs Agreement, and the SCM Agreement. Could Tawana rely on Article III:8(a), Article XX, or special and differential treatment?
2. Assess the legality of Measure 2. Does the export duty violate the GATT? Could the domestic supply obligation and regulated price be treated as a countervailable subsidy to fertiliser producers in Norland's investigation, in the light of the jurisprudence on export restraints and *EU — Biodiesel*?
3. Advise Tawana on its options in relation to CBAM, including the likely WTO arguments, the advantages and disadvantages of introducing a domestic carbon tax, and the use of the revenue.
4. Consider whether the AfCFTA could support an alternative strategy for developing a regional solar manufacturing value chain that complies with Tawana's WTO obligations. :::

CHAPTER SUMMARY

- Energy trade is characterised by geographical concentration, export-side concerns, network dependence, state ownership, and environmental externalities. Energy was historically on the margins of the GATT but has been brought within the WTO through the accession of major producers.
- Energy goods are governed by the GATT and associated agreements, and energy services by the GATS. Commitments on energy services are limited.
- The core WTO disciplines apply to energy: MFN, national treatment, the prohibition of quantitative restrictions, state trading, and freedom of transit. The general exceptions in Article XX are applied through a two-tier test established in *US — Gasoline*. The security exception in Article XXI is not wholly self-judging, following *Russia — Traffic in Transit*.
- Local content requirements in renewable energy support schemes violate Article III:4 and the TRIMs Agreement, as *Canada — Renewable Energy* and *India — Solar Cells* establish. Fossil fuel subsidies remain largely undisciplined, and trade remedies are widely used in clean energy sectors.
- Disciplines on export restrictions are weaker than those on imports, and export duties are disciplined only by specific commitments. The Chinese raw materials cases strictly construe the exceptions for export restrictions. OPEC production quotas and dual pricing remain contested.

- The TBT Agreement governs fuel standards, efficiency standards, and sustainability criteria. Measures based on processes and production methods raise unresolved questions.
- RTAs, including the USMCA, EU agreements, the Energy Community Treaty, the AfCFTA, and ECOWAS instruments, contain energy provisions that go beyond WTO rules.
- The EU CBAM imposes a carbon price on imports of carbon-intensive goods from 2026. Its compatibility with WTO law turns on national treatment, MFN, and the Article XX chapeau, and it raises questions under the principle of common but differentiated responsibilities.
- The Appellate Body crisis has left several energy disputes unresolved and encouraged unilateralism and alternative forums.

DOCTORAL RESEARCH QUESTIONS

1. Should WTO law accommodate local content requirements in clean energy support schemes? Evaluate proposals for a "climate waiver" or a revival of non-actionable subsidies for environmental purposes.
2. Do OPEC+ production quotas violate Article XI of the GATT? Assess the legal arguments and the reasons why no WTO member has challenged them.
3. Is the EU CBAM compatible with the chapeau of Article XX and with the principle of common but differentiated responsibilities? Propose design modifications that would enhance its legitimacy in the eyes of developing states.
4. To what extent does the jurisprudence on export restrictions constrain resource-based industrialisation strategies in Africa? Should special and differential treatment extend to export measures?
5. How has the paralysis of the WTO Appellate Body affected the governance of international energy trade, and what alternatives — the MPIA, regional dispute settlement, or plurilateral arrangements — offer the most promise?
6. Evaluate the potential of the AfCFTA to support the development of regional energy value chains in Africa, with particular reference to renewable energy manufacturing and critical minerals processing.

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14 CHAPTER

International Investment Law and Protection of Energy Investments

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the characteristics of foreign direct investment in the energy sector that make it particularly dependent on legal protection, and identify the domestic and international sources of that protection.
2. Describe the architecture of the international investment regime, including bilateral investment treaties, the Energy Charter Treaty, and regional instruments, and analyse the threshold questions of "investment", "investor", and nationality planning.
3. Analyse the fair and equitable treatment standard, including legitimate expectations, arbitrariness, transparency, and denial of justice, and the related standard of full protection and security.
4. Evaluate the national treatment and most-favoured-nation obligations as applied to energy investments.
5. Distinguish direct from indirect expropriation under investment treaties and assess the role of the police powers doctrine.
6. Analyse the interaction between stabilisation clauses, umbrella clauses, and treaty standards in protecting energy investors against regulatory change.
7. Assess the operation and legitimacy of investor–state dispute settlement in the energy sector.
8. Evaluate the reform of the investment treaty regime in the context of the energy transition, including the modernisation of and withdrawals from the Energy Charter Treaty.

Introduction

International investment law has been shaped, more than by any other sector, by disputes over energy and natural resources. The nationalisations of foreign oil concessions in Mexico in 1938, Iran in 1951, and Libya in the 1970s generated

the foundational controversies over the standard of compensation for expropriation, examined in Chapter 3. The arbitrations arising from those nationalisations — *Anglo-Iranian Oil Co*, *Aramco*, *Texaco*, *BP*, *LIAMCO*, and *Aminoil* — established principles on the internationalisation of contracts and stabilisation clauses that continue to influence the law. In the modern era of investment treaty arbitration, the energy sector accounts for a larger share of disputes than any other: oil, gas, and mining, together with electricity and other energy, account for a substantial proportion — around two-fifths in recent years — of the cases registered at the International Centre for Settlement of Investment Disputes (ICSID), and many of the largest awards in the history of investment arbitration, including the awards of around USD 50 billion against Russia in the *Yukos* cases and of several billion dollars against Venezuela, have concerned energy investments.

The reasons for this prominence lie in the characteristics of energy investment. Energy projects are capital-intensive, with high upfront costs and long payback periods; once constructed, they are immobile and cannot be relocated if conditions deteriorate. These features create what economists call the **"obsolescing bargain"**: before an investment is made, the investor has bargaining power, because the host state needs its capital and technology; once the investment is sunk, bargaining power shifts to the host state, which may be tempted to change the terms — through taxation, regulation, renegotiation, or expropriation — to capture a greater share of the returns, especially when commodity prices rise. Energy is also politically salient, closely connected with sovereignty and national identity, and subject to frequent regulatory intervention for reasons of security, affordability, and environmental protection. International investment law seeks to mitigate the obsolescing bargain by providing investors with enforceable international legal protections against certain forms of state conduct.

At the same time, investment law has become intensely contested. Critics argue that investment treaties constrain the regulatory autonomy of states, including their ability to pursue the energy transition, by exposing them to claims for compensation when they phase out fossil fuels, reform renewable support schemes, or strengthen environmental regulation; that investor–state arbitration lacks legitimacy, consistency, and accountability; and that the regime is asymmetrical, conferring rights on investors without corresponding obligations. These critiques have driven a wave of reform, including the termination of treaties, the renegotiation of treaty standards, proposals for a multilateral investment court, and the withdrawal of the European Union and many of its member states from the Energy Charter Treaty.

This chapter examines foreign direct investment in energy projects (section 14.1), bilateral and multilateral investment treaties (section 14.2), fair and equitable treatment (section 14.3), national treatment and most-favoured-nation

treatment (section 14.4), direct and indirect expropriation (section 14.5), stabilisation clauses and regulatory change (section 14.6), investor–state dispute settlement (section 14.7), and investment treaty reform and the energy transition (section 14.8). The customary international law of expropriation and compensation, and the doctrine of permanent sovereignty over natural resources, have been examined in Chapters 2 and 3; contractual stabilisation and change-in-law clauses are examined from a contract-drafting perspective in Chapter 15; and the procedure of investment treaty arbitration, including jurisdiction, evidence, damages, and enforcement, is examined in Chapter 26.

14.1 Foreign Direct Investment in Energy Projects

The Scale and Structure of Energy Investment

Global investment in energy amounts to around three trillion US dollars per year, of which an increasing proportion — now around two-thirds — is directed to clean energy, including renewable power, grids, storage, low-emission fuels, efficiency, and electrification, with the remainder directed to fossil fuel supply and power. Foreign direct investment (FDI) is a significant component of energy investment, particularly in developing countries that lack domestic capital and technology. However, the distribution of energy investment is highly uneven: Africa, which is home to around a fifth of the world's population, receives only a small fraction of global clean energy investment, owing in large part to the high cost of capital, which reflects perceived political, regulatory, currency, and offtaker risks.

The actors in energy FDI are diverse. **International oil companies** (IOCs), such as ExxonMobil, Shell, BP, Chevron, TotalEnergies, and Eni, have historically dominated upstream petroleum investment in developing countries. **National oil companies** (NOCs) of producing states, such as Saudi Aramco, Petrobras, and the Nigerian National Petroleum Company Limited (NNPC Ltd), increasingly invest abroad. **State-owned enterprises** of China, including China National Petroleum Corporation, Sinopec, CNOOC, State Grid, and China Three Gorges, together with Chinese policy banks, have become major investors and financiers in energy infrastructure in Africa, Asia, and Latin America. **Independent power producers** and **renewable energy developers**, often backed by private equity and infrastructure funds, invest in power generation. **Sovereign wealth funds** and **pension funds** invest in energy infrastructure as long-term assets. And **development finance institutions** — the World Bank Group (including the International Finance Corporation and the Multilateral Investment Guarantee Agency), the African Development Bank, and bilateral DFIs — provide finance, guarantees, and political risk insurance, examined in Chapter 16.

Domestic Legal Protection of Foreign Investment

Foreign investors in energy are protected, in the first instance, by the domestic law of the host state — its constitution, investment legislation, sectoral legislation, and contracts with the state. Many developing countries have enacted **investment codes** offering guarantees to foreign investors. In Nigeria, the **Nigerian Investment Promotion Commission Act 1995** established the Nigerian Investment Promotion Commission (NIPC) and provides, in section 25, that no enterprise shall be nationalised or expropriated by any government of the Federation, and that no person who owns, whether wholly or in part, the capital of any enterprise shall be compelled by law to surrender his interest in the capital to any other person, except where the acquisition is in the national interest or for a public purpose and under a law which makes provision for payment of fair and adequate compensation and a right of access to the courts for the determination of the investor's interest or right and the amount of compensation. The Act also guarantees the unconditional transferability of funds through an authorised dealer, and, in section 26, provides for the settlement of investment disputes by mutually agreed procedures, including arbitration under the ICSID rules or under a bilateral or multilateral investment protection agreement to which Nigeria is a party. Section 44 of the Constitution of the Federal Republic of Nigeria 1999 prohibits the compulsory acquisition of property except in the manner and for the purposes prescribed by a law that, among other things, requires prompt payment of compensation, while section 44(3) vests minerals, mineral oils, and natural gas in the Government of the Federation, as examined in Chapter 3.

Domestic law protections may, however, be changed by the host state, and may be enforced only in the host state's courts, which investors may not trust to be independent. Contracts with the state — concessions, production sharing contracts, host government agreements, and power purchase agreements — offer more specific protection, including stabilisation clauses and international arbitration clauses, but their enforcement depends on the availability of effective international remedies. International investment law supplements these protections with treaty-based standards enforceable through international arbitration.

Investment Screening and the Return of the State

While investment treaties protect investments once made, states have increasingly regulated the admission of foreign investment in the energy sector on national security grounds. The European Union's **FDI Screening Regulation** (Regulation (EU) 2019/452) established a framework for the screening of foreign direct investments into the Union on grounds of security or public order, identifying critical infrastructure, including energy, and critical inputs, including energy and raw materials, as relevant factors. The United States expanded the jurisdiction of the **Committee on Foreign Investment in the United States** (CFIUS)

through the Foreign Investment Risk Review Modernization Act of 2018, including over investments in critical infrastructure. The United Kingdom's **National Security and Investment Act 2021** requires mandatory notification of acquisitions in seventeen sensitive sectors, including energy and civil nuclear. These regimes reflect concerns about control of strategic energy infrastructure by foreign state-owned enterprises, and have been used to block or condition investments in energy networks and critical minerals. Most investment treaties apply only to investments that have been admitted in accordance with the host state's law, and do not confer a right of establishment, although some treaties, notably those of the United States and Canada, extend national treatment and MFN to the pre-establishment phase.

14.2 Bilateral and Multilateral Investment Treaties

Bilateral Investment Treaties

The first **bilateral investment treaty** (BIT) was concluded between Germany and Pakistan in 1959. BITs proliferated from the late 1980s, and there are now more than 2,000 BITs in force, together with several hundred other treaties with investment provisions, including free trade agreements with investment chapters. BITs typically define the investments and investors protected; provide substantive standards of protection — fair and equitable treatment, full protection and security, national treatment, MFN treatment, protection against expropriation without compensation, free transfer of funds, and, in some treaties, an umbrella clause; and provide for the settlement of disputes between the host state and investors of the other party through international arbitration, as well as state-to-state dispute settlement.

Nigeria has signed BITs with a number of states, of which a smaller number are in force. The **Morocco–Nigeria BIT** (2016) has attracted attention as an innovative treaty that includes obligations on investors — including to comply with environmental and social impact assessment requirements, to maintain an environmental management system, to uphold human rights in the host state, to act in accordance with core labour standards, and not to engage in corruption — and provides for the possibility of civil liability actions against investors in their home state for acts causing significant damage, personal injuries, or loss of life in the host state. It also preserves the right of states to regulate and requires the exhaustion or attempt of domestic remedies and amicable settlement before arbitration.

The Energy Charter Treaty

The **Energy Charter Treaty** (ECT), signed in Lisbon in 1994 and in force since 1998, is the only multilateral treaty dedicated to the energy sector. It originated in the European Energy Charter, a political declaration of 1991 intended to promote

energy cooperation between Western Europe and the states of the former Soviet Union and Central and Eastern Europe, which possessed vast energy resources but needed Western investment and technology. The ECT has had around fifty contracting parties, including the European Union and Euratom, most European states, Japan, Turkey, and the states of Central Asia and the Caucasus; Russia signed the Treaty and applied it provisionally until 2009, but never ratified it.

The ECT covers trade in energy materials and products (applying GATT rules among parties, including those not WTO members), transit (Article 7, examined in Chapter 6), energy efficiency and environmental aspects (Article 19 and the Protocol on Energy Efficiency and Related Environmental Aspects), and, most significantly in practice, investment. **Part III** of the ECT provides for the promotion and protection of investments in the energy sector: Article 10(1) requires each contracting party to encourage and create stable, equitable, favourable, and transparent conditions for investors of other contracting parties, to accord their investments fair and equitable treatment, the most constant protection and security, and treatment no less favourable than that required by international law, including treaty obligations, and contains an umbrella clause requiring each party to observe any obligations it has entered into with an investor or an investment. Article 13 prohibits expropriation except for a purpose in the public interest, not discriminatory, carried out under due process of law, and accompanied by the payment of prompt, adequate, and effective compensation, amounting to the fair market value of the investment immediately before the expropriation or before it became known. **Article 26** provides for the settlement of disputes between an investor and a contracting party by international arbitration under the ICSID Convention, the ICSID Additional Facility, the UNCITRAL Arbitration Rules, or the Arbitration Institute of the Stockholm Chamber of Commerce. Article 17 permits a contracting party to deny the advantages of Part III to a legal entity owned or controlled by nationals of a third state if that entity has no substantial business activities in the territory of the contracting party in which it is organised.

The ECT has been the most frequently invoked investment treaty in the world, with more than one hundred and fifty known cases, the majority of them brought by investors from EU member states against other EU member states, principally concerning renewable energy support schemes (examined in Chapter 10). The *Yukos* arbitrations, in which tribunals constituted under the ECT awarded the former majority shareholders of Yukos Oil Company around USD 50 billion against Russia on the basis of Russia's provisional application of the Treaty, are among the largest awards in the history of international arbitration, examined in section 14.5. The modernisation of the ECT and the withdrawal of the European Union and many member states are examined in section 14.8.

Regional Investment Instruments in Africa

Several regional instruments address investment in Africa. The **ECOWAS Supplementary Act on Investments** (A/SA.3/12/08), adopted in 2008, establishes common investment rules for ECOWAS member states, including both investor rights and investor obligations — such as obligations to conduct environmental and social impact assessments, to comply with labour and human rights standards, and to refrain from corruption — and the right of states to regulate. The **ECOWAS Energy Protocol** (2003), examined in Chapter 13, provides investment protection in the energy sector modelled on the ECT. The **SADC Protocol on Finance and Investment** (2006), as amended in 2016 to remove investor–state arbitration and to narrow the fair and equitable treatment standard, and the **SADC Model Bilateral Investment Treaty Template** (2012) reflect a "development-oriented" approach to investment protection. The **Protocol on Investment to the AfCFTA Agreement**, adopted in 2023, seeks to establish a continental framework for the promotion, facilitation, and protection of investment, emphasising sustainable development, the right to regulate, and investor obligations, with the question of investor–state dispute settlement left to a separate annex. The **Pan-African Investment Code** (2016), a non-binding instrument adopted under the auspices of the African Union, influenced the AfCFTA protocol. These instruments reflect a distinctive African approach to investment law, emphasising balance between investor rights and host state interests.

Threshold Questions: Investment and Investor

Before a tribunal can consider the substantive merits of a claim, it must be satisfied that the claimant is a protected **investor** and that the dispute concerns a protected **investment**. Investment treaties typically define "investment" broadly, as "every kind of asset", followed by an illustrative list including shares, loans, contractual rights, intellectual property, and concessions under public law, including concessions to search for, extract, or exploit natural resources. Article 1(6) of the ECT defines investment as every kind of asset, owned or controlled directly or indirectly by an investor, including any right conferred by law or contract or by virtue of any licences and permits granted pursuant to law to undertake any economic activity in the energy sector.

Where arbitration is brought under the ICSID Convention, Article 25(1) requires that the dispute arise "directly out of an investment", and tribunals have debated whether this imposes an objective requirement independent of the treaty definition. The *Salini* test, derived from *Salini Costruttori SpA v Morocco* (ICSID, 2001), identified as indicative criteria a contribution of money or assets, a certain duration, an element of risk, and, more controversially, a contribution to the economic development of the host state. Energy investments generally satisfy

these criteria without difficulty, but questions arise in relation to pre-investment expenditures, purely contractual arrangements such as sales contracts, and financial instruments.

The definition of "investor" turns principally on nationality, which for corporations is typically determined by incorporation or seat, sometimes with additional requirements of substantial business activity or control. Investors may structure their investments through holding companies in states with favourable treaties — a practice known as **nationality planning** or "treaty shopping". Tribunals have generally accepted nationality planning undertaken before a dispute arises, but have rejected as an **abuse of process** restructurings undertaken when a specific dispute was foreseeable. In *Philip Morris Asia Ltd v Australia* (PCA, 2015), the tribunal declined jurisdiction over a claim brought by a Hong Kong company that had acquired its Australian subsidiary after the Australian government had announced its intention to introduce plain packaging legislation, holding that the restructuring was an abuse of rights because the dispute was foreseeable at the time. In *Plama Consortium Ltd v Bulgaria* (ICSID, 2005), a leading ECT case, the tribunal held that the denial of benefits clause in Article 17 operates prospectively and must be exercised by the state, and that the state had not effectively exercised it before the claim was brought. Subsequent tribunals, including in the *Yukos* cases, considered whether the claimants — companies incorporated in Cyprus and the Isle of Man, ultimately owned by Russian nationals — were protected investors, and held that the ECT did not exclude claims by entities incorporated in a contracting party, even if ultimately controlled by nationals of the respondent state.

14.3 Fair and Equitable Treatment

The Centrality of the FET Standard

The **fair and equitable treatment** (FET) standard is the most frequently invoked and most frequently successful standard in investment treaty arbitration. Its open-textured language gives tribunals considerable latitude in its interpretation, and its scope has been the subject of extensive debate. In the energy sector, FET claims have been central to disputes arising from changes to regulatory regimes — including renewable energy support schemes, tariff regimes, fiscal terms, and environmental regulation — and from the conduct of host state authorities in granting, administering, and revoking licences and permits.

	Core content	Leading authorities
Fair and equitable treatment	Legitimate expectations; non-arbitrariness; due process; transparency; denial of justice	Tecmed; Saluka; Parkerings; Waste Management II; Chevron v Ecuador II
Full protection & security	Due diligence against physical harm, incl. by third parties	AAPL v Sri Lanka; Wena Hotels v Egypt
National treatment / MFN	No less favourable treatment in like circumstances	SD Myers; Occidental I; Methanex; Maffezini; Plama
Expropriation	Public purpose, non-discrimination, due process, compensation (FMV)	Metalclad; Methanex; Philip Morris v Uruguay; Yukos
Umbrella clause	Observance of obligations entered into with the investor	SGS v Pakistan; SGS v Philippines; ECT Art 10(1)

Figure 14.1 The principal standards of investment protection, their core content, and leading energy-related authorities. *Source:* prepared for this edition.

The Relationship with the Minimum Standard of Treatment

A preliminary question is whether the FET standard is an autonomous treaty standard or a reference to the **customary international minimum standard of treatment** of aliens. The classic formulation of the minimum standard is found in *Neer v Mexico* (1926), in which the United States–Mexico General Claims Commission held that the treatment of an alien, in order to constitute an international delinquency, should amount to an outrage, to bad faith, to wilful neglect of duty, or to an insufficiency of governmental action so far short of international standards that every reasonable and impartial person would readily recognise its insufficiency.

In the context of NAFTA, the Free Trade Commission issued a binding **Note of Interpretation** in 2001 providing that Article 1105(1) of NAFTA prescribes the customary international law minimum standard of treatment of aliens as the minimum standard to be afforded to investments, and that the concepts of "fair and equitable treatment" and "full protection and security" do not require treatment in addition to or beyond that which is required by the customary minimum standard. NAFTA tribunals have held that the customary standard has evolved since *Neer*, but that it imposes a relatively high threshold. In *Waste Management Inc v Mexico (No 2)* (ICSID, 2004), the tribunal held that the minimum standard

is infringed by conduct attributable to the state and harmful to the claimant if the conduct is arbitrary, grossly unfair, unjust, or idiosyncratic, is discriminatory and exposes the claimant to sectional or racial prejudice, or involves a lack of due process leading to an outcome which offends judicial propriety. In *Glamis Gold Ltd v United States* (UNCITRAL, 2009), concerning California measures restricting open-pit mining near Native American sacred sites, the tribunal held that the claimant bore the burden of proving the content of customary international law, which required conduct sufficiently egregious and shocking — a gross denial of justice, manifest arbitrariness, blatant unfairness, a complete lack of due process, evident discrimination, or a manifest lack of reasons — and dismissed the claim.

Most BITs and the ECT, however, contain unqualified FET clauses, which many tribunals have interpreted as autonomous standards, not limited to the customary minimum standard, although the practical difference between the two approaches has narrowed as tribunals interpreting the customary standard have recognised its evolution.

The Elements of FET

Arbitral practice has identified several elements of the FET standard.

Legitimate expectations. The protection of the investor's legitimate expectations is the most important and most controversial element of FET in energy disputes. In *Técnicas Medioambientales Tecmed SA v Mexico* (ICSID, 2003), the tribunal articulated an expansive conception, holding that FET requires the host state to provide treatment that does not affect the basic expectations taken into account by the foreign investor in making the investment, and that the investor expects the host state to act in a consistent manner, free from ambiguity and totally transparently, so that it may know beforehand any and all rules and regulations that will govern its investments. This formulation was criticised as setting an unattainable standard of perfect governance, and subsequent tribunals adopted more balanced approaches. In *Saluka Investments BV v Czech Republic* (UNCITRAL, 2006), the tribunal held that the determination of a breach of FET requires a weighing of the claimant's legitimate and reasonable expectations on the one hand and the respondent's legitimate regulatory interests on the other, and that no investor may reasonably expect that the circumstances prevailing at the time the investment is made remain totally unchanged.

The prevailing approach distinguishes between expectations arising from **specific commitments** or representations made by the state to the investor, which are more likely to be protected, and expectations arising from **general legislation**, which states are generally entitled to change. In *Parkerings-Compagniet AS v Lithuania* (ICSID, 2007), the tribunal held that it is each state's undeniable right

and privilege to exercise its sovereign legislative power, and that, in the absence of a stabilisation clause or a specific promise, an investor cannot legitimately expect that the legal framework will not change. As examined in Chapter 10, the renewable energy arbitrations against Spain, Italy, and the Czech Republic have extensively explored the circumstances in which legislation targeted at a specific class of investors, and administrative acts such as registration, can create legitimate expectations. Legitimate expectations must be assessed at the time of the investment, must be reasonable in the light of all the circumstances, including the political, socio-economic, cultural, and historical conditions prevailing in the host state, and must take into account the investor's due diligence.

Arbitrariness and discrimination. FET prohibits arbitrary conduct — conduct not based on legal standards but on discretion, prejudice, or personal preference — and unreasonable conduct lacking a rational relationship to a legitimate public policy. In *AES Summit Generation Ltd v Hungary* (ICSID, 2010), concerning the reintroduction of regulated electricity prices that reduced the profits of power generators with long-term PPAs, the tribunal held that a measure is reasonable if there is an appropriate correlation between the state's public policy objective and the measure adopted to achieve it, and found that Hungary's measures, adopted in response to political concern about excessive generator profits, were a reasonable response to a rational policy objective, dismissing the claim.

Transparency and due process. FET requires a measure of transparency and procedural fairness in administrative decision-making, including notice and an opportunity to be heard in proceedings affecting the investor's rights. In *Metalclad Corp v Mexico* (ICSID, 2000), the tribunal held that Mexico had failed to ensure a transparent and predictable framework for the investor's business planning and investment, when a municipality denied a construction permit for a hazardous waste facility that had been approved by federal authorities, a finding partially set aside by the Supreme Court of British Columbia on the ground that the tribunal had imported a transparency obligation not found in NAFTA's Chapter 11.

Consistency and good faith. FET prohibits conduct that is inconsistent, contradictory, or in bad faith, such as the use of regulatory powers for purposes other than those for which they were conferred. In *Bilcon of Delaware Inc v Canada* (UNCITRAL, 2015), the tribunal held, by majority, that Canada had breached the minimum standard when a joint federal–provincial environmental review panel recommended rejecting a quarry and marine terminal project on the basis of "community core values", a criterion not authorised by the applicable law, after the investor had been encouraged to invest. The dissenting arbitrator warned that the decision would chill environmental regulation, a concern widely echoed by commentators.

Denial of justice. FET encompasses the customary prohibition of **denial of justice**, which concerns the conduct of the host state's judicial system: the refusal of access to courts, undue delay, proceedings lacking fundamental fairness, and manifestly unjust judgments. Denial of justice generally requires the exhaustion of available local remedies, because the state is responsible only for the failure of its judicial system as a whole. In *Chevron Corp and Texaco Petroleum Co v Ecuador (II)* (PCA, 2018), the tribunal held that the judgment of the Ecuadorian courts in the Lago Agrio litigation — which had ordered Chevron to pay around USD 9.5 billion for environmental damage in the Ecuadorian Amazon attributed to Texaco's operations — had been procured through fraud, bribery, and corruption, including the ghost-writing of the judgment by the plaintiffs' representatives, and constituted a denial of justice in breach of the FET standard and the umbrella clause of the US–Ecuador BIT. The case, which is also examined in Chapter 27, illustrates the intersection of investment arbitration with transnational environmental litigation.

Full Protection and Security

The standard of **full protection and security** (FPS) requires the host state to exercise due diligence to protect the investor's investment against physical harm, including harm caused by third parties such as rioters, rebels, or criminals. In *Asian Agricultural Products Ltd v Sri Lanka* (ICSID, 1990), the first BIT arbitration, the tribunal held that FPS does not impose strict liability on the state, but requires the exercise of due diligence, and found Sri Lanka liable for the destruction of the claimant's shrimp farm during a counter-insurgency operation by security forces. In *Wena Hotels Ltd v Egypt* (ICSID, 2000), the tribunal found Egypt liable for failing to prevent the seizure of hotels by a state-owned company. Some tribunals have extended FPS beyond physical security to legal security — the stability of the legal framework — but the prevailing view, and the express text of many recent treaties, limits it to physical protection. FPS is of particular significance for energy investments in regions affected by armed conflict, civil unrest, or criminality, including oil theft, pipeline vandalism, and attacks on installations, such as those that have affected the Niger Delta, examined in Chapter 24.

::: casebox {title="Occidental Petroleum Corporation v Republic of Ecuador (ICSID, 2012; Annulment 2015)"} **Facts.** Occidental, a United States oil company, held a participation contract with Ecuador's state oil company, Petroecuador, for the exploration and exploitation of Block 15 in the Ecuadorian Amazon. The contract prohibited the transfer of rights under it without ministerial authorisation. In 2000, Occidental "farmed out" a forty per cent economic interest in Block 15 to a Canadian company, AEC (later EnCana), without seeking ministerial authorisation. In 2006, after the relationship between Ecuador and Occidental had deteriorated, Ecuador issued a *caducidad* (termination) decree.

The decree terminated the contract, and Block 15 and its associated assets were transferred to the state, on the ground that the unauthorised transfer violated the contract and Ecuadorian law.

Award. The tribunal held, by majority, that Occidental had breached the contract by failing to obtain authorisation for the farmout. It held, however, that the *caducidad* decree was a disproportionate response to the breach and that Ecuador had therefore breached the FET standard of the US–Ecuador BIT and customary international law. The decree also constituted an expropriation of Occidental's investment. The tribunal emphasised that Ecuador had alternatives, such as negotiating a transfer fee or imposing a lesser sanction. Applying the principle of proportionality, it found that the termination of a contract of such value was out of proportion to the wrongdoing and to the interest Ecuador sought to protect. The majority awarded around USD 1.77 billion plus interest, reduced by twenty-five per cent to reflect Occidental's contributory fault. One arbitrator dissented on the quantum, arguing that the farmout had transferred forty per cent of the investment to AEC, so that Occidental could not claim compensation for that share.

Annulment. In 2015, an ICSID *ad hoc* committee partially annulled the award, holding that the tribunal had manifestly exceeded its powers by awarding Occidental compensation for the forty per cent interest that had been transferred to AEC. It reduced the award by forty per cent, to around USD 1.06 billion plus interest. Ecuador subsequently agreed a settlement with Occidental.

Significance. The case illustrates the application of proportionality to the exercise of contractual and regulatory sanctions by host states against energy investors. It also illustrates the interaction between contract breach and treaty breach, the role of contributory fault in reducing compensation, and the significance of annulment as a corrective mechanism. :::

14.4 National Treatment and Most-Favoured-Nation Treatment

The Relative Standards

National treatment and MFN treatment are **relative** standards: they do not prescribe an absolute level of treatment, but require that foreign investors and their investments be treated no less favourably than domestic investors (national treatment) or investors of third states (MFN treatment) in like circumstances. They protect against discrimination on grounds of nationality. In the energy sector, discrimination claims arise from measures favouring national oil companies or state-owned utilities, from local content requirements, from preferential fiscal or regulatory treatment of domestic or third-state investors, and from the differential application of regulatory measures.

National Treatment

A typical national treatment clause requires each party to accord to investors of the other party and their investments treatment no less favourable than that which it accords, in like circumstances, to its own investors and their investments with respect to the establishment, acquisition, expansion, management, conduct, operation, and sale or other disposition of investments. The analysis typically involves three steps: identifying the appropriate comparator in "like circumstances"; determining whether the foreign investor was accorded less favourable treatment; and determining whether the difference in treatment is justified by a legitimate, non-discriminatory policy rationale.

The identification of **like circumstances** has been contested. In *SD Myers Inc v Canada* (UNCITRAL, 2000), concerning Canada's ban on the export of PCB waste, which prevented a US company from processing Canadian waste in the United States, the tribunal held that the assessment of like circumstances must take into account circumstances that would justify governmental regulations that treat investors differently in order to protect the public interest, and that the US investor was in like circumstances with Canadian waste processors because they were in the same economic or business sector and competed for the same business. It found a breach of national treatment, holding that Canada's measure was designed to protect the Canadian industry rather than to protect the environment. In *Occidental Exploration and Production Co v Ecuador* (LCIA, 2004), concerning the refusal of value added tax refunds to oil exporters, the tribunal adopted a broad comparison, holding that "in like situations" in the US–Ecuador BIT did not refer to the oil sector alone but to exporters generally, and that Occidental had been treated less favourably than exporters of other products, such as flowers and seafood, who received refunds; Ecuador's subsequent challenge to the award in the English courts was unsuccessful. In *Methanex Corp v United States* (UNCITRAL, 2005), concerning California's ban on the gasoline additive MTBE, the tribunal held that the appropriate comparator was domestic producers of methanol, not producers of the competing additive ethanol, and found no discrimination.

The jurisprudence suggests that tribunals will examine whether a difference in treatment between foreign and domestic investors is reasonably related to a legitimate public policy objective that does not distinguish, on its face or de facto, between foreign and domestic investors, and that the burden lies initially on the claimant to establish less favourable treatment of an investor in like circumstances, and then shifts to the state to justify the difference.

National Oil Companies and State Enterprises

The privileged position of national oil companies and state-owned utilities in many host states raises questions of national treatment. Host states commonly reserve exclusive rights or preferential terms to NOCs — for example, the right to participate in all licences on a carried basis, preferential access to acreage, or exemption from regulatory requirements. Most investment treaties permit such reservations, either through exceptions in the treaty text, as in the annexes of non-conforming measures in US and Canadian treaties, or because the NOC is not in like circumstances with private investors given its public functions. Under the USMCA, Mexico reserved its constitutional ownership of hydrocarbons. Under the ECT, contracting parties could list exceptions to national treatment in the pre-investment phase. Where no exception applies, discriminatory treatment favouring NOCs may breach national treatment, unless justified by a legitimate public policy. The prohibition of **performance requirements** in some treaties, such as Article 5 of the ECT (which incorporates the TRIMs Agreement) and the performance requirements articles of US and Canadian treaties, further constrains local content requirements imposed on foreign energy investors.

Most-Favoured-Nation Treatment

The MFN clause requires each party to accord to investors of the other party treatment no less favourable than that which it accords, in like circumstances, to investors of any third state. Its most significant and contested application concerns not discrimination in fact but the **importation of more favourable provisions** from other treaties concluded by the host state.

Procedural provisions. In *Emilio Agustín Maffezini v Spain* (ICSID, 2000), the tribunal held that the MFN clause in the Argentina–Spain BIT allowed the claimant to rely on the more favourable dispute settlement provisions of the Chile–Spain BIT, which did not contain the eighteen-month domestic litigation requirement of the Argentina–Spain BIT, while noting that the MFN clause could not be used to override fundamental public policy considerations that the parties had envisaged as conditions of their acceptance of arbitration. In *Plama Consortium Ltd v Bulgaria* (ICSID, 2005), the tribunal took the opposite view, holding that an MFN provision in a basic treaty does not incorporate by reference dispute settlement provisions set forth in another treaty, in whole or in part, unless the MFN provision leaves no doubt that the contracting parties intended to incorporate them. Arbitral practice has remained divided, and many recent treaties expressly exclude dispute settlement provisions from the scope of the MFN clause.

Substantive provisions. Tribunals have more readily permitted the use of MFN clauses to import substantive standards of protection from other treaties.

In *White Industries Australia Ltd v India* (UNCITRAL, 2011), the tribunal allowed the claimant to rely on the MFN clause in the Australia–India BIT to import the obligation to provide "effective means of asserting claims and enforcing rights" from the India–Kuwait BIT, and found that India had breached that obligation through the nine-year delay of its courts in dealing with the enforcement of an arbitral award. Some recent treaties exclude the importation of substantive standards from other treaties, providing that substantive obligations in other treaties do not in themselves constitute "treatment" for the purposes of MFN.

14.5 Direct and Indirect Expropriation

Expropriation in Treaty Law and Custom

The customary international law of expropriation, including the historical controversy over the standard of compensation and the nationalisations of the twentieth century, has been examined in Chapter 3. This section examines the treaty standard of protection against expropriation, focusing on the distinction between direct and indirect expropriation and the boundary between compensable expropriation and non-compensable regulation, which is the central issue in modern energy disputes.

Investment treaties typically prohibit expropriation or nationalisation of investments, directly or indirectly through measures having an effect equivalent to expropriation, except for a public purpose, on a non-discriminatory basis, in accordance with due process of law, and against the payment of compensation. Most treaties specify that compensation shall be prompt, adequate, and effective, and shall be equivalent to the fair market value of the investment immediately before the expropriation or before the impending expropriation became public knowledge — effectively incorporating the "Hull formula". An expropriation that satisfies all the conditions is lawful, and the state must pay the compensation specified in the treaty. An expropriation that fails to satisfy them — for example, because no compensation is paid, or because it is discriminatory — is unlawful, and the state's obligation is to make full reparation under customary international law, as articulated by the Permanent Court of International Justice in *Factory at Chorzów* (1928), which may result in higher compensation, including consequential losses and the value of the investment at the date of the award where that is higher than its value at the date of the expropriation, as held in *ADC Affiliate Ltd v Hungary* (ICSID, 2006). The valuation of damages is examined in Chapter 26.

Direct Expropriation

Direct expropriation involves the formal transfer of title to the investment to the state or a third party, or its outright physical seizure. Direct expropriations of energy investments were common in the era of the great nationalisations and have

recurred in the twenty-first century. In Venezuela, the government of President Chávez required foreign investors in the heavy oil projects of the Orinoco Belt to convert their interests into minority stakes in joint ventures controlled by the state oil company PDVSA in 2007, and expropriated the interests of those that refused, including ExxonMobil and ConocoPhillips. In *ConocoPhillips Petrozuata BV and others v Venezuela* (ICSID, 2019), the tribunal held that Venezuela had unlawfully expropriated ConocoPhillips' investments, having failed to negotiate compensation in good faith, and awarded around USD 8.7 billion; an annulment committee upheld the award in 2025. In *Venezuela Holdings BV (Mobil) v Venezuela* (ICSID, 2014), the tribunal found that the expropriation was not unlawful merely because the compensation offered was inadequate, and awarded around USD 1.6 billion, an amount substantially reduced on annulment in 2017. The enforcement of these and other awards against Venezuela, including through the court-ordered sale of shares in CITGO, a US refining subsidiary of PDVSA, is examined in Chapter 26.

Indirect Expropriation

Indirect expropriation occurs where the state, without formally transferring title, takes measures that substantially deprive the investor of the use, enjoyment, or value of its investment. Indirect expropriation may result from a single measure or from a series of measures ("creeping expropriation"). Its identification raises the central question of where the line is to be drawn between compensable expropriation and the non-compensable exercise of the state's regulatory powers.

Tribunals have adopted several approaches. Under the **sole effect doctrine**, the decisive criterion is the effect of the measure on the investor — whether it results in a substantial deprivation of the economic value or control of the investment — irrespective of the state's intention or purpose. In *Metalclad Corp v Mexico* (ICSID, 2000), the tribunal held that expropriation includes covert or incidental interference with the use of property which has the effect of depriving the owner, in whole or in significant part, of the use or reasonably expected economic benefit of property, even if not necessarily to the obvious benefit of the host state, and that it need not consider the motivation or intent of the measure. Under the **police powers doctrine**, by contrast, the state is not liable to pay compensation for the bona fide exercise of its regulatory powers in the public interest, such as the protection of health, safety, or the environment, even if the measure causes substantial economic harm. In *Methanex Corp v United States* (UNCITRAL, 2005), the tribunal held that, as a matter of general international law, a non-discriminatory regulation for a public purpose, which is enacted in accordance with due process and which affects, among others, a foreign investor or investment, is not deemed expropriatory and compensable unless specific commitments had been given by the regulating government to the investor con-

templating investment that the government would refrain from such regulation. In *Philip Morris Brands Sàrl v Uruguay* (ICSID, 2016), the tribunal held that Uruguay's tobacco control measures — a single presentation requirement and enlarged health warnings — were a valid exercise of the state's police powers for the protection of public health and did not constitute an expropriation.

Between these poles, many tribunals adopt a **balancing** or proportionality approach, considering the economic impact of the measure, its interference with the investor's reasonable investment-backed expectations, and its character, including its purpose and proportionality. In *Tecmed v Mexico* (ICSID, 2003), the tribunal held that, in determining whether regulatory measures are expropriatory, it must consider whether such measures are proportional to the public interest presumably protected thereby and to the protection legally granted to investments, taking into account that the significance of the impact has a key role in deciding the proportionality.

Modern treaties increasingly codify the analysis of indirect expropriation. The US Model BIT (2012), the Comprehensive Economic and Trade Agreement between Canada and the European Union (CETA), and many other treaties contain annexes providing that the determination of whether a measure constitutes an indirect expropriation requires a case-by-case, fact-based inquiry considering the economic impact of the measure, the extent to which it interferes with distinct, reasonable investment-backed expectations, and its character, and that, except in rare circumstances, non-discriminatory regulatory actions designed and applied to protect legitimate public welfare objectives, such as public health, safety, and the environment, do not constitute indirect expropriations. CETA adds that the rare circumstances include those where the impact of a measure is so severe in light of its purpose that it appears manifestly excessive.

Fiscal Measures and Expropriation

Fiscal measures — taxes, royalties, and windfall levies — are a common source of energy investment disputes, given the volatility of commodity prices and the pressure on host states to capture windfall profits. States are generally entitled to impose taxes, and the imposition of taxation is not, in itself, expropriatory. However, taxation may amount to expropriation where it is confiscatory, discriminatory, or an abuse of the taxing power. Many treaties, including Article 21 of the ECT, carve out taxation measures from most treaty obligations, while preserving the expropriation provision, often subject to a referral of the question whether the tax is expropriatory to the competent tax authorities of the parties.

In *Burlington Resources Inc v Ecuador* (ICSID, 2012), Ecuador had enacted Law 42, imposing a levy of first fifty and then ninety-nine per cent on oil companies' "extraordinary revenues" — revenues resulting from oil prices above

the reference price in their production sharing contracts. The tribunal held that the windfall levy at fifty per cent, and even at ninety-nine per cent, did not in itself constitute an expropriation, because the investment continued to be profitable and the investor retained control of it, although the subsequent physical takeover of the oil blocks constituted an unlawful expropriation. In *Perenco Ecuador Ltd v Ecuador* (ICSID, 2014), a tribunal found that the ninety-nine per cent levy breached the FET standard, given the specific commitments in Perenco's contracts. Both tribunals subsequently considered Ecuador's counterclaims for environmental damage, examined in section 14.7.

The Yukos Arbitrations

The *Yukos* arbitrations are the largest expropriation cases in the history of investment arbitration. Yukos Oil Company, Russia's largest oil producer in the early 2000s, was subjected from 2003 to a series of tax reassessments, asset freezes, and criminal proceedings against its senior executives, including its chief executive, Mikhail Khodorkovsky, culminating in the auction of its principal production subsidiary, Yuganskneftegaz, which was acquired by the state-owned Rosneft, and the company's bankruptcy in 2006. The former majority shareholders brought claims under the ECT, which Russia had signed and applied provisionally under Article 45 pending ratification. In 2014, tribunals constituted under the UNCITRAL Rules and administered by the Permanent Court of Arbitration held that Russia's measures, taken as a whole, had the effect of an expropriation of Yukos and that their primary objective was not to collect taxes but to bankrupt Yukos and appropriate its assets, and awarded around USD 50 billion in damages.

In 2016, the District Court of The Hague set aside the awards, holding that the tribunals lacked jurisdiction because Russia's provisional application of the ECT under Article 45 was limited to provisions consistent with its constitution, laws, and regulations, which did not permit investor–state arbitration. The Hague Court of Appeal reversed that judgment in 2020, holding that Article 45 required an assessment of whether the principle of provisional application as such was consistent with Russian law, rather than whether each provision was consistent. The Supreme Court of the Netherlands upheld the Court of Appeal's ruling on jurisdiction in 2021, but remitted the case for consideration of Russia's argument that the awards were procured by fraud, which the Amsterdam Court of Appeal rejected in 2024, subject to Russia's further appeal. The shareholders have pursued enforcement in several jurisdictions. The *Yukos* saga illustrates the interaction of investment arbitration with the provisional application of treaties under the Vienna Convention on the Law of Treaties, the role of national courts at the seat of arbitration, and the challenges of enforcing awards against a non-compliant state.

14.6 Stabilization Clauses and Regulatory Change

The Problem of Regulatory Change

The obsolescing bargain manifests itself most frequently not in outright expropriation but in regulatory and fiscal change: increases in taxes and royalties, the imposition of windfall levies, the reform of tariff regimes, the tightening of environmental and local content requirements, and the renegotiation of contracts. Such changes may be justified by legitimate public interests, by changes in economic circumstances, or by the inadequacy of the original bargain, but they may also be opportunistic attempts to appropriate the returns on sunk investments. Investors seek protection against adverse regulatory change through two principal instruments: **contractual stabilisation clauses**, and the **treaty standards** examined in sections 14.3 to 14.5, particularly FET and the umbrella clause. This section examines their interaction. The drafting and contractual operation of stabilisation and change-in-law clauses are examined from a contractual perspective in Chapter 15.

Types of Stabilisation Clauses

Stabilisation clauses in host government contracts take several forms. **Freezing clauses** provide that the law applicable to the contract, or to the investor's activities, is the law in force at the date of the contract, and that subsequent changes in law shall not apply to the investor without its consent. **Consistency clauses** provide that the investment is governed by the terms of the contract, and that legislation inconsistent with the contract shall not apply. **Economic equilibrium clauses** do not prevent changes in law from applying to the investor, but require the state to restore the economic equilibrium of the contract — through compensation, tariff adjustment, or other means — if changes in law adversely affect the investor's economic position. **Hybrid clauses** combine elements of these approaches. Contemporary practice has shifted from freezing clauses towards economic equilibrium clauses, which are regarded as more compatible with state sovereignty and with the state's ongoing need to regulate.

The legal effect of stabilisation clauses was considered in the classic oil nationalisation arbitrations, examined in Chapter 3. In *Texaco Overseas Petroleum Co v Libya* (1977), the sole arbitrator held that stabilisation clauses in Libya's concessions were binding, and that Libya's nationalisation breached them. In *AGIP SpA v People's Republic of the Congo* (ICSID, 1979), the tribunal held that the stabilisation clauses in the agreement were binding under international law and that their breach engaged the state's responsibility. In *Kuwait v American Independent Oil Co (Aminoil)* (1982), the tribunal held that the stabilisation clause in the concession did not prohibit nationalisation in the absence of an express prohibition

for a limited period, but that its existence was relevant to the determination of compensation, reflecting the investor's legitimate expectations.

Stabilisation Clauses and Treaty Standards

Investment treaties interact with stabilisation clauses in several ways.

Umbrella clauses. An umbrella clause — such as the last sentence of Article 10(1) of the ECT, requiring each contracting party to "observe any obligations it has entered into with an Investor or an Investment of an Investor" — may elevate the breach of a contractual obligation, including a stabilisation clause, into a breach of the treaty, enabling the investor to bring a treaty claim. The scope of umbrella clauses has been contested. In *SGS Société Générale de Surveillance SA v Pakistan* (ICSID, 2003), the tribunal adopted a narrow interpretation, holding that the umbrella clause did not elevate every breach of contract into a breach of the treaty. In *SGS v Philippines* (ICSID, 2004), a differently constituted tribunal adopted a broader interpretation, holding that the umbrella clause made it a breach of the BIT for the host state to fail to observe binding commitments, including contractual commitments, which it had assumed with regard to specific investments, although the investor could not rely on the clause to bypass the contractual forum selection clause. Subsequent tribunals have generally held that umbrella clauses cover obligations specifically entered into with the investor, including stabilisation commitments, but differ on whether they cover obligations entered into by state entities with separate legal personality and on the effect of contractual forum clauses.

Legitimate expectations. A stabilisation clause is the clearest form of specific commitment by a host state, and its existence may support the investor's legitimate expectation that the regulatory framework will not change to its detriment, so that a breach of the clause may also breach the FET standard. In *Duke Energy Electroquil Partners and Electroquil SA v Ecuador* (ICSID, 2008), the tribunal held that Ecuador had breached the FET standard by failing to honour guarantees in a power purchase agreement, including the guarantee of payment, on which the investors had legitimately relied. Conversely, as noted in section 14.3, in the absence of a stabilisation clause or specific commitment, tribunals have generally held that investors cannot legitimately expect that the legal framework will not change, as in *Parkerings*. The existence or absence of stabilisation therefore significantly affects the outcome of FET claims arising from regulatory change.

The Argentine Gas Cases and the Necessity Defence

The economic crisis in Argentina in 2001–2002 gave rise to a wave of investment arbitrations, many involving energy investors, particularly in the gas transportation and distribution sector, which had been privatised in the 1990s under a regulatory framework that guaranteed tariffs calculated in US dollars and adjusted

in accordance with the United States producer price index. In response to the crisis, Argentina's Emergency Law of January 2002 abolished the convertibility of the peso with the US dollar, converted tariffs into pesos at a rate of one to one, and froze tariff adjustments. Investors claimed that the measures breached the FET standard, the umbrella clause, and the expropriation provisions of the US–Argentina BIT and other treaties.

The tribunals generally held that Argentina had breached the FET standard by dismantling the tariff guarantees on which the investors had relied, but differed on whether Argentina could rely on the defence of necessity, either under customary international law, as codified in Article 25 of the International Law Commission's Articles on Responsibility of States for Internationally Wrongful Acts, or under Article XI of the US–Argentina BIT, which provided that the treaty did not preclude measures necessary for the maintenance of public order or the protection of a party's essential security interests. In *CMS Gas Transmission Co v Argentina* (ICSID, 2005), the tribunal rejected the necessity defence, holding that the crisis was not so severe as to threaten the state's existence, that Argentina had contributed to it, and that the measures adopted were not the only means available, and awarded around USD 133 million. In *LG&E Energy Corp v Argentina* (ICSID, 2006), by contrast, the tribunal accepted the necessity defence under Article XI for the period of the most acute crisis, from December 2001 to April 2003, holding that Argentina was exempted from liability for that period. The annulment committee in *CMS* (2007) criticised the tribunal's conflation of Article XI with customary necessity but declined to annul the award on that ground; in *Sempra Energy International v Argentina* (2010) and *Enron Corp v Argentina* (2010), annulment committees annulled the awards for failure to apply Article XI properly. In *Continental Casualty Co v Argentina* (ICSID, 2008), the tribunal interpreted Article XI by reference to the WTO jurisprudence on necessity, holding that most of Argentina's measures were justified.

The Argentine gas cases illustrate the significance of emergency and necessity defences in energy disputes arising from economic crises, and the inconsistency of arbitral reasoning, which has been a principal ground of criticism of the investment arbitration system.

Stabilisation and the Right to Regulate

Stabilisation clauses have been criticised for constraining the ability of host states — particularly developing states — to adopt new environmental, social, and human rights protections. A study conducted in 2008 for the International Finance Corporation and the United Nations Special Representative of the Secretary-General on Business and Human Rights, *Stabilization Clauses and Human Rights*, found that stabilisation clauses in contracts with non-OECD host states were more likely to be broad freezing clauses that could exempt

investors from new social and environmental laws, whereas those in OECD states more commonly limited stabilisation to fiscal matters or excluded laws of general application. The *Principles for Responsible Contracts*, issued by the Special Representative in 2011 and welcomed by the Human Rights Council, provide in Principle 4 that contractual stabilisation clauses, if used, should be carefully drafted so that any protections for investors against future changes in law do not interfere with the state's bona fide efforts to implement laws, regulations, or policies in a non-discriminatory manner in order to meet its human rights obligations.

Contemporary practice increasingly limits stabilisation to fiscal matters, excludes changes in law of general application relating to health, safety, environment, and human rights, provides for economic equilibrium rather than freezing, and limits the duration of stabilisation. The host government agreements for the Baku–Tbilisi–Ceyhan pipeline, concluded in 2000, contained broad economic equilibrium clauses which were criticised by civil society organisations; in response, the project sponsors executed a unilateral **Human Rights Undertaking** in 2003, committing not to invoke the economic equilibrium provisions in relation to the host states' regulation of health, safety, environment, and human rights in a manner consistent with international standards.

Windfall Taxes and the 2022 Energy Crisis

The energy price crisis of 2022 led many states to impose windfall taxes on the profits of energy companies, including the EU's temporary **solidarity contribution** on fossil fuel companies under Council Regulation (EU) 2022/1854, examined in Chapter 9, and the United Kingdom's Energy Profits Levy. These measures generated legal challenges both before national and EU courts — including actions by ExxonMobil subsidiaries before the General Court challenging the legality of the solidarity contribution — and under investment treaties, including ECT claims by the Klesch group against the European Union, Germany, and Denmark. These challenges raise questions concerning the competence of the European Union to impose such measures, the application of the ECT's taxation carve-out, the effect of stabilisation commitments, and the legitimacy of windfall taxation in the face of extraordinary profits.

14.7 Investor–State Dispute Settlement

Arbitration Without Privity

The distinctive feature of modern investment treaties is the provision for **investor–state dispute settlement** (ISDS), which enables an investor of one contracting party to bring a claim directly against the other contracting party before an international arbitral tribunal, without the need for the investor's home state to espouse the claim through diplomatic protection. The state's consent to

arbitration is expressed in the treaty as a standing offer to arbitrate with all qualifying investors; the investor accepts the offer by commencing arbitration, thereby perfecting the agreement to arbitrate. This model, described by Jan Paulsson as **"arbitration without privity"**, was first applied in *Asian Agricultural Products Ltd v Sri Lanka* (ICSID, 1990). It transformed investment protection from a matter of interstate diplomacy into a system of enforceable investor rights.

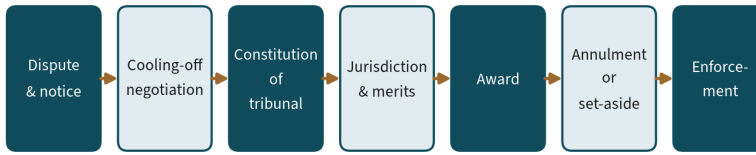


Figure 14.2 The principal stages of an investor–state dispute. *Source:* prepared for this edition.

Arbitral Forums

Investment treaties typically offer investors a choice among several arbitral forums. The **International Centre for Settlement of Investment Disputes** (ICSID), established by the Convention on the Settlement of Investment Disputes between States and Nationals of Other States (the ICSID Convention, 1965), is the principal institution, administering the majority of known investment treaty cases. ICSID arbitration is delocalised from national legal systems: ICSID awards are subject only to the Convention's internal annulment procedure (Article 52), not to review by national courts at the seat of arbitration, and must be recognised and enforced by all contracting states as if they were final judgments of their own courts (Article 54). The **ICSID Additional Facility** Rules apply where one of the parties is not a contracting state or a national of a contracting state. Arbitration under the **UNCITRAL Arbitration Rules**, often administered by the Permanent Court of Arbitration, and arbitration under the rules of the **Stockholm Chamber of Commerce** (SCC), which is expressly provided for in the ECT, are subject to the law of the seat of arbitration and to setting aside proceedings in its courts, and their awards are enforced under the **New York Convention** on the Recognition and Enforcement of Foreign Arbitral Awards (1958). The procedural frameworks of ICSID and UNCITRAL arbitration are examined in Chapter 26.

Preconditions and Parallel Proceedings

Investment treaties often impose preconditions on the investor's right to arbitrate, including a period of amicable negotiation (a "cooling-off" period, typically three to six months), the pursuit of domestic remedies for a specified period (as in the eighteen-month requirement at issue in *Maffezini*), or an election between domestic courts and international arbitration (a **"fork-in-the-road"** clause), under which the investor's choice to submit a dispute to domestic courts precludes subsequent arbitration of the same dispute. Tribunals have generally applied fork-in-the-road clauses narrowly, requiring that the dispute submitted to domestic courts be the same as that submitted to arbitration — in the sense of having the same fundamental basis, as held in *Pantechniki SA v Albania* (ICSID, 2009). Some treaties, notably the US and Canadian models and NAFTA, adopt a **"no U-turn"** or waiver approach, requiring the investor to waive the right to initiate or continue domestic proceedings with respect to the measure as a condition of arbitration.

Energy disputes frequently involve both contract claims, arising under a concession, PSC, or PPA with a forum selection clause, and treaty claims. In *Compañía de Aguas del Aconquija SA and Vivendi Universal v Argentina* (ICSID, annulment decision 2002), the annulment committee drew a fundamental distinction between contract claims and treaty claims, holding that whether there has been a breach of the BIT and whether there has been a breach of contract are different questions, each determined by reference to its own proper law, and that a contractual forum selection clause does not preclude a tribunal from considering a treaty claim, even where the treaty claim requires consideration of the contract. This distinction underpins the ability of energy investors to bring treaty claims notwithstanding contractual dispute resolution clauses.

State Counterclaims and Investor Obligations

The traditional asymmetry of investment treaties — which confer rights on investors and obligations on states — has been tempered by the increasing acceptance of **counterclaims** by host states. In the energy sector, Ecuador successfully brought counterclaims for environmental damage against oil companies in *Burlington Resources Inc v Ecuador* (ICSID, decision on counterclaims 2017), in which the tribunal awarded Ecuador around USD 40 million for environmental remediation and infrastructure damage, and in *Perenco Ecuador Ltd v Ecuador* (ICSID), in which the tribunal found Perenco liable for environmental harm in the blocks it had operated. In *Urbaser SA v Argentina* (ICSID, 2016), the tribunal accepted jurisdiction over a counterclaim alleging that the investor had violated the human right to water, holding that investors may be subject to obligations under international law, although it dismissed the counterclaim on the merits because the human right to water imposed obligations on states rather than investors

in the circumstances. The inclusion of investor obligations in recent treaties, such as the Morocco–Nigeria BIT and the ECOWAS Supplementary Act, may provide a firmer basis for counterclaims in the future.

Transparency and Third-Party Participation

Investment arbitration, modelled on confidential commercial arbitration, was initially criticised for its lack of transparency, given that it concerns public measures and public funds. The **UNCITRAL Rules on Transparency in Treaty-based Investor-State Arbitration** (2014) provide for the publication of documents, open hearings, and submissions by third parties (*amicus curiae*) and non-disputing treaty parties, and the **United Nations Convention on Transparency in Treaty-based Investor-State Arbitration** (the Mauritius Convention, 2014, in force 2017) extends their application to existing treaties. The ICSID Arbitration Rules, amended in 2022, provide for the publication of awards and the participation of non-disputing parties. Transparency has also been addressed through the disclosure of **third-party funding**, which is increasingly common in investment arbitration, including energy disputes, and raises issues of conflicts of interest and security for costs.

Legitimacy and State Responses

ISDS has been subjected to sustained criticism on several grounds: the inconsistency of arbitral decisions, illustrated by the divergent outcomes in the Argentine gas cases and the renewable energy cases; the lack of an appellate mechanism; concerns about the independence and impartiality of arbitrators, who are appointed by the parties and may act alternately as counsel and arbitrator ("double-hatting"), as illustrated by the annulment of the *Eiser* award, examined in Chapter 10; the costs of proceedings; the size of awards, which may represent a significant burden on the public finances of developing states; and the potential for "regulatory chill", whereby states refrain from adopting legitimate regulation for fear of claims.

States have responded in diverse ways. Bolivia (2007), Ecuador (2009), and Venezuela (2012) denounced the ICSID Convention, although Ecuador rejoined in 2021; Honduras denounced it in 2024. Ecuador's Constitution of 2008 prohibits the state from entering into treaties that cede sovereign jurisdiction to international arbitration in contractual or commercial disputes between the state and private parties, and a 2024 referendum rejected a proposal to remove the prohibition. South Africa terminated many of its BITs from 2012 and enacted the Protection of Investment Act 2015, which provides for domestic protection and state-to-state arbitration. India terminated most of its BITs in 2017 and adopted a new model BIT in 2015 that narrows the protections offered and requires the exhaustion of local remedies. The European Union has pursued the replacement

of ISDS by an **Investment Court System** in its trade agreements and the establishment of a multilateral investment court, examined in section 14.8.

14.8 Investment Treaty Reform and the Energy Transition

Investment Law and Climate Policy

The relationship between investment law and climate policy has become one of the most pressing issues in international economic law. The energy transition requires states to phase out fossil fuel infrastructure, to cancel or refuse permits for new fossil fuel projects, to reform energy markets, and to reallocate resources towards clean energy — measures that may adversely affect existing investments and give rise to claims under investment treaties. At the same time, the transition requires enormous private investment in clean energy, which investment protection may encourage by reducing political and regulatory risk. The challenge for reform is to preserve the protective function of investment law for clean energy investment, while ensuring that it does not impede, or impose unaffordable costs on, legitimate climate action.

Claims Arising from Fossil Fuel Phase-Out

Several investment claims have arisen from measures phasing out fossil fuels or restricting fossil fuel projects.

- In *RWE AG and RWE Eemshaven Holding II BV v Netherlands* and *Uniper SE and others v Netherlands*, brought under the ECT in 2021, German energy companies claimed compensation for the Dutch law prohibiting the use of coal for electricity generation from 2030. The claims were brought after the companies had recently commissioned new coal plants. The Dutch courts rejected the companies' domestic challenges to the law, and both claims were withdrawn in 2022, Uniper's in the context of its rescue by the German government.
- In *Rockhopper Italia SpA and others v Italy*, brought under the ECT, a British company claimed compensation for Italy's refusal to grant a production concession for the Ombrina Mare offshore oil field following the reintroduction of a ban on new oil and gas activities within twelve nautical miles of the coast. In 2022, the tribunal awarded the claimants around EUR 190 million plus interest. An ICSID annulment committee annulled the award in 2025.
- In *Westmoreland Coal Co v Canada*, a US coal company challenged Alberta's phase-out of coal-fired electricity generation, which provided transition payments to the owners of coal plants but not to the claimant, a supplier of coal. The claim was dismissed on jurisdictional grounds in 2022.
- In *TC Energy Corp and TransCanada Pipelines Ltd v United States*, the Canadian sponsor of the Keystone XL pipeline claimed more than USD 15

billion following the revocation of the pipeline's presidential permit in 2021. In 2024, the tribunal dismissed the claim for lack of jurisdiction, holding that the USMCA's legacy provisions did not extend to measures adopted after NAFTA's termination.

These cases illustrate the potential for fossil fuel investors to use investment treaties to challenge climate measures, and the debate about whether such claims are a legitimate protection of property rights against abrupt policy change or an illegitimate constraint on climate action. The Intergovernmental Panel on Climate Change noted in its Sixth Assessment Report that international investment agreements may be used by fossil fuel companies to block the phase-out of fossil fuel infrastructure, and the risk of "regulatory chill" has been widely discussed. On the other hand, tribunals have generally recognised the right of states to regulate in the public interest, and fossil fuel investors have not uniformly prevailed.

The Modernisation of the Energy Charter Treaty

The ECT, negotiated in the early 1990s with the primary objective of protecting Western investment in the energy sectors of the former Soviet Union, came to be regarded by many as inconsistent with the objectives of the Paris Agreement, because it protects investments in fossil fuels on the same terms as investments in clean energy, and because it has been used extensively to challenge regulatory changes in EU member states. In 2017, the Energy Charter Conference launched a process of modernisation of the Treaty, and negotiations began in 2020. The principal elements of the modernised Treaty, agreed in principle in 2022 and adopted by the Energy Charter Conference in December 2024, include:

- **Flexibility mechanism for fossil fuels.** Contracting parties may exclude from investment protection new investments in fossil fuels, and phase out protection for existing investments in fossil fuels after a transitional period. The European Union and the United Kingdom used this mechanism to exclude fossil fuel investments from protection in their territories.
- **Clarified standards.** The FET standard is defined by an exhaustive list of breaches, including denial of justice, fundamental breach of due process, manifest arbitrariness, targeted discrimination, and abusive treatment. Indirect expropriation is clarified, and the right to regulate in pursuit of legitimate policy objectives, including climate change mitigation and adaptation, is expressly affirmed.
- **Definitions.** The definitions of "investor" and "investment" are narrowed, including through a requirement of substantial business activities to address treaty shopping.

- **Expanded scope.** The Treaty's coverage is extended to new technologies, including hydrogen, ammonia, synthetic fuels, and carbon capture, utilisation, and storage.
- **Procedural reforms.** These include provisions on frivolous claims, security for costs, third-party funding, and transparency.

Amendments to the Treaty were provisionally applied from September 2025 by contracting parties that did not object.

Withdrawal from the Energy Charter Treaty

Before the modernisation was adopted, several states concluded that it did not go far enough and announced their withdrawal from the ECT. Italy withdrew in 2016, for reasons unrelated to climate policy. From 2022, France, Germany, Poland, Luxembourg, Slovenia, Spain, Portugal, the Netherlands, Denmark, Ireland, and the United Kingdom announced or completed their withdrawal. In 2024, the European Union and Euratom notified their withdrawal, which took effect in 2025, while the Council authorised the remaining EU member states that wished to do so to approve the modernised Treaty.

Withdrawal from the ECT does not immediately terminate its effect. Under Article 47(3), the Treaty's provisions continue to apply to investments made in the territory of a withdrawing party, or by investors of a withdrawing party in the territory of other parties, for a period of **twenty years** from the date when the withdrawal takes effect — the so-called **"sunset clause"**. Withdrawing states have therefore sought to neutralise the sunset clause. In 2024, the European Union and most of its member states signed an inter se agreement on the interpretation of the ECT, confirming their view, consistent with the judgment of the Court of Justice in *Komstroy* (2021), that Article 26 does not apply to intra-EU disputes and that the sunset clause does not apply to intra-EU relations. Whether such an inter se agreement can bind arbitral tribunals, or affect the rights of investors from third states, is contested.

UNCITRAL Working Group III and the Multilateral Investment Court

Since 2017, **UNCITRAL Working Group III** has considered the reform of investor–state dispute settlement, identifying concerns relating to the consistency, coherence, predictability, and correctness of arbitral decisions; the independence, impartiality, and diversity of arbitrators; and the costs and duration of proceedings. Its work has produced several instruments:

- a **Code of Conduct for Arbitrators** in International Investment Dispute Resolution, adopted in 2023, which addresses independence and impartiality, disclosure, and limits on double-hatting;
- a **Code of Conduct for Judges** in International Investment Dispute Resolution, adopted in 2023;

- a **Statute of the Advisory Centre on International Investment Dispute Resolution**, adopted in 2024, to provide legal assistance and capacity-building to developing states and least developed countries.

Discussions continue on more structural reforms, including a **standing multilateral investment court** with an appellate mechanism, championed by the European Union, and mechanisms for the selection and appointment of tribunal members. The EU has included a two-tier Investment Court System in its agreements with Canada (CETA), Viet Nam, Singapore, Mexico, and Chile.

New Generation Treaties

States have increasingly adopted "new generation" investment treaties and model treaties that seek to rebalance investor protection and the right to regulate. Common features include:

- preambular references to sustainable development and the right to regulate;
- narrower definitions of investment and investor;
- FET clauses limited to an exhaustive list of breaches or tied to the customary minimum standard;
- annexes clarifying indirect expropriation;
- general exceptions modelled on Article XX of the GATT;
- exclusion of MFN for dispute settlement;
- obligations on investors relating to corporate social responsibility, human rights, environment, and anti-corruption;
- limits on ISDS, including exhaustion of local remedies, short limitation periods, or the replacement of ISDS by state-to-state arbitration or domestic remedies.

Examples include the Netherlands Model BIT (2019), which provides that tribunals may take into account non-compliance by investors with the UN Guiding Principles on Business and Human Rights and the OECD Guidelines for Multinational Enterprises in determining compensation; the Indian Model BIT (2015); the SADC Model BIT (2012); and the Morocco–Nigeria BIT (2016). Some commentators have proposed "climate-aligned" investment treaties that would deny protection to new fossil fuel investments, provide enhanced protection for clean energy investments, and include obligations on home and host states to promote climate-friendly investment.

Implications for Africa

African states face distinctive choices in investment treaty reform. On the one hand, they need substantial foreign investment to develop their energy resources, expand electricity access, and capture the opportunities of the energy transition, including in critical minerals and green hydrogen. This creates an incentive to

offer credible protection to investors and reduce the cost of capital. On the other hand, African states have been respondents in significant energy and mining arbitrations — including the *Process and Industrial Developments (P&ID) v Nigeria* commercial arbitration, in which an award of around USD 11 billion arising from a gas processing contract was set aside by the English High Court in 2023 on the ground that it had been obtained by fraud, examined in Chapter 26 — and they are concerned about the constraints of investment treaties on their regulatory autonomy and the burden of awards on their public finances. The AfCFTA Protocol on Investment, the ECOWAS Supplementary Act, and new generation treaties such as the Morocco–Nigeria BIT represent an African approach that seeks to balance these concerns, emphasising investor obligations, the right to regulate, and sustainable development. The design of investment protection in Africa's energy transition is a subject of considerable importance for doctoral research.

::: exercise {title="Practical Exercise: The Republic of Azuri Coal Phase-Out and Gas Windfall Levy (HYPOTHETICAL)"} *This exercise is entirely hypothetical. The Republic of Azuri, the Kingdom of Westria, Westria Power AG, Meridian Gas BV, and all other entities named are fictional.*

The Republic of Azuri is a party to the ICSID Convention and to a 2005 BIT with the Kingdom of Westria. The BIT contains an unqualified FET clause, an umbrella clause, a standard expropriation clause, and ISDS under the ICSID Convention or UNCITRAL Rules, with a six-month cooling-off period and no exceptions for taxation or environmental measures. Azuri is also a contracting party to the original (unmodernised) Energy Charter Treaty. It has not ratified the modernised Treaty and has not notified withdrawal.

The coal plant. In 2016, Westria Power AG (HYPOTHETICAL) commissioned a 1,200 MW coal-fired power plant in Azuri. It operates under a twenty-five-year PPA with the state utility. The PPA contains an economic equilibrium clause covering "any change in law that materially increases the Seller's costs". In 2025, Azuri enacted the Clean Air and Climate Act, which prohibits the generation of electricity from coal after 2030. The Act offers compensation for unrecovered book value of investments, but not for lost future profits. It also exempts a coal plant owned by a domestic state-owned company until 2035 "for reasons of security of supply".

The gas levy. Meridian Gas BV (HYPOTHETICAL) is a company incorporated in Westria. Its ultimate parent is incorporated in a third state with no investment treaty with Azuri. Meridian holds a production sharing contract with a stabilisation clause freezing "the fiscal regime" for fifteen years from 2015. In 2022, Meridian's parent restructured its holding of the PSC through Meridian. In

the same year, Azuri imposed a "temporary solidarity levy" of 33 per cent on the profits of gas producers above 120 per cent of their 2018–2021 average profits.

1. Advise Westria Power on its potential claims under the BIT and the ECT, including FET, national treatment (considering the exemption for the state-owned plant), expropriation, and the umbrella clause. What is the significance of the economic equilibrium clause, and of the compensation offered under the Act?
2. Advise Azuri on its defences, including the police powers doctrine, the right to regulate, the relevance of its obligations under the Paris Agreement, and the reasonableness of the investor's expectations in 2016.
3. Advise Meridian on jurisdiction, including whether the 2022 restructuring could be characterised as an abuse of process, and on the merits of a claim relating to the solidarity levy, with reference to the stabilisation clause and the taxation provisions of the ECT.
4. Azuri's government is considering withdrawing from the ECT and terminating the Westria BIT. Advise it on the legal consequences, including the sunset clauses, and on alternative strategies, including renegotiation and an interpretive agreement with Westria. :::

CHAPTER SUMMARY

- Energy investment is capital-intensive, immobile, and politically salient, and is subject to the "obsolescing bargain". This explains the predominance of energy disputes in investment arbitration.
- Foreign investors are protected by host state law, by contracts with the state, and by more than 2,000 BITs, the ECT, and regional instruments. Threshold questions of "investment", "investor", and nationality planning determine access to treaty protection, and restructuring when a dispute is foreseeable may be an abuse of process.
- The FET standard protects legitimate expectations, particularly those based on specific commitments, against arbitrariness, lack of transparency and due process, bad faith, and denial of justice. Tribunals balance investor expectations against the state's right to regulate.
- National treatment and MFN protect against nationality-based discrimination in like circumstances. The use of MFN to import procedural provisions from other treaties remains contested.
- Treaties prohibit direct and indirect expropriation without compensation. The boundary between indirect expropriation and non-compensable regulation is drawn using effects-based, police powers, and proportionality approaches, increasingly codified in treaty annexes. *Yukos* and the Venezuelan cases illustrate the scale of modern expropriation disputes.

- Stabilisation clauses interact with treaty standards through umbrella clauses and legitimate expectations. The Argentine gas cases reveal the significance and inconsistency of the necessity defence, and contemporary practice limits stabilisation to protect the right to regulate.
- ISDS rests on "arbitration without privity". Preconditions, the distinction between contract and treaty claims, counterclaims, and transparency reforms shape its operation, and legitimacy concerns have prompted diverse state responses.
- The energy transition has generated claims challenging fossil fuel phase-outs. The ECT has been modernised and abandoned by the EU and many member states, subject to its sunset clause. UNCITRAL Working Group III and new generation treaties seek to rebalance the regime.

DOCTORAL RESEARCH QUESTIONS

1. Does investment treaty protection impede the energy transition? Evaluate the empirical evidence of "regulatory chill" and the outcomes of claims challenging fossil fuel phase-out measures.
2. Is the distinction between general legislation and specific commitments a satisfactory basis for the protection of legitimate expectations under the FET standard in the energy sector?
3. Critically assess the modernised Energy Charter Treaty. Does the flexibility mechanism for fossil fuels adequately align the Treaty with the Paris Agreement?
4. Can an inter se agreement among EU member states neutralise the ECT's sunset clause? Analyse the question under the Vienna Convention on the Law of Treaties.
5. Should investor obligations, such as those in the Morocco–Nigeria BIT and the ECOWAS Supplementary Act, be enforceable through counterclaims in investment arbitration? What are the doctrinal and practical obstacles?
6. Design a model investment treaty for an African energy-producing state that balances the attraction of clean energy investment with the preservation of regulatory space for the energy transition.

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15 CHAPTER

International Energy Contracts and Commercial Risk Allocation

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the principles governing the formation of international energy contracts and the determination of their applicable law, including party autonomy, mandatory rules, and the role of transnational rules.
2. Analyse the contractual architecture of concession and production sharing agreements as instruments of risk and reward allocation between host states and investors.
3. Evaluate the allocation of risk in engineering, procurement, and construction contracts, including liquidated damages, performance guarantees, and fitness for purpose obligations.
4. Assess the structure and governance of joint operating agreements and shareholder agreements in the energy sector.
5. Distinguish force majeure, hardship, and frustration, and analyse their application to energy contracts in the light of recent crises.
6. Draft and evaluate change-in-law and stabilisation provisions from the perspective of both investors and host states.
7. Analyse the allocation of liability through indemnities, exclusions, caps, guarantees, and insurance, including the "knock-for-knock" regime.
8. Explain the principles of contract interpretation, renegotiation, and termination as applied to long-term energy contracts.

Introduction

Contracts are the operating system of the energy industry. A single large energy project — an offshore oil field, an LNG export terminal, an independent power plant, or a solar park — is constituted by a web of interlocking contracts: a host government agreement or licence defining the investor's rights; a production sharing contract or concession allocating the resource and its proceeds; a joint

operating agreement governing the relationship among co-venturers; engineering, procurement, and construction contracts for the facilities; supply, transportation, and offtake agreements for inputs and outputs; operation and maintenance agreements; loan agreements, security documents, and direct agreements with lenders; and insurance policies. Each contract allocates risks — geological, construction, operational, market, political, regulatory, and environmental — among the parties, and the bankability and success of the project depend on the coherence of the overall allocation.

International energy contracts present distinctive legal features. Many involve states or state entities as parties, raising questions of sovereignty, the internationalisation of contracts, and the relationship between contract and public law. They are typically long-term — twenty, thirty, or more years — and must accommodate changes in economic, political, technological, and regulatory circumstances that cannot be foreseen at the time of contracting. They involve very large sums and complex technical subject matter. They are frequently negotiated on the basis of industry model forms, such as those of the Association of International Energy Negotiators (AIEN, formerly the AIPN) and the International Federation of Consulting Engineers (FIDIC), which have generated a body of transnational contractual practice. And their disputes are usually resolved by international arbitration rather than national courts.

This chapter examines contract formation and applicable law (section 15.1), concession and production sharing agreements as contractual instruments (section 15.2), engineering, procurement, and construction contracts (section 15.3), joint venture and shareholder agreements (section 15.4), force majeure and hardship (section 15.5), change-in-law and stabilisation provisions (section 15.6), indemnities, guarantees, and liability allocation (section 15.7), and contract interpretation, renegotiation, and termination (section 15.8). The licensing frameworks within which petroleum contracts operate have been examined in Chapter 5; gas sales and LNG contracts in Chapter 6; power purchase agreements in Chapter 9; the treaty-based protection of investors against regulatory change in Chapter 14; and the financing documents of energy projects in Chapter 16.

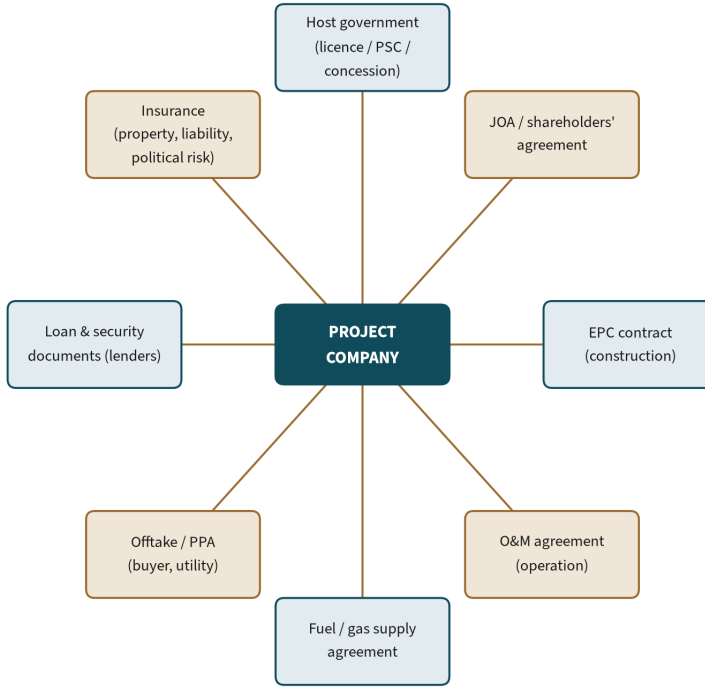


Figure 15.1 The contractual web of a typical energy project. *Source:* prepared for this edition.

15.1 Contract Formation and Applicable Law

Formation of International Energy Contracts

The formation of international energy contracts follows the general principles of the applicable law of contract, but the complexity and value of energy transactions give certain issues particular prominence.

Preliminary agreements. Energy transactions are typically negotiated over long periods, during which the parties may sign letters of intent, memoranda of understanding, heads of terms, term sheets, confidentiality agreements, and exclusivity agreements. The legal effect of these preliminary documents depends on the intention of the parties, as determined by the applicable law. Under English law, which governs a large proportion of international energy contracts, the question is whether the parties, objectively assessed, intended to create legal relations and had agreed all the terms they regarded, or the law requires, as essential. In *RTS Flexible Systems Ltd v Molkerei Alois Müller GmbH* [2010] UKSC 14, the Supreme Court held that a contract had been concluded on the terms agreed by the parties' conduct, notwithstanding that the parties had contemplated that

the contract would not become binding until executed, because they had subsequently agreed all the essential terms and acted on them, waiving the requirement of execution. The case illustrates the risk that parties proceeding with work before signing a formal contract may become bound on uncertain terms. Documents expressed to be "subject to contract" are generally not binding, but confidentiality and exclusivity undertakings in preliminary agreements are commonly intended to be binding. Civil law systems may impose pre-contractual duties of good faith, breach of which gives rise to liability (*culpa in contrahendo*), a concept also recognised in Article 2.1.15 of the UNIDROIT Principles of International Commercial Contracts, which provides that a party who negotiates or breaks off negotiations in bad faith is liable for the losses caused to the other party.

Standard terms and the battle of the forms. Trade in crude oil and petroleum products is conducted largely under standard terms, such as the general terms and conditions published by major trading companies (notably the BP Oil International General Terms and Conditions for sales and purchases of crude oil and petroleum products), and incorporating the International Chamber of Commerce's **Incoterms** rules, which define the delivery obligations, transfer of risk, and allocation of costs under terms such as FOB (free on board), CFR (cost and freight), CIF (cost, insurance, and freight), and DAP (delivered at place). Where parties exchange conflicting standard terms, the "battle of the forms" must be resolved by the applicable law.

Authority and capacity of state entities. Contracts with states and state entities raise questions of the authority of the officials who sign them and the capacity of state entities to conclude them, including whether necessary approvals — from ministries, cabinets, legislatures, or regulators — have been obtained. In some jurisdictions, petroleum and mining agreements require legislative ratification, which may give them the force of law. A failure to obtain required approvals may render the contract void or unenforceable under the law of the host state, although international tribunals have generally held that a state cannot rely on its own internal law to avoid obligations it has undertaken, a principle reflected in Article 27 of the Vienna Convention on the Law of Treaties and in arbitral jurisprudence. Corruption in the procurement of state contracts may render them unenforceable, as examined in Chapter 16.

Party Autonomy and Choice of Law

The principle of **party autonomy** — the freedom of the parties to choose the law governing their contract — is recognised by virtually all legal systems for international commercial contracts. In the European Union, Article 3 of the **Rome I Regulation** (Regulation (EC) No 593/2008) provides that a contract shall be governed by the law chosen by the parties, expressly or clearly demonstrated by the terms of the contract or the circumstances of the case. In the absence of choice,

Article 4 provides default rules, including that a contract for the sale of goods is governed by the law of the country of the seller's habitual residence and a contract for the provision of services by the law of the service provider's habitual residence, subject to an escape clause in favour of the law of the country with which the contract is manifestly more closely connected. The **Hague Principles on Choice of Law in International Commercial Contracts** (2015) provide a non-binding model of party autonomy that has influenced national laws.

English law is the most commonly chosen governing law for international energy contracts not involving host governments — including LNG sale and purchase agreements, EPC contracts, charterparties, and trading contracts — owing to its predictability, commercial orientation, respect for the literal terms of contracts, and the expertise of the London courts and arbitral community. New York law is also widely used, particularly in contracts involving US parties and financing. Contracts with host governments, such as petroleum agreements and concessions, are typically governed by the law of the host state, as host states generally insist that contracts relating to their natural resources be governed by their own law; in Nigeria, petroleum contracts with NNPC Ltd and the federal government are governed by Nigerian law.

Overriding Mandatory Rules and Public Policy

Party autonomy is limited by **overriding mandatory rules** — provisions regarded as crucial by a country for safeguarding its public interests, such as its political, social, or economic organisation, to such an extent that they apply irrespective of the law otherwise applicable (Article 9(1) of the Rome I Regulation). In the energy sector, overriding mandatory rules may include sanctions and export control legislation, competition law, environmental and safety legislation, local content requirements, foreign exchange controls, and rules on the ownership of natural resources. Under Article 9(3) of Rome I, effect may be given to the overriding mandatory provisions of the law of the country where the obligations arising out of the contract are to be or have been performed, in so far as those provisions render the performance of the contract unlawful. Sanctions regimes, such as those imposed on Russia following its invasion of Ukraine, have had a profound effect on energy contracts, rendering performance unlawful for parties subject to the sanctioning jurisdictions and giving rise to disputes over force majeure, frustration, and illegality.

The Vienna Convention on the International Sale of Goods

The **United Nations Convention on Contracts for the International Sale of Goods** (CISG, 1980) applies to contracts of sale of goods between parties whose places of business are in different contracting states, unless excluded by the parties. Article 2(f) excludes sales of electricity from its scope, but the CISG

applies, in principle, to sales of crude oil, petroleum products, natural gas, coal, and other energy commodities. In practice, the CISG is commonly excluded in international energy sales contracts, partly because English law, the most common choice, is not a party to the Convention (the United Kingdom has not ratified it), and partly because parties prefer the certainty of the national law with which they are familiar. Nigeria is not a party to the CISG.

State Contracts and Internationalisation

Contracts between states (or state entities) and foreign investors — **state contracts** — have raised the question whether they may be governed by international law or general principles of law, rather than by the law of the host state, thereby protecting the investor against unilateral changes by the host state to its own law. In the oil concession arbitrations of the mid-twentieth century, examined in Chapter 3, tribunals held that concessions could be "internationalised". In *Sapphire International Petroleums Ltd v National Iranian Oil Co* (1963), the arbitrator held that the concession was governed by general principles of law, given the absence of an express choice and the parties' evident intention to exclude the exclusive application of Iranian law. In *Texaco Overseas Petroleum Co v Libya* (1977), the sole arbitrator held that the concessions, which referred to the principles of Libyan law common to the principles of international law and, in their absence, to general principles of law, were internationalised contracts, and that Libya was bound by their stabilisation clauses under international law.

Modern practice frequently addresses the applicable law of state contracts through combined clauses — referring to the law of the host state supplemented by applicable principles of international law, or by the principles of law common to the host state and international law. Article 42(1) of the ICSID Convention provides that, in the absence of agreement by the parties, an ICSID tribunal shall apply the law of the contracting state party to the dispute and such rules of international law as may be applicable, which has been interpreted as giving international law a supplemental and corrective function.

Transnational Rules: The UNIDROIT Principles and Lex Mercatoria

The **UNIDROIT Principles of International Commercial Contracts**, first published in 1994 and revised in 2004, 2010, and 2016, are a non-binding restatement of general principles of international contract law, prepared by the International Institute for the Unification of Private Law. The Principles may apply where the parties have chosen them, where the parties have agreed that their contract shall be governed by general principles of law or the *lex mercatoria*, where the parties have not chosen any law, or as a means of interpreting or supplementing international uniform law instruments or domestic law. They are frequently referred to in international arbitration, including in energy disputes,

and contain provisions of particular relevance to long-term energy contracts, including those on hardship (Articles 6.2.1 to 6.2.3) and force majeure (Article 7.1.7). The 2016 edition was revised specifically to take into account the needs of long-term contracts, adding provisions on, among other things, contracts with open terms, cooperation, and termination for compelling reasons.

The concept of a ***lex mercatoria*** — an autonomous body of transnational commercial law, consisting of trade usages, general principles, and arbitral practice — has been invoked in energy arbitrations, although its existence and content are contested. Industry model contracts, such as the AIEN model forms for joint operating agreements, farmout agreements, gas sales agreements, and other transactions, contribute to a body of transnational contractual practice that informs the interpretation of energy contracts; some scholars describe this body of practice as a *lex petrolea*.

Arbitration Agreements and Forum Selection

International energy contracts almost invariably provide for dispute resolution by international arbitration, typically under the rules of the International Chamber of Commerce (ICC), the London Court of International Arbitration (LCIA), the Singapore International Arbitration Centre (SIAC), or the UNCITRAL Arbitration Rules, or, for contracts with states, ICSID. The arbitration agreement is separable from the main contract, so that its validity is not affected by the invalidity or termination of the contract. In Nigeria, the **Arbitration and Mediation Act 2023**, which replaced the Arbitration and Conciliation Act 1988, governs arbitration and mediation, adopting the UNCITRAL Model Law framework with modern features, including provisions on emergency arbitrators, third-party funding, and the consolidation of proceedings. Questions of arbitrability — whether a particular type of dispute, such as a tax dispute, may be resolved by arbitration — may be governed by the law of the seat or of the place of enforcement, and have arisen in Nigerian energy disputes, as examined in section 15.2. The international arbitration of energy disputes is examined in Chapter 26.

15.2 Concession and Production Sharing Agreements

Petroleum Agreements as Instruments of Risk Allocation

The principal forms of host government petroleum arrangement — concessions or licences, production sharing contracts (PSCs), service contracts, and joint ventures with state companies — and their place in national licensing regimes have been examined in Chapter 5. This section examines these arrangements from a contractual perspective, focusing on how their terms allocate risk and reward between the host state and the investor, and on the clauses that most frequently generate disputes.

Every petroleum agreement must allocate several fundamental risks. **Geological risk** — the risk that exploration will not discover commercial quantities of petroleum — is almost always borne by the investor, which funds exploration at its own risk and recovers nothing if no commercial discovery is made. **Price risk** — the risk that petroleum prices will fall or rise — is shared through the fiscal terms, which determine how the proceeds of production are divided at different price levels. **Cost risk** — the risk that development and operating costs will exceed expectations — is allocated through cost recovery provisions. **Political and regulatory risk** — the risk that the host state will change the terms unilaterally — is addressed through stabilisation provisions, examined in section 15.6, and treaty protections, examined in Chapter 14. The design of these provisions determines the "government take" — the share of the economic rent from petroleum production captured by the state — and the investor's incentives to explore, develop, and produce efficiently.

The Concession or Licence

Under a **concession** or **licence**, the host state grants the investor the exclusive right to explore for and produce petroleum in a defined area, and the investor acquires title to the petroleum produced at the wellhead or another delivery point, in exchange for the payment of royalties, taxes, and other fees. The concession is the oldest form of petroleum arrangement and remains the dominant model in many jurisdictions, including the United Kingdom, Norway, the United States, Australia, and Brazil (for most areas). In Nigeria, the licence model was historically used through oil prospecting licences and oil mining leases, often operated through joint ventures with the state oil company, and continues under the PIA through petroleum prospecting licences and petroleum mining leases.

The principal contractual or regulatory terms of a concession include the duration of the exploration and production phases; the minimum work programme and expenditure commitments; the relinquishment of portions of the area at specified intervals; the procedure for declaring a commercial discovery and approving a field development plan; the fiscal terms, including royalties and taxes; the state's right to participate; obligations relating to local content, training, and community development; environmental and decommissioning obligations; and provisions on assignment, change of control, termination, and dispute settlement.

The Production Sharing Contract

Under a **production sharing contract**, first used by Indonesia in the 1960s, the host state (usually through its national oil company) retains ownership of the petroleum, and the investor (the contractor) is engaged to conduct petroleum operations at its own risk and expense. If production is achieved, the contractor is entitled to recover its costs from a portion of production (**cost oil**), subject to a

limit, and the remaining production (**profit oil**) is shared between the state and the contractor according to an agreed formula. The contractor may also be liable to pay income tax, which in some PSCs is paid on its behalf by the national oil company out of the state's share of profit oil.

The key risk-allocation provisions of a PSC include the following.

Cost recovery limit. The **cost recovery limit** (or "cost stop") caps the proportion of production in any period that can be used for cost recovery, typically between fifty and eighty per cent, ensuring that the state receives a share of production from the outset. Unrecovered costs may be carried forward to subsequent periods. Disputes frequently arise over whether particular expenditures are recoverable, the allocation of overhead costs, and the treatment of costs incurred outside the contract area. Detailed **accounting procedures**, annexed to the PSC, define recoverable costs, and the state typically has audit rights.

Profit oil split. Profit oil may be split in fixed proportions or according to a sliding scale linked to production rates, cumulative production, the **R-factor** (the ratio of the contractor's cumulative revenues to cumulative costs), or the contractor's internal rate of return. Progressive splits, under which the state's share increases as profitability increases, allocate a greater share of windfall gains to the state and reduce the pressure for renegotiation when prices rise.

Ring-fencing. Ring-fencing provisions require costs to be recovered only from production in the contract area in which they were incurred, preventing the contractor from recovering exploration costs in one area from production in another.

Domestic market obligation. Some PSCs require the contractor to supply a proportion of its share of production to the domestic market, often at a discounted price. This obligation reflects the policy considerations examined in Chapter 13 in the context of dual pricing.

Title and lifting. PSCs specify the point at which title to the contractor's share of production passes, and contain lifting arrangements governing how each party takes its entitlement.

PSCs in Nigeria

Nigeria introduced PSCs in 1993 for deepwater and inland basin acreage, and the fiscal terms of PSCs for deep offshore and inland basin areas were set by the **Deep Offshore and Inland Basin Production Sharing Contracts Act 1999**. The Act provided for royalty rates that varied with water depth, falling to zero for production in water depths exceeding one thousand metres, and for a review of the terms when the price of oil exceeded twenty US dollars per barrel. The failure to implement the price review mechanism became a source of controversy, and in 2019 the National Assembly amended the Act to introduce a royalty

based on both water depth and price, with an additional royalty when oil prices exceed specified levels. The PIA has since introduced a new fiscal framework for new licences, while preserving existing PSCs subject to conversion options, as examined in Chapter 24.

Disputes under Nigerian PSCs have raised important questions of arbitrability and the relationship between contract and public law. In a dispute between Esso Exploration and Production Nigeria Ltd and the Nigerian National Petroleum Corporation concerning the allocation of crude oil under a PSC to satisfy royalty and tax obligations ("tax oil" and "royalty oil"), an arbitral tribunal rendered an award in favour of Esso in 2011. The Nigerian courts subsequently set aside the award, holding, among other things, that disputes concerning the assessment and collection of petroleum profits tax were not arbitrable because the relevant tax legislation vested jurisdiction in the Federal High Court and the tax authorities. The case illustrates the risk that disputes under host government contracts touching on taxation and other public law matters may be held non-arbitrable by the courts of the host state.

Service Contracts

Under a **service contract**, the contractor performs exploration and production services for the state or its national oil company in exchange for a fee, without acquiring title to petroleum or a share of production. Under a **risk service contract**, the contractor bears exploration risk and is remunerated, if production is achieved, by a fee per barrel or a share of revenues; under a **pure service contract**, the contractor is paid for its services regardless of the outcome. Service contracts are used in jurisdictions where constitutional or political constraints prohibit the grant of rights to petroleum to foreign companies, such as Iran (through its "buy-back" contracts and the subsequent Iran Petroleum Contract), Mexico (before its 2013 reform), and Iraq, where the technical service contracts awarded in 2009 for the redevelopment of major producing fields provided for a remuneration fee per barrel of incremental production, subject to production targets and cost recovery. Service contracts allocate most of the price risk and upside to the state, and have been criticised by investors for offering inadequate returns.

Clauses Frequently in Dispute

Beyond the fiscal terms, several clauses of petroleum agreements frequently give rise to disputes.

Assignment and change of control. Petroleum agreements typically prohibit the assignment of the contractor's interest without the consent of the state or the regulator, and many extend the requirement of consent to indirect transfers through a change of control of the contractor. Unauthorised transfers may lead to termination, as in *Occidental v Ecuador*, examined in Chapter 14. Host states

have increasingly sought to tax indirect transfers of interests, as in the disputes over capital gains tax on the sale of interests in Ugandan oil licences, examined in Chapter 16.

Commerciality and development obligations. Disputes may arise over whether a discovery is commercial, whether a development plan must be approved, and whether the contractor has diligently pursued development. Host states may seek to terminate agreements for failure to develop discoveries, while contractors may argue that development was not commercially viable or was delayed by the state.

Relinquishment and expiry. Questions arise as to the extent of the area to be relinquished, the treatment of discoveries in areas due for relinquishment, and the rights of the parties on expiry, including in relation to the renewal of licences. The renewal of Nigerian oil mining leases has generated disputes between licensees and the government.

Decommissioning. As examined in Chapter 5, the allocation of decommissioning liabilities between current and former licensees, and between the contractor and the state on the expiry of the agreement, is an increasingly significant source of disputes, particularly in mature basins and in onshore divestments in Nigeria.

15.3 Engineering, Procurement, and Construction Contracts

The EPC Model

Energy facilities — power plants, refineries, LNG trains, offshore platforms, pipelines, and renewable energy installations — are constructed under construction contracts that allocate the risks of design, procurement, construction, completion, and performance. The **engineering, procurement, and construction** (EPC) contract, also known as a **turnkey** contract, is the predominant model for project-financed energy facilities. Under an EPC contract, a single contractor undertakes responsibility for the design, procurement of equipment and materials, construction, commissioning, and testing of the facility, delivering a completed facility that meets specified performance requirements, for a fixed **lump-sum price** and by a fixed **completion date**. The EPC model offers the project company and its lenders a single point of responsibility, price certainty, and time certainty, which are essential to the financing of projects whose revenues will service debt only once the facility is operational. In exchange, the contractor assumes substantial risks, for which it charges a risk premium.

The **FIDIC Conditions of Contract for EPC/Turnkey Projects** (the "Silver Book", first published in 1999 and revised in 2017) is the most widely used international standard form for EPC contracts. It allocates most risks to the contractor, including the risk of unforeseeable physical conditions, which is allocated

to the employer under FIDIC's other standard forms. The FIDIC "Yellow Book" (Plant and Design-Build) and "Red Book" (Construction, for works designed by the employer) adopt more balanced risk allocations. Large energy projects frequently use bespoke contracts, adapted from standard forms, reflecting the parties' bargaining power and the requirements of lenders.

Allocation of Construction Risks

The key risk-allocation provisions of an EPC contract include the following.

Scope and design responsibility. The contractor is typically responsible for the design of the facility, based on the employer's requirements, and for any errors in the employer's requirements that an experienced contractor should have discovered. Disputes frequently arise over the scope of the works and whether particular work constitutes a variation entitling the contractor to additional payment and time.

Ground conditions. The risk of unforeseen ground or subsurface conditions is particularly significant for energy projects, including offshore wind foundations, tunnels, pipelines, and hydropower dams. Under the Silver Book, the contractor bears this risk, while other forms allocate it to the employer where conditions could not reasonably have been foreseen by an experienced contractor.

Time and delay. The contractor must complete the works by the completion date, subject to extensions of time for delays caused by the employer, variations, force majeure, and other specified events. If the contractor fails to complete on time, it must pay **delay liquidated damages** at a specified daily or weekly rate, usually subject to a cap, which are intended to compensate the employer for lost revenues and additional financing costs. Under English law, the **prevention principle** provides that an employer cannot recover liquidated damages for delay that it has caused, and an extension of time mechanism is necessary to preserve the employer's right to liquidated damages where it has caused delay. The treatment of **concurrent delay** — where delay is caused by both an employer risk event and a contractor risk event — is a frequent source of dispute; in *North Midland Building Ltd v Cyden Homes Ltd* [2018] EWCA Civ 1744, the Court of Appeal upheld a clause providing that concurrent delay would not entitle the contractor to an extension of time, holding that parties are free to allocate the risk of concurrent delay by clear words. **Time bar** provisions, which require the contractor to give notice of claims within a specified period as a condition of entitlement, are generally enforced by English courts; in *Obrascon Huarte Lain SA v Her Majesty's Attorney General for Gibraltar* [2014] EWHC 1028 (TCC), the court held that the twenty-eight-day notice requirement in the FIDIC Yellow Book was a condition precedent to the contractor's entitlement to an extension of time.

Performance. The contractor guarantees that the completed facility will meet specified performance criteria — such as output capacity, efficiency (heat rate), availability, and emissions — demonstrated through performance tests. If the facility fails to meet the guaranteed levels, the contractor must pay **performance liquidated damages**, typically calculated by reference to the net present value of the lost revenues or increased costs over the life of the facility, or, if performance falls below a minimum level, the employer may reject the facility.

Liquidated damages and penalties. Under English law, a liquidated damages clause is enforceable unless it is a **penalty**. In *Cavendish Square Holding BV v Makdessi; ParkingEye Ltd v Beavis* [2015] UKSC 67, the Supreme Court reformulated the penalty doctrine, holding that the test is whether the clause is a secondary obligation which imposes a detriment on the contract-breaker out of all proportion to any legitimate interest of the innocent party in the enforcement of the primary obligation. The decision makes it more difficult to challenge liquidated damages clauses negotiated by commercial parties of comparable bargaining power. In *Triple Point Technology Inc v PTT Public Co Ltd* [2021] UKSC 29, a case arising from a software contract with a Thai state-owned energy company, the Supreme Court held that, where a contract is terminated before completion, liquidated damages for delay accrue up to the date of termination, after which the employer's remedy is general damages for breach — confirming the orthodox position.

Limitation of liability. EPC contracts typically cap the contractor's aggregate liability (often at a percentage of, or the whole of, the contract price), with separate sub-caps for delay and performance liquidated damages, and exclude liability for indirect and consequential losses, subject to exceptions for fraud, wilful misconduct, and, in some contracts, gross negligence. Liability allocation is examined in section 15.7.

Security. The contractor's obligations are typically secured by performance bonds, advance payment guarantees, retention money, and parent company guarantees, examined in section 15.7.

::: casebox {title="MT Højgaard A/S v E.ON Climate & Renewables UK Robin Rigg East Ltd [2017] UKSC 59"} **Facts.** MT Højgaard contracted to design, fabricate, and install the foundations for the turbines of the Robin Rigg offshore wind farm in the Solway Firth. The technical requirements of the contract had two relevant elements. First, they required the foundations to be designed in accordance with an international standard for offshore wind turbine design, J101. Second, they provided that the design of the foundations "shall ensure a lifetime of 20 years in every aspect without planned replacement". The J101 standard contained an error, discovered after installation, that substantially overestimated the axial capacity of grouted connections. Shortly after completion,

the grouted connections began to fail. E.ON incurred remedial costs of around EUR 26 million and claimed them from MT Højgaard.

Decision. The Supreme Court held that MT Højgaard was liable. Its obligation to ensure that the foundations had a design life of twenty years was a free-standing obligation, not inconsistent with the obligation to comply with J101. Compliance with J101 was a minimum requirement and did not preclude a more onerous obligation to achieve a specified result. The Court observed that, where a contract contains two provisions, one requiring a product to be designed in accordance with a specified standard and the other requiring it to achieve a particular performance, the courts will generally treat the latter as an additional, more rigorous requirement. The contractor must comply with both.

Significance. The decision confirms that fitness for purpose and performance obligations in energy construction contracts may impose strict liability for achieving a specified result. The contractor may be liable even where it exercised reasonable skill and care and complied with the industry standard. The case is a leading authority on the interpretation of technical requirements in offshore energy contracts, and it emphasises the importance of careful drafting and of the contractor's insurance and liability caps. :::

EPC Structures for Energy Projects

Large energy projects often deviate from the single EPC model. **Split EPC contracts** divide the works into an offshore supply contract (for equipment supplied from outside the host country) and an onshore construction contract, to manage tax, customs, and licensing issues, with a **wrap-around guarantee** or coordination agreement ensuring single-point responsibility. **Multi-contract** structures, in which the employer contracts separately with suppliers of major equipment (such as turbines or gas compressors) and with construction contractors, may reduce costs but leave the employer with interface risk, which lenders may be reluctant to accept. LNG projects frequently use consortia of specialist contractors, and offshore wind projects typically use multi-contract structures, with separate contracts for turbines, foundations, cables, and installation vessels. The allocation of interface risk among multiple contractors is a recurrent source of disputes.

Large energy construction projects are prone to cost overruns and delays, often resulting in major disputes. The construction of LNG facilities in Australia and nuclear power plants in Europe and the United States has been marked by significant overruns, and project disputes are frequently resolved by international arbitration involving complex delay and quantum analysis, examined in Chapter 26.

15.4 Joint Venture and Shareholder Agreements

The Rationale for Joint Ventures

Energy projects are commonly undertaken by groups of companies acting together in joint ventures. The reasons include the sharing of risk and capital requirements, particularly in high-cost, high-risk exploration and development; the combination of complementary expertise, assets, and access to markets; compliance with legal requirements for state participation or local ownership; and the management of political risk through the participation of national oil companies, development finance institutions, or local partners. Joint ventures take two principal forms: **unincorporated joint ventures**, in which the parties hold undivided interests in the licence or contract and associated assets and conduct operations through a contractual arrangement, typically a **joint operating agreement** (JOA); and **incorporated joint ventures**, in which the parties hold shares in a jointly owned company, which holds the assets and conducts the business, governed by its constitutional documents and a **shareholders' agreement**. Upstream petroleum ventures are typically unincorporated, reflecting the tax advantages of direct ownership and the desire of each party to take and dispose of its share of production separately, while midstream, downstream, and power projects are more commonly incorporated.

The Joint Operating Agreement

The JOA governs the relationship among the holders of a petroleum licence or contract, who are usually described as the "parties" or "co-venturers". The **AIEN Model Form International Joint Operating Agreement**, first published in 1990 and most recently revised in 2012, is the most widely used international model, and national model forms exist in several jurisdictions, including the United Kingdom and Norway (where the state prescribes the JOA as a condition of the licence). The principal provisions of a JOA address the following matters.

Participating interests. Each party holds a specified participating interest, which determines its share of costs, production, and voting rights, and its share of liabilities.

The operator. One of the parties is appointed as **operator**, responsible for conducting joint operations on behalf of all the parties, in accordance with approved work programmes and budgets. The operator is typically not entitled to profit from its role as operator, recovering only its costs (including an overhead charge) from the joint account. The AIEN model provides that the operator shall not be liable to the non-operators for any losses arising from joint operations, except where the losses result from the **gross negligence or wilful misconduct** of the operator's senior supervisory personnel. This limitation reflects the "no gain, no loss" principle of operatorship — since the operator does not profit

from its role, it should not bear disproportionate liability — and has significant consequences for the allocation of loss in major incidents. The meaning of "gross negligence" is notoriously uncertain; English law does not recognise gross negligence as a distinct legal concept, and courts have interpreted it, in a contractual context, as connoting a serious disregard of or indifference to an obvious risk.

The operating committee. Decisions on joint operations, including approval of work programmes, budgets, and development plans, are made by an **operating committee** composed of representatives of the parties, voting according to their participating interests and specified pass marks. The pass mark determines whether a single party with a large interest can control decisions, or whether a minority can block them, and is a key subject of negotiation.

Sole risk and non-consent operations. Where the operating committee does not approve an operation that some parties wish to undertake, the JOA may permit those parties to conduct the operation at their sole risk and expense (**sole risk** or **exclusive operations**), with the non-participating parties having the option to join subsequently on payment of a premium, which compensates the participating parties for the risk they have borne. These provisions balance the interest of parties who wish to proceed with an opportunity against the interest of parties who do not wish to fund it.

Default. A party that fails to pay its share of costs (a **cash call**) is in default, and the JOA provides remedies, which may include the suspension of the defaulting party's rights, the payment of its share by the non-defaulting parties, the withholding of its share of production, and ultimately the **forfeiture** of its participating interest to the non-defaulting parties, with or without compensation. The enforceability of forfeiture provisions under the rules against penalties and relief against forfeiture has been debated. Following *Cavendish v Makdessi*, examined in section 15.3, forfeiture provisions in JOAs negotiated between sophisticated parties are likely to be upheld under English law, given the legitimate interest of the non-defaulting parties in the continuation of joint operations.

Transfers. JOAs restrict the transfer of participating interests, typically requiring the consent of the other parties or the host government and granting the other parties **pre-emption rights** (rights of first refusal) on transfers to third parties. Pre-emption provisions have generated substantial litigation, including over whether indirect transfers through changes of control trigger pre-emption, and over the valuation of interests in "package deals" in which the transferred interest is sold together with other assets.

Disposal of production. Each party is entitled and obliged to take in kind and separately dispose of its share of production, subject to lifting arrangements. Where the parties market gas jointly, as is common for gas projects, a separate gas marketing agreement may be required.

Joint Ventures with National Oil Companies

In many producing states, national oil companies participate in petroleum ventures alongside international companies. In Nigeria, the federal government has historically participated in onshore and shallow water petroleum operations through joint ventures between NNPC and international oil companies, under which NNPC typically held a majority participating interest of fifty-five or sixty per cent, governed by JOAs and memoranda of understanding. These **unincorporated joint ventures** imposed on the government an obligation to fund its share of costs through cash calls, and chronic underfunding by the government, leading to large arrears, constrained investment and production. Alternative funding arrangements, including modified carry agreements and incorporated joint ventures, were developed to address the problem. Under the PIA, NNPC was converted into a limited liability company, NNPC Ltd, operating on commercial principles, and the Act provides for the conversion of existing joint ventures into incorporated joint ventures, as examined in Chapter 24. The participation of a state company in a joint venture raises distinctive issues, including the potential conflict between the company's commercial role as a co-venturer and its role as an instrument of state policy, the application of public procurement and transparency rules, and the risk that the state company's funding obligations will be subordinated to other demands on the public budget.

Shareholders' Agreements in Incorporated Ventures

In incorporated joint ventures — common in LNG, refining, pipelines, and power projects — the relationship among the shareholders is governed by a shareholders' agreement and the company's articles of association. Key provisions include:

- **reserved matters**, which require the unanimous or super-majority approval of shareholders or directors, protecting minority shareholders against decisions on fundamental matters, such as changes to the business, major capital expenditure, borrowing, related-party transactions, and the admission of new shareholders;
- **deadlock** provisions, which provide mechanisms for resolving disagreements on reserved matters, including escalation to senior management, mediation, expert determination, and, ultimately, "Russian roulette" or "Texas shoot-out" buy-sell mechanisms, or winding up;
- **transfer restrictions**, including pre-emption rights, lock-up periods, and **tag-along** and **drag-along** rights;
- **funding** obligations, including commitments to provide equity and shareholder loans and the consequences of failure to fund;

- **offtake** and **supply** arrangements between the joint venture company and its shareholders, which may raise conflicts of interest;
- **dispute resolution**, typically by international arbitration.

Good Faith and Relational Contracts

Whether parties to joint venture agreements owe each other duties of good faith or fiduciary duties varies across legal systems. Civil law systems generally recognise a duty of good faith in the performance of contracts, as in Article 1104 of the French Civil Code and § 242 of the German Civil Code (BGB), and Article 1.7 of the UNIDROIT Principles provides that each party must act in accordance with good faith and fair dealing in international trade. English law has traditionally refused to recognise a general duty of good faith in commercial contracts, but in *Yam Seng Pte Ltd v International Trade Corp Ltd* [2013] EWHC 111 (QB), the High Court suggested that a duty of good faith may be implied in "relational contracts" — long-term contracts involving a high degree of communication, cooperation, and predictable performance based on mutual trust and confidence — and the concept has been applied in subsequent cases, including *Bates v Post Office Ltd (No 3)* [2019] EWHC 606 (QB). Long-term energy joint ventures and supply agreements may be characterised as relational contracts, although English courts have emphasised that express terms excluding or limiting such duties will be respected. Parties to unincorporated joint ventures do not generally owe each other fiduciary duties under English law merely by virtue of their relationship, although an operator may owe fiduciary duties in relation to specific functions, such as the handling of joint funds, and parties may agree to assume such duties expressly.

15.5 Force Majeure and Hardship

Supervening Events in Long-Term Contracts

Long-term energy contracts are particularly exposed to supervening events that prevent, delay, or make more onerous the performance of contractual obligations: natural disasters, armed conflict, terrorism, epidemics, strikes, government action, sanctions, and extreme market disruptions. The 2020 COVID-19 pandemic, which disrupted energy demand, supply chains, and construction; the 2022 disruption of Russian gas supplies to Europe; the attacks on shipping in the Red Sea from late 2023; and armed insurgency affecting LNG projects in northern Mozambique, have each tested the legal mechanisms for dealing with supervening events. Three principal mechanisms are relevant: contractual **force majeure** clauses; contractual **hardship** clauses and their civil law counterparts; and the common law doctrine of **frustration**.

Force Majeure Clauses

A **force majeure** clause excuses a party from liability for failure or delay in performing its obligations to the extent that the failure or delay is caused by an event beyond its reasonable control, as defined in the contract. The common law does not recognise force majeure as a general doctrine, so that the scope and effect of force majeure in common law contracts depend entirely on the terms of the clause. Civil law systems recognise force majeure as a general excuse for non-performance — under Article 1218 of the French Civil Code, for example, force majeure is an event beyond the debtor's control, which could not reasonably have been foreseen at the time of contracting and whose effects cannot be avoided by appropriate measures, which prevents the debtor from performing — but contractual clauses are typically used to define its scope precisely. Article 79 of the CISG and Article 7.1.7 of the UNIDROIT Principles provide similar excuses for impediments beyond a party's control.

Force majeure clauses in energy contracts typically contain several elements: a definition of force majeure, usually combining a general description (an event beyond the reasonable control of the affected party, which it could not have avoided or overcome by the exercise of reasonable diligence) with a non-exhaustive list of specific events; exclusions, such as financial inability to perform and changes in market conditions; requirements of causation, notice, and mitigation; the consequences of force majeure, including suspension of affected obligations and extension of time; and the right to terminate if the force majeure continues beyond a specified period. In PPAs and other contracts with state counterparties, force majeure is often divided into **natural force majeure** (events such as natural disasters, which are generally borne by each party without compensation) and **political force majeure** (events such as war, expropriation, and changes in law attributable to the host government, which may entitle the private party to compensation or to require the government to purchase the project).

The Interpretation of Force Majeure Clauses

English courts interpret force majeure clauses according to their terms, with several recurring principles.

Market changes are not force majeure. In *Thames Valley Power Ltd v Total Gas & Power Ltd* [2005] EWHC 2208 (Comm), the court held that a substantial increase in the market price of gas, which made the supplier's performance of a long-term gas supply contract uneconomic, did not constitute force majeure, because the supplier was not prevented from performing — it could buy gas at the higher price — and a clause excusing performance for events beyond a party's control does not generally cover an increase in the cost of performance.

Causation. In *Classic Maritime Inc v Limbungan Makmur Sdn Bhd* [2019] EWCA Civ 1102, a case arising from the collapse of the Fundão tailings dam in Brazil, examined in Chapter 7, which halted the supply of iron ore pellets for shipment under a contract of affreightment, the Court of Appeal held that the charterer could not rely on the force majeure clause because it would not have performed even if the dam had not collapsed, owing to the lack of demand; the clause required that the force majeure event be the cause of the failure to perform, applying a "but for" test.

Reasonable endeavours and non-contractual performance. In *RTI Ltd v MUR Shipping BV* [2024] UKSC 18, the Supreme Court held that a force majeure clause requiring that the event "cannot be overcome by reasonable endeavours from the party affected" did not require the affected party to accept an offer of non-contractual performance — in that case, payment in euros rather than the contractually stipulated US dollars, where sanctions against the charterer's parent company impeded payment in dollars — because the principle of freedom of contract entitles a party to insist on performance in accordance with the contract's terms. The decision is significant for the operation of force majeure clauses in the context of sanctions.

Hardship

Hardship clauses address events that do not make performance impossible but make it excessively onerous, fundamentally altering the equilibrium of the contract. Article 6.2.2 of the UNIDROIT Principles defines hardship as occurring where the occurrence of events fundamentally alters the equilibrium of the contract, either because the cost of a party's performance has increased or because the value of the performance a party receives has diminished, and the events occur or become known after the conclusion of the contract, could not reasonably have been taken into account by the disadvantaged party, are beyond its control, and the risk of the events was not assumed by it. Under Article 6.2.3, the disadvantaged party is entitled to request renegotiation, and if the parties fail to reach agreement within a reasonable time, either party may resort to the court, which may terminate the contract or adapt it with a view to restoring its equilibrium. The **ICC Hardship Clause 2020**, published by the International Chamber of Commerce together with the ICC Force Majeure Clause 2020, provides model clauses that parties may incorporate.

Several civil law systems recognise hardship as a general doctrine. Article 1195 of the French Civil Code, introduced by the reform of the law of obligations in 2016, provides that if a change of circumstances unforeseeable at the time of the conclusion of the contract renders performance excessively onerous for a party who had not accepted the risk of such a change, that party may ask the other party to renegotiate the contract, and, if renegotiation fails, the court may, at the request

of a party, revise or terminate the contract. Section 313 of the German Civil Code provides for the adaptation of a contract where circumstances that became the basis of the contract have significantly changed, a provision that was invoked in the context of long-term gas contracts during the 2022 energy crisis. In long-term gas supply contracts, the price review clauses examined in Chapter 6 function as a specialised form of hardship mechanism, permitting the adjustment of prices in response to significant changes in market conditions.

Frustration

The common law doctrine of **frustration** discharges a contract where, without the default of either party, a supervening event renders performance impossible, illegal, or radically different from that contemplated by the parties. The doctrine was established in *Taylor v Caldwell* (1863), in which the destruction by fire of a music hall discharged a contract for its hire. In *Davis Contractors Ltd v Fareham Urban District Council* [1956] AC 696, the House of Lords held that frustration occurs whenever the law recognises that, without default of either party, a contractual obligation has become incapable of being performed because the circumstances in which performance is called for would render it a thing radically different from that which was undertaken by the contract, and that the fact that performance has become more onerous or expensive is not sufficient. In *Tsakiroglou & Co Ltd v Noblee Thorl GmbH* [1962] AC 93, the closure of the Suez Canal during the 1956 crisis did not frustrate a contract for the shipment of groundnuts from Sudan to Hamburg, because the goods could be shipped via the Cape of Good Hope, although at greater cost and over a longer period. Frustration is applied narrowly; it does not apply where the contract makes provision for the event, where the event was foreseeable and the parties could have provided for it, or where the event was caused by the party seeking to rely on it (self-induced frustration, as in *J Lauritzen AS v Wijsmuller BV (The Super Servant Two)* [1990] 1 Lloyd's Rep 1). The narrowness of frustration underscores the importance of carefully drafted force majeure and hardship clauses in long-term energy contracts.

Recent Crises and Energy Contracts

The disruption of Russian gas supplies to Europe in 2022 generated extensive disputes under long-term gas contracts. Gazprom invoked force majeure in respect of reductions in deliveries, citing the unavailability of turbines for the Nord Stream pipeline owing to sanctions, an argument rejected by European buyers. As noted in Chapter 6, an arbitral tribunal awarded the German company Uniper more than EUR 13 billion in damages against Gazprom Export for non-delivery, and other European buyers obtained awards. In northern Mozambique, TotalEnergies declared force majeure on the Mozambique LNG project in April 2021

following an attack by insurgents on the town of Palma, suspending construction for an extended period; the declaration affected contractors, lenders, and the host government, and illustrates the use of force majeure in the context of armed conflict. The COVID-19 pandemic led to numerous force majeure declarations, including by Chinese LNG buyers in early 2020, which were largely rejected by sellers, and construction delays, whose treatment under EPC contracts depended on the precise wording of force majeure and change-in-law clauses.

15.6 Change-in-Law and Stabilisation Clauses: Drafting and Operation

From Investment Protection to Contractual Drafting

Section 14.6 examined stabilisation clauses from the perspective of international investment law: their legal effect, their relationship with the sovereign's right to regulate, and their treatment by investment tribunals. This section examines change-in-law and stabilisation provisions from the perspective of the drafter and the contract manager: how the clauses are structured, what events they cover, what thresholds and procedures apply, and what remedies they provide. In practice, the effectiveness of contractual protection against regulatory change depends far more on careful drafting than on abstract debates about the validity of freezing clauses under international law.

Defining "Change in Law"

The definition of the triggering event is the foundation of any change-in-law clause. Key drafting issues include the following.

What counts as "law". The definition typically covers legislation, subordinate legislation, regulations, decrees, and binding decisions of governmental authorities, and may extend to changes in the interpretation or application of law by courts or competent authorities, to the imposition of new requirements in permits, and to changes in tax law. Whether the definition includes changes in **industry standards**, **international obligations**, or the **policies** of regulators that are not legally binding is frequently contested.

Which jurisdiction. The clause usually covers changes in the law of the host state, but may also extend to changes in other relevant jurisdictions, such as the jurisdiction of equipment suppliers or financiers.

Discriminatory and general changes. Many PPAs and concession agreements distinguish between **discriminatory** (or specific) changes in law, which affect the project, the project company, or projects of the same type specifically, and **general** changes in law, which affect all businesses equally. Discriminatory changes are typically borne by the state counterparty, while general changes may be borne by the private party, at least within limits, on the theory that a general

change is a normal business risk which the private party can price into its tariff and which affects all competitors equally. This distinction reflects the influence of the regulatory taking and non-discrimination doctrines examined in Chapter 14.

Foreseeability. Changes that were announced, published in draft, or reasonably foreseeable at the date of the contract may be excluded, placing on the private party the risk of changes it should have anticipated.

Exclusions. Changes in the rate of general income or corporate taxes are sometimes excluded, as are changes required by international obligations or by the private party's own breach.

Thresholds and Procedures

Change-in-law clauses typically apply only where the change has a material effect, defined by reference to a financial threshold — for example, an increase in costs or decrease in revenue exceeding a specified amount per year, or a percentage of the project's revenue — so as to avoid disputes over trivial effects. The affected party must give notice of the change and its effect, supported by evidence, within specified time limits, and the parties are required to negotiate in good faith to agree on the appropriate adjustment, with recourse to expert determination or arbitration if they fail to agree. Mitigation obligations require the affected party to take reasonable steps to minimise the effect of the change.

Remedies

The remedies available under change-in-law clauses take several forms, corresponding to the categories of stabilisation clause identified in section 14.6:

- **compensation** by the state or state-owned counterparty, through a lump sum payment or periodic payments;
- **tariff adjustment**, under which the tariff payable under a PPA or concession is adjusted to restore the private party's economic position;
- **extension** of the term of the concession or licence, to allow the private party to recover additional costs over a longer period;
- **relief** from the obligation to comply with the change, which is unusual and difficult to enforce;
- **termination**, with a termination payment, where the change is so severe that the project cannot continue.

The standard of restoration is critical. Clauses may require that the private party be restored to the position it would have occupied had the change not occurred, or to the same **economic equilibrium**, measured by reference to the project's financial model, its equity internal rate of return, or its debt service coverage ratios. Lenders typically require that change-in-law protection preserve the project's ability to service debt, and the financial model agreed at financial close is frequently used as the reference point for determining compensation.

Political Force Majeure and Buy-Out Obligations

In PPAs and concession agreements with state counterparties in developing countries, change-in-law provisions are often combined with **political force majeure** provisions and **put option** or **buy-out** obligations, under which the host government is obliged to purchase the project for a specified price — usually sufficient to repay outstanding debt and, depending on the event, to provide a return on equity — if a political force majeure event continues beyond a specified period or a change in law makes the project unviable. These provisions are typically supported by government guarantees or put and call option agreements, and their credibility depends on the fiscal capacity of the host government, an issue examined further in Chapter 16.

Guidance and Best Practice

The World Bank's **Guidance on PPP Contractual Provisions**, published in 2017 and revised in 2019, provides recommended approaches to change-in-law, force majeure, termination payments, and related provisions in public-private partnership contracts, including power and energy projects. The Guidance recommends that the allocation of change-in-law risk reflect the capacity of each party to control and manage the risk, that discriminatory changes generally be borne by the public party, and that the allocation of general change-in-law risk depend on the maturity of the market and the capacity of the private party to pass costs through to users. The **Principles for Responsible Contracts**, developed by the UN Special Representative on Business and Human Rights in 2011 and endorsed by the Human Rights Council, provide that stabilisation clauses should not prevent the state from implementing laws, regulations, or policies in a non-discriminatory manner in order to meet its human rights obligations, and that any compensation provisions should be carefully drafted to ensure that the protection of investors does not interfere with the state's legitimate efforts to fulfil those obligations. This principle reflects the shift in drafting practice, noted in section 14.6, from freezing clauses towards economic equilibrium clauses that preserve regulatory space while protecting the investor's economic position.

Climate Policy and Change-in-Law Risk

The energy transition has brought change-in-law clauses to the centre of contractual risk allocation. Carbon taxes, emissions trading schemes, accelerated phase-out dates for coal and other fossil fuel plants, methane regulations, and the European Union's carbon border adjustment mechanism, examined in Chapter 13, may all impose substantial costs on energy projects and trigger change-in-law protections. For long-term contracts entered into today, the foreseeability of climate policy measures is a key issue: given the commitments made under the Paris Agreement, examined in Chapter 18, a tribunal may consider that the intro-

duction of carbon pricing or fossil fuel phase-out was reasonably foreseeable and therefore excluded from change-in-law protection, or that the investor assumed the risk. Drafters increasingly address climate measures expressly, either allocating their cost to a particular party or providing for specific adjustment mechanisms, rather than leaving the issue to general change-in-law provisions.

15.7 Liability Allocation, Indemnities, and Security Instruments

Knock-for-Knock Indemnities

The offshore petroleum industry has developed a distinctive approach to the allocation of liability for personal injury and property damage among the many contractors and subcontractors working on an installation: the **knock-for-knock** (or reciprocal) indemnity. Under a knock-for-knock arrangement, each party bears responsibility for injury to its own personnel and damage to its own property, regardless of fault, and indemnifies the other parties against claims in respect of them. Thus, if a drilling contractor's employee is injured through the negligence of the operator, the drilling contractor indemnifies the operator against the employee's claim. The rationale is that the arrangement avoids the cost and complexity of determining fault among multiple parties in a complex operating environment, allows each party to insure its own people and property without duplicating cover, and reduces disputes. Knock-for-knock indemnities are usually extended to the parties' respective "groups", including affiliates, co-venturers, and subcontractors, through mutual hold-harmless deeds, and may cover pollution and well control liabilities, typically allocating to the operator responsibility for pollution emanating from the reservoir and to the contractor responsibility for pollution emanating from its own equipment above the surface.

The enforceability of indemnities that cover the indemnified party's own negligence depends on clear drafting. Under English law, applying the principles derived from *Canada Steamship Lines Ltd v The King* [1952] AC 192, an indemnity will cover the indemnified party's negligence only if it does so expressly or by necessary implication, although modern courts approach the issue as one of ordinary contractual interpretation. In the United States, several states have enacted **anti-indemnity statutes** limiting indemnities for the indemnified party's own negligence in oilfield contracts, which may affect knock-for-knock arrangements.

The Deepwater Horizon Litigation

The 2010 Deepwater Horizon disaster in the Gulf of Mexico, which killed eleven workers and caused the largest marine oil spill in United States history, tested the indemnity provisions of the drilling contract between BP and Transocean, the owner and operator of the drilling rig. The contract allocated to BP responsibility

for pollution originating below the surface of the water, and to Transocean responsibility for pollution originating above the surface. In 2012, the federal district court held that BP was obliged to indemnify Transocean for third-party compensatory damages arising from pollution originating below the surface, even if the pollution resulted from Transocean's own gross negligence, although the indemnity did not extend to punitive damages or civil penalties under the Clean Water Act, which public policy prevented from being indemnified. The decision illustrates both the strength of carefully drafted pollution indemnities and their limits where public policy intervenes. The litigation also demonstrated the scale of liability that can arise from a major incident: BP's total costs exceeded USD 60 billion, including civil and criminal penalties, compensation, and clean-up costs.

Exclusion of Consequential Loss

Energy contracts commonly exclude liability for "consequential" or "indirect" losses, often defined to include loss of production, loss of profit, and business interruption. The interpretation of such exclusions has generated difficulties. Under the traditional English approach, derived from *Hadley v Baxendale* (1854), "consequential loss" was interpreted as referring to losses falling within the second limb of the remoteness rule — losses that do not arise naturally from the breach, but are recoverable only if they were within the contemplation of the parties at the time of contracting because of special circumstances. This interpretation meant that an exclusion of "consequential loss" might not exclude loss of profit or loss of production arising directly from the breach. More recent decisions have emphasised that the meaning of such exclusions depends on the ordinary meaning of the words in their context, and the industry practice is to define the excluded losses expressly, listing loss of production, loss of profit, and loss of use, and to apply the exclusion on a mutual basis, so that each party bears its own consequential losses regardless of fault. In *Transocean Drilling UK Ltd v Providence Resources plc* [2016] EWCA Civ 372, the Court of Appeal upheld a mutual exclusion of consequential loss in a drilling contract, which excluded the operator's claim for the spread costs incurred during a period of delay caused by the contractor's equipment failure, emphasising that the clause formed part of a coherent allocation of risk agreed between commercial parties of equal bargaining power, and should be given its natural meaning.

Liability Caps and Carve-Outs

Energy contracts frequently limit each party's aggregate liability to a specified amount, such as a percentage of the contract price, the value of the work, or available insurance, with exceptions for liabilities that the parties consider should be unlimited. Typical carve-outs from liability caps and consequential loss exclusions include liability for fraud, **gross negligence or wilful misconduct**, death

and personal injury, breaches of confidentiality, and the knock-for-knock indemnities themselves. The scope of these carve-outs is often heavily negotiated, and their interpretation, particularly the meaning of "gross negligence", frequently becomes the central issue in disputes following major incidents.

Security Instruments

Energy contracts rely on various security instruments to secure performance and payment obligations.

Parent company guarantees. Where a contracting party is a subsidiary or special purpose vehicle, the counterparty may require a guarantee from its parent company, under which the parent undertakes to perform or procure performance of the subsidiary's obligations. Host governments commonly require parent company guarantees from licensees and contractors under PSCs to secure work programme commitments and decommissioning obligations, as examined in Chapter 5.

Performance bonds and on-demand guarantees. EPC contracts and other construction contracts typically require contractors to provide performance bonds, advance payment guarantees, and retention bonds issued by banks. The critical distinction is between **conditional** (or accessory) guarantees, under which the guarantor is liable only if the principal is in breach of the underlying contract, and **on-demand** (or independent) guarantees, under which the guarantor must pay on presentation of a complying demand, without inquiring into whether the underlying obligation has been breached. In *Edward Owen Engineering Ltd v Barclays Bank International Ltd* [1978] QB 159, the Court of Appeal held that a bank which has given an on-demand performance guarantee must honour it according to its terms, and is not concerned with whether the beneficiary is entitled to the payment as between itself and the principal, subject only to the exception of clear fraud of which the bank has notice. The autonomy of on-demand guarantees makes them a powerful tool for beneficiaries, but also exposes contractors to the risk of abusive calls, and courts in some jurisdictions, such as Singapore, permit injunctions restraining calls on grounds of unconscionability as well as fraud.

Standby letters of credit. Standby letters of credit perform a similar function to on-demand guarantees and are common in United States practice and in energy trading. They are typically governed by the ICC's **Uniform Customs and Practice for Documentary Credits (UCP 600)** or the **International Standby Practices (ISP98)**, while on-demand guarantees are commonly subject to the ICC's **Uniform Rules for Demand Guarantees (URDG 758)**, which set out the rules governing demands, examination, and payment.

Payment security under PPAs. Independent power producers selling to state-owned utilities in developing countries frequently require payment security in the form of letters of credit, escrow accounts, sovereign guarantees, or partial risk guarantees from multilateral institutions, as examined in Chapter 16.

Insurance

Insurance complements contractual risk allocation. Energy projects commonly carry construction all risks insurance, operational property damage and business interruption insurance, third-party liability insurance, control of well insurance (for drilling operations), marine insurance, and political risk insurance. The interaction between insurance and contractual indemnities is critical: indemnities are only as valuable as the financial capacity of the indemnifying party, and insurance provides the funds to meet them. Contracts typically require each party to insure the risks it has assumed, to name the other parties as co-insureds, and to waive the insurers' rights of subrogation against them, thereby preventing an insurer that has paid a claim from recovering it from another party whom the indemnity scheme was designed to protect.

15.8 Interpretation, Renegotiation, and Termination of Energy Contracts

Principles of Contractual Interpretation

The meaning of contractual provisions is frequently the central issue in energy disputes, and the approach to interpretation varies across legal systems. English law, which governs a large proportion of international energy contracts, has developed its principles of interpretation through a series of leading decisions. In *Investors Compensation Scheme Ltd v West Bromwich Building Society* [1998] 1 WLR 896, Lord Hoffmann stated that interpretation is the ascertainment of the meaning which the document would convey to a reasonable person having all the background knowledge which would reasonably have been available to the parties in the situation in which they were at the time of the contract, an approach that emphasised context. In *Arnold v Britton* [2015] UKSC 36, the Supreme Court emphasised the primacy of the natural and ordinary meaning of the words, cautioning that commercial common sense should not be invoked to undervalue the language of the provision or to rescue a party from a bad bargain. In *Wood v Capita Insurance Services Ltd* [2017] UKSC 24, the Court reconciled these approaches, describing interpretation as a unitary exercise in which textual analysis and contextual analysis are tools to be used in different proportions depending on the nature, formality, and quality of drafting of the contract — sophisticated, professionally drafted contracts, such as most energy agreements, being more likely to be interpreted principally by textual analysis.

The implication of terms is approached restrictively. In *Marks and Spencer plc v BNP Paribas Securities Services Trust Co (Jersey) Ltd* [2015] UKSC 72, the Supreme Court held that a term will be implied only if it is necessary for business efficacy or so obvious that it goes without saying, and not merely because it would be reasonable or fair. The restrictive approach to implication reinforces the importance of comprehensive drafting.

Civil law systems give greater weight to the common intention of the parties, which may be ascertained from pre-contractual negotiations and subsequent conduct, both of which English law generally excludes from consideration. Article 4.1 of the UNIDROIT Principles provides that a contract shall be interpreted according to the common intention of the parties, or, if such an intention cannot be established, according to the meaning that reasonable persons of the same kind as the parties would give to it in the same circumstances, and Article 4.3 permits consideration of preliminary negotiations, practices established between the parties, and subsequent conduct. International arbitral tribunals, which frequently include arbitrators from different legal traditions, may be influenced by these differing approaches even where the governing law is English law, a phenomenon that experienced counsel take into account in presenting their cases.

Renegotiation and Adaptation

Long-term energy contracts frequently require adaptation over their life, as market conditions, technology, regulation, and the parties' circumstances change. Several mechanisms facilitate adaptation.

Price review clauses. Long-term gas supply contracts typically contain price review clauses permitting either party to request a review of the contract price at specified intervals or upon a significant change in market circumstances, with recourse to arbitration if the parties fail to agree, as examined in Chapter 6. Price review arbitrations, which require tribunals to determine a revised price in light of changed market conditions, are among the most commercially significant forms of energy dispute.

Renegotiation clauses. Many state contracts contain clauses obliging the parties to renegotiate in good faith upon specified events, such as fundamental changes in circumstances. The effect of such clauses is limited where the parties fail to agree, unless the contract provides for a third party to adapt the contract. In *Kuwait v American Independent Oil Co (Aminoil)* (1982), the tribunal recognised that the parties' long-term relationship, and the evolution of their contractual arrangements over time, implied an obligation to negotiate in good faith, while holding that the government's nationalisation of the concession was lawful subject to the payment of appropriate compensation, as examined in Chapter 14.

Adaptation by tribunals. Whether an arbitral tribunal may adapt a contract to changed circumstances depends on the governing law and the parties' agreement. Under the UNIDROIT Principles and French law, a tribunal may adapt the contract in cases of hardship, while English law does not empower courts or tribunals to rewrite contracts. Parties may expressly empower tribunals to adapt contracts, as price review clauses do.

Government-Initiated Contract Review

Host governments have periodically undertaken comprehensive reviews of existing contracts with investors, particularly in the extractive sector, on the grounds that the contracts were negotiated under unfavourable conditions, procured through corruption, or became unbalanced as commodity prices rose. The **Democratic Republic of Congo** launched a review of sixty-one mining contracts in 2007–2008, following concerns that contracts concluded during the conflict period were inequitable; the review recommended that a number of contracts be renegotiated or cancelled. Several other African states, including Guinea, Zambia, and Liberia, have undertaken similar reviews of mining contracts.

The most far-reaching recent example is **Tanzania**, which in 2017 enacted the Natural Wealth and Resources (Permanent Sovereignty) Act and the Natural Wealth and Resources Contracts (Review and Re-Negotiation of Unconscionable Terms) Act. The first statute declared the permanent sovereignty of the people of Tanzania over natural resources, prohibited the export of raw minerals, and provided that disputes over natural resources should be resolved by Tanzanian judicial bodies, purporting to exclude international arbitration. The second empowered Parliament to review contracts and to direct the government to renegotiate terms considered unconscionable, failing which the terms would be treated as ceasing to have effect. The legislation led to extensive renegotiations with mining companies, including the settlement of a major dispute with Acacia Mining (later acquired by Barrick Gold), under which the government obtained a share of the economic benefits of the mines and a minority interest in the operating companies, and to investment arbitration claims by other investors.

Government-initiated contract reviews raise the issues of sanctity of contract and sovereign regulatory authority examined throughout Part IV. Under international law, a state's unilateral modification or termination of a contract with a foreign investor may constitute a breach of fair and equitable treatment or an expropriation if it involves the exercise of sovereign power, and may be a breach of an umbrella clause, as examined in Chapter 14. At the same time, contracts procured by corruption may be unenforceable, as examined in Chapter 16, and the legitimacy of renegotiation is enhanced where it follows transparent processes and offers fair compensation for the investor's legitimate expectations.

Termination and Termination Payments

Energy contracts contain detailed provisions on termination, specifying the grounds on which each party may terminate — such as material breach, insolvency, prolonged force majeure, and change of control — and the consequences of termination, including **termination payments**. In PPAs and concession agreements, the termination payment typically depends on the cause of termination:

- termination for **government default** or political force majeure typically entitles the private party to a payment covering outstanding debt, equity, and a return on equity, reflecting the principle that the government should bear the consequences of its own default or of events within its sphere of control;
- termination for **natural force majeure** typically entitles the private party to a payment covering outstanding debt, but not necessarily equity returns, reflecting the sharing of the risk of events outside either party's control;
- termination for **private party default** may entitle the private party to a reduced payment, often limited to outstanding senior debt or the market value of the project, so as to protect lenders while penalising the defaulting sponsor.

Upon termination of a concession or PPA, the project assets are commonly transferred to the government or its designee, either under build-operate-transfer arrangements or on payment of the termination amount. The adequacy and enforceability of termination payments are critical to the bankability of energy projects, and lenders scrutinise these provisions closely. Termination payment provisions also affect the calculation of damages in investment arbitration, where tribunals may use the contractual termination payment as a reference point for compensation, as examined in Chapter 26.

Decommissioning and Post-Termination Obligations

Termination or expiry of an upstream contract or licence triggers obligations to decommission facilities and restore sites, as examined in Chapter 5. Contracts typically address the allocation of decommissioning costs among co-venturers, the establishment of decommissioning funds or security, and the continuing liability of former participants. The obligations of parties that have transferred their interests may continue under the contract, the JOA, or applicable legislation, as illustrated by the United Kingdom's regime under which the regulator may require former licensees to fund decommissioning if current licensees default. Post-termination obligations relating to confidentiality, data, environmental liabilities, and dispute resolution also survive termination.

::: exercise {title="Practical Exercise 15: Negotiating the Contract Web (HYPOTHETICAL)"} **This exercise is entirely hypothetical. All parties, places,**

contracts, and events are fictitious and do not refer to any real persons, companies, or states.

The Republic of Kelora has awarded a gas-to-power project to a consortium consisting of Northbridge Energy (fifty-one per cent), the Kelora National Petroleum Company (KNPC, thirty per cent), and Tamari Infrastructure Fund (nineteen per cent). The project comprises the development of an offshore gas field under a production sharing contract, a pipeline to shore, and a 600 MW combined-cycle power plant, which will sell electricity to the state-owned Kelora Power Distribution Company (KPDC) under a twenty-year PPA. The plant is to be built by a Korean EPC contractor under a lump-sum turnkey contract.

The following issues have arisen during negotiations and early implementation:

1. KNPC insists that the joint operating agreement for the upstream field require unanimous approval of all work programmes and budgets. Northbridge wishes to retain control as operator. Tamari is concerned about funding defaults by KNPC, which has a history of arrears on cash calls in other ventures.
2. The draft PPA defines "Change in Law" to cover only changes that discriminate specifically against the project. Tamari's lenders insist that the definition extend to general changes in tax law and to any carbon tax. KPDC refuses.
3. Eighteen months into construction, an insurgent attack on a coastal town fifty kilometres from the plant site leads the EPC contractor to withdraw its expatriate staff and declare force majeure. The consortium doubts that the attack prevents construction, since the site itself has not been attacked.
4. The EPC contract contains a mutual exclusion of "consequential loss", defined to include "loss of revenue, loss of profit, and loss of production". The contractor's delay will cause the project company to lose revenue under the PPA and to pay liquidated damages to KPDC for late commissioning.
5. The government of Kelora announces a review of all energy contracts concluded in the past ten years, on the ground that several were "unconscionable".

Tasks.

(a) Draft a voting and default regime for the JOA that balances the interests of the three parties.

(b) Prepare a negotiation brief for the consortium on the change-in-law definition, proposing a compromise acceptable to the lenders and KPDC.

(c) Advise the project company on whether the EPC contractor's force majeure declaration is valid, identifying the information needed and the relevant authorities.

(d) Advise whether the project company can recover from the contractor its lost PPA revenue and the liquidated damages payable to KPDC, in light of the consequential loss exclusion and the EPC delay damages regime.

(e) Advise the consortium on its legal position, under the contracts and international law, in the event that the contract review leads to a demand for renegotiation of the PPA tariff. :::

CHAPTER SUMMARY

- Energy projects rest on a web of interrelated contracts, whose terms must be coordinated to ensure that risks allocated under one contract are matched by corresponding protections under others.
- The formation and governing law of international energy contracts raise issues of pre-contractual liability, choice of law, the role of the UNIDROIT Principles and lex petrolea, and the internationalisation of state contracts.
- Host government agreements take the form of concessions, production sharing contracts, and service contracts, which allocate risk and reward between the state and investors in distinct ways.
- EPC contracts allocate construction risk to the contractor through fixed prices, completion dates, performance guarantees, and liquidated damages, whose enforceability is governed by the modern penalty doctrine.
- Joint operating agreements and shareholders' agreements govern relations among co-venturers, addressing operatorship, decision-making, sole risk, default, transfers, and deadlock.
- Force majeure clauses, hardship mechanisms, and the doctrine of frustration address supervening events, with English courts interpreting force majeure clauses strictly according to their terms.
- Change-in-law clauses require careful drafting of definitions, thresholds, and remedies, and increasingly address climate policy measures expressly.
- Liability allocation through knock-for-knock indemnities, consequential loss exclusions, liability caps, security instruments, and insurance determines who bears the cost of failures and incidents.
- Interpretation, renegotiation, government-initiated contract review, and termination payments determine how contracts respond to change over their long life.

DOCTORAL RESEARCH QUESTIONS

1. To what extent has the AIEN Model JOA and other model forms created a body of transnational commercial practice that functions as a "lex petrolea", and how should arbitral tribunals use such practice in interpreting energy contracts?

2. Evaluate whether the English law approach to force majeure and frustration, which emphasises the parties' express allocation of risk, is appropriate for long-term energy contracts, compared with the civil law and UNIDROIT approaches to hardship and adaptation.
3. How should change-in-law clauses in long-term energy contracts allocate the risk of climate policy measures, in light of the Paris Agreement and the foreseeability of decarbonisation policies?
4. Assess the legitimacy and effectiveness of government-initiated contract reviews in the extractive sector, with reference to the experiences of Tanzania and the Democratic Republic of Congo.
5. To what extent do knock-for-knock indemnities and the gross negligence carve-out achieve an efficient and fair allocation of liability for major offshore incidents?
6. How can the Principles for Responsible Contracts be operationalised in the drafting of host government agreements, and what would be their effect on bankability?

FURTHER READING

- Peter Roberts, *Joint Operating Agreements: A Practical Guide* (Globe Law and Business, latest edn).
- Peter Cameron, *International Energy Investment Law: The Pursuit of Stability* (2nd edn, Oxford University Press 2021).
- Edwin Peel, *Treitel on The Law of Contract* (15th edn, Sweet & Maxwell 2020), chapters on frustration and interpretation.
- Ewan McKendrick (ed), *Force Majeure and Frustration of Contract* (2nd edn, Lloyd's of London Press 1995).
- Klaus Peter Berger, *The Creeping Codification of the New Lex Mercatoria* (2nd edn, Kluwer Law International 2010).
- World Bank, *Guidance on PPP Contractual Provisions* (2019 edn).
- John Ruggie, *Principles for Responsible Contracts: Integrating the Management of Human Rights Risks into State–Investor Contract Negotiations* (UN Doc A/HRC/17/31/Add.3, 2011).

16 CHAPTER

Energy Project Finance, Taxation, and Fiscal Governance

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the structure and legal architecture of limited-recourse project finance as applied to energy and extractive projects.
2. Analyse the allocation of risk among sponsors, lenders, contractors, offtakers, and host governments through the debt and equity structure and the security package.
3. Evaluate sovereign guarantees, government support agreements, and multilateral risk mitigation instruments, and assess the contingent fiscal liabilities they create.
4. Compare the principal petroleum and mining fiscal regimes and the criteria by which they are evaluated.
5. Distinguish royalties, resource rent taxes, windfall taxes, and production levies, and assess their economic and legal implications.
6. Analyse the application of transfer pricing rules, tax treaties, and the OECD/G20 BEPS framework to multinational energy and mining enterprises, including the taxation of indirect transfers.
7. Assess the legal frameworks for revenue transparency and the prevention of corruption in the extractive sector, and the consequences of corruption for the enforceability of contracts and awards.
8. Explain the legal tools available for managing project financial distress, utility insolvency, and sovereign debt linked to natural resources.

Introduction

Energy is capital. A large hydroelectric dam, an LNG export terminal, a deepwater oil development, a transmission interconnector, or a utility-scale solar programme requires capital investment measured in hundreds of millions or billions of dollars, committed years before the first revenue is earned and recovered over decades.

The International Energy Agency has estimated that global energy investment now exceeds USD 3 trillion a year, with clean energy accounting for roughly two-thirds of the total, and that the investment required to meet climate and access goals in emerging and developing economies — outside China — must increase several-fold. How that capital is raised, secured, priced, and protected is therefore a central question of energy law, and one in which legal design directly affects the cost of capital and hence the price that consumers pay for energy.

At the same time, energy and natural resources are a principal source of public revenue for many states. Petroleum and mineral revenues finance a large share of government budgets in Nigeria, Angola, Algeria, Saudi Arabia, Norway, Chile, Zambia, the Democratic Republic of Congo, and many other states. The design of fiscal regimes — the balance between royalties, taxes, production sharing, and state participation — determines how the economic rent generated by natural resources is divided between the state and investors, and the effectiveness of tax administration, transparency, and anti-corruption controls determines whether that rent reaches the public treasury and is used for the benefit of the population. The phenomenon frequently described as the "resource curse" — the tendency of resource-rich states to experience slower growth, weaker institutions, greater inequality, and more conflict than resource-poor states — has been attributed in significant part to failures of fiscal governance.

This chapter brings together these two dimensions — finance and fiscal governance — because they are inseparable in practice. Lenders assess the fiscal regime and its stability when deciding whether to finance a project; host governments assess the financing structure when negotiating fiscal terms; and sovereign guarantees, tax holidays, and contingent liabilities link the private balance sheet of the project to the public balance sheet of the state. The chapter examines project finance structures (section 16.1), debt, equity, and risk allocation (section 16.2), sovereign guarantees and government support (section 16.3), petroleum and mining fiscal regimes (section 16.4), royalties, resource rent taxes, and production levies (section 16.5), transfer pricing and international taxation (section 16.6), revenue transparency and anti-corruption controls (section 16.7), and debt restructuring and financial distress (section 16.8). It builds on the analysis of host government agreements and contractual risk allocation in Chapter 15 and of investment protection in Chapter 14, and anticipates the treatment of Nigeria's petroleum fiscal regime in Chapter 24 and of damages and enforcement in Chapter 26.

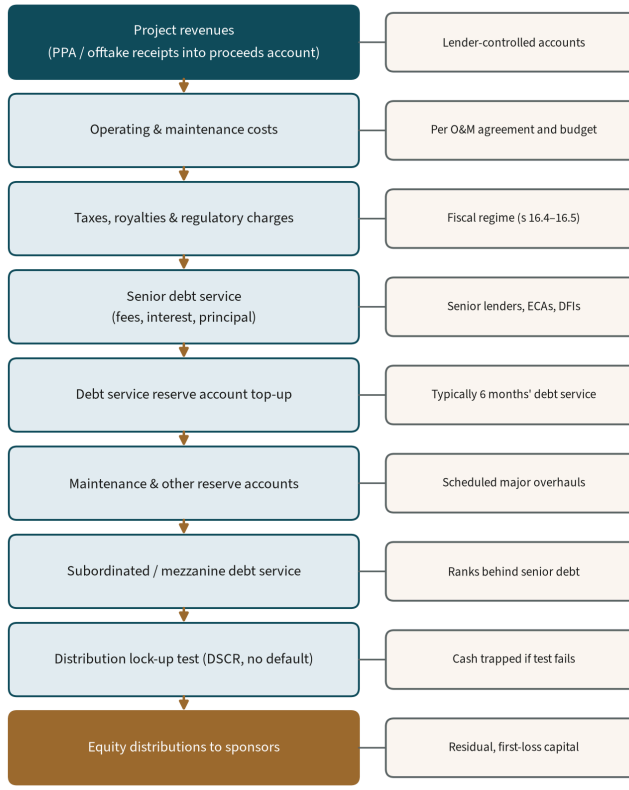


Figure 16.1 The cash-flow waterfall of a project-financed energy project. *Source:* prepared for this edition.

16.1 Project Finance Structures

The Concept of Project Finance

Project finance is a method of financing a discrete, capital-intensive asset in which lenders look primarily to the cash flows generated by the project, and to the project's assets and contracts, for repayment, rather than to the general balance sheet of the project's sponsors. It is commonly described as **non-recourse** or, more accurately, **limited-recourse** finance, because lenders usually have some recourse to the sponsors during the construction phase or in specified circumstances, but not general recourse for the repayment of the debt. Project finance contrasts with **corporate finance**, in which a company borrows on the strength of its overall balance sheet and credit rating, and the lenders have recourse to all its assets.

The essential features of a project financing are:

- a **special purpose vehicle** (SPV), or project company, established solely to develop, own, and operate the project, with no other business, so that the project's assets, liabilities, and cash flows are ring-fenced from those of the sponsors;
- **high leverage**, with debt typically representing seventy to eighty per cent or more of project costs, and equity the balance;
- a **contractual structure** that allocates the project's risks to the parties best able to bear them, through the web of contracts examined in Chapter 15 — construction risk to the EPC contractor, operating risk to the operator, fuel supply risk to the supplier, market risk to the offtaker, and political risk to the host government or insurers;
- a **security package** giving lenders security over all the project company's assets, contracts, accounts, and shares, and rights to step in and cure defaults under key project contracts;
- **cash flow controls**, under which project revenues are paid into accounts controlled by the lenders and applied in a strict order of priority — the **cash-flow waterfall** illustrated in Figure 16.1 — with distributions to equity permitted only if financial ratios are satisfied.

The Rationale for Project Finance

Sponsors choose project finance, despite its higher transaction costs and complexity, for several reasons. It allows sponsors to undertake projects that are large in relation to their balance sheets, without exposing their other assets to the risks of the project. It permits the sharing of risk with lenders and other participants. It may allow the debt to be kept off the sponsors' balance sheets, subject to accounting rules. It enables participation by parties, such as national oil companies, local partners, and infrastructure funds, which could not otherwise finance their share of a large project. And it may reduce political risk, because the involvement of multilateral lenders, export credit agencies, and international banks increases the reputational and financial cost to a host government of interfering with the project — the so-called "halo effect" or "umbrella" of multilateral participation.

For host governments and the public, project finance has advantages and disadvantages. It mobilises private capital for infrastructure that the state could not otherwise finance, and the discipline imposed by lenders' due diligence may improve project selection and design. But the cost of capital is typically higher than for public borrowing, the complexity of the contractual structure creates information asymmetries, and the risk allocation often leaves the state bearing significant contingent liabilities through offtake commitments, guarantees, and termination payments, as examined in section 16.3.

Historical Development

Modern project finance has its roots in the energy sector. Oil production payments in the United States in the early twentieth century, under which lenders were repaid from the proceeds of oil produced from specific properties, were an early form of limited-recourse lending. The financing of the development of the North Sea in the 1970s, including BP's financing of the Forties field, established the modern model of limited-recourse lending against petroleum reserves. In the 1980s and 1990s, the liberalisation of electricity markets and the emergence of independent power producers (IPPs) in the United States, the United Kingdom, and developing countries — including Pakistan, India, Indonesia, the Philippines, and China — led to the widespread use of project finance for power generation, and the financing of LNG projects, such as those in Qatar, demonstrated its use for the largest energy developments. Since the 2000s, project finance has become the principal method of financing utility-scale wind and solar projects worldwide.

The experience of the 1990s also revealed the vulnerabilities of project finance in emerging markets. The Asian financial crisis of 1997–1998 left several Indonesian power projects with dollar-denominated tariffs that the state utility could no longer afford, leading to the suspension and renegotiation of numerous PPAs and to arbitration, as illustrated by the *Karaha Bodas* dispute examined in section 16.3. The Dabhol power project in India, sponsored by Enron and supplying the Maharashtra State Electricity Board, collapsed in 2001 after the board refused to pay tariffs it considered excessive, generating extensive litigation and arbitration. These experiences shaped the subsequent design of risk allocation and credit enhancement in emerging market energy projects.

Project Finance in Africa and Nigeria

Project finance has been used for a growing number of energy projects in Africa, including the Lake Turkana wind farm in Kenya, South Africa's Renewable Energy Independent Power Producer Procurement Programme (REIPPPP), and gas-fired IPPs in Ghana, Senegal, and Côte d'Ivoire. In Nigeria, the **Azura-Edo** independent power project, an open-cycle gas turbine plant of approximately 460 MW near Benin City, reached financial close in 2015 and commenced commercial operation in 2018. It was one of the first privately financed IPPs following the privatisation of the Nigerian power sector, and was supported by a PPA with the Nigerian Bulk Electricity Trading Plc (NBET), a put and call option agreement with the Federal Government, a World Bank partial risk guarantee, and political risk insurance from the Multilateral Investment Guarantee Agency (MIGA). The project illustrates both the possibilities and the difficulties of project finance in a market with severe liquidity problems, as payment shortfalls in the Nigerian electricity market subsequently placed pressure on the credit support arrangements. The reform of the Nigerian electricity market under the Electricity Act 2023,

examined in Chapters 9 and 24, has significant implications for the creditworthiness of offtakers and the bankability of future IPPs.

Variants and Emerging Structures

Several variants of the basic project finance model are used in the energy sector.

Reserve-based lending (RBL) is a form of limited-recourse finance used in the upstream petroleum sector, under which a revolving credit facility is secured on a portfolio of producing or near-producing fields, and the amount available — the "borrowing base" — is periodically redetermined by reference to the estimated value of the reserves, using agreed price assumptions. RBL facilities have been widely used in the North Sea and West Africa, including by Nigerian independents acquiring onshore assets divested by international oil companies.

Portfolio financing and **holding company financing** are increasingly used for renewable energy, where individual projects are too small to justify separate financing, and developers aggregate projects into portfolios financed on a combined basis.

Green, social, and sustainability bonds allow project companies and sponsors to raise debt in capital markets for eligible environmental projects. The International Capital Market Association's Green Bond Principles, first published in 2014, provide voluntary guidance on the use of proceeds, project evaluation, management of proceeds, and reporting, and the European Union's **European Green Bond Standard**, established by Regulation (EU) 2023/2631, provides a voluntary standard aligned with the EU taxonomy. **Sustainability-linked loans and bonds**, whose pricing varies according to the borrower's performance against sustainability targets, are also widely used.

Blended finance combines concessional finance from public or philanthropic sources with commercial capital to improve the risk-return profile of projects in challenging markets. The **Just Energy Transition Partnerships** (JETPs) announced from 2021 — with South Africa (initially USD 8.5 billion), Indonesia (USD 20 billion), Viet Nam (USD 15.5 billion), and Senegal (EUR 2.5 billion) — combine grants, concessional loans, guarantees, and commercial finance to support the transition from coal, although their implementation has been slower than anticipated, and the United States withdrew from the partnerships in 2025. Just transition is examined further in Chapter 18.

Environmental and Social Standards in Project Finance

The financing of energy projects is subject to environmental and social standards imposed by lenders. The **IFC Performance Standards on Environmental and Social Sustainability** (2012) cover assessment and management of environmental and social risks, labour, resource efficiency and pollution prevention, community health and safety, land acquisition and involuntary resettlement, biodiversity,

indigenous peoples, and cultural heritage. The **Equator Principles**, a voluntary framework adopted by more than one hundred financial institutions, the fourth version of which (EP4) took effect in 2020, require participating banks to apply the IFC Performance Standards to project finance and related transactions in non-designated countries, and require climate change risk assessment for projects with significant emissions. The World Bank's Environmental and Social Framework (2018) applies to World Bank-financed projects. These standards function as a form of private or transnational regulation, often more demanding than host state law, and are enforced through loan covenants, independent monitoring, and, in the case of the multilateral institutions, independent accountability mechanisms such as the IFC's Compliance Advisor Ombudsman and the World Bank Inspection Panel. Their legal significance, and the immunity of international financial institutions from suit — addressed by the United States Supreme Court in *Jam v International Finance Corp* (2019), examined in Chapter 7 — are considered further in Chapters 21 and 22.

16.2 Debt, Equity, and Risk Allocation

The Capital Structure

The capital structure of a project-financed energy project comprises equity contributed by the sponsors and debt provided by lenders. The ratio of debt to equity — the **gearing** or **leverage** — is determined by the project's risk profile and the predictability of its cash flows. Projects with long-term contracted revenues from creditworthy offtakers, such as renewable energy projects with government-backed PPAs, may achieve gearing of eighty per cent or more, while projects exposed to market or commodity price risk, such as merchant power plants or upstream petroleum developments, typically have lower gearing. Lenders size the debt by reference to the projected cash flows and the required **debt service coverage ratio** (DSCR) — the ratio of cash flow available for debt service to scheduled debt service in a given period — and other ratios such as the **loan life coverage ratio** (LLCR).

Equity is the first-loss capital, bearing the residual risk of the project and receiving distributions only after operating costs, taxes, debt service, and reserve accounts have been funded. Sponsors may contribute equity in the form of share capital or subordinated shareholder loans, which provide tax and repatriation advantages and rank behind senior debt. Lenders typically require that equity be contributed before or pro rata with debt drawdowns, and may require sponsors to provide **equity contribution guarantees** or letters of credit to secure their commitments.

Senior debt is provided by commercial banks, export credit agencies, development finance institutions, and capital markets investors, and ranks ahead of all

other claims on the project's cash flows and assets. **Mezzanine** or **subordinated debt** ranks between senior debt and equity, and carries higher returns to compensate for its greater risk.

Sources of Debt

Commercial banks have historically been the principal providers of project finance debt, through syndicated loans arranged by lead banks and distributed among participating banks. Regulatory capital requirements under the Basel framework have constrained the willingness of banks to provide long-tenor loans, increasing the role of other sources.

Export credit agencies (ECAs), such as the United States Export-Import Bank, the Export-Import Bank of China, China Export and Credit Insurance Corporation (Sinosure), UK Export Finance, Germany's Euler Hermes, and Japan's JBIC and NEXI, provide loans, guarantees, and insurance to support exports of goods and services from their countries, and play a major role in financing energy projects that use imported equipment. The terms of officially supported export credits from OECD members are governed by the **OECD Arrangement on Officially Supported Export Credits**, a "gentlemen's agreement" that sets minimum interest rates, maximum repayment terms, and minimum premium rates, with sector understandings for specific industries. In 2021, participants agreed to end officially supported export credits for unabated coal-fired power plants, reflecting the influence of climate policy on export finance. Chinese state banks and ECAs, which are not bound by the Arrangement, became major financiers of energy infrastructure in Africa, Asia, and Latin America during the 2000s and 2010s, although China announced in 2021 that it would not build new coal-fired power projects abroad.

Development finance institutions (DFIs) — multilateral institutions such as the IFC, the African Development Bank, the Asian Development Bank, and the European Bank for Reconstruction and Development, and bilateral institutions such as British International Investment, the Netherlands' FMO, Germany's DEG, France's Proparco, and the United States International Development Finance Corporation — provide long-term loans, equity, and guarantees for projects in developing countries, often on terms unavailable from commercial lenders. Their participation provides the political risk mitigation and environmental and social standards described in section 16.1. Under the **A/B loan** structure, a multilateral institution acts as lender of record for a loan whose "B" portion is funded by commercial banks, extending to the commercial participants the multilateral institution's preferred creditor status in practice.

Capital markets provide finance through project bonds, which may be issued under United States Rule 144A to institutional investors or listed on international

exchanges, and which offer longer tenors and fixed rates but less flexibility than bank loans in the event of restructuring. Institutional investors, such as pension funds and insurance companies, are increasingly significant investors in operational energy infrastructure.

Islamic finance is used for energy projects in the Gulf states, Malaysia, and elsewhere, through structures compliant with Shariah principles prohibiting interest (*riba*) and excessive uncertainty (*gharar*), such as *istisna'a* (construction financing), *ijara* (leasing), and *sukuk* (asset-based certificates). Islamic tranches are often combined with conventional tranches in multi-source financings, requiring intercreditor arrangements to coordinate the rights of different classes of lenders.

The Finance Documents

The principal finance documents in a project financing include:

- the **facility agreement** or common terms agreement, setting out the terms of the loans, the conditions precedent to drawdown, representations and warranties, covenants, events of default, and remedies, often based on the Loan Market Association (LMA) recommended forms for loans governed by English law;
- the **intercreditor agreement**, governing the relationship among different classes of lenders, including voting, ranking, enforcement, and the sharing of proceeds;
- the **security documents**, creating security over the project company's assets, contracts, accounts, and shares;
- the **direct agreements** (also called consent agreements or tripartite agreements) between the lenders, the project company, and the counterparties to key project contracts, under which the counterparties agree to give the lenders notice of defaults and an opportunity to cure them, and permit the lenders to step in and assume the project company's rights and obligations, or to transfer the contract to a substitute entity;
- the **accounts agreement**, establishing the project accounts and the cash-flow waterfall;
- **hedging agreements**, protecting against interest rate, currency, and commodity price risks.

The transition from the London Interbank Offered Rate (LIBOR) to risk-free reference rates, such as the Secured Overnight Financing Rate (SOFR) for US dollars and SONIA for sterling, completed for most US dollar settings in June 2023, required the amendment of thousands of project finance documents and hedging agreements.

Conditions Precedent, Covenants, and Events of Default

Before the lenders advance funds, the project company must satisfy **conditions precedent**, including the execution of all project contracts and finance documents, the grant of all necessary permits and licences, the delivery of legal opinions, the satisfaction of environmental and social requirements, the injection of equity, and confirmation by independent engineers, insurance advisers, and model auditors of the technical, insurance, and financial assumptions. Lenders' due diligence, conducted by legal, technical, environmental, insurance, and market advisers, is extensive and costly.

Throughout the life of the loan, the project company is bound by **covenants**: positive covenants, such as obligations to maintain permits, insurance, and books, and to comply with law and environmental and social action plans; negative covenants, restricting the project company from incurring additional debt, granting security, disposing of assets, amending project contracts, or changing its business; and financial covenants, requiring the maintenance of specified DSCRs and other ratios. **Distribution lock-up** tests prevent distributions to equity unless historical and projected DSCRs exceed specified levels and no default is continuing.

Events of default include non-payment, breach of covenants, misrepresentation, insolvency, loss of key permits, termination of key project contracts, expropriation, and failure to achieve completion by a long-stop date. On an event of default, the lenders may accelerate the loan, enforce their security, and exercise step-in rights, although in practice lenders generally prefer to negotiate waivers, amendments, or restructurings rather than enforce, as examined in section 16.8.

Risk Allocation from the Lenders' Perspective

Lenders evaluate each category of risk and require that it be mitigated through contractual, financial, or insurance arrangements. **Completion risk** — the risk that the project will not be completed on time, on budget, and to specification — is mitigated by the fixed-price, date-certain EPC contract examined in Chapter 15, supported by liquidated damages and performance security, and frequently by **completion support** from the sponsors, such as completion guarantees, cost overrun facilities, or contingent equity, which falls away once the project passes completion tests. **Operating risk** is mitigated by experienced operators, O&M contracts with performance incentives, and reserve accounts. **Supply risk** is mitigated by long-term fuel supply agreements, and for upstream projects by independent reserves reports. **Market risk** is mitigated by long-term offtake agreements, such as take-or-pay PPAs and gas sales agreements, or, in merchant projects, by hedging and conservative debt sizing. **Currency risk** arises where project revenues are denominated in local currency while debt is denominated in

foreign currency, and is mitigated by indexing tariffs to foreign currency, hedging, or local currency financing; currency convertibility and transfer risk is mitigated by offshore accounts and political risk insurance. **Political and regulatory risk** is mitigated by government support agreements, stabilisation and change-in-law provisions, political risk insurance, and the participation of multilateral institutions. **Environmental and social risk** is mitigated by compliance with the standards described in section 16.1.

The allocation of these risks is not merely a technical exercise but a negotiation in which the bargaining power of the parties, the state of the market, and the availability of credit enhancement determine the outcome. Where the host government or state-owned offtaker assumes risks that private parties will not bear, those risks become contingent liabilities of the state, to which we now turn.

16.3 Sovereign Guarantees and Government Support Agreements

Why Government Support Is Required

In many developing and emerging markets, the offtaker of a power project is a state-owned utility or single buyer whose creditworthiness is weak, the regulatory framework is untested, and currency convertibility cannot be assumed. In such conditions, lenders and investors will not finance projects on the strength of the project contracts alone, and require the host government to provide additional support. The forms of support, and the extent of the contingent liabilities they create for the state, are among the most important and contested issues in energy project finance.

Forms of Government Support

Government support takes several forms, which are frequently combined.

Implementation agreements and government support agreements. An implementation agreement (IA) or government support agreement (GSA) is a contract between the host government and the project company under which the government undertakes obligations to facilitate the project: granting permits and licences, providing land, ensuring the availability of foreign exchange and the right to repatriate profits, granting tax and customs exemptions, protecting the project against discriminatory changes in law, and agreeing to international arbitration. Pakistan's IPP programme of the 1990s, under which implementation agreements were concluded with each IPP, established a widely copied template.

Sovereign guarantees of payment obligations. The government may guarantee the payment obligations of the state-owned offtaker under the PPA or of a state-owned fuel supplier, so that the project company can claim against the

government if the offtaker fails to pay. Such guarantees transfer the offtaker's credit risk directly to the sovereign.

Put and call option agreements. A put and call option agreement (PCOA) gives the project company the right to require the government to purchase the project (the put option) on the occurrence of specified events — such as prolonged political force majeure, government or offtaker default, or change in law — at a price usually sufficient to repay outstanding debt and, depending on the event, to compensate equity. The government in turn has the right to purchase the project (the call option) in specified circumstances. PCOAs were used in Nigeria's Azura-Edo project and other IPPs, and are functionally similar to termination payment obligations in concession agreements, examined in Chapter 15.

Letters of support or comfort. A government may provide a letter of support expressing its support for the project and its intention to ensure that the state-owned offtaker meets its obligations, without assuming a legally binding guarantee. The legal effect of such letters depends on their wording; English courts have generally treated comfort letters as statements of present intention rather than contractual promises, unless the language indicates an intention to create legal relations.

Liquidity support. To address short-term payment delays, offtakers may be required to provide letters of credit, escrow accounts funded from designated revenue streams, or payment security mechanisms, often supported by multilateral guarantees, covering several months of payments.

Multilateral and Bilateral Risk Mitigation

Multilateral and bilateral institutions provide instruments that cover risks which governments cannot credibly guarantee alone.

The Multilateral Investment Guarantee Agency. MIGA was established by the Convention Establishing the Multilateral Investment Guarantee Agency, opened for signature in Seoul in 1985 and in force since 1988, as a member of the World Bank Group with the purpose of encouraging the flow of investment to developing member countries by issuing guarantees against non-commercial risks. Under Article 11 of the Convention, MIGA may cover the risks of currency transfer restriction, expropriation and similar measures, breach of contract by the host government (where the investor has no recourse to a judicial or arbitral forum, faces unreasonable delay, or cannot enforce a favourable decision), and war and civil disturbance; it also offers cover for the non-honouring of sovereign financial obligations. Upon paying a claim, MIGA is subrogated to the investor's rights against the host state, and disputes between MIGA and the host state are subject to negotiation and arbitration under the Convention. The deterrent effect of MIGA's involvement, arising from the host government's relationship with

the World Bank Group, is a significant component of its value, and MIGA has historically paid relatively few claims, resolving many disputes through mediation.

World Bank guarantees. The International Bank for Reconstruction and Development and the International Development Association provide project-based guarantees — formerly described as partial risk guarantees and partial credit guarantees — which cover private lenders against the failure of a government or state entity to meet its contractual obligations, such as payment obligations under a PPA or termination payments, supported by a counter-guarantee and indemnity from the host government to the World Bank. The guarantees have been used for power projects in Nigeria, Ghana, Kenya, Côte d'Ivoire, and elsewhere. In 2024, the World Bank Group announced the consolidation of its guarantee instruments into a single platform housed at MIGA, with a view to expanding their use.

Other providers. The African Development Bank provides partial risk guarantees; the African Trade and Investment Development Insurance (ATIDI, formerly the African Trade Insurance Agency) provides political risk insurance and liquidity products for IPPs in member states; GuarantCo, part of the Private Infrastructure Development Group, provides local currency guarantees; and national investment insurers and the private political risk insurance market, centred on Lloyd's of London, provide additional cover.

Contingent Liabilities and Fiscal Risk

Government support creates **contingent liabilities** — obligations that crystallise only on the occurrence of future events — which may be large in relation to the state's fiscal capacity and are often not recorded in government budgets or debt statistics. Take-or-pay PPAs commit the state-owned offtaker to pay capacity charges whether or not the electricity is needed, and where demand projections prove optimistic, the resulting excess capacity charges become a substantial fiscal burden. Several states have experienced the consequences. In **Pakistan**, the accumulation of unpaid obligations through the electricity supply chain, known as "circular debt", reached very large proportions, leading to negotiations with IPPs in 2020 on tariff reductions and, in 2024, to the negotiated early termination of a number of older IPP contracts. In **Ghana**, the conclusion of numerous take-or-pay PPAs during a period of power shortages in the 2010s led to payments for capacity that was not used, and the government sought to renegotiate or terminate several contracts. Indian state distribution companies accumulated large arrears to generators, leading to the adoption of payment security mechanisms and rules on late payment surcharges.

The management of fiscal risks from energy projects has become a focus of international guidance. The IMF and the World Bank developed the **PPP Fiscal Risk Assessment Model** (P-FRAM) to help governments assess the fiscal costs

and risks of public-private partnership projects, and the IMF's fiscal transparency standards call for the disclosure and management of contingent liabilities. Many states have adopted public finance legislation requiring the approval of guarantees by the ministry of finance or the legislature, the recording of contingent liabilities, and limits on the aggregate value of guarantees. In Nigeria, the Fiscal Responsibility Act 2007 and the Debt Management Office Establishment Act 2003 regulate public borrowing and guarantees, and the issuance of PCOAs and guarantees for IPPs has been subject to approval by the Federal Executive Council.

Enforcing Government Support

The enforcement of sovereign guarantees and government support agreements raises the question of sovereign immunity, examined in Chapter 26. Government support agreements typically contain waivers of immunity from suit and from execution, and submission to international arbitration, often under the ICSID Convention. Under the United Kingdom's State Immunity Act 1978 and the United States' Foreign Sovereign Immunities Act of 1976, states are not immune in respect of commercial transactions and where they have submitted to arbitration, but immunity from execution protects state property not in use for commercial purposes, and central bank assets enjoy special protection. Investors holding awards against states must therefore identify commercial assets available for execution, a process that may be protracted, as illustrated by the *Karaha Bodas* case.

::: casebox {title="Karaha Bodas Co v Pertamina (1998–2007): Enforcing a Power Project Award Against a State Company"} **Background.** In 1994, Karaha Bodas Company (KBC), a project company sponsored by United States investors, concluded a joint operation contract with Pertamina, the Indonesian state oil and gas company, for the development of a geothermal field in West Java, and an energy sales contract with PLN, the state electricity company, for the sale of electricity. During the Asian financial crisis, the Indonesian government issued presidential decrees in 1997 and 1998 suspending the project, together with numerous other infrastructure projects.

The arbitration. KBC commenced arbitration against Pertamina and PLN under the UNCITRAL Arbitration Rules, seated in Geneva. In December 2000, the tribunal held that the suspension did not constitute force majeure excusing Pertamina and PLN, since the contracts allocated the risk of government action to the state companies, and awarded KBC approximately USD 261 million, comprising its expenditure and lost profits.

Annulment and enforcement. Pertamina's challenge to the award in Switzerland, the seat, was unsuccessful. Pertamina then obtained a judgment from a Jakarta district court in 2002 purporting to annul the award. KBC sought

enforcement under the New York Convention in several jurisdictions, including the United States, Hong Kong, and Singapore. In the United States, the Court of Appeals for the Fifth Circuit upheld the enforcement of the award in 2003, holding that only the courts of the seat, Switzerland, had primary jurisdiction to set aside the award, and that the Indonesian annulment did not prevent enforcement; the court also held that the alleged irregularities did not justify refusing enforcement under Article V of the Convention. KBC subsequently pursued Pertamina's assets in several jurisdictions, and the award was ultimately satisfied.

Significance. The case illustrates the allocation of political risk to state-owned counterparties in energy contracts, the narrowness of the force majeure defence where the risk is within the state's sphere, the distinction between primary and secondary jurisdiction under the New York Convention, and the practical difficulties of enforcing awards against state entities. It also illustrates why lenders to emerging market energy projects place great weight on multilateral support, which provides a remedy that does not depend on the pursuit of state assets. :::

16.4 Petroleum and Mining Fiscal Regimes

Objectives of Fiscal Design

A **fiscal regime** for the extractive sector comprises the set of legal instruments through which the state captures a share of the value of its natural resources: bonuses, rentals, royalties, income taxes, resource rent taxes, production sharing, state participation, withholding taxes, customs duties, and other charges. Fiscal design involves a balance among several objectives, which are frequently in tension:

- **revenue** — maximising the present value of government revenue over the life of the resource, rather than in a single year;
- **attractiveness** — ensuring that the regime provides investors with a return sufficient to attract the investment required, in competition with other jurisdictions;
- **progressivity** — increasing the government's share as the profitability of a project increases, so that the state shares in the upside of high prices or rich deposits;
- **neutrality** — minimising distortion of investment and production decisions, so that projects that are economic before tax remain economic after tax;
- **risk-sharing** — allocating the risks of price volatility and geological uncertainty between the state and investors;
- **stability and predictability** — providing a credible and stable framework for long-term investment;

- **administrative simplicity** — limiting the scope for disputes, avoidance, and corruption, and matching the capacity of the tax administration.

The economic justification for special fiscal treatment of the extractive sector lies in the existence of **economic rent** — the surplus of revenue over all costs of production, including a normal return on capital — which is generated by the scarcity and quality of natural resources. Because the state is usually the owner of subsoil resources, as examined in Chapter 2, it is entitled to capture a substantial share of this rent, and in principle a tax on pure economic rent does not affect investment decisions. In practice, however, rent is difficult to measure, investors and governments disagree about the appropriate normal return, and the time-inconsistency problem — the incentive for governments to increase their share once investments are sunk — creates the demand for stability examined in Chapters 14 and 15.

The Principal Families of Fiscal Regime

Petroleum fiscal regimes are conventionally divided into two families, corresponding to the forms of host government agreement examined in Chapter 15.

Concessionary (tax and royalty) regimes, used in the United States, the United Kingdom, Norway, Canada, Australia, and many other states, grant the investor ownership of the petroleum produced, subject to the payment of royalties and taxes. The government's share is captured through royalties, corporate income tax, and frequently special petroleum taxes or resource rent taxes.

Contractual regimes, comprising production sharing contracts and service contracts, provide for the state to retain ownership of the petroleum, with the contractor receiving a share of production (under PSCs) or a fee (under service contracts) as compensation for its services and risk. The government's share is captured through royalties (where applicable), the state's share of profit oil, corporate income tax, and state participation.

The differences between these families are, in economic terms, less significant than they appear: any combination of fiscal instruments can be designed to produce a similar division of rent, and the choice between them is often driven by constitutional, political, and historical considerations, including the symbolic importance of state ownership of the resource in post-colonial states. The principal features are compared in Figure 16.2.

	Concession (tax & royalty)	Production sharing contract	Service contract
Ownership of production	Investor at wellhead	State; contractor takes cost & profit oil	State; contractor paid a fee
Main state instruments	Bonus, royalty, income tax, resource rent tax	Royalty, profit-oil split, income tax, state participation	Retention of all production; fee set by contract
Cost recovery	Through tax deductions	Cost oil, subject to a recovery limit	Reimbursed with fee (risk or pure service)
Risk bearer	Investor bears most risk	Contractor bears exploration risk	State bears most market risk
Examples	UK, Norway, USA, Australia	Indonesia, Angola, Nigeria (deep offshore)	Iran, Iraq (TSCs), pre-2013 Mexico

Figure 16.2 The principal families of petroleum fiscal regime compared. *Source:* prepared for this edition.

Measuring and Comparing Fiscal Regimes

Fiscal regimes are compared using several measures. **Government take** is the government's share of the pre-tax net cash flow of a project, expressed as a percentage, and typically ranges from around fifty per cent to more than ninety per cent for petroleum projects. The **average effective tax rate** (AETR) measures the present value of government revenue as a share of the present value of pre-tax cash flow, while the **marginal effective tax rate** (METR) measures the effect of the regime on the return to marginal investment. Measures of **progressivity** examine how government take varies with prices and project profitability, and the **front-loading** of the regime measures how early in the project life the government receives revenue. Front-loaded instruments, such as bonuses and royalties, provide early and secure revenue for the state, but discourage marginal projects and transfer more risk to investors; profit-based instruments, such as resource rent taxes, are more neutral and progressive, but defer government revenue and expose the state to price and cost risks and to the manipulation of costs. The IMF's **Fiscal Analysis of Resource Industries** (FARI) model and the IMF's work on fiscal regimes for extractive industries provide widely used analytical frameworks.

Comparative Petroleum Regimes

Norway applies the ordinary corporate income tax of twenty-two per cent together with a special petroleum tax, yielding a combined marginal rate of

seventy-eight per cent, but allows generous deductions for investment costs and, since a reform adopted in 2022, operates the special tax as a cash-flow tax, under which investment costs are deducted immediately and the tax value of losses is refunded by the state. The regime is designed to be close to neutral, so that the state effectively shares in both the costs and the revenues of petroleum development, and is combined with direct state participation through the State's Direct Financial Interest, managed by Petoro, and the state's majority shareholding in Equinor.

The United Kingdom taxes petroleum production through ring-fence corporation tax and a supplementary charge on ring-fence profits. Petroleum revenue tax, which applied to older fields, has been set at zero since 2016. In response to high prices following the Russian invasion of Ukraine, the government introduced an **Energy Profits Levy** in 2022, initially at twenty-five per cent and subsequently increased, which was extended to 2030 and is accompanied by an investment allowance. The levy raised questions about the stability of the regime and its effect on investment, and the government has consulted on a permanent mechanism to respond to future price shocks.

Nigeria, under the Petroleum Industry Act 2021, replaced the petroleum profits tax for new and converted licences and leases with a combination of companies income tax, a **hydrocarbon tax** applicable to crude oil production from onshore and shallow water areas (but not to deep offshore and frontier acreage), royalties varying by terrain, production volume, and price, and limits on deductible costs through a cost price ratio. The Act provides incentives for gas development and for frontier exploration, and allows holders of existing licences to convert to the new regime on specified terms, as examined in Chapters 5 and 24. Production sharing contracts in the deep offshore remain subject to the Deep Offshore and Inland Basin Production Sharing Contracts Act, as amended in 2019, examined in Chapter 15.

Australia applies a profit-based Petroleum Resource Rent Tax to offshore projects, at a rate of forty per cent on profits above a threshold return, together with corporate income tax. The PRRT's treatment of LNG projects and the accumulation of large deductible carry-forward expenditures led to concerns that the tax was yielding little revenue from major LNG projects, and to reforms in 2023 that limited the use of deductions.

Mining Fiscal Regimes

Mining fiscal regimes typically rely more heavily on royalties and corporate income tax than petroleum regimes, with resource rent taxes and state participation used in some jurisdictions. Mining royalties are often ad valorem — a percentage of the value of the mineral, such as three to five per cent for gold and copper —

although profit-based and sliding-scale royalties linked to prices are increasingly common. The volatility of mineral prices and the long life of mines have led to frequent changes in mining fiscal regimes, and to disputes with investors.

Several recent reforms illustrate the trend towards increasing the state's share. The **Democratic Republic of Congo** adopted a revised Mining Code in 2018, which increased royalty rates, permitted the designation of "strategic substances" — including cobalt — subject to a royalty of ten per cent, introduced a tax on super-profits, and reduced the stability period for existing mining conventions, provoking strong objections from mining companies that had relied on the stability guarantees of the 2002 Code. **Chile**, the world's largest copper producer, enacted a new mining royalty law in 2023, applicable to large copper producers, combining an ad valorem component with a margin-based component, subject to a cap on the total tax burden. **Zambia** has repeatedly revised its mining royalty regime, oscillating between higher and lower rates in response to price changes and investor reactions, and in 2022–2023 adopted measures, including restoring the deductibility of mineral royalties for income tax purposes, intended to attract new investment. The **African Mining Vision**, adopted by African Union heads of state in 2009, calls for fiscal regimes that capture a fair share of resource rents for development, and the Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development has published guidance on mining fiscal policy.

Fiscal Stability

Fiscal stability is the issue at which fiscal design and investment law intersect. Investors seek stability through stabilisation clauses in host government agreements and through the protection of investment treaties, while governments seek to preserve the ability to adjust fiscal terms in response to changing circumstances. The most common sources of dispute are increases in royalty or tax rates, the introduction of windfall taxes, and the withdrawal of incentives. Investment tribunals have generally treated the general power to tax as an incident of sovereignty, and many investment treaties exclude or limit the application of their protections to taxation measures, as in Article 21 of the Energy Charter Treaty. Nevertheless, tax measures may breach treaty obligations where they are confiscatory, discriminatory, or contrary to specific commitments, as in *Occidental v Ecuador* and the claims arising from Ecuador's windfall tax, examined in Chapter 14, and in the *Yukos* awards, where the tribunals held that the Russian tax measures against Yukos were not a bona fide exercise of the taxation power but a means of expropriating the company.

16.5 Royalties, Resource Rent Taxes, and Production Levies

Royalties

A **royalty** is a payment to the resource owner — usually the state — calculated by reference to the volume or value of the resource produced, regardless of the profitability of production. Royalties are the oldest and most widespread extractive fiscal instrument, reflecting the owner's entitlement to compensation for the depletion of a non-renewable asset. They may be **specific** (a fixed amount per unit of production, such as per tonne or per barrel) or **ad valorem** (a percentage of the value of production). Ad valorem royalties are more common because they automatically adjust for price changes.

From the state's perspective, royalties have the advantages of providing revenue from the start of production, being relatively simple to administer, and being less vulnerable than profit-based taxes to the manipulation of costs. From the investor's perspective, royalties are regressive, because they are payable regardless of profitability and therefore represent a higher proportion of profits for marginal projects, and may cause premature abandonment of fields or mines when costs rise towards revenue. To address these effects, many regimes use **sliding-scale royalties**, whose rates vary with production volume, water depth, or price. Nigeria's Petroleum Industry Act 2021 combines production-based royalty rates that vary by terrain with an additional price-based royalty applicable when oil prices exceed specified thresholds, and provides lower rates for gas and for frontier and deep offshore acreage. Chile's 2023 mining royalty and Zambia's sliding-scale mineral royalty illustrate price-sensitive designs in mining.

The **valuation** of production for royalty purposes is a frequent source of dispute. Royalties may be calculated on the value at the wellhead or mine gate, requiring the deduction of transportation, processing, and marketing costs from the sale price — the **netback** method — or on the value at the point of sale. Where production is sold to affiliates, the sale price may not reflect market value, and royalty legislation commonly provides for valuation by reference to benchmark or reference prices, such as Brent or other published crude prices adjusted for quality and location, or the London Metal Exchange price for metals. In the history of the petroleum industry, the disputes between producing states and the international oil companies over **posted prices** — the reference prices used for royalty and tax purposes — were central to the shift of power to OPEC in the early 1970s, as examined in Chapter 4.

Resource Rent Taxes

A **resource rent tax** (RRT) is a tax on the economic rent generated by a resource project, levied on cash flow after the investor has recovered its costs together with a threshold rate of return. The concept was proposed by the economists Ross

Garnaut and Anthony Clunies Ross in the 1970s as a means of capturing a high share of rent from rich projects without deterring marginal ones. Because the tax applies only once the investor has earned a threshold return, it is in principle neutral and highly progressive, and allocates more of the risk of poor outcomes to the investor, who bears the full cost of failure until the threshold is reached, while sharing the upside with the state.

Australia's **Petroleum Resource Rent Tax**, examined in section 16.4, is the best-known example. The attempt to extend the resource rent concept to Australian mining illustrates the political difficulties of such taxes: the Resource Super Profits Tax proposed in 2010 following the Henry Tax Review provoked an intense campaign by mining companies and contributed to a change of Prime Minister; it was replaced by a narrower **Minerals Resource Rent Tax**, applicable to iron ore and coal, which took effect in 2012, raised far less revenue than projected, and was repealed in 2014. RRTs have been adopted in several developing countries, including in petroleum legislation in Ghana and Timor-Leste, but their effectiveness depends on the capacity of tax administrations to audit costs, since the tax base is vulnerable to the inflation of costs through transfer pricing and other means.

PSCs perform a similar function through **R-factor** or rate-of-return-based profit oil splits, examined in Chapter 15, under which the state's share of profit oil increases as the contractor recovers its investment and earns specified returns.

Windfall Taxes

Windfall taxes are taxes imposed on profits that are regarded as unexpectedly high, usually as a result of a sharp increase in commodity prices. They are typically introduced on a temporary basis, outside the ordinary fiscal regime, and frequently apply to existing projects. Examples include the 2006 Mongolian windfall profits tax on copper and gold, subsequently repealed; the Ecuadorian law of 2006, increased in 2007, which claimed fifty and then ninety-nine per cent of oil revenues above a reference price, generating the investment claims examined in Chapter 14; the United Kingdom's Energy Profits Levy of 2022; and the European Union's **solidarity contribution**, introduced by Council Regulation (EU) 2022/1854 on an emergency intervention to address high energy prices, which required Member States to levy a temporary contribution of at least thirty-three per cent on surplus taxable profits of companies active in the crude petroleum, natural gas, coal, and refinery sectors, and imposed a cap on the market revenues of inframarginal electricity generators. The regulation, adopted under Article 122 of the Treaty on the Functioning of the European Union as an emergency measure, was challenged before the General Court by several energy companies on the ground, among others, that Article 122 did not permit the adoption of fiscal measures by qualified majority.

Windfall taxes are popular politically, particularly when consumers face high energy prices, but raise significant legal and economic questions. They may undermine fiscal stability and investor confidence, breach stabilisation clauses, give rise to investment treaty claims, and discourage the investment needed to increase supply. Economists generally prefer permanent progressive fiscal regimes, which capture a rising share of rent automatically when prices rise, over ad hoc windfall taxes.

Production Levies and Earmarked Charges

In addition to royalties and taxes, extractive companies frequently pay levies earmarked for particular purposes. In Nigeria, petroleum companies are subject to contributions to the **Niger Delta Development Commission**, established in 2000 to promote development in the oil-producing region; the **Tertiary Education Trust Fund**; and, under the Petroleum Industry Act 2021, contributions of three per cent of their actual annual operating expenditure to **Host Communities Development Trusts**, established for the benefit of communities in the areas of operations, examined in Chapters 21 and 24. The Act also established a Midstream and Downstream Gas Infrastructure Fund and a Frontier Exploration Fund. Earmarked levies respond to legitimate concerns about the distribution of resource benefits to affected regions and communities, but they fragment public revenues, complicate the assessment of the overall fiscal burden, and raise questions of governance and accountability in the management of the funds.

State Participation and Carried Interests

Many states take an equity interest in extractive projects, either through a national resource company or directly. **Carried interests**, under which the investor funds the state's share of costs — wholly (a "free" carried interest) or until a specified stage, with the costs recoverable from the state's share of production — are common in mining and petroleum. Ghana's mining law provides for a free carried interest of ten per cent in mining companies, and Tanzania's 2017 legislation provided for a non-dilutable free carried interest of not less than sixteen per cent in mining operations. State participation is a fiscal instrument — since it entitles the state to dividends — but also an instrument of control, information, and technology transfer, and its value depends on the governance of the state company.

Export Taxes, Export Restrictions, and Local Processing

Some resource-rich states impose export taxes or restrictions on unprocessed minerals to promote domestic processing and capture a larger share of the value chain. Indonesia's ban on the export of nickel ore, reinstated in 2020, contributed to large-scale investment in domestic smelting, principally by Chinese companies, and was found inconsistent with the General Agreement on Tariffs and Trade by a WTO panel in 2022 in a dispute brought by the European Union, a decision

appealed by Indonesia "into the void" in the absence of a functioning Appellate Body, as examined in Chapter 13. Zimbabwe banned the export of unprocessed lithium ore in 2022. Such measures raise questions under international trade law, investment treaties, and stabilisation clauses, and their effectiveness depends on the competitiveness of domestic processing.

Revenue Management and Sovereign Wealth Funds

The capture of resource revenues is only the first step; their management determines whether they contribute to sustainable development. The volatility of resource revenues, the risk of "Dutch disease" — the appreciation of the real exchange rate and the decline of other tradable sectors — and the exhaustibility of resources have led many states to adopt fiscal rules and to establish **sovereign wealth funds**. Norway's Government Pension Fund Global, financed from petroleum revenues and governed by a fiscal rule limiting the use of fund returns in the budget, is the largest sovereign wealth fund in the world. Ghana's Petroleum Revenue Management Act 2011 allocates petroleum revenues among the national budget, a Ghana Stabilisation Fund, and a Ghana Heritage Fund, with oversight by a Public Interest and Accountability Committee. Nigeria established the Nigeria Sovereign Investment Authority in 2011, with a stabilisation fund, a future generations fund, and an infrastructure fund, succeeding the Excess Crude Account, whose management had been controversial. Timor-Leste's Petroleum Fund, established in 2005, finances a large proportion of government spending. The **Santiago Principles** — the Generally Accepted Principles and Practices for sovereign wealth funds, adopted in 2008 by the International Working Group of Sovereign Wealth Funds and promoted by its successor, the International Forum of Sovereign Wealth Funds — provide voluntary standards on governance, accountability, and investment practices.

16.6 Transfer Pricing and International Taxation

The International Tax Framework

The taxation of multinational energy and mining enterprises operating across borders is governed by national tax laws and a network of more than three thousand bilateral **double taxation treaties**, most of which are based on the **OECD Model Tax Convention on Income and on Capital** or the **United Nations Model Double Taxation Convention between Developed and Developing Countries**. The treaties allocate taxing rights between the state of residence of the taxpayer and the state of source of the income, and provide mechanisms for relieving double taxation. The UN Model generally preserves greater taxing rights for source states, which is significant for capital-importing resource-rich states. Both models permit the source state to tax the business profits of a non-resident enterprise only if the enterprise has a **permanent establishment** in the source

state, and both treat a mine, an oil or gas well, a quarry, or any other place of extraction of natural resources as a permanent establishment. Offshore activities and short-duration service contracts raise specific issues, which some treaties address through special provisions for offshore exploration and exploitation. The United Nations Committee of Experts on International Cooperation in Tax Matters published a **Handbook on Selected Issues for Taxation of the Extractive Industries by Developing Countries** in 2017, which addresses treaty, transfer pricing, and other issues specific to the sector.

The Arm's Length Principle

Multinational enterprises conduct a large proportion of their transactions with related parties — affiliates within the same group — in the course of selling production, purchasing equipment and services, licensing technology, providing management services, and lending money. The prices charged in these transactions, known as **transfer prices**, determine how the group's profits are allocated among the jurisdictions in which it operates, and therefore how much tax is paid in each. Article 9 of the OECD and UN Models establishes the **arm's length principle**: the profits of associated enterprises may be adjusted if the conditions of their commercial or financial relations differ from those that would be made between independent enterprises. The **OECD Transfer Pricing Guidelines for Multinational Enterprises and Tax Administrations**, most recently consolidated in 2022, set out the methods for applying the principle: the comparable uncontrolled price method, the resale price method, the cost plus method, the transactional net margin method, and the transactional profit split method.

Transfer pricing is especially significant in the extractive sector because of the scale of related-party sales of commodities, the use of marketing and trading hubs in low-tax jurisdictions such as Switzerland, Singapore, and Dubai, the prevalence of intra-group services and financing, and the limited capacity of many developing country tax administrations to audit complex transactions. Studies by international organisations and civil society groups have suggested that resource-rich developing countries lose significant revenue through mispricing of commodity exports and inflated costs, although estimates vary widely and are contested.

Commodity Transactions

For commodity sales, the comparable uncontrolled price method, using quoted prices on commodity exchanges or published price reporting agencies, is generally the most appropriate. Following the BEPS project, the OECD added guidance to the Transfer Pricing Guidelines in 2015 confirming that quoted prices may be used as a reference for commodity transactions between associated enterprises, subject to comparability adjustments, and addressing the **pricing date** — since the ability to select the pricing date with hindsight would permit the manipula-

tion of prices — by allowing tax administrations to impute the shipment date where the taxpayer cannot prove the agreed pricing date. Several Latin American states, beginning with Argentina in 2003, adopted a so-called "sixth method", under which commodity exports to related parties are priced by reference to quoted market prices on the date of shipment, an approach that influenced the OECD guidance. The Platform for Collaboration on Tax — a joint initiative of the IMF, OECD, UN, and World Bank — and the Intergovernmental Forum on Mining have produced additional guidance and toolkits on the pricing of minerals.

Intra-Group Financing and Interest Deductions

Multinational groups may reduce their taxable profits in high-tax resource jurisdictions by financing operations with intra-group loans carrying high interest rates, the interest on which is deductible in the source state, while the corresponding income is received in a low-tax jurisdiction. States respond through **thin capitalisation** rules, which limit the deduction of interest where debt exceeds a specified ratio to equity, and through **earnings stripping** rules, which limit net interest deductions to a percentage of earnings. BEPS Action 4 recommended a fixed ratio rule limiting net interest deductions to between ten and thirty per cent of earnings before interest, taxes, depreciation, and amortisation (EBITDA), which has been implemented in the European Union by the Anti-Tax Avoidance Directive (Directive (EU) 2016/1164), and in many other states. Nigeria's tax law introduced a limit on the deductibility of interest on related-party debt at thirty per cent of EBITDA, following the BEPS recommendation. The application of the arm's length principle to intra-group loans was tested in a leading Australian case concerning an energy multinational.

::: casebox {title="Chevron Australia Holdings Pty Ltd v Commissioner of Taxation [2017] FCAFC 62"} **Facts.** In 2003, Chevron Australia Holdings Pty Ltd (CAHPL), the Australian holding company of the Chevron group's Australian operations, which included interests in major LNG projects, borrowed approximately USD 2.5 billion (in Australian dollar equivalent) from a newly established United States subsidiary, Chevron Texaco Funding Corporation, under a credit facility agreement. The funding company raised the money in the United States commercial paper market at an interest rate of approximately 1.2 per cent, with the benefit of a guarantee from the ultimate parent company, and lent it to CAHPL at an interest rate of approximately nine per cent, without security or a parent guarantee. The interest paid by CAHPL was deductible in Australia, while the funding company's interest income was effectively not taxed in the United States, and was returned to CAHPL as dividends that were exempt from Australian tax.

The dispute. The Commissioner of Taxation issued amended assessments under the Australian transfer pricing provisions, disallowing a substantial part of the interest deductions on the ground that the interest exceeded an arm's length amount, with resulting tax and penalties of approximately AUD 340 million.

The decision. The Full Court of the Federal Court of Australia unanimously upheld the assessments. The Court held that the arm's length price had to be determined by reference to what an independent borrower in CAHPL's position, as a member of a multinational group, would have paid to an independent lender for the same loan, and that in determining this, it was appropriate to take into account that an independent borrower in CAHPL's position would ordinarily have provided security and obtained a guarantee from its parent company in order to secure a lower rate. The Court rejected Chevron's argument that the loan should be priced as if CAHPL were a stand-alone company without the support of the group, and held that the absence of security and covenants, which would have been present in an arm's length transaction, could not be used to justify a higher interest rate. An application for special leave to appeal to the High Court of Australia was subsequently discontinued.

Significance. The case is a leading authority on the pricing of intra-group loans in the extractive sector and on the relevance of group membership and implicit parental support in applying the arm's length principle. It has been cited in the OECD's 2020 guidance on financial transactions and in transfer pricing disputes in other jurisdictions, and illustrates the scale of revenue at stake in the financing structures of major energy projects. :::

Indirect Transfers of Interests

A distinctive issue in the extractive sector is the taxation of **indirect transfers** — sales of shares in a company, often incorporated in a low-tax jurisdiction, that directly or indirectly holds a licence, concession, or other interest in a resource project in the host state. Because the asset sold is a share in a foreign company rather than the resource interest itself, the host state may not have the right under its domestic law or its tax treaties to tax the gain, although the value of the shares derives from the resource interest located in its territory.

The issue has generated major disputes. In **India**, the Supreme Court held in *Vodafone International Holdings BV v Union of India* (2012) that the gain on the sale of shares of a Cayman Islands company that indirectly held an Indian telecommunications business was not taxable in India. The government responded by amending the Income Tax Act with retrospective effect to tax such indirect transfers. The retrospective tax was applied to Vodafone and to the British energy company Cairn Energy in respect of an internal reorganisation of its Indian oil and gas business, leading to investment treaty arbitrations in which tribunals

ruled in favour of Vodafone in 2020 and of Cairn in December 2020, awarding Cairn approximately USD 1.2 billion plus interest and costs. Cairn commenced enforcement proceedings against Indian state assets in several jurisdictions, and India subsequently enacted the Taxation Laws (Amendment) Act 2021, which withdrew the retrospective demands in return for the withdrawal of claims, and refunded the taxes collected.

In **Uganda**, the sale by Heritage Oil and Gas of its interests in oil exploration areas in the Lake Albert basin to Tullow Oil in 2010, for approximately USD 1.45 billion, led the Uganda Revenue Authority to assess capital gains tax of approximately USD 404 million. Tullow paid a portion of the tax to the government as a condition of obtaining approval of the transaction, and successfully claimed an indemnity from Heritage in the English courts in *Tullow Uganda Operations Pty Ltd v Heritage Oil and Gas Ltd* [2014] EWCA Civ 1048. The case illustrates the leverage that host governments derive from the requirement of consent to the transfer of licences, and the importance of tax indemnities in acquisition agreements.

Many resource-rich states have since amended their laws to tax indirect transfers of interests in domestic resource assets, and the Platform for Collaboration on Tax published a toolkit on the taxation of offshore indirect transfers in 2020. The UN Model was amended in 2017 to permit source state taxation of gains from the alienation of shares deriving their value principally from immovable property, including resource rights, a provision also reflected in Article 13(4) of the OECD Model.

BEPS and the Global Minimum Tax

The OECD/G20 **Base Erosion and Profit Shifting** (BEPS) project, whose fifteen action reports were delivered in 2015, addressed the strategies by which multinational enterprises shift profits to low-tax jurisdictions, including hybrid mismatches, treaty abuse, the artificial avoidance of permanent establishment status, transfer pricing of intangibles and risk, and interest deductions. The BEPS measures were implemented through national legislation, revisions to the OECD Model and Guidelines, and the **Multilateral Convention to Implement Tax Treaty Related Measures to Prevent BEPS** (the MLI), signed in 2017, which introduced a **principal purpose test** to deny treaty benefits where obtaining them was one of the principal purposes of an arrangement. These measures have restricted the use of treaty shopping through conduit jurisdictions, such as Mauritius and the Netherlands, which had been widely used to structure investment in African resource projects.

The **Two-Pillar Solution**, agreed by the OECD/G20 Inclusive Framework in October 2021, comprises Pillar One, which would reallocate a portion of

the residual profits of the largest and most profitable multinational enterprises to market jurisdictions, and from which extractive activities are excluded, and **Pillar Two**, the Global Anti-Base Erosion (GloBE) rules, which establish a global minimum effective tax rate of fifteen per cent for multinational groups with annual revenues of at least EUR 750 million. Pillar Two has been implemented in the European Union by Council Directive (EU) 2022/2523 and in numerous other jurisdictions from 2024. Pillar Two has significant implications for resource-rich states: tax incentives, such as tax holidays granted under stabilised investment agreements, may result in the profits being taxed by a "top-up tax" in the parent company's jurisdiction, so that the incentive transfers revenue to the residence state rather than benefiting the investor, and host states may respond by adopting a **qualified domestic minimum top-up tax** to collect the top-up tax themselves. The interaction between Pillar Two and stabilisation clauses that freeze the tax regime is an emerging legal issue. The United States' position on the application of Pillar Two to United States-parented groups changed in 2025, creating uncertainty about the global implementation of the rules.

Developing countries have argued that the OECD-led process has not adequately reflected their interests. In December 2023, the United Nations General Assembly decided to begin work on a **United Nations Framework Convention on International Tax Cooperation**, and terms of reference for the negotiation of the convention and early protocols were adopted in 2024, with negotiations scheduled to continue until 2027. The African Tax Administration Forum and initiatives such as Tax Inspectors Without Borders, launched by the OECD and UNDP in 2015, support capacity building in tax administration, including in the extractive sector.

16.7 Revenue Transparency and Anti-Corruption Controls

Corruption and the Resource Sector

The extractive sector is widely regarded as one of the sectors most exposed to corruption. The concentration of large rents in a small number of transactions, the discretionary power of officials who allocate licences and approve transfers, the involvement of state-owned enterprises, the complexity of fiscal arrangements, and the opacity of commodity trading combine to create opportunities for bribery, embezzlement, and the misappropriation of public revenues. The OECD's Foreign Bribery Report of 2014, which analysed concluded foreign bribery enforcement actions, found that the extractive sector accounted for the largest share of cases. Corruption in the sector diverts public resources, distorts investment decisions, undermines the legitimacy of contracts, and contributes to the institutional weaknesses associated with the resource curse. The legal response comprises three overlapping elements: transparency regimes, which require the

disclosure of payments, contracts, and beneficial ownership; anti-bribery laws, which criminalise the bribery of foreign and domestic officials; and the treatment of corruption in contract law and arbitration.

The Extractive Industries Transparency Initiative

The **Extractive Industries Transparency Initiative** (EITI), launched in 2002 and formally established in 2003, is a multi-stakeholder initiative involving governments, companies, and civil society, which promotes the disclosure of information on the governance of the extractive sector. Implementing countries commit to publishing information in accordance with the **EITI Standard**, most recently revised in 2023, which requires disclosure across the value chain: the legal and institutional framework, including the allocation of contracts and licences; exploration and production; revenue collection, including the reconciliation of payments reported by companies with revenues reported by governments; revenue allocation; and social and economic spending. The Standard requires the disclosure of the **beneficial owners** of companies holding extractive rights, the publication of contracts and licences entered into or amended from 2021, and information on the sale of the state's share of production, environmental payments, gender, and energy transition issues. Implementation is overseen by a national multi-stakeholder group in each country and assessed through a process of **validation** by the EITI International Board, which may suspend countries that fail to make progress. More than fifty countries implement the Standard.

Nigeria was among the first countries to implement the EITI and was the first to give it a statutory basis, through the **Nigeria Extractive Industries Transparency Initiative (NEITI) Act 2007**, which established NEITI as a statutory body with a mandate to conduct audits of the oil, gas, and mining sectors. NEITI's audit reports have documented discrepancies between company payments and government receipts, unremitted revenues by state entities, crude oil losses through theft, and weaknesses in metering and cost recovery, and have contributed to public debate and reform, although their recommendations have not always been implemented.

Mandatory Payment Disclosure

In addition to the voluntary EITI, several jurisdictions have adopted mandatory requirements for extractive companies to disclose payments to governments on a country-by-country and project-by-project basis, following campaigns by the Publish What You Pay coalition. Section 1504 of the United States **Dodd-Frank Wall Street Reform and Consumer Protection Act** of 2010 required the Securities and Exchange Commission to adopt rules requiring resource extraction issuers to disclose payments to governments. The SEC's first rule was vacated by a federal court in 2013, and a second rule was disapproved by Congress in

2017 under the Congressional Review Act; a third rule was adopted in 2020. In the European Union, Chapter 10 of the **Accounting Directive** (Directive 2013/34/EU) and the amended Transparency Directive require large extractive and logging companies and listed issuers to publish annual reports on payments to governments, implemented in the United Kingdom by the Reports on Payments to Governments Regulations 2014. Canada's **Extractive Sector Transparency Measures Act**, enacted in 2014 and in force from 2015, imposes similar requirements. Norway has also adopted country-by-country reporting rules for extractive companies.

Contract Transparency and Beneficial Ownership

The disclosure of the contracts between governments and extractive companies has become an important norm. Historically, petroleum and mining contracts were treated as confidential, often at the insistence of both parties, but the confidentiality of contracts conceals their terms from citizens and parliaments, impedes accountability, and facilitates corruption. Many states now publish extractive contracts, and the EITI Standard, the IMF's Guide on Resource Revenue Transparency, the Natural Resource Charter — a set of principles for governments and societies on how best to harness the opportunities created by extractive resources, whose second edition was published in 2014 — and the IFC's policy on disclosure of extractive contracts support contract transparency. The online repository ResourceContracts.org, developed by the Natural Resource Governance Institute and partners, makes thousands of contracts publicly available. Chapter 24 examines the extent to which Nigeria's PIA requires the publication of licences and contracts.

Beneficial ownership transparency addresses the use of anonymous companies to conceal the identity of those who benefit from extractive rights, including politically exposed persons. The EITI requires implementing countries to maintain public registers of beneficial owners of extractive companies, and many countries, including Nigeria through the Companies and Allied Matters Act 2020, have established beneficial ownership registers. The European Union's anti-money laundering framework required public access to beneficial ownership registers, although the Court of Justice held in *WM and Sovim SA v Luxembourg Business Registers* (2022) that unrestricted public access was incompatible with the rights to privacy and data protection under the Charter of Fundamental Rights, requiring access to be limited to persons with a legitimate interest.

Anti-Bribery Law

The bribery of foreign public officials is prohibited by national laws with extraterritorial reach and by international conventions.

The United States **Foreign Corrupt Practices Act** of 1977 (FCPA) prohibits issuers of securities in the United States, domestic concerns, and persons acting in the United States from making corrupt payments to foreign officials to obtain or retain business, and requires issuers to maintain accurate books and records and adequate internal controls. The FCPA has been enforced vigorously by the Department of Justice and the SEC, and a large proportion of its most significant enforcement actions have involved the energy and extractive sector, including actions against oil companies, oilfield service companies, and commodity traders. In February 2025, the United States President issued an executive order pausing new FCPA enforcement pending the review of enforcement guidelines, and revised guidelines were subsequently issued focusing enforcement on specified priorities, creating uncertainty about the future intensity of enforcement.

The United Kingdom's **Bribery Act 2010** criminalises the bribery of foreign public officials and creates a corporate offence of **failure to prevent bribery** by persons associated with a commercial organisation, to which it is a defence that the organisation had adequate procedures in place to prevent bribery. The failure to prevent model has influenced legislation elsewhere and has been extended in the United Kingdom to the facilitation of tax evasion and, by the Economic Crime and Corporate Transparency Act 2023, to fraud.

At the international level, the **OECD Convention on Combating Bribery of Foreign Public Officials in International Business Transactions** of 1997 requires parties to criminalise foreign bribery and is monitored through peer review by the OECD Working Group on Bribery. The **United Nations Convention against Corruption** (UNCAC) of 2003, which has near-universal participation, addresses prevention, criminalisation, international cooperation, and asset recovery. Regional instruments include the **African Union Convention on Preventing and Combating Corruption** (2003) and the **ECOWAS Protocol on the Fight against Corruption** (2001). Nigeria's domestic framework includes the Corrupt Practices and Other Related Offences Act 2000, the Economic and Financial Crimes Commission (Establishment) Act 2004, and the Money Laundering (Prevention and Prohibition) Act 2022.

Deferred prosecution agreements (DPAs), available in the United States and, since 2014, in England and Wales, permit companies to resolve corruption charges through the payment of penalties and the implementation of compliance programmes without conviction. Major resolutions in the energy and commodities sector include the 2022 guilty pleas by the commodity trader Glencore in the United States and the United Kingdom, in respect of bribery in several African and Latin American states, involving combined penalties exceeding USD 1 billion; resolutions involving the Brazilian state oil company Petrobras and the

construction company Odebrecht arising from the *Lava Jato* investigation; and actions involving oilfield service and offshore contractors.

The World Bank and other multilateral development banks operate **sanctions systems** under which firms found to have engaged in fraud or corruption in bank-financed projects may be debarred from future contracts, and the banks' agreement on mutual enforcement of debarment decisions, concluded in 2010, extends the effect of debarment across institutions.

Corruption in Contract and Arbitration

Corruption affects the validity and enforceability of contracts and the admissibility of claims in arbitration. Arbitral tribunals have held that contracts procured by bribery are contrary to international public policy and unenforceable. In *World Duty Free Co Ltd v Republic of Kenya* (2006), an ICSID tribunal held that a claimant which had obtained a contract by paying a bribe to the head of state could not maintain claims based on the contract, because bribery is contrary to international public policy. In *Metal-Tech Ltd v Republic of Uzbekistan* (2013), an ICSID tribunal declined jurisdiction over a claim under a bilateral investment treaty, finding that the investment had been made through payments to officials in breach of Uzbek law, so that it did not qualify for protection under the treaty's legality requirement, although the tribunal noted that the state's participation in the corruption was reflected in the allocation of costs.

Corruption may also be raised at the stage of enforcement or set-aside. In *Federal Republic of Nigeria v Process & Industrial Developments Ltd* [2023] EWHC 2638 (Comm), the English Commercial Court set aside arbitral awards worth approximately USD 11 billion, including interest, on the ground that they had been obtained by fraud, including bribery of a Nigerian official and the improper retention of Nigeria's privileged documents, a case examined in detail in Chapter 26. The decision illustrates the role of national courts in policing the integrity of arbitration involving state energy contracts.

The scope of corruption allegations in the allocation of petroleum rights is illustrated by the controversy surrounding Nigeria's **OPL 245**, an offshore oil block allocated in 1998 to a company associated with a former petroleum minister and later sold to Shell and Eni in 2011 in a transaction in which payments by the companies passed through the Nigerian government to the original holder. Criminal prosecutions of the companies and individuals in Italy ended in acquittal in 2021, a result that became final when the prosecution abandoned its appeal, while proceedings in other jurisdictions and civil claims raised continuing questions about the transaction. The case illustrates the difficulty of proving corruption in complex transactions and the multiplicity of forums in which such allegations may be pursued.

Towards Integrated Governance

Transparency and anti-corruption measures are necessary but not sufficient conditions for good governance of resource revenues. Disclosure has limited effect unless citizens, parliaments, the media, and oversight institutions have the capacity and freedom to use the information to hold governments and companies accountable, and unless law enforcement institutions are able to investigate and prosecute wrongdoing. Research on the effectiveness of the EITI suggests that its effects on corruption and governance outcomes have been modest, although it has contributed to the production of information and the institutionalisation of public debate in implementing countries. The integration of transparency into national law, as in the NEITI Act, the publication of contracts and beneficial ownership, the strengthening of supreme audit institutions and parliamentary oversight, the reform of state-owned enterprises, and the protection of civil society space are complementary elements of an effective framework. For legal practitioners, anti-corruption compliance — including due diligence on agents, partners, and intermediaries, contractual anti-corruption representations and audit rights, and internal controls — has become a core element of transactional practice in the energy sector, as examined further in Chapter 22.

16.8 Debt Restructuring and Project Financial Distress

Sources of Financial Distress

Energy projects and companies experience financial distress for many reasons: construction cost overruns and delays; operational underperformance; collapses in commodity prices, as in the oil price falls of 2014–2016 and 2020; the failure of offtakers to pay; currency devaluation that makes foreign currency debt unaffordable; changes in law or regulation that reduce revenues; and, increasingly, the stranding of fossil fuel assets as a result of climate policy and technological change. Distress may affect individual projects, entire sectors — such as state-owned utilities whose tariffs do not cover costs — and sovereign states whose public finances depend on resource revenues or which have borrowed against future resource production. The legal tools for managing distress differ at each level.

Project-Level Distress: Waivers, Standstills, and Restructuring

When a project-financed energy project encounters difficulties, the finance documents examined in section 16.2 give lenders extensive rights: to block distributions, to declare events of default, to accelerate the debt, to enforce their security, and to step in under the direct agreements. In practice, however, enforcement is rarely the preferred option. The value of a project lies in its continued operation under its contracts and permits, which may be lost or impaired on enforcement; the market for distressed energy assets in emerging markets may be thin; and enforcement may require the consent of the host government to the transfer of

licences or shares. Lenders therefore usually prefer consensual solutions: **waivers** of defaults; **standstill agreements**, under which lenders agree not to enforce for a period while the parties negotiate; **amendments** to the repayment profile, covenants, and financial ratios; injections of new equity or subordinated debt by sponsors; and, in more serious cases, comprehensive **restructurings**, which may involve the conversion of debt into equity, the transfer of ownership of the project to the lenders, or the sale of the project to a new sponsor.

Restructuring requires the cooperation of multiple classes of creditors with different interests — commercial banks, ECAs, DFIs, bondholders, and hedge counterparties — whose relationships are governed by the intercreditor agreement. Unanimity requirements for fundamental changes may allow individual creditors to hold out, and bonds are often more difficult to restructure than loans because of the dispersed ownership of bondholders. Lenders' **step-in rights** under direct agreements allow them to assume control of the project's contractual relationships, but lenders are generally reluctant to exercise them because doing so may expose them to the project company's liabilities.

Restructuring Procedures and Insolvency

Where consensual restructuring fails, formal procedures may be used. In the United States, **Chapter 11** of the Bankruptcy Code allows a debtor to reorganise under court supervision, with an automatic stay on enforcement, the ability to obtain new financing with priority, and a plan of reorganisation that may be imposed on dissenting classes of creditors ("cram-down"), subject to statutory protections. Chapter 11 was extensively used by United States shale producers during the oil price collapses of 2015–2016 and 2020, when dozens of exploration and production companies filed for bankruptcy. A significant legal issue in these cases was whether debtors could reject long-term gathering and transportation agreements with midstream companies under the Bankruptcy Code's power to reject executory contracts, which depended on whether the dedication of production under those agreements constituted a covenant running with the land under state property law. Courts reached differing conclusions on the facts of different agreements, and the issue illustrates the interaction between insolvency law, property law, and long-term energy contracts.

In the United Kingdom, the **restructuring plan** introduced by the Corporate Insolvency and Governance Act 2020, as Part 26A of the Companies Act 2006, allows a company in financial difficulty to propose a compromise with its creditors which the court may sanction even if one or more classes of creditors vote against it, subject to conditions — a form of cross-class cram-down. Restructuring plans and the older schemes of arrangement have been used by companies in the energy and offshore services sectors, including foreign companies with English law-governed debt. Many other jurisdictions, including those of the European Union

under the Directive on preventive restructuring frameworks (Directive (EU) 2019/1023), have adopted similar procedures. The **UNCITRAL Model Law on Cross-Border Insolvency** (1997), adopted by numerous states, facilitates the recognition of foreign insolvency proceedings and cooperation among courts, which is often essential in the restructuring of multinational energy groups. Nigeria's Companies and Allied Matters Act 2020 introduced company voluntary arrangements and administration procedures intended to promote corporate rescue.

Utility Insolvency and Sector-Wide Distress

The financial distress of state-owned and privatised utilities poses distinctive problems because of their essential public service role. Where tariffs set by governments or regulators do not cover costs, and where losses through theft, non-payment, and technical inefficiency are high, utilities accumulate debts to generators, fuel suppliers, and lenders, which may cascade through the energy supply chain.

South Africa's state utility **Eskom** illustrates the scale of the problem. Operational failures, cost overruns at new coal plants, and accumulated debt led to prolonged load-shedding and a debt burden that threatened the solvency of the utility and the fiscal position of the state. In 2023, the government enacted legislation providing debt relief to Eskom of approximately ZAR 254 billion, subject to conditions, and the utility is being restructured into separate generation, transmission, and distribution entities. In **Nigeria**, the electricity distribution companies created by the 2013 privatisation accumulated large shortfalls in remittances to the market, owing to tariffs below cost, high losses, and weak metering, leading to repeated government interventions, the takeover of several distribution companies by lenders, and the reforms of the Electricity Act 2023, examined in Chapters 9 and 24. In **Ghana**, energy sector arrears, including arrears to IPPs and gas suppliers, became a significant component of the public debt addressed in the country's IMF-supported programme and debt restructuring from 2022.

Utility distress raises legal questions concerning the enforceability of PPAs and guarantees, the priority of payments within the supply chain, the regulatory obligation to set cost-reflective tariffs, the protection of vulnerable consumers, and the potential application of insolvency law to state-owned enterprises, which are frequently excluded from ordinary insolvency procedures.

Sovereign Debt and Natural Resources

Resource-rich states are particularly exposed to sovereign debt distress because of the volatility of resource revenues, and natural resources are directly linked to sovereign debt through **resource-backed loans** — loans repaid in, or secured on, future revenues from resource production. Such loans, provided by Chinese

policy banks, commodity traders, and other lenders, have been used by Angola, the Republic of Congo, Chad, Venezuela, Ecuador, and others. They may enable states with limited market access to finance infrastructure, but they are often opaque, may carry high effective costs, and can complicate debt restructuring because the lenders have a privileged claim on resource revenues. Studies by the Natural Resource Governance Institute have documented the risks associated with resource-backed lending and called for greater transparency.

The restructuring of sovereign debt lacks a statutory framework comparable to corporate insolvency. Sovereign bonds increasingly contain **collective action clauses** allowing a qualified majority of bondholders to bind dissenting holders, and the **G20 Common Framework for Debt Treatments beyond the DSSI**, agreed in November 2020, provides a framework for coordinated restructuring of the official bilateral debt of low-income countries, including debt owed to China and other non-Paris Club creditors, with comparable treatment required from private creditors. Chad, Zambia, Ethiopia, and Ghana requested treatment under the Common Framework. Chad's restructuring, completed in 2022–2023, was notable for involving oil-backed debt owed to the commodity trader Glencore, and for incorporating a mechanism under which creditors would share in the upside of high oil prices. The slowness of Common Framework restructurings, particularly that of Zambia, which defaulted in 2020 and reached agreements with official and bondholder creditors only in 2023–2024, has led to calls for reform.

The relationship between sovereign debt and investment arbitration is complex. Bondholders have brought investment treaty claims arising from sovereign debt restructurings, as in *Abadat v Argentina*, in which an ICSID tribunal accepted jurisdiction over a mass claim by Italian bondholders in 2011. Holders of arbitral awards against states — including energy investors — may compete with other creditors for the state's assets, as in the protracted enforcement proceedings against Venezuela and its state oil company PDVSA, which culminated in the court-supervised sale of shares in the parent company of CITGO Petroleum in the United States, examined in Chapter 26. Debt-for-nature and debt-for-climate swaps, under which debt is reduced in return for commitments to environmental expenditure, as in transactions involving Ecuador, Belize, and Gabon, illustrate the growing linkage between sovereign debt and environmental objectives, and the broader agenda of reform of the international financial architecture to address climate vulnerability, examined in Chapter 18.

Stranded Assets and Transition Risk

The energy transition creates a distinctive form of financial distress: the **stranding** of fossil fuel assets whose economic life is cut short by climate policy, technological change, or shifts in demand. Coal-fired power plants financed on

the assumption of thirty or forty years of operation may be forced to close early, and upstream petroleum reserves may become uneconomic to develop. Stranding raises questions of who should bear the losses — investors, lenders, consumers, or taxpayers — and whether compensation is payable, examined in Chapter 14 in the context of coal phase-out claims under the Energy Charter Treaty. Innovative financial mechanisms have been developed to accelerate the retirement of coal plants while managing financial losses, such as the Asian Development Bank's **Energy Transition Mechanism**, which proposes to use concessional finance to refinance or acquire coal plants and retire them earlier than planned, and transactions under the JETPs examined in section 16.1. Financial regulators, through the Network for Greening the Financial System and supervisory expectations on climate-related financial risk, increasingly require banks and insurers to assess their exposure to transition risk, an issue examined in Chapter 30.

::: exercise {title="Practical Exercise 16: Structuring and Rescuing a Gas-to-Power Project (HYPOTHETICAL)"} **This exercise is entirely hypothetical. All parties, places, contracts, and events are fictitious and do not refer to any real persons, companies, or states.**

The Republic of Sabelia, a lower-middle-income West African state with a nascent gas industry, has selected a consortium led by Coastline Power Holdings to build a 450 MW gas-fired power plant. The total project cost is USD 600 million. The single buyer, Sabelia Bulk Power Company (SBPC), a state-owned entity with a history of payment arrears, will purchase the plant's output under a twenty-year take-or-pay PPA with tariffs denominated in US dollars but payable in the local currency, the Sabelian dinar. Gas will be supplied by the national oil company under a fuel supply agreement. Proposed lenders include a commercial bank syndicate, an export credit agency, and a multilateral DFI.

Sabelia's petroleum law imposes a royalty of ten per cent, corporate income tax of thirty per cent, and a production levy of two per cent of operating expenditure for a host communities fund. The Ministry of Finance has proposed a ten-year tax holiday for the project. Coastline proposes to finance part of its equity through a shareholder loan from a Mauritius-incorporated affiliate at an interest rate of twelve per cent.

Five years after commercial operation, the Sabelian dinar has lost sixty per cent of its value against the US dollar; SBPC has accumulated eighteen months of arrears; the project company is in breach of its DSCR covenants; and the government, facing a sovereign debt crisis, has requested treatment under an international debt restructuring framework. Meanwhile, a newspaper has alleged that a consultant engaged by Coastline during the procurement process made payments to an official of the Ministry of Energy.

Tasks.

(a) Prepare a term sheet outlining the credit support package that the lenders are likely to require from the government and multilateral institutions, and assess the contingent liabilities it would create for Sabelia.

(b) Advise the Ministry of Finance on whether the proposed tax holiday is advisable, taking into account the global minimum tax and the interests of the state.

(c) Advise the Sabelian Revenue Authority on how it should approach the deductibility of interest on the shareholder loan.

(d) Advise the lenders on the options available to them following the currency devaluation and arrears, and on the risks of each option.

(e) Advise Coastline on the potential consequences of the corruption allegation for the project contracts, any future arbitration, and the company's exposure under anti-bribery laws, and recommend immediate steps. :::

CHAPTER SUMMARY

- Project finance enables the financing of large energy projects on a limited-recourse basis through special purpose vehicles, high leverage, contractual risk allocation, comprehensive security, and cash-flow controls.
- The capital structure and finance documents allocate risks among sponsors and diverse lenders — commercial banks, ECAs, DFIs, capital markets, and Islamic financiers — and impose environmental and social standards such as the IFC Performance Standards and the Equator Principles.
- Government support agreements, sovereign guarantees, put and call options, and multilateral instruments such as MIGA cover and World Bank guarantees mitigate political and credit risks, but create contingent liabilities that must be managed transparently.
- Petroleum and mining fiscal regimes combine bonuses, royalties, income taxes, resource rent taxes, production sharing, and state participation, and are evaluated by reference to revenue, progressivity, neutrality, risk-sharing, stability, and administrability.
- Royalties provide early and secure revenue but are regressive; resource rent taxes are neutral and progressive but depend on administrative capacity; windfall taxes respond to price shocks but undermine stability.
- Transfer pricing, intra-group financing, indirect transfers, BEPS, and the global minimum tax are central to the taxation of multinational energy enterprises and to the revenue of resource-rich states.
- Revenue transparency through the EITI, mandatory payment disclosure, contract transparency, and beneficial ownership registers, together with anti-bribery laws, address the high corruption risk of the sector, and corruption may render contracts and awards unenforceable.

- Financial distress at the levels of projects, utilities, and sovereigns is managed through consensual restructuring, insolvency procedures, sector reform, and sovereign debt frameworks, with the energy transition adding the challenge of stranded assets.

DOCTORAL RESEARCH QUESTIONS

1. To what extent do government support agreements and multilateral guarantees transfer the risks of energy projects from private investors to the public, and how should the law of public finance constrain the creation of contingent liabilities?
2. Evaluate the relative merits of royalties and resource rent taxes for a developing country with limited tax administration capacity, with reference to comparative experience in Australia, Ghana, and Nigeria.
3. How will the global minimum tax affect the design and legal effect of fiscal stabilisation clauses and tax incentives in extractive agreements in Africa?
4. Has the EITI been effective in reducing corruption and improving the governance of resource revenues? Assess the evidence with reference to the experience of Nigeria under the NEITI Act 2007.
5. Should arbitral tribunals and courts refuse to enforce contracts or awards tainted by corruption where the host state's own officials were complicit? Analyse with reference to *World Duty Free*, *Metal-Tech*, and *P&ID*.
6. What legal mechanisms would enable the orderly early retirement of coal-fired power plants financed under long-term PPAs, and who should bear the associated losses?

FURTHER READING

- E R Yescombe and Edward Farquharson, *Public-Private Partnerships for Infrastructure: Principles of Policy and Finance* (2nd edn, Butterworth-Heinemann 2018).
- Graham Vinter and Gareth Price, *Project Finance: A Legal Guide* (Sweet & Maxwell, latest edn).
- Philip Daniel, Michael Keen and Charles McPherson (eds), *The Taxation of Petroleum and Minerals: Principles, Problems and Practice* (Routledge 2010).
- Daniel Johnston, *International Petroleum Fiscal Systems and Production Sharing Contracts* (PennWell 1994).
- Thomas Baunsgaard, *A Primer on Mineral Taxation* (IMF Working Paper WP/01/139, 2001).
- United Nations, *Handbook on Selected Issues for Taxation of the Extractive Industries by Developing Countries* (2017).
- EITI International Secretariat, *The EITI Standard 2023* (EITI 2023).



PART V

Environmental Law, Climate Governance, and Resource Sustainability

The international and national legal frameworks that govern the environmental consequences of energy and resource development — from the foundational principles of international environmental law, through the climate change regime and the law of shared waters, to the law of the sea and the protection of the marine environment.

17 CHAPTER

International Environmental Law and Natural Resource Protection

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the sources, institutions, and historical development of international environmental law and its relationship with the law of natural resources.
2. Analyse the principles of prevention, precaution, and polluter-pays, and assess their legal status and application to energy and extractive activities.
3. Evaluate the legal requirements for environmental impact assessment under international and national law, including the assessment of downstream greenhouse gas emissions.
4. Apply the rules of international law governing transboundary environmental harm to energy and resource projects.
5. Assess the legal frameworks for the conservation of biodiversity and their interaction with resource extraction, including protected areas and "no-go" commitments.
6. Analyse the international and regional legal regimes governing pollution from offshore petroleum operations.
7. Explain the principles and mechanisms of environmental liability and remediation at the international and national levels, with particular reference to oil pollution in the Niger Delta.
8. Evaluate the mechanisms for securing compliance with, and enforcement of, environmental obligations.

Introduction

Every stage of the energy and natural resource cycle has environmental consequences. Exploration disturbs land, seabed, and wildlife; extraction generates waste rock, tailings, produced water, and emissions; transport by pipeline and tanker risks spills; refining, processing, and combustion pollute air and water and emit greenhouse gases; and decommissioning and mine closure leave legacies that

may persist for generations. Even renewable energy, as examined in Chapter 10, requires land, water, and minerals, and may affect biodiversity, landscapes, and communities. International environmental law, and the national environmental laws through which it is largely implemented, therefore form an indispensable part of the legal framework of energy and natural resources.

The relationship between environmental law and resource law has historically been one of tension. The principle of permanent sovereignty over natural resources, examined in Chapter 2, affirms the right of states to exploit their resources in accordance with their own development policies, while international environmental law imposes limits on that right in the interests of other states, the global commons, future generations, and the environment itself. The formula adopted in Principle 21 of the Stockholm Declaration of 1972 and repeated in Principle 2 of the Rio Declaration of 1992 captures this tension: states have the sovereign right to exploit their own resources pursuant to their own environmental and developmental policies, and the responsibility to ensure that activities within their jurisdiction or control do not cause damage to the environment of other states or of areas beyond the limits of national jurisdiction. The concept of **sustainable development** — development that meets the needs of the present without compromising the ability of future generations to meet their own needs — was developed to reconcile these imperatives, and has been described by the International Court of Justice as expressing the need to reconcile economic development with protection of the environment.

This Part of the book examines the environmental dimension of energy and resource law. The present chapter sets out the general framework of international environmental law as applied to energy and natural resources: its foundations (section 17.1); the principles of prevention, precaution, and polluter-pays (section 17.2); environmental impact assessment (section 17.3); transboundary harm (section 17.4); biodiversity conservation and resource extraction (section 17.5); marine pollution from offshore operations (section 17.6); environmental liability and remediation (section 17.7); and compliance and enforcement (section 17.8). Chapter 18 examines the climate change regime and the energy transition; Chapter 19, the law of freshwater resources and transboundary watercourses; and Chapter 20, the law of the sea, including the protection of the marine environment under the United Nations Convention on the Law of the Sea and the regulation of deep-seabed mining.

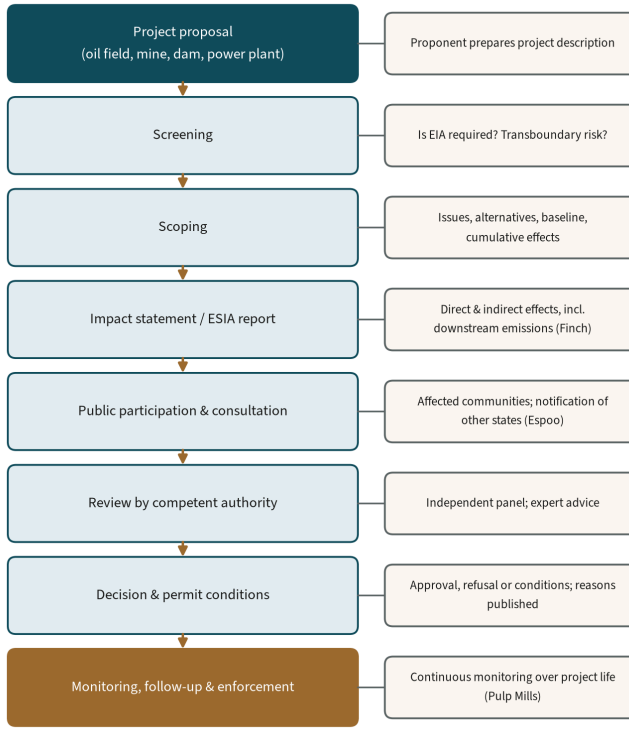


Figure 17.1 The environmental impact assessment process for an energy project. *Source:* prepared for this edition.

17.1 Foundations of International Environmental Law

Origins

International environmental law is a relatively recent branch of public international law. Before the 1970s, environmental questions were addressed in a limited number of treaties concerned principally with the protection of particular species or with the use of shared resources, such as fisheries and boundary waters, and in a handful of arbitral and judicial decisions. Two decisions laid the foundations of the modern law.

In the **Trail Smelter** arbitration between the United States and Canada (awards of 1938 and 1941), concerning damage to crops and forests in the State of Washington caused by sulphur dioxide emissions from a zinc and lead smelter at Trail, British Columbia, the tribunal stated that, under the principles of international law, no state has the right to use or permit the use of its territory in such a manner as to cause injury by fumes in or to the territory of another or the properties or persons therein, when the case is of serious consequence and

the injury is established by clear and convincing evidence. The tribunal awarded compensation and prescribed a regime to control the smelter's future emissions.

In the **Corfu Channel** case (*United Kingdom v Albania*, 1949), the International Court of Justice affirmed, in a different context, that every state is under an obligation not knowingly to allow its territory to be used for acts contrary to the rights of other states — a principle that underpins the modern obligation of prevention.

The Stockholm and Rio Conferences

The **United Nations Conference on the Human Environment**, held in Stockholm in 1972, marked the emergence of international environmental law as a distinct field. The conference adopted the Stockholm Declaration, whose Principle 21 has been examined above, an Action Plan, and the recommendation to establish the **United Nations Environment Programme** (UNEP), which was created by the General Assembly later that year with its headquarters in Nairobi. The following two decades saw the conclusion of numerous multilateral environmental agreements (MEAs), on subjects including marine pollution, trade in endangered species, long-range air pollution, the ozone layer, and hazardous wastes.

The **World Commission on Environment and Development**, chaired by Gro Harlem Brundtland, published its report, *Our Common Future*, in 1987, popularising the concept of sustainable development. The **United Nations Conference on Environment and Development**, held in Rio de Janeiro in 1992, adopted the Rio Declaration on Environment and Development, whose twenty-seven principles articulate the core principles of the field, and Agenda 21, a programme of action, and opened for signature the United Nations Framework Convention on Climate Change and the Convention on Biological Diversity. Subsequent conferences — the World Summit on Sustainable Development in Johannesburg in 2002 and the United Nations Conference on Sustainable Development (Rio+20) in 2012 — reaffirmed and developed these commitments. The **2030 Agenda for Sustainable Development**, adopted by the General Assembly in 2015, set seventeen Sustainable Development Goals, including Goal 7 on affordable and clean energy, Goal 13 on climate action, and Goals 14 and 15 on life below water and on land. Rio+20 led to the strengthening of UNEP and the establishment of the **United Nations Environment Assembly**, with universal membership, which first met in 2014.

Sources

The sources of international environmental law are those of public international law generally, as listed in Article 38(1) of the Statute of the International Court of Justice.

Treaties are the principal source. There are several hundred MEAs, global and regional, which establish specific obligations and, frequently, institutional machinery in the form of conferences of the parties, secretariats, scientific bodies, and compliance mechanisms. Many MEAs are **framework conventions**, which establish general principles and institutions and are supplemented by protocols containing more detailed obligations, as in the cases of the Vienna Convention for the Protection of the Ozone Layer (1985) and its Montreal Protocol (1987), and the Framework Convention on Climate Change and its Kyoto Protocol and Paris Agreement, examined in Chapter 18.

Customary international law includes the obligation to prevent significant transboundary harm, the obligation to conduct environmental impact assessment for activities likely to have significant adverse transboundary effects, and procedural obligations of notification, consultation, and cooperation, as examined in sections 17.2 to 17.4. The customary status of other principles, such as precaution and polluter-pays, remains debated.

General principles of law and **judicial decisions** have played an important role in developing the field, through the decisions of the International Court of Justice, the International Tribunal for the Law of the Sea, arbitral tribunals, and regional human rights courts.

Soft law — declarations, resolutions, guidelines, and codes of conduct that are not formally binding — is particularly significant in international environmental law. Instruments such as the Stockholm and Rio Declarations have shaped the development of treaty and customary law, guided the interpretation of binding obligations, and influenced national legislation.

The Expanding Jurisprudence

The International Court of Justice has developed the law through a series of decisions. In its Advisory Opinion on the **Legality of the Threat or Use of Nuclear Weapons** (1996), the Court recognised that the existence of the general obligation of states to ensure that activities within their jurisdiction and control respect the environment of other states or of areas beyond national control is now part of the corpus of international law relating to the environment. In the **Gabčíkovo-Nagymaros Project** case (*Hungary v Slovakia*, 1997), concerning a treaty for the construction of a system of dams on the Danube, examined further in Chapter 19, the Court referred to the concept of sustainable development and to the need to take account of new environmental norms in the implementation of long-term projects. In **Pulp Mills on the River Uruguay** (*Argentina v Uruguay*, 2010), the Court held that the obligation to conduct an environmental impact assessment where there is a risk of significant transboundary harm is a requirement of general international law, and in **Certain Activities Carried Out by Nicaragua in the**

Border Area and **Construction of a Road in Costa Rica along the San Juan River** (*Costa Rica v Nicaragua*; *Nicaragua v Costa Rica*, 2015), it extended this reasoning to activities beyond industrial projects. In 2018, in the compensation phase of the *Certain Activities* case, the Court for the first time awarded compensation for environmental damage, as examined in section 17.7.

Most significantly, international courts and tribunals have recently addressed the obligations of states in relation to climate change in a series of advisory opinions: the opinion of the International Tribunal for the Law of the Sea on climate change and the marine environment (May 2024), the opinion of the International Court of Justice on the **Obligations of States in respect of Climate Change** (July 2025), and the opinion of the Inter-American Court of Human Rights on the climate emergency and human rights (2025). These opinions, examined in Chapter 18, confirm that the general principles of international environmental law — including the obligation of prevention, the standard of due diligence, and the requirement of environmental impact assessment — apply to greenhouse gas emissions, and have significant implications for energy policy.

Environmental Protection and Human Rights

The relationship between environmental protection and human rights has become increasingly prominent. Regional human rights courts have recognised that environmental degradation may violate rights to life, health, private and family life, and property, and the African Charter on Human and Peoples' Rights expressly recognises, in Article 24, the right of all peoples to a general satisfactory environment favourable to their development — a provision applied by the African Commission in *SERAC v Nigeria* (2001), concerning the environmental and human rights impacts of oil production in Ogoniland, examined in Chapter 21. In 2021, the UN Human Rights Council recognised the human right to a clean, healthy, and sustainable environment, and the General Assembly did so in Resolution 76/300 of July 2022. More than one hundred and fifty states recognise a right to a healthy environment in their constitutions, legislation, or regional treaties. Section 20(1) of the Constitution of the Federal Republic of Nigeria 1999, which provides that the state shall protect and improve the environment and safeguard the water, air, land, forest, and wildlife of Nigeria, is located among the non-justiciable Fundamental Objectives and Directive Principles of State Policy, but Nigerian courts have recognised environmental protection as an aspect of the right to life, as in *Gbemre v Shell Petroleum Development Co of Nigeria Ltd* (2005), in which the Federal High Court held that gas flaring in the Niger Delta violated the applicants' constitutional rights to life and dignity. The human rights dimension is examined further in Part VI.

Institutions

The institutional landscape of international environmental law is fragmented. UNEP and the UN Environment Assembly provide a general forum for environmental policy, but lack the powers of a global environmental organisation. Most MEAs have their own conferences of the parties and secretariats, creating a complex system of treaty bodies. Other organisations with significant environmental roles include the International Maritime Organization, examined in section 17.6; the International Seabed Authority, examined in Chapter 20; the Food and Agriculture Organization; the World Health Organization; the World Bank and regional development banks, through their environmental and social standards; and the Global Environment Facility, established in 1991, which finances environmental projects in developing countries and serves as a financial mechanism for several MEAs. A French initiative for a **Global Pact for the Environment**, which would have codified the principles of international environmental law in a binding treaty, was considered by a UN working group but failed to secure agreement in 2019, reflecting the continuing resistance of many states to comprehensive codification.

17.2 Prevention, Precaution, and the Polluter-Pays Principle

The Role of Principles

International environmental law is characterised by the prominence of principles — general norms that guide the interpretation and development of more specific rules, and that may themselves have binding force. The Rio Declaration articulates many of these principles, including the sovereign right to exploit resources and the responsibility not to cause transboundary harm (Principle 2), the right to development (Principle 3), the integration of environmental protection into development (Principle 4), common but differentiated responsibilities (Principle 7), public participation and access to information and justice (Principle 10), the precautionary approach (Principle 15), the internalisation of environmental costs and the polluter-pays principle (Principle 16), and environmental impact assessment (Principle 17). The legal status of these principles varies: some, such as prevention and environmental impact assessment, are now established rules of customary international law, while others, such as precaution and polluter-pays, are more contested as general obligations, although they are incorporated in numerous treaties and national laws. The principle of common but differentiated responsibilities, which is central to the climate regime, is examined in Chapter 18.

The Principle of Prevention

The **principle of prevention** requires states to take measures to prevent significant harm to the environment, both within and beyond their territories. In its transboundary dimension, derived from *Trail Smelter* and Principle 21 of the

Stockholm Declaration, it is firmly established as customary international law. In *Pulp Mills*, the International Court of Justice described the principle as having its origins in the due diligence that is required of a state in its territory, and held that a state is obliged to use all the means at its disposal to avoid activities which take place in its territory, or in any area under its jurisdiction, causing significant damage to the environment of another state.

The obligation of prevention is an obligation of **due diligence** — of conduct rather than of result. A state does not breach the obligation merely because harm occurs; it breaches the obligation if it fails to take the measures that could reasonably be expected of it to prevent the harm. In *Pulp Mills*, the Court explained that due diligence entails not only the adoption of appropriate rules and measures, but also a certain level of vigilance in their enforcement and the exercise of administrative control applicable to public and private operators, such as the monitoring of activities undertaken by such operators. The standard of due diligence is variable: it depends on the nature of the activity, the degree of risk, the state of scientific knowledge and technology, and the capacity of the state, and it becomes more demanding as the risk increases. In its 2011 Advisory Opinion on the responsibilities of states sponsoring activities in the Area, the Seabed Disputes Chamber of the International Tribunal for the Law of the Sea described due diligence as a variable concept which may change over time as measures considered sufficiently diligent at a certain moment become not diligent enough in light of new scientific or technological knowledge. In its 2024 Advisory Opinion on climate change, the Tribunal held that the due diligence required of states in relation to greenhouse gas emissions is "stringent", given the high risks of serious and irreversible harm.

The **International Law Commission's Draft Articles on Prevention of Transboundary Harm from Hazardous Activities** (2001) codify the obligation of prevention and its procedural corollaries: the obligation to take all appropriate measures to prevent significant transboundary harm or to minimise the risk thereof; the requirement of prior authorisation of hazardous activities; the assessment of risk; notification and information of potentially affected states; consultation on preventive measures; and the exchange of information. These obligations apply directly to energy and extractive activities with transboundary risks, such as nuclear power plants near borders, oil and gas installations in shared seas, mines and tailings facilities on shared rivers, and hydroelectric dams.

The Precautionary Principle

The **precautionary principle**, or precautionary approach, addresses the problem of scientific uncertainty. Principle 15 of the Rio Declaration provides that, in order to protect the environment, the precautionary approach shall be widely applied by states according to their capabilities, and that where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as

a reason for postponing cost-effective measures to prevent environmental degradation. The principle thus permits — and in stronger formulations requires — action to prevent harm before the harm is scientifically established.

The precautionary approach is incorporated in numerous treaties, including the UNFCCC (Article 3(3)), the Convention on Biological Diversity (preamble), the Cartagena Protocol on Biosafety (2000), the UN Fish Stocks Agreement (1995), and regional agreements such as the OSPAR Convention, examined in section 17.6. In the European Union, Article 191(2) of the Treaty on the Functioning of the European Union provides that Union policy on the environment shall be based on the precautionary principle, and the Court of Justice has applied the principle extensively, holding that where there is uncertainty as to the existence or extent of risks to human health or the environment, protective measures may be taken without having to wait until the reality and seriousness of those risks become fully apparent.

The status of the principle in general international law remains debated. The International Tribunal for the Law of the Sea, in the *Southern Bluefin Tuna* provisional measures order (1999), held that the parties should act with "prudence and caution", and in its 2011 Advisory Opinion, the Seabed Disputes Chamber observed that the precautionary approach had been incorporated into a growing number of international instruments, initiating a trend towards making it part of customary international law, and held that it is an integral part of the general obligation of due diligence. The International Court of Justice, in *Pulp Mills*, accepted that a precautionary approach may be relevant in the interpretation and application of a treaty, but held that it does not operate as a reversal of the burden of proof. Within the World Trade Organization, the Appellate Body in *EC – Hormones* (1998) declined to decide whether the principle had become customary law and held that it did not override the specific provisions of the Agreement on Sanitary and Phytosanitary Measures, although the SPS Agreement itself reflects precautionary elements.

In the energy sector, the precautionary principle has been invoked in support of restrictions on hydraulic fracturing (fracking), deep-sea mining, and the siting of nuclear facilities. France prohibited hydraulic fracturing for the exploration and production of hydrocarbons by a law of 2011, whose constitutionality was upheld by the Conseil constitutionnel in 2013; several other European states and sub-national governments adopted moratoria or bans; and the United Kingdom imposed a moratorium on fracking in England in 2019, following seismic events at a site in Lancashire, which was briefly lifted and then reinstated in 2022. The calls by numerous states for a precautionary pause on deep-sea mining, examined in Chapter 20, are a further application of the principle.

The Polluter-Pays Principle

The **polluter-pays principle** requires that the costs of pollution prevention, control, and remediation be borne by the polluter rather than by society at large. It was first articulated as an economic principle by the OECD in its 1972 Recommendation on Guiding Principles concerning International Economic Aspects of Environmental Policies, which aimed to prevent governments from subsidising pollution control in ways that would distort international trade. Principle 16 of the Rio Declaration provides that national authorities should endeavour to promote the internalisation of environmental costs and the use of economic instruments, taking into account the approach that the polluter should, in principle, bear the cost of pollution, with due regard to the public interest and without distorting international trade and investment.

The polluter-pays principle is applied through several legal mechanisms: environmental taxes and charges, such as carbon taxes and effluent charges; emissions trading schemes, examined in Chapter 18; liability regimes that require polluters to pay for remediation and compensation, examined in section 17.7; financial security requirements, such as decommissioning and mine closure bonds; and the removal of subsidies for polluting activities, including fossil fuel subsidies. In the European Union, the principle is enshrined in Article 191(2) TFEU and implemented through the Environmental Liability Directive (Directive 2004/35/EC). In *Raffinerie Mediterranee (ERG) SpA v Ministero dello Sviluppo economico* (Case C-378/08, 2010), the Court of Justice held that, under the Directive, the competent authority may presume a causal link between operators and pollution on the basis of the proximity of their installations to the pollution and the correspondence between the pollutants found and the substances used by the operators, provided it has plausible evidence.

In the energy sector, the principle underlies obligations to decommission offshore installations and to rehabilitate mine sites at the operator's expense, examined in Chapters 5 and 7, and the debate over whether fossil fuel producers should contribute to the costs of climate change, examined in Chapter 18. Its application in developing countries is complicated by weak enforcement, the insolvency or departure of responsible operators, and the existence of historical or "orphan" pollution for which no solvent polluter can be identified.

Other Principles

Several other principles are relevant to energy and resource activities. The principle of **intergenerational equity** — that the present generation holds the natural environment in trust for future generations — has been invoked in national courts, notably by the Supreme Court of the Philippines in *Minors Oposa v Secretary of the Department of Environment and Natural Resources* (1993), in which

the Court recognised the standing of children to sue on behalf of themselves and future generations to challenge timber licences. The principle of **integration**, reflected in Principle 4 of the Rio Declaration, requires environmental protection to constitute an integral part of the development process. The principle of **public participation**, reflected in Principle 10, has been given binding form in the UNECE **Aarhus Convention** on Access to Information, Public Participation in Decision-making and Access to Justice in Environmental Matters (1998) and the Latin American and Caribbean **Escazú Agreement** (2018), which entered into force in 2021 and is the first treaty to include specific provisions on the protection of environmental human rights defenders. The principle of **non-regression**, which holds that environmental protection should not be weakened once achieved, has been recognised in some national and regional legal systems.

17.3 Environmental Impact Assessment

The Function of Environmental Impact Assessment

Environmental impact assessment (EIA) is a procedure for identifying, predicting, and evaluating the likely environmental effects of a proposed activity before a decision is taken on whether, and on what conditions, to authorise it. EIA originated in the United States with the **National Environmental Policy Act of 1969** (NEPA), which required federal agencies to prepare an environmental impact statement for major federal actions significantly affecting the quality of the human environment, and has since been adopted in virtually every legal system. EIA is the principal legal tool through which environmental considerations are integrated into decisions on energy and extractive projects, and it frequently becomes the focus of litigation by communities and environmental organisations seeking to challenge projects.

The EIA process typically comprises the following stages, illustrated in Figure 17.1: **screening**, to determine whether an activity requires EIA; **scoping**, to identify the issues and alternatives to be addressed; the preparation of an **environmental impact statement** or report by the proponent, describing the project, the baseline environment, the likely effects, alternatives, and mitigation measures; **public participation** and consultation with affected communities and authorities; **review** of the statement by the competent authority, sometimes with the assistance of an independent panel; the **decision** whether to authorise the project, with conditions; and **monitoring** and follow-up of the project's actual effects. Many jurisdictions now require **environmental and social impact assessment** (ESIA), which integrates the assessment of social effects, including effects on livelihoods, health, cultural heritage, and human rights, and international lenders require ESIA in accordance with IFC Performance Standard 1 for the projects they finance, as examined in Chapter 16.

EIA in International Law

At the international level, Principle 17 of the Rio Declaration provides that EIA, as a national instrument, shall be undertaken for proposed activities that are likely to have a significant adverse impact on the environment and are subject to a decision of a competent national authority. The **Espoo Convention on Environmental Impact Assessment in a Transboundary Context** (1991), adopted under the auspices of the UN Economic Commission for Europe, requires parties to assess the environmental impact of listed activities — including crude oil refineries, large thermal power stations, nuclear installations, large dams, oil and gas pipelines, and major mining — likely to cause significant adverse transboundary impact, to notify and consult potentially affected states, and to allow the public of affected states to participate. Its **Protocol on Strategic Environmental Assessment** (Kyiv, 2003) extends assessment to plans and programmes, including energy plans. The Espoo Convention's Implementation Committee has addressed numerous cases concerning energy projects, including nuclear power plants and the lifetime extension of nuclear reactors, which it has held may require transboundary assessment.

In *Pulp Mills*, the International Court of Justice held that it may now be considered a requirement under general international law to undertake an environmental impact assessment where there is a risk that the proposed industrial activity may have a significant adverse impact in a transboundary context, in particular on a shared resource. The Court held that the content of the assessment is to be determined by each state in its domestic legislation or in the authorisation process, having regard to the nature and magnitude of the proposed development and its likely adverse impact on the environment, as well as to the need to exercise due diligence; that the assessment must be conducted prior to the implementation of the project; and that, once operations have started, continuous monitoring of the project's effects on the environment shall be undertaken throughout its life. In *Certain Activities* and *Construction of a Road* (2015), the Court clarified that the obligation applies generally to proposed activities that may have a significant adverse impact in a transboundary context, and that a state must first ascertain, through a preliminary assessment, whether there is such a risk; if so, it must conduct a full EIA and, where the EIA confirms the risk, notify and consult the potentially affected state in good faith. The African Convention on the Conservation of Nature and Natural Resources, as revised at Maputo in 2003, also requires parties to ensure that activities likely to affect natural resources are subject to adequate impact assessment.

EIA in National Law

In the **European Union**, the EIA Directive (Directive 2011/92/EU, as amended by Directive 2014/52/EU) requires assessment of projects listed in Annex I —

including crude oil refineries, thermal power stations of 300 MW or more, nuclear power stations, the extraction of petroleum and natural gas above specified thresholds, large pipelines, and large dams — and screening of projects listed in Annex II. The 2014 amendments required the assessment of effects on climate, including greenhouse gas emissions and vulnerability to climate change, and on biodiversity. The Strategic Environmental Assessment Directive (Directive 2001/42/EC) applies to plans and programmes, including energy plans.

In **Nigeria**, the **Environmental Impact Assessment Act** of 1992 requires the assessment of the environmental effects of public or private projects before decisions are taken, and lists mandatory study activities, including petroleum exploration and production, refineries, pipelines, power stations, and mining. The Act is administered by the Federal Ministry of Environment, and provides that no project to which it applies shall be undertaken without the prior consideration of its environmental effects. The Court of Appeal's decision in *Oronto Douglas v Shell Petroleum Development Co Ltd* (1999), recognising the standing of an environmental activist to challenge a failure to comply with the Act in relation to an LNG project, was an early landmark in the enforcement of EIA requirements in Nigeria, although enforcement of the Act has remained uneven, and the relationship between federal EIA requirements and the environmental provisions of the Petroleum Industry Act 2021, including the environmental management plans required by the Act, raises questions of coordination.

In the **United States**, the scope of NEPA review has been narrowed. In *Seven County Infrastructure Coalition v Eagle County* (2025), concerning a railway line intended to transport crude oil from Utah's Uinta Basin, the Supreme Court held that courts should afford substantial deference to agencies in determining the scope of NEPA review, and that an agency is not required to assess the environmental effects of separate upstream or downstream projects that it does not regulate, such as oil production and refining, merely because they are foreseeable consequences of the project under review. The decision, together with the rescission of the regulations of the Council on Environmental Quality implementing NEPA in 2025, has reduced the role of NEPA in the review of fossil fuel infrastructure.

Assessing Downstream Emissions

The question whether the EIA of a fossil fuel extraction project must assess the greenhouse gas emissions that will result from the combustion of the extracted product — so-called "downstream" or "scope 3" emissions — has become one of the most significant issues in energy law. The United Kingdom Supreme Court answered it in the affirmative.

::: casebox {title="R (Finch) v Surrey County Council [2024] UKSC 20"}
Facts. Surrey County Council granted planning permission for the expansion of an onshore oil extraction site at Horse Hill, near Gatwick Airport, permitting the drilling of four new wells and the production of oil over a period of twenty-five years. The environmental statement assessed the greenhouse gas emissions from the construction and operation of the site, but not the emissions from the combustion of the oil once refined and used, on the ground that those emissions were not effects of the project. Sarah Finch, on behalf of a local campaign group, sought judicial review of the permission.

The issue. The question was whether, under the EIA Regulations implementing the EIA Directive, which require the assessment of the "direct and indirect significant effects" of a project, the combustion emissions were effects of the project that had to be assessed.

The decision. By a majority of three to two, the Supreme Court held that the permission was unlawful. Lord Leggatt, giving the leading judgment, held that it was inevitable that the oil extracted would be refined and burned, and that the resulting emissions were therefore effects of the project. The question whether an effect is an effect of the project is a question of causation, and the combustion emissions were clearly caused by the extraction, since the oil would not otherwise be produced; the fact that the oil would be refined elsewhere did not break the chain of causation. The council's conclusion that the combustion emissions did not need to be assessed was therefore legally flawed. The Court emphasised that the purpose of EIA is to ensure that decisions are taken with full knowledge of their environmental consequences, and that the assessment of combustion emissions did not determine whether the project should be approved, which remained a matter for the decision-maker. The dissenting justices considered that the extent of the project's effects was a matter of evaluative judgment for the planning authority.

Aftermath. The decision has had major consequences for oil and gas development in the United Kingdom. In January 2025, the Court of Session in Scotland held that the consents granted for the Rosebank and Jackdaw offshore fields were unlawful because the assessments had not considered combustion emissions, while permitting preparatory work to continue pending fresh decisions, and the government subsequently issued guidance on the assessment of such emissions. Courts in other jurisdictions have reached similar conclusions; in Norway, the Oslo District Court held in 2024 that approvals of several offshore oil field developments were invalid for failure to assess combustion emissions, and the EFTA Court subsequently confirmed that such emissions fall within the scope of the EIA Directive as applied in the European Economic Area. The contrast with

the narrowing of NEPA review in the United States illustrates divergent judicial approaches to the role of EIA in addressing climate change. :::

Strategic Environmental Assessment

Strategic environmental assessment (SEA) applies assessment to policies, plans, and programmes, such as national energy plans, petroleum licensing rounds, and renewable energy zoning, before individual projects are proposed. SEA allows the cumulative effects of multiple projects, alternatives at the level of policy, and the consistency of plans with climate and biodiversity objectives to be considered at a stage when the range of options is still open. Several resource-rich developing countries, including Uganda for the Albertine Graben petroleum region, have undertaken SEAs of their extractive sectors, and SEA is increasingly used in the planning of offshore wind zones, examined in Chapter 20.

Effectiveness and Limitations

EIA is an essentially procedural tool: it requires that environmental effects be identified and considered, but it does not in itself require that projects with adverse effects be refused. Its effectiveness depends on the quality and independence of the assessment, which is usually prepared by consultants engaged by the proponent; the genuineness of public participation; the capacity of the reviewing authority; and the enforcement of conditions and monitoring. Common weaknesses in practice include inadequate baseline data, the failure to consider alternatives or cumulative effects, consultation that is late or superficial, and the absence of follow-up. Courts have played an important role in enforcing EIA requirements, and the developments in the assessment of downstream emissions illustrate how judicial interpretation can transform the significance of a procedural requirement.

17.4 Transboundary Environmental Harm

The Nature of the Problem

Many energy and resource activities have the potential to cause harm beyond the borders of the state in which they take place. Air pollution from power plants and smelters may travel hundreds or thousands of kilometres; oil spills from offshore installations and pipelines may drift into neighbouring waters; tailings dam failures and industrial accidents may pollute shared rivers; nuclear accidents may contaminate entire regions; and dams may alter the flow of rivers downstream. The law of transboundary harm addresses the rights and obligations of states in relation to such harm, and is a central element of the international law governing energy and natural resources. The specific regime governing shared watercourses and hydropower is examined in Chapter 19, and the marine environment in section 17.6 and Chapter 20.

The Substantive Obligation

The customary obligation of prevention, examined in section 17.2, requires states to exercise due diligence to prevent activities within their jurisdiction or control from causing **significant** harm to the environment of other states or of areas beyond national jurisdiction. Several elements of the obligation require explanation.

Significant harm. The obligation does not prohibit all transboundary effects, which would be impossible in an interdependent world, but only harm that is "significant" — a threshold described by the International Law Commission as something more than "detectable" but not necessarily "serious" or "substantial", which must lead to a real detrimental effect on matters such as human health, industry, property, the environment, or agriculture in other states, measurable by factual and objective standards.

Jurisdiction or control. The obligation applies to activities within the territory of a state and to activities under its jurisdiction or control elsewhere, such as those of vessels flagged in the state or of installations on its continental shelf. The extent to which home states are responsible for the extraterritorial activities of their corporations is debated, and is examined in Chapters 21 and 22.

Due diligence. As examined in section 17.2, the obligation is one of conduct, requiring the state to take appropriate legislative, administrative, and enforcement measures, proportionate to the risk.

Areas beyond national jurisdiction. The obligation extends to the global commons — the high seas, the deep seabed, Antarctica, outer space, and the atmosphere — whose protection is of concern to the international community as a whole. The obligations of states in respect of the climate system, as clarified by the 2025 advisory opinion of the International Court of Justice, are examined in Chapter 18.

Procedural Obligations

The substantive obligation is accompanied by procedural obligations, which the International Court of Justice has treated as distinct from, though closely connected with, the obligation of prevention. They include the obligation to conduct an EIA where there is a risk of significant transboundary harm, examined in section 17.3; the obligation to **notify** potentially affected states of planned activities that may cause significant harm, providing them with available technical and other information; the obligation to **consult** with those states in good faith, with a view to achieving acceptable solutions regarding the measures to be adopted; and the obligation to notify affected states without delay of **emergencies** that may cause harm to them, and to cooperate in responding to them. In the **Lac Lanoux** arbitration (*Spain v France*, 1957), concerning a French hydroelectric project that

diverted waters of a river flowing into Spain, the tribunal held that France was obliged to consult Spain in good faith and to take Spain's interests into account, but that it was not required to obtain Spain's prior consent, which would amount to a right of veto. In *Pulp Mills*, the Court held that Uruguay had breached its procedural obligations under the bilateral treaty governing the River Uruguay by authorising the construction of pulp mills without notifying the joint administrative commission established under the treaty, but that it had not breached its substantive obligations, since Argentina had not proved that the mills had caused significant pollution.

Industrial Accidents and Emergencies

Several major industrial accidents have shaped the law of transboundary harm. The **Chernobyl** nuclear accident of 1986, which spread radioactive contamination across Europe, led to the rapid adoption, under the auspices of the International Atomic Energy Agency, of the Convention on Early Notification of a Nuclear Accident and the Convention on Assistance in the Case of a Nuclear Accident or Radiological Emergency, both of 1986, examined in Chapter 11; notably, no state brought a claim against the Soviet Union for the damage caused. The **Sandoz** chemical warehouse fire near Basel in the same year, which released toxic chemicals into the Rhine and killed fish along hundreds of kilometres of the river, contributed to the development of cooperation in the Rhine basin and, together with the Seveso disaster of 1976, to the adoption of the UNECE **Convention on the Transboundary Effects of Industrial Accidents** (Helsinki, 1992), which requires parties to identify hazardous activities, to take preventive and preparedness measures, and to notify and assist each other in the event of an accident. The **Baia Mare** cyanide spill of 2000, in which a tailings dam at a gold mine in Romania failed, releasing cyanide-contaminated water that flowed into Hungary and Serbia via the Tisza and Danube rivers, illustrated the transboundary risks of mining and led to the strengthening of European Union law on mining waste, through the Mining Waste Directive (Directive 2006/21/EC).

Long-Range Air Pollution

The UNECE **Convention on Long-Range Transboundary Air Pollution** (1979), adopted in response to evidence that acid rain in Scandinavia was caused by emissions of sulphur dioxide from power plants and industry in other European states, was the first multilateral treaty to address air pollution on a regional basis. Its eight protocols, including the Gothenburg Protocol (1999, amended 2012), set emission ceilings for sulphur, nitrogen oxides, volatile organic compounds, ammonia, and fine particulate matter, and have contributed to substantial reductions in emissions from the energy sector in Europe and North America. The convention is frequently cited as an example of effective

science-based cooperation. In South-East Asia, the **ASEAN Agreement on Transboundary Haze Pollution** (2002) addresses the smoke haze caused by land and forest fires, associated with the clearing of land for plantations, including on peatlands.

Hazardous Wastes

The transboundary movement of hazardous wastes, including wastes generated by the petroleum and mining industries, is regulated by the **Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal** (1989), which requires the prior informed consent of importing and transit states and the environmentally sound management of wastes, and whose Ban Amendment, prohibiting exports of hazardous wastes from OECD to non-OECD states, entered into force in 2019. African states, concerned that the Basel Convention did not go far enough, adopted the **Bamako Convention** on the Ban of the Import into Africa and the Control of Transboundary Movement and Management of Hazardous Wastes within Africa (1991). The dumping in Abidjan, Côte d'Ivoire, in 2006 of toxic waste generated on board the vessel *Probo Koala*, chartered by the commodity trader Trafigura, from the processing of petroleum products, which caused deaths and widespread illness, illustrated the risks associated with the waste streams of the petroleum industry and led to criminal proceedings in the Netherlands and civil litigation in England, which was settled. The waste generated by the energy transition, including end-of-life solar panels, wind turbine blades, and batteries, presents new challenges, examined in Chapters 8 and 10.

State Responsibility and Remedies

A state that breaches its obligations in relation to transboundary harm incurs **state responsibility** under the general rules of international law, codified in the International Law Commission's **Articles on Responsibility of States for Internationally Wrongful Acts** (2001), and is obliged to cease the wrongful act and to make full reparation for the injury caused, in the form of restitution, compensation, or satisfaction. The difficulties of proving causation and damage, and the reluctance of states to bring claims against each other, have meant that interstate claims for transboundary environmental harm have been rare, and that the law has developed principally through civil liability regimes and the procedural obligations examined above.

Where harm results from lawful activities conducted with due diligence, state responsibility does not arise. To address the resulting gap, the International Law Commission adopted **Principles on the Allocation of Loss in the Case of Transboundary Harm Arising out of Hazardous Activities** (2006), which provide that states should ensure that prompt and adequate compensation is

available for victims of transboundary damage caused by hazardous activities, primarily through the imposition of liability on the operator, supported by insurance and, where necessary, by additional funding from industry or state sources. These principles are reflected in the civil liability conventions for oil pollution and nuclear damage examined in section 17.7 and Chapter 11.

17.5 Biodiversity Conservation and Resource Extraction

Biodiversity Loss and the Energy Sector

Biodiversity — the variability among living organisms and the ecosystems of which they are part — is declining at an unprecedented rate. The Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES), in its Global Assessment of 2019, estimated that around one million animal and plant species are threatened with extinction, and identified changes in land and sea use, direct exploitation of organisms, climate change, pollution, and invasive species as the principal drivers. Energy and extractive activities contribute to each of these drivers: mining, oil and gas development, and the associated roads and infrastructure fragment and destroy habitats; hydropower dams transform river ecosystems; oil spills and mining waste pollute land and water; and the combustion of fossil fuels drives climate change. At the same time, the minerals required for the energy transition are frequently located in biodiversity-rich areas, including tropical forests and the deep seabed, creating new tensions between climate and biodiversity objectives.

The Convention on Biological Diversity

The **Convention on Biological Diversity** (CBD), opened for signature at Rio in 1992 and ratified by almost all states (with the notable exception of the United States), has three objectives: the conservation of biological diversity, the sustainable use of its components, and the fair and equitable sharing of the benefits arising from the utilisation of genetic resources. It reaffirms the sovereign rights of states over their biological resources, while recognising that the conservation of biodiversity is a common concern of humankind. Parties are required to develop national biodiversity strategies and action plans, to integrate biodiversity into sectoral policies, to establish systems of protected areas, to rehabilitate degraded ecosystems, and, under Article 14, to introduce appropriate procedures requiring EIA of proposed projects likely to have significant adverse effects on biodiversity. The **Nagoya Protocol** (2010) addresses access to genetic resources and benefit-sharing, and the **Cartagena Protocol** (2000) addresses biosafety.

The CBD's obligations are framed in general and qualified terms — many are to be implemented "as far as possible and as appropriate" — and implementation has been weak; none of the twenty Aichi Biodiversity Targets adopted for the period 2011–2020 was fully achieved. In December 2022, the fifteenth Con-

ference of the Parties adopted the **Kunming-Montreal Global Biodiversity Framework**, which sets twenty-three targets for 2030, including the conservation of at least thirty per cent of terrestrial, inland water, and coastal and marine areas through protected areas and other effective area-based conservation measures (the "30 by 30" target); the restoration of thirty per cent of degraded ecosystems; the reduction of harmful subsidies; and, under Target 15, measures to encourage and enable large companies and financial institutions to monitor, assess, and disclose their risks, dependencies, and impacts on biodiversity. The Framework is significant for the extractive sector because the expansion of protected areas may restrict access to resources, and because the disclosure requirements are reflected in initiatives such as the **Taskforce on Nature-related Financial Disclosures** (TNFD), whose recommendations were published in 2023.

Protected Areas, World Heritage, and "No-Go" Areas

The relationship between protected areas and resource extraction is a recurrent source of conflict. Many national laws prohibit or restrict mining and petroleum activities in national parks and other protected areas, but governments may de-gazette or reduce protected areas to permit extraction, and the boundaries between protection and development are frequently contested.

The **Convention concerning the Protection of the World Cultural and Natural Heritage** (World Heritage Convention, 1972) requires parties to protect and conserve sites of outstanding universal value inscribed on the World Heritage List, and the World Heritage Committee may inscribe sites on the List of World Heritage in Danger where they are threatened, including by extractive activities. The Committee has consistently stated that mineral, oil, and gas exploration and exploitation are incompatible with World Heritage status. In 2003, the International Council on Mining and Metals (ICMM) committed its members not to explore or mine in World Heritage sites — the first "no-go" commitment by an industry body — and several oil and gas companies have made similar commitments. The commitment was tested in **Virunga National Park** in the Democratic Republic of Congo, Africa's oldest national park and a World Heritage site, where the grant of oil exploration rights to SOCO International provoked an international campaign, leading the company in 2014 to agree not to undertake drilling in the park unless the government and UNESCO agreed that such activities were compatible with its World Heritage status. In 2022, the government of the DRC included blocks overlapping Virunga and other sensitive areas in a petroleum licensing round, renewing the controversy. In Tanzania, the construction of a large hydroelectric dam at Stiegler's Gorge within the **Selous Game Reserve**, a World Heritage site, proceeded despite the objections of the World Heritage Committee, illustrating the limits of the Convention in the face of national development priorities.

Wetlands and Migratory Species

The **Ramsar Convention on Wetlands of International Importance** (1971) requires parties to designate wetlands for inclusion in a List of Wetlands of International Importance and to promote their conservation and "wise use". Wetlands such as the Niger Delta, the Sudd in South Sudan, and the Okavango Delta in Botswana are threatened by oil development, drainage, and changes in water flows. The **Convention on the Conservation of Migratory Species of Wild Animals** (CMS, 1979) protects migratory species, which may be affected by wind farms, power lines, and offshore energy installations, and its parties have adopted guidance on the deployment of renewable energy technologies in ways compatible with the conservation of migratory species, examined in Chapter 10.

Biodiversity Standards in Project Finance

The most demanding biodiversity standards applicable to extractive projects are frequently those imposed by lenders. **IFC Performance Standard 6** on Biodiversity Conservation and Sustainable Management of Living Natural Resources requires projects to apply the mitigation hierarchy — avoidance, minimisation, restoration, and, as a last resort, offsetting — and prohibits activities in **critical habitats** unless, among other conditions, the project does not lead to measurable adverse impacts on the biodiversity values for which the habitat was designated and achieves **net gains** in those values. **Biodiversity offsets**, which compensate for residual impacts by conserving or restoring biodiversity elsewhere, are increasingly required, but are controversial because of the difficulty of establishing equivalence between lost and gained biodiversity and the risk that offsets legitimise destructive projects.

Contested Extraction in Biodiverse Areas

Several cases illustrate the political and legal dynamics of extraction in biodiverse areas. In **Ecuador**, the Yasuní-ITT initiative, launched in 2007, proposed that Ecuador would leave the oil reserves beneath the Yasuní National Park — one of the most biodiverse areas on earth and the home of indigenous peoples living in voluntary isolation — undeveloped, in return for international contributions of half the value of the forgone revenue. The initiative was abandoned in 2013 after failing to attract sufficient contributions, and drilling commenced. In August 2023, following a decision of the Constitutional Court permitting a popular consultation, Ecuadorian voters approved by a substantial majority a proposal to halt oil production in Block 43 (ITT) and leave the oil underground; the implementation of the result, which required the decommissioning of existing production, proceeded slowly and was contested. In **Uganda and Tanzania**, the development of oil fields in and near Murchison Falls National Park and the construction of the **East African Crude Oil Pipeline** (EACOP), which

crosses important ecosystems, have been opposed by environmental and human rights organisations and have been the subject of litigation, including in the East African Court of Justice and the French courts under the duty of vigilance law, examined in Chapter 22. In **Botswana and Namibia**, petroleum exploration in the Kavango basin, upstream of the Okavango Delta World Heritage site, raised concerns about effects on water resources and wildlife.

These cases illustrate the tension between the sovereign right of states to develop their resources, which is often exercised in the name of poverty reduction and development, and the international interest in conserving biodiversity of global significance. They also raise the question of who should bear the cost of conservation: if the international community values the conservation of biodiversity in developing countries, there is a strong argument that it should contribute to the opportunity costs of forgoing development, through mechanisms such as payments for ecosystem services, debt-for-nature swaps, and biodiversity finance, as recognised in Target 19 of the Kunming-Montreal Framework.

17.6 Marine Pollution and Offshore Operations

Scope of this Section

The protection of the marine environment is governed by Part XII of the United Nations Convention on the Law of the Sea (UNCLOS), which establishes the general obligation of states to protect and preserve the marine environment and specific obligations in respect of different sources of pollution. The general framework of Part XII, the new Agreement on Marine Biological Diversity of Areas beyond National Jurisdiction, and the environmental regulation of deep-seabed mining are examined in Chapter 20. This section focuses on a narrower question of particular importance to energy law: the international, regional, and national rules governing pollution arising from offshore petroleum operations — exploration, production, and the associated transport of hydrocarbons — and the lessons drawn from major offshore incidents.

The Absence of a Global Offshore Regime

Article 208 of UNCLOS requires coastal states to adopt laws and regulations to prevent, reduce, and control pollution of the marine environment arising from or in connection with seabed activities subject to their jurisdiction and from artificial islands, installations, and structures under their jurisdiction, which shall be no less effective than international rules, standards, and recommended practices, and requires states to endeavour to harmonise their policies at the appropriate regional level. Unlike shipping, however, offshore petroleum activities are not governed by a comprehensive global convention. The International Maritime Organization's conventions apply principally to ships, and their application to offshore installations is partial.

The **International Convention for the Prevention of Pollution from Ships** (MARPOL 73/78) applies to fixed and floating platforms in certain respects, including their discharges of machinery-space oily water and garbage, but does not regulate discharges arising directly from the exploration, exploitation, and associated offshore processing of seabed mineral resources. The **International Convention on Oil Pollution Preparedness, Response and Co-operation** (OPRC, 1990) requires operators of offshore units to have oil pollution emergency plans, coordinated with national systems, and to report oil pollution incidents without delay, and requires parties to establish national systems for responding to oil pollution incidents and to cooperate in response. The **London Convention** (1972) and its **Protocol** (1996) regulate the dumping of wastes at sea, including the disposal of offshore platforms, and the Protocol was amended in 2006 to permit the storage of carbon dioxide in sub-seabed geological formations, as examined in Chapter 12.

The **civil liability conventions** for oil pollution — the International Convention on Civil Liability for Oil Pollution Damage (CLC, 1992), the International Oil Pollution Compensation Fund Convention (1992) and its Supplementary Fund Protocol (2003), and the Bunkers Convention (2001) — apply to pollution from ships, including tankers carrying oil in bulk, but not, in general, to pollution from fixed offshore installations, although their application to floating production, storage, and offloading units (FPSOs) depends on whether such units are "ships" within the meaning of the conventions, a question that has been debated. Following the Montara blowout in 2009, Indonesia proposed that the IMO develop a convention on liability and compensation for transboundary pollution damage from offshore oil exploration and exploitation, but the IMO Legal Committee concluded that there was no compelling need for a global convention and instead developed guidance to assist states wishing to conclude bilateral or regional agreements.

Regional Regimes

In the absence of a global regime, the most developed rules governing offshore pollution are found in regional seas agreements.

The North-East Atlantic. The **OSPAR Convention** for the Protection of the Marine Environment of the North-East Atlantic (1992) contains, in Annex III, specific provisions on the prevention and elimination of pollution from offshore sources, and the OSPAR Commission has adopted binding decisions and recommendations on matters including the discharge of oil in produced water, the use and discharge of offshore chemicals, and drill cuttings. **OSPAR Decision 98/3** on the Disposal of Disused Offshore Installations, adopted following the controversy over Shell's proposal in 1995 to dispose of the Brent Spar oil storage buoy in the deep Atlantic, prohibits the dumping and leaving wholly or partly in

place of disused offshore installations, subject to derogations for certain large steel structures and concrete installations, and is a key instrument in decommissioning law, as examined in Chapter 5.

The Mediterranean. The **Offshore Protocol** to the Barcelona Convention, adopted in 1994 and in force since 2011, regulates exploration and exploitation of the continental shelf and seabed, including authorisation, the use and discharge of harmful substances, safety measures, contingency planning, and the removal of installations.

The Gulf. The Kuwait Regional Convention for Cooperation on the Protection of the Marine Environment from Pollution (1978) is supplemented by a Protocol concerning Marine Pollution resulting from Exploration and Exploitation of the Continental Shelf (1989).

West and Central Africa. The **Abidjan Convention** for Cooperation in the Protection, Management and Development of the Marine and Coastal Environment of the Atlantic Coast of the West, Central and Southern Africa Region (1981), to which Nigeria is a party, has been supplemented by an Additional Protocol concerning Environmental Norms and Standards for Offshore Oil and Gas Exploration and Exploitation Activities, adopted at Malabo in 2019, which addresses EIA, the management of discharges and wastes, emergency preparedness, decommissioning, and liability, and represents an important development for a region in which offshore production is expanding rapidly, including in the Gulf of Guinea and off Senegal, Mauritania, and Namibia.

Lessons from Major Offshore Incidents

Deepwater Horizon (2010). The blowout of the Macondo well in the Gulf of Mexico, examined in Chapter 15 in relation to contractual indemnities, killed eleven workers and released several million barrels of oil over eighty-seven days. The National Commission on the BP Deepwater Horizon Oil Spill and Offshore Drilling found that the disaster resulted from systemic failures of risk management by the companies involved and of regulatory oversight by the federal regulator, the Minerals Management Service, whose functions as a revenue collector, a promoter of leasing, and a safety regulator were in conflict. The United States responded by dividing those functions among separate agencies — the Bureau of Ocean Energy Management, the Bureau of Safety and Environmental Enforcement, and the Office of Natural Resources Revenue — and by adopting new rules on well control and safety management systems. BP pleaded guilty to criminal charges and, in 2015, agreed to a settlement with the federal government and the Gulf states of approximately USD 20.8 billion, including civil penalties under the Clean Water Act and natural resource damages under the Oil Pollution Act of 1990.

Montara (2009). The blowout of a well on the Montara platform in the Timor Sea, off the north-west coast of Australia, released oil for more than ten weeks, some of which reached waters off Indonesia. A Commission of Inquiry found serious deficiencies in the operator's well integrity practices and in regulatory oversight, leading to the establishment of a national offshore petroleum safety and environmental regulator (NOPSEMA). In 2021, in a class action brought by Indonesian seaweed farmers, the Federal Court of Australia held in *Sanda v PTTEP Australasia (Ashmore Cartier) Pty Ltd* that the operator owed a duty of care to the farmers and had breached it, causing damage to their seaweed crops, and the case was subsequently settled. The case is significant as an example of transboundary civil liability for offshore pollution pursued in the courts of the operator's state.

The European Union response. In the light of Deepwater Horizon, the European Union adopted the **Offshore Safety Directive** (Directive 2013/30/EU), which requires operators to prepare reports on major hazards and internal emergency response plans, to demonstrate their technical and financial capacity, including the capacity to meet liabilities, before being licensed, and requires Member States to establish independent competent authorities separate from the authorities responsible for licensing and economic development — reflecting the lesson of the conflict of functions in the United States.

Offshore Pollution in Nigeria

Nigeria's offshore sector has experienced significant pollution incidents. In December 2011, an oil spill at the Bonga deepwater field, operated by Shell Nigeria Exploration and Production Company, released an estimated forty thousand barrels of oil during a cargo transfer operation to a tanker. The National Oil Spill Detection and Response Agency (NOSDRA) proposed a very large fine and sought compensation, and the matter became the subject of protracted litigation and regulatory dispute. The incident highlighted gaps in the legal framework for offshore pollution, including the limited penalties under NOSDRA's legislation and the absence of clear standards for natural resource damage assessment.

Operational Discharges, Gas Flaring, and Methane

Beyond accidents, offshore and onshore petroleum operations generate routine discharges and emissions that are regulated by national law and regional standards: produced water, drilling muds and cuttings, chemicals, sewage, and air emissions. **Gas flaring** — the burning of associated gas produced with oil, for which no market or infrastructure exists — wastes a valuable resource, emits greenhouse gases and pollutants, and has long been a major environmental and health problem in the Niger Delta. Nigeria has prohibited routine flaring since the Associated Gas Re-injection Act of 1979, but successive deadlines for ending

flaring were not met, and penalties were long set at levels too low to deter the practice. The **Flare Gas (Prevention of Waste and Pollution) Regulations 2018** established a commercial framework for the capture and sale of flare gas and increased penalties, and the Petroleum Industry Act 2021 provides for the elimination of routine flaring subject to exemptions, and for penalties to be paid into an environmental remediation fund. The World Bank's **Zero Routine Flaring by 2030** initiative, launched in 2015, has been endorsed by numerous governments and companies.

Methane emissions from oil and gas operations — through venting, flaring, and leaks — are a major contributor to climate change, given methane's high short-term warming potential. The **Global Methane Pledge**, launched at COP26 in 2021, commits participants to reduce global methane emissions by at least thirty per cent from 2020 levels by 2030. The European Union's **Methane Regulation** (Regulation (EU) 2024/1787) requires operators to measure, report, and verify methane emissions, to detect and repair leaks, and to cease routine venting and flaring, and will progressively apply methane intensity requirements to imports of oil, gas, and coal into the Union — an extraterritorial dimension with significant implications for exporting states, including Nigeria, Algeria, and other African suppliers.

17.7 Environmental Liability and Remediation

State Responsibility for Environmental Damage

Under the law of state responsibility, a state that commits an internationally wrongful act causing environmental damage is obliged to make full reparation. The assessment of compensation for environmental damage, which may not have a market value, presents particular difficulties, which the International Court of Justice addressed for the first time in 2018.

In the compensation phase of *Certain Activities Carried Out by Nicaragua in the Border Area* (*Costa Rica v Nicaragua*), the Court held that damage to the environment, and the consequent impairment or loss of the ability of the environment to provide goods and services, is compensable under international law, and that compensation may include indemnification for the impairment or loss of environmental goods and services in the period prior to recovery and payment for the restoration of the damaged environment. Rejecting both the parties' proposed methodologies, the Court adopted an "overall assessment" of the impairment of environmental goods and services, and awarded Costa Rica compensation totalling approximately USD 379,000, including USD 120,000 for the impairment or loss of environmental goods and services. The amount was modest, but the recognition of pure environmental damage as compensable was a significant development. In *Armed Activities on the Territory of the Congo* (*DRC v*

Uganda), the Court in 2022 awarded the DRC reparations of USD 325 million, including USD 60 million for damage related to natural resources, including the illegal exploitation of minerals, timber, and wildlife.

The **United Nations Compensation Commission**, established by the Security Council in 1991 to process claims arising from Iraq's invasion and occupation of Kuwait, awarded more than USD 5 billion for environmental damage, including damage caused by the deliberate release of oil and the setting on fire of more than seven hundred oil wells, and established a follow-up programme to monitor the use of the awards for environmental remediation. Its practice remains the most extensive international precedent for the assessment of large-scale environmental damage arising from the energy sector.

Civil Liability Regimes

Because the activities causing environmental damage are usually conducted by private operators, the principal international mechanism for compensation is the imposition of civil liability on operators through treaties that harmonise national law. The civil liability conventions for oil pollution from ships, examined in section 17.6, and for nuclear damage, examined in Chapter 11, share common features: **strict liability** of a designated person (the shipowner or the operator of the nuclear installation), regardless of fault, subject to limited defences; **channelling** of liability to that person; **limitation** of liability to specified amounts; **compulsory insurance** or other financial security; and, in the case of oil pollution and nuclear damage, additional tiers of compensation from industry-funded or state-funded sources. Civil liability conventions for other hazardous activities, including the 1993 Lugano Convention on Civil Liability for Damage resulting from Activities Dangerous to the Environment and the 1999 Basel Protocol on Liability and Compensation, have not entered into force, reflecting the reluctance of states to accept harmonised liability for activities other than shipping and nuclear energy.

National Liability Regimes

National environmental liability regimes take several forms. In the **United States**, the Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (CERCLA, or "Superfund") imposes strict, joint and several, and retroactive liability for the costs of cleaning up hazardous substance releases on current and former owners and operators of facilities, generators of hazardous substances, and transporters, and the Oil Pollution Act of 1990 imposes strict liability on responsible parties for oil spills, including liability for **natural resource damages** assessed by federal and state trustees. In the **European Union**, the Environmental Liability Directive (Directive 2004/35/EC) establishes an administrative regime under which operators of listed dangerous activities are strictly liable to prevent

and remedy environmental damage to protected species and habitats, water, and land, through **primary remediation** (restoring the damaged resources to baseline condition), **complementary remediation** (compensating for incomplete primary remediation), and **compensatory remediation** (compensating for interim losses pending recovery). Unlike CERCLA, the Directive does not give private parties a right to compensation, but permits affected persons and environmental organisations to request action by the competent authority.

Oil Pollution in the Niger Delta

The Niger Delta has experienced decades of pollution from oil spills, caused by equipment failure and corrosion of ageing pipelines, operational failures, sabotage, and the theft of crude oil ("bunkering") and illegal refining. The legal framework for liability and remediation comprises several elements.

Statutory compensation. The **Oil Pipelines Act** of 1956 requires the holder of a pipeline licence to pay compensation to any person suffering damage as a consequence of a breakage of or leakage from the pipeline, except where the damage is caused by the claimant's own default or the malicious act of a third person — an exception that has generated extensive dispute about whether spills were caused by sabotage. Nigerian courts have also applied common law torts, including negligence, nuisance, and the rule in *Rylands v Fletcher*.

Regulation. The **National Oil Spill Detection and Response Agency (Establishment) Act** 2006 established NOSDRA, with responsibility for the detection of and response to oil spills, and regulations made under it govern spill reporting, clean-up, remediation, and damage assessment. The **Environmental Guidelines and Standards for the Petroleum Industry in Nigeria** (EGASPIN), first issued by the Department of Petroleum Resources in 1991 and revised subsequently, set standards for discharges and remediation, including target and intervention values for soil and groundwater contamination. The Petroleum Industry Act 2021 requires holders of licences and leases to contribute to an **environmental remediation fund** for the rehabilitation of environmental damage, and to establish decommissioning and abandonment funds, as examined in Chapters 5 and 24.

The Ogoniland assessment and HYPREP. In 2011, the United Nations Environment Programme published its **Environmental Assessment of Ogoniland**, following an invitation from the Nigerian government. The report found widespread and severe contamination of land, groundwater, and surface water, including drinking water contaminated with benzene at levels far exceeding World Health Organization guidelines, and concluded that the restoration of the environment could take twenty-five to thirty years. It recommended the establishment of an Ogoniland Environmental Restoration Authority and an

initial capital fund of USD 1 billion, contributed by the oil industry operators and the government, for the first five years. The government established the **Hydrocarbon Pollution Remediation Project** (HYPREP) to implement the recommendations, relaunched in 2016, but progress in remediation has been slow and has been criticised by communities and civil society.

Transnational litigation. The difficulty of obtaining redress in Nigerian courts has led communities to bring claims against Shell and its Nigerian subsidiary in the courts of the United Kingdom and the Netherlands. In *Bodo Community v Shell Petroleum Development Co of Nigeria Ltd*, concerning two major spills from the Trans-Niger Pipeline in 2008, Shell's Nigerian subsidiary admitted liability under the Oil Pipelines Act and settled the claims in 2015 for GBP 55 million. In *Okpabi v Royal Dutch Shell plc* [2021] UKSC 3, the United Kingdom Supreme Court held that it was arguable that the parent company owed a duty of care to communities affected by spills from its subsidiary's operations, allowing the claims to proceed in England. In *Oguru and Milieudefensie v Royal Dutch Shell plc*, the Court of Appeal of The Hague held in January 2021 that Shell's Nigerian subsidiary was liable for spills affecting two villages, rejecting the defence of sabotage as unproven, and that the parent company was obliged to ensure the installation of a leak detection system; the claims were subsequently settled. These cases, which raise questions of parent company liability and jurisdiction examined in Chapters 22 and 27, have been accompanied by the divestment of Shell's onshore Nigerian business to a consortium of predominantly Nigerian companies, completed in 2025, raising concerns about responsibility for legacy pollution and the adequacy of funds for remediation and decommissioning.

Remediation Standards and Financial Assurance

The effectiveness of liability regimes depends on remediation standards and the availability of funds. Key legal issues include the standard to which contaminated sites must be remediated — whether to background conditions, to standards appropriate for the intended use, or to risk-based targets; the allocation of responsibility for historical pollution where responsible parties no longer exist or are insolvent; and the requirement of **financial assurance** — bonds, guarantees, insurance, or dedicated funds — to ensure that funds are available for remediation, decommissioning, and mine closure, even if the operator becomes insolvent or divests its assets. The divestment of ageing onshore and shallow water petroleum assets by international oil companies to smaller operators, in Nigeria and elsewhere, has heightened concern that liabilities may be transferred to entities lacking the resources to meet them, and has led regulators to scrutinise the decommissioning and remediation provisions of transactions. The catastrophic failures of tailings dams at Mariana (2015) and Brumadinho (2019) in Brazil, examined in Chapter 7, and the resulting settlements — including a settlement concluded in

October 2024 of approximately BRL 170 billion in respect of the Mariana disaster — illustrate the scale of liabilities that may arise from mining and the importance of financial assurance and robust safety regulation.

17.8 Environmental Compliance and Enforcement

The Implementation Gap

The proliferation of environmental treaties and national laws since the 1970s has not been matched by corresponding improvements in environmental outcomes. The first global assessment of the environmental rule of law, published by UNEP in 2019, found that while environmental laws had increased thirty-eight-fold since 1972, their implementation and enforcement were frequently weak, owing to inadequate institutional capacity, insufficient funding, poor coordination among agencies, lack of political will, corruption, and limited public access to information and justice. This "implementation gap" is particularly pronounced in the energy and extractive sectors in many developing countries, where regulators may lack the technical capacity and independence to monitor powerful operators, and where the state's dual role as resource owner and regulator — often as a partner in joint ventures, as examined in Chapter 15 — creates conflicts of interest. This section examines the mechanisms by which compliance with international and national environmental obligations is promoted and enforced.

Compliance with Multilateral Environmental Agreements

Theories of compliance with international environmental law have emphasised two broad approaches. The **enforcement** approach emphasises the deterrent effect of sanctions for non-compliance. The **managerial** approach, associated with the work of Abram and Antonia Chayes, holds that non-compliance with environmental treaties usually results not from deliberate defiance but from ambiguity in treaty obligations, limited capacity, and the time required for implementation, and that compliance is best promoted through transparency, reporting, review, capacity building, and assistance. Most MEAs combine elements of both approaches, with a strong emphasis on managerial techniques.

The principal compliance mechanisms of MEAs include the following.

Reporting and review. Most MEAs require parties to report periodically on their implementation, and many provide for the review of reports by expert bodies or by the conference of the parties. The Paris Agreement's enhanced transparency framework, examined in Chapter 18, is the most elaborate example.

Non-compliance procedures. The **Montreal Protocol** on Substances that Deplete the Ozone Layer pioneered a non-compliance procedure, adopted in 1992, under which an Implementation Committee considers cases of possible non-compliance referred by parties, the secretariat, or the party concerned, and recommends measures to the Meeting of the Parties, which may include

assistance, cautions, and the suspension of rights under the Protocol. The procedure is complemented by the Protocol's **Multilateral Fund**, which finances the incremental costs of compliance for developing countries — a combination of facilitation and potential sanction that is widely credited with the Protocol's success in phasing out ozone-depleting substances. The **Kyoto Protocol**'s compliance mechanism comprised a facilitative branch and an enforcement branch, the latter empowered to determine non-compliance with emission targets and to impose consequences; it suspended the eligibility of several parties to participate in the flexibility mechanisms for failure to meet reporting requirements. The compliance committee of the Paris Agreement, established under Article 15, is expressly facilitative, non-adversarial, and non-punitive.

Public communications. The **Aarhus Convention Compliance Committee** is distinctive in allowing members of the public to submit communications alleging non-compliance by a party, and has made findings of non-compliance against numerous parties, including the European Union and the United Kingdom, in respect of matters such as public participation in decisions on energy projects and access to justice. The Espoo Convention's Implementation Committee similarly considers information from the public, and has examined many cases concerning nuclear power plants and other energy installations.

Reactive monitoring. Under the World Heritage Convention, the World Heritage Committee may examine the state of conservation of sites threatened by extractive or energy projects, request information and missions, and inscribe sites on the List of World Heritage in Danger, as examined in section 17.5.

International Dispute Settlement

Inter-state litigation and arbitration concerning environmental matters have increased, though they remain relatively rare. Cases before the International Court of Justice have addressed transboundary harm and environmental impact assessment, as examined in sections 17.1 to 17.4, and in *Whaling in the Antarctic* (*Australia v Japan*, 2014), the Court held that Japan's whaling programme was not conducted for purposes of scientific research within the meaning of the International Convention for the Regulation of Whaling. The International Tribunal for the Law of the Sea and Annex VII arbitral tribunals under UNCLOS have addressed marine environmental issues, including in the *South China Sea Arbitration* (2016), in which the tribunal found that China had breached its obligations under Part XII of UNCLOS through land reclamation and the harvesting of endangered species, examined in Chapter 20. The advisory opinions on climate change, examined in Chapter 18, have clarified states' obligations in a manner that may be relied upon in future contentious proceedings. Human rights courts and treaty bodies have become important forums for environmental claims, as examined in Part VI.

National Enforcement

The primary responsibility for enforcing environmental law lies with national authorities. Enforcement tools include the following.

Administrative enforcement. Environmental agencies issue permits and licences with conditions, conduct inspections, require monitoring and reporting, issue compliance and stop-work orders, impose administrative penalties, and suspend or revoke permits. In Nigeria, environmental regulation of the petroleum industry is distributed among several bodies: the **Nigerian Upstream Petroleum Regulatory Commission** (NUPRC) and the **Nigerian Midstream and Downstream Petroleum Regulatory Authority** (NMDPRA), established under the Petroleum Industry Act 2021, which regulate environmental aspects of upstream and midstream and downstream operations; NOSDRA, responsible for oil spill detection and response; the Federal Ministry of Environment, responsible for EIA; and the **National Environmental Standards and Regulations Enforcement Agency** (NESREA), established in 2007, whose enforcement mandate expressly excludes the oil and gas sector. The fragmentation of responsibilities, overlapping mandates, and limited resources of these agencies have been identified as obstacles to effective enforcement.

Criminal enforcement. Serious environmental violations are increasingly treated as crimes. The European Union's revised **Environmental Crime Directive** (Directive (EU) 2024/1203) expanded the list of environmental offences, introduced a "qualified offence" for conduct causing catastrophic damage comparable to ecocide, and harmonised minimum penalties for individuals and companies. At the international level, an Independent Expert Panel convened by the Stop Ecocide Foundation proposed in 2021 a definition of **ecocide** as unlawful or wanton acts committed with knowledge that there is a substantial likelihood of severe and either widespread or long-term damage to the environment, and in 2024 Vanuatu, Fiji, and Samoa formally proposed an amendment to the Rome Statute of the International Criminal Court to include ecocide as a crime. The Office of the Prosecutor of the International Criminal Court has also indicated that it will give particular consideration to crimes committed by means of, or resulting in, environmental destruction and the illegal exploitation of natural resources.

Civil enforcement and public interest litigation. Citizen suit provisions, such as those of the United States Clean Water Act and Clean Air Act, allow private persons to sue violators and regulators. In many common law jurisdictions, courts have relaxed traditional standing requirements to allow public interest litigation on environmental matters. In Nigeria, the Supreme Court held in *Centre for Oil Pollution Watch v Nigerian National Petroleum Corporation* (2018) that a non-governmental organisation committed to environmental pro-

tection had standing to bring an action concerning an oil spill that had polluted a community's water sources, without having to show that it had suffered a personal injury, a decision that significantly expanded access to justice in environmental matters. The Fundamental Rights (Enforcement Procedure) Rules 2009 also permit public interest litigation for the enforcement of fundamental rights.

Specialised environmental courts. Specialised environmental courts and tribunals have been established in many jurisdictions, including India's **National Green Tribunal**, established in 2010, Kenya's Environment and Land Court, and the Land and Environment Court of New South Wales. Such courts bring expertise and efficiency to environmental disputes, and the Indian Supreme Court's development of environmental jurisprudence, including the application of the precautionary and polluter-pays principles and the public trust doctrine, has been influential across the common law world.

Transparency, Technology, and Private Governance

Compliance is increasingly promoted through transparency and technology. Satellite monitoring enables the detection of gas flaring, methane leaks, oil spills, and deforestation, and the public availability of such data allows civil society, investors, and regulators to identify non-compliance. Corporate sustainability reporting requirements — including the European Union's **Corporate Sustainability Reporting Directive** (Directive (EU) 2022/2464), whose scope and timetable were revised by simplification measures adopted in 2025, and the sustainability disclosure standards issued by the International Sustainability Standards Board (IFRS S1 and S2) in 2023 — require companies to disclose environmental risks and impacts. Environmental management systems certified to standards such as ISO 14001, lender covenants enforcing environmental and social action plans, and supply chain due diligence obligations, examined in Chapter 22, extend environmental governance beyond the reach of public authorities.

Conclusion

International environmental law has developed from a handful of arbitral decisions into a dense body of principles, treaties, and institutions that shape every stage of energy and resource development. Its core principles — prevention, precaution, polluter-pays, and environmental impact assessment — have been progressively hardened into binding obligations, interpreted by courts with increasing rigour, and extended to new challenges such as the assessment of downstream emissions. Yet the gap between the law's aspirations and environmental outcomes remains wide, particularly in resource-rich developing countries, where the effectiveness of environmental law depends on institutional capacity, regulatory independence, transparency, and access to justice. For the energy lawyer,

environmental compliance is no longer a peripheral concern but a central determinant of the legality, financing, and social acceptance of projects.

::: exercise {title="Practical Exercise 17: Environmental Liability Assessment of an Offshore Spill (HYPOTHETICAL)"} **This exercise is entirely hypothetical. All parties, places, installations, and events are fictitious and do not refer to any real persons, companies, or states.**

The Republic of Coralia and the Federal Republic of Marenia are neighbouring coastal states in a semi-enclosed gulf. Both are parties to UNCLOS, the OPRC Convention, and a regional seas convention modelled on the Abidjan Convention, but neither is party to the Additional Protocol on offshore oil and gas activities.

Deltamar Petroleum, a company incorporated in Coralia and wholly owned by Atlantica Energy plc, a company listed in London, operates the Stella field on Coralia's continental shelf, twenty kilometres from the maritime boundary with Marenia. Coralia's petroleum regulator approved Deltamar's field development plan and environmental impact assessment in 2019. The assessment did not consider transboundary effects, and Coralia did not notify Marenia of the project. Coralia's regulator, which is also responsible for maximising petroleum revenues, conducted no inspections of the installation after 2021, citing budgetary constraints.

In March of the present year, a well control incident at Stella released oil for forty days. Oil reached Marenia's coast, damaging mangroves, fisheries, and a Ramsar-listed wetland, and affecting the livelihoods of an estimated twelve thousand artisanal fishers. Deltamar claims that the incident was caused by unforeseeable geological conditions. Coralia's petroleum law caps the operator's liability for pollution damage at USD 50 million. Deltamar's assets in Coralia are worth approximately USD 80 million, and Atlantica has announced that it intends to sell Deltamar to a small independent company.

Tasks.

(a) Advise the government of Marenia on whether Coralia has breached its obligations under international law, identifying the substantive and procedural obligations engaged and the evidence required.

(b) Identify the forums in which Marenia might bring a claim against Coralia and assess the remedies available, including the methodology that a tribunal might use to assess compensation for environmental damage.

(c) Advise a group of Marenian fishers on the options for civil claims against Deltamar and Atlantica, including possible forums, causes of action, and obstacles, drawing on the case law discussed in this chapter.

(d) Prepare a memorandum for Coralia's regulator on the measures it should take in relation to the proposed sale of Deltamar to protect the state's interests in remediation and decommissioning.

(e) Draft three reforms to Coralia's legal framework to reduce the risk of a recurrence, with reference to the lessons of Deepwater Horizon and Montara. :::

CHAPTER SUMMARY

- International environmental law developed from *Trail Smelter* and *Corfu Channel*, through the Stockholm (1972) and Rio (1992) conferences, into a body of treaties, customary rules, principles, and institutions centred on sustainable development.
- The principle of prevention, an obligation of due diligence, is established customary law; the precautionary principle addresses scientific uncertainty and is widely incorporated in treaties; the polluter-pays principle underlies liability, decommissioning, and environmental taxation.
- Environmental impact assessment is a requirement of general international law where there is a risk of significant transboundary harm, and is required by national law for most energy projects; *Finch* establishes in the United Kingdom that downstream combustion emissions of extracted fossil fuels must be assessed.
- States must prevent significant transboundary harm and notify and consult potentially affected states; major accidents have shaped regimes on nuclear emergencies, industrial accidents, mining waste, and hazardous wastes.
- The Convention on Biological Diversity and the Kunming-Montreal Framework, World Heritage, Ramsar, and lender standards such as IFC Performance Standard 6 constrain extraction in biodiverse areas, generating contests such as Virunga, Yasuní, and EACOP.
- Offshore petroleum pollution lacks a global regime; regional instruments such as OSPAR, the Barcelona Offshore Protocol, and the Abidjan Additional Protocol, and national reforms following Deepwater Horizon and Montara, fill the gap.
- Environmental damage is compensable under international law, as recognised in *Costa Rica v Nicaragua*; civil liability regimes, national statutes, and transnational litigation address oil pollution, including in the Niger Delta.
- Compliance relies on reporting, non-compliance procedures, public communications, litigation, administrative and criminal enforcement, standing reform, specialised courts, and transparency technologies, yet a substantial implementation gap remains.

DOCTORAL RESEARCH QUESTIONS

1. To what extent has the due diligence standard in the obligation of prevention become more demanding in the light of the climate change advisory opinions, and what are the implications for the authorisation of new fossil fuel projects?
2. Evaluate the significance of *R (Finch) v Surrey County Council* for the regulation of fossil fuel extraction, and compare the approaches of courts in the United Kingdom, Norway, and the United States to the assessment of downstream emissions.
3. Is there a case for a global convention on liability and compensation for pollution from offshore petroleum activities? Assess the arguments in the light of the Montara and Deepwater Horizon incidents and the Abidjan Additional Protocol.
4. How should the international community share the costs of conserving globally significant biodiversity located above valuable mineral and petroleum resources in developing countries?
5. Assess the effectiveness of the Nigerian legal and institutional framework for the remediation of oil pollution in the Niger Delta, with reference to the UNEP Ogoniland assessment and the transnational litigation against Shell.
6. Should ecocide be recognised as an international crime, and what would be the implications for corporate actors in the energy and extractive sectors?

FURTHER READING

- Philippe Sands, Jacqueline Peel, Adriana Fabra and Ruth MacKenzie, *Principles of International Environmental Law* (4th edn, Cambridge University Press 2018).
- Alan Boyle and Catherine Redgwell, *Birnie, Boyle, and Redgwell's International Law and the Environment* (4th edn, Oxford University Press 2021).
- Pierre-Marie Dupuy and Jorge E Viñuales, *International Environmental Law* (2nd edn, Cambridge University Press 2018).
- Daniel Bodansky, *The Art and Craft of International Environmental Law* (Harvard University Press 2010).
- Neil Craik, *The International Law of Environmental Impact Assessment: Process, Substance and Integration* (Cambridge University Press 2008).
- Abram Chayes and Antonia Handler Chayes, *The New Sovereignty: Compliance with International Regulatory Agreements* (Harvard University Press 1995).
- United Nations Environment Programme, *Environmental Assessment of Ogoniland* (UNEP 2011).

18 CHAPTER

Climate Change Law and the Global Energy Transition

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the structure, principles, and institutions of the United Nations Framework Convention on Climate Change and its relationship with the science of climate change.
2. Compare the architecture of the Kyoto Protocol and the Paris Agreement, and analyse the legal character of the obligations under the Paris Agreement.
3. Evaluate the role, content, and legal status of nationally determined contributions and national climate framework legislation, with particular reference to Nigeria.
4. Analyse the obligations of states in respect of mitigation, adaptation, loss and damage, and climate finance, as clarified by the advisory opinions of international courts and tribunals.
5. Assess the legal instruments used to regulate fossil fuels and drive the energy transition, including carbon pricing, carbon markets, coal phase-out, supply-side measures, and subsidy reform.
6. Analyse the principal strands of climate litigation against states and corporations and their implications for the energy sector.
7. Explain the concept of just transition and the legal and financial instruments designed to give it effect.
8. Evaluate proposals for reform of international climate governance.

Introduction

Climate change is the defining challenge for energy law in the twenty-first century. The combustion of fossil fuels for energy accounts for roughly three-quarters of global greenhouse gas emissions, and the transformation of the energy system — from one dominated by coal, oil, and gas to one based on renewable energy, electrification, efficiency, and low-carbon fuels — is therefore at the heart of the

global response to climate change. That transformation is reshaping every area of energy and natural resources law examined in this book: the licensing of petroleum and mining, the regulation of electricity markets, the design of fiscal regimes, the protection of investments, the content of contracts, the financing of projects, and the resolution of disputes.

The scientific basis for action is well established. The Intergovernmental Panel on Climate Change (IPCC), in its Sixth Assessment Report completed in 2023, concluded that human activities, principally through emissions of greenhouse gases, have unequivocally caused global warming, with global surface temperature having reached about 1.1°C above pre-industrial levels in 2011–2020; that widespread and rapid changes in the atmosphere, ocean, cryosphere, and biosphere have occurred; and that limiting warming to 1.5°C with no or limited overshoot requires deep, rapid, and sustained reductions in greenhouse gas emissions, with global net zero carbon dioxide emissions reached in the early 2050s. The World Meteorological Organization reported that 2024 was the first calendar year in which the global average temperature exceeded 1.5°C above pre-industrial levels, although the Paris Agreement's temperature goal refers to long-term averages rather than individual years.

The legal response to climate change operates at multiple levels: the international regime established by the United Nations Framework Convention on Climate Change and the Paris Agreement; the general rules of international law, including the law of state responsibility, the law of the sea, and human rights law, as clarified by recent advisory opinions; regional law, notably that of the European Union; national climate legislation and energy regulation; and the increasingly important role of courts, investors, and private standards. This chapter examines the UNFCCC (section 18.1), the Kyoto Protocol and the Paris Agreement (section 18.2), nationally determined contributions (section 18.3), mitigation and adaptation obligations (section 18.4), the regulation of fossil fuels and transition policies (section 18.5), climate litigation and corporate accountability (section 18.6), just transition (section 18.7), and the reform of climate governance (section 18.8). The procedural dimensions of climate litigation in domestic courts — standing, justiciability, and remedies — are examined in Chapter 27, and the theoretical foundations of energy justice in Chapter 23.

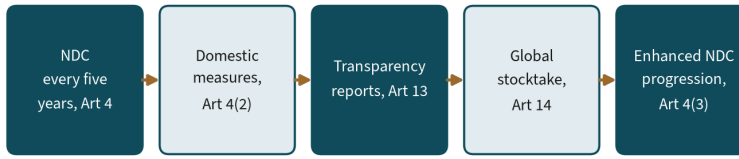


Figure 18.1 The ambition cycle of the Paris Agreement. *Source:* prepared for this edition.

18.1 United Nations Framework Convention on Climate Change

Origins

Scientific concern about the effects of carbon dioxide emissions on the climate dates to the nineteenth century, but climate change emerged as an issue of international policy only in the 1980s. In 1988, the World Meteorological Organization and UNEP established the **Intergovernmental Panel on Climate Change** to assess the scientific, technical, and socio-economic information relevant to understanding climate change, and the United Nations General Assembly declared in Resolution 43/53 that climate change is a "common concern of mankind". The IPCC's First Assessment Report of 1990 provided the scientific basis for negotiations, which culminated in the adoption of the **United Nations Framework Convention on Climate Change** (UNFCCC) in May 1992 and its opening for signature at the Rio Conference in June 1992. The Convention entered into force in 1994 and has near-universal participation, with 198 parties.

The Objective

Article 2 sets out the ultimate objective of the Convention: to achieve stabilisation of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system, within a timeframe sufficient to allow ecosystems to adapt naturally to climate change, to ensure that food production is not threatened, and to enable economic development to proceed in a sustainable manner. The objective does not define the level of "dangerous" interference, a question that was subsequently addressed by the political agreement on the 2°C limit in the Copenhagen Accord of 2009 and the Cancún Agreements of 2010, and by the temperature goal of the Paris Agreement.

The Principles

Article 3 sets out principles to guide the parties' actions. They include the principle that parties should protect the climate system for the benefit of present and future generations of humankind, on the basis of **equity** and in accordance with their **common but differentiated responsibilities and respective capabilities** (CBDR-RC), with developed country parties taking the lead in combating climate change (Article 3(1)); the principle that the specific needs and special circumstances of developing countries, especially those particularly vulnerable to climate change, should be given full consideration (Article 3(2)); the **precautionary** principle, providing that parties should take precautionary measures and that lack of full scientific certainty should not be used as a reason for postponing such measures (Article 3(3)); the right to and duty to promote **sustainable development** (Article 3(4)); and the principle that measures taken to combat climate change should not constitute a means of arbitrary or unjustifiable discrimination or a disguised restriction on international trade (Article 3(5)).

The principle of CBDR-RC has been the most contested element of the regime. It reflects two ideas: that developed countries bear greater responsibility for the accumulated stock of greenhouse gases in the atmosphere, as a result of their historical emissions since the industrial revolution; and that they have greater financial and technological capacity to respond. Developing countries have relied on the principle to resist binding emission limits and to demand finance and technology transfer, while developed countries have argued that the rapid growth of emissions from large emerging economies, notably China, which became the world's largest emitter in the mid-2000s, requires all major economies to act. The evolution of the principle — from a binary division between developed and developing countries under the Kyoto Protocol to a more nuanced formulation under the Paris Agreement, which adds the words "in the light of different national circumstances" — has been central to the development of the regime.

Commitments and Annexes

Article 4(1) sets out general commitments applicable to all parties, including the preparation of national inventories of emissions and removals, the formulation and implementation of national programmes containing measures to mitigate climate change and facilitate adaptation, the promotion of technology transfer, the conservation of sinks, and cooperation in research, education, and public awareness. Article 4(2) imposed more specific commitments on developed country parties listed in **Annex I** — the members of the OECD in 1992 and countries with economies in transition — to adopt policies and measures to mitigate climate change, with the aim of returning their emissions to 1990 levels, a formulation that was too weak to constitute a binding target. Developed country parties listed in **Annex II** — the OECD members — undertook to provide new and additional

financial resources to meet the costs incurred by developing countries in complying with their obligations, and to promote technology transfer (Articles 4(3)–(5)). Article 4(7) provides that the extent to which developing country parties will effectively implement their commitments will depend on the effective implementation by developed country parties of their commitments related to financial resources and transfer of technology. Article 4(8) and (10) require consideration of the needs and concerns of developing countries arising from the impact of response measures, including countries whose economies are highly dependent on income from fossil fuels — a provision on which oil-exporting states, including Saudi Arabia and other OPEC members, have relied.

Institutions

The **Conference of the Parties** (COP), meeting annually, is the supreme body of the Convention, and reviews its implementation and takes decisions to promote its effective implementation. The COP is assisted by two subsidiary bodies — the Subsidiary Body for Scientific and Technological Advice (SBSTA) and the Subsidiary Body for Implementation (SBI) — and by a secretariat based in Bonn. The **financial mechanism** of the Convention is operated by the Global Environment Facility and, since 2010, the **Green Climate Fund**, headquartered in Songdo, Republic of Korea. The COP also serves as the meeting of the parties to the Kyoto Protocol (CMP) and the Paris Agreement (CMA).

A distinctive feature of the regime is that the parties have never adopted the rules of procedure of the COP, because of disagreement over the voting rule; as a result, decisions are taken by consensus, which gives each party a potential veto and has frequently led to lowest-common-denominator outcomes and protracted negotiations. The annual COPs have grown into very large events, attended by tens of thousands of delegates, observers, and representatives of business and civil society, and the hosting of recent COPs by major oil and gas producers — the United Arab Emirates in 2023 and Azerbaijan in 2024 — and the participation of large numbers of fossil fuel industry representatives have prompted debate about the integrity of the process, examined in section 18.8.

The Convention and the Energy Sector

The Convention does not refer expressly to fossil fuels, energy, or the energy transition, reflecting the political sensitivity of these issues for fossil fuel producing states. Its obligations are framed in terms of greenhouse gas emissions generally, leaving to parties the choice of means. Only in the decisions of the COP and the CMA in recent years — the Glasgow Climate Pact of 2021, which called for the phasedown of unabated coal power and the phase-out of inefficient fossil fuel subsidies, and the outcome of the first global stocktake at Dubai in 2023, which called on parties to contribute to "transitioning away from fossil fuels in energy

systems, in a just, orderly and equitable manner" — has the regime addressed fossil fuels explicitly. This evolution, examined in section 18.2, reflects the growing recognition that the objectives of the Convention cannot be achieved without a fundamental transformation of the energy system.

The Position of the United States

The participation of the United States, historically the largest cumulative emitter, has fluctuated with changes of administration. The United States signed but never ratified the Kyoto Protocol; joined the Paris Agreement in 2016; withdrew in 2020; rejoined in 2021; and, following a further change of administration, notified its withdrawal again in January 2025, effective one year later. The United States ratified the UNFCCC itself in 1992 with the advice and consent of the Senate; the extent of its continued engagement in the wider regime, and the constitutional question whether a President may withdraw from a Senate-approved treaty without congressional involvement, have since become matters of debate. The repeated reversals illustrate the vulnerability of international climate commitments to domestic politics, and the significance of sub-national governments, businesses, and courts in sustaining climate action in the absence of federal leadership.

18.2 Kyoto Protocol and Paris Agreement

The Kyoto Protocol

The **Kyoto Protocol**, adopted at COP3 in 1997 and in force from 2005, gave effect to the "top-down" approach of setting legally binding, quantified emission limitation and reduction commitments for developed countries. Under Article 3 and Annex B, developed country parties undertook to reduce their aggregate emissions of six greenhouse gases by at least five per cent below 1990 levels in the first commitment period of 2008 to 2012, with individual targets ranging from an eight per cent reduction for the European Community to a ten per cent increase for Iceland. Developing countries, including China, India, Brazil, and the members of OPEC, had no quantified commitments.

The Protocol introduced three **flexibility mechanisms** to enable parties to meet their targets at lower cost: **international emissions trading** among Annex B parties (Article 17); **joint implementation**, under which Annex B parties could earn credits from emission reduction projects in other Annex B parties (Article 6); and the **Clean Development Mechanism** (CDM), under which Annex B parties could earn certified emission reductions from projects in developing countries that contributed to sustainable development (Article 12). The CDM registered several thousand projects, many in the energy sector, including renewable energy, energy efficiency, and the capture of industrial gases and landfill methane, and established an international infrastructure for the monitoring, reporting, and

verification of emission reductions. It was also criticised for the questionable additionality of many projects, the concentration of projects in a few large developing countries, and the environmental integrity of credits from some project types. The Protocol's compliance mechanism, examined in Chapter 17, was the most rigorous of any MEA.

The Protocol's effectiveness was limited by its architecture. The United States did not ratify it; Canada withdrew in 2011; and because developing countries had no targets, the Protocol covered a declining share of global emissions as emissions from emerging economies grew. The **Doha Amendment** of 2012, establishing a second commitment period from 2013 to 2020, attracted limited participation and entered into force only on the last day of the period, at the end of 2020.

From Copenhagen to Paris

The failure of the Copenhagen conference of 2009 to agree a comprehensive successor to Kyoto marked the end of the top-down approach. The **Copenhagen Accord**, which was "taken note of" rather than adopted by the COP, introduced a "bottom-up" approach under which countries — developed and developing — would pledge their own mitigation targets and actions, and articulated the goal of limiting warming to below 2°C and the commitment of developed countries to mobilise USD 100 billion a year by 2020 for climate action in developing countries. The **Cancún Agreements** of 2010 formalised these elements within the UNFCCC, and the **Durban Platform** of 2011 launched negotiations on "a protocol, another legal instrument or an agreed outcome with legal force under the Convention applicable to all Parties", to be adopted in 2015. In the lead-up to Paris, parties submitted "intended nationally determined contributions", which formed the basis of the new agreement.

The Paris Agreement

The **Paris Agreement** was adopted at COP21 on 12 December 2015 and entered into force on 4 November 2016, after reaching the threshold of ratification by at least fifty-five parties accounting for at least fifty-five per cent of global emissions. It is a treaty within the meaning of the Vienna Convention on the Law of Treaties, but it combines legally binding procedural obligations with non-binding or discretionary substantive elements, in a design intended to secure universal participation — including that of the United States, whose administration sought to avoid the need for Senate approval — while creating a framework for progressively increasing ambition.

The purpose and goals (Article 2). The Agreement aims to strengthen the global response to climate change by holding the increase in the global average temperature to well below 2°C above pre-industrial levels and pursuing efforts to limit the increase to 1.5°C (Article 2(1)(a)); increasing the ability to adapt to the

adverse impacts of climate change and fostering climate resilience and low greenhouse gas emissions development (Article 2(1)(b)); and making finance flows consistent with a pathway towards low greenhouse gas emissions and climate-resilient development (Article 2(1)(c)). The Agreement is to be implemented to reflect equity and the principle of CBDR-RC, "in the light of different national circumstances" (Article 2(2)). Subsequent decisions of the CMA, notably the Glasgow Climate Pact of 2021, resolved to pursue efforts to limit the temperature increase to 1.5°C, and the International Court of Justice in its 2025 advisory opinion described 1.5°C as having become the parties' agreed primary temperature goal.

Mitigation (Article 4). To achieve the temperature goal, parties aim to reach global peaking of greenhouse gas emissions as soon as possible and to undertake rapid reductions thereafter, so as to achieve a balance between anthropogenic emissions by sources and removals by sinks — net zero — in the second half of this century (Article 4(1)). Each party "shall prepare, communicate and maintain successive nationally determined contributions that it intends to achieve", and "shall pursue domestic mitigation measures, with the aim of achieving the objectives of such contributions" (Article 4(2)). Each successive NDC "will represent a progression beyond the Party's then current nationally determined contribution and reflect its highest possible ambition, reflecting its common but differentiated responsibilities and respective capabilities, in the light of different national circumstances" (Article 4(3)). Developed countries "should continue taking the lead by undertaking economy-wide absolute emission reduction targets", while developing countries "should continue enhancing their mitigation efforts" and are encouraged to move over time towards economy-wide targets (Article 4(4)). NDCs are communicated every five years (Article 4(9)) and recorded in a public registry (Article 4(12)).

Sinks and forests (Article 5) and **cooperative approaches (Article 6)**, which provide for international transfers of mitigation outcomes and a new crediting mechanism, examined in section 18.5.

Adaptation (Article 7) establishes the global goal on adaptation of enhancing adaptive capacity, strengthening resilience, and reducing vulnerability to climate change, and requires each party to engage in adaptation planning processes.

Loss and damage (Article 8) recognises the importance of averting, minimising, and addressing loss and damage associated with the adverse effects of climate change, while the decision adopting the Agreement provides that Article 8 does not involve or provide a basis for any liability or compensation — a provision insisted upon by developed countries.

Finance, technology, and capacity building (Articles 9–11) provide that developed country parties "shall provide financial resources to assist developing

country Parties with respect to both mitigation and adaptation", and encourage other parties to provide support voluntarily.

Transparency (Article 13) establishes an **enhanced transparency framework**, under which each party must regularly provide a national inventory report and the information necessary to track progress in implementing and achieving its NDC, subject to technical expert review and a facilitative, multilateral consideration of progress. Parties submitted their first biennial transparency reports under the framework by the end of 2024.

The global stocktake (Article 14) requires the CMA periodically to take stock of the implementation of the Agreement to assess the collective progress towards achieving its purpose and long-term goals, in order to inform parties in updating and enhancing their actions and support. The first global stocktake was concluded in 2023, and subsequent stocktakes will take place every five years.

Compliance (Article 15) establishes a mechanism to facilitate implementation and promote compliance, which is expert-based, facilitative, transparent, non-adversarial, and non-punitive.

The Legal Character of the Paris Agreement

The Paris Agreement's legal character has been described as a "hybrid" of binding and non-binding elements. Its procedural obligations — to prepare, communicate, and maintain NDCs, to report on progress, and to participate in review — are legally binding obligations of result, expressed in mandatory language ("shall"). By contrast, the content of NDCs is nationally determined, and there is no legal obligation to achieve them; Article 4(2) requires parties to "pursue" domestic measures "with the aim of" achieving them, an obligation of conduct rather than result. The expectations of progression and highest possible ambition in Article 4(3) are expressed in the language of "will", which is less clearly mandatory. The Agreement relies on the combination of transparency, peer pressure, the global stocktake, and the five-yearly cycle of NDCs — the **ambition cycle** or "ratchet" mechanism illustrated in Figure 18.1 — to drive increasing ambition over time.

The question whether the Agreement's substantive provisions impose enforceable standards on parties' discretion in setting their NDCs was addressed by the International Court of Justice in its 2025 advisory opinion, examined in section 18.4, which held that the discretion of parties in preparing their NDCs is limited, and that NDCs must, taken together, be capable of achieving the temperature goal.

The Paris Rulebook and Subsequent Decisions

The detailed rules for implementing the Agreement — the "Paris Rulebook" — were largely agreed at COP24 in **Katowice** in 2018, with the rules for Article 6

completed at **Glasgow** in 2021 and further operational details agreed at **Baku** in 2024. Key decisions since Paris include the following:

- the **Glasgow Climate Pact** (2021), which called on parties to accelerate efforts towards the phasedown of unabated coal power and the phase-out of inefficient fossil fuel subsidies — the first explicit reference to fossil fuels in a COP decision — and requested parties to revisit and strengthen their 2030 targets;
- the **Sharm el-Sheikh** decisions (2022), which established new funding arrangements, including a fund, for responding to loss and damage;
- the **UAE Consensus** following the first global stocktake at Dubai (2023), which called on parties to contribute to global efforts including tripling renewable energy capacity globally and doubling the global average annual rate of energy efficiency improvements by 2030, accelerating efforts towards the phasedown of unabated coal power, and "transitioning away from fossil fuels in energy systems, in a just, orderly and equitable manner, accelerating action in this critical decade, so as to achieve net zero by 2050 in keeping with the science", and recognised that transitional fuels can play a role in facilitating the energy transition while ensuring energy security;
- the **New Collective Quantified Goal** on climate finance adopted at Baku (2024), under which developed countries are to take the lead in mobilising at least USD 300 billion a year by 2035 for developing countries, within a broader call for all actors to scale up finance to at least USD 1.3 trillion a year by 2035 — a figure widely criticised by developing countries as inadequate.

COP30, held in Belém, Brazil, in November 2025, was intended to mark the submission of a new round of NDCs and a shift towards implementation. Its formal outcome did not include an agreed roadmap for transitioning away from fossil fuels, despite the support of a substantial group of parties, and the presidency indicated that work on such roadmaps would be taken forward outside the formal negotiations. The difficulty of translating the Dubai language on fossil fuels into concrete commitments illustrates the continuing influence of producer states and the limits of consensus decision-making.

18.3 Nationally Determined Contributions

Content and Form

Nationally determined contributions are the central instrument of the Paris Agreement. Each party determines the content of its NDC, which may take various forms: economy-wide absolute emission reduction targets relative to a base year (the form expected of developed countries); targets relative to a "business as usual" projection; emissions intensity targets, relative to gross domestic product; peaking years; or sectoral policies and actions, such as renewable energy or forestry

targets. Many developing country NDCs distinguish between **unconditional** contributions, which the party will implement with its own resources, and **conditional** contributions, which depend on the provision of international finance, technology, and capacity building. Decisions of the CMA require parties to provide information to facilitate clarity, transparency, and understanding (ICTU) of their NDCs, including the reference point, time frames, scope and coverage, planning processes, assumptions and methodological approaches, and how the party considers that its NDC is fair and ambitious.

Parties submitted their first NDCs around the time of the Agreement's entry into force, updated or enhanced NDCs in 2020–2021, and were expected to submit a new round of NDCs, containing targets for 2035, by early 2025, although many parties submitted late. The synthesis reports prepared by the UNFCCC secretariat and the annual Emissions Gap Reports of UNEP have consistently found that the aggregate effect of NDCs falls far short of what is required to meet the Paris temperature goal; UNEP's 2024 report estimated that full implementation of then-current unconditional NDCs would limit warming to around 2.6–2.8°C this century, and that continuation of current policies would result in warming of around 3°C — the so-called **ambition gap** and **implementation gap**.

The Energy Content of NDCs

Because energy accounts for the largest share of emissions, NDCs are in substance energy policy documents. Typical energy measures include targets for the share of renewable energy in electricity generation, energy efficiency targets, coal phase-out dates, the electrification of transport, the reduction of gas flaring and methane emissions, and the expansion of clean cooking. NDCs therefore provide a reference point for the interpretation of national energy legislation and for the assessment of the consistency of energy projects with national climate commitments, including in EIA, as examined in Chapter 17, and in litigation, as examined in section 18.6.

Nigeria's NDC and Climate Legislation

Nigeria submitted its first NDC in 2015 and an updated NDC in 2021, which committed to an unconditional reduction of twenty per cent in greenhouse gas emissions below business-as-usual levels by 2030, and a conditional reduction of forty-seven per cent subject to international support, with measures including the reduction of gas flaring, the expansion of off-grid solar, improved energy efficiency, clean cooking, and reductions in methane emissions from the oil and gas sector. Nigeria prepared a further NDC in 2025, setting targets for 2035, as part of the third round of contributions.

In 2021, Nigeria enacted the **Climate Change Act**, which provides a framework for achieving low greenhouse gas emissions and climate-resilient

development. The Act sets a target date for achieving net zero emissions between 2050 and 2070; establishes the **National Council on Climate Change**, chaired by the President, with responsibility for coordinating climate policy and administering a Climate Change Fund; requires the Federal Ministry of Environment to set a carbon budget for successive five-year periods; requires the development of a National Climate Change Action Plan; imposes climate-related obligations on ministries, departments, agencies, and private entities with fifty or more employees; and provides for the development of mechanisms for carbon emission trading and carbon taxation. The **Energy Transition Plan**, launched by the federal government in 2021, sets out a pathway to net zero emissions in the energy sector by 2060, emphasising the role of natural gas as a transition fuel, the expansion of renewable energy, and universal access to modern energy by 2030. The relationship between these instruments and the Petroleum Industry Act 2021, which promotes the development of petroleum resources, illustrates the tension that many developing producer states face between climate commitments and the development of hydrocarbon resources, examined further in section 18.7 and Chapter 24.

National Climate Framework Laws

Climate framework laws, which establish long-term targets, carbon budgets, institutional arrangements, and accountability mechanisms, have been adopted by a growing number of states. The **United Kingdom's Climate Change Act 2008**, the first such law, originally set a target of an eighty per cent reduction in emissions below 1990 levels by 2050, amended in 2019 to net zero; established a system of five-yearly legally binding carbon budgets set fifteen years in advance; created the independent Climate Change Committee to advise on budgets and report on progress; and requires the government to publish proposals and policies for meeting the budgets. In *R (Friends of the Earth Ltd) v Secretary of State for Business, Energy and Industrial Strategy* [2022] EWHC 1841 (Admin), the High Court held that the government's net zero strategy did not comply with its statutory obligations because it lacked sufficient information on the quantified effects of its policies, and in a further decision in 2024 the court held that the replacement Carbon Budget Delivery Plan was also unlawful, as the minister had not been given sufficient information about the risks to the delivery of policies.

The **European Climate Law** (Regulation (EU) 2021/1119) makes binding the Union's objective of climate neutrality by 2050 and its target of a net domestic reduction of at least fifty-five per cent below 1990 levels by 2030, implemented through the "Fit for 55" package of legislation, including reforms of the EU Emissions Trading System, the Renewable Energy Directive, and the Energy Efficiency Directive; the Commission has proposed a 2040 target of a ninety per cent net reduction. Germany's Federal Climate Change Act of 2019

was amended in 2021 following the decision of the Federal Constitutional Court in *Neubauer*, examined in section 18.6. In Africa, **Kenya's Climate Change Act 2016**, amended in 2023 to regulate carbon markets, and **South Africa's Climate Change Act 2024** provide framework legislation, the latter providing for sectoral emission targets and carbon budgets for companies.

18.4 Climate Mitigation and Adaptation Obligations

The Sources of States' Climate Obligations

The obligations of states in relation to climate change arise from several sources: the climate treaties — the UNFCCC, the Kyoto Protocol, and the Paris Agreement; other treaties, including UNCLOS and human rights treaties; and customary international law, including the obligation to prevent significant transboundary harm and the duty to cooperate. For many years, the relationship between these sources was uncertain. Some states argued that the climate treaties constitute a *lex specialis* that exclusively governs states' obligations in relation to climate change, displacing general international law; others argued that general international law, including the prevention principle and human rights law, applies concurrently. Between 2024 and 2025, three international courts and tribunals delivered advisory opinions that substantially clarified these questions.

The ITLOS Advisory Opinion (2024)

In response to a request from the Commission of Small Island States on Climate Change and International Law, the International Tribunal for the Law of the Sea delivered a unanimous advisory opinion on 21 May 2024. The Tribunal held that anthropogenic greenhouse gas emissions into the atmosphere constitute "pollution of the marine environment" within the meaning of Article 1(1)(4) of UNCLOS, because they introduce substances or energy into the marine environment — through ocean warming, sea level rise, and ocean acidification — resulting in deleterious effects. Consequently, states parties to UNCLOS have specific obligations under Article 194 to take all necessary measures to prevent, reduce, and control marine pollution from anthropogenic greenhouse gas emissions, and to endeavour to harmonise their policies. The Tribunal held that this is an obligation of due diligence, the standard of which is "stringent" given the high risks of serious and irreversible harm, and that the obligations under UNCLOS are not satisfied simply by complying with the Paris Agreement, since UNCLOS and the Paris Agreement are separate agreements with separate sets of obligations, although the Paris Agreement's temperature goal and the best available science are relevant to determining the necessary measures. The opinion is examined further in Chapter 20.

The Advisory Opinion of the International Court of Justice (2025)

The most authoritative statement of states' climate obligations is the advisory opinion of the International Court of Justice requested by the General Assembly.

::: casebox {title="Obligations of States in respect of Climate Change, Advisory Opinion (ICJ, 23 July 2025)"} **The request.** In March 2023, following a campaign initiated by students in the Pacific and led diplomatically by Vanuatu, the United Nations General Assembly adopted by consensus Resolution 77/276, requesting an advisory opinion from the International Court of Justice on two questions: what are the obligations of states under international law to ensure the protection of the climate system and other parts of the environment from anthropogenic greenhouse gas emissions for states and for present and future generations; and what are the legal consequences under these obligations for states where they, by their acts and omissions, have caused significant harm to the climate system and other parts of the environment. The proceedings attracted the largest number of written and oral submissions in the Court's history.

The applicable law. The Court unanimously held that the obligations of states are found in the climate treaties, in UNCLOS and other environmental treaties, in customary international law, and in international human rights law. It rejected the argument that the climate treaties constitute a *lex specialis* excluding the application of other rules, holding that the various sources apply concurrently and inform each other.

The climate treaties. The Court held that the 1.5°C goal has become the primary temperature goal agreed by the parties to the Paris Agreement. While the content of NDCs is nationally determined, the discretion of parties is limited: their NDCs must reflect their highest possible ambition and must, taken together, be capable of achieving the temperature goal; and parties have an obligation of conduct, assessed by a stringent standard of due diligence, to pursue measures capable of achieving their NDCs.

Customary law. The Court held that states have customary obligations to prevent significant harm to the climate system and other parts of the environment, by acting with due diligence, and to cooperate in good faith. The standard of due diligence is stringent, and requires states to use all means at their disposal, including the adoption of regulatory and legislative measures and their enforcement, in the light of the best available science and taking into account the principle of common but differentiated responsibilities.

Fossil fuels. In a passage of particular significance for energy law, the Court stated that a state's failure to take appropriate action to protect the climate system from greenhouse gas emissions — including through fossil fuel production, fossil fuel consumption, the granting of fossil fuel exploration licences, or the provision

of fossil fuel subsidies — may constitute an internationally wrongful act attributable to that state.

Human rights. The Court held that the protection of the environment is a precondition for the enjoyment of human rights, that the human right to a clean, healthy, and sustainable environment is essential for the enjoyment of other human rights, and that states must take measures to protect human rights against the adverse effects of climate change.

Legal consequences. The Court held that the breach of these obligations entails the full range of legal consequences under the law of state responsibility: cessation of the wrongful conduct, assurances and guarantees of non-repetition, and full reparation, in the form of restitution, compensation, and satisfaction, subject to the establishment of a sufficiently direct and certain causal nexus between the wrongful act and the injury, which may be established in the climate context even though climate change results from the cumulative emissions of many states. The obligations concerned are owed *erga omnes*, so that all states have a legal interest in their performance.

Significance. Although advisory opinions are not binding, the opinion represents the most authoritative statement of the law by the principal judicial organ of the United Nations, and will be relied upon in domestic and international litigation, in negotiations, and in the assessment of national energy policies, including decisions to license new fossil fuel production. Its implications for producer states, and for the relationship between climate obligations and investment protection, will be debated for years to come. :::

The Inter-American Court and the European Court of Human Rights

The **Inter-American Court of Human Rights**, in its Advisory Opinion OC-32/25 on the climate emergency and human rights, delivered in 2025 at the request of Chile and Colombia, held that states have obligations under the American Convention on Human Rights to mitigate climate change and to adapt to its effects, applying a standard of enhanced due diligence, and recognised the right to a healthy climate as part of the right to a healthy environment. In *Verein KlimaSeniorinnen Schweiz v Switzerland*, examined in section 18.6, the Grand Chamber of the European Court of Human Rights held in April 2024 that Article 8 of the European Convention on Human Rights encompasses a right to effective protection by the state authorities from the serious adverse effects of climate change on life, health, well-being, and quality of life.

Mitigation Obligations in Practice

Taken together, these developments indicate that states' mitigation obligations involve: an obligation to prepare and implement NDCs reflecting their highest possible ambition, consistent with the 1.5°C goal and their fair share of the

global effort; an obligation to adopt and enforce effective domestic regulatory frameworks; an obligation to regulate private actors, including energy companies, within their jurisdiction; and an obligation to assess the climate effects of projects and policies. The extent to which these obligations constrain particular energy decisions — such as the licensing of new oil and gas fields, the construction of coal plants, or the provision of fossil fuel subsidies — remains contested, and will depend on the application of the due diligence standard to particular facts. Producer states and developing countries have emphasised the principle of CBDR-RC, their right to development, and the need for international finance as conditions for ambitious action.

Adaptation

Adaptation — adjustment in natural or human systems in response to actual or expected climatic effects — has assumed growing importance as the effects of climate change have intensified. Article 7 of the Paris Agreement establishes the global goal on adaptation and requires parties to engage in adaptation planning, including through national adaptation plans, vulnerability assessments, and monitoring. The **UAE Framework for Global Climate Resilience**, adopted in 2023, set targets for adaptation in areas including water, food, health, ecosystems, infrastructure, poverty, and cultural heritage, and work continues on indicators to measure progress.

The energy sector is both a driver of adaptation needs and vulnerable to climate impacts. Hydropower is affected by droughts and changes in river flows, as illustrated by power shortages in Zambia and Zimbabwe resulting from low water levels at the Kariba dam in 2015–2016 and 2024; thermal power plants depend on cooling water; coastal energy infrastructure, including refineries, LNG terminals, and power stations, is exposed to sea level rise and storms; and transmission networks are vulnerable to heat, wildfires, and extreme weather. Legal frameworks increasingly require climate risk assessment in the planning, permitting, and financing of energy infrastructure, and the EIA Directive, as amended, requires assessment of a project's vulnerability to climate change.

Loss and Damage

Loss and damage refers to the adverse effects of climate change that cannot be avoided through mitigation or adaptation, including both extreme weather events and slow-onset events such as sea level rise and desertification. The Warsaw International Mechanism for Loss and Damage was established in 2013, and the Santiago Network, established in 2019, provides technical assistance. At COP27 in 2022, after decades of advocacy by vulnerable developing countries, parties agreed to establish new funding arrangements for loss and damage, including a dedicated fund, which was operationalised at COP28 in 2023, hosted initially

by the World Bank as a financial intermediary fund, with initial pledges of several hundred million dollars — a small fraction of the estimated needs. The relationship between loss and damage and legal liability remains contested; the decision adopting the Paris Agreement excludes Article 8 as a basis for liability or compensation, but the ICJ's advisory opinion confirms that the general law of state responsibility, including the obligation to make reparation, applies to breaches of climate obligations.

Climate Finance

Climate finance — the flow of financial resources for mitigation and adaptation, particularly from developed to developing countries — is central to the bargain of the climate regime. The commitment of developed countries to mobilise USD 100 billion a year by 2020 was, according to the OECD, met only in 2022, two years late, and its accounting — including the counting of loans at face value and the reclassification of existing development aid — has been criticised. The New Collective Quantified Goal agreed in 2024, examined in section 18.2, sets a higher target for 2035. The adequacy, predictability, and accessibility of climate finance are critical to the ability of developing countries to implement the conditional elements of their NDCs, and to the transformation of their energy systems, as examined in Chapter 16 in the context of blended finance and JETPs.

18.5 Fossil Fuel Regulation and Transition Policies

The Policy Toolkit

States use a wide range of legal and policy instruments to reduce emissions from the energy sector: carbon pricing through taxes and emissions trading; regulatory standards for power plants, vehicles, buildings, and appliances; support for renewable energy and low-carbon technologies, examined in Chapters 10 and 12; the phase-out of coal and other high-emitting technologies; restrictions on the supply of fossil fuels; the reform of fossil fuel subsidies; and the regulation of methane, examined in Chapter 17. The choice and design of instruments raises legal questions under constitutional law, administrative law, trade law, and investment law.

Carbon Pricing

Carbon pricing places a price on greenhouse gas emissions, internalising their external costs in accordance with the polluter-pays principle. According to the World Bank, carbon taxes and emissions trading systems now cover roughly a quarter of global emissions, although prices in most systems remain below the levels considered necessary to achieve the Paris goals.

Emissions trading systems (ETS) set a cap on total emissions from covered sectors and issue allowances, which may be traded, so that emissions reductions occur where they are cheapest. The **European Union Emissions Trading**

System, established by Directive 2003/87/EC and operating since 2005, is the largest and longest-established system, covering power generation, energy-intensive industry, aviation within Europe, and, since 2024, maritime transport. After a period of low prices caused by an oversupply of allowances, the introduction of the **Market Stability Reserve** in 2019 and the tightening of the cap under the Fit for 55 package raised prices substantially. A separate system for emissions from buildings and road transport fuels ("ETS2") is due to begin operation later in the decade, and the **Carbon Border Adjustment Mechanism** (Regulation (EU) 2023/956), examined in Chapter 13, applies a carbon price to imports of specified goods to prevent carbon leakage. **China's national ETS**, which began trading in 2021 covering the power sector, is the largest by covered emissions, and is being extended to other sectors. Other systems operate in the United Kingdom, Korea, New Zealand, California, the north-eastern United States, and elsewhere.

Carbon taxes impose a price per tonne of emissions. Sweden introduced a carbon tax in 1991, which now has one of the highest rates in the world. In Africa, **South Africa's Carbon Tax Act 2019** imposes a tax on fossil fuel combustion and industrial process emissions, with substantial allowances in its initial phases, and Nigeria's Climate Change Act provides for the development of carbon tax mechanisms.

Carbon Markets under Article 6

Article 6 of the Paris Agreement provides for voluntary cooperation among parties in implementing their NDCs. Under **Article 6.2**, parties may engage in cooperative approaches involving the use of **internationally transferred mitigation outcomes** (ITMOs) towards their NDCs, subject to guidance on robust accounting, including **corresponding adjustments** to avoid the double counting of emission reductions by both the transferring and acquiring parties. Bilateral agreements under Article 6.2 have been concluded by states such as Switzerland, Japan, Singapore, and Sweden with host countries including Ghana, Senegal, Kenya, and others. **Article 6.4** establishes a centralised mechanism — the **Paris Agreement Crediting Mechanism** — supervised by a body under the CMA, as a successor to the CDM, whose methodological standards and procedures were substantially agreed at Baku in 2024. The **voluntary carbon market**, in which companies purchase credits to offset their emissions, has been subject to criticism about the quality of credits, particularly those from forest conservation projects, leading to initiatives such as the Integrity Council for the Voluntary Carbon Market's Core Carbon Principles (2023). African governments have sought to expand their participation in carbon markets through the **Africa Carbon Markets Initiative**, launched in 2022, while also asserting control over the benefits through legislation, such as Kenya's Climate Change (Amendment) Act 2023 and

regulations in Zimbabwe, Nigeria, and other states, which require benefit-sharing with communities and government.

Coal Phase-Out

Coal is the most carbon-intensive fossil fuel, and its phase-out from electricity generation is a central element of climate policy. The **Powering Past Coal Alliance**, launched by the United Kingdom and Canada in 2017, brings together governments, businesses, and organisations committed to phasing out unabated coal power. The United Kingdom closed its last coal-fired power station, at Ratcliffe-on-Soar, in September 2024, completing a transition from coal providing the majority of electricity within living memory. Germany's **Coal Phase-Out Act** of 2020 provides for the closure of all coal-fired power plants by 2038 at the latest, with compensation for operators of lignite plants and support for affected regions, the compensation for which required approval by the European Commission under the state aid rules. The legal design of coal phase-out — whether through regulation, auctions for compensated closure, carbon pricing, or negotiated agreements — has implications under constitutional property protections and investment treaties, as illustrated by the claims of RWE and Uniper against the Netherlands under the Energy Charter Treaty, examined in Chapter 14.

Supply-Side Measures

Traditionally, climate policy has focused on reducing the demand for fossil fuels. Increasingly, attention has turned to **supply-side** measures that restrict the exploration and production of fossil fuels. France enacted a law in 2017 ending the grant of new exploration permits and providing for the end of hydrocarbon production in its territory by 2040. Denmark decided in 2020 to cancel future licensing rounds and to end oil and gas extraction in the North Sea by 2050, and with Costa Rica launched the **Beyond Oil and Gas Alliance** at COP26 in 2021. New Zealand banned new offshore oil and gas exploration permits in 2018, but legislation enacted in 2025 reversed the ban. A group of states, led by Pacific island nations, has called for the negotiation of a **Fossil Fuel Non-Proliferation Treaty**, which would end new fossil fuel exploration and production and manage a global transition. The International Energy Agency's Net Zero by 2050 roadmap, published in 2021, concluded that no new oil and gas fields would need to be approved for development in its net zero pathway — a finding frequently cited in litigation and policy debates, and contested by producer states and companies.

Supply-side measures raise the problem of "leakage" — production restricted in one country may be replaced by production elsewhere — and concerns about equity, since developing producer states argue that they should not forgo the development of their resources when developed countries built their prosperity

on fossil fuels and continue to produce them. The 2025 advisory opinion of the ICJ, in recognising that the granting of exploration licences may in principle constitute an internationally wrongful act, has given new legal significance to supply-side policy.

Fossil Fuel Subsidies

Fossil fuel subsidies — whether consumer subsidies that reduce the price of fuels below market levels, or producer subsidies such as tax incentives for exploration — encourage consumption and production of fossil fuels, burden public finances, and frequently benefit wealthier households disproportionately. The G20 committed in 2009 to phase out inefficient fossil fuel subsidies over the medium term, and the commitment was repeated in the Glasgow Climate Pact. The IMF has estimated that explicit fossil fuel subsidies — undercharging for supply costs — amounted to more than USD 1 trillion in 2022, and that total subsidies, including the failure to price environmental costs, were several times higher. Reform is politically difficult, as illustrated by protests following attempts to remove fuel subsidies in Nigeria in 2012, Ecuador in 2019, and Kazakhstan in 2022. Nigeria removed its petrol subsidy in May 2023, with significant effects on prices, inflation, and public finances, examined in Chapter 24. Within the World Trade Organization, a group of members has sought to advance disciplines on fossil fuel subsidies, and the WTO Agreement on Fisheries Subsidies illustrates the possibility of trade disciplines on environmentally harmful subsidies.

Industrial Policy and the Energy Transition

The energy transition has been accompanied by a resurgence of industrial policy, in which governments provide large subsidies and incentives for clean energy manufacturing and deployment. The United States **Inflation Reduction Act** of 2022 provided tax credits for renewable energy, electric vehicles, hydrogen, carbon capture, and clean manufacturing, many of which were curtailed or accelerated to expiry by legislation enacted in July 2025. The European Union responded with the Green Deal Industrial Plan and the **Net-Zero Industry Act** (Regulation (EU) 2024/1735), examined in Chapter 12, and China's dominance of clean energy supply chains, examined in Chapter 8, reflects decades of state support. These policies raise questions under the WTO rules on subsidies and local content, examined in Chapter 13, and reflect the intersection of climate policy with geopolitics and energy security, examined in Chapter 25.

18.6 Climate Litigation and Corporate Accountability

The Growth of Climate Litigation

Climate litigation has become a significant driver of climate law and policy. Databases maintained by the Sabin Center for Climate Change Law at Columbia

University and the Grantham Research Institute at the London School of Economics record more than two thousand climate cases worldwide, the majority in the United States but with a rapidly increasing number in other jurisdictions, including in the Global South. This section examines the substantive legal theories on which the principal strands of climate litigation rest, and their implications for states and energy companies. The procedural dimensions of such litigation — standing, justiciability, the separation of powers, evidence, and remedies — are examined in Chapter 27.

Climate cases may be classified by their targets and objectives: **framework cases** against governments, challenging the adequacy of their overall climate policies; **project cases**, challenging the authorisation of particular fossil fuel projects, typically on the basis of environmental assessment or planning law, as in *Finch*, examined in Chapter 17; **corporate cases**, seeking to hold energy companies liable for their contribution to climate change or to require them to reduce their emissions; **disclosure and greenwashing cases**, concerning the accuracy of companies' climate-related statements; and **investor-state cases**, in which investors challenge climate measures, as examined in Chapter 14.

Framework Cases against States

Urgenda. The foundational framework case is *State of the Netherlands v Urgenda Foundation*, in which the Supreme Court of the Netherlands held in December 2019 that the Dutch state was obliged, under Articles 2 and 8 of the European Convention on Human Rights, to reduce greenhouse gas emissions by at least twenty-five per cent below 1990 levels by the end of 2020, on the basis that the state has a positive obligation to take reasonable and appropriate measures to protect its residents against the real and immediate risk of dangerous climate change. The Court rejected the argument that the Netherlands' small share of global emissions relieved it of responsibility, holding that each state is responsible for its share. The decision led the Dutch government to adopt measures including the accelerated closure of a coal-fired power plant.

Neubauer. In *Neubauer v Germany*, the German Federal Constitutional Court held in March 2021 that the Federal Climate Change Act was partly unconstitutional because it did not make adequate provision for emission reductions after 2030, thereby placing a disproportionate burden of reductions on future periods and threatening the future freedoms of the young complainants — an "intertemporal" protection of freedom. The legislature amended the Act within months, bringing forward the target date for climate neutrality to 2045 and increasing the 2030 target.

KlimaSeniorinnen. In *Verein KlimaSeniorinnen Schweiz v Switzerland*, the Grand Chamber of the European Court of Human Rights held in April 2024 that

Switzerland had violated Article 8 of the Convention by failing to adopt an adequate regulatory framework for mitigating climate change, including by failing to quantify a national carbon budget or emission limits, and had violated Article 6 because the domestic courts had not seriously examined the applicants' claims. The Court recognised the standing of an association representing older women particularly affected by heatwaves, while holding the individual applicants' claims inadmissible, and in two parallel cases — *Duarte Agostinho v Portugal and 32 Others* and *Carême v France* — declared the applications inadmissible for lack of exhaustion of domestic remedies and lack of victim status, respectively. The judgment establishes that states have a positive obligation under the Convention to adopt and effectively apply regulations and measures capable of mitigating the existing and potentially irreversible future effects of climate change.

Other jurisdictions. Framework cases have been brought in many other jurisdictions. In *Leghari v Federation of Pakistan* (2015), the Lahore High Court ordered the government to implement its climate policy and established a Climate Change Commission. In *Future Generations v Ministry of the Environment* (2018), the Supreme Court of Colombia ordered the government to address deforestation in the Amazon, recognising the Colombian Amazon as a subject of rights. In the United States, *Juliana v United States*, a constitutional claim by young people, was dismissed for lack of standing by the Court of Appeals for the Ninth Circuit in 2020 and finally ended in 2025, but in *Held v State of Montana*, the Supreme Court of Montana held in December 2024 that provisions of state law preventing the consideration of greenhouse gas emissions in environmental reviews of energy projects violated the right to a clean and healthful environment guaranteed by the Montana Constitution. In Australia, the Full Court of the Federal Court in *Minister for the Environment v Sharma* (2022) reversed a first-instance finding that the environment minister owed a duty of care to young people in approving a coal mine extension, and in 2025 the Federal Court dismissed the negligence claim brought by Torres Strait Islanders in *Pabai v Commonwealth*, while acknowledging the gravity of the harm they faced.

Corporate Cases

Claims against energy companies seek either to require companies to reduce their emissions or to hold them liable for the costs of climate change.

::: casebox {title="Milieudefensie v Royal Dutch Shell plc (District Court of The Hague 2021; Court of Appeal of The Hague 2024)"} **The claim.** Milieudefensie (Friends of the Earth Netherlands), with other organisations and thousands of individual claimants, sued Royal Dutch Shell plc, the parent company of the Shell group, then headquartered in The Hague, seeking an order requiring Shell to reduce the carbon dioxide emissions of the group and of the products it sells. The claim was based on the unwritten standard of care in the Dutch Civil Code,

interpreted in the light of human rights, the Paris Agreement, and soft law instruments, including the UN Guiding Principles on Business and Human Rights.

First instance. In May 2021, the District Court of The Hague held that Shell was obliged, through its corporate policy, to reduce the net carbon dioxide emissions of the Shell group's activities and of the end-use of its products (Scope 1, 2, and 3 emissions) by forty-five per cent by 2030, relative to 2019 levels. The court characterised the obligation as an obligation of result in respect of the group's own emissions and a "significant best-efforts obligation" in respect of the emissions of suppliers and customers. It was the first time a court had ordered a private company to reduce its emissions in line with the Paris Agreement.

Appeal. In November 2024, the Court of Appeal of The Hague overturned the order. The court confirmed that Shell, like other companies, has a duty of care to contribute to the protection of people against dangerous climate change, informed by human rights and the Paris Agreement, and that companies such as Shell, which have contributed substantially to the problem and are able to contribute to its solution, have a special responsibility. However, the court held that it could not determine a specific reduction percentage applicable to Shell, because the forty-five per cent figure was a global average that could not be applied to individual companies or sectors without more specific scientific consensus, and because a court-ordered reduction of Shell's Scope 3 emissions might not be effective if other suppliers filled the gap. The court observed that Shell's investments in new oil and gas fields might be at odds with the energy transition, but did not need to decide that question. A further appeal to the Supreme Court of the Netherlands was lodged.

Significance. Although the order was overturned, the appellate judgment affirms that private companies owe a legally enforceable duty to contribute to climate mitigation, a principle with implications for corporate governance, disclosure, and future litigation, including claims targeting specific investments in new fossil fuel production. The case also illustrates the difficulty of translating global carbon budgets into obligations for individual companies. :::

Liability for climate damage. In *Lliuya v RWE AG*, a Peruvian farmer sued the German energy company RWE, claiming that its historical emissions had contributed to the melting of a glacier above his home town, increasing the risk of a glacial lake outburst flood, and seeking a contribution proportionate to RWE's share of historical global emissions — approximately 0.5 per cent — towards the costs of protective measures. In May 2025, the Higher Regional Court of Hamm dismissed the claim on the facts, finding that the risk to the claimant's property was too low, but accepted in principle that a major emitter could be liable under the German Civil Code for its contribution to climate-related risks. In *Smith v Fonterra Co-operative Group Ltd* [2024] NZSC 5, the Supreme Court of New

Zealand allowed a claim by a Māori elder against seven major New Zealand emitters, including an energy company, in public nuisance, negligence, and a proposed novel tort, to proceed to trial, declining to strike it out.

In the United States, numerous states and municipalities have sued major oil companies, alleging that they knew of the dangers of their products and engaged in campaigns of deception, seeking damages for the costs of adaptation under state tort and consumer protection laws. The companies have sought, largely unsuccessfully, to move these cases to federal courts, where they argue that the claims are displaced by federal law; the Court of Appeals for the Second Circuit held in *City of New York v Chevron Corp* (2021) that claims based on global emissions were displaced by the Clean Air Act and foreign policy considerations, but other courts have allowed state-law deception claims to proceed in state courts. Several states, including Vermont and New York, have enacted "climate superfund" laws imposing liability on major fossil fuel producers for a share of the costs of climate adaptation, which have been challenged by industry groups and the federal government.

Disclosure and Greenwashing

Claims concerning the accuracy of companies' climate-related statements have increased with the growth of mandatory climate disclosure, examined in Chapters 17 and 22. Regulators and private parties have challenged advertising and corporate statements alleged to overstate companies' climate credentials or net zero commitments, under consumer protection and securities laws. In *ClientEarth v Shell plc* [2023] EWHC 1897 (Ch), the High Court refused permission for a shareholder to bring a derivative claim against Shell's directors for alleged breaches of their duties under the Companies Act 2006 in managing climate risk, holding that the directors' management of the company's transition strategy was a matter of commercial judgment with which the court would not interfere on the evidence presented — a decision that illustrates the limits of directors' duty claims, examined in Chapter 22. The European Union's Directive on empowering consumers for the green transition (Directive (EU) 2024/825) prohibits generic environmental claims and claims based on offsetting that are not substantiated.

Assessment

Climate litigation has achieved significant successes, particularly against governments in jurisdictions with strong human rights protections, and has contributed to the development of legal principles — such as the duty of states to protect against climate harm and the duty of care of companies — that are now reflected in the advisory opinions of international courts. Its limitations are also apparent: courts are reluctant to prescribe specific emission reductions for companies or to award damages for climate harm, causation remains a significant obstacle, and

litigation outcomes depend heavily on the legal culture and institutional context of each jurisdiction. For energy companies, climate litigation is a material legal and financial risk, which must be managed through governance, disclosure, and transition planning.

18.7 Just Transition and Distributional Justice

The Concept

The concept of a **just transition** originated in the labour movement in the United States in the 1980s and 1990s, as a demand that workers in polluting industries affected by environmental regulation should receive support. It has since broadened to encompass the fair distribution of the costs and benefits of the transition to a low-carbon economy among workers, communities, regions, and countries. The preamble to the Paris Agreement refers to "the imperatives of a just transition of the workforce and the creation of decent work and quality jobs in accordance with nationally defined development priorities". The **International Labour Organization's Guidelines for a just transition towards environmentally sustainable economies and societies for all** (2015) provide a framework for policies on social dialogue, social protection, skills development, and active labour market policies. The **Solidarity and Just Transition Silesia Declaration**, adopted at COP24 in Katowice in 2018, and the **Just Transition Work Programme**, established at COP27 in 2022, have further developed the concept within the climate regime. The theoretical foundations of energy justice — distributional, procedural, and recognition justice — are examined in Chapter 23; this section focuses on the legal and financial instruments through which just transition is pursued.

Just Transition in Coal Regions

The phase-out of coal raises the most acute just transition challenges, because coal mining and coal-fired power are often concentrated in particular regions, where they provide employment, tax revenues, and community identity. The European Union's **Just Transition Fund**, established by Regulation (EU) 2021/1056, supports regions most affected by the transition, in accordance with territorial just transition plans; the **Social Climate Fund**, established by Regulation (EU) 2023/955, will use revenues from ETS2 to support vulnerable households, micro-enterprises, and transport users. Germany's coal phase-out was accompanied by legislation providing approximately EUR 40 billion in structural support for coal regions, as well as adjustment payments for older workers. In South Africa, where coal provides most electricity and supports substantial employment in Mpumalanga province, a Presidential Climate Commission was established in 2020 to oversee a just transition, and adopted a Just Transition Framework in 2022, and the JETP examined in Chapter 16 aims to finance the transition.

The International Dimension

At the international level, just transition concerns the distribution of the burden of the energy transition between developed and developing countries. Around 600 million people in sub-Saharan Africa lack access to electricity, and many more lack access to clean cooking, as examined in Chapter 23. African governments have argued that their countries, which have contributed least to climate change and have the lowest per capita emissions, should not be denied the opportunity to use their natural gas and other resources to achieve universal energy access and industrialisation. The **African Common Position on Energy Access and Just Energy Transition**, adopted by the African Union Executive Council in 2022, affirms that African countries will continue to deploy all forms of their abundant energy resources, including renewable and non-renewable energy, to address energy demand, and calls for the use of natural gas as a transition fuel, combined with the expansion of renewable energy. This position contrasts with calls for an end to new fossil fuel investment and with the policies of some development finance institutions and banks that have restricted financing for upstream oil and gas, generating debate about "energy colonialism" and the fairness of international climate finance. Nigeria's Energy Transition Plan, with its emphasis on gas as a transition fuel and a net zero target of 2060, reflects this perspective.

Workers, Communities, and Consumers

A just transition also involves the protection of workers in fossil fuel industries through retraining, redeployment, and social protection; the involvement of affected communities in decision-making; the fair distribution of the costs of the transition, including the protection of low-income consumers from increases in energy prices resulting from carbon pricing and subsidy reform; and the fair distribution of the benefits of new energy industries, including through community ownership of renewable energy, local content, and benefit-sharing, as examined in Chapters 10 and 21. The mining of critical minerals for the energy transition raises its own justice concerns, including the rights of communities and workers in mining regions and the risk of reproducing patterns of extraction that have characterised the fossil fuel economy, as examined in Chapter 8.

Legal Instruments

The legal instruments of just transition remain under-developed compared with mitigation instruments. They include national climate framework laws that incorporate just transition principles, such as Scotland's Climate Change (Emissions Reduction Targets) (Scotland) Act 2019, which requires climate policy to be consistent with just transition principles, and South Africa's Climate Change Act 2024; sectoral legislation providing for structural support, retraining, and social protection; the conditions attached to climate finance, including JETPs;

and the incorporation of social criteria in the procurement of renewable energy. The question whether just transition is merely a policy objective or is capable of generating enforceable rights — for example, as an element of the human rights obligations of states, or as a constraint on the design of climate measures — is an emerging area of legal scholarship and litigation.

18.8 Climate Governance and International Legal Reform

The Limits of the Current Architecture

The international climate regime has achieved universal participation, a shared long-term goal, and an architecture for transparency and progressive ambition. Yet global emissions have continued to rise for most of the period since the UNFCCC was adopted, the world remains on a path towards warming well above the Paris goals, and the adverse effects of climate change are intensifying. The gap between commitments and outcomes has generated extensive debate about the reform of climate governance. The principal proposals concern the decision-making processes of the UNFCCC, the integrity of carbon markets, climate finance and the international financial architecture, the role of other international regimes, the governance of fossil fuel production, and the accountability of non-state actors.

Reforming the UNFCCC Process

The consensus rule, examined in section 18.1, allows a small number of states to block or dilute decisions, and has been blamed for weak outcomes on fossil fuels. Proposals for reform include the adoption of rules of procedure permitting majority voting on certain matters, as a last resort; the strengthening of the role of science in negotiations; criteria for the selection of COP hosts and presidencies; rules on conflicts of interest and the participation of fossil fuel industry representatives; and a greater focus on implementation, accountability, and the review of whether states are delivering their commitments. A group of eminent persons convened by the Club of Rome argued in 2024 that the COP process was "no longer fit for purpose" and called for smaller, more frequent, solution-oriented meetings. Others have defended the UNFCCC as the only forum in which all states, including the most vulnerable, have a voice, and argue that its weaknesses reflect the political realities of climate action rather than institutional design.

Minilateralism and Clubs

In parallel with the universal regime, smaller groups of states have formed "clubs" and partnerships to accelerate action in particular sectors. Examples include the **Climate Club** launched by the G7 in 2022 to promote the decarbonisation of industry; the Powering Past Coal Alliance and the Beyond Oil and Gas Alliance, examined in section 18.5; the Global Methane Pledge, examined in Chapter 17;

the Breakthrough Agenda, launched at COP26, which promotes cooperation to make clean technologies affordable and accessible; and the JETPs, examined in Chapter 16. Such initiatives can move faster than the UNFCCC and can align trade, finance, and technology policies among like-minded states, but they risk fragmenting the regime, excluding developing countries from decisions that affect them, and creating trade frictions, as illustrated by the controversy over the EU's CBAM, examined in Chapter 13.

Climate Finance and the International Financial Architecture

The scale of investment required for the energy transition in developing countries, examined in Chapter 16, has led to calls for the reform of the international financial architecture. The **Bridgetown Initiative**, launched by the Prime Minister of Barbados in 2022, proposes measures including the expansion of lending by multilateral development banks, the use of special drawing rights for climate finance, the inclusion of natural disaster clauses in debt instruments, and new international taxes or levies on sectors such as fossil fuels, shipping, and aviation to fund climate action. The World Bank and other multilateral development banks have undertaken reforms to expand their lending capacity and integrate climate considerations, and the IMF established the **Resilience and Sustainability Trust** in 2022 to provide long-term financing for climate resilience and pandemic preparedness. A Global Solidarity Levies Task Force, launched in 2023, has explored options for levies on fossil fuels, aviation, shipping, and financial transactions. The high cost of capital in developing countries — often several times higher than in developed countries, reflecting perceptions of currency, political, and regulatory risk — is a major barrier to renewable energy investment, and reducing it through guarantees, currency risk instruments, and regulatory reform is a central policy objective.

Other International Regimes

Climate change cuts across many areas of international law, and the reform of climate governance involves the integration of climate objectives into other regimes. **International shipping and aviation**, whose emissions are not allocated to national NDCs, are addressed by the International Maritime Organization and the International Civil Aviation Organization. The IMO adopted a revised strategy in 2023 aiming for net zero emissions from international shipping by or around 2050, and in April 2025 its Marine Environment Protection Committee approved a draft "Net-Zero Framework" combining a global fuel standard with greenhouse gas pricing, the formal adoption of which was postponed in October 2025 after opposition from a number of states. ICAO's Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA) requires airlines to offset emissions growth above a baseline. The **trade regime** must accommodate

climate measures such as carbon border adjustments and green subsidies, as examined in Chapter 13. The **investment regime**, as examined in Chapter 14, is being reformed to protect the right to regulate for climate purposes, including through the modernisation of and withdrawals from the Energy Charter Treaty. **Human rights** law, through the decisions examined in section 18.6 and the work of the UN Special Rapporteur on climate change and human rights, increasingly frames climate obligations in terms of rights.

Governing Fossil Fuel Production

The most significant gap in the international climate regime has been its silence, until recently, on the production of fossil fuels. The UNEP **Production Gap Report**, published periodically since 2019 by research institutes and UNEP, has found that governments plan to produce more than double the amount of fossil fuels in 2030 than would be consistent with limiting warming to 1.5°C. The Dubai language on transitioning away from fossil fuels, the ICJ's recognition that fossil fuel licensing and subsidies may engage state responsibility, and the proposal for a Fossil Fuel Non-Proliferation Treaty all point towards greater international attention to supply. Any international framework for the managed decline of fossil fuel production would have to address the equitable allocation of the remaining production among states, taking into account their dependence on fossil fuel revenues, their capacity to diversify, and their historical responsibility — questions that connect climate law with the law of permanent sovereignty over natural resources, examined in Chapter 2, and with the fiscal and economic diversification challenges of producer states examined in Chapter 16.

Non-State Actors and Accountability

Cities, regions, companies, and financial institutions have made a large number of net zero pledges. The UN Secretary-General's High-Level Expert Group on the Net Zero Emissions Commitments of Non-State Entities reported in 2022, recommending that net zero pledges should include interim targets, cover all emissions, avoid reliance on low-quality offsets, and exclude new investment in fossil fuel supply, and that pledges should be subject to transparency and accountability. The **Recognition and Accountability Framework** established by the UN to track non-state commitments, mandatory climate disclosure requirements, examined in Chapter 22, and greenwashing litigation, examined in section 18.6, have increased the accountability of non-state actors, while the withdrawal of a number of financial institutions from voluntary net zero alliances in 2024–2025 illustrated the fragility of voluntary commitments in the face of political pressure.

Geoengineering

As the prospect of exceeding 1.5°C has grown, attention has turned to **geoengineering** — deliberate large-scale intervention in the climate system —

including carbon dioxide removal, examined in Chapter 12, and solar radiation modification, which would reduce warming by reflecting sunlight. Solar radiation modification raises profound governance questions, including the risks of unintended effects, the distribution of benefits and harms, the possibility of unilateral deployment, and the moral hazard of reducing incentives for mitigation. The Conference of the Parties to the Convention on Biological Diversity adopted a decision in 2010 calling for no climate-related geoengineering activities that may affect biodiversity until there is an adequate scientific basis, and the parties to the London Protocol adopted an amendment in 2013 regulating marine geoengineering, including ocean fertilisation. An international framework for the governance of solar radiation modification does not exist, and proposals range from a non-use agreement to frameworks for research governance.

Conclusion

Climate change law has evolved from a framework convention with weak commitments, through the binding but narrow Kyoto Protocol, to the universal but largely procedural Paris Agreement, and is now being reshaped by the advisory opinions of international courts, national framework laws, carbon markets, and litigation. For energy law, the consequences are profound: the licensing of fossil fuel production, the design of electricity markets, the allocation of investment risk, the content of contracts, and the duties of company directors are all being reconfigured by the imperative of decarbonisation. At the same time, the transition raises fundamental questions of justice — between developed and developing countries, between producer and consumer states, and between present and future generations — that the law has only begun to address.

::: exercise {title="Practical Exercise 18: Drafting a Climate Change and Energy Transition Bill (HYPOTHETICAL)"} **This exercise is entirely hypothetical. The state, institutions, and facts described are fictitious and do not refer to any real country.**

The Republic of Tamara is a lower-middle-income country with a population of forty million. It produces 250,000 barrels of oil per day, mostly for export, and petroleum revenues account for half of government revenue. It has substantial undeveloped gas reserves. Sixty per cent of the population has access to electricity, which is generated principally from diesel and heavy fuel oil, and gas flaring is widespread. Tamara is highly vulnerable to coastal flooding and drought. Its updated NDC commits to an unconditional reduction of fifteen per cent below business-as-usual emissions by 2030, and a conditional reduction of forty per cent.

The government has decided to enact a Climate Change and Energy Transition Act and has asked you to prepare the first draft. The Minister of Petroleum insists

that the Act must not prevent the development of the country's gas reserves or the award of new oil licences. The Minister of Finance is concerned about the fiscal effects of a carbon tax and of the removal of fuel subsidies. Civil society organisations demand a ban on routine gas flaring within three years, a just transition for oil workers, and the right of citizens to challenge government decisions that are inconsistent with the country's climate commitments.

Tasks.

(a) Prepare a structure for the Act, setting out its parts and principal provisions, including the long-term target, carbon budgets, institutional arrangements, and accountability mechanisms.

(b) Draft the provisions of the Act dealing with (i) the assessment of the climate effects of new petroleum licences and projects, including downstream emissions, and (ii) the elimination of routine gas flaring.

(c) Draft a provision on just transition, addressing workers, communities, and low-income consumers.

(d) Advise the government on how the Act should be designed to be consistent with the obligations identified in the ICJ's 2025 advisory opinion while preserving its policy space to develop its gas reserves, and on the risks of investment treaty claims by existing licence holders.

(e) Prepare a short memorandum on the options for using Article 6 of the Paris Agreement to finance the conditional element of Tamara's NDC, identifying the legal safeguards required. :::

CHAPTER SUMMARY

- The UNFCCC (1992) establishes the objective of preventing dangerous anthropogenic interference with the climate system and the principles of equity, CBDR-RC, and precaution, with institutions operating by consensus.
- The Kyoto Protocol imposed binding targets on developed countries only; the Paris Agreement combines universal participation, nationally determined contributions, binding procedural obligations, and a five-yearly ambition cycle, with a temperature goal of well below 2°C and efforts towards 1.5°C.
- NDCs are in substance energy policy documents, but in aggregate fall far short of the Paris goals; national framework laws, including Nigeria's Climate Change Act 2021, establish targets, carbon budgets, and institutions.
- The advisory opinions of ITLOS (2024), the ICJ (2025), and the Inter-American Court (2025) confirm that states have stringent due diligence obligations to prevent climate harm, that the climate treaties do not displace general international law, and that fossil fuel production, licensing, and subsidies may engage state responsibility.

- Transition policies include carbon pricing, Article 6 carbon markets, coal phase-out, supply-side restrictions, subsidy reform, and industrial policy, each raising legal issues under domestic, trade, and investment law.
- Climate litigation against states (*Urgenda*, *Neubauer*, *KlimaSeniorinnen*) and companies (*Milieudefensie v Shell*, *Lliuya v RWE*) has established duties to mitigate, although courts are cautious about prescribing company-specific reductions or awarding damages.
- Just transition addresses the fair distribution of the costs and benefits of decarbonisation among workers, communities, regions, and countries, including the energy access needs of Africa.
- Reform proposals address UNFCCC decision-making, climate finance and the international financial architecture, the integration of climate objectives into other regimes, the governance of fossil fuel production, and non-state accountability.

DOCTORAL RESEARCH QUESTIONS

1. Assess the legal significance of the ICJ's 2025 advisory opinion for the licensing of new oil and gas production by developing producer states such as Nigeria.
2. Is the "hybrid" architecture of the Paris Agreement capable of delivering the emission reductions required to meet its temperature goal? Evaluate the effectiveness of the ambition cycle in the light of the first global stocktake.
3. To what extent can the principle of common but differentiated responsibilities and respective capabilities justify differentiated obligations with respect to fossil fuel production?
4. Compare the approaches of the Dutch courts in *Urgenda* and *Milieudefensie v Shell* to the translation of global carbon budgets into obligations for states and companies.
5. How should Article 6 of the Paris Agreement be implemented in African states to ensure environmental integrity, benefit-sharing with communities, and contribution to national climate goals?
6. Design a legal framework for a just transition in a petroleum-dependent economy, addressing workers, communities, public finances, and international support.

FURTHER READING

- Daniel Bodansky, Jutta Brunnée and Lavanya Rajamani, *International Climate Change Law* (Oxford University Press 2017).
- Benoit Mayer, *The International Law on Climate Change* (Cambridge University Press 2018).

- Lavanya Rajamani and Jacqueline Peel (eds), *The Oxford Handbook of International Environmental Law* (2nd edn, Oxford University Press 2021).
- Meinhard Doelle and Sara Seck (eds), *Research Handbook on Climate Change Law and Loss and Damage* (Edward Elgar 2021).
- Jacqueline Peel and Hari M Osofsky, *Climate Change Litigation: Regulatory Pathways to Cleaner Energy* (Cambridge University Press 2015).
- Intergovernmental Panel on Climate Change, *Climate Change 2023: Synthesis Report* (IPCC 2023).
- United Nations Environment Programme, *Emissions Gap Report* (annual).

19 CHAPTER

Water Resources Law and Transboundary Environmental Governance

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the legal status of freshwater resources under international and national law, and the principal doctrines of water rights.
2. Analyse the development and content of the law of international watercourses, including the 1997 UN Watercourses Convention and the UNECE Water Convention.
3. Apply the principle of equitable and reasonable utilisation and the obligation not to cause significant harm to disputes over shared rivers.
4. Evaluate the institutional and treaty arrangements for transboundary water cooperation, with particular reference to African river basins.
5. Assess the legal framework governing hydropower development in shared river basins, including the Nile, the Niger, the Indus, and the Mekong.
6. Explain the emerging international law of transboundary aquifers and the national regulation of groundwater.
7. Analyse the human right to water and sanitation and its relationship with water allocation for energy and extractive uses.
8. Evaluate the mechanisms for the prevention and resolution of international water disputes.

Introduction

Water and energy are inseparable. Water is used to generate electricity directly through hydropower, which supplies around one-sixth of global electricity and is the largest source of renewable electricity, and indirectly as a coolant in thermal and nuclear power stations, which are among the largest users of freshwater in many countries. Water is essential to the extraction and processing of petroleum

and minerals — in drilling, hydraulic fracturing, enhanced oil recovery, ore processing, and tailings management — and to the production of biofuels and green hydrogen. Conversely, energy is required to pump, treat, desalinate, and distribute water. This "water-energy nexus" means that decisions about energy are frequently decisions about water, and that the law governing water resources is an essential part of the legal framework of energy and natural resources.

Freshwater is also a quintessentially shared resource. There are more than three hundred international river basins and a comparable number of transboundary aquifers, shared by two or more states, which together account for a large share of the world's freshwater flows and are home to a large proportion of the world's population. In Africa, nearly every major river — the Nile, the Niger, the Congo, the Zambezi, the Senegal, the Orange-Senqu, the Volta, and the Lake Chad basin — is shared by several states. The development of hydropower and other water uses in one state can therefore profoundly affect the interests of others, and disputes over shared waters, including those concerning large dams, are among the most persistent sources of tension in international relations. Climate change, which is altering rainfall patterns, increasing the frequency of droughts and floods, and reducing glacier-fed flows, is intensifying these pressures.

This chapter examines the legal status of freshwater resources (section 19.1), the law of international watercourses (section 19.2), the principle of equitable and reasonable utilisation (section 19.3), transboundary water cooperation (section 19.4), hydropower development and river basin management (section 19.5), groundwater and aquifer governance (section 19.6), water rights, human rights, and public access (section 19.7), and the resolution of international water disputes (section 19.8). Chapter 10 has examined hydropower as a renewable energy technology, including its licensing and social impacts at the project level; the present chapter focuses on the transboundary and basin-level dimensions of hydropower and on the law of water resources more generally.

	Core content	Sources and authorities
Equitable & reasonable utilisation	Each state entitled to an equitable share; all factors weighed; vital human needs	UNWC Arts 5, 6, 10; Gabčíkovo-Nagymaros; Helsinki Rules
No significant harm	Due diligence to prevent harm; mitigation; compensation where appropriate	UNWC Art 7; Trail Smelter; Pulp Mills
Notification & consultation	Prior notice of planned measures; six-month reply; good-faith negotiation	UNWC Arts 11–19; Lac Lanoux; Silala
Cooperation & data exchange	Joint bodies; regular exchange of hydrological data; flood warning	UNWC Arts 8–9; Water Convention; OMVS; NBA
Ecosystem protection	Environmental flows; pollution control; emergencies	UNWC Arts 20–28; Kishenganga awards

Figure 19.1 The core principles of international watercourse law. *Source:* prepared for this edition.

19.1 Legal Status of Freshwater Resources

Water as a Resource and as a Right

Freshwater occupies a distinctive position among natural resources. Like petroleum and minerals, it is a resource that can be allocated, used, and, to some extent, traded, and its use for hydropower, irrigation, industry, and mining generates substantial economic value. Unlike most other resources, however, it is essential to life, to human health and dignity, and to ecosystems; it is renewable but finite, and its availability varies greatly across space and time; and it flows across boundaries, linking upstream and downstream users in relationships of interdependence. These characteristics have shaped the legal regimes that govern it, which must reconcile the treatment of water as an economic good with its recognition as a human right and an essential component of the environment.

Sovereignty over Shared Waters

In international law, the question of the rights of states over shared waters has been answered by several competing theories.

Absolute territorial sovereignty, associated with the "Harmon Doctrine" — named after an opinion of the United States Attorney General in 1895 concerning the diversion of the Rio Grande, which asserted that the United States

had the right to use the waters within its territory without regard to the effects on Mexico — holds that a state may do as it wishes with waters within its territory, regardless of the effects on other states. This position, which favours upstream states, has been rejected in state practice and judicial decisions, and the United States itself did not adhere to it in its subsequent treaty practice with Mexico and Canada.

Absolute territorial integrity holds that a downstream state is entitled to the continued natural flow of a river, undiminished in quantity and quality, by upstream states. This position, which favours downstream states and would effectively prohibit upstream development, has likewise been rejected, although elements of it are reflected in the claims of downstream states to "historic" or "acquired" rights.

Limited territorial sovereignty — the position that has prevailed in modern international law — holds that each state has the right to use the waters within its territory, but must exercise that right in a manner that respects the corresponding rights of other riparian states. This position is expressed in the principles of equitable and reasonable utilisation and the obligation not to cause significant harm, examined in section 19.3.

A further approach, the **community of interest** in shared waters, was articulated by the Permanent Court of International Justice in the *Territorial Jurisdiction of the International Commission of the River Oder* case (1929), in which the Court stated that the community of interest in a navigable river becomes the basis of a common legal right, the essential features of which are the perfect equality of all riparian states in the use of the whole course of the river and the exclusion of any preferential privilege of any one riparian state in relation to the others. The International Court of Justice endorsed this statement in the *Gabčíkovo-Nagymaros* case and extended it to non-navigational uses. The community of interest approach underpins the integrated management of river basins through joint institutions, examined in section 19.4.

National Ownership and Control of Water

Within states, legal systems have adopted different approaches to the ownership and control of water. In many civil law and post-colonial systems, water resources are declared to be public property or vested in the state. In Nigeria, the **Water Resources Act** of 1993 vests the right to the use and control of all surface and groundwater, and of any watercourse affecting more than one state, in the Federal Government, and provides for the licensing of water uses by the federal authorities, subject to exceptions for domestic and certain other uses. The constitutional distribution of competences over water between the federal government and the states, and the role of river basin development authorities established under

federal legislation, have complicated water governance, and successive attempts since 2017 to enact a comprehensive National Water Resources Bill, intended to establish a regulatory commission and a framework for integrated water resources management, have been controversial, owing in part to concerns among some states about federal control of water resources.

South Africa's **National Water Act** of 1998, adopted after the end of apartheid, abolished the system of private water rights linked to riparian land ownership and declared the national government to be the public trustee of the nation's water resources. It established a **Reserve** — comprising a basic human needs reserve and an ecological reserve — which has priority over all other uses, and requires all other significant uses to be authorised by licence. The Act is frequently cited as a model of water law reform based on principles of equity, sustainability, and public trusteeship.

Doctrines of Water Rights

In common law jurisdictions, rights to use surface water have traditionally been based on two principal doctrines. Under the **riparian doctrine**, derived from English common law and applied in the eastern United States and other common law jurisdictions, owners of land adjoining a watercourse have the right to make reasonable use of the water, subject to the equal rights of other riparian owners. Under the **prior appropriation doctrine**, developed in the arid western United States during the nineteenth century to meet the needs of miners and farmers, water rights are acquired by diverting water and putting it to beneficial use, and rights are ranked by priority in time — "first in time, first in right" — so that in times of shortage, senior rights holders are entitled to their full allocation before junior rights holders receive any water. The prior appropriation doctrine was originally developed in mining camps in California, and illustrates the historical connection between water law and the extractive industries.

Many jurisdictions have replaced or supplemented these doctrines with **administrative permit systems**, under which the use of water above specified thresholds requires a permit issued by a public authority, which may specify the quantity, purpose, location, and duration of the use, and may be revised in response to changing conditions. Permit systems allow governments to allocate water in accordance with policy priorities, to protect environmental flows, and to respond to scarcity, but raise questions about the security of rights required for investment, including in hydropower and mining.

The Public Trust Doctrine

The **public trust doctrine** holds that certain natural resources, including navigable waters, are held by the state in trust for the public and cannot be alienated in a manner that would defeat the public interest. Originating in Roman law

and English common law, the doctrine was applied by the United States Supreme Court in *Illinois Central Railroad Co v Illinois* (1892) and extended by the Supreme Court of California in *National Audubon Society v Superior Court* (1983), the Mono Lake case, to require the state to reconsider water diversions by the City of Los Angeles that were damaging a lake ecosystem. The Supreme Court of India adopted the doctrine in *MC Mehta v Kamal Nath* (1997), holding that the state, as trustee of natural resources, could not grant a lease to a private company that interfered with the natural flow of a river. The doctrine has been incorporated into the constitutions and water legislation of several states, including South Africa and Kenya, and provides a basis for judicial review of decisions allocating water to energy and extractive projects.

19.2 International Watercourse Law

Historical Development

International law relating to rivers developed first in relation to **navigation**. The Final Act of the Congress of Vienna of 1815 proclaimed the principle of freedom of navigation on international rivers for riparian states, and international commissions were established for the Rhine (1815, reorganised by the Mannheim Convention of 1868) and the Danube (1856), which are among the oldest international organisations. The Barcelona Convention and Statute on the Regime of Navigable Waterways of International Concern of 1921 sought to generalise these principles. During the twentieth century, as the use of rivers for hydropower, irrigation, and industry grew, attention shifted to **non-navigational uses**, which are the principal subject of modern watercourse law.

The **Helsinki Rules on the Uses of the Waters of International Rivers**, adopted by the International Law Association in 1966, articulated the principle of equitable utilisation of the waters of an international drainage basin and identified factors relevant to its application. The ILA's **Berlin Rules on Water Resources** (2004) sought to update the Helsinki Rules in the light of developments in international environmental law and human rights, giving greater emphasis to the obligation not to cause significant harm, ecological integrity, and the human right to water, although they were controversial among some members of the Association.

The UN Watercourses Convention

In 1970, the UN General Assembly requested the International Law Commission to study the law of the non-navigational uses of international watercourses. After more than two decades of work, the Commission adopted draft articles in 1994, on the basis of which the General Assembly adopted the **Convention on the Law of the Non-Navigational Uses of International Watercourses** in 1997, by a vote of 103 to 3 with 27 abstentions. The states voting against were Burundi,

China, and Turkey — all significant upstream states — and the abstentions included Egypt, Ethiopia, France, India, and Pakistan. The Convention entered into force in 2014, after the deposit of the thirty-fifth instrument of ratification, and has some forty parties, including Nigeria, which acceded in 2010, and several other West African states, but not most of the major upstream states.

The Convention applies to **international watercourses** — systems of surface waters and groundwaters constituting by virtue of their physical relationship a unitary whole and normally flowing into a common terminus, parts of which are situated in different states. It is a framework convention, which does not displace existing watercourse agreements and encourages states to conclude agreements applying and adjusting its provisions to the characteristics of particular watercourses (Article 3). Its principal provisions are:

- the principle of **equitable and reasonable utilisation and participation** (Article 5), with the factors relevant to its application (Article 6), examined in section 19.3;
- the **obligation not to cause significant harm** (Article 7), requiring watercourse states to take all appropriate measures to prevent the causing of significant harm to other watercourse states, and, where harm is caused, to take measures to eliminate or mitigate it and, where appropriate, to discuss compensation;
- the **general obligation to cooperate** (Article 8) and the regular exchange of data and information (Article 9);
- the principle that no use enjoys inherent **priority** over other uses, subject to special regard to the requirements of vital human needs (Article 10);
- **procedural obligations** relating to **planned measures** that may have a significant adverse effect on other watercourse states (Articles 11–19), including prior notification, a six-month period for reply during which the notifying state may not implement the measures without the consent of the notified states, consultation and negotiation, and procedures in the absence of notification;
- obligations to **protect and preserve ecosystems** (Article 20), to prevent, reduce, and control pollution (Article 21), and to take measures concerning harmful conditions and emergencies (Articles 27–28);
- provisions on the protection of watercourses and installations in time of armed conflict (Article 29); and
- **dispute settlement** (Article 33), including a compulsory fact-finding procedure at the request of any party, and optional submission to the International Court of Justice or arbitration.

Although the Convention has attracted relatively limited participation, many of its core provisions are regarded as reflecting customary international law, as confirmed by the International Court of Justice in *Gabčíkovo-Nagymaros* (which referred to the draft articles before the Convention was adopted) and in the *Silala* case (*Chile v Bolivia*, 2022), in which the parties agreed, and the Court accepted, that the customary principles of equitable and reasonable utilisation, prevention of significant harm, and notification and consultation applied to the Silala, a watercourse that rises in Bolivia and flows into Chile.

The UNECE Water Convention

The **Convention on the Protection and Use of Transboundary Watercourses and International Lakes**, adopted under the auspices of the UN Economic Commission for Europe at Helsinki in 1992 (the "Water Convention"), emphasises the prevention, control, and reduction of transboundary impacts, and requires riparian parties to conclude bilateral or multilateral agreements and to establish joint bodies for the management of shared waters. Its **Protocol on Water and Health** (1999) addresses access to safe drinking water and sanitation. Amendments that entered into force in 2013 opened the Convention to accession by all UN member states, and since 2016 a number of states outside the UNECE region, including Chad, Senegal, Ghana, Togo, Guinea-Bissau, Cameroon, and others in Africa, have acceded, making it, alongside the UN Watercourses Convention, a global framework for transboundary water cooperation. The two conventions are regarded as compatible and complementary, and the secretariats cooperate in promoting both.

Regional and Basin Agreements

The most detailed rules governing shared waters are found in regional and basin agreements. In southern Africa, the **SADC Revised Protocol on Shared Watercourses** (2000) closely follows the UN Watercourses Convention and provides the framework for basin agreements such as the agreement establishing the Zambezi Watercourse Commission (ZAMCOM, 2004). In West Africa, the **Water Charter of the Niger Basin** (2008) and the conventions establishing the Senegal River Basin Development Organisation (OMVS) and the Volta Basin Authority provide detailed frameworks for cooperation, examined in section 19.4. In Europe, the **Water Framework Directive** (Directive 2000/60/EC) requires the management of water at the level of river basin districts, including international districts, with the objective of achieving good status for all waters. Hundreds of bilateral treaties govern specific rivers, lakes, and projects.

19.3 Equitable and Reasonable Utilization

The Principle

The principle of **equitable and reasonable utilisation** is the cornerstone of international watercourse law. Article 5(1) of the UN Watercourses Convention provides that watercourse states shall in their respective territories utilise an international watercourse in an equitable and reasonable manner, and in particular shall use and develop it with a view to attaining optimal and sustainable utilisation thereof and benefits therefrom, taking into account the interests of the watercourse states concerned, consistent with adequate protection of the watercourse. Article 5(2) adds the principle of **equitable participation**: watercourse states shall participate in the use, development, and protection of an international watercourse in an equitable and reasonable manner, including both the right to utilise the watercourse and the duty to cooperate in its protection and development.

The principle does not require an equal division of waters, but an equitable one, determined by reference to all relevant factors. It confers on each riparian state a right to an equitable and reasonable share of the uses and benefits of the watercourse, and correspondingly limits each state's use to its equitable share. The application of the principle is inherently contextual, and its indeterminacy is both its strength — allowing it to accommodate the diversity of watercourses and the evolution of needs — and its weakness, since it provides limited guidance to states in dispute.

Relevant Factors

Article 6(1) of the Convention lists factors and circumstances relevant to equitable and reasonable utilisation:

- geographic, hydrographic, hydrological, climatic, ecological, and other factors of a natural character;
- the social and economic needs of the watercourse states concerned;
- the population dependent on the watercourse in each watercourse state;
- the effects of the use or uses of the watercourse in one watercourse state on other watercourse states;
- existing and potential uses of the watercourse;
- conservation, protection, development, and economy of use of the water resources of the watercourse and the costs of measures taken to that effect; and
- the availability of alternatives, of comparable value, to a particular planned or existing use.

Article 6(3) provides that the weight to be given to each factor is to be determined by its importance in comparison with that of other relevant factors, and that in determining what is a reasonable and equitable use, all relevant factors are to be considered together and a conclusion reached on the basis of the whole. Article 10 provides that, in the absence of agreement or custom to the contrary, no use of an international watercourse enjoys inherent priority over other uses, but that in the event of a conflict, special regard is to be given to the requirements of **vital human needs** — understood as sufficient water to sustain human life, including drinking water and water for the production of food to prevent starvation.

The inclusion of "existing and potential uses" among the factors is significant for disputes between states that have historically developed a river — often downstream states — and states that seek to develop it for the first time, often upstream. Existing uses are relevant but not decisive; they do not confer an absolute right to the continuation of historic shares, but must be weighed against the potential uses of other states. This issue lies at the heart of the dispute over the Nile, examined in section 19.5.

The Relationship with the No-Harm Rule

The relationship between the principle of equitable utilisation and the obligation not to cause significant harm, in Article 7, was the most contentious issue in the negotiation of the Convention. Downstream states generally favoured the primacy of the no-harm rule, which protects existing uses against interference, while upstream states favoured the primacy of equitable utilisation, which permits new uses that may cause harm to others if they are equitable. The compromise reached in the Convention subordinates the no-harm rule to equitable utilisation in substance, while preserving its procedural significance: Article 7(1) requires watercourse states to take "all appropriate measures" to prevent the causing of significant harm — an obligation of due diligence rather than an absolute prohibition — and Article 7(2) provides that where significant harm is nevertheless caused, the state whose use causes it shall, in consultation with the affected state, take all appropriate measures, having due regard to Articles 5 and 6, to eliminate or mitigate the harm and, where appropriate, to discuss the question of compensation. The prevailing interpretation is that a use causing significant harm may nevertheless be permissible if it is equitable and reasonable in the light of all relevant factors, but that the causing of significant harm is a factor weighing heavily against the equitableness of a use, and that harm to vital human needs is particularly difficult to justify.

Equitable Utilisation in the Jurisprudence

The leading judicial application of the principle is the decision of the International Court of Justice in the *Gabčíkovo-Nagymaros* case, which concerned a project for the joint development of hydropower on the Danube.

::: casebox {title="Gabčíkovo-Nagymaros Project (Hungary/Slovakia) [1997] ICJ Rep 7"} **Background.** In 1977, Hungary and Czechoslovakia concluded a treaty providing for the construction and joint operation of a system of locks, comprising a barrage and hydropower plant at Gabčíkovo in Czechoslovakia and another at Nagymaros in Hungary, on the stretch of the Danube forming their common border. The project was designed to produce hydroelectricity, improve navigation, and protect against floods. In the late 1980s, amid growing environmental concerns and political change, Hungary suspended and then abandoned work on its part of the project. Czechoslovakia proceeded with a unilateral alternative, known as "Variant C", which involved the diversion of the Danube into a bypass canal on Czechoslovak territory, putting it into operation in 1992 and diverting the majority of the river's flow. Hungary purported to terminate the treaty in 1992. After the dissolution of Czechoslovakia, Slovakia succeeded to its rights and obligations.

The decision. The Court held that Hungary was not entitled to suspend and abandon the works, rejecting its plea of a "state of ecological necessity", because the alleged environmental perils were not sufficiently certain and imminent and Hungary had other means of responding to them. The Court also held that Hungary's notification of termination did not terminate the treaty, rejecting arguments based on necessity, impossibility of performance, fundamental change of circumstances, material breach, and the development of new norms of environmental law. Conversely, the Court held that Czechoslovakia was entitled to proceed with preparatory works for Variant C, but not to put it into operation, because by unilaterally assuming control of a shared resource and depriving Hungary of its right to an equitable and reasonable share of the natural resources of the Danube, Czechoslovakia had failed to respect the proportionality required by international law. Referring to the community of interest in an international river and to the draft articles of the International Law Commission, the Court described Hungary's right to an equitable and reasonable share as a basic right.

Sustainable development and evolving norms. The Court observed that, owing to new scientific insights and a growing awareness of the risks for mankind of interventions in nature, new norms and standards had been developed, which must be taken into consideration and given proper weight not only when states contemplate new activities but also when continuing activities begun in the past; this need to reconcile economic development with protection of the environment, it said, is aptly expressed in the concept of sustainable development. The

Court held that the treaty remained in force, and directed the parties to negotiate in good faith, taking account of the objectives of the treaty and the norms of international environmental law, to find an agreed solution, including on the environmental consequences of the operation of the Gabčíkovo plant.

Significance. The judgment remains the leading authority on equitable utilisation, the community of interest in international rivers, the stability of treaties governing energy infrastructure in the face of changed circumstances, and the relevance of evolving environmental norms to long-term projects. Notably, the parties have not, more than a quarter of a century later, reached a full agreement implementing the judgment, illustrating the limits of judicial settlement in complex water disputes. :::

The principle of equitable utilisation has also been applied in arbitral decisions, including the *Lac Lanoux* arbitration, examined in Chapter 17, and the *Indus Waters Kishenganga* arbitration, examined in section 19.5, and in state practice through the allocation of waters in basin treaties.

19.4 Transboundary Water Cooperation

The Duty to Cooperate

Article 8 of the UN Watercourses Convention provides that watercourse states shall cooperate on the basis of sovereign equality, territorial integrity, mutual benefit, and good faith in order to attain optimal utilisation and adequate protection of an international watercourse, and that they may consider the establishment of joint mechanisms or commissions. Cooperation is both a legal obligation and a practical necessity: the optimal development of a river basin, including the siting of dams where they will produce the most power with the least harm, often requires that states plan jointly rather than individually, and the management of floods, droughts, and pollution requires the exchange of data and coordinated responses. The duty to cooperate is given content by the procedural obligations of notification, consultation, and the exchange of information, examined in Chapter 17, and by the establishment of joint institutions.

River Basin Organisations

Joint institutions for the management of shared waters — river basin organisations (RBOs) — vary widely in their mandates and powers. At one end of the spectrum are bodies with limited functions, such as the exchange of data and the coordination of national plans; at the other are organisations with powers to plan, build, own, and operate infrastructure, and to allocate waters among member states. Prominent examples include:

- the **International Joint Commission**, established by the Boundary Waters Treaty of 1909 between the United States and the United Kingdom (on

behalf of Canada), which approves projects affecting boundary waters and investigates matters referred to it by the governments, and is often cited as a model of long-standing bilateral cooperation;

- the **International Commission for the Protection of the Rhine** and the **International Commission for the Protection of the Danube River**, which coordinate environmental protection in two of Europe's most heavily used rivers;
- the **Mekong River Commission**, established by the 1995 Agreement on the Cooperation for the Sustainable Development of the Mekong River Basin among Cambodia, Laos, Thailand, and Viet Nam, whose effectiveness is limited by the non-membership of China, the upstream state, and by the weakness of its procedures for prior consultation on mainstream dams, as illustrated by the construction of the Xayaburi dam in Laos despite the objections of downstream states; and
- the **Nile Basin Initiative**, launched in 1999 as a transitional partnership among Nile riparian states pending the conclusion of a permanent legal and institutional framework, examined in section 19.5.

African River Basin Organisations

Africa has a rich tradition of basin institutions, reflecting the large number of shared rivers on the continent.

The Senegal River Basin Development Organisation (Organisation pour la mise en valeur du fleuve Sénégal, OMVS), established in 1972 by Mali, Mauritania, and Senegal, and joined by Guinea in 2006, is frequently cited as one of the most advanced examples of basin cooperation in the world. Under the conventions governing the OMVS, the Senegal River is declared an international river, and common infrastructure — including the Manantali dam and hydropower plant in Mali and the Diama anti-salt barrage near the river's mouth — is the **joint property** of the member states, which share the costs and benefits according to an agreed formula, including the allocation of electricity from Manantali among Mali, Mauritania, and Senegal. The **Senegal River Water Charter** (2002) establishes principles for the allocation of water among uses, including the priority of drinking water, and procedures for approving new projects. The OMVS model illustrates the concept of **benefit-sharing** — the allocation among states of the benefits derived from the use of a river, rather than of the water itself — which can facilitate cooperation by allowing states to focus on the optimal development of the basin as a whole.

The Niger Basin Authority (NBA), established by the Convention Creating the Niger Basin Authority of 1980 as successor to the Niger River Commission of 1964, comprises nine member states: Benin, Burkina Faso, Cameroon, Chad,

Côte d'Ivoire, Guinea, Mali, Niger, and Nigeria. Its **Water Charter**, adopted in 2008, establishes principles of equitable and reasonable utilisation, the obligation not to cause significant harm, prior notification of planned measures, environmental protection, and the participation of water users, and provides for the allocation of water among uses and states, the maintenance of minimum flows, and the joint review of new infrastructure. The NBA has developed a Shared Vision and Sustainable Development Action Plan identifying priority infrastructure, including dams such as Kandadji in Niger, Taoussa in Mali, and Fomi in Guinea, whose effects on downstream flows — including the inflows to Nigeria's Kainji and Jebba hydropower plants and the Inner Niger Delta wetland in Mali — require coordinated management.

The Lake Chad Basin Commission (LCBC), established in 1964 by Cameroon, Chad, Niger, and Nigeria, later joined by the Central African Republic and Libya, addresses a basin in which the surface area of the lake has shrunk dramatically since the 1960s, owing to drought, climate variability, and increased water abstraction, with severe consequences for livelihoods and contributing to insecurity in the region. The LCBC adopted a Water Charter in 2012, and has studied the controversial "Transaqua" proposal for an inter-basin transfer of water from the Congo basin to replenish the lake.

The Zambezi River Authority, established by Zambia and Zimbabwe in 1987, owns and operates the Kariba dam on their common border, while the hydropower stations on each bank are owned by the national utilities, and the **Zambezi Watercourse Commission** (ZAMCOM), whose agreement was concluded in 2004 and entered into force in 2011, provides a framework for basin-wide cooperation among the eight Zambezi riparian states. The **Orange-Senqu River Commission** (ORASECOM), established in 2000 by Botswana, Lesotho, Namibia, and South Africa, and the bilateral treaty of 1986 between Lesotho and South Africa establishing the **Lesotho Highlands Water Project**, under which water is transferred from the highlands of Lesotho to South Africa's industrial heartland in return for royalties and hydropower for Lesotho, illustrate further models of cooperation.

Data Sharing, Flood Management, and Dam Operations

The exchange of data and information, required by Article 9 of the UN Watercourses Convention, is a fundamental element of cooperation, particularly in relation to the operation of dams. The release of water from upstream reservoirs during periods of heavy rainfall can cause severe flooding downstream, and the timely notification of releases is critical to allow downstream authorities to warn and evacuate populations. Nigeria has experienced severe floods, notably in 2012 and 2022, which were exacerbated by releases of water from the **Lagdo dam** on the Benue River in Cameroon, during periods of exceptional rainfall, causing

loss of life, the displacement of large numbers of people, and extensive damage to property and agriculture. The floods have led to calls for improved notification and coordination between Cameroon and Nigeria, including through the Nigeria Hydrological Services Agency and the Niger Basin Authority, and for the construction of the long-planned Dasin Hausa dam in Nigeria to buffer releases from Lagdo. The case illustrates the importance of procedural obligations of notification and the exchange of data, and of agreed operating rules for dams on shared rivers.

Benefit-Sharing and the Political Economy of Cooperation

Cooperation over shared waters is shaped by asymmetries of power, geography, and development. Upstream states are often in a stronger position to develop rivers, but downstream states may be more powerful militarily or economically, as in the cases of Egypt on the Nile and India on certain rivers shared with Nepal and Bangladesh. The concept of benefit-sharing has been promoted as a means of moving beyond zero-sum disputes over water quantities, by identifying the benefits that cooperative development can generate — hydropower, irrigation, flood control, navigation, environmental protection — and negotiating their distribution, including through the trade of electricity across borders. The development of regional power pools, such as the West African Power Pool and the Southern African Power Pool, examined in Chapter 9, creates opportunities for benefit-sharing from hydropower, since electricity generated in one state can be exported to others.

19.5 Hydropower Development and River Basin Management

Hydropower in Shared Basins

Hydropower is the form of energy most directly governed by the law of shared waters. Large hydropower projects on international rivers alter the timing and volume of flows downstream, trap sediment, block fish migration, affect water quality, and change the risk of floods and droughts, and their reservoirs may inundate land in neighbouring states. They may also provide significant benefits to downstream states, including flood control, regulated dry-season flows, and electricity for export. The law governing hydropower in shared basins therefore draws on all the principles examined in this chapter: equitable and reasonable utilisation, the obligation not to cause significant harm, prior notification and consultation, environmental impact assessment, and cooperation through joint institutions. Chapter 10 has examined the project-level regulation of hydropower, including licensing, dam safety, resettlement, and the recommendations of the World Commission on Dams; this section examines hydropower at the level of the river basin, through four major basins that illustrate the legal issues.

The Nile

The Nile basin, shared by eleven states, is the setting of the most prominent contemporary dispute over hydropower on a shared river. The legal regime has its origins in the colonial period. Under the **1929 Exchange of Notes** between the United Kingdom (on behalf of Sudan and its East African territories) and Egypt, Egypt's "natural and historic rights" to the waters of the Nile were recognised, and no irrigation or power works were to be constructed on the Nile or its tributaries in territories under British administration that would reduce the volume of water reaching Egypt, without Egypt's agreement. Under the **1959 Agreement for the Full Utilization of the Nile Waters** between Egypt and Sudan, concluded in connection with the construction of the Aswan High Dam, the two states allocated the entire average annual flow of the Nile, as measured at Aswan, between themselves — 55.5 billion cubic metres to Egypt and 18.5 billion cubic metres to Sudan — with no allocation to the upstream states, which were not parties. Egypt has relied on these agreements and on its historic uses, while upstream states, including Ethiopia — the source of the Blue Nile, which contributes the majority of the Nile's flow at Aswan — reject them as colonial instruments to which they are not bound, invoking the "clean slate" principle of state succession and the principle of equitable utilisation.

The **Nile Basin Initiative**, launched in 1999, sought to negotiate a permanent framework. The resulting **Cooperative Framework Agreement** (CFA), opened for signature at Entebbe in 2010, establishes principles of equitable and reasonable utilisation and the obligation not to cause significant harm, and would establish a Nile River Basin Commission. Egypt and Sudan refused to sign, objecting to the omission of a provision protecting their existing uses and rights, and to the replacement of the obligation of prior notification by a weaker formulation. The CFA entered into force in October 2024 following its ratification by a sixth state, South Sudan, among upstream riparians, but without the participation of the downstream states, so that the basin now has a framework agreement that binds only part of the basin.

Ethiopia began construction of the **Grand Ethiopian Renaissance Dam** (GERD) on the Blue Nile near the border with Sudan in 2011. With an installed capacity of more than five thousand megawatts, it is the largest hydropower plant in Africa, intended to supply electricity to Ethiopia, where a large proportion of the population lacked access, and for export to neighbouring states. Egypt regards the dam as an existential threat to its water security, particularly during the filling of the reservoir and in periods of drought, while Sudan has expressed concern about dam safety and the coordination of operations with its own dams, while acknowledging potential benefits from regulated flows and reduced sedimentation. In 2015, the three states signed a **Declaration of Principles** at Khartoum,

which affirmed principles of cooperation, equitable and reasonable utilisation, and the obligation not to cause significant harm, and committed the parties to agree guidelines and rules on the first filling and annual operation of the dam. Negotiations, facilitated at various times by the United States, the World Bank, and the African Union, failed to produce a binding agreement, and Ethiopia proceeded to fill the reservoir in stages from 2020. The United Nations Security Council considered the matter in 2020 and 2021, encouraging the parties to resume negotiations under the auspices of the African Union. Ethiopia formally inaugurated the dam in 2025. The dispute illustrates the difficulty of reconciling historic uses with the development needs of upstream states, the importance of procedural obligations and agreed operating rules, and the limits of the existing legal framework in the absence of a basin-wide agreement and a mechanism for binding dispute settlement.

The Indus

The Indus basin, shared principally by India and Pakistan, is governed by the **Indus Waters Treaty** of 1960, concluded with the assistance of the World Bank, which is also a signatory for certain purposes. The treaty allocates the waters of the three "eastern rivers" (the Ravi, Beas, and Sutlej) to India for unrestricted use, and the waters of the three "western rivers" (the Indus, Jhelum, and Chenab) to Pakistan, subject to India's right to make limited uses of the western rivers, including the generation of hydroelectric power through run-of-river plants meeting detailed design criteria set out in an annex. The treaty establishes a Permanent Indus Commission, with a commissioner from each state, and a graduated dispute settlement mechanism: "questions" are addressed by the Commission, "differences" of a technical nature may be referred to a **neutral expert**, and "disputes" may be referred to a **court of arbitration**. The treaty survived several wars between the two states and was long regarded as one of the most successful water treaties.

Several disputes over Indian hydropower projects on the western rivers have been resolved under the treaty's mechanisms. In 2007, a neutral expert largely upheld the design of India's Baglihar project on the Chenab, subject to modifications. The *Kishenganga* arbitration addressed a project involving the diversion of water between tributaries of the Jhelum.

::: casebox {title="Indus Waters Kishenganga Arbitration (Pakistan v India), Partial Award (2013) and Final Award (2013)"} **Background.** India planned the Kishenganga Hydro-Electric Project, which would divert water from the Kishenganga (known in Pakistan as the Neelum) river, a tributary of the Jhelum, through a tunnel to a power station on another tributary, before returning it to the Jhelum. Pakistan objected that the diversion would reduce flows in the Neelum, interfering with its own planned Neelum-Jhelum hydroelectric project downstream and

causing environmental harm, and that India's proposed technique of drawing down the reservoir below dead storage level for sediment flushing was prohibited by the treaty. Pakistan initiated arbitration in 2010 before a Court of Arbitration constituted under the treaty, administered by the Permanent Court of Arbitration.

Partial Award. In its Partial Award of February 2013, the Court held that the project was a run-of-river plant permitted by the treaty, and that India was entitled to divert water for power generation, but that India was under an obligation to construct and operate the project in such a way as to maintain a minimum flow in the Kishenganga/Neelum downstream of the diversion, in accordance with the treaty and with customary international environmental law. Referring to *Pulp Mills* and *Gabčíkovo-Nagymaros*, the Court held that states are required under contemporary customary international law to take environmental protection into consideration when planning and developing projects that may cause injury to a bordering state, and that the treaty should be interpreted in the light of such principles. The Court also held that India could not use drawdown flushing for sediment control at the project, and that the prohibition applied to future run-of-river plants on the western rivers.

Final Award. In its Final Award of December 2013, the Court determined that India must release a minimum flow of nine cubic metres per second into the Kishenganga/Neelum downstream of the diversion at all times, balancing India's right to generate hydropower against the need to mitigate adverse effects on Pakistan's hydroelectric use and on the environment, and providing for reconsideration of the minimum flow after seven years.

Significance. The awards demonstrate the integration of customary environmental law into the interpretation of a water treaty concluded before the emergence of modern environmental norms, the use of environmental flows as a legal tool, and the capacity of treaty-based arbitration to resolve technical disputes over hydropower on shared rivers. :::

Subsequent disputes over the Kishenganga and Ratle projects led, from 2016, to parallel proceedings, with India requesting the appointment of a neutral expert and Pakistan the constitution of a court of arbitration, and the World Bank eventually appointing both. India declined to participate in the court of arbitration. In April 2025, following a terrorist attack in Indian-administered Kashmir, India announced that it would hold the treaty "in abeyance", a step without basis in the treaty's provisions, which do not provide for unilateral suspension; the court of arbitration subsequently issued an award holding that the treaty could not be suspended unilaterally and that its own competence was unaffected. The developments placed in jeopardy one of the most durable water treaties in the world and illustrate the vulnerability of water cooperation to wider political conflict.

The Mekong

The Mekong rises on the Tibetan Plateau in China, where it is known as the Lancang, and flows through Myanmar, Laos, Thailand, Cambodia, and Viet Nam. China has constructed a cascade of large dams on the upper Mekong, and Laos — which aspires to become the "battery of South-East Asia" by exporting hydropower to Thailand and other neighbours — has developed numerous dams on the lower mainstream and tributaries, including the Xayaburi and Don Sahong dams. The 1995 Mekong Agreement requires prior consultation for mainstream projects during the dry season, through the Procedures for Notification, Prior Consultation and Agreement, but does not give other members a veto, and the consultations on the Xayaburi and Don Sahong projects were widely regarded as having failed to address the concerns of downstream states and communities about effects on fisheries — which provide the principal source of protein for millions of people — and on sediment flows to the Mekong Delta in Viet Nam. China, which is not a member of the Mekong River Commission, established the Lancang-Mekong Cooperation framework in 2016, and has agreed to share year-round hydrological data. The Mekong illustrates the challenges of governing cumulative effects from multiple dams in a basin where the most powerful state is upstream and outside the principal basin institution.

The Niger, the Congo, and the Zambezi

In Nigeria, the **Kainji**, **Jebba**, and **Shiroro** hydropower stations on the Niger and Kaduna rivers, constructed between the 1960s and 1990s, were privatised under concession arrangements in 2013. The **Zungeru** hydropower project on the Kaduna river, with an installed capacity of 700 MW, built with Chinese financing, began generating in 2023 and has been placed under a private concession. The proposed **Mambilla** hydropower project on the Donga river in Taraba State, with a planned capacity of over 3,000 MW, has been delayed for decades and became the subject of an ICC arbitration brought by a project developer against the Federal Government. The performance of Nigeria's hydropower stations depends on flows from upstream states in the Niger basin, and the planned dams in Guinea, Mali, and Niger, examined in section 19.4, will affect them; the coordination of reservoir operations through the Niger Basin Authority is therefore of direct importance to Nigeria's electricity supply.

The **Congo** River, with the second largest discharge of any river in the world, has enormous hydropower potential, concentrated at the Inga Falls in the Democratic Republic of Congo, where the proposed **Grand Inga** scheme could provide tens of thousands of megawatts. The DRC and South Africa concluded a treaty in 2013 on the development of the Grand Inga project, under which South Africa would purchase a significant share of the power from the first phase, Inga 3, but the project has faced repeated delays owing to its cost, governance concerns, and

the withdrawal of the World Bank in 2016. On the **Zambezi**, the Kariba dam, operated by the Zambezi River Authority, has required extensive rehabilitation of its plunge pool to ensure its structural safety, and severe droughts in 2015–2016, 2019, and 2024 reduced generation at Kariba and Cahora Bassa, causing prolonged power cuts in Zambia and Zimbabwe — illustrating the vulnerability of hydropower-dependent electricity systems to climate variability.

Basin-Level Planning and Environmental Flows

The experience of these basins suggests several principles for the governance of hydropower at the basin level. First, **cumulative impact assessment** and strategic planning at the basin level are necessary to identify the combination of projects that will maximise benefits and minimise harm, since project-by-project assessment cannot capture cumulative effects. Second, **environmental flows** — the quantity, timing, and quality of water flows required to sustain freshwater ecosystems and the livelihoods that depend on them — should be defined and protected, as recognised in the *Kishenganga* awards and in the Brisbane Declaration on environmental flows (2007, revised 2018). Third, agreed **operating rules** for dams, particularly in periods of drought and flood and during reservoir filling, are essential to avoid disputes. Fourth, **dam safety** requires robust regulation and transboundary cooperation, as tragically illustrated by the collapse of two dams above the city of Derna in Libya during a storm in September 2023, which killed thousands of people. Fifth, financing institutions play a significant role: the World Bank's operational policy on Projects on International Waterways requires that, before financing a project on an international waterway, the Bank must ensure that the riparian states have been notified and have had an opportunity to raise objections, and similar requirements are applied by other development banks, giving financiers influence over the observance of procedural obligations. Finally, the **Hydropower Sustainability Standard**, launched in 2021 by the International Hydropower Association with the participation of industry and civil society, provides a certification scheme for the environmental, social, and governance performance of hydropower projects.

19.6 Groundwater Regulation and Aquifer Governance

The Importance of Groundwater

Groundwater constitutes the vast majority of the world's liquid freshwater, and provides drinking water for around half of the world's population and a substantial share of water for irrigation. In many arid and semi-arid regions, including large parts of Africa and the Middle East, groundwater is the principal or only reliable source of water. Groundwater is used extensively in the energy and extractive sectors: for cooling, for hydraulic fracturing and enhanced oil recovery, for mineral processing, and, increasingly, for the production of green hydrogen

in water-scarce regions. Mining and petroleum operations can also contaminate aquifers and lower water tables through dewatering, and the injection of fluids and the storage of carbon dioxide in geological formations, examined in Chapter 12, require the protection of overlying aquifers.

Groundwater presents particular governance challenges. Aquifers are invisible and poorly understood; their recharge may be very slow, so that many aquifers in arid regions contain "fossil" water that is effectively non-renewable; over-abstraction can cause irreversible depletion, land subsidence, and saline intrusion; and pollution can persist for decades. The **International Groundwater Resources Assessment Centre** has identified several hundred transboundary aquifers.

International Law of Transboundary Aquifers

The UN Watercourses Convention applies to groundwater only where it forms part of a system of surface waters and groundwaters constituting a unitary whole and flowing into a common terminus; it does not apply to "confined" aquifers unrelated to surface waters. To fill this gap, the International Law Commission adopted **Draft Articles on the Law of Transboundary Aquifers** in 2008, which the General Assembly took note of in Resolution 63/124, commending them to states for guidance in bilateral and regional arrangements, while deferring the question of their final form. The Draft Articles affirm the sovereignty of each aquifer state over the portion of a transboundary aquifer located within its territory, to be exercised in accordance with international law; establish the principles of equitable and reasonable utilisation — with specific attention to the recharge of the aquifer and, for non-recharging aquifers, to maximising long-term benefits — and the obligation not to cause significant harm; and set out obligations of cooperation, the exchange of data, the protection of recharge and discharge zones, the prevention of pollution, monitoring, and prior notification of planned activities. The emphasis on sovereignty reflected the concerns of states and was controversial among some commentators, who favoured a stronger emphasis on the shared nature of the resource. The UNECE **Model Provisions on Transboundary Groundwaters** (2012) provide additional guidance.

Aquifer Agreements and Cooperation

Formal agreements on transboundary aquifers remain few. The **Guarani Aquifer Agreement**, concluded in 2010 by Argentina, Brazil, Paraguay, and Uruguay, which entered into force in 2020, applies principles derived from the ILC Draft Articles to one of the world's largest aquifers. In North Africa, Chad, Egypt, Libya, and Sudan established a Joint Authority for the Study and Development of the **Nubian Sandstone Aquifer System** in the early 1990s, and later adopted a regional strategy for its management; the system, a largely non-renewable fossil aquifer, supplies Libya's "Great Man-Made River" project, one

of the largest water transfer projects in the world. Algeria, Libya, and Tunisia established a consultation mechanism for the **North Western Sahara Aquifer System**. In West Africa, the states sharing the **Iullemeden** and associated aquifer systems, including Mali, Niger, and Nigeria, have cooperated in assessing the aquifers through projects coordinated by the Sahara and Sahel Observatory. In the *Silala* case, the International Court of Justice addressed a watercourse with a significant groundwater component, confirming the application of customary principles to such systems.

National Groundwater Regulation

At the national level, groundwater was historically treated as an incident of land ownership in many legal systems, with landowners entitled to abstract water beneath their land with few restrictions — a regime that encouraged over-abstraction. Most modern legal systems now subject significant groundwater abstraction to permits, require the registration of boreholes, and provide for the designation of protected zones. In Nigeria, the Water Resources Act vests control of groundwater in the Federal Government, and the **Nigeria Hydrological Services Agency**, established in 2010, is responsible for the assessment of surface and groundwater resources, but the regulation of borehole drilling, which is widespread owing to the inadequacy of public water supply, remains weak.

In the energy and extractive sectors, specific regulations address the protection of groundwater. In the United Kingdom, the Infrastructure Act 2015 prohibited hydraulic fracturing in protected groundwater source areas; in the United States, the exemption of hydraulic fracturing from the Safe Drinking Water Act's underground injection control program, enacted in 2005, has been controversial, and states regulate well construction, fluid disclosure, and wastewater disposal. Mining legislation and the EIA process typically require the assessment of effects on groundwater, the management of dewatering and acid mine drainage, and long-term monitoring after closure. The protection of groundwater in carbon storage regulation, including under the EU CCS Directive, is examined in Chapter 12.

19.7 Water Rights, Human Rights, and Public Access

The Human Right to Water and Sanitation

Access to safe drinking water and sanitation is fundamental to human life, health, and dignity, yet according to the World Health Organization and UNICEF, around two billion people lack safely managed drinking water services and several billion lack safely managed sanitation. The international recognition of a human right to water has developed progressively. Although the International Covenant on Economic, Social and Cultural Rights (1966) does not mention water expressly, the Committee on Economic, Social and Cultural Rights, in its **General Comment No 15** (2002), interpreted the rights to an adequate standard

of living (Article 11) and to health (Article 12) as including the right to water, defined as the right of everyone to sufficient, safe, acceptable, physically accessible, and affordable water for personal and domestic uses. Water is expressly mentioned in the Convention on the Elimination of All Forms of Discrimination against Women (Article 14(2)(h)), the Convention on the Rights of the Child (Article 24(2)(c)), and the Convention on the Rights of Persons with Disabilities (Article 28(2)(a)). In 2010, the UN General Assembly adopted **Resolution 64/292**, recognising the right to safe and clean drinking water and sanitation as a human right that is essential for the full enjoyment of life and all human rights, by a vote of 122 to none with 41 abstentions, and the Human Rights Council affirmed the right in Resolution 15/9 later that year. Since 2015, the General Assembly and the Human Rights Council have recognised the rights to water and to sanitation as distinct rights. Sustainable Development Goal 6 aims to ensure availability and sustainable management of water and sanitation for all by 2030.

The right entails obligations to **respect** (refraining from interfering with existing access, for example through arbitrary disconnections or the pollution of water sources), to **protect** (preventing third parties, including companies, from interfering with the right), and to **fulfil** (taking steps, to the maximum of available resources, to achieve progressively the full realisation of the right). General Comment No 15 identifies a minimum core obligation to ensure access to the minimum essential amount of water sufficient for personal and domestic uses and to prevent disease, and emphasises that priority in the allocation of water must be given to the right to water for personal and domestic uses, and to the water resources required to prevent starvation and disease.

Constitutional Rights to Water

Numerous states have recognised the right to water in their constitutions. Section 27(1)(b) of the **Constitution of South Africa** provides that everyone has the right to have access to sufficient water, and requires the state to take reasonable legislative and other measures, within its available resources, to achieve the progressive realisation of the right. In *Mazibuko v City of Johannesburg* [2009] ZACC 28, the Constitutional Court upheld the City's policy of providing a free basic water supply of six kilolitres per household per month and the installation of prepaid water meters in Soweto, holding that the right did not entitle the applicants to a specific quantity of water and that the City's policy was reasonable, while emphasising the importance of the right and the obligation of the state to review its policies — a decision criticised by some commentators for its deference to the executive. Article 43(1)(d) of the **Constitution of Kenya** 2010 guarantees the right to clean and safe water in adequate quantities, and the constitutions of Uganda, Ecuador, Bolivia, Uruguay, and many other states contain comparable provisions. In Nigeria, the Constitution does not expressly guarantee a right to

water, but the Fundamental Objectives and Directive Principles of State Policy direct the state to ensure suitable and adequate shelter, food, and other necessities, and courts and commentators have argued that access to water is an aspect of the rights to life and dignity, as examined in Chapter 17 in relation to environmental rights.

Water for Energy and the Right to Water

The allocation of water to energy and extractive uses may conflict with the human right to water, particularly in water-scarce regions. Mining and petroleum operations may compete with communities for limited water resources, contaminate drinking water sources, or lower water tables; hydropower reservoirs may displace communities and alter downstream flows on which riverine populations depend; and large-scale green hydrogen production, which requires substantial quantities of water for electrolysis, has raised concerns in arid regions, including in Namibia, Morocco, and parts of the Middle East, where proposals for desalination have been advanced to avoid competition with domestic and agricultural uses. The UN Special Rapporteur on the human rights to safe drinking water and sanitation has reported on the effects of extractive industries and large infrastructure on the rights to water and sanitation, emphasising the priority of personal and domestic uses, the need for human rights impact assessment, and the participation of affected communities. In Chile, the extensive use of water by the copper and lithium mining industries in the arid north has been contested by indigenous communities and farmers, and reforms of the Water Code in 2022 established that water is a national public good for public use, prioritised human consumption, and limited the duration of new water rights. The **UN Guiding Principles on Business and Human Rights**, examined in Chapter 21, require companies to respect the right to water, including through due diligence on their water use and pollution.

Indigenous Peoples and Water

Indigenous peoples frequently have special relationships with water bodies, which may be of spiritual, cultural, and economic significance. The **UN Declaration on the Rights of Indigenous Peoples** (2007) recognises the right of indigenous peoples to maintain and strengthen their distinctive spiritual relationship with their traditionally owned or otherwise occupied and used lands, territories, waters, and coastal seas (Article 25), and requires states to consult and cooperate in good faith with indigenous peoples to obtain their free and informed consent prior to the approval of any project affecting their lands or territories and other resources, particularly in connection with the development, utilisation, or exploitation of mineral, water, or other resources (Article 32(2)). In some jurisdictions, rivers have been recognised as legal persons, in part to give effect to in-

digenous relationships with them: New Zealand's **Te Awa Tupua (Whanganui River Claims Settlement) Act 2017** recognised the Whanganui River as a legal person, with guardians appointed by the Crown and the Whanganui iwi, and courts in Colombia, India, and Bangladesh have recognised rivers as subjects of rights. The opposition of the Standing Rock Sioux Tribe to the Dakota Access oil pipeline in the United States, on the ground that it threatened the tribe's water supply and sacred sites, illustrated the intersection of indigenous rights, water, and energy infrastructure; in 2020, a federal court vacated an easement for the pipeline for failure to prepare an adequate environmental impact statement, although the pipeline continued to operate. Free, prior, and informed consent is examined in detail in Chapter 21.

Public Access and Participation

The governance of water resources increasingly emphasises public participation and access to information, in line with Principle 10 of the Rio Declaration and the Aarhus and Escazú Conventions, examined in Chapter 17. **Integrated water resources management** (IWRM), endorsed by the Dublin Statement on Water and Sustainable Development (1992), promotes the coordinated development and management of water, land, and related resources, and the participation of users, planners, and policy-makers at all levels, including through catchment management agencies and water user associations. The Dublin Statement's principle that water has an economic value in all its competing uses, and should be recognised as an economic good, has been influential but controversial, particularly where it has been associated with the privatisation of water services and the introduction of cost-recovery tariffs. The tension between water as an economic good and water as a human right — reflected in disputes over water privatisation in Cochabamba, Bolivia, in 2000, and in investment arbitrations arising from water concessions in Argentina and Tanzania, examined in Chapter 14 — remains a central theme of water governance.

19.8 International Water Dispute Resolution

The Nature of Water Disputes

Disputes over shared waters arise from competing claims to the quantity of water, from the effects of infrastructure such as dams and diversions, from pollution, from the operation of reservoirs, and from boundaries defined by rivers. They are frequently embedded in broader political relationships and may be linked to questions of national security, food security, and energy security. Despite the frequent prediction of "water wars", research has found that interstate cooperation over shared waters has historically been far more common than violent conflict, and that states have concluded hundreds of water treaties. Nevertheless, water

disputes can be protracted and bitter, and the prevention and resolution of such disputes is a central function of international water law.

Treaty-Based Mechanisms

Water treaties commonly provide graduated mechanisms for dispute resolution, beginning with consultation and negotiation within joint commissions, and proceeding to more formal mechanisms. The **Indus Waters Treaty**, examined in section 19.5, provides a three-tier system distinguishing questions, differences, and disputes, with recourse to a neutral expert or a court of arbitration. Article 33 of the **UN Watercourses Convention** provides that, if the parties cannot reach agreement by negotiation, they may jointly seek the good offices of, or request mediation or conciliation by, a third party, or submit the dispute to arbitration or the International Court of Justice; if, after six months, they have not settled the dispute, any party may require the establishment of an impartial **fact-finding commission**, whose report the parties must consider in good faith. Parties may declare their acceptance of the compulsory jurisdiction of the ICJ or arbitration. Basin agreements in Africa, including the Niger Basin Water Charter and the SADC Revised Protocol, provide for dispute resolution within their institutions, with recourse to regional courts or arbitration; the SADC Protocol provides for disputes to be referred to the SADC Tribunal, whose functioning was suspended in 2010 and subsequently restricted to disputes between states.

International Courts and Tribunals

The International Court of Justice has decided several water disputes: the *Gabčíkovo-Nagymaros* case (1997), examined in section 19.3; the *Kasikili/Sedudu Island* case (*Botswana/Namibia*, 1999), concerning a boundary defined by the main channel of the Chobe River; *Pulp Mills* (2010), examined in Chapter 17; the *Certain Activities* and *Construction of a Road* cases (2015) concerning the San Juan River; the *Dispute regarding Navigational and Related Rights* (*Costa Rica v Nicaragua*, 2009); and the *Silala* case (2022), in which the Court found that many of the parties' claims had become moot because their positions had converged during the proceedings, while confirming the applicable customary principles. Arbitral tribunals, often administered by the Permanent Court of Arbitration, have decided disputes including *Lac Lanoux* (1957), the *Indus Waters Kishenganga* arbitration (2013), and the arbitration concerning the *Iron Rhine* railway (*Belgium/Netherlands*, 2005), which applied principles of environmental law to the reactivation of a railway through a nature reserve.

Diplomatic and Facilitated Processes

Many water disputes are addressed through diplomatic and facilitated processes rather than adjudication. The World Bank played a central role in the negotiation of the Indus Waters Treaty and in the appointment of the neutral expert and court

of arbitration under it; the African Union, the United States, and the World Bank facilitated negotiations on the GERD; and the Nile Basin Initiative provided a forum for dialogue among Nile riparians. Basin organisations, such as the OMVS and the Niger Basin Authority, provide institutional settings for the prevention of disputes through joint planning, data sharing, and the joint review of proposed projects. Mediation and conciliation, which allow parties to explore solutions based on benefit-sharing rather than legal entitlements alone, are well suited to water disputes, in which the parties' long-term relationship and interdependence make purely adversarial outcomes unattractive.

Water and Investment Disputes

Water disputes also arise in investor-state arbitration, where investors in water concessions or hydropower projects have challenged government measures affecting their investments. Disputes arising from the privatisation of water and sanitation services in Argentina, including *Azurix Corp v Argentina* (2006), *Suez, Sociedad General de Aguas de Barcelona SA and Vivendi Universal SA v Argentina* (2010), and *Urbaser SA v Argentina* (2016) — in which the tribunal accepted that it had jurisdiction over a counterclaim by Argentina based on the human right to water, although it rejected the counterclaim on the merits — and *Biwater Gauff (Tanzania) Ltd v Tanzania* (2008), illustrate the interaction between investment protection and the human right to water, examined in Chapter 14. Hydropower investors have brought claims arising from tariff disputes, delays, and terminations, and the *Mambilla* arbitration in Nigeria, examined in section 19.5, illustrates the significance of commercial arbitration in hydropower development.

Water, Armed Conflict, and Infrastructure

The protection of water resources and installations in armed conflict is addressed by international humanitarian law, including Article 54 of Additional Protocol I to the Geneva Conventions (1977), which prohibits attacking, destroying, removing, or rendering useless objects indispensable to the survival of the civilian population, including drinking water installations and supplies and irrigation works, and Article 56, which provides special protection for works and installations containing dangerous forces, namely dams, dykes, and nuclear electrical generating stations. The destruction of the **Kakhovka dam** on the Dnipro River in Ukraine in June 2023, during the Russian occupation of the area, caused extensive flooding, loss of life, environmental damage, and the loss of water supply for irrigation and for the cooling of the Zaporizhzhia nuclear power plant, illustrating the catastrophic consequences of attacks on water and energy infrastructure. The ILC's Principles on the Protection of the Environment in Relation to Armed Conflicts (2022) and the **Geneva List of Principles on the Protection**

of Water Infrastructure (2019) provide further guidance. The protection of energy infrastructure in armed conflict is examined further in Chapter 25.

Conclusion

International water law has developed a coherent set of principles — equitable and reasonable utilisation, the obligation not to cause significant harm, prior notification and consultation, environmental protection, and cooperation — which are reflected in the UN Watercourses Convention, the UNECE Water Convention, and customary international law. Its effectiveness depends on their application through basin agreements and institutions, and on the willingness of states to cooperate. For energy law, water law matters because hydropower, thermal generation, petroleum, mining, and hydrogen production all depend on water, and because the development of shared rivers for hydropower is among the most consequential and contentious decisions that states make. In a warming world, in which the availability of water is becoming less predictable, the importance of effective legal frameworks for shared waters will only increase.

::: exercise {title="Practical Exercise 19: Simulated Arbitration over a Dam on a Shared River (HYPOTHETICAL)"} **This exercise is entirely hypothetical. The states, river, institutions, projects, and evidence described are fictitious and do not refer to any real countries or disputes.**

The Sarana River rises in the highlands of the Republic of Kesh, flows through the Republic of Valoria, and enters the sea in the Federal Republic of Ondo-Mare. All three states are parties to the UN Watercourses Convention. Kesh and Ondo-Mare are also parties to a 1962 bilateral agreement, concluded before Valoria's independence, which allocates seventy per cent of the river's average annual flow to Ondo-Mare and thirty per cent to Kesh, without any allocation to the territory that became Valoria. Valoria has never acceded to the 1962 agreement. The three states are members of the Sarana Basin Commission, established in 2005, whose charter requires prior notification of planned measures and the joint review of major projects, but provides no binding dispute settlement procedure.

Valoria, where only thirty-five per cent of the population has access to electricity, has begun constructing the Valoria Unity Dam, a 3,200 MW hydropower project with a reservoir that will take between four and nine years to fill, depending on the filling strategy and rainfall. Valoria notified the Commission of the project after construction had begun, provided a summary environmental assessment, and has declined to share its detailed reservoir operation plans, citing national security. Ondo-Mare, which depends on the Sarana for ninety per cent of its freshwater and irrigates large areas of farmland in the river's delta, claims that rapid filling would reduce its water supply by up to twenty-five per cent in drought years and threaten the delta's ecosystem. Kesh is planning its own

irrigation scheme upstream. Valoria has offered to sell electricity from the dam to Ondo-Mare at concessional prices.

The three states have agreed, by special agreement, to submit to an arbitral tribunal the questions whether Valoria has breached its obligations under international law in respect of the dam and, if so, what consequences follow; and to request the tribunal to indicate principles for the filling and operation of the reservoir.

Tasks. Divide into teams representing Valoria, Ondo-Mare, and Kesh, and a tribunal of three members.

(a) Each party team should prepare a written memorial (of no more than 3,000 words) addressing: the legal effect of the 1962 agreement on Valoria; the application of equitable and reasonable utilisation and the no-harm rule; Valoria's compliance with procedural obligations; and the remedies sought.

(b) Each party should identify the expert evidence it would adduce (hydrological, ecological, economic) and prepare questions for the cross-examination of the opposing parties' experts.

(c) The tribunal should conduct a two-hour oral hearing and deliver a reasoned award, addressing in particular whether a failure of prior notification renders the project unlawful, and indicating principles for the filling and annual operation of the reservoir, including in drought years.

(d) All participants should reflect, in a short note, on whether a benefit-sharing arrangement — involving electricity trade, joint operation of the dam, or compensation — would provide a better solution than adjudication, and how it might be negotiated. :::

CHAPTER SUMMARY

- Freshwater is both an economic resource and a human right; modern international law has rejected absolute territorial sovereignty and absolute territorial integrity in favour of limited territorial sovereignty and a community of interest in shared waters.
- National legal systems govern water through state ownership or public trusteeship, riparian and prior appropriation doctrines, administrative permits, and the public trust doctrine; Nigeria's Water Resources Act 1993 vests control of water in the Federal Government.
- The UN Watercourses Convention (1997, in force 2014), to which Nigeria is a party, codifies equitable and reasonable utilisation, the obligation not to cause significant harm, cooperation, and the procedural obligations relating to planned measures, much of which reflects customary law; the UNECE Water Convention is now open globally.

- Equitable and reasonable utilisation is determined by weighing all relevant factors, with special regard for vital human needs; *Gabčíkovo-Nagymaros* remains the leading authority.
- River basin organisations range from advisory bodies to institutions owning joint infrastructure, such as the OMVS; the Niger Basin Authority and the Lake Chad Basin Commission are central to West African water governance, and the Lagdo dam floods illustrate the importance of notification and data sharing.
- Hydropower disputes in the Nile, Indus, and Mekong basins illustrate the tensions between historic uses and new development, the significance of procedural obligations and environmental flows, and the vulnerability of cooperation to political conflict.
- The ILC Draft Articles on Transboundary Aquifers and a small number of aquifer agreements address groundwater, which is critical in arid regions and affected by energy and extractive activities.
- The human right to water, recognised in 2010, requires priority for personal and domestic uses and constrains allocations to energy and extractive uses; indigenous relationships with water and public participation are increasingly recognised.
- Water disputes are addressed through treaty mechanisms, international courts and tribunals, facilitated diplomacy, and investment arbitration, while international humanitarian law protects water and energy infrastructure in armed conflict.

DOCTORAL RESEARCH QUESTIONS

1. Has the principle of equitable and reasonable utilisation provided adequate guidance for the resolution of disputes over hydropower on shared rivers? Evaluate with reference to the Nile and the Indus.
2. Assess the legal effect of colonial-era water agreements on post-colonial upstream states, with reference to the 1929 and 1959 Nile agreements and the law of state succession.
3. To what extent can the benefit-sharing model of the Senegal River Basin Development Organisation be replicated in the Niger basin to optimise hydropower development and protect downstream users, including Nigeria?
4. Evaluate the adequacy of the legal and institutional framework for managing the transboundary flood risks associated with the Lagdo dam on the Benue River.
5. How should the human right to water be balanced against the water requirements of mining, petroleum, and green hydrogen projects in water-scarce regions?

6. Should the ILC Draft Articles on the Law of Transboundary Aquifers be converted into a binding convention, and how should they treat non-recharging aquifers?

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- Inga T Winkler, *The Human Right to Water: Significance, Legal Status and Implications for Water Allocation* (Hart Publishing 2012).
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20 CHAPTER

Marine Resources, Offshore Energy, and the Law of the Sea

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the structure, institutions, and dispute settlement system of the United Nations Convention on the Law of the Sea and its implementing agreements.
2. Analyse the rights and obligations of coastal states and other states in the territorial sea and the exclusive economic zone, including in relation to offshore renewable energy.
3. Explain the legal regime of the continental shelf, including the delineation of the outer continental shelf and the payments regime of Article 82.
4. Evaluate the international legal framework governing offshore petroleum and gas installations, offshore wind, and other marine energy activities, including their security.
5. Analyse the law governing submarine pipelines and cables and its adequacy in the face of recent incidents of damage.
6. Assess the regime for deep-seabed mining under Part XI of UNCLOS and the controversy over the adoption of exploitation regulations.
7. Explain the obligations to protect and preserve the marine environment under Part XII of UNCLOS and the Agreement on Marine Biological Diversity of Areas beyond National Jurisdiction.
8. Apply the law of maritime delimitation and the rules governing activities in disputed areas, joint development, and the unitisation of cross-border fields.

Introduction

The oceans cover more than seventy per cent of the earth's surface and are among the most important sources and conduits of energy. Offshore fields account for around thirty per cent of global oil production and a similar share of gas production; offshore wind is one of the fastest-growing sources of renewable electricity; and the seabed hosts the pipelines that carry gas between continents,

the cables that transmit electricity between national grids and nearly all intercontinental data, and potentially vast deposits of the minerals required for the energy transition. The sea also carries the tankers and LNG carriers through which most internationally traded oil and much internationally traded gas moves, as examined in Chapter 6. The law of the sea, which allocates jurisdiction over marine spaces and their resources among states and the international community, is therefore a foundational element of international energy and natural resources law.

The modern law of the sea is codified principally in the **United Nations Convention on the Law of the Sea** (UNCLOS), often described as a "constitution for the oceans". It divides the oceans into maritime zones — internal waters, the territorial sea, the contiguous zone, the exclusive economic zone, the continental shelf, the high seas, and the Area — in each of which coastal states and other states have different rights and obligations, as illustrated in Figure 20.1. The extension of coastal state jurisdiction over offshore resources, which began with the Truman Proclamation on the continental shelf in 1945, was driven to a large extent by the desire of states to control offshore petroleum, and the rules governing the continental shelf and the exclusive economic zone remain central to the development of offshore energy.

This chapter examines UNCLOS (section 20.1), the territorial sea and exclusive economic zone (section 20.2), the continental shelf (section 20.3), offshore petroleum and gas development and other marine energy activities (section 20.4), submarine pipelines and cables (section 20.5), deep-seabed minerals (section 20.6), marine environmental protection (section 20.7), and maritime boundaries and offshore resource disputes (section 20.8). The regulation of pollution from offshore petroleum operations under regional seas agreements and national law has been examined in Chapter 17, the licensing of offshore petroleum in Chapter 5, and decommissioning in Chapters 5 and 17.

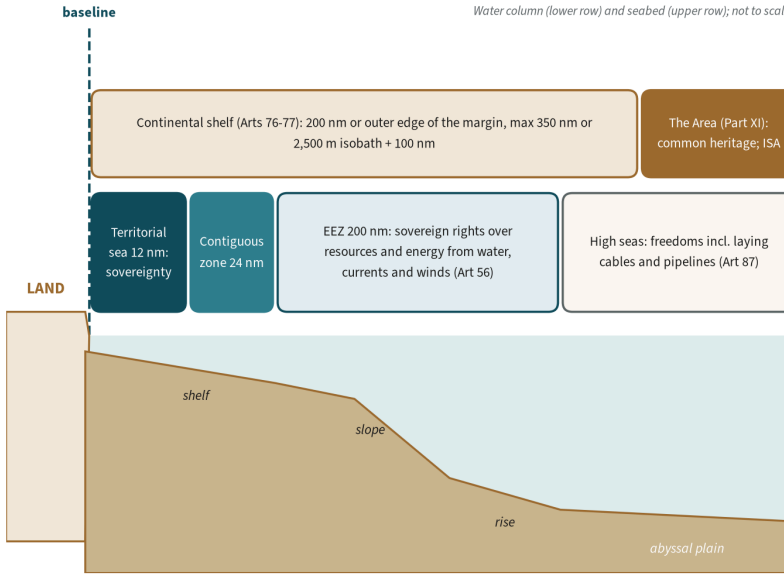


Figure 20.1 Maritime zones under UNCLOS and their significance for offshore energy (schematic, not to scale). *Source:* prepared for this edition.

20.1 United Nations Convention on the Law of the Sea

From Freedom of the Seas to UNCLOS

For centuries, the law of the sea was dominated by the doctrine of the **freedom of the seas**, articulated by Hugo Grotius in *Mare Liberum* (1609), under which the oceans beyond a narrow coastal belt were open to all states for navigation and fishing. Coastal state jurisdiction was confined to a territorial sea, whose breadth — often linked to the range of a cannon shot and conventionally fixed at three nautical miles — was contested.

The transformation of this regime began with the **Truman Proclamation** of 28 September 1945, in which the United States declared that it regarded the natural resources of the subsoil and seabed of the continental shelf beneath the high seas but contiguous to its coasts as appertaining to the United States, subject to its jurisdiction and control. The proclamation, motivated principally by the prospect of offshore oil production in the Gulf of Mexico, was rapidly followed by similar claims by other states, some of which — notably Chile, Ecuador, and Peru in the Santiago Declaration of 1952 — claimed sovereignty and jurisdiction over a maritime zone extending two hundred nautical miles from their coasts, primarily to protect fisheries.

The **First United Nations Conference on the Law of the Sea**, held in Geneva in 1958, adopted four conventions: on the Territorial Sea and the Con-

tiguous Zone, the High Seas, Fishing and Conservation of the Living Resources of the High Seas, and the **Continental Shelf**. The Convention on the Continental Shelf defined the shelf by reference to a depth of two hundred metres or, beyond that limit, to where the depth of the superjacent waters admitted of the exploitation of the natural resources — the "exploitability" criterion, which threatened to allow technologically advanced states to extend their jurisdiction indefinitely. The conference failed to agree on the breadth of the territorial sea, as did a second conference in 1960.

In 1967, Ambassador Arvid Pardo of Malta proposed to the UN General Assembly that the seabed beyond national jurisdiction should be declared the **common heritage of mankind**, and in 1970 the General Assembly adopted a Declaration of Principles Governing the Sea-Bed and the Ocean Floor. The **Third United Nations Conference on the Law of the Sea**, which met from 1973 to 1982, negotiated a comprehensive convention by consensus as a "package deal", balancing the interests of coastal states, maritime powers, landlocked and geographically disadvantaged states, and developing countries. UNCLOS was adopted on 10 December 1982 at Montego Bay, Jamaica, and entered into force on 16 November 1994.

Structure and Content

UNCLOS comprises 320 articles in seventeen parts and nine annexes. Its principal parts address the territorial sea and contiguous zone (Part II); straits used for international navigation (Part III); archipelagic states (Part IV); the exclusive economic zone (Part V); the continental shelf (Part VI); the high seas (Part VII); the regime of islands (Part VIII); enclosed or semi-enclosed seas (Part IX); the rights of landlocked states (Part X); the Area (Part XI); the protection and preservation of the marine environment (Part XII); marine scientific research (Part XIII); the development and transfer of marine technology (Part XIV); and the settlement of disputes (Part XV). Article 309 prohibits reservations, reflecting the package-deal character of the Convention.

Implementing Agreements

Three implementing agreements supplement UNCLOS. The **Agreement relating to the Implementation of Part XI** (1994) substantially modified the deep-seabed mining regime of Part XI, which had been unacceptable to the United States and other industrialised states, in order to secure their participation, as examined in section 20.6. The **UN Fish Stocks Agreement** (1995) addresses straddling and highly migratory fish stocks. The **Agreement under UNCLOS on the Conservation and Sustainable Use of Marine Biological Diversity of Areas beyond National Jurisdiction** (BBNJ Agreement), adopted in June

2023, reached the sixty ratifications required for its entry into force in September 2025, and entered into force in January 2026, as examined in section 20.7.

Participation

UNCLOS has near-universal participation, with around 170 parties, including the European Union. Nigeria ratified the Convention in 1986. The most notable non-party is the **United States**, which signed the 1994 Implementation Agreement but has not ratified either instrument, owing to opposition in the Senate, principally to the deep-seabed mining regime. The United States nevertheless accepts most of the Convention's provisions, other than Part XI, as reflecting customary international law, as declared in President Reagan's Ocean Policy Statement of 1983 and subsequent practice, and has proclaimed a territorial sea, a contiguous zone, and an exclusive economic zone consistent with the Convention. In 2023, the United States published the outer limits of its extended continental shelf on the basis of the criteria of Article 76, without making a submission to the Commission on the Limits of the Continental Shelf. In 2025, the United States took steps to authorise deep-seabed mining beyond national jurisdiction under its domestic legislation, outside the framework of Part XI, as examined in section 20.6.

Institutions

UNCLOS established three institutions. The **International Seabed Authority** (ISA), based in Kingston, Jamaica, is the organisation through which states parties organise and control activities in the Area, in particular with a view to administering its resources. The **Commission on the Limits of the Continental Shelf** (CLCS), a body of twenty-one experts in geology, geophysics, and hydrography, considers the data submitted by coastal states concerning the outer limits of the continental shelf beyond two hundred nautical miles and makes recommendations. The **International Tribunal for the Law of the Sea** (ITLOS), based in Hamburg, is a standing tribunal of twenty-one judges with jurisdiction over disputes concerning the interpretation or application of the Convention, and its **Seabed Disputes Chamber** has jurisdiction over disputes relating to activities in the Area and may give advisory opinions at the request of the ISA. The **Meeting of States Parties** elects the members of ITLOS and the CLCS and addresses budgetary matters.

Dispute Settlement

Part XV establishes a system of compulsory dispute settlement, which was an essential element of the package deal. States parties must settle disputes concerning the interpretation or application of the Convention by peaceful means, and if a dispute is not settled by means chosen by the parties, it must, at the request of any party, be submitted to a court or tribunal having jurisdiction under Section

2 of Part XV. Under Article 287, states may choose one or more of four forums: ITLOS, the International Court of Justice, an arbitral tribunal constituted under Annex VII, and a special arbitral tribunal under Annex VIII for certain categories of disputes. If the parties have not accepted the same procedure, the dispute is submitted to Annex VII arbitration, which is therefore the default procedure. The system is subject to limitations and exceptions: Article 297 excludes certain disputes concerning the exercise of coastal state sovereign rights in the EEZ, and Article 298 permits states to declare that they do not accept compulsory procedures for disputes concerning maritime boundary delimitation, historic bays or titles, military activities, and matters before the Security Council. States that make a declaration excluding boundary disputes remain subject to compulsory **conciliation** under Annex V for disputes arising after the Convention's entry into force — a procedure first used in the conciliation between **Timor-Leste and Australia** concerning the Timor Sea, which led to a maritime boundary treaty in 2018, examined in section 20.8. The dispute settlement system has been used extensively, including in cases with significant energy dimensions such as the *South China Sea Arbitration* (2016) and the *Ghana/Côte d'Ivoire* case (2017).

20.2 Territorial Seas and Exclusive Economic Zones

Baselines and Internal Waters

Maritime zones are measured from **baselines**. The normal baseline is the low-water line along the coast as marked on large-scale charts officially recognised by the coastal state (Article 5). Where the coastline is deeply indented or fringed with islands, the coastal state may draw **straight baselines** joining appropriate points (Article 7), a method approved by the International Court of Justice in the *Anglo-Norwegian Fisheries* case (1951), provided the lines do not depart to any appreciable extent from the general direction of the coast. Archipelagic states may draw archipelagic baselines enclosing their islands (Article 47). Waters on the landward side of the baseline are **internal waters**, over which the coastal state has full sovereignty, including in ports and many estuaries where energy terminals, refineries, and LNG facilities are located.

The Territorial Sea

Every state has the right to establish a **territorial sea** of up to twelve nautical miles from its baselines (Article 3), over which it has sovereignty extending to the airspace, seabed, and subsoil (Article 2), subject to the right of **innocent passage** of foreign ships (Articles 17–19). The coastal state has exclusive authority over the exploration and exploitation of resources in its territorial sea, including oil and gas, minerals, and the generation of renewable energy, and may adopt laws relating to innocent passage concerning, among other matters, the protection of navigational aids and facilities and other facilities or installations, and the protection of

cables and pipelines (Article 21(1)). Nigeria's **Territorial Waters Act** establishes a territorial sea of twelve nautical miles. Much offshore wind development in Europe began in the territorial sea, close to shore, before moving further offshore into the EEZ.

The Contiguous Zone

In a **contiguous zone** extending up to twenty-four nautical miles from the baselines, the coastal state may exercise the control necessary to prevent and punish infringement of its customs, fiscal, immigration, or sanitary laws within its territory or territorial sea (Article 33). The zone has limited direct significance for energy activities, although customs enforcement against smuggled fuel and stolen crude oil — a serious problem in the Gulf of Guinea — may rely on it.

The Exclusive Economic Zone

The **exclusive economic zone** (EEZ), which may extend up to two hundred nautical miles from the baselines (Article 57), is a *sui generis* zone in which the coastal state has specified sovereign rights and jurisdiction, and other states enjoy specified freedoms. Under Article 56(1), the coastal state has:

- **sovereign rights** for the purpose of exploring and exploiting, conserving and managing the natural resources, whether living or non-living, of the waters superjacent to the seabed and of the seabed and its subsoil, and with regard to other activities for the economic exploitation and exploration of the zone, **such as the production of energy from the water, currents and winds**; and
- **jurisdiction** with regard to the establishment and use of artificial islands, installations, and structures; marine scientific research; and the protection and preservation of the marine environment.

The express reference to the production of energy from water, currents, and winds — included at a time when offshore renewable energy was largely theoretical — provides the legal basis for coastal state jurisdiction over offshore wind, wave, tidal, and ocean thermal energy conversion projects in the EEZ. Article 56(3) provides that the rights with respect to the seabed and subsoil are to be exercised in accordance with Part VI on the continental shelf, so that offshore petroleum and mineral rights within two hundred nautical miles rest on both regimes.

Under Article 58, all states enjoy in the EEZ the freedoms of navigation, overflight, and the **laying of submarine cables and pipelines**, and other internationally lawful uses of the sea related to these freedoms. Both the coastal state and other states must have **due regard** to each other's rights and duties (Articles 56(2) and 58(3)). Disputes over the attribution of rights not expressly allocated

by the Convention are to be resolved on the basis of equity and in the light of all relevant circumstances (Article 59).

Article 60 grants the coastal state the exclusive right to construct and to authorise and regulate the construction, operation, and use of artificial islands, installations, and structures for economic purposes in the EEZ, and exclusive jurisdiction over them, including with regard to customs, fiscal, health, safety, and immigration laws. The coastal state may establish reasonable **safety zones** around installations, not exceeding five hundred metres, unless authorised by generally accepted international standards or recommended by the competent international organisation (Article 60(4)–(5)). Installations may not be established where they may cause interference with the use of recognised sea lanes essential to international navigation (Article 60(7)), and abandoned or disused installations must be removed to ensure safety of navigation, taking into account generally accepted international standards (Article 60(3)). Installations do not possess the status of islands and have no territorial sea of their own (Article 60(8)).

Nigeria established its EEZ by the **Exclusive Economic Zone Act** of 1978, which asserts sovereign rights for the exploration and exploitation of the natural resources of the zone. Nigeria's deepwater oil fields, including Bonga, Agbami, Akpo, Egina, and Erha, lie within its EEZ and on its continental shelf, more than one hundred kilometres offshore.

Offshore Wind and Marine Spatial Planning

The rapid development of offshore wind in the EEZs of European states, China, the United States, and increasingly other states has raised legal questions about the interaction between wind farms and other uses of the sea, including navigation, fishing, cables and pipelines, defence, and environmental protection. States have developed legal frameworks for designating zones for offshore wind, leasing the seabed, and permitting projects. In the United Kingdom, the Energy Act 2004 established a Renewable Energy Zone corresponding to the EEZ, and seabed leases are granted by the Crown Estate. In the United States, the Outer Continental Shelf Lands Act of 1953, as amended in 2005, authorises the Bureau of Ocean Energy Management to lease areas of the outer continental shelf for offshore wind; in 2025, the federal government suspended new offshore wind leasing and issued orders halting work on certain projects under construction, leading to litigation by developers and states. The European Union's **Offshore Renewable Energy Strategy** (2020) set an objective of at least 300 GW of offshore wind by 2050, and in the **Ostend Declaration** of 2023, nine North Seas states committed to developing the North Sea as a "green power plant" with a combined 300 GW by 2050. **Marine spatial planning**, required in the European Union by Directive 2014/89/EU, allocates marine space among competing uses and is increasingly used to designate zones for offshore wind while protecting navigation, fisheries,

and ecosystems. The "due regard" obligation and the restrictions on interference with sea lanes in Article 60(7) provide the international framework within which such planning operates.

Islands and Rocks

Under Article 121, islands generate the same maritime zones as land territory, but rocks which cannot sustain human habitation or economic life of their own have no EEZ or continental shelf (Article 121(3)). The interpretation of this provision has major implications for offshore resources, since a single feature may generate an EEZ of more than 125,000 square nautical miles. In the *South China Sea Arbitration* (*Philippines v China*, 2016), an Annex VII tribunal held that none of the high-tide features in the Spratly Islands was capable of sustaining human habitation or economic life of their own in their natural condition, so that none generated an EEZ or continental shelf, and that China's claims to historic rights within the "nine-dash line" were without legal basis to the extent that they exceeded the limits of its maritime zones under the Convention. The tribunal also found that China had violated the Philippines' sovereign rights in its EEZ, including by interfering with oil and gas exploration at Reed Bank. China rejected the award, which it regards as null and void, and the dispute over resources in the South China Sea, which is believed to contain significant hydrocarbon reserves, continues.

20.3 Continental Shelf Resource Rights

The Legal Concept of the Continental Shelf

The continental shelf has both a geological and a legal meaning. Geologically, the continental margin — the submerged prolongation of the land mass — comprises the shelf, a gently sloping platform; the slope, where the seabed descends more steeply; and the rise, which merges with the deep ocean floor. Legally, under Article 76(1) of UNCLOS, the continental shelf of a coastal state comprises the seabed and subsoil of the submarine areas that extend beyond its territorial sea throughout the **natural prolongation** of its land territory to the outer edge of the continental margin, or to a distance of **two hundred nautical miles** from the baselines where the outer edge of the margin does not extend up to that distance.

The definition therefore combines two criteria. Every coastal state has a continental shelf of at least two hundred nautical miles, regardless of the geology of the seabed — a distance criterion that coincides with the EEZ. Where the continental margin extends further, the coastal state's shelf extends to the outer edge of the margin, determined by one of two formulae in Article 76(4): a line connecting points at which the thickness of sedimentary rocks is at least one per cent of the shortest distance to the foot of the continental slope (the "Gardiner" or sediment-thickness formula), or a line connecting points not more than sixty nautical miles

from the foot of the slope (the "Hedberg" formula). The outer limit is subject to maximum constraints: it may not exceed **350 nautical miles** from the baselines, or **100 nautical miles from the 2,500-metre isobath** (Article 76(5)), with special rules for submarine ridges.

The Nature of Coastal State Rights

Under Article 77, the coastal state exercises over the continental shelf **sovereign rights for the purpose of exploring it and exploiting its natural resources**. These rights are **exclusive** in the sense that, if the coastal state does not explore or exploit the shelf, no one may undertake these activities without its express consent (Article 77(2)). They **do not depend on occupation**, effective or notional, or on any express proclamation (Article 77(3)). The natural resources comprise the mineral and other non-living resources of the seabed and subsoil together with sedentary species (Article 77(4)).

These provisions codify the principle stated by the International Court of Justice in the *North Sea Continental Shelf* cases (1969), in which the Court held that the rights of the coastal state in respect of the shelf exist *ipso facto* and *ab initio*, by virtue of its sovereignty over the land, as an extension of it in the form of an exercise of sovereign rights for the purpose of exploring the seabed and exploiting its natural resources. The rights of the coastal state over the continental shelf do not affect the legal status of the superjacent waters or airspace (Article 78), and the exercise of these rights must not infringe or result in unjustifiable interference with navigation and other rights and freedoms of other states.

Under Article 81, the coastal state has the **exclusive right to authorise and regulate drilling** on the continental shelf for all purposes. Article 80 applies the regime of Article 60 on artificial islands, installations, and structures *mutatis mutandis* to those on the continental shelf. Article 85 preserves the right of the coastal state to exploit the subsoil by means of tunnelling, irrespective of the depth of water. Together, these provisions constitute the international legal foundation for national petroleum licensing systems offshore, examined in Chapter 5, and for the regulation of offshore installations, examined in section 20.4.

Delineation of the Outer Continental Shelf

A coastal state intending to establish the outer limits of its continental shelf beyond two hundred nautical miles must submit information on the limits, with supporting scientific and technical data, to the **Commission on the Limits of the Continental Shelf** within ten years of the entry into force of the Convention for that state (Article 76(8) and Annex II, Article 4) — a deadline modified by a decision of the Meeting of States Parties in 2001, which allowed states to submit preliminary information to satisfy it. The Commission makes recommendations on matters related to the establishment of the outer limits, and the limits estab-

lished by a coastal state on the basis of these recommendations are **final and binding** (Article 76(8)). The coastal state must deposit charts and information permanently describing the outer limits with the UN Secretary-General (Article 76(9)).

The Commission has received close to one hundred submissions, and its workload is such that some submissions will not be considered for many years. Under its Rules of Procedure, the Commission will not consider a submission concerning an area subject to a land or maritime dispute without the prior consent of all states that are parties to the dispute. The delineation of the outer limits, which concerns the boundary between the continental shelf and the Area, is distinct from **delimitation** between the continental shelves of opposite or adjacent states, addressed in section 20.8, and Article 76(10) provides that the provisions of the article are without prejudice to delimitation.

Many African coastal states, including Nigeria, have made submissions to the Commission, often with technical support from international programmes. **Nigeria** submitted information on the outer limits of its continental shelf beyond two hundred nautical miles in 2009, claiming an extended shelf in the Gulf of Guinea, and the submission was taken up for consideration by the Commission. The **Gulf of Guinea** is a region in which extended continental shelf claims overlap, requiring coordination among Nigeria, Benin, Togo, Ghana, São Tomé and Príncipe, Equatorial Guinea, and Cameroon. Since rights to the shelf exist *ipso facto*, a coastal state has sovereign rights over its extended continental shelf whether or not its outer limits have been established, but the establishment of final and binding limits provides legal certainty for investors in deepwater exploration.

Relationship between the Continental Shelf and the EEZ

Within two hundred nautical miles, the seabed and subsoil are subject to both the continental shelf regime and the EEZ regime, with Article 56(3) providing that seabed rights in the EEZ are exercised in accordance with Part VI. The continental shelf regime is not dependent on a claim, while the EEZ must be proclaimed. Beyond two hundred nautical miles, the coastal state has rights over the seabed and subsoil of the extended continental shelf, but the superjacent waters are high seas. This raises particular issues where the extended continental shelf of one state lies within the EEZ of another — the "grey area" created by maritime delimitation in the *Bay of Bengal* cases (*Bangladesh/Myanmar*, ITLOS 2012; *Bangladesh v India*, Annex VII 2014), in which tribunals delimited the continental shelf beyond two hundred nautical miles in a manner that created an area where the shelf belonged to one state and the water column to another, and called on the parties to cooperate. In *Nicaragua v Colombia* (Question of the Delimitation of the Continental Shelf beyond 200 Nautical Miles, 2023), the International Court of Justice held that under customary international law a state's entitlement to a

continental shelf beyond two hundred nautical miles from its baselines may not extend within two hundred nautical miles of the baselines of another state.

Payments for Exploitation beyond 200 Nautical Miles

Article 82, a compromise between broad-margin states and states that favoured a narrower shelf, requires the coastal state to make **payments or contributions in kind** in respect of the exploitation of the non-living resources of the continental shelf beyond two hundred nautical miles. Payments are made annually with respect to all production at a site after the first five years of production. In the sixth year, the rate is one per cent of the value or volume of production at the site, rising by one per cent each year until the twelfth year, after which it remains at seven per cent. Production does not include resources used in connection with exploitation. A developing state that is a net importer of a mineral resource produced from its continental shelf is exempt in respect of that resource. Payments are made through the ISA, which distributes them to states parties on the basis of equitable sharing criteria, taking into account the interests and needs of developing states, particularly the least developed and landlocked among them.

Article 82 has yet to be applied in practice, but offshore production beyond two hundred nautical miles is approaching in several regions, and the ISA has undertaken work on its implementation, including on the definition of the "value" of production, the identification of the production site, and the procedures for distribution. Article 82 is significant for petroleum fiscal regimes, since it imposes on the coastal state an international obligation that will be passed through, in practice, to licensees in the design of production sharing contracts and licences for extended continental shelf areas.

20.4 Offshore Petroleum and Gas Development

The Layered Legal Framework

Offshore petroleum and gas development is governed by several layers of law: the international law of the sea, which allocates jurisdiction and imposes obligations relating to navigation, safety, and environmental protection; international standards adopted by the International Maritime Organization and other bodies; regional agreements, particularly regional seas conventions; national petroleum legislation and licences or contracts; and, for floating facilities, the law of the flag state. Offshore operations present distinctive legal issues relating to the status and safety of installations, the jurisdiction applicable to persons and activities on them, the security of operations, and their decommissioning.

National Legislation

Coastal states regulate offshore petroleum operations under legislation that applies to their territorial sea and continental shelf. The United States **Outer**

Continental Shelf Lands Act of 1953 asserts federal jurisdiction over the outer continental shelf beyond state waters, applies federal law and, as surrogate federal law, the law of the adjacent state, and governs leasing by the Bureau of Ocean Energy Management and safety regulation by the Bureau of Safety and Environmental Enforcement — a separation of functions introduced after the *Deepwater Horizon* disaster of 2010, examined in Chapter 17. The United Kingdom's **Petroleum Act 1998** vests petroleum in the Crown and provides for licensing of the continental shelf, and the Continental Shelf Act 1964 applies civil and criminal law to installations. Norway's **Petroleum Activities Act** of 1996 governs licensing and operations on the Norwegian continental shelf. In **Nigeria**, the **Petroleum Industry Act 2021** governs upstream operations onshore, in shallow water, and in deep offshore areas, with the Nigerian Upstream Petroleum Regulatory Commission as regulator, and provides distinct fiscal terms for deep offshore and frontier acreage. The Deep Offshore and Inland Basin Production Sharing Contract Act, as amended in 2019, governs the royalty terms of deep offshore production sharing contracts.

Installations: Status, Safety, and Jurisdiction

Offshore installations range from fixed steel or concrete platforms to floating production, storage, and offloading vessels (FPSOs), drilling rigs, and subsea production systems connected to onshore or floating facilities. Under Articles 60 and 80, the coastal state has exclusive jurisdiction over installations on its continental shelf and in its EEZ, including with regard to safety, health, customs, fiscal, and immigration laws. Floating installations raise questions about the relationship between coastal state jurisdiction and **flag state** jurisdiction, since FPSOs and mobile drilling units are often registered as ships. The International Maritime Organization's **Code for the Construction and Equipment of Mobile Offshore Drilling Units** (MODU Code) establishes international standards, while coastal states apply their own safety regimes. The Nigerian fleet of FPSOs, including those at Bonga, Agbami, Akpo, and Egina, operates under Nigerian regulatory oversight while also subject to classification society and flag state requirements.

The **Piper Alpha** disaster in the United Kingdom North Sea in 1988, in which 167 people died, led to the Cullen Report and a fundamental shift to a goal-setting "safety case" regime, under which operators must demonstrate to the regulator that major accident hazards have been identified and risks reduced to as low as reasonably practicable. The *Deepwater Horizon* disaster led to similar reforms in the United States and to the European Union's **Offshore Safety Directive** (2013/30/EU), which requires reports on major hazards, independent verification, and the assessment of the financial capacity of licensees to meet liabilities. These regimes, examined further in Chapter 17, illustrate the transfer of regulatory models among jurisdictions.

Safety Zones and Navigation

Coastal states establish **safety zones** of up to five hundred metres around installations, within which entry by unauthorised vessels is prohibited. In the *Arctic Sunrise* arbitration (*Netherlands v Russia*, Annex VII, 2015), arising from the boarding and detention by Russia of a Greenpeace vessel and its crew after a protest at the Prirazlomnaya oil platform in the Russian EEZ in the Pechora Sea in 2013, the tribunal held that Russia had breached its obligations to the Netherlands as flag state. It recognised that a coastal state may take measures to protect its sovereign rights and to prevent interference with installations, including the enforcement of safety zones, and that protest at sea is an internationally lawful use of the sea related to the freedom of navigation, but held that the boarding, seizure, and detention of the vessel were not justified on any of the grounds advanced by Russia. The award illustrates the balance between coastal state rights over offshore installations and the freedoms of other states.

Maritime Security in the Gulf of Guinea

The Gulf of Guinea has been one of the world's principal areas of maritime insecurity, with piracy and armed robbery against ships, kidnapping of crews, crude oil theft, and attacks on offshore installations. Under UNCLOS, piracy is defined as illegal acts of violence or detention committed for private ends on the high seas or in the EEZ by the crew or passengers of a private ship against another ship (Article 101), and all states may seize pirate ships on the high seas and exercise jurisdiction over pirates (Article 105). Acts committed in the territorial sea are "armed robbery at sea", within the exclusive jurisdiction of the coastal state.

The regional response is founded on the **Yaoundé Code of Conduct** of 2013, adopted by ECOWAS, the Economic Community of Central African States, and the Gulf of Guinea Commission, which established an architecture of regional and zonal maritime security centres for information sharing and coordination. **Nigeria** enacted the **Suppression of Piracy and Other Maritime Offences Act 2019**, the first standalone anti-piracy legislation in West and Central Africa, which criminalises piracy and armed robbery at sea and provides for universal jurisdiction over piracy; prosecutions and convictions under the Act followed within a few years of its enactment. Nigeria also implemented the **Integrated National Security and Waterways Protection Infrastructure** (the "Deep Blue Project"), launched in 2021, comprising aircraft, vessels, and command and control facilities. Reported incidents of piracy in the Gulf of Guinea declined substantially from their peak in 2020, although insecurity, including crude oil theft from pipelines and offshore facilities, remains a serious concern for the petroleum industry. The protection of energy infrastructure against security threats is examined further in Chapter 25.

Decommissioning

Article 60(3) requires that abandoned or disused installations be removed to ensure the safety of navigation, taking into account generally accepted international standards established by the competent international organisation, and that appropriate publicity be given to any installation not entirely removed. The **IMO Guidelines and Standards for the Removal of Offshore Installations and Structures** (1989) require the removal of installations in water depths of less than seventy-five metres and weighing less than four thousand tonnes in air, and of all installations placed after 1998 in water depths of less than one hundred metres and weighing less than four thousand tonnes, while permitting partial removal of larger structures subject to conditions. In the North-East Atlantic, **OSPAR Decision 98/3** prohibits the dumping or leaving wholly or partly in place of disused offshore installations, subject to derogations for the footings of large steel jackets and concrete installations — a regime that followed the controversy over the proposed deep-sea disposal of the *Brent Spar* storage buoy in 1995. The regulation of decommissioning, including its financial security, has been examined in Chapters 5 and 17. A growing question is whether offshore installations and pipelines may be repurposed for carbon capture and storage, hydrogen transport, or offshore renewable energy, rather than removed — an approach that raises issues of liability transfer and long-term stewardship.

Offshore Carbon Storage and Other Marine Energy Activities

The legal framework for offshore activities is being extended to new uses. The storage of carbon dioxide in sub-seabed geological formations is regulated under the **London Protocol** (1996) to the London Dumping Convention, which was amended in 2006 to permit such storage subject to conditions and in 2009 to allow the cross-border export of carbon dioxide for storage — an amendment provisionally applied by agreement of the parties from 2019 pending its entry into force. National legislation, such as the EU CCS Directive (2009/31/EC), the UK Energy Act 2008, and Norway's regulations, governs the licensing of offshore storage sites; Norway's Northern Lights project began injecting carbon dioxide from European emitters in 2025. **Offshore wind**, including floating wind in deeper waters, **wave and tidal energy**, **ocean thermal energy conversion**, and **offshore hydrogen production** are governed by the coastal state's rights under Article 56(1)(a), national leasing regimes, and the due regard and navigation-related constraints of UNCLOS. **Hybrid projects**, connecting offshore wind farms in the EEZs of more than one state to multiple national grids, raise novel questions of jurisdiction, market regulation, and the allocation of costs and benefits, which the North Seas states and the European Union are addressing through cooperation agreements and the EU's Trans-European Networks for Energy Regulation.

20.5 Submarine Pipelines and Cables

Significance

Submarine pipelines and cables form the hidden infrastructure of the global energy and information economy. Subsea gas pipelines connect Norway to continental Europe and the United Kingdom, North Africa to Southern Europe, Russia to Germany and Turkey, and Azerbaijan to Europe through the Southern Gas Corridor. **High-voltage direct current (HVDC) interconnectors** link national electricity systems across seas — for example, NorNed between Norway and the Netherlands, NordLink between Norway and Germany, North Sea Link between Norway and the United Kingdom, and Viking Link between the United Kingdom and Denmark, which entered into operation in 2023 as one of the world's longest land and subsea interconnectors. Export cables connect offshore wind farms to shore. Telecommunications cables carry the overwhelming majority of intercontinental data traffic, on which the digital management of energy systems and markets depends. Proposals for much longer interconnectors — such as the Xlinks project to bring solar and wind power from Morocco to the United Kingdom, and the Great Sea Interconnector between Greece, Cyprus, and Israel — illustrate the ambitions and difficulties of such projects.

The UNCLOS Framework

The right to lay submarine cables and pipelines is one of the traditional freedoms of the high seas, recognised in the 1958 Geneva Conventions and in UNCLOS. Under **Article 87**, the freedom of the high seas includes the freedom to lay submarine cables and pipelines, subject to Part VI. Under **Article 58**, all states enjoy this freedom in the EEZ of other states. Under **Article 79**, which applies to the continental shelf:

- all states are entitled to lay submarine cables and pipelines on the continental shelf (Article 79(1));
- subject to its right to take reasonable measures for the exploration of the continental shelf, the exploitation of its natural resources, and the prevention, reduction, and control of pollution from pipelines, the coastal state may not impede the laying or maintenance of such cables or pipelines (Article 79(2));
- the **delineation of the course** for the laying of pipelines on the continental shelf is subject to the consent of the coastal state (Article 79(3)) — a requirement that does not apply to cables;
- nothing in Part VI affects the right of the coastal state to establish conditions for cables or pipelines entering its territory or territorial sea, or its jurisdiction over cables and pipelines constructed or used in connection with the exploration of its continental shelf or exploitation of its resources (Article 79(4)); and

- states laying cables or pipelines must have due regard to cables or pipelines already in position, and in particular must not prejudice the possibilities of repairing existing ones (Article 79(5)).

The distinction between pipelines and cables in Article 79(3) reflects the greater environmental risk posed by pipelines. In practice, coastal states also require environmental impact assessments and permits for pipelines and cables crossing their continental shelves, relying on their jurisdiction over the protection of the marine environment in the EEZ and their rights to take reasonable measures against pollution. The extent to which coastal states may regulate cables that are not connected with the exploitation of their resources — for example, international telecommunications cables or interconnectors that merely cross their shelves — is a matter of debate, and practice varies.

In the territorial sea, the coastal state's sovereignty means that cables and pipelines may be laid only with its consent and subject to its conditions. In archipelagic waters, archipelagic states must respect existing submarine cables laid by other states and permit their maintenance and replacement (Article 51(2)).

Protection against Damage

Articles 113 to 115 of UNCLOS, building on the **Convention for the Protection of Submarine Telegraph Cables** of 1884, require states to adopt laws and regulations providing that the breaking or injury, by a ship flying their flag or by a person subject to their jurisdiction, of a submarine cable or pipeline beneath the high seas, done wilfully or through culpable negligence, is a punishable offence (Article 113); that owners who cause damage to other cables or pipelines in laying or repairing their own bear the cost of repairs (Article 114); and that ship owners who sacrifice anchors or nets to avoid injuring a cable or pipeline are indemnified by the owner of the cable or pipeline (Article 115). These provisions focus on the obligations of flag states and states of nationality and do not themselves confer enforcement jurisdiction on coastal states over foreign ships that damage cables or pipelines in their EEZs or on their continental shelves beyond the territorial sea — a significant gap in the light of recent incidents.

The overwhelming majority of damage to submarine cables results from accidental causes, principally fishing and anchoring, with the industry's International Cable Protection Committee estimating that between 150 and 200 faults occur each year. However, a series of incidents has raised concern about deliberate attacks on submarine infrastructure:

- In **September 2022**, explosions ruptured three of the four strings of the **Nord Stream** and Nord Stream 2 gas pipelines in the Baltic Sea, in the EEZs of Denmark and Sweden. Danish and Swedish investigations concluded that the pipelines had been deliberately sabotaged but were closed without iden-

tifying those responsible, while a German investigation continued and led to the arrest of suspects.

- In **October 2023**, the Balticconnector gas pipeline between Finland and Estonia and telecommunications cables were damaged; Finnish investigators attributed the damage to the dragging of an anchor by a Hong Kong-flagged container ship.
- In **November and December 2024**, further cables in the Baltic Sea and the Estlink 2 power interconnector between Finland and Estonia were damaged, the latter allegedly by the anchor of a tanker associated with the "shadow fleet" carrying Russian oil, which Finnish authorities detained. In January 2025, NATO launched the "Baltic Sentry" activity to enhance the surveillance and protection of critical undersea infrastructure.

These incidents raise difficult legal questions: whether coastal states may board and detain foreign ships suspected of damaging infrastructure in their EEZs; whether the deliberate destruction of infrastructure may constitute a use of force or an armed attack, and how attribution may be established; the adequacy of criminal laws under Article 113; and the responsibility of flag states. States and the European Union have responded with action plans on cable security, enhanced surveillance, and proposals for stronger legal frameworks, and some commentators have proposed a new international instrument on the protection of critical submarine infrastructure. The security of energy infrastructure, including its protection in armed conflict and against hybrid threats, is examined in Chapter 25.

Pipeline Treaties and Regional Regimes

Transboundary submarine pipelines are often governed by **intergovernmental agreements** between the states concerned, supplemented by host government agreements with project sponsors, as examined in Chapter 6. Examples include the agreements between Norway and the United Kingdom concerning the Frigg and Langeled pipelines, and the intergovernmental agreements for the Trans Adriatic Pipeline among Greece, Albania, and Italy. Such agreements address jurisdiction over the pipeline, the applicable technical and safety standards, taxation, access, and dispute settlement.

The **West African Gas Pipeline**, which transports Nigerian gas to Benin, Togo, and Ghana along a largely offshore route of around 680 kilometres, is governed by a **Treaty** concluded in 2003 among the four states, which established the West African Gas Pipeline Authority, together with an International Project Agreement with the West African Gas Pipeline Company. The pipeline has suffered from interruptions of supply, including damage by a ship's anchor off Lomé in 2012, which illustrated its vulnerability. The proposed **Nigeria–**

Morocco Gas Pipeline, intended to run around 5,600 kilometres or more, largely offshore along the West African coast and connecting to Europe, has been the subject of memoranda of understanding between Nigeria, Morocco, ECOWAS, and transit states since 2016, and of front-end engineering studies, but its feasibility and financing remain uncertain. Such a project would require a multilateral legal framework addressing transit through many maritime zones, as examined in Chapter 6.

The **Convention on the Legal Status of the Caspian Sea**, signed in Aktau in 2018 by the five littoral states, provides that states may lay submarine cables and pipelines on the seabed, provided that their projects comply with environmental standards, and that the route is to be determined in agreement with the state through whose seabed sector the pipeline is to be laid — a provision relevant to proposals for a Trans-Caspian gas pipeline from Turkmenistan to Azerbaijan.

The Nord Stream 2 Saga

The legal history of the **Nord Stream 2** pipeline illustrates the interaction of the law of the sea, EU law, investment law, and geopolitics. Because the pipeline crossed the territorial seas and EEZs of Russia, Finland, Sweden, Denmark, and Germany, it required the consent of each state to its route. Denmark, which had refused a permit for the route through its territorial sea, adopted legislation in 2017 allowing the refusal of permits on foreign policy and security grounds, and ultimately permitted a route through its EEZ in 2019. The European Union amended the Gas Directive in 2019 (Directive (EU) 2019/692) to extend its unbundling and third-party access requirements to gas pipelines between member states and third countries, within the territory and territorial sea of the member state. Nord Stream 2 AG challenged the amendment before the EU courts and brought a claim against the European Union under the Energy Charter Treaty, alleging discrimination. The pipeline was completed in 2021 but never entered into operation: Germany suspended its certification in February 2022 before the Russian invasion of Ukraine, and one of its strings was destroyed in the September 2022 explosions. The project illustrates how the freedom to lay pipelines is conditioned by coastal state consent, regional regulation, and political considerations.

20.6 Deep-Seabed Minerals Regulation

The Area and the Common Heritage of Mankind

The **Area** comprises the seabed and ocean floor and subsoil beyond the limits of national jurisdiction (Article 1(1)(1)), that is, beyond the outer limits of the continental shelves of coastal states. Under Article 136, the Area and its resources are the **common heritage of mankind**. No state may claim or exercise sovereignty or sovereign rights over any part of the Area or its resources, and no state or person may appropriate any part of them (Article 137(1)). All rights in the resources of

the Area are vested in mankind as a whole, on whose behalf the **International Seabed Authority** acts (Article 137(2)). Activities in the Area are to be carried out for the benefit of mankind as a whole, taking into particular consideration the interests and needs of developing states, and the ISA must provide for the equitable sharing of financial and other economic benefits derived from activities in the Area (Article 140). The Area is to be used exclusively for peaceful purposes (Article 141).

The resources of the Area include three principal types of mineral deposit, each of interest as a potential source of minerals for the energy transition:

- **polymetallic nodules**, potato-sized concretions lying on the abyssal plains at depths of four to six thousand metres, containing manganese, nickel, copper, and cobalt, most notably in the **Clarion-Clipperton Zone** of the Pacific Ocean;
- **polymetallic sulphides**, formed at hydrothermal vents along mid-ocean ridges, containing copper, zinc, gold, and silver; and
- **cobalt-rich ferromanganese crusts**, formed on the flanks of seamounts, containing cobalt, nickel, manganese, and rare earth elements.

The Original Part XI and the 1994 Agreement

The original Part XI established a parallel system under which mining would be undertaken both by the ISA's operating arm, **the Enterprise**, and by states and private entities sponsored by states, with mandatory transfer of technology to the Enterprise and developing states, production limitations to protect land-based producers, and a review conference. These provisions, reflecting the New International Economic Order, were unacceptable to the United States, the United Kingdom, Germany, and other industrialised states, which refused to ratify the Convention.

The **1994 Agreement relating to the Implementation of Part XI**, negotiated under the auspices of the UN Secretary-General, substantially modified the regime, providing that its provisions and Part XI are to be interpreted and applied together as a single instrument and that, in the event of inconsistency, the Agreement prevails. It abolished mandatory technology transfer and production limitations, provided that the Enterprise would conduct its initial operations through joint ventures and operate in accordance with sound commercial principles, established a Finance Committee, gave major contributors a decisive role in decision-making through a chambered voting system in the Council, and provided for the application of market principles. The 1994 Agreement enabled the entry into force of UNCLOS with broad participation.

The ISA and the Mining Code

The ISA, whose organs include the **Assembly** of all members, the thirty-six-member **Council**, which is its executive organ, the **Legal and Technical Commission**, the **Finance Committee**, and the **Secretariat**, regulates prospecting, exploration, and exploitation in the Area through rules, regulations, and procedures collectively known as the **Mining Code**. The ISA has adopted regulations on prospecting and exploration for polymetallic nodules (2000, revised 2013), polymetallic sulphides (2010), and cobalt-rich ferromanganese crusts (2012), and has granted around thirty exploration contracts, each for fifteen years with possible extensions, to state enterprises and private companies sponsored by states including China, Russia, India, the Republic of Korea, Japan, France, Germany, the United Kingdom, Belgium, and several Pacific island states — Nauru, Tonga, Kiribati, and the Cook Islands — in partnership with private companies.

Exploitation, however, requires the adoption of **exploitation regulations**, which have been under negotiation in the ISA Council since 2014. The negotiations address environmental protection, including environmental thresholds and regional environmental management plans; the payment mechanism and royalty rates; the equitable sharing of benefits; inspection, compliance, and enforcement; and the role of the Enterprise.

The Two-Year Rule and the Moratorium Debate

In June 2021, **Nauru**, which sponsors Nauru Ocean Resources Inc., a subsidiary of The Metals Company, invoked **Section 1, paragraph 15 of the Annex to the 1994 Agreement**, which provides that where a member state notifies the Council that an entity it sponsors intends to apply for approval of a plan of work for exploitation, the Council must complete the adoption of the relevant rules, regulations, and procedures within two years, and that if it fails to do so, it must nonetheless consider and provisionally approve a plan of work based on the provisions of the Convention and any rules, regulations, and procedures that the Council may have adopted provisionally. The two-year period expired in July 2023 without the adoption of exploitation regulations, and the Council decided that the commercial exploitation of mineral resources should not be carried out until such regulations had been adopted, while continuing negotiations with a view to their adoption.

Meanwhile, a growing number of states — more than thirty by 2025, including France, Germany, the United Kingdom, Canada, Brazil, Chile, and several Pacific states — called for a **moratorium**, precautionary pause, or ban on deep-seabed mining until its environmental impacts are adequately understood and effective regulation is in place. Scientists have warned that mining would cause the irreversible loss of poorly understood deep-sea biodiversity and ecosystem

functions, including through the destruction of habitats, sediment plumes, and noise. Proponents argue that seabed minerals could supply critical minerals with lower social and environmental impacts than land-based mining, as examined in Chapter 8.

The Responsibilities of Sponsoring States

In its advisory opinion on the ***Responsibilities and Obligations of States Sponsoring Persons and Entities with respect to Activities in the Area*** (2011), requested by the ISA Council following a proposal from Nauru, the ITLOS Seabed Disputes Chamber held that sponsoring states have two kinds of obligation: a "responsibility to ensure" that sponsored contractors comply with the terms of their contracts and the Convention, which is an obligation of **due diligence** — an obligation of conduct requiring the adoption of laws, regulations, and administrative measures that are reasonably appropriate; and **direct obligations**, including to apply a precautionary approach, to apply best environmental practices, to ensure the availability of recourse for compensation, and to conduct environmental impact assessments. The Chamber held that these obligations apply equally to developed and developing sponsoring states, to prevent the emergence of "sponsoring states of convenience". The opinion was significant not only for seabed mining but also for the development of the law on due diligence and the precautionary approach, and it prompted states, including the United Kingdom, Belgium, Germany, Singapore, Nauru, and Tonga, to adopt sponsoring state legislation.

The United States and Unilateral Licensing

The United States, as a non-party to UNCLOS, has maintained under the **Deep Seabed Hard Mineral Resources Act** of 1980 a domestic system for licensing exploration and permitting commercial recovery by US citizens beyond national jurisdiction, administered by the National Oceanic and Atmospheric Administration. In April 2025, the President issued an executive order directing the expedited processing of applications under the Act and the development of seabed mineral resources within and beyond US jurisdiction, and The Metals Company, through a US subsidiary, applied for a commercial recovery permit in respect of an area of the Clarion-Clipperton Zone. These steps were criticised by the ISA, by many states parties, and by commentators as inconsistent with the common heritage principle, which many regard as reflecting customary international law binding on all states. The ISA Council took steps to examine the position of contractors that might pursue exploitation outside the ISA framework. The controversy raises fundamental questions about the authority of the ISA, the customary status of the common heritage principle, and the relationship between treaty regimes and unilateral action by non-parties.

Seabed Mining within National Jurisdiction

Seabed mining may also take place within national jurisdiction, under national law. The failure of the **Solwara 1** project in the territorial waters of Papua New Guinea, whose developer, Nautilus Minerals, became insolvent in 2019, illustrated the commercial and regulatory risks. The **Cook Islands** has granted exploration licences for nodules in its EEZ. **Norway** decided in January 2024 to open part of its continental shelf in the Norwegian and Greenland Seas to mineral exploration, but the first licensing round was postponed in December 2024 as part of a political agreement on the budget. Seabed mining within national jurisdiction is subject to the obligations of Part XII of UNCLOS, including Article 208, which requires coastal states to adopt laws to prevent, reduce, and control pollution from seabed activities subject to their jurisdiction that are no less effective than international rules, standards, and recommended practices.

20.7 Marine Environmental Protection

Scope of this Section

Chapter 17 examined the regulation of marine pollution from offshore petroleum operations and shipping, including the MARPOL Convention, the civil liability and compensation regime for oil pollution damage, the regional seas conventions applicable to offshore operations, and national liability regimes. This section focuses on the general framework of **Part XII of UNCLOS** as the overarching treaty basis for the protection of the marine environment, its interpretation by international courts and tribunals — in particular in the ITLOS advisory opinion on climate change — and the **BBNJ Agreement**, with particular attention to their implications for offshore energy activities and the energy transition.

The General Obligation and its Structure

Article 192 provides, in deceptively simple terms, that "States have the obligation to protect and preserve the marine environment". This is the first provision of any global treaty to impose a general obligation of environmental protection on states, and it applies to all maritime zones and to all activities, including those within national jurisdiction. Article 193 affirms the sovereign right of states to exploit their natural resources pursuant to their environmental policies, but "in accordance with their duty to protect and preserve the marine environment" — formulating sovereignty over resources as subject to environmental obligations.

Article 194 requires states to take, individually or jointly as appropriate, all measures consistent with the Convention that are necessary to prevent, reduce, and control pollution of the marine environment from any source, using for this purpose the best practicable means at their disposal and in accordance with their capabilities (Article 194(1)), and to take all measures necessary to ensure that activities under their jurisdiction or control are so conducted as not to cause damage

by pollution to other states and their environment (Article 194(2)). The measures must deal with all sources of pollution, including pollution from installations and devices used in the exploration or exploitation of the natural resources of the seabed and subsoil, in particular measures for preventing accidents and dealing with emergencies, ensuring the safety of operations at sea, and regulating the design, construction, equipment, operation, and manning of such installations (Article 194(3)(c)). They must include measures necessary to protect and preserve rare or fragile ecosystems and the habitat of depleted, threatened, or endangered species (Article 194(5)).

"Pollution of the marine environment" is defined broadly in Article 1(1)(4) as the introduction by man, directly or indirectly, of substances or energy into the marine environment which results or is likely to result in deleterious effects such as harm to living resources and marine life, hazards to human health, hindrance to marine activities, impairment of the quality of sea water, and reduction of amenities. The inclusion of "energy" covers noise and heat, relevant to seismic surveys, pile-driving for offshore wind foundations, and thermal discharges.

Part XII then addresses particular sources of pollution: land-based sources (Article 207); pollution from seabed activities subject to national jurisdiction (Article 208), which requires coastal states to adopt laws no less effective than international rules and standards; activities in the Area (Article 209); dumping (Article 210); vessels (Article 211); and pollution from or through the atmosphere (Article 212). It establishes enforcement jurisdiction of flag states, port states, and coastal states (Articles 213–222), provides safeguards (Articles 223–233), and contains a special provision for ice-covered areas (Article 234). Article 235 provides that states are responsible for the fulfilment of their international obligations concerning the protection and preservation of the marine environment and are liable in accordance with international law, and requires them to ensure that recourse is available under their legal systems for prompt and adequate compensation for damage caused by pollution by persons under their jurisdiction.

Cooperation, Monitoring, and Environmental Impact Assessment

Section 2 of Part XII requires states to cooperate on a global and regional basis in formulating international rules and standards (Article 197), to notify other states of imminent or actual damage (Article 198), and to develop contingency plans against pollution (Article 199). Section 4 requires states to monitor the risks or effects of pollution (Article 204) and, under **Article 206**, when they have reasonable grounds for believing that planned activities under their jurisdiction or control may cause substantial pollution of or significant and harmful changes to the marine environment, to assess, as far as practicable, the potential effects of such activities and to communicate reports of the results. Article 206 is the treaty basis for **environmental impact assessment** of offshore energy projects with

potential transboundary or high-seas effects, and international courts have treated the obligation as one of due diligence requiring an assessment before activities are authorised.

Judicial Interpretation

International courts and tribunals have interpreted Part XII expansively. In the *South China Sea Arbitration* (2016), the tribunal held that Article 192 imposes a positive obligation to take active measures to protect and preserve the marine environment and a negative obligation not to degrade it, that Article 194(5) extends to the protection of fragile ecosystems from harvesting of endangered species, and that China had breached Part XII through its land reclamation and construction of artificial islands, which caused severe harm to coral reef ecosystems, and through its toleration of harmful harvesting by its fishing vessels. In the *Chagos Marine Protected Area Arbitration* (*Mauritius v United Kingdom*, 2015), the tribunal held that the United Kingdom had breached its obligation to have due regard to the rights of Mauritius in declaring a marine protected area. In the *Seabed Disputes Chamber's* advisory opinion of 2011, examined in section 20.6, the Chamber described the obligation to conduct an environmental impact assessment as a general obligation under customary international law.

The ITLOS Advisory Opinion on Climate Change

In its unanimous advisory opinion of **21 May 2024**, requested by the Commission of Small Island States on Climate Change and International Law, ITLOS addressed the obligations of states parties to UNCLOS to prevent, reduce, and control pollution of the marine environment in relation to the deleterious effects that result or are likely to result from climate change and ocean acidification caused by anthropogenic greenhouse gas emissions, and to protect and preserve the marine environment in relation to climate change impacts. The Tribunal held, in summary, that:

- anthropogenic greenhouse gas emissions into the atmosphere constitute **pollution of the marine environment** within the meaning of Article 1(1)(4);
- under Article 194(1), states have a specific obligation to take all necessary measures to prevent, reduce, and control marine pollution from such emissions, and to endeavour to harmonise their policies; this is an obligation of **due diligence**, whose standard is **stringent** given the high risks of serious and irreversible harm, and which may vary according to states' capabilities and available resources;
- the obligation under UNCLOS is not satisfied simply by compliance with the obligations and commitments under the Paris Agreement, although the

temperature goal and the timelines of the IPCC are relevant to determining the necessary measures;

- under Article 194(2), states must take all measures necessary to ensure that emissions under their jurisdiction or control do not cause damage by pollution to other states, an obligation applicable in a transboundary context;
- the obligations to conduct environmental impact assessments under Article 206 and to monitor under Article 204 apply to planned activities that may cause substantial pollution through greenhouse gas emissions; and
- Articles 192 and 194(5) require states to protect and preserve the marine environment from climate change impacts and ocean acidification, including by restoring degraded habitats and ecosystems.

The opinion, together with the advisory opinion of the International Court of Justice of 23 July 2025, examined in Chapter 18, has potential implications for offshore petroleum development: it suggests that the authorisation of new offshore fields and the conduct of environmental impact assessments under Article 206 must take account of greenhouse gas emissions, including potentially downstream emissions, a question also raised in domestic litigation such as *Finch*, examined in Chapter 17. The opinion also supports offshore renewable energy as a means of fulfilling the obligation to reduce marine pollution, while reminding states that such projects are themselves subject to the obligations of Part XII.

Environmental Dimensions of Offshore Renewable Energy

The expansion of offshore wind and other marine renewable energy raises environmental issues that must be addressed consistently with Part XII and regional and national law: underwater noise from pile-driving during construction, which can harm marine mammals; collision risks for birds and the displacement of bird and marine species; electromagnetic fields from cables; changes to seabed habitats; and cumulative effects. Wind farm foundations may also create artificial reefs and exclusion areas that benefit some species. Regulatory responses include environmental impact assessments, seasonal restrictions on piling, noise mitigation measures such as bubble curtains, marine spatial planning, and monitoring. The balance between accelerating offshore renewable deployment, as required by climate objectives, and protecting marine biodiversity is a recurring theme in the permitting of projects, reflected in the European Union's provisions for renewables acceleration areas under the revised Renewable Energy Directive, examined in Chapter 10.

The BBNJ Agreement

The **Agreement under UNCLOS on the Conservation and Sustainable Use of Marine Biological Diversity of Areas beyond National Jurisdiction** was adopted on 19 June 2023 after nearly two decades of discussion, opened for

signature in September 2023, and entered into force in January 2026, following the deposit of the sixtieth instrument of ratification in September 2025. It applies to areas beyond national jurisdiction — the high seas and the Area — which cover nearly two-thirds of the ocean. The Agreement addresses four elements:

- **marine genetic resources**, including digital sequence information, with obligations of notification and fair and equitable sharing of monetary and non-monetary benefits;
- **area-based management tools, including marine protected areas**, which the Conference of the Parties may establish in areas beyond national jurisdiction, providing a mechanism to contribute to the global target of protecting thirty per cent of the ocean by 2030 under the Kunming-Montreal Global Biodiversity Framework;
- **environmental impact assessments**, requiring parties to assess planned activities under their jurisdiction or control in areas beyond national jurisdiction that may cause more than minor or transitory effects, and to consider cumulative impacts and impacts in areas within national jurisdiction; and
- **capacity building and the transfer of marine technology** to developing states.

The Agreement establishes a Conference of the Parties, a Scientific and Technical Body, a Secretariat, a clearing-house mechanism, and a funding mechanism. It must be interpreted and applied in a manner that does not undermine relevant legal instruments and frameworks and relevant global, regional, subregional, and sectoral bodies — a provision designed to address its relationship with the ISA, regional fisheries management organisations, and the International Maritime Organization. For the energy sector, the Agreement is relevant to the environmental assessment of activities in areas beyond national jurisdiction, such as the laying of cables and pipelines on the high seas, offshore installations beyond national jurisdiction, and potentially marine geoengineering and carbon dioxide removal techniques, and the establishment of marine protected areas on the high seas may constrain such activities. Its relationship with deep-seabed mining, which remains within the competence of the ISA, will be a significant issue in its implementation.

Marine Geoengineering and Ocean-Based Carbon Removal

Ocean-based approaches to carbon dioxide removal, such as ocean fertilisation, ocean alkalinity enhancement, and the sinking of biomass, are attracting commercial and research interest. The London Protocol was amended in 2013 to regulate marine geoengineering, prohibiting ocean fertilisation other than legitimate scientific research authorised under a permit — an amendment that has not yet entered into force — and the parties to the London Convention and Protocol have continued to evaluate other techniques. The regulation of such activities

under Part XII, the London Protocol, the Convention on Biological Diversity, and the BBNJ Agreement is an emerging area of law at the intersection of climate and marine environmental governance.

20.8 Maritime Boundaries and Offshore Resource Disputes

The Problem of Overlapping Entitlements

Because coastal states are entitled to territorial seas of twelve nautical miles and EEZs and continental shelves of at least two hundred nautical miles, their entitlements overlap wherever their coasts are opposite or adjacent at distances less than twice these breadths. Hundreds of potential maritime boundaries exist worldwide, of which a substantial proportion remain undelimited. Since many of the world's offshore petroleum provinces lie in areas of overlapping entitlement, maritime boundary disputes have been closely connected with offshore resources: disputes often arise or intensify when one state licenses exploration in an area claimed by another, and the prospect of discoveries can either motivate agreement or entrench disagreement.

The Law of Delimitation

UNCLOS provides that, failing agreement, neither state may extend its **territorial sea** beyond the median line every point of which is equidistant from the nearest points on the baselines, except where it is necessary by reason of historic title or other special circumstances to delimit in a different way (Article 15). For the **EEZ** and **continental shelf**, Articles 74(1) and 83(1) provide that delimitation between states with opposite or adjacent coasts is to be effected by agreement on the basis of international law, as referred to in Article 38 of the Statute of the International Court of Justice, **in order to achieve an equitable solution**. These provisions, deliberately vague as a result of the division at the Third Conference between proponents of equidistance and proponents of equitable principles, have been given content by international jurisprudence.

In the *North Sea Continental Shelf* cases (1969), the International Court of Justice rejected the argument that the equidistance principle of the 1958 Convention was binding as customary law and held that delimitation was to be effected by agreement in accordance with equitable principles, taking account of all relevant circumstances. Subsequent cases, including *Tunisia/Libya* (1982), *Gulf of Maine* (*Canada/United States*, Chamber, 1984), *Libya/Malta* (1985), and *Jan Mayen* (*Denmark v Norway*, 1993), refined the approach, and in *Maritime Delimitation in the Black Sea* (*Romania v Ukraine*, 2009), the Court consolidated a **three-stage methodology**:

1. the construction of a **provisional equidistance line** (or median line for opposite coasts), using appropriate base points on the relevant coasts;

2. the consideration of whether **relevant circumstances** call for the adjustment or shifting of the provisional line to achieve an equitable result — such as the configuration of the coasts, the presence of islands, and significant disparities in coastal lengths; and
3. a **disproportionality test**, to verify that the line does not lead to an inequitable result by reason of a marked disproportion between the ratio of the respective coastal lengths and the ratio of the maritime areas allocated to each state.

This methodology has been followed by ITLOS and Annex VII tribunals, including in *Bangladesh/Myanmar* (2012), *Bangladesh v India* (2014), *Ghana/Côte d'Ivoire* (2017), and *Mauritius/Maldives* (2023), and by the International Court of Justice in *Nicaragua v Colombia* (2012), *Peru v Chile* (2014), and *Somalia v Kenya* (2021). In some cases of unstable or concave coasts, tribunals have used an angle-bisector method instead of equidistance, as in *Nicaragua v Honduras* (2007) and in part of *Bangladesh v India*.

A consistent theme of the jurisprudence is that **the location of natural resources** is generally not a relevant circumstance, except in exceptional cases. In *Barbados v Trinidad and Tobago* (Annex VII, 2006), the tribunal observed that resource-related criteria have been treated more cautiously and have not generally been applied as relevant circumstances, and in *Ghana/Côte d'Ivoire*, the Special Chamber declined to adjust the line to take account of the location of oil fields. Similarly, **oil practice** — the pattern of concessions granted by the parties — has been relevant only where it evidences a tacit agreement between the parties on a boundary, which tribunals have been reluctant to find, as in *Cameroon v Nigeria* (2002) and *Ghana/Côte d'Ivoire*.

::: casebox {title="Ghana/Côte d'Ivoire: Oil Fields in a Disputed Area"}
Dispute concerning Delimitation of the Maritime Boundary between Ghana and Côte d'Ivoire in the Atlantic Ocean (ITLOS Special Chamber, Judgment of 23 September 2017).

Facts. Ghana and Côte d'Ivoire had not delimited their maritime boundary. Ghana granted petroleum licences in waters that it regarded as being on its side of an equidistance line that it claimed had been tacitly accepted by both states through decades of oil practice, and the TEN (Tweneboa, Enyenra, Ntomme) fields were discovered and developed close to the claimed boundary. Côte d'Ivoire contested this boundary, claiming a line based on the angle-bisector method that would have placed the TEN fields on its side. In 2014, the parties submitted the dispute to a Special Chamber of ITLOS. In 2015, at the request of Côte d'Ivoire, the Special Chamber prescribed provisional measures requiring Ghana to ensure that no new drilling took place in the disputed area, while declining to suspend

ongoing activities, given the risk of irreparable prejudice to Ghana and the environment.

Judgment. The Special Chamber rejected Ghana's claim that the parties had tacitly agreed on a boundary based on their oil practice, holding that the evidence of a tacit agreement must be compelling and that oil practice, however consistent, did not by itself establish a maritime boundary. It applied the three-stage methodology, constructed an equidistance line, found no relevant circumstances justifying its adjustment, and confirmed the absence of disproportionality. The resulting boundary placed the TEN fields on Ghana's side. The Chamber also rejected Côte d'Ivoire's claim that Ghana's activities in the disputed area had violated its sovereign rights, holding that, since the area was found to belong to Ghana, its activities had not breached Côte d'Ivoire's rights; that Ghana had not violated the obligation to negotiate in good faith; and that Ghana had not breached the obligation under Article 83(3) not to jeopardise or hamper the reaching of an agreement, since that obligation was triggered only by activities after the claim had been made known, and Ghana had suspended new drilling following the provisional measures order.

Significance. The case is the leading African maritime delimitation decision with direct consequences for petroleum operations. It confirms that oil practice is not a reliable substitute for a delimitation agreement, illustrates the use of provisional measures to manage petroleum activities in disputed areas, and leaves open the extent of liability for unilateral activities in areas that are ultimately found to belong to the other party. It also illustrates that boundary uncertainty imposes costs on investors and states, which are best avoided by early agreement or joint development. :::

Cameroon v Nigeria

The *Land and Maritime Boundary between Cameroon and Nigeria* case (ICJ, 2002) is of particular significance for Nigeria. Cameroon brought proceedings in 1994 following armed clashes on the **Bakassi Peninsula**, a potentially oil-rich area administered by Nigeria. The Court held, on the basis of the Anglo-German Agreement of 1913, that sovereignty over Bakassi lay with Cameroon, and it delimited the maritime boundary, confirming the effect of the Maroua Declaration of 1975 and drawing an equidistance line further offshore, while declining to adjust it for the configuration of the coast or the parties' oil practice. Nigeria's oil concessions on Cameroon's side of the line did not constitute evidence of an agreed boundary. Following the judgment, a Cameroon–Nigeria Mixed Commission, established under UN auspices, oversaw implementation, and in the **Greentree Agreement** of 2006, Nigeria agreed to withdraw from the Bakassi Peninsula, which it completed in 2008, with protections for the Nigerian population remaining in the area. The peaceful implementation of the judgment

is often cited as a model, although the resettlement and livelihoods of affected communities have remained contested.

Activities in Disputed Areas: Articles 74(3) and 83(3)

Pending agreement on delimitation, Articles 74(3) and 83(3) require the states concerned, in a spirit of understanding and cooperation, to **make every effort to enter into provisional arrangements of a practical nature** and, during this transitional period, **not to jeopardise or hamper the reaching of the final agreement**. In *Guyana v Suriname* (Annex VII, 2007), which arose from the forcible removal by Surinamese naval vessels of a drilling rig licensed by Guyana from a disputed area in 2000, the tribunal held that both states had breached their obligations under these provisions: Guyana by authorising exploratory drilling, which could cause permanent physical change to the marine environment, without consulting Suriname; and Suriname by threatening force. The tribunal distinguished activities that cause permanent physical change, such as exploratory drilling and exploitation, which jeopardise or hamper a final agreement, from activities such as seismic surveys, which do not. Suriname's threat of force was also held to violate the Convention and the UN Charter. The tribunal's distinction has influenced the practice of states and companies in managing petroleum activities in disputed areas, although it has been criticised as both too permissive and too restrictive.

Joint Development Agreements

Where states cannot agree on a boundary, or where resources straddle a boundary, they may conclude **joint development agreements**, under which they jointly manage the exploration and exploitation of resources in a defined area and share the revenues, without prejudice to their boundary claims. Examples include the Malaysia–Thailand Joint Development Area established by a memorandum of understanding of 1979 and an agreement of 1990; the Timor Gap Treaty between Australia and Indonesia (1989) and its successor, the Timor Sea Treaty between Australia and Timor-Leste (2002); and the arrangements between Saudi Arabia and Kuwait for the former Neutral Zone.

The **Nigeria–São Tomé and Príncipe Joint Development Zone**, established by a **Treaty** concluded in 2001, is the leading African example. Rather than delimit their overlapping EEZ claims, the two states established a Joint Development Zone, managed by a **Joint Ministerial Council** and a **Joint Development Authority**, in which resources are shared in the proportion of sixty per cent to Nigeria and forty per cent to São Tomé and Príncipe. Licensing rounds were held, and exploration wells were drilled in the 2000s, but no commercial discoveries were made, and the Zone's activities have been limited. The JDZ has been studied for its institutional design and for the governance challenges arising from the

disparity in size and capacity between the partners, including concerns over the transparency of licence allocations.

The **Timor Sea** illustrates the evolution from joint development to delimitation. Following the first-ever compulsory conciliation under Annex V of UNCLOS, initiated by Timor-Leste in 2016, Australia and Timor-Leste concluded the **Treaty Establishing their Maritime Boundaries in the Timor Sea** in March 2018, which delimited permanent boundaries and established a special regime for the Greater Sunrise gas fields, under which revenues are shared seventy to thirty or eighty to twenty in favour of Timor-Leste, depending on the location of processing. The development of Greater Sunrise remains under negotiation.

Unitisation of Cross-Border Fields

Where a petroleum deposit straddles an agreed boundary, international practice favours **unitisation** — the joint development of the field as a single unit, with production allocated between the states and their licensees according to an agreed formula, subject to periodic redetermination. Unitisation prevents wasteful competitive drilling and maximises recovery, and its obligations may reflect a duty of cooperation in respect of shared resources under customary law, although the existence of a customary obligation to unitise is debated. Unitisation clauses are common in boundary agreements: the United Kingdom–Norway agreement of 1965 on the delimitation of the continental shelf provided for agreement on the exploitation of straddling fields, implemented for the Frigg, Statfjord, and Murchison fields, and the **Framework Agreement concerning Cross-Boundary Petroleum Co-operation** of 2005 provides a general regime. Nigeria and Equatorial Guinea concluded a maritime boundary treaty in 2000 and subsequently agreed on the unitised development of the Ekanga–Zafiro field straddling their boundary. Within national systems, unitisation is required by petroleum legislation, including Nigeria's Petroleum Industry Act 2021.

Other Resource-Related Boundary Disputes

Other disputes with significant energy dimensions include *Somalia v Kenya* (ICJ, 2021), in which the Court rejected Kenya's claim that Somalia had acquiesced in a boundary along a parallel of latitude and drew an adjusted equidistance line, with consequences for offshore blocks licensed by both states, while rejecting Somalia's claim that Kenya had violated its sovereign rights; the dispute between **Guyana and Venezuela** over the Essequibo region, before the International Court of Justice since 2018, which has intensified since major oil discoveries offshore Guyana from 2015, and in which the Court has indicated provisional measures requiring Venezuela to refrain from altering the existing situation; the **Israel–Lebanon** maritime boundary agreement of October 2022, mediated by the United States, which resolved a dispute over gas prospects including the Qana prospect; and dis-

putes in the **Eastern Mediterranean**, including those involving Cyprus, Turkey, and Greece, in which gas discoveries have intersected with unresolved questions of sovereignty and the entitlement of islands. The interaction between maritime boundary disputes and geopolitics is examined further in Chapter 25, and the procedural aspects of inter-state litigation in Chapter 27.

Conclusion

The law of the sea provides the jurisdictional architecture within which offshore energy and marine resources are developed. UNCLOS, building on the Truman Proclamation and the *North Sea Continental Shelf* cases, grants coastal states sovereign rights over the resources of the continental shelf and EEZ — including, prophetically, the production of energy from water, currents, and winds — while preserving the freedoms of other states, including the freedom to lay submarine cables and pipelines. The Convention's institutions and its compulsory dispute settlement system have generated a rich jurisprudence on maritime delimitation, activities in disputed areas, and the protection of the marine environment, culminating in the ITLOS advisory opinion of 2024, which brought greenhouse gas emissions within the scope of Part XII.

The energy transition is reshaping the law of the sea. Offshore wind is turning EEZs into spaces of intensive industrial use, requiring marine spatial planning and new cooperative arrangements for hybrid projects; HVDC interconnectors and pipelines are becoming targets in hybrid conflicts, exposing gaps in the legal protection of submarine infrastructure; the demand for critical minerals has brought deep-seabed mining to the threshold of commercial exploitation, testing the common heritage principle and the authority of the ISA; and the BBNJ Agreement extends environmental governance to areas beyond national jurisdiction. For coastal states such as Nigeria, the law of the sea offers both opportunities — in deepwater petroleum, offshore renewable energy, and an extended continental shelf — and responsibilities, in maritime security, environmental protection, and the peaceful resolution of disputes. For the doctoral researcher, these developments offer fertile ground for examining how a treaty regime negotiated in the 1970s can adapt to the demands of the twenty-first century.

::: exercise {title="Practical Exercise 20: Negotiating a Joint Development Agreement in a Disputed Maritime Area (HYPOTHETICAL)"} **This exercise is entirely hypothetical. The states, companies, geological features, and events described are fictitious and do not refer to any real countries, companies, or disputes.**

The Republic of Azuria and the Republic of Corvenne are adjacent coastal states on a gulf. Both are parties to UNCLOS; Corvenne has made a declaration under Article 298 excluding maritime boundary disputes from compulsory

procedures entailing binding decisions. Their maritime boundary has never been delimited. Azuria claims a boundary based on an equidistance line; Corvenne claims a boundary based on a perpendicular to the general direction of the coast, which would give it a substantially larger area. The overlapping claims define a triangular disputed area of around 9,000 square kilometres.

Five years ago, Azuria granted an exploration licence covering part of the disputed area to Meridian Offshore Ltd, a company incorporated in a third state. Corvenne protested diplomatically and has itself offered blocks overlapping the same area in a licensing round. Meridian acquired seismic data across the area and recently drilled an exploration well, the Halcyon-1 well, which discovered a significant gas field that appears, on the basis of the seismic data, to extend into areas claimed by both states and possibly beyond the equidistance line. Corvenne's navy has since shadowed Meridian's support vessels and warned that further drilling will be prevented. Both governments, under pressure from their populations, which suffer from frequent power shortages, wish to develop the gas for domestic power generation and export. A regional economic community has offered to facilitate negotiations.

Tasks.

(a) As legal adviser to the Ministry of Foreign Affairs of Corvenne, prepare a memorandum assessing whether Azuria and Meridian have breached international law by acquiring seismic data and drilling Halcyon-1, with reference to Articles 74(3) and 83(3) of UNCLOS and the relevant jurisprudence, and advising on the options available to Corvenne, including the possibility of compulsory conciliation, and the legal constraints on the use of naval force.

(b) As counsel to Meridian, prepare an advice on the company's legal and commercial risks, including the validity of its licence, the risk of a future boundary placing the field in Corvenne's area, investment treaty protection, and the implications for its financing.

(c) In teams representing Azuria and Corvenne, negotiate the heads of terms of a **joint development agreement**, addressing: the definition of the joint development zone; the institutional structure (a joint authority, joint ministerial council, or other arrangement); the revenue-sharing ratio and its justification; the treatment of Meridian's existing licence and Corvenne's offered blocks; the applicable law and fiscal regime; environmental protection and decommissioning; a without-prejudice clause preserving boundary claims; dispute settlement; and the relationship between the agreement and any future boundary.

(d) As an independent expert, write a short reflection comparing the joint development model with the alternatives of adjudication and the delimita-

tion of a boundary followed by a unitisation agreement, with reference to the experience of the Nigeria–São Tomé and Príncipe Joint Development Zone and the Timor Sea. :::

CHAPTER SUMMARY

- UNCLOS, adopted in 1982 and in force since 1994, is the comprehensive framework for the law of the sea, supplemented by the 1994 Part XI Agreement, the Fish Stocks Agreement, and the BBNJ Agreement; it established the ISA, the CLCS, and ITLOS and a system of compulsory dispute settlement.
- The coastal state has sovereignty over its territorial sea of up to twelve nautical miles and sovereign rights in its EEZ of up to two hundred nautical miles over natural resources and the production of energy from water, currents, and winds, together with jurisdiction over installations, subject to the freedoms of other states and the obligation of due regard.
- The continental shelf extends to at least two hundred nautical miles and, where the margin extends further, to its outer edge within maximum limits; coastal state rights exist *ipso facto* and *ab initio*; outer limits established on the basis of CLCS recommendations are final and binding; and Article 82 requires payments for exploitation beyond two hundred nautical miles.
- Offshore petroleum is governed by national licensing and safety regimes within the UNCLOS framework of coastal state jurisdiction over installations, safety zones, and decommissioning; maritime security, especially in the Gulf of Guinea, and new uses such as offshore wind, carbon storage, and hydrogen raise further legal issues.
- All states may lay submarine cables and pipelines on the continental shelf, subject to coastal state consent to the course of pipelines and reasonable measures; recent incidents of damage in the Baltic Sea have exposed gaps in the protection of submarine infrastructure.
- The Area and its resources are the common heritage of mankind administered by the ISA; exploitation regulations have not yet been adopted, a growing number of states support a moratorium, and unilateral licensing by the United States has challenged the regime.
- Part XII of UNCLOS imposes a general obligation to protect and preserve the marine environment; the ITLOS advisory opinion of 2024 held that greenhouse gas emissions constitute marine pollution subject to stringent due diligence obligations; and the BBNJ Agreement extends governance to areas beyond national jurisdiction.
- Maritime delimitation follows a three-stage methodology to achieve an equitable solution; resources and oil practice rarely affect the boundary; states must make every effort to agree provisional arrangements and must not jeopardise or hamper agreement; and joint development and unitisation provide cooperative solutions for disputed and straddling resources.

DOCTORAL RESEARCH QUESTIONS

1. Has the three-stage methodology of maritime delimitation achieved predictability at the cost of equity? Consider the treatment of natural resources and oil practice as relevant circumstances.
2. Evaluate the adequacy of the international legal framework for the protection of submarine pipelines, power interconnectors, and cables against deliberate damage, and propose reforms.
3. Does the common heritage of mankind principle, as implemented by the ISA, provide an adequate basis for the governance of deep-seabed mining in the context of demand for transition minerals? Assess the legality of unilateral licensing by non-parties to UNCLOS.
4. What are the implications of the ITLOS advisory opinion of 2024 for the licensing and environmental impact assessment of new offshore oil and gas projects by coastal states, including Nigeria?
5. Evaluate the institutional design and performance of the Nigeria–São Tomé and Príncipe Joint Development Zone, and identify lessons for future joint development arrangements in the Gulf of Guinea.
6. How should the law of the sea accommodate the large-scale development of offshore wind and hybrid interconnector projects in the EEZ, and is the "due regard" standard sufficient to manage conflicts among marine uses?

FURTHER READING

- Donald R Rothwell and Tim Stephens, *The International Law of the Sea* (3rd edn, Hart Publishing 2023).
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- Douglas R Burnett, Robert C Beckman and Tara M Davenport (eds), *Submarine Cables: The Handbook of Law and Policy* (Martinus Nijhoff 2014).
- Hazel Fox (ed), *Joint Development of Offshore Oil and Gas* (British Institute of International and Comparative Law 1989).
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PART VI

Human Rights, Corporate Responsibility, and Resource Justice

The human dimensions of energy and natural resource development — the rights of affected communities and indigenous peoples, the governance and accountability of the enterprises that develop resources, and the demands of justice and equity in the distribution of energy and resource wealth.

21 CHAPTER

Human Rights, Indigenous Peoples, and Extractive Industry Accountability

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Identify the international and regional human rights obligations of states that are engaged by energy and natural resource development, including the right to a clean, healthy, and sustainable environment.
2. Analyse the rights of indigenous peoples under international law and the duty of states to consult them, with reference to comparative jurisprudence.
3. Evaluate the content, legal status, and practical implementation of the principle of free, prior, and informed consent.
4. Explain the legal standards governing land acquisition, resettlement, and compensation for resource projects, including under Nigerian law and the standards of international financial institutions.
5. Assess the protection of labour rights and occupational safety in the energy and extractive industries, including forced and child labour in supply chains.
6. Evaluate the design and effectiveness of community grievance mechanisms and the accountability mechanisms of international financial institutions.
7. Explain the corporate responsibility to respect human rights and the emergence of mandatory human rights due diligence legislation.
8. Analyse the judicial and non-judicial remedies available to victims of human rights abuses connected with resource development, both in host states and in the home states of multinational enterprises.

Introduction

The development of energy and natural resources has profound consequences for the people who live where resources are found. Oil fields, mines, pipelines, dams, wind farms, and solar parks require land, water, and the cooperation of local populations; they bring employment, revenues, infrastructure, and access to energy, but they may also displace communities, pollute their environment,

disrupt their livelihoods and cultures, and expose workers to danger. Where governance is weak, resource development has been associated with repression, violence, and corruption. The history of oil production in the Niger Delta, examined in Chapters 17 and 24, of mining in the Democratic Republic of the Congo, of hydropower in the Amazon and the Mekong, and of resource extraction on indigenous lands throughout the world illustrates that the benefits and burdens of resource development are often distributed unequally, and that the people who bear the burdens are frequently those with the least voice in decisions.

International human rights law provides a framework for addressing these consequences. It imposes obligations on states to respect, protect, and fulfil the human rights of persons within their jurisdiction, including in the context of resource development undertaken by state enterprises or private companies. Over the past three decades, it has been supplemented by a body of law and practice addressing the specific rights of **indigenous peoples**, including the principle of **free, prior, and informed consent**, and by the framework of **business and human rights**, under which business enterprises have a responsibility to respect human rights and to conduct human rights due diligence — a responsibility increasingly given legal force by national and regional legislation. Victims of abuses have sought remedies in national courts, regional human rights systems, and non-judicial mechanisms, including through transnational litigation against parent companies in their home states.

This chapter examines the human rights obligations engaged by resource development (section 21.1), indigenous peoples' rights and the duty to consult (section 21.2), free, prior, and informed consent (section 21.3), land acquisition, resettlement, and compensation (section 21.4), labour rights and occupational safety (section 21.5), community grievance mechanisms (section 21.6), business and human rights due diligence (section 21.7), and judicial and non-judicial remedies (section 21.8). Chapter 22 examines corporate governance, compliance, and anti-corruption, including the governance dimensions of due diligence, and Chapter 23 examines energy justice and the equitable distribution of resource benefits.

	What it requires	Common failures in practice
Free	No coercion, intimidation, manipulation or undue pressure; community sets the process	Security presence; selective payments; dividing communities
Prior	Sought before concessions are granted and before activities begin	Consultation after licensing or construction
Informed	Full, accessible information on scope, risks, impacts and benefits; independent advice	Technical documents only; no local language; no time to deliberate
Consent	Collective decision through the people's own institutions; may be given, conditioned or withheld	Hand-picked representatives; treating consultation as consent
Ongoing agreement	Benefit-sharing, mitigation, monitoring and grievance redress over the project life	One-off sign-off; no follow-up on changes

Figure 21.1 The elements of free, prior, and informed consent and common failures in practice. *Source:* prepared for this edition.

21.1 Human Rights Obligations in Resource Development

The Sources of Obligation

The human rights obligations of states engaged by resource development derive from universal treaties, regional treaties, and customary international law. The **International Covenant on Civil and Political Rights** (ICCPR) protects, among other rights, the right to life (Article 6), freedom from torture and inhuman treatment (Article 7), the right to liberty and security of the person (Article 9), the right to privacy and home (Article 17), freedoms of expression, assembly, and association (Articles 19, 21, and 22), the right to participate in public affairs (Article 25), and the rights of members of ethnic, religious, and linguistic minorities to enjoy their own culture (Article 27). The **International Covenant on Economic, Social and Cultural Rights** (ICESCR) protects the rights to work and to just and favourable conditions of work (Articles 6 and 7), to an adequate standard of living, including adequate food, clothing, and housing (Article 11), to the highest attainable standard of physical and mental health, including through the improvement of environmental and industrial hygiene (Article 12), and to take part in cultural life (Article 15). Common Article 1 of both Covenants recognises the right of all peoples to self-determination, including the right freely

to dispose of their natural wealth and resources, and provides that in no case may a people be deprived of its own means of subsistence — a provision linked to the principle of permanent sovereignty over natural resources examined in Chapter 1.

Other universal instruments relevant to resource development include the **International Convention on the Elimination of All Forms of Racial Discrimination** (ICERD), whose monitoring committee has addressed the impact of extractive projects on indigenous peoples; the **Convention on the Elimination of All Forms of Discrimination against Women**; the **Convention on the Rights of the Child**; and the conventions of the International Labour Organization examined in section 21.5.

The African Charter

The **African Charter on Human and Peoples' Rights** (1981), to which all members of the African Union except Morocco are parties and which Nigeria has incorporated into domestic law by the African Charter on Human and Peoples' Rights (Ratification and Enforcement) Act, is distinctive in protecting collective "peoples' rights" alongside individual rights. **Article 21** provides that all peoples shall freely dispose of their wealth and natural resources, that this right shall be exercised in the exclusive interest of the people, and that in no case shall a people be deprived of it; that in case of spoliation, the dispossessed people shall have the right to lawful recovery of its property and adequate compensation; and that states undertake to eliminate all forms of foreign economic exploitation, particularly that practised by international monopolies, so as to enable their peoples to benefit fully from the advantages derived from their national resources. **Article 24** provides that all peoples shall have the right to a general satisfactory environment favourable to their development — the first binding recognition of a right to environment in an international human rights treaty. Article 22 recognises the right of all peoples to their economic, social, and cultural development.

::: casebox {title="SERAC v Nigeria: Oil, Rights, and the Ogoni People"}
Social and Economic Rights Action Center and Center for Economic and Social Rights v Nigeria (African Commission on Human and Peoples' Rights, Communication 155/96, decided October 2001).

Facts. The communication alleged that the military government of Nigeria had been directly involved in oil production through the Nigerian National Petroleum Corporation, the majority shareholder in a consortium with Shell Petroleum Development Company, and that their operations had caused environmental degradation and health problems among the Ogoni people through the contamination of soil, water, and air. It further alleged that the government had failed to monitor operations, to require environmental and social impact assessments, and to provide information to communities, and that its security forces

had attacked, burned, and destroyed Ogoni villages and homes in response to non-violent protests led by the Movement for the Survival of the Ogoni People.

Decision. The Commission found violations of Articles 2, 4, 14, 16, 18(1), 21, and 24 of the African Charter. It held that states have duties to **respect, protect, promote, and fulfil** all rights, and that Nigeria had violated its duty to protect by failing to protect the Ogoni from harmful acts of private parties, in particular the oil companies. It held that Article 24 imposes obligations on states to take reasonable measures to prevent pollution and ecological degradation, to promote conservation, and to secure ecologically sustainable development and use of natural resources, including by ordering independent scientific monitoring, requiring environmental and social impact studies before industrial development, providing information to affected communities, and giving them meaningful opportunities to be heard. It found a violation of Article 21 because the government had given the green light to private actors, and the oil companies in particular, to devastatingly affect the wellbeing of the Ogonis. Reading the rights to health, property, and family protection together, it found implied rights to **food** and **housing**. It appealed to the government to stop all attacks on Ogoni communities, conduct an investigation, compensate victims, ensure environmental and social impact assessments for future operations, and undertake a comprehensive clean-up.

Significance. *SERAC* is a landmark in the protection of economic, social, and cultural rights and in the application of human rights law to extractive industries. It established the state's duty to protect against harm by companies, gave content to the right to a satisfactory environment, and has been widely cited by courts and commentators. Its recommendations on clean-up resonate with the UNEP Environmental Assessment of Ogoniland (2011) and the subsequent Hydrocarbon Pollution Remediation Project, examined in Chapter 17. :::

The African Commission and the **African Court on Human and Peoples' Rights** have developed this jurisprudence in subsequent cases. In ***Centre for Minority Rights Development and Minority Rights Group International on behalf of the Endorois Welfare Council v Kenya*** (2010), the Commission found that the eviction of the Endorois from their ancestral lands around Lake Bogoria for the creation of a game reserve, without adequate consultation or compensation, violated their rights to property, religion, culture, natural resources, and development, and held that the right to development requires that development be both equitable and participatory and that, for development projects with a major impact, the state must obtain the free, prior, and informed consent of the community. In ***African Commission on Human and Peoples' Rights v Kenya*** (the *Ogiek* case, 2017), the African Court held that the eviction of the Ogiek from the Mau Forest violated their rights, including under Articles

14, 21, and 22, and in its reparations judgment of 2022 ordered Kenya to take measures to delimit, demarcate, and title Ogiek ancestral land and to consult the Ogiek on development projects in that land. The **ECOWAS Community Court of Justice**, which has jurisdiction over human rights cases brought by individuals against ECOWAS member states without a requirement to exhaust domestic remedies, held in ***SERAP v Nigeria*** (2012) that Nigeria had violated Articles 1 and 24 of the African Charter by failing to prevent and remedy oil pollution in the Niger Delta and to hold perpetrators accountable.

The Right to a Clean, Healthy, and Sustainable Environment

The recognition of a human right to a healthy environment has advanced rapidly. Beyond Article 24 of the African Charter, the right is recognised in the **San Salvador Protocol** to the American Convention on Human Rights, the **Arab Charter on Human Rights**, the **Aarhus Convention** (in its procedural dimensions), and the **Escazú Agreement** (2018), the first regional environmental treaty in Latin America and the Caribbean, which also contains specific provisions on the protection of environmental human rights defenders. More than a hundred and fifty states recognise the right in their constitutions, legislation, or regional treaties. In October 2021, the UN Human Rights Council adopted **Resolution 48/13**, recognising the right to a clean, healthy, and sustainable environment, and in July 2022, the UN General Assembly adopted **Resolution 76/300** to the same effect by 161 votes to none, with eight abstentions. Although these resolutions are not binding, they reflect a broad consensus and have been relied upon by courts and in the advisory opinions on climate change examined in Chapter 18. The Inter-American Court of Human Rights, in its **Advisory Opinion OC-23/17** (2017), recognised the right to a healthy environment as an autonomous right under the American Convention, and in ***Lhaka Honhat v Argentina*** (2020), it found for the first time a violation of the right to a healthy environment in a contentious case, together with the rights to food, water, and cultural identity of indigenous communities.

The **Constitution of Nigeria** includes environmental protection among the Fundamental Objectives and Directive Principles of State Policy in section 20, which are not directly justiciable under section 6(6)(c), but Nigerian courts have interpreted the rights to life and dignity in Chapter IV of the Constitution and the African Charter to protect against environmental harm. In ***Gbemre v Shell Petroleum Development Company of Nigeria Ltd*** (Federal High Court, Benin, 2005), the court held that gas flaring in the applicant's community violated the rights to life and dignity of the human person under sections 33 and 34 of the Constitution and Articles 4, 16, and 24 of the African Charter, and ordered that flaring be stopped — an order that was not complied with. The decision, although

of limited precedential weight, is often cited as an early example of rights-based environmental litigation.

The Tripartite Typology of Obligations

Human rights treaty bodies and regional mechanisms have articulated a tripartite typology of state obligations: to **respect** rights, by refraining from interfering with their enjoyment; to **protect** rights, by preventing, investigating, punishing, and redressing their violation by third parties, including business enterprises; and to **fulfil** rights, by taking positive measures to facilitate and provide for their enjoyment. In resource development, the obligation to respect is engaged when the state, or a state-owned enterprise, directly causes harm — for example, through forced evictions or violent responses to protests. The obligation to protect is engaged when private companies cause harm, requiring the state to regulate, monitor, and enforce, as in *SERAC*. The obligation to fulfil may require the state to ensure that resource revenues are used for the realisation of rights, as examined in Chapter 23.

Extraterritorial Obligations

Whether states have human rights obligations with respect to the conduct of companies domiciled in their territory that cause harm abroad is a contested question of importance for the regulation of multinational extractive enterprises. The **Maastricht Principles on Extraterritorial Obligations of States in the area of Economic, Social and Cultural Rights** (2011), adopted by experts, articulate obligations to regulate companies over which states are in a position to exercise influence. The UN Committee on Economic, Social and Cultural Rights, in **General Comment No. 24** (2017) on state obligations in the context of business activities, stated that states parties are required to take the necessary steps to prevent human rights violations abroad by corporations domiciled in their territory or jurisdiction, without infringing the sovereignty of host states. The UN Guiding Principles on Business and Human Rights, examined in section 21.7, take a more cautious position, stating that states are not generally required under international human rights law to regulate the extraterritorial activities of businesses domiciled in their territory, but are not generally prohibited from doing so, and that they should set out clearly the expectation that such businesses respect human rights throughout their operations. The mandatory due diligence legislation and transnational litigation examined in sections 21.7 and 21.8 give practical effect to home state regulation.

Human Rights Defenders and Civic Space

Resource development is associated with serious risks to **environmental and land defenders**. Global Witness has recorded the killing of more than two thousand land and environmental defenders worldwide between 2012 and 2023,

many in connection with mining, agribusiness, hydropower, and logging, with the largest numbers in Latin America and the Philippines. The execution of Ken Saro-Wiwa and eight other Ogoni leaders by the Nigerian military government in November 1995, following a trial widely condemned as unfair, remains one of the most prominent cases connecting resource development with the suppression of dissent; it led to Nigeria's suspension from the Commonwealth and, later, to litigation in the United States examined in section 21.8. The **UN Declaration on Human Rights Defenders** (1998), the Escazú Agreement, and the guidance of the UN Special Rapporteur on the situation of human rights defenders require states to protect defenders, and companies are expected under the UNGPs to avoid contributing to attacks on them. The **Voluntary Principles on Security and Human Rights** (2000), a multi-stakeholder initiative of governments, extractive companies, and non-governmental organisations, provide guidance on risk assessment and on the relationship between companies and public and private security forces, an issue of particular importance in conflict-affected areas such as the Niger Delta and the eastern Democratic Republic of the Congo.

21.2 Indigenous Peoples' Rights and Consultation

Who Are Indigenous Peoples?

International law contains no formal definition of indigenous peoples. The working definition proposed by José Martínez Cobo in his study for the UN Sub-Commission in the 1980s emphasised historical continuity with pre-invasion and pre-colonial societies, distinctiveness from other sectors of society, non-dominance, and a determination to preserve, develop, and transmit ancestral territories and ethnic identity. ILO Convention No. 169 applies to tribal peoples whose social, cultural, and economic conditions distinguish them from other sections of the national community, and to peoples regarded as indigenous on account of their descent from the populations which inhabited the country at the time of conquest or colonisation, and treats **self-identification** as a fundamental criterion. The UN Declaration on the Rights of Indigenous Peoples deliberately does not define indigenous peoples.

In Africa, where most of the population can claim to be "indigenous" in the sense of pre-colonial descent, the concept has been contested. The African Commission's **Working Group on Indigenous Populations/Communities**, in its report adopted in 2005, emphasised that the concept in Africa is not about first occupancy but about groups whose cultures and ways of life differ considerably from the dominant society, which are often marginalised and discriminated against, and which have a special attachment to and use of their traditional land and territories — such as hunter-gatherers and pastoralists. The Commission's **Advisory Opinion on the UN Declaration on the Rights of Indigenous**

Peoples (2007) sought to address the concerns of African states that recognising indigenous peoples would threaten national unity. The *Endorois* and *Ogiek* decisions applied the concept to particular communities in Kenya. In Nigeria, the federal government has generally taken the position that all Nigerians are indigenous, and the claims of groups such as the Ogoni and the Ijaw have been framed principally in terms of minority rights, peoples' rights under the African Charter, and resource control, rather than indigenous rights in the international sense — although the Ogoni Bill of Rights of 1990 invoked self-determination and the right to control a fair share of the resources of their land.

ILO Convention No. 169 and UNDRIP

The **ILO Indigenous and Tribal Peoples Convention, 1989 (No. 169)** is the only binding international treaty devoted specifically to indigenous peoples' rights. It has been ratified by around twenty-four states, most of them in Latin America, together with Denmark, Norway, Spain, the Netherlands, Luxembourg, Germany, Fiji, Nepal, and the Central African Republic, the only African state party. It requires governments to **consult** the peoples concerned, through appropriate procedures and in particular through their representative institutions, whenever consideration is being given to legislative or administrative measures which may affect them directly, in good faith and with the objective of achieving agreement or consent (Article 6). It recognises rights of ownership and possession over the lands which they traditionally occupy (Article 14), and provides that where the state retains ownership of mineral or subsurface resources, governments must establish procedures to consult the peoples before undertaking or permitting any programmes for the exploration or exploitation of such resources, and that the peoples shall wherever possible participate in the benefits and receive fair compensation for any damages (Article 15). Relocation may take place only with their free and informed consent or, where consent cannot be obtained, following appropriate procedures (Article 16).

The **UN Declaration on the Rights of Indigenous Peoples** (UNDRIP), adopted by the General Assembly in September 2007 by 144 votes to four (Australia, Canada, New Zealand, and the United States), with eleven abstentions, sets out a comprehensive catalogue of rights, including the rights to self-determination (Article 3), to autonomy or self-government in internal and local affairs (Article 4), to lands, territories, and resources which they have traditionally owned, occupied, or otherwise used or acquired (Article 26), to redress for lands taken without their free, prior, and informed consent (Article 28), and to determine and develop priorities and strategies for the development or use of their lands or territories and other resources (Article 32). The four states that voted against later endorsed the Declaration. Although UNDRIP is not a treaty, some of its provisions are regarded as reflecting customary international law or

general principles, and it has been relied upon by courts and treaty bodies. Several states have given it domestic legal effect — notably **Canada**, through the United Nations Declaration on the Rights of Indigenous Peoples Act 2021 at the federal level and the Declaration on the Rights of Indigenous Peoples Act 2019 in British Columbia, and **Bolivia**, which incorporated the Declaration into law in 2007.

The Inter-American System

The Inter-American Court of Human Rights has developed the most extensive jurisprudence on indigenous land and resource rights, interpreting the right to property in Article 21 of the American Convention to protect the communal property of indigenous and tribal peoples. In ***Mayagna (Sumo) Awas Tingni Community v Nicaragua*** (2001), the Court held that the grant of a logging concession on indigenous land without demarcation or consultation violated the community's right to property. In ***Saramaka People v Suriname*** (2007), examined in section 21.3, it held that the state must consult and, for large-scale projects, obtain the consent of tribal peoples before granting concessions affecting their territories. In ***Kichwa Indigenous People of Sarayaku v Ecuador*** (2012), it held that Ecuador had violated the rights to consultation, property, and cultural identity by permitting an oil company to undertake seismic exploration, including the placement of explosives, on Sarayaku territory without prior consultation, and described the obligation to consult as a general principle of international law. In *Kaliña and Lokono Peoples v Suriname* (2015), it addressed the impact of bauxite mining and protected areas on indigenous lands.

The Duty to Consult in National Law: Comparative Perspectives

Several national legal systems have developed a duty to consult indigenous peoples on resource development.

Canada. Section 35 of the Constitution Act 1982 recognises and affirms existing Aboriginal and treaty rights. In ***Haida Nation v British Columbia (Minister of Forests)*** (2004), the Supreme Court of Canada held that the Crown has a duty to consult and, where appropriate, accommodate Aboriginal peoples when it has knowledge, real or constructive, of the potential existence of an Aboriginal right or title and contemplates conduct that might adversely affect it. The duty, grounded in the honour of the Crown, varies with the strength of the claim and the seriousness of the potential impact, along a spectrum from notice to deep consultation. In ***Tsilhqot'in Nation v British Columbia*** (2014), the Court for the first time recognised Aboriginal title to a specific area and held that, once title is established, the Crown must obtain the consent of the titleholders or justify an infringement under a stringent test. In ***Clyde River (Hamlet) v Petroleum Geo-Services Inc*** (2017), the Court quashed an authorisation for offshore seismic testing in Nunavut because the regulator's consultation had been inadequate,

while in the companion case of *Chippewas of the Thames First Nation v Enbridge Pipelines Inc* (2017), it upheld the approval of a pipeline modification, finding consultation adequate. The Federal Court of Appeal quashed the approval of the **Trans Mountain Expansion** pipeline in *Tsleil-Waututh Nation v Canada* (2018) for inadequate consultation; after renewed consultation, the project was re-approved, the renewed approval was upheld in 2020, and the pipeline entered service in 2024.

Australia. In ***Mabo v Queensland (No 2)*** (1992), the High Court rejected the doctrine of *terra nullius* and recognised native title. The **Native Title Act 1993** provides for the recognition of native title and gives registered native title claimants and holders a **right to negotiate** over the grant of mining tenements, with recourse to the National Native Title Tribunal in the absence of agreement, and provides for **Indigenous Land Use Agreements**. In *Northern Territory v Griffiths* (the *Timber Creek* case, 2019), the High Court addressed the assessment of compensation for the extinguishment of native title, including for cultural loss.

Latin America. Many Latin American states have enacted prior consultation laws or developed constitutional jurisprudence. **Peru**'s Law on the Right to Prior Consultation of Indigenous Peoples (2011) requires consultation on measures affecting indigenous peoples, although it does not require consent. The **Constitutional Court of Colombia** has developed extensive jurisprudence, holding in Judgment T-129 of 2011 that consent is required for projects that would involve displacement, the storage of toxic waste, or high social, cultural, and environmental impacts threatening the community's survival.

Norway. In the ***Fosen*** case, decided in October 2021 and examined in Chapter 10, the Supreme Court of Norway held that the licences for two wind farms on the Fosen peninsula violated the rights of Sámi reindeer herders under Article 27 of the ICCPR, rendering the licences invalid; agreements between the herders and the developers on continued operation, with compensation and mitigation, were concluded in 2023 and 2024 after prolonged protest. The case demonstrates that renewable energy projects, no less than fossil fuel projects, must respect indigenous rights.

21.3 Free, Prior, and Informed Consent

The Elements of FPIC

The principle of **free, prior, and informed consent** (FPIC) expresses the idea that indigenous peoples should be able to give or withhold their consent to projects and measures affecting their lands, territories, and resources, through their own representative institutions. Each element has a recognised content, articulated in UN and regional practice, including the reports of the UN Special Rapporteur on the rights of indigenous peoples and the Expert Mechanism on

the Rights of Indigenous Peoples, whose study of 2018 provides a comprehensive analysis, and summarised in Figure 21.1:

- **Free** implies the absence of coercion, intimidation, manipulation, or undue pressure — including from security forces, threats of withdrawal of services, bribery, or the division of communities — and requires that the community itself determines the process, timeline, and decision-making structure.
- **Prior** implies that consent is sought sufficiently in advance of any authorisation or commencement of activities, respecting the time requirements of indigenous consultation and consensus processes, and before key decisions — such as the grant of a concession — have been taken.
- **Informed** implies that the community receives full, objective, and accessible information, in a language and form it understands, on the nature, size, pace, reversibility, and scope of the proposed project; its purpose and duration; the areas affected; a preliminary assessment of its economic, social, cultural, and environmental impacts and risks; the personnel likely to be involved; and the procedures the project may entail — and has access to independent advice.
- **Consent** implies a collective decision reached through the community's customary decision-making processes and representative institutions, which may be given, given subject to conditions, or withheld, and which may need to be revisited as the project evolves.

Legal Status and Content: Consultation or Consent?

The central controversy surrounding FPIC is whether it confers on indigenous peoples a **right to veto** projects affecting them, or merely a right to be consulted in good faith with the objective of obtaining consent. The instruments differ. ILO Convention No. 169 requires consultation with the objective of achieving agreement or consent (Article 6(2)) and requires consent only for relocation, subject to an exception (Article 16). UNDRIP refers to FPIC in several contexts: it provides that indigenous peoples shall not be forcibly removed from their lands without their FPIC (Article 10); that no storage or disposal of hazardous materials shall take place on their lands without their FPIC (Article 29(2)); and that redress is due for lands taken without their FPIC (Article 28). In the provisions most relevant to resource development, however, UNDRIP requires states to "consult and cooperate in good faith with the indigenous peoples concerned through their own representative institutions **in order to obtain** their free, prior and informed consent" before adopting legislative or administrative measures that may affect them (Article 19) and "prior to the approval of any project affecting their lands or territories and other resources, particularly in connection with the development, utilization or exploitation of mineral, water or other resources" (Article 32(2)).

Former UN Special Rapporteur James Anaya argued that FPIC should not be regarded as a general veto power, but that the strength of the requirement varies with the nature and severity of the impact: as a general rule, extractive activities should not take place within the territories of indigenous peoples without their consent, and consent is required where a project would have a significant, direct impact on their lives or territories, while states may proceed without consent in limited circumstances where the rights limitation is necessary, proportionate, and in pursuit of a valid public purpose within a human rights framework. Many indigenous organisations and scholars reject this approach and maintain that FPIC is an expression of self-determination that necessarily includes the right to say no. States and companies have tended to emphasise the consultative dimension.

::: casebox {title="Saramaka People v Suriname: Consent for Large-Scale Projects"} **Saramaka People v Suriname (Inter-American Court of Human Rights, Judgment of 28 November 2007; Interpretation Judgment of 12 August 2008).**

Facts. The Saramaka, a tribal people descended from Africans who escaped slavery in the seventeenth and eighteenth centuries and established autonomous communities in the interior of Suriname, claimed that the state had granted logging and mining concessions on their traditional territory without consultation, and that the Afobaka dam, built in the 1960s to supply power to an aluminium smelter, had flooded part of their territory and displaced communities. Suriname's legal system did not recognise collective property rights of indigenous or tribal peoples, and Suriname owned all subsoil resources.

Judgment. The Court held that the Saramaka, as a tribal people sharing the distinctive characteristics of indigenous peoples, were entitled to the protection of Article 21 of the American Convention (right to property) in respect of the lands they had traditionally used and occupied, and to the natural resources on and within the land that were necessary for their survival. The state could restrict their property rights to grant concessions only if it complied with three safeguards: it must ensure the **effective participation** of the Saramaka, in conformity with their customs and traditions, in any development, investment, exploration, or extraction plan within their territory; it must guarantee that they receive a **reasonable benefit** from such plans; and it must ensure that no concession is issued until independent and technically capable entities, under the state's supervision, perform a prior **environmental and social impact assessment**. The Court further held that, regarding **large-scale development or investment projects that would have a major impact** within Saramaka territory, the state has a duty not only to consult but also to **obtain their free, prior, and informed consent**, according to their customs and traditions. It ordered Suriname to

delimit, demarcate, and grant collective title over Saramaka territory, to review existing concessions, and to adopt legislation recognising collective rights.

Significance. *Saramaka* is the leading international judicial authority for the proposition that consent, and not merely consultation, is required for large-scale projects with a major impact on indigenous or tribal territories. Its three safeguards — participation, benefit-sharing, and impact assessment — have been widely adopted, including by the African Commission in *Endorois*. Its implementation by Suriname has been slow, illustrating the gap between judicial pronouncement and practice. :::

FPIC in the Standards of Financial Institutions and Industry

FPIC has been incorporated into the standards of international financial institutions and industry bodies, giving it practical effect through financing conditions. The **International Finance Corporation's Performance Standard 7** on Indigenous Peoples, as revised in 2012, requires the client to obtain the FPIC of affected indigenous communities in specified circumstances: impacts on lands and natural resources subject to traditional ownership or under customary use; relocation of indigenous peoples from such lands; and significant impacts on critical cultural heritage, or the use of cultural heritage for commercial purposes. PS7 defines FPIC as established through good faith negotiation and states that it does not necessarily require unanimity and may be achieved even when individuals or groups within the community explicitly disagree. The **Equator Principles**, adopted by financial institutions to manage environmental and social risk in project finance, incorporate the IFC Performance Standards and, in their fourth iteration (2020), require FPIC in the circumstances identified in PS7 regardless of the project's location. The **World Bank's Environmental and Social Standard 7** under its Environmental and Social Framework (2016, in effect 2018) similarly requires FPIC in specified circumstances.

Among industry bodies, the **International Council on Mining and Metals** (ICMM) adopted a position statement on indigenous peoples and mining in 2013, committing its members to work to obtain the consent of indigenous peoples for new projects and changes to existing projects located on their lands and likely to have significant adverse impacts, while acknowledging that it would be for governments to decide whether a project proceeds where consent is withheld. Standards such as the Initiative for Responsible Mining Assurance and the Roundtable on Sustainable Palm Oil also incorporate FPIC.

The Practice of FPIC

The practical implementation of FPIC raises numerous issues: identifying the communities entitled to give consent and their legitimate representatives; the risk of manipulation by companies or governments through the selection of represen-

tatives or the payment of benefits to sections of a community; the role of women, young people, and other groups within communities; the relationship between FPIC and the state's ownership of subsoil resources; the treatment of projects that predate the recognition of FPIC; and whether consent, once given, may be withdrawn. Many indigenous peoples have developed their own **FPIC protocols**, setting out how they expect to be consulted, which some courts and governments have recognised.

National implementation varies. The **Philippines**' Indigenous Peoples' Rights Act 1997 requires FPIC, certified by the National Commission on Indigenous Peoples, for the grant of concessions on ancestral domains — one of the few statutory consent requirements — although its implementation has been criticised for manipulation of consent processes. In Australia, the destruction by Rio Tinto in May 2020 of the 46,000-year-old rock shelters at **Juukan Gorge** in the Pilbara, which had been lawfully authorised under Western Australian heritage legislation dating from 1972, despite the objections of the Puutu Kunti Kurrama and Pinikura peoples, led to a parliamentary inquiry whose report *Never Again* (2021) recommended reform of cultural heritage law and the adoption of FPIC-based standards, the resignation of Rio Tinto's chief executive and other senior executives, and a reappraisal of agreement-making practices in the mining industry. Western Australia enacted a new Aboriginal Cultural Heritage Act in 2021, but repealed it in 2023 following opposition from landholders, reverting to amended earlier legislation.

FPIC and the Energy Transition

The energy transition has brought FPIC to the centre of debates about renewable energy and transition minerals. A large proportion of the known reserves of lithium, nickel, cobalt, copper, and other minerals, and of the land suitable for wind and solar projects, lies on or near the lands of indigenous and peasant communities. The Business and Human Rights Resource Centre's Transition Minerals Tracker has recorded hundreds of allegations of human rights abuses linked to the mining of transition minerals. Disputes have arisen over lithium extraction in the salt flats of Argentina, Bolivia, and Chile, affecting indigenous communities' access to water; over the Thacker Pass lithium mine in Nevada, where indigenous groups claimed inadequate consultation; and over wind farms on indigenous lands in Mexico's Isthmus of Tehuantepec, Kenya's Lake Turkana region, and Norway's Fosen peninsula. Advocates argue that a **just transition** must respect FPIC and ensure that indigenous peoples share in the benefits of renewable energy, including through equity participation and community ownership, a theme developed in Chapter 23.

21.4 Land Acquisition, Resettlement, and Compensation

The Need for Land and the Risk of Displacement

Resource projects require land — for mines, wells, pipelines, processing facilities, reservoirs, transmission lines, solar parks, and wind farms — and its acquisition often involves the physical displacement of people from their homes, or the economic displacement of people from their sources of livelihood, such as farmland, fishing grounds, and forests. Large dams alone are estimated to have displaced between forty and eighty million people in the twentieth century, according to the World Commission on Dams. Displacement carries well-documented risks of impoverishment — which the sociologist Michael Cernea identified as landlessness, joblessness, homelessness, marginalisation, food insecurity, increased morbidity, loss of access to common property, and social disarticulation — and has disproportionate effects on women, indigenous peoples, and the poor, particularly those whose rights to land are customary or informal.

International Standards

International human rights law prohibits **forced evictions**, defined by the Committee on Economic, Social and Cultural Rights in **General Comment No. 7** (1997) as the permanent or temporary removal against their will of individuals, families, or communities from the homes or land which they occupy, without the provision of, and access to, appropriate forms of legal or other protection. Evictions may be justified only in exceptional circumstances, in accordance with law and the principles of reasonableness and proportionality, and subject to procedural protections, including genuine consultation, adequate and reasonable notice, information, legal remedies, and legal aid. The **UN Basic Principles and Guidelines on Development-based Evictions and Displacement** (2007) elaborate these requirements, including the obligation to explore all feasible alternatives, to ensure that evictions do not result in homelessness, and to provide just compensation and adequate resettlement.

The standards of international financial institutions are the most detailed and widely applied. The **IFC's Performance Standard 5** on Land Acquisition and Involuntary Resettlement and the **World Bank's Environmental and Social Standard 5** require clients to avoid or minimise displacement by exploring alternative project designs; to avoid forced eviction; to mitigate adverse impacts by providing **compensation for loss of assets at full replacement cost** and other assistance to help displaced persons improve or at least restore their livelihoods and standards of living; to improve the living conditions of physically displaced persons through the provision of adequate housing with security of tenure; to provide compensation and assistance to persons without formal legal rights to the land they occupy; to ensure informed participation and consultation; and to

prepare a resettlement action plan or livelihood restoration plan. The **Voluntary Guidelines on the Responsible Governance of Tenure of Land, Fisheries and Forests** (2012), endorsed by the Committee on World Food Security, provide guidance on the recognition and protection of legitimate tenure rights, including customary rights.

Nigerian Law

The acquisition of land for resource development in Nigeria is governed principally by the **Land Use Act 1978**, which vests all land comprised in the territory of each state in the Governor of the state, to be held in trust and administered for the use and common benefit of all Nigerians. The Governor grants statutory rights of occupancy in urban areas, and local governments grant customary rights of occupancy in non-urban areas. Under section 28, the Governor may revoke a right of occupancy for **overriding public interest**, which includes the requirement of the land for mining purposes or oil pipelines or for any purpose connected therewith. Compensation under section 29 is payable for the value of unexhausted improvements — buildings, crops, and other developments — and, in the case of land required for mining or oil pipelines, the holder and occupier are entitled to compensation under the appropriate provisions of the minerals or oil pipelines legislation. The Act does not provide compensation for the value of the land itself, reflecting its vesting in the state, a feature that has been criticised as inadequate in the light of international standards requiring compensation at replacement cost. The Act is entrenched in the Constitution and can be amended only by the procedure for constitutional amendment, which has impeded reform.

Section 44(3) of the Constitution vests the entire property in and control of all minerals, mineral oils, and natural gas in, under, or upon any land in Nigeria, its territorial waters, and its EEZ in the Government of the Federation. Section 44(1) prohibits the compulsory acquisition of property except in the manner and for the purposes prescribed by a law that requires prompt payment of compensation and gives the owner a right of access to a court for the determination of the amount, although section 44(2) excepts certain cases. The **Oil Pipelines Act** provides for the grant of permits and licences for oil pipelines and for compensation to owners and occupiers for disturbance and for damage caused by the construction or operation of pipelines, including by leaks. The **Petroleum Industry Act 2021** addresses access to land for petroleum operations and requires the payment of fair and adequate compensation for the disturbance of surface rights and for damage. The **Nigerian Minerals and Mining Act 2007** provides for compensation to owners and occupiers of land for disturbance of surface rights and damage, and requires mining lease holders to conclude a **Community Development Agreement** with host communities, addressing matters such as employment, education, infrastructure, and the protection of the environment. The PIA's

provisions on **host communities development trusts**, which channel three per cent of the actual annual operating expenditure of upstream operators to trusts for host communities, are examined in Chapter 24.

Comparative Perspectives

Other African jurisdictions have strengthened the protection of customary land rights in the context of resource development. **Kenya**'s Constitution of 2010 recognises community land as a category of land tenure, and the **Community Land Act 2016** provides for its registration and requires that any agreement relating to the investment in community land be made with the free, prior, and informed consent of the community. **Mozambique**'s Land Law of 1997 recognises customary rights acquired through occupation and requires community consultation before the grant of rights to investors. **South Africa**'s Interim Protection of Informal Land Rights Act 1996 protects holders of informal rights, and in *Maledu v Itereleng Bakgatla Mineral Resources* (2018), the Constitutional Court held that a mining right did not entitle the holder to evict a community holding informal land rights without their consent or expropriation under the Act.

Resettlement in Practice

The resettlement associated with large energy projects in Africa illustrates both the application of international standards and their limitations. The **Mozambique LNG** project in Cabo Delgado required the resettlement of several hundred households from the Afungi peninsula and affected the livelihoods of many more, including fishers who lost access to the sea; the project was suspended in 2021 following an insurgent attack on the town of Palma, and the resumption of construction was announced in 2025. The **East African Crude Oil Pipeline** and associated **Tilenga** and **Kingfisher** projects in Uganda and Tanzania have required the acquisition of land affecting more than a hundred thousand people, with complaints concerning delays in compensation, inadequate valuation, and restrictions on land use during the period between valuation and payment; the projects have been the subject of litigation in France under the Duty of Vigilance Law, examined in Chapter 22, and of campaigns that have led many banks and insurers to decline involvement. The **Lake Turkana Wind Power** project in Kenya, Africa's largest wind farm, faced litigation by communities claiming that the land had been acquired without their consent; in 2021, the Environment and Land Court held that the acquisition of the land had been irregular, while declining to order the removal of the project. These cases illustrate the importance of secure tenure, early and meaningful consultation, fair valuation, prompt payment, livelihood restoration, and independent monitoring.

21.5 Labour Rights and Occupational Safety

Labour in the Energy and Extractive Industries

The energy and extractive industries employ tens of millions of people worldwide, from highly skilled engineers on offshore platforms to artisanal miners digging by hand. They are among the most hazardous sectors of the economy: mining, in particular, accounts for a disproportionate share of occupational fatalities, and petroleum operations carry risks of fires, explosions, blowouts, and exposure to toxic substances. Workers in remote locations, on offshore installations, and in fly-in, fly-out arrangements face distinctive pressures, and migrant and contract workers are often vulnerable to exploitation. The energy transition is creating new forms of employment in renewable energy and transition mineral supply chains, while threatening jobs in coal and other fossil fuel industries — a challenge addressed through the concept of just transition, examined in Chapter 18.

The International Labour Framework

The **International Labour Organization** (ILO), founded in 1919, adopts conventions and recommendations establishing international labour standards. The **ILO Declaration on Fundamental Principles and Rights at Work** (1998) commits all member states, by virtue of their membership and regardless of ratification, to respect, promote, and realise principles concerning **freedom of association and the effective recognition of the right to collective bargaining**, the **elimination of all forms of forced or compulsory labour**, the **effective abolition of child labour**, and the **elimination of discrimination in respect of employment and occupation**. In June 2022, the International Labour Conference amended the Declaration to add a fifth category, **a safe and healthy working environment**, and designated the Occupational Safety and Health Convention, 1981 (No. 155) and the Promotional Framework for Occupational Safety and Health Convention, 2006 (No. 187) as fundamental conventions.

Conventions of particular relevance to energy and extractive industries include the **Safety and Health in Mines Convention, 1995 (No. 176)**, which requires states to adopt and enforce legislation on mine safety, imposes duties on employers to eliminate or minimise risks, and grants workers rights, including the right to remove themselves from danger; the **Worst Forms of Child Labour Convention, 1999 (No. 182)**, which classifies hazardous work, including underground mining, among the worst forms of child labour; the **Minimum Age Convention, 1973 (No. 138)**; the **Forced Labour Convention, 1930 (No. 29)** and its **Protocol of 2014**; the **Violence and Harassment Convention, 2019 (No. 190)**; and the **Maritime Labour Convention, 2006**, which applies to sea-

farers, including those on vessels serving offshore installations. Nigeria has ratified the core conventions, including Conventions Nos. 29, 87, 98, 138, and 182.

Occupational Safety in Petroleum and Mining

The regulation of occupational safety in petroleum operations has been shaped by major disasters. As noted in Chapter 20, the **Piper Alpha** disaster in 1988 led to the adoption of the safety case regime in the United Kingdom, which has been widely emulated, and the *Deepwater Horizon* disaster, in which eleven workers died, led to reforms in the United States and the European Union. In mining, disasters such as the Soma mine explosion in Turkey in 2014, which killed 301 miners, and the Brumadinho tailings dam collapse in Brazil in 2019, which killed 270 people, most of them workers of Vale, have prompted reforms of mine safety and tailings management, including the **Global Industry Standard on Tailings Management** (2020), developed by the ICMM, UNEP, and the Principles for Responsible Investment.

The **Marikana** massacre in South Africa in August 2012, in which police shot dead thirty-four striking mineworkers at Lonmin's platinum mine, after ten other people had been killed in the preceding days, illustrated the intersection of labour rights, living conditions, the role of security forces, and corporate responsibility in the extractive sector. The **Marikana Commission of Inquiry**, chaired by Judge Ian Farlam, reported in 2015, criticising the police, the unions, and Lonmin, including for its failure to implement the housing obligations of its social and labour plan, a requirement under South Africa's mining legislation. Civil claims by the families of those killed against the state were subsequently settled in part.

In **Nigeria**, labour relations are governed by the **Labour Act**, the **Trade Unions Act**, the **Factories Act**, and the **Employee's Compensation Act 2010**, which provides for compensation for workplace injuries, diseases, and death through a fund administered by the Nigeria Social Insurance Trust Fund. Occupational safety in petroleum operations is regulated under petroleum legislation and regulations, with the Nigerian Upstream Petroleum Regulatory Commission and the Nigerian Midstream and Downstream Petroleum Regulatory Authority exercising regulatory functions under the PIA, while mine safety falls under the Nigerian Minerals and Mining Act 2007 and its regulations. The petroleum sector's unions — the **Petroleum and Natural Gas Senior Staff Association of Nigeria** (PENGASSAN) and the **National Union of Petroleum and Natural Gas Workers** (NUPENG) — have historically played a significant role in industrial relations and in national politics, including through strikes concerning fuel subsidies and the casualisation of labour through contract staffing. The Nigerian Oil and Gas Industry Content Development Act 2010, examined in Chapter 7, promotes the employment and training of Nigerians in the petroleum industry.

Forced and Child Labour in Supply Chains

The globalisation of energy supply chains has brought attention to forced and child labour in the production of minerals and components for the energy transition. Two cases illustrate the issues.

Cobalt in the Democratic Republic of the Congo. The DRC produces around seventy per cent of the world's cobalt, a mineral used in lithium-ion batteries. A significant share is produced by **artisanal and small-scale miners**, in conditions that have been documented as dangerous and involving child labour. Amnesty International's report *This Is What We Die For* (2016) traced cobalt mined by children through supply chains to major electronics and electric vehicle manufacturers, prompting due diligence initiatives such as the **Responsible Minerals Initiative**, the **Fair Cobalt Alliance**, and the DRC government's establishment of the Entreprise Générale du Cobalt to formalise artisanal cobalt production. The **OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas** (2011, revised 2016) provides the principal framework for supply chain due diligence in the minerals sector.

Polysilicon and forced labour. Investigations by researchers and journalists, and subsequent findings by governments, have alleged that a significant share of the world's polysilicon, the key material in solar panels, is produced in the Xinjiang Uyghur Autonomous Region of China in circumstances involving state-sponsored labour transfer programmes that may amount to forced labour. In 2022, the UN High Commissioner for Human Rights published an assessment finding that serious human rights violations had been committed in Xinjiang. The **Uyghur Forced Labor Prevention Act** of the United States (2021), in force since June 2022, establishes a rebuttable presumption that goods mined, produced, or manufactured wholly or in part in Xinjiang are made with forced labour and are prohibited from importation, and US Customs and Border Protection has detained solar panel shipments under the Act. The European Union adopted the **Forced Labour Regulation** (Regulation (EU) 2024/3015), which will prohibit the placing on the EU market, and the export, of products made with forced labour, irrespective of origin, and will apply from December 2027. These measures illustrate the tension between the rapid and low-cost deployment of renewable energy and the respect for labour rights — a tension that also engages the trade law issues examined in Chapter 13.

Artisanal and Small-Scale Mining

Artisanal and small-scale mining (ASM) employs an estimated forty million or more people worldwide, many times the number employed in large-scale mining, and supports the livelihoods of many more. In Africa, ASM is a major

source of gold, cobalt, tin, tantalum, tungsten, diamonds, and gemstones. It is frequently informal or illegal, and associated with dangerous working conditions, child labour, mercury use in gold processing, environmental degradation, and, in conflict-affected areas, the financing of armed groups. In Nigeria, artisanal gold mining in Zamfara State led in 2010 to a lead poisoning epidemic that killed more than four hundred children, according to Médecins Sans Frontières, and illegal mining has been linked to banditry in the north-west. Legal approaches to ASM include formalisation through simplified licensing, the creation of designated artisanal mining zones, the establishment of cooperatives, and support for responsible sourcing; the **Minamata Convention on Mercury** (2013) requires parties with more than insignificant artisanal gold mining to develop national action plans to reduce and, where feasible, eliminate mercury use. The **Nigerian Minerals and Mining Act 2007** provides for small-scale mining leases and the formation of mining cooperatives, and the federal government has pursued formalisation programmes, including through the Artisanal and Small-Scale Mining Department of the Ministry of Solid Minerals Development.

21.6 Community Grievance Mechanisms

The Role of Grievance Mechanisms

Resource projects inevitably generate grievances among affected communities and workers — about land, compensation, employment, pollution, noise, dust, traffic, security, and the distribution of benefits. If these grievances are not addressed promptly and fairly, they may escalate into conflict, protests, blockades, sabotage, and litigation, with consequences for the human rights of affected people and the viability of projects. **Grievance mechanisms** — procedures through which individuals and communities can raise concerns and seek remedies — therefore serve both a human rights function, as a means of access to remedy, and a risk management function, as an early warning system for problems.

The Effectiveness Criteria of the UNGPs

The UN Guiding Principles on Business and Human Rights, examined in section 21.7, provide that business enterprises should establish or participate in effective **operational-level grievance mechanisms** for individuals and communities who may be adversely impacted (Principle 29). Principle 31 sets out **effectiveness criteria** for non-judicial grievance mechanisms, both state-based and non-state-based. They should be:

- **legitimate**: enabling trust from the stakeholder groups for whose use they are intended, and being accountable for the fair conduct of grievance processes;

- **accessible**: being known to all stakeholder groups for whose use they are intended, and providing adequate assistance for those who may face barriers to access;
- **predictable**: providing a clear and known procedure with an indicative time frame for each stage, and clarity on the types of process and outcome available and means of monitoring implementation;
- **equitable**: seeking to ensure that aggrieved parties have reasonable access to sources of information, advice, and expertise necessary to engage in a grievance process on fair, informed, and respectful terms;
- **transparent**: keeping parties informed about the progress of the grievance, and providing sufficient information about the mechanism's performance to build confidence in its effectiveness;
- **rights-compatible**: ensuring that outcomes and remedies accord with internationally recognised human rights;
- **a source of continuous learning**: drawing on relevant measures to identify lessons for improving the mechanism and preventing future grievances and harms; and, for operational-level mechanisms,
- **based on engagement and dialogue**: consulting the stakeholder groups for whose use they are intended on their design and performance, and focusing on dialogue as the means to address and resolve grievances.

Operational-Level Grievance Mechanisms

Operational-level grievance mechanisms are now standard in large resource projects, required by the IFC Performance Standards (PS1 requires a grievance mechanism for affected communities and PS2 for workers) and by many national laws and contracts. They range from simple complaint-logging systems to multi-tiered processes involving community liaison officers, joint committees, independent mediators, and appeal to external bodies. Experience has revealed several challenges: mechanisms controlled by companies may lack legitimacy and independence; communities may fear retaliation or the loss of benefits; the most serious allegations, such as those involving security forces or sexual violence, may be beyond the capacity of operational mechanisms; and mechanisms may be used to require complainants to waive their legal rights.

The **Porgera** gold mine in Papua New Guinea illustrates these difficulties. Following reports of sexual violence by mine security personnel, Barrick Gold, the mine's operator, established a remedy framework in 2012 under which women who had been subjected to sexual violence could receive individual reparations packages. The framework was criticised by human rights organisations because recipients were required to sign waivers of their right to sue the company, because the packages were considered inadequate, and because of a lack of independent

oversight; subsequently, some claimants who had pursued legal action outside the framework obtained larger settlements. The UN High Commissioner for Human Rights, in a letter of 2013, addressed the compatibility of such legal waivers with the UNGPs, indicating that they should be avoided where possible and, if used, subject to safeguards, including access to independent legal advice.

Community Development Agreements and Host Community Structures

In some jurisdictions, legal frameworks establish structures that combine benefit-sharing with dispute resolution. Nigeria's Nigerian Minerals and Mining Act 2007 requires that **Community Development Agreements** include procedures for the resolution of disputes between the mining lease holder and the community. The **Petroleum Industry Act 2021** requires each **host communities development trust** to have a management committee and an advisory committee, and provides for the resolution of disputes, while also providing that where acts of vandalism, sabotage, or civil unrest occur that cause damage to petroleum facilities in a host community, the community forfeits its entitlement to the extent of the cost of repairs — a provision that has been criticised for imposing collective punishment and for failing to recognise that vandalism may be committed by outsiders. These provisions are examined further in Chapter 24.

Accountability Mechanisms of International Financial Institutions

International financial institutions have established independent accountability mechanisms through which people affected by projects they finance can raise complaints:

- The **World Bank Inspection Panel**, established in 1993, investigates complaints from people who claim to have been harmed by projects financed by the International Bank for Reconstruction and Development or the International Development Association as a result of the Bank's failure to comply with its own policies and procedures. In 2020, the Bank established an Accountability Mechanism, including a Dispute Resolution Service, alongside the Panel.
- The **Compliance Advisor Ombudsman** (CAO), established in 1999, addresses complaints concerning projects of the **International Finance Corporation** and the **Multilateral Investment Guarantee Agency** through dispute resolution and compliance investigations. The CAO's compliance investigation into the Tata Mundra coal-fired power plant in India found that the IFC had failed to ensure adequate protection of local fishing communities, which led to the litigation in *Jam v International Finance Corporation*, examined in Chapter 7, in which the United States Supreme Court held in 2019 that international organisations enjoy only the restrictive

immunity accorded to foreign governments. Following consultations that began in 2022, the IFC and MIGA adopted an approach to remedial action, under which they may contribute to remedy where they have contributed to harm.

- Regional development banks have equivalent mechanisms, including the **Independent Recourse Mechanism** of the African Development Bank, the **Accountability Mechanism** of the Asian Development Bank, the **Independent Project Accountability Mechanism** of the European Bank for Reconstruction and Development, and the **Independent Consultation and Investigation Mechanism** of the Inter-American Development Bank.

These mechanisms have provided communities with avenues for redress and have influenced the policies of the institutions, but they cannot award compensation, and their findings are often not fully implemented. Their effectiveness depends on the willingness of the institutions' boards and management to act on findings and to use their leverage over clients.

State-Based Non-Judicial Mechanisms

State-based non-judicial mechanisms include **national human rights institutions**, such as Nigeria's National Human Rights Commission, established under the National Human Rights Commission Act 1995 as amended in 2010, which may receive and investigate complaints of human rights violations, including those involving companies; ombudsman institutions; labour inspectorates; environmental regulators; and the **National Contact Points** established under the OECD Guidelines for Multinational Enterprises on Responsible Business Conduct. National Contact Points, which every adhering government must establish, handle "specific instances" — complaints that an enterprise has not observed the Guidelines — through good offices, mediation, and, where agreement is not reached, final statements that may include recommendations and determinations. Numerous specific instances have concerned energy and extractive companies, including complaints concerning oil pollution in the Niger Delta, mining in Africa and Latin America, and hydropower and coal projects in Asia. Nigeria is not an adherent to the OECD Guidelines, but complaints concerning the conduct of companies from adhering states in Nigeria may be brought before the National Contact Points of those states. Specific instances have produced mediated agreements in some cases, but the National Contact Points vary considerably in their independence, capacity, and willingness to make determinations, and their outcomes are not binding.

Nigerian Mechanisms for Oil Spill Complaints

In Nigeria, complaints concerning oil spills are addressed through the **National Oil Spill Detection and Response Agency** (NOSDRA), established by the Na-

tional Oil Spill Detection and Response Agency (Establishment) Act 2006, which is responsible for the detection of and response to oil spills and the coordination of joint investigation visits to determine their cause and volume. The integrity of the **joint investigation visit** process, in which company representatives, regulators, and community representatives jointly inspect spill sites, has been contested, since the determination that a spill was caused by sabotage, rather than operational failure, relieves the operator of liability to pay compensation under its policies and Nigerian law. The reliability of these determinations was a significant issue in the litigation in the Netherlands and the United Kingdom examined in section 21.8 and in Chapter 17. The **Hydrocarbon Pollution Remediation Project** (HYPREP), established in 2016 to implement the recommendations of UNEP's Environmental Assessment of Ogoniland, provides another, albeit imperfect, institutional channel for addressing the legacy of pollution.

21.7 Business and Human Rights Due Diligence

From Voluntary Codes to the UN Guiding Principles

The question of how international law should address the human rights impacts of business enterprises, which are not generally subjects of international human rights obligations, has been debated since the 1970s. Early efforts included the abortive UN Draft Code of Conduct on Transnational Corporations, negotiated from 1977 to the early 1990s; the **ILO Tripartite Declaration of Principles concerning Multinational Enterprises and Social Policy** (1977); and the **OECD Guidelines for Multinational Enterprises** (1976). The **UN Global Compact**, launched in 2000, invited companies to commit to ten principles on human rights, labour, environment, and anti-corruption. In 2003, the UN Sub-Commission on the Promotion and Protection of Human Rights adopted the **Norms on the Responsibilities of Transnational Corporations and Other Business Enterprises with regard to Human Rights**, which would have imposed direct human rights obligations on companies, but the Commission on Human Rights declined to adopt them.

In 2005, the UN Secretary-General appointed Professor John Ruggie as Special Representative on business and human rights. His work, informed by extensive consultations and research — including on the extractive sector, which accounted for the largest share of alleged corporate human rights abuses in his early surveys — led to the **"Protect, Respect and Remedy" Framework** (2008) and the **UN Guiding Principles on Business and Human Rights** (UNGPs), unanimously endorsed by the Human Rights Council in Resolution 17/4 in June 2011.

The Three Pillars

The UNGPs rest on three pillars:

1. **The state duty to protect** against human rights abuse within their territory and/or jurisdiction by third parties, including business enterprises, through appropriate policies, legislation, regulations, and adjudication (Principles 1–10). This includes ensuring policy coherence, including in investment treaties and contracts with investors — a point developed in Ruggie's **Principles for Responsible Contracts** (2011), which address the integration of human rights risk management into state–investor contract negotiations, examined in Chapter 15.
2. **The corporate responsibility to respect** human rights, meaning that business enterprises should avoid infringing on the human rights of others and should address adverse human rights impacts with which they are involved (Principles 11–24). The responsibility exists independently of states' abilities or willingness to fulfil their own obligations and over and above compliance with national laws. It refers, at a minimum, to the rights in the International Bill of Human Rights and the ILO Declaration on Fundamental Principles and Rights at Work.
3. **Access to remedy** for victims of business-related human rights abuses, through judicial, state-based non-judicial, and non-state-based grievance mechanisms (Principles 25–31), examined in sections 21.6 and 21.8.

Human Rights Due Diligence

The core of the corporate responsibility to respect is **human rights due diligence** (Principles 17–21). To identify, prevent, mitigate, and account for how they address their adverse human rights impacts, business enterprises should carry out due diligence, which should include:

- **assessing actual and potential human rights impacts**, drawing on internal and independent expertise and involving meaningful consultation with potentially affected groups and other stakeholders;
- **integrating and acting upon the findings**, assigning responsibility within the enterprise and allocating resources;
- **tracking the effectiveness of the response**, based on qualitative and quantitative indicators and feedback from affected stakeholders; and
- **communicating** how impacts are addressed, particularly when concerns are raised by or on behalf of affected stakeholders.

The UNGPs distinguish three modes of involvement in adverse impacts. Where an enterprise **causes** or may cause an impact, it should take the necessary steps to cease or prevent it and provide for remediation. Where it **contributes** or may contribute to an impact, it should cease or prevent its contribution, use its leverage to mitigate any remaining impact, and contribute to remediation. Where an impact is **directly linked** to its operations, products, or services by a business

relationship, without the enterprise having caused or contributed to it, the enterprise should use its leverage to prevent or mitigate the impact and consider whether to terminate the relationship, taking into account the severity of the impact and the consequences of termination. Due diligence should be ongoing, recognising that risks change over time, and should be prioritised according to the **severity** of impacts — their scale, scope, and irremediable character.

The UNGPs have been incorporated into the **OECD Guidelines for Multinational Enterprises**, updated in 2011 and again in 2023 as the OECD Guidelines for Multinational Enterprises on Responsible Business Conduct, which extend the due diligence approach to environmental matters, including climate change and biodiversity, and to the use of technology. The OECD has issued sector-specific guidance, including the **Due Diligence Guidance for Meaningful Stakeholder Engagement in the Extractive Sector** (2017) and the Minerals Guidance noted in section 21.5. The IFC Performance Standards, the Equator Principles, and the standards of industry bodies such as ICMM and IPIECA, the global oil and gas industry association for advancing environmental and social performance, also reflect the UNGPs.

Human Rights Due Diligence in the Extractive Sector

Human rights due diligence in the extractive sector must address a characteristic range of risks: impacts on land and livelihoods, including resettlement; impacts on indigenous peoples; environmental impacts on health, water, and food; the conduct of public and private security forces; labour conditions, including among contractors and suppliers; the treatment of human rights defenders; in-migration and its social effects, including on women and girls; corruption and the misuse of revenues; and operations in conflict-affected areas, where the UNGPs call for **heightened due diligence** (Principle 7 and Principle 23) and where companies risk complicity in gross human rights abuses, which the UNGPs indicate should be treated as a legal compliance issue. The energy transition adds risks in transition mineral supply chains and renewable energy projects, as examined in sections 21.3 and 21.5.

Mandatory Human Rights Due Diligence

The UNGPs were initially implemented through voluntary measures and **national action plans** on business and human rights, adopted by around thirty states, including, in Africa, Kenya (2019) and Uganda (2021); Nigeria has developed a draft national action plan with the support of the National Human Rights Commission. However, growing recognition of the limits of voluntary approaches led to a trend towards **mandatory human rights due diligence** legislation in home states of multinational enterprises, which forms part of the "smart mix" of measures that the UNGPs recommend (Principle 3).

Transparency and reporting laws were the first step. The **California Transparency in Supply Chains Act** (2010), the **UK Modern Slavery Act 2015** (section 54), and the **Australian Modern Slavery Act 2018** require large companies to report on steps taken to address modern slavery in their operations and supply chains, without requiring them to take any particular steps. Section 1502 of the **Dodd-Frank Act** (2010) in the United States requires listed companies to disclose whether they use conflict minerals — tin, tantalum, tungsten, and gold — originating in the DRC or adjoining countries, and the due diligence they have undertaken. The **EU Conflict Minerals Regulation** (Regulation (EU) 2017/821), applicable since January 2021, requires EU importers of these minerals above specified volume thresholds to conduct supply chain due diligence in accordance with the OECD Guidance.

General due diligence laws followed. The **French Duty of Vigilance Law** (Loi relative au devoir de vigilance, 2017) requires large French companies to establish, publish, and effectively implement a **vigilance plan** identifying risks and preventing serious violations of human rights and fundamental freedoms, health and safety, and the environment resulting from their activities and those of companies they control, subcontractors, and suppliers, and provides for injunctions and civil liability. It has been invoked in litigation against TotalEnergies concerning its projects in Uganda and Tanzania and its climate strategy, examined in Chapter 22. The **German Supply Chain Due Diligence Act** (Lieferkettensorgfaltspflichtengesetz), in force since 2023, requires large companies to conduct due diligence in their own operations and with respect to direct suppliers and, on substantiated knowledge, indirect suppliers, enforced by a federal agency through administrative fines, but does not create a new basis for civil liability. The **Norwegian Transparency Act** (2022) requires large enterprises to carry out due diligence in accordance with the OECD Guidelines and to respond to requests for information.

The **EU Corporate Sustainability Due Diligence Directive** (Directive (EU) 2024/1760), adopted in 2024 and examined in Chapter 8, requires large EU and non-EU companies meeting specified thresholds to conduct risk-based human rights and environmental due diligence across their own operations, those of their subsidiaries, and their "chain of activities", to adopt and put into effect a transition plan for climate change mitigation, and to provide for civil liability for damage caused by intentional or negligent failure to comply with prevention and remediation obligations. In 2025, as part of a broader simplification initiative, the EU postponed the Directive's application and proposed amendments to narrow its scope and requirements, reflecting political contestation over the burden of due diligence obligations. Sector-specific due diligence obligations also apply under the **EU Batteries Regulation** (Regulation (EU) 2023/1542), which

requires economic operators placing batteries on the market to conduct supply chain due diligence on cobalt, natural graphite, lithium, and nickel, and the **EU Deforestation Regulation** (Regulation (EU) 2023/1115).

A Binding Treaty?

In 2014, the Human Rights Council adopted **Resolution 26/9**, establishing an **open-ended intergovernmental working group** with the mandate to elaborate an international legally binding instrument to regulate, in international human rights law, the activities of transnational corporations and other business enterprises. The initiative, led by Ecuador and South Africa, was opposed by most Western states, which argued that it would undermine the UNGPs. Successive drafts, beginning with a "zero draft" in 2018, have addressed the prevention of human rights abuses through mandatory due diligence, legal liability, access to remedy, jurisdiction, applicable law, and mutual legal assistance. Negotiations have continued through annual sessions, but significant disagreements remain over the scope of the instrument — whether it should apply to all business enterprises or only those with transnational activities — and over the extent of corporate liability. Whatever its outcome, the process has influenced national and regional legislation.

21.8 Judicial and Non-Judicial Remedies

The Right to a Remedy and its Barriers

The third pillar of the UNGPs provides that, as part of their duty to protect against business-related human rights abuse, states must take appropriate steps to ensure that those affected have access to effective remedy through judicial, administrative, legislative, or other appropriate means (Principle 25). The right to an effective remedy is recognised in human rights treaties, including Article 2(3) of the ICCPR and Article 7 of the African Charter. Remedies may include apologies, restitution, rehabilitation, financial or non-financial compensation, punitive sanctions, and guarantees of non-repetition. In practice, victims of human rights abuses connected with resource development face formidable barriers: weak and under-resourced courts in host states; corruption and political interference; the cost of litigation and the lack of legal aid; difficulties in obtaining evidence held by companies; limitation periods; the separate legal personality of corporate group members; and, in transnational litigation, doctrines such as *forum non conveniens* and the presumption against extraterritoriality.

Remedies in Host States

The host state is the natural forum for most claims, and Nigerian courts have adjudicated a substantial body of litigation concerning oil pollution, under the law of tort (negligence, nuisance, trespass, and the rule in *Rylands v Fletcher*),

the Oil Pipelines Act, and constitutional and African Charter rights, as noted in Chapter 17. The decisions in *Gbemre*, noted in section 21.1, and in cases awarding compensation for oil spills illustrate the potential of domestic litigation. Nigerian courts have also, on occasion, awarded substantial damages against oil companies, but the duration of litigation, which may take decades, difficulties of proof, problems of enforcement, and the attribution of spills to sabotage have limited the effectiveness of domestic remedies. The **Fundamental Rights (Enforcement Procedure) Rules 2009** liberalised standing and procedure in human rights cases, allowing public interest litigation.

Regional Human Rights Mechanisms

Regional mechanisms have provided important remedies. The African Commission's decisions in *SERAC* and *Endorois*, the African Court's judgments in *Ogiek*, and the ECOWAS Court's judgment in *SERAP v Nigeria*, noted in section 21.1, illustrate the potential of the African system, although compliance with decisions has been uneven. The Inter-American Court's judgments, including *Saramaka* and *Sarayaku*, have been accompanied by detailed reparations orders, including collective reparations and guarantees of non-repetition, and supervised through compliance proceedings.

Home-State Litigation: The United States

For three decades, the United States **Alien Tort Statute** of 1789 (ATS), which grants federal district courts jurisdiction over civil actions by aliens for torts committed in violation of the law of nations or a treaty of the United States, was the principal vehicle for transnational human rights litigation against corporations. Following *Filártiga v Peña-Irala* (1980), plaintiffs brought numerous claims against extractive companies for complicity in human rights abuses abroad. In ***Wiwa v Royal Dutch Petroleum Co***, concerning Shell's alleged complicity in the execution of Ken Saro-Wiwa and other Ogoni leaders, Shell settled in 2009 for USD 15.5 million, without admitting liability. In *Doe v Unocal*, concerning forced labour in the construction of the Yadana gas pipeline in Myanmar, the parties settled in 2005.

The Supreme Court progressively restricted the ATS. In ***Kiobel v Royal Dutch Petroleum Co*** (2013), a claim by Ogoni plaintiffs alleging that Shell had aided and abetted human rights abuses by the Nigerian military, the Court held that the **presumption against extraterritoriality** applies to the ATS and that, where all the relevant conduct took place outside the United States, the claims must "touch and concern" the territory of the United States with sufficient force to displace the presumption; mere corporate presence was insufficient. In *Jesner v Arab Bank* (2018), the Court held that foreign corporations may not be defendants in ATS suits, and in *Nestlé USA Inc v Doe* (2021), it held that allegations

of general corporate activity, such as decision-making in the United States, were insufficient to establish domestic application. The ATS is now of limited utility for claims against extractive companies concerning conduct abroad.

Home-State Litigation: The United Kingdom, the Netherlands, and Canada

As the ATS route narrowed, claimants turned to the courts of the home states of parent companies, relying on ordinary tort law and the duty of care of parent companies. In the **United Kingdom**, the Supreme Court in ***Lungowe v Vedanta Resources plc*** [2019] UKSC 20, concerning pollution from the Nchanga copper mine in Zambia, and ***Okpabi v Royal Dutch Shell plc*** [2021] UKSC 3, concerning oil pollution in the Niger Delta, held that it was arguable that the parent company owed a duty of care to persons affected by its subsidiary's operations, where the parent had intervened in, controlled, supervised, or advised on the relevant operations, or held itself out as exercising such control, including through group-wide policies, and allowed the claims to proceed. The jurisdictional rules on which these claims rested, and the subsequent litigation, including the *Bodo* settlement of £55 million in 2015, are examined in Chapters 17 and 27. In the litigation arising from the collapse of the Fundão tailings dam in Brazil in 2015, *Município de Mariana v BHP Group*, the Court of Appeal allowed a claim by more than six hundred thousand claimants to proceed in England in 2022, and a liability trial took place in 2024 and 2025, illustrating the scale that such litigation may attain.

In the **Netherlands**, in ***Oguru, Efanga and others v Shell***, the Court of Appeal of The Hague held in January 2021 that Shell's Nigerian subsidiary was liable for oil spills from pipelines near the villages of Oruma and Goi, and that the parent company had breached a duty of care by failing to install a leak detection system on a pipeline; in 2022, Shell agreed to pay €15 million to the affected communities. The litigation is examined in Chapter 17.

::: casebox {title="Nevsun Resources Ltd v Araya: Customary International Law in Domestic Tort Claims"} **Nevsun Resources Ltd v Araya, 2020 SCC 5 (Supreme Court of Canada, 28 February 2020).**

Facts. Three Eritrean workers brought proceedings in British Columbia against Nevsun Resources, a Canadian company that owned, through subsidiaries, a majority interest in the Bisha gold, copper, and zinc mine in Eritrea, in partnership with the Eritrean state mining company. They alleged that they had been conscripted under Eritrea's national service programme and forced to work at the mine in harsh and dangerous conditions, by state-controlled contractors, and that Nevsun was liable for breaches of customary international law prohibitions on forced labour, slavery, cruel, inhuman, or degrading treatment, and

crimes against humanity, as well as for domestic torts such as battery and unlawful confinement. Nevsun applied to strike the claims, arguing that they were barred by the act of state doctrine and that customary international law could not ground a civil claim against a corporation.

Judgment. A majority of the Supreme Court of Canada (five to four on the customary international law issue) dismissed Nevsun's application. It held that the act of state doctrine does not form part of Canadian law, having been subsumed into the principles of conflict of laws and judicial restraint. It further held that customary international law is automatically adopted into Canadian common law in the absence of conflicting legislation, and that it was not "plain and obvious" that the prohibitions on forced labour, slavery, and other norms invoked, some of which are *jus cogens*, could not apply to corporations or ground a civil remedy in Canadian law. The claims were allowed to proceed to trial.

Subsequent events. The parties settled the claims later in 2020 on undisclosed terms.

Significance. *Nevsun* opened the possibility, in Canada and potentially in other common law jurisdictions, of civil claims against corporations based directly on breaches of customary international law, alongside conventional tort claims. It rejected the act of state doctrine as a bar to claims concerning the conduct of foreign states involved in resource projects and is regarded as a significant development in the law of corporate accountability for human rights abuses abroad. Canada has also established a **Canadian Ombudsperson for Responsible Enterprise** (2019), with a mandate to review complaints concerning Canadian companies in the mining, oil and gas, and garment sectors abroad, although its investigative powers have been criticised as limited. :::

Criminal Accountability

Criminal accountability of companies and their officers for complicity in human rights abuses abroad is emerging, though rare. In France, the cement company **Lafarge** was charged with complicity in crimes against humanity and financing terrorism in connection with payments to armed groups, including the Islamic State, to keep a plant operating in Syria; its parent company pleaded guilty in the United States in 2022 to providing material support to terrorist organisations and paid USD 778 million in penalties. In Sweden, executives of **Lundin Oil** (later Lundin Energy) were charged in 2021 with complicity in war crimes in connection with the company's operations in southern Sudan between 1999 and 2003, in one of the longest trials in Swedish history. Such proceedings, which depend on the willingness of prosecutors to pursue complex cases, may have significant deterrent effects.

Non-Judicial Remedies and Their Interaction with Litigation

Non-judicial mechanisms — the operational-level grievance mechanisms, accountability mechanisms of international financial institutions, National Contact Points, and national human rights institutions examined in section 21.6 — complement judicial remedies. They can provide faster, cheaper, and more flexible solutions, including dialogue-based outcomes and remedies tailored to communities' needs, but they lack binding force and may be used to deflect litigation. The UNGPs envisage a "remedy ecosystem" in which judicial and non-judicial mechanisms work together, and the effectiveness of remedies depends on the interaction among them, the leverage of financial institutions and home states, and the capacity of affected communities and their advocates to use them strategically. The procedural dimensions of transnational litigation, including jurisdiction, *forum non conveniens*, applicable law, and the enforcement of judgments — including the saga of the *Lago Agrio* litigation against Chevron — are examined in Chapter 27.

Conclusion

Human rights law has become an essential framework for the governance of energy and natural resource development. The obligations of states to respect, protect, and fulfil human rights, articulated in universal and regional instruments and given concrete content by decisions such as *SERAC*, *Endorois*, *Saramaka*, and *Sarayaku*, require states to regulate resource development so as to prevent harm to affected communities, to ensure their participation, and to remedy violations. The rights of indigenous peoples, including the principle of free, prior, and informed consent, have moved from the margins to the centre of resource law, influencing the standards of financial institutions and companies, although the scope of consent remains contested. International standards on land acquisition, resettlement, and compensation, and on labour rights and occupational safety, establish benchmarks against which national laws — including Nigeria's Land Use Act — and project practices may be measured.

The framework of business and human rights has transformed the expectations placed on companies. The UNGPs' articulation of the corporate responsibility to respect and of human rights due diligence has been embedded in the OECD Guidelines, financial institution standards, and industry practice, and is increasingly backed by mandatory legislation in Europe, although the political commitment to such legislation has been tested. Victims of abuses have pursued remedies through grievance mechanisms, regional human rights systems, and transnational litigation, with notable successes in the courts of the United Kingdom, the Netherlands, and Canada. Yet the gap between norms and practice remains wide, and the energy transition, with its demands for land, minerals, and

water, will test whether the lessons of the fossil fuel era have been learned. For legal scholars and practitioners in resource-rich states such as Nigeria, the challenge is to translate these international norms into effective domestic law and institutions that protect the rights of affected people while enabling resource development to contribute to national development.

::: exercise {title="Practical Exercise 21: A Lithium Mine on Customary Lands (HYPOTHETICAL)"} **This exercise is entirely hypothetical. The state, communities, companies, and events described are fictitious and do not refer to any real country, people, company, or dispute.**

The Federal Republic of Mbandara, a party to the ICCPR, the ICESCR, and the African Charter, but not to ILO Convention No. 169, has discovered substantial lithium deposits on the Kalunde plateau. The plateau is the customary land of the Wasila, a semi-nomadic pastoralist people of around 30,000 who identify as indigenous, whose identity has been recognised in a report of the African Commission's Working Group, and who use the plateau for seasonal grazing, water sources, and sacred sites. Mbandara's Constitution vests all minerals in the state, and its Land Act vests all land in the President in trust for the people, providing for compensation only for "unexhausted improvements" upon revocation of rights of occupancy. Mbandara's government has declared that it does not recognise any group as "indigenous" in the international sense.

The Ministry of Mines granted an exploration licence, and subsequently a mining lease, to Kalunde Lithium Ltd, a subsidiary of Veridian Metals plc, a company listed in London and headquartered in a European Union member state, which has a supply agreement with a European electric vehicle manufacturer. Before the grant, the Ministry held two public meetings in the district headquarters, conducted in the national language, which few Wasila elders speak, and obtained the signature of a "community representative" appointed by the district commissioner. The project will require the relocation of four Wasila settlements, the fencing of 8,000 hectares of grazing land, and the abstraction of significant volumes of groundwater. The company has offered cash compensation for houses and crops and a community development fund equal to 0.5 per cent of revenues. Wasila youth have blockaded the access road; the police have arrested twenty protesters, and two have been injured. The project is seeking financing from a development finance institution that applies the IFC Performance Standards.

Tasks.

(a) As counsel to the Wasila Council of Elders, prepare an opinion identifying the rights of the Wasila under international and regional human rights law, the breaches committed by Mbandara, and the forums available, including the domestic courts, the African Commission and Court, and the ECOWAS-type regional court of which Mbandara is a member (assume

it has jurisdiction over human rights claims by individuals). Advise on the prospects of a claim against Veridian Metals plc in its home jurisdiction.

(b) As head of human rights at Veridian Metals plc, prepare a memorandum to the board assessing the company's responsibilities under the UNGPs, the OECD Guidelines, the applicable EU and member state due diligence legislation, and the IFC Performance Standards, and recommending a course of action, including whether the project should be paused, how a genuine FPIC process could be conducted, how resettlement and compensation should be redesigned, and how the company should respond to the arrests of protesters.

(c) As adviser to the development finance institution, identify the conditions that should be satisfied before financing is approved.

(d) In groups, design an operational-level grievance mechanism for the project that satisfies the effectiveness criteria of Principle 31 of the UNGPs, specifying its governance, accessibility for a semi-nomadic population, procedures, remedies, and relationship with judicial remedies. :::

CHAPTER SUMMARY

- Resource development engages a wide range of state human rights obligations under universal and regional instruments, including the rights to life, health, food, water, housing, culture, and property, and the peoples' rights to dispose of natural resources and to a satisfactory environment under Articles 21 and 24 of the African Charter; the right to a clean, healthy, and sustainable environment was recognised by the UN General Assembly in 2022.
- *SERAC v Nigeria* established the state's duty to protect against harm by oil companies and gave content to the right to environment; the African system has developed the rights of indigenous peoples in *Endorois* and *Ogiek*.
- Indigenous peoples' rights are protected by ILO Convention No. 169, UNDRIP, and regional jurisprudence, notably of the Inter-American Court; national systems in Canada, Australia, Latin America, and Norway have developed duties to consult, and the concept of indigenous peoples in Africa emphasises marginalisation and attachment to land rather than first occupancy.
- Free, prior, and informed consent requires consultation free of coercion, before decisions are taken, with full information, and through the people's own institutions; its status as a right of veto is contested, but *Saramaka* requires consent for large-scale projects with major impacts, and FPIC is required in specified circumstances by IFC PS7 and the Equator Principles.
- International standards prohibit forced evictions and require resettlement and compensation at full replacement cost and livelihood restoration; Nigeria's Land Use Act provides compensation only for unexhausted improvements, falling short of these standards.

- ILO fundamental principles, now including a safe and healthy working environment, and sectoral conventions govern labour in the energy and extractive industries; forced and child labour in transition mineral and solar supply chains have prompted import bans and due diligence obligations.
- Grievance mechanisms should satisfy the effectiveness criteria of the UNGPs; accountability mechanisms of international financial institutions and OECD National Contact Points provide non-judicial avenues for redress.
- The UNGPs established the corporate responsibility to respect human rights through human rights due diligence, now increasingly mandatory under national and EU legislation; negotiations on a binding treaty continue.
- Victims face significant barriers to remedy; after the narrowing of the US Alien Tort Statute, litigation against parent companies in the United Kingdom, the Netherlands, and Canada has opened new avenues, alongside regional human rights mechanisms and emerging criminal accountability.

DOCTORAL RESEARCH QUESTIONS

1. Does the principle of free, prior, and informed consent entail a right of veto? Evaluate the competing positions with reference to UNDRIP, the jurisprudence of the Inter-American Court and the African Commission, and the standards of financial institutions.
2. Assess the application of the concept of "indigenous peoples" in Africa, and its implications for the protection of communities affected by resource development in Nigeria.
3. Evaluate the compatibility of Nigeria's Land Use Act 1978 and the compensation provisions of its petroleum and mining legislation with international standards on resettlement and compensation, and propose reforms.
4. To what extent has mandatory human rights due diligence legislation in Europe improved the human rights performance of extractive companies operating in Africa? Consider the French Duty of Vigilance Law and the EU Corporate Sustainability Due Diligence Directive.
5. Evaluate the effectiveness of transnational litigation against parent companies as a means of securing remedies for communities affected by oil pollution in the Niger Delta.
6. How can the human rights risks associated with transition minerals and renewable energy projects be addressed without unduly delaying the energy transition?

FURTHER READING

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22 CHAPTER

Corporate Governance, Compliance, and Anti-Corruption in Energy Industries

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the principal models and instruments of corporate governance applicable to energy enterprises, including national oil companies and other state-owned enterprises.
2. Analyse the duties of directors of energy companies, including in relation to the oversight of compliance, safety, and climate-related risks.
3. Evaluate the international, regional, and national legal frameworks against bribery and corruption, and their enforcement in the energy and extractive sectors.
4. Explain the rules on beneficial ownership transparency and payments-to-government reporting, and their role in combating corruption in resource sectors.
5. Assess the legal framework of economic sanctions and export controls and its implications for energy trade, investment, and contracts.
6. Analyse the emerging regimes for environmental, social, and governance disclosure and the legal risks of greenwashing.
7. Explain the legal protection of whistleblowers and the conduct of corporate investigations, including issues of privilege and cooperation with authorities.
8. Design the elements of an effective compliance programme for an energy enterprise and evaluate how enforcement authorities assess such programmes.

Introduction

The energy and extractive industries are characterised by features that make governance and compliance especially challenging: very large capital investments and revenue streams; the central role of governments as owners of resources, licensors, regulators, partners, and purchasers; the prominence of state-owned enterprises; operations in jurisdictions with weak institutions and high levels of corruption; complex chains of contractors, agents, and joint venture partners; exposure

to geopolitical risks, including sanctions; and growing scrutiny of their environmental and social impacts by investors, regulators, and civil society. The sector has been the setting for some of the largest corruption scandals and enforcement actions in history, from the bribery of Nigerian officials in connection with the Bonny Island LNG project to the *Lava Jato* investigation surrounding Petrobras in Brazil, and for leading cases on the duties of directors in relation to climate change.

This chapter examines the legal frameworks governing the conduct of energy enterprises and the individuals who direct them. It begins with corporate governance in energy enterprises, including national oil companies (section 22.1), and the responsibilities and fiduciary duties of boards (section 22.2). It then examines anti-bribery and anti-corruption frameworks (section 22.3), beneficial ownership and corporate transparency (section 22.4), sanctions compliance and export controls (section 22.5), environmental, social, and governance disclosure (section 22.6), whistleblower protection and corporate investigations (section 22.7), and the design and enforcement of compliance programmes (section 22.8). The chapter builds on the treatment of human rights due diligence in Chapter 21: whereas Chapter 21 examined due diligence from the perspective of affected rights-holders and remedies, this chapter examines the governance structures, legal duties, and compliance systems through which companies organise their response to legal and ethical risks.

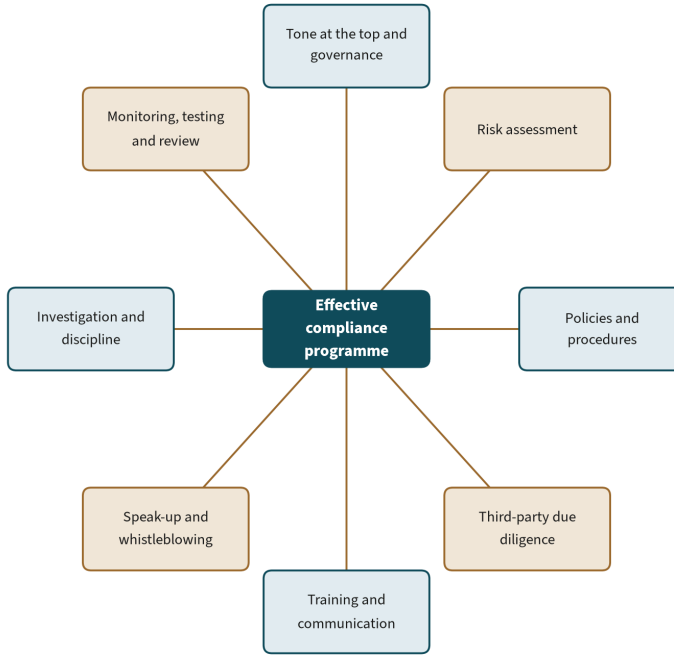


Figure 22.1 The elements of an effective compliance programme. *Source:* prepared for this edition.

22.1 Corporate Governance in Energy Enterprises

The Concept and Purpose of Corporate Governance

Corporate governance refers to the system by which companies are directed and controlled — in the formulation of the Cadbury Report of 1992 in the United Kingdom — and involves the relationships among a company's management, its board, its shareholders, and other stakeholders. It addresses the "agency problem" arising from the separation of ownership and control: the risk that managers will act in their own interests rather than those of the shareholders, and, in companies with controlling shareholders, the risk that controlling shareholders will extract private benefits at the expense of minority shareholders. Good governance is associated with better access to capital, lower costs of capital, better performance, and reduced risks of misconduct.

Two broad models have shaped corporate governance. The **shareholder primacy** model, dominant in the United States and the United Kingdom, holds that the purpose of the company is to maximise value for its shareholders, with the interests of other stakeholders protected principally by contract and regulation. The **stakeholder** model, associated with continental Europe and Japan, recog-

nises the interests of employees, creditors, communities, and others in corporate decision-making, reflected for example in employee representation on the supervisory boards of large German companies under co-determination legislation. The distinction has become less sharp: the United Kingdom's "enlightened shareholder value" approach requires directors to have regard to stakeholder interests in promoting the success of the company, examined in section 22.2, and debates about corporate purpose, stakeholder capitalism, and ESG have brought stakeholder considerations into the mainstream, though they have also provoked a political backlash, particularly in the United States.

Sources of Governance Standards

Corporate governance standards derive from company law, securities regulation and listing rules, codes of corporate governance, and the policies of institutional investors. The **G20/OECD Principles of Corporate Governance**, revised in 2023 to address sustainability and resilience, provide the leading international benchmark, addressing the rights of shareholders, the role of institutional investors, disclosure and transparency, and the responsibilities of the board. National codes, such as the **UK Corporate Governance Code**, operate principally on a "comply or explain" basis.

In **Nigeria**, the principal company law is the **Companies and Allied Matters Act 2020** (CAMA), which replaced the Act of 1990 and introduced significant reforms, including provisions on the disclosure of persons with significant control, examined in section 22.4, the codification of directors' duties, and enhanced protection of minority shareholders. The **Nigerian Code of Corporate Governance 2018**, issued by the Financial Reporting Council of Nigeria, applies to public companies, private companies that are holding companies of public companies, concessioned or privatised companies, and regulated private companies, on an "apply and explain" basis. Sector regulators, including the Central Bank of Nigeria for banks, the Securities and Exchange Commission, and the National Insurance Commission, issue sector-specific governance codes. The Nigerian Upstream Petroleum Regulatory Commission and the Nigerian Electricity Regulatory Commission impose governance-related requirements on licensees through regulations and licence conditions.

National Oil Companies and State-Owned Enterprises

A distinctive feature of the energy sector is the prominence of **state-owned enterprises** (SOEs), including national oil companies (NOCs), which control the majority of the world's proven oil and gas reserves, and state-owned electricity utilities. NOCs vary widely: some, such as Saudi Aramco, Equinor, and Petronas, are commercially sophisticated and internationally active; others serve primarily as instruments of government policy, revenue collection, and patronage. The

Natural Resource Governance Institute has estimated that NOCs collectively control assets worth trillions of dollars and that their governance has major consequences for public finances in resource-rich states. NOCs face distinctive governance challenges: multiple and sometimes conflicting objectives (commercial returns, revenue for the treasury, energy security, employment, and social programmes); political interference in appointments and investment decisions; conflicts of interest where the NOC also regulates the sector or manages the state's participation; soft budget constraints; and weak transparency.

The **OECD Guidelines on Corporate Governance of State-Owned Enterprises**, first adopted in 2005 and revised in 2015 and 2024, recommend that the state act as an informed and active owner, ensuring that SOE governance is carried out in a transparent and accountable manner with professionalism and effectiveness; that there be a clear separation between the state's ownership function and its other functions, particularly market regulation; that SOEs compete on a level playing field with private enterprises; that their objectives, including any public policy objectives, be clearly defined and disclosed and their costs funded transparently; that their boards have the authority, competencies, and objectivity to carry out their functions; and that they observe high standards of transparency and disclosure, including in relation to sustainability. For **sovereign wealth funds**, the **Santiago Principles** (Generally Accepted Principles and Practices, 2008), developed by the International Working Group of Sovereign Wealth Funds, address governance, accountability, and investment practices. Nigeria's sovereign wealth fund, the Nigeria Sovereign Investment Authority, established in 2011, is a member of the International Forum of Sovereign Wealth Funds.

The **Norwegian model**, which separates the commercial NOC (Equinor, formerly Statoil, partially listed since 2001), the regulator (the Norwegian Offshore Directorate, formerly the Norwegian Petroleum Directorate), the policy ministry, and the state's direct financial interest (managed by Petoro), is often cited as a benchmark of NOC governance, although commentators caution that it depends on institutional preconditions not easily replicated. The listing of shares in NOCs — such as the partial listing of Petrobras, Equinor, Ecopetrol, and Saudi Aramco, whose initial public offering in December 2019 was the largest in history — subjects them to securities regulation and market discipline, while also exposing them, as in the case of Petrobras, to securities litigation by investors.

NNPC Limited and the Petroleum Industry Act

The governance of Nigeria's national oil company was among the central concerns of the **Petroleum Industry Act 2021**. The Nigerian National Petroleum Corporation, established in 1977 as a statutory corporation, had been criticised for decades for opacity, inefficiency, its dual role as commercial operator and quasi-regulator, the remittance of revenues to the Federation Account, and the

subsidisation of petroleum products. The PIA required the incorporation of a successor company, **NNPC Limited**, under the Companies and Allied Matters Act, with its shares held by the Government of the Federation through the Ministry of Finance Incorporated and the Ministry of Petroleum Incorporated, and the transfer to it of the Corporation's assets, interests, and liabilities as determined by the Minister. NNPC Limited was incorporated and commenced operations in July 2022. The PIA requires NNPC Limited to operate on a commercial basis, without recourse to government funds, to declare dividends, and to be subject to the same company law and governance requirements as other limited liability companies, and it separated regulatory functions, transferring them to the Nigerian Upstream Petroleum Regulatory Commission and the Nigerian Midstream and Downstream Petroleum Regulatory Authority. The PIA also contemplates the eventual offering of shares in NNPC Limited to the public. The transformation is a significant test of whether legal restructuring can change the governance and performance of an NOC, and early experience has highlighted continuing challenges, including transparency in relation to crude-for-loan arrangements and the cost of fuel subsidies before their removal in 2023, examined in Chapter 24.

Governance Failures in Energy SOEs

Several high-profile cases illustrate the consequences of governance failures in energy SOEs. The ***Lava Jato*** ("Car Wash") investigation, launched in Brazil in 2014, revealed a scheme in which construction and engineering companies colluded to inflate contracts with **Petrobras** and paid bribes to Petrobras executives and politicians. The investigation led to the convictions of numerous executives and politicians, although some convictions were later annulled on procedural grounds, and to enforcement actions in Brazil, the United States, and elsewhere, including Petrobras's resolution with US authorities in 2018 for USD 1.78 billion and Odebrecht's and Braskem's global resolution in 2016 of at least USD 3.5 billion. In **South Africa**, the Judicial Commission of Inquiry into State Capture (the Zondo Commission), which reported in 2022, documented the "capture" of **Eskom** and other SOEs by networks connected with the Gupta family, including through irregular coal supply contracts and board and management appointments, contributing to the utility's financial and operational crisis examined in Chapter 16. In **Angola**, the "Luanda Leaks" investigation published in 2020 raised allegations concerning the conduct of Isabel dos Santos, daughter of the former president, who had chaired the NOC **Sonangol**; Angolan authorities froze assets and brought charges. These cases demonstrate that governance failures in energy SOEs can have systemic consequences for public finances, energy security, and political stability.

Governance of Joint Ventures

Much energy investment is undertaken through **joint ventures** — unincorporated joint operating agreements in upstream petroleum, examined in Chapter 7, and incorporated joint ventures in midstream, power, and mining projects. Joint venture governance raises distinctive issues: the allocation of decision-making between the operator and the operating committee; voting thresholds and veto rights; the information and audit rights of non-operators; the management of conflicts of interest when the operator also provides services; and the responsibility of non-operating partners for compliance failures by the operator or the joint venture. Anti-corruption and sanctions risks are acute in joint ventures with state entities and local partners, and companies increasingly insist on compliance covenants, audit rights, and termination rights in joint venture agreements, examined in section 22.8.

22.2 Board Responsibilities and Fiduciary Duties

The Role of the Board

The **board of directors** is responsible for the overall direction and supervision of the company: setting strategy, appointing and overseeing senior management, overseeing risk management and internal control, ensuring the integrity of financial reporting, and accounting to shareholders. In energy enterprises, the board's oversight extends to major operational risks, including safety and environmental risks, whose materialisation can threaten the survival of the company, as the *Deepwater Horizon* disaster demonstrated for BP. Boards typically operate through committees — audit, risk, remuneration, nomination, and, increasingly, safety, sustainability, or ethics committees.

Directors' Duties in English and Nigerian Law

In English law, the **Companies Act 2006** codifies the general duties of directors, which are owed to the company. Section 171 requires directors to act within their powers; **section 172** requires a director to act in the way he or she considers, in good faith, would be most likely to promote the success of the company for the benefit of its members as a whole, and in doing so to have regard (among other matters) to the likely consequences of any decision in the long term, the interests of the company's employees, the need to foster business relationships with suppliers, customers, and others, the impact of the company's operations on the community and the environment, the desirability of maintaining a reputation for high standards of business conduct, and the need to act fairly as between members; section 173 requires independent judgment; **section 174** requires a director to exercise reasonable care, skill, and diligence, measured by both an objective standard (the knowledge, skill, and experience that may reasonably be expected of a person carrying out the director's functions) and a subjective

standard (the director's actual knowledge, skill, and experience); and sections 175 to 177 address conflicts of interest. The **strategic report** required of large companies must include a statement describing how directors have had regard to the matters in section 172.

Nigerian law follows a similar structure. CAMA 2020 provides that directors stand in a fiduciary relationship towards the company and must observe the utmost good faith in any transaction with it or on its behalf, must act in what they believe to be the best interests of the company as a whole so as to preserve its assets, further its business, and promote the purposes for which it was formed, and must exercise the powers and discharge the duties of their office honestly, in good faith, and in the best interests of the company, with the degree of care, diligence, and skill that a reasonably prudent director would exercise in comparable circumstances. The Act provides that the matters to which directors must have regard include the interests of the company's employees in general, as well as the interests of its members, and contains provisions on conflicts of interest, secret profits, and the misuse of corporate opportunities. Directors who breach their duties may be liable to the company, and the Act provides for derivative actions by shareholders.

The Duty of Oversight

The duty of care requires directors to oversee the company's compliance with law and the management of risk. In the United States, the Delaware Court of Chancery in ***In re Caremark International Inc Derivative Litigation*** (1996) held that directors have a duty to attempt in good faith to assure that a corporate information and reporting system, which the board concludes is adequate, exists, and that failure to do so may, in some circumstances, render directors liable for losses caused by non-compliance with applicable legal standards. Liability under *Caremark* requires that directors utterly failed to implement any reporting or information system or controls, or, having implemented such a system, consciously failed to monitor or oversee its operations, as clarified in *Stone v Ritter* (2006). Although historically difficult to establish, *Caremark* claims have been allowed to proceed in cases involving "mission critical" risks, such as food safety in ***Marchand v Barnhill*** (Delaware Supreme Court, 2019), and aircraft safety in the *Boeing* derivative litigation (Court of Chancery, 2021). For energy companies, safety, environmental compliance, and, increasingly, climate risks may be regarded as mission-critical, requiring board-level reporting systems.

Directors' Duties and Climate Change

Whether, and how, directors' duties require directors to consider climate-related risks has been the subject of extensive analysis. In Australia, legal opinions by Noel Hutley SC and Sebastian Hartford Davis, commissioned by the Centre for Policy Development in 2016 and updated in 2019 and 2021, concluded

that directors who fail to consider climate change risks that are foreseeable and material to their company could be found liable for breach of the duty of care and diligence. Similar analyses have been published in many jurisdictions under the auspices of the Commonwealth Climate and Law Initiative. Regulators, including the Australian Securities and Investments Commission and the Bank of England's Prudential Regulation Authority, have endorsed the view that climate risk is a financial risk that boards must address. In *McVeigh v Retail Employees Superannuation Pty Ltd*, a member of an Australian pension fund claimed that the trustee had failed to manage climate risk; the case settled in 2020 with the fund committing to net zero by 2050 and to enhanced climate risk management and disclosure.

::: casebox {title="ClientEarth v Shell plc: Climate Strategy and the Derivative Claim"} **ClientEarth v Shell plc [2023] EWHC 1897 (Ch) (High Court of England and Wales, 24 July 2023).**

Facts. ClientEarth, an environmental law organisation holding a small number of shares in Shell plc, sought permission under Part 11 of the Companies Act 2006 to bring a **derivative claim** on behalf of the company against its directors, alleging that they had breached their duties under sections 172 and 174 by failing to adopt and implement an energy transition strategy that properly managed the risks to the company from climate change, including by failing to set adequate interim emission reduction targets, and by failing to comply with the order of the District Court of The Hague in *Milieudefensie v Royal Dutch Shell* (2021), which had required Shell to reduce the emissions of the Shell group by forty-five per cent by 2030 relative to 2019 levels (an order later overturned on appeal in November 2024, as examined in Chapter 18). ClientEarth sought declarations and an order requiring the directors to adopt a strategy consistent with the Paris Agreement and to comply with the Dutch order.

Decision. Trower J refused permission, first on the papers in May 2023 and then, after an oral hearing, in July 2023. He held that ClientEarth had not established a *prima facie* case. The directors' duties under sections 172 and 174 are owed to the company, and the management of climate risk involves commercial judgments on which directors must balance a range of competing considerations; it is for the directors, not the court, to determine how best to promote the success of the company, and the court will not interfere with the directors' judgment in matters of commercial management absent evidence that they have acted irrationally or in bad faith. The evidence did not establish that the directors' approach to climate risk fell outside the range of reasonable approaches, and the general duties do not impose specific, absolute obligations of the kind alleged. The judge also considered that the small size of ClientEarth's shareholding and its broader

purposes suggested that the claim was brought to advance its own policy agenda rather than in the interests of the company.

Significance. The decision indicates the difficulty of using directors' duties to compel a particular climate strategy through litigation, in light of the deference accorded to directors' business judgment and the hurdles of the derivative action. It does not, however, preclude liability where directors ignore climate risk altogether or act irrationally, and climate risk remains a matter that directors must consider in discharging their duties of care and to promote the success of the company. Shareholder engagement, disclosure regulation, and litigation against companies themselves have proved more effective avenues for influencing corporate climate strategies. :::

Shareholder Engagement and Activism

Shareholders exercise influence over the governance of energy companies through voting, engagement, and resolutions. In **May 2021**, the activist hedge fund **Engine No. 1**, holding a very small stake in ExxonMobil, succeeded with the support of major institutional investors in electing three of its four nominees to ExxonMobil's board, arguing that the company lacked a credible strategy for the energy transition — a landmark in climate-related activism. Shareholder resolutions calling on oil and gas companies to set emission reduction targets, including Scope 3 targets, have been filed regularly by organisations such as Follow This, with varying levels of support. "**Say on climate**" votes, in which companies submit their climate transition plans to an advisory shareholder vote, have been adopted by several energy companies. Since 2022, however, support for climate-related resolutions has declined amid political opposition to ESG in the United States, concerns about energy security, and the reassertion by some companies, including BP and Shell, of strategies emphasising oil and gas returns.

Parent Company Responsibility

The board of a parent company may also face legal exposure for harms caused by subsidiaries. As examined in Chapters 17 and 21, the UK Supreme Court in *Vedanta* and *Okpabi* recognised that a parent company may owe a duty of care to persons affected by its subsidiary's operations where it has assumed responsibility for relevant aspects of those operations, including through group-wide policies that it implements, supervises, or holds out as applying. This creates a governance dilemma: group-wide policies on safety, environment, and human rights may be necessary to manage risks effectively and to satisfy due diligence obligations, but they may also be cited as evidence of the parent's assumption of responsibility. The emerging mandatory due diligence laws, examined in Chapter 21, resolve this dilemma in favour of active parent company oversight, by requiring it. Boards should therefore ensure that group policies are not merely promulgated but

implemented and monitored, since a gap between policy and practice is likely to be the most damaging scenario both legally and reputationally.

22.3 Anti-Bribery and Anti-Corruption Frameworks

Corruption in the Energy and Extractive Sectors

Corruption — the abuse of entrusted power for private gain — is a pervasive risk in the energy and extractive sectors. The OECD's *Foreign Bribery Report* (2014), analysing concluded enforcement actions under the OECD Anti-Bribery Convention, found that the extractive sector accounted for the largest share of foreign bribery cases, around one-fifth of the total, followed by construction, transportation and storage, and information and communication. The risks arise at every stage of the value chain: in the allocation of licences and contracts, where discretionary decisions by officials are worth billions; in the award of service and supply contracts; in the obtaining of permits, customs clearances, and visas; in the marketing of state crude oil and the import of petroleum products; in the administration of fuel subsidies; in local content requirements and the selection of local partners; and in the management of revenues by NOCs and governments. The consequences include the diversion of public revenues, the distortion of investment decisions, the undermining of regulation, including safety and environmental regulation, and the entrenchment of political elites — contributing to the "resource curse" examined in Chapter 23.

International and Regional Conventions

The international legal framework against corruption comprises several treaties:

- The **OECD Convention on Combating Bribery of Foreign Public Officials in International Business Transactions** (1997), in force since 1999, with forty-six parties including all OECD members and several non-members such as Argentina, Brazil, Bulgaria, Peru, Russia (whose participation was suspended in 2022), and South Africa — the only African party. It requires parties to criminalise the bribery of foreign public officials, to establish the liability of legal persons, and to impose effective, proportionate, and dissuasive sanctions. Implementation is monitored through peer review by the OECD Working Group on Bribery, and the **2021 Recommendation** of the OECD Council on further combating bribery addresses the demand side, whistleblower protection, and non-trial resolutions.
- The **United Nations Convention against Corruption** (UNCAC, 2003), in force since 2005, with around 190 parties, including Nigeria, is the only global legally binding anti-corruption instrument. It addresses prevention (including codes of conduct, public procurement, and the independence of anti-corruption bodies), criminalisation (including bribery of national and

foreign public officials, embezzlement, trading in influence, abuse of functions, illicit enrichment, and money laundering), international cooperation, and **asset recovery**, establishing the return of assets as a fundamental principle (Chapter V). An Implementation Review Mechanism, established in 2009, conducts peer reviews.

- The **African Union Convention on Preventing and Combating Corruption** (2003), in force since 2006, requires parties to criminalise a range of corrupt practices, including illicit enrichment, and to adopt measures on political party funding and access to information. The **ECOWAS Protocol on the Fight against Corruption** (2001) addresses corruption in the West African region.
- The **Inter-American Convention against Corruption** (1996) and the Council of Europe's **Criminal Law Convention on Corruption** (1999), monitored by the Group of States against Corruption (GRECO), are regional instruments of similar purpose.

The US Foreign Corrupt Practices Act

The **Foreign Corrupt Practices Act** of 1977 (FCPA), enacted in the aftermath of revelations of widespread corporate bribery abroad, was the first national law to criminalise the bribery of foreign public officials, and its vigorous enforcement since the mid-2000s has made it the most significant anti-bribery law in practice. It has two sets of provisions:

- The **anti-bribery provisions** prohibit issuers of securities registered in the United States, domestic concerns, and other persons acting within US territory from corruptly offering, paying, promising, or authorising the payment of money or anything of value to a foreign official, foreign political party, or candidate, or to any person knowing that it will be passed on to such persons, in order to influence an official act, induce a breach of lawful duty, secure an improper advantage, or obtain or retain business. Officials of state-owned enterprises, including NOCs, are "foreign officials". The provisions contain a narrow exception for **facilitating payments** to expedite routine governmental action, and affirmative defences for payments lawful under the written laws of the foreign country and for reasonable and bona fide expenditure directly related to product promotion or contract performance.
- The **accounting provisions** require issuers to make and keep books and records that accurately and fairly reflect their transactions and to devise and maintain a system of internal accounting controls. These provisions, enforced by the Securities and Exchange Commission, apply regardless of whether bribery is proved and have been used extensively.

The FCPA is enforced criminally by the Department of Justice and civilly by the SEC. Its jurisdictional reach is broad, extending to foreign companies listed in the United States and to conduct involving the use of US interstate commerce, including US dollar transactions and emails passing through US servers. Many of the largest resolutions have involved energy and resource companies. In **February 2025**, the President issued an executive order pausing new FCPA investigations and enforcement actions pending a review of enforcement policy, and the Department of Justice issued revised guidelines in June 2025, prioritising cases connected with cartels and transnational criminal organisations, cases that harm the competitiveness of US companies, and cases involving serious misconduct rather than routine business practices. The long-term effects of the change on enforcement in the energy sector remain to be seen, but the statute itself, and the limitation periods applicable to past conduct, remain in force, and other jurisdictions have continued to enforce their own laws.

::: casebox {title="The Bonny Island LNG Bribery Scheme"} **Enforcement actions concerning the TSKJ joint venture and Nigeria LNG (United States and other jurisdictions, 2008–2011).**

Facts. Between 1995 and 2004, a joint venture known as **TSKJ**, comprising Technip of France, Snamprogetti Netherlands (then a subsidiary of Eni of Italy), Kellogg Brown & Root (KBR, then a subsidiary of Halliburton of the United States), and JGC Corporation of Japan, was awarded engineering, procurement, and construction contracts worth more than USD 6 billion to build liquefied natural gas facilities on Bonny Island, Nigeria, for Nigeria LNG Limited, a company in which the Nigerian National Petroleum Corporation held a forty-nine per cent interest. US authorities established that the joint venture had paid around USD 180 million in bribes to Nigerian government officials, at various levels, to obtain the contracts, channelled through two agents — a United Kingdom solicitor and a Japanese trading company — under sham consultancy agreements.

Resolutions. In 2009, KBR pleaded guilty to FCPA violations, and KBR and Halliburton paid a combined total of USD 579 million to resolve criminal and civil charges — then the largest FCPA resolution by a US company. Technip paid USD 338 million in 2010, Snamprogetti and Eni paid USD 365 million in 2010, and JGC paid USD 218.8 million in 2011. The UK solicitor who acted as an agent was extradited to the United States, pleaded guilty, and was sentenced to imprisonment, and a former KBR chief executive pleaded guilty and was sentenced to imprisonment. In Nigeria, the federal government brought charges against companies and individuals and reached settlements with several of the companies, which agreed to pay fines in Nigeria.

Significance. The case — with total resolutions in the United States of around USD 1.5 billion — illustrated the vulnerability of large energy infrastructure

projects to corruption, the use of agents and consultancy arrangements to disguise bribes, the exposure of joint venture partners to liability for a common scheme, and the extraterritorial reach of US enforcement. It also prompted the Nigerian authorities to seek accountability at home, raising questions about the allocation of penalties between the home states of the companies and the state whose officials were bribed — a theme in debates about the return of penalties to affected countries. :::

The UK Bribery Act 2010

The **Bribery Act 2010** of the United Kingdom, in force since July 2011, is widely regarded as among the strictest anti-bribery laws. It creates offences of bribing another person (section 1), being bribed (section 2), and bribing a foreign public official (section 6), which apply to private-to-private bribery as well as bribery of officials and contain no exception for facilitating payments. Section 7 creates a strict liability corporate offence of **failure to prevent bribery**: a relevant commercial organisation — a body incorporated in or carrying on business in the United Kingdom — is guilty of an offence if a person associated with it (including employees, agents, and subsidiaries performing services for it) bribes another person intending to obtain or retain business or an advantage in the conduct of business for the organisation. The only defence is to prove that the organisation had in place **adequate procedures** designed to prevent persons associated with it from undertaking such conduct. The Ministry of Justice's Guidance (2011) identifies six principles for adequate procedures, examined in section 22.8.

The Bribery Act has been enforced principally through **deferred prosecution agreements** (DPAs), introduced by the Crime and Courts Act 2013, examined in section 22.7. In 2022, the Serious Fraud Office secured a conviction of **Glencore Energy UK** for bribery in connection with oil operations in Africa, including in Nigeria, Cameroon, Côte d'Ivoire, Equatorial Guinea, and South Sudan, resulting in a penalty of around £281 million, following Glencore's guilty pleas in the United States and Brazil earlier that year in resolutions exceeding USD 1.1 billion. The **Economic Crime and Corporate Transparency Act 2023** extended the "failure to prevent" model to a new offence of **failure to prevent fraud** by large organisations, in force from September 2025, and reformed the identification doctrine for attributing criminal liability to companies, making it easier to attribute the conduct of senior managers to the company for economic crimes.

Other Jurisdictions

France's **Sapin II Law** (2016) established the French Anti-Corruption Agency (AFA), which oversees mandatory compliance programmes for large companies, and introduced the *convention judiciaire d'intérêt public* (CJIP), a non-trial reso-

lution similar to a DPA. Brazil's **Clean Company Act** (2013), adopted in the context of the *Lava Jato* investigation, imposes strict civil and administrative liability on companies for corrupt acts against foreign or domestic public administrations. Commodity trading companies have been the subject of a series of enforcement actions for bribery in connection with the purchase of crude oil and products from NOCs, including resolutions by **Vitol** (2020), **Glencore** (2022), **Trafigura** (2024), and **Gunvor** (2024) in the United States, Brazil, Switzerland, and the United Kingdom, highlighting the risks of the oil trading business.

Nigerian Anti-Corruption Law

Nigeria's anti-corruption framework comprises several statutes and institutions. The **Corrupt Practices and Other Related Offences Act 2000** established the **Independent Corrupt Practices and Other Related Offences Commission** (ICPC) and criminalises bribery, the corrupt demanding of benefits by officials, fraudulent acquisition of property, and related offences. The **Economic and Financial Crimes Commission (Establishment) Act 2004** established the **Economic and Financial Crimes Commission** (EFCC), with broad powers to investigate and prosecute economic and financial crimes, including money laundering and advance fee fraud. The **Code of Conduct Bureau and Tribunal Act** and the Fifth Schedule to the Constitution require public officers to declare their assets and prohibit conflicts of interest. The **Money Laundering (Prevention and Prohibition) Act 2022** and the **Proceeds of Crime (Recovery and Management) Act 2022** strengthened the anti-money laundering and asset recovery framework, partly in response to evaluations by the Financial Action Task Force, which placed Nigeria on its list of jurisdictions under increased monitoring in 2023 — a listing from which Nigeria was removed in October 2025 after completing its action plan.

Corruption in Nigeria's petroleum sector has been documented in numerous official reports, including the reports of the **Nigeria Extractive Industries Transparency Initiative** (NEITI), the Ribadu-led Petroleum Revenue Special Task Force (2012), and the investigations into the administration of the fuel subsidy regime, which revealed large-scale fraud in subsidy claims in 2011–2012. The **OPL 245** affair — concerning the award of a prolific offshore block to Shell and Eni in 2011 for USD 1.3 billion, most of which was paid via a government escrow account to Malabu Oil and Gas, a company associated with a former petroleum minister — led to prosecutions in Italy, in which the companies and individual defendants were acquitted in 2021, a decision that became final in 2022, and to proceedings in Nigeria and an arbitration brought by Eni against Nigeria under the Netherlands–Nigeria bilateral investment treaty. The affair illustrates the complexity of corruption cases involving licence awards, the difficulty of proof across jurisdictions, and the interaction of criminal, civil, and

investment proceedings. A former minister of petroleum resources was charged in the United Kingdom in 2023 with bribery offences in connection with the award of oil contracts, in proceedings that were continuing.

Asset Recovery

The recovery and return of the proceeds of corruption is an important dimension of anti-corruption law for resource-rich states. Nigeria has been a leading participant in asset recovery, notably with respect to funds attributed to the former military head of state General Sani Abacha. Under agreements with Switzerland, the United States, Jersey, and others, and with the involvement of the World Bank in monitoring, several hundred million dollars have been returned to Nigeria since the early 2000s, including USD 322 million returned by Switzerland under an agreement of 2017 and disbursed through a conditional cash transfer programme, and USD 308 million returned by the United States and Jersey under an agreement of 2020 for infrastructure projects. The **Global Forum on Asset Recovery Principles** (2017) provide guidance on the transparent and accountable return of assets.

22.4 Beneficial Ownership and Corporate Transparency

The Problem of Anonymous Ownership

Anonymous companies, trusts, and other legal arrangements are used to conceal the identity of those who ultimately own or control assets, facilitating corruption, money laundering, tax evasion, and sanctions evasion. In the extractive sector, investigations have repeatedly found that licences and contracts were awarded to companies whose undisclosed beneficial owners included politically exposed persons — public officials, their family members, and close associates. The Natural Resource Governance Institute's analysis of such cases has identified the use of secret ownership as a central mechanism of extractive sector corruption. **Beneficial ownership transparency** — the identification and disclosure of the natural persons who ultimately own or control legal entities — is therefore a key anti-corruption measure.

International Standards

The **Financial Action Task Force** (FATF), the intergovernmental body that sets international standards on anti-money laundering and countering the financing of terrorism, requires countries to ensure that competent authorities have access to adequate, accurate, and up-to-date information on the beneficial ownership and control of legal persons (**Recommendation 24**) and legal arrangements such as trusts (**Recommendation 25**). Recommendation 24 was revised in 2022 to require countries to use a multi-pronged approach including a public authority or body holding beneficial ownership information, such as a central register, and

Recommendation 25 was revised in 2023. Financial institutions and designated non-financial businesses and professions must identify and verify the beneficial owners of their customers as part of customer due diligence, and apply enhanced due diligence to politically exposed persons.

The **Extractive Industries Transparency Initiative** (EITI), launched in 2003, is a global standard for the disclosure of information along the extractive industry value chain, implemented by more than fifty countries through multi-stakeholder groups of government, companies, and civil society. Under the **EITI Standard**, implementing countries must disclose information on the legal framework and fiscal regime, licence allocations, contracts, production, exports, state participation, revenue collection, revenue allocation, and social and economic spending. Since January 2020, the Standard has required implementing countries to request, and companies to disclose, the **beneficial owners** of companies that apply for or hold participating interests in exploration or production licences or contracts, including the identity of any politically exposed persons. Since January 2021, it has required the disclosure of the full text of contracts and licences granted, entered into, or amended from that date. The 2023 EITI Standard added requirements concerning the energy transition, gender, and anti-corruption.

Nigeria was among the first countries to implement the EITI, enacting the **Nigeria Extractive Industries Transparency Initiative Act 2007**, which made NEITI a statutory body and requires audits of the oil, gas, and mining sectors — the first EITI implementing country to give the initiative a legal foundation. NEITI's audits have identified significant revenue discrepancies and unremitted funds and have contributed to policy reform. NEITI has published a register of beneficial owners in the extractive sector and has advocated contract transparency. **CAMA 2020** requires companies to disclose **persons with significant control** — those holding, directly or indirectly, at least a prescribed percentage of shares or voting rights, or otherwise exercising significant influence or control — to the Corporate Affairs Commission, and to maintain a register of such persons, and the Commission launched a public electronic register of persons with significant control. The **Petroleum Industry Act 2021** contains requirements for the disclosure of information on the beneficial owners of applicants for and holders of petroleum licences and leases, and for the publication of licences, leases, and contracts, building on NEITI's work.

National Registers and the Debate over Public Access

The **United Kingdom** introduced a register of **people with significant control** in 2016, making it the first G20 country to establish a public central register of beneficial ownership, and later a register of overseas entities owning UK property under the Economic Crime (Transparency and Enforcement) Act 2022. The Economic Crime and Corporate Transparency Act 2023 introduced identity

verification for directors and persons with significant control. The **European Union**'s Fifth Anti-Money Laundering Directive (2018) required member states to provide public access to beneficial ownership registers, but in ***WM and Sovim SA v Luxembourg Business Registers*** (Joined Cases C-37/20 and C-601/20, November 2022), the Court of Justice of the European Union held that general public access constituted a serious interference with the rights to privacy and the protection of personal data under the Charter of Fundamental Rights that was not limited to what was strictly necessary. The EU's anti-money laundering package of 2024 provides for access by persons with a legitimate interest, including journalists and civil society organisations. In the **United States**, the **Corporate Transparency Act** of 2021 required many companies to report beneficial ownership information to the Financial Crimes Enforcement Network, but following litigation, FinCEN issued an interim final rule in March 2025 exempting US companies and US persons from reporting requirements, limiting the obligation to certain foreign companies registered to do business in the United States.

Payments-to-Governments Reporting

Complementing EITI, several jurisdictions require extractive companies to publish the payments they make to governments on a country-by-country and project-by-project basis, enabling citizens to follow the money. The **EU Accounting Directive** (2013/34/EU, Chapter 10) and the **Transparency Directive**, as amended in 2013, require large EU and EU-listed extractive and logging companies to report payments to governments above a threshold, broken down by country and project, including taxes, royalties, dividends, signature and production bonuses, licence fees, and infrastructure payments. The United Kingdom implemented these requirements through the Reports on Payments to Governments Regulations 2014, and Norway and Canada adopted similar rules — Canada through the Extractive Sector Transparency Measures Act 2015. In the United States, **section 1504 of the Dodd-Frank Act** directed the SEC to adopt payment disclosure rules; after its first rule was vacated by a court in 2013 and its second was disapproved by Congress in 2017 under the Congressional Review Act, the SEC adopted a third rule in December 2020, with a narrower definition of "project" and broader exemptions.

Commodity Trading Transparency

Payments by commodity traders to governments and NOCs for the purchase of crude oil, which may represent the largest single revenue stream for some producer governments, were initially outside the scope of payment reporting rules. The EITI encourages the disclosure of revenues from the sale of the state's share of production, and the **Swiss** government, home to many major commodity traders, has considered but not yet adopted mandatory disclosure of payments by traders.

Several trading companies voluntarily disclose payments to governments. The series of enforcement actions against traders for bribery, noted in section 22.3, has increased pressure for greater transparency in this segment of the oil value chain.

22.5 Sanctions Compliance and Export Controls

Sanctions as Instruments of Energy Geopolitics

Economic sanctions — measures restricting trade, investment, and financial transactions with targeted states, entities, and individuals — have become a central instrument of foreign policy, and the energy sector, as a principal source of revenue for many targeted states, is among their primary targets. Sanctions against Iraq, Iran, Libya, Venezuela, Syria, and Russia have profoundly affected global oil and gas markets, investment flows, and the operations of energy companies. For energy enterprises, sanctions compliance has become a core legal function, since violations may result in severe criminal and civil penalties, exclusion from the US financial system, and reputational damage. The geopolitical dimensions of sanctions are examined in Chapter 25; this section addresses their legal framework and compliance implications.

Sources of Sanctions

United Nations sanctions are adopted by the Security Council under Chapter VII of the UN Charter and are binding on all member states under Article 25. Energy-related UN sanctions have included the comprehensive sanctions on Iraq from 1990, administered from 1996 to 2003 through the **Oil-for-Food Programme**, whose administration was marred by corruption documented by the Independent Inquiry Committee chaired by Paul Volcker in 2005; measures concerning illicit petroleum exports from Libya; and the sanctions on Iran relating to its nuclear programme, which were lifted in 2016 under Security Council **Resolution 2231** endorsing the Joint Comprehensive Plan of Action. In August 2025, France, Germany, and the United Kingdom triggered the "snapback" mechanism of Resolution 2231, alleging significant non-performance by Iran, and the earlier UN sanctions were reimposed at the end of September 2025, a step whose validity Iran, Russia, and China disputed.

Autonomous sanctions are imposed by individual states and regional organisations without Security Council authorisation. The **United States** imposes sanctions under the International Emergency Economic Powers Act of 1977 and specific statutes, administered principally by the Treasury Department's **Office of Foreign Assets Control** (OFAC), which maintains the **Specially Designated Nationals and Blocked Persons List** (SDN List). US persons are prohibited from dealing with designated persons, whose property within US jurisdiction is blocked; under OFAC's **fifty per cent rule**, entities owned fifty per cent or more, directly or indirectly, by blocked persons are themselves blocked. The United

States also imposes **secondary sanctions**, which threaten non-US persons with penalties — such as exclusion from the US financial system — for engaging in specified transactions with targeted countries or persons, even where no US nexus exists. The **European Union** adopts restrictive measures through decisions of the Council under the Common Foreign and Security Policy and regulations under Article 215 of the Treaty on the Functioning of the European Union, which apply within the EU, to EU nationals and companies, and to business done in the EU. The **United Kingdom**, after leaving the EU, imposes sanctions under the Sanctions and Anti-Money Laundering Act 2018, administered by the Office of Financial Sanctions Implementation.

Sanctions on Russian Energy

The sanctions imposed on Russia following its annexation of Crimea in 2014 and, far more extensively, its full-scale invasion of Ukraine in February 2022, are the most significant energy sanctions in history, given Russia's position as one of the world's largest exporters of oil, gas, and coal. The 2014 sectoral sanctions restricted financing for Russian energy companies and the export of technology for deepwater, Arctic offshore, and shale oil projects. From 2022, Western states imposed sweeping measures, including:

- an **EU embargo** on seaborne imports of Russian crude oil from December 2022 and of petroleum products from February 2023, with temporary exemptions for pipeline imports to certain member states, and a ban on imports of Russian coal from August 2022;
- a **price cap** agreed by the G7, the EU, and Australia, under which Western service providers — shipping, insurance, finance, and brokering — may support the maritime transport of Russian crude oil and products to third countries only if they are sold at or below a specified price, initially set at USD 60 per barrel for crude oil; in 2025, the EU and the United Kingdom lowered the cap and introduced a dynamic mechanism linking it to market prices;
- the designation of large numbers of vessels of the "**shadow fleet**" of ageing tankers, often with opaque ownership and inadequate insurance, used to transport Russian oil outside the price cap regime;
- restrictions on new investment in the Russian energy sector and the export of energy-related technology and equipment;
- measures against Russian LNG, including a ban on transhipment in EU ports from 2025 and steps taken by the EU from 2025 towards a legal framework to phase out imports of Russian gas, including LNG; and
- the designation by the United States, in October 2025, of the major Russian oil companies **Rosneft** and **Lukoil**, with consequences for their international assets and trading partners.

Russia responded with counter-measures, including decrees restricting the exit of Western investors, the placing of assets of Western companies under temporary administration, and prohibitions on compliance with the price cap. These measures led many Western energy companies to exit Russia, incurring substantial write-downs, and generated a wave of disputes, including claims in arbitration and national courts concerning the termination of contracts, the non-delivery of gas by Gazprom to European buyers, and the expropriation of assets, examined in Chapters 6 and 26.

Export Controls

Export controls restrict the export, re-export, and transfer of goods, software, and technology for reasons of national security and foreign policy. They apply to military goods and to **dual-use** items, which have both civilian and military applications. The United States administers export controls principally under the **Export Administration Regulations**, overseen by the Department of Commerce's Bureau of Industry and Security, with extraterritorial reach to foreign-made items containing more than a specified proportion of US-origin content and items produced with certain US technology. The European Union's **Dual-Use Regulation** (Regulation (EU) 2021/821) establishes a common regime, and the multilateral **Wassenaar Arrangement** coordinates national lists. Export controls on energy-related technology have been used against Russia, Iran, and Venezuela.

Export controls have also become instruments of competition over the minerals and technologies of the energy transition. **China**, which dominates the processing of many critical minerals, imposed export controls on gallium and germanium in 2023, on graphite in December 2023, on technologies for rare earth processing, and on antimony in 2024, and in 2025 announced broad controls on rare earth elements, magnets, and related technologies, prompting negotiations with the United States and other trading partners. The United States has imposed controls on the export to China of advanced semiconductors and manufacturing equipment. These measures, discussed in relation to critical minerals in Chapter 8 and to trade law in Chapter 13, raise questions about their consistency with the security exceptions of the GATT and their implications for supply chain resilience.

Sanctions and Contracts

Sanctions frequently disrupt existing contracts, raising questions of contract law that have been litigated extensively in the English courts. Parties may rely on **sanctions clauses**, which typically permit a party to suspend or terminate performance where performance would expose it to sanctions; on **force majeure** clauses; or on the doctrine of **frustration**. In ***Lamesa Investments Ltd v***

Cynergy Bank Ltd [2020] EWCA Civ 821, the Court of Appeal held that a clause permitting a borrower to withhold payment in order to comply with any mandatory provision of law extended to US secondary sanctions, which would expose the borrower to serious penalties even though they did not directly prohibit payment. In ***MUR Shipping BV v RTI Ltd*** [2024] UKSC 18, concerning a shipping contract affected by US sanctions on the charterer's parent company, the Supreme Court held that a force majeure clause requiring the affected party to use reasonable endeavours to overcome the force majeure event did not require it to accept performance in a currency other than that provided for in the contract — reinforcing the principle of freedom of contract and commercial certainty. Sanctions may also affect the enforcement of arbitral awards and the payment of judgments, and the European Union's **Blocking Statute** (Council Regulation (EC) No 2271/96), which prohibits EU persons from complying with specified extraterritorial US sanctions, creates potential conflicts of obligation, as considered by the Court of Justice in *Bank Melli Iran v Telekom Deutschland GmbH* (Case C-124/20, 2021).

Sanctions Compliance

An effective sanctions compliance programme for an energy enterprise requires screening of counterparties, vessels, and transactions against applicable sanctions lists, including the identification of beneficial owners to apply ownership rules; due diligence on the origin and destination of cargoes, including the detection of ship-to-ship transfers, manipulation of vessel location data, and falsified documentation that characterise sanctions evasion in oil trading; contractual protections, including sanctions representations, warranties, and termination rights; controls over the export of technology and equipment; and training and escalation procedures. OFAC's *Framework for OFAC Compliance Commitments* (2019) identifies five essential components: management commitment, risk assessment, internal controls, testing and auditing, and training. Enforcement actions by OFAC and the Department of Justice against energy and shipping companies demonstrate that sanctions violations are treated as serious offences, and the criminalisation of sanctions violations has been extended within the European Union by Directive (EU) 2024/1226.

22.6 Environmental, Social, and Governance Disclosure

The Rise of ESG Disclosure

Investors, lenders, and other stakeholders increasingly demand information on companies' environmental, social, and governance performance and risks, on the basis that such factors may affect financial performance and that investors have an interest in the impact of their investments. Disclosure began as a voluntary practice, guided by frameworks such as the **Global Reporting Initiative** (GRI),

established in 1997, the **Sustainability Accounting Standards Board** (SASB), and the **CDP** (formerly the Carbon Disclosure Project). The proliferation of frameworks led to calls for consolidation and for mandatory, comparable, and decision-useful disclosure, particularly in relation to climate change.

From the TCFD to the ISSB

In 2017, the **Task Force on Climate-related Financial Disclosures** (TCFD), established by the Financial Stability Board, published recommendations for disclosure of climate-related financial risks and opportunities, organised around four pillars — governance, strategy, risk management, and metrics and targets — and recommending the use of scenario analysis, including a 2°C or lower scenario. The TCFD recommendations were widely adopted and incorporated into mandatory requirements in several jurisdictions, including the United Kingdom, New Zealand, Japan, and Switzerland.

In 2021, the IFRS Foundation established the **International Sustainability Standards Board** (ISSB), which in June 2023 issued its first two standards: **IFRS S1** (General Requirements for Disclosure of Sustainability-related Financial Information) and **IFRS S2** (Climate-related Disclosures), which incorporate the TCFD recommendations and require disclosure of Scope 1, 2, and 3 greenhouse gas emissions, climate-related transition plans, and climate resilience assessed through scenario analysis, with industry-specific guidance derived from SASB. The ISSB standards, effective for annual periods beginning on or after 1 January 2024 where adopted by jurisdictions, provide a "global baseline" for investor-focused sustainability disclosure. Many jurisdictions have announced adoption or alignment, including **Nigeria**, where the Financial Reporting Council announced its adoption of the ISSB standards in 2023 with a phased roadmap — making Nigeria among the first countries to commit to the standards — and other African jurisdictions such as Ghana and Kenya.

The European Union

The European Union has adopted the most extensive mandatory sustainability disclosure regime. The **Corporate Sustainability Reporting Directive** (Directive (EU) 2022/2464, CSRD), replacing the Non-Financial Reporting Directive of 2014, requires large companies and listed companies, including non-EU companies with significant EU activities, to report on sustainability matters in accordance with **European Sustainability Reporting Standards** (ESRS), on the basis of **double materiality** — covering both the financial risks and opportunities for the company arising from sustainability matters (financial materiality) and the company's impacts on people and the environment (impact materiality) — with mandatory limited assurance. In 2025, as part of an "Omnibus" simplification package, the EU postponed the application of the CSRD for companies

in later waves and proposed to reduce its scope substantially and simplify the ESRS. The **EU Taxonomy Regulation** (Regulation (EU) 2020/852) establishes a classification system for environmentally sustainable economic activities, with associated disclosure obligations; its inclusion, through a Complementary Delegated Act of 2022, of certain natural gas and nuclear activities as transitional activities was controversial and was challenged before the EU courts by Austria and environmental organisations. The **Sustainable Finance Disclosure Regulation** (Regulation (EU) 2019/2088) imposes disclosure requirements on financial market participants.

The United States

In the United States, the **Securities and Exchange Commission** adopted rules in March 2024 requiring registrants to disclose climate-related risks with a material impact on their business, governance and risk management processes, material climate-related targets, and, for larger companies, Scope 1 and 2 emissions where material. The rules were challenged by business groups and states, stayed pending judicial review, and in March 2025 the Commission voted to end its defence of the rules. **California** enacted legislation in 2023 — the Climate Corporate Data Accountability Act (SB 253) and the Climate-Related Financial Risk Act (SB 261) — requiring large companies doing business in the state to disclose their greenhouse gas emissions, including Scope 3 emissions, and climate-related financial risks, which became the most significant climate disclosure requirements in the United States, subject to ongoing litigation.

Disclosure Obligations of Energy Companies

For energy companies, ESG disclosure raises specific issues: the disclosure of **Scope 3 emissions** from the use of sold products, which account for the great majority of the emissions associated with oil and gas companies; the disclosure of **reserves** and the risk of "stranded assets" under transition scenarios; the credibility of **net zero targets** and **transition plans**; the disclosure of methane emissions, now subject to measurement, reporting, and verification requirements under the EU Methane Regulation (Regulation (EU) 2024/1787), which will extend to imports of oil and gas; and the interaction between sustainability disclosures and financial statements, including the assumptions used in impairment testing and decommissioning provisions. Disclosure obligations also exist under securities laws of general application: misstatements in disclosure may give rise to liability to investors, as illustrated by securities litigation against energy companies concerning safety, reserves, and climate risks.

Greenwashing

As companies make increasing claims about their environmental credentials, **greenwashing** — misleading claims about environmental performance — has

become a significant source of legal risk. Regulators and courts have acted against misleading claims in advertising and in investment products. In the Netherlands, the District Court of Amsterdam held in March 2024 that advertisements by the airline KLM suggesting that its flights were sustainable were misleading. Advertising standards authorities in the United Kingdom and elsewhere have ruled against advertisements by energy companies that emphasised renewable energy investments while omitting the predominance of fossil fuels in their activities. In Australia, the Federal Court imposed a penalty on a superannuation fund manager in 2024 for misleading claims that its "sustainable" investment options excluded fossil fuel investments, and the Australasian Centre for Corporate Responsibility brought proceedings against the oil and gas producer **Santos**, alleging that its claims about natural gas as "clean energy" and its net zero plan were misleading. The European Union adopted the **Empowering Consumers for the Green Transition Directive** (Directive (EU) 2024/825), which prohibits generic environmental claims that cannot be substantiated and claims based on carbon offsetting that a product has a neutral, reduced, or positive environmental impact. Energy companies must therefore ensure that their sustainability communications are accurate, substantiated, and consistent with their disclosures and business strategies.

22.7 Whistleblower Protection and Corporate Investigations

The Importance of Whistleblowers

Whistleblowers — employees, contractors, and others who report wrongdoing they have observed in the course of their work — are among the most important sources of information on corruption, fraud, safety violations, environmental offences, and sanctions breaches. Many of the major enforcement actions in the energy sector originated in disclosures by insiders. Yet whistleblowers often face retaliation, including dismissal, demotion, harassment, blacklisting, legal action, and, in some contexts, physical danger. Legal protection of whistleblowers, and the establishment of effective internal reporting channels, are therefore essential components of both public enforcement and corporate compliance.

Legal Frameworks for Whistleblower Protection

The **United Kingdom's Public Interest Disclosure Act 1998**, which inserted provisions into the Employment Rights Act 1996, protects workers who make "qualifying disclosures" of information which, in their reasonable belief, is made in the public interest and tends to show that a criminal offence, breach of a legal obligation, miscarriage of justice, danger to health and safety, environmental damage, or concealment of any of these has occurred, is occurring, or is likely to occur. Protection depends on the disclosure being made to an appropriate recipient — the employer, a prescribed regulator, or, under stricter conditions, a wider

audience. Workers are protected against dismissal and detriment, with uncapped compensation for unfair dismissal.

The **EU Whistleblower Protection Directive** (Directive (EU) 2019/1937) requires member states to protect persons who report breaches of EU law in specified areas, including public procurement, financial services, anti-money laundering, product safety, transport safety, protection of the environment, nuclear safety, and public health. Private entities with fifty or more workers must establish internal reporting channels; member states must establish external reporting channels; and reporting persons are protected against retaliation, with a reversal of the burden of proof in retaliation proceedings, provided they had reasonable grounds to believe that the information was true at the time of reporting.

In the **United States**, the Sarbanes-Oxley Act of 2002 protects employees of public companies who report securities fraud, and the Dodd-Frank Act of 2010 established the **SEC Whistleblower Program**, which awards whistleblowers who voluntarily provide original information leading to successful enforcement actions resulting in sanctions exceeding USD 1 million between ten and thirty per cent of the sanctions collected. The programme has made awards totalling more than two billion dollars, including in FCPA cases, and is open to non-US persons. The Department of Justice launched a **Corporate Whistleblower Awards Pilot Program** in 2024, covering foreign corruption and other corporate crimes. The False Claims Act provides for *qui tam* actions in which private persons may sue on behalf of the government for fraud against federal programmes, including in the energy sector concerning royalties on federal lands.

In **Nigeria**, the Federal Ministry of Finance introduced a **Whistleblower Policy** in December 2016, which provides for rewards of between 2.5 and 5 per cent of recovered funds to whistleblowers who provide information leading to the voluntary return of stolen or concealed public funds or assets, and for protection against retaliation. The policy led to significant recoveries in its early years, but it is an executive policy rather than legislation, and bills to establish a comprehensive statutory whistleblower protection regime have been considered by the National Assembly. Whistleblowers are also protected to some extent under the Freedom of Information Act 2011, the Money Laundering (Prevention and Prohibition) Act 2022, and the Nigerian Code of Corporate Governance 2018, which requires companies to have whistleblowing frameworks. The absence of comprehensive statutory protection and concerns about the confidentiality of reports and the security of whistleblowers remain obstacles.

Internal Reporting Channels

Effective internal reporting — "speak-up" — channels allow employees and third parties to raise concerns confidentially and, where permitted, anonymously,

and are a central element of compliance programmes. Enforcement authorities evaluate whether companies have such channels, whether employees know about and trust them, whether reports are investigated promptly and independently, and whether retaliation is prohibited and punished. The **ISO 37002** standard on whistleblowing management systems (2021) provides guidance on their design. In the energy sector, reporting channels should be accessible to contractors and workers at remote sites and in joint ventures, and should cover safety and environmental concerns as well as corruption and fraud, given the importance of a "safety culture" in which workers feel able to raise concerns, a lesson emphasised by the investigations into the Piper Alpha and *Deepwater Horizon* disasters.

Corporate Investigations

When allegations of misconduct arise — from a whistleblower, an audit, a regulator's inquiry, or media reports — companies must decide whether and how to investigate. An internal investigation may be conducted by in-house lawyers, compliance or audit functions, or external counsel and forensic accountants, and may be overseen by the board or a committee of independent directors. A well-conducted investigation enables the company to establish the facts, stop misconduct, discipline those responsible, remediate control failures, and decide whether to self-report to authorities. Key issues include:

- **Scope and independence.** The investigation's terms of reference should be defined in light of the allegations, and the investigators should be independent of those whose conduct is under examination, particularly where senior management is implicated.
- **Legal privilege.** The protection of investigation materials from disclosure depends on the rules of privilege of the relevant jurisdictions, which differ significantly. In the United States, *Upjohn Co v United States* (1981) established that communications between corporate counsel and employees for the purpose of providing legal advice to the company are protected by attorney-client privilege, and lawyers interviewing employees give "*Upjohn* warnings" that the privilege belongs to the company. In England, legal advice privilege is narrowly construed to cover only communications between lawyers and those authorised to seek and receive legal advice on behalf of the company, as held in *Three Rivers District Council v Bank of England (No 5)* (2003) and reaffirmed in later cases, while **litigation privilege** protects communications made for the dominant purpose of conducting adversarial litigation that is reasonably in contemplation. In ***Director of the Serious Fraud Office v Eurasian Natural Resources Corporation Ltd*** [2018] EWCA Civ 2006, concerning an internal investigation into allegations of corruption at the company's mining operations in Kazakhstan and Africa,

the Court of Appeal held that documents generated in the investigation were protected by litigation privilege, since a criminal prosecution was reasonably in contemplation once the SFO had become involved, and that the avoidance of such litigation could be the dominant purpose of the investigation. The decision is significant for companies investigating misconduct in the shadow of potential enforcement.

- **Data protection and employment law.** Investigations involving the collection and review of emails and other personal data must comply with data protection laws, such as the EU General Data Protection Regulation and Nigeria's **Nigeria Data Protection Act 2023**, and with employment law and contractual obligations concerning interviews and disciplinary processes. Cross-border investigations must navigate blocking statutes, bank secrecy laws, and state secrecy laws that may restrict the transfer of documents to foreign authorities.
- **Reporting and remediation.** The investigation's findings must be reported to the board, and decisions taken on remediation, discipline, and disclosure.

Self-Reporting, Cooperation, and Non-Trial Resolutions

Enforcement authorities in many jurisdictions offer incentives to companies that voluntarily self-report misconduct, cooperate fully with investigations, and remediate. The US Department of Justice's **Corporate Enforcement and Voluntary Self-Disclosure Policy**, revised in 2023 and 2025, provides that companies that voluntarily self-disclose misconduct, fully cooperate, and timely and appropriately remediate will generally receive a declination, absent aggravating circumstances, subject to disgorgement. Resolutions may take the form of **non-prosecution agreements** or **deferred prosecution agreements**, under which prosecution is deferred or not pursued in exchange for the payment of penalties, cooperation, and compliance undertakings, sometimes including the appointment of an independent **compliance monitor**.

In the United Kingdom, **deferred prosecution agreements**, introduced by Schedule 17 to the Crime and Courts Act 2013, may be entered into by the SFO or the Crown Prosecution Service with organisations (not individuals) for specified economic offences, subject to approval by a judge who must be satisfied that the agreement is in the interests of justice and that its terms are fair, reasonable, and proportionate. The SFO's DPA with **Rolls-Royce** in 2017, concerning bribery in several countries including in the energy sector, involved payments of around £497 million plus costs. France's CJIP and similar mechanisms in other jurisdictions reflect a global trend towards negotiated corporate resolutions. Nigeria's **Administration of Criminal Justice Act 2015** provides for plea bargaining,

which has been used in corruption cases, and anti-corruption agencies have entered into negotiated settlements with companies, including in the TSKJ case.

Negotiated resolutions have been criticised for allowing companies to "buy their way out" of prosecution, for the failure to prosecute individuals, and for the limited participation and compensation of the victims of corruption, including the populations of the states whose officials were bribed. Responding to the last concern, some resolutions have included compensation for affected countries, and the OECD's 2021 Recommendation encourages consideration of the interests of affected states.

22.8 Compliance Programme Design and Enforcement

Why Compliance Programmes Matter

A **compliance programme** is the set of policies, procedures, controls, and organisational arrangements through which a company seeks to prevent, detect, and respond to violations of law and its own ethical standards. Compliance programmes matter legally because the existence of an effective programme may provide a **defence** — as under section 7 of the UK Bribery Act 2010 and the corporate offence of failure to prevent fraud — or may be a significant factor in decisions whether to prosecute and in the determination of penalties, as under the US Sentencing Guidelines and the Department of Justice's policies. They may also be required by law, as under France's Sapin II Law for large companies, and by contracts with lenders, partners, and governments. Beyond their legal significance, effective compliance programmes protect companies from the financial, operational, and reputational consequences of misconduct.

Authoritative Frameworks

Several authoritative frameworks guide the design of compliance programmes:

- **The UK Ministry of Justice Guidance on the Bribery Act** (2011) identifies **six principles** for adequate procedures: **proportionate procedures** (procedures proportionate to the bribery risks faced and to the nature, scale, and complexity of the organisation's activities); **top-level commitment** (by the board and senior management to preventing bribery, fostering a culture in which bribery is never acceptable); **risk assessment** (periodic, informed, and documented assessment of the nature and extent of exposure to external and internal risks of bribery); **due diligence** (proportionate and risk-based due diligence in respect of persons who perform or will perform services for or on behalf of the organisation); **communication**, including training (ensuring that prevention policies and procedures are embedded and understood throughout the organisation); and **monitoring and review** (of procedures, and improvement where necessary).

- **The US Department of Justice's *Evaluation of Corporate Compliance Programs*** (first published in 2017 and updated in 2019, 2020, 2023, and 2024) guides prosecutors in assessing programmes through three fundamental questions: whether the programme is **well designed**; whether it is **adequately resourced and empowered to function effectively**; and whether it **works in practice**. The 2024 update addressed the management of risks arising from emerging technologies, including artificial intelligence, and the use of data analytics, lessons learned from misconduct, and protection of whistleblowers.
- **International standards**, including **ISO 37001** (anti-bribery management systems, 2016, revised in 2025) and **ISO 37301** (compliance management systems, 2021), provide certifiable frameworks, although certification does not guarantee effectiveness.
- **The World Bank Group's Integrity Compliance Guidelines** set out the expectations for companies that have been sanctioned by the Bank and seek release from debarment.

The Elements of an Effective Programme

Drawing on these frameworks, the elements of an effective compliance programme for an energy enterprise, illustrated in Figure 22.1, include:

1. **Governance and tone at the top.** Board oversight of compliance, a senior compliance officer with adequate authority, independence, resources, and direct access to the board or audit committee, and visible commitment by leadership, reinforced by the "tone in the middle" set by line managers.
2. **Risk assessment.** Periodic assessment of compliance risks by country, business line, transaction type, and third party, taking account of the specific risks of the energy sector: interactions with government officials and NOCs in licensing, permitting, and customs; local content and community development obligations; agents and intermediaries; joint venture partners; commodity trading; and sanctions exposure.
3. **Policies and procedures.** A code of conduct and specific policies on anti-bribery, gifts and hospitality, facilitation payments, charitable donations and community investments, political contributions, conflicts of interest, sanctions and export controls, anti-money laundering, competition, and data protection, with clear procedures for approvals.
4. **Third-party due diligence and management.** Risk-based due diligence on agents, consultants, contractors, suppliers, joint venture partners, and local content partners, including the identification of beneficial owners and politically exposed persons; contractual protections (compliance representations, audit rights, and termination rights); and ongoing monitoring of

payments and performance. Third parties are the source of the great majority of foreign bribery cases, and the TSKJ case illustrates the risk of sham consultancy arrangements.

5. **Training and communication.** Risk-based training tailored to roles, with particular attention to employees who interact with officials, and communication of policies to third parties.
6. **Speak-up and whistleblowing.** Confidential reporting channels, protection against retaliation, and prompt follow-up, as examined in section 22.7.
7. **Investigation and discipline.** Procedures for investigating allegations, consistent disciplinary measures for misconduct, regardless of seniority, and incentives — including in remuneration — that reward ethical conduct.
8. **Monitoring, testing, and review.** Internal audit and compliance testing, data analytics to detect anomalies in payments, root cause analysis of incidents, and periodic review and improvement of the programme.

Compliance in Joint Ventures and Local Content Arrangements

Compliance in energy joint ventures and local content arrangements deserves particular attention. Non-operating partners may be exposed to liability for corrupt payments made by the operator or the joint venture, especially where they have knowledge of red flags, and should secure compliance undertakings, information and audit rights, and the right to participate in the selection of agents and contractors. Local content requirements, examined in Chapter 7, may require the engagement of local partners or subcontractors, whose ownership must be scrutinised for links to officials. Community development obligations, such as contributions to host communities development trusts under the PIA, examined in Chapter 24, and social investment programmes must be structured and documented to avoid their use as conduits for improper payments to officials or their associates.

Debarment by Multilateral Development Banks

Companies participating in projects financed by multilateral development banks are subject to the banks' **sanctions systems**, under which companies found to have engaged in fraud, corruption, collusion, coercion, or obstruction may be debarred from future bank-financed contracts. Under the **Agreement for Mutual Enforcement of Debarment Decisions** of 2010 among the African Development Bank, the Asian Development Bank, the European Bank for Reconstruction and Development, the Inter-American Development Bank, and the World Bank Group, debarments of more than one year by one bank are generally enforced by the others. Debarred companies may be released subject to conditions, often including the implementation of an integrity compliance

programme. Debarment is a significant risk for contractors in energy and infrastructure projects in developing countries.

Enforcement and the Measurement of Effectiveness

Enforcement authorities increasingly emphasise that compliance programmes must be effective in practice, not merely on paper — "paper programmes" receive no credit. Indicators of effectiveness include the use of data to monitor risks, the investigation of and response to reported concerns, the integration of compliance into business processes such as procurement and mergers and acquisitions, the consequences imposed for misconduct, compensation structures that reward compliance, and the lessons learned from incidents, including at peer companies. For energy enterprises operating in high-risk environments, compliance is not merely a legal function but a strategic necessity, closely related to the management of safety, environmental, human rights, and sanctions risks and to the company's licence to operate.

Conclusion

The governance and compliance frameworks examined in this chapter shape how energy enterprises are directed and how they respond to the legal and ethical risks inherent in their activities. Corporate governance standards, including those for state-owned enterprises, seek to ensure that enterprises are managed in the interests of their owners and stakeholders; Nigeria's transformation of NNPC into a limited liability company under the PIA is a significant test of whether legal restructuring can improve the governance of a national oil company. Directors' duties require boards to oversee compliance and manage material risks, including climate risks, although courts have been reluctant to second-guess the commercial judgments of directors, as *ClientEarth v Shell* illustrates.

The international anti-corruption framework, led by the OECD Convention, UNCAC, and the extraterritorial enforcement of the FCPA and the UK Bribery Act, has produced landmark enforcement actions in the energy sector, from Bonny Island to the commodity trading cases, although the political commitment to enforcement has fluctuated. Beneficial ownership transparency, EITI, and payments-to-governments reporting address the secrecy that facilitates corruption. Sanctions and export controls have become pervasive features of energy trade, especially since 2022, requiring sophisticated compliance systems and raising difficult contractual issues. ESG disclosure has moved from voluntary practice to mandatory regulation, though its scope is contested, and greenwashing has become a significant legal risk. Whistleblower protection, sound investigations, and well-designed compliance programmes are the instruments through which companies translate legal obligations into practice. For lawyers advising energy

enterprises — whether in Lagos, London, or Houston — mastery of these frameworks is indispensable.

::: exercise {title="Practical Exercise 22: A Compliance Crisis in an Offshore Joint Venture (HYPOTHETICAL)"} **This exercise is entirely hypothetical. The companies, persons, states, and events described are fictitious and do not refer to any real company, individual, country, or matter.**

Harrowgate Energy plc is an oil and gas company incorporated in England, with shares listed in London and American depositary receipts listed in New York. It holds a forty per cent non-operated interest in an offshore production sharing contract in the Republic of Tessaly, alongside Tessaly's national oil company (thirty per cent) and Bluewater Petroleum Inc, a privately owned company incorporated in a third state, which is the operator (thirty per cent). Tessaly is a party to UNCAC and has a statutory host community trust regime and local content legislation requiring that a specified proportion of contracts be awarded to Tessalian-owned companies.

Harrowgate's regional finance manager has reported anonymously through the company's speak-up hotline that: (i) Bluewater has awarded a USD 90 million logistics contract to Coastline Services Ltd, a recently incorporated Tessalian company with no track record, whose shareholders, according to the corporate registry, are two nominee companies in a secrecy jurisdiction; (ii) a local newspaper has reported that the brother-in-law of Tessaly's deputy minister of petroleum is a director of Coastline; (iii) Harrowgate's share of the contract costs has been charged to the joint account and paid; (iv) the joint venture has made a "community development contribution" of USD 4 million to a foundation chaired by the governor of the coastal province, outside the statutory host community trust; and (v) one of the joint venture's support vessels, chartered from a third party, appears from vessel tracking data to have conducted a ship-to-ship transfer of fuel with a tanker designated by OFAC as part of a sanctions evasion network.

Harrowgate's board has also received a letter from an investor coalition demanding that it explain how its board oversees climate risk, after Harrowgate announced a new offshore exploration programme, and asking it to submit its transition plan to an advisory shareholder vote.

Tasks.

(a) As general counsel of Harrowgate, prepare a memorandum to the board identifying the potential legal exposures of the company and its directors under the UK Bribery Act 2010, the FCPA (including its accounting provisions), the Economic Crime and Corporate Transparency Act 2023, UK and US sanctions law, and Tessalian law, and the contractual rights of Harrowgate under the joint operating agreement (assume a standard agreement containing audit rights and anti-corruption covenants).

(b) Design an internal investigation plan, addressing who should oversee and conduct the investigation, its scope, the preservation of documents, interviews (including of employees seconded to the operator), legal privilege in the relevant jurisdictions, data protection, and the handling of the whistleblower.

(c) Advise the board on whether, when, and to whom Harrowgate should self-report, weighing the incentives under the policies of the relevant authorities against the risks.

(d) Prepare a short paper for the board on the remediation of Harrowgate's compliance programme, with particular reference to non-operated joint ventures, local content partners, community investments, and maritime sanctions risks.

(e) Draft a response to the investor coalition explaining how the board oversees climate risk and setting out the considerations relevant to the decision whether to hold a "say on climate" vote, with reference to the directors' duties under sections 172 and 174 of the Companies Act 2006. :::

CHAPTER SUMMARY

- Corporate governance addresses the direction and control of companies; energy enterprises face distinctive challenges arising from state ownership, joint ventures, and high-risk operating environments. The OECD Guidelines on SOEs and the Santiago Principles provide benchmarks for state enterprises, and the PIA transformed NNPC into a limited liability company.
- Directors owe duties to promote the success of the company and to exercise care, skill, and diligence, including the oversight of compliance and mission-critical risks; climate risk must be considered, but courts defer to directors' business judgment, as *ClientEarth v Shell* shows.
- The OECD Anti-Bribery Convention, UNCAC, the AU Convention, and national laws, including the FCPA, the UK Bribery Act's failure to prevent offence, France's Sapin II, and Nigeria's ICPC and EFCC legislation, prohibit corruption; the extractive sector accounts for the largest share of foreign bribery cases, illustrated by the Bonny Island LNG and commodity trading cases.
- Beneficial ownership transparency, required by FATF standards, the EITI Standard, and national registers such as Nigeria's under CAMA 2020, and payments-to-governments reporting address the secrecy that facilitates corruption.
- UN and autonomous sanctions, including the unprecedented sanctions on Russian energy since 2022, and export controls, including on critical minerals, require rigorous compliance and raise difficult contractual issues.
- ESG disclosure has moved from voluntary frameworks to mandatory regimes, including the ISSB standards, adopted by Nigeria, and the EU CSRD; greenwashing is a growing source of legal risk.

- Whistleblower protection, internal reporting channels, and properly conducted investigations, with attention to legal privilege, are central to detecting misconduct; self-reporting and cooperation may lead to negotiated resolutions such as DPAs.
- Effective compliance programmes, assessed by authorities on whether they are well designed, resourced, and effective in practice, comprise governance, risk assessment, policies, third-party due diligence, training, speak-up channels, investigation and discipline, and monitoring and review.

DOCTORAL RESEARCH QUESTIONS

1. Has the transformation of NNPC into a limited liability company under the Petroleum Industry Act 2021 improved its governance and accountability? Evaluate against the OECD Guidelines on Corporate Governance of State-Owned Enterprises.
2. To what extent do directors' duties under English and Nigerian law require directors of energy companies to align corporate strategy with the goals of the Paris Agreement?
3. Evaluate the effectiveness of extraterritorial anti-bribery enforcement by home states in addressing corruption in the Nigerian petroleum sector, and consider whether penalties should be shared with or returned to the states whose officials were bribed.
4. Assess the contribution of NEITI and beneficial ownership disclosure to the reduction of corruption in Nigeria's extractive sector.
5. How should energy companies and their lawyers manage conflicts between sanctions regimes, blocking statutes, and contractual obligations? Evaluate with reference to recent English case law.
6. Does the global baseline of the ISSB standards adequately serve the needs of developing countries such as Nigeria, or is a double materiality approach preferable for extractive industries?

FURTHER READING

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- Colin Nicholls and others, *Corruption and Misuse of Public Office* (3rd edn, Oxford University Press 2017).

- OECD, *OECD Foreign Bribery Report: An Analysis of the Crime of Bribery of Foreign Public Officials* (OECD Publishing 2014).
- Paul L Davies, Sarah Worthington and Christopher Hare, *Gower Principles of Modern Company Law* (11th edn, Sweet & Maxwell 2021).
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23 CHAPTER

Energy Justice, Resource Equity, and Sustainable Development

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the theoretical foundations of energy justice and its relationship to theories of distributive justice, environmental justice, and climate justice.
2. Apply the distributional, procedural, and recognition tenets of energy justice, together with restorative and intergenerational dimensions, to the analysis of energy law and policy.
3. Analyse the scale and causes of energy poverty and evaluate legal and policy instruments for achieving universal access to electricity and clean cooking.
4. Assess the gender and social equity dimensions of energy governance and the legal instruments designed to address them.
5. Evaluate legal frameworks for the distribution of natural resource revenues, including fiscal federalism, derivation, and sovereign wealth funds, with particular reference to Nigeria.
6. Explain the principle of intergenerational equity and its application to the management of non-renewable resources and the energy transition.
7. Analyse the role of the Sustainable Development Goals, especially SDG 7, in shaping energy law and policy.
8. Apply principles of energy justice to the design of equitable regulatory systems, including tariff design, universal service, community energy, and just transition mechanisms.

Introduction

The preceding chapters have examined the legal frameworks governing the production, trade, and regulation of energy and natural resources, and the protection of the environment and human rights. This chapter asks a more fundamental question: are the benefits and burdens of energy and resource systems distributed justly, and how can law contribute to making them more just? The question is

pressing. Hundreds of millions of people, most of them in sub-Saharan Africa, lack access to electricity; more than two billion lack access to clean cooking fuels and technologies, with grave consequences for health, especially for women and children; and many households, in rich and poor countries alike, cannot afford the energy services they need. At the same time, the communities that host oil fields, mines, and power plants often bear their environmental and social costs without sharing equitably in their benefits, and the revenues from natural resources have frequently enriched elites rather than improving the lives of citizens — the phenomenon known as the "resource curse". The energy transition, while necessary to avert dangerous climate change, raises its own questions of justice: who will pay for it, who will lose their livelihoods, who will benefit from new industries, and whose lands and waters will be used for renewable energy and transition minerals?

Energy justice has emerged over the past decade and a half as a framework for addressing these questions, drawing on philosophy, law, economics, and the social sciences. It complements the more established concepts of environmental justice and climate justice and offers analytical tools — the "tenets" of distributional, procedural, and recognition justice, supplemented by restorative and intergenerational dimensions — that can be applied to evaluate and design energy law and policy. This chapter examines the theoretical foundations of energy justice (section 23.1) and its principal tenets (section 23.2), before applying them to energy poverty and universal access (section 23.3), gender and social equity (section 23.4), resource revenue distribution (section 23.5), intergenerational equity (section 23.6), the Sustainable Development Goals (section 23.7), and the design of equitable energy regulatory systems (section 23.8).

	Core question	Illustrative legal and policy instruments
Distributional	Who bears the burdens and who receives the benefits of energy systems?	Lifeline tariffs; targeted subsidies; revenue sharing; benefit-sharing funds
Procedural	Who participates in decisions, with what information and remedies?	EIA and public hearings; access to information; FPIC; access to justice
Recognition	Whose identities, needs and knowledge are acknowledged or ignored?	Gender assessments; indigenous rights; disaggregated data
Restorative	How are past and present harms repaired?	Remediation funds; compensation; clean-up mandates
Intergenerational & cosmopolitan	What is owed to future generations and to people beyond borders?	Sovereign wealth funds; climate finance; future-generations bodies

Figure 23.1 The tenets of energy justice and illustrative legal instruments. *Source:* prepared for this edition.

23.1 Theoretical Foundations of Energy Justice

The Emergence of Energy Justice Scholarship

Concerns about the fairness of energy systems are not new — debates about the affordability of electricity, the rights of communities affected by dams and mines, and the distribution of oil revenues are as old as the modern energy industry — but the articulation of **energy justice** as a distinct conceptual framework dates from the early 2010s. Scholars such as Kirsten Jenkins, Darren McCauley, Raphael Heffron, Benjamin Sovacool, and Michael Dworkin developed the concept to bring the insights of justice theory to bear on energy policy, which had been dominated by technical and economic analysis focused on security, cost, and, increasingly, carbon emissions. In an influential formulation, McCauley, Heffron, Stephan, and Jenkins (2013) proposed a "triumvirate of tenets" — distributional, procedural, and recognition justice — drawn from environmental justice scholarship. Sovacool and Dworkin, in *Global Energy Justice* (2014), defined energy justice as a global energy system that fairly disseminates both the benefits and costs of energy services and one that has representative and impartial energy decision-making, and proposed a set of principles — availability, affordability, due process, transparency and accountability, sustainability, intragenera-

tional equity, intergenerational equity, and responsibility — to guide policy. Later scholarship added **restorative justice**, concerned with repairing harms caused by energy activities, and **cosmopolitan justice**, concerned with obligations that extend beyond national borders.

Energy justice is both a **conceptual tool** for analysing how energy systems distribute benefits and burdens, an **analytical tool** for researchers to understand how values are incorporated into energy systems, and a **decision-making tool** for policymakers and lawyers seeking to design fairer laws and institutions. Its relationship to law is direct: the legal rules governing access to energy, tariff-setting, licensing, land acquisition, environmental assessment, revenue distribution, and participation are among the principal means by which the distribution of benefits and burdens is determined, and legal processes are among the principal means by which injustice may be contested.

Philosophical Foundations

Energy justice draws on several traditions of political philosophy.

Distributive justice theories address the fair distribution of goods and burdens in society. John Rawls's *A Theory of Justice* (1971) proposed that principles of justice are those that free and rational persons would choose behind a "veil of ignorance", not knowing their place in society, and argued that they would choose equal basic liberties for all and the "difference principle", under which social and economic inequalities are justified only if they benefit the least advantaged members of society. Applied to energy, a Rawlsian approach would prioritise the energy needs of the poorest and evaluate energy policies by their effects on the least advantaged. Utilitarian approaches, by contrast, evaluate policies by their aggregate welfare effects, underlying much cost-benefit analysis in energy policy, while libertarian approaches emphasise property rights and voluntary exchange, informing market-based models of energy regulation.

The **capabilities approach**, developed by Amartya Sen and Martha Nussbaum, focuses not on resources or welfare as such but on what people are actually able to do and to be — their capabilities to achieve functionings they have reason to value, such as being well nourished, healthy, educated, and able to participate in social life. Energy is not valued for its own sake but for the capabilities it enables: cooking, heating, lighting, refrigeration, communication, mobility, education, and productive work. The capabilities approach has been influential in understanding energy poverty as a deprivation of capabilities rather than merely a lack of access to particular fuels or a high ratio of expenditure to income, and it informs the multidimensional measurement of energy access.

The **politics of recognition**, associated with Nancy Fraser, Axel Honneth, and Iris Marion Young, holds that justice requires not only the fair distribution

of goods but also the recognition of the equal status and distinct identities of social groups, and that misrecognition — cultural domination, non-recognition, and disrespect — is itself a form of injustice. Fraser's account of justice as **parity of participation** integrates redistribution, recognition, and representation. Applied to energy, recognition justice draws attention to groups whose needs and perspectives are ignored or devalued in energy decision-making — indigenous peoples, women, the rural poor, people with disabilities, and minorities.

Environmental Justice

Environmental justice emerged as a social movement in the United States in the 1980s, when civil rights activists protested against the disproportionate siting of hazardous waste facilities in African American and low-income communities — notably in Warren County, North Carolina, in 1982. The United Church of Christ's report *Toxic Wastes and Race in the United States* (1987) documented a strong correlation between race and the location of hazardous waste sites. Executive Order 12898 of 1994 directed federal agencies to identify and address the disproportionately high and adverse human health or environmental effects of their programmes on minority and low-income populations. In 2021 and 2023, the federal government adopted further environmental justice measures, including the **Justice40 Initiative**, which aimed to direct forty per cent of the overall benefits of certain federal investments in climate and clean energy to disadvantaged communities; these measures were revoked in January 2025. Environmental justice scholarship, notably the work of Robert Bullard and David Schlosberg, has developed the distributional, procedural, and recognition dimensions on which energy justice draws.

In the Global South, environmental justice movements have often arisen in response to resource extraction — the "environmentalism of the poor" described by Joan Martínez-Alier — including the struggles of the Ogoni and other communities of the Niger Delta against the impacts of oil production, examined in Chapters 17 and 21. The **Environmental Justice Atlas**, a global database, has documented thousands of environmental conflicts, a large proportion of them related to fossil fuels, mining, and energy infrastructure.

Climate Justice

Climate justice addresses the distribution of the burdens of climate change and of the costs of mitigation and adaptation. It highlights that those who have contributed least to greenhouse gas emissions — the poor, developing countries, small island states, and future generations — are often the most vulnerable to climate impacts, while the historical responsibility for emissions lies predominantly with industrialised countries. Climate justice is reflected in the principle of common but differentiated responsibilities and respective capabilities, in the

provisions on climate finance and loss and damage, and in the concept of just transition, examined in Chapter 18. The tension between the development needs of energy-poor countries and the imperative of decarbonisation — the "energy trilemma" of security, equity, and sustainability, as framed by the World Energy Council — is a central concern of energy justice in Africa, where proponents of the right to development argue that countries with low historical emissions should be permitted to develop their fossil fuel resources, including natural gas, to meet their energy needs, while others argue that renewable energy offers a more just and sustainable path.

Energy Justice and Law

The relationship between energy justice and law operates at several levels. Law **embodies** distributional choices — for example, in the allocation of resource ownership, the design of tariffs and subsidies, and the sharing of revenues. Law **structures** procedures — determining who participates in decisions, what information is available, and what remedies exist. Law **recognises** or fails to recognise particular groups and interests — for example, through the recognition of customary land rights, indigenous peoples, and gender equality. And law provides **avenues for contestation** — through litigation, complaints mechanisms, and political advocacy. Scholars such as Heffron have argued that energy law should be understood as a discipline whose purpose includes the achievement of energy justice, and have proposed that the principles of energy justice be incorporated into legislation, regulatory mandates, and judicial reasoning.

23.2 Distributional, Procedural, and Recognition Justice

Distributional Justice

Distributional justice concerns the allocation of the benefits and burdens of energy systems. The benefits include access to energy services, employment, revenues, infrastructure, and economic development; the burdens include pollution, health risks, landscape impacts, displacement, the costs of energy, and the risks of accidents. Distributional injustice occurs when these are allocated unequally in ways that cannot be justified — for example, when pollution from power plants and refineries is concentrated in poor or minority neighbourhoods, when communities hosting oil fields remain without electricity or clean water, or when the costs of the energy transition fall disproportionately on low-income households.

Distributional questions arise at multiple scales. **Globally**, energy consumption per capita varies enormously: the average American consumes many times as much electricity as the average Nigerian, and the richest tenth of the world's population is responsible for around half of consumption-based emissions. **Nationally**, energy access, quality, and affordability vary between urban and rural areas, regions, and income groups; in Nigeria, for example, access to grid electricity

is concentrated in urban areas and the south, and the quality of supply varies widely among tariff bands, examined in section 23.8. **Locally**, the siting of energy infrastructure determines which communities bear its impacts.

Legal instruments that address distributional justice include tariff structures and subsidies, examined in section 23.8; universal service obligations and rural electrification programmes, examined in section 23.3; revenue-sharing arrangements, examined in section 23.5; benefit-sharing mechanisms for host communities, such as Nigeria's host communities development trusts, examined in Chapter 24; siting regulations and environmental standards; and compensation regimes. Distributional analysis is also increasingly incorporated into environmental and regulatory impact assessment, for example through distributional impact assessments of carbon pricing and other climate policies.

Procedural Justice

Procedural justice concerns the fairness of the processes by which energy decisions are made. It requires that affected people have access to information, opportunities to participate meaningfully in decision-making, impartial decision-makers, and access to remedies. Procedural injustice occurs when decisions are taken without consultation, when information is withheld, when participation is tokenistic, when decision-makers are captured by interested parties, or when affected people lack access to justice.

The legal foundations of procedural justice in environmental and energy matters are well established. **Principle 10 of the Rio Declaration** (1992) provides that environmental issues are best handled with the participation of all concerned citizens at the relevant level, and that each individual shall have appropriate access to information, the opportunity to participate in decision-making processes, and effective access to judicial and administrative proceedings, including redress and remedy. These "three pillars" of environmental democracy are given binding form in the **Aarhus Convention** (1998), which applies in Europe and Central Asia and whose Compliance Committee has addressed numerous energy-related cases, including those concerning nuclear power, renewable energy planning, and national energy policies, and the **Escazú Agreement** (2018) in Latin America and the Caribbean. Procedural justice is also embodied in the requirements of environmental impact assessment, examined in Chapter 17, the principle of free, prior, and informed consent, examined in Chapter 21, the procedural requirements of administrative law governing energy regulators, and rights of access to information, such as Nigeria's **Freedom of Information Act 2011**.

Procedural justice in energy regulation involves the transparency and participation of regulatory processes, such as tariff reviews. The **Electricity Act 2023** of Nigeria and its predecessor require the Nigerian Electricity Regulatory Commis-

sion to consult stakeholders in tariff-setting and rule-making, and NERC holds public hearings and consultations on major regulatory decisions; consumers may lodge complaints with distribution companies and, on appeal, with NERC's **Forum Offices**, consumer dispute resolution bodies located throughout the country. The effectiveness of such procedures depends on the capacity of consumers and civil society to participate, the availability of information in accessible forms, and the responsiveness of regulators.

Recognition Justice

Recognition justice concerns whether the identities, needs, values, and knowledge of different groups are acknowledged and respected in energy systems. Misrecognition may take the form of the failure to recognise that particular groups exist or are affected, the disregard of their distinctive needs, the devaluation of their knowledge and cultural values, or stereotyping and discrimination. It is often a precondition of distributional and procedural injustice: groups that are not recognised are unlikely to be consulted or to have their needs met.

Examples of misrecognition in energy systems include the failure to recognise the customary land rights of rural communities and indigenous peoples in the siting of energy projects, examined in Chapter 21; the neglect of the distinct energy needs of women, who in many developing countries bear the burden of collecting fuel and cooking with polluting fuels, examined in section 23.4; the failure to account for the needs of people with disabilities, older people, and people with medical conditions dependent on electricity in the design of disconnection rules and tariffs; and the treatment of informal settlements as "illegal" and therefore ineligible for formal electricity connections. Legal instruments addressing recognition include the recognition of customary and indigenous rights, anti-discrimination law, the collection of disaggregated data, gender impact assessments, and protections for vulnerable customers in energy regulation, such as rules prohibiting the disconnection of vulnerable customers during winter or extreme heat.

Restorative Justice

Restorative justice, a later addition to the energy justice framework, concerns the repair of harms caused by energy activities to people and the environment. It draws on the restorative justice tradition in criminal law, which emphasises repairing the harm caused by wrongdoing and restoring relationships, rather than merely punishing offenders. In energy, restorative justice is concerned with the remediation of pollution, the compensation of victims, the rehabilitation of mining and oil-producing areas, and the transition of communities dependent on declining industries. Legal instruments include environmental liability regimes, remediation obligations, decommissioning and mine closure requirements, com-

pensation funds, and just transition programmes. The legacy of oil pollution in the Niger Delta, and the slow progress of the remediation of Ogoniland under the **Hydrocarbon Pollution Remediation Project** established in 2016 following the UNEP Environmental Assessment of 2011, examined in Chapter 17, illustrates both the need for restorative justice and the difficulties of achieving it.

Applying the Tenets: The Energy Transition

The tenets of energy justice can be applied together to evaluate energy policies and projects. Consider the deployment of a large solar park in a rural area. A **distributional** analysis would ask who benefits from the electricity and revenues — whether local communities gain access to electricity, employment, and a share of revenues, or whether the power is exported to urban centres or other countries — and who bears the burdens, such as the loss of grazing or agricultural land. A **procedural** analysis would ask whether affected communities were consulted, given adequate information, and able to influence the project's design and location, and whether they have access to grievance mechanisms. A **recognition** analysis would ask whether the customary land rights of local people, including women and pastoralists, were recognised, and whether their knowledge of the land was respected. A **restorative** analysis would ask how any harms will be remedied and how the site will be restored at the end of its life. The analysis may reveal that a project that appears beneficial in aggregate terms — reducing emissions and expanding generation capacity — nonetheless perpetuates injustice, and may suggest legal and design changes, such as community benefit-sharing, local ownership, or alternative sites.

The energy transition makes such analysis especially important. Studies of the transition have identified risks of "green grabbing" — the appropriation of land and resources for environmental ends — and of "sacrifice zones" in regions where transition minerals are extracted; of regressive effects of carbon pricing and of the costs of renewable energy support schemes passed through to electricity bills; of the concentration of the benefits of rooftop solar, electric vehicles, and energy efficiency subsidies among wealthier households; and of the displacement of workers and communities dependent on fossil fuel industries. A just transition, examined in Chapter 18 and in section 23.8, requires that these risks be identified and addressed through law and policy.

23.3 Energy Poverty and Universal Access

The Scale of the Challenge

Access to modern energy services is a precondition for the realisation of many human rights and for economic and social development. Electricity powers lighting, communications, refrigeration of food and medicines, water pumping, health facilities, schools, and productive enterprises; clean cooking fuels and

technologies protect health and save time. Yet, according to the annual *Tracking SDG 7* reports prepared by the International Energy Agency, the International Renewable Energy Agency, the UN Statistics Division, the World Bank, and the World Health Organization, around 666 million people lacked access to electricity in 2023 — a figure that had declined from more than a billion a decade earlier but remained far from the goal of universal access by 2030. Around eighty-five per cent of those without access live in **sub-Saharan Africa**, where population growth has outpaced new connections in many countries. Many more people have connections that are unreliable, of poor quality, or unaffordable, so that the binary measure of "access" understates the extent of energy deprivation.

Access to **clean cooking** is an even greater challenge. Around 2.1 billion people — more than a quarter of the world's population — continue to cook with polluting fuels and technologies, such as firewood, charcoal, crop residues, animal dung, and kerosene, mostly in sub-Saharan Africa and Asia. The World Health Organization estimates that household air pollution from the use of polluting fuels and technologies for cooking causes around 3.2 million premature deaths each year, predominantly among women and children, from pneumonia, stroke, ischaemic heart disease, chronic obstructive pulmonary disease, and lung cancer. The collection of fuelwood imposes a heavy time burden, predominantly on women and girls, and contributes to deforestation.

Nigeria has the largest electricity access deficit of any country in the world, with an estimated eighty-five to ninety million people — around forty per cent of the population — lacking access to grid electricity, and many of those connected receiving only a few hours of supply each day, examined in Chapter 24. Most Nigerian households cook with firewood or charcoal, and Nigeria is among the countries with the highest burden of deaths attributable to household air pollution. The paradox of energy poverty in a major oil and gas producer, where many households and businesses rely on expensive and polluting petrol and diesel generators, is a central concern of Nigerian energy law and policy.

Defining and Measuring Energy Poverty

Energy poverty has been defined in different ways. In developing countries, it is commonly understood as the lack of access to modern energy services — electricity and clean cooking — and measured by access rates. The World Bank's **Multi-Tier Framework** for measuring energy access, developed under the Sustainable Energy for All initiative, defines access along a spectrum of tiers from zero to five, based on attributes including capacity, availability (hours of supply), reliability, quality, affordability, legality, health and safety, and convenience — recognising that a solar lantern and a reliable grid connection provide very different levels of service. The **Modern Energy Minimum**, proposed by a coalition of researchers and organisations, argues for a higher threshold of around one thousand kilowatt-

hours per person per year, including productive uses, as the minimum consistent with economic development.

In developed countries, energy poverty, or **fuel poverty**, is generally understood as the inability to afford adequate energy services, particularly heating and cooling, at reasonable cost, and measured by indicators such as the share of income spent on energy, the inability to keep the home adequately warm, and arrears on utility bills. The United Kingdom's Warm Homes and Energy Conservation Act 2000 required the government to adopt a strategy to end fuel poverty. In the European Union, the recast **Energy Efficiency Directive** (Directive (EU) 2023/1791) defines energy poverty as a household's lack of access to essential energy services that underpin a decent standard of living and health, including adequate warmth, cooling, lighting, and energy to power appliances, caused by a combination of factors including non-affordability, insufficient disposable income, high energy expenditure, and poor energy efficiency of homes, and requires member states to implement energy efficiency measures as a priority among people affected by energy poverty and vulnerable customers. The **Social Climate Fund** (Regulation (EU) 2023/955) will finance measures to support vulnerable households, microenterprises, and transport users affected by the extension of emissions trading to buildings and road transport. The energy price crisis of 2022 brought energy affordability to the forefront of policy throughout Europe, prompting emergency measures including price caps, windfall taxes, and support payments.

Energy Access as a Human Right?

Whether there is a **human right to energy** or to electricity is debated. No international human rights treaty expressly recognises such a right, with the partial exception of Article 14(2)(h) of the Convention on the Elimination of All Forms of Discrimination against Women, which requires states to ensure to rural women the right to enjoy adequate living conditions, particularly in relation to housing, sanitation, electricity and water supply, transport, and communications. However, the UN Committee on Economic, Social and Cultural Rights has recognised that the right to adequate housing includes sustainable access to energy for cooking, heating, and lighting (General Comment No. 4, 1991), and access to energy is instrumental to the realisation of the rights to health, education, food, water, and work. Some national courts have recognised a right to electricity as a component of constitutional rights to life and dignity, notably in India and South Africa, where courts have addressed disconnections of electricity supply; in *Joseph v City of Johannesburg* (2009), the Constitutional Court of South Africa held that tenants were entitled to procedural fairness before the disconnection of electricity supply by the municipality, recognising electricity as a basic municipal service. Scholars have argued for the recognition of energy access as an emerging human

right, while others caution that rights-based framing must be accompanied by the resources and institutions to deliver access.

Instruments for Universal Access

Achieving universal access requires a combination of grid extension, mini-grids, and stand-alone systems, supported by legal and regulatory frameworks, financing, and institutions. The IEA and the World Bank estimate that, in sub-Saharan Africa, the least-cost path to universal access involves a large share of new connections through **decentralised solutions** — mini-grids and solar home systems — particularly in rural areas distant from the grid.

Rural electrification agencies and funds. Many countries have established dedicated agencies and funds to finance rural electrification, often supported by levies on electricity consumers or allocations from the national budget. In Nigeria, the **Rural Electrification Agency** (REA) and the **Rural Electrification Fund**, originally established under the Electric Power Sector Reform Act 2005 and continued under the Electricity Act 2023, promote and finance rural electrification. The REA implements the **Nigeria Electrification Project**, supported by the World Bank and the African Development Bank, which provides grants to private developers of mini-grids and solar home systems, and the **Distributed Access through Renewable Energy Scale-up** (DARES) project, approved by the World Bank in 2023 with USD 750 million in financing, which aims to provide electricity to millions of Nigerians through distributed renewable energy.

Mini-grid regulation. Mini-grids — small, isolated electricity networks typically powered by solar photovoltaics with battery storage, serving villages or clusters of customers — require a regulatory framework addressing licensing, tariff-setting, quality standards, and, critically, what happens when the main grid arrives. Nigeria's **NERC Regulations for Mini-Grids**, first issued in 2016, were among the first comprehensive mini-grid regulations in Africa: they distinguish between isolated and interconnected mini-grids, establish a light-touch permitting regime for small mini-grids, allow developers to charge cost-reflective tariffs agreed with communities, and provide options when the grid arrives, including conversion to a distribution franchise or compensation for the developer's assets. Under the Electricity Act 2023, which devolves power to states to regulate electricity markets within their territories, states that have established their own regulators may regulate mini-grids within the state, examined in Chapters 9 and 24. The framework has supported the development of hundreds of mini-grids, although challenges remain in affordability, demand, and the financing of the sector.

Off-grid solar. Solar home systems, ranging from small lanterns to systems capable of powering lights, phones, televisions, and fans, have been distributed

to tens of millions of households in Africa, often through **pay-as-you-go** business models enabled by mobile money, under which customers pay in small instalments. Legal and regulatory issues include product quality standards (such as the *Lighting Global* quality standards), consumer protection in the financing of systems, import duties and value added tax, and electronic waste management.

Clean cooking. The promotion of clean cooking involves policies on liquefied petroleum gas (LPG), including subsidies, cylinder recirculation models, and safety regulation; improved biomass cookstoves; electric cooking; biogas; and ethanol. Nigeria's **National LPG Expansion Plan** sought to increase LPG penetration, and the removal of value added tax on domestic LPG was implemented to reduce costs. The IEA's **Summit on Clean Cooking in Africa**, held in Paris in May 2024, mobilised pledges of around USD 2.2 billion and proposed that clean cooking be given greater priority in national policies and international finance. Carbon finance, through the generation of carbon credits from clean cookstove projects, has been a significant source of funding, though concerns about the over-crediting of cookstove projects have prompted methodological reforms, examined in Chapter 18.

Mission 300. In 2024, the World Bank Group and the African Development Bank launched **Mission 300**, an initiative to connect 300 million people in Africa to electricity by 2030, through grid extension and distributed renewable energy. At the **Mission 300 Africa Energy Summit** in Dar es Salaam in January 2025, African heads of state adopted the **Dar es Salaam Energy Declaration**, and an initial group of countries, including Nigeria, presented **national energy compacts** setting out policy reforms and investment plans, including commitments on utility reform, cost-reflective tariffs with targeted support for poor consumers, private investment, and regional integration. The compacts illustrate the linkage between international finance and domestic legal reform in the pursuit of universal access.

Affordability and Subsidies

Access is meaningless if energy is unaffordable. Many governments have sought to make energy affordable through subsidies, either on fuels — such as petrol, diesel, kerosene, and LPG — or on electricity tariffs. However, **fossil fuel subsidies** are often regressive, benefiting wealthier households that consume more energy, and fiscally costly, consuming resources that could be used for health, education, and targeted social protection. The IMF has estimated explicit fossil fuel subsidies globally at more than a trillion dollars in 2022. Nigeria's **petrol subsidy**, which had cost the federal government many trillions of naira annually and was associated with large-scale fraud and smuggling, was removed in May 2023, examined in Chapters 18 and 24. Subsidy reform raises acute questions of energy justice: while removal may free fiscal resources and improve efficiency, it

imposes immediate costs on households, including through increases in transport and food prices, and requires accompanying measures — such as targeted cash transfers, investment in public transport, and the expansion of alternatives — to protect the poor. International experience, documented by the IMF and the World Bank, suggests that successful subsidy reform depends on public communication, gradual implementation, and credible compensation measures.

23.4 Gender and Social Equity in Energy Governance

Why Gender Matters in Energy

Energy systems are not gender-neutral. In many developing countries, women and girls bear primary responsibility for collecting fuel and cooking, so that energy poverty imposes disproportionate burdens on them — in time spent collecting firewood (often several hours a day), in exposure to household air pollution, and in risks of violence when collecting fuel in insecure areas. Women's lack of access to electricity constrains their opportunities for education, income-generating activities, and participation in public life. At the same time, women are underrepresented in the energy workforce, especially in technical and leadership roles, and in energy decision-making: the IEA has estimated that women account for only around sixteen per cent of employment in the traditional energy sector, though their share is higher in renewable energy, which IRENA has estimated at around a third. Women's land rights are often weaker than men's, which affects their entitlement to compensation and benefits when land is acquired for energy projects, and large projects may bring increased risks of gender-based violence associated with influxes of workers.

International Legal Framework

The international legal framework for gender equality in energy includes the **Convention on the Elimination of All Forms of Discrimination against Women** (CEDAW), whose Article 14 requires states to ensure that rural women participate in and benefit from rural development and enjoy adequate living conditions, including electricity, and whose Committee has addressed energy and climate in **General Recommendation No. 34** (2016) on the rights of rural women and **General Recommendation No. 37** (2018) on the gender-related dimensions of disaster risk reduction in a changing climate; the **Beijing Declaration and Platform for Action** (1995); **SDG 5** on gender equality, whose targets include ensuring women's full and effective participation and equal opportunities for leadership and undertaking reforms to give women equal rights to economic resources, including land; and the **Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa** (Maputo Protocol, 2003), which provides for women's right to a healthy and sustainable environment and to participate in the planning, management, and preservation

of the environment and the sustainable use of natural resources (Article 18), and to sustainable development, including access to land (Article 19). Under the UNFCCC, the **Lima Work Programme on Gender** (2014) and the **Gender Action Plan**, enhanced in 2019 and renewed in subsequent years, promote gender-responsive climate policy.

Regional Instruments: ECOWAS

The **Economic Community of West African States** has been a pioneer in addressing gender in energy. The **ECOWAS Policy for Gender Mainstreaming in Energy Access**, adopted by ECOWAS energy ministers in 2015 and endorsed by heads of state in 2017, developed by the ECOWAS Centre for Renewable Energy and Energy Efficiency, aims to address barriers to the equal participation of women and men in expanding energy access, with objectives on gender-sensitive policies, women's participation in energy-sector employment and decision-making, and women's entrepreneurship. It was followed by an **ECOWAS Directive on Gender Assessments in Energy Projects**, which requires gender assessments to be conducted for energy projects in member states, making it among the first legally binding regional instruments to require gender assessment in the energy sector. The implementation of these instruments in national law, including in Nigeria, remains a work in progress.

Gender in National Energy Law and Practice

Gender considerations are increasingly incorporated into national energy policies, project requirements, and the standards of financial institutions. The IFC Performance Standards and the World Bank's Environmental and Social Framework require consideration of gender in environmental and social assessments, resettlement, and stakeholder engagement, and the World Bank has adopted good practice notes on addressing gender-based violence in projects involving major civil works. Energy access programmes increasingly target women as customers and entrepreneurs — for example, through programmes supporting women-led clean energy enterprises and the recruitment of women as sales agents for solar products and clean cookstoves. Nigeria's **National Gender Policy**, revised in 2021, and the Nigeria Energy Transition Plan recognise the gender dimensions of energy, and the Rural Electrification Agency has implemented programmes to promote women's participation in the off-grid sector. The **Discrimination Against Persons with Disabilities (Prohibition) Act 2018** of Nigeria addresses the accessibility of public facilities and services, relevant to the design of energy services for people with disabilities.

Broader Social Equity

Beyond gender, energy justice requires attention to other dimensions of social equity: age, disability, ethnicity, indigeneity, income, geographic location, and

migration status. Older people and people with disabilities may be particularly vulnerable to energy poverty and to the health effects of extreme heat and cold; people dependent on electrically powered medical equipment require reliable supply and protection from disconnection; informal settlement residents may be excluded from formal connections; and young people are both disproportionately affected by unemployment in fossil fuel transitions and well placed to participate in new energy industries. **Intersectionality** — the recognition that different dimensions of identity and disadvantage interact — implies that energy law and policy should be informed by disaggregated data and by the participation of diverse groups. Legal instruments include anti-discrimination law, consumer protection rules for vulnerable customers, social tariffs, targeted support, accessibility standards, and requirements for disaggregated monitoring and reporting.

23.5 Resource Revenue Distribution

The Resource Curse

The paradox that countries rich in natural resources, particularly oil, gas, and minerals, have often experienced slower economic growth, greater inequality, more corruption, weaker democratic institutions, and a higher risk of civil conflict than countries with fewer resources is known as the **resource curse**. The term was popularised by the economist Richard Auty in 1993, and empirical work by Jeffrey Sachs and Andrew Warner in the 1990s found a negative association between natural resource dependence and economic growth. Michael Ross's *The Oil Curse* (2012) documented the association between oil wealth and authoritarianism, conflict, and gender inequality, while Paul Collier's work linked resource dependence to the risk of civil war.

Several mechanisms have been identified: **Dutch disease**, in which a resource boom causes the real exchange rate to appreciate, harming the competitiveness of agriculture and manufacturing; **revenue volatility**, as commodity price swings produce boom-and-bust cycles in public spending; **rent-seeking and corruption**, as elites compete for control of resource rents; the **weakening of state–society accountability**, since governments funded by resource rents rather than taxation have less need to respond to citizens; and **conflict**, as resources finance armed groups or become objects of contestation between regions and groups. The resource curse is not inevitable: countries such as Norway, Botswana, Chile, and Malaysia have managed resource wealth relatively successfully, and the literature increasingly emphasises that the quality of institutions — including legal institutions governing revenue management — determines whether resources are a blessing or a curse.

The Natural Resource Charter

The **Natural Resource Charter**, developed by economists and lawyers and now stewarded by the Natural Resource Governance Institute, sets out twelve precepts for the governance of natural resources for the benefit of citizens, covering the whole decision chain: the development strategy and the role of institutions; accountability and transparency; the exploration and licensing of resources; taxation; local impacts; the role of national resource companies; investment of revenues; smoothing of expenditure; the efficiency of public spending; private sector development; the role of companies; and the role of the international community. The second edition (2014) emphasises that the government should ensure that resource extraction yields the greatest possible benefit for citizens, taking into account the interests of future generations, and that the management of resource revenues should be transparent, accountable, and subject to public oversight.

Fiscal Federalism and Derivation in Nigeria

In federal states, the distribution of resource revenues between the central government and sub-national units is a central question of constitutional law and politics, and in Nigeria it has been among the most contentious issues since independence. Under section 44(3) of the **Constitution of the Federal Republic of Nigeria 1999**, the entire property in and control of all minerals, mineral oils, and natural gas is vested in the Government of the Federation. Revenues from petroleum accrue principally to the **Federation Account**, established by section 162(1), into which all revenues collected by the Government of the Federation are paid, subject to specified exceptions.

The Federation Account is shared among the federal, state, and local governments according to a **revenue allocation formula** prescribed by the National Assembly on the advice of the **Revenue Mobilisation Allocation and Fiscal Commission** (RMAFC). The vertical allocation formula in operation since the early 2000s allocates approximately 52.68 per cent to the federal government, 26.72 per cent to the states, and 20.60 per cent to local governments, with horizontal allocation among states and local governments based on factors including equality, population, landmass and terrain, social development, and internally generated revenue. The RMAFC has proposed revisions to the formula, which have been the subject of continuing debate.

Section 162(2) provides the **derivation principle**: the President shall table before the National Assembly proposals for revenue allocation, and in determining the formula, the National Assembly shall take into account the allocation principles especially those of population, equality of states, internal revenue generation, landmass, terrain, and population density, provided that the principle

of derivation shall be constantly reflected in any approved formula as being **not less than thirteen per cent** of the revenue accruing to the Federation Account directly from any natural resources. The thirteen per cent derivation is paid to the oil-producing states. The history of the derivation principle reflects shifting political balances: under the regional constitutions of the 1950s and 1960s, derivation was fifty per cent or more, when the dominant export revenues came from agricultural produce of the regions; it was reduced progressively under military rule as oil became the dominant source of revenue, reaching as low as 1.5 per cent in the 1980s, before the 1999 Constitution restored a minimum of thirteen per cent.

The application of the derivation principle to offshore resources was addressed by the Supreme Court in ***Attorney-General of the Federation v Attorney-General of Abia State and 35 Others (No 2)*** (2002). The Court held that, for the purposes of derivation, the seaward boundary of a littoral state is the low-water mark of the land surface of the state or the seaward limit of inland waters within the state, so that revenues from offshore resources beyond that boundary do not accrue to the littoral states and are not subject to derivation. Following political agitation by the littoral states, the National Assembly enacted the **Allocation of Revenue (Abolition of Dichotomy in the Application of the Principle of Derivation) Act 2004**, which provides that the two-hundred-metre water depth isobath contiguous to a state is deemed to be part of that state for the purposes of computing derivation revenue, extending derivation to shallow offshore production but not to deep offshore production.

Oil-producing communities and states in the Niger Delta have long argued that the derivation principle does not adequately compensate them for the environmental and social costs of oil production or reflect their claims to the resources of their lands, giving rise to demands for **"resource control"** — the right of states and communities to own or control the resources in their territories, with payment of taxes or royalties to the federal government. The resource control agitation, advanced by governors of oil-producing states, ethnic nationality movements, and militant groups, has been a recurring theme of Nigerian politics and was debated in successive constitutional conferences, including the National Conference of 2014, without resolution.

Special Institutions for the Niger Delta

In addition to derivation, the federal government has established special institutions to address the development of the Niger Delta. The **Niger Delta Development Commission** (NDDC), established by the Niger Delta Development Commission (Establishment, etc) Act 2000, is funded by contributions from the federal government, oil and gas companies (originally three per cent of their total annual budgets), and a share of the ecological fund, and is charged with formulating policies and implementing projects for the sustainable development of the

region. The NDDC has been criticised for corruption, poor project execution, and political interference, documented in forensic audits. The **Ministry of Niger Delta Affairs**, established in 2008, and the **Presidential Amnesty Programme**, established in 2009 to disarm, demobilise, and reintegrate militants, are further elements of the institutional response. The **Petroleum Industry Act 2021** added **host communities development trusts**, funded by three per cent of the actual annual operating expenditure of upstream operators, to provide direct benefits to host communities, examined in Chapter 24. These overlapping mechanisms raise questions about coordination, accountability, and the direct benefit to citizens.

Comparative Models of Revenue Sharing

Other resource-rich federations and decentralised states have adopted different models. In **Canada**, provinces own natural resources within their territories under section 109 of the Constitution Act 1867, as confirmed by the Natural Resources Transfer Agreements of 1930 and section 92A, added in 1982, so that resource royalties accrue to the provinces, with equalisation payments from the federal government to less wealthy provinces. In the **United States**, resources on private land belong to the landowner, and royalties from federal lands are shared with states. In **Indonesia**, decentralisation laws allocate shares of resource revenues to producing regions, with special autonomy arrangements for Aceh and Papua. In **Iraq**, the Constitution of 2005 provides that oil and gas are owned by all the people of Iraq in all the regions and governorates, but the division of powers between the federal government and the Kurdistan Regional Government over oil has been a source of persistent dispute. The comparative experience suggests that the design of revenue-sharing arrangements must balance national cohesion, regional equity, compensation for the costs borne by producing areas, and incentives for good governance.

Revenue Management Laws and Sovereign Wealth Funds

Many resource-rich states have adopted legal frameworks to manage resource revenues, including fiscal rules, stabilisation funds, savings funds, and transparency and oversight mechanisms.

Norway's **Government Pension Fund Global**, which originated in the Petroleum Fund established in 1990, receives the state's net cash flow from petroleum activities and invests it abroad in equities, fixed income, real estate, and renewable energy infrastructure. Its value exceeds the equivalent of USD 1.8 trillion, making it the world's largest sovereign wealth fund. A **fiscal rule** limits the structural non-oil deficit financed from the fund to the expected real return, currently estimated at three per cent of the fund's value, so that petroleum wealth is converted into financial wealth that benefits future generations. The fund's ethical guidelines, administered by a Council on Ethics, provide for the exclusion

of companies involved in specified activities, including thermal coal mining and power production above thresholds, and of companies whose conduct involves an unacceptable risk of serious human rights violations or severe environmental damage.

Ghana's **Petroleum Revenue Management Act 2011** is widely regarded as a model in Africa. It requires all petroleum revenues to be paid into a Petroleum Holding Fund at the Bank of Ghana and allocates them according to a formula among the Ghana National Petroleum Corporation, the **Annual Budget Funding Amount** for public spending in priority areas, the **Ghana Stabilisation Fund**, to cushion the budget against revenue shortfalls, and the **Ghana Heritage Fund**, an endowment for future generations. It establishes the **Public Interest and Accountability Committee** (PIAC), a body representing civil society and professional groups, to monitor compliance and report publicly. Implementation has faced challenges, including the capping of the Stabilisation Fund and the use of revenues for debt service, but the Act's transparency and oversight provisions have been praised.

Nigeria's experience has been more troubled. The **Excess Crude Account**, established administratively in 2004 to save oil revenues above a benchmark price, lacked a clear legal basis and was depleted through withdrawals agreed by the federal and state governments, particularly during the oil price collapse of 2014–2016. The **Nigeria Sovereign Investment Authority (Establishment, etc) Act 2011** established the **Nigeria Sovereign Investment Authority** (NSIA) to manage three funds: the **Stabilisation Fund**, the **Future Generations Fund**, and the **Nigeria Infrastructure Fund**. The NSIA has developed a reputation for professional management and transparency, but its capitalisation has been modest relative to the scale of Nigeria's oil revenues, reflecting the political difficulty of saving revenues in a federation with pressing current needs and constitutional obligations to share revenues among all tiers of government. The **Fiscal Responsibility Act 2007** introduced fiscal rules, including an oil price-based fiscal rule, though compliance has been inconsistent.

Other models include **Botswana**'s Pula Fund, built from diamond revenues, and its Sustainable Budget Index; **Chile**'s Economic and Social Stabilisation Fund, based on copper revenues and a structural balance rule; **Timor-Leste**'s Petroleum Fund, which finances most of the state budget; and the **Alaska Permanent Fund**, which pays an annual dividend to every resident of the state — a model that has inspired proposals for direct distribution of resource revenues to citizens ("oil-to-cash") as a means of strengthening accountability and reducing poverty.

The **Chad–Cameroon pipeline** project provides a cautionary tale. The World Bank's support for the project in 2000 was conditioned on Chad's

adoption of a petroleum revenue management law (Law No. 001 of 1999), which required most direct revenues to be allocated to priority sectors and a Future Generations Fund and established an oversight committee. In 2006, Chad amended the law to increase discretionary spending and abolish the Future Generations Fund, and in 2008, after Chad failed to comply with its obligations, the World Bank withdrew from the project. The experience demonstrated the limits of externally imposed revenue management frameworks where domestic political commitment is lacking.

23.6 Intergenerational Equity

The Principle

Intergenerational equity holds that each generation holds the earth and its resources in trust for future generations and has obligations to pass on a planet and resource base that is in no worse condition than it received. The principle was given its most influential legal articulation by Edith Brown Weiss in *In Fairness to Future Generations* (1989), which proposed three principles: **conservation of options**, requiring each generation to conserve the diversity of the natural and cultural resource base so as not to restrict the options available to future generations; **conservation of quality**, requiring each generation to maintain the quality of the planet so that it is passed on in no worse condition; and **conservation of access**, requiring each generation to provide its members with equitable rights of access to the legacy of past generations and to conserve this access for future generations.

The principle is reflected in the Brundtland Commission's definition of **sustainable development** as development that meets the needs of the present without compromising the ability of future generations to meet their own needs (1987); in Principle 3 of the Rio Declaration (1992), which provides that the right to development must be fulfilled so as to equitably meet developmental and environmental needs of present and future generations; in Article 3(1) of the UNFCCC, which requires parties to protect the climate system for the benefit of present and future generations; and in the preamble to the Paris Agreement. Its legal status remains debated: it is widely invoked as a guiding principle, but its content and justiciability are uncertain, since future generations cannot be represented in the conventional sense, and their interests must be articulated by present persons and institutions.

Judicial Recognition

Courts have increasingly invoked intergenerational equity, particularly in environmental and climate litigation. In ***Minors Oposa v Secretary of the Department of Environment and Natural Resources*** (1993), the Supreme Court of the Philippines recognised the standing of children to sue on their own behalf

and on behalf of generations yet unborn to challenge timber licences, on the basis of the constitutional right to a balanced and healthful ecology and the concept of intergenerational responsibility. In ***Neubauer v Germany*** (2021), examined in Chapter 18, the Federal Constitutional Court held that Germany's Climate Protection Act violated fundamental rights by failing to provide sufficient specifications for emission reductions after 2030, thereby unconstitutionally shifting the burden of reductions to the future — an "intertemporal guarantee of freedom". In ***Future Generations v Ministry of the Environment*** (2018), the Supreme Court of Colombia recognised the Colombian Amazon as a subject of rights and ordered the government to adopt an intergenerational pact to reduce deforestation. The **advisory opinion of the International Court of Justice** of 23 July 2025, examined in Chapter 18, referred to intergenerational equity as a consideration relevant to the interpretation of states' climate obligations.

Institutions for Future Generations

Several states have established institutions to represent the interests of future generations. **Wales**'s **Well-being of Future Generations (Wales) Act 2015** requires public bodies to carry out sustainable development, defined by reference to seven well-being goals, and establishes a **Future Generations Commissioner** to promote the sustainable development principle and to review the work of public bodies. **Hungary** established a Parliamentary Commissioner for Future Generations in 2008, later incorporated as a deputy of the ombudsman. In 2023, a group of experts adopted the **Maastricht Principles on the Human Rights of Future Generations**, which articulate the obligations of states to respect, protect, and fulfil the human rights of future generations. At the international level, the **Pact for the Future**, adopted by the UN General Assembly in September 2024, included a **Declaration on Future Generations**, committing states to consider the interests of future generations in decision-making.

Application to Non-Renewable Resources

The extraction of non-renewable resources raises distinctive questions of intergenerational equity, since resources consumed by one generation are unavailable to the next. Economists have proposed the **Hartwick rule**, under which intergenerational equity in consumption can be maintained if the rents from the extraction of non-renewable resources are invested in reproducible capital — physical, human, or financial — so that the total stock of capital is maintained. Sovereign wealth funds, such as Norway's fund and Ghana's Heritage Fund, examined in section 23.5, are institutional expressions of this approach, converting depleting natural assets into financial assets that benefit future generations. Investment in education, health, and infrastructure may serve the same purpose.

The energy transition adds a further dimension. Intergenerational equity requires not only that the wealth from resources be shared with future generations, but also that the present generation not impose on future generations the catastrophic costs of climate change. This implies that a substantial proportion of known fossil fuel reserves should remain unextracted — studies have estimated that large shares of oil, gas, and coal reserves would need to remain in the ground to meet the Paris temperature goals — and raises the prospect of **stranded assets** for producers. For resource-rich developing countries, this creates a dilemma: the window for monetising hydrocarbon resources may be closing, but their development may lock in emissions and expose future generations to the financial risks of stranded assets and the physical risks of climate change. In August 2023, voters in **Ecuador** approved, by around fifty-nine per cent, a referendum requiring the cessation of oil extraction in Block 43 (ITT) within the Yasuní National Park, an area of exceptional biodiversity and home to indigenous peoples living in voluntary isolation — a notable example of a democratic decision to leave proven reserves in the ground, although its implementation has been delayed. Proposals for a **Fossil Fuel Non-Proliferation Treaty**, endorsed by a number of small island and other states, and the **Beyond Oil and Gas Alliance**, launched by Costa Rica and Denmark in 2021, seek to manage the phase-down of fossil fuel production in an equitable manner, including through support for developing producer states.

Intergenerational equity thus points in two directions for resource-rich developing countries such as Nigeria: towards the prudent management and saving of resource revenues for future generations, and towards a transition away from dependence on fossil fuel revenues in a manner that protects future generations from climate harm and fiscal shocks. Reconciling these demands, while meeting the urgent needs of the present generation — including the millions without access to electricity — is among the most difficult challenges of energy justice.

23.7 Sustainable Development Goals and Energy Policy

From the MDGs to the SDGs

The **Millennium Development Goals** (2000–2015) did not include a goal on energy, an omission widely regarded as a significant gap given the role of energy in achieving the goals on poverty, health, education, and gender equality. In 2011, the UN Secretary-General launched the **Sustainable Energy for All** (SEforALL) initiative, with three objectives for 2030: ensuring universal access to modern energy services, doubling the global rate of improvement in energy efficiency, and doubling the share of renewable energy in the global energy mix. The UN General Assembly declared 2012 the International Year of Sustainable Energy for All and 2014–2024 the UN Decade of Sustainable Energy for All. These efforts laid the

foundation for the inclusion of energy in the **2030 Agenda for Sustainable Development**, adopted by the General Assembly in September 2015 (Resolution 70/1), comprising seventeen **Sustainable Development Goals** (SDGs) and 169 targets.

SDG 7: Affordable and Clean Energy

SDG 7 aims to "ensure access to affordable, reliable, sustainable and modern energy for all". Its targets are:

- **Target 7.1**: by 2030, ensure universal access to affordable, reliable, and modern energy services;
- **Target 7.2**: by 2030, increase substantially the share of renewable energy in the global energy mix;
- **Target 7.3**: by 2030, double the global rate of improvement in energy efficiency;
- **Target 7.a**: by 2030, enhance international cooperation to facilitate access to clean energy research and technology, and promote investment in energy infrastructure and clean energy technology; and
- **Target 7.b**: by 2030, expand infrastructure and upgrade technology for supplying modern and sustainable energy services for all in developing countries, in particular least developed countries, small island developing states, and landlocked developing countries.

Progress is monitored through indicators, including the proportion of the population with access to electricity (7.1.1), the proportion of the population with primary reliance on clean fuels and technology (7.1.2), the renewable energy share in total final energy consumption (7.2.1), energy intensity (7.3.1), international financial flows to developing countries in support of clean energy (7.a.1), and installed renewable energy-generating capacity in developing countries per capita (7.b.1). The annual *Tracking SDG 7* reports have consistently found that the world is **off track** to achieve SDG 7 by 2030: on current trends, hundreds of millions of people, mostly in sub-Saharan Africa, will still lack electricity in 2030, and around 1.8 billion will lack clean cooking; renewable energy deployment, while growing rapidly in electricity, is slower in heating and transport; and international public financial flows to developing countries for clean energy have declined from their peak in 2016 and are highly concentrated in a few countries.

The SDGs as an Integrated Framework

The SDGs are designed to be "integrated and indivisible", and SDG 7 is closely linked to other goals. Energy access contributes to **ending poverty** (SDG 1), **food security** (SDG 2, through irrigation, processing, and cold storage), **health** (SDG 3, through clean cooking and electrified health facilities), **education** (SDG 4, through lighting and digital learning), **gender equality** (SDG 5), **clean water**

and sanitation (SDG 6, through pumping and treatment), **decent work and economic growth** (SDG 8), **industry, innovation, and infrastructure** (SDG 9), **reduced inequalities** (SDG 10), **sustainable cities** (SDG 11), and **climate action** (SDG 13). The relationships are not always synergistic: the expansion of energy access through fossil fuels may conflict with climate action; large hydropower may conflict with the protection of ecosystems and the rights of displaced people (SDGs 6, 15, and 16); and the extraction of transition minerals may conflict with goals on health, water, and peace. The concept of **policy coherence for sustainable development** (SDG target 17.14) requires governments to identify and manage such trade-offs.

The Legal Status of the SDGs

The SDGs are not legally binding. They are political commitments adopted by a General Assembly resolution, to be implemented through national policies and voluntary reporting — through **Voluntary National Reviews** presented at the UN **High-Level Political Forum on Sustainable Development**. Nigeria has presented Voluntary National Reviews, including in 2017 and 2020, which addressed energy access among national priorities. Nonetheless, the SDGs have legal significance in several respects: they inform the interpretation of legal obligations, including human rights obligations, whose realisation the SDGs are expressly intended to advance; they have been incorporated into national development plans, legislation, and regulatory mandates; they are used as benchmarks by international financial institutions, development partners, and investors, including through sustainable finance taxonomies and SDG-linked bonds; and they influence the conduct of companies through sustainability reporting and voluntary commitments. Courts have occasionally referred to the SDGs in interpreting constitutional and statutory provisions on development and the environment.

International Commitments after the SDGs

Subsequent international commitments have built on SDG 7. At the **UN High-Level Dialogue on Energy** in September 2021, the first summit-level meeting on energy under the UN General Assembly in four decades, governments and other stakeholders announced **Energy Compacts** — voluntary commitments to accelerate progress on SDG 7. At **COP28** in Dubai in 2023, parties to the Paris Agreement, in the outcome of the first global stocktake, called on parties to contribute to global efforts including **tripling renewable energy capacity globally and doubling the global average annual rate of energy efficiency improvements by 2030**, and **transitioning away from fossil fuels in energy systems, in a just, orderly and equitable manner**, examined in Chapter 18. The **African Union's Agenda 2063** identifies energy as a priority for the continent's development, and the **Africa Renewable Energy Initiative**, launched at COP21, aimed

to install substantial new renewable capacity in Africa. The **Nairobi Declaration** adopted at the Africa Climate Summit in 2023 called for increasing Africa's renewable generation capacity from 56 GW in 2022 to at least 300 GW by 2030.

SDG 7 and Nigerian Energy Policy

Nigeria's energy policies are framed by reference to SDG 7. The **Energy Transition Plan** (2021) aims to achieve universal access to modern energy by 2030 and net zero emissions in the energy sector by 2060, with natural gas playing a transitional role, including in clean cooking through LPG. The **National Renewable Energy and Energy Efficiency Policy** (2015) set targets for renewable energy and efficiency. The Electricity Act 2023 requires the federal government to adopt a national electricity policy addressing, among other matters, renewable energy and rural electrification, and contains provisions on renewable energy, energy efficiency, and the obligations of licensees in respect of renewable energy. The **Mission 300 national energy compact** presented by Nigeria in 2025 sets out commitments to expand access, including through distributed renewable energy, and to improve the financial viability of the electricity sector. The gap between these commitments and outcomes — in a country where access has grown slowly and the grid remains unreliable — underscores the importance of implementation, institutional capacity, financing, and governance, examined in Chapter 24.

23.8 Designing Equitable Energy Regulatory Systems

Principles of Equitable Regulatory Design

How can the principles of energy justice be translated into the design of energy regulatory systems? Drawing on the preceding analysis, several principles may be identified:

1. **Universal service**: regulatory frameworks should aim to ensure that all people have access to affordable, reliable, and modern energy services, through obligations on licensees, dedicated funds, and enabling frameworks for decentralised solutions.
2. **Affordability with financial sustainability**: tariffs should reflect the efficient costs of supply, to ensure the financial viability of the sector and attract investment, while protecting poor and vulnerable consumers through targeted measures.
3. **Fair allocation of costs and benefits**: the costs of policies, including the energy transition, should be allocated in ways that do not unduly burden the poor, and the benefits should be shared broadly, including with host communities.
4. **Meaningful participation**: regulatory processes should be transparent and open to the participation of consumers, communities, and civil society, with access to information and remedies.

5. **Recognition of vulnerability and diversity**: regulation should identify and protect vulnerable groups and take account of gender and other dimensions of disadvantage.
6. **Accountability**: regulators, utilities, and governments should be accountable for their performance, through independent regulation, performance standards, reporting, and judicial review.

Tariff Design

Tariff design is perhaps the most consequential regulatory instrument for energy justice. Regulators must balance the objective of **cost-reflectivity** — tariffs that recover the efficient costs of generation, transmission, and distribution, including a reasonable return on investment — with the objective of **affordability**. Where tariffs are set below cost for political reasons, utilities accumulate losses, investment stagnates, supply deteriorates, and the burden falls on the state budget or, ultimately, on consumers through unreliable supply and costly self-generation — an outcome that is often more regressive than cost-reflective tariffs with targeted support.

Instruments for protecting the poor within cost-reflective tariff structures include **lifeline tariffs**, under which a minimum block of consumption is provided at a low price, with higher prices for consumption above that block (**increasing block tariffs**); **social tariffs** for eligible low-income households, identified through social registries; **cross-subsidies** from commercial and industrial consumers or high-consumption households to low-consumption households; **connection subsidies** to reduce the upfront cost of connection, which is often a greater barrier to access than the tariff itself; and **direct cash transfers** to compensate households for tariff increases. Each instrument has advantages and disadvantages: lifeline tariffs may be poorly targeted where poor households share connections, and cross-subsidies may burden industries and encourage them to leave the grid. Experience suggests that the design of support should be based on data about the consumption patterns and characteristics of poor households.

In **Nigeria**, NERC sets tariffs through a **Multi-Year Tariff Order** (MYTO) methodology. Since 2020, Nigeria has adopted a **service-based tariff** system, under which customers are classified into bands (A to E) according to the average hours of supply they receive, with higher tariffs for customers receiving more hours of supply. In April 2024, NERC approved a substantial increase in the tariff for **Band A** customers, receiving at least twenty hours of supply per day, removing the subsidy for this band, while maintaining subsidised tariffs for other bands; the tariff was subsequently adjusted. The approach seeks to align tariffs with service quality and to target subsidies at customers receiving poorer service, but it has raised concerns about the accuracy of band classifications, the enforcement of supply commitments, and the burden on small businesses. Metering remains a

significant challenge: a large proportion of customers are unmetered and subject to **estimated billing**, which has been a major source of consumer grievance and which NERC has sought to address through capped estimated billing and metering programmes, including the Meter Asset Provider regulations and the National Mass Metering Programme.

Universal Service Obligations and Rural Electrification

Regulatory frameworks may impose **universal service obligations** on distribution licensees, requiring them to connect customers within their franchise areas on request, subject to conditions, and to meet quality-of-service standards. Where grid extension is uneconomic, frameworks should enable decentralised solutions through mini-grid and off-grid regulation, as examined in section 23.3, and provide public funding through rural electrification funds, results-based financing, and concessional finance. The integration of grid and off-grid planning, through least-cost electrification plans, helps to avoid conflicts between grid extension and mini-grid investments. Under Nigeria's Electricity Act 2023, the Rural Electrification Agency, the Rural Electrification Fund, and the power of states to regulate electricity within their territories create a multilevel framework for universal access, whose effectiveness will depend on coordination among federal and state institutions.

Community Energy and Benefit-Sharing

Community energy — renewable energy projects owned or controlled by local communities, cooperatives, or municipalities — offers a means of sharing the benefits of the energy transition more widely and increasing public acceptance. **Germany**'s energy transition was driven in its early stages by citizens' energy cooperatives and local ownership, supported by feed-in tariffs. In **Denmark**, legislation required that a share of new wind projects be offered to local residents. The European Union's revised **Renewable Energy Directive** (Article 22) and the **Electricity Market Directive** (Directive (EU) 2019/944, Article 16) require member states to provide enabling frameworks for **renewable energy communities** and **citizen energy communities**, respectively, including rights to produce, consume, store, and sell renewable energy and to share it within the community. **Scotland** has adopted good practice principles under which developers of onshore wind projects are expected to provide community benefits of £5,000 per megawatt per year, together with opportunities for shared ownership.

In developing countries, benefit-sharing mechanisms for energy projects include community development agreements, revenue-sharing with local governments and communities, community equity in projects, local employment and procurement requirements, and community funds. South Africa's **Renewable Energy Independent Power Producer Procurement Programme**

(REIPPPP), examined in Chapter 10, required bidders to commit to economic development obligations, including local community ownership of at least a minimum percentage of the project company and spending on socio-economic and enterprise development in communities within a specified radius of the project. Such mechanisms can contribute to distributional justice, but their effectiveness depends on governance, transparency, and the capacity of community structures to manage benefits, as the experience of community trusts in the REIPPPP and of Nigeria's host communities development trusts, examined in Chapter 24, illustrates.

Just Transition Mechanisms

The **just transition** — ensuring that the transition to a low-carbon economy is fair to workers, communities, and countries dependent on fossil fuels — requires legal and policy mechanisms to manage its distributional consequences, as examined in Chapter 18. These include social dialogue with workers and communities; retraining and redeployment programmes; social protection; economic diversification in affected regions; the remediation and repurposing of fossil fuel sites; and the allocation of carbon pricing revenues to support affected households and regions. The **Just Energy Transition Partnerships** (JETPs) — financing arrangements between groups of developed countries and multilateral institutions and emerging economies, beginning with **South Africa** in 2021 (an initial USD 8.5 billion), followed by **Indonesia** (USD 20 billion) and **Viet Nam** (USD 15.5 billion) in 2022 and **Senegal** (EUR 2.5 billion) in 2023 — illustrate an international dimension of just transition, though questions have been raised about the balance between grants and loans, the participation of affected communities, and the withdrawal of the United States from these partnerships in 2025. South Africa's **Presidential Climate Commission** adopted a **Just Transition Framework** in 2022, setting out principles of distributive, restorative, and procedural justice to guide the transition — a notable example of the incorporation of energy justice concepts into national policy.

The **revenue recycling** of carbon pricing illustrates how regulatory design can address distributional concerns. Canada's federal carbon pricing system returned most of the proceeds of its fuel charge directly to households through quarterly rebates, so that most lower-income households received more in rebates than they paid in increased costs; nonetheless, the consumer fuel charge became politically contentious and was set to zero in April 2025. The European Union's **Social Climate Fund** is designed to address the distributional effects of its second emissions trading system for buildings and road transport. These examples illustrate that the political sustainability of climate policies depends on their perceived fairness.

Regulatory Institutions and Accountability

Finally, the achievement of energy justice depends on the quality of regulatory institutions. **Independent regulators**, insulated from short-term political pressures and subject to transparent procedures, can set cost-reflective tariffs while protecting consumers, enforce quality-of-service standards, and resolve disputes, but they require adequate resources, technical capacity, and legitimacy. Regulatory decisions should be subject to **accountability** through legislative oversight, judicial review, and public reporting. Consumer advocacy bodies, such as the United Kingdom's Citizens Advice, which has statutory functions as the energy consumer advocate, can strengthen the representation of consumers, especially vulnerable consumers, in regulatory processes. In Nigeria, the **Electricity Act 2023** reinforces NERC's consumer protection mandate and requires state electricity regulators to observe similar principles, while NERC's Forum Offices and customer service standards provide mechanisms for consumer redress. Strengthening these institutions, and ensuring that energy justice considerations are embedded in their mandates, is among the most important tasks for energy law reform in Nigeria and elsewhere.

Conclusion

Energy justice offers a framework for evaluating whether energy and resource systems distribute benefits and burdens fairly, whether decisions are made through fair processes, whether the needs and identities of all groups are recognised, and whether harms are remedied. Its theoretical foundations in distributive justice, the capabilities approach, the politics of recognition, and environmental and climate justice provide a rich vocabulary for legal analysis and reform. Applied to the realities of the energy sector, it highlights the scandal of energy poverty, which leaves hundreds of millions without electricity and billions without clean cooking; the gendered burdens of energy deprivation; the distribution of resource revenues, which in Nigeria has been shaped by constitutional battles over derivation and resource control; and the obligations owed to future generations, both to share resource wealth and to avert catastrophic climate change.

The Sustainable Development Goals, and SDG 7 in particular, provide a widely accepted policy framework, but progress remains off track, especially in sub-Saharan Africa. Achieving energy justice requires the translation of its principles into the concrete design of legal and regulatory instruments: tariffs that are financially sustainable while protecting the poor; universal service frameworks that enable grid and off-grid solutions; benefit-sharing and community energy mechanisms; just transition policies that protect workers and communities; and regulatory institutions that are independent, competent, and accountable. For Nigeria, where the paradox of energy poverty amid resource abundance is acute,

energy justice is not an abstract ideal but a practical imperative of law and policy, to which Chapter 24 returns in examining the Nigerian energy and resource legal framework in comparative perspective.

::: exercise {title="Practical Exercise 23: An Energy Justice Review of a State Electrification Law (HYPOTHETICAL)"} **This exercise is entirely hypothetical. The state, legislation, institutions, and data described are fictitious and do not refer to any real jurisdiction, statute, or body.**

Following the devolution of electricity regulation to the states of the Federal Republic of Amara, the government of Riverbend State — a state with a population of eight million, of whom forty-five per cent lack access to grid electricity, concentrated in rural and riverine areas — has published a draft **Riverbend State Electricity and Universal Access Bill** for public comment. The Bill's principal features are as follows:

- It establishes the Riverbend Electricity Regulatory Commission, whose members are appointed by the Governor and may be removed "at the Governor's pleasure".
- It requires all tariffs to be fully cost-reflective within twelve months, with no provision for lifeline or social tariffs, but authorises the Governor to direct "such subsidies as may be appropriate from time to time".
- It grants an exclusive twenty-year distribution franchise over the whole state to a single company, which has no obligation to connect customers more than two kilometres from its existing network.
- It permits mini-grids only in areas that the franchisee certifies it does not intend to serve within five years, and provides that when the grid arrives, mini-grid assets transfer to the franchisee without compensation.
- It establishes a Rural Electrification Fund financed by a levy of two per cent on the bills of all customers, to be administered by the Ministry of Energy, without provision for reporting.
- It requires consultation on tariff changes "where the Commission considers it appropriate" and provides no dedicated consumer complaints mechanism.
- It contains no provisions on gender, disability, or vulnerable customers, and no data collection or reporting obligations.

Riverbend State is also an oil-producing state, receiving derivation revenues, and hosts several oil fields whose host communities lack electricity. A coalition of women's cooperatives, mini-grid developers, and community organisations has asked for your assistance.

Tasks.

(a) Using the distributional, procedural, recognition, and restorative tenets of energy justice, prepare a written submission (of no more than 3,000 words)

to the State House of Assembly evaluating the Bill and identifying its principal deficiencies.

(b) Propose specific amendments to the Bill, drafted in legislative language, addressing at least: the independence and accountability of the regulator; tariff design and protection of poor and vulnerable customers; universal service obligations; the framework for mini-grids and off-grid solutions, including the arrival of the grid; the governance and transparency of the Rural Electrification Fund; consumer participation and complaints; gender and social inclusion; and data and reporting.

(c) Prepare a short note on how the state could use a portion of its derivation revenues to accelerate electrification in host communities, consistently with the federal constitutional framework and principles of sound revenue management.

(d) In a group discussion, debate whether the Bill's emphasis on rapid cost-reflectivity is compatible with energy justice, and how the tension between financial sustainability and affordability should be resolved. :::

CHAPTER SUMMARY

- Energy justice, which emerged in the 2010s, applies principles of justice to energy systems, drawing on distributive justice, the capabilities approach, the politics of recognition, and environmental and climate justice; it functions as a conceptual, analytical, and decision-making tool.
- Its core tenets are distributional justice (the fair allocation of benefits and burdens), procedural justice (fair, transparent, and participatory decision-making with access to remedies), and recognition justice (respect for the identities and needs of all groups), supplemented by restorative, intergenerational, and cosmopolitan dimensions.
- Around 666 million people lack electricity and around 2.1 billion lack clean cooking, mostly in sub-Saharan Africa; Nigeria has the world's largest electricity access deficit. Instruments for universal access include rural electrification agencies and funds, mini-grid and off-grid frameworks, clean cooking programmes, and initiatives such as Mission 300; subsidy reform must be accompanied by protection for the poor.
- Energy systems are gendered: women bear disproportionate burdens of energy poverty and are underrepresented in the energy workforce and decision-making; CEDAW, the Maputo Protocol, and ECOWAS instruments on gender in energy provide a legal framework.
- The resource curse highlights the importance of institutions for revenue management; Nigeria's revenue allocation system, including the thirteen per cent derivation principle and the onshore/offshore dichotomy, has been deeply

contested; revenue management laws and sovereign wealth funds, such as those of Norway and Ghana, provide models.

- Intergenerational equity requires conservation of options, quality, and access for future generations, is increasingly recognised by courts, and has implications both for saving resource revenues and for limiting fossil fuel extraction in the energy transition.
- SDG 7 provides a widely accepted but non-binding framework for universal access, renewable energy, and efficiency; progress is off track, and the SDGs are integrated with commitments made at COP28 and in African initiatives.
- Equitable regulatory systems require universal service, tariffs that balance cost-reflectivity with protection of the poor, fair allocation of transition costs, community energy and benefit-sharing, just transition mechanisms, and independent and accountable regulators.

DOCTORAL RESEARCH QUESTIONS

1. Can energy justice serve as a legally operative principle, rather than merely an analytical framework? Evaluate proposals to incorporate energy justice into energy legislation and regulatory mandates, with reference to Nigeria's Electricity Act 2023.
2. Is there, or should there be, a human right to energy? Assess the legal basis for such a right in international and African human rights law and in national constitutional jurisprudence.
3. Evaluate the service-based tariff system adopted in the Nigerian electricity sector from the perspective of distributional and procedural justice.
4. Has the thirteen per cent derivation principle, together with the NDDC and host communities development trusts, delivered distributional justice to oil-producing communities in the Niger Delta? Propose reforms.
5. How should resource-rich developing countries reconcile intergenerational equity in the management of resource wealth with the imperative of limiting fossil fuel extraction to protect future generations from climate change?
6. Assess the effectiveness of regional instruments on gender and energy, such as those of ECOWAS, and identify the legal measures required to implement them in national law.

FURTHER READING

- Benjamin K Sovacool and Michael H Dworkin, *Global Energy Justice: Problems, Principles, and Practices* (Cambridge University Press 2014).
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- Michael L Ross, *The Oil Curse: How Petroleum Wealth Shapes the Development of Nations* (Princeton University Press 2012).
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PART VII

Comparative Law, Geopolitics, and International Dispute Resolution

The comparative study of national energy and resource legal systems, the geopolitical and security context in which energy law operates, and the arbitral and judicial mechanisms through which energy and resource disputes are resolved.

24 CHAPTER

Comparative Energy and Natural Resources Law

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the purposes, methods, and pitfalls of comparative legal research in energy and natural resources law, including the transplantation of legal models.
2. Analyse the constitutional, legislative, institutional, and fiscal framework of the Nigerian petroleum, gas, electricity, and mining sectors, with particular reference to the Petroleum Industry Act 2021 and the Electricity Act 2023.
3. Explain the principal features of United States energy regulation, including the division of federal and state authority and recent changes in administrative law.
4. Evaluate the energy law of the United Kingdom and the European Union, including the liberalisation of energy markets and the legal framework of the energy transition.
5. Compare the models of petroleum governance in major Middle Eastern producing states.
6. Analyse the energy and mineral resource frameworks of China, India, Indonesia, Japan, and Australia.
7. Assess the evolution of resource sovereignty and regulation in Latin America, including cycles of liberalisation and nationalisation.
8. Draw comparative lessons from the experience of African resource-rich states for the design of resource legal frameworks.

Introduction

The preceding chapters have examined international energy and natural resources law and, at many points, compared the national laws through which international norms are implemented and resource activities regulated. This chapter undertakes a more systematic comparison. Comparative law is indispensable to the energy and resource lawyer for several reasons. Most resource activities are governed principally by national law, so that the practitioner advising on a transaction or dispute

must understand the legal framework of the host state. Legal reform in the energy sector has been marked by extensive borrowing of legal models — production sharing contracts from Indonesia, the regulatory model of the independent utility regulator from the United States and the United Kingdom, the sovereign wealth fund model from Norway, the safety case regime from the United Kingdom, and many others — so that understanding the origins and contexts of these models is necessary to evaluate their suitability elsewhere. Comparative analysis reveals the range of choices available to legislators and the consequences of different choices, enabling evidence-based reform. And comparative research is a mainstay of doctoral scholarship in the field, examined in Chapter 28.

This chapter begins with comparative legal methodology (section 24.1). It then examines, in turn, energy and petroleum law in Nigeria (section 24.2), which receives extended treatment as the jurisdiction of primary interest to many readers of this book; United States energy regulation (section 24.3); United Kingdom and European Union energy law (section 24.4); Middle Eastern petroleum governance (section 24.5); Asian energy and mineral resource frameworks (section 24.6); Latin American resource sovereignty and regulation (section 24.7); and comparative lessons from African resource-rich states (section 24.8). The treatment of each jurisdiction is necessarily selective, emphasising features of comparative significance; many aspects have been examined in earlier chapters and are cross-referenced.

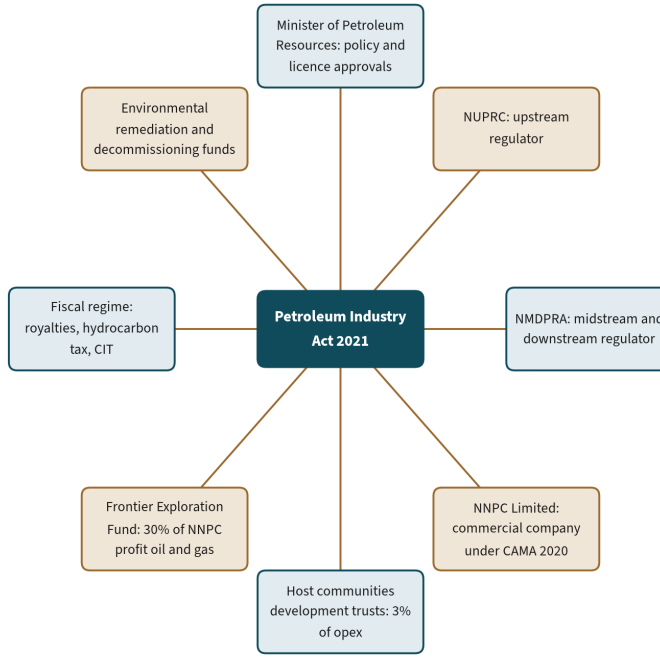


Figure 24.1 The institutional and fiscal architecture of Nigeria's Petroleum Industry Act 2021. *Source:* prepared for this edition.

24.1 Comparative Legal Methodology

Purposes of Comparative Energy Law

Comparative law serves several purposes in the energy and resource field. It is a tool of **knowledge**, deepening understanding of one's own legal system by contrasting it with others. It is a tool of **law reform**, enabling legislators and regulators to learn from the experience of other jurisdictions. It is a tool of **practice**, enabling lawyers to advise on transactions and disputes involving multiple legal systems, to draft contracts that will operate effectively in different legal environments, and to anticipate the approach of foreign courts and regulators. It contributes to **harmonisation** and the development of model laws, standards, and contracts, such as the model production sharing contracts and petroleum laws promoted by international institutions and the model contracts of the Association of International Energy Negotiators examined in Chapter 7. And it contributes to the identification of **general principles of law**, a source of international law under Article 38(1)(c) of the Statute of the International Court of Justice, which arbitral tribunals have drawn upon in petroleum disputes, as examined in Chapters 1 and 15.

Methods

Comparative law scholarship has developed several methodological approaches. The **functional method**, articulated by Konrad Zweigert and Hein Kötz in their *Introduction to Comparative Law*, proceeds from the premise that legal systems face similar social problems and solve them by different means, often with similar results; the comparatist should therefore identify a problem and examine how different systems address it, rather than comparing legal rules or institutions bearing the same name. In energy law, a functional approach might ask how different systems allocate the right to explore for and produce petroleum — through concessions, licences, production sharing contracts, or service contracts, as examined in Chapter 5 — or how they ensure third-party access to networks, rather than comparing institutions in the abstract.

The functional method has been criticised for assuming that problems are universal and for neglecting the cultural, historical, and political contexts that shape law. **Contextual** and **cultural** approaches emphasise that law is embedded in society and that legal rules cannot be understood apart from their institutional, political, economic, and cultural settings. Pierre Legrand has argued, provocatively, that legal transplants are impossible, since the meaning of a rule changes when it is transferred to a different culture. The **law and development** and **institutional economics** traditions examine how legal institutions affect economic outcomes, and the "legal origins" literature associated with Rafael La Porta, Florencio López-de-Silanes, and Andrei Shleifer argued, controversially, that common law systems are associated with stronger investor protection and financial development than civil law systems. In energy and resource law, **socio-legal** approaches examining how laws operate in practice — including the gap between law on the books and law in action — are particularly important, since many resource-rich states have sophisticated laws that are weakly implemented.

Legal Families and Energy Law

Comparative lawyers have traditionally classified legal systems into **legal families** — the common law, the civil law (with Romanistic, Germanic, and Nordic branches), Islamic law, socialist or post-socialist law, and customary law — although the classification is increasingly regarded as simplistic, given the hybrid character of many systems. Energy and resource law is influenced by legal family in several respects: the conception of property and ownership of resources; the role of codes and statutes versus case law; the approach to administrative law and state contracts, including the French doctrine of the *contrat administratif* with its powers of unilateral variation, which influenced petroleum contracts in francophone Africa and the Middle East; and, in Islamic jurisdictions, the influence of Sharia principles, for example on interest and on certain forms of contract. Many African states have **pluralistic** legal systems in which statutory law coexists

with customary law, which governs much land tenure and thereby affects the acquisition of land for resource projects, as examined in Chapter 21. Nigeria's legal system combines English common law, statute, customary law, and, in the northern states, Islamic law.

Ownership Models

A fundamental comparative distinction concerns the **ownership of subsurface resources**. In most of the world, petroleum and minerals are owned by the state, regardless of the ownership of the surface — the **domanial** or **regalian** system, derived from Roman and Spanish colonial traditions and adopted in nearly all states in Africa, Latin America, the Middle East, and Asia, as well as in the United Kingdom for petroleum (under the Petroleum (Production) Act 1934 and now the Petroleum Act 1998) and Norway. In the **accession** system, the owner of the surface owns the minerals beneath it, subject to the possibility of severance; the principal example is the United States, where private landowners own most onshore petroleum on private land, which has contributed to the development of a large independent oil and gas industry and to the "shale revolution". Ownership models have consequences for the distribution of resource revenues, the bargaining relationship between companies and landowners, and the political economy of resource development.

Legal Transplants in Energy Law

The history of energy law is in large part a history of **legal transplants** — the borrowing of rules and institutions from one system by another, a phenomenon analysed by Alan Watson. Notable examples include the production sharing contract, pioneered by Indonesia in the 1960s and adopted across Africa, Asia, and the Middle East; the independent regulatory commission, modelled on United States public utility commissions and United Kingdom regulators such as Ofgem and adopted throughout the world in the wave of electricity reforms promoted by the World Bank in the 1990s; the "standard market design" and wholesale electricity markets; the Norwegian model of NOC governance, examined in Chapter 22; the Ghana model of petroleum revenue management, examined in Chapter 23; and the safety case regime for offshore petroleum. The success of transplants depends on their fit with the institutional, economic, and political context of the receiving state. The **standard model** of electricity reform — unbundling, privatisation, independent regulation, and competitive markets — has had mixed results in developing countries, where weak institutions, low tariffs, and political interference have often frustrated reform, as the experience of the Nigerian electricity privatisation of 2013 illustrates, examined in section 24.2. Comparative energy lawyers therefore emphasise the need to adapt models to local conditions,

to sequence reforms appropriately, and to build the institutions necessary for their implementation.

Pitfalls of Comparison

Comparative energy research faces several pitfalls: the difficulty of obtaining accurate and up-to-date information on foreign law, which changes rapidly in the energy sector; the risk of comparing formal law without understanding its implementation; linguistic barriers and the risk of mistranslation; the tendency to treat Western models as benchmarks against which other systems are measured; and the danger of selecting jurisdictions for comparison without a clear rationale. A sound comparative study defines its research question clearly, justifies its selection of jurisdictions, uses primary sources wherever possible, examines law in context and in practice, and is cautious about drawing prescriptive conclusions. These methodological considerations are developed further in Chapter 28.

24.2 Energy and Petroleum Law in Nigeria

The Constitutional Framework

Nigeria is a federation of thirty-six states and the Federal Capital Territory, governed by the **Constitution of the Federal Republic of Nigeria 1999**, as amended. The Constitution allocates legislative powers between the National Assembly and the State Houses of Assembly through the **Exclusive Legislative List** and the **Concurrent Legislative List** in the Second Schedule. Matters on the Exclusive List — on which only the National Assembly may legislate — include "mines and minerals, including oil fields, oil mining, geological surveys and natural gas" (item 39) and "nuclear energy" (item 41). Section 44(3) vests the entire property in and control of all minerals, mineral oils, and natural gas in, under, or upon any land in Nigeria, its territorial waters, and its EEZ in the Government of the Federation. Electricity was formerly largely a federal matter, subject to limited state powers under the Concurrent List, but the **Constitution (Fifth Alteration) Act** of 2023 amended the Concurrent List to empower states to legislate on electricity generation, transmission, and distribution within the state, a change implemented by the **Electricity Act 2023**, examined below. The distribution of petroleum revenues, governed by section 162, has been examined in Chapter 23.

The Evolution of Nigerian Petroleum Law

Commercial oil production in Nigeria began in 1958 following Shell-BP's discovery at Oloibiri in 1956. The colonial **Mineral Oils Ordinance** of 1914 governed early exploration, and after independence the **Petroleum Act 1969** became the principal statute, vesting petroleum in the state and providing for the grant of oil exploration licences, oil prospecting licences, and oil mining leases by the Minis-

ter. Nigeria joined OPEC in 1971, and the government acquired participating interests in the operations of the international oil companies, rising to sixty per cent by 1979, through **joint ventures** with the companies operating under **joint operating agreements**, examined in Chapter 7. The Nigerian National Petroleum Corporation was established in 1977. From 1993, the government granted deep offshore acreage under **production sharing contracts**, governed by the **Deep Offshore and Inland Basin Production Sharing Contract Act 1999** (amended in 2019). Other features of the legal framework included the Petroleum Profits Tax Act, the Associated Gas Re-Injection Act 1979, which sought to end gas flaring, and, from 2010, the Nigerian Oil and Gas Industry Content Development Act, examined in Chapter 7.

The legal framework was widely criticised: it was fragmented and outdated; it concentrated discretionary power in the Minister; the regulator, the Department of Petroleum Resources, lacked independence; NNPC combined commercial and regulatory roles and was opaque; the fiscal regime was complex and, for deep offshore PSCs, generous to investors; the joint venture funding arrangements, under which NNPC was obliged to fund its share of costs through "cash calls", resulted in chronic underfunding; and the regime failed to address the grievances of host communities. Reform efforts began with the Oil and Gas Sector Reform Implementation Committee in 2000, and a comprehensive **Petroleum Industry Bill** was first introduced in the National Assembly in 2008. After more than a decade of failed attempts, during which the Bill was split into components and a Petroleum Industry Governance Bill was passed in 2018 but not assented to, the **Petroleum Industry Act 2021** (PIA) was signed into law in August 2021.

The Petroleum Industry Act 2021: Structure and Institutions

The PIA is a comprehensive statute of more than three hundred sections, divided into five chapters: **governance and institutions** (Chapter 1), **administration** (Chapter 2), **host communities development** (Chapter 3), **petroleum industry fiscal framework** (Chapter 4), and **miscellaneous provisions** (Chapter 5). Its institutional architecture, illustrated in Figure 24.1, separates policy, regulation, and commercial functions.

- The **Minister of Petroleum Resources** is responsible for the formulation, monitoring, and administration of government policy in the petroleum industry, and retains powers to grant licences and leases on the recommendation of the regulator, and to exercise general supervisory functions. The President has held the petroleum portfolio in several administrations, assisted by a Minister of State.
- The **Nigerian Upstream Petroleum Regulatory Commission** (NUPRC), a body corporate, is responsible for the technical and commer-

cial regulation of upstream petroleum operations, including the conduct of licensing rounds, the recommendation of licences and leases, the determination of royalties, the approval of field development plans, the enforcement of health, safety, and environmental standards, and the management of the National Data Repository.

- The **Nigerian Midstream and Downstream Petroleum Regulatory Authority** (NMDPRA) regulates midstream and downstream operations, including refining, gas processing, transportation and distribution, storage, and the wholesale and retail supply of petroleum products and natural gas, and administers the Midstream and Downstream Gas Infrastructure Fund.
- **NNPC Limited**, the successor to the Corporation, operates as a commercial company under the Companies and Allied Matters Act 2020, as examined in Chapter 22.

The separation of the regulators from the NOC and from each other, and the establishment of their governing boards with specified qualifications, were designed to address the conflicts of interest and lack of independence that characterised the previous regime. The regulators are funded by levies and fees, including a share of royalties and rents for the NUPRC and specified levies for the NMDPRA, rather than by direct budgetary allocations, enhancing their financial autonomy.

Licensing under the PIA

The PIA introduced a new licensing framework for upstream operations. The NUPRC may grant **petroleum exploration licences** (non-exclusive, for data acquisition), **petroleum prospecting licences** (exclusive, for exploration drilling), and **petroleum mining leases** (for development and production), on the recommendation of the Commission and with the approval of the Minister, through a transparent competitive bid process, with exceptions specified in the Act. Petroleum prospecting licences have defined terms that vary with the terrain (onshore, shallow water, deep offshore, and frontier basins), and petroleum mining leases are granted for a term of up to twenty years, renewable. The Act introduced **"drill or drop"** provisions requiring the relinquishment of undeveloped acreage, to address the long-standing problem of acreage held without development, and provides for the conversion of existing oil prospecting licences and oil mining leases into the new licences and leases on a voluntary basis, with incentives for conversion, while existing arrangements continue to be governed by the prior law until conversion or expiry. The NUPRC has conducted licensing rounds under the new framework, including the marginal field bid round concluded in 2021 and the licensing round launched in 2024 and continued in 2025.

The PIA requires the **disclosure** of licences, leases, and contracts, and provides for the publication of information on the beneficial owners of licensees, as noted

in Chapter 22. It also addresses the **divestment** of onshore and shallow water assets by international oil companies, a trend that accelerated in the 2020s as companies such as Shell, ExxonMobil, Eni, Equinor, and TotalEnergies sold onshore and shallow water interests to Nigerian independent companies, requiring the approval of the Minister on the recommendation of the NUPRC. The regulatory review of these divestments — including the assessment of the technical and financial capacity of acquirers and the treatment of decommissioning and environmental liabilities — became a significant test of the PIA's framework, given concerns raised by communities and civil society about the legacy of pollution in the Niger Delta.

The Fiscal Framework of the PIA

Chapter 4 of the PIA established a new fiscal regime for petroleum operations, replacing the Petroleum Profits Tax for converted and new acreage. Its principal elements are:

- **Royalties**, payable at rates that vary according to the terrain and the level of production, with lower rates for deep offshore and frontier acreage and for smaller volumes of production, and price-based royalties at higher oil prices; royalties on gas are generally lower than on oil, to promote gas development.
- **Hydrocarbon tax**, payable on profits from crude oil (but not from associated or non-associated gas) produced from onshore and shallow water acreage, at rates of thirty per cent for petroleum mining leases and fifteen per cent for petroleum prospecting licences, with deep offshore acreage exempt.
- **Companies income tax**, payable at the general rate on the profits of upstream companies.
- **Cost price ratio limits** on the deductibility of costs, and production allowances for new acreage and converted acreage.
- **Rents** on licences and leases.

The PIA also requires the payment of a contribution to the **Frontier Exploration Fund**, equal to thirty per cent of NNPC Limited's share of profit oil and profit gas under production sharing contracts, to finance exploration in frontier basins — a provision criticised by governors of northern and southern states alike for diverting revenues that would otherwise accrue to the Federation Account, and by commentators as a potential source of inefficient spending on exploration in basins of uncertain prospectivity. The fiscal regime was designed to make Nigeria more competitive in attracting investment in the face of declining investment in its petroleum sector, while increasing government take on onshore and shallow water production. The comparative design of petroleum fiscal regimes has been examined in Chapter 16. Subsequently, the federal government introduced fiscal incentives for non-associated gas and deepwater projects by executive orders in

2024 and pursued broader tax reform through legislation enacted in 2025, which consolidated aspects of the tax framework, including for the petroleum industry.

Host Communities Development

Chapter 3 of the PIA established a regime for **host communities development**, intended to address the grievances of communities in oil-producing areas and to secure their cooperation in the protection of petroleum facilities. Every holder of a petroleum prospecting licence or petroleum mining lease (a "settlor") must incorporate a **host communities development trust** for the benefit of the host communities in which its operations take place. The settlor must make an annual contribution to the trust equal to **three per cent of its actual annual operating expenditure** in the preceding financial year in upstream petroleum operations affecting the host communities. The trust's funds are allocated to a capital fund (for capital projects), a reserve fund (for future use when contributions cease), and, where determined, an endowment fund. The trust is administered by a **board of trustees**, a **management committee**, which includes representatives of the host communities, and an **advisory committee**, and must prepare community needs assessments and development plans.

The regime has been both praised and criticised. Supporters regard it as an advance on the previous arrangements, which relied on voluntary memoranda of understanding between companies and communities, and as providing a predictable source of funding for community development. Critics argue that three per cent of operating expenditure is inadequate, compared with the five per cent or more demanded by community representatives during the legislative process; that the trusts are controlled by settlors, who appoint board members; that the definition of host communities may exclude communities affected by pollution but not hosting facilities; and that the provision under which communities forfeit their entitlement to the extent of the cost of repairing damage caused by vandalism, sabotage, or civil unrest imposes collective punishment for acts that communities may not be able to control, as examined in Chapter 21. The implementation of the regime — including the incorporation of trusts, the selection of community representatives, and the execution of projects — has been uneven, and its success will depend on its governance and transparency.

Environmental Provisions

The PIA contains environmental provisions, including requirements for environmental management plans approved by the relevant regulator; the establishment by licensees of an **environmental remediation fund** for the rehabilitation or management of negative environmental impacts; the establishment of **decommissioning and abandonment funds**, to ensure that the costs of decommissioning are provided for; and provisions on the elimination of **gas flaring**, under

which flaring or venting is prohibited except in limited circumstances, such as emergencies or with an exemption, and flare penalties are payable, the proceeds of which are credited to the Midstream and Downstream Gas Infrastructure Fund. Nigeria has been among the world's largest gas flarers, and its gas flaring regime is examined in Chapter 17. The **Nigerian Gas Flare Commercialisation Programme** sought to auction flare sites to third parties for the commercial use of flared gas.

Gas Policy and the Midstream and Downstream Sectors

Nigeria has the largest proven gas reserves in Africa, more than two hundred trillion cubic feet, and the federal government declared 2021–2030 the **"Decade of Gas"**, seeking to use gas as a transition fuel for power generation, industry, and clean cooking. The PIA provides for the regulation of gas transportation through **network codes** and third-party access, the establishment of a **domestic gas delivery obligation** requiring producers to supply a specified volume of gas to the domestic market, and a **domestic base price** for gas supplied to the power sector and strategic industries, with a transition to willing-buyer, willing-seller pricing. Major gas infrastructure projects include the **Ajaokuta–Kaduna–Kano (AKK) pipeline**, intended to transport gas from the south to northern Nigeria, and the expansion of **Nigeria LNG**, whose Train 7 project was under construction. Nigeria also participates in the West African Gas Pipeline and has promoted the proposed Nigeria–Morocco Gas Pipeline, examined in Chapter 20.

In the downstream sector, the PIA provided for the deregulation of the petroleum products market, with prices to be determined by market forces. The most significant reform was the **removal of the petrol subsidy**, announced by the incoming President on 29 May 2023. The subsidy, under which the government had fixed the retail price of petrol below the import parity price and compensated NNPC (and, earlier, marketers) for the difference, had cost trillions of naira annually, constituted a growing share of federal revenue, and was associated with smuggling to neighbouring countries and large-scale fraud. Its removal led to a sharp increase in petrol prices, contributing, together with the unification and devaluation of the exchange rate in June 2023, to high inflation and a cost-of-living crisis. The government introduced palliative measures, including cash transfers to vulnerable households, the provision of compressed natural gas buses and conversion kits under the **Presidential Compressed Natural Gas Initiative**, and wage increases, but the social impacts have been severe, illustrating the energy justice challenges of subsidy reform examined in Chapter 23.

The commissioning of the **Dangote Petroleum Refinery** in the Lekki Free Zone near Lagos, with a nameplate capacity of 650,000 barrels per day — the largest single-train refinery in the world — which began producing petroleum products in 2024, has the potential to transform the downstream sector by reduc-

ing Nigeria's dependence on imported products. Its start-up gave rise to disputes with the NMDPRA, NNPC, and marketers over crude oil supply, product quality, import licences, and pricing, including litigation, illustrating the regulatory challenges of transition from an import-dependent, state-dominated market to a deregulated market with a dominant private refiner. The federal government introduced a "naira-for-crude" arrangement under which crude oil would be sold to domestic refiners in naira. NNPC's own refineries at Port Harcourt, Warri, and Kaduna underwent rehabilitation, with mixed results.

The Electricity Sector

Nigeria's electricity sector was reformed under the **Electric Power Sector Reform Act 2005** (EPSRA), which unbundled the vertically integrated National Electric Power Authority into eighteen successor companies — six generation companies, eleven distribution companies, and the Transmission Company of Nigeria — and established the **Nigerian Electricity Regulatory Commission** (NERC). In November 2013, the generation and distribution companies were privatised, while transmission remained in public ownership. The privatisation has been widely regarded as having failed to deliver its objectives: the sector has suffered from a lack of cost-reflective tariffs, high aggregate technical, commercial, and collection losses, inadequate transmission capacity, gas supply constraints, liquidity shortfalls requiring repeated government interventions, and frequent collapses of the national grid. Installed generation capacity exceeds 13,000 MW, but available capacity has generally been around 4,000 to 5,000 MW for a population of more than 200 million — among the lowest per capita electricity supplies in the world.

The **Electricity Act 2023**, signed in June 2023, repealed EPSRA and established a new legal framework. Its key features include:

- **Decentralisation**: implementing the constitutional amendment of 2023, the Act empowers states to make laws regulating electricity generation, transmission, and distribution within the state, and to establish state electricity regulatory commissions. NERC transfers regulatory oversight of the intrastate electricity market to a state once it has established its own market and regulator. Several states, including Enugu, Ekiti, Ondo, Imo, Oyo, Edo, Kogi, Lagos, and Niger, enacted electricity laws and established regulators, and NERC began transferring regulatory oversight to them from 2024, while the Act continues to govern the national grid and interstate and international electricity.
- **Renewable energy**: the Act requires the promotion of renewable energy and contains provisions on renewable energy obligations, licensing of renew-

able generation, and incentives, and requires the government to adopt an integrated resource plan.

- **Market development**: the Act provides for the evolution of the electricity market, including the establishment of an independent system operator separate from the Transmission Company of Nigeria, and the development of a competitive market.
- **Consumer protection and enforcement**: the Act strengthens consumer protection, criminalises electricity theft and vandalism of electricity infrastructure, and requires licensees to meet performance standards.
- **Rural electrification**: the Act continues the Rural Electrification Agency and the Rural Electrification Fund, examined in Chapter 23.

The Act represents a significant shift towards a multilevel, decentralised electricity governance model, raising questions of coordination between federal and state regulators, the design of state markets, the financial viability of distribution companies operating across state boundaries, and the capacity of state institutions. Tariff reform, including the service-based tariff system and the April 2024 increase in Band A tariffs, has been examined in Chapter 23. Electricity sector law is examined from a comparative perspective in Chapter 9.

The Mining Sector

The solid minerals sector is governed by the **Nigerian Minerals and Mining Act 2007**, which vests minerals in the federal government and establishes the **Mining Cadastre Office**, which administers mineral titles on a first-come, first-served basis — reconnaissance permits, exploration licences, small-scale mining leases, mining leases, quarry leases, and water use permits. The Act requires mining lease holders to conclude community development agreements with host communities, as examined in Chapter 21, and provides for royalties, environmental obligations, and the rehabilitation of mined land. The sector contributes a small share of GDP, but the federal government has sought to develop it as a source of diversification, including through the promotion of **critical minerals** such as lithium, examined in Chapter 8. In 2022, Nigeria restricted the export of unprocessed solid minerals, and the Ministry of Solid Minerals Development has promoted local processing, including through agreements with investors to build lithium processing plants in Kaduna and Nasarawa States. Illegal mining, the involvement of armed groups, and the environmental and health impacts of artisanal mining remain significant challenges.

Climate Law and the Tension with Petroleum Development

Nigeria's **Climate Change Act 2021**, examined in Chapter 18, provides a framework for achieving net zero emissions between 2050 and 2070 and establishes the National Council on Climate Change. The **Energy Transition Plan**

targets net zero in the energy sector by 2060. These commitments coexist with the PIA, whose central objective is to attract investment in petroleum production, including frontier exploration, and with the Decade of Gas. The tension between climate commitments and the development of hydrocarbon resources — a dilemma faced by many producing states in the Global South, examined in Chapters 18 and 23 — is managed in Nigerian policy through the positioning of gas as a transition fuel, the reduction of gas flaring and methane emissions, the use of carbon markets, and the argument that Nigeria, with low per capita emissions and widespread energy poverty, is entitled to develop its resources. Critics argue that investments in new oil and gas production risk creating stranded assets and locking in emissions, and that Nigeria should prioritise renewable energy and economic diversification. The legal and policy framework does not yet fully reconcile these objectives.

Assessment

The Nigerian legal framework for energy and resources has undergone its most significant transformation in decades, with the PIA, the Electricity Act 2023, the Climate Change Act 2021, and the removal of the petrol subsidy. These reforms address long-standing weaknesses: the conflation of policy, regulatory, and commercial functions; the lack of a framework for host community benefits; the centralisation of electricity governance; and the fiscal burden of subsidies. Their success depends, however, on implementation: the independence and capacity of the regulators; the commercial discipline of NNPC Limited; the governance of host community trusts; the security of oil infrastructure against theft and vandalism, which has severely curtailed production; the financial viability of the electricity sector; and the coordination of federal and state institutions. Comparative experience, examined in the remainder of this chapter, suggests that legal reform is a necessary but not sufficient condition for the effective governance of energy and resources.

24.3 United States Energy Regulation

Federalism and the Division of Authority

United States energy law is shaped by the **federal structure** of the Constitution, which divides authority between the federal government and the states. The federal government's authority over energy derives principally from the **Commerce Clause** (Article I, section 8), which empowers Congress to regulate commerce among the states, and from the Property Clause, which grants Congress power over federal lands. States retain authority over matters within their borders, including the siting of power plants and transmission lines, retail electricity sales, the regulation of local distribution utilities, and the regulation of oil and gas production on state and private lands. The boundary between federal and state

jurisdiction has been the subject of extensive litigation, and the resulting division is complex.

Ownership and the Oil and Gas Industry

As noted in section 24.1, the United States follows the **accession** principle: the owner of the surface owns the minerals beneath, unless they have been severed. Most onshore oil and gas production on private land is governed by private **oil and gas leases** between mineral owners and producers, providing for bonuses, royalties, and the obligations of the lessee, and by state law, including the **rule of capture**, under which a landowner owns the oil and gas produced from wells on its land even if drained from beneath neighbouring land, tempered by state **conservation laws** regulating well spacing, preventing waste, and providing for pooling and unitisation. This framework, combined with private ownership, well-developed capital markets, a large independent sector, and technological innovation in horizontal drilling and hydraulic fracturing, enabled the **shale revolution** from the late 2000s, which made the United States the world's largest producer of oil and gas and, from 2023, the largest exporter of LNG. Hydraulic fracturing is regulated principally by states, as the Energy Policy Act of 2005 exempted it from the underground injection control provisions of the Safe Drinking Water Act (except where diesel fuels are used), as noted in Chapter 19.

Oil and gas on **federal lands** and the **outer continental shelf** are owned by the federal government and leased under the **Mineral Leasing Act** of 1920 and the **Outer Continental Shelf Lands Act** of 1953, respectively, administered by the Bureau of Land Management and the Bureau of Ocean Energy Management. Federal leasing policy has varied with administrations: a pause on new leasing was announced in 2021, constrained by litigation and by the Inflation Reduction Act of 2022, which conditioned renewable energy leasing on continued oil and gas lease sales, and in 2025 the federal government adopted an "energy dominance" policy expanding leasing and reducing royalty rates under legislation enacted in July 2025.

The Federal Energy Regulatory Commission

The **Federal Energy Regulatory Commission** (FERC), an independent agency established in 1977 as the successor to the Federal Power Commission, regulates the interstate transmission and wholesale sale of electricity under the **Federal Power Act** of 1935, the interstate transportation and sale for resale of natural gas under the **Natural Gas Act** of 1938, the siting of interstate natural gas pipelines and LNG terminals, the licensing of non-federal hydropower projects, and the interstate transportation of oil by pipeline. Under both Acts, FERC must ensure that rates are **"just and reasonable"** and not unduly discriminatory, a standard rooted in the public utility tradition of cost-of-service regulation.

FERC has been the principal architect of the restructuring of the US gas and electricity industries. In the gas sector, **Order No. 636** (1992) required interstate pipelines to unbundle their transportation services from gas sales, enabling a competitive market in gas supply. In the electricity sector, building on the **Public Utility Regulatory Policies Act** of 1978 (PURPA), which required utilities to purchase power from qualifying cogeneration and small renewable facilities, and the **Energy Policy Act** of 1992, which expanded FERC's authority to order transmission access, **Order No. 888** (1996) required transmission-owning utilities to provide **open access** transmission service on non-discriminatory terms under a pro forma tariff, and **Order No. 2000** (1999) encouraged the formation of **Regional Transmission Organisations** (RTOs). RTOs and Independent System Operators, such as PJM, MISO, CAISO, NYISO, ISO New England, and SPP, operate transmission systems and wholesale markets covering much of the country, while the Southeast and much of the West retain vertically integrated utilities. Texas's grid, operated by **ERCOT**, is largely isolated from interstate commerce and therefore outside FERC's jurisdiction — a factor in the regulatory response to **Winter Storm Uri** in February 2021, in which prolonged outages led to hundreds of deaths and billions of dollars in losses, prompting reforms of Texas law on weatherisation and market design.

More recent FERC orders address the integration of new resources: **Order No. 1000** (2011) on transmission planning and cost allocation; **Order No. 841** (2018) on the participation of electric storage in wholesale markets; **Order No. 2222** (2020) on the participation of aggregations of distributed energy resources; **Order No. 2023** (2023) on the reform of generator interconnection procedures, to address queues of thousands of projects awaiting connection; and **Order No. 1920** (2024) on long-term regional transmission planning. The Supreme Court has addressed the boundary between federal and state jurisdiction in ***FERC v Electric Power Supply Association*** (2016), upholding FERC's authority to regulate demand response in wholesale markets, and ***Hughes v Talen Energy Marketing*** (2016), striking down a Maryland programme that set the rate received by a new generator in the federal wholesale market, while leaving room for state measures not tethered to wholesale market participation.

State Regulation and Policy

States regulate retail electricity rates and the planning of vertically integrated utilities through **public utility commissions**, and many have adopted their own energy and climate policies. **Renewable portfolio standards**, requiring utilities to obtain a specified share of electricity from renewable sources, have been adopted by around thirty states, and many states have adopted targets for 100 per cent clean electricity. **California** has been a leader in climate policy, with a cap-and-trade programme linked to Québec's and, as examined in Chapter 22,

climate disclosure laws. The **Regional Greenhouse Gas Initiative** is a cap-and-trade programme for the power sector among north-eastern states. Around half of the states have restructured their electricity markets to allow retail competition. State and local regulation of siting is a significant constraint on the deployment of renewable energy and transmission.

Environmental Regulation and Administrative Law

Energy projects in the United States are subject to environmental laws including the **Clean Air Act**, the **Clean Water Act**, and the **National Environmental Policy Act** of 1969 (NEPA), which requires federal agencies to prepare environmental impact statements for major federal actions significantly affecting the environment. NEPA reviews have been a source of delay for both fossil fuel and clean energy projects, and Congress amended NEPA in the **Fiscal Responsibility Act** of 2023 to impose page and time limits. In ***Seven County Infrastructure Coalition v Eagle County*** (2025), concerning a railway to transport crude oil from Utah, the Supreme Court held that courts must afford substantial deference to agencies' determinations of the scope of NEPA review, and that agencies need not consider the environmental effects of separate upstream and downstream projects — a decision that significantly narrowed NEPA's reach.

The Supreme Court has reshaped the administrative law governing energy regulation. In ***West Virginia v Environmental Protection Agency*** (2022), the Court applied the **major questions doctrine** to hold that the Clean Air Act did not authorise the EPA to adopt emission caps based on "generation shifting" from coal to gas and renewables, as contemplated by the Clean Power Plan of 2015, stating that agencies must point to clear congressional authorisation for regulations of vast economic and political significance. In ***Loper Bright Enterprises v Raimondo*** (2024), the Court overruled the *Chevron* doctrine, under which courts deferred to reasonable agency interpretations of ambiguous statutes, holding that courts must exercise independent judgment in determining the meaning of statutory provisions. These decisions constrain the ability of agencies such as the EPA and FERC to adopt ambitious climate and energy regulations without express statutory authority, increasing the importance of legislation and the scope for litigation.

Federal Energy Legislation and the Energy Transition

The **Inflation Reduction Act** of 2022 (IRA) was the most significant climate legislation in US history, providing tax credits for clean electricity, clean hydrogen, carbon capture, electric vehicles, and domestic manufacturing of clean energy components, many with domestic content and labour requirements, as examined in Chapters 10, 12, and 13. In July 2025, Congress enacted legislation that substantially curtailed the IRA's tax credits, phasing out credits for wind and solar

projects placed in service after specified dates, ending consumer credits for electric vehicles, and imposing restrictions related to foreign entities of concern, while largely preserving credits for nuclear, geothermal, storage, and carbon capture. The executive branch also withdrew from the Paris Agreement, as examined in Chapter 18, lifted a pause on approvals of LNG exports to non-free trade agreement countries that had been imposed in January 2024, and sought to reverse regulations limiting emissions from power plants and vehicles. The volatility of US federal energy policy across administrations illustrates the importance of legislative entrenchment and the limits of executive action in securing long-term investment.

24.4 United Kingdom and European Union Energy Law

The United Kingdom: From Nationalisation to Liberalisation

The United Kingdom's energy sector was largely nationalised after the Second World War, with state-owned enterprises in coal (the National Coal Board), electricity (the Central Electricity Generating Board and area boards), and gas (British Gas). From the 1980s, the Conservative government pursued **privatisation and liberalisation**, which became a model for reforms worldwide: British Gas was privatised in 1986 under the Gas Act 1986; the electricity industry was restructured and privatised under the **Electricity Act 1989**, which separated generation, transmission, distribution, and supply and established an independent regulator; and British Coal was privatised in 1994. The regulators for gas and electricity were merged in 2000 to form the **Office of Gas and Electricity Markets** (Ofgem), governed by the Gas and Electricity Markets Authority, whose principal objective is to protect the interests of existing and future consumers, including their interests in the reduction of greenhouse gas emissions. Network companies are regulated under the **RIIO** (Revenue = Incentives + Innovation + Outputs) framework of performance-based regulation, a successor to the RPI-X price cap model pioneered in the United Kingdom.

The **Energy Act 2013** introduced **Electricity Market Reform**, including **contracts for difference** (CfDs) for low-carbon generation, under which generators receive the difference between a strike price and a market reference price, and a capacity market, examined in Chapter 10. The CfD model has supported the expansion of offshore wind, making the United Kingdom a world leader in the technology, and has been widely emulated. The **Energy Act 2023** provided for an independent **National Energy System Operator**, established in 2024, licensing of carbon dioxide transport and storage and hydrogen transport, and the regulation of heat networks. The **Great British Energy Act 2025** established a publicly owned company, Great British Energy, to invest in clean energy projects.

UK Petroleum Law

The United Kingdom vests petroleum in the Crown under the **Petroleum Act 1998** and licenses exploration and production on the UK continental shelf through licensing rounds. The **North Sea Transition Authority** (formerly the Oil and Gas Authority), established as an independent regulator under the Energy Act 2016 following the Wood Review of 2014, regulates licensing and the stewardship of resources. The principal objective of **maximising economic recovery** (MER UK), introduced in 2015, was revised in 2021 to require the industry to support the achievement of net zero, including by reducing emissions from production. The fiscal regime comprises ring-fence corporation tax, a supplementary charge, and, from 2022, the **Energy Profits Levy**, a windfall tax introduced in response to high prices following Russia's invasion of Ukraine and extended to 2030. The government elected in 2024 committed not to issue new licences to explore new fields and to end the Energy Profits Levy's investment allowance, while maintaining production from existing fields, and following the Supreme Court's decision in *Finch* (2024), examined in Chapter 17, issued guidance requiring assessment of the scope 3 emissions of new field developments. Decommissioning on the UK continental shelf is governed by the Petroleum Act 1998, OSPAR Decision 98/3, and tax relief arrangements, as examined in Chapters 5 and 17.

The European Union: Legal Basis and Liberalisation

The European Union's energy competence was historically founded on the internal market and environmental provisions of the treaties, together with the Euratom and the (expired) European Coal and Steel Community Treaties. The **Treaty of Lisbon** (2009) introduced an express energy competence in **Article 194 TFEU**, under which EU energy policy aims, in a spirit of solidarity, to ensure the functioning of the energy market, security of supply, energy efficiency and the development of new and renewable forms of energy, and the interconnection of energy networks — without affecting a member state's right to determine the conditions for exploiting its energy resources, its choice between different energy sources, and the general structure of its energy supply. The tension between EU-level integration and national sovereignty over the energy mix is a recurring theme of EU energy law.

The **liberalisation** of EU gas and electricity markets proceeded through successive legislative packages: the First Package (1996 for electricity and 1998 for gas), which introduced gradual market opening; the Second Package (2003), which required legal unbundling of network operators and full market opening by 2007; the **Third Package** (2009), which introduced **ownership unbundling** of transmission system operators (with alternatives of an independent system operator or independent transmission operator), strengthened national regula-

tors, and established the **Agency for the Cooperation of Energy Regulators** (ACER) and the European Networks of Transmission System Operators (ENTSO-E and ENTSOG); and the **Clean Energy for All Europeans Package** (2018–2019), which revised the electricity market design to integrate renewables, empower consumers, and establish citizen energy communities, and adopted the **Governance Regulation** (Regulation (EU) 2018/1999), requiring member states to adopt integrated national energy and climate plans. The Court of Justice has enforced the internal market and competition rules in the energy sector, including in cases on the "third country clause" and Nord Stream 2, examined in Chapter 20, and on the principle of energy solidarity in ***Germany v Poland*** (Case C-848/19 P, 2021), in which the Court held that energy solidarity under Article 194 is a legally binding principle requiring the EU institutions and member states to take into account the interests of other member states in energy security.

The European Green Deal and the Energy Crisis

The **European Green Deal** (2019) and the **European Climate Law** (Regulation (EU) 2021/1119), which made climate neutrality by 2050 and a fifty-five per cent net emissions reduction by 2030 legally binding, led to the **"Fit for 55"** package, which revised the EU Emissions Trading System, established a second emissions trading system for buildings and road transport, adopted the Carbon Border Adjustment Mechanism, raised the binding renewable energy target to at least 42.5 per cent by 2030 under the revised Renewable Energy Directive (RED III), and raised energy efficiency targets, as examined in Chapters 10, 13, and 18. The Russian invasion of Ukraine in 2022 triggered an energy crisis in Europe, leading to **REPowerEU** (2022), which aimed to end dependence on Russian fossil fuels through diversification, efficiency, and accelerated renewables deployment, and a series of emergency regulations on gas storage, demand reduction, joint purchasing of gas, and a market correction mechanism capping gas prices. The **electricity market design reform** of 2024 (Regulation (EU) 2024/1747 and Directive (EU) 2024/1711) responded to the price crisis by requiring that direct price support for new low-carbon generation take the form of **two-sided contracts for difference**, promoting power purchase agreements, and strengthening consumer protection. The **Hydrogen and Decarbonised Gas Market Package** (2024) established a framework for hydrogen networks. In 2025, the EU proposed a target of a ninety per cent reduction in net emissions by 2040, with flexibilities including the limited use of international carbon credits, which was the subject of agreement among member states later that year.

The EU's withdrawal from the **Energy Charter Treaty**, approved in 2024 and effective in 2025, and the modernisation of the Treaty, examined in Chapter

14, illustrate the EU's reassessment of investment protection in the energy sector in light of its climate objectives.

The UK after Brexit

Following its withdrawal from the European Union, the United Kingdom left the EU internal energy market, and electricity and gas trading across interconnectors is governed by the energy provisions of the **EU–UK Trade and Cooperation Agreement** (2020), which provide for cooperation on interconnector trading arrangements, security of supply, and offshore renewable energy in the North Seas. The UK established its own emissions trading scheme in 2021 and agreed with the EU in 2025 to work towards linking the UK and EU emissions trading systems, with a view to mutual exemption from carbon border adjustment mechanisms. The UK's experience after Brexit illustrates the costs of disintegration from a regional energy market and the incentives for continued cooperation, relevant to regional integration efforts in Africa examined in Chapter 9.

24.5 Middle Eastern Petroleum Governance

From Concessions to National Control

The Middle East holds around half of the world's proven oil reserves and a large share of its gas reserves, and its petroleum governance has shaped the international petroleum industry and its law. The early concessions — the D'Arcy concession granted by Persia in 1901, the concessions granted in Iraq to the Turkish (later Iraq) Petroleum Company, the Saudi concession granted to Standard Oil of California in 1933, and the Kuwaiti concession of 1934 — granted foreign companies exclusive rights over vast areas for very long periods in return for modest royalties, as examined in Chapter 1. Mid-century, producing states renegotiated these terms, introducing fifty-fifty profit sharing from 1950, and then, following the creation of **OPEC** in 1960 and the assertion of permanent sovereignty over natural resources, acquired control of their industries through participation agreements and nationalisation in the 1970s. The arbitral awards arising from these processes — *Aramco* (1958), the Libyan nationalisation cases (*BP*, *Texaco*, and *LIAMCO*, 1973–1977), *Aminoil* (1982) — are foundational texts of international petroleum law, examined in Chapters 1 and 15.

Saudi Arabia

Saudi Arabia's petroleum sector is dominated by **Saudi Aramco**, the successor to the Arabian American Oil Company, which the state acquired in stages between 1973 and 1980, and which holds exclusive rights over the Kingdom's hydrocarbon resources under its concession, as restated by royal decree. Saudi Arabia has no comprehensive petroleum statute comparable to those of many other producers; the sector is governed by royal decrees, the Basic Law of Governance

(1992), which declares natural resources to be the property of the state, the policies of the Ministry of Energy, and Aramco's governance documents. Aramco's partial listing on the Saudi stock exchange in December 2019, followed by a secondary offering in 2024, required the adoption of a new concession agreement in 2019 that clarified the fiscal relationship between the company and the state, including a royalty structure, and subjected the company to securities regulation and enhanced disclosure. Saudi Arabia does not generally permit foreign equity participation in upstream oil, although foreign companies participate in gas, downstream, and petrochemical projects through joint ventures. As the leading member of OPEC and of the **OPEC+** grouping with Russia and other producers, established through the **Declaration of Cooperation** of 2016 and the **Charter of Cooperation** of 2019, Saudi Arabia plays a pivotal role in managing global oil supply. Its **Vision 2030** programme seeks to diversify the economy, including through renewable energy, hydrogen, and mining.

Iran

Iran's nationalisation of the Anglo-Iranian Oil Company in 1951, led by Prime Minister Mohammad Mosaddegh, was a landmark in the history of resource sovereignty, and its aftermath, including the *Anglo-Iranian Oil Co* case before the International Court of Justice (1952), in which the Court held that it lacked jurisdiction, and the coup of 1953, shaped attitudes to foreign investment for decades. The Constitution of the Islamic Republic (1979) prohibits the granting of concessions to foreigners, and foreign participation in the upstream sector has taken the form of **buy-back contracts**, a type of service contract under which the contractor finances development and is repaid from production, without acquiring ownership of reserves. In 2016, Iran introduced the **Iranian Petroleum Contract** (IPC), a longer-term contract model with features of production sharing, intended to attract foreign investment after the lifting of nuclear-related sanctions; the reimposition of US sanctions in 2018 and of UN sanctions in 2025, as examined in Chapter 22, largely frustrated this objective.

Iraq

Iraq's petroleum governance illustrates the interaction of constitutional federalism, resource ownership, and contract models in a post-conflict state. The **Constitution of 2005** provides that oil and gas are owned by all the people of Iraq in all the regions and governorates (Article 111), and that the federal government, with the producing governorates and regional governments, shall undertake the management of oil and gas extracted from present fields, with revenues distributed fairly in proportion to the population distribution (Article 112). The ambiguity of these provisions — particularly concerning "present" versus future fields and the respective roles of the federal government and the **Kurdistan**

Regional Government (KRG) — has been the source of persistent dispute. A federal oil and gas law has been under discussion since 2007 without being enacted. The KRG enacted its own **Oil and Gas Law** in 2007 and concluded production sharing contracts with international companies, which the federal government regarded as unlawful. In **February 2022**, the Federal Supreme Court of Iraq held that the KRG's Oil and Gas Law was unconstitutional and that the KRG must hand over oil production to the federal government. In **March 2023**, an ICC arbitral tribunal, in proceedings brought by Iraq against Turkey under the **Iraq–Turkey Pipeline Agreement** of 1973, held that Turkey had breached the agreement by allowing the export of Kurdish oil without the federal government's authorisation, leading to the suspension of exports through the pipeline; exports resumed in September 2025 under an arrangement among the federal government, the KRG, and international producers.

In federal Iraq, the Ministry of Oil awarded **technical service contracts** to international companies in licensing rounds beginning in 2009, under which contractors receive a fixed remuneration fee per barrel of incremental production, rather than a share of production or profits, reflecting political sensitivities about foreign ownership of resources. The model enabled a substantial increase in production but was criticised by companies for its low returns and rigidity, and Iraq has since moved towards profit-sharing terms in certain contracts, including a major agreement with TotalEnergies in 2023 for an integrated gas, solar, and water project in the south.

Qatar and the United Arab Emirates

Qatar has built one of the world's largest LNG industries on the North Field, the world's largest non-associated gas field, which it shares with Iran's South Pars. Its national company, **QatarEnergy** (formerly Qatar Petroleum), partners with international companies in joint ventures for LNG production, under development and production sharing agreements and long-term LNG sales and purchase agreements, as examined in Chapter 6. The **North Field expansion** projects, with partners selected in 2022 and 2023, will substantially increase Qatar's LNG export capacity by the late 2020s. Qatar's model combines state control through QatarEnergy with long-term partnerships with international companies holding minority interests.

The **United Arab Emirates** — particularly Abu Dhabi — has retained a **concession** model, under which the **Abu Dhabi National Oil Company** (ADNOC) grants concessions to joint ventures in which ADNOC holds a majority and international companies hold minority interests, with concessions renewed through competitive processes since 2015. ADNOC has listed several subsidiaries on the Abu Dhabi Securities Exchange and has expanded internationally, including through its investment arm, XRG. The UAE's **Masdar**, a renewable energy

company, has invested in renewable projects worldwide, including in Africa. The UAE hosted COP28 in 2023, as examined in Chapter 18, highlighting the dual role of Gulf states as fossil fuel producers and investors in clean energy.

Comparative Observations

Middle Eastern petroleum governance displays a spectrum of models, from the exclusive control of the NOC in Saudi Arabia, through the service contracts of Iran and federal Iraq and the PSCs of the KRG, to the concession and joint venture models of the UAE and Qatar. Common features include the dominance of NOCs, the importance of OPEC and OPEC+ in market management, the absence or limited role of comprehensive petroleum legislation in some states, reliance on executive and contractual instruments, and, increasingly, diversification strategies in the face of the energy transition. The region's experience demonstrates that effective resource governance does not depend on a single legal model, but on the coherence of the institutional framework, the commercial capacity of the NOC, and the stability of the relationship between the state and its partners.

24.6 Asian Energy and Mineral Resource Frameworks

China

China is the world's largest energy consumer, the largest emitter of greenhouse gases, the largest producer and consumer of coal, the largest importer of oil, and the dominant manufacturer of solar panels, wind turbines, batteries, and electric vehicles, and it controls a large share of the processing of critical minerals. Its energy governance combines state ownership, central planning, and markets. Mineral resources, including oil and gas, are owned by the state under the Constitution and the **Mineral Resources Law** of 1986, which was comprehensively revised in 2024 with effect from July 2025, introducing competitive allocation of mining rights, strengthening the security of supply of strategic minerals, and adding provisions on ecological restoration. The upstream petroleum sector is dominated by three state-owned companies — **China National Petroleum Corporation** (CNPC/PetroChina), **China Petroleum and Chemical Corporation** (Sinopec), and **China National Offshore Oil Corporation** (CNOOC) — which are also major international investors. Policy is set by the **National Development and Reform Commission** and the **National Energy Administration**.

In November 2024, the National People's Congress Standing Committee adopted the **Energy Law of the People's Republic of China**, in force from January 2025, China's first comprehensive framework law for the energy sector, which addresses energy planning, the development of renewable energy and the "new energy system", energy reserves and emergency response, and the obligations of energy enterprises, and confirms the priority of energy security

alongside the low-carbon transition. China's **Renewable Energy Law** of 2005, amended in 2009, introduced feed-in tariffs and mandatory purchase obligations that supported the rapid expansion of wind and solar, followed by a transition to market-based pricing. China launched its **national emissions trading system** for the power sector in 2021, expanding it to steel, cement, and aluminium from 2025, and has pledged to peak emissions before 2030 and achieve carbon neutrality before 2060. Its updated NDC, announced in 2025, committed to reduce net emissions by seven to ten per cent from peak levels by 2035. China's external energy and mineral investments, including through the **Belt and Road Initiative**, and its export controls on critical minerals, examined in Chapters 8 and 22, give its legal and policy choices global significance.

India

India, the world's most populous country and third-largest energy consumer, governs energy through a federal system in which the Union has legislative competence over the regulation and development of oil fields and mineral oil resources (Union List, Entry 53) and of mines and mineral development to the extent declared by Parliament to be in the public interest (Entry 54), while states have competence over the regulation of mines and mineral development subject to Union law (State List, Entry 23), and electricity is on the Concurrent List. In the petroleum sector, the **Hydrocarbon Exploration and Licensing Policy** (HELP) of 2016 introduced a revenue-sharing model, a uniform licence for conventional and unconventional hydrocarbons, and open acreage licensing, replacing the profit-sharing PSCs of the earlier New Exploration Licensing Policy. The Oilfields (Regulation and Development) Amendment Act 2025 sought to modernise the legal framework for petroleum operations, including by introducing a petroleum lease and strengthening investor protections.

In the mining sector, the **Mines and Minerals (Development and Regulation) Act 1957** (MMDR Act) was amended in 2015 to require the allocation of mineral concessions through **auctions**, following the Supreme Court's decision in ***Manohar Lal Sharma v Principal Secretary*** (2014), which held that the allocation of coal blocks by the government without competitive bidding since 1993 was arbitrary and illegal and cancelled the allocations of 214 blocks. The **Coal Mines (Special Provisions) Act 2015** provided for the re-allocation of these blocks by auction. A 2023 amendment of the MMDR Act empowered the Union government to auction concessions for specified **critical minerals**, including lithium, and introduced exploration licences for deep-seated and critical minerals. The **Electricity Act 2003** unbundled state electricity boards and established central and state electricity regulatory commissions, while the **Energy Conservation (Amendment) Act 2022** provided for a domestic carbon credit trading scheme. India's **National Green Hydrogen Mission** (2023) and its

target of 500 GW of non-fossil electricity capacity by 2030 reflect its ambitions in the energy transition, while coal remains dominant in power generation.

Indonesia

Indonesia, the birthplace of the production sharing contract, is a major producer of coal, nickel, and gas. Article 33 of its **1945 Constitution** provides that the land, waters, and natural resources within them are controlled by the state and shall be used for the greatest prosperity of the people — a provision that the Constitutional Court has applied to strike down legislation, including, in 2012, the provisions of the Oil and Gas Law of 2001 establishing the upstream regulator BPMIGAS, on the ground that the arrangement undermined the state's control over oil and gas resources; BPMIGAS was replaced by SKK Migas. In the mining sector, the **Mining Law** of 2009 (Law No. 4/2009), amended in 2020 (Law No. 3/2020), replaced the contract of work system with licences and required the domestic processing of minerals. Indonesia's ban on the export of **nickel ore** from January 2020, designed to promote domestic smelting and the development of a battery industry, attracted massive investment, principally from Chinese companies, and made Indonesia the world's leading producer of nickel. The European Union challenged the ban at the WTO, and in 2022 a panel found it inconsistent with Article XI of the GATT and not justified under the exceptions, but Indonesia appealed into the void created by the paralysis of the Appellate Body, as examined in Chapter 13. The nickel boom has raised serious environmental and social concerns, including deforestation, pollution, coal-fired captive power, and labour conditions. Indonesia's **Just Energy Transition Partnership**, examined in Chapter 23, aims to accelerate the retirement of coal power.

Japan

Japan, which has very limited domestic energy resources, relies heavily on imports and has pursued energy security through diversification, long-term LNG contracts, strategic reserves, and nuclear power. The **Fukushima Daiichi** nuclear accident of 2011, examined in Chapter 11, led to the shutdown of all nuclear reactors, the establishment of the independent **Nuclear Regulation Authority** in 2012, and stricter safety standards, under which reactors have been restarted gradually. Japan liberalised its retail electricity market in 2016 and required the legal unbundling of transmission and distribution from 2020. In 2023, Japan enacted the **GX (Green Transformation) Promotion Act**, providing for the issuance of "climate transition bonds" to finance investment in decarbonisation and the introduction of carbon pricing, including an emissions trading system to become mandatory from 2026 and a carbon levy on fossil fuel importers, and legislation allowing the operation of nuclear reactors beyond sixty years by excluding

periods of shutdown. Japan's support for hydrogen and ammonia, including co-firing in coal plants, has been controversial internationally.

Australia

Australia, a federation in which the states own most onshore mineral and petroleum resources and the Commonwealth regulates offshore resources beyond state coastal waters under the **Offshore Petroleum and Greenhouse Gas Storage Act 2006**, is one of the world's largest exporters of coal, LNG, iron ore, and lithium. The **National Electricity Market**, covering the eastern and southern states, is governed by the National Electricity Law, a model law applied by participating jurisdictions, with rules made by the Australian Energy Market Commission, regulation by the Australian Energy Regulator, and operation by the Australian Energy Market Operator. Australia reformed its **Safeguard Mechanism** in 2023 to require large industrial emitters to reduce their emissions baselines by around five per cent per year, and legislated targets of a forty-three per cent emissions reduction by 2030 and net zero by 2050 in the Climate Change Act 2022. The Federal Court's decision in ***Santos NA Barossa Pty Ltd v Tipakalippa*** (Full Federal Court, 2022), which upheld the setting aside of the approval of an environment plan for offshore drilling because the operator had failed to consult traditional owners of the Tiwi Islands, illustrates the significance of consultation obligations in Australian offshore regulation. Australia's native title regime and the Juukan Gorge controversy have been examined in Chapter 21.

24.7 Latin American Resource Sovereignty and Regulation

The Tradition of Resource Sovereignty

Latin America has been a crucible of resource sovereignty. The **Calvo doctrine**, articulated by the Argentine jurist Carlos Calvo in the nineteenth century, held that foreign investors should be subject to the jurisdiction of host state courts and should not seek the diplomatic protection of their home states — a principle reflected in "Calvo clauses" in contracts and constitutions and, for much of the twentieth century, in the reluctance of Latin American states to accept international arbitration. The **Mexican expropriation** of foreign oil companies in March 1938, under President Lázaro Cárdenas, and the creation of **Pemex**, was the first large-scale nationalisation of a petroleum industry and an enduring symbol of national sovereignty, commemorated as a national holiday. The **Hull formula** of "prompt, adequate and effective" compensation was articulated in the correspondence between US Secretary of State Cordell Hull and Mexico concerning agrarian expropriations, as examined in Chapter 14. Venezuela's nationalisation of its oil industry in 1976 and the creation of **Petróleos de Venezuela** (PDVSA), and Bolivia's repeated nationalisations of its hydrocarbons, most recently in 2006, continued this tradition.

Latin American resource governance has oscillated between cycles of **liberalisation**, in which states opened their resource sectors to foreign investment and signed bilateral investment treaties, often in the 1990s, and cycles of **resource nationalism**, in which states reasserted control, increased their fiscal take, renegotiated contracts, and in some cases withdrew from investment arbitration, notably in the 2000s during the commodity boom. These cycles reflect commodity prices, political ideologies, and the bargaining power of states and investors — illustrating the "obsolescing bargain" described by Raymond Vernon, under which the bargaining power of the host state increases once the investor has sunk its capital, as examined in Chapter 15.

Mexico

Mexico's Constitution of 1917 vested ownership of subsoil resources in the nation (Article 27), and constitutional amendments after 1938 reserved the petroleum industry exclusively to the state through Pemex and the electricity industry to the state through the **Comisión Federal de Electricidad** (CFE). In 2013, a landmark **energy reform** amended Articles 25, 27, and 28 of the Constitution to open the upstream petroleum sector to private investment through licences, production sharing contracts, and profit-sharing contracts, awarded through competitive rounds by the National Hydrocarbons Commission, and to open electricity generation and supply to competition, with an independent system operator. The reform attracted substantial investment commitments. From 2018, the administration of President López Obrador sought to reverse aspects of the reform, favouring Pemex and the CFE through regulatory and legislative measures, including an amendment to the Electricity Industry Law in 2021 to prioritise CFE dispatch, which was challenged before the Supreme Court and by private generators. In 2024, constitutional amendments redefined Pemex and the CFE as "public state companies" and gave the CFE priority in electricity, and in 2025 a new package of secondary energy laws replaced the legislation of the 2014 reform, dissolving the independent energy regulators and reorganising the sector around the state companies while preserving a role for private investment under state planning. The Mexican experience illustrates the volatility of resource governance and its implications for investors, including under the USMCA and investment treaties.

Venezuela

Venezuela, which holds the world's largest proven oil reserves, consisting largely of extra-heavy crude in the Orinoco Belt, opened its sector to foreign investment in the 1990s through the *apertura petrolera*, including strategic associations for the Orinoco projects. The **Hydrocarbons Law of 2001**, enacted under President Chávez, required that primary activities be conducted by the state or

by **mixed companies** (*empresas mixtas*) in which the state holds more than fifty per cent, and raised royalties. In 2007, Venezuela required the migration of the Orinoco associations and operating agreements to mixed companies, leading to the exit of ExxonMobil and ConocoPhillips and to investment arbitrations, in which ICSID tribunals found that Venezuela had unlawfully expropriated ConocoPhillips's investments, awarding around USD 8.7 billion in 2019, a decision upheld in annulment proceedings. Venezuela's production collapsed amid mismanagement, the deterioration of PDVSA, and US sanctions imposed from 2017 and 2019. The enforcement of awards against Venezuelan assets, including the court-supervised sale of shares in CITGO's parent company in the United States, examined in Chapter 26, illustrates the long tail of resource nationalism.

Brazil

Brazil broke the state monopoly of Petrobras with a constitutional amendment of 1995 and the **Petroleum Law** (Law No. 9,478 of 1997), which established the **National Agency of Petroleum, Natural Gas and Biofuels** (ANP) and introduced a concession regime under which blocks are awarded in competitive bidding rounds. Following the discovery of the giant **pre-salt** fields offshore in 2006–2007, Brazil adopted a **production sharing regime** for the pre-salt area under Law No. 12,351 of 2010, initially requiring Petrobras to be the operator with a minimum thirty per cent interest in all pre-salt blocks; Law No. 13,365 of 2016 removed the mandatory operatorship, giving Petrobras a right of preference. Brazil thus operates a mixed regime, with concessions outside the pre-salt and strategic areas and production sharing within them, and has become one of the world's largest oil producers, with Petrobras a leader in deepwater technology. The *Lava Jato* corruption scandal at Petrobras has been examined in Chapter 22. Brazil's leadership in biofuels, its large hydropower sector, and its expanding wind and solar capacity, together with its hosting of COP30 in Belém in 2025, illustrate its significance in the energy transition, while its plans to expand oil production, including in the Equatorial Margin near the mouth of the Amazon, illustrate the tensions examined in Chapter 18.

Argentina

Argentina's Constitution, as amended in 1994, provides that the provinces have original ownership of the natural resources in their territories (Article 124), so that provinces grant hydrocarbon permits and concessions under the federal Hydrocarbons Law (Law No. 17,319 of 1967, as amended, including in 2014 to accommodate unconventional resources). In 2012, Argentina expropriated fifty-one per cent of the shares of **YPF**, the formerly state-owned oil company privatised in the 1990s, from Repsol, which received compensation of around USD 5 billion under a settlement in 2014. Minority shareholders, whose rights

under YPF's by-laws to a tender offer had not been honoured, sued in the United States, and in September 2023 the US District Court for the Southern District of New York entered judgment against Argentina for around USD 16.1 billion in ***Petersen Energía Inversora v Argentine Republic***; in 2025 the court ordered Argentina to turn over its YPF shares in partial satisfaction of the judgment, an order that was stayed pending appeal. The development of the **Vaca Muerta** shale formation, one of the largest unconventional resources outside North America, has transformed Argentina into a growing oil and gas exporter. The **Incentive Regime for Large Investments** (RIGI), enacted in 2024, offers tax, customs, and foreign exchange incentives and stability guarantees, with access to international arbitration, for large investments in sectors including energy and mining.

The Lithium Triangle and Resource Sovereignty in the Transition

The **"lithium triangle"** of Argentina, Bolivia, and Chile holds a majority of the world's identified lithium resources, in brine deposits in high-altitude salt flats. The three states have adopted contrasting legal models. **Chile**, where lithium has been designated a strategic mineral not subject to ordinary mining concessions since 1979, has granted rights through contracts with the state development agency CORFO; its **National Lithium Strategy** (2023) provides for state participation in lithium projects, including through a partnership between the state copper company Codelco and SQM in the Salar de Atacama, under which Codelco acquired a majority interest from 2031. **Bolivia** reserves lithium development to the state through **Yacimientos de Litio Bolivianos** (YLB), which has concluded agreements with Chinese and Russian partners for direct lithium extraction projects, subject to legislative approval. **Argentina**'s provinces grant mining concessions to private companies under the federal Mining Code, which has attracted significant investment, while some provinces have sought greater participation through provincial companies. These models illustrate different approaches to capturing value from transition minerals, examined in Chapter 8, and their interaction with the rights of indigenous communities, examined in Chapter 21.

Rights of Nature and Environmental Constitutionalism

Latin America has also pioneered **environmental constitutionalism**. **Ecuador**'s Constitution of 2008 recognises the **rights of nature** (*Pacha Mama*), including the right to the integral respect for its existence and the maintenance and regeneration of its life cycles (Articles 71–74). In the ***Los Cedros*** case (2021), the Constitutional Court of Ecuador held that mining concessions in the Los Cedros protected forest violated the rights of nature and the right to environmental consultation, revoking the concessions. Colombia's courts have recognised rivers and the Amazon as subjects of rights. The **Escazú Agreement**, examined

in Chapters 21 and 23, establishes procedural environmental rights across the region. These developments have influenced debates about the legal status of nature in other regions, including Africa.

Guyana

Guyana has become one of the world's newest oil producers since ExxonMobil's discovery of the Liza field in the Stabroek Block in 2015, with production beginning in 2019 and rising rapidly. The **Stabroek production sharing agreement** of 2016, criticised for its fiscal terms, including a two per cent royalty, a fifty-fifty profit oil split, and the payment of the contractor's taxes from the government's share, became the subject of public controversy and litigation. Guyana enacted the **Natural Resource Fund Act 2021**, establishing a sovereign wealth fund, and the **Petroleum Activities Act 2023**, which modernised the legal framework, including by providing for a new model PSA with higher royalties for future contracts. The maritime and territorial dispute with Venezuela over the Essequibo region, examined in Chapter 20, overshadows Guyana's development. Guyana's experience offers lessons for new producers, including in Africa, on the importance of negotiating capacity, transparency, and revenue management.

24.8 Comparative Lessons from African Resource-Rich States

Diversity of Experience

Africa's resource-rich states display a wide diversity of legal frameworks and outcomes. This section surveys selected experiences, emphasising features of comparative significance for the design of resource legal frameworks, including in Nigeria.

Botswana: Institutions and Partnership

Botswana is frequently cited as Africa's foremost example of successful resource governance. Since the discovery of diamonds shortly after independence, Botswana has used diamond revenues to finance infrastructure, education, and health, and has maintained macroeconomic stability and democratic governance. Its legal framework vests minerals in the state under the Mines and Minerals Act, and its diamond production is conducted principally through **Debswana**, a fifty-fifty joint venture between the government and De Beers, under agreements that have evolved to increase the government's share of revenues and its role in sorting, valuation, and sales. In 2023 and 2025, the government and De Beers agreed on new sales and licence arrangements, extending Debswana's mining licences and increasing the share of rough diamonds sold through the state-owned Okavango Diamond Company. Botswana's fiscal rules, including the **Sustainable Budget Index**, under which mineral revenues are to be used only for investment spending, and the **Pula Fund**, examined in Chapter 23, have contributed to prudent

revenue management. Botswana's success has been attributed to the quality of its institutions, an inclusive political settlement, a long-term partnership with a single major investor, and conservative fiscal management — factors that may be difficult to replicate, and which have been tested by the decline in diamond demand in the 2020s.

Ghana: Revenue Management and Transparency

Ghana, which became an oil producer with the Jubilee field in 2010, adopted the **Petroleum Revenue Management Act 2011**, examined in Chapter 23, and the **Petroleum (Exploration and Production) Act 2016** (Act 919), which modernised its upstream legal framework, introducing competitive licensing, a public register of petroleum agreements and licences, beneficial ownership disclosure, and strengthened regulation by the Petroleum Commission. Ghana's local content regulations of 2013 set ambitious targets for Ghanaian participation. Ghana's experience demonstrates the value of early adoption of transparency and revenue management frameworks by a new producer, although the country's debt crisis and restructuring in 2022–2024 showed that sound petroleum laws cannot alone ensure macroeconomic stability.

Tanzania: Resource Nationalism and Renegotiation

Tanzania enacted in 2017 a package of laws asserting sovereignty over natural resources: the **Natural Wealth and Resources (Permanent Sovereignty) Act 2017**, which declared permanent sovereignty over natural resources, required that resource agreements be approved by Parliament, and prohibited the settlement of disputes concerning natural resources by international arbitration, requiring adjudication in Tanzanian courts; the **Natural Wealth and Resources Contracts (Review and Re-Negotiation of Unconscionable Terms) Act 2017**, which empowered Parliament to review and require renegotiation of agreements containing "unconscionable" terms; and amendments to the Mining Act increasing royalties and requiring a free carried interest of at least sixteen per cent for the government. The government's dispute with Acacia Mining (a subsidiary of Barrick Gold) over alleged under-declaration of export revenues, which led to an assessment of more than USD 190 billion in unpaid taxes and penalties, was resolved in 2019–2020 through a framework agreement under which Barrick paid USD 300 million and established a joint venture, Twiga Minerals, with the government holding a sixteen per cent interest and sharing economic benefits equally. Tanzania's approach illustrates the tools of resource nationalism and their costs: while the government obtained improved terms, investment in the sector declined, and later administrations sought to restore investor confidence, including by amending the restrictions on arbitration.

The Democratic Republic of the Congo: Strategic Minerals

The **Democratic Republic of the Congo**, the world's leading producer of cobalt and a major producer of copper, revised its **Mining Code** in 2018, increasing royalties, introducing a fifty per cent tax on "super-profits", reducing the stability period for existing mining rights from ten to five years, and authorising the designation of **strategic substances**, subject to a royalty of ten per cent; cobalt and germanium were so designated. The revision, adopted despite the opposition of major mining companies, which argued that it violated the stability provisions of the 2002 Code, illustrates the use of fiscal reform to increase government take during a commodity boom. In 2025, faced with low cobalt prices due to oversupply, the DRC suspended cobalt exports for several months and then introduced an export quota system, using its market position to influence prices. The DRC's renegotiation of the *Sicomines* infrastructure-for-minerals agreement with Chinese partners in 2024, which increased the infrastructure commitment, illustrates the review of resource-for-infrastructure deals. The DRC's mineral wealth is also intertwined with conflict in the east of the country, examined in Chapter 25, and with concerns about artisanal mining and child labour, examined in Chapter 21.

Senegal, Mozambique, and Uganda: New Producers

New producers have sought to learn from earlier experience. **Senegal**, which began oil production from the offshore Sangomar field in June 2024 and gas production from the Greater Tortue Ahmeyim LNG project, shared with Mauritania, adopted a new **Petroleum Code** in 2019, a **local content law** in 2019, and a law on the distribution and management of petroleum revenues in 2022, and its constitution, as amended in 2016, provides that natural resources belong to the people. The government elected in 2024 launched an audit and renegotiation of oil, gas, and mining contracts. **Mozambique**, which holds vast gas reserves in the Rovuma Basin, adopted a Petroleum Law in 2014 and special decree-laws for the LNG projects, but its development has been disrupted by the insurgency in Cabo Delgado, examined in Chapter 21. **Uganda**, which discovered oil in the Albertine Graben in 2006, adopted the **Petroleum (Exploration, Development and Production) Act 2013** and the **Public Finance Management Act 2015**, establishing a Petroleum Fund, and is developing its oil fields with TotalEnergies and CNOOC, together with the East African Crude Oil Pipeline, examined in Chapters 6 and 21.

South Africa: Transformation and Electricity Reform

South Africa's **Mineral and Petroleum Resources Development Act 2002** (MPRDA) vested mineral and petroleum resources in the custodianship of the state for the benefit of all South Africans, abolishing private mineral rights and

requiring holders to convert old-order rights into new-order rights. The **Mining Charter**, issued under the MPRDA and revised several times, set targets for black economic empowerment in ownership, management, procurement, and community development, and its legal status was addressed by the High Court in 2021, which held that the Charter is a policy instrument rather than binding law. In the electricity sector, the crisis of **Eskom**, examined in Chapter 16, led to the **Electricity Regulation Amendment Act 2024**, which provides for the establishment of an independent transmission system operator and a competitive wholesale electricity market, and to the liberalisation of private generation, including the removal of licensing requirements for embedded generation in 2022. The REIPPPP, examined in Chapters 10 and 23, has been an influential model of renewable energy procurement.

Comparative Lessons

Several lessons emerge from this comparative survey for the design of resource legal frameworks in Africa, including in Nigeria:

1. **Institutions matter more than models.** Similar legal models — production sharing contracts, NOCs, sovereign wealth funds, independent regulators — have produced very different outcomes depending on the quality of institutions, the rule of law, and political commitment. Legal reform must therefore be accompanied by investment in institutional capacity and accountability.
2. **Transparency and accountability are foundational.** The disclosure of contracts, licences, beneficial owners, and payments, and independent oversight, as in Ghana's PIAC and Nigeria's NEITI, constrain corruption and build public trust.
3. **Stability and flexibility must be balanced.** Resource nationalism, as in Tanzania and the DRC, may increase government take in the short term but may deter investment; excessive stability guarantees may prevent states from capturing a fair share of windfalls or responding to changing circumstances. Fiscal regimes with progressive features that respond automatically to prices and profitability may reduce the pressure for renegotiation, as examined in Chapter 16.
4. **Revenue management requires rules and political consensus.** Fiscal rules and savings funds work only when supported by political consensus and legal safeguards, as the contrast between Botswana and Ghana on the one hand and Chad on the other illustrates.
5. **Communities must benefit.** Sustainable resource development requires that host communities share in the benefits and that their rights be respected, through mechanisms such as community development agreements and host community trusts, whose governance must be credible.

6. **The energy transition is reshaping the agenda.** Africa's resource-rich states must manage the risks of stranded fossil fuel assets while capturing value from transition minerals, including through local processing, and must expand energy access for their own populations — challenges addressed by the **Africa Mining Vision** (2009), adopted by the African Union to promote transparent, equitable, and optimal exploitation of mineral resources for broad-based sustainable growth, and by subsequent continental strategies on green minerals.

Conclusion

Comparative energy and natural resources law reveals both the diversity of legal approaches to common problems and the recurring patterns that shape the sector: the tension between state control and private investment, between stability and adaptability, between national revenues and local benefits, and, increasingly, between the development of fossil fuel resources and the energy transition. Nigeria's recent reforms — the Petroleum Industry Act 2021, the Electricity Act 2023, the Climate Change Act 2021, and the removal of the petrol subsidy — draw on international models while responding to the country's distinctive federal structure, history, and challenges. The experience of the United States, the United Kingdom, and the European Union illustrates the evolution of liberalised energy markets and the legal frameworks of the energy transition; the Middle East, Asia, and Latin America illustrate the range of models of petroleum and mineral governance and the cycles of liberalisation and resource nationalism; and Africa's resource-rich states offer lessons about the primacy of institutions, transparency, and inclusive benefit-sharing. For the doctoral researcher, comparative law offers a powerful method for evaluating and improving legal frameworks, provided that it is conducted with methodological rigour and sensitivity to context — themes to which Chapter 28 returns.

::: exercise {title="Practical Exercise 24: A Comparative Law Reform Brief for a New Producer (HYPOTHETICAL)"} **This exercise is entirely hypothetical. The state, discovery, institutions, and political circumstances described are fictitious and do not refer to any real country or event.**

The Republic of Lundara, a coastal West African state with a population of fifteen million, a civil law tradition inherited from its colonial period, and a unitary constitution that vests all natural resources in "the people, represented by the State", has announced a major offshore oil and gas discovery. The discovery lies in a block granted in 2014 under a production sharing contract governed by Lundara's Hydrocarbons Code of 1998, which provides for a ten per cent royalty, a profit oil split that rises from forty to sixty per cent for the state according to cumulative production, a ten per cent carried interest for the state oil company,

a twenty-year stability clause, and ICC arbitration. Lundara's state oil company, Petrolund, both regulates the sector and holds the state's participating interests. Lundara has no sovereign wealth fund, no local content law, and no legal framework for community benefits. Forty per cent of its population lacks access to electricity, and the government has pledged in its NDC to reach net zero by 2060. A newly elected government, which campaigned on "sovereignty over our resources", has established a Law Reform Commission to recommend a new legal framework before the first production, expected in five years.

Tasks. You are appointed as comparative law adviser to the Law Reform Commission.

(a) Prepare a methodological note (of no more than 1,000 words) identifying the jurisdictions you propose to examine and justifying your selection, explaining how you will address the risks of inappropriate legal transplantation.

(b) Prepare a comparative brief (of no more than 4,000 words) addressing the following questions, drawing on the jurisdictions examined in this chapter: (i) whether, and how, regulatory and commercial functions should be separated, with reference to Nigeria's PIA and Norway; (ii) the design of a petroleum revenue management law and sovereign wealth fund, with reference to Ghana, Norway, Nigeria, and Chad; (iii) the design of a host community benefit-sharing regime, with reference to Nigeria's host communities development trusts and other models; (iv) whether the existing PSC should be renegotiated, and the legal constraints on doing so, with reference to the experience of Tanzania, Senegal, and Guyana and the stability clause; (v) how the new framework should address domestic gas supply and electricity access; and (vi) how the framework should reconcile petroleum development with the government's climate commitments.

(c) Draft the headings and principal provisions of a proposed Petroleum Governance Bill, indicating for each major provision its comparative source of inspiration.

(d) In a group, debate the motion: "Lundara should adopt the Norwegian model of petroleum governance." :::

CHAPTER SUMMARY

- Comparative law serves knowledge, law reform, practice, harmonisation, and the identification of general principles; functional, contextual, and socio-legal methods each have strengths and limitations, and the transplantation of energy law models requires attention to local institutions and context.
- Nigeria's constitution vests petroleum in the federation; the Petroleum Industry Act 2021 separated policy, regulation (NUPRC and NMDPRA), and

commercial functions (NNPC Limited), introduced a new licensing and fiscal regime, host communities development trusts, and environmental funds; the Electricity Act 2023 decentralised electricity regulation to the states; and the removal of the petrol subsidy in 2023 and the Dangote refinery are transforming the downstream sector.

- United States energy law is shaped by federalism, private mineral ownership, FERC's regulation of interstate gas and electricity, state regulation, and environmental review; recent Supreme Court decisions (*West Virginia v EPA*, *Loper Bright*, *Seven County*) have constrained agency power, and federal energy policy has been volatile.
- The United Kingdom pioneered privatisation, independent regulation, and contracts for difference; the European Union has built an internal energy market through successive liberalisation packages and is implementing the Green Deal, REPowerEU, and electricity market reform.
- Middle Eastern petroleum governance ranges from exclusive NOC control (Saudi Arabia) through service contracts (Iran, federal Iraq) to concession and joint venture models (UAE, Qatar), with Iraq illustrating federal disputes over resources.
- Asian frameworks include China's state-led system and new Energy Law, India's federal competences and auction-based allocation, Indonesia's constitutional resource control and nickel export ban, Japan's post-Fukushima reforms, and Australia's federal resource ownership and market institutions.
- Latin America has oscillated between liberalisation and resource nationalism, from Mexico's 1938 expropriation and 2013 reform and counter-reform to Venezuela's nationalisations, Brazil's mixed regime, Argentina's YPF expropriation and Vaca Muerta, divergent lithium models, and environmental constitutionalism.
- African experience — Botswana's institutions, Ghana's revenue management, Tanzania's and the DRC's resource nationalism, new producers such as Senegal, and South Africa's reforms — shows that institutions, transparency, balanced stability, revenue management rules, community benefits, and attention to the energy transition are critical.

DOCTORAL RESEARCH QUESTIONS

1. To what extent has the Petroleum Industry Act 2021 addressed the governance failures of Nigeria's previous petroleum legal framework? Evaluate its implementation in comparative perspective.
2. Assess the legal and institutional challenges of the decentralisation of electricity regulation under Nigeria's Electricity Act 2023, drawing on comparative experience of multilevel electricity governance in the United States, India, and Australia.

3. Is resource nationalism a rational strategy for African resource-rich states? Evaluate with reference to Tanzania, the DRC, and Latin American experience.
4. Can the Norwegian model of petroleum governance be successfully transplanted to African producers? Assess using comparative legal methodology.
5. Compare the legal approaches of Chile, Bolivia, and Argentina to lithium and evaluate their lessons for African producers of transition minerals, including Nigeria.
6. How have recent changes in United States administrative law affected the capacity of federal agencies to regulate the energy transition, and what lessons do they hold for the design of regulatory mandates elsewhere?

FURTHER READING

- Konrad Zweigert and Hein Kötz, *An Introduction to Comparative Law* (Tony Weir tr, 3rd edn, Oxford University Press 1998).
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- Martha M Roggenkamp, Catherine Redgwell, Anita Rønne and Iñigo del Guayo (eds), *Energy Law in Europe: National, EU and International Regulation* (3rd edn, Oxford University Press 2016).
- Joseph P Tomain and Richard D Cudahy, *Energy Law in a Nutshell* (West Academic, latest edition).
- Yinka Omorogbe, *Oil and Gas Law in Nigeria: Simplified* (Malthouse Press 2003), read together with subsequent commentary on the Petroleum Industry Act 2021.
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25 CHAPTER

Energy Geopolitics, National Security, and International Legal Order

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the concept of energy security, its dimensions, and its relationship to state sovereignty and international law.
2. Evaluate the legality of economic sanctions affecting energy trade under the UN Charter, general international law, and the law of the World Trade Organization.
3. Analyse the legal framework of strategic petroleum and gas reserves, including the International Energy Program and EU stockholding obligations.
4. Assess the legal dimensions of pipeline politics and cross-border energy dependencies.
5. Apply international humanitarian law to the protection of energy infrastructure and natural resources in armed conflict, and evaluate the legal responses to resource-related conflict.
6. Analyse the legal framework of maritime security as it affects energy supply, including chokepoints and attacks on shipping.
7. Evaluate national and regional emergency powers used to respond to energy supply disruptions.
8. Assess the legal responses to geopolitical risk in the energy transition, including in relation to critical minerals, cyber security, and investment screening.

Introduction

Energy has always been entangled with power. The control of coal shaped the industrial and military strength of nations in the nineteenth century; the conversion of the Royal Navy from coal to oil before the First World War, and the struggle for oil in the Second World War, made petroleum a strategic commodity; the oil embargo of 1973–1974 demonstrated the vulnerability of importing states and the power of producers; and the Russian invasion of Ukraine in 2022 precipitated

the most severe energy crisis in Europe in half a century and a fundamental reorientation of global energy flows. The energy transition is reshaping, rather than ending, energy geopolitics: as dependence on oil and gas declines, the control of critical minerals, clean energy manufacturing, technology, and electricity networks is becoming a new focus of strategic competition.

International law operates within, and seeks to structure, this geopolitical context. It establishes the sovereignty of states over their resources and their freedom to determine their energy policies; it constrains the use of force and economic coercion; it regulates trade and transit; it establishes cooperative mechanisms for managing supply emergencies; it protects energy infrastructure in armed conflict; and it provides frameworks for maritime security and cyber security. Yet energy security has also been invoked to justify measures that test the limits of international law — unilateral sanctions with extraterritorial effects, export restrictions, the weaponisation of supply, and attacks on infrastructure. This chapter examines energy security and state sovereignty (section 25.1), the international legal framework of sanctions affecting energy trade (section 25.2), strategic petroleum reserves (section 25.3), pipeline politics and cross-border dependencies (section 25.4), resource conflict and international humanitarian law (section 25.5), maritime security and energy infrastructure (section 25.6), energy supply disruptions and emergency powers (section 25.7), and geopolitical risk and international legal responses (section 25.8). Sanctions compliance by companies has been examined in Chapter 22, and the law of the sea governing submarine pipelines and cables in Chapter 20; this chapter focuses on the inter-state and systemic dimensions.

	Meaning	Illustrative legal instruments
Availability	Physical existence of energy resources and supply capacity	Licensing and investment law; strategic reserves; resource diversification
Accessibility	Ability to obtain supply across borders and networks despite geopolitical obstacles	Transit rules (GATT Art V, ECT Art 7); pipeline agreements; maritime security
Affordability	Supply at prices that households and economies can bear	Price regulation; subsidies and social tariffs; emergency market measures
Acceptability	Environmental and social sustainability of supply	Climate and environmental law; human rights due diligence; transition policies
Resilience	Capacity to withstand and recover from shocks, including cyber and physical attacks	Critical infrastructure and cyber laws; emergency powers; solidarity clauses

Figure 25.1 Dimensions of energy security and illustrative legal instruments. *Source:* prepared for this edition, adapting the "four As" framework of the Asia Pacific Energy Research Centre.

25.1 Energy Security and State Sovereignty

The Concept of Energy Security

Energy security is a contested and multidimensional concept. The International Energy Agency defines it as the uninterrupted availability of energy sources at an affordable price, distinguishing **long-term energy security**, concerned with timely investments to supply energy in line with economic developments and environmental needs, and **short-term energy security**, concerned with the ability of the energy system to react promptly to sudden changes in the supply-demand balance. The Asia Pacific Energy Research Centre proposed in 2007 an influential framework of **"four As"**: **availability** (the geological existence of resources), **accessibility** (the ability to obtain them, given geopolitical and infrastructural constraints), **affordability** (prices), and **acceptability** (environmental and social sustainability). Later scholarship has added **resilience** — the capacity of systems to withstand and recover from shocks — and dimensions such as governance and technology. Figure 25.1 relates these dimensions to illustrative legal instruments.

Energy security means different things to different states. For **importing states**, it is principally security of supply — the reliable availability of energy

at reasonable prices — pursued through diversification of sources and routes, strategic stocks, domestic production, efficiency, and international cooperation. For **exporting states**, it is principally **security of demand** — stable markets and revenues — pursued through long-term contracts, market management (as through OPEC), and investment in downstream markets. For **transit states**, it involves the revenues and leverage derived from transit and the risks of disruption. For **energy-poor states**, energy security is inseparable from energy access, as examined in Chapter 23. The energy transition is adding new dimensions: the security of supply of critical minerals and clean energy technologies, the stability of electricity systems with high shares of variable renewables, and climate security, including the vulnerability of energy infrastructure to extreme weather.

Energy Security and Sovereignty

International law affirms the **sovereignty of states over their natural resources** and their freedom to determine their energy policies, as examined in Chapter 1. The principle of **permanent sovereignty over natural resources**, articulated in General Assembly Resolution 1803 (XVII) of 1962 and reflected in common Article 1 of the human rights Covenants, entitles states to control the exploration, development, and disposition of their resources. The Charter of Economic Rights and Duties of States (1974) proclaimed that every state has and shall freely exercise full permanent sovereignty, including possession, use, and disposal, over all its wealth, natural resources, and economic activities. At the regional level, Article 194(2) of the TFEU preserves a member state's right to determine the conditions for exploiting its energy resources, its choice between energy sources, and the general structure of its energy supply, while Article 18 of the **Energy Charter Treaty** recognises state sovereignty over energy resources, examined in Chapter 14.

Sovereignty over resources implies the freedom of producing states to decide whether, when, and how to develop and export their resources — including the freedom to restrict exports, subject to their international obligations, notably under the WTO. It also implies the freedom of importing states to choose their suppliers and to pursue energy independence. These freedoms are constrained by international law in several respects: by the prohibition of the use of force and of intervention; by treaty obligations on trade, transit, and investment; by obligations of cooperation, for example under the IEA's International Energy Program and the EU's principle of energy solidarity; and by environmental and human rights obligations. The tension between sovereignty and interdependence is the central theme of the international law of energy security.

Energy as a Weapon

The use of energy supply as an instrument of coercion — the "energy weapon" — has a long history. The **Arab oil embargo** of October 1973, imposed by Arab members of OPEC against the United States, the Netherlands, and other states supporting Israel in the Yom Kippur War, together with production cuts, quadrupled oil prices and led to the creation of the International Energy Agency. Russia's interruptions of gas supplies to Ukraine in 2006 and 2009, which affected supplies to Europe, and its reduction and eventual cessation of most pipeline gas supplies to Europe in 2022 are more recent examples. The use of energy as a weapon raises questions about the legality of economic coercion under international law. The principle of non-intervention, as articulated in the **Friendly Relations Declaration** (General Assembly Resolution 2625 (XXV), 1970), provides that no state may use or encourage the use of economic, political, or any other type of measures to coerce another state in order to obtain from it the subordination of the exercise of its sovereign rights and to secure from it advantages of any kind. In ***Military and Paramilitary Activities in and against Nicaragua*** (1986), the International Court of Justice held that intervention is wrongful when it uses methods of coercion in regard to matters in which each state is permitted to decide freely, but it found that the economic measures taken by the United States against Nicaragua, including the cessation of economic aid and a trade embargo, did not constitute a breach of the customary principle of non-intervention. The threshold at which economic pressure becomes unlawful coercion remains contested, particularly where supply is cut off in breach of contract or as part of a broader campaign of aggression.

Energy Security in Africa and Nigeria

For Africa, energy security has distinctive features. Many African states are energy exporters, yet face domestic energy insecurity in the form of low access rates, unreliable electricity, and dependence on imported refined products — the paradox of exporting crude oil while importing petrol, which characterised Nigeria for decades until the start-up of the Dangote refinery, examined in Chapter 24. Energy insecurity in Africa is also linked to physical insecurity: crude oil theft and pipeline vandalism in the Niger Delta, which at times reduced Nigeria's production substantially below its OPEC quota; attacks on energy infrastructure by insurgents in the Sahel and Mozambique; and the vulnerability of electricity grids to vandalism and sabotage. Regional integration through the **West African Power Pool**, examined in Chapter 9, and regional gas infrastructure, such as the West African Gas Pipeline, can enhance energy security by diversifying supply and sharing reserves, while creating new interdependencies. The **ECOWAS Energy Protocol** (2003), modelled on the Energy Charter Treaty, provides a legal frame-

work for investment and transit in the region, though its implementation has been limited.

25.2 International Sanctions and Energy Trade

Scope of this Section

Chapter 22 examined sanctions from the perspective of corporate compliance and contract law. This section examines the international legal framework governing the imposition of sanctions affecting energy trade: their legality under the UN Charter and general international law, their consistency with WTO law and investment treaties, and the controversy over unilateral and extraterritorial sanctions.

UN Sanctions

Sanctions adopted by the **UN Security Council** under Chapter VII of the Charter are binding on all member states, and under Article 103 of the Charter, obligations under the Charter prevail over conflicting obligations under other international agreements. Energy-related UN sanctions have included the oil embargo on Southern Rhodesia (1966), the comprehensive sanctions on Iraq (1990–2003), the measures against illicit petroleum exports from Libya under Resolution 2146 (2014), the prohibition of trade in oil with the Islamic State and other terrorist groups under Resolution 2199 (2015), the sanctions on Iran's nuclear programme, and the restrictions on North Korea's imports of refined petroleum products and exports of coal. The Security Council has also addressed the role of natural resources in financing conflict, through measures against "conflict diamonds" from Angola and Sierra Leone and natural resources in the Democratic Republic of the Congo, examined in section 25.5. The UN sanctions regime is subject to concerns about its humanitarian consequences, which led to the shift from comprehensive to "targeted" or "smart" sanctions after the experience of Iraq, and to due process concerns about the listing of individuals, addressed in part by the Office of the Ombudsperson for the ISIL and Al-Qaida sanctions regime and, in the European Union, by the *Kadi* judgments of the Court of Justice.

Unilateral Sanctions and International Law

Unilateral or **autonomous sanctions**, imposed by states or regional organisations without Security Council authorisation, are more controversial. Their legality under general international law depends on their character. Measures that are within a state's sovereign discretion — such as the refusal to trade, the suspension of aid, or the freezing of assets of individuals within its jurisdiction — are generally regarded as **retorsion**, lawful acts that are unfriendly but do not breach any international obligation. Measures that would otherwise breach

the imposing state's international obligations — such as obligations under trade agreements, investment treaties, or rules on state immunity — may be justified as **countermeasures** under the law of state responsibility, if they respond to a prior internationally wrongful act of the target state and comply with the conditions in Articles 49 to 54 of the ILC Articles on State Responsibility (2001), including proportionality and the requirement that they be directed at inducing compliance. Whether states not directly injured by a breach of an obligation owed to the international community as a whole, such as the prohibition of aggression, may take countermeasures ("third-party" or "collective" countermeasures) is unsettled, although Article 54 of the ILC Articles reserves the possibility of lawful measures and state practice — including the sanctions against Russia since 2022 — has been invoked as supporting such a right.

Many states of the Global South, together with Russia and China, regard **unilateral coercive measures** as unlawful. The UN General Assembly has adopted resolutions condemning such measures, and the Human Rights Council established in 2014 the mandate of a **Special Rapporteur on the negative impact of unilateral coercive measures on the enjoyment of human rights**, whose reports have criticised the humanitarian effects of sanctions, including on access to energy, medicines, and food, and the "over-compliance" by banks and companies that extends their impact beyond their formal scope. The General Assembly has also, since 1992, annually called for an end to the US embargo against Cuba, by overwhelming majorities.

Extraterritorial and Secondary Sanctions

The **extraterritorial** application of sanctions — particularly US **secondary sanctions**, which threaten non-US persons with penalties for dealing with targeted states even in the absence of any US connection — raises questions of jurisdiction under international law. Jurisdiction to prescribe is generally based on territoriality, nationality, the protective principle, or universality, and the application of sanctions to foreign persons for conduct abroad with no connection to the regulating state has been widely criticised as exceeding these bases. The **European Union** responded to the extraterritorial application of US sanctions against Cuba, Iran, and Libya in the 1990s by adopting the **Blocking Statute** (Council Regulation (EC) No 2271/96), updated in 2018 following the US withdrawal from the Iran nuclear agreement, and the US–EU dispute over the extraterritorial reach of the *Helms-Burton Act* and the *Iran and Libya Sanctions Act* was referred to the WTO before a political understanding was reached in 1998. The extraterritorial sanctions imposed by the United States on companies involved in the construction of the **Nord Stream 2** pipeline, under the Protecting Europe's Energy Security Act of 2019, prompted objections from Germany and the EU. In 2023, the European Union adopted the **Anti-Coercion Instrument**

(Regulation (EU) 2023/2675), which empowers the EU to respond to economic coercion by third countries — defined as measures affecting trade or investment to prevent or obtain the cessation, modification, or adoption of a particular act by the EU or a member state — through countermeasures, including restrictions on trade, investment, and access to public procurement.

Sanctions and WTO Law

Sanctions restricting trade in energy goods may be inconsistent with obligations under the **General Agreement on Tariffs and Trade**, including the most-favoured-nation obligation (Article I), the prohibition of quantitative restrictions (Article XI), and the freedom of transit (Article V), as examined in Chapter 13. States imposing sanctions typically invoke the **security exceptions** of **Article XXI**, which provide that nothing in the GATT shall be construed to prevent any contracting party from taking any action which it considers necessary for the protection of its essential security interests, relating to fissionable materials, traffic in arms, or "taken in time of war or other emergency in international relations", or to prevent any contracting party from taking action in pursuance of its obligations under the UN Charter for the maintenance of international peace and security.

In ***Russia — Measures concerning Traffic in Transit*** (2019), the first panel decision interpreting Article XXI, a WTO panel rejected Russia's argument that Article XXI is entirely self-judging. It held that the existence of an "emergency in international relations" is an objective fact subject to review, while the determination of the essential security interests and the necessity of the measures is for the invoking state, subject to the obligation of good faith, which requires that the measures meet a minimum requirement of plausibility in relation to the proffered security interests. The panel found that the situation between Russia and Ukraine since 2014 constituted an emergency in international relations and that Russia's transit restrictions satisfied the requirements of Article XXI. Subsequent panels, including in *United States — Steel and Aluminium Products* (2022), followed this approach, finding that US tariffs justified by national security did not fall within the exception, a conclusion the United States rejected, maintaining that Article XXI is self-judging. The trade measures adopted against Russia since 2022, including the suspension of most-favoured-nation treatment by many WTO members, have been justified by reference to Article XXI. The paralysis of the WTO Appellate Body, examined in Chapter 13, limits the possibility of authoritative resolution of these questions.

Sanctions and Investment Law

Sanctions may also interfere with foreign investments, potentially breaching obligations under bilateral investment treaties and the Energy Charter Treaty, including obligations of fair and equitable treatment, protection against expro-

priation, and free transfer of funds. Many investment treaties contain **security exceptions** or "non-precluded measures" clauses, whose interpretation — including whether they are self-judging — has been addressed in arbitral awards arising from the Argentine financial crisis, such as *CMS v Argentina* (2005) and *LG&E v Argentina* (2006), examined in Chapter 14. The Energy Charter Treaty contains, in Article 24, exceptions for measures necessary for the protection of essential security interests, including those relating to the supply of energy materials and products to military establishments and those taken in time of war, armed conflict, or other emergency in international relations. The wave of claims arising from sanctions against Russia and Russian countermeasures since 2022, including claims by Western investors against Russia for the seizure of their assets and claims by Russian investors against Western states, will test the application of these exceptions.

Sanctions and Inter-State Litigation

Inter-state litigation has addressed sanctions in several cases. In ***Alleged Violations of the 1955 Treaty of Amity, Economic Relations, and Consular Rights*** (*Iran v United States*), the International Court of Justice indicated provisional measures in October 2018 requiring the United States to remove impediments arising from the reimposed sanctions to the free exportation to Iran of medicines, foodstuffs, agricultural commodities, and goods and services necessary for the safety of civil aviation, and in 2021 rejected the US preliminary objections. In ***Certain Iranian Assets*** (*Iran v United States*, 2023), the Court found that the United States had violated the Treaty of Amity by measures allowing the attachment of the assets of Iranian state-owned companies to satisfy judgments in terrorism-related cases, but held that it lacked jurisdiction over the claims concerning the assets of the Central Bank of Iran. These cases illustrate the potential, and the limits, of treaty-based adjudication of sanctions disputes.

25.3 Strategic Petroleum Reserves and Supply Security

The Rationale for Emergency Stocks

Emergency oil stocks are the principal instrument by which importing states protect themselves against sudden supply disruptions. By holding reserves of crude oil and refined products that can be released in an emergency, states can replace lost supply, dampen price spikes, and reduce the leverage of suppliers who might use supply interruption as a weapon. Stocks may be held by governments (**public stocks**), by specialised agencies (**agency stocks**), or by industry under legal obligation (**obligated industry stocks**). The legal framework of emergency stocks operates at three levels: the international obligations of members of the International Energy Agency, regional obligations in the European Union, and national legislation establishing and governing reserves.

The International Energy Program

The **International Energy Agency** (IEA) was established in November 1974 by the **Agreement on an International Energy Program** (IEP Agreement), adopted within the framework of the OECD in response to the oil crisis of 1973–1974. The IEP Agreement requires each participating country to maintain emergency reserves sufficient to sustain consumption for at least sixty days with no net oil imports, an obligation increased to **ninety days of net oil imports** in 1976. Net exporters, such as Canada, Mexico, and Norway, are not subject to the stockholding obligation. The Agreement establishes an **emergency sharing system**, under which, in the event of a supply reduction of at least seven per cent of normal supply for the group or any member, participating countries are to restrain demand and share available oil according to a prescribed formula, with an elaborate allocation mechanism. The emergency sharing system has never been formally activated; instead, the IEA developed the more flexible **Coordinated Emergency Response Measures** (CERM) in 1984, under which members may agree on coordinated stock releases and demand restraint in response to actual or imminent disruptions without triggering the formal sharing mechanism.

The IEA has coordinated **collective actions** on five occasions: during the Gulf War in 1991; after Hurricanes Katrina and Rita damaged US Gulf of Mexico production and refining in 2005; during the Libyan civil war in 2011; and twice in 2022, in March and April, in response to the Russian invasion of Ukraine, when members agreed to release a total of more than 180 million barrels — the largest collective actions in the IEA's history, with the United States separately announcing the release of 180 million barrels from its Strategic Petroleum Reserve over six months. The IEA's membership comprises some thirty OECD states, and it has developed **association** arrangements with major emerging economies, including China, India, Indonesia, and South Africa, and has expanded its mandate to encompass gas and electricity security, critical minerals, and clean energy transitions, under a ministerial mandate of 2022 reflecting the evolution of energy security concerns.

The European Union

The European Union's stockholding obligations are set out in **Council Directive 2009/119/EC**, which requires member states to maintain total oil stocks corresponding to at least ninety days of average daily net imports or sixty-one days of average daily inland consumption, whichever is greater. Member states may hold stocks through a **central stockholding entity**, through obligated economic operators, or both, and stocks may be held in other member states under bilateral arrangements. The Commission monitors compliance and may convene a coordination group in the event of a supply disruption.

The European Union's response to the gas crisis of 2022 extended the logic of strategic reserves to **natural gas**. **Regulation (EU) 2022/1032** amended the Gas Security of Supply Regulation to require member states to fill their underground gas storage facilities to at least ninety per cent of capacity by 1 November each year, with intermediate targets, and introduced the certification of storage system operators to address the risk of control by third-country entities — a response to the low filling levels of storage facilities controlled by Gazprom in 2021. In 2025, the regulation was extended for two years with greater flexibility, allowing member states to meet the ninety per cent target at any time between 1 October and 1 December and to deviate from it under specified market conditions. The EU also established the **AggregateEU** mechanism for the joint purchasing of gas, under Council Regulation (EU) 2022/2576, to leverage the collective bargaining power of member states.

National Reserves

The **United States Strategic Petroleum Reserve** (SPR) was established by the **Energy Policy and Conservation Act of 1975** and is held in salt caverns along the Gulf Coast, with a capacity of more than 700 million barrels. The President may order a drawdown upon a finding that a "severe energy supply interruption" exists or is imminent or that a drawdown is required by the IEP Agreement, and Congress has also mandated sales from the SPR to raise revenue, a practice criticised as undermining its security purpose. The SPR's level fell substantially after the 2022 releases, and replenishment has been an ongoing policy issue. Other importing states maintain significant reserves: **Japan** holds government and industry stocks covering well over the IEA minimum; **China** has built substantial strategic and commercial reserves since the 2000s, whose size it does not fully disclose; and **India** has established strategic reserves in underground rock caverns through the Indian Strategic Petroleum Reserves Limited, including partnerships with producing states such as the United Arab Emirates under which the partner stores oil in Indian facilities with India having priority access in an emergency.

Stocks in Africa and Nigeria

Few African states maintain strategic stocks meeting international standards, leaving them vulnerable to supply disruptions and price spikes, as experienced during the crises of 2022. **South Africa** maintains strategic crude oil stocks through the **Strategic Fuel Fund**, whose controversial sale of a large part of its reserves in 2015 was challenged by the Minister of Energy and declared unlawful by the Western Cape High Court, which set aside the sale contracts as having been concluded without proper authorisation. **Nigeria** has historically held limited strategic reserves of refined products, despite periodic proposals to establish a strategic petroleum reserve, and episodes of fuel scarcity have exposed the vulner-

ability of supply. The Petroleum Industry Act 2021 empowers the NMDPRA to regulate the maintenance of stocks, including strategic stocks, and the increase in domestic refining capacity may facilitate the development of national reserves. At the regional level, proposals for ECOWAS or African Union strategic stocks have been discussed, and the African Refiners and Distributors Association has promoted harmonised fuel specifications and supply security.

The Transition and New Reserves

The energy transition is generating debate about **reserves of critical minerals**. The United States maintains the **National Defense Stockpile**, which holds strategic materials, including certain critical minerals, for defence purposes, and Japan's JOGMEC manages stockpiles of rare metals. The **European Critical Raw Materials Act** (Regulation (EU) 2024/1252), examined in Chapter 8, requires the Commission and member states to coordinate strategic stockpiling of critical raw materials and provides for joint purchasing, and the EU has developed a **RESourceEU** plan, announced in 2025, which includes pilot stockpiling and a European critical raw materials centre. China maintains state reserves of critical minerals. These developments extend the logic of the IEA's emergency stocks system to the materials of the energy transition, though the heterogeneity of mineral markets complicates its application.

25.4 Pipeline Politics and Cross-Border Energy Dependencies

The Geopolitics of Fixed Infrastructure

Pipelines create **mutual dependence** between producers, transit states, and consumers. Unlike oil shipped by tanker, which can be redirected to alternative markets, gas transported by pipeline is tied to the route and its counterparties, so that pipelines lock in long-term commercial and political relationships. This interdependence can promote cooperation, as the proponents of the Soviet–West European gas trade argued during the Cold War, but it can also create vulnerabilities and leverage, which may be used for political ends. The legal framework of transit, including Article V of the GATT, Article 7 of the Energy Charter Treaty, and intergovernmental and host government agreements, has been examined in Chapter 6; this section examines the geopolitical dimensions of pipeline dependencies and the legal responses to them.

Russia, Ukraine, and Europe

The history of Russian gas supplies to Europe through Ukraine illustrates the interaction of commerce, politics, and law in pipeline relationships. Disputes between Gazprom and Naftogaz over prices, transit fees, and debts led to interruptions of supplies in January 2006 and, more severely, in January 2009, when supplies to south-eastern Europe were cut off for about two weeks in mid-winter.

The 2009 crisis prompted the European Union to strengthen its security of supply legislation, investing in interconnectors and reverse-flow capacity, and to support Ukraine's integration into the EU energy market through the **Energy Community**. The 2009 contracts between Gazprom and Naftogaz, concluded in the aftermath of the crisis, became the subject of arbitrations at the Arbitration Institute of the Stockholm Chamber of Commerce, which in 2017 and 2018 resulted in awards that, on balance, required Gazprom to pay Naftogaz approximately USD 2.56 billion, as noted in Chapter 6. A new five-year transit contract was concluded in December 2019, under EU mediation. Russia's construction of pipelines bypassing Ukraine — **Nord Stream** (2011) under the Baltic Sea to Germany, **TurkStream** (2020) under the Black Sea to Turkey, and **Nord Stream 2**, completed in 2021 but never brought into operation — reduced Ukraine's role as a transit state and was opposed by Ukraine, Poland, the Baltic states, and the United States as increasing European dependence on Russia.

Following the invasion of Ukraine in February 2022, Russia progressively reduced pipeline gas supplies to Europe, halting supplies to several states that refused to pay in roubles and reducing flows through Nord Stream before stopping them altogether in August 2022, citing technical problems. In **September 2022**, explosions destroyed three of the four strings of the Nord Stream pipelines in the exclusive economic zones of Denmark and Sweden, an act of sabotage whose perpetrators remain the subject of criminal investigation and contested attribution; Sweden and Denmark closed their investigations, while Germany continued its investigation and, in 2025, sought the extradition of a Ukrainian national arrested in Italy. The **transit contract** through Ukraine expired at the end of 2024 and was not renewed, ending Russian gas transit through Ukraine on 1 January 2025. The European Union, through the **REPowerEU** plan and subsequent legislation, has committed to phase out Russian gas imports, agreeing in 2025 on legislation to prohibit imports of Russian pipeline gas and LNG under new contracts and to phase out imports under existing contracts by 2027. The Russian share of EU gas imports fell from around forty per cent in 2021 to below a fifth, with supplies replaced by Norwegian pipeline gas and LNG, particularly from the United States.

Russia's Pivot to Asia

Russia has sought to redirect its gas exports to Asia. The **Power of Siberia** pipeline, which began deliveries to China in December 2019 under a thirty-year contract between Gazprom and CNPC, has a capacity of thirty-eight billion cubic metres per year. In September 2025, Gazprom and CNPC signed a legally binding memorandum on the construction of the **Power of Siberia 2** pipeline through Mongolia, though key commercial terms, including price, were reported to remain unresolved. The shift illustrates how pipeline politics follows geopolitical

realignment, and how the bargaining power of the parties shifts when a supplier loses its principal market.

Other Pipeline Dependencies

Pipeline politics is not confined to Russia. The **Baku–Tbilisi–Ceyhan** oil pipeline (2006) and the **Southern Gas Corridor**, comprising the South Caucasus Pipeline, the Trans-Anatolian Pipeline (TANAP), and the Trans-Adriatic Pipeline (TAP), were developed with Western support to transport Caspian resources to Western markets while bypassing Russia and Iran, under intergovernmental and host government agreements examined in Chapter 6. The **Druzhba** oil pipeline continues to supply Russian crude to Hungary and Slovakia under exemptions from the EU embargo, and has been subject to periodic interruptions and to Ukrainian attacks on infrastructure in Russia. The **Iraq–Turkey Pipeline**, examined in Chapter 24, illustrates the interaction of federal disputes and transit agreements. In the Middle East, the **Dolphin** gas pipeline from Qatar to the United Arab Emirates and Oman continued to operate during the blockade of Qatar by Saudi Arabia, the UAE, Bahrain, and Egypt between 2017 and 2021, illustrating the resilience of some energy relationships in political crises. The **East Mediterranean** gas discoveries have led to new interdependencies, including exports from Israel to Egypt and Jordan, expanded under a major supply agreement concluded in 2025, and to disputes over maritime boundaries, examined in Chapter 20.

African Pipelines

In Africa, pipelines have both connected and divided. The **West African Gas Pipeline**, which has transported Nigerian gas to Benin, Togo, and Ghana since 2010 under a treaty of 2003 among the four states, has suffered from supply interruptions and payment disputes, illustrating the challenges of regional gas trade. The **Nigeria–Morocco Gas Pipeline**, a proposed offshore and onshore pipeline traversing up to thirteen West African states, and the **Trans-Saharan Gas Pipeline**, proposed to connect Nigeria to Algeria through Niger, have been promoted as means of supplying gas to West Africa and Europe; both face formidable financial, technical, and security challenges, and their realisation would require complex intergovernmental and host government agreements. The pipeline transporting **South Sudan**'s oil through Sudan to Port Sudan illustrates the vulnerability of landlocked producers: South Sudan shut down production in 2012 in a dispute with Sudan over transit fees, and exports were disrupted again in 2024 by the war in Sudan. The **East African Crude Oil Pipeline**, examined in Chapters 6 and 21, will make Uganda's oil exports dependent on transit through Tanzania under an intergovernmental agreement and host government agreements.

Legal Responses to Pipeline Dependence

The legal responses to pipeline dependence include: (a) **diversification requirements**, such as the EU's infrastructure standard under the Gas Security of Supply Regulation (Regulation (EU) 2017/1938), which requires member states to ensure that, in the event of a disruption of the single largest gas infrastructure, the remaining infrastructure can satisfy total gas demand on a day of exceptionally high demand (the "N-1" standard); (b) **solidarity mechanisms**, under which member states must assist each other in severe emergencies, as provided in the same regulation and given legal force by the principle of energy solidarity in Article 194 TFEU, applied by the Court of Justice in *Germany v Poland* (2021), examined in Chapter 24; (c) **regulation of third-country pipelines**, through the extension of EU unbundling and third-party access rules to pipelines from third countries by Directive (EU) 2019/692, which was challenged by Nord Stream 2 AG before the EU courts and in an Energy Charter Treaty arbitration; (d) **investment screening** of foreign acquisitions of energy infrastructure, examined in section 25.8; and (e) **transit dispute resolution**, including the conciliation mechanism of Article 7(7) of the Energy Charter Treaty, which has never been used.

25.5 Resource Conflicts and International Humanitarian Law

Natural Resources and Armed Conflict

Natural resources are implicated in armed conflict in several ways. They may be a **cause** of conflict, when groups compete for control of valuable resources or when grievances over the distribution of resource revenues and environmental harm fuel rebellion, as in the Niger Delta. They may **finance** conflict, when armed groups exploit "lootable" resources such as alluvial diamonds, gold, timber, and coltan, or extract revenues from oil through theft and illicit trade, as in Sierra Leone, Liberia, Angola, the eastern Democratic Republic of the Congo, and the territories controlled by the Islamic State in Iraq and Syria. They may be **targets** of military operations, when energy infrastructure is attacked to weaken an adversary's military capacity or civilian morale, as in the Russian attacks on Ukraine's electricity system since 2022. And they may be **victims** of conflict, when the environment is damaged by warfare, as in the burning of Kuwait's oil wells by Iraqi forces in 1991. The United Nations Environment Programme has estimated that at least forty per cent of intrastate conflicts in the past sixty years have had a link to natural resources. International law addresses these dimensions through international humanitarian law, international criminal law, Security Council measures, and supply chain regulation.

Pillage and the Exploitation of Resources in Occupied Territory

International humanitarian law prohibits **pillage** — the appropriation of property for private or personal use without the consent of the owner — in all circumstances. The prohibition appears in the **Hague Regulations** of 1907 (Articles 28 and 47), the **Fourth Geneva Convention** of 1949 (Article 33), and **Additional Protocol II** (Article 4(2)(g)), and is a war crime under the **Rome Statute of the International Criminal Court** (Article 8(2)(b)(xvi) and (e)(v)). After the Second World War, the Nuremberg Military Tribunals convicted German industrialists for the plunder of resources in occupied territories, including in the *Krupp* and *Flick* cases, and the Dutch courts held in the case of the *Singapore Oil Stocks* (1956) that oil resources seized by Japan during the occupation of the Dutch East Indies had been unlawfully appropriated. The **exploitation of natural resources in occupied territory** is governed by **Article 55** of the Hague Regulations, under which the occupying state is to be regarded only as **administrator and usufructuary** of public buildings, real estate, forests, and agricultural estates belonging to the hostile state, and must safeguard the capital of these properties. The usufruct principle is generally interpreted as permitting the occupant to continue existing levels of exploitation, such as production from existing oil wells, for the benefit of the occupied territory and the needs of the occupation, but not to open new mines or wells or exploit resources beyond the existing rate.

In ***Armed Activities on the Territory of the Congo*** (*Democratic Republic of the Congo v Uganda*, 2005), the International Court of Justice held that Uganda was internationally responsible for acts of looting, plundering, and exploitation of the DRC's natural resources committed by members of the Ugandan armed forces in the territory of the DRC, and for failing, as an occupying power in the Ituri district, to prevent such acts. In its **reparations judgment of February 2022**, the Court awarded the DRC a total of USD 325 million, including USD 60 million for damage related to natural resources, determined as a global sum in the light of the evidence of the exploitation of gold, diamonds, coltan, and timber. The judgment illustrates both the availability of state responsibility for resource exploitation in armed conflict and the evidentiary difficulties of quantifying it. Questions concerning the exploitation of resources in occupied territory have also arisen in relation to Western Sahara, examined in Chapter 21, the occupied Palestinian territories, and the territories of Ukraine occupied by Russia, including Crimea, where Ukrainian companies have obtained investment treaty awards against Russia for the expropriation of their energy assets, including in *Naftogaz v Russia* (2023), in which a tribunal under the Ukraine–Russia BIT awarded around USD 5 billion.

Corporate Complicity in Resource-Related Conflict

International criminal law extends to individuals, including corporate executives, who aid and abet war crimes. The trial of two former executives of **Lundin Oil** (later Lundin Energy) before the Stockholm District Court, which began in 2023, concerning alleged complicity in war crimes committed by the Sudanese government and allied militias in clearing populations from the company's oil concession area in southern Sudan between 1999 and 2003, is the first prosecution of corporate executives for complicity in war crimes since the post-Second World War trials, and illustrates the potential criminal liability of companies operating in conflict zones, as examined in Chapter 22. The **OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas**, examined in Chapter 8, and the EU Conflict Minerals Regulation provide frameworks for corporate due diligence to avoid contributing to conflict through mineral sourcing.

Security Council Measures and Certification Schemes

The Security Council has adopted measures to sever the link between natural resources and conflict. It imposed embargoes on diamonds from areas controlled by UNITA in Angola (Resolution 1173 (1998)), on rough diamonds from Sierra Leone not certified by the government (Resolution 1306 (2000)), and on diamonds and timber from Liberia. It has established groups of experts to investigate the illegal exploitation of natural resources in the DRC, whose reports named companies and individuals involved in the trade. The concern over **"conflict diamonds"** led to the **Kimberley Process Certification Scheme**, launched in 2003 with the support of the General Assembly, under which participating states undertake to trade in rough diamonds only with other participants and to accompany shipments with certificates guaranteeing that they are conflict-free. The scheme has been credited with reducing the trade in conflict diamonds, but criticised for its narrow definition of conflict diamonds (confined to diamonds used by rebel movements to finance wars against legitimate governments), which excludes human rights abuses by state forces, as in the Marange fields of Zimbabwe, and for its consensus-based decision-making, which has blocked reform. In relation to oil, the Security Council authorised measures against vessels carrying illicitly exported crude oil from Libya (Resolution 2146 (2014)) and condemned trade with the Islamic State (Resolution 2199 (2015)), which had derived substantial revenues from oil fields under its control.

Protection of Energy Infrastructure in Armed Conflict

The protection of energy infrastructure in armed conflict is governed by the fundamental principles of international humanitarian law: **distinction**, **proportionality**, and **precaution**. Under **Additional Protocol I** (1977), attacks may

be directed only against **military objectives**, defined in Article 52(2) as objects which by their nature, location, purpose, or use make an effective contribution to military action and whose total or partial destruction, capture, or neutralisation, in the circumstances ruling at the time, offers a definite military advantage. Civilian objects may not be the object of attack. Energy infrastructure poses particular difficulties because it is typically **dual-use**: electricity grids, fuel depots, and refineries serve both civilian populations and military forces. Some military manuals and commentators have treated electricity generation and distribution facilities as military objectives when they supply military consumers, while others insist that the contribution to military action must be effective and the military advantage definite, and that the destruction of entire grids to coerce civilian populations cannot be justified. Even where a facility qualifies as a military objective, an attack is prohibited if it may be expected to cause incidental civilian harm excessive in relation to the concrete and direct military advantage anticipated (Article 51(5)(b)), and attackers must take feasible precautions to minimise civilian harm (Article 57). The **reverberating effects** of attacks on energy infrastructure — on water supply, sanitation, hospitals, and heating — must be considered in the proportionality assessment.

Additional protections apply to specific categories. **Article 54** of Additional Protocol I prohibits attacking, destroying, removing, or rendering useless objects indispensable to the survival of the civilian population, such as drinking water installations and supplies and irrigation works. **Article 56** provides special protection for **works and installations containing dangerous forces** — dams, dykes, and nuclear electrical generating stations — which shall not be made the object of attack, even where they are military objectives, if such attack may cause the release of dangerous forces and consequent severe losses among the civilian population, subject to narrow exceptions; Article 15 of Additional Protocol II contains a similar rule for non-international armed conflicts. **Articles 35(3) and 55** prohibit methods or means of warfare intended, or which may be expected, to cause widespread, long-term, and severe damage to the natural environment. In 2022, the International Law Commission adopted the **Principles on the Protection of the Environment in Relation to Armed Conflicts**, which address the protection of the environment before, during, and after armed conflict, including in situations of occupation, and the obligations of states and business enterprises.

The War in Ukraine

The war in Ukraine has brought these rules into sharp focus. Russia's systematic missile and drone attacks on **Ukraine's electricity system** since October 2022, which destroyed or damaged a large proportion of Ukraine's generating capacity and caused prolonged blackouts in winter, have been condemned by the UN

Human Rights Monitoring Mission and the Independent International Commission of Inquiry on Ukraine as potentially constituting war crimes and, in some assessments, crimes against humanity. In March 2024, the **International Criminal Court** issued arrest warrants for two senior Russian commanders for the war crimes of directing attacks at civilian objects and causing excessive incidental harm, and the crime against humanity of inhumane acts, in connection with strikes on Ukrainian electricity infrastructure, and in June 2024 it issued further warrants against the former Russian defence minister and the chief of the general staff on similar grounds. Ukraine has also attacked Russian oil refineries and energy infrastructure, justified by Ukraine as targeting the revenues and fuel supplies sustaining Russia's war effort.

The occupation of the **Zaporizhzhia Nuclear Power Plant**, Europe's largest, by Russian forces since March 2022 has created unprecedented nuclear safety risks. The International Atomic Energy Agency has maintained a continuous presence at the plant since September 2022, and its Director General articulated **seven indispensable pillars** of nuclear safety and security during armed conflict and, in May 2023, **five concrete principles** for the protection of the plant, presented to the Security Council, including that there should be no attack from or against the plant and that it should not be used as a base for heavy weapons. The IAEA General Conference and Board of Governors have adopted resolutions calling on Russia to withdraw. The destruction of the **Kakhovka dam** in June 2023, which caused extensive flooding and environmental damage, raised questions under Article 56 of Additional Protocol I and the prohibition of environmental damage. The **International Claims Commission for Ukraine**, established by a convention concluded under the auspices of the Council of Europe in December 2025, together with the **Register of Damage** established in 2023, is intended to address claims for damage caused by Russia's aggression, including damage to energy infrastructure.

Conflict and Energy in Africa

In Africa, conflict and energy have been intertwined in the Niger Delta, where militant groups attacked oil installations and kidnapped workers in the 2000s, until the **Presidential Amnesty Programme** of 2009, and where crude oil theft and illegal refining continue to cause economic and environmental harm; in Libya, where competing authorities have fought over oil fields and terminals and blockaded production; in South Sudan and Sudan; in the eastern DRC, where the conflict involving the M23 armed group, which captured Goma and Bukavu in early 2025, has been linked to the control of mineral resources and trade routes, and where a peace agreement between the DRC and Rwanda was signed in Washington in June 2025, accompanied by discussions of a regional economic integration framework involving minerals; and in northern Mozambique, where

the insurgency in Cabo Delgado led to the suspension of the Mozambique LNG project. The **African Union's Agenda 2063** aspiration of "silencing the guns" and the **Africa Mining Vision** both recognise the importance of breaking the link between resources and conflict.

25.6 Maritime Security and Energy Infrastructure

Chokepoints and the Freedom of Navigation

A large share of the world's oil and LNG is transported by sea, through a small number of **maritime chokepoints**. The **Strait of Hormuz**, between Iran and Oman, through which around a fifth of global oil consumption and a significant share of LNG exports, notably from Qatar, pass each year, is the most important. Others include the **Strait of Malacca**, between Malaysia, Singapore, and Indonesia, the principal route for oil imports to East Asia; the **Bab el-Mandeb**, between Yemen and Djibouti and Eritrea, linking the Red Sea and the Gulf of Aden; the **Suez Canal** and the SUMED pipeline; the **Turkish Straits**, governed by the **Montreux Convention** of 1936; and the **Danish Straits**. The **Panama Canal**, whose capacity was restricted by drought in 2023–2024, illustrates the vulnerability of chokepoints to climate change.

The legal regime of passage through straits used for international navigation is set out in **Part III of UNCLOS**, examined in Chapter 20, which establishes the right of **transit passage** — the exercise of the freedom of navigation and overflight solely for the purpose of continuous and expeditious transit — which shall not be impeded or suspended by the states bordering the strait. Iran, which has signed but not ratified UNCLOS, has maintained that the right of transit passage is a contractual right available only to parties, and that non-parties enjoy only the more limited right of innocent passage, a position contested by the United States and other maritime states, which regard transit passage as reflecting customary international law. Iran has repeatedly threatened to close the Strait of Hormuz in response to sanctions or military pressure, and has seized foreign tankers, including the British-flagged *Stena Impero* in July 2019, two weeks after British forces assisted the Gibraltar authorities in detaining the Iranian tanker *Grace 1* on suspicion of carrying oil to Syria in breach of EU sanctions.

The Tanker War and the *Oil Platforms* Case

During the Iran–Iraq War (1980–1988), both belligerents attacked merchant shipping in the Persian Gulf in the so-called **"Tanker War"**, leading the United States and other states to reflag and escort Kuwaiti tankers. In ***Oil Platforms*** (*Iran v United States*, 2003), the International Court of Justice considered US attacks on Iranian offshore oil platforms in 1987 and 1988, which the United States justified as self-defence in response to attacks on US-flagged and US warships attributed to Iran. The Court held that the attacks could not be justified as

measures necessary to protect the essential security interests of the United States under the 1955 Treaty of Amity, because the United States had not discharged the burden of proving that it had been the victim of armed attacks by Iran and the attacks on the platforms were not necessary or proportionate. The Court also held, however, that the attacks did not breach the treaty's provision on freedom of commerce between the parties, because oil from the platforms was not at the time being exported to the United States owing to a US embargo. The case is significant for the law of self-defence and for the treatment of energy infrastructure as a target.

Red Sea Attacks

From November 2023, the **Houthi** movement in Yemen attacked commercial shipping in the Red Sea and Gulf of Aden, ostensibly in solidarity with Palestinians in Gaza, using missiles, drones, and the hijacking of the vessel *Galaxy Leader*. The attacks caused major shipping lines and many tankers to divert around the Cape of Good Hope, increasing costs and transit times and reducing traffic through the Suez Canal, a significant source of revenue for Egypt. The Security Council, in **Resolution 2722 (2024)**, condemned the attacks and demanded that they cease, noting the right of member states to defend their vessels in accordance with international law. The United States launched **Operation Prosperity Guardian** in December 2023 and, with the United Kingdom, conducted strikes on Houthi targets from January 2024, justified as self-defence; the European Union launched the defensive naval operation **EUNAVFOR Aspides** in February 2024. A ceasefire between the United States and the Houthis was announced in May 2025, though attacks on shipping linked to Israel continued. The Red Sea crisis demonstrated the vulnerability of global energy and trade flows to non-state actors and the limits of naval protection.

Piracy and Maritime Crime in the Gulf of Guinea

The **Gulf of Guinea**, which encompasses the oil-producing states of West and Central Africa, including Nigeria, Angola, Equatorial Guinea, and Ghana, became in the 2010s the world's most dangerous area for piracy and armed robbery at sea, with attacks on tankers, kidnapping of crew for ransom, and theft of oil cargoes ("bunkering"). The legal framework for maritime security in the region, examined in Chapter 20, includes the definition of **piracy** in Article 101 of UNCLOS, which applies only on the high seas and in the exclusive economic zone, and the concept of **armed robbery at sea**, which covers equivalent acts in territorial waters. The **Yaoundé Code of Conduct** (2013), adopted by ECOWAS, ECCAS, and the Gulf of Guinea Commission, established a regional framework for cooperation in the repression of piracy, armed robbery, and illicit maritime activity, and Nigeria's **Suppression of Piracy and Other Maritime**

Offences Act 2019 — the first stand-alone anti-piracy law in West and Central Africa — enabled the prosecution of pirates, with the first convictions obtained in 2020. Combined with the **Deep Blue Project** (Integrated National Security and Waterways Protection Infrastructure) launched in 2021 and international naval cooperation, these measures contributed to a significant decline in incidents in the region.

Protection of Offshore Installations

Offshore oil and gas installations, LNG terminals, and wind farms are potential targets of terrorism, sabotage, and military attack. The **Protocol for the Suppression of Unlawful Acts against the Safety of Fixed Platforms Located on the Continental Shelf** (1988), amended in 2005, requires parties to criminalise acts such as seizing or damaging fixed platforms and to prosecute or extradite offenders. Coastal states may establish **safety zones** of up to 500 metres around artificial islands, installations, and structures in their exclusive economic zones and on their continental shelves under Articles 60 and 80 of UNCLOS. The protection of **critical undersea infrastructure**, including pipelines, power cables, and data cables, has become a priority after the Nord Stream sabotage and repeated incidents of damage to cables and pipelines in the Baltic Sea, including the damage to the Balticconnector gas pipeline between Finland and Estonia in October 2023 by the anchor of a Chinese-owned vessel and to the Estlink 2 power cable in December 2024, after which NATO launched the **Baltic Sentry** mission in January 2025. The limitations of the legal framework — including the limited jurisdiction of coastal states over foreign vessels in the exclusive economic zone, the outdated 1884 Convention for the Protection of Submarine Telegraph Cables, and the difficulties of attribution — have been examined in Chapter 20.

25.7 Energy Supply Disruptions and Emergency Powers

Emergency Powers in Energy Law

Energy supply disruptions — whether caused by geopolitical events, natural disasters, technical failures, or market dysfunction — require rapid governmental responses that may exceed the ordinary powers of regulators. Most legal systems therefore confer **emergency powers** in the energy sector, enabling governments to direct the allocation of supplies, ration consumption, control prices, requisition facilities, suspend market rules, and override contractual arrangements. Such powers raise important questions of constitutional and administrative law: the conditions for their activation, their scope and duration, the safeguards against abuse, the compensation of affected parties, and their compatibility with international obligations under trade and investment law.

National Emergency Legislation

In the **United Kingdom**, the **Energy Act 1976** empowers the Secretary of State, by order, to regulate or prohibit the production, supply, acquisition, or use of fuels and electricity, and to give directions to energy undertakings, when it appears that an actual or threatened emergency affecting fuel or electricity supplies makes it necessary. In the electricity sector, the system operator's emergency procedures, including **demand control** through rota disconnections, are governed by the Grid Code and the **Electricity Supply Emergency Code**. In the **United States**, the **Federal Power Act** (section 202(c)) authorises the Secretary of Energy, in an emergency, to order the temporary generation, delivery, interchange, or transmission of electricity — a power used in 2025 to require the continued operation of certain fossil fuel power plants scheduled for retirement, a controversial use that was challenged by states and environmental groups. The **Defense Production Act 1950** has been invoked to support domestic production of critical minerals and clean energy technologies. The **Energy Policy and Conservation Act 1975** governs drawdowns of the Strategic Petroleum Reserve, as noted in section 25.3.

In **South Africa**, the government declared a **national state of disaster** under the **Disaster Management Act 2002** in February 2023 in response to the electricity crisis and severe load-shedding, enabling exemptions from procurement and other rules to accelerate new generation capacity, and appointed a Minister of Electricity in the Presidency. The declaration was challenged in court and was terminated in April 2023, illustrating the controversy surrounding the use of general emergency legislation to address structural energy problems. In **Nigeria**, the Electricity Act 2023 and the Petroleum Industry Act 2021 confer powers on the regulators to issue orders and directives in the interest of security of supply, and the President's powers under the Constitution and emergency legislation may be invoked in extreme situations; episodes of fuel scarcity and grid collapse — the national grid suffered repeated collapses in 2024 — have prompted calls for clearer emergency response frameworks, including for the restoration of the grid and the protection of critical infrastructure.

European Union Emergency Measures in 2022

The energy crisis of 2022 led the European Union to adopt unprecedented emergency measures on the basis of **Article 122(1) TFEU**, which permits the Council, on a proposal from the Commission, to decide, in a spirit of solidarity between member states, upon measures appropriate to the economic situation, in particular if severe difficulties arise in the supply of certain products, notably in the area of energy. Because Article 122 does not require the participation of the European Parliament, its use was criticised as bypassing democratic procedures. The measures included:

1. **Council Regulation (EU) 2022/1369** on coordinated demand-reduction measures for gas, which established a voluntary target for member states to reduce their gas consumption by fifteen per cent between August 2022 and March 2023 compared with the average of the previous five years, with the possibility of a mandatory target in a "Union alert", and which was extended and later converted into a recommendation;
2. **Council Regulation (EU) 2022/1854** on an emergency intervention to address high energy prices, which required a mandatory reduction of electricity consumption at peak hours, imposed a cap on the market revenues of "inframarginal" generators (such as renewables and nuclear) at EUR 180 per megawatt-hour, with the excess revenues to be redistributed to consumers, and introduced a temporary **solidarity contribution** on the surplus profits of fossil fuel companies;
3. **Council Regulation (EU) 2022/2576** on enhancing solidarity through better coordination of gas purchases, reliable price benchmarks, and exchanges of gas across borders; and
4. **Council Regulation (EU) 2022/2578** establishing a **market correction mechanism** to limit excessive gas prices on the Dutch TTF trading hub.

The solidarity contribution was challenged before the EU General Court by energy companies, including ExxonMobil, which argued that Article 122 could not serve as a legal basis for a tax measure, which would require unanimity under Article 113 or 115 TFEU; the litigation raised fundamental questions about the scope of emergency powers in the EU legal order. Member states also adopted national measures, including windfall taxes, price caps, the nationalisation or bailout of energy companies — notably Germany's nationalisation of **Uniper**, the largest importer of Russian gas, in 2022 — and the use of state aid under the Commission's **Temporary Crisis Framework**. The reform of the electricity market design in 2024, examined in Chapter 24, includes a mechanism for the Council to declare a regional or Union-wide **electricity price crisis**, enabling member states to set regulated prices below cost for households and small businesses.

Electricity System Emergencies

Electricity systems face distinctive emergencies because supply and demand must be balanced instantaneously. Major blackouts — such as the **North-East blackout** in the United States and Canada in August 2003, the **Indian blackout** of July 2012, which affected more than 600 million people, the **Texas** crisis during Winter Storm Uri in February 2021, examined in Chapter 24, and the **Iberian blackout** of 28 April 2025, which left Spain and Portugal without power for most of a day — have led to reviews of reliability standards, system operation, and emergency procedures. In the European Union, **Regulation (EU)**

2019/941 on risk-preparedness in the electricity sector requires member states to identify regional and national electricity crisis scenarios, prepare **risk-preparedness plans**, and cooperate in a spirit of solidarity, including through regional and bilateral measures to assist each other in crises. In the United States, the North American Electric Reliability Corporation develops mandatory reliability standards enforced by FERC, including standards on cold-weather preparedness adopted after Winter Storm Uri. In Africa, the frequent collapse of national grids, as in Nigeria, and the prolonged load-shedding in South Africa between 2007 and 2024 illustrate the consequences of underinvestment and weak maintenance.

Force Majeure and Contractual Responses

Energy supply disruptions have significant contractual consequences, examined in Chapters 6 and 16. Suppliers may invoke **force majeure** clauses to suspend performance, and buyers may invoke them to excuse non-acceptance; the interpretation of such clauses in the context of sanctions, wars, and pandemics has generated substantial litigation and arbitration. Following its reduction of supplies to European customers in 2022, Gazprom invoked force majeure, a claim rejected in several arbitrations. In **June 2024**, an arbitral tribunal awarded **Uniper** damages of more than EUR 13 billion against Gazprom Export for non-delivery of gas, and other European buyers, including RWE, OMV, and Ørsted, brought claims or obtained awards. Gazprom obtained **anti-suit and anti-arbitration injunctions** from Russian courts under Article 248.1 of the Russian Arbitrazh Procedure Code, introduced in 2020, which confers exclusive jurisdiction on Russian courts over disputes involving sanctioned Russian persons, and obtained judgments against European companies for continuing arbitrations. The conflict between Russian courts and international arbitration, examined in Chapter 26, illustrates the fragmentation of the legal order governing energy trade in times of geopolitical confrontation.

Price Controls and Export Restrictions

Energy emergencies also prompt **price controls** and **export restrictions**. Price controls on fuel and electricity, while politically attractive, may cause shortages and deter investment, as Nigeria's long experience with petrol subsidies, examined in Chapter 24, illustrates. Export restrictions, such as **Indonesia's ban on coal exports in January 2022**, imposed to secure supplies for domestic power plants, and China's restrictions on exports of refined products and critical minerals, may conflict with obligations under Article XI of the GATT, subject to the exception in Article XI:2(a) for temporary restrictions applied to prevent or relieve critical shortages of foodstuffs or other products essential to the exporting party, and the general exceptions in Article XX, as examined in Chapter 13. The interpretation of these exceptions in the context of energy emergencies remains largely untested.

25.8 Geopolitical Risk and International Legal Responses

The Geopolitics of the Energy Transition

The energy transition is reshaping energy geopolitics. As the role of oil and gas declines, the geopolitical significance of petrostates may diminish, raising risks of economic and political instability in states dependent on hydrocarbon revenues, including in Africa, as examined in Chapter 23. At the same time, new dependencies are emerging. The supply chains of clean energy technologies are highly concentrated: China accounts for the majority of global refining of lithium, cobalt, and graphite, almost all rare earth processing, and the large majority of solar panel, battery, and electrolyser manufacturing. The **weaponisation** of these dependencies has become a concern: China imposed export controls on gallium and germanium in 2023, graphite in late 2023, antimony in 2024, and certain rare earth elements and related technologies in 2025, in the context of trade and technological rivalry with the United States, as examined in Chapter 22. Clean energy also creates new forms of interdependence through **electricity interconnections** and **hydrogen trade**, which may raise security issues similar to those of gas pipelines.

Critical Minerals Diplomacy

States have responded to the concentration of critical mineral supply chains with a range of legal and policy instruments, examined in Chapter 8. These include industrial policies, such as the US **Inflation Reduction Act**, whose clean vehicle tax credit included critical mineral sourcing requirements and excluded "foreign entities of concern", and the EU **Critical Raw Materials Act**, which sets benchmarks for domestic extraction, processing, and recycling and limits dependence on any single third country to sixty-five per cent of annual consumption of each strategic raw material; international partnerships, such as the **Minerals Security Partnership**, launched by the United States and partners in 2022, the **G7 Critical Minerals Action Plan** adopted in 2025, and bilateral critical minerals agreements, including the agreements of the United States with Japan (2023), Ukraine (April 2025, establishing a joint reconstruction investment fund financed in part by revenues from new natural resource projects), and Australia (October 2025); and strategic partnerships between the European Union and resource-rich states, including memoranda of understanding with the DRC, Zambia, Namibia, and Rwanda. Resource-rich developing states, including in Africa, are seeking to use their mineral endowments to capture more value through local processing and manufacturing, as in the export bans of Indonesia and Zimbabwe, examined in Chapters 8 and 24, and the **Green Minerals Strategy** developed under the auspices of the African Union. The tension between the supply security of importing

states and the developmental aspirations of producing states is a defining feature of critical minerals geopolitics.

Cyber Security of Energy Infrastructure

Energy infrastructure is increasingly exposed to **cyber attacks**. The attacks on **Ukraine's electricity distribution** in December 2015, which caused outages affecting around 225,000 customers, and in December 2016, using the "Industroyer" malware, were the first confirmed cyber attacks causing power outages, and were attributed to Russian military intelligence. The **ransomware attack on Colonial Pipeline** in May 2021, which led the operator of the largest refined products pipeline in the United States to shut down its operations for several days, causing fuel shortages along the East Coast, demonstrated the vulnerability of energy systems to criminal as well as state actors. The **Stuxnet** worm, discovered in 2010, which damaged centrifuges at Iran's Natanz uranium enrichment facility, illustrated the potential for state-sponsored cyber operations to cause physical damage to energy-related infrastructure.

International law applies to cyber operations, as affirmed by the UN Group of Governmental Experts in 2013 and 2015 and the Open-Ended Working Group. The **2015 report of the UN Group of Governmental Experts**, endorsed by the General Assembly, articulated voluntary norms of responsible state behaviour, including that states should not conduct or knowingly support ICT activity contrary to their obligations that intentionally damages critical infrastructure or otherwise impairs its use and operation to provide services to the public, and should take appropriate measures to protect their critical infrastructure. The **Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations** (2017), prepared by an international group of experts, applies the rules on sovereignty, non-intervention, the use of force, and international humanitarian law to cyber operations, concluding, for example, that a cyber operation causing physical damage comparable to a kinetic attack may constitute a use of force. At the regional level, the European Union's **NIS2 Directive** (Directive (EU) 2022/2555) imposes cyber security risk management and incident reporting obligations on entities in critical sectors, including energy, and the **Critical Entities Resilience Directive** (Directive (EU) 2022/2557) requires member states to identify critical entities and ensure their resilience against physical and other threats. In the United States, the Transportation Security Administration issued mandatory **security directives** for pipeline operators after the Colonial Pipeline attack, and NERC's **Critical Infrastructure Protection** standards apply to the bulk electric system. In Nigeria, the **Cybercrimes (Prohibition, Prevention, etc.) Act 2015**, amended in 2024, provides for the designation and protection of critical national information infrastructure, which may include energy systems.

Investment Screening

Concern about foreign control of critical energy infrastructure and resources has led to the expansion of **foreign direct investment screening** regimes. The **Committee on Foreign Investment in the United States** (CFIUS), whose powers were expanded by the Foreign Investment Risk Review Modernization Act of 2018, reviews transactions that could result in foreign control of US businesses or certain non-controlling investments in critical technology, critical infrastructure, and sensitive personal data, including energy infrastructure. The **EU Foreign Direct Investment Screening Regulation** (Regulation (EU) 2019/452), which has been the subject of a revision intended to require all member states to maintain screening mechanisms covering minimum sectors, establishes a cooperation mechanism among member states and the Commission. The **United Kingdom's National Security and Investment Act 2021** requires mandatory notification of acquisitions in seventeen sensitive sectors, including energy and civil nuclear, and has been used to block or condition transactions. Similar regimes exist in Australia, Canada, Germany, and other states. These regimes reflect the securitisation of energy investment and may conflict with investment treaty obligations, although screening measures are generally accompanied by security exceptions. Producing states have also imposed restrictions: Canada announced in 2022 that it would require the divestment by Chinese companies of interests in Canadian critical minerals companies.

International Energy Institutions

The institutional architecture of international energy governance, examined in Chapter 2, plays a role in managing geopolitical risk. The **International Energy Agency** has evolved from an oil-consumers' club into a broader forum for energy security and the energy transition; **OPEC** and **OPEC+** manage oil supply; the **Gas Exporting Countries Forum** brings together major gas producers; the **International Energy Forum** promotes dialogue between producers and consumers; the **International Renewable Energy Agency** promotes renewable energy; the **International Atomic Energy Agency** addresses nuclear safety and security; and the **Energy Charter Treaty**, despite the withdrawals of the European Union and several member states, continues to provide a framework for investment protection and transit among its remaining parties, following the adoption of modernised treaty provisions by the Energy Charter Conference in December 2024. Africa's institutional architecture includes the **African Energy Commission** (AFREC), the African Petroleum Producers' Organization (APPO), the regional power pools, and ECOWAS's Centre for Renewable Energy and Energy Efficiency.

The Fragmentation of the International Energy Order

The geopolitical turbulence of the 2020s has accentuated the **fragmentation** of the international energy order. The sanctions against Russia have created parallel markets, with Russian oil redirected to China, India, and Turkey, often through a "shadow fleet" and payment systems outside Western financial institutions, examined in Chapter 22. The rivalry between the United States and China is leading to the creation of competing supply chains for clean energy technologies and critical minerals. The withdrawal of the United States from the Paris Agreement, effective in January 2026, as examined in Chapter 18, and the use of tariffs and export controls as instruments of economic statecraft have weakened multilateral cooperation. The paralysis of the WTO's dispute settlement mechanism has limited the enforceability of trade rules. Against this background, international law faces the challenge of maintaining a rules-based framework for energy trade, investment, and security that accommodates legitimate security concerns without becoming a cover for protectionism and coercion.

For African states, including Nigeria, the fragmentation of the international order presents both risks and opportunities: risks of being caught in great-power rivalries, of exposure to the extraterritorial effects of sanctions, and of volatility in commodity markets; and opportunities to diversify partnerships, attract investment in critical minerals and gas from competing partners, and leverage their resource endowments for development. The legal frameworks examined in this book — investment law, contract law, trade law, regional integration through the AfCFTA, and the protection of human rights and the environment — provide tools for navigating these choices.

::: casebox {title="Case Study: Russia — Measures concerning Traffic in Transit (WTO Panel, 2019)"} **Background.** From 2014, following the events in Crimea and eastern Ukraine, Russia imposed restrictions on the transit through its territory of goods, including road and rail traffic, from Ukraine destined for Kazakhstan and the Kyrgyz Republic, and bans on the transit of certain goods subject to Russian import bans. Ukraine challenged the measures as inconsistent with the freedom of transit under Article V of the GATT and Russia's accession commitments.

Russia's defence. Russia invoked Article XXI(b)(iii) of the GATT, arguing that the measures were taken in time of an emergency in international relations and were necessary for the protection of its essential security interests, and that Article XXI is entirely self-judging, so that the panel lacked jurisdiction to review its invocation.

Decision. The panel rejected the argument that Article XXI is entirely self-judging. It held that whether a measure is "taken in time of war or other emergency in international relations" is a matter for objective determination by

the panel. An emergency in international relations refers generally to a situation of armed conflict, latent armed conflict, heightened tension or crisis, or general instability engulfing or surrounding a state; political or economic differences between members are not sufficient. The determination of what constitutes the invoking member's "essential security interests" is left to that member, but the obligation of **good faith** requires it to articulate those interests sufficiently to demonstrate their veracity, and requires that the measures meet a **minimum requirement of plausibility** in relation to those interests. Applying these standards, the panel found that the situation between Ukraine and Russia since 2014 constituted an emergency in international relations, and that the transit restrictions were not implausible as measures protective of Russia's essential security interests. Russia's measures were therefore justified.

Significance. The decision established that security exceptions in trade law are subject to a limited but real degree of international review — a principle with significant implications for energy trade restrictions justified on security grounds, including sanctions, export controls, and transit restrictions. Subsequent panels have applied the same approach, but its authority has been contested by the United States. The case illustrates the tension between the sovereignty of states over security matters and the integrity of rules-based trade, which lies at the heart of energy geopolitics. :::

::: exercise {title="Practical Exercise 25: Crisis in the Kessara Corridor (HYPOTHETICAL)"} **This exercise is entirely hypothetical. All states, companies, infrastructure, and events described are fictitious and do not refer to any real country, company, or incident.**

The Kessara Gas Pipeline transports natural gas from the Republic of Ostrava, a major producer, through the Federal Republic of Tamsin, a transit state, to the Union of Belmore, a regional economic organisation of six importing states. The pipeline is governed by an intergovernmental agreement among Ostrava, Tamsin, and Belmore's member states, which provides for uninterrupted transit, prohibits interruption of transit in the event of a dispute, and refers disputes to arbitration. All three states are WTO members and parties to a treaty containing provisions equivalent to Articles 7 and 24 of the Energy Charter Treaty.

Following a political dispute over Tamsin's application for membership of Belmore, Ostrava reduces gas flows through the pipeline by seventy per cent, citing "technical maintenance". Belmore's member states consider the reduction a deliberate act of coercion. In response: (i) Belmore imposes a ban on imports of Ostravan oil and a freeze on the assets of Ostrava's state gas company, OstraGaz, and its directors; (ii) Belmore's largest member state, Varnia, nationalises the local gas trading subsidiary of OstraGaz to secure supplies, offering compensation at "book value"; (iii) Belmore's Council adopts an emergency regulation, without

the involvement of its parliament, imposing a mandatory fifteen per cent reduction in gas consumption and a cap on the revenues of electricity generators; and (iv) an unidentified cyber attack disrupts the control systems of a Tamsin compressor station, halting all flows for three days, which Belmore attributes to Ostravan military intelligence. Ostrava denies responsibility. Meanwhile, an Ostravan court issues an anti-arbitration injunction prohibiting Belmorean buyers from pursuing arbitration against OstraGaz under their long-term supply contracts.

Tasks.

(a) Advise Belmore on whether Ostrava's reduction of gas flows breaches international law, including the principle of non-intervention and the transit obligations, and on the remedies available.

(b) Assess the legality of Belmore's measures in (i) under WTO law and general international law, including the law of countermeasures.

(c) Advise OstraGaz on any claim it may bring in respect of the nationalisation in (ii) and the defences available to Varnia.

(d) Evaluate the legality of the emergency regulation in (iii) under Belmore's constitutional framework (assume it is modelled on the EU Treaties) and under the investment protection treaty.

(e) Analyse the application of international law to the cyber attack in (iv), including attribution, sovereignty, non-intervention, and the use of force.

(f) Advise the Belmorean buyers on the effect of the anti-arbitration injunction and the steps they can take to protect their arbitral rights. :::

Conclusion

Energy geopolitics is the terrain on which the sovereignty of states over their resources and energy policies meets the realities of interdependence, rivalry, and conflict. International law structures this terrain through the principles of sovereignty and non-intervention, the law of sanctions and countermeasures, the trade and investment regimes and their security exceptions, cooperative mechanisms for emergency stocks and solidarity, the protection of energy infrastructure in armed conflict, the law of the sea, and the emerging law of cyber security. The crises of the 2020s — the Russian invasion of Ukraine, the sabotage of Nord Stream, the attacks on Ukraine's grid and the occupation of Zaporizhzhia, the Red Sea attacks, and the weaponisation of critical mineral supply chains — have tested this framework and exposed its gaps: the contested legality of unilateral sanctions, the uncertain limits of security exceptions, the weakness of enforcement against aggressors, the inadequate protection of undersea infrastructure, and the absence of binding rules on cyber attacks against critical infrastructure. At the same time, they have prompted legal innovation, from the EU's emergency measures

and Anti-Coercion Instrument to the ICC's warrants for attacks on energy infrastructure and the IAEA's principles for nuclear safety in armed conflict. For Africa and Nigeria, energy security remains inseparable from domestic energy access, physical security, and regional integration, and the fragmentation of the international order calls for strategic use of the legal tools available. The resolution of the disputes generated by energy geopolitics is the subject of Chapter 26.

CHAPTER SUMMARY

- Energy security is multidimensional — availability, accessibility, affordability, acceptability, and resilience — and its meaning differs for importing, exporting, transit, and energy-poor states; international law affirms state sovereignty over resources, constrained by obligations of non-intervention, trade, investment, and cooperation.
- UN Security Council sanctions are binding and prevail under Article 103 of the Charter; unilateral sanctions may be lawful as retorsion or countermeasures, but their extraterritorial application and humanitarian effects are contested; WTO panels have held that the security exception in Article XXI of the GATT is subject to limited review for good faith and plausibility.
- The IEP Agreement requires IEA members to hold ninety days of net oil imports and has supported five collective actions; EU law requires oil stocks and, since 2022, minimum gas storage levels; Nigeria and most African states lack adequate strategic stocks; stockpiling is being extended to critical minerals.
- Pipelines create mutual dependence; the history of Russian gas supplies to Europe, ending with the cessation of transit through Ukraine in 2025, illustrates the risks; legal responses include diversification standards, solidarity, regulation of third-country pipelines, investment screening, and transit dispute resolution.
- International humanitarian law prohibits pillage, limits the exploitation of resources in occupied territory to usufruct, protects civilian objects and works containing dangerous forces, and requires proportionality in attacks on dual-use energy infrastructure; the ICJ's DRC v Uganda judgments, the ICC's warrants concerning Ukraine's grid, and the IAEA's principles for Zaporizhzhia illustrate the application of these rules.
- Maritime chokepoints, the Tanker War, the Red Sea attacks, and piracy in the Gulf of Guinea demonstrate the vulnerability of seaborne energy trade; UNCLOS transit passage, regional codes, and national anti-piracy laws provide the legal framework.
- Emergency powers enable governments to respond to supply disruptions, but raise questions of legality and accountability, as in the EU's use of Article 122 TFEU in 2022; force majeure, anti-suit injunctions, and arbitration have followed Russian supply cuts.

- The energy transition creates new geopolitical risks in critical minerals, clean technology supply chains, and cyber security, addressed by industrial policies, mineral partnerships, cyber regulation, and investment screening, in an increasingly fragmented international order.

DOCTORAL RESEARCH QUESTIONS

1. Is there a customary international law prohibition of economic coercion through the interruption of energy supplies? Evaluate in the light of state practice from 1973 to the present.
2. Assess the legality of collective countermeasures in response to violations of obligations *erga omnes*, with reference to the energy sanctions against Russia since 2022.
3. To what extent does international humanitarian law adequately protect electricity infrastructure in armed conflict? Evaluate in the light of the war in Ukraine and propose reforms.
4. Should African states establish regional strategic petroleum reserves? Evaluate the legal and institutional options, drawing on the IEA and EU models.
5. Critically assess the adequacy of international law governing the protection of critical undersea energy infrastructure.
6. How should international economic law accommodate the security concerns of states regarding critical mineral supply chains without legitimising protectionism?

FURTHER READING

- Daniel Yergin, *The Prize: The Epic Quest for Oil, Money, and Power* (Simon & Schuster 1991).
- Daniel Yergin, *The New Map: Energy, Climate, and the Clash of Nations* (Penguin Press 2020).
- Barry Barton, Catherine Redgwell, Anita Rønne and Donald N Zillman (eds), *Energy Security: Managing Risk in a Dynamic Legal and Regulatory Environment* (Oxford University Press 2004).
- Michael N Schmitt (ed), *Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations* (Cambridge University Press 2017).
- Meghan L O'Sullivan, *Windfall: How the New Energy Abundance Upends Global Politics and Strengthens America's Power* (Simon & Schuster 2017).
- International Law Commission, *Principles on the Protection of the Environment in Relation to Armed Conflicts* (2022).
- International Energy Agency, *Agreement on an International Energy Program* (1974, as amended), and the IEA's annual *World Energy Outlook*.

26 CHAPTER

International Arbitration and Energy Dispute Resolution

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Classify energy disputes by parties, subject matter, legal basis, and forum, and explain why arbitration predominates in the energy sector.
2. Analyse the legal framework of international commercial arbitration in energy disputes, including arbitration agreements, institutional rules, the seat, and the role of national courts.
3. Explain the procedural conduct of investment treaty arbitration in energy disputes, from notice of dispute to award.
4. Compare the ICSID and UNCITRAL frameworks, including their rules on jurisdiction, procedure, annulment, and enforcement.
5. Evaluate the principles governing jurisdiction, admissibility, and applicable law in energy arbitration.
6. Assess the role of evidence, expert testimony, and the principles and methods of quantifying damages in energy disputes.
7. Analyse the recognition and enforcement of arbitral awards, including against sovereign states and their assets, with particular reference to Nigeria.
8. Evaluate mediation, negotiation, expert determination, and other alternative methods for resolving energy disputes.

Introduction

The energy sector generates disputes of exceptional value, complexity, and duration. Long-term contracts for the exploration and production of petroleum, the construction of power plants and LNG terminals, the sale of gas and electricity, and the transit of energy across borders are exposed to shifts in prices, technology, regulation, and politics over decades. Investments are large, immobile, and vulnerable to host state action once made. Parties come from different legal systems and are often reluctant to submit to each other's courts; states and state-owned

enterprises are frequently involved. These features explain why **international arbitration** has become the predominant method for resolving energy disputes. Energy and natural resources disputes consistently account for a substantial share of the caseloads of the major arbitral institutions: cases concerning oil, gas, and mining, together with electric power and other energy, account for around two-fifths of the historical caseload of the International Centre for Settlement of Investment Disputes (ICSID), and they represent and a significant proportion of cases at the International Chamber of Commerce (ICC), the London Court of International Arbitration (LCIA), and the Stockholm Chamber of Commerce (SCC). Many of the largest awards in the history of arbitration — *Yukos*, *ConocoPhillips v Venezuela*, *Process and Industrial Developments v Nigeria* (subsequently set aside), and the Gazprom-related awards — have arisen from energy disputes.

Earlier chapters have examined the substantive law applied in energy disputes: the protection of investments under treaties, including the institutional overview of investor–state dispute settlement and its reform in Chapter 14; the drafting and interpretation of energy contracts, including dispute resolution clauses, in Chapter 15; and the state–state disputes over maritime boundaries and shared resources in Chapter 20. This chapter focuses on the **process** of energy dispute resolution, from the perspective of the practitioner and the scholar of procedure. It examines the nature and classification of energy disputes (section 26.1), international commercial arbitration (section 26.2), the conduct of investment treaty arbitration (section 26.3), the ICSID and UNCITRAL frameworks (section 26.4), jurisdiction, admissibility, and applicable law (section 26.5), evidence, expert testimony, and damages (section 26.6), the recognition and enforcement of awards (section 26.7), and mediation, negotiation, and alternative dispute resolution (section 26.8). Litigation before national courts, including the jurisdiction of courts over transnational energy disputes and the enforcement of foreign judgments, is examined in Chapter 27.

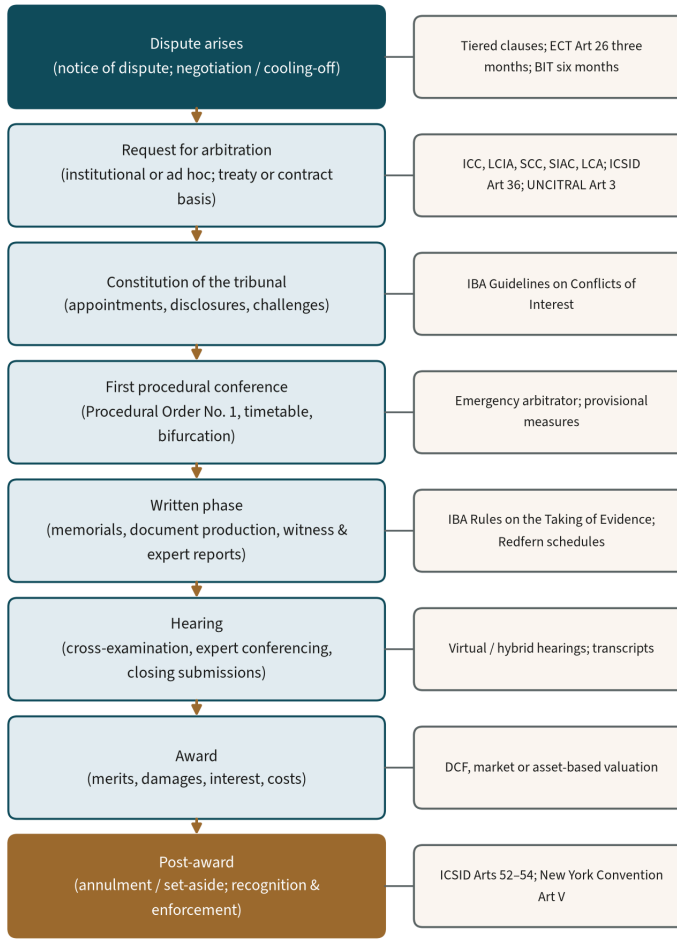


Figure 26.1 The life cycle of an international energy arbitration, with illustrative governing rules. *Source:* prepared for this edition.

26.1 Nature and Classification of Energy Disputes

Why Energy Disputes Are Distinctive

Energy disputes share characteristics that shape the choice and design of dispute resolution mechanisms. First, they typically arise from **long-term relationships** — production sharing contracts of twenty to thirty years, gas sales agreements of twenty years or more, power purchase agreements of fifteen to twenty-five years — in which the parties may need to continue performing while the dispute is resolved, and in which the preservation of the relationship has value. Second, they involve **high stakes**: the amounts in dispute frequently run to hundreds of mil-

lions or billions of dollars, justifying substantial investment in legal representation and expert evidence. Third, they are **technically complex**, involving questions of geology, reservoir engineering, construction, pricing formulae, accounting, and valuation that require specialist expertise. Fourth, they frequently involve **states and state entities**, raising issues of sovereign immunity, public policy, and the interaction of contract and public law. Fifth, they are **politically sensitive**, touching on national sovereignty over resources, public revenues, energy prices, and environmental and social concerns, and attracting public and media attention. Sixth, they are **multi-jurisdictional**, involving parties, assets, and operations in several countries and requiring enforcement across borders.

Classification by Parties

Energy disputes may be classified by the parties involved:

1. **Commercial disputes between private parties**, such as disputes between joint venture partners under joint operating agreements, between buyers and sellers under LNG or crude oil sales contracts, between project owners and EPC contractors, or between operators and service providers. These are generally resolved by international commercial arbitration or national litigation.
2. **Disputes between investors and states or state entities**, which may be contractual (arising from a concession, PSC, or PPA with the state or a state entity), treaty-based (arising from the alleged breach of an investment treaty), or both. These may be resolved by commercial arbitration under the contract, investment treaty arbitration under ICSID or UNCITRAL rules, or national courts.
3. **Disputes between states**, such as disputes over maritime boundaries and shared resources, transit, or the interpretation of treaties, resolved by negotiation, the International Court of Justice, the International Tribunal for the Law of the Sea, arbitration under Annex VII of UNCLOS, inter-state arbitration under bilateral treaties, or the WTO dispute settlement system.
4. **Disputes involving communities and civil society**, such as claims by communities affected by oil pollution against companies, or challenges to project approvals, generally resolved by national courts, non-judicial grievance mechanisms, or human rights bodies, as examined in Chapters 21 and 27.

Classification by Subject Matter

By subject matter, recurring categories of energy disputes include: (a) **price review and price reopener disputes** under long-term gas contracts, examined in Chapter 6, which became the most common category of gas disputes in Europe from the late 2000s as oil-indexed contract prices diverged from hub prices; (b) **take-or-pay and delivery disputes**; (c) **cost recovery and profit oil disputes**

under PSCs, including disputes over the allocation of costs and the computation of the state's share; (d) **fiscal and tax disputes**, including challenges to windfall taxes and the application of stabilisation clauses, examined in Chapters 14 and 16; (e) **expropriation and nationalisation**; (f) **termination and licence revocation disputes**; (g) **construction disputes** over delay, defects, variations, and cost overruns in energy infrastructure projects; (h) **joint venture disputes** over operatorship, sole risk, default, and pre-emption rights; (i) **unitisation and boundary disputes**; (j) **regulatory disputes**, including challenges to tariff decisions and the withdrawal of renewable energy incentives, of which the Spanish renewable energy arbitrations under the Energy Charter Treaty are the leading example; and (k) **environmental and climate-related disputes**, including claims arising from fossil fuel phase-out measures and, increasingly, counterclaims by states for environmental harm.

Classification by Legal Basis and Forum

Energy disputes may also be classified by their legal basis — contract, treaty, statute, or general international law — and by the forum: national courts, international commercial arbitration, investment treaty arbitration, inter-state adjudication, or alternative mechanisms such as expert determination and mediation. The same dispute may involve several legal bases and fora. The expropriation of an oil investment, for example, may give rise to a contractual claim under the PSC's arbitration clause, a treaty claim under a BIT, proceedings in the host state's courts, and enforcement proceedings in several jurisdictions, raising questions about parallel proceedings, *lis pendens*, *res judicata*, and "fork-in-the-road" clauses, addressed in section 26.5.

Why Arbitration Predominates

Arbitration's predominance in energy disputes reflects its advantages over national litigation for transnational disputes: **neutrality**, allowing parties to avoid the courts of either party's state; **party autonomy**, allowing parties to choose arbitrators with relevant expertise, the procedure, the seat, and the language; **enforceability**, through the New York Convention of 1958, which has more than 170 contracting states, and the ICSID Convention, which provides for the enforcement of ICSID awards as if they were final judgments of national courts; **confidentiality**, which is valued in commercial disputes although increasingly limited in investor–state disputes; and **finality**, with limited grounds for challenge. Its disadvantages include cost, duration, the limited availability of appeal, the difficulty of joining third parties, and, in investor–state arbitration, concerns about legitimacy examined in Chapter 14.

26.2 International Commercial Arbitration

The Legal Framework

International commercial arbitration rests on a layered legal framework: the **arbitration agreement**, which is the foundation of the tribunal's jurisdiction; the **arbitration rules** chosen by the parties, whether institutional (such as the ICC, LCIA, SCC, Singapore International Arbitration Centre (SIAC), or Hong Kong International Arbitration Centre rules) or ad hoc (such as the UNCITRAL Arbitration Rules); the **law of the seat** (*lex arbitri*), which governs the arbitral procedure and the supervisory jurisdiction of the courts; the **law governing the arbitration agreement**; the **law governing the substance** of the dispute; and the **international conventions** governing recognition and enforcement, principally the **New York Convention**.

Many states have adopted arbitration legislation based on the **UNCITRAL Model Law on International Commercial Arbitration** (1985, amended 2006), which establishes principles including the separability of the arbitration agreement, the competence of the tribunal to rule on its own jurisdiction (*Kompetenz-Kompetenz*), the limitation of court intervention, equal treatment of the parties, and exhaustive grounds for setting aside awards. **Nigeria** was among the first states to adopt the Model Law, in the **Arbitration and Conciliation Act 1988**, which was repealed and replaced by the **Arbitration and Mediation Act 2023**. The 2023 Act modernises Nigerian arbitration law, introducing provisions on emergency arbitrators, third-party funding (with disclosure obligations), the joinder and consolidation of proceedings, the immunity of arbitrators, an **Award Review Tribunal** that parties may opt into to review awards on the grounds for setting aside, limitation periods that exclude the period of the arbitration, and mediation, and it confirms that awards made in Nigeria shall be recognised and enforced. The **English Arbitration Act 1996**, amended by the **Arbitration Act 2025** — which introduced a default rule that the law governing the arbitration agreement is the law of the seat, a statutory duty of disclosure for arbitrators, a power to make awards on a summary basis, and clarified the framework for jurisdictional challenges — governs arbitrations seated in London, the most popular seat for energy arbitrations, alongside Paris, Geneva, Stockholm, Singapore, and New York.

The Arbitration Agreement

The arbitration agreement must be valid, cover the dispute, and be capable of being performed. In energy contracts, arbitration agreements are often embedded in **multi-tiered dispute resolution clauses**, which require the parties to negotiate, escalate the dispute to senior management, or submit it to mediation or expert determination before commencing arbitration, as examined in Chapter

15. The enforceability of such pre-arbitral steps has been addressed by courts in several jurisdictions: the English courts have held that a clause requiring "friendly discussions" for a specified period before arbitration was enforceable as a condition precedent (*Emirates Trading Agency v Prime Mineral Exports*, 2014), while later holding that non-compliance with pre-arbitral steps goes to admissibility rather than jurisdiction, and is therefore for the tribunal rather than the courts (*NWA v NVF*, 2021; *Republic of Sierra Leone v SL Mining*, 2021). The doctrine of **separability**, recognised in *Fiona Trust & Holding Corp v Privalov* (House of Lords, 2007), which also established a presumption that rational businessmen intend all disputes arising out of their relationship to be decided by the same tribunal, ensures that the arbitration agreement survives allegations that the main contract is invalid.

The **law governing the arbitration agreement** may differ from the law of the main contract and the law of the seat. In ***Enka Insaat ve Sanayi AS v OOO Insurance Company Chubb*** (UK Supreme Court, 2020), a dispute arising from the construction of a power plant in Russia, the Supreme Court held that, where the parties have not specified the law of the arbitration agreement, the law chosen for the main contract will generally apply to the arbitration agreement, and in the absence of such a choice, the arbitration agreement is governed by the law with which it is most closely connected, generally the law of the seat. The Arbitration Act 2025 has since reversed this approach for arbitrations seated in England, providing that the law of the seat applies unless the parties expressly agree otherwise. In ***Kabab-Ji SAL v Kout Food Group*** (UK Supreme Court, 2021), the Supreme Court held that the validity of an arbitration agreement at the enforcement stage is governed by the law chosen by the parties for the main contract, which in that case was English law, and refused enforcement of a French-seated award, while the French courts upheld it — illustrating the potential for divergent outcomes.

Arbitral Institutions and the Energy Sector

The **ICC International Court of Arbitration**, founded in 1923, administers the largest number of international commercial arbitrations, and energy and construction disputes form a significant share of its caseload. Distinctive features of ICC arbitration include the **Terms of Reference**, which define the issues and procedure at the outset, and the Court's **scrutiny** of draft awards. The **LCIA** is favoured for disputes under English law, including many energy and commodity disputes; the **SCC** has historically been prominent in East–West energy disputes, including those involving Russia and the former Soviet states, and in Energy Charter Treaty cases; **SIAC** and the **HKIAC** have grown rapidly as Asian seats; and the **Permanent Court of Arbitration** (PCA) in The Hague administers many investor–state and inter-state arbitrations under the UNCITRAL Rules.

In Africa, the **Lagos Court of Arbitration**, the **Cairo Regional Centre for International Commercial Arbitration**, the **Kigali International Arbitration Centre**, the **Arbitration Foundation of Southern Africa**, and the **Common Court of Justice and Arbitration** (CCJA) of OHADA, which serves the seventeen member states of the Organisation for the Harmonisation of Business Law in Africa, are developing as regional fora. The under-representation of African seats, institutions, and arbitrators in disputes involving African parties — despite the large number of energy and mining disputes involving African states — has been a persistent concern, addressed by initiatives such as the African Promise, launched in 2019 to improve the representation of African arbitrators.

The Role of National Courts

National courts at the seat of arbitration play a **supportive** and **supervisory** role. In their supportive role, courts may appoint arbitrators where the parties' mechanism fails, grant interim measures in support of arbitration, compel the attendance of witnesses, and enforce arbitration agreements by staying court proceedings brought in breach of them, as required by Article II(3) of the New York Convention. Courts may also grant **anti-suit injunctions** restraining a party from pursuing proceedings in breach of an arbitration agreement, a power exercised by the English courts, including in support of foreign-seated arbitrations. In ***UniCredit Bank GmbH v RusChemAlliance LLC*** (UK Supreme Court, 2024), the Supreme Court upheld an anti-suit injunction restraining a Russian company from pursuing proceedings in Russia, under the exclusive jurisdiction conferred on Russian courts in sanctions-related disputes by Article 248.1 of the Russian Arbitrazh Procedure Code, in breach of a Paris-seated ICC arbitration agreement governed by English law, illustrating the role of courts in protecting arbitration from the fragmentation described in Chapter 25. In their supervisory role, courts at the seat may set aside awards on limited grounds, generally those in Article 34 of the Model Law, which mirror the grounds for refusing enforcement in Article V of the New York Convention, examined in section 26.7.

Confidentiality and Transparency

Commercial arbitration is generally **private**, in that hearings are not open to the public, and in many jurisdictions it is also **confidential**, in that the parties are under an implied or express obligation not to disclose the proceedings and award. The scope of confidentiality varies: English law recognises an implied duty of confidentiality subject to exceptions, while Australian law, in *Esso Australia Resources v Plowman* (High Court of Australia, 1995), a case arising from gas supply arbitrations, held that confidentiality is not an essential attribute of arbitration, and that the public interest may require disclosure where a public authority is a party. The tension between confidentiality and the public interest is particularly acute

in energy disputes involving states and public revenues, and the transparency of contract-based disputes involving states has been increasingly questioned, as the *P&ID* case, examined in section 26.7, illustrates.

26.3 Investment Treaty Arbitration

Scope of this Section

Chapter 14 examined the substantive standards of investment protection and gave an institutional overview of investor–state dispute settlement (ISDS), including the concept of "arbitration without privity", the available forums, state counterclaims, transparency, and the reform debate. This section addresses investment treaty arbitration from the procedural perspective of the parties and counsel, following the life cycle of a typical energy investment arbitration, shown in Figure 26.1, and highlighting the strategic decisions that arise at each stage.

Pre-Arbitral Stage: Notice and Cooling-Off

Most investment treaties require the investor to notify the host state of the dispute and to attempt to settle it amicably for a specified **cooling-off period** — typically three to six months — before commencing arbitration. Article 26 of the **Energy Charter Treaty**, for example, permits the investor to submit the dispute to arbitration if it has not been settled amicably within three months of a request for amicable settlement. Some treaties also require the investor to pursue local remedies for a specified period, typically eighteen months, before resorting to arbitration. Tribunals have differed on the consequences of non-compliance with cooling-off periods: some have treated them as jurisdictional requirements whose breach precludes the claim (as in *Burlington Resources v Ecuador*, 2010, in relation to the specific claim not notified), while others have regarded them as procedural requirements whose non-observance may be excused where negotiation would be futile (as in *SGS v Pakistan*, 2003). The prudent course is to comply strictly with the notice requirements and to document the attempts at settlement, which may also lead to a negotiated outcome, examined in section 26.8.

The pre-arbitral stage also involves strategic decisions on **structuring and standing**: whether the claimant is a protected investor under the treaty, examined in section 26.5; which treaty and forum to invoke where several are available; whether to bring contract claims, treaty claims, or both; whether to seek **third-party funding**; and how to preserve evidence and manage the continuing relationship with the host state. Investors must also consider the effect of **fork-in-the-road** clauses, under which the choice of domestic courts may preclude subsequent treaty arbitration, and **waiver** requirements, such as Article 1121 of the former NAFTA and its successors, which require the investor to waive the right to pursue domestic proceedings concerning the same measure.

Commencement and Constitution of the Tribunal

Arbitration is commenced by a **request for arbitration** under Article 36 of the ICSID Convention, which the Secretary-General of ICSID registers unless the dispute is manifestly outside the jurisdiction of the Centre, or by a **notice of arbitration** under Article 3 of the UNCITRAL Arbitration Rules. Tribunals usually comprise three arbitrators, one appointed by each party and the president by agreement or by an appointing authority — the Chair of the ICSID Administrative Council in ICSID cases, or the Secretary-General of the PCA or an authority designated by him in UNCITRAL cases.

The **independence and impartiality** of arbitrators are fundamental. Arbitrators must disclose circumstances that may give rise to justifiable doubts about their independence or impartiality, and parties may challenge arbitrators on such grounds. The **IBA Guidelines on Conflicts of Interest in International Arbitration** (2014, revised 2024) classify circumstances into red, orange, and green lists according to their seriousness, and are widely used as a reference. Concerns about **repeat appointments** of the same arbitrators by the same parties or law firms, and about **"double-hatting"** — the practice of individuals acting as arbitrators in some cases and as counsel in others raising similar issues — have prompted the development of the **Code of Conduct for Arbitrators in International Investment Dispute Resolution**, adopted by UNCITRAL in 2023, which limits double-hatting and strengthens disclosure obligations. Under the ICSID Convention, a challenge must show a "manifest lack" of the qualities required by Article 14(1), which has been interpreted as requiring the lack to be evident or obvious, and is decided by the unchallenged arbitrators or, if they are equally divided or the challenge concerns a majority, by the Chair of the Administrative Council.

The First Session and Procedural Framework

Following its constitution, the tribunal holds a **first session** or case management conference, typically by video conference, to establish the procedural framework, recorded in **Procedural Order No. 1**. The order addresses the applicable rules, the language and place of proceedings, the timetable for written submissions, the rules on document production and evidence (usually by reference to the **IBA Rules on the Taking of Evidence in International Arbitration**, as guidelines), confidentiality and transparency, the organisation of hearings, and the allocation of costs. A key early decision is whether to **bifurcate** the proceedings — addressing jurisdiction, or liability, separately from quantum — or to hear all issues together. Bifurcation may save costs if a preliminary issue disposes of the case, but may lengthen the proceedings if it does not; under Rule 44 of the 2022 ICSID Arbitration Rules, the tribunal considers all relevant circum-

stances, including whether bifurcation would materially reduce time and cost and whether the issues are so intertwined as to make bifurcation impractical.

States may also file **preliminary objections** that a claim is manifestly without legal merit, under Rule 41 of the 2022 ICSID Arbitration Rules (formerly Rule 41(5)), which must be decided expeditiously, and many modern treaties provide for similar expedited objections. **Security for costs** — an order requiring the claimant to provide security for the respondent's potential costs — has become an important issue in cases funded by third parties, where an impecunious claimant may be unable to pay an adverse costs award; Rule 53 of the 2022 ICSID Rules expressly empowers tribunals to order security for costs, taking into account the party's ability to comply with an adverse costs decision, the existence of third-party funding, and other relevant circumstances.

Provisional Measures

Tribunals may recommend or order **provisional measures** to preserve the rights of the parties pending the award. Under Article 47 of the ICSID Convention, a tribunal may "recommend" provisional measures, but tribunals have consistently held that such recommendations are binding. Provisional measures in energy disputes have included orders to preserve the status quo, to refrain from enforcing tax assessments or criminal proceedings that would aggravate the dispute, and to preserve evidence. In ***Perenco Ecuador v Ecuador*** (2009), the tribunal recommended that Ecuador refrain from enforcing a windfall levy against the investor's oil assets pending the arbitration; Ecuador's subsequent seizure of the investor's oil production in disregard of the recommendation was later taken into account in the tribunal's assessment. In ***Burlington v Ecuador***, the tribunal similarly recommended that Ecuador refrain from seizing the investor's oil. Tribunals have also considered requests to suspend domestic criminal proceedings against investors' personnel, granting such measures only in exceptional circumstances, out of respect for the state's sovereign prerogative to prosecute crimes.

The Written and Oral Phases

The written phase typically involves two rounds of submissions — the claimant's **memorial** and the respondent's **counter-memorial**, followed by a **reply** and a **rejoinder** — each accompanied by witness statements, expert reports, legal authorities, and documentary exhibits. Between the rounds, the parties exchange requests for the production of documents, usually in the form of a **Redfern schedule** setting out the documents requested, the justification for each request, the objections, and the tribunal's decision. The oral **hearing**, which in an energy arbitration may last two or three weeks, involves opening statements, the examination and cross-examination of fact and expert witnesses, and closing statements, followed by post-hearing briefs and submissions on costs. Since the COVID-19

pandemic, **virtual and hybrid hearings** have become common, and tribunals have generally rejected objections that virtual hearings violate the right to be heard.

Non-Disputing Parties and Transparency

Investment treaty arbitration differs from commercial arbitration in its increasing **transparency**. The **UNCITRAL Rules on Transparency in Treaty-based Investor-State Arbitration** (2014), which apply to UNCITRAL arbitrations under treaties concluded after April 2014 and to earlier treaties under the **Mauritius Convention on Transparency** (2014), require the publication of key documents and awards, open hearings, and the possibility of submissions by non-disputing parties. The 2022 ICSID Rules provide for the publication of awards (with the parties' consent deemed given unless they object within a specified period), the participation of **non-disputing parties** such as NGOs and community representatives through *amicus curiae* submissions, and submissions by the non-disputing treaty party. In energy disputes with significant environmental or community dimensions, such as disputes involving mining projects in Latin America, *amicus* submissions by affected communities and NGOs have been admitted, though their influence on outcomes is debated.

Third-Party Funding

Third-party funding, in which a funder with no other interest in the dispute finances the claimant's costs in return for a share of any recovery, has become common in investment arbitration, including in energy disputes, enabling claimants to pursue claims they could not otherwise afford and allowing companies to manage the financial risk of litigation. Funding raises concerns about conflicts of interest between funders and arbitrators, the encouragement of speculative claims, and the claimant's ability to pay an adverse costs award. Rule 14 of the 2022 ICSID Rules requires parties to disclose the name and address of any third-party funder, and many treaties and institutional rules now contain similar provisions; Nigeria's Arbitration and Mediation Act 2023 also requires disclosure of funding arrangements. The P&ID case, examined in section 26.7, illustrates the role of funders in high-value disputes involving states.

26.4 ICSID and UNCITRAL Arbitration Frameworks

The ICSID Convention

The **Convention on the Settlement of Investment Disputes between States and Nationals of Other States** (ICSID Convention), concluded in 1965 under the auspices of the World Bank and in force since 1966, established the **International Centre for Settlement of Investment Disputes** as an autonomous international institution. It has more than 150 contracting states, including

Nigeria, which signed in 1965 and ratified in the same year, and which gave the Convention force of law through the **International Centre for Settlement of Investment Disputes (Enforcement of Awards) Act**. The Convention provides a self-contained, **delocalised** system of arbitration, insulated from the law and courts of any national seat: ICSID awards are subject only to the remedies provided in the Convention itself, and contracting states must recognise and enforce the pecuniary obligations imposed by awards as if they were final judgments of their own courts (Article 54). The Convention's procedural framework is supplemented by the **ICSID Arbitration Rules**, comprehensively revised in 2022 to improve efficiency and transparency, with timelines for procedural steps and awards, provisions on expedited arbitration, disclosure of third-party funding, and security for costs.

Jurisdiction of the Centre

Under **Article 25(1)** of the ICSID Convention, the jurisdiction of the Centre extends to any **legal dispute arising directly out of an investment**, between a **contracting state** (or a constituent subdivision or agency designated to the Centre) and a **national of another contracting state**, which the parties have **consented in writing** to submit to the Centre. Each element has generated jurisprudence. The term "investment" is not defined in the Convention, and tribunals have debated whether it has an objective meaning; the **Salini test**, derived from *Salini Costruttori v Morocco* (2001), identifies the characteristics of an investment as a contribution, a certain duration, the assumption of risk, and, in some formulations, a contribution to the host state's development, while other tribunals have treated these as indicative features rather than jurisdictional requirements. Energy projects generally satisfy any formulation of the test, though questions have arisen concerning pre-investment expenditures, sale contracts, and financial instruments. The requirement that the investor be a **national of another contracting state** excludes nationals of the host state, subject to Article 25(2)(b), which allows a locally incorporated company under foreign control to be treated as a foreign national if the parties so agree — a provision of particular importance in the energy sector, where investments are often made through local subsidiaries, and which states may agree to in the investment contract. Consent may be given in an investment contract, in domestic investment legislation, or in a treaty; once given, it cannot be withdrawn unilaterally (Article 25(1)).

Annulment

ICSID awards are not subject to appeal or to setting aside by national courts, but may be **annulled** by an *ad hoc* committee of three persons appointed by the Chair of the Administrative Council, on the exhaustive grounds in **Article 52(1)**: that the tribunal was not properly constituted; that it **manifestly exceeded its**

powers; that there was corruption on the part of a member; that there has been a **serious departure from a fundamental rule of procedure**; or that the award has **failed to state the reasons** on which it is based. Annulment is not an appeal: a committee may not review the merits of the award or substitute its own view of the law or facts. Early annulment decisions, including *Klöckner v Cameroon* (1985) and *Amco v Indonesia* (1986), were criticised for intrusive review, but subsequent committees have generally adopted a more restrained approach. Failure to apply the applicable law may constitute a manifest excess of powers, but an error in its application does not. Notable energy-related annulment decisions include the partial annulment of the award in *Occidental v Ecuador* (2015), in which the committee reduced the damages of around USD 1.8 billion by around forty per cent on the ground that the tribunal had manifestly exceeded its powers by awarding compensation for a share of the investment that had been transferred to a third party in breach of Ecuadorian law, and the annulment of awards in several Spanish renewable energy cases, including *Eiser v Spain* (2020), on the ground that an arbitrator had failed to disclose a relationship with the claimant's damages experts, while most other Spanish awards survived annulment. Where an award is annulled, either party may resubmit the dispute to a new tribunal. Annulment proceedings may result in a **stay of enforcement** of the award, which the committee may condition on the provision of security.

Recognition and Enforcement under ICSID

Article 54 requires all contracting states to recognise ICSID awards as binding and to enforce their pecuniary obligations as if they were final judgments of their courts, with no grounds for refusal comparable to those in Article V of the New York Convention. Article 53 provides that awards are binding and not subject to appeal or any other remedy except those provided for in the Convention. However, **Article 55** preserves the law of each contracting state on the immunity of states from execution, so that a successful investor may face obstacles in executing an award against state assets, examined in section 26.7. The European Commission's position that intra-EU awards are contrary to EU law, following *Achmea* (2018) and *Komstroy* (2021), examined in Chapter 14, has led to conflicts between the obligation to enforce ICSID awards and EU law, as courts in the United States, the United Kingdom, and Australia have enforced intra-EU awards against Spain while EU courts have refused to do so. In ***Spain v Infrastructure Services Luxembourg*** (2023), the High Court of Australia held that Spain's agreement to Article 54 amounted to a waiver of its immunity from recognition and enforcement proceedings, though not from execution, and in 2024 the English Court of Appeal reached a similar conclusion in *Infrastructure Services Luxembourg v Spain*, a question subsequently taken to the UK Supreme Court.

The ICSID Additional Facility

The **ICSID Additional Facility Rules**, revised in 2022, provide for the administration by ICSID of proceedings that fall outside the scope of the Convention, including disputes where either the host state or the investor's home state is not a party to the Convention, and disputes that do not arise directly out of an investment. Additional Facility awards are not governed by the Convention's annulment and enforcement regime, but are subject to the law of the seat and to the New York Convention. The Additional Facility was used extensively in disputes against Mexico and Canada under NAFTA, since neither was a party to the ICSID Convention at the time.

The UNCITRAL Arbitration Rules

The **UNCITRAL Arbitration Rules**, adopted in 1976 and revised in 2010, 2013 (to incorporate the transparency rules), and 2021 (adding expedited arbitration rules), provide a framework for ad hoc arbitration that may be used in both commercial and investor–state disputes. Many investment treaties, including the Energy Charter Treaty, give investors a choice between ICSID and UNCITRAL arbitration. UNCITRAL arbitrations are often administered by the PCA, which also acts as appointing authority. Unlike ICSID arbitrations, UNCITRAL arbitrations have a **legal seat**, whose law and courts govern the procedure and supervise the award; the award may be set aside by the courts of the seat and is enforced under the New York Convention. The choice of seat in investor–state UNCITRAL arbitration therefore has important consequences, as the *Yukos* litigation in the Dutch courts — in which the Hague District Court set aside the awards in 2016, the Hague Court of Appeal reinstated them in 2020, and, after the Dutch Supreme Court remitted the fraud allegation in 2021, the Amsterdam Court of Appeal rejected Russia's remaining arguments in 2024 — illustrates.

Choosing Between ICSID and UNCITRAL

Investors and states choosing between ICSID and UNCITRAL consider several factors: the **delocalised** character of ICSID arbitration, which excludes review by national courts and provides for enforcement without the New York Convention defences, against the possibility of annulment by an ICSID committee; the **transparency** of ICSID proceedings and the publication of awards; the **institutional support** and fixed fee structure of ICSID; the **jurisdictional requirements** of Article 25, which do not apply in UNCITRAL arbitration; the **supervisory role of the seat** in UNCITRAL arbitration, which may be advantageous or disadvantageous depending on the seat; and the **enforcement** strategy, including the location of the state's assets and the attitude of the courts there to sovereign immunity.

26.5 Jurisdiction, Admissibility, and Applicable Law

Jurisdiction and Admissibility Distinguished

Objections to the hearing of a claim may go to **jurisdiction** — the power of the tribunal to decide the dispute — or to **admissibility** — whether the claim, although within the tribunal's jurisdiction, should be heard. The distinction, developed in inter-state litigation and adopted in investment arbitration, has practical consequences: decisions on jurisdiction are subject to review by courts and annulment committees, whereas decisions on admissibility are generally regarded as matters for the tribunal; jurisdictional objections must generally be raised at an early stage; and a lack of jurisdiction cannot be cured by the parties' conduct in the same way as an admissibility defect. Objections concerning the existence and scope of consent, the nationality of the investor, the existence of a protected investment, and the temporal scope of the treaty are jurisdictional; objections concerning the ripeness of a claim, compliance with cooling-off periods (on some views), the exhaustion of local remedies where required, abuse of process, and the effect of parallel proceedings are commonly treated as going to admissibility.

Consent and its Limits

Jurisdiction in arbitration rests on **consent**. In contract-based arbitration, consent is expressed in the arbitration agreement; in treaty-based arbitration, the state's standing offer to arbitrate in the treaty is accepted by the investor when it commences arbitration. The scope of consent depends on the terms of the offer, which may be limited to certain disputes (for example, disputes concerning the amount of compensation for expropriation, as in some older Soviet-era and Chinese treaties), to investments made in accordance with host state law, or to disputes arising after the treaty's entry into force. The **Energy Charter Treaty** contains an unconditional consent in Article 26(3), subject to the possibility for states listed in Annex ID to exclude disputes previously submitted to other fora, and for states listed in Annex IA to exclude claims under the last sentence of Article 10(1) (the "umbrella clause"). The ECT's **provisional application** under Article 45, examined in Chapter 14, was the basis for the *Yukos* tribunals' jurisdiction over Russia, which had signed but not ratified the Treaty, and the interpretation of the "limitation clause" in Article 45(1) — under which provisional application applies to the extent that it is not inconsistent with the signatory's constitution, laws, or regulations — was central to the litigation in the Dutch courts.

Protected Investors: Nationality and Corporate Structuring

Investment treaties protect investors who are nationals of the other contracting state. For natural persons, nationality is determined by the law of the state concerned, and dual nationals of the host state may be excluded, as under

Article 25(2)(a) of the ICSID Convention. For companies, most treaties define nationality by reference to the place of incorporation, sometimes combined with a requirement of seat or substantial business activities, or subject to a **denial of benefits** clause allowing the host state to deny treaty protection to companies owned or controlled by nationals of third states or of the host state that have no substantial business activities in the home state. Article 17 of the Energy Charter Treaty contains such a clause, which the *Plama v Bulgaria* tribunal (2005) held operates prospectively from the time it is invoked, rather than retrospectively — a conclusion rejected by later treaties that expressly allow retrospective denial.

The protection of investors who channel their investments through holding companies in states with favourable treaties — **treaty planning** or "nationality planning" — is a recurring issue in energy disputes, where investments are commonly held through intermediate companies in jurisdictions such as the Netherlands, Luxembourg, Cyprus, and Mauritius. Tribunals have generally accepted that investors may structure their investments to obtain treaty protection, as in *Aguas del Tunari v Bolivia* (2005), but have held that a restructuring undertaken when a dispute is foreseeable, for the purpose of gaining access to treaty protection, constitutes an **abuse of process**. In ***Philip Morris Asia v Australia*** (2015), the tribunal declined jurisdiction because the restructuring had occurred when the dispute was reasonably foreseeable; in ***Mobil v Venezuela*** (2010), arising from the nationalisation of oil investments, the tribunal held that it had jurisdiction over disputes arising after the restructuring of the investment through a Dutch holding company, but not over disputes that already existed at the time of the restructuring; and in ***Pac Rim Cayman v El Salvador*** (2012), arising from a gold mining project, the tribunal found that a change of nationality to gain access to CAFTA arbitration was an abuse of process as regards disputes that had already arisen. The **"shell company"** problem has led to the inclusion in new treaties of substantial business activity requirements and denial of benefits clauses.

Protected Investments: Legality and Origin

Many treaties protect only investments made **"in accordance with the law"** of the host state, and tribunals have held, even in the absence of such a clause, that investments procured through **corruption or fraud** are not protected. In ***World Duty Free v Kenya*** (2006), the tribunal held that a contract procured by a bribe to the President of Kenya was unenforceable as contrary to international public policy. In ***Metal-Tech v Uzbekistan*** (2013), the tribunal declined jurisdiction because the investment had been established through corrupt payments, notwithstanding that the state's officials had been the recipients of the bribes. In the energy sector, allegations of corruption in the procurement of licences and contracts are frequently raised by states as defences, as examined in Chapter 22,

raising difficult questions about the standard and burden of proof — tribunals have variously applied a heightened standard of "clear and convincing evidence" or the ordinary balance of probabilities, sometimes drawing inferences from "red flags" — and about the fairness of allowing a state to rely on the corruption of its own officials. The P&ID case, examined in section 26.7, illustrates the consequences of corruption for the enforceability of awards.

Parallel Proceedings, Contract Claims, and Treaty Claims

Energy investments typically involve both a **contract** with the state or a state entity, often containing its own dispute resolution clause, and **treaty** protection. The relationship between contract and treaty claims was clarified in ***Compañía de Aguas del Aconquija and Vivendi Universal v Argentina*** (*Vivendi I* annulment decision, 2002), in which the committee held that whether there has been a breach of a treaty and whether there has been a breach of contract are different questions, each determined by reference to its own proper law, and that a contractual forum selection clause does not preclude a treaty tribunal from deciding a treaty claim, even if it requires consideration of the contract. **Umbrella clauses**, under which the host state undertakes to observe any obligation it has entered into with regard to investments, may elevate contractual breaches to treaty breaches, though tribunals have differed on their scope, as in the contrasting decisions in *SGS v Pakistan* (2003) and *SGS v Philippines* (2004), examined in Chapter 14.

Parallel proceedings may also arise when an investor, or its affiliates at different levels of the corporate chain, bring claims under different treaties, or when the investor pursues both domestic litigation and international arbitration. The doctrines of ***lis pendens*** and ***res judicata***, the **fork-in-the-road** clause, and the prohibition of **abuse of process** may limit duplicative claims. In *Orascom v Algeria* (2017), the tribunal held that the claimant's pursuit of claims at several levels of its corporate structure, under different treaties, in relation to the same measures and harm, was an abuse of rights, and declined to hear the claim. The **consolidation** of related proceedings, available under some treaties and institutional rules, may reduce the risk of inconsistent decisions.

Applicable Law

The law applicable to the substance of the dispute depends on the basis of the claim and the forum. In **commercial arbitration**, the tribunal applies the law chosen by the parties or, in the absence of a choice, the law or rules of law it determines to be appropriate, as provided in the applicable arbitration rules and the *lex arbitri*; energy contracts commonly choose English or New York law, the law of the host state, or the law of the host state combined with international law, as examined in Chapter 15. In **ICSID arbitration**, **Article 42(1)** of the

Convention provides that the tribunal shall decide the dispute in accordance with the rules of law agreed by the parties, and, in the absence of agreement, the law of the contracting state party to the dispute (including its rules on the conflict of laws) and such rules of international law as may be applicable. Tribunals have interpreted the second sentence as giving international law a **supplemental and corrective** role, so that host state law inconsistent with international law will not be applied, as the annulment committees in *Klöckner* and *Amco* held, and as later tribunals, including the *Wena Hotels v Egypt* annulment committee (2002), affirmed. In **treaty arbitration**, the treaty itself is the primary source of applicable law, interpreted in accordance with the **Vienna Convention on the Law of Treaties** (Articles 31 to 33), supplemented by general international law; Article 26(6) of the Energy Charter Treaty provides that the tribunal shall decide the issues in accordance with the Treaty and applicable rules and principles of international law. Host state law remains relevant to incidental questions, such as the existence and scope of the investor's property rights, the legality of the investment, and the content of contractual rights.

The Interaction of International Law and Other Regimes

Energy arbitrations increasingly raise questions about the interaction between investment law and other areas of international law, including human rights, environmental law, and climate change. States have invoked their environmental and human rights obligations as defences or as the basis for **counterclaims**. In ***Burlington v Ecuador*** (2017) and ***Perenco v Ecuador*** (2015 interim decision and 2019 award), the tribunals upheld Ecuador's environmental counterclaims and awarded it compensation for environmental damage caused by the investors' oil operations in the Amazon, applying Ecuadorian environmental law. In ***Urbaser v Argentina*** (2016), a water concession case, the tribunal accepted jurisdiction over a human rights counterclaim, though it rejected it on the merits. The relevance of the **Paris Agreement** and the obligations of states in respect of climate change, clarified by the International Court of Justice's advisory opinion of July 2025, examined in Chapter 18, to the interpretation of investment treaties and the assessment of claims arising from climate measures is likely to be a major issue in the coming decade, as in the claims brought by RWE and Uniper against the Netherlands in respect of its coal phase-out, which were withdrawn in 2022–2023 after the Court of Justice's *Komstroy* judgment, and the claims against Canada and Slovenia concerning restrictions on fossil fuel projects.

26.6 Evidence, Expert Testimony, and Damages

Evidence in International Arbitration

International arbitration combines elements of the **common law** and **civil law** traditions of evidence. Tribunals generally have broad discretion over the admis-

sibility, relevance, materiality, and weight of evidence, under Article 34 of the ICSID Arbitration Rules (2022 Rule 36) and Article 27(4) of the UNCITRAL Rules. The **IBA Rules on the Taking of Evidence in International Arbitration** (2010, revised 2020) reflect a compromise between traditions: they provide for the production of documents, but only of narrow and specific categories that are relevant to the case and material to its outcome, rejecting the broad discovery of US litigation; they treat written witness statements as the evidence-in-chief, with oral evidence focused on cross-examination; and they provide for party-appointed and tribunal-appointed experts. The **Prague Rules on the Efficient Conduct of Proceedings in International Arbitration** (2018) offer an alternative, more inquisitorial model, emphasising the active role of the tribunal and limiting document production.

The **burden of proof** generally lies on the party asserting a fact (*actori incumbit probatio*), and the standard of proof is usually the balance of probabilities, although tribunals have applied heightened standards to allegations of fraud and corruption, as noted in section 26.5. Tribunals may draw **adverse inferences** from a party's failure to produce documents or witnesses within its control. In energy disputes, key evidence often includes technical data on reserves and production, contractual and regulatory correspondence, internal valuation documents, board minutes, and, in disputes involving states, government records, which states may withhold on grounds of political or institutional sensitivity or confidentiality, as recognised in Article 9 of the IBA Rules.

Expert Evidence

Expert evidence is central to energy arbitration. Experts address technical issues — the geology and reserves of a petroleum field, the design and construction of a power plant, the causes of delay or defects, the operation of pricing mechanisms — and, most importantly, the **quantum** of damages. In the common practice of international arbitration, each party appoints its own experts, who submit reports and are cross-examined at the hearing. The tendency of party-appointed experts to reach widely divergent conclusions, which may undermine their credibility and the tribunal's confidence, has led to procedural innovations. **Expert conferencing** or **"witness conferencing"** (sometimes called "hot-tubbing"), in which experts in the same discipline give evidence together and are questioned by the tribunal and counsel, can narrow the issues and expose the reasons for disagreement. **Joint expert reports**, prepared after meetings between the experts, identify the matters on which they agree and disagree. Tribunals may also appoint their own experts, under Article 6 of the IBA Rules, Article 29 of the UNCITRAL Rules, or the ICSID Rules, though this practice is less common in investment arbitration. The **CIArb Protocol for the Use of Party-Appointed**

Expert Witnesses in International Arbitration and the IBA Rules emphasise the expert's overriding duty of independence and objectivity.

Principles of Compensation

The principles governing compensation differ depending on whether the claim is for a lawful expropriation or for an internationally wrongful act. For **lawful expropriation**, investment treaties typically require compensation equivalent to the **fair market value** of the investment immediately before the expropriation or before it became publicly known, often expressed in the **Hull formula** of "prompt, adequate and effective" compensation, as examined in Chapter 14. For **unlawful acts**, including unlawful expropriation and breaches of other treaty standards, the customary standard is **full reparation**, as articulated by the Permanent Court of International Justice in ***Factory at Chorzów*** (1928): reparation must, as far as possible, wipe out all the consequences of the illegal act and re-establish the situation which would, in all probability, have existed if that act had not been committed — a principle codified in Articles 31 and 34 to 36 of the ILC Articles on State Responsibility. The practical difference is that, for unlawful expropriation, the investor may be entitled to the value of the investment at the date of the award, if higher than at the date of expropriation, together with consequential losses. In ***ADC v Hungary*** (2006), the tribunal applied this principle to award the value of the investment at the date of the award, which had increased substantially, and in ***ConocoPhillips v Venezuela*** (2019), the tribunal, having found that Venezuela had expropriated unlawfully by failing to negotiate in good faith for compensation, valued the investments at the date of the award, taking into account subsequent oil price movements.

For breaches of standards other than expropriation, such as fair and equitable treatment, tribunals generally award damages measured by the difference between the investor's position with and without the breach, which may require the valuation of the investment in the "but-for" scenario and the "actual" scenario. **Causation** must be established: the investor must show that the loss was caused by the state's breach rather than by other factors, such as market conditions or the investor's own conduct. **Contributory fault** may reduce the award, as in *Occidental v Ecuador* (2012), in which the tribunal reduced the damages by twenty-five per cent on account of the investor's breach of its contract by transferring an interest without authorisation, and in *Yukos*, in which the tribunals reduced the damages by twenty-five per cent on account of the claimants' contributory fault, including abuse of tax regimes.

Valuation Methods

The principal methods of valuation used in energy arbitrations, summarised in Figure 26.2, are:

1. **Income-based methods**, principally the **discounted cash flow** (DCF) method, which calculates the present value of the expected future net cash flows of the investment, discounted at a rate reflecting the time value of money and the risks of the investment. DCF is the preferred method for valuing going concerns with a record of profitability, such as producing oil fields and operating power plants, and is the standard approach of the energy industry. It is, however, highly sensitive to assumptions about future prices, production volumes, costs, and the discount rate — including the **country risk premium** — and disputes over these assumptions frequently account for most of the difference between the parties' valuations.
2. **Market-based methods**, which value the investment by reference to **comparable transactions**, trading multiples of comparable listed companies, or bona fide offers for the investment, adjusted for differences in timing and characteristics.
3. **Asset-based or cost methods**, which value the investment by reference to the amount invested (**sunk costs**), the book value, or the replacement cost. Tribunals have preferred these methods where the investment had no record of profitability, as in early-stage projects, where future cash flows are too speculative to be estimated with reasonable certainty.
4. **Loss of opportunity**, which values the chance of obtaining a benefit, discounted by the probability of its realisation, and has been applied by some tribunals to exploration-stage investments where the existence of commercially viable reserves was uncertain, although others have rejected it.

The **"reasonable certainty"** standard for future profits has been decisive in mining and exploration disputes. In ***Tethyan Copper v Pakistan*** (2019), arising from the denial of a mining lease for the Reko Diq copper and gold deposit, the tribunal accepted a DCF valuation of the undeveloped project, on the basis that the deposit had been extensively explored and a feasibility study completed, awarding around USD 5.9 billion including interest — an award subsequently resolved by a settlement under which the project was reconstituted with Barrick Gold and Pakistani state entities, as noted in Chapter 7. By contrast, tribunals in other mining cases, such as *Crystallex v Venezuela* (2016) and *Rusoro v Venezuela* (2016), used a range of methods, including market-based approaches, owing to the lack of production history.

Interest and Costs

Interest may account for a substantial part of the total award in energy disputes, which often last for many years. Tribunals award **pre-award interest** from the date of the breach or valuation to the date of the award, and **post-award interest** until payment, and modern practice favours **compound interest**, as reflecting commercial reality, rather than the simple interest traditionally awarded. The

choice of rate — a risk-free rate, the investor's cost of borrowing, or a commercial rate — can have dramatic effects on the total award. **Costs**, including the fees of the tribunal and the institution and the parties' legal and expert costs, which may run to tens of millions of dollars in major energy arbitrations, are allocated at the tribunal's discretion; the trend has been towards "costs follow the event", under which the losing party pays a significant proportion of the successful party's costs, adjusted for the parties' conduct and relative success.

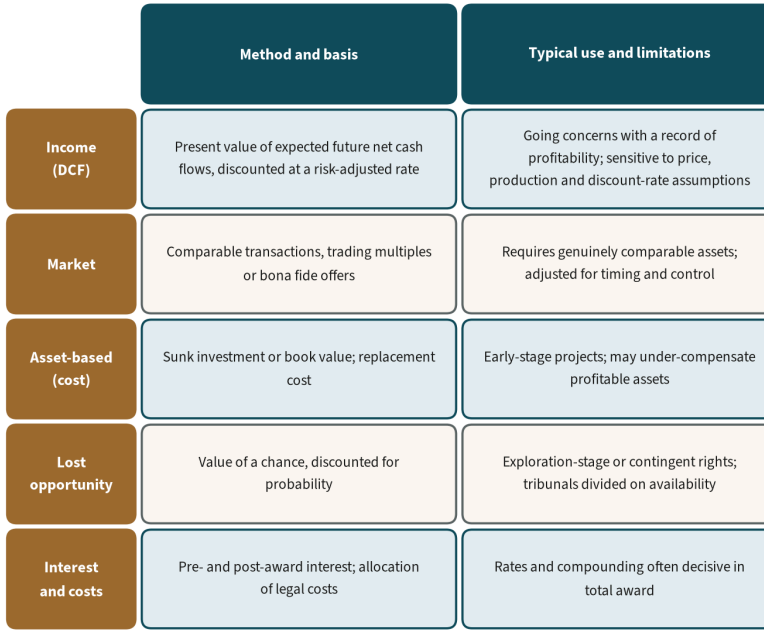


Figure 26.2 Principal approaches to the valuation of damages in energy arbitration. *Source:* prepared for this edition.

26.7 Recognition and Enforcement of Arbitral Awards

The New York Convention

The **Convention on the Recognition and Enforcement of Foreign Arbitral Awards** (New York Convention), concluded in 1958, is the cornerstone of the international enforcement of non-ICSID awards. Its contracting states, now numbering more than 170, including Nigeria (which acceded in 1970 and gave effect to the Convention in the Arbitration and Conciliation Act 1988, now the Arbitration and Mediation Act 2023), undertake to recognise arbitration agreements (Article II) and to recognise and enforce foreign arbitral awards (Article III), subject only to the grounds for refusal in **Article V**. Many states, including

Nigeria, have made the **reciprocity reservation**, applying the Convention only to awards made in the territory of other contracting states, and the **commercial reservation**, applying it only to differences arising out of legal relationships considered commercial under national law.

Under **Article V(1)**, recognition and enforcement may be refused, at the request of the party against whom the award is invoked, only if that party proves: (a) the **incapacity** of a party or the **invalidity of the arbitration agreement**; (b) that it was not given proper **notice** of the appointment of the arbitrator or the proceedings or was otherwise **unable to present its case**; (c) that the award deals with matters **beyond the scope** of the submission to arbitration; (d) that the composition of the tribunal or the procedure was **not in accordance with the agreement** of the parties or, failing such agreement, the law of the seat; or (e) that the award has **not yet become binding**, or has been **set aside or suspended** by a competent authority of the country in which, or under the law of which, it was made. Under **Article V(2)**, recognition and enforcement may also be refused if the competent authority finds that the subject matter is **not arbitrable** under its law or that recognition or enforcement would be contrary to its **public policy**. The grounds are exhaustive, the burden of proof lies on the resisting party, and courts have a discretion ("may be refused") to enforce even where a ground is established. The **pro-enforcement bias** of the Convention has been recognised by courts in many jurisdictions, and the public policy ground has generally been interpreted narrowly, as referring to the forum's most basic notions of morality and justice, or "international public policy".

Enforcement of Annulled Awards

The effect of the setting aside of an award at the seat on its enforcement elsewhere is contested. Most jurisdictions treat an award set aside at the seat as having no legal existence, so that it cannot be enforced, applying Article V(1)(e). French courts, however, have held that an international award is not anchored in the legal order of the seat, and may be enforced in France notwithstanding its annulment at the seat, as in *Hilmarton* (1994) and *Putrabali* (2007). The US courts, in ***Corporación Mexicana de Mantenimiento Integral v Pemex-Exploración y Producción*** (Second Circuit, 2016), a dispute concerning offshore oil platform construction contracts, upheld the enforcement of an award set aside in Mexico on the ground that the annulment, based on legislation retroactively depriving the claimant of its remedy, violated basic notions of justice. The English courts, in *Yukos Capital v Rosneft* (2014), held that an English court is not bound to recognise a foreign annulment decision that offends basic principles of honesty, natural justice, and domestic concepts of public policy. The approach of these courts reflects concern about annulments procured through partiality or abuse by the courts of a state party.

Sovereign Immunity from Enforcement and Execution

The enforcement of awards against **states and state entities** confronts the doctrine of **state immunity**. Under the **restrictive theory** of state immunity, now prevailing in most jurisdictions and reflected in the **UN Convention on Jurisdictional Immunities of States and Their Property** (2004, not yet in force), the US **Foreign Sovereign Immunities Act 1976**, and the UK **State Immunity Act 1978**, states are not immune from jurisdiction in respect of commercial transactions and in proceedings relating to arbitration agreements to which they have agreed. However, immunity from **execution** against state property is broader: property used or intended for use for **governmental (non-commercial) purposes** — including diplomatic premises, military property, and, under many statutes, the property of central banks — is generally immune from measures of constraint, unless the state has expressly waived immunity from execution in respect of that property. The identification of commercial assets of a state that are available for execution, and the attribution of the assets of state-owned enterprises to the state, are therefore crucial to the enforcement of awards against states.

Award creditors in energy disputes have pursued creative enforcement strategies. The creditors of the *Yukos* awards sought to attach Russian state assets in several jurisdictions, including France, Belgium, the Netherlands, and the United States, with mixed results. The creditors of Venezuela, including ConocoPhillips and Crystallex, obtained from the US courts an order permitting the sale of the shares in **PDV Holding**, the parent company of **CITGO**, owned by PDVSA, on the basis that PDVSA was the **alter ego** of Venezuela under the US Supreme Court's decision in *First National City Bank v Banco Para El Comercio Exterior de Cuba* (*Bancec*, 1983), which establishes a presumption of the separate juridical status of state instrumentalities that may be rebutted where the state exercises extensive control over the entity; the court-supervised auction process resulted in the approval of a sale in 2025. Award creditors of Argentina, Kazakhstan, Spain, and other states have also pursued assets across jurisdictions. The ***Stati v Kazakhstan*** litigation, arising from an SCC award of around USD 500 million under the Energy Charter Treaty concerning oil and gas investments, involved attachment of assets in several countries, including assets of Kazakhstan's National Fund held by a custodian in the Netherlands and Belgium, while Kazakhstan's allegations that the award had been procured by fraud generated further litigation in several jurisdictions.

Enforcement in Nigeria

In **Nigeria**, foreign arbitral awards may be enforced under the Arbitration and Mediation Act 2023, which incorporates the New York Convention, by application to the Federal High Court or a State High Court, and ICSID awards

under the ICSID (Enforcement of Awards) Act, by registration in the Supreme Court. The grounds for refusing recognition and enforcement under the 2023 Act mirror those in Article V of the New York Convention. Nigerian courts have generally adopted a supportive attitude to arbitration, though concerns persist about delays in court proceedings and the use of domestic litigation to frustrate arbitration, as in the *NNPC v Lutin Investments* (2006) and *Statoil v NNPC* (2013) litigation over the enforceability of arbitral awards and the jurisdiction of the Federal High Court under section 251 of the Constitution. Enforcement against the Federal Government and its agencies is further subject to section 84 of the **Sheriffs and Civil Process Act**, which requires the consent of the Attorney-General before execution against public funds in the custody of a public officer — a requirement that the Supreme Court has upheld, though its application to arbitral awards and to state-owned enterprises such as NNPC Limited, incorporated as a company under the PIA, has been debated. The P&ID case, examined in the casebox below, is the most significant arbitration involving Nigeria and has had profound implications for the country's approach to arbitration.

::: casebox {title="Case Study: The Federal Republic of Nigeria v Process & Industrial Developments Ltd (English Commercial Court, 2023)"} **Background.** In 2010, the Nigerian Ministry of Petroleum Resources entered into a **Gas Supply and Processing Agreement** (GSPA) with Process & Industrial Developments Ltd (P&ID), a company incorporated in the British Virgin Islands, under which the government was to supply wet gas to a processing facility to be constructed by P&ID, which would return lean gas to the government free of charge for power generation. The GSPA provided for arbitration seated in London. The government did not supply the gas, and P&ID did not construct the facility. P&ID commenced arbitration, and in **2017** a tribunal awarded it damages of around **USD 6.6 billion**, representing the profits it would have earned over the twenty-year life of the agreement, with interest at seven per cent, so that by 2023 the total exceeded **USD 11 billion** — a sum representing a substantial share of Nigeria's foreign exchange reserves.

Enforcement and challenge. P&ID sought to enforce the award in England, and in 2019 the Commercial Court granted leave to enforce. Nigeria applied for an extension of time to challenge the award under section 68 of the Arbitration Act 1996 (serious irregularity), alleging that the GSPA had been procured by bribery, that P&ID had continued to bribe Nigerian officials during the arbitration, and that P&ID had improperly obtained Nigeria's privileged legal documents during the arbitration. In 2020, the court granted the extension, finding a strong prima facie case of fraud.

Decision. In **October 2023**, after a trial in which extensive evidence was heard, Robin Knowles J set aside the award under **section 68(2)(g)** of the

Arbitration Act 1996, on the ground that it had been **obtained by fraud** and in a way **contrary to public policy**. The court found that P&ID had paid bribes to a Nigerian official involved in negotiating the GSPA, and continued to make payments during the arbitration to ensure the official's silence; that P&ID had presented false evidence to the tribunal; and that P&ID had improperly retained and used Nigeria's internal legal documents, which it had obtained during the arbitration. The court was also highly critical of the conduct of the arbitration process and the shortcomings of Nigeria's own legal representation and the failure of the tribunal and the arbitral process to detect the fraud, observing that the case raised serious questions about the capacity of arbitration to address fraud and corruption in disputes involving states. In 2024, the court ordered P&ID to pay a substantial part of Nigeria's costs.

Significance. The P&ID case is a landmark in the law of arbitration and corruption. It demonstrates that the courts of the seat will set aside awards obtained by fraud and corruption, even after long delay; it highlights the vulnerability of states with weak contract governance and inadequate legal representation to oppressive claims; and it has prompted reflection on the responsibilities of arbitrators, counsel, and institutions in detecting corruption, the role of third-party funders, the importance of transparency in state contracting, and the need for states to invest in their own capacity to negotiate contracts and conduct arbitrations. For Nigeria, the case has led to closer scrutiny of contracts entered into by ministries and agencies and of the dispute resolution clauses they contain. :::

26.8 Mediation, Negotiation, and Alternative Dispute Resolution

The Spectrum of Dispute Resolution

Arbitration is not the only, nor always the most appropriate, method for resolving energy disputes. The **spectrum** of dispute resolution methods ranges from **negotiation**, in which the parties resolve the dispute themselves; through **mediation** or **conciliation**, in which a neutral third party assists the parties to reach a settlement; **expert determination**, in which an expert decides a technical issue, usually on a binding basis; and **dispute boards**, which resolve disputes during the performance of a construction contract; to **adjudicative** methods, including arbitration and litigation. The appropriate method depends on the nature of the dispute, the relationship between the parties, the importance of confidentiality and speed, and the need for a binding and enforceable outcome. Energy contracts commonly combine several methods in **multi-tiered clauses**, examined in Chapter 15.

Negotiation and Renegotiation

Most energy disputes are resolved by **negotiation**, often in the shadow of arbitration. Long-term contracts frequently contain **renegotiation** or **hardship** clauses requiring the parties to negotiate in good faith to restore the economic equilibrium of the contract upon the occurrence of specified events, and **price review** clauses requiring negotiation of the contract price at specified intervals, with arbitration as a fallback if negotiation fails, as examined in Chapters 6 and 15. Investor–state disputes are also frequently settled: a significant proportion of ICSID cases are discontinued, many following settlement. Negotiated outcomes in energy disputes have included the restructuring of the Reko Diq project following the *Tethyan* award, the settlement between Barrick and Tanzania examined in Chapter 24, and the settlements between Repsol and Argentina over the expropriation of YPF. The structure of a negotiated settlement may include the payment of compensation in cash, bonds, or oil; the restructuring of the investment; the grant of new rights; or the withdrawal of claims in exchange for regulatory changes.

Mediation and Conciliation

Mediation has become increasingly important in international commercial and investor–state disputes, though its use in energy disputes remains less common than arbitration. The advantages of mediation include its flexibility, confidentiality, lower cost, speed, and capacity to preserve relationships and produce creative solutions that are not available in adjudication. The **United Nations Convention on International Settlement Agreements Resulting from Mediation** (**Singapore Convention on Mediation**), adopted in 2018 and in force since September 2020, provides a framework for the cross-border enforcement of settlement agreements resulting from mediation, analogous to the New York Convention for arbitral awards, subject to limited grounds for refusal. Nigeria signed the Singapore Convention in 2019, and the Arbitration and Mediation Act 2023 contains provisions on mediation and the enforcement of mediated settlement agreements.

In the investor–state context, the **ICSID Mediation Rules** (2022) and the **ICSID Conciliation Rules** provide frameworks for mediation and conciliation, the **IBA Rules for Investor-State Mediation** (2012) offer a model procedure, and the **Energy Charter Secretariat** has promoted mediation of investment disputes, publishing a **Guide on Investment Mediation** in 2016. The **UNCITRAL Working Group III** on ISDS reform has developed provisions on mediation, including model clauses for treaties, as part of the reform package examined in Chapter 14. Some newer investment treaties, including the EU's treaties with Canada (CETA), Singapore, and Vietnam, and the **AfCFTA Investment Protocol** (2023), provide for or encourage mediation before or during arbitration.

Expert Determination

Expert determination is widely used in the energy sector for technical disputes, such as disputes over the redetermination of tract participations under unitisation agreements, the measurement of gas quality and quantity, the calculation of price adjustments, the valuation of assets in pre-emption and default provisions, and the determination of reserves for financing purposes. The expert, who is selected for his or her technical expertise, decides the issue, often within a short timeframe, and the determination is usually final and binding, subject to limited challenge for fraud, manifest error, or the expert exceeding his or her mandate. The English courts have held that an expert determination may be challenged where the expert has departed from the instructions in a material respect, but not merely because the determination is wrong (*Jones v Sherwood Computer Services*, 1992). Expert determination is not governed by arbitration legislation or the New York Convention, so the determination must be enforced as a contractual obligation. The **ICC Rules for Expertise Administration** and the **ICC International Centre for ADR** provide institutional frameworks for expert determination.

Dispute Boards

Dispute boards, consisting of one or three independent experts appointed at the beginning of a construction project to resolve disputes as they arise during performance, are widely used in large energy infrastructure projects, particularly under the **FIDIC** contracts, examined in Chapter 15. **Dispute Adjudication Boards** issue decisions that are binding unless and until revised by arbitration, and **Dispute Avoidance/Adjudication Boards** under the 2017 FIDIC suite also assist in avoiding disputes. The binding force of dispute board decisions pending arbitration has been addressed in arbitral awards and by the courts, notably in the Singapore Court of Appeal's decisions in *PT Perusahaan Gas Negara (Persero) TBK v CRW Joint Operation* (2011 and 2015), arising from a gas pipeline project in Indonesia, in which the court ultimately upheld the enforcement of an interim award giving effect to a binding but non-final DAB decision.

Inter-State Energy Disputes and Diplomatic Means

Inter-state energy disputes are often resolved by **negotiation** and **diplomatic means**. The settlement of maritime boundary disputes by negotiation and the conclusion of joint development agreements, examined in Chapter 20, the mediation of the EU in Russia–Ukraine gas transit negotiations in 2014 and 2019, and the facilitation by the United States of the maritime boundary agreement between Israel and Lebanon in 2022 illustrate the role of diplomacy. The conciliation mechanism of Article 7(7) of the Energy Charter Treaty for transit disputes, and **compulsory conciliation** under Annex V of UNCLOS, used for the first time in the *Timor Sea Conciliation* between Timor-Leste and Australia (2016–2018),

which resulted in the Treaty establishing maritime boundaries in the Timor Sea in 2018, including provisions on the Greater Sunrise gas fields, demonstrate the potential of structured conciliation in resource disputes.

Community Grievance Mechanisms

Disputes between energy companies and communities are increasingly addressed through **non-judicial grievance mechanisms**, including operational-level grievance mechanisms established by companies, the **National Contact Points** under the OECD Guidelines for Multinational Enterprises, the **Compliance Advisor Ombudsman** of the International Finance Corporation and MIGA, and the **Independent Accountability Mechanisms** of development banks, as examined in Chapters 21 and 22. The **UN Guiding Principles on Business and Human Rights** set effectiveness criteria for such mechanisms: legitimacy, accessibility, predictability, equitability, transparency, rights-compatibility, being a source of continuous learning, and, for operational-level mechanisms, being based on engagement and dialogue. The **Hague Rules on Business and Human Rights Arbitration** (2019) offer a framework for the arbitration of business-related human rights disputes, including between companies and affected communities, though their use remains limited.

Choosing and Designing Dispute Resolution Mechanisms

The design of dispute resolution mechanisms for energy contracts and treaties requires careful attention to the likely types of disputes and the interests of the parties. Practitioners advising states and investors should consider: (a) the **types of dispute** likely to arise and the methods suited to each — expert determination for technical and valuation issues, dispute boards for construction disputes, mediation for relational disputes, and arbitration for disputes requiring a binding and enforceable decision; (b) the **sequencing** of methods in multi-tiered clauses, with clear time limits and triggers to avoid uncertainty about whether conditions precedent have been satisfied; (c) the **seat and institution** of arbitration, taking into account the supportiveness of the courts of the seat, the neutrality of the venue, and the cost; (d) the **enforceability** of the outcome, including the waiver of sovereign immunity from jurisdiction and execution in contracts with states; (e) the **joinder and consolidation** of related disputes under interconnected contracts, such as the PSC, the JOA, and the gas sales agreement, through compatible dispute resolution clauses; (f) **transparency** and the public interest in disputes involving states and public resources; and (g) the **capacity** of the parties, particularly states, to participate effectively — a lesson starkly illustrated by the P&ID case. For African states, investment in specialised government legal teams, the development of local arbitral institutions and expertise, and participation in regional initiatives to strengthen arbitration capacity are essential.

::: exercise {title="Practical Exercise 26: The Ndara Gas Processing Arbitration (HYPOTHETICAL)"} **This exercise is entirely hypothetical. All states, companies, persons, agreements, and events are fictitious and do not refer to any real country, company, person, or dispute.**

In 2016, the Ministry of Energy of the Republic of Ndara, a West African state, entered into a **Gas Processing Agreement** (GPA) with Kestrel Gas Processing BV (Kestrel), a company incorporated in the Netherlands and wholly owned by Kestrel Holdings Ltd, a company incorporated in Ndara and owned by Ndaran nationals. Under the GPA, Ndara was to supply 200 million standard cubic feet per day of associated gas to a processing plant to be built by Kestrel, which would return processed gas to Ndara's state power company. The GPA is governed by Ndaran law, and provides for "friendly negotiations for sixty days" followed by ICC arbitration seated in London. Kestrel was incorporated in the Netherlands in 2017, after the GPA was signed by Kestrel's Ndaran parent, which assigned the GPA to Kestrel with the Ministry's consent. The Netherlands–Ndara BIT (2002) provides for ICSID arbitration after a six-month cooling-off period, and contains a fair and equitable treatment clause, an umbrella clause, and a clause limiting protection to investments "made in accordance with the laws of the host State". Both states are parties to the ICSID Convention and the New York Convention.

Ndara never supplied gas, citing the failure of its upstream partners to build gathering facilities, and Kestrel never began construction of the plant, beyond preliminary engineering work costing USD 30 million. In 2021, a new government cancelled the GPA, alleging that it had been signed without the approval of the Federal Executive Council required by Ndaran law, and that a former Ministry official received USD 2 million from a consultant retained by Kestrel's parent shortly before the GPA was signed. Kestrel commenced ICC arbitration ten days after the cancellation and, simultaneously, served a notice of dispute under the BIT. It claims USD 4.2 billion, representing the net present value of the processing fees it would have earned over twenty years.

Tasks.

(a) Advise Ndara on its jurisdictional and admissibility objections in the ICC arbitration and in any ICSID arbitration Kestrel may bring, including on the cooling-off periods, the nationality of Kestrel, the legality of the investment, and the relationship between the contract and treaty claims.

(b) Advise Kestrel on the most appropriate forum or fora for its claims, and on the risks of parallel proceedings.

(c) Evaluate the appropriate method for quantifying Kestrel's damages if liability is established, explaining why a DCF valuation may or may not be appropriate.

(d) Assume that an ICC tribunal awards Kestrel USD 3.1 billion, and that Ndara subsequently obtains evidence that the consultant's payment was a bribe and that Kestrel presented false evidence to the tribunal. Advise Ndara on the available remedies at the seat and at the enforcement stage, and Kestrel on its enforcement strategy against Ndaran assets in England, the United States, and Ndara.

(e) Prepare a memorandum to the Attorney-General of Ndara recommending reforms to the government's contracting and dispute management practices to prevent a recurrence. :::

Conclusion

International arbitration is the backbone of energy dispute resolution, providing a neutral, flexible, and enforceable forum for disputes of exceptional value and complexity. Its effectiveness depends on a robust legal infrastructure — the New York and ICSID Conventions, modern arbitration laws such as Nigeria's Arbitration and Mediation Act 2023 and England's Arbitration Act 2025, institutional rules, and supportive national courts — and on the competence and integrity of arbitrators, counsel, and experts. The process raises distinctive issues: the strategic structuring of investments and claims, the limits of consent and the rejection of abuse of process, the interaction of contract and treaty, the quantification of damages in long-lived energy projects, and the enforcement of awards against sovereign assets. Recent developments — the revision of the ICSID Rules in 2022, the UNCITRAL Code of Conduct for arbitrators, the growth of third-party funding and transparency, the conflict between intra-EU awards and EU law, the fragmentation resulting from sanctions and Russian anti-arbitration injunctions, and the setting aside of the P&ID award for fraud — show a system in evolution. Alongside arbitration, negotiation, mediation, expert determination, and dispute boards offer valuable tools, increasingly supported by the Singapore Convention. For African states and practitioners, the challenge is to build the capacity to participate effectively and on equal terms in the resolution of the disputes arising from their resources. Chapter 27 turns to the role of national and international courts in energy and environmental litigation.

CHAPTER SUMMARY

- Energy disputes are distinctive in their long-term relational context, high stakes, technical complexity, state involvement, political sensitivity, and multi-jurisdictional character; they may be classified by parties, subject matter, legal basis, and forum, and arbitration predominates because of its neutrality, flexibility, and enforceability.

- International commercial arbitration rests on the arbitration agreement, institutional or ad hoc rules, the law of the seat, and the New York Convention; Nigeria's Arbitration and Mediation Act 2023 and England's Arbitration Act 2025 modernise the framework, and national courts support and supervise arbitration, including through anti-suit injunctions.
- Investment treaty arbitration proceeds through notice and cooling-off, commencement, constitution of the tribunal, procedural orders, written and oral phases, and award, with distinctive issues of arbitrator independence, bifurcation, preliminary objections, provisional measures, transparency, and third-party funding.
- The ICSID Convention provides a delocalised system with jurisdiction under Article 25, annulment under Article 52, and enforcement under Article 54 subject to immunity from execution; UNCITRAL arbitration has a legal seat and relies on the New York Convention.
- Jurisdiction depends on consent, protected investors, and protected investments; abuse of process limits treaty planning; corruption may defeat claims; contract and treaty claims are distinct; and applicable law depends on the forum and basis of the claim, with growing attention to environmental counterclaims and climate obligations.
- Evidence is governed by tribunal discretion guided by the IBA Rules; expert evidence is central; compensation follows the Chorzów Factory principle of full reparation for unlawful acts, using DCF, market, asset-based, or loss-of-opportunity methods, with interest and costs often decisive.
- Awards are enforced under the New York Convention, subject to exhaustive grounds for refusal, and under the ICSID Convention; enforcement against states confronts immunity from execution; the P&ID case illustrates the setting aside of awards obtained by fraud.
- Negotiation, mediation (supported by the Singapore Convention), expert determination, dispute boards, conciliation, and grievance mechanisms complement arbitration, and dispute resolution clauses should be designed to match methods to disputes.

DOCTORAL RESEARCH QUESTIONS

1. Has the P&ID judgment exposed systemic weaknesses in international arbitration's capacity to detect fraud and corruption in disputes involving states? Propose reforms.
2. Evaluate the arbitral jurisprudence on corporate restructuring and abuse of process in energy investment disputes, and assess whether the current approach strikes an appropriate balance between legitimate treaty planning and abuse.

3. Is the DCF method appropriate for valuing undeveloped natural resource projects? Critically assess with reference to *Tethyan Copper v Pakistan* and other mining and petroleum awards.
4. To what extent should environmental and human rights counterclaims be permitted in energy investment arbitration, and what legal bases support them?
5. Assess the prospects for African arbitral seats and institutions to become preferred venues for energy disputes involving African parties.
6. How should international arbitration respond to the obligations of states under the Paris Agreement as clarified by the ICJ advisory opinion of 2025?

FURTHER READING

- Nigel Blackaby, Constantine Partasides and Alan Redfern, *Redfern and Hunter on International Arbitration* (7th edn, Oxford University Press 2023).
- Gary B Born, *International Commercial Arbitration* (3rd edn, Kluwer Law International 2021).
- Christoph H Schreuer and others, *The ICSID Convention: A Commentary* (Cambridge University Press, latest edition).
- Rudolf Dolzer, Ursula Kriebaum and Christoph Schreuer, *Principles of International Investment Law* (3rd edn, Oxford University Press 2022).
- Sergey Ripinsky with Kevin Williams, *Damages in International Investment Law* (British Institute of International and Comparative Law 2008).
- Irmgard Marboe, *Calculation of Compensation and Damages in International Investment Law* (2nd edn, Oxford University Press 2017).
- Federal Republic of Nigeria v Process & Industrial Developments Ltd [2023] EWHC 2638 (Comm).

27 CHAPTER

Energy Litigation, Judicial Review, and Regulatory Enforcement

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the role of domestic courts in resolving energy disputes, including questions of jurisdiction, justiciability, and specialised courts.
2. Analyse administrative and regulatory litigation in the energy sector, including appeals against tariff and regulatory decisions.
3. Apply the grounds of judicial review to licensing and consent decisions for energy and resource projects.
4. Evaluate environmental and climate litigation strategies, including standing, evidence, attribution, and remedies.
5. Assess contractual and commercial litigation in the energy sector, including the exclusive jurisdiction of the Federal High Court of Nigeria.
6. Compare class actions and other mechanisms of collective redress for mass harms arising from energy activities.
7. Analyse injunctions, remedies, and regulatory enforcement tools in energy law.
8. Evaluate cross-border litigation against energy companies, including parent company liability, forum non conveniens, and the enforcement of foreign judgments.

Introduction

Although arbitration predominates in transnational commercial and investor–state energy disputes, as examined in Chapter 26, **courts** remain central to energy law. National courts review the legality of licensing decisions, regulatory determinations, and environmental approvals; they adjudicate contractual disputes not subject to arbitration and supervise arbitration; they provide remedies for communities harmed by oil spills, gas flaring, and mining pollution; they hear climate cases that have reshaped the obligations of governments and companies; they enforce regulatory requirements through civil and criminal proceedings; and

they determine questions of jurisdiction and enforcement in transnational litigation. Regional and international courts — the International Court of Justice, the European Court of Human Rights, the Court of Justice of the European Union, the ECOWAS Community Court of Justice, and the African Court on Human and Peoples' Rights — add further layers of adjudication.

Energy litigation is distinctive in several respects. It frequently involves the review of complex technical and economic judgments by regulators, raising questions about the appropriate intensity of judicial scrutiny. It engages fundamental questions of the separation of powers, as courts are asked to adjudicate matters of energy and climate policy traditionally reserved to the executive and legislature. It often involves claimants with limited resources confronting well-resourced companies and governments, raising questions of access to justice, standing, costs, and collective redress. And it increasingly crosses borders, as claimants harmed by the activities of multinational companies in one country seek remedies in the courts of the companies' home states.

This chapter examines domestic courts and energy disputes (section 27.1), administrative and regulatory litigation (section 27.2), judicial review of licensing decisions (section 27.3), environmental and climate litigation (section 27.4), contractual and commercial litigation (section 27.5), class actions and collective redress (section 27.6), injunctions, remedies, and enforcement orders (section 27.7), and cross-border litigation and jurisdictional conflicts (section 27.8). The substantive principles of climate law and litigation have been examined in Chapter 18 and environmental liability in Chapter 17; this chapter focuses on litigation as a process and on the procedural and remedial questions that shape its outcomes.

27.1 Domestic Courts and Energy Disputes

The Functions of Courts in Energy Governance

Domestic courts perform several functions in energy governance. They **resolve disputes** between private parties, between private parties and the state, and among public authorities. They **review the legality** of governmental and regulatory action, ensuring that authorities act within their powers, follow fair procedures, and respect fundamental rights. They **interpret** constitutional provisions on resource ownership, the division of powers in federal states, and environmental rights, as in the Nigerian Supreme Court's decisions on the ownership of offshore resources and the Indian Supreme Court's decision on coal block allocations, examined in Chapter 24. They **enforce** regulatory requirements through civil and criminal proceedings. And they **develop the law**, particularly in common law systems, through the elaboration of principles of tort, contract, and administrative law in the energy context — as in the development of the law of nuisance

and negligence in oil pollution cases and the duty of care of parent companies, examined in Chapter 17.

Jurisdiction and Specialised Courts

The allocation of jurisdiction over energy disputes varies among legal systems. In **Nigeria**, section 251(1) of the Constitution of 1999 confers on the **Federal High Court** exclusive jurisdiction in civil causes and matters relating to, among other things, the revenue of the federal government, the operation of companies, and "mines and minerals (including oil fields, oil mining, geological surveys and natural gas)", as well as actions for declarations or injunctions affecting the validity of executive or administrative actions of the federal government or its agencies. The Supreme Court and the Court of Appeal have interpreted this provision as conferring on the Federal High Court exclusive jurisdiction over disputes relating to oil and gas operations, including, in many cases, claims for compensation for oil pollution against oil companies — though the precise scope of the provision, particularly its application to simple contract and tort claims between private parties connected with the oil industry, has generated inconsistent decisions. In ***Shell Petroleum Development Co v Isaiah*** and later cases, the courts addressed compensation claims for oil spills, and the question whether State High Courts have jurisdiction over such claims has been the subject of extensive litigation, with the trend favouring the exclusive jurisdiction of the Federal High Court where the claim arises from oil mining operations.

Several jurisdictions have established **specialised courts** for environmental and energy matters. **Environmental courts and tribunals** exist in more than fifty countries, including the **Land and Environment Court of New South Wales**, established in 1980; India's **National Green Tribunal**, established in 2010, which has decided many cases concerning power plants, mining, and pollution; Kenya's **Environment and Land Court**; and the environmental courts of Chile, Brazil, and other Latin American states. Specialised courts can develop expertise in technical matters, apply more flexible procedures, and provide faster and more accessible justice, but may also be criticised for their limited powers or capture by particular interests. In the United Kingdom, the **Planning Court** and the **Competition Appeal Tribunal** hear energy-related planning and competition matters, and the **Commercial Court** and **Technology and Construction Court** are leading fora for energy contract disputes.

Justiciability and the Separation of Powers

A recurring question in energy litigation is whether disputes involving energy and climate policy are **justiciable** — that is, suitable for judicial resolution — or whether they raise **political questions** that should be left to the elected branches. In the **United States**, the political question doctrine and standing requirements

have been invoked to dismiss climate cases against the federal government. In ***Juliana v United States***, in which youth plaintiffs alleged that the federal government's promotion of fossil fuels violated their constitutional rights to life, liberty, and property, the Ninth Circuit held in 2020 that the plaintiffs lacked standing because the relief sought — an order requiring the government to prepare and implement a plan to phase out fossil fuel emissions — was beyond the power of an Article III court to provide, and the case was finally dismissed, with the Supreme Court declining review in 2025. In ***American Electric Power v Connecticut*** (2011), the Supreme Court held that the Clean Air Act and the EPA's authority to regulate greenhouse gas emissions displaced federal common law nuisance claims against power companies.

Courts in other jurisdictions have been more willing to adjudicate climate and energy policy. In ***Urgenda Foundation v State of the Netherlands*** (2019), the Dutch Supreme Court held that the state's duty of care under Articles 2 and 8 of the European Convention on Human Rights required it to reduce greenhouse gas emissions by at least twenty-five per cent from 1990 levels by 2020, rejecting the argument that the case raised political questions beyond judicial competence: the courts' task was to protect human rights, and the determination of the minimum required reduction did not intrude on the political choice of the means. In ***Neubauer v Germany*** (2021), the German Federal Constitutional Court held that the Federal Climate Change Act violated the fundamental rights of young people by failing to specify emission reductions after 2030, thereby placing a disproportionate burden of future reductions on the younger generation. In ***Verein KlimaSeniorinnen Schweiz v Switzerland*** (2024), the Grand Chamber of the European Court of Human Rights held that Article 8 of the Convention encompasses a right to effective protection by the state from the serious adverse effects of climate change, and that Switzerland had failed to comply with its positive obligations, while recognising the standing of an association but not of individual applicants who did not meet a high threshold of victim status. These cases, examined in Chapter 18, demonstrate the varying approaches of courts to the separation of powers in climate matters.

Courts in Nigeria's Energy Sector

Nigerian courts have played an important role in the energy sector, though their effectiveness has been constrained by delays, the complexity of jurisdictional rules, and enforcement difficulties. Landmark decisions include ***Gbemre v Shell Petroleum Development Co*** (Federal High Court, Benin, 2005), in which the court held that gas flaring in the claimant's community violated the constitutional rights to life and dignity and the right to a healthy environment under the African Charter, as domesticated by the African Charter on Human and Peoples' Rights (Ratification and Enforcement) Act, and ordered that flaring cease — an order

not implemented; ***Centre for Oil Pollution Watch v NNPC*** (Supreme Court, 2018), in which the Supreme Court held that a non-governmental organisation had standing to bring an action to compel the remediation of oil pollution in the public interest, liberalising the restrictive approach to standing in environmental matters; and the many decisions awarding compensation for oil spills under statute and the common law. The **National Industrial Court**, the **Investments and Securities Tribunal**, and the tribunals established under the Petroleum Industry Act and the Electricity Act add further fora. The **Fundamental Rights (Enforcement Procedure) Rules 2009** provide an expedited procedure for the enforcement of fundamental rights, which has been used in environmental cases, and expressly encourage public interest litigation by providing that human rights activists, advocates, and NGOs may institute proceedings on behalf of applicants.

27.2 Administrative and Regulatory Litigation

Regulatory Decisions and the Courts

Energy regulators make decisions with significant economic consequences: determining tariffs for network services and regulated supply, granting and revoking licences, approving mergers and acquisitions, imposing penalties, and setting market rules. The legal mechanisms for challenging such decisions include **appeals** on the merits, available where legislation provides for them, and **judicial review** of the lawfulness of the decision. The design of these mechanisms involves a trade-off between **accountability** and **regulatory certainty**: broad rights of appeal enhance the accountability of regulators and protect the rights of regulated entities, but may undermine regulatory independence and predictability, encourage strategic litigation, and transfer technical decisions to generalist judges.

Appeals Against Regulatory Decisions

In the **United Kingdom**, decisions of the Gas and Electricity Markets Authority (**Ofgem**) on price controls under the RIIO framework may be appealed to the **Competition and Markets Authority** (CMA), which reviews whether the decision was wrong on specified grounds, including errors of fact, law, or the exercise of discretion. In its decision on the appeals against the **RIIO-2** price control in 2021, the CMA upheld most of Ofgem's decisions but found that Ofgem had erred in its approach to the "outperformance wedge", which reduced the allowed return of network companies. Appeals against Ofgem's licence modification decisions and penalties are also available. In the **United States**, decisions of the **Federal Energy Regulatory Commission** are subject to rehearing before the Commission and then to review in the federal courts of appeals, particularly the D.C. Circuit, under the **Administrative Procedure Act** standard of whether the decision was "arbitrary, capricious, an abuse of discretion, or otherwise not in accordance with law", and whether the Commission's findings of fact are

supported by substantial evidence. In ***Allegheny Defense Project v FERC*** (D.C. Circuit en banc, 2020), the court held that FERC could not indefinitely delay rehearing requests through "tolling orders" while allowing pipeline construction to proceed, protecting landowners' access to judicial review. In the **European Union**, decisions of the Agency for the Cooperation of Energy Regulators (ACER) may be appealed to the ACER Board of Appeal and then to the EU General Court.

In **Nigeria**, the **Electricity Act 2023** provides for the resolution of disputes and the review of NERC's decisions, including through NERC's own review processes and appeals to the courts, and for the establishment of state electricity regulatory commissions, whose decisions will raise new questions of jurisdiction and review in the decentralised regime examined in Chapter 24. Under the **Petroleum Industry Act 2021**, the decisions of the NUPRC and the NMDPRA are subject to internal review and judicial review, and the Act provides for the resolution of certain disputes through arbitration or expert determination. The **Federal Competition and Consumer Protection Act 2018** established the Federal Competition and Consumer Protection Commission and the **Competition and Consumer Protection Tribunal**, which hears appeals against the Commission's decisions, including in the energy sector, where the Commission has investigated pricing practices of electricity distribution companies and fuel marketers.

The Intensity of Review and Regulatory Deference

The intensity of judicial review of regulatory decisions — the degree of **deference** courts accord to the expertise of regulators — varies among jurisdictions and is the subject of continuing debate. In the **United States**, the ***Chevron*** doctrine (*Chevron USA v Natural Resources Defense Council*, 1984), which required courts to defer to reasonable agency interpretations of ambiguous statutes, was **overruled** in ***Loper Bright Enterprises v Raimondo*** (2024), in which the Supreme Court held that the Administrative Procedure Act requires courts to exercise their independent judgment in deciding whether an agency has acted within its statutory authority. Together with the **major questions doctrine** articulated in *West Virginia v EPA* (2022), examined in Chapter 24, *Loper Bright* has significantly expanded the scope of judicial scrutiny of energy and environmental regulation in the United States, though courts continue to defer to agencies' technical and factual judgments under the arbitrary and capricious standard. In the **United Kingdom**, courts accord considerable respect to the expertise of regulators on technical and economic matters, intervening on the grounds of illegality, irrationality, or procedural impropriety, while the CMA's appellate jurisdiction allows more intensive review of price control decisions. The approach of courts in other common law jurisdictions, including Nigeria, generally follows the English model, though the extent of deference to specialist regulators is less developed.

Litigation Between Public Authorities

Energy litigation also arises between public authorities, particularly in federal systems. In **Nigeria**, the federal government and the states have litigated over the ownership of offshore resources in ***Attorney-General of the Federation v Attorney-General of Abia State*** (2001 and 2002), in which the Supreme Court held that, for the purposes of the derivation principle, the seaward boundary of littoral states is the low-water mark, so that revenues from offshore resources were not subject to derivation — a decision reversed by the **Allocation of Revenue (Abolition of Dichotomy in the Application of the Principle of Derivation) Act 2004**. States have also challenged federal laws on electricity, environmental regulation, and value added tax, and the Electricity Act 2023 may generate further intergovernmental litigation over the boundaries of federal and state competence. In the **United States**, states have repeatedly sued federal agencies over energy and environmental regulation, as in *Massachusetts v EPA* (2007), in which the Supreme Court held that Massachusetts had standing to challenge the EPA's refusal to regulate greenhouse gas emissions from vehicles and that greenhouse gases were air pollutants under the Clean Air Act, and in the numerous challenges by states to federal power plant and methane regulations and to federal efforts to curtail renewable energy.

27.3 Judicial Review of Licensing Decisions

Licensing Decisions as a Focus of Litigation

The grant, renewal, transfer, and revocation of licences, leases, and consents for energy and resource projects are among the most frequently litigated decisions in energy law. Licensing decisions determine who may explore for and produce resources, build power plants and pipelines, and operate in regulated markets; they shape the environmental and social impacts of projects; and they allocate valuable rights. They may be challenged by unsuccessful applicants, by licence holders whose licences are revoked or conditioned, by communities affected by projects, and by environmental organisations seeking to prevent fossil fuel development. The grounds of review — summarised in Figure 27.1 — and the procedures for challenge determine the extent to which courts can hold licensing authorities to account.

	Core question	Energy illustrations
Illegality	Did the decision-maker act within its statutory powers and consider relevant matters?	Failure to assess downstream (scope 3) emissions: Finch (UKSC 2024)
Irrationality	Was the decision one no reasonable authority could reach?	Unreasoned tariff or licence refusals; reliance on irrelevant considerations
Procedural impropriety	Were consultation, fairness and statutory procedures followed?	Inadequate consultation of affected communities and traditional owners (Tipakalippa)
Legitimate expectation	Was a clear, unqualified representation frustrated without justification?	Retroactive withdrawal of renewable feed-in tariffs
Rights and proportionality	Is interference with protected rights justified and proportionate?	Climate and environmental rights claims (Neubauer; KlimaSeniorinnen)

Figure 27.1 Principal grounds of judicial review applied to energy licensing decisions. *Source:* prepared for this edition.

The Grounds of Judicial Review

In the common law tradition, the grounds of judicial review were classified by Lord Diplock in ***Council of Civil Service Unions v Minister for the Civil Service*** (1985) as **illegality**, **irrationality**, and **procedural impropriety**, with the possibility of adding **proportionality**. **Illegality** encompasses acting outside the powers conferred by statute, misdirection in law, failing to take into account relevant considerations or taking into account irrelevant ones, and using powers for an improper purpose. **Irrationality**, or *Wednesbury* unreasonableness (*Associated Provincial Picture Houses v Wednesbury Corporation*, 1948), refers to a decision so unreasonable that no reasonable authority could have reached it. **Procedural impropriety** covers failure to observe statutory procedures and breaches of the rules of natural justice, including the right to be heard and the rule against bias. The doctrine of **legitimate expectation** protects expectations generated by a public authority's clear and unambiguous representations or established practice, procedurally and, in some circumstances, substantively, unless frustrating the expectation is justified by an overriding public interest. Where fundamental rights are engaged, courts apply a **proportionality** analysis. Civil law systems apply

analogous grounds through administrative courts, often with a more developed proportionality doctrine.

In **Nigeria**, judicial review of administrative action is available in the Federal High Court and State High Courts, and the courts apply grounds derived from English administrative law, as well as constitutional grounds, including the right to fair hearing under section 36 of the Constitution. The revocation of oil mining leases and licences has been challenged in several cases, and the Petroleum Industry Act 2021 specifies the grounds and procedures for the revocation of licences and leases, including the requirement of notice and an opportunity to remedy defaults, providing a statutory basis for review. The controversy over the **OPL 245** block, examined in Chapter 22, generated litigation in Nigeria, Italy, the United Kingdom, and the United States, as well as arbitration, illustrating the multiple legal consequences of a contested licensing decision.

Environmental Assessment and Licensing

Many of the most significant licensing challenges concern the adequacy of **environmental impact assessment**, examined in Chapter 17. Courts have quashed consents where the assessment failed to consider relevant environmental effects, alternatives, or cumulative impacts, or where the public was not adequately consulted. The question whether the assessment of fossil fuel projects must include **downstream (scope 3) emissions** from the combustion of the extracted fuel has become a central issue in climate-related licensing litigation.

In ***R (Finch) v Surrey County Council*** (UK Supreme Court, June 2024), the Supreme Court held, by a majority of three to two, that the environmental impact assessment for a project to extend oil production from a well site in Surrey was required, under the Town and Country Planning (Environmental Impact Assessment) Regulations 2017, implementing the EU EIA Directive, to assess the greenhouse gas emissions from the combustion of the oil to be produced, because those emissions were an inevitable consequence of the project and their extent could be readily estimated. The grant of planning permission was therefore unlawful. *Finch* has had far-reaching consequences. In September 2024, the government conceded a challenge to the consent for a new coal mine in **Cumbria** (Whitehaven), and the High Court quashed it. In **January 2025**, the Court of Session in Scotland held, in *Greenpeace Ltd v Advocate General* and a related case brought by Uplift, that the development consents for the **Rosebank** and **Jackdaw** oil and gas fields in the North Sea were unlawful because the environmental assessments had not addressed scope 3 emissions, and set them aside, while permitting work to continue pending reconsideration. The government issued new guidance in June 2025 on the assessment of downstream emissions for offshore projects, under which the operators submitted revised environmental statements for reconsideration.

Courts in other jurisdictions have reached similar conclusions. In ***Gloucester Resources Ltd v Minister for Planning*** (Land and Environment Court of New South Wales, 2019), Preston CJ upheld the refusal of consent for the Rocky Hill open-cut coal mine, finding that its scope 1, 2, and 3 emissions would contribute to climate change and that the mine was in the "wrong place at the wrong time". In ***Earthlife Africa Johannesburg v Minister of Environmental Affairs*** (High Court of South Africa, 2017), the court held that the environmental authorisation for the Thabametsi coal-fired power station was unlawful because the authority had failed to consider a climate change impact assessment, which it held to be a relevant consideration under the National Environmental Management Act. In Norway, the Oslo District Court in 2024 invalidated the approval of three oil and gas fields for failure to assess combustion emissions, and the question was referred to the EFTA Court, which in 2025 confirmed that combustion emissions must be assessed under the EIA Directive. By contrast, courts in the United States have reached varying conclusions on the scope of NEPA reviews, and the Supreme Court in *Seven County Infrastructure Coalition v Eagle County* (2025), examined in Chapter 24, emphasised deference to agencies in defining the scope of their environmental reviews and held that NEPA does not require agencies to assess the effects of separate upstream and downstream projects.

Consultation and Indigenous Rights

Licensing decisions may also be challenged for failure to consult affected communities and indigenous peoples. In ***Santos NA Barossa v Tipakalippa*** (Full Federal Court of Australia, 2022), the court upheld the decision setting aside the approval of an environment plan for offshore drilling because the regulator could not have been satisfied that the operator had consulted all relevant persons, including the traditional owners of the Tiwi Islands, whose functions, interests, or activities might be affected. In ***Sustaining the Wild Coast NPC v Minister of Mineral Resources and Energy*** (High Court of South Africa, Makhanda, 2022), the court set aside the exploration right granted to Shell and Impact Africa for seismic surveys off the Wild Coast of the Eastern Cape, on the grounds that the consultation process had been inadequate, particularly with the affected communities whose customary rights and spiritual and cultural relationship with the ocean were at stake, and that the decision-maker had failed to consider relevant factors, including the climate impacts of oil and gas development. Courts in Canada have repeatedly quashed approvals of pipelines and energy projects for failure to fulfil the Crown's **duty to consult and accommodate** indigenous peoples, as in the Federal Court of Appeal's decision in ***Tsleil-Waututh Nation v Canada*** (2018) quashing the approval of the Trans Mountain pipeline expansion, which was subsequently re-approved after further consultation. These cases,

examined in Chapter 21, demonstrate the importance of consultation as a ground of review in resource licensing.

Remedies in Licensing Challenges

The remedies available in licensing challenges include **quashing orders** setting aside the decision, **mandatory orders** requiring the authority to perform its duty, **prohibiting orders**, **declarations**, and, in some systems, **suspension** of the decision pending judgment. Courts have discretion to refuse relief, for example where the outcome would inevitably have been the same, or to suspend the effect of a quashing order to allow the authority to remedy the defect, as in the Rosebank and Jackdaw litigation. The **timing** of challenges is important: judicial review claims must generally be brought promptly, and in England within six weeks for planning decisions, and courts may refuse relief if a claimant has delayed. Licensing challenges also raise issues of **costs**, which may deter claimants; the **Aarhus Convention**, examined in Chapter 21, requires that procedures in environmental matters be not "prohibitively expensive", leading to costs-capping rules in the United Kingdom and other parties.

27.4 Environmental and Climate Litigation

Scope of this Section

Chapter 18 examined climate litigation as a source and driver of climate law, surveying the landmark cases against governments and corporations and their substantive holdings. This section focuses on the **litigation process** — the procedural and evidential questions that determine whether environmental and climate claims succeed: standing, the choice of legal basis and forum, evidence and causation, the role of science, and litigation strategy.

Typology of Environmental and Climate Cases

Environmental and climate cases in the energy sector may be classified as: (a) **framework cases** challenging the overall adequacy of governmental climate policies, such as *Urgenda*, *Neubauer*, and *KlimaSeniorinnen*; (b) **project cases** challenging the approval of specific fossil fuel projects, such as *Finch* and the Rosebank litigation; (c) **corporate cases** seeking to impose obligations on companies to reduce emissions, such as ***Milieudefensie v Royal Dutch Shell***, in which the Hague District Court ordered Shell in 2021 to reduce its global emissions by forty-five per cent by 2030, a decision reversed by the Hague Court of Appeal in November 2024, which held that Shell has a duty to reduce emissions but that there was insufficient consensus to impose a specific percentage reduction; (d) **damages cases** seeking compensation for climate harms from fossil fuel producers, such as the claims brought by US states and municipalities against oil companies under state law, and ***Lliuya v RWE***, in which a Peruvian farmer

sought a contribution from RWE towards the costs of protecting his home from a glacial lake outburst flood; (e) **"greenwashing" cases** challenging misleading environmental claims, examined in Chapter 22; (f) **disclosure and fiduciary duty cases**, such as *ClientEarth v Shell's Directors* (2023), examined in Chapter 22; and (g) **pollution cases** seeking remedies for local environmental harm, such as the oil pollution cases in the Niger Delta.

Standing

Standing (*locus standi*) — the right to bring a claim — is a threshold barrier in environmental litigation, because environmental harms are often diffuse, affecting the public at large rather than particular individuals. Approaches to standing vary widely. In the **United States**, Article III of the Constitution requires a plaintiff to show an injury in fact that is concrete and particularised, fairly traceable to the defendant's conduct, and likely to be redressed by a favourable decision (*Lujan v Defenders of Wildlife*, 1992), a requirement that has defeated many climate claims, although in *Massachusetts v EPA* (2007) the Supreme Court accepted that a state had standing given its "special solicitude" as a sovereign and the injury to its coastal land. In the **United Kingdom**, a claimant for judicial review must have a "sufficient interest" in the matter, which courts have interpreted liberally to allow environmental organisations to bring public interest claims (*R v Inspectorate of Pollution, ex p Greenpeace (No 2)*, 1994). The **Aarhus Convention** requires parties to ensure that members of the public concerned, including environmental NGOs meeting requirements under national law, have access to justice in environmental matters. The European Court of Human Rights in *KlimaSeniorinnen* developed a distinctive approach, recognising the standing of associations representing persons affected by climate change while imposing stringent requirements for individual victim status.

In **Nigeria**, the traditional restrictive approach to standing, articulated in ***Adesanya v President of the Federal Republic of Nigeria*** (1981), required a claimant to show that his civil rights and obligations had been or were in danger of being violated, which often defeated public interest environmental claims. The Supreme Court's decision in ***Centre for Oil Pollution Watch v NNPC*** (2018) marked a decisive shift, holding that an NGO dedicated to environmental protection had standing to sue for the remediation of an oil spill affecting a community's water supply, and emphasising that environmental protection is a matter of public interest and that the restrictive approach would leave environmental wrongs without remedy. The Fundamental Rights (Enforcement Procedure) Rules 2009 also liberalise standing in human rights matters. At the regional level, the **ECOWAS Community Court of Justice** held in ***SERAP v Nigeria*** (2012) that an NGO could bring a claim concerning the failure of the Nigerian government to prevent and remedy oil pollution in the Niger Delta, finding a violation of the right to a

satisfactory environment under Article 24 of the African Charter and ordering the government to take measures to restore the environment and prevent future damage.

Evidence, Causation, and Attribution Science

Environmental and climate litigation raises formidable questions of **evidence and causation**. In pollution cases, claimants must prove that the defendant's activities caused the pollution and that the pollution caused the claimed harm — often difficult where there are multiple potential sources, as in the Niger Delta, where spills may result from operational failures, corrosion, or sabotage and theft. The **burden of proof** of the cause of spills has been a central issue: under the Nigerian Oil Pipelines Act, an operator is not liable for damage caused by the malicious act of a third person, and companies have frequently attributed spills to sabotage, a claim contested by communities. In ***Four Nigerian Farmers and Milieudefensie v Shell*** (Hague Court of Appeal, January 2021), examined in section 27.8, the court held that, where Shell's Nigerian subsidiary alleged sabotage, it bore the burden of proving sabotage beyond reasonable doubt, and that it had failed to do so in respect of two of the spills. Joint fact-finding and independent investigations of spills, as recommended by the UNEP **Environmental Assessment of Ogoniland** (2011), can reduce disputes over causation.

In climate cases, the causal chain between a defendant's emissions and particular harms is long and diffuse. **Attribution science** — the scientific study of the extent to which climate change has influenced specific extreme weather events and the contribution of particular emitters to climate change — has advanced rapidly and is increasingly invoked in litigation. Studies attributing a quantifiable share of historical emissions to specific fossil fuel producers ("carbon majors"), and studies estimating the influence of climate change on the probability and intensity of particular events, provide a basis for claims that particular companies have contributed to particular harms. In ***Lliuya v RWE***, the Higher Regional Court of Hamm held in **May 2025**, after taking expert evidence including a site inspection in Peru, that the claim failed because the risk to the claimant's property from a glacial lake outburst flood was too low to justify relief, but it accepted in principle that a major emitter could be liable under German law for its proportionate contribution to climate-related risks — a significant recognition of the legal relevance of attribution. Courts in the United States have grappled with the removal of state law climate claims to federal court and with defences of federal displacement and pre-emption, and the progress of these cases through state courts will test the use of attribution evidence at trial.

The Role of Science and Expert Evidence

Environmental and climate litigation relies heavily on **scientific evidence** — on the reports of the Intergovernmental Panel on Climate Change, which courts in *Urgenda*, *Neubauer*, and *KlimaSeniorinnen* treated as authoritative statements of the scientific consensus; on carbon budgets; on environmental monitoring and modelling; and on expert testimony on pollution, health impacts, and remediation. The treatment of scientific evidence raises questions about the expertise of courts, the admissibility of expert evidence (governed in the United States by the *Daubert* standard), and the use of court-appointed experts. The **precautionary principle**, examined in Chapter 17, may affect the burden and standard of proof where there is scientific uncertainty about environmental risks.

Litigation Strategy

Strategic environmental and climate litigation is often pursued by NGOs and communities as part of broader campaigns for legal and policy change, with objectives that include not only winning particular cases but also establishing legal principles, generating public attention, influencing corporate and governmental behaviour, and building movements. Litigation strategy involves choices about the **claimants** (individuals, communities, NGOs, youth, or local governments), the **defendants** (governments, regulators, parent companies, subsidiaries, financiers, or directors), the **legal basis** (constitutional rights, human rights treaties, tort, administrative law, company law, consumer protection, or securities law), the **forum** (domestic courts, regional human rights courts, UN treaty bodies, OECD National Contact Points, or international courts), and the **remedies** (declarations, injunctions, damages, or orders for policy change). The proliferation of climate cases — more than 2,500 cases have been recorded worldwide by the Sabin Center for Climate Change Law at Columbia University — and their diffusion across jurisdictions reflect a transnational network of litigants, lawyers, and scientists. At the same time, companies and governments have also used litigation strategically, including through **strategic lawsuits against public participation** (SLAPPs) against activists and NGOs, as in the defamation and racketeering claims brought by Energy Transfer against Greenpeace in North Dakota, which resulted in a jury verdict of more than USD 660 million against Greenpeace entities in March 2025, later reduced by the court, and which Greenpeace International has challenged in the Netherlands under the EU's **Anti-SLAPP Directive** (Directive (EU) 2024/1069).

27.5 Contractual and Commercial Litigation

Courts and Energy Contracts

Despite the prevalence of arbitration clauses in international energy contracts, a large volume of energy contract disputes is litigated in national courts. Domestic

energy contracts — between distribution companies and customers, between suppliers and marketers, between operators and local contractors — frequently provide for litigation rather than arbitration. Many international contracts, particularly in finance, trading, and shipping, choose the courts of England or New York. Courts also hear disputes arising from contracts with arbitration clauses, when the validity or scope of the clause is contested, when interim relief is sought, or when the parties waive arbitration. And courts develop the principles of contract law applied in arbitration, as the English courts have done through a substantial body of case law on energy contracts.

The English Commercial Court and Energy Contracts

The **Commercial Court** in London is among the leading fora for international energy contract litigation, owing to the popularity of English law in energy and commodities contracts, the expertise of its judges, and the enforceability of its judgments. Its decisions on energy contracts have shaped the law in areas examined in Chapter 15, including the interpretation of force majeure clauses (as in ***Classic Maritime v Limbungan Makmur***, Court of Appeal 2019, on the "but for" causation requirement, and ***RTI Ltd v MUR Shipping BV***, UK Supreme Court 2024, which held that a "reasonable endeavours" proviso in a force majeure clause did not require the affected party to accept non-contractual performance, such as payment in a different currency, to overcome a sanctions-related obstacle); the interpretation of take-or-pay clauses (*M&J Polymers v Imerys*, 2008, holding that take-or-pay clauses are not penalties, a conclusion strengthened by the reformulation of the penalty doctrine in *Cavendish Square Holding v Makdessi*, UK Supreme Court 2015); the implication of terms and the duty of good faith in long-term relational contracts (*Bates v Post Office*, 2019, and its application in energy contracts); and the operation of pre-emption, default, and forfeiture provisions in joint operating agreements. The Commercial Court has also heard major disputes arising from Russian sanctions, including claims by aircraft lessors and energy traders, and disputes arising from the collapse of energy trading companies.

Commercial Litigation in Nigeria

In **Nigeria**, the exclusive jurisdiction of the **Federal High Court** over matters relating to mines and minerals, including oil fields and natural gas, under section 251(1)(n) of the Constitution, examined in section 27.1, channels much energy contract litigation to that court. Recurring categories of commercial litigation in the Nigerian energy sector include disputes between **joint venture partners**, including claims concerning cash calls, the allocation of production, and the exercise of pre-emption rights on the sale of interests; disputes over the **assignment and transfer** of interests in oil mining leases, which require ministerial consent, as illustrated by the litigation surrounding several divestments by international

companies examined in Chapter 24; disputes between operators and **marginal field** owners; disputes between **service contractors** and operators over payment, termination, and local content obligations; disputes arising from the **privatised electricity sector**, including between generation companies, distribution companies, and the Nigerian Bulk Electricity Trading Plc over payment shortfalls and the enforcement of power purchase agreements; and disputes between **banks and borrowers** in oil and gas financing, including the enforcement of security over oil assets. The delays in Nigerian courts, the possibility of interlocutory appeals, and the difficulty of enforcing judgments against government entities have encouraged parties to prefer arbitration, while also leading to the use of the courts to obstruct arbitration through anti-arbitration injunctions — a practice that Nigerian appellate courts have increasingly discouraged, emphasising the policy of the Arbitration and Mediation Act 2023 to limit court intervention.

Commercial Litigation Involving States and State Entities

Commercial litigation against states and state-owned enterprises raises questions of **state immunity**, examined in Chapter 26 in the context of the enforcement of awards. Under the restrictive theory, states are not immune from jurisdiction in respect of **commercial transactions**, including contracts for the supply of goods or services and loans. The characterisation of energy transactions as commercial or sovereign acts can be contested: the sale of oil by a national oil company is generally commercial, but the grant of a petroleum licence or the imposition of a tax is sovereign. In ***Republic of Argentina v Weltover*** (1992), the US Supreme Court held that the issuance of government bonds was a commercial activity because it was the type of activity in which private parties engage, regardless of the governmental purpose, a test applied to energy transactions in subsequent cases. Litigation against states has also involved disputes over **sovereign debt** backed by oil revenues, such as the litigation concerning bonds issued by PDVSA and the Republic of Congo's oil-backed loans, and disputes concerning the **Act of State doctrine**, under which US courts decline to adjudicate the validity of the public acts of a foreign sovereign within its own territory, as applied to nationalisations of oil interests in *Banco Nacional de Cuba v Sabbatino* (1964).

Commodity Trading and Price Disputes

Energy commodity trading generates litigation over the performance of trading contracts, the interpretation of standard terms (such as the **ISDA Master Agreement** for derivatives, and the BP, Shell, and other standard terms for crude oil and products trading), the validity of termination and close-out under derivatives contracts, and market manipulation. The **"negative oil prices"** of April 2020, when the price of the West Texas Intermediate futures contract fell below zero, led to litigation over losses incurred by investors and the conduct of trading

platforms. The collapse of commodity traders and fraud in commodity trade finance, as in the collapse of **Hin Leong Trading** in Singapore in 2020 and the nickel trading crisis on the **London Metal Exchange** in March 2022 — when the LME suspended nickel trading and cancelled trades executed on the morning of 8 March after prices doubled in a day, a decision upheld by the English High Court against judicial review claims by hedge funds in ***R (Elliott Associates) v London Metal Exchange*** (2023) — illustrate the interaction of commercial and regulatory law in energy and metals markets.

27.6 Class Actions and Collective Redress

The Problem of Mass Harm

Energy and resource activities can cause **mass harm** affecting large numbers of people — oil spills affecting entire communities, as in the Niger Delta and the *Deepwater Horizon* disaster; dam failures, as in the collapse of the Fundão tailings dam at Mariana in Brazil in 2015 and the Brumadinho dam in 2019; mine-related diseases, such as silicosis among gold miners in South Africa; air pollution from power plants; and consumer harms from energy pricing and mis-selling. Individual claims in such cases may be too small to justify the cost of litigation, and the duplication of individual proceedings would be inefficient and inconsistent. **Collective redress** mechanisms — class actions, representative actions, group litigation, and mass claims settlement mechanisms — allow claims to be aggregated, increasing access to justice and the efficiency of proceedings, while raising concerns about abusive litigation, the role of litigation funders, and the protection of absent class members.

Class Actions in the United States

The **US class action** under **Rule 23 of the Federal Rules of Civil Procedure** is the most developed model. A class action may be certified if the class is so numerous that joinder is impracticable, there are questions of law or fact common to the class, the claims of the representative parties are typical of those of the class, and the representatives will fairly and adequately protect the interests of the class; and, for damages class actions under Rule 23(b)(3), if common questions predominate over individual questions and a class action is superior to other methods. US class actions operate on an **opt-out** basis, binding all class members who do not exclude themselves, and are supported by contingency fees and the absence of the "loser pays" rule. The *Deepwater Horizon* litigation, consolidated in multidistrict litigation in New Orleans, resulted in a court-approved economic and property damages settlement and a medical benefits settlement with BP, under which billions of dollars were paid to claimants, as part of total costs to BP of more than USD 65 billion, including the settlement with the federal and Gulf states governments of around USD 20.8 billion, examined in Chapter 17. The

US Supreme Court's decision in *Wal-Mart Stores v Dukes* (2011), requiring that common questions generate common answers apt to drive the resolution of the litigation, has made certification more demanding.

Group Litigation in England and Wales

In **England and Wales**, collective redress is available principally through **Group Litigation Orders** (GLOs) under the Civil Procedure Rules, which allow the court to manage claims giving rise to common or related issues of fact or law, on an **opt-in** basis, and through **representative actions** under CPR 19.8, which allow one or more persons with the "same interest" to represent others, on an opt-out basis. In ***Lloyd v Google*** (2021), the Supreme Court held that a representative action for damages could not proceed where damages required individualised assessment, limiting the use of representative actions for damages, though it suggested that a bifurcated approach, with common issues determined in a representative action and individual damages in subsequent proceedings, could be appropriate. Opt-out collective proceedings are available for competition law claims before the **Competition Appeal Tribunal** under the Consumer Rights Act 2015, and have been brought against energy and other companies.

English group litigation has become a major forum for claims arising from the overseas activities of UK-domiciled energy and mining companies. The ***Bodo Community v Shell Petroleum Development Co*** litigation, concerning two oil spills in 2008 in the Bodo community in Ogoniland, was settled in 2015 for £55 million, paid to more than 15,000 individual claimants and the community, as noted in Chapter 17; a further claim concerning the remediation of the spills proceeded to a trial in 2025. In ***Okpabi v Royal Dutch Shell*** (2021), examined in section 27.8, the Supreme Court allowed claims by more than 40,000 Nigerians from the Ogale and Bille communities to proceed against Shell's parent company. In the ***Município de Mariana v BHP Group*** litigation, more than 600,000 Brazilian claimants, including municipalities, businesses, and indigenous communities, brought claims against BHP in respect of the Fundão dam collapse; after the Court of Appeal allowed the claim to proceed in 2022, reversing a decision to strike it out as an abuse of process, the High Court held in **November 2025**, following a trial of liability, that BHP was liable under Brazilian law as a polluter. The case is reportedly the largest group action in English legal history and illustrates the potential and the challenges of group litigation for mass harms abroad, including its interaction with settlements in the home state — in October 2024, BHP, Vale, and their joint venture Samarco agreed a settlement with Brazilian authorities of around BRL 170 billion.

Collective Redress in the European Union and Other Jurisdictions

In the **European Union**, the **Representative Actions Directive** (Directive (EU) 2020/1828), applicable since June 2023, requires member states to provide for representative actions by qualified entities for the protection of the collective interests of consumers, including for redress measures such as compensation, in areas including energy. The **Netherlands** has become a leading European forum for collective actions under the **Act on Redress of Mass Damages in a Collective Action** (WAMCA) of 2020, which allows opt-out collective damages actions by representative foundations, and under which claims have been brought against energy companies, including in relation to the earthquakes caused by gas production in the **Groningen** field, which led to extensive litigation and compensation schemes administered under public law. **Canada**'s provincial class action regimes, **Australia**'s Part IVA representative proceedings under the Federal Court of Australia Act 1976, which have been used in claims over the Montara oil spill in the Timor Sea (*Sanda v PTTEP Australasia*, 2021, in which the Federal Court found the operator liable in negligence to Indonesian seaweed farmers), and **South Africa**'s judicially developed class action procedure, used in the silicosis litigation against gold mining companies, which resulted in a settlement of around ZAR 5 billion in 2018, provide other models.

Collective Redress in Nigeria

In **Nigeria**, collective claims are brought principally through **representative actions** under the rules of the Federal High Court and State High Courts, which allow one or more persons to sue on behalf of persons having the same interest, and through the joinder of numerous claimants. Oil pollution claims are frequently brought by community leaders on behalf of their communities, and the courts have addressed questions about the authority of representatives, the identification of the persons represented, and the distribution of compensation. The absence of a developed class action regime, combined with the cost and delay of litigation and the difficulty of enforcing judgments, has led many Niger Delta communities to pursue claims in the courts of England and the Netherlands, as examined in section 27.8. The Petroleum Industry Act's provisions on **host community development trusts**, examined in Chapter 24, and on compensation for environmental damage, may influence the future of community claims, as may the development of **non-judicial grievance mechanisms** and mass claims settlement procedures such as the claims process established in the Bodo settlement.

Litigation Funding and Costs

Collective redress depends on **funding**. In the United States, contingency fees and the absence of costs-shifting make class actions viable. In England, the combination of conditional fee agreements, damages-based agreements, after-the-

event insurance, and **third-party litigation funding** supports group litigation, though the costs-shifting rule exposes unsuccessful claimants to adverse costs. The UK Supreme Court's decision in ***R (PACCAR Inc) v Competition Appeal Tribunal*** (2023), which held that litigation funding agreements under which the funder's return is calculated as a percentage of damages are damages-based agreements subject to statutory regulation, disrupted the funding market, prompting proposals for legislation to reverse it, which the government announced in 2025 it would introduce following a review by the Civil Justice Council. The availability and regulation of litigation funding significantly affect access to justice for claimants harmed by energy activities, particularly in transnational cases.

27.7 Injunctions, Remedies, and Enforcement Orders

The Range of Remedies

The effectiveness of energy litigation depends on the **remedies** available. Courts may award **damages** to compensate for loss; grant **injunctions** prohibiting or requiring conduct; make **declarations** of rights or of the unlawfulness of conduct; issue **public law orders** quashing decisions or compelling authorities to act; order **restitution** or the **account of profits**; and, in environmental cases, order **remediation** and restoration. Regulatory authorities have an additional range of **enforcement tools**, from warnings and compliance notices to civil penalties, licence suspension and revocation, and criminal prosecution. The choice of remedy reflects the nature of the harm, the objectives of the claimant, and the constitutional limits of judicial power.

Damages

Damages are the principal remedy for breach of contract and tort. In contract, damages aim to put the claimant in the position it would have been in had the contract been performed, subject to rules of remoteness (*Hadley v Baxendale*, 1854), mitigation, and contractual limitations of liability, which are ubiquitous in energy contracts, as examined in Chapter 15. In tort, damages aim to restore the claimant to the position it would have been in had the tort not been committed. In oil pollution cases, damages may cover loss of property, loss of income from fishing and farming, the cost of remediation, personal injury, and, in some jurisdictions, general damages for inconvenience and loss of amenity. Nigerian courts have awarded damages for oil pollution under the common law and statute, including under the **Oil Pipelines Act**, which provides for compensation for damage caused by the bursting or leakage of pipelines, and have occasionally awarded substantial sums, though the enforcement of large judgments has often been delayed by appeals. **Exemplary or punitive damages**, which punish egregious conduct, are available in the United States — as in the

Exxon Valdez litigation, in which the Supreme Court in ***Exxon Shipping Co v Baker*** (2008) reduced a punitive damages award of USD 2.5 billion to around USD 507 million, holding that under maritime law punitive damages should not exceed compensatory damages — and, more restrictively, in England and Nigeria.

Injunctions

Injunctions are particularly important in energy and environmental litigation, because damages may be an inadequate remedy for irreversible environmental harm, and because claimants often seek to prevent harm rather than to be compensated for it. **Interim injunctions** may be granted to preserve the status quo pending trial. In England and in Nigeria, which follows the English approach, the principles in ***American Cyanamid Co v Ethicon*** (1975) require the claimant to show a serious question to be tried, after which the court considers whether damages would be an adequate remedy and where the **balance of convenience** lies, and claimants are usually required to give an **undertaking in damages** to compensate the defendant if the injunction proves to have been wrongly granted — a requirement that may deter community and NGO claimants, although courts may dispense with it in public interest cases, and the Aarhus Convention requires that injunctive relief in environmental cases be available on terms that are not prohibitively expensive. **Final injunctions** may prohibit continuing nuisances, such as pollution, or require positive action, such as remediation. **Mandatory injunctions**, requiring the defendant to take positive steps, are granted more cautiously than prohibitory ones, because they may require continuing supervision by the court.

Energy-related injunctions illustrate the possibilities and limits of the remedy. In ***Gbemre v Shell***, the Federal High Court ordered the cessation of gas flaring, but the order was not complied with and contempt proceedings stalled. In ***Milieudefensie v Shell***, the Hague District Court's order requiring Shell to reduce its emissions was a novel form of **structural injunction**, setting an outcome to be achieved rather than prescribing specific measures, before it was reversed on appeal. In the United States, injunctions have halted pipeline construction pending the completion of environmental reviews, as in the litigation over the **Dakota Access Pipeline**, in which the District Court for the District of Columbia ordered in 2020 that the pipeline be shut down pending a new environmental impact statement — an order reversed by the D.C. Circuit in 2021, which upheld the requirement of an EIS but held that the district court had not made the findings required for an injunction. The US Supreme Court's decision in ***Winter v Natural Resources Defense Council*** (2008) requires a plaintiff seeking a preliminary injunction to establish that it is likely to succeed on the merits, likely to suffer irreparable harm, that the balance of equities tips in its favour, and that an

injunction is in the public interest, and *Monsanto v Geertson Seed Farms* (2010) held that an injunction does not follow automatically from a NEPA violation.

Injunctions are also used **against protesters**. Energy companies and governments have obtained injunctions restraining protests at energy facilities, including **"persons unknown" injunctions** in England against climate protesters blocking oil terminals and motorways, the legitimacy of which was addressed by the UK Supreme Court in ***Wolverhampton City Council v London Gypsies and Travellers*** (2023), which held that courts may grant injunctions against "newcomers" — persons unknown who have not yet committed the prohibited acts — subject to safeguards. The balance between the protection of energy infrastructure and the rights of freedom of expression and assembly is a recurring issue, as examined in Chapter 21.

Declarations and Public Law Remedies

Declarations — statements of the legal position — are often the primary remedy sought in public law and climate litigation, because they establish the unlawfulness of governmental conduct while leaving the choice of means to the authorities, respecting the separation of powers. In *Urgenda*, the court ordered the state to achieve a minimum reduction target, leaving the means to the government; in *Neubauer*, the Federal Constitutional Court declared the Climate Change Act partly unconstitutional and required the legislature to amend it, which it did within months, adopting more ambitious targets. In *KlimaSeniorinnen*, the European Court of Human Rights found a violation and left the measures to be taken to Switzerland under the supervision of the Committee of Ministers, which has been criticised by Switzerland's parliament as an overreach. Courts in some jurisdictions have adopted **supervisory** or **structural** remedies, retaining jurisdiction to monitor compliance — as in the Indian Supreme Court's **continuing mandamus** in environmental cases, the Colombian Supreme Court's orders in ***Future Generations v Ministry of Environment*** (2018), which required the government to formulate plans to reduce deforestation in the Amazon, and the orders of the South African courts in the *Deadly Air* case (*Trustees for the time being of groundWork Trust v Minister of Environmental Affairs*, 2022), in which the High Court declared that the poor air quality in the Highveld Priority Area, caused largely by coal-fired power stations and a coal-to-liquids plant, violated the constitutional right to an environment not harmful to health and well-being, and ordered the minister to make regulations to implement the area's air quality management plan.

Regulatory Enforcement

Regulatory authorities enforce energy law through administrative, civil, and criminal tools, arranged in what Ayres and Braithwaite called an **"enforcement**

pyramid", in which regulators begin with persuasion and escalate to more severe sanctions in response to non-compliance, as examined in Chapter 3. In the **United Kingdom**, **Ofgem** may impose financial penalties of up to ten per cent of a licensee's turnover for breaches of licence conditions and statutory requirements, and has secured substantial penalties and redress payments from energy suppliers for mis-selling, billing failures, and breaches of consumer protection rules, often through negotiated settlements that direct payments to consumer redress funds. The **Health and Safety Executive** and the **North Sea Transition Authority** have enforcement powers in their areas. In the **United States**, the **Environmental Protection Agency** enforces environmental laws through administrative orders, civil actions, and **consent decrees** requiring defendants to install pollution controls, pay penalties, and fund environmental projects, and **FERC** has authority to impose civil penalties of more than USD one million per day for violations of the Federal Power Act and the Natural Gas Act, including market manipulation. Criminal prosecutions followed the *Deepwater Horizon* disaster, with BP pleading guilty to felony charges and paying around USD 4 billion in criminal penalties.

In **Nigeria**, the **NUPRC** and the **NMDPRA** have powers under the Petroleum Industry Act 2021 to impose administrative penalties, suspend and revoke licences, and refer matters for prosecution; NUPRC regulations provide for penalties for gas flaring, under which flaring penalties are payable per thousand standard cubic feet flared, and for failure to report spills. The **National Oil Spill Detection and Response Agency** (NOSDRA), established in 2006, is responsible for detecting and responding to oil spills and enforcing compliance with environmental legislation in the oil sector, and has imposed fines on companies for failure to report or clean up spills, though its powers to impose penalties have been contested in court. **NERC** enforces the Electricity Act through licence conditions, orders, and penalties, including against distribution companies for failure to meet performance standards and metering obligations. The effectiveness of regulatory enforcement in Nigeria has been limited by inadequate resources, regulatory capture, political interference, and the difficulty of monitoring operations in remote and insecure areas — challenges that the institutional reforms of the PIA seek to address.

Enforcement of Judgments and Contempt

The enforcement of judgments is the final stage of litigation, and often the most difficult. Judgments for money may be enforced through execution against the debtor's assets, garnishee proceedings, and charging orders; injunctions are enforced through **contempt of court**, which may lead to fines, sequestration of assets, or imprisonment. In **Nigeria**, the enforcement of judgments against government agencies is subject to the requirement of the Attorney-General's consent

under section 84 of the Sheriffs and Civil Process Act, noted in Chapter 26, and the enforcement of judgments against companies is often delayed by appeals and applications for stays of execution. The challenges of enforcing environmental judgments are illustrated by the slow progress of the clean-up of Ogoniland under the **Hydrocarbon Pollution Remediation Project** (HYPREP), established in 2016 to implement the UNEP report, which demonstrates that even consensual and well-funded remediation obligations face severe implementation difficulties.

27.8 Cross-Border Litigation and Jurisdictional Conflicts

Transnational Litigation Against Energy Companies

The activities of multinational energy and mining companies in host states have generated **transnational litigation**, in which victims of harm in one state sue the parent company, its subsidiaries, or both, in the courts of the parent company's home state. Claimants pursue such litigation because home state courts may offer greater independence, more effective procedures (including disclosure of documents), access to funding and legal expertise, and the ability to enforce judgments against the parent company's assets. Such litigation raises three principal sets of questions: whether the home state courts have **jurisdiction**, and whether they should decline it in favour of the host state's courts; whether the **parent company** can be liable for the conduct of its subsidiary; and what **law** applies to the claim. The substantive question of parent company liability has been examined in Chapters 17 and 22; this section focuses on jurisdiction and enforcement.

Jurisdiction in the United Kingdom and the European Union

In the **European Union**, the **Brussels I Regulation (Recast)** (Regulation (EU) No 1215/2012) provides that persons domiciled in a member state shall be sued in the courts of that state (Article 4), and the Court of Justice held in ***Owusu v Jackson*** (2005) that a court of a member state may not decline jurisdiction over a defendant domiciled in that state on the ground of **forum non conveniens**, even if the courts of a non-member state would be more appropriate. This enables claimants to sue EU-domiciled parent companies in their home courts as of right. Claims against foreign subsidiaries may be joined, under national rules, as **necessary or proper parties** to the claim against the parent. Following Brexit, the Brussels I Regulation no longer applies in the **United Kingdom**, and English courts have reverted to common law rules for defendants not domiciled in the EU, under which the courts may decline jurisdiction on the ground of forum non conveniens where another forum is clearly more appropriate (*Spiliada Maritime Corp v Cansulex*, House of Lords 1987), though jurisdiction over UK-domiciled defendants remains established as of right by service within the jurisdiction, subject to the discretion to stay.

In ***Lubbe v Cape plc*** (House of Lords, 2000), the House of Lords refused to stay claims by South African asbestos victims against an English parent company on the ground of forum non conveniens, because the claimants would be denied **substantial justice** in South Africa owing to the lack of legal aid and of procedures for group litigation. In ***Vedanta Resources plc v Lungowe*** (UK Supreme Court, 2019), concerning pollution from the Nchanga copper mine in Zambia, operated by Vedanta's Zambian subsidiary Konkola Copper Mines, the Supreme Court held that the claims against the English parent company raised a real issue to be tried, as the parent might have assumed a duty of care through its group-wide environmental policies and their implementation, and that, although Zambia was the proper place for the claims, there was a real risk that the claimants would not obtain **substantial justice** in Zambia, owing to the lack of funding and of suitably experienced legal teams, so that the claims against both companies could proceed in England. The claim was settled in 2021. In ***Okpabi v Royal Dutch Shell plc*** (UK Supreme Court, 2021), the Supreme Court applied *Vedanta* to allow claims by the Ogale and Bille communities against Shell's parent company and its Nigerian subsidiary to proceed, holding that the Court of Appeal had erred by conducting a "mini-trial" at the jurisdictional stage and by failing to recognise that the parent company's control and oversight of its subsidiary, including through group-wide standards, could give rise to a duty of care, and that the claimants should have the opportunity to obtain disclosure of internal documents.

The Netherlands: Four Nigerian Farmers

In ***Four Nigerian Farmers and Milieudefensie v Royal Dutch Shell and Shell Petroleum Development Company of Nigeria***, Nigerian farmers from the villages of Oruma, Goi, and Ikot Ada Udo, supported by Milieudefensie, sued Shell's parent company and its Nigerian subsidiary in the Netherlands for damage caused by oil spills. In **January 2021**, after thirteen years of litigation, the **Hague Court of Appeal**, applying Nigerian law, held SPDC liable for spills at Oruma and Goi, ordering it to pay compensation, and held that the parent company had breached a duty of care by failing to ensure that a leak detection system was installed on the Oruma pipeline. In 2022, Shell agreed to pay EUR 15 million to the communities in settlement. The case was the first in which a Dutch court held a Dutch parent company to a duty of care in relation to the activities of its foreign subsidiary, and demonstrated the role of the Dutch courts, under the Brussels regime, as a forum for transnational claims.

The United States: The Alien Tort Statute and Forum Non Conveniens

In the **United States**, the **Alien Tort Statute** (ATS) of 1789, which grants the district courts jurisdiction over civil actions by aliens for torts committed

in violation of the law of nations, became a vehicle for human rights claims against corporations in the 1990s and 2000s, including claims arising from energy operations in Nigeria, Myanmar, and elsewhere. In ***Kiobel v Royal Dutch Petroleum Co*** (2013), arising from the alleged complicity of Shell in human rights abuses against the Ogoni people in the 1990s, the Supreme Court held that the **presumption against extraterritoriality** applies to the ATS, so that claims must "touch and concern" the territory of the United States with sufficient force to displace the presumption, and that mere corporate presence is insufficient. In ***Jesner v Arab Bank*** (2018), the Court held that foreign corporations may not be sued under the ATS, and in ***Nestlé USA v Doe*** (2021) it held that general corporate activity in the United States, such as decision-making, is insufficient to establish domestic application. These decisions have largely closed the US federal courts to transnational human rights claims against energy companies, shifting litigation to state courts and to other jurisdictions. The US courts also apply a robust doctrine of **forum non conveniens**, as in ***Aguinda v Texaco*** (Second Circuit, 2002), in which claims by Ecuadorian and Peruvian residents for pollution caused by Texaco's oil operations in the Ecuadorian Amazon were dismissed in favour of the Ecuadorian courts — setting the stage for the Lago Agrio litigation examined in the casebox below.

The Enforcement of Foreign Judgments

A judgment obtained in one state must often be enforced in another, where the defendant's assets are located. The recognition and enforcement of foreign judgments are governed by national law and by treaties. The **Hague Convention on the Recognition and Enforcement of Foreign Judgments in Civil or Commercial Matters** (2019), which entered into force in September 2023 between the European Union and Ukraine, and to which the United Kingdom acceded with effect from July 2025, provides a framework for the circulation of judgments among its parties, subject to grounds for refusal including fraud, lack of notice, and public policy. In the absence of a treaty, common law courts enforce foreign money judgments that are final and conclusive, given by a court of competent jurisdiction (judged by the enforcing court's rules), and not obtained by fraud or in breach of natural justice or contrary to public policy. In **Nigeria**, the **Foreign Judgments (Reciprocal Enforcement) Act 1990** provides for the registration of judgments from countries extended by ministerial order on the basis of reciprocity, and the **Reciprocal Enforcement of Judgments Ordinance 1922** continues to apply to judgments of certain Commonwealth countries; judgments from other countries may be enforced by action at common law. Within ECOWAS, the enforcement of judgments of the Community Court of Justice depends on national implementation, which has been weak.

Inter-State Litigation in Energy and Resource Matters

Energy and resource disputes between states may be adjudicated by the **International Court of Justice**, the **International Tribunal for the Law of the Sea**, and inter-state arbitral tribunals. The procedure of the ICJ — which requires the consent of both states, through special agreement, compromissory clauses in treaties, or declarations under the optional clause in Article 36(2) of the Statute — and its powers to indicate **provisional measures** under Article 41 have been applied in resource disputes including ***Gabčíkovo-Nagymaros Project*** (*Hungary/Slovakia*, 1997), concerning a treaty for the joint construction of a hydroelectric dam system on the Danube, in which the Court emphasised the need to reconcile economic development with environmental protection and the concept of sustainable development; ***Pulp Mills on the River Uruguay*** (*Argentina v Uruguay*, 2010), in which the Court held that the obligation to conduct an environmental impact assessment for industrial activities that may have a significant adverse impact in a transboundary context is a requirement of general international law; and the maritime boundary cases examined in Chapter 20, including ***Land and Maritime Boundary between Cameroon and Nigeria*** (2002), which determined sovereignty over the oil-rich Bakassi Peninsula and the maritime boundary. Inter-state cases face distinctive procedural questions, including the admissibility of scientific evidence, the use of court-appointed experts (as in the reparations phase of *Armed Activities on the Territory of the Congo* (2022), in which the Court appointed experts on, among other things, the exploitation of natural resources, and in the maritime delimitation case between Costa Rica and Nicaragua (2018)), and the compliance of states with judgments, which, in the *Bakassi* case, was facilitated by the **Greentree Agreement** of 2006 under UN auspices.

::: casebox {title="Case Study: The Lago Agrio Litigation and Chevron v Ecuador"} **Background.** From 1964 to 1992, Texaco Petroleum Company (TexPet) operated oil fields in the Oriente region of the Ecuadorian Amazon in a consortium with the state oil company, later Petroecuador. Residents alleged that the operations caused extensive contamination through the discharge of produced water and the use of unlined waste pits. In 1993, residents sued Texaco in New York (*Aguinda v Texaco*), but the claims were dismissed in 2002 on the ground of forum non conveniens, on condition that Texaco submit to the jurisdiction of the Ecuadorian courts. Meanwhile, in 1995 and 1998, TexPet concluded settlement and release agreements with the government of Ecuador under which it carried out a remediation programme in exchange for a release from environmental claims by the state. Chevron acquired Texaco in 2001.

The Lago Agrio judgment. In 2003, residents sued Chevron in the Provincial Court of Sucumbíos in Lago Agrio, Ecuador. In **February 2011**, the court

entered judgment against Chevron for around USD 18 billion, including punitive damages that would be removed if Chevron apologised; the National Court of Justice reduced the judgment to around **USD 9.5 billion** in 2013, and the Constitutional Court upheld it in 2018. Chevron, which had no significant assets in Ecuador, refused to pay.

The RICO judgment. Chevron sued the plaintiffs' lead US lawyer, Steven Donziger, and others in New York under the Racketeer Influenced and Corrupt Organizations Act. In **March 2014**, the District Court for the Southern District of New York held that the Lago Agrio judgment had been procured by fraud, including the bribery of the judge and the ghostwriting of the judgment by the plaintiffs' team, and imposed an injunction preventing Donziger and others from profiting from the judgment in the United States; the Second Circuit affirmed in 2016.

The investment arbitration. Chevron and TexPet commenced UNCITRAL arbitration under the US–Ecuador BIT, administered by the PCA. In **August 2018**, the tribunal held that Ecuador had committed a **denial of justice** under customary international law and breached the fair and equitable treatment and other obligations of the BIT, because the Lago Agrio judgment was procured through fraud and corruption, and that the 1995 settlement had released TexPet and its affiliates from diffuse environmental claims of the kind pursued in Lago Agrio. It ordered Ecuador to take steps to prevent the enforcement of the judgment. Ecuador's challenge to the award in the Dutch courts was rejected.

Enforcement attempts. The plaintiffs sought to enforce the judgment against Chevron's subsidiaries in **Canada**, **Brazil**, and **Argentina**. In *Chevron Corp v Yaiguaje* (2015), the Supreme Court of Canada held that the Ontario courts had jurisdiction to hear the enforcement action, as no real and substantial connection between the defendant and the forum was required in an action to enforce a foreign judgment; but the Ontario courts subsequently held that Chevron's Canadian subsidiary was a separate legal entity whose assets could not be seized to satisfy a judgment against its indirect parent, and the enforcement action failed. Enforcement actions in Brazil and Argentina also failed.

Significance. The saga illustrates the full complexity of transnational energy litigation: the consequences of forum non conveniens dismissals, the conflict between national courts and investment arbitration, the role of fraud and corruption, the limits of the enforcement of foreign judgments against corporate groups, and the question whether the communities affected by the contamination have obtained any remedy at all. It remains a central reference point in debates about corporate accountability, access to justice, and the integrity of litigation. :::

::: exercise {title="Practical Exercise 27: Judicial Decision Analysis and Litigation Strategy — The Imara Creek Spill (HYPOTHETICAL)"} **This exercise**

is entirely hypothetical. All communities, companies, regulators, courts' decisions, and events described are fictitious and do not refer to any real place, company, or case.

In March 2025, a pipeline operated by Delta Imara Petroleum Ltd (DIP), a Nigerian company wholly owned by Imara Energy plc, a company incorporated and headquartered in England, ruptured near the Imara Creek community in a Niger Delta state, spilling an estimated 8,000 barrels of crude oil into the creek and surrounding mangroves. The community's 22,000 residents depend on fishing and farming. DIP attributes the spill to sabotage, citing a hacksaw cut found on the pipeline; the community contends that the pipeline was corroded, citing an internal integrity report, leaked to journalists, which recommended replacement of the section in 2019. Imara Energy plc publishes group-wide standards on pipeline integrity and spill response and has a group health, safety, and environment committee that reviews the performance of subsidiaries. NOSDRA's joint investigation visit report records the cause as "under dispute". The regulator has not imposed any sanction. A licence renewal for DIP's oil mining lease is pending before the relevant authority.

Tasks.

(a) **Decision analysis.** Write a critical analysis (of no more than 2,000 words) of the reasoning in *Okpabi v Royal Dutch Shell* (2021) and *Vedanta v Lungowe* (2019), identifying the principles relevant to a claim by the Imara Creek community against Imara Energy plc.

(b) **Forum and strategy.** Advise the Imara Creek community on the options for litigation in Nigeria (including the court with jurisdiction, standing, representative proceedings, and the fundamental rights procedure), in England, and before the ECOWAS Community Court of Justice, evaluating the advantages and risks of each and the possibility of combining them.

(c) **Evidence.** Identify the evidence the community will need to establish causation, including on the cause of the spill and the burden of proof regarding sabotage, and explain how it might be obtained.

(d) **Remedies.** Advise on the remedies the community should seek, including interim relief to compel clean-up, damages, and remediation orders, and on the enforcement of any judgment.

(e) **Judicial review.** Advise a Nigerian environmental NGO on whether and how it might challenge the regulator's failure to sanction DIP and any decision to renew DIP's lease.

(f) **Defence.** Advise Imara Energy plc on its defences, including on jurisdiction, and on the risk management lessons for its group governance. :::

Conclusion

Courts are indispensable actors in energy law. They review the legality of regulatory and licensing decisions, with recent decisions such as *Finch* reshaping the assessment of fossil fuel projects; they adjudicate commercial disputes and develop the principles of energy contract law; they provide remedies for communities harmed by pollution and, increasingly, for the harms of climate change; they supervise regulatory enforcement; and they determine the reach of national jurisdiction over transnational corporate activity. Energy litigation is shaped by procedural questions — jurisdiction, justiciability, standing, evidence, causation, collective redress, funding, remedies, and enforcement — as much as by substantive law. Recent developments have expanded access to courts in some respects, as in the liberalisation of standing in Nigeria after *Centre for Oil Pollution Watch*, the recognition of parent company duties in *Vedanta*, *Okpabi*, and the Dutch Shell litigation, the BHP judgment, and the climate jurisprudence of the European Court of Human Rights, while narrowing it in others, as in the US Supreme Court's restrictions on the Alien Tort Statute and on agency deference. The Lago Agrio saga demonstrates that litigation can also become a theatre of fraud and conflict, and that remedies may remain elusive even after decades of proceedings. For the doctoral researcher and practitioner, the challenge is to understand litigation not only as a means of resolving disputes but as an instrument of governance and accountability, whose effectiveness depends on institutional design, access to justice, and the capacity to enforce outcomes. Part VIII turns to the role of scholarship, law reform, and professional leadership in shaping the future of energy and natural resources law.

CHAPTER SUMMARY

- Domestic courts resolve disputes, review governmental action, interpret constitutional resource provisions, enforce regulation, and develop the law; Nigeria's Federal High Court has exclusive jurisdiction over matters relating to mines, minerals, oil fields, and natural gas; specialised environmental courts exist in many jurisdictions; and justiciability doctrines vary, with US courts more restrictive than European courts in climate cases.
- Regulatory decisions may be challenged through merits appeals and judicial review, balancing accountability and regulatory certainty; the overruling of *Chevron* deference in *Loper Bright* has intensified judicial scrutiny of US energy regulation.
- Licensing decisions are reviewed for illegality, irrationality, procedural impropriety, legitimate expectation, and proportionality; *Finch* (2024) and subsequent decisions require assessment of downstream emissions, and consultation failures have invalidated approvals in Australia, South Africa, and Canada.

- Environmental and climate litigation turns on standing, liberalised in Nigeria by *Centre for Oil Pollution Watch v NNPC* (2018), evidence and causation, including the burden of proving sabotage in oil spill cases, attribution science, recognised in principle in *Lliuya v RWE* (2025), and strategic choices of claimants, defendants, legal bases, fora, and remedies.
- Commercial litigation in energy contracts is significant in the English Commercial Court and the Nigerian Federal High Court, raising issues of contract interpretation, joint venture disputes, state immunity, and commodity trading.
- Collective redress mechanisms — US class actions, English group litigation and representative actions, EU representative actions, and the Dutch WAMCA — enable mass claims, as in the *Deepwater Horizon*, *Bodo*, and BHP Mariana litigation; Nigerian collective claims rely on representative actions.
- Remedies include damages, interim and final injunctions, declarations, structural orders, and remediation; regulators enforce through penalties, licence sanctions, and prosecution; enforcement of judgments, especially against public bodies and in environmental cases, remains difficult.
- Transnational litigation against parent companies depends on jurisdictional rules (Brussels I *bis*, forum non conveniens, substantial justice), as in *Lubbe*, *Vedanta*, *Okpabi*, and the Dutch Shell case; the ATS has been curtailed; the enforcement of foreign judgments faces corporate separateness, as in the Lago Agrio saga; and inter-state adjudication addresses transboundary resource disputes.

DOCTORAL RESEARCH QUESTIONS

1. How should courts calibrate the intensity of review of technical and economic decisions by energy regulators? Compare the approaches in the United States after *Loper Bright*, the United Kingdom, and Nigeria.
2. Evaluate the implications of *R (Finch) v Surrey County Council* for the licensing of fossil fuel projects, and assess whether comparable obligations exist or should be developed under Nigerian environmental assessment law.
3. Has the jurisprudence on parent company duties of care in *Vedanta*, *Okpabi*, and the Dutch Shell litigation created an effective remedy for victims of corporate environmental harm in the Global South? Assess its limits.
4. Propose a model for collective redress for oil pollution claims in Nigeria, drawing on comparative experience.
5. Can attribution science provide a sufficient basis for establishing the civil liability of fossil fuel producers for climate harms? Evaluate in the light of *Lliuya v RWE* and other recent cases.
6. Assess the lessons of the Lago Agrio litigation for the design of transnational mechanisms for corporate accountability and the enforcement of judgments.

FURTHER READING

- Jacqueline Peel and Hari M Osofsky, *Climate Change Litigation: Regulatory Pathways to Cleaner Energy* (Cambridge University Press 2015).
- Ivano Alogna, Christine Bakker and Jean-Pierre Gauci (eds), *Climate Change Litigation: Global Perspectives* (Brill Nijhoff 2021).
- Liesbeth Enneking, *Foreign Direct Liability and Beyond* (Eleven International Publishing 2012).
- Harry Woolf, Jeffrey Jowell, Catherine Donnelly and Ivan Hare, *De Smith's Judicial Review* (9th edn, Sweet & Maxwell 2023).
- United Nations Environment Programme, *Environmental Assessment of Ogoniland* (UNEP 2011).
- Sabin Center for Climate Change Law, Climate Change Litigation Databases (Columbia Law School, continuously updated).



PART VIII

Doctoral Scholarship, Legal Reform, and Professional Leadership

The methods of advanced legal research, the design and reform of energy legal systems, and the future of international energy and natural resources law, framed as an agenda for the next generation of doctoral scholars and professional leaders.

28 CHAPTER

Advanced Legal Research Methodology in Energy and Natural Resources Law

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the principal epistemological positions in legal scholarship and their implications for research in energy and natural resources law.
2. Conduct rigorous doctrinal legal research and evaluate its strengths and limitations.
3. Design comparative and transnational legal research projects with methodological awareness.
4. Apply socio-legal and empirical approaches to questions of energy and resource governance.
5. Select and combine qualitative and quantitative research methods appropriate to the research question.
6. Identify and address the ethical obligations of legal researchers, including research integrity, plagiarism, and the responsible use of artificial intelligence.
7. Use legal databases, primary authorities, and citation systems effectively and accurately.
8. Formulate and defend an original contribution to knowledge at doctoral level.

Introduction

The preceding twenty-seven chapters have surveyed the substantive law of international energy and natural resources. This final Part turns from what the law *is* to how it is studied, reformed, and led. Doctoral research differs from professional legal work and from taught study not merely in its length but in its purpose: the doctoral researcher must make an **original contribution to knowledge** — establishing something that was not known, explaining something that was not understood, or proposing something that had not been conceived — and must do so through a method that others can scrutinise and evaluate. The quality of a

doctoral thesis therefore depends as much on the rigour of its methodology as on the significance of its conclusions.

Energy and natural resources law presents distinctive methodological challenges. It is **multi-level**, spanning international treaties, regional instruments, national legislation, subnational regulation, and private contracts. It is **interdisciplinary**, intersecting with economics, engineering, geology, environmental science, political science, and international relations. It is **dynamic**, as the energy transition, technological change, and geopolitical upheaval transform the legal landscape faster than the literature can follow. Much of its most important law is **private or confidential** — petroleum agreements, arbitral awards, and regulatory correspondence — or exists in the practice of institutions rather than in published texts. And it is **normatively charged**, raising questions of justice, sovereignty, and sustainability on which researchers may hold strong views. These features require the researcher to make deliberate methodological choices and to justify them.

This chapter examines the epistemological foundations of legal scholarship (section 28.1), doctrinal legal research (section 28.2), comparative and transnational legal research (section 28.3), socio-legal and empirical research (section 28.4), qualitative and quantitative research approaches (section 28.5), research ethics and academic integrity (section 28.6), legal databases, primary authorities, and citation methods (section 28.7), and the development of original doctoral contributions (section 28.8). Figure 28.1 presents the research process as a sequence of stages, each requiring specific methodological decisions. Appendices N, O, and P provide a doctoral research proposal template, a model dissertation structure and evaluation criteria, and a guide to legal research, citation, and case analysis.

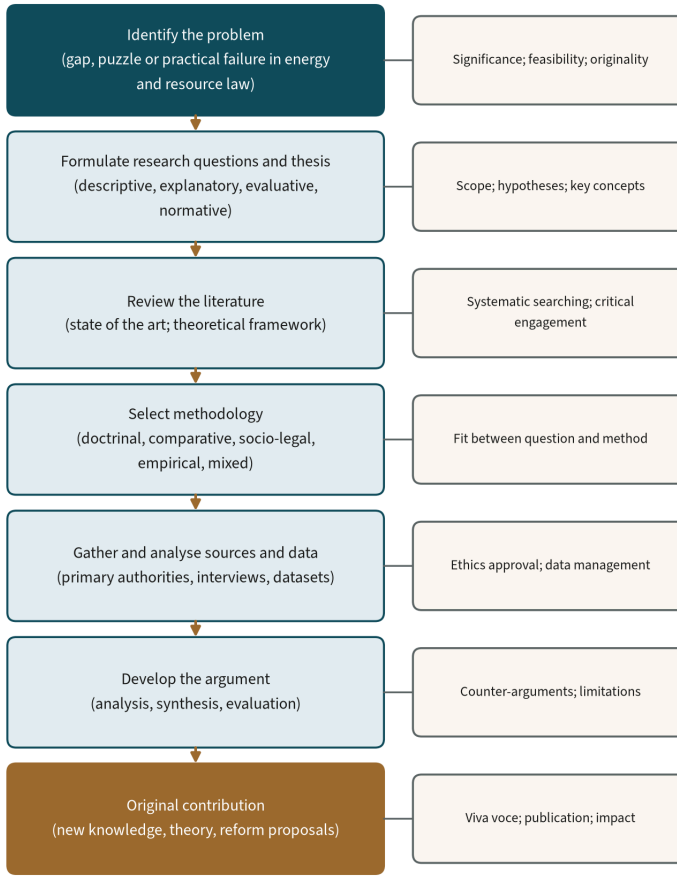


Figure 28.1 The doctoral research process in energy and natural resources law. *Source:* prepared for this edition.

28.1 Epistemological Foundations of Legal Scholarship

What Kind of Knowledge is Legal Knowledge?

Epistemology is the branch of philosophy concerned with the nature, sources, and limits of knowledge. Every research project rests, explicitly or implicitly, on epistemological assumptions: about what counts as knowledge, how it can be obtained, and how claims to knowledge can be justified. In legal scholarship, these questions are particularly contested, because law is simultaneously a body of authoritative norms, a social practice, a system of institutions, and an object of moral evaluation. The researcher who asks "what is the law governing the decommissioning of offshore installations?" is asking a different kind of question

from one who asks "why do states fail to enforce decommissioning obligations?" or "what should the law on decommissioning be?", and each question calls for a different kind of knowledge and a different method of inquiry.

The Internal Perspective: Law as a Normative System

The dominant tradition of legal scholarship adopts what H.L.A. Hart called the **internal point of view** — the perspective of participants in a legal system who accept its rules as standards for guiding and evaluating conduct. From this perspective, legal knowledge consists of knowledge of valid legal norms and their correct interpretation, and legal scholarship is a normative and interpretive discipline, closely allied with the practice of judges and lawyers. **Legal positivism**, associated with Hart, Hans Kelsen, and Joseph Raz, holds that the existence and content of law depend on social facts — such as enactment by a legislature or recognition by courts according to a rule of recognition — rather than on its moral merits. Positivism supports the conception of doctrinal scholarship as the systematic exposition of valid law, which can be undertaken without endorsing the law's moral value. **Natural law** theories, by contrast, hold that the validity or authority of law depends at least in part on its conformity with moral principles, and **interpretivism**, associated with Ronald Dworkin, holds that legal propositions are true if they figure in or follow from the principles of justice, fairness, and procedural due process that provide the best constructive interpretation of the community's legal practice. On interpretivist views, legal scholarship is inescapably evaluative, because determining what the law requires involves judgments about which interpretation shows the legal practice in its best light. These theories, examined in relation to energy law in Chapter 1, have practical consequences for research: a positivist researcher examining the legality of a stabilisation clause will focus on the sources and their authoritative interpretation; an interpretivist will ask how the clause fits within the principles that best justify the legal order, including principles of sovereignty and the right to regulate.

The External Perspective: Law as a Social Phenomenon

An alternative tradition adopts the **external point of view**, studying law as a social phenomenon whose causes, functions, and effects can be investigated by the methods of the social sciences. **Legal realism**, in its American form associated with Oliver Wendell Holmes, Karl Llewellyn, and Jerome Frank, emphasised that law is what courts and officials actually do, rather than what rules say, and directed attention to the social and psychological factors that shape judicial decisions. Scandinavian realism, associated with Alf Ross, sought to reduce legal concepts to empirically observable phenomena. The **law and society** movement and **socio-legal studies** investigate the operation of law in society — how legal rules are implemented, used, avoided, and experienced — drawing on sociology,

anthropology, and political science. **Law and economics** applies the tools of economic analysis to explain and evaluate legal rules, asking whether they promote efficiency, as in the analysis of regulatory regimes, property rights over resources, and liability rules for environmental harm examined in Chapters 3 and 17. From the external perspective, legal knowledge is a species of social scientific knowledge, to be established through empirical evidence and theory testing.

Critical Perspectives

Critical approaches challenge the claims of both internal and external perspectives to neutrality and objectivity. **Critical legal studies** argued that legal doctrine is indeterminate and that legal reasoning conceals political choices that serve dominant interests. **Feminist legal theory** examines how law reflects and reinforces gender hierarchies, including in the distribution of the benefits and burdens of resource development — as in the gendered effects of displacement, compensation, and employment in extractive projects. **Critical race theory** and **Third World Approaches to International Law** (TWAIL), associated with scholars such as Antony Anghie, B.S. Chimni, and Makau Mutua, examine how international law, including the law of resources and investment, was shaped by colonialism and continues to reproduce inequalities between the Global North and South — a perspective of obvious relevance to the history of concessions, the doctrine of permanent sovereignty, and the contemporary governance of critical minerals examined throughout this book. **Environmental and ecological** approaches, including "Earth jurisprudence" and the rights of nature examined in Chapter 24, question the anthropocentrism of legal thought. Critical perspectives emphasise **reflexivity**: the researcher's awareness of his or her own position, assumptions, and interests and their influence on the research.

Paradigms and Methodological Pluralism

Social science research methodology distinguishes **research paradigms** — sets of assumptions about the nature of reality (**ontology**) and knowledge (**epistemology**). **Positivist** (in the social scientific rather than legal sense) or **post-positivist** paradigms assume that there is an objective reality that can be known, at least approximately, through observation and measurement, and favour quantitative methods and hypothesis testing. **Interpretivist** or **constructivist** paradigms hold that social reality is constructed through human meaning and interaction, and favour qualitative methods that seek to understand the meanings actors attach to their actions. **Critical** paradigms seek not only to understand but to transform social reality, exposing power relations. **Pragmatist** paradigms, associated with philosophers such as John Dewey, judge methods by their usefulness in answering the research question, and support **mixed methods** research.

For the doctoral researcher in energy law, the lesson is not that one epistemology is correct, but that the choice of research question entails epistemological commitments that must be recognised and made consistent with the chosen methods. A thesis that asks a doctrinal question but answers it with anecdotal empirical claims, or that asks an empirical question but answers it through doctrinal analysis alone, suffers from a mismatch between question and method. **Methodological pluralism** — the combination of different approaches within a single project — can enrich legal research, provided that the researcher is explicit about how the approaches relate to each other and what each contributes.

Types of Research Questions

Research questions in law may be classified by the kind of knowledge they seek:

1. **Descriptive** or **expository** questions ask what the law is: "What obligations does the Petroleum Industry Act 2021 impose on settlors of host community development trusts?"
2. **Explanatory** questions ask why the law is as it is, or why it has particular effects: "Why have host community development trusts failed to reduce conflict in some Niger Delta communities?"
3. **Evaluative** questions ask how well the law achieves its objectives or conforms to a standard: "Do host community development trusts satisfy the international standards for benefit-sharing with affected communities?"
4. **Normative** or **prescriptive** questions ask what the law should be: "How should Nigerian law be reformed to ensure equitable benefit-sharing with host communities?"
5. **Comparative** questions ask how legal systems differ and what can be learned from the differences: "What can Nigeria learn from the benefit-sharing regimes of Ghana, Canada, and Australia?"
6. **Historical** questions ask how the law developed and what explains its trajectory: "How did the concept of host community benefits evolve in Nigerian petroleum legislation from 1969 to 2021?"

Most doctoral theses combine several types — typically moving from description and explanation through evaluation to normative proposals — and the research design should make clear how each type of question will be answered.

28.2 Doctrinal Legal Research

The Nature of Doctrinal Research

Doctrinal or **"black-letter"** legal research is the traditional and still predominant method of legal scholarship. It involves the identification, analysis, and synthesis of the **authoritative sources** of law — constitutions, treaties, legislation, subsidiary instruments, judicial decisions, and, in international law, custom and general principles — in order to state the law on a particular question, explain its

structure and rationale, identify inconsistencies and gaps, and propose solutions. In a widely cited formulation derived from the Pearce Committee's review of Australian law schools (1987), and popularised by Terry Hutchinson and Nigel Duncan, doctrinal research provides a systematic exposition of the rules governing a particular legal category, analyses the relationship between rules, explains areas of difficulty and, perhaps, predicts future development. It is the method of most of the analysis in this book.

Doctrinal research is sometimes dismissed as merely descriptive, but at its best it is a sophisticated intellectual activity involving **interpretation** (determining the meaning of texts in context), **analysis** (breaking down complex legal materials into their components), **synthesis** (constructing coherent principles from diverse sources), **criticism** (identifying inconsistencies, errors, and injustices), and **prescription** (proposing reforms or preferred interpretations). It is particularly well suited to questions about the content and coherence of the law and is indispensable as a foundation for other methods: an empirical study of the implementation of a law, or a comparative study of several legal systems, presupposes an accurate understanding of the law itself.

The Process of Doctrinal Research

The doctrinal research process typically involves the following steps:

1. **Identifying the legal issue** precisely, including its jurisdictional context (international, regional, national, or subnational) and the relevant area of law.
2. **Locating the primary sources**: the relevant treaties and their *travaux préparatoires*, constitutional provisions, statutes and subsidiary legislation, judicial decisions and arbitral awards, and, in international law, evidence of state practice and *opinio juris*.
3. **Locating the secondary sources**: textbooks, treatises, commentaries, journal articles, and reports that analyse the primary sources, which assist in understanding the law and identifying debates, but are not themselves authoritative (except, in international law, the "teachings of the most highly qualified publicists" as a subsidiary means for the determination of rules under Article 38(1)(d) of the Statute of the International Court of Justice).
4. **Reading and analysing the sources**, applying the canons of interpretation appropriate to each type of source.
5. **Synthesising** the sources into a coherent statement of the law, identifying principles, exceptions, and areas of uncertainty.
6. **Evaluating** the law against criteria — internal coherence, consistency with higher norms, fidelity to purpose, justice, efficiency, or international standards — which must be made explicit.
7. **Drawing conclusions** and, where appropriate, proposing reforms.

Interpretation of Legal Sources

Each type of source has its own interpretive conventions. **Treaties** are interpreted in accordance with Articles 31 to 33 of the **Vienna Convention on the Law of Treaties** (1969), which require interpretation in good faith in accordance with the ordinary meaning of the terms in their context and in the light of the treaty's object and purpose, taking into account subsequent agreements and practice and relevant rules of international law applicable between the parties (Article 31(3)(c)) — a provision of growing importance for the interpretation of investment treaties in the light of environmental and human rights obligations — with recourse to supplementary means, including the preparatory work, to confirm the meaning or resolve ambiguity. **Statutes** in common law systems are interpreted according to principles that have evolved from the literal, golden, and mischief rules towards a **purposive** approach that reads the words in context in the light of the statute's purpose, as the UK Supreme Court has emphasised, and Nigerian courts apply similar principles, together with the rules of interpretation of the Constitution, which favour a broad and liberal construction. **Judicial decisions** are analysed to identify the *ratio decidendi* — the principle necessary for the decision — as distinct from *obiter dicta*, and their authority depends on the hierarchy of courts and the doctrine of precedent. **Arbitral awards** have no formal precedential value, but investment tribunals frequently refer to earlier awards, giving rise to a *jurisprudence constante* in some areas and to persistent divergence in others; the researcher must evaluate awards critically, considering the composition of the tribunal, the quality of reasoning, and subsequent reception. **Custom** is identified by evidence of general state practice accepted as law, in accordance with the International Law Commission's **Conclusions on Identification of Customary International Law** (2018).

Strengths and Limitations

Doctrinal research has significant strengths: it engages directly with the materials that legal actors use; it produces knowledge directly useful to courts, practitioners, and legislators; and its methods are well understood and transparent. Its limitations are equally significant. It tends to focus on **law in the books** rather than **law in action**, and may therefore miss the gap between formal rules and their implementation — a gap that is often wide in the energy and resource sector, where, as observed throughout this book, impressive legislation frequently coexists with weak enforcement. It may be **parochial**, confined to a single legal system. Its evaluative criteria may be implicit and unexamined. And it may struggle with questions that require empirical evidence, such as the effects of a law on investment, revenues, or environmental outcomes. The doctoral researcher using doctrinal methods should therefore articulate the evaluative framework explicitly,

acknowledge the limits of doctrinal knowledge, and consider whether the research question requires supplementation by other methods.

Doctrinal Research in Energy Law: Distinctive Challenges

Doctrinal research in energy law faces specific challenges. The **multi-level** structure of energy law requires the researcher to trace the interaction of international, regional, national, and contractual norms — for example, how a host state's obligations under an investment treaty interact with a stabilisation clause in a petroleum agreement and with domestic environmental legislation. The **confidentiality** of key sources — petroleum agreements, commercial arbitral awards, and regulatory decisions — limits the materials available, although contract disclosure initiatives such as ResourceContracts.org, the EITI's contract transparency requirements, and the publication of investment awards by ICSID and the PCA have improved access. The **technicality** of energy law requires an understanding of the industry's operations, economics, and terminology. And the **rapidity of change** requires constant updating; a thesis submitted several years after its research began may describe law that has since changed, and the researcher should state the date at which the law is stated.

28.3 Comparative and Transnational Legal Research

The Uses of Comparative Research

Chapter 24 introduced comparative law as a method and applied it to the energy and resource laws of major jurisdictions. This section considers comparative research from the perspective of research design. Comparative research may serve several purposes in a doctoral thesis: to **understand** one's own legal system better by seeing it in the light of others; to **identify solutions** to legal problems by examining how other systems have addressed them; to **explain** differences and similarities among legal systems by reference to historical, economic, political, and cultural factors; to **test hypotheses** about the relationship between legal rules and outcomes; and to contribute to the **harmonisation** or **transnationalisation** of law.

Designing a Comparative Study

The design of a comparative study involves several key decisions:

1. **The comparative question.** The study must ask a question that can be answered by comparison, such as how different systems address a common problem, and the problem must be defined in functional terms that do not presuppose the concepts of any one system — for example, "how do legal systems allocate the risk of regulatory change in long-term petroleum investments?" rather than "how do legal systems treat stabilisation clauses?", since some systems address the risk through other mechanisms.

2. **The selection of jurisdictions.** Jurisdictions must be selected on a principled basis related to the research question. The **"most similar systems"** design compares jurisdictions that are similar in most respects but differ in the variable of interest, allowing differences in outcomes to be attributed to that variable; the **"most different systems"** design compares jurisdictions that differ in most respects but share an outcome, allowing the common factor to be identified. Researchers may also select jurisdictions as **exemplars** of particular models (for example, Norway as an exemplar of the separation of commercial and regulatory functions), as **sources of influence** on the home jurisdiction, or as **leading or innovative** systems. The number of jurisdictions involves a trade-off between depth and breadth: a study of two or three jurisdictions permits contextual depth; a study of many permits broader generalisation but risks superficiality.
3. **The *tertium comparationis*.** The common point of reference — the problem, function, or standard against which the jurisdictions are compared — must be clearly defined.
4. **The method of analysis.** The **functional method**, associated with Zweigert and Kötz, compares the solutions that different systems provide to the same social problem, on the assumption that legal systems face similar problems even if they solve them differently. **Contextual** or **cultural** approaches, associated with scholars such as Pierre Legrand, emphasise that legal rules are embedded in cultures and cannot be understood or transplanted in isolation. **Legal origins** research, associated with economists such as La Porta, Lopez-de-Silanes, and Shleifer, uses quantitative methods to test the relationship between legal traditions (common law or civil law) and economic outcomes, though its methods and conclusions have been heavily criticised.
5. **Access to sources.** The researcher must have access to the primary sources of each jurisdiction, in the original language or reliable translation, and an understanding of their context; reliance on secondary sources in English may lead to errors.

Transnational Legal Research

Energy law is increasingly **transnational**, involving norms that cross the boundaries between national and international law and between public and private ordering: model contracts such as the AIPN model joint operating agreement; industry standards developed by bodies such as the American Petroleum Institute and the International Organization for Standardization; soft law instruments such as the UN Guiding Principles on Business and Human Rights and the OECD Guidelines; the standards of international financial institutions, such as the IFC Performance Standards, examined in Chapter 21; certification schemes

such as the Kimberley Process and responsible mining standards; and arbitral practice, sometimes described as a *lex petrolea*, examined in Chapter 1. **Transnational legal research** studies how these norms are created, diffused, and enforced, and how they interact with state law. Concepts such as **transnational legal orders** (Halliday and Shaffer), **global administrative law** (Kingsbury, Krisch, and Stewart), and **legal pluralism** provide frameworks for such research. Methods include the tracing of norm diffusion across jurisdictions, the analysis of the "life cycle" of transnational norms, and the study of the interaction between transnational standards and national law in specific contexts, such as the incorporation of the IFC Performance Standards into project finance agreements and their effect on community outcomes.

Pitfalls of Comparative and Transnational Research

The doctoral researcher should be alert to the pitfalls identified in Chapter 24: superficial comparison that juxtaposes descriptions without analysis; the assumption that a foreign model will work in the home jurisdiction regardless of context ("legal transplant" fallacies); ethnocentric bias that evaluates other systems by the standards of one's own; the selection of jurisdictions to confirm a preconceived conclusion; and neglect of the gap between formal law and practice in the jurisdictions compared. A comparative thesis should explicitly address the transferability of lessons, considering the institutional, economic, and political conditions on which the success of a foreign model depends.

28.4 Socio-Legal and Empirical Research

Law in Action

Socio-legal research studies law as a social phenomenon, investigating how legal rules and institutions operate in practice, how they are experienced by those subject to them, and what effects they have. It responds to the limitation of doctrinal research identified above: the gap between **law in the books** and **law in action**, a distinction drawn by Roscoe Pound in 1910. In the energy and resource sector, this gap is often the most important object of inquiry. Nigeria's prohibition of routine gas flaring dates from 1984, yet flaring continued for decades; environmental impact assessment has been mandatory since 1992, yet its quality and influence vary widely; host community development trusts were established by the Petroleum Industry Act 2021, but their effects on community welfare and conflict depend on their implementation. Understanding why laws succeed or fail requires evidence of how they operate in practice, which doctrinal analysis alone cannot provide.

Themes of Socio-Legal Research in Energy and Resources

Socio-legal research in energy and natural resources has addressed themes including: **implementation and compliance** — the factors that explain whether companies comply with environmental, local content, and transparency obligations, and whether regulators enforce them; **regulatory capture and governance** — the relationships between regulators, companies, and political actors; **access to justice** — the experience of communities seeking remedies for pollution through courts and grievance mechanisms; **legal pluralism** — the interaction of state law with customary land tenure and community norms in resource areas, as examined in Chapter 21; **the resource curse** — the relationship between resource wealth, institutions, and development outcomes, examined in Chapter 23; **energy justice** — the distribution of the benefits and burdens of energy systems, also examined in Chapter 23; **the political economy of reform** — why some reforms, such as the Petroleum Industry Act, take decades to enact, and how interests shape their content; and **the effects of transparency** — whether initiatives such as the EITI change the behaviour of governments and companies.

Research Designs

Socio-legal research employs a range of **research designs**:

1. **Case studies** examine a single case or a small number of cases in depth — a particular community, project, regulatory agency, or legal reform — using multiple sources of evidence. Case studies are well suited to understanding complex processes in context and generating theory, and they predominate in socio-legal research on energy and resources. Their generalisability is limited, but can be enhanced by careful case selection (typical, deviant, critical, or most-likely and least-likely cases) and by **analytic generalisation** to theory rather than statistical generalisation to populations, as Robert Yin has emphasised.
2. **Comparative case studies** examine several cases to identify the conditions associated with particular outcomes — for example, comparing host communities where development trusts have and have not reduced conflict.
3. **Process tracing** reconstructs the causal processes linking causes and outcomes within a case, using detailed evidence of sequences of events and decisions — for example, tracing how the political bargaining over the Petroleum Industry Bill shaped the fiscal provisions of the final Act.
4. **Surveys** collect standardised data from samples of respondents — for example, surveys of community members' perceptions of benefit-sharing, or of investors' perceptions of regulatory risk.
5. **Experimental and quasi-experimental designs** estimate causal effects by comparing outcomes between groups exposed and not exposed to an intervention; true experiments are rare in legal research, but **natural exper-**

iments and quasi-experimental methods such as difference-in-differences and regression discontinuity are increasingly used to evaluate the effects of laws and policies, such as the effect of mining royalties or community development agreements on local welfare.

6. **Ethnography** involves the researcher's immersion in a social setting over an extended period, observing and participating in its life, to understand the meanings and practices of its members — as in anthropological studies of oil communities in the Niger Delta and of corporate social responsibility practices in mining companies.

Data Sources

Socio-legal researchers draw on diverse **data sources**: **documents**, including legislation, regulatory decisions, court records, company reports, EITI reports, parliamentary debates, and media reports; **interviews** with officials, company representatives, community members, lawyers, and NGOs; **observation** of hearings, consultations, and community meetings; **surveys**; and **administrative and statistical data**, such as data on production, revenues, spills, flaring, electricity access, and court caseloads. Energy researchers increasingly use **satellite and remote sensing data** — for example, satellite observations of gas flaring by the US National Oceanic and Atmospheric Administration and the World Bank's Global Gas Flaring Reduction Partnership, of night-time lights as a proxy for electricity access, and of deforestation and land use change around mines — which provide independent evidence that can be compared with official reports.

Challenges of Empirical Research in the Energy Sector

Empirical research in the energy and resource sector faces practical challenges. **Access** to companies and government officials may be difficult, as these actors may be reluctant to discuss sensitive matters; researchers may need to build relationships of trust, use intermediaries, and offer confidentiality. **Security** risks may arise in conflict-affected resource areas, such as parts of the Niger Delta, the Sahel, or the eastern DRC, requiring careful risk assessment and planning. **Power imbalances** between researchers and participants, particularly vulnerable communities, raise ethical issues addressed in section 28.6. **Data quality** may be poor, with incomplete or unreliable official statistics, and **positionality** — the researcher's identity and relationship to the research setting — may influence access and interpretation. These challenges do not preclude empirical research, but require careful design, ethical reflection, and transparency about limitations.

28.5 Qualitative and Quantitative Research Approaches

The Distinction

Research methods may be broadly classified as **qualitative** or **quantitative**. **Qualitative** methods collect and analyse non-numerical data — texts, interview transcripts, observations — to understand meanings, processes, and contexts, and are associated with interpretivist paradigms. **Quantitative** methods collect and analyse numerical data to measure variables, describe distributions, and test hypotheses about relationships among variables, and are associated with post-positivist paradigms. **Mixed methods** research combines qualitative and quantitative methods within a single study. The choice of approach should be driven by the research question: questions about meanings, experiences, and processes generally call for qualitative methods; questions about prevalence, magnitude, and statistical relationships call for quantitative methods; and many questions in energy law benefit from both.

Qualitative Methods

The principal qualitative methods in legal research include:

1. **Semi-structured interviews**, which follow an interview guide but allow flexibility to explore issues raised by the interviewee, and are widely used to obtain the perspectives of officials, practitioners, and community members. Interviewees may be selected by **purposive sampling** (selecting those with relevant knowledge), **snowball sampling** (asking interviewees to suggest others), or **theoretical sampling** (selecting cases to develop emerging theory). The number of interviews is typically determined by the principle of **saturation** — the point at which further interviews yield no new insights.
2. **Focus groups**, which bring together small groups of participants to discuss a topic, useful for exploring collective views, for example within host communities.
3. **Documentary analysis**, the systematic analysis of documents, which may include **qualitative content analysis**, identifying themes and categories in texts, and **discourse analysis**, examining how language constructs meaning and power — for example, how corporate sustainability reports or regulatory documents frame the energy transition.
4. **Thematic analysis**, a widely used method for identifying, analysing, and reporting patterns in qualitative data, through coding of data, development of themes, and their interpretation, as systematised by Virginia Braun and Victoria Clarke.
5. **Grounded theory**, associated with Barney Glaser and Anselm Strauss, which develops theory inductively from data through iterative coding and comparison.

Qualitative analysis may be supported by **computer-assisted qualitative data analysis software** (CAQDAS), such as NVivo or ATLAS.ti, which assists in organising and coding data but does not replace the researcher's interpretation. The quality of qualitative research is assessed by criteria such as **credibility**, **transferability**, **dependability**, and **confirmability** (Lincoln and Guba), supported by practices such as triangulation of sources, member checking, thick description, and an audit trail of analytical decisions.

Quantitative Methods

Quantitative methods in legal research include:

1. **Descriptive statistics**, summarising data on legal phenomena — for example, the number and outcomes of investment arbitrations in the energy sector, the duration of oil spill litigation in Nigerian courts, or the frequency of particular clauses in petroleum contracts.
2. **Systematic content analysis** of legal texts, coding features of large numbers of documents — such as the presence of stabilisation clauses in petroleum agreements, the content of investment treaties, or the reasoning of arbitral awards — to identify patterns and trends. Datasets of investment treaties and awards, such as those compiled by UNCTAD and academic projects, have enabled quantitative studies of treaty design and arbitral outcomes.
3. **Inferential statistics and econometrics**, testing hypotheses about relationships among variables using regression analysis and related techniques — for example, whether investment treaties increase foreign investment in the energy sector, whether resource wealth is associated with weaker institutions, or whether arbitral outcomes are associated with the characteristics of arbitrators.
4. **Computational legal analysis**, using techniques of natural language processing and machine learning to analyse large corpora of legal texts — for example, to classify clauses in contracts or to predict outcomes of cases — an area transformed by advances in artificial intelligence, discussed in Chapter 30.

Quantitative research requires attention to **validity** (whether the measures capture the concepts of interest), **reliability** (whether measurements are consistent), the representativeness of samples, and the distinction between **correlation and causation**. Legal phenomena are often difficult to quantify without losing nuance, and quantitative legal research has been criticised for reducing complex norms to crude variables, as in the debates over the legal origins literature and the World Bank's former *Doing Business* indicators, discontinued in 2021 after an investigation into data irregularities.

Mixed Methods

Mixed methods research combines qualitative and quantitative approaches to exploit their complementary strengths. Common designs include the **convergent** design, in which qualitative and quantitative data are collected in parallel and compared; the **explanatory sequential** design, in which quantitative results are followed by qualitative research to explain them; and the **exploratory sequential** design, in which qualitative findings inform the development of quantitative instruments. A doctoral study of local content regulation in Nigeria might, for example, combine a doctrinal analysis of the Nigerian Oil and Gas Industry Content Development Act 2010, a quantitative analysis of data on Nigerian participation in contracts, and qualitative interviews with officials and contractors to explain the patterns observed. The integration of methods must be planned, and the thesis should explain how the different strands contribute to answering the research question.

Legal Research Combining Doctrinal and Empirical Methods

For many doctoral researchers in law, the most productive design combines **doctrinal analysis** with **targeted empirical inquiry**. The doctrinal component establishes the legal framework and its interpretation; the empirical component investigates its operation and effects; and the evaluative and normative components draw on both to assess the law and propose reforms. This combination, sometimes called **"empirical legal research"** or **"law in context"**, is particularly well suited to energy and resource law, where the effectiveness of legal frameworks is often the central question. It requires the researcher to acquire competence in empirical methods — through training, collaboration, or supervision — and to be realistic about the scale of empirical work feasible within a doctoral programme.

28.6 Research Ethics and Academic Integrity

The Foundations of Research Integrity

Research integrity refers to the adherence to ethical principles and professional standards in the conduct of research. Its core values, as articulated in instruments such as the **Singapore Statement on Research Integrity** (2010) and the **European Code of Conduct for Research Integrity** (revised 2023), include **honesty** in all aspects of research; **accountability** for the conduct of research; **professional courtesy and fairness** in working with others; **good stewardship** of research on behalf of others; **reliability** in ensuring the quality of research; and **respect** for colleagues, research participants, society, ecosystems, and the environment. Universities and research funders typically require doctoral researchers to comply with institutional codes of research integrity and to complete training in research ethics.

Research Misconduct

Research misconduct is conventionally defined as **fabrication**, **falsification**, and **plagiarism** in proposing, performing, reviewing, or reporting research. **Fabrication** is making up data or results; **falsification** is manipulating research materials, processes, or data, or changing or omitting results, so that the research is not accurately represented. In legal research, falsification may take subtler forms, such as misrepresenting the holding of a case, quoting selectively from a source so as to distort its meaning, citing a source for a proposition it does not support, or ignoring contrary authority. **Plagiarism** is the appropriation of another person's ideas, processes, results, or words without giving appropriate credit. It includes verbatim copying without quotation marks and attribution, close paraphrasing without attribution, the use of another's ideas or structure without acknowledgement, and **self-plagiarism** — the reuse of one's own previously published or submitted work without disclosure. Other unacceptable practices include the misrepresentation of credentials, the manipulation of authorship, and the citation of sources one has not read.

The consequences of misconduct for a doctoral researcher can be severe, including failure of the thesis, revocation of the degree, and lasting damage to professional reputation; for legal practitioners, misrepresentation of authorities to a court is also a serious breach of professional duty. Universities use text-matching software to detect plagiarism, but integrity depends ultimately on the researcher's own practices: keeping careful records of sources, distinguishing clearly in notes between quotation, paraphrase, and one's own ideas, and citing every source on which the work relies.

Artificial Intelligence and Research Integrity

The emergence of **generative artificial intelligence** tools capable of producing fluent text, summarising sources, and answering legal questions has created new challenges for research integrity. These tools can assist researchers in searching, summarising, translating, and editing, but they also present serious risks. They may generate **fabricated authorities** — plausible but non-existent cases, statutes, and articles — and misstate the content of real sources, as several courts have discovered when lawyers submitted AI-generated briefs citing fictitious cases: in ***Mata v Avianca*** (2023), a US federal court sanctioned lawyers for submitting a brief containing fictitious citations generated by an AI tool, and in ***R (Ayinde) v London Borough of Haringey*** (2025), the Divisional Court of the High Court of England and Wales warned that lawyers who cite fictitious AI-generated authorities may face regulatory and even criminal consequences. The use of AI to generate text presented as one's own may constitute plagiarism or misrepresentation under institutional policies, and the use of AI tools may raise issues of confidentiality, copyright, and bias.

Universities, publishers, and courts have issued **guidance** on the use of AI. Common principles include that researchers remain fully responsible for the accuracy and integrity of their work; that AI tools cannot be credited as authors; that the use of AI in research and writing must be disclosed in accordance with institutional and publisher policies; that all authorities and factual claims must be verified against primary sources; and that confidential or personal data must not be entered into AI tools without appropriate safeguards. The doctoral researcher should consult and comply with the policies of his or her institution, which vary considerably, and should treat AI outputs as unverified leads rather than authorities. The principle that every proposition of law must be checked against the authoritative source is the foundation of reliable legal research, whatever tools are used.

Ethics of Research Involving Human Participants

Research involving **human participants** — interviews, surveys, focus groups, observation — is subject to ethical principles derived from instruments such as the **Belmont Report** (1979), which identified the principles of **respect for persons**, **beneficence**, and **justice**, and the **Declaration of Helsinki** (in medical research). Universities typically require such research to be approved by a **research ethics committee** or institutional review board before it begins. Key ethical requirements include:

1. **Informed consent**: participants must be informed of the purpose of the research, what participation involves, any risks, how the data will be used and stored, and their right to withdraw, and must consent freely. In communities with low literacy or different cultural norms, consent processes must be adapted, for example by oral consent and explanation in local languages, and community-level consent may be appropriate in addition to individual consent, particularly in research with indigenous peoples, consistent with the principle of free, prior, and informed consent examined in Chapter 21.
2. **Confidentiality and anonymity**: researchers must protect the identity of participants where promised, which may be crucial for officials, employees, and community members who could face retaliation for speaking about corruption, pollution, or conflict.
3. **Avoidance of harm**: researchers must assess and minimise risks to participants, including physical, psychological, legal, social, and economic risks, and risks to themselves, particularly in conflict-affected areas.
4. **Data protection**: personal data must be collected, stored, and processed in accordance with data protection law — such as the EU General Data Protection Regulation and Nigeria's **Nigeria Data Protection Act 2023** — and institutional data management plans.

5. **Fairness and reciprocity**: researchers should consider how the research can benefit the communities studied, share findings with participants in accessible forms, and avoid "extractive" research that takes knowledge from communities without return — a concern that parallels the critique of extractive industries themselves.

Conflicts of Interest and Funding

Research in energy law is often funded by, or conducted in association with, governments, companies, international organisations, or NGOs with interests in the outcome. Researchers must **disclose conflicts of interest**, including funding sources, consultancy relationships, and personal interests, and must preserve their **independence** in designing, conducting, and reporting research. The influence of industry funding on research on fossil fuels and climate change has been the subject of scholarly and public concern, and researchers should be transparent about funding and resist pressure to suppress or shape findings.

Normative Commitments and Objectivity

Energy and resource law raises questions on which researchers may hold strong normative commitments — to climate action, to resource sovereignty, to community rights, or to investment protection. Such commitments are not incompatible with rigorous scholarship, and critical scholarship explicitly embraces normative goals. But the researcher must distinguish between advocacy and scholarship: scholarly work must engage fairly with opposing arguments and evidence, acknowledge uncertainty and limitations, and allow conclusions to be determined by the analysis rather than by prior commitment. Being explicit about one's normative framework is a mark of rigour, not bias.

28.7 Legal Databases, Primary Authorities, and Citation Methods

The Research Environment

Legal research has been transformed by digital technology. The doctoral researcher in energy law now has access to vast collections of primary and secondary materials through commercial databases, free public resources, and institutional repositories. The challenge is less access than **navigation**: identifying the authoritative and relevant sources efficiently, evaluating their reliability, and keeping up to date in a rapidly changing field.

Primary Sources

International law sources include: the **United Nations Treaty Collection**, which publishes the texts and status of treaties registered with the UN; the websites of treaty bodies and international organisations, such as the UNFCCC, the International Seabed Authority, the IEA, IRENA, and the Energy Charter

Secretariat; the judgments, advisory opinions, and pleadings of the **International Court of Justice**, the **International Tribunal for the Law of the Sea**, and the **Permanent Court of Arbitration**; the documents of the **International Law Commission**; the WTO's dispute settlement database; and the decisions of regional human rights bodies, such as the HUDOC database of the European Court of Human Rights and the decisions of the African Commission and Court on Human and Peoples' Rights.

Investment arbitration sources include the **ICSID** website, which publishes cases and many awards; **italaw**, a free repository of investment treaty awards and documents; the **UNCTAD Investment Policy Hub**, which provides databases of investment treaties and investor–state cases; and commercial services such as Investment Arbitration Reporter and Global Arbitration Review. **Commercial arbitration** awards are less accessible, but the ICC and other institutions publish redacted extracts, and the Kluwer Arbitration database provides extensive materials.

National law sources vary by jurisdiction. For **Nigeria**, sources include the official **Federal Government Gazette**; the **Laws of the Federation of Nigeria** (2004 revision, with subsequent legislation); the websites of the National Assembly, regulators (NUPRC, NMDPRA, NERC, NEITI), and ministries; and law reports including the **Nigerian Weekly Law Reports** (NWLR), the **Supreme Court of Nigeria Judgments** (SCJ), and the **Law Pavilion** and **LawNigeria** electronic databases. For the **United Kingdom**, **legislation.gov.uk** and the **National Archives' Find Case Law** service provide free access to legislation and judgments, and **BAILII** provides free access to judgments of the UK and Ireland, with Westlaw and LexisNexis offering comprehensive commercial coverage. For the **United States**, sources include the US Code, the Code of Federal Regulations, the Federal Register, and court decisions on PACER, CourtListener, and commercial databases. The **EUR-Lex** database provides EU legislation and case law, and the **CURIA** website the judgments of the Court of Justice. **ResourceContracts.org**, developed by the Natural Resource Governance Institute, the World Bank, and the Columbia Center on Sustainable Investment, provides a repository of publicly available oil, gas, and mining contracts, with annotations; the **EITI** publishes country reports and contract disclosures.

Secondary Sources

Secondary sources include **treatises and textbooks**; **journals**, including specialised energy and resource law journals such as the *Journal of Energy & Natural Resources Law*, the *Journal of World Energy Law & Business*, *Oil, Gas & Energy Law Intelligence* (OGEL), the *Journal of World Investment & Trade*, *ICSID Review*, *Transnational Environmental Law*, and the *Review of European, Comparative & International Environmental Law*, as well as general international law

and law and development journals; **working papers** on SSRN and institutional repositories; **reports** of international organisations, think tanks, and NGOs; and **news and specialist services**. Researchers should evaluate secondary sources critically, considering the author's expertise and potential interests, the quality of peer review, and the currency of the analysis.

Search Strategies

Effective legal research requires systematic **search strategies**: defining key concepts and synonyms; using Boolean operators and proximity searches; following citation trails backwards (to sources cited by a key work) and forwards (to later works citing it, using citator services); monitoring new developments through alerts; and maintaining a **research log** recording searches and sources. For literature reviews, some researchers adopt **systematic review** methods, which use explicit, reproducible procedures for identifying, selecting, and appraising literature, adapted from the health and social sciences. **Reference management software** (such as Zotero, EndNote, or Mendeley) assists in organising sources and formatting citations.

Citation Methods

Accurate **citation** serves several purposes: it acknowledges the work of others, allows readers to verify claims and locate sources, demonstrates the scholarly foundations of the research, and avoids plagiarism. Legal citation follows specialised conventions. The **Oxford University Standard for the Citation of Legal Authorities** (**OSCOLA**), which is revised periodically by the Oxford Law Faculty, is widely used in the United Kingdom and many Commonwealth jurisdictions, including in Nigerian universities, and is the style used in this book; it uses footnotes, minimal punctuation, and specific formats for cases, legislation, treaties, and secondary sources. **The Bluebook: A Uniform System of Citation** is the standard in the United States; the **Australian Guide to Legal Citation** (AGLC) in Australia; and the **McGill Guide** in Canada. Nigerian legal writing commonly cites cases by party names, year, and law report series (for example, NWLR with part and page numbers). The doctoral researcher should follow the style required by his or her institution consistently, cite primary sources wherever possible, provide pinpoint references to specific pages or paragraphs, and verify every citation against the original source. Appendix P provides a guide to citation and case analysis.

Stating the Law as at a Date

Because energy law changes rapidly, a thesis or publication should state the **date at which the law is stated**, and the researcher should update the analysis as close as possible to submission, noting significant developments after that date where feasible. Pending legislation, ongoing litigation, and recent awards should

be described with appropriate caution, distinguishing between what has been decided and what is proposed or contested.

28.8 Developing Original Doctoral Contributions

What is an Original Contribution?

The defining requirement of a doctorate is an **original contribution to knowledge**. University regulations typically require that a doctoral thesis demonstrate the creation and interpretation of new knowledge through original research or other advanced scholarship, of a quality to satisfy peer review, extend the forefront of the discipline, and merit publication. Originality in legal research does not require a revolutionary discovery. Estelle Phillips and Derek Pugh, in their widely used guide to doctoral study, identify many forms of originality, including: setting down a major piece of new information in writing for the first time; continuing a previously original piece of work; carrying out original work designed by others; providing a single original technique, observation, or result in an otherwise unoriginal but competent piece of research; having many original ideas, methods, and interpretations under the guidance of a supervisor; showing originality in testing someone else's idea; carrying out empirical work that has not been done before; making a synthesis that has not been made before; using already known material but with a new interpretation; trying out something in one country that has previously only been done in another; applying a particular technique to a new area; bringing new evidence to bear on an old issue; being cross-disciplinary and using different methodologies; and looking at areas that people in the discipline have not looked at before.

In energy and natural resources law, original contributions may take many forms: the first comprehensive doctrinal analysis of a new statute, such as the Petroleum Industry Act 2021 or the Electricity Act 2023, or of a body of arbitral awards; a new interpretation of an established doctrine, such as permanent sovereignty or stabilisation, in the light of the energy transition; a comparative study yielding new insights into the design of resource legal frameworks; an empirical study of the implementation of a legal regime in a particular context, such as host community trusts in a specific state; a theoretical framework explaining patterns of compliance or reform; the application of methods from another discipline, such as economics or political science, to a legal question; or a reasoned proposal for legal reform grounded in rigorous analysis.

Identifying a Research Gap

An original contribution begins with the identification of a **research gap** — a question that has not been answered, or not answered adequately, in the existing literature. Gaps may be **empirical** (a phenomenon that has not been studied), **theoretical** (a phenomenon that existing theories fail to explain), **methodolog-**

ical (a question that has not been approached with appropriate methods), **contextual** (a question studied in some jurisdictions but not others, such as an issue studied extensively in Europe but not in Africa), or **practical** (a problem in practice for which the law provides no adequate solution). Identifying a gap requires a thorough and critical **literature review**, which maps the state of knowledge, identifies debates and limitations, and situates the proposed research. The literature review is not a summary of everything written on a topic but a critical analysis organised around the research question, demonstrating why the question matters and how the research will advance knowledge.

African energy and resource law offers abundant opportunities for original research. Many legal regimes in African states have been studied only superficially or not at all; recent reforms — Nigeria's PIA and Electricity Act, the mining codes of the DRC and other states, the petroleum laws of new producers, the AfCFTA and its investment protocol — await rigorous analysis; and the perspectives of African states, communities, and scholars remain under-represented in the international literature. Doctoral research by African scholars, grounded in local knowledge and access, can make distinctive contributions to both African and global debates.

Formulating the Research Question and Thesis

The **research question** is the foundation of the doctoral project. A good research question is **significant** (addressing an important problem), **original** (addressing a gap), **focused** (narrow enough to be answered thoroughly within the constraints of a doctorate), **researchable** (capable of being answered with available sources and methods), and **clear**. Many doctoral projects begin with questions that are too broad — "how should Africa govern its natural resources?" — and must be narrowed through successive refinement to a question that can be answered in depth — "to what extent does the legal framework for host community development trusts under the Petroleum Industry Act 2021 conform to international standards of benefit-sharing, and how can it be reformed?" The main research question is typically supported by **sub-questions**, each addressed in one or more chapters of the thesis.

A doctoral thesis is not merely a report of research but an **argument**: it advances a **thesis** in the sense of a central claim or position, which the research supports and which the work defends against counter-arguments. The thesis statement should be capable of being stated concisely, and the structure of the dissertation should be organised to develop and support it. Appendix O sets out a model structure.

The Research Proposal

Most doctoral programmes require a **research proposal**, which sets out the research question, its significance and originality, the literature review, the theoretical framework, the methodology, the ethical considerations, the expected contribution, a timeline, and a preliminary bibliography. The proposal is both a planning document and a test of the feasibility of the project. Appendix N provides a template. A strong proposal demonstrates a clear and focused question, a command of the literature, a justified methodology appropriate to the question, awareness of limitations and risks, and a realistic plan.

Writing, Supervision, and the Viva

Doctoral research is a **process** of several years, during which the research question, arguments, and structure typically evolve. Effective doctoral researchers write regularly from the outset, rather than leaving writing until the end, since writing is itself a means of thinking and clarifying arguments. The relationship with the **supervisor** or supervisory team is central: supervisors provide guidance on the direction, methodology, and quality of the research, and doctoral researchers should seek regular feedback, agree expectations, and take responsibility for their own progress. Presenting work at **seminars and conferences** and publishing articles during the doctorate test ideas, build networks, and develop a scholarly profile.

The doctorate is typically examined through the submission of the thesis and an **oral examination** (*viva voce*), in which examiners — including at least one external examiner — question the candidate on the thesis. The examiners assess whether the thesis makes an original contribution, demonstrates a command of the field and of appropriate methods, presents a coherent and well-supported argument, and is presented to a satisfactory standard. The candidate should be able to explain and defend the research question, methodology, findings, and contribution, acknowledge limitations, and discuss implications and future research. Common outcomes include award without corrections, award with minor or major corrections, resubmission, or the award of a lower degree.

Impact and Dissemination

Doctoral research in energy and natural resources law can have significant **impact** beyond academia — informing legislative reform, regulatory practice, judicial decisions, corporate policy, and civil society advocacy. Researchers can enhance the impact of their work by publishing in accessible forms, including policy briefs and articles in practitioner and general media, as well as in peer-reviewed journals and monographs; engaging with policymakers, regulators, and practitioners; contributing to consultations on legislative and regulatory reform, examined in Chapter 29; and sharing findings with the communities studied. The conversion

of a doctoral thesis into a published monograph or a series of journal articles is an important step in an academic career. In all these activities, the scholar's credibility depends on the rigour, honesty, and independence of the underlying research.

Disclaimer on Doctoral Status

As stated in the preliminary pages, this book is a scholarly resource intended to support advanced study and doctoral research. Reading it, or completing the exercises it contains, does not confer a doctorate or any other academic qualification, which can be awarded only by a duly authorised institution on the basis of its own requirements and examination, and nothing in this book constitutes legal advice.

::: exercise {title="Practical Exercise 28: Designing a Doctoral Research Project (Comparative Legal Research Exercise)"} **This exercise concerns the design of research and does not involve any hypothetical parties. Where the candidate uses illustrative scenarios, they must be identified as hypothetical.**

(a) Select one of the doctoral research questions listed at the end of any chapter of this book, or formulate your own question in energy and natural resources law.

(b) Refine the question into a focused main research question and three to five sub-questions, explaining the type of each question (descriptive, explanatory, evaluative, normative, comparative, or historical).

(c) Write a literature review outline (of no more than 1,500 words) identifying the main bodies of literature relevant to your question, the principal debates, and the gap your research will address.

(d) Prepare a methodology chapter outline (of no more than 2,000 words) justifying your choice of methods — doctrinal, comparative, socio-legal, empirical, or mixed — explaining how each sub-question will be answered, and, for a comparative study, justifying the selection of jurisdictions.

(e) Identify the ethical issues your research raises, including any involving human participants, data protection, conflicts of interest, and the use of artificial intelligence tools, and explain how you will address them.

(f) State, in no more than 300 words, the original contribution your research will make.

(g) Using Appendix N, prepare a complete research proposal and present it to a peer group for critique, as in a doctoral progression review. :::

Conclusion

Research methodology is the foundation of doctoral scholarship. The questions a researcher asks entail epistemological commitments — to law as a normative system, as a social phenomenon, or as an object of critique — that must be recognised and matched by appropriate methods. Doctrinal research, the tradi-

tional method of legal scholarship, remains indispensable for understanding the content and coherence of energy law, but its focus on law in the books must often be complemented by comparative, transnational, socio-legal, and empirical research that investigates the law in action, its effects, and its context. Qualitative and quantitative methods offer complementary tools, and mixed methods designs can combine their strengths. All research must be conducted with integrity — honesty in the use of sources and data, respect for participants, transparency about interests, and, in the age of artificial intelligence, rigorous verification of every authority. Effective use of legal databases and accurate citation underpin credible scholarship. And the ultimate goal — an original contribution to knowledge — requires the identification of a genuine gap, a focused question, a coherent argument, and a sustained process of research, writing, and critical engagement. For researchers in African energy and resource law, the opportunities for original and impactful scholarship are exceptional. Chapter 29 considers how such scholarship can inform the reform of energy law and the design of regulatory institutions.

CHAPTER SUMMARY

- Legal research rests on epistemological assumptions: internal perspectives (positivism, natural law, interpretivism) treat law as a normative system; external perspectives (realism, socio-legal studies, law and economics) treat it as a social phenomenon; critical perspectives (CLS, feminist theory, TWAIL, ecological approaches) expose power and bias; research questions may be descriptive, explanatory, evaluative, normative, comparative, or historical, and methods must fit the question.
- Doctrinal research identifies, interprets, and synthesises authoritative sources using source-specific interpretive conventions; it is indispensable but limited in addressing law in action, and faces distinctive challenges in multi-level, confidential, technical, and rapidly changing energy law.
- Comparative research requires a functional question, principled selection of jurisdictions, a clear *tertium comparationis*, and attention to context and transferability; transnational research studies norms crossing public–private and national–international boundaries.
- Socio-legal and empirical research investigates the implementation and effects of law through case studies, process tracing, surveys, quasi-experiments, and ethnography, using documents, interviews, observation, statistics, and remote sensing, subject to practical challenges of access, security, and data quality.
- Qualitative methods (interviews, focus groups, documentary, thematic, and discourse analysis) and quantitative methods (descriptive statistics, content analysis, econometrics, computational analysis) can be combined in mixed methods designs, and doctrinal and empirical methods are often productively combined.

- Research integrity requires honesty, accountability, and respect; fabrication, falsification, and plagiarism are misconduct; AI tools must be used transparently and every authority verified; research with human participants requires ethics approval, informed consent, confidentiality, harm avoidance, and data protection.
- Legal research relies on international, arbitral, national, and contract databases, critical use of secondary sources, systematic search strategies, and consistent, accurate citation, such as OSCOLA.
- An original contribution may take many forms; it requires the identification of a research gap, a focused research question, a defensible thesis, a sound proposal, sustained writing and supervision, and a successful oral examination, with impact achieved through dissemination and engagement.

DOCTORAL RESEARCH QUESTIONS

1. Is doctrinal research a sufficient method for doctoral research in energy and natural resources law? Defend a position with reference to specific research questions.
2. What can Third World Approaches to International Law contribute to the study of the international law of natural resources and the energy transition?
3. Design an empirical study to evaluate the effectiveness of host community development trusts under the Petroleum Industry Act 2021, addressing methodological and ethical challenges.
4. How should legal scholars respond to the opportunities and risks of artificial intelligence for legal research and research integrity?
5. Evaluate the use of quantitative methods in the study of investment treaty arbitration in the energy sector. What can they reveal, and what do they miss?
6. How can doctoral research in African energy and resource law achieve impact on law reform and policy?

FURTHER READING

- H L A Hart, *The Concept of Law* (3rd edn, Oxford University Press 2012).
- Mike McConville and Wing Hong Chui (eds), *Research Methods for Law* (2nd edn, Edinburgh University Press 2017).
- Terry Hutchinson, *Researching and Writing in Law* (4th edn, Thomson Reuters 2018).
- Robert K Yin, *Case Study Research and Applications: Design and Methods* (6th edn, SAGE 2018).
- Peter Cane and Herbert M Kritzer (eds), *The Oxford Handbook of Empirical Legal Research* (Oxford University Press 2010).

- Estelle M Phillips and Derek S Pugh, *How to Get a PhD: A Handbook for Students and Their Supervisors* (Open University Press, latest edition).
- Antony Anghie, *Imperialism, Sovereignty and the Making of International Law* (Cambridge University Press 2005).
- Oxford Faculty of Law, *OSCOLA: The Oxford University Standard for the Citation of Legal Authorities* (current edition).

29 CHAPTER

Energy Law Reform, Regulatory Design, and Policy Leadership

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the principal theories of legal and regulatory reform and apply them to energy and natural resources law.
2. Conduct and critically evaluate a regulatory impact assessment for an energy law reform proposal.
3. Apply the principles and techniques of legislative drafting to energy governance.
4. Evaluate the design of energy regulatory institutions, including their independence, accountability, and capacity.
5. Design effective public consultation and stakeholder participation processes for energy reforms.
6. Apply methods of policy evaluation to energy laws and regulatory programmes.
7. Analyse the political economy of governance reform in resource-rich economies, with particular reference to Nigeria.
8. Identify the features of future-ready energy legal systems capable of adapting to the energy transition.

Introduction

Energy law is constantly being reformed. Over the past two decades, states across the world have liberalised electricity markets, restructured their petroleum sectors, introduced renewable energy support schemes, enacted climate legislation, revised mining codes, and created new regulatory institutions. Nigeria alone has, since 2005, enacted the Electric Power Sector Reform Act, the Nigerian Oil and Gas Industry Content Development Act, the Nigeria Extractive Industries Transparency Initiative Act, the Petroleum Industry Act, the Climate Change Act, the Electricity Act, and a major package of tax reforms. The energy transition will demand further and deeper reforms: to accommodate new technologies such

as storage, hydrogen, and carbon capture; to manage the decline of fossil fuel industries; to mobilise investment in clean energy; to protect consumers and communities; and to meet international climate commitments.

Reform is difficult. Energy reforms affect powerful interests and large revenues; they are technically complex; they often have long gestation periods and uncertain effects; and they require the cooperation of multiple institutions. Many reforms fail — they are never enacted, as Nigeria's Petroleum Industry Bill was not for nearly two decades; or they are enacted but not implemented; or they are implemented but fail to achieve their objectives, or produce unintended consequences. The quality of reform depends on the quality of the processes by which problems are diagnosed, options are assessed, laws are drafted, stakeholders are consulted, institutions are designed, and outcomes are evaluated.

This chapter examines these processes from the perspective of the scholar and practitioner who seeks not only to analyse energy law but to improve it. It considers theories of legal and regulatory reform (section 29.1), regulatory impact assessment (section 29.2), legislative drafting for energy governance (section 29.3), institutional capacity and regulatory independence (section 29.4), public consultation and stakeholder participation (section 29.5), policy evaluation and implementation (section 29.6), governance reform in resource-rich economies (section 29.7), and the design of future-ready energy legal systems (section 29.8). Figure 29.1 presents the reform cycle that structures the chapter. The chapter builds on the theories of regulation examined in Chapter 3 and the comparative analysis in Chapter 24, and is intended to equip doctoral graduates for roles of leadership in law reform.

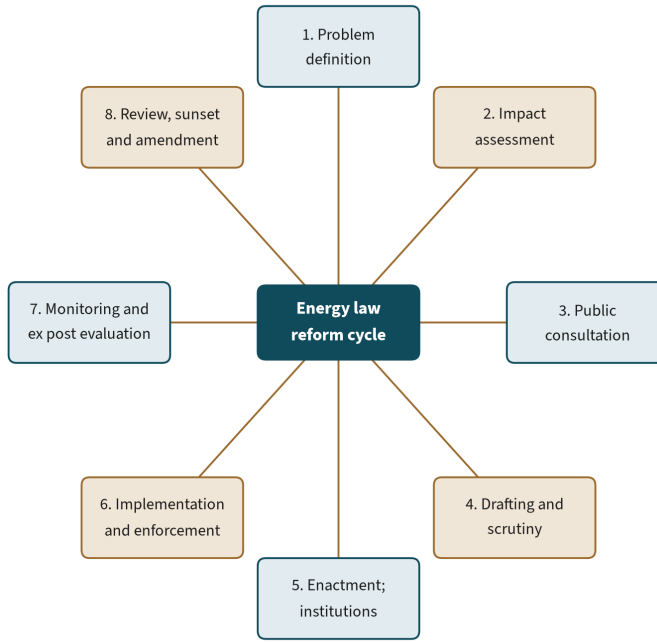


Figure 29.1 The energy law reform cycle. *Source:* prepared for this edition.

29.1 Theories of Legal and Regulatory Reform

Why Reform? Rationales for Regulatory Intervention

Theories of regulation, examined in Chapter 3, provide the starting point for understanding why reforms are undertaken. **Public interest** theories hold that regulation corrects **market failures** — natural monopoly in networks, externalities such as pollution and climate change, information asymmetries, public goods such as energy security, and the coordination problems of complex systems — and pursues social objectives such as universal access and distributive justice. On this view, reform is warranted where existing regulation fails to correct market failures efficiently, where circumstances or technologies change, or where social objectives evolve. **Public choice** and **interest group** theories, associated with George Stigler's theory of economic regulation (1971) and the work of Mancur Olson, hold that regulation is supplied in response to the demands of organised interests, and that concentrated interests — such as incumbent energy companies — are better able to organise and influence regulation than diffuse interests such as consumers, leading to **regulatory capture**. On this view, reform may be driven by shifts in the balance of interests, and successful reform requires strategies to overcome the resistance of concentrated interests. **Institutionalist** theories em-

phasise the role of institutions — formal rules, organisations, and informal norms — in shaping behaviour and outcomes, and the **path dependence** of institutional arrangements, which makes reform difficult once institutions are established.

Models of the Policy Process

Several models of the **policy process** help explain how and when reforms occur. The **policy cycle** or "stages" model divides the process into agenda setting, policy formulation, decision-making, implementation, and evaluation, as in Figure 29.1; though criticised as an idealisation of a messier reality, it provides a useful framework for analysis and practice. John Kingdon's **multiple streams** framework holds that policy change occurs when three independent streams — **problems** (conditions recognised as requiring action), **policies** (solutions developed by experts and advocates), and **politics** (public mood, elections, and interest group pressure) — converge in a **policy window**, often opened by a crisis or a change of government, which **policy entrepreneurs** exploit to couple solutions to problems. The convergence of fiscal crisis, the availability of a long-debated policy proposal, and a new administration's political commitment helps explain the removal of Nigeria's petrol subsidy in May 2023, examined in Chapter 24, and the enactment of the PIA in 2021 after nearly two decades of debate.

Frank Baumgartner and Bryan Jones's **punctuated equilibrium** theory holds that policy is characterised by long periods of stability, maintained by policy monopolies and incremental change, punctuated by short bursts of radical change when issues are redefined and new actors mobilise — a pattern evident in energy policy following crises such as the oil shocks of the 1970s, the Fukushima accident, and the 2022 energy crisis. Paul Sabatier's **advocacy coalition framework** holds that policy change results from competition among coalitions of actors sharing core beliefs, and from policy-oriented learning within and between coalitions over periods of a decade or more, as in the contest between coalitions favouring and opposing fossil fuel development. **Policy transfer** and **diffusion** theories, examined in Chapter 24 in relation to legal transplants, explain how policies spread among jurisdictions through learning, emulation, competition, and coercion — as in the diffusion of electricity market liberalisation through the conditions of international financial institutions in the 1990s, and the diffusion of feed-in tariffs and renewable auctions.

Theories of Legal Change

Legal scholarship offers further perspectives on reform. **Responsive law**, as conceptualised by Philippe Nonet and Philip Selznick, describes a legal order that responds to social needs and purposes rather than adhering rigidly to formal rules. **Reflexive law**, developed by Gunther Teubner, holds that law in complex societies cannot directly control social subsystems such as the economy, but can

structure their self-regulation by establishing procedures and incentives — an insight reflected in disclosure requirements, environmental management systems, and corporate due diligence obligations, examined in Chapter 22. **Responsive regulation**, associated with Ian Ayres and John Braithwaite, examined in Chapter 3, proposes that regulators respond to the conduct of regulated entities, escalating from persuasion to sanctions through an enforcement pyramid. **Smart regulation** (Gunningham and Grabosky) advocates the use of a mix of instruments and actors, including market-based instruments, self-regulation, and third parties. **Experimentalist governance**, associated with Charles Sabel and Jonathan Zeitlin, describes a recursive process of setting provisional framework goals, allowing local units discretion to pursue them, reporting and peer review of performance, and revision of goals and procedures in the light of experience — a model reflected in the Paris Agreement's cycle of nationally determined contributions and in the EU's governance of the energy union.

The Political Economy of Reform

Reform is ultimately a **political** process. Its success depends on the alignment of interests, the distribution of gains and losses, the timing and sequencing of measures, and the capacity of reformers to build coalitions and manage opposition. Reforms with **concentrated costs and diffuse benefits** — such as the removal of fuel subsidies, which imposes immediate costs on consumers and transporters while the benefits of fiscal savings are diffuse and delayed — are particularly difficult. Strategies for overcoming resistance include **compensation** of losers (such as cash transfers to poor households accompanying subsidy reform, as Nigeria attempted after 2023), **sequencing** reforms to build support, **communication** to build public understanding, **bundling** reforms so that gains in one area offset losses in another, and **credible commitment** through legislation or international agreements. The International Monetary Fund's studies of fuel subsidy reform across many countries identify the importance of comprehensive reform plans, communication, phasing, compensation, improving the efficiency of state-owned enterprises, and depoliticising pricing through automatic mechanisms.

Reform in Resource-Rich States

Reform in resource-rich states faces distinctive obstacles, examined in section 29.7: the concentration of rents creates incentives for elites to resist reforms that would reduce their control over resources; patronage networks depend on the discretionary allocation of licences, contracts, and subsidies; and the resource revenues that accrue directly to the state reduce governments' dependence on taxation and thus their accountability to citizens. These dynamics help explain why legal reforms in resource-rich states often take the form of impressive legislation weakly implemented — what has been called **"isomorphic mimicry"**

(Pritchett, Woolcock, and Andrews), in which states adopt the forms of successful institutions without their functions — and why successful reform often requires sustained domestic coalitions supported by transparency and international engagement.

29.2 Regulatory Impact Assessment

Purpose and Development

Regulatory impact assessment (RIA) — also called regulatory impact analysis — is a systematic process for identifying and assessing the expected effects of proposed regulation, and of alternative options, before decisions are taken. Its purposes are to improve the quality of regulation by ensuring that it is based on evidence, that alternatives are considered, and that benefits justify costs; to enhance transparency and accountability by documenting the rationale for regulatory decisions; and to provide a basis for consultation and later evaluation. RIA originated in the United States, where executive orders since the 1970s and 1980s, notably **Executive Order 12291** (1981) and **Executive Order 12866** (1993), required federal agencies to analyse the costs and benefits of significant regulations, subject to review by the **Office of Information and Regulatory Affairs** (OIRA), and has been adopted by OECD members and many developing countries, promoted by the **OECD Recommendation of the Council on Regulatory Policy and Governance** (2012). The European Commission conducts **impact assessments** of significant legislative proposals under its Better Regulation guidelines, scrutinised by the Regulatory Scrutiny Board, and the United Kingdom requires impact assessments for government regulatory proposals, scrutinised by the Regulatory Policy Committee.

The Stages of RIA

A standard RIA involves the following stages:

1. **Problem definition**: identifying the problem to be addressed, its causes, its magnitude, and why government intervention is needed — for example, market failure, regulatory failure, or a distributional concern. A clear problem definition is the foundation of good regulation; many poor regulations result from poorly defined problems.
2. **Objectives**: specifying the objectives of intervention, ideally in measurable terms, such as a target for electricity access, a reduction in gas flaring, or an increase in investment.
3. **Options**: identifying a range of options, including the **"do nothing"** or baseline option, non-regulatory options (such as information, voluntary agreements, or market-based instruments), and different regulatory designs. The consideration of genuine alternatives is the core of RIA.

4. **Impact analysis**: assessing the expected costs, benefits, and distributional effects of each option, including economic, social, and environmental impacts, using quantitative methods such as **cost-benefit analysis** and **cost-effectiveness analysis** where feasible, and qualitative assessment where not. Impacts on particular groups, such as small businesses, low-income households, women, and affected communities, and on competition, should be considered.
5. **Comparison and recommendation**: comparing the options against the objectives and recommending a preferred option, with reasons.
6. **Implementation and compliance**: considering how the preferred option will be implemented, enforced, and complied with, and its administrative costs.
7. **Monitoring and evaluation**: specifying how the regulation's effects will be monitored and when it will be reviewed.
8. **Consultation**: consulting stakeholders on the problem, options, and impacts, examined in section 29.5.

Cost-Benefit Analysis and its Limits

Cost-benefit analysis (CBA) monetises the expected costs and benefits of a regulatory option, discounts them to present values, and compares them. In the energy sector, CBA is used to assess, for example, the benefits of network investments, the costs and benefits of renewable support schemes, and the health benefits of air pollution regulation. CBA raises difficult methodological and ethical questions. The valuation of non-market goods — human health and life (through the **value of a statistical life**), ecosystem services, biodiversity, and cultural heritage — is contested. The choice of **discount rate** profoundly affects the valuation of long-term effects such as climate change: a high discount rate reduces the present value of future climate damages, while a low rate gives greater weight to future generations, as the debate between the *Stern Review* (2006) and its critics illustrated. The **social cost of carbon** — the monetised value of the damages caused by emitting an additional tonne of carbon dioxide — is a key parameter in the assessment of energy and climate regulation; in the United States, its value has been revised repeatedly by successive administrations, rising substantially in 2023 before being withdrawn from federal regulatory analysis in 2025, illustrating the political sensitivity of the parameter. CBA may also neglect **distributional** effects, since it aggregates costs and benefits regardless of who bears them, and may therefore need to be complemented by distributional analysis and consideration of energy justice, examined in Chapter 23.

RIA in Developing Countries and Nigeria

Many developing countries have adopted RIA requirements, often with the support of international organisations, but their implementation has frequently been limited by capacity constraints, lack of data, political resistance, and the treatment of RIA as a formality conducted after decisions have been taken. In **Nigeria**, there is no general statutory requirement of RIA for legislation, although the **National Policy on Regulatory Reform** and the work of the Presidential Enabling Business Environment Council have promoted better regulation, and some regulators, including NERC, conduct consultations and publish analyses supporting their regulatory decisions, such as the Multi-Year Tariff Order. The Electricity Act 2023 and the Petroleum Industry Act 2021 require regulators to consult stakeholders before making regulations, which provides an opportunity for evidence-based analysis. The institutionalisation of RIA for significant energy legislation and regulations, proportionate to the importance of the measure and supported by capacity-building, would improve the quality of energy regulation in Nigeria and other African states.

Climate and Sustainability Assessments

Increasingly, RIA is complemented by specialised assessments of the effects of regulation on climate change and sustainability. The European Union applies a **"do no significant harm"** principle and a **climate consistency** assessment to legislative proposals under the European Climate Law. The United Kingdom's Climate Change Act 2008 and the advisory role of the Climate Change Committee encourage the assessment of policies against carbon budgets. Nigeria's **Climate Change Act 2021**, examined in Chapter 18, requires ministries and agencies to integrate climate considerations into their plans and established the National Council on Climate Change, whose role could include reviewing the climate implications of energy legislation. **Strategic environmental assessment**, examined in Chapter 17, extends environmental assessment from projects to policies, plans, and programmes, and is particularly relevant to energy sector plans, such as the integrated resource plans that determine the future mix of generation.

29.3 Legislative Drafting for Energy Governance

The Importance of Drafting

The quality of legislation depends not only on the policy it embodies but on the clarity, coherence, and effectiveness of its drafting. Poorly drafted legislation creates uncertainty, invites litigation, and may fail to achieve its purposes. Energy legislation poses particular drafting challenges: it must establish complex institutional arrangements, allocate powers among multiple bodies and levels of government, regulate technical matters, create fiscal regimes with precise definitions and formulae, provide for transitions from earlier regimes, and remain adaptable to

technological and market change. The Petroleum Industry Act 2021, with its five chapters, hundreds of sections, and multiple schedules, and the Electricity Act 2023, illustrate the scale and complexity of modern energy legislation.

Principles of Good Legislative Drafting

The principles of good legislative drafting, as articulated in the guidance of drafting offices and the scholarship of drafting experts such as Helen Xanthaki, include:

1. **Clarity**: legislation should be expressed in clear language that conveys the intended meaning to its users, avoiding ambiguity, unnecessary complexity, and archaic expressions. The **plain language** movement has promoted the use of shorter sentences, active voice, everyday words, and logical structure.
2. **Precision**: legislation must be sufficiently precise to determine rights and obligations and to be applied consistently, particularly in fiscal provisions, where small differences in definitions can have large revenue consequences.
3. **Unambiguity**: terms should be used consistently, and key terms defined where necessary, avoiding the use of different words for the same concept or the same word for different concepts.
4. **Coherence**: legislation should be internally consistent and consistent with the wider legal system, including the constitution, other legislation, and international obligations, with explicit provisions for repeals, amendments, and the relationship with other laws.
5. **Effectiveness**: legislation should be capable of achieving its objectives, which requires attention to implementation, enforcement, and institutional capacity — what Xanthaki describes as the ultimate goal of drafting, with clarity, precision, and unambiguity as means to effectiveness.
6. **Accessibility**: legislation should be organised logically, with clear headings, explanatory notes, and consolidated versions available to the public.

Structure of Energy Legislation

A typical energy statute is structured as follows: a **preliminary part** stating the title, commencement, objectives, scope, and application, and sometimes guiding principles; **definitions**, either at the beginning or in a schedule; **institutional provisions** establishing the ministry's policy role, the regulator, and other bodies, and their functions, powers, governance, funding, and accountability; **substantive provisions** regulating the activities within its scope — licensing, technical and safety standards, market rules, tariffs, consumer protection, environmental obligations, community obligations, and fiscal terms; **enforcement provisions** setting out inspection powers, offences, penalties, and administrative sanctions; **dispute resolution and appeals**; **transitional provisions** governing the conversion of existing rights, the transfer of assets and staff, and the continuation of

existing proceedings; **regulations-making powers** delegating detailed rules to the minister or regulator; and **repeals and consequential amendments**.

Framework Legislation and Delegated Rule-Making

A key design choice is the balance between **primary legislation** and **delegated** or **subsidiary legislation**. Energy statutes increasingly take the form of **framework legislation** that establishes institutions, principles, and broad powers, delegating detailed rules to regulators through regulations, codes, licence conditions, and guidelines, which can be adapted more readily to technical and market change. The Petroleum Industry Act and the Electricity Act both confer extensive regulation-making powers on the NUPRC, the NMDPRA, and NERC. Delegation enhances flexibility and draws on regulatory expertise, but raises concerns about democratic accountability and legal certainty, which can be addressed through clear statutory objectives and limits on delegated powers, mandatory consultation, parliamentary scrutiny of delegated legislation, publication requirements, and judicial review. The US Supreme Court's major questions doctrine and the overruling of *Chevron* deference, examined in Chapter 27, illustrate the constitutional tensions that delegation can create, and suggest that legislatures should express clearly any intention to confer broad regulatory powers on agencies.

Drafting for the Energy Transition

Drafting energy legislation for the energy transition requires particular techniques: **technology neutrality**, defining obligations by reference to outcomes (such as emissions or reliability) rather than particular technologies, so that legislation does not become obsolete as technologies evolve; **adaptive provisions**, such as periodic review clauses, sunset clauses, and powers to adjust parameters (such as targets and support levels) within defined limits; **targets and pathways**, as in climate framework laws that set long-term targets and require periodic plans and budgets; **grandfathering and transitional arrangements** that balance the protection of existing investments with the need for change, examined in Chapter 14 in relation to investment law; and **integration** across sectors, since the transition links electricity, transport, heating, industry, and land use. The drafting of **definitions** of emerging technologies — such as "green hydrogen", "energy storage", "carbon capture and storage", and "critical minerals" — has significant consequences for the scope of regulation and incentives, as the debates over the definition of "clean hydrogen" in the United States and "renewable fuels of non-biological origin" in the European Union illustrate.

The Drafting Process

Legislative drafting is a collaborative process involving policy officials, who define the policy; **legislative drafters** (in Nigeria, the Legal Drafting Department of the Federal Ministry of Justice and the legal departments of the National Assembly),

who translate policy into legal text; subject-matter experts; and, increasingly, external stakeholders through consultation on draft bills. Drafting instructions should set out the policy objectives, the intended legal effect, and the relevant background clearly. In Nigeria, many energy bills have originated as private members' bills or executive bills subject to extensive amendment in the National Assembly, and the coordination of drafting across the executive and legislature has been a recurrent challenge, as the protracted history of the Petroleum Industry Bill, which was split into several bills in 2016–2018 before being reconsolidated, illustrates. Pre-legislative scrutiny of draft bills and public hearings by legislative committees can improve the quality of legislation before enactment.

29.4 Institutional Capacity and Regulatory Independence

The Rationale for Independent Regulators

The creation of **independent regulatory agencies** was a central feature of energy sector reforms from the 1990s. The rationale for independence is that regulators insulated from short-term political pressure can make credible long-term commitments to investors, apply technical expertise, and protect consumers from both monopolistic exploitation and politically motivated tariff decisions. Independence addresses the **time inconsistency** problem: once investors have sunk capital in immobile assets such as networks and power plants, governments face a temptation to reduce tariffs below cost for political gain, and investors who anticipate this will under-invest unless the government can credibly commit not to do so. An independent regulator, with a statutory mandate and secure tenure, provides such a commitment. The European Union's energy directives require member states to ensure that national regulatory authorities are legally distinct and functionally independent from any other public or private entity, that their staff and management act independently of market interests and do not seek or take instructions from any government or other public or private entity, and that they have separate annual budget allocations and adequate resources, as interpreted by the Court of Justice in ***Commission v Germany*** (2021), which held that Germany had infringed EU law by conferring on the federal government powers to determine tariff methodologies that belonged exclusively to the regulator.

Dimensions of Independence

Regulatory independence has several dimensions: **legal independence** (a statutory basis, a clear mandate, and protection from direction by ministers on individual decisions); **appointment and tenure** (transparent, merit-based appointment of commissioners for fixed terms, with removal only for cause); **financial independence** (secure funding, often through levies on regulated entities, not dependent on annual political approval); **operational independence** (control

over staff, procedures, and decisions); and **independence from regulated entities** (rules on conflicts of interest and post-employment restrictions to prevent capture). Independence is never absolute: regulators operate within the policy framework set by government and legislature, and in many systems ministers retain the power to issue general policy directions, as under Nigeria's Electricity Act and PIA. The appropriate balance between independence and political accountability is a central question of regulatory design.

Accountability

Independence must be balanced by **accountability**, which legitimises the exercise of regulatory power by unelected bodies. Mechanisms of accountability include: **statutory objectives** that define and limit the regulator's mandate; **transparency** in decision-making, through the publication of decisions, reasons, consultation responses, and analysis; **procedural requirements**, such as consultation and RIA; **parliamentary oversight**, through annual reports, committee hearings, and budget scrutiny; **judicial review and appeals**, examined in Chapter 27; **performance evaluation**, including independent reviews; and **ombudsman** schemes for consumer complaints. The challenge is to design accountability mechanisms that ensure regulators serve the public interest without undermining their independence and capacity to make difficult decisions.

Institutional Capacity

Independence without **capacity** is ineffective. Regulators require adequate **human resources** — economists, engineers, lawyers, accountants, and environmental and social specialists — with competitive remuneration to recruit and retain skilled staff; adequate **financial resources**; **information** about the regulated industry, including powers to require information from regulated entities and systems to analyse it; and **organisational systems** for planning, decision-making, and enforcement. Capacity constraints are acute in many developing countries, where regulators may lack skilled staff, data, and resources relative to the companies they regulate, creating information asymmetries that undermine effective regulation. Capacity-building through training, twinning arrangements with experienced regulators, regional cooperation (as through the ECOWAS Regional Electricity Regulatory Authority and the African Forum for Utility Regulators), and technical assistance can help address these constraints.

Regulatory Institutions in Nigeria

Nigeria's energy regulatory architecture has been transformed by recent legislation. In the **electricity** sector, the **Nigerian Electricity Regulatory Commission**, established by the Electric Power Sector Reform Act 2005 and continued by the Electricity Act 2023, regulates the national market and, under the 2023 Act, transfers regulatory oversight to state regulators as states establish their own

electricity markets, as examined in Chapter 24. In the **petroleum** sector, the **Petroleum Industry Act 2021** replaced the Department of Petroleum Resources and the Petroleum Products Pricing Regulatory Agency with two regulators: the **Nigerian Upstream Petroleum Regulatory Commission** (NUPRC) and the **Nigerian Midstream and Downstream Petroleum Regulatory Authority** (NMDPRA), each governed by a board and funded by statutory fees and levies, and separated the commercial functions of NNPC Limited from regulation. The PIA's provisions on appointment, funding, and accountability of the regulators represent significant progress towards independence, though concerns persist about political influence on appointments, the capacity of the new bodies, and the relationship between the regulators, the Minister of Petroleum Resources, and NNPC Limited. The effectiveness of these institutions will be decisive for the success of the PIA.

Multi-Level and Regional Regulation

Energy regulation increasingly operates at **multiple levels**. In federal systems, the allocation of regulatory competence between federal and state regulators raises questions of coordination and consistency, as in the United States, examined in Chapter 24, and now in Nigeria under the Electricity Act 2023, where the emergence of state regulators — beginning with states such as Enugu, Ekiti, Ondo, Imo, and Oyo — requires mechanisms for coordinating interstate transactions, the national grid, and the protection of investors' rights across jurisdictions. At the regional level, the **ECOWAS Regional Electricity Regulatory Authority** (ERERA) regulates cross-border electricity trade in the West African Power Pool, and the European Union's **ACER** coordinates national regulators. Regional regulation can promote market integration and share scarce expertise, but depends on the willingness of states to cede authority.

29.5 Public Consultation and Stakeholder Participation

Why Consult?

Public consultation and **stakeholder participation** in the making of energy laws and regulations serve several purposes. They improve the **quality** of decisions by drawing on the knowledge and expertise of those affected, identifying problems and unintended consequences, and testing evidence and assumptions. They enhance **legitimacy** and public acceptance, by giving affected persons a voice and demonstrating that their views have been considered. They promote **transparency** and accountability, by exposing proposals and their rationale to public scrutiny. And they give effect to **rights** of participation in environmental decision-making, recognised in Principle 10 of the Rio Declaration (1992), the **Aarhus Convention** (1998), and the **Escazú Agreement** (2018), examined in

Chapter 21, and to the rights of indigenous peoples to free, prior, and informed consent.

Principles of Effective Consultation

Effective consultation should be **timely**, taking place when proposals are still at a formative stage and can genuinely be influenced; **inclusive**, reaching all affected groups, including those that are less organised or marginalised, such as host communities, women, rural consumers, and small businesses; **informed**, providing sufficient information about the proposal, its rationale, and its expected impacts to allow intelligent responses; of **adequate duration**, allowing sufficient time for consideration and response; **responsive**, with the decision-maker genuinely considering responses and explaining how they have been taken into account; and **accessible**, using languages, formats, and channels appropriate to the audiences, including public hearings, community meetings, and digital platforms. In English administrative law, the ***Gunning* principles**, endorsed by the UK Supreme Court in ***R (Moseley) v Haringey LBC*** (2014), require that consultation take place when proposals are at a formative stage, that sufficient reasons be given to permit intelligent consideration and response, that adequate time be given for consideration and response, and that the product of consultation be conscientiously taken into account. The consultation on the UK's nuclear power policy was held unlawful in ***R (Greenpeace) v Secretary of State for Trade and Industry*** (2007) because the information provided was inadequate and the process seriously flawed.

Consultation in Energy Regulation

Energy regulators commonly use formal consultation procedures: publishing **consultation papers** on proposed regulations, tariff methodologies, and market rules; holding **public hearings**, particularly on tariff applications; establishing **stakeholder forums** and advisory panels; and publishing **responses** to consultations explaining how comments were addressed. In Nigeria, NERC holds public hearings on tariff reviews and consults on regulations, and the PIA requires the NUPRC and NMDPRA to consult before making regulations. The quality and influence of consultation vary, and consultations are sometimes criticised as formalities, conducted with short deadlines and limited responsiveness, or dominated by organised industry interests.

Participation of Host Communities

The participation of **host communities** in decisions about resource projects and legislation affecting them is a recurring theme of this book. The Petroleum Industry Act 2021 requires settlors of host community development trusts to conduct needs assessments in consultation with host communities and to involve them in the governance of the trusts, examined in Chapter 24. The effectiveness

of community participation depends on the representativeness of community representatives, the inclusion of women and youth, access to information and independent advice, the capacity of communities to engage, and the willingness of companies and governments to share power. Mechanisms for meaningful participation include community consultation protocols, community development agreements, participatory monitoring, and grievance mechanisms.

Participation in Legislative Processes

At the legislative level, participation takes place through **public hearings** by legislative committees, submissions on bills, and engagement by civil society organisations, industry associations, professional bodies, and academics. The Nigerian National Assembly holds public hearings on major energy bills, including the Petroleum Industry Bill and the Electricity Bill, at which stakeholders present memoranda. The role of **academic expertise** in legislative processes — through evidence to committees, advisory roles, and published research — offers an avenue for doctoral graduates to contribute to reform. The legitimacy and quality of legislation can be enhanced by publishing draft bills for comment before formal introduction, as many jurisdictions do through white papers and exposure drafts.

Risks of Participation

Participation also carries risks: **capture** of consultation processes by well-resourced interests; **consultation fatigue**; the raising of expectations that cannot be met; the use of consultation to legitimise decisions already made; and the exclusion of marginalised voices. Designing participation to mitigate these risks requires attention to the representation of diffuse interests, for example through consumer advocates and funding for public interest participants, and to the transparency of how input is used.

29.6 Policy Evaluation and Implementation

The Implementation Gap

The gap between the objectives of energy laws and their actual effects — the **implementation gap** — is a pervasive problem, particularly in developing countries. Laws may fail to be implemented because of inadequate institutional capacity, lack of funding, conflicting incentives among implementing agencies, resistance from affected interests, poor design that ignores implementation realities, or lack of political commitment. Implementation research, from Jeffrey Pressman and Aaron Wildavsky's classic study *Implementation* (1973) onwards, has emphasised the difficulty of translating policy into practice through complex chains of actors and decisions, the importance of "street-level" implementers (Michael Lipsky), and the need to design policies with implementation in mind.

Planning for Implementation

Effective implementation requires planning from the outset of the reform process: identifying the institutions responsible for implementation and ensuring they have the necessary powers, resources, and capacity; developing **implementation plans** with clear milestones, responsibilities, and timelines; adopting the necessary **subsidiary legislation** promptly, since many energy laws depend on regulations for their operation; establishing **transitional arrangements**; communicating the reform to affected parties; and monitoring progress. The implementation of the Petroleum Industry Act, which required the incorporation of NNPC Limited, the establishment of the new regulators, the adoption of numerous regulations, the conversion of licences and leases, and the establishment of host community trusts within specified periods, illustrates the scale of the implementation task for major energy reforms.

Policy Evaluation

Policy evaluation assesses whether laws and policies are achieving their objectives, at what cost, and with what unintended consequences, in order to support accountability, learning, and improvement. Evaluations may be **ex ante** (assessing likely effects before adoption, as in RIA), **interim** (during implementation), or **ex post** (after implementation). They may be **process evaluations**, examining how a policy is being implemented; **outcome evaluations**, examining whether intended outcomes have been achieved; or **impact evaluations**, estimating the causal effect of the policy on outcomes, using experimental or quasi-experimental methods examined in Chapter 28. The OECD's Development Assistance Committee criteria — **relevance**, **coherence**, **effectiveness**, **efficiency**, **impact**, and **sustainability** — provide a widely used framework.

Evaluation of energy laws faces methodological challenges: outcomes such as investment, prices, and emissions are influenced by many factors besides the law, making attribution difficult; data may be unavailable or unreliable; and the effects of laws may take years to materialise. Techniques for addressing these challenges include the development of a **theory of change** or **logic model**, which specifies the causal chain from inputs and activities through outputs to outcomes and impacts, and the identification of indicators for each link; the establishment of **baselines** and **counterfactuals**; and the use of **mixed methods**. Legislatures can institutionalise evaluation through **review clauses**, which require the government or regulator to review the operation of a law after a specified period and report to the legislature, and **sunset clauses**, which provide for a law to expire unless renewed. The European Union's Better Regulation agenda requires evaluations of existing legislation and "fitness checks" of policy areas, and the United Kingdom requires post-implementation reviews of significant regulations.

Learning from Failure

Evaluation should support **learning**, including from failure. The history of energy reform is rich in instructive failures: the California electricity crisis of 2000–2001, which revealed flaws in market design; the Nigerian power sector privatisation of 2013, whose disappointing results reflected flaws in sequencing, tariffs, and the condition of the assets, as examined in Chapter 24; the retroactive cuts to renewable energy support in Spain, which generated dozens of investment arbitrations, as examined in Chapter 14; and the difficulties of many African community development schemes. A culture of evaluation and learning, in which failures are acknowledged and analysed rather than concealed, is a hallmark of effective governance.

29.7 Governance Reform in Resource-Rich Economies

The Governance Challenge

The **resource curse** literature, examined in Chapter 23, has established that the economic and political outcomes of resource wealth depend critically on the quality of governance. States with strong institutions — accountable governments, effective bureaucracies, the rule of law, and control of corruption — have generally converted resource wealth into development, as Norway, Botswana, Chile, and Malaysia illustrate, while states with weak institutions have often suffered from volatility, corruption, conflict, and stagnation. Governance reform is therefore at the heart of the resource development agenda. Yet it is also especially difficult in resource-rich states, because resource rents create powerful incentives to preserve the status quo.

Frameworks for Resource Governance

Several frameworks articulate the components of good resource governance. The **Natural Resource Charter**, developed by an independent group of experts and hosted by the Natural Resource Governance Institute (NRGI), sets out twelve precepts covering the **decision chain** of resource management: strategy, consultation, and institutions; accountability; the exploration and award of rights; taxation; local effects; state-owned enterprises; investing for sustained development; revenue smoothing; government spending; private sector development; the roles of companies; and international cooperation. The NRGI's **Resource Governance Index**, which assesses the quality of resource governance in many countries across value realisation, revenue management, and the enabling environment, has consistently found significant governance deficits in many resource-rich states, including in Africa. The **Africa Mining Vision** (2009) and its **Country Mining Vision** guidelines, the **EITI Standard**, examined in Chapter 22, and the **IMF's Fiscal Transparency Code** with its resource revenue pillar provide further frameworks. The World Bank's **Extractive Industries**

Value Chain framework describes the sequence from the award of contracts and licences through regulation and monitoring of operations, collection of taxes and royalties, and revenue distribution and management, to the implementation of sustainable development policies.

Key Areas of Reform

Drawing on these frameworks and the analysis throughout this book, the key areas of governance reform in resource-rich economies include:

1. **The award of rights**: replacing discretionary allocation of licences and contracts with transparent, competitive procedures, published criteria, and disclosure of contracts and beneficial owners, as examined in Chapters 5 and 22.
2. **Fiscal regimes**: designing fiscal regimes that capture a fair share of resource rents through progressive instruments, minimise discretionary incentives, and are administered by capable tax authorities with powers to audit costs and address transfer pricing, as examined in Chapter 16.
3. **State-owned enterprises**: clarifying the mandates of national oil and mining companies, separating their commercial and regulatory functions, subjecting them to commercial governance, transparency, and audit, and limiting quasi-fiscal activities, as the PIA sought to do for NNPC Limited.
4. **Revenue management**: adopting fiscal rules, stabilisation and savings funds, and transparent budgeting to manage volatility and ensure that revenues benefit current and future generations, as examined in Chapter 23.
5. **Regulatory institutions**: establishing independent, capable, and accountable regulators, as discussed in section 29.4.
6. **Local benefits and community rights**: ensuring that host communities share in benefits and that their rights are respected, through benefit-sharing mechanisms, consultation, and grievance mechanisms, as examined in Chapters 21 and 24.
7. **Environmental protection**: effective environmental regulation, remediation of legacy pollution, and decommissioning security.
8. **Transparency and accountability**: disclosure of payments, contracts, and beneficial owners; independent audit; parliamentary oversight; freedom of information; and space for civil society and the media.
9. **Anti-corruption**: effective investigation and prosecution of corruption, international cooperation on asset recovery, and integrity in public procurement, as examined in Chapter 22.

Reform in Nigeria: Lessons and Challenges

Nigeria's experience offers rich material for the study of governance reform in a resource-rich economy. The **Petroleum Industry Act 2021** addressed long-

standing governance problems: the conflict of roles within the old NNPC, the opacity of the fiscal regime, the absence of a legal framework for host community benefits, and the fragmentation of regulation. The **Nigeria Extractive Industries Transparency Initiative Act 2007** made Nigeria the first EITI implementing country to give the initiative a statutory basis, and NEITI's audits have exposed significant revenue losses and governance failures. The **removal of the petrol subsidy** in 2023 and the **unification of exchange rates** addressed major fiscal distortions. The **Electricity Act 2023** opened the electricity sector to state-level markets. The **tax reform legislation of 2025** sought to modernise and harmonise the tax system.

Significant challenges remain: the implementation of the PIA, including the effectiveness of host community trusts and the commercial transformation of NNPC Limited; **crude oil theft** and pipeline vandalism, which have at times severely reduced production; the **remediation** of legacy pollution in the Niger Delta; the **financial viability** of the electricity sector; the management of **resource revenues** across the federation, including the depletion of the Excess Crude Account and the limited size of the Nigeria Sovereign Investment Authority's funds; the **energy transition**, including the role of gas, the development of renewable energy, and the implications of global decarbonisation for oil revenues; and the broader challenges of corruption and institutional capacity. The Nigerian experience demonstrates that legislative reform is a necessary but not sufficient condition of governance reform, and that implementation, enforcement, and the sustained engagement of citizens and institutions are equally important.

International Support and Its Limits

International actors — multilateral development banks, the IMF, bilateral donors, the EITI, and international NGOs — have supported governance reform in resource-rich states through technical assistance, conditionality, transparency standards, and advocacy. Their contributions have included model laws and contracts, capacity-building for regulators and tax authorities, support for revenue management frameworks, and pressure for transparency. However, externally driven reforms may lack domestic ownership, reflect inappropriate models, or produce isomorphic mimicry. The lesson of the governance literature is that sustainable reform depends on **domestic coalitions** for reform — reformist officials, legislators, civil society, the media, business, and citizens — and that international support is most effective when it strengthens such coalitions rather than substituting for them. The approach of **problem-driven iterative adaptation** (PDIA), developed by Matt Andrews, Lant Pritchett, and Michael Woolcock, emphasises locally defined problems, iterative experimentation, and broad engagement, rather than the imposition of best-practice solutions.

29.8 Designing Future-Ready Energy Legal Systems

The Challenge of Change

Energy legal systems designed for the twentieth century — centralised, fossil-fuel-based, and organised around large utilities and national oil companies — must be redesigned for a twenty-first century characterised by decarbonisation, decentralisation, digitalisation, and deepening geopolitical uncertainty. A **future-ready** energy legal system is one capable of guiding and accommodating this transformation: providing the stability investors need while adapting to rapid change, protecting the vulnerable while enabling innovation, and integrating energy, climate, environmental, and social objectives. This section identifies the design principles of such systems, drawing on the analysis throughout this book.

Design Principles

1. **Clear long-term direction.** Future-ready systems establish a clear, legally anchored long-term direction — such as a net zero target and pathway in a climate framework law, with interim targets and carbon budgets — which provides a framework for investment and policy, as in the United Kingdom's Climate Change Act 2008 and Nigeria's Climate Change Act 2021. Long-term direction reduces policy uncertainty and the risk of stranded assets.
2. **Adaptive regulation.** Systems must be able to adapt to technological and market change without constant legislative amendment, through framework legislation, delegated rule-making with accountability, periodic review, **regulatory sandboxes** that allow innovators to test new business models under relaxed rules and close supervision (as introduced by Ofgem, the Australian Energy Market Commission, and regulators in Kenya and other African states), and **experimental** approaches that learn from evaluation.
3. **Technology neutrality with strategic choice.** Regulation should generally be technology-neutral, setting outcomes rather than prescribing technologies, while allowing targeted support for strategic technologies at early stages of development.
4. **Integrated governance.** The transition requires integration across sectors (electricity, transport, heating, industry), levels of government (international, national, subnational), and policy domains (energy, climate, environment, industry, trade, and social policy), through coordinating institutions and integrated planning, such as the integrated national energy and climate plans required of EU member states and Nigeria's Energy Transition Plan.

5. **Investment-enabling stability.** Future-ready systems balance adaptability with the stability investors require, through credible and independent institutions, transparent rules, protection against arbitrary change, and mechanisms for managing unavoidable change fairly, such as transitional arrangements and compensation for early closure where appropriate — lessons drawn from the investment disputes examined in Chapters 14 and 26.
6. **Just transition.** Systems must protect workers, communities, and consumers affected by the transition, through social protection, retraining, regional development, affordable energy, and the participation of affected groups in transition planning, as examined in Chapter 23.
7. **Access and affordability.** In Africa, a future-ready energy legal system must above all achieve universal access to modern energy, through frameworks for grid extension, mini-grids, off-grid solutions, and clean cooking, and through tariff and subsidy designs that protect the poor without undermining financial viability.
8. **Resilience and security.** Systems must be resilient to physical, cyber, climate, and geopolitical shocks, through diversification, reserves, critical infrastructure protection, and emergency frameworks, as examined in Chapter 25.
9. **Digital and data governance.** As energy systems become digitalised, legal systems must govern data ownership, access, privacy, cyber security, and the use of artificial intelligence, as examined in Chapter 30.
10. **Accountability and participation.** Future-ready systems are transparent, accountable, and participatory, enabling citizens, communities, and civil society to hold governments and companies to account and to shape the transition.

The Role of Legal Leaders

The design and reform of energy legal systems requires **leadership** — from legislators, regulators, judges, practitioners, academics, and civil society. Legal leaders in the energy sector need not only technical legal expertise but also an understanding of economics, technology, and politics; the ability to communicate across disciplines and with the public; the integrity to resist capture and corruption; and the vision to see beyond immediate interests to long-term public goods. Doctoral graduates in energy and natural resources law are well placed to exercise such leadership — as advisers to governments and regulators, as drafters of legislation, as judges and arbitrators, as counsel to companies and communities, as officials of international organisations, and as scholars whose research informs reform. Chapter 30 considers the challenges that this generation of legal leaders will face.

::: casebox {title="Case Study: The Long Road to the Petroleum Industry Act 2021"} **Background.** By the early 2000s, Nigeria's petroleum legislation — principally the Petroleum Act 1969, the NNPC Act 1977, and numerous subsidiary instruments — was widely regarded as outdated. The NNPC combined commercial, regulatory, and policy functions; the fiscal regime was complex and opaque; host communities had no statutory share in petroleum benefits; and the industry was beset by funding problems in the joint ventures, gas flaring, and conflict in the Niger Delta.

Reform efforts. In 2000, the government established the Oil and Gas Sector Reform Implementation Committee (OGIC), whose work led to a National Oil and Gas Policy (2004) and, in 2008, to the first **Petroleum Industry Bill**, an omnibus bill consolidating the legal framework. The bill was introduced in successive National Assemblies but not enacted, as disagreements over fiscal terms, the powers of the minister, the structure of NNPC, and host community benefits, together with the opposition of affected interests and the lobbying of international companies concerned about fiscal terms, stalled its progress. In 2016, the National Assembly adopted a strategy of splitting the bill into several components, beginning with the **Petroleum Industry Governance Bill**, which was passed in 2018 but not signed by the President, reportedly owing to concerns about the powers of the regulator and revenue retention.

Enactment. In 2020, the executive submitted a new consolidated Petroleum Industry Bill, which, after public hearings and amendment in the National Assembly, including contentious debates over the percentage of operating expenditure payable to host community trusts (eventually set at three per cent) and the allocation of thirty per cent of NNPC's profit oil and gas to a Frontier Exploration Fund, was passed and signed into law in **August 2021**.

Analysis. The PIA's history illustrates many themes of this chapter: the difficulty of reforms that redistribute rents among powerful interests; the role of policy windows — in this case the oil price collapse of 2020, the decline in investment, and the government's need for reform to attract capital; the trade-off between comprehensive and incremental reform strategies; the importance of legislative drafting and consultation; and the reality that enactment is the beginning, not the end, of reform. Its implementation — the commercialisation of NNPC Limited, the effectiveness of the new regulators, the functioning of host community trusts, and the response of investors — will determine whether the long road was worth travelling. :::

::: exercise {title="Practical Exercise 29: Legislative Drafting — The Riverine State Mini-Grid and Clean Cooking Bill (HYPOTHETICAL)"} **This exercise is entirely hypothetical. The state, institutions, data, and political circum-**

stances described are fictitious and do not refer to any real state or legislation.

The (fictitious) Riverine State, a state in a federal West African republic whose federal electricity law (modelled on Nigeria's Electricity Act 2023) permits states to regulate electricity markets within their territory, has two million residents, of whom fifty-five per cent lack access to grid electricity and eighty per cent cook with firewood or charcoal. Its government has resolved to establish a state electricity market focused on **mini-grids** and **off-grid solar**, and to promote **clean cooking**. It has commissioned you, as a doctoral graduate in energy law, to lead the preparation of legislation.

Tasks.

(a) **Problem definition and RIA.** Prepare a regulatory impact assessment (of no more than 2,500 words) defining the problem, setting measurable objectives, identifying at least three options (including a non-legislative option), and assessing their expected costs, benefits, and distributional impacts, including on women and low-income households.

(b) **Drafting instructions.** Prepare drafting instructions setting out the policy objectives and intended legal effect of the bill.

(c) **Drafting.** Draft the following provisions of a Riverine State Electricity Market and Clean Cooking Bill: (i) the objectives and guiding principles; (ii) the establishment, functions, independence, funding, and accountability of a Riverine State Electricity Regulatory Commission; (iii) a light-handed licensing or registration regime for mini-grids below a specified capacity, including provisions on what happens when the main grid arrives; (iv) tariff principles for mini-grids balancing cost recovery and affordability; (v) a clean cooking promotion framework; (vi) consultation requirements for regulations; and (vii) a review clause.

(d) **Consultation plan.** Design a consultation plan for the bill that reaches rural communities, women's groups, mini-grid developers, the distribution company currently serving the state, and the federal regulator.

(e) **Evaluation framework.** Develop a theory of change and a set of indicators for evaluating the law five years after its enactment. :::

Conclusion

Energy law reform is a continuous, demanding, and essential task. Theories of regulation and the policy process explain why reforms occur, why they often fail, and how policy windows, coalitions, and entrepreneurs can drive change. The quality of reform depends on disciplined processes: regulatory impact assessment that defines problems, considers alternatives, and weighs impacts, including distributional and climate impacts; legislative drafting that achieves

clarity, precision, coherence, and, above all, effectiveness, with appropriate use of framework legislation and delegated rule-making; regulatory institutions that combine independence, accountability, and capacity; consultation that is timely, inclusive, and responsive; and implementation planning and evaluation that close the gap between law and outcomes and enable learning from failure. In resource-rich economies, governance reform must confront the political economy of rents, and its success depends on domestic coalitions supported by transparency and international engagement. Nigeria's recent reforms — the PIA, the Electricity Act, subsidy removal, and tax reform — illustrate both the possibilities and the difficulties. Future-ready energy legal systems will combine long-term direction with adaptability, stability with justice, and security with access. Designing and leading such systems is a central vocation for the doctoral graduates for whom this book is written.

CHAPTER SUMMARY

- Theories of regulation (public interest, public choice, institutionalist) and of the policy process (policy cycle, multiple streams, punctuated equilibrium, advocacy coalitions, policy transfer) explain the drivers of and obstacles to energy law reform; theories of legal change (responsive, reflexive, smart, experimentalist) inform reform design; the political economy of concentrated costs and diffuse benefits shapes outcomes.
- Regulatory impact assessment involves problem definition, objectives, options, impact analysis, comparison, implementation planning, monitoring, and consultation; cost-benefit analysis is valuable but raises difficulties in valuing non-market goods, choosing discount rates, setting the social cost of carbon, and addressing distribution; Nigeria lacks a general RIA requirement.
- Good legislative drafting achieves clarity, precision, unambiguity, coherence, effectiveness, and accessibility; energy statutes follow a recognisable structure; framework legislation and delegated rule-making enhance flexibility but require accountability; drafting for the transition calls for technology neutrality, adaptive provisions, and careful definitions.
- Independent regulators address time inconsistency and capture, but require accountability and capacity; Nigeria's PIA and Electricity Act have restructured regulatory institutions, including at state level.
- Consultation improves quality and legitimacy and gives effect to participation rights; it should be timely, inclusive, informed, adequate, responsive, and accessible, as reflected in the *Gunning* principles.
- The implementation gap requires implementation planning, and policy evaluation — process, outcome, and impact — supported by theories of change, review and sunset clauses, and a culture of learning from failure.

- Governance reform in resource-rich economies addresses the award of rights, fiscal regimes, state enterprises, revenue management, regulation, community benefits, environment, transparency, and anti-corruption; sustainable reform depends on domestic coalitions.
- Future-ready energy legal systems provide long-term direction, adaptive regulation, technology neutrality, integration, investment stability, just transition, universal access, resilience, data governance, and accountability, requiring legal leadership.

DOCTORAL RESEARCH QUESTIONS

1. Applying Kingdon's multiple streams framework, explain the enactment of the Petroleum Industry Act 2021 after two decades of failed attempts.
2. Should Nigeria adopt a statutory requirement of regulatory impact assessment for energy legislation and regulations? Design a proportionate framework.
3. How can the independence and accountability of energy regulators be reconciled in federal systems with emerging state-level regulators? Evaluate with reference to Nigeria's Electricity Act 2023.
4. Assess the role of regulatory sandboxes in enabling innovation in African electricity markets, and the legal safeguards they require.
5. To what extent do resource governance frameworks such as the Natural Resource Charter and the EITI Standard produce real governance improvements? Evaluate the evidence.
6. What legal design features make an energy legal system "future-ready"? Develop and apply an evaluative framework to one African jurisdiction.

FURTHER READING

- Robert Baldwin, Martin Cave and Martin Lodge, *Understanding Regulation: Theory, Strategy, and Practice* (2nd edn, Oxford University Press 2012).
- John W Kingdon, *Agendas, Alternatives, and Public Policies* (2nd edn, Longman 2003).
- Helen Xanthaki, *Drafting Legislation: Art and Technology of Rules for Regulation* (Hart Publishing 2014).
- Ian Ayres and John Braithwaite, *Responsive Regulation: Transcending the Deregulation Debate* (Oxford University Press 1992).
- Matt Andrews, Lant Pritchett and Michael Woolcock, *Building State Capability: Evidence, Analysis, Action* (Oxford University Press 2017).
- Natural Resource Governance Institute, *Natural Resource Charter* (2nd edn, 2014).
- OECD, *Recommendation of the Council on Regulatory Policy and Governance* (OECD 2012).

30 CHAPTER

The Future of International Energy and Natural Resources Law

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Analyse the forces reshaping the architecture of global energy governance.
2. Evaluate how decarbonisation is transforming the law of natural resources, from fossil fuels to critical minerals.
3. Assess the legal implications of artificial intelligence and automation for energy regulation.
4. Analyse the legal questions raised by digitalisation, data ownership, and new energy market models.
5. Evaluate the legal frameworks governing climate-related financial and legal risks.
6. Anticipate emerging categories of international energy disputes and regulatory conflicts.
7. Identify unresolved questions in global resource governance and evaluate competing approaches to them.
8. Formulate a doctoral research agenda responsive to the challenges facing the next generation of energy and resource lawyers.

Introduction

This book has traced the development of international energy and natural resources law from the concessions of the early twentieth century, through the assertion of permanent sovereignty and the rise of national oil companies, the liberalisation of energy markets and the proliferation of investment treaties, to the climate regime, the energy transition, and the geopolitical upheavals of the 2020s. This final chapter looks forward. It does not attempt to predict the future — energy forecasts have a notoriously poor record — but to identify the forces that are reshaping the field, the legal questions they raise, and the research agenda they generate for the next generation of scholars and practitioners.

Eight forces, depicted in Figure 30.1, appear likely to define the field in the coming decades: **decarbonisation**, as the world seeks to limit global warming by transforming energy systems built on fossil fuels; **digitalisation and artificial intelligence**, which are transforming the operation of energy systems and the practice of regulation; **decentralisation**, as distributed generation, storage, and prosumers challenge centralised utility models; **geopolitical fragmentation**, as rivalry among major powers reshapes trade, investment, and security; the rise of **critical minerals** as strategic resources; the recognition of **climate risk** as a financial and legal risk; the growing salience of **justice and rights** in energy governance; and **new frontiers** of resource development, from the deep seabed to hydrogen and beyond. These forces interact: decarbonisation drives demand for critical minerals, which intensifies geopolitical competition and raises questions of justice for producing communities; digitalisation enables decentralisation while creating new security risks. The chapter examines the changing architecture of global energy governance (section 30.1), decarbonisation and the transformation of resource law (section 30.2), artificial intelligence and automated energy regulation (section 30.3), digitalisation, data ownership, and energy markets (section 30.4), climate-related financial and legal risks (section 30.5), emerging international disputes and regulatory conflicts (section 30.6), unresolved questions in global resource governance (section 30.7), and a doctoral research agenda for the next generation (section 30.8).

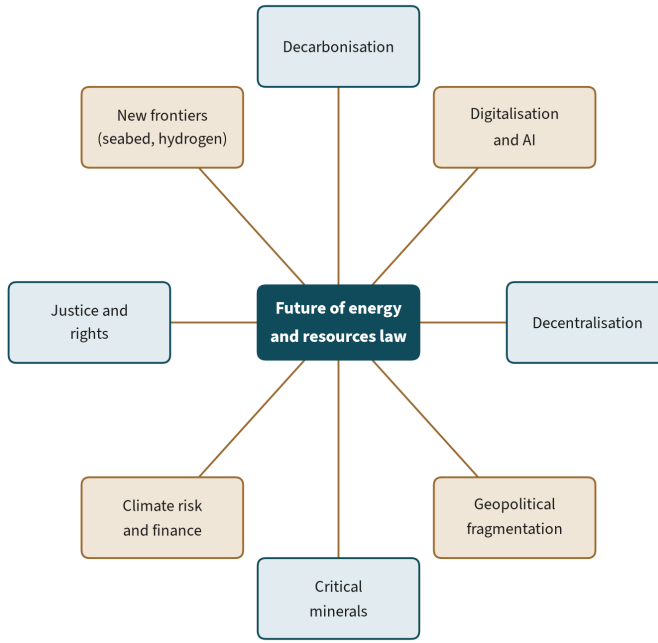


Figure 30.1 Forces shaping the future of international energy and natural resources law. *Source:* prepared for this edition.

30.1 The Changing Architecture of Global Energy Governance

A Fragmented Architecture

Unlike trade, which has the WTO, or climate change, which has the UNFCCC, energy has never had a single, comprehensive international regime. Global energy governance, examined in Chapter 2, is a **regime complex** — a loosely coupled set of institutions with overlapping mandates and memberships: the **International Energy Agency**, founded as a consumers' organisation and now a leading source of energy data and transition analysis; **OPEC** and **OPEC+**, representing major oil producers; the **Gas Exporting Countries Forum**; the **International Energy Forum**, a producer–consumer dialogue; the **International Renewable Energy Agency**; the **International Atomic Energy Agency**; the **Energy Charter Treaty** and its Conference; the climate regime; the WTO; the multilateral development banks; the G7 and G20; initiatives such as **Sustainable Energy for All**; regional organisations, including the European Union, ECOWAS, and the African Union; and a growing array of plurilateral clubs and partnerships. This fragmentation reflects the strategic importance of energy, the divergent interests

of producers and consumers, and the reluctance of states to cede sovereignty over their resources.

Forces of Change

Several forces are reshaping this architecture. First, the **energy transition** is shifting the centre of gravity from institutions focused on oil and gas towards those concerned with renewables, electricity, efficiency, and critical minerals. The IEA's mandate has broadened accordingly, IRENA's membership has become near-universal, and new initiatives on critical minerals — the Minerals Security Partnership, the G7 Critical Minerals Action Plan, and the UN Secretary-General's **Panel on Critical Energy Transition Minerals**, which in 2024 proposed guiding principles for the responsible development of transition minerals, including respect for human rights, benefit-sharing, and value addition in producing countries — have emerged. Second, the **shift in energy demand** to emerging economies, particularly China, India, and South-East Asia, and the projected growth of demand in Africa, are challenging the dominance of OECD-centred institutions. The IEA's association with major emerging economies and the expansion of the **BRICS** grouping in 2024 to include major energy producers such as Saudi Arabia (whose membership status has remained ambiguous), the UAE, and Iran, as well as Egypt and Ethiopia, and later Indonesia, illustrate the diversification of energy diplomacy. Third, **geopolitical fragmentation**, examined in Chapter 25, is producing competing blocs and supply chains, undermining universal institutions. Fourth, the **climate regime's** increasing engagement with energy — in the COP28 decision of 2023 calling for a transition away from fossil fuels in energy systems and a tripling of renewable capacity and doubling of efficiency by 2030, examined in Chapter 18 — is making the UNFCCC an increasingly important forum for energy governance.

Clubs, Partnerships, and Minilateralism

In the face of the difficulties of universal agreement, energy governance is increasingly conducted through **clubs** and **minilateral** arrangements among smaller groups of like-minded states. These include **Just Energy Transition Partnerships**, examined in Chapter 23; **climate clubs**, such as the G7-initiated Climate Club launched in 2023 to promote industrial decarbonisation; the **Powering Past Coal Alliance**; the **Beyond Oil and Gas Alliance**, launched in 2021 by Denmark and Costa Rica to promote the managed phase-out of oil and gas production; sectoral initiatives on methane, such as the **Global Methane Pledge** (2021) to reduce methane emissions by at least thirty per cent from 2020 levels by 2030; and bilateral and plurilateral critical minerals agreements. Clubs can achieve deeper cooperation among willing participants and may expand over time, but raise concerns about exclusion, legitimacy, and the fragmentation of rules,

particularly where they impose standards or trade measures on non-members, as in the case of carbon border adjustment mechanisms.

Proposals for Reform

Scholars and policymakers have proposed various reforms of global energy governance: the transformation of the IEA into a genuinely global energy agency including major emerging economies; a **Fossil Fuel Non-Proliferation Treaty**, promoted by a coalition of states (initially Pacific island states, led by Vanuatu and Tuvalu), cities, and civil society, to end the expansion of fossil fuel production and manage a global transition — a proposal that gained momentum when Colombia and the Netherlands announced at COP30 in 2025 that they would co-host a first international conference on the transition away from fossil fuels in 2026; a multilateral framework for critical minerals; the reform or replacement of the Energy Charter Treaty; and the integration of energy access and transition finance into the reform of the international financial architecture. The **ICJ advisory opinion of July 2025** on the obligations of states in respect of climate change, which affirmed that the failure of a state to take appropriate action to protect the climate system from greenhouse gas emissions — including through fossil fuel production, the granting of exploration licences, or fossil fuel subsidies — may constitute an internationally wrongful act, as examined in Chapter 18, may influence the evolution of energy governance by clarifying the legal obligations that constrain state energy policies.

Africa in the Emerging Architecture

Africa's place in the emerging architecture of energy governance is pivotal. The continent holds a large share of the world's critical mineral reserves, vast renewable energy potential, and significant oil and gas resources, and is projected to account for a growing share of global energy demand and population. Yet Africa has historically been a rule-taker in global energy governance, under-represented in its institutions. The **African Union's** Agenda 2063, the Africa Mining Vision, the Green Minerals Strategy, the **African Single Electricity Market** initiative, and the **AfCFTA** offer frameworks for a more coherent African voice, and the **Africa Climate Summit** declarations of Nairobi (2023) and Addis Ababa (2025) articulated African positions on climate finance and the transition. The challenge for African states is to translate their resource endowments into bargaining power in global forums and value addition at home, while ensuring universal energy access and a just transition — a challenge in which legal scholarship has an important role.

30.2 Decarbonisation and the Transformation of Resource Law

From Scarcity to Stranding

For most of the twentieth century, the law of natural resources was shaped by a paradigm of **scarcity**: the assumption that resources were finite and valuable, that demand would grow, and that the central legal questions concerned the allocation of rights to exploit them, the division of rents between states and investors, and the protection of investments. Decarbonisation is challenging this paradigm for fossil fuels. If the world is to meet the temperature goals of the Paris Agreement, a large share of known fossil fuel reserves must remain unburned — studies have estimated that, for a fifty per cent chance of limiting warming to 1.5°C, nearly sixty per cent of oil and fossil methane gas and ninety per cent of coal reserves would need to remain unextracted by 2050 — and the IEA's Net Zero Emissions scenario envisaged no new long-lead-time oil and gas fields approved for development beyond those already committed as of 2021. Whether or not these pathways are followed, the prospect of declining demand transforms the legal questions surrounding fossil fuel resources: from how to allocate scarce resources to how to manage **stranded assets**, decline, and exit.

The Law of Managed Decline

A body of law is emerging to govern the **managed decline** of fossil fuel production. Its elements include: **restrictions on new exploration and production**, such as the bans on new oil and gas licensing adopted by Denmark (which set an end date of 2050 for production), France, Ireland, Costa Rica, and New Zealand (which subsequently reversed its ban on new offshore exploration), and the United Kingdom's policy, examined in Chapter 24, of not issuing new licences to explore new fields; **coal phase-out laws**, such as Germany's Coal Phase-out Act 2020, which sets an end date of 2038 at the latest and provides compensation to operators, and the Netherlands' ban on coal-fired electricity from 2030; **phase-out of fossil fuel subsidies**, a long-standing G20 commitment rarely fulfilled; **decommissioning** obligations and financial security, examined in Chapter 7, whose importance grows as fields reach the end of their lives; **just transition** frameworks for affected workers and regions, examined in Chapter 23; and the treatment of **compensation** for early closure, which raises questions under investment law, as the RWE and Uniper claims against the Netherlands and the Westmoreland claim against Canada's coal phase-out, examined in Chapter 26, illustrate.

Producer Economies and the Transition

The managed decline of fossil fuels poses profound challenges for **producer economies**, particularly developing countries such as Nigeria, Angola, Mozam-

bique, Senegal, and Uganda, which depend on, or hope to benefit from, fossil fuel revenues. These states argue that they bear little historical responsibility for climate change, that their populations lack access to modern energy, and that they should be permitted to develop their resources, particularly gas, as a **transition fuel** — a position reflected in the African Union's common position on energy access and just transition (2022) and in Nigeria's Energy Transition Plan, which envisages gas as a bridge to net zero by 2060. Others argue that new fossil fuel investments risk becoming stranded assets, locking producer states into dependence on declining revenues, and that the principle of **common but differentiated responsibilities** requires that wealthy producers phase out first and finance the transition of developing producers. The legal questions include how international law should allocate the remaining carbon budget among producers, whether there is an emerging obligation to limit production — a question to which the ICJ's 2025 advisory opinion has given new salience — and how transition finance can be mobilised to support economic diversification. Proposals for an equitable **production phase-out**, in which wealthy, diversified producers phase out first, have been advanced by scholars and civil society, but have not been accepted by major producers.

The Rise of Transition Minerals

While the transition threatens fossil fuel resources with stranding, it creates soaring demand for **transition minerals** — lithium, cobalt, nickel, graphite, copper, manganese, and rare earth elements — used in batteries, electric vehicles, wind turbines, solar panels, and electricity networks, as examined in Chapter 8. The IEA has projected that demand for some of these minerals could increase several-fold by 2040 in scenarios consistent with climate goals. The law of transition minerals is being shaped by the same questions that once shaped the law of oil — sovereignty, rent capture, value addition, investment protection, and the rights of affected communities — but in a new context characterised by heightened geopolitical competition, stronger human rights and environmental norms, and the assertiveness of producer states seeking to avoid the mistakes of the past. The **export bans** of Indonesia and Zimbabwe, the **strategic minerals** regimes of the DRC and Chile, the **local processing** requirements of several African states, and the **critical minerals** strategies of consumer states, examined in Chapters 8, 24, and 25, are elements of this emerging law.

A central question is whether the rush for transition minerals will repeat the patterns of the resource curse — enclave development, environmental devastation, human rights abuses, and the capture of rents by elites and foreign investors — or whether it can be governed to promote broad-based development. The UN Secretary-General's Panel on Critical Energy Transition Minerals, the OECD due diligence guidance, the EU's Battery Regulation (Regulation (EU)

2023/1542), with its due diligence and battery passport requirements, the EU Corporate Sustainability Due Diligence Directive (whose obligations were scaled back and postponed by the "Omnibus" simplification package agreed in 2025), and the **Initiative for Responsible Mining Assurance** standard are among the instruments that seek to embed responsible practices. The legal frameworks of producer states — their mining codes, fiscal regimes, community benefit-sharing, and environmental regulation — will be decisive.

Deep Seabed Minerals

The **deep seabed** has become a new frontier of transition minerals. Polymetallic nodules on the abyssal plains of the Pacific Ocean contain nickel, cobalt, copper, and manganese, and the **International Seabed Authority**, examined in Chapter 20, has been negotiating a **Mining Code** of exploitation regulations since 2014. The invocation by Nauru in 2021 of the "two-year rule" in the 1994 Implementation Agreement, which was intended to require the Authority to complete the regulations within two years, did not result in their adoption, and a growing number of states have called for a **moratorium or precautionary pause** on deep seabed mining pending better scientific understanding of its environmental effects. In 2025, the United States, which is not a party to UNCLOS, issued an executive order seeking to accelerate deep seabed mining under its **Deep Seabed Hard Mineral Resources Act 1980**, including in areas beyond national jurisdiction, and a company sought a commercial recovery permit from the US authorities — prompting objections from the ISA and many states that such unilateral action would violate the principle that the resources of the Area are the **common heritage of mankind**. The outcome of this contest will test the resilience of the UNCLOS regime and shape the law of resources beyond national jurisdiction.

New Resources: Hydrogen, Carbon Storage, and Beyond

Decarbonisation is also creating **new categories of resources** requiring legal frameworks: **geological storage space** for captured carbon dioxide and hydrogen, which raises questions of ownership of pore space, long-term liability, and cross-border transport, examined in Chapter 12; **natural (geological) hydrogen**, whose discovery in several countries has prompted questions about whether it falls within existing mining or petroleum legislation; **offshore wind** sites, allocated through seabed leasing rounds; **geothermal** resources; and the **radio spectrum** and **data** on which digital energy systems depend. Each requires the law to define property rights, allocate access, capture public value, and manage environmental risks — the classic tasks of resource law, applied to novel resources.

30.3 Artificial Intelligence and Automated Energy Regulation

AI in Energy Systems

Artificial intelligence is transforming energy systems. Machine learning algorithms are used to forecast renewable generation and demand, optimise the dispatch of generation and storage, detect faults and predict maintenance needs in grids and power plants, manage distributed energy resources and electric vehicle charging, trade electricity in wholesale markets, explore for oil, gas, and minerals by analysing seismic and geological data, and detect methane leaks and illegal activities through satellite imagery. At the same time, AI is becoming a major source of **energy demand**: the rapid expansion of data centres to train and run AI models has increased electricity demand in several countries, prompting concerns about grid capacity, emissions, and water consumption, and leading technology companies to contract for new nuclear, gas, and renewable capacity. The IEA has estimated that electricity consumption by data centres could roughly double by 2030, with AI the most important driver.

Legal Questions Raised by AI in Energy

The use of AI in energy systems raises several legal questions. **Liability**: who is liable when an AI system causes harm — for example, a blackout caused by an erroneous algorithmic dispatch decision, or a trading loss caused by an autonomous trading algorithm? Existing liability rules, based on fault and causation, may be difficult to apply to opaque, adaptive systems; the European Union's revised **Product Liability Directive** (Directive (EU) 2024/2853) extends product liability to software, including AI systems. **Market conduct**: algorithmic trading in energy markets raises risks of market manipulation and **tacit algorithmic collusion**, in which pricing algorithms learn to coordinate on supra-competitive prices without explicit agreement, challenging competition law's reliance on proof of agreement. The EU's **REMIT** regime, revised in 2024, and FERC's anti-manipulation rules apply to algorithmic trading, but their enforcement against AI-driven strategies is untested. **Safety and reliability**: AI systems controlling critical infrastructure must be safe, robust, and secure against cyber attack, raising questions of certification and standards. **Transparency and accountability**: the opacity of complex AI models ("black boxes") may make it difficult for regulators and affected persons to understand and challenge decisions. **Discrimination and fairness**: AI systems used in customer management — for example, to assess creditworthiness, target disconnections, or set personalised tariffs — may produce discriminatory outcomes.

Regulating AI

The European Union's **Artificial Intelligence Act** (Regulation (EU) 2024/1689), the first comprehensive legal framework for AI, adopts a risk-based

approach, imposing strict requirements on **high-risk** AI systems, which include AI systems intended to be used as safety components in the management and operation of critical infrastructure, including the supply of electricity, gas, and heating. Providers of such systems must implement risk management systems, ensure data quality, maintain technical documentation and logs, ensure transparency and human oversight, and meet standards of accuracy, robustness, and cyber security, with obligations phased in from 2025 to 2027. Other jurisdictions have adopted different approaches: the United States has relied principally on sectoral regulation and executive policy, which shifted in 2025 towards promoting AI development and infrastructure, including the acceleration of energy supply for data centres; China has adopted regulations on algorithmic recommendations, deep synthesis, and generative AI; and the **Council of Europe Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law** (2024) is the first international treaty on AI. In Africa, the African Union adopted a **Continental Artificial Intelligence Strategy** in 2024, and Nigeria has developed a National AI Strategy.

AI in Regulation: "RegTech" and Automated Regulation

AI is also being used **by regulators**: to monitor compliance through the analysis of large datasets, satellite imagery, and smart meter data; to detect market manipulation; to process licence applications; and to draft and analyse regulations. Such "**RegTech**" and "**SupTech**" applications can enhance the effectiveness and efficiency of regulation, particularly for under-resourced regulators. They also raise concerns about **automated decision-making** in public administration: the requirements of administrative law — that decisions be made by the authorised decision-maker, on relevant considerations, with reasons, and subject to review — may be difficult to satisfy where decisions are made or heavily influenced by opaque algorithms. Data protection law, including Article 22 of the EU's General Data Protection Regulation, which gives individuals a right not to be subject to decisions based solely on automated processing that produce legal or similarly significant effects, and Nigeria's Data Protection Act 2023, which contains a similar provision, impose limits. The idea of "**regulation by code**", in which compliance is built into the technical architecture of energy systems — for example, through smart contracts that automatically settle peer-to-peer energy trades in accordance with market rules — raises deeper questions about the relationship between law and technology, which Lawrence Lessig anticipated in his observation that "code is law".

30.4 Digitalisation, Data Ownership, and Energy Markets

The Digital Energy System

Energy systems are being transformed by **digitalisation** — the deployment of smart meters, sensors, communication networks, and data platforms that enable the real-time monitoring and control of energy flows. Digitalisation enables **decentralisation**: households and businesses become **prosumers** who generate electricity from rooftop solar, store it in batteries, and sell it to the grid or to neighbours; electric vehicles become mobile storage devices; and **aggregators** combine the flexibility of many small resources into **virtual power plants** that can participate in electricity markets. It enables new market models, including **peer-to-peer trading**, **local energy markets**, **energy communities**, and dynamic tariffs that reflect real-time prices. In Africa, digital technologies — including mobile money and **pay-as-you-go** (PAYG) business models, in which customers pay for solar home systems in small instalments through mobile phones, with remote disabling of systems for non-payment — have enabled the rapid expansion of off-grid solar access, as examined in Chapter 23.

Legal Questions of Digital Energy Markets

Digitalisation raises legal questions that existing energy law, designed for centralised systems, does not adequately address. **Market access**: can prosumers, aggregators, and energy communities participate in markets on equal terms with incumbent generators? The European Union's Electricity Directive (Directive (EU) 2019/944) gives final customers the right to act as **active customers**, to participate in **citizen energy communities**, and to contract with independent aggregators, and requires member states to remove barriers to demand response; in the United States, **FERC Order No. 2222** (2020), examined in Chapter 24, requires regional markets to allow aggregations of distributed energy resources to participate. **Network tariffs**: as prosumers generate more of their own electricity, the allocation of fixed network costs becomes contested, since customers who self-generate may pay less towards networks they continue to rely on, shifting costs to others — a debate reflected in the reform of net metering schemes in several US states and elsewhere. **The role of distribution system operators**, which must evolve from passive network operators to active managers of distributed resources, raising questions about their neutrality and unbundling. **Consumer protection**: dynamic tariffs, automated control of appliances, and PAYG remote disabling raise issues of transparency, fairness, and the protection of vulnerable consumers, including in relation to disconnection.

Data Ownership and Access

Digital energy systems generate vast quantities of **data** — on consumption patterns, generation, network conditions, and market transactions — whose

ownership, control, and use raise fundamental legal questions. Smart meter data can reveal intimate details of household life, such as when occupants are at home, asleep, or away, raising **privacy** concerns addressed by data protection law, including the EU's GDPR and Nigeria's **Data Protection Act 2023**. Data are also economically valuable: access to consumption data enables new services, such as energy efficiency advice and flexibility aggregation, and control over data may confer market power. The law generally does not recognise **property** in data as such; instead, data are governed by a combination of data protection law, contract, intellectual property (including database rights), competition law, and sector-specific rules on data access. The EU's Electricity Directive requires member states to ensure that final customers can access their consumption data and authorise third parties to access it, and the **EU Data Act** (Regulation (EU) 2023/2854), applicable from September 2025, gives users of connected products, including smart energy devices, rights to access and share the data they generate. The concept of **data spaces** — governed frameworks for data sharing — is being developed for the energy sector in the European Union's **Common European Energy Data Space**.

In developing countries, questions of data governance intersect with concerns about **data sovereignty** and the control of data by foreign technology companies. Data generated by PAYG solar systems and smart meters in Africa are often held by foreign companies, raising questions about who benefits from their value and how they may be used. The African Union's **Data Policy Framework** (2022) and national data protection laws seek to balance the free flow of data with sovereignty and protection. The governance of **geological and resource data** — the seismic and geological information on which resource exploration depends — is also significant: under most petroleum and mining laws, such data are the property of the state, and their management through national data repositories, such as Nigeria's National Data Repository administered by the NUPRC, is an important function of resource governance.

Cyber Security

Digitalisation increases the exposure of energy systems to **cyber attacks**, as examined in Chapter 25. The proliferation of connected devices — smart meters, inverters, electric vehicle chargers, and home energy management systems — multiplies the points of vulnerability, and concerns have been raised about the security of devices manufactured in states regarded as strategic competitors, leading some jurisdictions to restrict the use of certain foreign components in critical energy infrastructure. Legal responses include cyber security obligations for operators of essential services (such as under the EU's NIS2 Directive), security standards for connected devices (such as the EU's **Cyber Resilience Act**, Regulation (EU) 2024/2847, which imposes cyber security requirements on products with digital

elements), network codes on cyber security for cross-border electricity flows, and the screening of foreign suppliers.

Blockchain and Smart Contracts

Distributed ledger technologies, including blockchain, have been proposed for applications in energy markets, including peer-to-peer trading, the tracking of renewable energy certificates and guarantees of origin, the traceability of critical minerals through supply chains, and the settlement of transactions through **smart contracts** — self-executing code that automatically performs contractual obligations when specified conditions are met. Legal questions include the enforceability of smart contracts and their relationship with traditional contract law — the Law Commission of England and Wales concluded in 2021 that the existing law is capable of accommodating smart legal contracts — the allocation of liability for errors in code, the application of financial and energy regulation to tokenised energy assets, and the governance of decentralised platforms. Many early blockchain energy projects have not progressed beyond pilots, but the technology's potential for supply chain **traceability**, for example in battery passports required under the EU Battery Regulation, may prove more durable.

30.5 Climate-Related Financial and Legal Risks

Climate Change as Financial Risk

Climate change is now widely recognised as a source of **financial risk**. Following the influential speech by Mark Carney, then Governor of the Bank of England, on the "tragedy of the horizon" (2015), financial regulators have identified two principal categories of climate-related financial risk: **physical risks**, arising from the physical effects of climate change, such as damage to assets from extreme weather and chronic changes in temperature and sea levels; and **transition risks**, arising from the process of adjustment to a low-carbon economy, including changes in policy, technology, and market preferences that may reduce the value of carbon-intensive assets. A third category, **liability risks**, arises from claims against companies and their directors and insurers for contributing to or failing to manage climate change, examined in Chapters 18, 22, and 27. Energy companies are particularly exposed to transition risks, as the value of fossil fuel reserves and infrastructure depends on assumptions about future demand and prices that decarbonisation may undermine.

Disclosure

The first legal response to climate-related financial risk has been **disclosure**. The **Task Force on Climate-related Financial Disclosures** (TCFD), established by the Financial Stability Board in 2015, published recommendations in 2017 for disclosure of climate-related risks and opportunities across four pillars: gover-

nance, strategy, risk management, and metrics and targets, including scenario analysis. The TCFD recommendations have been incorporated into the **IFRS S1 and S2** sustainability disclosure standards issued by the **International Sustainability Standards Board** (ISSB) in June 2023, which a growing number of jurisdictions are adopting. Nigeria was among the early adopters, with the **Financial Reporting Council of Nigeria** announcing a roadmap for the adoption of the ISSB standards, with voluntary adoption from 2024 and mandatory adoption for public interest entities thereafter. The European Union's **Corporate Sustainability Reporting Directive** (Directive (EU) 2022/2464) requires extensive sustainability reporting under the European Sustainability Reporting Standards, although its scope and timetable were reduced by the 2025 Omnibus simplification package. In the United States, the Securities and Exchange Commission adopted climate disclosure rules in March 2024, which were stayed pending litigation and which the Commission voted in 2025 to stop defending, while California enacted climate disclosure laws applicable to large companies doing business in the state. Disclosure aims to enable investors to price climate risk and to encourage companies to manage it, but its effectiveness depends on the quality and comparability of disclosures and the response of markets.

Prudential Regulation

Central banks and financial supervisors have begun to integrate climate risk into **prudential regulation**. The **Network of Central Banks and Supervisors for Greening the Financial System** (NGFS), established in 2017 and comprising well over one hundred members, has developed **climate scenarios** used by supervisors and financial institutions to assess climate risk, and guidance on integrating climate risk into supervision. Supervisors in the United Kingdom, the European Union, and elsewhere have conducted **climate stress tests** of banks and insurers, and the **Basel Committee on Banking Supervision** issued principles for the effective management and supervision of climate-related financial risks in 2022. These measures may affect the cost and availability of finance for fossil fuel projects, particularly if supervisors impose capital requirements reflecting climate risk. The **Central Bank of Nigeria** has issued Nigerian Sustainable Banking Principles and has engaged with the NGFS, and African central banks are increasingly attentive to climate risk, though concerns have been raised that climate-related financial regulation in developed countries could raise the cost of capital for energy projects in developing countries.

Sustainable Finance Taxonomies

Sustainable finance taxonomies classify economic activities according to their environmental sustainability, to guide investment and prevent greenwashing. The **EU Taxonomy Regulation** (Regulation (EU) 2020/852) establishes criteria for

activities that contribute substantially to environmental objectives without doing significant harm to others; its inclusion of certain gas and nuclear activities as transitional activities in 2022 was controversial and was challenged by Austria before the EU General Court, which dismissed the challenge in 2025. Taxonomies have also been adopted or developed by China, South Africa (whose Green Finance Taxonomy was published in 2022), and other jurisdictions, and the interoperability of taxonomies is a concern for cross-border finance. The design of taxonomies has significant implications for the financing of gas, nuclear, and transition activities in developing countries.

Directors' Duties and Fiduciary Duties

Climate risk also raises questions about the **duties of company directors** and **institutional investors**. Legal opinions in several jurisdictions, including Australia (the Hutley opinions, from 2016) and the United Kingdom, have concluded that directors who fail to consider foreseeable climate-related risks may breach their duties of care and diligence. The *ClientEarth v Shell's Directors* case (2023), examined in Chapter 22, illustrated the difficulty of enforcing such duties through derivative actions. For pension funds and other institutional investors, the question whether **fiduciary duties** permit or require the consideration of climate risk has been contested, particularly in the United States, where some states have enacted laws restricting the consideration of environmental, social, and governance (ESG) factors in public investment, while others require it. The legal treatment of climate risk in fiduciary duties is likely to remain a significant and politically contested area.

30.6 Emerging International Disputes and Regulatory Conflicts

The Shifting Landscape of Energy Disputes

The categories of energy disputes examined in Chapter 26 — price reviews, expropriations, fiscal disputes, construction claims — will persist, but the forces examined in this chapter are generating new categories of disputes and regulatory conflicts. Anticipating these disputes is important for practitioners advising clients, for states designing laws and treaties, and for scholars analysing the evolution of the field.

Transition-Related Investment Disputes

Disputes arising from climate and transition measures are likely to increase. Investors in fossil fuel assets may bring claims against states that phase out coal, ban new oil and gas exploration, revoke licences, or impose windfall taxes, as in the claims under the Energy Charter Treaty and investment treaties examined in Chapters 14 and 26, including *Rockhopper v Italy* (2022), in which a tribunal

awarded compensation for the refusal of an offshore production concession following a ban on drilling near the coast, an award subsequently annulled in 2025. Investors in renewable energy may bring claims when states reduce incentives, as in the Spanish cases. States may increasingly invoke their climate obligations, clarified by the ICJ's 2025 advisory opinion, as defences, and the interpretation of investment treaties in the light of the Paris Agreement and the right to regulate will be contested. The withdrawal of the EU and several member states from the Energy Charter Treaty, and the modernisation of its investment provisions to allow the exclusion of fossil fuel investments, examined in Chapter 14, will reduce but not eliminate such claims, given the treaty's twenty-year sunset clause.

Trade Disputes over Green Industrial Policy

Trade disputes over green industrial policies and carbon border measures are likely to multiply. The EU's **Carbon Border Adjustment Mechanism** (Regulation (EU) 2023/956), whose definitive period began in January 2026, examined in Chapter 13, has been criticised by India, China, South Africa, and other states as protectionist and inconsistent with the principle of common but differentiated responsibilities, and its consistency with WTO law may be tested. The local content and subsidy provisions of the US **Inflation Reduction Act**, partially curtailed in 2025, China's support for its clean energy industries, and the countervailing and anti-dumping duties imposed by the United States, the European Union, and others on Chinese solar panels, batteries, and electric vehicles — including the EU's countervailing duties on Chinese electric vehicles imposed in 2024, which China challenged at the WTO — illustrate the tension between the WTO's rules on subsidies and non-discrimination and states' pursuit of green industrialisation. **Export restrictions** on critical minerals, examined in Chapter 25, may also generate disputes, as the EU's challenge to Indonesia's nickel export ban illustrates. The paralysis of the WTO Appellate Body and the rise of national security justifications complicate the resolution of these disputes.

Critical Minerals Disputes

The scramble for **critical minerals** is generating disputes between investors and host states over resource nationalism, licence revocations, and renegotiations, as in the disputes arising from Mali's 2023 mining code and its conflict with Barrick over the Loulo-Gounkoto complex, which led to the suspension of operations, the detention of employees, and ICSID proceedings before a settlement in late 2025; Panama's closure of the Cobre Panamá copper mine following the Supreme Court's declaration in November 2023 that the law approving the concession contract was unconstitutional, which led to arbitration claims by First Quantum; and the review of mining contracts in Niger, Burkina Faso, and the DRC. Disputes between consumer states over **supply chain** measures, including export

controls and investment screening, and disputes concerning **deep seabed mining** under UNCLOS are also likely.

Climate Liability and Loss and Damage

Climate liability disputes — claims against fossil fuel companies for climate damages, against states for inadequate climate action, and potentially between states for transboundary climate harm — are likely to grow, building on the jurisprudence examined in Chapters 18 and 27, the ICJ advisory opinion, and the advisory opinions of the **International Tribunal for the Law of the Sea** (May 2024), which held that anthropogenic greenhouse gas emissions constitute pollution of the marine environment under UNCLOS, and of the **Inter-American Court of Human Rights** (July 2025), which recognised a right to a healthy climate. The operationalisation of the **Fund for responding to Loss and Damage**, established at COP27 and COP28, may provide a non-adversarial channel for addressing climate harms, but will not eliminate litigation. **Inter-state** climate litigation, long discussed, may become a reality, as small island states and others explore claims against major emitters in the light of the ICJ's clarification of the law.

Disputes over Digital Energy and AI

Disputes concerning **data, AI, and digital energy systems** are emerging: disputes over access to energy data, liability for algorithmic failures, cyber incidents and their insurance coverage, intellectual property in energy technologies, and the treatment of digital assets in energy markets. Disputes may also arise over the **electricity supply** of data centres, including the allocation of grid capacity and the costs of network reinforcement between data centre operators and other consumers, and over co-location arrangements in which data centres are connected directly to power plants, an issue addressed by FERC in proceedings concerning co-located load in 2024–2025.

Regulatory Conflicts in a Fragmented Order

Beyond formal disputes, the coming decades are likely to see increasing **regulatory conflicts** — clashes between the regulatory requirements of different jurisdictions that impose inconsistent obligations on companies operating across borders. Examples include conflicts between sanctions regimes and blocking statutes; between the extraterritorial due diligence obligations of EU law and the laws of producer states; between ESG requirements in some US states and anti-ESG laws in others; between data localisation requirements and cross-border data flows; and between the EU's carbon border measures and the carbon pricing policies of trading partners. Managing such conflicts will require international cooperation, mutual recognition, and the development of principles of **comity**

and **interoperability**, and will place demands on lawyers capable of navigating multiple legal orders.

30.7 Unresolved Questions in Global Resource Governance

The Enduring Questions

Despite a century of development, international energy and natural resources law leaves fundamental questions unresolved. Some are perennial; others are new; all will shape the field's future. This section identifies a selection of these questions, not to answer them, but to frame them for the research agenda that follows.

Sovereignty and Global Commons

1. **Sovereignty and the climate.** How should the permanent sovereignty of states over their natural resources be reconciled with the global imperative to limit the extraction and combustion of fossil fuels? Does the principle of permanent sovereignty now carry duties as well as rights, including duties to use resources in a manner consistent with the protection of the climate system? The ICJ's 2025 advisory opinion suggests that it does, but the content of these duties and their relationship with the right to development remain contested.
2. **Common heritage beyond national jurisdiction.** How should the resources of areas beyond national jurisdiction — the deep seabed, the high seas, outer space, and perhaps the Arctic and Antarctic — be governed? Will the common heritage principle survive unilateral challenges, and how will the benefits of these resources be shared?

Equity and Justice

3. **Equity in the transition.** How should the burdens and benefits of the energy transition be shared among states — between historical emitters and developing countries, between fossil fuel producers and consumers, and between mineral-producing and technology-producing states? What does "common but differentiated responsibilities and respective capabilities, in the light of different national circumstances" require in relation to fossil fuel production and transition minerals?
4. **Communities and benefit-sharing.** How can the rights of indigenous peoples and local communities, including free, prior, and informed consent, be given effect in the development of transition minerals and renewable energy projects, which increasingly affect their lands? Is there an emerging right to share in the benefits of resource development?
5. **Energy access as a right.** Is there, or should there be, a human right to energy, and what would it require of states and companies?

Investment and Regulation

6. **Investment protection and the right to regulate.** How should investment law balance the protection of investors against arbitrary state conduct with states' freedom to pursue climate and transition policies? Will investor–state arbitration be reformed, replaced, or abandoned?
7. **Stability and change.** How can legal frameworks provide the stability investors need while adapting to rapid technological, market, and climate change? Are stabilisation clauses compatible with the energy transition?

Institutions and Order

8. **The architecture of governance.** Will global energy governance converge towards more inclusive multilateral institutions, or fragment further into competing blocs and clubs? What role should international law play in a multipolar energy order?
9. **Corporate accountability.** How should transnational corporations in the energy and mining sectors be held accountable for human rights and environmental harms across their value chains? Will a binding international instrument on business and human rights, negotiated under the auspices of the UN Human Rights Council since 2014, be concluded and, if so, with what effect?
10. **Technology and law.** How should law govern the technologies of the transition — AI, digital platforms, hydrogen, carbon removal, nuclear innovation, and perhaps geoengineering — whose risks and benefits are uncertain?

30.8 A Doctoral Research Agenda for the Next Generation

Framing the Agenda

The unresolved questions identified above, and the analysis throughout this book, suggest a research agenda for the next generation of doctoral scholars in international energy and natural resources law. The agenda below is organised by theme and is intended to stimulate, not to limit, the formulation of original research questions. Each chapter of this book concludes with specific doctoral research questions, and Appendix T provides further research topics and emerging scholarly debates.

Themes for Doctoral Research

1. The law of managed decline and just transition. Research is needed on the legal frameworks for phasing out fossil fuel production and use, including restrictions on new licensing, decommissioning, compensation, and just transition measures for workers and communities, with particular attention to producer

economies in Africa. How can legal frameworks enable producer states such as Nigeria to diversify their economies while managing declining revenues?

2. Transition minerals and resource sovereignty. The governance of critical minerals offers a fertile field for research on value addition, export restrictions, fiscal regimes, community rights, environmental regulation, supply chain due diligence, and the interaction of producer and consumer state regulation. Comparative studies of African mining codes, empirical studies of their implementation, and analyses of the compatibility of resource nationalist measures with trade and investment law are needed.

3. Investment law and the climate. Research on the reform of investment treaties, the interpretation of investment obligations in the light of climate obligations, the treatment of transition measures in arbitration, and alternatives to investor–state arbitration remains urgent.

4. Energy access and energy justice in Africa. Legal research on frameworks for universal energy access — mini-grids, off-grid solar, clean cooking, tariff design, subsidy reform, and the decentralisation of electricity regulation, as under Nigeria's Electricity Act 2023 — can make a direct contribution to development.

5. The implementation of Nigerian energy reforms. The Petroleum Industry Act 2021, the Electricity Act 2023, the Climate Change Act 2021, and the tax reforms of 2025 require doctrinal analysis, empirical evaluation, and comparative assessment, including studies of host community development trusts, the commercialisation of NNPC Limited, the new regulators, and state electricity markets.

6. Digital energy law. The legal implications of AI, data, platforms, and cyber security for energy systems are an emerging field, in which doctoral research can help shape nascent regulation.

7. Climate liability and litigation. Research on attribution, causation, corporate and state liability, remedies, and the role of courts in climate governance will remain central, as will research on the ICJ advisory opinion and its implementation.

8. Resources beyond national jurisdiction. The governance of deep seabed minerals, the implementation of the BBNJ Agreement, the Arctic, and outer space resources raise questions of common heritage, environmental protection, and benefit-sharing.

9. Regional integration and African energy law. The AfCFTA and its investment protocol, regional power pools, the African Single Electricity Market, ECOWAS energy instruments, and regional approaches to minerals and climate offer opportunities for research on regional integration as a tool of energy governance.

10. Theory and method. The field needs theoretical work that explains and evaluates its evolution — drawing on TWAIL, energy justice, political economy, regulatory theory, and transnational law — and methodological innovation, including empirical and interdisciplinary research.

The Responsibility of the Scholar

The next generation of scholars in energy and natural resources law will work at the intersection of some of the most consequential questions of the century: how to power the world's growing population while protecting the climate; how to govern the resources on which the transition depends; how to share the burdens and benefits of change fairly; and how to maintain a rules-based international order in a fragmented world. Their research will be read by courts, legislatures, regulators, companies, and communities. This places on them a responsibility to conduct research of the highest rigour and integrity, to engage with the realities of the sector and the experiences of those affected by it, to speak truthfully about evidence and uncertainty, and to bring the insights of scholarship to bear on the design of better laws and institutions.

::: casebox {title="Case Study: The ICJ Advisory Opinion on Obligations of States in respect of Climate Change (2025) and its Implications for Resource Law"} **Background.** In March 2023, the UN General Assembly, by consensus, adopted Resolution 77/276, initiated by Vanuatu and supported by a large group of states, requesting an advisory opinion from the International Court of Justice on the obligations of states under international law to ensure the protection of the climate system and other parts of the environment from anthropogenic greenhouse gas emissions, and the legal consequences for states that, by their acts and omissions, have caused significant harm to the climate system. The proceedings attracted an unprecedented number of written and oral submissions from states and international organisations.

The opinion. In its unanimous opinion delivered on **23 July 2025**, the Court held, among other things, that states have obligations under the climate change treaties, including to prepare and pursue nationally determined contributions that collectively are capable of achieving the 1.5°C temperature goal, which the Court identified as the primary goal under the Paris Agreement; that states have obligations under customary international law, including the duty to prevent significant harm to the environment, which requires due diligence of a stringent standard in the context of climate change, and the duty to cooperate; that these obligations are informed by human rights law, the protection of the climate system being a precondition for the enjoyment of human rights; and that a breach of these obligations constitutes an internationally wrongful act entailing the full range of legal consequences, including cessation, guarantees of non-repetition, and full reparation, subject to the establishment of a sufficiently direct and certain

causal nexus. The Court stated that the failure of a state to take appropriate action to protect the climate system, including through fossil fuel production, fossil fuel consumption, the granting of fossil fuel exploration licences, or the provision of fossil fuel subsidies, may constitute an internationally wrongful act attributable to that state.

Implications for resource law. Although advisory opinions are not binding, the opinion carries great authority and is likely to influence national courts, investment tribunals, and law-making. For energy and natural resources law, it raises fundamental questions: how the permanent sovereignty of states over their resources is constrained by climate obligations; whether and how states must assess the climate effects of licensing decisions, a question already addressed in national courts in *Finch* and related cases examined in Chapter 27; how the principle of common but differentiated responsibilities applies to developing country producers; whether fossil fuel subsidies are compatible with international law; how investment tribunals should treat climate measures; and what reparation may be owed by major emitters. The opinion thus marks a potential turning point in the relationship between climate law and resource law, whose consequences will be worked out over the coming decades. :::

::: exercise {title="Practical Exercise 30: The Future Energy Law Commission (HYPOTHETICAL Simulation)"} **This exercise is entirely hypothetical. The commission, states, and scenarios described are fictitious; where the exercise refers to real instruments, participants must use accurate sources.**

The (fictitious) Pan-African Future Energy Law Commission has been established by a group of African states to recommend a legal agenda for the period 2030–2050. Participants are divided into five working groups, each representing a fictitious stakeholder:

- **Group A** represents the Republic of Okanta, an oil and gas producer whose government derives seventy per cent of its revenue from hydrocarbons and which has recently discovered a large offshore gas field;
- **Group B** represents the Federal Republic of Mavuno, a major producer of cobalt and copper seeking to develop domestic battery precursor manufacturing;
- **Group C** represents the Coastal States Alliance, a group of small coastal and island states highly vulnerable to sea level rise;
- **Group D** represents an association of independent power producers, mini-grid developers, and technology companies;
- **Group E** represents a coalition of host community organisations, labour unions, and environmental NGOs.

Tasks.

(a) Each group prepares a position paper (of no more than 2,500 words) setting out its priorities for the legal agenda, addressing at least four of the forces examined in this chapter, and grounding its positions in relevant international law, including the ICJ's 2025 advisory opinion, the Paris Agreement, investment and trade law, human rights law, and the law of the sea.

(b) The Commission convenes in plenary to negotiate a draft **Declaration of Principles on the Future of African Energy and Resources Law**, comprising no more than fifteen principles. Each group must seek to include its priorities while building consensus.

(c) Following the negotiation, each participant writes an individual reflective essay (of no more than 2,000 words) evaluating the legal, political, and ethical tensions revealed by the exercise, and identifying one doctoral research question arising from it. :::

Conclusion to the Book

International energy and natural resources law stands at a turning point. The legal order that governed the extraction and trade of fossil fuels for a century — built on concessions and contracts, permanent sovereignty and national oil companies, investment treaties and arbitration, and liberalised markets — is being transformed by the imperatives of decarbonisation, the rise of critical minerals, the digitalisation and decentralisation of energy systems, the recognition of climate risk, the claims of justice and human rights, and the fragmentation of the international order. The law that emerges will shape whether the world can meet its climate goals, whether resource-rich developing countries — including Nigeria and its African neighbours — can convert their resources into broad-based prosperity, whether the hundreds of millions of people without modern energy gain access to it, and whether the communities on the frontlines of resource development share in its benefits rather than bearing its burdens.

This book has sought to equip its readers with the knowledge, analytical skills, and methodological tools to engage with these challenges at the highest level: an understanding of the jurisprudential and institutional foundations of the field; command of the law governing petroleum, gas, mining, electricity, renewable energy, and emerging technologies; mastery of the international trade, investment, and contractual frameworks; a grounding in environmental, climate, and human rights law; comparative and geopolitical perspective; familiarity with dispute resolution; and the methods of doctoral research and law reform. Throughout, it has emphasised the importance of rigour, integrity, and attention to context — and, in particular, to the perspectives and experiences of Africa, whose resources and people will be central to the energy future.

The questions that remain unresolved are many and profound. Answering them will be the work of a generation. It is the hope of this book that its readers will be among those who answer them — as scholars, practitioners, regulators, judges, legislators, and leaders — and that they will do so with the knowledge, wisdom, and integrity the task demands.

CHAPTER SUMMARY

- Global energy governance is a fragmented regime complex being reshaped by the energy transition, the shift of demand to emerging economies, geopolitical fragmentation, the climate regime's engagement with energy, and the rise of clubs and minilateralism; Africa's role is pivotal but under-represented.
- Decarbonisation is shifting resource law from a paradigm of scarcity to one of stranding and managed decline, raising questions of licensing restrictions, phase-outs, compensation, decommissioning, and just transition, while creating demand for transition minerals, contests over deep seabed mining, and new resources such as storage space and natural hydrogen.
- AI transforms energy systems and drives demand for electricity, raising questions of liability, market conduct, safety, transparency, and fairness, addressed by instruments such as the EU AI Act; AI in regulation raises questions of automated decision-making.
- Digitalisation enables prosumers, aggregators, energy communities, and new market models, raising questions of market access, network tariffs, consumer protection, data ownership and access, data sovereignty, cyber security, and smart contracts.
- Climate change is a source of physical, transition, and liability risks, addressed through disclosure (TCFD, ISSB, CSRD), prudential regulation (NGFS, stress tests), taxonomies, and directors' and fiduciary duties.
- Emerging disputes include transition-related investment claims, trade disputes over green industrial policy and carbon border measures, critical minerals disputes, climate liability claims, disputes over digital energy and AI, and regulatory conflicts in a fragmented order.
- Unresolved questions concern sovereignty and the climate, the common heritage, equity in the transition, community rights, a right to energy, investment protection and the right to regulate, stability and change, the architecture of governance, corporate accountability, and the governance of technology.
- A doctoral research agenda for the next generation encompasses managed decline and just transition, transition minerals, investment law and climate, energy access and justice in Africa, Nigerian energy reforms, digital energy law, climate liability, resources beyond national jurisdiction, regional integration, and theory and method.

DOCTORAL RESEARCH QUESTIONS

1. Does the principle of permanent sovereignty over natural resources now entail duties to limit fossil fuel extraction? Evaluate in the light of the ICJ's 2025 advisory opinion and the right to development.
2. Design a legal framework for the managed decline of oil and gas production in a developing producer economy, addressing licensing, fiscal, decommissioning, and just transition dimensions.
3. Can the governance of transition minerals avoid the resource curse? Evaluate the legal strategies of African producer states and the role of consumer state regulation.
4. How should liability for harm caused by AI systems used in the operation of electricity networks be allocated? Propose a legal framework.
5. Evaluate the legality and legitimacy of unilateral deep seabed mining outside the framework of the International Seabed Authority.
6. To what extent should prudential regulation of climate-related financial risk take account of the development needs of energy-poor countries?

FURTHER READING

- International Energy Agency, *World Energy Outlook* (IEA, annual).
- International Energy Agency, *The Role of Critical Minerals in Clean Energy Transitions* (IEA 2021), and subsequent *Global Critical Minerals Outlook* reports.
- Thijs Van de Graaf and Benjamin K Sovacool, *Global Energy Politics* (Polity 2020).
- Benjamin K Sovacool and Michael H Dworkin, *Global Energy Justice: Problems, Principles, and Practices* (Cambridge University Press 2014).
- Raphael J Heffron, *Energy Law: An Introduction* (Springer, latest edition).
- Network for Greening the Financial System, *NGFS Climate Scenarios* (NGFS, periodically updated).
- International Court of Justice, *Obligations of States in respect of Climate Change*, Advisory Opinion of 23 July 2025.

APPENDIX A

Selected International Energy and Natural Resources Treaties

This appendix lists the principal treaties and other international instruments examined in this book, with their dates of adoption and entry into force and a short note of their relevance to energy and natural resources. It is a finding aid, not a substitute for the authoritative texts, which are available from the depositaries (principally the United Nations Treaty Collection and the depositary organisations named). Dates of entry into force refer to the instrument's general entry into force; readers should check the status of ratification of each instrument for any particular state, together with any reservations, declarations, amendments, or withdrawals.

General International Law and Resource Sovereignty

Table A.1 General international law and resource sovereignty instruments.

Instrument	Adopted	In force	Relevance
Charter of the United Nations	1945	1945	Sovereign equality; peaceful settlement; foundation of the UN system
UNGA Resolution 1803 (XVII), Permanent Sovereignty over Natural Resources	1962	— (resolution)	Classic statement of permanent sovereignty; compensation on expropriation
UNGA Resolution 3281 (XXIX), Charter of Economic Rights and Duties of States	1974	— (resolution)	New International Economic Order; contested standard of compensation
Vienna Convention on the Law of Treaties	1969	1980	Treaty interpretation (arts 31–33); termination and withdrawal
International Covenant on Economic, Social and Cultural Rights; International Covenant on Civil and Political Rights	1966	1976	Common article 1(2): peoples' free disposal of natural wealth and resources
African Charter on Human and Peoples' Rights	1981	1986	Article 21 (free disposal of wealth and resources); article 24 (satisfactory environment)

Instrument	Adopted	In force	Relevance
ILO Convention No 169 on Indigenous and Tribal Peoples	1989	1991	Consultation and participation regarding resource exploitation

Investment, Trade, and Dispute Settlement

Table A.2 Investment, trade, and dispute settlement instruments.

Instrument	Adopted	In force	Relevance
Convention on the Recognition and Enforcement of Foreign Arbitral Awards (New York Convention)	1958	1959	Enforcement of commercial arbitral awards
Convention on the Settlement of Investment Disputes between States and Nationals of Other States (ICSID Convention)	1965	1966	Investor–state arbitration; self-contained annulment
Energy Charter Treaty	1994	1998	Energy investment protection, trade, and transit; withdrawal by the EU, Euratom, and several member states; modernisation adopted 2024
Marrakesh Agreement Establishing the WTO (incorporating GATT 1994, SCM Agreement, TBT Agreement, and others)	1994	1995	Trade disciplines on energy goods, subsidies, export restrictions, and standards
UN Convention on Transparency in Treaty-based Investor-State Arbitration (Mauritius Convention)	2014	2017	Transparency in investor–state arbitration
UN Convention on International Settlement Agreements Resulting from Mediation (Singapore Convention)	2019	2020	Enforcement of mediated settlements
Agreement Establishing the African Continental Free Trade Area	2018	2019	Continental market; protocols on investment and competition
ECOWAS Revised Treaty	1993	1995	Regional integration in West Africa, including cooperation in energy
ECOWAS Energy Protocol (A/P4/1/03)	2003	—	Regional energy investment and trade framework modelled on the ECT (check current status with ECOWAS)

Environment and Climate

Table A.3 Environmental and climate instruments.

Instrument	Adopted	In force	Relevance
Vienna Convention for the Protection of the Ozone Layer; Montreal Protocol	1985; 1987	1988; 1989	Phase-out of ozone-depleting substances; Kigali Amendment (2016, in force 2019) on HFCs
Basel Convention on Transboundary Movements of Hazardous Wastes	1989	1992	Waste from energy and mining activities
Bamako Convention	1991	1998	African regime on hazardous waste imports
Espoo Convention on EIA in a Transboundary Context	1991	1997	Transboundary environmental impact assessment
UN Framework Convention on Climate Change	1992	1994	Framework of climate regime; CBDR-RC
Convention on Biological Diversity	1992	1993	Biodiversity conservation; Kunming-Montreal Global Biodiversity Framework (2022)
UNECE Water Convention	1992	1996	Transboundary watercourses; opened globally in 2016
Kyoto Protocol	1997	2005	Quantified emission commitments for developed countries; flexible mechanisms
UN Watercourses Convention	1997	2014	Equitable and reasonable utilisation; no significant harm
Aarhus Convention	1998	2001	Access to information, participation, and justice in environmental matters
Stockholm Convention on Persistent Organic Pollutants	2001	2004	Chemicals including those used in energy equipment
Minamata Convention on Mercury	2013	2017	Mercury in artisanal gold mining and coal combustion
Paris Agreement	2015	2016	Temperature goals; NDCs; article 6 markets; transparency
Escazú Agreement	2018	2021	Environmental democracy and environmental defenders in Latin America and the Caribbean

Law of the Sea and Marine Pollution

Table A.4 Law of the sea and marine pollution instruments.

Instrument	Adopted	In force	Relevance
International Convention for the Prevention of Pollution from Ships (MARPOL)	1973/1978	1983	Operational pollution from ships, including tankers

Instrument	Adopted	In force	Relevance
UN Convention on the Law of the Sea	1982	1994	Maritime zones; continental shelf; the Area; marine environment
Agreement relating to the Implementation of Part XI of UNCLOS	1994	1996	Deep seabed mining regime
International Convention on Civil Liability for Oil Pollution Damage (1992 Protocol) and Fund Convention (1992 Protocol)	1992	1996	Tanker oil pollution liability and compensation
International Convention on Civil Liability for Bunker Oil Pollution Damage	2001	2008	Bunker oil pollution
Agreement on Marine Biological Diversity of Areas beyond National Jurisdiction (BBNJ)	2023	2026	Marine genetic resources; area-based management; EIA beyond national jurisdiction

Nuclear Energy

Table A.5 Nuclear energy instruments.

Instrument	Adopted	In force	Relevance
Paris Convention on Third Party Liability in the Field of Nuclear Energy	1960	1968	OECD nuclear liability regime (amended 2004)
Vienna Convention on Civil Liability for Nuclear Damage	1963	1977	Global nuclear liability regime (1997 Protocol)
Treaty on the Non-Proliferation of Nuclear Weapons	1968	1970	Article IV right to peaceful uses; safeguards
Convention on the Physical Protection of Nuclear Material (and 2005 Amendment)	1980	1987 (Amendment 2016)	Nuclear security
Convention on Early Notification of a Nuclear Accident; Assistance Convention	1986	1986–1987	Emergency notification and assistance
Convention on Nuclear Safety	1994	1996	Safety of civil nuclear power plants
Joint Convention on the Safety of Spent Fuel and Radioactive Waste Management	1997	2001	Waste and spent fuel
Convention on Supplementary Compensation for Nuclear Damage	1997	2015	Bridge between liability regimes

Anti-Corruption and Institutions

Table A.6 Anti-corruption and institutional instruments.

Instrument	Adopted	In force	Relevance
Agreement on an International Energy Program (IEA)	1974	1976	Emergency oil stocks and co-ordinated response
OECD Anti-Bribery Convention	1997	1999	Criminalisation of foreign bribery
AU Convention on Preventing and Combating Corruption	2003	2006	Regional anti-corruption obligations
UN Convention against Corruption	2003	2005	Prevention, criminalisation, asset recovery
Statute of the International Renewable Energy Agency	2009	2010	Intergovernmental renewable energy agency

Using Treaties in Research and Practice

When working with any of these instruments, the researcher or practitioner should: (1) obtain the authentic text from the depositary; (2) confirm the status of the relevant state (signature, ratification, accession, succession, withdrawal) and the date from which obligations apply; (3) identify reservations, declarations, and objections; (4) check for amendments, protocols, and decisions of the treaty bodies (such as decisions of the Conference of the Parties); (5) consult the *travaux préparatoires* where relevant under article 32 VCLT; (6) consider how the treaty has been interpreted by courts, tribunals, and treaty bodies; and (7) consider the relationship between the treaty and domestic law, including whether domestication is required — as under section 12 of the Constitution of the Federal Republic of Nigeria 1999, which provides that a treaty does not have the force of law in Nigeria unless enacted into law by the National Assembly.

APPENDIX B

Model Petroleum Exploration and Production Agreement

The following model sets out the structure and key clauses of an exploration and production agreement of the licence-plus-agreement or concession type, with drafting notes. It is an **illustrative teaching model**, prepared for this book. It is not modelled on any particular state's form and must not be used in any transaction without adaptation to the applicable law and professional advice. Bracketed text indicates matters for negotiation or national choice.

Parties and Recitals

THIS AGREEMENT is made on [date] between:

(1) **THE GOVERNMENT OF [STATE]**, acting through the [Ministry responsible for petroleum] (the "**Government**"); [and, where applicable, the national oil company, [NOC]]; and

(2) **[CONTRACTOR COMPANY]**, a company incorporated under the laws of [jurisdiction] (the "**Contractor**").

WHEREAS (A) all petroleum existing in its natural condition in the territory, territorial sea, exclusive economic zone, and continental shelf of [State] is vested in [the State / the people of the State, held in trust by the State] under [constitutional or statutory provision]; (B) the Government wishes to promote the exploration and development of petroleum resources in the Contract Area for the benefit of [the State and its people]; and (C) the Contractor has the financial capability, technical competence, and professional skills necessary to carry out Petroleum Operations.

Drafting note. The recitals should accurately reflect the constitutional or statutory ownership regime (Chapter 3). In Nigeria, for instance, section 44(3) of the Constitution vests the entire property in and control of all minerals, mineral oils, and natural gas in the Government of the Federation, and the Petroleum Industry Act 2021 provides that petroleum is vested in the Government of the Federation for the benefit of Nigerians.

Article 1 — Definitions and Interpretation

Key definitions should include: Affiliate; Appraisal Well; Associated Gas; Barrel; Commercial Discovery; Contract Area; Contract Year; Crude Oil; Decommissioning; Development Plan; Discovery; Effective Date; Exploration Period; Good International Petroleum Industry Practice ("**GIPIP**"); Natural Gas; Non-Associated Gas; Petroleum; Petroleum Operations; Production Period; Work Programme and Budget.

Drafting note. The definition of GIPIP is a frequent source of dispute. A definition referring to "practices, methods, standards, and procedures generally accepted and followed by prudent, diligent, skilled, and experienced operators in the international petroleum industry in similar circumstances" is common; parties may refer to specific standards (such as those of the International Association of Oil and Gas Producers) to reduce uncertainty.

Article 2 — Grant of Rights and Scope

2.1 The Government grants to the Contractor the exclusive right to conduct Petroleum Operations in the Contract Area during the term of this Agreement, subject to the Petroleum Law and the terms of this Agreement.

2.2 The Contractor shall bear all risks and costs of Petroleum Operations. [If no Commercial Discovery is made, the Contractor shall not be entitled to any reimbursement.]

2.3 [Title to petroleum passes to the Contractor at the wellhead / at the Delivery Point, subject to payment of royalty.]

Article 3 — Term, Phases, and Relinquishment

3.1 The Exploration Period shall be [six] years, divided into an Initial Phase of [three] years and [one/two] Extension Phases, each conditional on fulfilment of the Minimum Work Obligations of the preceding phase.

3.2 At the end of each phase, the Contractor shall relinquish [25–50] per cent of the original Contract Area, excluding Development Areas and areas subject to an approved appraisal programme.

3.3 Upon declaration of a Commercial Discovery and approval of a Development Plan, the Contractor shall be entitled to a Production Period of [20–25] years for the relevant Development Area, [extendable by agreement].

Article 4 — Minimum Work Obligations and Guarantees

4.1 The Contractor shall carry out, as a minimum, the following work: [Initial Phase: acquisition and processing of [x] km² of 3D seismic; drilling of [one] exploration well to a minimum depth of [x] metres or to [named formation]].

4.2 The Contractor shall provide a [bank guarantee / parent company guarantee] in the amount of [USD x] securing the Minimum Work Obligations; the amount shall reduce as obligations are performed.

4.3 If the Contractor fails to complete the Minimum Work Obligations, it shall pay to the Government an amount equal to the estimated cost of the unperformed work.

Article 5 — Work Programmes, Budgets, and Operating Committee

5.1 The Contractor shall submit annual Work Programmes and Budgets to the [Regulator] for [review/approval] within [x] days before the start of each Contract Year.

5.2 [Where the NOC participates, an Operating Committee comprising representatives of the parties shall supervise Petroleum Operations; decisions shall be taken by [voting rules], with [specified matters] reserved to unanimity.]

Article 6 — Discovery, Appraisal, and Development

6.1 The Contractor shall notify the Regulator of any Discovery within [30] days and submit an appraisal programme within [x] months.

6.2 If the Contractor considers a Discovery commercial, it shall submit a Development Plan containing: reservoir and production profiles; development concept and facilities; cost estimates; gas utilisation plan (including measures to eliminate routine flaring); local content plan; environmental and social management plan; health and safety plan; and preliminary decommissioning plan and cost estimate.

6.3 The Regulator shall approve the Development Plan [within x days], approval not to be unreasonably withheld.

Article 7 — Royalty, Taxes, and Fiscal Terms

7.1 The Contractor shall pay royalty at the rates set out in [Annex / the Petroleum Law], [on a sliding scale by production or price].

7.2 The Contractor shall be subject to [petroleum income tax / corporate income tax and any resource rent tax] in accordance with the applicable fiscal legislation.

7.3 [Signature bonus of USD x payable within 30 days of the Effective Date; production bonuses on reaching specified cumulative production.]

7.4 [Annual surface rentals per km² of the Contract Area.]

Drafting note. Fiscal terms are increasingly set by legislation rather than negotiated contract (see Chapter 16). Stabilisation of fiscal terms (Article 18) should be considered alongside.

Article 8 — State Participation

8.1 The Government shall have the right, exercisable through [NOC], to acquire a participating interest of up to [x] per cent in any Development Area, [carried through exploration / reimbursed from production].

Article 9 — Domestic Supply Obligation and Gas

9.1 The Contractor shall supply [a proportion] of its share of crude oil [and natural gas] to the domestic market at [prices set by formula].

9.2 The Contractor shall not flare or vent natural gas except as permitted by the Petroleum Law for safety reasons or with the Regulator's approval, and shall pay any applicable flaring penalties.

Article 10 — Local Content, Employment, and Training

10.1 The Contractor shall give preference to [nationals / national companies] in employment and procurement, in accordance with [local content legislation], and shall implement an approved training programme.

Article 11 — Host Community and Social Obligations

11.1 The Contractor shall [establish / contribute to] [a host community development trust or community development agreement] in accordance with [applicable law] and shall consult affected communities in good faith throughout Petroleum Operations.

Article 12 — Environment, Health, and Safety

12.1 The Contractor shall conduct Petroleum Operations in accordance with applicable environmental law and GIPIP, shall carry out an environmental and social impact assessment before commencing any significant operations, and shall prevent, minimise, and remedy pollution.

12.2 The Contractor shall be liable for pollution damage caused by Petroleum Operations [without prejudice to fault-based or strict liability under applicable law] and shall maintain insurance [and financial security] adequate to meet such liability.

Article 13 — Decommissioning

13.1 The Contractor shall decommission all facilities in accordance with an approved decommissioning plan and shall make contributions to a decommissioning fund [from the start of production / when [x] per cent of reserves have been produced].

Article 14 — Information, Data, and Confidentiality

14.1 All data acquired in the course of Petroleum Operations shall be the property of the State; the Contractor may use the data for the purposes of Petroleum Operations.

14.2 [This Agreement shall be published.] Subject thereto, the parties shall keep confidential information obtained under this Agreement, with exceptions for disclosure required by law, stock exchange rules, or transparency commitments (including the Extractive Industries Transparency Initiative).

Article 15 — Assignment and Change of Control

15.1 The Contractor shall not assign any interest in this Agreement, or undergo a change of control, without the prior written consent of the Government, [not to be unreasonably withheld], [and subject to payment of any applicable tax on gains].

Article 16 — Force Majeure

16.1 Neither party shall be liable for failure to perform an obligation (other than an obligation to pay money) to the extent that performance is prevented by an event beyond its reasonable control which could not reasonably have been foreseen or avoided. The affected party shall notify promptly and use reasonable endeavours to mitigate. Periods for performance shall be extended accordingly.

Article 17 — Anti-Corruption

17.1 Each party represents and undertakes that it has not made and will not make any payment or offer of anything of value, directly or indirectly, to any public official or other person in violation of applicable anti-corruption law, [and the Government may terminate this Agreement in the event of material breach].

Article 18 — Stabilisation

18.1 [If, after the Effective Date, a change in law materially and adversely affects the economic benefits of the Contractor under this Agreement, the parties shall negotiate in good faith such amendments as are necessary to restore the economic equilibrium, [provided that this clause shall not apply to laws of general application adopted in good faith for the protection of health, safety, the environment, or human rights].]

Drafting note. Freezing clauses have largely given way to economic equilibrium clauses; the proviso reflects the concern, expressed in the UN Principles for Responsible Contracts (2011), that stabilisation should not impede legitimate regulation (see Chapters 14 and 15).

Article 19 — Termination

19.1 The Government may terminate this Agreement upon [material breach not remedied within x days of notice; insolvency; failure to pay amounts due; failure to commence production by a specified date], subject to the dispute resolution procedure.

Article 20 — Governing Law and Dispute Resolution

20.1 This Agreement shall be governed by the laws of [State] [and such rules of international law as may be applicable].

20.2 Disputes shall be referred first to [senior management negotiation for x days], then [to an expert in the case of technical disputes], and finally to arbitration under [the ICSID Convention / the UNCITRAL Arbitration Rules / the ICC Rules], seated in [neutral seat], before three arbitrators, in the English language. See Appendix I.

Annexes

Annex 1: Map and coordinates of the Contract Area. Annex 2: Minimum Work Obligations. Annex 3: Accounting Procedure. Annex 4: Form of Guarantee. Annex 5: Fiscal terms (if not in legislation). Annex 6: Local content requirements.

APPENDIX C

Annotated Production Sharing Contract

This appendix presents the core economic and governance clauses of a production sharing contract (PSC), with annotations explaining their purpose, the main negotiating issues, and the points at which disputes most commonly arise. The clauses are an **illustrative teaching text** prepared for this book; they do not reproduce any particular state's model contract. The PSC form originated in Indonesia in the 1960s and has since been adopted, in many variants, across Africa, Asia, the Middle East, and Latin America. Nigeria has used PSCs extensively, particularly for deep offshore acreage under the Deep Offshore and Inland Basin Production Sharing Contracts Act (as amended in 2019), and the Petroleum Industry Act 2021 provides for production sharing contracts among the forms of petroleum mining lease arrangements available for new acreage.

Clause 1 — Nature of the Contract

> 1.1 The Contractor is appointed as the exclusive contractor of the State [acting through the National Oil Company] to conduct Petroleum Operations in the Contract Area. Title to Petroleum shall remain with the State until the Delivery Point, at which point title to the Contractor's share of Cost Petroleum and Profit Petroleum shall pass to the Contractor.

Annotation. The defining feature of the PSC is that the state retains ownership of the resource and the contractor is remunerated in kind with a share of production. This distinguishes it from a concession (in which the licensee owns the petroleum produced, subject to royalty and tax) and from a service contract (in which the contractor is paid a fee in cash or kind without acquiring title). The legal distinction may matter for accounting (booking of reserves), for constitutional ownership provisions, and for the characterisation of the contractor's investment under investment treaties. In economic substance, however, a well-designed PSC and a concession with royalty and tax can produce similar outcomes.

Clause 2 — Risk and Financing

> 2.1 The Contractor shall provide all funds and bear all risks of Petroleum Operations. The Contractor shall recover its costs only from Cost Petroleum in accordance with Clause 6. If no Commercial Discovery is made, the Contractor shall bear all costs incurred without right of recovery.

Annotation. Exploration risk sits wholly with the contractor. The state (or NOC) may be "carried" through exploration — that is, its share of costs is paid by the contractor and recovered later from production — so that the state participates only once a commercial discovery has been made.

Clause 3 — Royalty

> 3.1 The Contractor shall pay royalty on the gross production of Crude Oil and Natural Gas at the rates specified in [Annex / the Petroleum Law], before the allocation of Cost Petroleum and Profit Petroleum.

Annotation. Not all PSCs impose a royalty; some rely wholly on the profit split. Royalty assures the state an early revenue stream regardless of profitability ("front-loading"), but, being insensitive to costs, can deter development of marginal fields. Sliding-scale royalties by water depth, production rate, or price (as in Nigeria's PIA regime) seek to mitigate this.

Clause 4 — Measurement and Valuation

> 4.1 Petroleum shall be measured at the Delivery Point in accordance with GIPIP using metering systems approved by the Regulator. 4.2 Crude Oil shall be valued at the arm's length market price realised in sales to third parties or, where sales are not at arm's length, by reference to [a basket of comparable crudes / a formula published by the Regulator].

Annotation. Valuation provisions are vital to prevent transfer mispricing, for instance through sales to affiliates at below-market prices. Regulators increasingly publish official selling prices or reference prices.

Clause 5 — Cost Recovery Limit

5.1 In each calendar quarter, after deduction of royalty, up to [60] per cent of the remaining Crude Oil produced ("**Cost Oil**") shall be allocated to the Contractor for the recovery of Recoverable Costs. Any Recoverable Costs not recovered in a quarter shall be carried forward to subsequent quarters [without uplift / with an uplift of x per cent].

Annotation. The cost recovery limit ("cost stop") guarantees the state a share of production (profit oil) from first oil, even if the contractor's costs are high. A low limit delays cost recovery and reduces the contractor's internal rate of return; a high limit (or none) accelerates recovery but defers state revenue. Disputes frequently concern whether particular costs are recoverable.

Clause 6 — Recoverable Costs and the Accounting Procedure

6.1 Recoverable Costs shall comprise exploration, appraisal, development, and operating costs incurred in Petroleum Operations, determined in accordance with the Accounting Procedure in Annex [x]. 6.2 The following shall not be Recoverable Costs: signature and production bonuses; costs of arbitration between the parties; fines and penalties; financing costs [unless approved]; costs of marketing beyond the Delivery Point; donations [other than approved community projects]; costs incurred as a result of the Contractor's gross negligence or wilful misconduct; and costs not incurred in accordance with an approved Work Programme and Budget.

Annotation. The Accounting Procedure is often the most contested part of the PSC in practice. Key issues include allocation of overheads from the parent company (frequently capped as a percentage of expenditure), the treatment of related-party services, depreciation schedules for capital costs, the ring-fencing of costs by contract area, and the state's audit rights (Clause 13). Weak audit capacity can lead to significant revenue loss through cost inflation ("gold-plating").

Clause 7 — Profit Petroleum Split

7.1 Crude Oil remaining after royalty and Cost Oil ("**Profit Oil**") shall be shared between the State and the Contractor according to the [R-factor / daily production / cumulative production] tranches in the following table:

Table C.1 Illustrative R-factor profit oil split (hypothetical figures).

R-factor (cumulative revenue ÷ cumulative costs)	State share	Contractor share
Less than 1.0	40%	60%
1.0 to less than 1.5	50%	50%
1.5 to less than 2.0	60%	40%
2.0 and above	70%	30%

Annotation. Progressive splits allow the state's share to rise with profitability, providing some protection against the need to renegotiate when prices rise sharply. The R-factor links the split to the contractor's recovery of costs; production-based tranches are simpler to administer but less responsive to costs and prices. Some PSCs use an internal rate of return trigger.

Clause 8 — Income Tax

> 8.1 The Contractor shall be liable to income tax on its income from Petroleum Operations [at the rate of x per cent] / [The State shall pay the Contractor's income tax on its behalf out of the State's share of Profit Oil ("tax paid" PSC)].

Annotation. Under the "tax paid" variant, the contractor's tax is discharged by the state from its share of profit oil, and the contractor receives a tax receipt — relevant to whether it can claim foreign tax credits in its home jurisdiction.

Clause 9 — Natural Gas

> 9.1 Associated Natural Gas shall be used, in priority, for Petroleum Operations, reinjection, or domestic supply. Flaring shall not be permitted except as authorised. 9.2 Where a Non-Associated Gas Discovery is made, the parties shall negotiate in good faith specific terms for gas development, including cost recovery, profit gas split, and marketing arrangements.

Annotation. Many older PSCs gave little attention to gas, contributing to flaring. Modern PSCs and legislation (including Nigeria's PIA, with its emphasis on gas utilisation and flare penalties) address gas explicitly.

Clause 10 — State Participation

> 10.1 The NOC shall have the option to participate in any Development Area with an interest of up to [x] per cent, its share of past costs being recovered by the Contractor from the NOC's share of Cost Oil.

Clause 11 — Ownership of Assets

11.1 Ownership of all fixed and movable assets acquired for Petroleum Operations shall pass to the State [upon acquisition / upon full cost recovery / upon termination], the Contractor having the exclusive right to use such assets for Petroleum Operations.

Annotation. Transfer of asset ownership to the state has implications for decommissioning liability. Contracts should clarify who bears decommissioning obligations for assets owned by the state but used by the contractor.

Clause 12 — Decommissioning Fund

12.1 From the date on which [50] per cent of the reserves under an approved Development Plan have been produced, the Contractor shall make contributions to an escrow account calculated on a unit-of-production basis to fund the estimated cost of Decommissioning, such contributions being Recoverable Costs.

Clause 13 — Accounts and Audit

13.1 The Contractor shall maintain accounts in accordance with the Accounting Procedure. 13.2 The State may audit the Contractor's accounts within [x] years after the end of each Calendar Year, with access to all relevant records, including those of Affiliates providing services. Exceptions not contested within [x] days shall be deemed accepted.

Clause 14 — Lifting and Marketing

14.1 The parties shall agree a Lifting Procedure governing nominations, scheduling, and the correction of overlifts and underlifts. Each party shall be entitled to lift and market separately its share of Petroleum.

Clause 15 — Stabilisation, Disputes, and Governing Law

See Appendix B, Articles 18 and 20, and Appendix I.

Worked Example (HYPOTHETICAL figures)

This example uses invented figures for teaching purposes only. In a quarter, a field produces 1,000,000 barrels, sold at USD 70 per barrel (gross revenue USD 70 million). Royalty is 10 per cent; the cost recovery limit is 60 per cent of post-royalty production; recoverable costs carried into the quarter total USD 50

million; the R-factor is 1.2; income tax is 30 per cent on the contractor's profit oil; and the state does not participate.

Table C.2 Illustrative PSC revenue allocation for one quarter (hypothetical figures).

Step	Calculation	Barrels	USD million
Gross production	—	1,000,000	70.0
Royalty (10%)	1,000,000 × 10%	100,000	7.0
Net production	—	900,000	63.0
Cost oil available (60%)	900,000 × 60%	540,000	37.8
Cost oil taken (costs USD 50m exceed limit)	—	540,000	37.8
Costs carried forward	50.0 − 37.8	—	12.2
Profit oil	900,000 − 540,000	360,000	25.2
State profit oil (50% at R 1.2)	—	180,000	12.6
Contractor profit oil (50%)	—	180,000	12.6
Income tax (30% of contractor profit oil)	—	—	3.78
State take for quarter	7.0 + 12.6 + 3.78	—	**23.38**
Contractor net cash flow (before costs)	37.8 + 12.6 − 3.78	—	46.62

Questions. (1) How would the state's take change if the cost recovery limit were 50 per cent? (2) What incentives does the cost recovery mechanism create for the contractor in relation to cost control? (3) How would you design audit provisions to protect against inflated costs?

APPENDIX D

Model International Gas Sales Agreement

This appendix outlines the principal provisions of a long-term international gas sales agreement (GSA) for pipeline gas or LNG, with drafting notes. It is an **illustrative teaching model** and should be read with Chapter 6 (gas law and markets) and Chapter 15 (contracts and risk allocation). Published models, such as those of the AIPN, are widely used as starting points in practice.

Part 1 — Parties, Definitions, and Conditions Precedent

The GSA is concluded between a **Seller** (often a project company or upstream consortium) and a **Buyer** (often a utility, aggregator, or trading company). Definitions of key quantities and units are critical, including: Contract Year; Day (often defined as a gas day beginning at a specified hour); Annual Contract Quantity ("**ACQ**"); Daily Contract Quantity ("**DCQ**"); Maximum Daily Quantity; Delivery Point; British thermal unit or gigajoule; and Specifications.

Conditions precedent typically include: final investment decision on the supply project; regulatory approvals and export or import licences; completion of transportation or regasification arrangements; and financing. A long-stop date allows termination if they are not satisfied.

Part 2 — Term and Build-Up

The term may run for fifteen to twenty-five years from the start of deliveries, with a build-up period during which quantities ramp up, and provisions for the start date ("window" narrowing). Delays in start-up must be allocated: the Seller may pay liquidated damages for late delivery and the Buyer for late readiness to receive.

Part 3 — Quantities and Nominations

The Buyer commits to take, and the Seller to deliver, the ACQ, with flexibility through: (a) downward quantity tolerance (for example, up to 10 per cent of ACQ); (b) daily swing above DCQ up to the Maximum Daily Quantity; and (c) nomination procedures (annual programmes, monthly and daily nominations). For LNG, an annual delivery programme schedules cargoes, and provisions address cargo size, vessel compatibility, and laytime.

Part 4 — Take-or-Pay and Make-Up

> If, in any Contract Year, the quantity taken by the Buyer is less than the Minimum Annual Quantity (ACQ less permitted reductions, Seller shortfalls, and force majeure quantities), the Buyer shall pay for the shortfall at [the Contract Price / a percentage of the Contract Price] ("**Take-or-Pay Quantity**"). The Buyer may take the Take-or-Pay Quantity in subsequent Contract Years ("**Make-Up Gas**") after taking the Minimum Annual Quantity for that year, paying only the difference (if any) between the then-current price and the price paid.

Drafting note. Take-or-pay underpins project finance by assuring revenue. Its enforceability as a primary obligation rather than a penalty has generally been upheld in English law (see *M & J Polymers Ltd v Imerys Minerals Ltd* [2008] EWHC 344 (Comm)), and the reformulated penalty doctrine in *Cavendish Square Holding BV v Makdessi* [2015] UKSC 67 reinforces the position for commercially justified primary obligations. Carry-forward provisions may allow the Buyer to credit quantities taken above the ACQ against future obligations.

Part 5 — Delivery, Title, and Risk

The Delivery Point determines where title and risk pass. For pipeline gas, this is often the outlet flange of a processing plant or a border interconnection point. For LNG, sales may be FOB (title and risk pass at the loading port, Buyer arranges shipping), DES/DAP (passing at the discharge port, Seller arranges shipping), or ex-ship variants. Destination restrictions in LNG contracts have been challenged on competition grounds, notably in the European Union and Japan, and newer contracts tend to allow greater destination flexibility.

Part 6 — Quality and Measurement

Gas must meet specifications (gross calorific value, Wobbe index, hydrogen sulphide, water dew point, hydrocarbon dew point, carbon dioxide, oxygen, and, increasingly, hydrogen blending limits). Off-specification gas may be rejected or accepted at a discount. Measurement provisions specify metering, calibration, testing, and resolution of disputes by independent expert.

Part 7 — Price

> The Contract Price for each Month shall be calculated as: $P = P_0 + a \times (\text{Brent}_3 - B_0)$ [oil indexation], or $P = HH \times [1.15] + [\text{constant}]$ [hub indexation], or a hybrid formula, as specified in Annex [x].

Drafting note. Long-term gas prices have traditionally been indexed to oil products (common in Asia and historically in Europe), increasingly replaced in Europe by hub prices (such as the Dutch TTF) and in US LNG exports by Henry Hub. Floors, ceilings, and S-curves dampen volatility.

Part 8 — Price Review

> Either party may, [once every three years] and additionally [once] at any time during the term, request a review of the Contract Price if: (a) there has been a significant change in economic circumstances in the Buyer's market, beyond the control of the parties, compared with those prevailing at the time of the last price determination; and (b) as a result, the Contract Price no longer reflects the value of gas in the Buyer's market [for a buyer acting prudently and efficiently]. Failing agreement within [x] months, either party may refer the matter to arbitration, and the tribunal shall determine the revised Contract Price in accordance with this Clause.

Drafting note. Price review clauses have generated many high-value arbitrations. Key issues are the threshold for review, the definition of the relevant market and comparators, whether the tribunal may revise the structure of the price formula or only its parameters, and the date from which a revised price applies. See Chapter 6, section 6.6, and Chapter 26.

Part 9 — Seller Shortfall and Remedies

If the Seller fails to deliver nominated quantities (other than for force majeure or Buyer's fault), the Buyer may receive a discount on shortfall quantities or recover cover costs, subject to caps. The Seller's liability is often limited to these remedies.

Part 10 — Force Majeure and Hardship

Force majeure provisions excuse performance of non-payment obligations for events beyond reasonable control, with specific lists (including, for example, pipeline failure, sanctions, and governmental action) and exclusions (such as lack of funds and changes in market conditions). Hardship provisions may require renegotiation. The treatment of sanctions has become a central issue since 2022.

Part 11 — Invoicing, Payment, and Security

Monthly invoicing, payment within a specified period, interest on late payment, rights to suspend delivery for non-payment, and credit support (parent guarantees or letters of credit) are standard.

Part 12 — Liability, Assignment, Confidentiality

Mutual exclusion of consequential losses; caps on liability; restrictions on assignment; confidentiality subject to legal and regulatory disclosure.

Part 13 — Governing Law and Dispute Resolution

English or New York law is frequently chosen for international GSAs. Disputes are referred to arbitration (commonly ICC, LCIA, or SIAC rules, or UNCITRAL Rules), with expert determination for technical matters such as measurement and quality. See Appendix I.

Part 14 — Decarbonisation Provisions (Emerging Practice)

Newer GSAs may include provisions on methane emissions reporting and intensity, verification under recognised frameworks (such as the Oil and Gas Methane Partnership 2.0), the treatment of carbon costs, the sharing of costs of compliance with the EU Methane Regulation, and the delivery of "carbon-neutral" or "certified" cargoes, together with rights to adjust or terminate if regulatory requirements cannot be met.

APPENDIX E

Sample Renewable Energy Power Purchase Agreement

This appendix sets out the structure of a long-term power purchase agreement (PPA) for a grid-connected solar or wind project sold to a state-owned or regulated off-taker, with drafting notes on bankability. It is an **illustrative teaching model** and should be read with Chapter 9, section 9.5, Chapter 10, and Chapter 16. Note the regulatory context in Nigeria, where the Electricity Act 2023 allows states to regulate intra-state electricity markets and where the Nigerian Bulk Electricity Trading Plc has historically acted as bulk purchaser.

Clause 1 — Parties and Recitals

The PPA is concluded between [Project Company] (the "**Seller**"), a special purpose vehicle that will build, own, and operate a [x] MW solar photovoltaic plant at [site], and [Off-taker] (the "**Buyer**"). Recitals record the Seller's generation licence, the grid connection agreement, the implementation agreement or government support arrangements (if any), and the Buyer's authority to purchase electricity.

Clause 2 — Term and Conditions Precedent

The term runs for [20–25] years from the Commercial Operation Date ("**COD**"). Conditions precedent include financial close, permits, land rights, the grid connection agreement, government support agreement, and, where applicable, the provision of payment security by the Buyer. A long-stop date permits termination if they are not satisfied.

Clause 3 — Construction and Commissioning

The Seller shall achieve COD by the Scheduled COD, subject to extension for Buyer risk events and force majeure. Commissioning tests (including capacity tests and grid code compliance tests) must be witnessed by the Buyer and the system operator. Delay liquidated damages may be payable by the Seller for late COD, and deemed energy payments by the Buyer where completion is delayed by the Buyer or by the unavailability of the grid.

Clause 4 — Sale and Purchase

> 4.1 From COD, the Seller shall sell and deliver, and the Buyer shall purchase and accept, all Net Electrical Output delivered at the Delivery Point, together with all environmental attributes associated with it [unless otherwise agreed].

Drafting note. Variable renewable PPAs are typically "as generated" (take-and-pay for all output) rather than capacity-based. The treatment of environmental attributes (renewable energy certificates, carbon credits) must be explicit (Chapter 10, section 10.7).

Clause 5 — Tariff

> 5.1 The Buyer shall pay for Net Electrical Output at the Tariff of [USD/kWh x] [indexed annually at [x] per cent / to US CPI for the foreign currency component], determined [through competitive auction / by the regulator].

Drafting note. Currency risk is a critical bankability issue. Tariffs may be denominated in foreign currency, in local currency indexed to the exchange rate, or partly in each. Nigerian practice has involved dollar-indexed tariffs, and the impact of naira devaluation has been a significant source of financial strain in the sector.

Clause 6 — Curtailment and Deemed Energy

> 6.1 If the Buyer or the System Operator instructs curtailment of the Facility, or the Facility is unable to deliver due to unavailability of the Grid, other than for reasons attributable to the Seller, emergency, or [x hours per year of permitted curtailment], the Buyer shall pay for the Deemed Energy that would have been delivered, calculated by reference to measured irradiance [or wind speed] and the Facility's performance model.

Drafting note. In weak grids, curtailment risk is high; lenders require deemed energy protection. The calculation methodology must be robust and verifiable.

Clause 7 — Metering, Billing, and Payment

Metering at the Delivery Point; monthly invoicing; payment within [30] days; interest on late payment; and a dispute procedure in which undisputed amounts must be paid.

Clause 8 — Payment Security

> 8.1 The Buyer shall provide and maintain [a standby letter of credit / a payment escrow] equal to [x] months of estimated payments. [The Government shall provide a guarantee / put-call option agreement under the Implementation Agreement.] [Partial risk guarantees from multilateral development institutions may support the Buyer's obligations.]

Drafting note. Off-taker creditworthiness is the single most important issue for many renewable projects in emerging markets. Instruments such as World Bank Group partial risk guarantees and the liquidity facilities developed for African IPPs have been used to mitigate it.

Clause 9 — Operation and Dispatch

The Seller shall operate the Facility in accordance with Prudent Utility Practice, the Grid Code, and the instructions of the System Operator; provide forecasts of output; and maintain availability. Provisions address planned outages and the provision of ancillary services (where required, such as reactive power).

Clause 10 — Change in Law

> 10.1 If a Change in Law results in an increase in the Seller's costs or a reduction in its revenues exceeding [threshold], the Tariff shall be adjusted to place the Seller in the same economic position as if the Change in Law had not occurred. A "Change in Law" includes the introduction, repeal, or change of any law, or change in the interpretation of any law by a competent authority, after [the bid date], [including changes in taxes specific to the electricity sector], but excluding [changes in general income tax].

Clause 11 — Force Majeure

Distinguishes natural force majeure (extended time, no payment) from political force majeure (including expropriation, change in law, political violence, and governmental action), in respect of which the Buyer or Government may be required to continue payments or compensate the Seller.

Clause 12 — Termination and Compensation

Events of default by each party, cure periods, and termination payments. Upon termination for Buyer default or political force majeure, the Buyer (or Government under the put option) must purchase the Facility at a price covering

outstanding debt, equity, and a return; upon Seller default, the price may be limited to outstanding debt (or less).

Clause 13 — Lender Protections

Consent to assignment by way of security; direct agreement giving lenders step-in rights and notice of default; and extended cure periods.

Clause 14 — Environmental and Social Compliance

The Seller shall comply with applicable environmental and social law and [the IFC Performance Standards / Equator Principles], including land acquisition and resettlement requirements and grievance mechanisms (see Appendix G).

Clause 15 — Disputes and Governing Law

Technical disputes to an expert; other disputes to arbitration under [ICC / LCIA / UNCITRAL Rules], seated in [London / a neutral seat / Lagos], with recourse to investment treaty protection where available (see Chapter 14 and Appendix I).

Corporate PPAs: Key Differences

A corporate PPA between a generator and a commercial or industrial customer may be physical (requiring delivery through the network or private wire, subject to the regulatory regime for supply) or virtual/financial (a contract for difference against a market reference price, with no physical delivery). Corporate PPAs raise particular issues of credit support from the corporate buyer, shape and balancing risk, the allocation of guarantees of origin, and, in virtual PPAs, accounting treatment as derivatives.

APPENDIX F

Mining Licence and Regulatory Compliance Checklist

This checklist is designed to assist in reviewing the regulatory compliance of a mining project across its life cycle. It is general in form, drawing on the issues examined in Chapter 7 and Chapter 8, and must be adapted to the applicable law. Where Nigerian law applies, the principal statute is the Nigerian Minerals and Mining Act 2007, administered by the Ministry of Solid Minerals Development and the Mining Cadastre Office, together with the Nigerian Minerals and Mining Regulations 2011 and relevant environmental legislation, including the Environmental Impact Assessment Act.

Part 1 — Title and Tenure

Table F.1 Mining compliance checklist — title and tenure.

No.	Item	Yes/No/N/A	Notes and evidence
1.1	Is the type of title held (reconnaissance permit, exploration licence, small-scale mining lease, mining lease, quarry lease, water use permit) appropriate to the activity?		
1.2	Has the title been granted by the competent authority, registered in the cadastre, and are the coordinates correct?		
1.3	Is the title current; are renewal or extension deadlines diarised?		
1.4	Have annual service fees and surface rents been paid?		
1.5	Are there overlapping titles, protected areas, or forest reserves within the area?		
1.6	Have relinquishment obligations been complied with?		
1.7	Have all transfers, assignments, and changes of control been approved?		
1.8	Are surface rights (land access, compensation to occupiers, consent of land owners, and, where applicable, statutory land use rights) secured?		

Part 2 — Work Obligations and Reporting

Table F.2 Mining compliance checklist — work obligations and reporting.

No.	Item	Yes/No/N/A	Notes and evidence
2.1	Have minimum work and expenditure obligations been met?		
2.2	Have periodic reports (exploration results, production, sales, and employment) been filed on time?		
2.3	Have reserves and resources been reported in accordance with a recognised code (JORC, NI 43-101, SAMREC, CRIRSCO template)?		
2.4	Has a feasibility study and mine plan been approved where required?		

Part 3 — Environmental and Social

Table F.3 Mining compliance checklist — environmental and social.

No.	Item	Yes/No/N/A	Notes and evidence
3.1	Has an EIA/ESIA been approved before commencement, and are its conditions being met?		
3.2	Is there an environmental management plan, monitoring programme, and reporting?		
3.3	Are permits held for water use, discharges, waste, explosives, and air emissions?		
3.4	Is tailings management consistent with the Global Industry Standard on Tailings Management (2020)?		
3.5	Has a community development agreement been concluded with host communities and is it being implemented?		
3.6	Have resettlement and compensation been carried out consistent with law and international standards?		
3.7	Is there an accessible, effective grievance mechanism?		
3.8	Has free, prior and informed consent been sought where indigenous peoples are affected (where applicable)?		
3.9	Are cultural heritage sites identified and protected?		

Part 4 — Health, Safety, and Labour

Table F.4 Mining compliance checklist — health, safety, and labour.

No.	Item	Yes/No/N/A	Notes and evidence
4.1	Is there a safety management system meeting legal requirements and recognised standards?		
4.2	Are incidents and fatalities reported and investigated?		
4.3	Is there compliance with labour law, including prohibitions of child and forced labour?		
4.4	Are security arrangements consistent with the Voluntary Principles on Security and Human Rights?		
4.5	Is the relationship with artisanal and small-scale miners on or near the concession managed lawfully and safely?		

Part 5 — Fiscal, Transparency, and Integrity

Table F.5 Mining compliance checklist — fiscal, transparency, and integrity.

No.	Item	Yes/No/N/A	Notes and evidence
5.1	Are royalties calculated and paid correctly (rate, base, valuation point)?		
5.2	Are corporate taxes, withholding taxes, and customs duties paid; are incentives (such as tax holidays) properly claimed?		
5.3	Are related-party sales and services priced at arm's length (transfer pricing documentation)?		
5.4	Are payments to government disclosed under EITI and any mandatory disclosure laws (such as the UK Reports on Payments to Governments Regulations 2014 or the EU Accounting Directive)?		
5.5	Is beneficial ownership disclosed as required?		
5.6	Are anti-bribery policies, due diligence on agents and partners, and training in place?		
5.7	Are export permits and mineral export documentation (including certificates of origin) in order?		

Part 6 — Supply Chain and Market Access

Table F.6 Mining compliance checklist — supply chain and market access.

No.	Item	Yes/No/N/A	Notes and evidence
6.1	Does the operation meet buyers' due diligence requirements under the OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas?		

No.	Item	Yes/No/N/A	Notes and evidence
6.2	Are obligations under the EU Conflict Minerals Regulation (2017/821), EU Batteries Regulation (2023/1542), or other import-market regimes relevant to the product?		
6.3	Is the operation participating in recognised assurance schemes (such as IRMA, the Responsible Minerals Initiative, or industry standards)?		
6.4	Are export restrictions or local beneficiation requirements applicable?		

Part 7 — Closure

Table F.7 Mining compliance checklist — closure.

No.	Item	Yes/No/N/A	Notes and evidence
7.1	Is there an approved closure and rehabilitation plan, updated periodically?		
7.2	Is financial assurance for closure (bond, trust fund, or guarantee) provided and adequate?		
7.3	Has progressive rehabilitation been undertaken?		
7.4	Are post-closure monitoring and community transition arrangements planned?		

Using the Checklist

Each item should be supported by documentary evidence and, where possible, verified by site visit and interviews. Non-compliance should be classified by severity (critical, major, minor), with recommended corrective actions, responsibilities, and deadlines. The checklist is a tool for structured inquiry, not a substitute for legal analysis of the applicable statutes, regulations, and licence conditions.

APPENDIX G

Environmental and Social Impact Assessment Framework

This appendix provides a framework for the conduct and review of an environmental and social impact assessment (ESIA) for an energy or natural resources project. It draws on the general principles of international environmental law examined in Chapter 17 — including the customary obligation to conduct an environmental impact assessment where there is a risk of significant transboundary harm, recognised by the International Court of Justice in *Pulp Mills on the River Uruguay* (*Argentina v Uruguay*) (2010) and *Certain Activities Carried Out by Nicaragua in the Border Area / Construction of a Road in Costa Rica along the San Juan River* (2015) — on national EIA legislation (in Nigeria, the Environmental Impact Assessment Act, Cap E12 Laws of the Federation of Nigeria 2004), and on international lender standards, in particular the IFC Performance Standards on Environmental and Social Sustainability (2012).

Stage 1 — Screening

Determine whether an ESIA is required and at what level, by reference to: (a) statutory lists of mandatory projects (for example, oil and gas fields, pipelines, refineries, power stations, large dams, and mines above specified thresholds); (b) the sensitivity of the location (protected areas, wetlands, indigenous territories, cultural heritage, densely populated areas); (c) the potential for transboundary impacts, triggering obligations under customary law and, where applicable, the Espoo Convention; and (d) lender categorisation (for example, Category A for projects with potential significant adverse impacts that are diverse, irreversible, or unprecedented).

Stage 2 — Scoping

Identify the key issues, alternatives, study area, baseline data needs, methods, and stakeholders. Scoping should be consultative and documented in terms of reference approved by the competent authority. Consider: project alternatives (including location, technology, and the "no project" option); associated facilities; cumulative impacts with other existing and planned projects; and climate change

(both the project's emissions and its vulnerability to climate impacts). The UK Supreme Court's decision in *R (Finch) v Surrey County Council* [2024] UKSC 20 — holding that the downstream (scope 3) combustion emissions of extracted oil were effects of the project that had to be assessed — illustrates the expanding scope of climate assessment.

Stage 3 — Baseline Studies

Collect data on the existing physical, biological, and social environment, including: geology, soils, hydrology, groundwater, air quality, noise, and climate; flora, fauna, habitats, and ecosystem services; land use and tenure; demography, livelihoods, health, education, and gender; vulnerable groups; indigenous peoples; and cultural heritage. Baseline data must be adequate in duration and coverage (for example, seasonal variation) to allow meaningful prediction.

Stage 4 — Impact Identification, Prediction, and Evaluation

For each project phase (construction, operation, decommissioning), identify impacts, predict their magnitude, extent, duration, reversibility, and likelihood, and evaluate their significance against legal standards and the sensitivity of receptors. Social impacts must be assessed with attention to differential effects on women, children, the elderly, persons with disabilities, and other vulnerable groups, and to human rights (see Appendix L).

Stage 5 — Mitigation and the Mitigation Hierarchy

Apply the mitigation hierarchy: avoid, minimise, restore or rehabilitate, and, as a last resort, offset or compensate. For biodiversity, aim for no net loss (or net gain) of natural habitat and net gains for critical habitat, consistent with IFC Performance Standard 6.

Stage 6 — Public Participation and Disclosure

Ensure meaningful consultation throughout, with: disclosure of information in accessible languages and formats; adequate time for comment; public hearings; specific engagement with affected communities, women, and vulnerable groups; free, prior and informed consent where indigenous peoples are affected in circumstances identified in IFC Performance Standard 7 and international human rights law; and responses to comments. The Aarhus and Escazú Conventions set out regional standards of access to information, participation, and justice.

Stage 7 — The ESIA Report and Environmental and Social Management Plan

The ESIA report should contain: a non-technical summary; project description; legal and institutional framework; baseline; alternatives; impacts; mitigation;

residual impacts; and an Environmental and Social Management Plan (ESMP) setting out mitigation measures, monitoring, responsibilities, timelines, budgets, and reporting.

Stage 8 — Review and Decision

The competent authority reviews the ESIA, often with an independent panel and public hearing, and decides whether to approve the project, with or without conditions, or to refuse it. The decision should be reasoned and published, and subject to administrative appeal and judicial review.

Stage 9 — Implementation, Monitoring, and Compliance

Implement the ESMP; monitor environmental and social performance; report to the authority and the public; conduct audits; and take corrective action. Monitoring obligations continue through the life of the project, as the ICJ emphasised in *Pulp Mills*.

Stage 10 — Decommissioning and Closure

Plan and assess decommissioning or closure impacts and rehabilitation, with financial assurance.

The IFC Performance Standards at a Glance

Table G.1 IFC Performance Standards on Environmental and Social Sustainability (2012).

Standard	Subject
PS1	Assessment and Management of Environmental and Social Risks and Impacts
PS2	Labor and Working Conditions
PS3	Resource Efficiency and Pollution Prevention
PS4	Community Health, Safety, and Security
PS5	Land Acquisition and Involuntary Resettlement
PS6	Biodiversity Conservation and Sustainable Management of Living Natural Resources
PS7	Indigenous Peoples
PS8	Cultural Heritage

Review Checklist for an ESIA Report

1. Was the ESIA prepared by competent, independent experts?
2. Were alternatives genuinely considered, including the "no project" option?
3. Are baseline data adequate in scope and duration?
4. Are direct, indirect, cumulative, and transboundary impacts assessed, including greenhouse gas emissions and climate vulnerability?

5. Are social and human rights impacts, including on vulnerable groups, assessed?
6. Is the significance evaluation transparent and justified?
7. Is the mitigation hierarchy properly applied?
8. Was public participation meaningful, and were comments addressed?
9. Is the ESMP specific, budgeted, and enforceable?
10. Are monitoring, reporting, grievance, and closure arrangements adequate?

APPENDIX H

Model Energy Project Risk Allocation Matrix

A risk allocation matrix identifies the risks of a project, allocates each to the party best able to manage it (or to bear it at least cost), and records the contractual and other instruments through which the allocation is effected. The following matrix is a **model for a hypothetical independent power project or upstream development** financed on a limited-recourse basis. Allocations vary by project, market, and bargaining power; this matrix illustrates a typical starting point and should be read with Chapter 15 and Chapter 16.

Table H.1 Model risk allocation matrix for a limited-recourse energy project (illustrative).

Risk	Description	Primary bearer	Mitigation / instrument
Resource / reserves	Insufficient resource (reserves, wind, solar irradiance, hydrology)	Sponsor / project company	Independent resource studies; P90 financing case; reserve tail
Site and land	Title defects, access, resettlement, archaeological finds	Project company (with host state support)	Due diligence; land acquisition by state; ESIA; insurance
Permitting	Delay or refusal of consents	Project company; state for changes after award	Conditions precedent; change in law clause; government support agreement
Construction — cost overrun	Costs exceed budget	EPC contractor (fixed price) / sponsors	Lump-sum turnkey EPC; contingency; sponsor completion support
Construction — delay	Late completion	EPC contractor	Delay liquidated damages; performance bonds; insurance
Technology / performance	Plant underperforms	EPC contractor; equipment suppliers	Performance tests; performance liquidated damages; warranties
Operation and maintenance	Poor operation, availability	O&M contractor	O&M agreement with KPIs, bonuses, and penalties

Risk	Description	Primary bearer	Mitigation / instrument
Fuel / input supply	Unavailability or price of fuel	Off-taker (pass-through) or fuel supplier	Fuel supply agreement; tariff pass-through
Market / demand	Insufficient demand or price	Off-taker under PPA or GSA	Take-or-pay; capacity payments; long-term contract
Off-taker credit	Non-payment by off-taker	Lenders/sponsors, mitigated by state	Letters of credit; escrow; sovereign guarantee; partial risk guarantee
Grid / transport	Grid unavailability, curtailment, pipeline outages	Off-taker / grid operator	Deemed energy; ship-or-pay transport agreements
Currency convertibility and transfer	Inability to convert or transfer funds	State / political risk insurers	Government support; offshore accounts; MIGA or private political risk insurance
Exchange rate	Devaluation of local currency	Off-taker (indexation) / project	Foreign currency tariff or indexation; hedging
Interest rate	Rising rates	Project company	Swaps and hedging
Inflation	Rising costs	Shared	Tariff indexation
Change in law	New taxes or regulations	State / off-taker	Change in law clause; stabilisation; economic equilibrium
Expropriation	Taking or creeping expropriation	State	Termination compensation; investment treaty; political risk insurance
Political violence	War, terrorism, civil disturbance	State (political force majeure) / insurers	Political force majeure clauses; insurance
Natural force majeure	Floods, earthquakes, storms	Shared	Force majeure relief; property insurance; business interruption insurance
Environmental and social	Pollution, community opposition, human rights impacts	Project company	ESIA; ESMP; community agreements; grievance mechanism; lender standards
Climate — physical	Increased extreme weather, water scarcity	Project company / shared	Climate risk assessment; design standards; insurance
Climate — transition	Carbon pricing, phase-out, demand decline	Project company / sponsors	Scenario analysis; contractual protections; diversification
Corruption and integrity	Bribery, fraud	Project company / all parties	Anti-corruption clauses; due diligence; compliance programmes

Risk	Description	Primary bearer	Mitigation / instrument
Sanctions	Imposition of sanctions affecting parties or supply chain	Shared	Sanctions clauses; diversification; termination rights
Cyber security	Attack on control systems	Project company	Cyber security standards; insurance
Decommissioning	Cost of decommissioning exceeds provision	Project company / owners	Decommissioning fund; security; regulator approval

Using the Matrix

1. **Identify** risks comprehensively, by project phase.
2. **Assess** each risk's likelihood and impact, ideally quantitatively.
3. **Allocate** each risk to the party best able to control it or to bear it at the lowest cost — a principle that lenders and development institutions often state as the "golden rule" of project finance risk allocation.
4. **Document** the allocation in the relevant contracts and ensure consistency across the contract suite (for example, that force majeure definitions in the EPC contract, PPA, and loan agreement align, avoiding "gaps" in which the project company bears risk it cannot pass on).
5. **Monitor** risks through the life of the project and adjust mitigation as necessary.

Exercise. For a hypothetical 100 MW solar project in a West African country with a state-owned distribution utility as off-taker, identify the three risks most likely to determine bankability and propose instruments to mitigate them, explaining which parties should bear them and why.

APPENDIX I

Sample International Arbitration Clause and Procedural Documents

This appendix provides model arbitration clauses and templates of key procedural documents for international energy arbitration. It should be read with Chapter 26. The model clauses of arbitral institutions are reproduced in substance from their published recommended forms; readers should always check the institution's current recommended wording before use.

Institutional Model Clauses

ICC. The ICC recommends the following standard clause: "All disputes arising out of or in connection with the present contract shall be finally settled under the Rules of Arbitration of the International Chamber of Commerce by one or more arbitrators appointed in accordance with the said Rules." The ICC suggests that parties add the place of arbitration, the governing law, the number of arbitrators, and the language.

LCIA. The LCIA's recommended clause for future disputes provides: "Any dispute arising out of or in connection with this contract, including any question regarding its existence, validity or termination, shall be referred to and finally resolved by arbitration under the LCIA Rules, which Rules are deemed to be incorporated by reference into this clause." Parties then specify the number of arbitrators, the seat, the language, and the governing law.

UNCITRAL. The model clause annexed to the UNCITRAL Arbitration Rules provides: "Any dispute, controversy or claim arising out of or relating to this contract, or the breach, termination or invalidity thereof, shall be settled by arbitration in accordance with the UNCITRAL Arbitration Rules," with a note that the parties should consider adding the appointing authority, the number of arbitrators, the place of arbitration, and the language.

ICSID. For investment contracts with a state party, ICSID publishes model clauses by which the parties consent to submit disputes to the Centre for settlement by arbitration under the ICSID Convention, including provisions on nationality (for locally incorporated companies under foreign control, by agreement under article 25(2)(b)) and on the applicable law.

A Tiered Dispute Resolution Clause for an Energy Contract (Model)

20.1 Negotiation. Any dispute arising out of or in connection with this Agreement (a "**Dispute**") shall first be referred, by written notice, to senior executives of the Parties, who shall meet within fifteen (15) days and attempt in good faith to resolve the Dispute.

20.2 Expert determination. Any Dispute concerning [measurement, quality, reserves determination, cost recovery accounting, or other technical matters specified in Annex X] not resolved within thirty (30) days of the notice may be referred by either Party to an independent expert appointed [by agreement / by the ICC International Centre for ADR]. The expert shall act as expert and not as arbitrator, and the expert's determination shall be final and binding save for fraud or manifest error.

20.3 Mediation. [Any other Dispute not resolved within forty-five (45) days may, by agreement, be referred to mediation under [the ICC Mediation Rules / CEDR / other rules].]

20.4 Arbitration. Any Dispute not resolved under Clauses 20.1 to 20.3 within [sixty (60)] days of the notice shall be finally settled by arbitration under the [Rules], by three arbitrators. The seat of arbitration shall be [London / Paris / Singapore / Geneva / Lagos]. The language shall be English. [The Parties waive any right of appeal on a point of law to the extent permitted by the law of the seat.]

20.5 Sovereign immunity. [The State/NOC] irrevocably waives any immunity from jurisdiction and [to the extent permitted by law] from execution in respect of any arbitration and the enforcement of any award, [except in respect of assets used for diplomatic, military, or central bank purposes].

20.6 Continuing performance. Pending resolution of a Dispute, the Parties shall continue to perform their obligations under this Agreement.

Drafting notes. (1) Ensure that pre-arbitral steps are clearly defined with fixed time limits, to avoid disputes over whether they are conditions precedent to arbitration (see *Emirates Trading Agency LLC v Prime Mineral Exports Pte Ltd* [2014] EWHC 2104 (Comm), enforcing a time-limited friendly discussion requirement). (2) Choose the seat carefully: it determines the curial law, the courts with supervisory jurisdiction, and the grounds for setting aside. Note the English Arbitration Act 2025, including the default rule on the law governing the arbitration agreement, and Nigeria's Arbitration and Mediation Act 2023. (3) Consider joinder and consolidation provisions in multi-contract projects. (4) Consider emergency arbitrator provisions and interim relief.

Template: Request for Arbitration (Outline)

1. Names, addresses, and representatives of the parties.
2. The arbitration agreement invoked (with copy).
3. The contract(s) or other legal instrument(s) out of which the dispute arises.
4. A description of the nature and circumstances of the dispute.
5. The relief sought, including amounts claimed (to the extent possible).
6. Proposals concerning the number of arbitrators, nomination of arbitrator (if three), seat, applicable rules of law, and language.
7. Fulfilment of pre-arbitral steps.
8. Payment of the filing fee.

Template: Procedural Order No. 1 (Outline)

1. Parties and representatives; tribunal; administrative secretary (if any) and their role.
2. Applicable rules; seat; language; place of hearings (including virtual hearings).
3. Communications and filing.
4. Procedural timetable (Annex).
5. Written submissions: number, sequence, and content (memorial and counter-memorial; reply and rejoinder).
6. Document production: scope and procedure, by reference to the IBA Rules on the Taking of Evidence in International Arbitration (2020) as guidelines; use of Redfern schedule.
7. Witness statements and expert reports; expert conferencing ("hot-tubbing").
8. Pre-hearing conference; hearing arrangements; time allocation (chess clock).
9. Post-hearing submissions and costs submissions.
10. Confidentiality and transparency; data protection and cyber security.
11. Tribunal secretary's role and fees; deposits.
12. Third-party funding disclosure.

Template: Redfern Schedule

Table I.1 Redfern schedule template for document production requests.

No.	Documents or category of documents requested	Relevance and materiality (with references to submissions)	Objections	Reply to objections	Tribunal's decision
1					
2					

Template: Witness Statement (Structure)

1. Identity, position, and relationship to the parties.
2. Source of knowledge, distinguishing personal knowledge from information and belief.
3. Chronological narrative of relevant facts, with references to exhibits.
4. Statement of truth, date, and signature.

Template: Expert Report (Structure)

1. Expert's qualifications, independence, and instructions.
2. Issues addressed and documents relied upon.
3. Methodology and assumptions.
4. Analysis and conclusions (for example, quantum by discounted cash flow, with sensitivity analysis).
5. Declaration of independence and duty to the tribunal.

Investment Treaty Arbitration: Additional Checks

Before commencing an investment treaty claim, counsel should check: (a) the nationality of the investor and any planning or restructuring concerns (see *Philip Morris Asia v Australia*, PCA, 2015, on abuse of process); (b) the definition of investment; (c) temporal scope and any sunset clause (relevant, for example, to the twenty-year sunset clause of the Energy Charter Treaty following withdrawal); (d) cooling-off and local remedies requirements; (e) fork-in-the-road and waiver clauses; (f) denial of benefits; (g) for intra-EU disputes, the implications of *Achmea* (2018) and *Komstroy* (2021); and (h) enforcement prospects, including immunity from execution.

APPENDIX J

Comparative Energy and Natural Resources Regulatory Frameworks

This appendix summarises the principal legal and institutional frameworks for petroleum, mining, and electricity in selected jurisdictions, as a starting point for comparative research (Chapter 24). The tables are necessarily simplified; laws and institutions change frequently, and readers must verify the current position from primary sources.

Petroleum

Table J.1 Comparative petroleum regulatory frameworks (simplified).

Jurisdiction	Ownership of petroleum	Principal legislation	Regulator(s)	National company	Main contractual forms
Nigeria	Federal Government (Constitution s 44(3))	Petroleum Industry Act 2021	NUPRC (upstream); NMDPRA (midstream/downstream)	NNPC Ltd (commercial company since 2022)	Licences and leases; PSCs; joint ventures
Ghana	Republic, held in trust for the people	Petroleum (Exploration and Production) Act 2016 (Act 919); Petroleum Revenue Management Act 2011 (Act 815)	Petroleum Commission	GNPC	Petroleum agreements (hybrid)
Norway	State	Petroleum Act 1996	Ministry of Energy; Norwegian Offshore Directorate; Petroleum Safety Authority	Equinor (partly state-owned); Petoro (SDFI)	Production licences; tax-based system
United Kingdom	Crown (Petroleum Act 1998)	Petroleum Act 1998; Energy Act 2016	North Sea Transition Authority; OPRED	None	Licences; tax-based system

Jurisdiction	Ownership of petroleum	Principal legislation	Regulator(s)	National company	Main contractual forms
United States	Private on private land; federal and state on public land and offshore	Outer Continental Shelf Lands Act 1953; Mineral Leasing Act 1920; state laws	BOEM; BSEE; state commissions	None	Leases
Brazil	Union	Petroleum Law (Law 9.478/1997); Pre-Salt Law (Law 12.351/2010, as amended)	ANP	Petrobras	Concessions; production sharing (pre-salt); transfer of rights
Indonesia	State	Oil and Gas Law (Law 22/2001, as modified by Constitutional Court rulings)	Ministry of Energy and Mineral Resources; SKK Migas	Pertamina	PSCs (cost recovery and gross split)

Mining

Table J.2 Comparative mining regulatory frameworks (simplified).

Jurisdiction	Principal legislation	Administering authority	Notable features
Nigeria	Nigerian Minerals and Mining Act 2007	Ministry of Solid Minerals Development; Mining Cadastre Office	Cadastre-based "first come, first served" titles; community development agreements
Ghana	Minerals and Mining Act 2006 (Act 703)	Minerals Commission	State free-carried interest; stability and development agreements
South Africa	Mineral and Petroleum Resources Development Act 2002	Department of Mineral Resources and Energy	State custodianship; transformation (Mining Charter)
Democratic Republic of the Congo	Mining Code 2002, as revised in 2018	Ministry of Mines; Cadastre Minier	Higher royalties; "strategic substances" (including cobalt)
Chile	Constitution; Mining Code 1983; Constitutional Organic Law on Mining Concessions 1982	SERNAGEOMIN; Ministry of Mining	Judicial concessions; mining royalty law of 2023

Jurisdiction	Principal legislation	Administering authority	Notable features
Australia (Western Australia)	Mining Act 1978 (WA)	Department responsible for mines (WA)	State ownership; native title interface under the Native Title Act 1993 (Cth)
Canada (Ontario)	Mining Act (Ontario)	Ministry responsible for mines (Ontario)	Online staking; duty to consult indigenous peoples

Electricity

Table J.3 Comparative electricity regulatory frameworks (simplified).

Jurisdiction	Market structure	Principal legislation	Regulator	Notable features
Nigeria	Unbundled; privatised generation and distribution; state-owned transmission; emerging state markets	Electricity Act 2023	NERC; state regulatory commissions	Decentralisation to states; mini-grid regulation; Multi-Year Tariff Order
South Africa	Vertically integrated utility undergoing restructuring; IPP procurement	Electricity Regulation Act 2006, as amended (2024)	NERSA	REIPPPP; wholesale market reform; transmission unbundling
Kenya	Single buyer; IPPs	Energy Act 2019	EPRA	Geothermal leadership; feed-in tariff and auctions
United Kingdom (GB)	Liberalised wholesale and retail	Electricity Act 1989; Energy Act 2023	Ofgem	Contracts for difference; independent system operator (NESO)
European Union	Integrated internal market	Electricity Directive (EU) 2019/944; Regulation (EU) 2019/943, as reformed 2024	National regulators; ACER	Market coupling; unbundling; two-sided CfDs
United States	Mixed: organised markets (RTOs/ISOs) and vertically integrated utilities	Federal Power Act 1935; state laws	FERC; state public utility commissions	Federal–state division; renewable portfolio standards
India	Partly liberalised; state distribution utilities	Electricity Act 2003	CERC; state commissions	Open access; renewable purchase obligations

Regional Frameworks in Africa

Table J.4 Regional frameworks relevant to energy and resources in Africa.

Framework	Members	Relevance
ECOWAS Energy Protocol and ECOWAS Regional Electricity Regulatory Authority (ERERA)	ECOWAS member states	Regional investment protection and electricity market regulation
West African Power Pool (WAPP)	Utilities of ECOWAS states	Cross-border interconnection and power trade
Southern African Power Pool (SAPP)	SADC utilities	Competitive regional electricity market
African Continental Free Trade Area	AU member states that have ratified	Trade in goods and services; investment protocol
Africa Mining Vision (2009)	AU	Resource-based industrialisation and governance
African Commission on Human and Peoples' Rights; African Court	AU member states	Rights relating to resources and environment

APPENDIX K

Energy Project Finance and Fiscal Evaluation Templates

This appendix provides templates and formulas for the financial and fiscal evaluation of energy and resource projects, to support the legal analysis in Chapter 16. Lawyers need not be financial modellers, but they must understand the metrics used by lenders, sponsors, and governments, because these metrics drive negotiation, are embedded in contractual covenants, and are central to quantum in disputes. **All numerical examples are hypothetical.**

Core Investment Metrics

Net present value (NPV). The sum of a project's discounted net cash flows:

$$NPV = \Sigma\,(CF_t \div (1 + r)^t), \text{ for } t = 0 \text{ to } n$$

where CF_t is the net cash flow in year t and r is the discount rate. A positive NPV at the investor's cost of capital indicates value creation.

Internal rate of return (IRR). The discount rate at which NPV equals zero. Sponsors compare project or equity IRR against hurdle rates reflecting their cost of capital and country risk.

Payback period. The time required for cumulative cash flows to become positive.

Levelised cost of electricity (LCOE). The constant real price per unit of electricity at which the present value of revenues equals the present value of costs over the project's life:

$$LCOE = \Sigma\,((I_t + O\&M_t + F_t) \div (1 + r)^t) \div \Sigma\,(E_t \div (1 + r)^t)$$

where I is investment, O&M is operating and maintenance cost, F is fuel cost, and E is electricity generated.

Debt Metrics Used in Loan Covenants

Debt service coverage ratio (DSCR). Cash flow available for debt service in a period divided by debt service (principal and interest) due in that period. Lenders typically require minimum DSCRs (for example, 1.20–1.40× for contracted power projects, higher for merchant or resource-exposed projects); a DSCR below a "lock-up" level may block distributions to equity, and below a lower level may constitute an event of default.

Loan life coverage ratio (LLCR). The NPV of cash flow available for debt service over the remaining loan term divided by outstanding debt.

Reserve tail. In upstream reserve-based lending, the proportion of reserves remaining after the scheduled repayment of debt.

Template: Simplified Project Cash Flow (HYPOTHETICAL)

Table K.1 Template for a simplified project cash flow.

Year	0	1	2	3	…	20
Revenue (output × tariff)	—					
Less operating costs	—					
EBITDA	—					
Less tax	—					
Less capital expenditure	(I_0)					
Less change in working capital						
Cash flow available for debt service (CFADS)						
Less interest and principal						
Cash flow to equity						
DSCR (CFADS ÷ debt service)						

Worked Example: Solar IPP (HYPOTHETICAL figures)

A 50 MW solar plant costs USD 40 million, financed 70 per cent debt and 30 per cent equity. Annual output is 90 GWh, sold at USD 0.07/kWh (revenue USD 6.3 million). Operating costs are USD 0.8 million per year. Annual debt service on an 18-year loan is USD 2.9 million. Ignoring tax for simplicity, CFADS is USD 5.5 million and DSCR is 5.5 ÷ 2.9 ≈ 1.90×. If curtailment reduced output by 15 per cent without deemed energy compensation, revenue would fall to USD 5.355 million, CFADS to USD 4.555 million, and DSCR to about 1.57×. The example illustrates why lenders insist on deemed energy and curtailment protections (Appendix E, Clause 6).

Fiscal Evaluation: Government Take

Government take is the share of a project's pre-tax net cash flow (revenue less exploration, development, and operating costs) received by the government through all fiscal instruments — royalties, taxes, production shares, bonuses, state participation (net of the state's contributions), and other levies:

> Government take (%) = Government's undiscounted (or discounted) net cash flow ÷ Project's pre-tax net cash flow × 100

Effective royalty rate. The minimum share of gross revenue the government receives in any period — revealing the degree to which the regime front-loads government revenue.

Progressivity. Whether government take rises with project profitability (for instance, as prices rise or costs fall). Instruments such as resource rent taxes, R-factor PSC splits, and windfall taxes increase progressivity; royalties and bonuses are regressive with respect to profitability.

Template: Fiscal Regime Comparison

Table K.2 Template for comparing petroleum fiscal regimes.

Indicator	Regime A	Regime B	Regime C
Royalty rate and base			
Corporate / petroleum income tax rate			
Resource rent tax or profit oil split			
Cost recovery limit and uplift			
Ring-fencing			
State participation (carried / paid)			
Bonuses and rentals			
Stability provisions			
Government take at USD 50, 70, 90 per barrel (undiscounted)			
Contractor IRR at USD 50, 70, 90 per barrel			
Effective royalty rate			
Progressivity (yes/no; how)			

Nigeria: Key Fiscal Features under the Petroleum Industry Act 2021

The PIA introduced a fiscal framework comprising royalties (with rates varying by terrain and production levels, and price-based royalties), hydrocarbon tax

(applicable to crude oil from onshore and shallow water areas), and companies income tax, together with cost price ratio limits on deductible costs, gas-related incentives, and contributions to the Host Communities Development Trusts (3 per cent of the previous year's actual annual operating expenditure) and the Frontier Exploration Fund (30 per cent of NNPC Ltd's profit oil and profit gas). Readers should consult the Act, subsequent fiscal and tax legislation (including the tax reform statutes of 2025), and NUPRC regulations for current rates and conditions.

Quantum in Disputes: Valuation Methods

Table K.3 Valuation methods in energy disputes.

Method	Description	Typical use
Discounted cash flow (income approach)	Present value of projected future cash flows	Going concerns with record of profitability or reliable contracted revenues
Market approach (comparable transactions or trading multiples)	Value by reference to prices paid for similar assets	Where reliable comparables exist
Asset-based / cost approach (sunk costs)	Amounts invested, sometimes with interest	Early-stage projects with speculative future profits
Liquidation value	Value on sale of assets	Non-going concerns

Tribunals have often been reluctant to award DCF valuations for projects without a track record of profitability, preferring sunk costs, as in several ICSID cases involving early-stage projects. Interest, the valuation date (date of the measure versus date of the award, particularly in unlawful expropriation following the principle of full reparation in the *Factory at Chorzów* case), and the treatment of country risk in discount rates are frequently decisive issues.

APPENDIX L

Corporate Human Rights Due Diligence Checklist

This checklist supports the design and review of human rights due diligence (HRDD) by companies in the energy and extractive sectors. It is structured around the UN Guiding Principles on Business and Human Rights (2011), the OECD Guidelines for Multinational Enterprises on Responsible Business Conduct (2023 update) and the OECD Due Diligence Guidance for Responsible Business Conduct (2018), and the mandatory due diligence regimes examined in Chapters 21 and 22, including the EU Corporate Sustainability Due Diligence Directive (2024/1760) — whose scope and timetable were subsequently revised through the EU's 2025 "Omnibus" simplification process, so that readers should verify the current requirements — the French Duty of Vigilance Law (2017), and the German Supply Chain Due Diligence Act (2021).

Step 1 — Policy Commitment and Governance (UNGP 16)

- ☐ A human rights policy approved at the most senior level, publicly available, and communicated internally and to business partners.
- ☐ Commitment references the International Bill of Human Rights, the ILO Declaration on Fundamental Principles and Rights at Work, and, where relevant, UNDRIP and the Voluntary Principles on Security and Human Rights.
- ☐ Board oversight and senior management responsibility assigned; human rights integrated into risk management, incentives, and procurement.
- ☐ Resources and expertise allocated.

Step 2 — Identify and Assess Actual and Potential Adverse Impacts (UNGP 18)

- ☐ Mapping of operations, joint ventures, and value chain (suppliers, contractors, security providers, customers).
- ☐ Assessment of salient human rights risks in each context, prioritised by severity (scale, scope, irremediability) and likelihood.

- ☐ Meaningful consultation with potentially affected groups and their representatives, including women, indigenous peoples, workers, and human rights defenders.
- ☐ Specific risks in energy and extractives assessed: land acquisition and resettlement; FPIC; security forces and use of force; water and pollution; health and safety; labour rights (including in contractors and the supply chain, and forced and child labour risks in mineral supply chains); gender-based violence; conflict and fragility; corruption; climate impacts; and impacts on human rights defenders.
- ☐ Heightened due diligence in conflict-affected and high-risk areas.

Step 3 — Integrate and Act to Cease, Prevent, and Mitigate (UNGP 19)

- ☐ Findings integrated into decision-making (investment, project design, procurement, contracting).
- ☐ Actions to cease impacts the company causes, prevent and mitigate those to which it contributes, and use leverage regarding those directly linked to it through business relationships.
- ☐ Contractual clauses with suppliers and partners, with verification and support (not merely transfer of responsibility).
- ☐ Responsible disengagement considered where leverage fails, taking account of the impacts of withdrawal.

Step 4 — Track Effectiveness (UNGP 20)

- ☐ Qualitative and quantitative indicators.
- ☐ Feedback from affected stakeholders.
- ☐ Internal and independent audits; review of grievance data.

Step 5 — Communicate (UNGP 21)

- ☐ Public reporting on how impacts are addressed, sufficient to allow evaluation (for example, under CSRD/ESRS, where applicable, or the UNGP Reporting Framework).
- ☐ Communication with affected stakeholders in accessible forms.
- ☐ Protection of confidentiality and safety of informants.

Step 6 — Remediation and Grievance Mechanisms (UNGPs 22, 29, 31)

- ☐ Process to provide for or cooperate in remediation of impacts caused or contributed to.

- ☐ Operational-level grievance mechanism meeting the effectiveness criteria of UNGP 31: legitimate, accessible, predictable, equitable, transparent, rights-compatible, a source of continuous learning, and based on engagement and dialogue.
- ☐ Non-retaliation and protection of complainants.
- ☐ Mechanism does not preclude access to judicial or other non-judicial remedies (including OECD National Contact Points).

Step 7 — Climate and Environment

- ☐ Assessment of environmental impacts that affect human rights (water, health, livelihoods).
- ☐ Climate transition plan, where required (for example, under the CSDDD as adopted, or applicable disclosure rules), consistent with the 1.5°C goal.

Step 8 — Legal Risk Review

- ☐ Exposure to parent company liability in home-state courts (see *Vedanta Resources plc v Lungowe* [2019] UKSC 20; *Okpabi v Royal Dutch Shell plc* [2021] UKSC 3; *Four Nigerian Farmers and Milieudefensie v Shell* (Court of Appeal of The Hague, January 2021)).
- ☐ Compliance with mandatory due diligence and reporting laws applicable to the company.
- ☐ Import restrictions on goods made with forced labour (for example, the EU Forced Labour Regulation (2024/3015) and United States customs measures).
- ☐ Anti-corruption and sanctions compliance in connection with security and community payments.

Red Flags Requiring Escalation

1. Allegations of killings, torture, or arbitrary detention linked to security forces protecting operations.
2. Forced eviction without due process or adequate compensation.
3. Absence of consent from indigenous peoples where required.
4. Child or forced labour in operations or the supply chain.
5. Serious pollution affecting drinking water or health.
6. Threats or attacks against human rights defenders who oppose the project.
7. Operations in areas of armed conflict without heightened due diligence.

APPENDIX M

Climate and Energy Transition Legal Compliance Framework

This framework assists organisations — energy companies, financial institutions, public authorities, and their advisers — to map and manage legal obligations and risks arising from climate change and the energy transition. It draws on Chapters 18, 27, and 30. It is general in nature; applicable obligations depend on jurisdiction, sector, size, and listing status, and many regimes have been revised recently, so that current texts must always be checked.

Module 1 — Mapping Applicable Obligations

Table M.1 Mapping climate and transition legal obligations.

Category	Examples	Applies?	Owner
National climate framework law	Nigeria Climate Change Act 2021 (including duties on public and private entities with 50 or more employees to implement measures to reduce emissions, and the National Council on Climate Change); UK Climate Change Act 2008; EU Climate Law (Regulation (EU) 2021/1119)		
Carbon pricing	EU ETS; UK ETS; national carbon taxes; Article 6 Paris Agreement transactions		
Carbon border measures	EU CBAM (Regulation (EU) 2023/956), definitive period from 2026		
Sectoral emissions rules	EU Methane Regulation (2024/1787); flaring and venting rules (including Nigerian gas flare regulations under the PIA framework)		
Licensing and planning	Climate assessment in EIA (*Finch*); climate tests for new licences		
Disclosure	ISSB IFRS S1 and S2 as adopted in national law; EU CSRD and ESRS (as revised by the Omnibus process); UK requirements; stock exchange rules		
Due diligence	EU CSDDD (as revised); national due diligence laws		

Category	Examples	Applies?	Owner
Financial regulation	Prudential expectations of central banks and supervisors; EU Taxonomy and SFDR; green bond standards		
Consumer and competition law	Rules on green claims and greenwashing; competition law treatment of sustainability agreements		
Human rights and torts	Climate-related human rights obligations (*KlimaSeniorinnen*, ECtHR 2024); tort claims against major emitters		
International obligations of the state	Paris Agreement NDCs; ICJ Advisory Opinion (2025)		

Module 2 — Governance

- Board-level responsibility for climate strategy and risk; competence and training of directors.
- Integration of climate considerations into directors' duties, having regard to legal opinions and guidance in relevant jurisdictions and to litigation such as *ClientEarth v Shell plc* [2023] EWHC 1897 (Ch), in which permission to bring a derivative claim against directors was refused.
- Executive remuneration linked to climate targets, where appropriate.
- Clear accountability for compliance with each obligation identified in Module 1.

Module 3 — Risk Assessment

- Physical risk assessment (acute and chronic) for assets and supply chains.
- Transition risk assessment: policy, legal, technology, market, and reputational.
- Scenario analysis, including scenarios consistent with 1.5°C, using recognised scenarios (such as the IEA's and the NGFS's).
- Assessment of stranded asset risk and impairment, and of decommissioning and closure liabilities.
- Litigation risk assessment: exposure to claims for misleading disclosure, greenwashing, failure to assess or disclose, breach of duties, and contribution to climate harm.

Module 4 — Targets and Transition Plan

- Emissions inventory (scopes 1, 2, and, where material, 3) using the GHG Protocol.
- Science-based targets (interim and long-term), with clarity on reliance on offsets and carbon removals.
- A credible transition plan setting out actions, capital expenditure, milestones, and governance, consistent with applicable frameworks (such as the ESRS E1 or national guidance).
- Just transition considerations for workers and communities.

Module 5 — Disclosure and Claims

- Disclosures consistent with the applicable standard and with financial statements.
- Substantiation of all environmental claims (for example, "net zero", "carbon neutral", "green") with robust, verifiable evidence; avoidance of misleading omissions.
- Assurance of sustainability information where required.
- Review by legal counsel of all public climate statements.

Module 6 — Contracts and Transactions

- Climate and transition clauses in supply, offtake, and financing agreements (emissions data, carbon cost allocation, change in law, methane verification, termination rights).
- Due diligence in mergers and acquisitions on climate liabilities and transition risk.
- Treatment of carbon credits (quality, integrity, corresponding adjustments).

Module 7 — Monitoring, Review, and Reporting

- Annual review of the obligations map and risk assessment.
- Internal audit of climate data and controls.
- Reporting to the board and to regulators.

Self-Assessment Scorecard

Table M.2 Climate and transition compliance self-assessment scorecard.

Module	0 — Not started	1 — Developing	2 — Established	3 — Leading
Obligations mapping				
Governance				

Module	0 — Not started	1 — Developing	2 — Established	3 — Leading
Risk assessment				
Targets and transition plan				
Disclosure and claims				
Contracts and transactions				
Monitoring and review				

APPENDIX N

Doctoral Research Proposal Template

This template supports the preparation of a doctoral research proposal in energy and natural resources law, complementing Chapter 28. Requirements vary between institutions — in length (often 3,000 to 8,000 words), structure, and emphasis — and candidates must follow the rules of their own institution and the advice of their prospective supervisors.

1. Title

A concise, informative working title that identifies the subject, scope, and (where appropriate) the approach — for example, "Managing Decline: A Comparative Legal Analysis of Petroleum Licensing Restrictions in Producer States, with Particular Reference to Nigeria". Avoid titles that are too broad ("Energy Law in Africa") or rhetorical without content.

2. Abstract (200–300 words)

State the problem, the research question(s), the method, the expected contribution, and the significance of the research.

3. Introduction and Background

Introduce the subject, explaining the legal, policy, and practical context and why the topic matters now. Identify the "puzzle" or "problem" that the research addresses — a gap, a contradiction, an unresolved question, or a practical difficulty that existing law or scholarship does not adequately address.

4. Research Questions

State one main research question and two to five subsidiary questions that, together, will answer it. Questions should be clear, focused, answerable within the time and resources available, and open (not answerable simply "yes" or "no"). For example:

- *Main question:* To what extent do the legal frameworks of selected producer states enable the managed decline of petroleum production consistently with their international climate obligations and the right to development?

- *Subsidiary questions:* (1) What do international climate law and the ICJ's 2025 advisory opinion require of producer states with respect to new licensing? (2) What legal instruments have producer states used to restrict or phase out licensing? (3) How do investment protections constrain such measures? (4) What lessons arise for Nigeria?

5. Literature Review

Critically review the principal scholarship, demonstrating command of the field and identifying the gap the research will fill. Organise thematically, not as a list of summaries. Show how the proposed research engages with and advances existing debates.

6. Theoretical or Conceptual Framework

Identify the theoretical perspective(s) that will inform the research — for example, legal positivism and doctrinal analysis, theories of resource sovereignty, law and development, energy justice, transnational legal ordering, Third World Approaches to International Law, or regulatory theory — and explain why they are appropriate.

7. Methodology

Explain and justify the methods: doctrinal analysis of primary sources; comparative method (with criteria for selection of jurisdictions and attention to functional equivalence); empirical methods (interviews, surveys, case file analysis), with consideration of access, sampling, and analysis; historical method; law and economics; or mixed methods. Identify the sources to be used and how they will be accessed. Address limitations.

8. Ethical Considerations

Where research involves human participants (interviews, surveys, fieldwork in affected communities), explain how informed consent, confidentiality, data protection (for example, under the Nigeria Data Protection Act 2023 or the GDPR), risk to participants and researcher, and institutional ethics approval will be addressed.

9. Originality and Contribution

State clearly the original contribution to knowledge — new knowledge, new interpretation, new synthesis, new method, or application to a new context — and its significance for scholarship, policy, and practice.

10. Structure of the Thesis (Provisional)

Provide a provisional chapter outline showing how the research questions will be addressed.

11. Timetable

Table N.1 Indicative timetable for a three-year full-time doctorate.

Period	Activities	Milestones
Year 1 (months 1–6)	Literature review; refinement of questions; methods training	Confirmation of research plan
Year 1 (months 7–12)	Doctrinal research; ethics approval; pilot fieldwork	Upgrade/progression review
Year 2	Main research and fieldwork; draft chapters	Draft core chapters
Year 3 (months 1–8)	Analysis; complete drafts	Full draft
Year 3 (months 9–12)	Revision; submission	Submission; preparation for viva

12. Resources and Feasibility

Identify resources needed (library and database access, travel, fieldwork costs, language skills) and how they will be secured.

13. Preliminary Bibliography

List key primary and secondary sources in a consistent citation style (for example, OSCOLA).

Self-Check Before Submission

1. Is the research question clear, focused, and significant?
2. Is the gap in existing knowledge convincingly identified?
3. Are the methods appropriate and justified?
4. Is the project feasible in the time available?
5. Is the expected original contribution clearly stated?
6. Are ethical issues identified and addressed?
7. Is the proposal well written, accurately referenced, and within the word limit?
8. Have all sources been verified, and has any use of generative AI tools complied with institutional policy (see Chapter 28)?

APPENDIX O

Doctoral Dissertation Structure and Evaluation Criteria

This appendix sets out a typical structure for a doctoral dissertation (thesis) in energy and natural resources law and the criteria commonly applied by examiners. It complements Chapter 28 and Appendix N. Institutional regulations — on length (commonly 80,000 to 100,000 words in law, inclusive or exclusive of footnotes and bibliography), format, and examination procedure — prevail and must be consulted.

Typical Structure

Preliminary pages. Title page; declaration of originality; abstract (300–500 words); acknowledgements; table of contents; tables of cases, legislation, and treaties; list of abbreviations; and, where applicable, lists of figures and tables.

Chapter 1 — Introduction. The problem and its significance; the research questions; the argument (thesis) in summary; the scope and limitations; the methodology and theoretical framework (or these may form a separate chapter); the original contribution; and a roadmap of the chapters.

Chapter 2 — Literature, theory, and method (optional separate chapter). A critical review of the scholarship, the conceptual framework, and a fuller account of methods, including any empirical work and ethics.

Substantive chapters (typically three to five). Each addresses one or more subsidiary research questions, builds the argument, and ends with a conclusion that links back to the main question. Comparative theses may devote chapters to individual jurisdictions or, preferably, to themes analysed comparatively.

Concluding chapter. Answers to the research questions; synthesis of the argument; contribution to knowledge; implications for law, policy, and practice (including recommendations for reform, where appropriate); limitations; and avenues for further research.

Bibliography and appendices. Full bibliography of primary and secondary sources; appendices with interview schedules, consent forms, datasets, or other supporting material.

Writing the Argument

A doctoral thesis is an extended argument, not a report. Each chapter should advance the thesis, and the reader should be able to trace the argument from introduction to conclusion. Signposting at the start and end of chapters and sections, consistent terminology, and careful structuring are essential. The writer should engage critically with opposing views, acknowledge limitations, and avoid both overclaiming and excessive description.

Common Evaluation Criteria

Table O.1 Common criteria for the evaluation of a doctoral thesis.

Criterion	What examiners look for
Originality and contribution	A significant, original contribution to knowledge, worthy of publication in whole or in part
Research question and argument	A clear, significant question and a coherent, sustained, persuasive argument
Knowledge of the field	Comprehensive, critical command of the relevant law and literature
Methodology	Appropriate, justified, rigorously applied methods; awareness of limitations
Use of sources	Accurate, authoritative, and current primary and secondary sources; correct interpretation
Critical analysis	Depth of analysis; evaluation, not mere description; engagement with counter-arguments
Structure and presentation	Logical organisation; clear, precise writing; accurate citation and referencing; compliance with regulations
Integrity	Independence of the work; absence of plagiarism; transparent and ethical research practice; compliance with institutional policies on generative AI
Defence (viva voce)	Ability to explain, justify, and defend the thesis and to situate it in the field

Typical Examination Outcomes

Outcomes vary by institution but commonly include: (a) award of the degree without amendments; (b) award subject to minor corrections (often within three to six months); (c) award subject to major corrections or revisions (often within twelve months), sometimes with a further oral examination; (d) resubmission for re-examination; (e) award of a lower degree (such as an MPhil); and (f) failure.

Preparing for the Viva Voce

1. Re-read the thesis carefully, noting errors and points of weakness.

2. Prepare a short (five-minute) summary of the thesis, its contribution, and its significance.
3. Anticipate questions: Why this topic? What is original? Why these methods and jurisdictions? What are the limitations? How does the work relate to [leading scholar]'s argument? What would you do differently? What are the implications for policy? What next?
4. Check developments since submission (new cases, legislation, or scholarship) and consider their implications.
5. Hold a mock viva with supervisors or colleagues.

APPENDIX P

Legal Research, Citation, and Case Analysis Guide

This guide summarises techniques for legal research, verification, citation, and case analysis in energy and natural resources law. It complements Chapter 28.

Research Strategy

1. **Define the question** precisely, identifying the jurisdiction(s), the time period, and the legal issues.
2. **Begin with secondary sources** — textbooks, treatises, encyclopaedias, and review articles — to understand the framework and identify key authorities.
3. **Locate primary sources**: constitutions, treaties, legislation, regulations, cases, and awards.
4. **Update and verify** every authority: check amendments, repeals, appeals, annulment and set-aside proceedings, and subsequent treatment.
5. **Record** sources and search methods systematically, using reference management software.

Key Sources for Energy and Natural Resources Law

Table P.1 Key sources for research in energy and natural resources law.

Type	Sources
Treaties	United Nations Treaty Collection; depositary websites (IAEA, IMO, Energy Charter Secretariat)
International courts	ICJ, ITLOS, PCA, ECtHR (HUDOC), IACtHR, African Court, and African Commission websites
Investment arbitration	ICSID case database; italaw; UNCTAD Investment Policy Hub (treaties and ISDS navigator)
Trade	WTO dispute settlement database and documents online
Nigerian law	Official Gazette; Laws of the Federation of Nigeria; National Assembly; law reports (Nigerian Weekly Law Reports, Supreme Court reports); regulators' websites (NUPRC, NMDPRA, NERC)
English law	legislation.gov.uk; National Archives Find Case Law; BAILII; law reports
EU law	EUR-Lex; CURIA

Type	Sources
Contracts	ResourceContracts.org (published extractive industry contracts)
Trans-parency data	EITI and national EITI reports (including NEITI)
Data and policy	IEA, IRENA, World Bank, IMF, and UNEP reports; OPEC Annual Statistical Bulletin
Climate litigation	Sabin Center climate litigation databases; Grantham Research Institute climate laws database

Verification Protocol

Before relying on any authority:

1. Obtain the authoritative text from an official or reputable source — never rely solely on a secondary description, a summary, or an AI-generated output.
2. Confirm the citation details (name, year, court, report, paragraph).
3. Confirm the proposition attributed to the authority by reading the relevant passage in context.
4. Check subsequent history (appeal, annulment, set aside, reversal, overruling, distinguishing) and amendments to legislation.
5. Record the date of verification.

The judgments in *Mata v Avianca, Inc* (SDNY 2023) and *R (Ayinde) v London Borough of Haringey* [2025] EWHC 1383 (Admin) illustrate the serious professional consequences of citing non-existent authorities generated by AI tools.

Citation: OSCOLA Examples

The Oxford University Standard for the Citation of Legal Authorities (OSCOLA) is widely used in English-language legal scholarship. Illustrative formats:

- **UK case:** *Okpabi v Royal Dutch Shell plc* [2021] UKSC 3.
- **ICJ judgment:** *Pulp Mills on the River Uruguay (Argentina v Uruguay)* (Judgment) [2010] ICJ Rep 14.
- **ICJ advisory opinion:** *Obligations of States in respect of Climate Change* (Advisory Opinion) ICJ, 23 July 2025.
- **Investment award:** *Yukos Universal Ltd (Isle of Man) v Russian Federation*, PCA Case No AA 227, Final Award (18 July 2014).
- **Treaty:** Paris Agreement (adopted 12 December 2015, entered into force 4 November 2016) (with the United Nations Treaty Series reference, verified from the UN Treaty Collection).
- **UK statute:** Climate Change Act 2008, s 1.
- **Nigerian statute:** Petroleum Industry Act 2021, s [x].

- **EU regulation:** Regulation (EU) 2023/956 of the European Parliament and of the Council of 10 May 2023 establishing a carbon border adjustment mechanism [2023] OJ L130/52.
- **Book:** Author, *Title* (edn, Publisher Year) page.
- **Journal article:** Author, 'Title' (Year) Volume Journal Abbreviation First page, pinpoint.

Footnotes should be used for citations; a bibliography lists secondary sources (and, in a thesis, may separately list primary sources). Use pinpoint references wherever possible. Institutions may prescribe other styles (such as the Bluebook or Chicago); consistency is essential.

Case Analysis Template

1. **Citation and court/tribunal.** Full citation; level of court; composition.
2. **Facts.** Material facts only.
3. **Procedural history.** How the case reached this forum.
4. **Issues.** The legal questions to be decided.
5. **Arguments.** The principal arguments of each party.
6. **Holding.** The decision on each issue.
7. **Reasoning.** The court's reasoning, including the authorities relied on and the interpretive approach.
8. **Separate and dissenting opinions.** Their reasoning and significance.
9. **Ratio and obiter.** Distinguish binding reasoning from observations (noting that arbitral awards and international decisions do not create binding precedent in the common law sense, although they may be highly persuasive).
10. **Subsequent history and treatment.** Appeals, annulment, set aside, citation in later cases.
11. **Evaluation.** Is the reasoning sound? Is the decision consistent with other authorities? What are its implications for law, policy, and practice — particularly in energy and resources?

Treaty Interpretation Template (VCLT arts 31–33)

1. Ordinary meaning of the terms, in their context, and in the light of the object and purpose of the treaty.
2. Context: the text, preamble, annexes, and related agreements and instruments.
3. Subsequent agreements and practice of the parties.
4. Relevant rules of international law applicable between the parties (systemic integration, art 31(3)(c)).
5. Special meaning, if established.

6. Supplementary means (art 32): *travaux préparatoires* and circumstances of conclusion — to confirm the meaning or where the meaning is ambiguous, obscure, or leads to a manifestly absurd or unreasonable result.
7. Treaties authenticated in more than one language (art 33).

APPENDIX Q

Selected International and Regional Judicial Decisions

This appendix summarises selected decisions of international and regional courts, tribunals, and quasi-judicial bodies, and leading national decisions with international significance, that are central to energy and natural resources law. Summaries are necessarily brief; readers should consult the full texts (Appendix P) and the chapters indicated.

International Court of Justice and Law of the Sea Tribunals

Table Q.1 Selected decisions of the ICJ and law of the sea tribunals.

Decision	Year	Key points	Chapter
Anglo-Iranian Oil Co (United Kingdom v Iran) (Preliminary Objection)	1952	ICJ lacked jurisdiction over dispute arising from nationalisation of oil concession; concession not a treaty	1, 3
North Sea Continental Shelf (Germany/Denmark; Germany/Netherlands)	1969	Continental shelf as natural prolongation; equidistance not customary; formation of custom	2, 20
Barcelona Traction (Belgium v Spain)	1970	Diplomatic protection of shareholders; obligations *erga omnes*	14
Gabčíkovo-Nagymaros Project (Hungary/Slovakia)	1997	Treaty law and state of necessity; sustainable development; joint regime for hydropower	2, 17, 19
Land and Maritime Boundary between Cameroon and Nigeria	2002	Bakassi Peninsula; maritime boundary in a hydrocarbon-rich area	3, 20, 24
Pulp Mills on the River Uruguay (Argentina v Uruguay)	2010	Customary duty to conduct EIA; procedural and substantive obligations; due diligence	17, 19
Certain Activities / Construction of a Road (Costa Rica v Nicaragua)	2015; 2018	EIA in transboundary context; first ICJ award of compensation for environmental damage (2018)	17
Armed Activities on the Territory of the Congo (DRC v Uganda) (Reparations)	2022	Reparation including for exploitation of natural resources (total of USD 325 million)	3, 25

Decision	Year	Key points	Chapter
Obligations of States in respect of Climate Change (Advisory Opinion)	2025	Climate obligations under treaties and custom; stringent due diligence; fossil fuel licensing and subsidies may constitute wrongful acts	18, 27, 30
ITLOS, *Responsibilities and Obligations of States with respect to Activities in the Area* (Seabed Disputes Chamber)	2011	Sponsoring state obligations of due diligence; precautionary approach; EIA	20
ITLOS Special Chamber, *Ghana/Côte d'Ivoire*	2017	Maritime delimitation; no tacit agreement; Ghana's oil activities did not breach Côte d'Ivoire's rights	20
ICJ, *Maritime Delimitation in the Indian Ocean (Somalia v Kenya)*	2021	Delimitation of maritime boundary in area of hydrocarbon interest	20
ITLOS, *Climate Change and International Law* (Advisory Opinion)	2024	GHG emissions as marine pollution under UNCLOS; stringent due diligence	18, 20

Investment and State Contract Arbitration

Table Q.2 Selected investment and state contract arbitration decisions.

Decision	Year	Key points	Chapter
Texaco Overseas Petroleum Co v Libya	1977	Internationalisation of state contracts; restitution; status of UNGA resolutions	1, 14
Kuwait v Aminoil	1982	Nationalisation lawful; stabilisation clause did not prohibit nationalisation; compensation based on legitimate expectations	14, 15
Methanex v United States (NAFTA)	2005	Non-discriminatory regulation for a public purpose not expropriation absent specific commitments	14
Occidental v Ecuador (ICSID)	2012 (annulment 2015)	Termination of participation contract disproportionate; damages reduced on partial annulment	14
Yukos Universal v Russia (PCA)	2014	Largest award (about USD 50 billion); ECT provisional application; awards set aside in 2016 and reinstated in 2020 by Dutch courts	14, 26
Chevron and TexPet v Ecuador (PCA)	2018	Lago Agrio judgment procured by fraud; denial of justice	26, 27
ConocoPhillips v Venezuela (ICSID)	2019	Unlawful expropriation of heavy oil projects; very large award	14
Eiser v Spain and related renewable energy cases (ECT)	2017 onwards	Radical changes to renewable support schemes breached FET; *Eiser* annulled in 2020 for an arbitrator's undisclosed relationships	10, 14

Decision	Year	Key points	Chapter
CJEU, *Slovak Republic v Achmea* (C-284/16)	2018	Intra-EU BIT arbitration clauses incompatible with EU law	14, 26
CJEU, *Republic of Moldova v Komstroy* (C-741/19)	2021	ECT arbitration incompatible with EU law in intra-EU disputes	14, 26
Rockhopper v Italy (ECT, ICSID)	2022 (annulled 2025)	Offshore drilling ban: award for investor later annulled	14, 20
TC Energy v United States (USMCA/NAFTA legacy)	2024	Keystone XL claim dismissed on jurisdiction	14, 25

Regional Human Rights Bodies

Table Q.3 Selected decisions of regional human rights bodies.

Decision	Year	Key points	Chapter
African Commission, *SERAC and CESR v Nigeria* (Communication 155/96)	2001	Violation of rights to health, environment, food, housing, and free disposal of resources in Ogoniland	21, 23
African Commission, *Endorois v Kenya* (Communication 276/2003)	2010	Indigenous land rights; benefit-sharing; consultation and consent	21
African Court, *African Commission v Kenya* (Ogiek)	2017; reparations 2022	Indigenous peoples' rights to land and resources	21
IACtHR, *Saramaka People v Suriname*	2007	Consultation, benefit-sharing, and prior impact assessment for concessions on indigenous lands	21
IACtHR, *Kichwa Indigenous People of Sarayaku v Ecuador*	2012	Oil exploration without prior consultation violated rights	21
IACtHR, Advisory Opinion OC-23/17	2017	Right to a healthy environment; extraterritorial obligations	17, 21
ECtHR, *Verein KlimaSeniorinnen Schweiz v Switzerland*	2024	Inadequate climate measures violated article 8 ECHR	18, 27
ECtHR, *Greenpeace Nordic and Others v Norway*	2025	Article 8 procedural requirements for assessment of petroleum licensing; no violation found on the facts	18, 27
IACtHR, Advisory Opinion on the Climate Emergency (OC-32/25)	2025	State obligations regarding climate emergency under the American Convention	18, 30

National Decisions of International Significance

Table Q.4 Selected national decisions of international significance.

Decision	Year	Key points	Chapter
Nigeria, *Gbemre v Shell Petroleum Development Co* (Federal High Court, Benin)	2005	Gas flaring violated constitutional rights to life and dignity	21, 27
Netherlands, *Urgenda Foundation v State of the Netherlands* (Supreme Court)	2019	State's duty of care under ECHR arts 2 and 8 to reduce emissions	18, 27
UK, *Vedanta Resources plc v Lungowe* [2019] UKSC 20	2019	Parent company duty of care; jurisdiction over claims by Zambian villagers	21, 27
Canada, *Nevsun Resources Ltd v Araya* 2020 SCC 5	2020	Claims based on customary international law may proceed against corporation	21
UK, *Okpabi v Royal Dutch Shell plc* [2021] UKSC 3	2021	Arguable parent company duty of care for Niger Delta pollution	21, 27
Netherlands, *Four Nigerian Farmers and Milieudefensie v Shell* (Court of Appeal, The Hague)	2021	Shell Nigeria liable for oil spills; parent duty to install leak detection	21, 27
Netherlands, *Milieudefensie v Royal Dutch Shell*	2021; appeal 2024	District Court ordered 45% emissions cut; Court of Appeal (Nov 2024) recognised duty but overturned specific reduction order	18, 27
UK, *R (Finch) v Surrey County Council* [2024] UKSC 20	2024	Downstream combustion emissions must be assessed in EIA for oil extraction	17, 27
Germany, *Lliuya v RWE* (Higher Regional Court, Hamm)	2025	Major emitter may in principle be liable under civil law for climate risks; claim dismissed on facts	27
UK, *Municipio de Mariana v BHP Group* (High Court)	2025	BHP held liable in relation to the Fundão dam collapse in Brazil	27

APPENDIX R

Professional Energy Law Drafting and Advisory Exercises

All exercises in this appendix are hypothetical. The states, companies, communities, and facts are invented for teaching purposes, and any resemblance to real persons, companies, or events is coincidental. Where an exercise refers to real legal instruments, participants must rely on accurate, verified sources.

Exercise R1 — Advice on Stabilisation (Advisory Memorandum)

The (fictitious) Republic of Tavora is negotiating a mining development agreement with Coreline Minerals plc for a lithium project. Coreline seeks a clause freezing the fiscal and regulatory regime for thirty years. The Ministry of Mines asks for advice. Prepare a memorandum (2,000 words) advising on: the types of stabilisation clauses; their enforceability and treatment in arbitral practice; their compatibility with Tavora's human rights and environmental obligations; and a recommended clause, with drafting.

Exercise R2 — Drafting a Host Community Benefit Clause

Draft a clause (with explanatory notes) for inclusion in a petroleum licence in the fictitious Federal Republic of Ondara, providing for a community development trust funded by the licensee, with governance arrangements that ensure representation of women and youth, transparency, audit, and grievance procedures. Compare your clause with the host community provisions of Nigeria's Petroleum Industry Act 2021, identifying any improvements you propose.

Exercise R3 — Force Majeure Notice and Response

The fictitious LNG buyer Eastgate Energy has received a force majeure notice from its seller, Coral Bay LNG, citing the imposition of sanctions affecting a shipping company it had chartered. Draft (a) Eastgate's response reserving its rights and requesting information, and (b) a memorandum analysing whether the notice is valid under a clause in the form set out in Appendix D, Part 10.

Exercise R4 — Regulatory Submission on a Tariff Review

The (fictitious) Sunrise Distribution Company seeks a tariff increase from the state electricity regulator of the fictitious Kanem State, which has assumed regulatory responsibility under national legislation modelled on Nigeria's Electricity Act 2023. Prepare a submission on behalf of a consumer association, addressing cost reflectivity, affordability, service quality, lifeline tariffs, and the legal obligations of the regulator.

Exercise R5 — Due Diligence Report

A (fictitious) private equity fund proposes to acquire a 40 per cent interest in Deltaflow Pipelines Ltd, which operates a crude oil pipeline in a fictitious West African state. Prepare a due diligence questions list and a report outline covering title and licences, regulatory compliance, environmental liabilities (including historical spills), host community relations, human rights, anti-corruption, contracts, disputes, and climate transition risk.

Exercise R6 — Notice of Dispute under an Investment Treaty

The fictitious Solaris Renewables GmbH, a national of a fictitious state party to a bilateral investment treaty with the fictitious Republic of Mirandia, has seen its feed-in tariff retroactively reduced by 40 per cent. Draft the notice of dispute required by the treaty, and a memorandum assessing the strength of claims for breach of fair and equitable treatment and indirect expropriation in light of the jurisprudence on renewable energy support schemes.

Exercise R7 — Legislative Drafting: Decommissioning

Draft the key provisions (no more than 1,500 words) of a decommissioning part of a petroleum bill for the fictitious Republic of Okanta, covering decommissioning plans, financial security, residual liability, the transfer of assets to the state, and re-use of infrastructure for carbon storage.

Exercise R8 — Opinion on a Greenwashing Allegation

The fictitious Northwind Petroleum plc has advertised its LNG as "carbon neutral" on the basis of offset purchases. A consumer group has complained to the advertising regulator. Prepare an opinion on the legal risks and recommendations for Northwind's future communications.

Assessment Criteria for Drafting and Advisory Exercises

Table R.1 Assessment criteria for drafting and advisory exercises.

Criterion	Description
Legal accuracy	Correct identification and application of relevant law, verified from primary sources
Analysis	Identification of issues, risks, and options; balanced evaluation
Practicality	Advice and drafting that are workable, commercially realistic, and responsive to the client's objectives
Drafting quality	Clarity, precision, consistency, and absence of ambiguity
Ethics and integrity	Attention to professional duties, human rights, environmental and anti-corruption considerations
Presentation	Logical structure; concise and professional style

APPENDIX S

Energy Law Terminology and Technical Glossary

This glossary defines legal, commercial, and technical terms frequently used in energy and natural resources law. Definitions are general; the meaning of a term in a particular statute or contract depends on its definition there.

Abandonment. The plugging of wells and removal or making safe of facilities at the end of their life; often used interchangeably with decommissioning.

Acreage. An area offered for or covered by a petroleum licence or contract.

Ancillary services. Services supporting the stable operation of an electricity system, such as frequency response, reserve, voltage control, and black start.

Annual Contract Quantity (ACQ). The quantity of gas a buyer is contracted to take in a year under a gas sales agreement.

Appraisal. Drilling and studies following a discovery to determine its size and commerciality.

Associated gas. Natural gas found in association with crude oil, either dissolved in the oil or as a gas cap.

Barrel (bbl). Standard unit of crude oil volume, equal to 42 US gallons (about 159 litres).

Benefit-sharing. Arrangements by which communities or states share in the benefits of resource development, through revenues, employment, services, or equity.

Bonus. A lump-sum payment to the state on signature of a contract, discovery, or attainment of production levels.

Capacity factor. The ratio of actual electricity output over a period to the maximum possible output at full nameplate capacity.

Carbon capture, utilisation and storage (CCUS). Technologies that capture carbon dioxide from emission sources or the atmosphere and either use it or store it permanently in geological formations.

Carried interest. An interest whose share of costs is paid by other parties (typically until a commercial discovery), to be recovered later from production.

Commercial discovery. A discovery that the contractor (and, where required, the state) considers economically viable to develop.

Concession. A grant by the state of the right to explore for and exploit resources, under which the concessionaire typically acquires title to the resources produced, subject to royalty and tax.

Contract for difference (CfD). An arrangement under which a generator receives (or pays) the difference between a strike price and a market reference price.

Cost oil / cost petroleum. Under a PSC, the share of production allocated to the contractor to recover its costs.

Critical minerals. Minerals essential to economic or national security and subject to supply risk, as defined by the relevant jurisdiction's lists.

Curtailment. Instruction to a generator to reduce output, typically because of network constraints or oversupply.

Decommissioning. The process of ending operations at a facility and removing, making safe, or re-using its infrastructure and restoring the site.

Deemed energy. Energy a generator is deemed to have delivered (and is paid for) when prevented from delivering by curtailment or grid unavailability for which it is not responsible.

Delivery point. The point at which product is delivered and title and risk pass.

Distribution. The transport of electricity or gas over lower-voltage or lower-pressure networks to end users.

Domestic market obligation. A requirement to supply a portion of production to the domestic market, sometimes at regulated prices.

Economic equilibrium clause. A contractual provision requiring adjustment of terms to restore the parties' economic balance after specified changes, such as changes in law.

Energy community. A legal entity, often citizen-controlled, engaged in energy generation, supply, aggregation, storage, or related services for members or the local community.

Energy justice. A framework concerned with fair distribution of the benefits and burdens of energy systems, fair procedures, and recognition of affected groups.

EPC contract. Engineering, procurement and construction contract, typically lump-sum and turnkey.

Expropriation. The taking of property by the state, either directly (by transfer of title) or indirectly (by measures having equivalent effect).

Fair and equitable treatment (FET). An investment treaty standard protecting investors against arbitrary, discriminatory, grossly unfair, or non-transparent treatment, and, in some formulations, frustration of legitimate expectations.

Farm-in / farm-out. A transaction in which a party acquires an interest in a licence (farm-in) from the holder (farm-out), usually in return for paying a share of costs.

Feed-in tariff. A guaranteed price for renewable electricity fed into the grid, typically for a fixed period.

Final investment decision (FID). The decision by project sponsors to proceed with construction.

Flaring. The controlled burning of natural gas, often associated gas, at production sites.

Force majeure. An event beyond a party's reasonable control that prevents performance, excusing the affected party from liability to the extent provided in the contract.

Free, prior and informed consent (FPIC). The principle that indigenous peoples should be able to give or withhold consent to projects affecting their lands, territories, and resources, through processes free from coercion, before authorisation, and with full information.

Gas-to-power. The use of natural gas to generate electricity.

Government take. The share of a project's net cash flow received by government through all fiscal instruments.

Green hydrogen. Hydrogen produced by electrolysis of water using renewable electricity.

Grid code. Technical rules governing connection to and operation of the electricity network.

Guarantee of origin / renewable energy certificate. An instrument certifying that a unit of electricity was produced from a particular source.

Host government agreement (HGA). An agreement between a host state and investors governing a major project, often a cross-border pipeline.

Independent power producer (IPP). A privately owned generator selling electricity to utilities or customers.

Interconnector. A transmission line or pipeline connecting the networks of different jurisdictions or zones.

Joint operating agreement (JOA). An agreement among co-venturers governing the conduct of operations under a licence, including the role of the operator, voting, and default.

Levelised cost of electricity (LCOE). The constant price per unit of electricity at which a project's revenues would equal its costs over its life in present value terms.

Liquefied natural gas (LNG). Natural gas cooled to about −162°C to become liquid for transport by ship.

Local content. Requirements or policies promoting the use of domestic labour, goods, services, and companies in resource projects.

Mini-grid. An isolated or grid-connected small-scale electricity network serving a community.

Nationally determined contribution (NDC). A state's climate action plan under the Paris Agreement.

Net metering / net billing. Arrangements under which consumers who generate electricity are credited for power exported to the grid.

Non-associated gas. Natural gas found in reservoirs without significant crude oil.

Off-taker. The purchaser of a project's output under a long-term contract.

Operator. The party responsible for conducting day-to-day operations on behalf of co-venturers.

Permanent sovereignty over natural resources (PSNR). The principle that peoples and states have sovereignty over the natural resources within their territory.

Power purchase agreement (PPA). A long-term contract for the sale of electricity.

Price review. A contractual mechanism allowing periodic adjustment of the price in a long-term contract.

Production sharing contract (PSC). A contract under which the state retains title to resources and the contractor recovers costs and receives a share of profit from production.

Profit oil / profit petroleum. Production remaining after royalty and cost oil, shared between the state and contractor under a PSC.

Prosumer. A consumer who also produces energy, typically through rooftop solar.

Regulatory sandbox. A framework allowing innovators to test products or business models under relaxed or tailored regulatory requirements, subject to safeguards.

Relinquishment. The surrender of parts of a licence area at specified times.

Reserves. Quantities of petroleum or minerals that are anticipated to be commercially recoverable, classified by certainty (for example, proved, probable, possible) under systems such as the Petroleum Resources Management System.

Ring-fencing. Limitation of the deduction of costs to the project or licence area that generated them.

Royalty. A payment to the resource owner calculated as a share of production or its value.

Service contract. A contract under which a contractor performs exploration or production services for a fee, without acquiring title to production.

Ship-or-pay. A transport contract obligation to pay for reserved capacity whether or not used.

Stabilisation clause. A contractual provision protecting an investor against adverse changes in law, through freezing, economic equilibrium, or allocation of burden.

Stranded asset. An asset that suffers premature write-down or conversion to a liability, for example because of climate policy or market changes.

Take-or-pay. An obligation to pay for a minimum quantity whether or not taken.

Third-party access. The right of third parties to use network infrastructure (pipelines, transmission) owned by another, on regulated or negotiated terms.

Transit. The carriage of energy across the territory of a state that is neither the origin nor the destination.

Transmission. The transport of electricity at high voltage, or gas at high pressure, over long distances.

Unbundling. Separation of network activities (transmission, distribution) from competitive activities (generation, supply), legally, functionally, or by ownership.

Unitisation. Joint development of a reservoir straddling licence or international boundaries as a single unit.

Windfall tax. A tax on unexpectedly high profits, often in response to price spikes.

Work programme. The exploration or development activities a licensee commits to carry out.

APPENDIX T

Further Research Topics and Emerging Scholarly Debates

This appendix supplements the doctoral research agenda in Chapter 30, section 30.8, and the doctoral research questions at the end of each chapter, by setting out a wider range of research topics and the scholarly debates to which they relate. Topics are intended as starting points; any doctoral project must be refined through a literature review, methodological design, and discussion with supervisors (Chapter 28; Appendix N).

Debate 1: Sovereignty, Development, and Climate Obligations

The debate. Do producer states, particularly developing states, retain a right to develop their fossil fuel resources, or do climate obligations require them to leave resources in the ground? How should burdens be shared?

Research topics. (1) The implications of the ICJ's 2025 advisory opinion for petroleum licensing in Africa. (2) The legal status of the "phase-out" and "transition away" language in COP decisions. (3) Compensation for forgone extraction: legal bases and precedents. (4) The proposed Fossil Fuel Non-Proliferation Treaty: legal design and prospects. (5) The right to development and the energy transition.

Debate 2: Investment Law after the Energy Charter Treaty

The debate. Is investment treaty protection an obstacle to climate action, a necessary safeguard for investment in the transition, or both? Can it be reformed?

Research topics. (1) The effect of the sunset clause on withdrawals from the ECT. (2) Climate carve-outs in new-generation investment treaties. (3) Counterclaims by states for environmental harm. (4) The future of ISDS in Africa under the AfCFTA Investment Protocol. (5) Compensation standards for climate-motivated measures.

Debate 3: Critical Minerals and Value Capture

The debate. Can resource-rich developing states capture greater value from transition minerals through export restrictions, local processing, and state participation, consistently with trade and investment law? Will the critical minerals boom repeat the "resource curse"?

Research topics. (1) Export bans and WTO law: lessons from *China — Raw Materials* and *Indonesia — Raw Materials*. (2) Critical minerals agreements and partnerships: legal nature and equity. (3) Mineral producer cooperation (cartels or coordination) and international law. (4) Due diligence regulation in consumer markets and its effects on producer states. (5) Artisanal cobalt mining and formalisation.

Debate 4: Deep Seabed Mining and the Common Heritage

The debate. Should deep seabed mining proceed, and if so under what rules? What does the common heritage of mankind require?

Research topics. (1) The legality of unilateral authorisation of seabed mining outside the ISA framework. (2) The "two-year rule" and the ISA Mining Code negotiations. (3) Benefit-sharing under article 140 UNCLOS. (4) The relationship between the BBNJ Agreement and the ISA. (5) Precaution and moratoria.

Debate 5: Justice and Energy Access

The debate. How should the law address the needs of those without modern energy? Is there a right to energy? How should justice be integrated into transition law?

Research topics. (1) A right to electricity under African human rights law. (2) Legal frameworks for mini-grids and distributed energy in Nigeria's state electricity markets. (3) Clean cooking regulation. (4) Just transition law for coal and oil-dependent regions. (5) Legal design of Just Energy Transition Partnerships.

Debate 6: Corporate Accountability for Transnational Harm

The debate. How far should parent companies and value chain actors be liable for harm caused abroad? What role for mandatory due diligence and home-state litigation?

Research topics. (1) The impact of the EU's Omnibus revisions on the CSDDD. (2) Parent company liability after *Okpabi* and the Dutch Shell cases. (3) Remediation of Niger Delta pollution: legal frameworks and implementation. (4) A binding treaty on business and human rights: progress and prospects. (5) Corporate climate liability after *Lliuya v RWE*.

Debate 7: Nigeria's Energy Reforms

The debate. Will the PIA and the Electricity Act 2023 achieve their objectives of investment, efficiency, accountability, and access? What are the constitutional and federal dimensions?

Research topics. (1) The commercialisation of NNPC Ltd and corporate governance. (2) Implementation of the host community provisions of the PIA. (3) The Frontier Exploration Fund and the future of frontier exploration. (4) State

electricity markets: constitutional allocation, regulatory capacity, and investment. (5) Gas commercialisation and the "decade of gas". (6) Refining policy and the domestic crude supply obligation after the commissioning of the Dangote refinery.

Debate 8: Digital, AI, and Energy

The debate. How should law govern the digitalisation of energy systems and the energy demands of digital technologies?

Research topics. (1) Liability for AI in grid operations. (2) Data centres and grid connection queues: legal responses. (3) Energy data governance and consumer rights. (4) Cyber security regulation of energy infrastructure in Africa. (5) Smart contracts and peer-to-peer energy trading.

Debate 9: Climate Finance and Financial Regulation

The debate. How should financial regulation address climate risk, and how can climate finance flows to developing countries be scaled and governed?

Research topics. (1) The legal architecture of the New Collective Quantified Goal. (2) Climate risk in African financial regulation. (3) Sovereign debt and climate: debt-for-nature swaps and climate-resilient debt clauses. (4) Green taxonomies in emerging markets. (5) Article 6 carbon markets: legal integrity and host country frameworks.

Debate 10: Theory and Method

The debate. What is energy and natural resources law as a field, and how should it be studied?

Research topics. (1) Energy law as a discipline: coherence and boundaries. (2) TWAIL perspectives on resource law. (3) Empirical legal studies of energy regulation in Africa. (4) Legal histories of the oil concession. (5) Interdisciplinary methods combining law with energy systems modelling.

Identifying a Research Gap: Practical Steps

1. Read the most recent review articles and leading monographs on the topic.
2. Search databases (Appendix P) for recent articles, theses, and reports.
3. Identify questions that existing works pose but do not answer, contradictions between authorities, jurisdictions or contexts not studied, and new developments not yet analysed.
4. Discuss emerging ideas with supervisors, practitioners, and peers.
5. Test feasibility: access to sources, methods, time, and resources.
6. Formulate a precise question and articulate the expected contribution (Appendix N, sections 4 and 9).

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